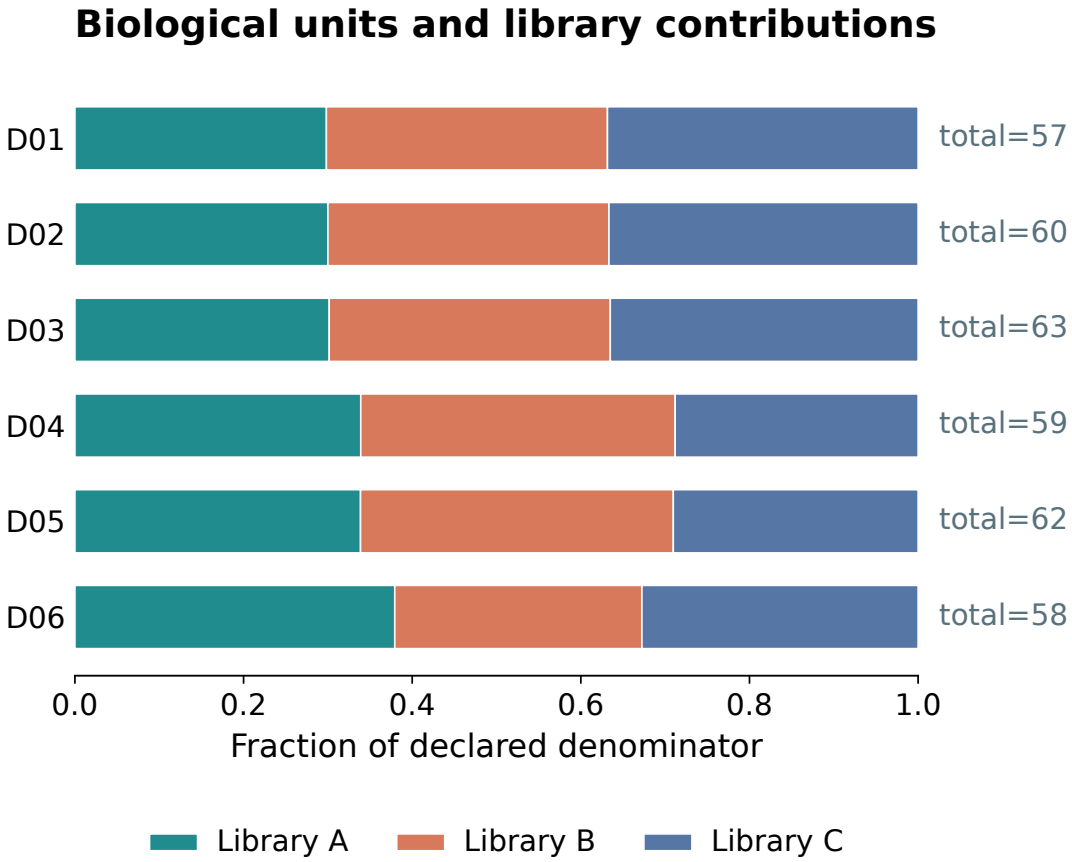
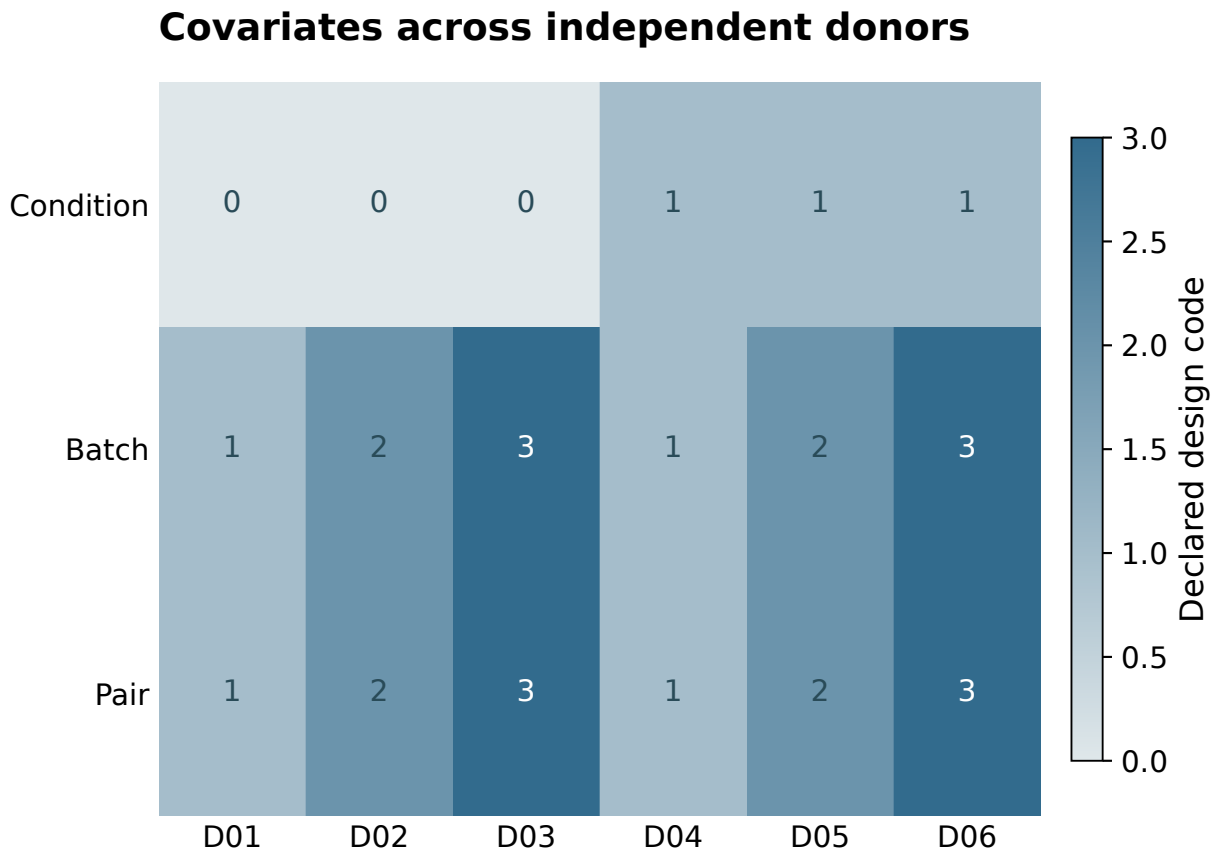
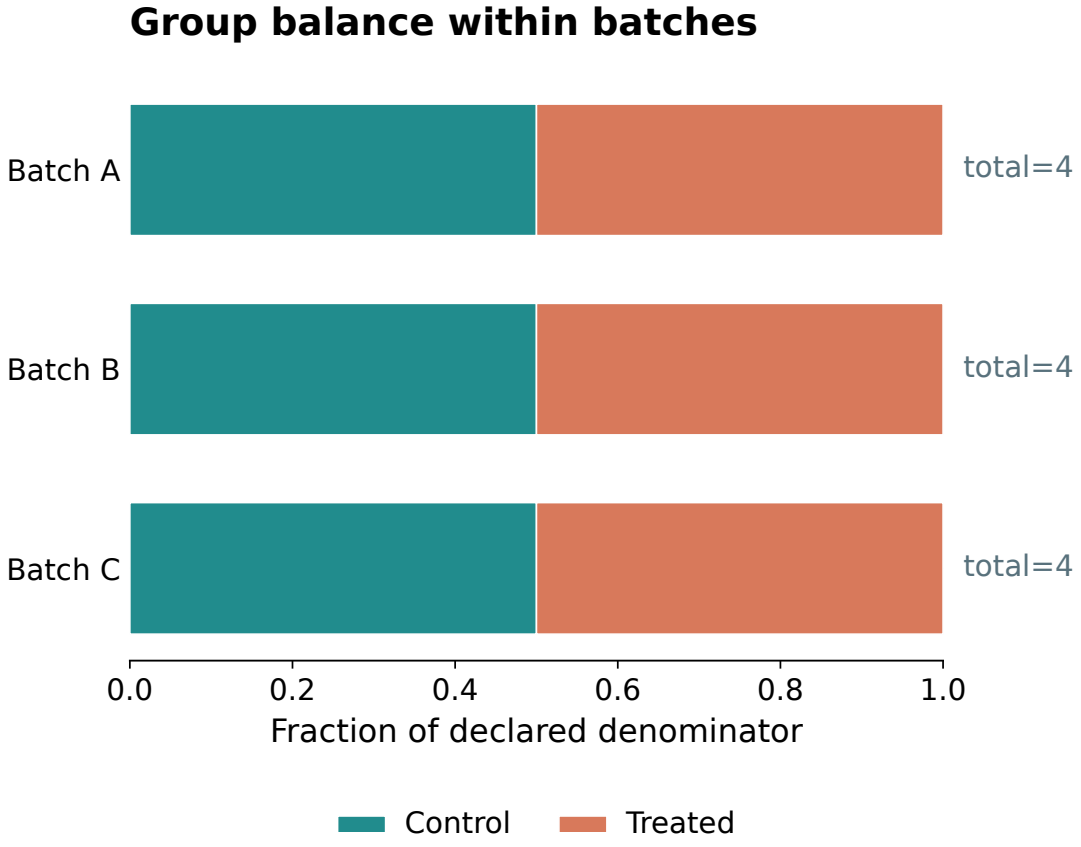
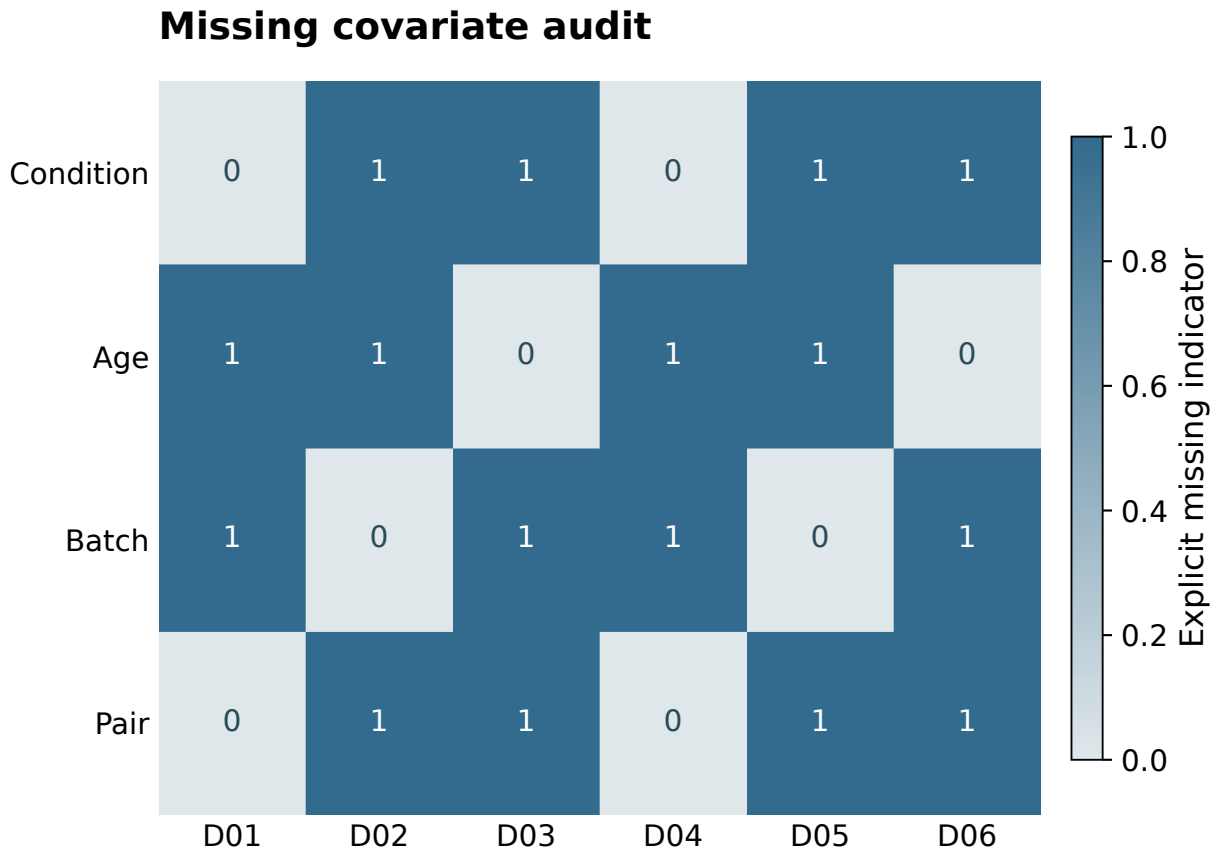
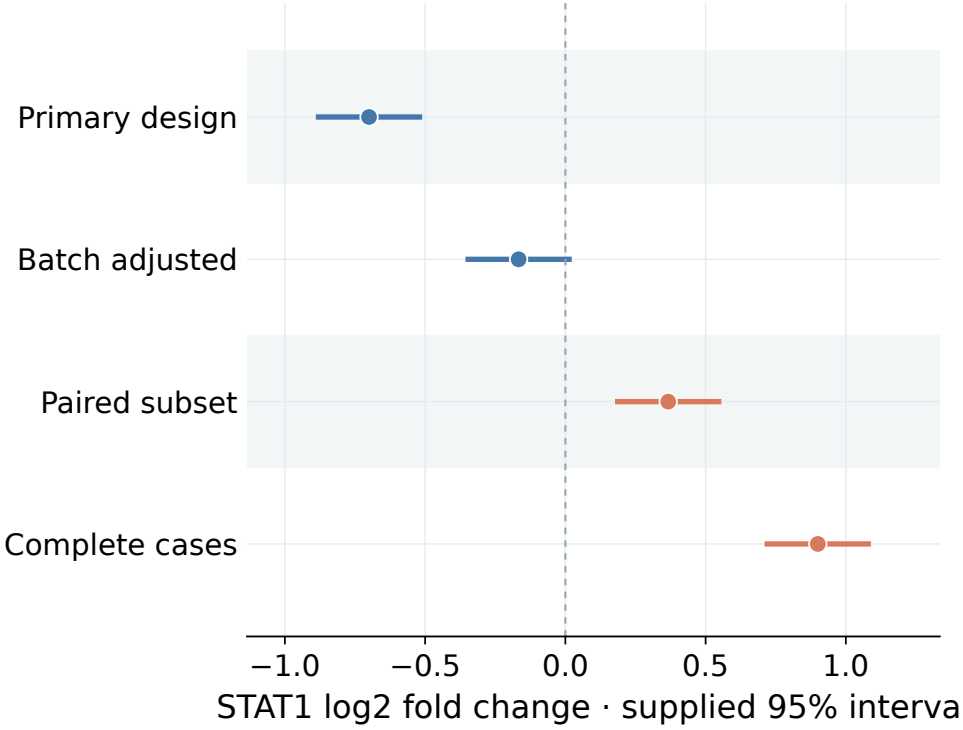

Biological units and library contributions
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Library A
Library B
Library C


Covariates across independent donors
Condition
Batch
Pair
0
0
0
1
1
1
1
2
3
1
2
3
1
2
3
1
2
3
D01
D02
D03
D04
D05
D06
Declared design code
3.0
2.5
2.0
1.5
1.0
0.5
0.0


Group balance within batches
Batch A
Batch B
Batch C
total=4
total=4
total=4
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Control
Treated


Missing covariate audit
Condition
Age
Batch
Pair
0 1 1 0 1 1
1 1 0 1 1 0
1 0 1 1 0 1
0 1 1 0 1 1
D01
D02
D03
D04
D05
D06
Explicit missing indicator
1.0
0.8
0.6
0.4
0.2
0.0

Prespecified design sensitivity



Estimate [interval] · q

-0.70 [-0.88, -0.52] 0.002 n=12

-0.17 [-0.35, 0.01] 0.65 n=12

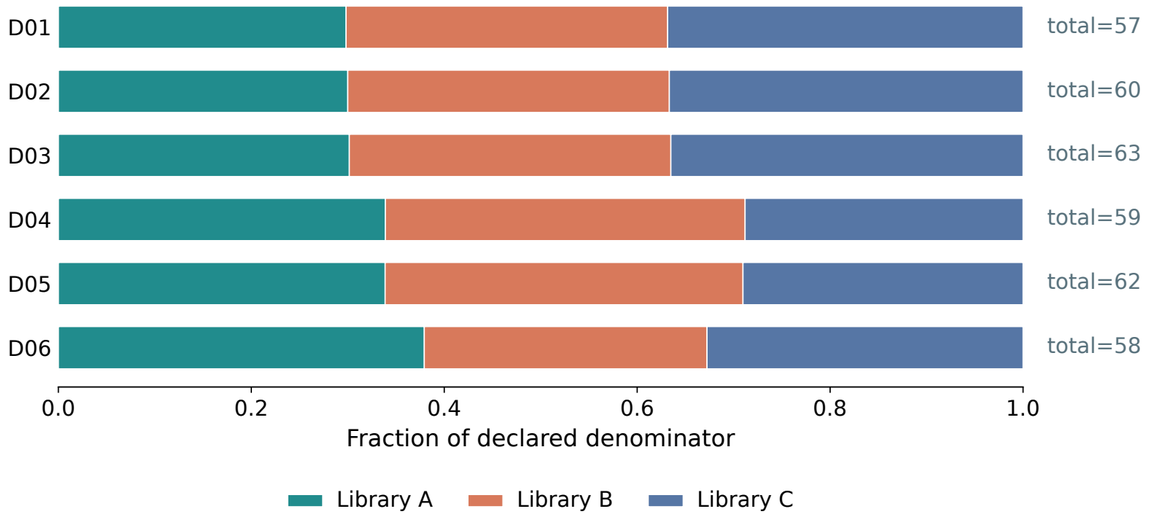
0.37 [0.19, 0.55] 0.038 n=6

0.90 [0.72, 1.08] 0.056 n=10

Bulk · design — complementary evidence

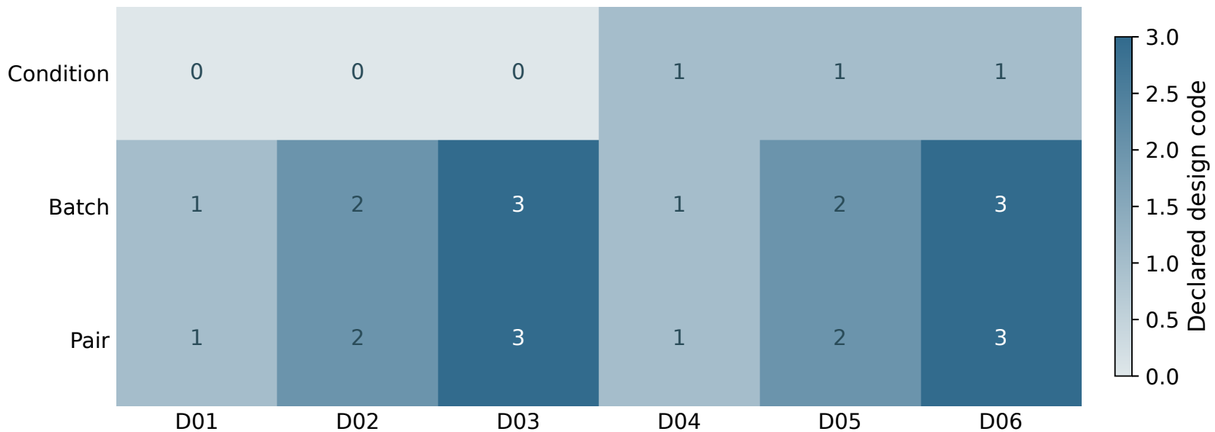
A

Biological units and library contributions



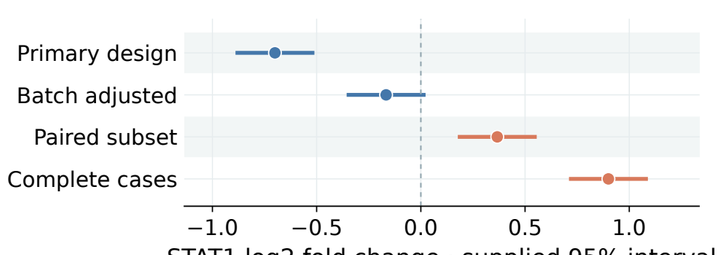
B

Covariates across independent donors



C

Prespecified design sensitivity



Estimate [interval] · q

-0.70 [-0.88, -0.52] 0.002 n=12

-0.17 [-0.35, 0.01] 0.65 n=12

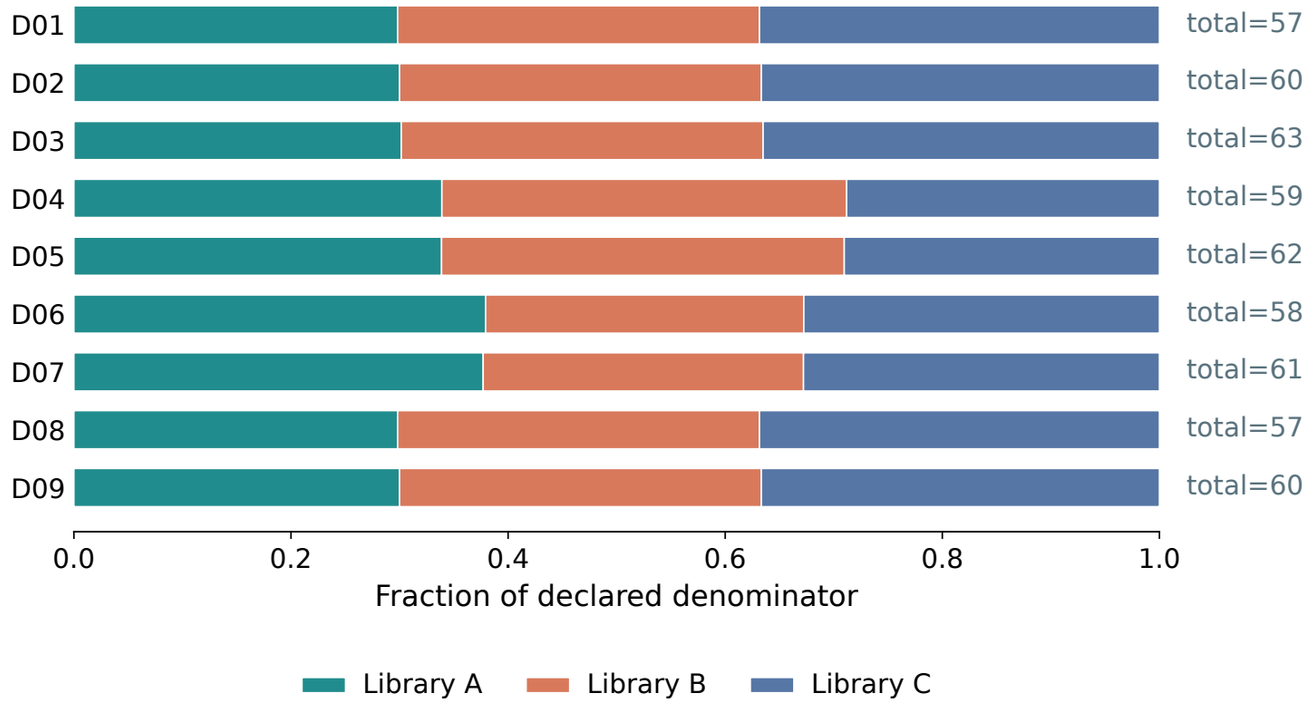
0.37 [0.19, 0.55] 0.038 n=6

0.90 [0.72, 1.08] 0.056 n=10

Bulk · design — complementary evidence

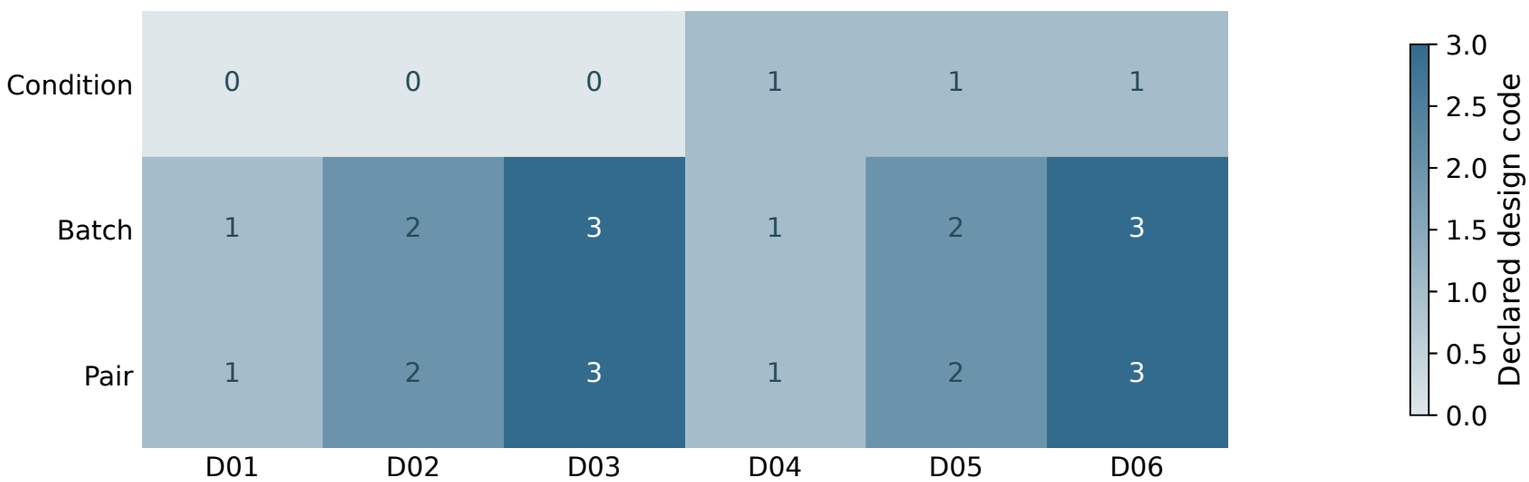
A

Biological units and library contributions



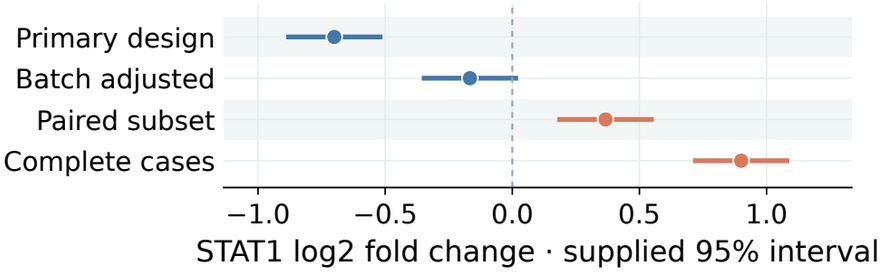
B

Covariates across independent donors



C

Prespecified design sensitivity



Estimate [interval] · q

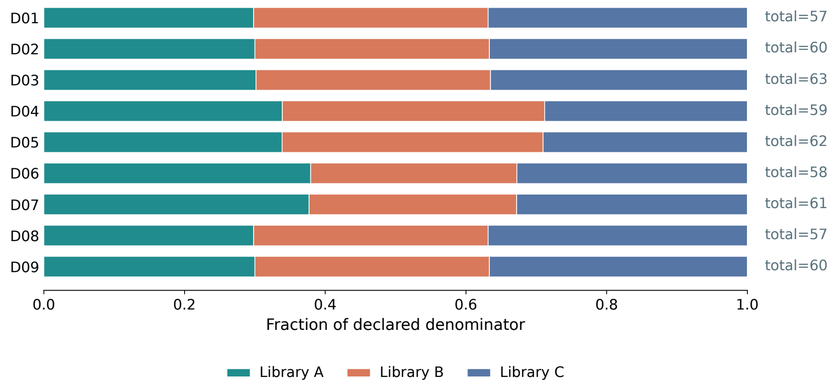
-0.70 [-0.88, -0.52] 0.002 n=24
-0.17 [-0.35, 0.01] 0.65 n=24
0.37 [0.19, 0.55] 0.038 n=12
0.90 [0.72, 1.08] 0.056 n=22

Software test fixture; not biological evidence

Bulk · design — complementary evidence

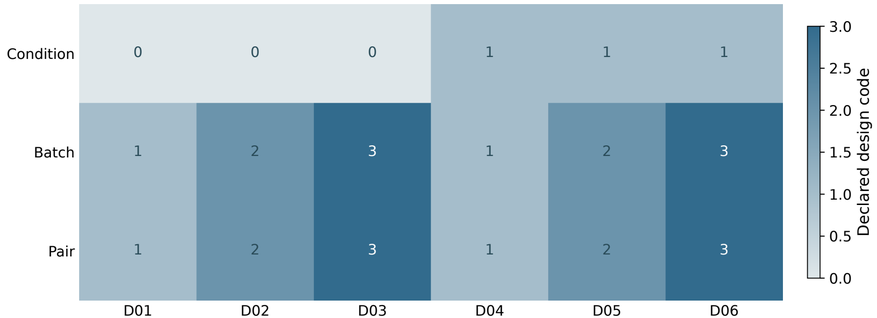
A

Biological units and library contributions



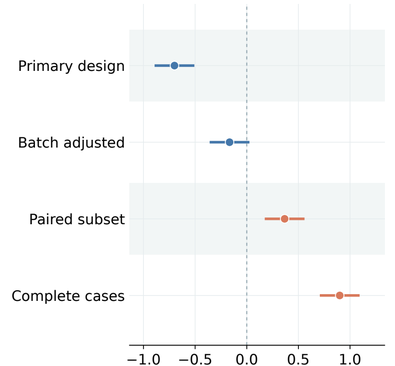
B

Covariates across independent donors



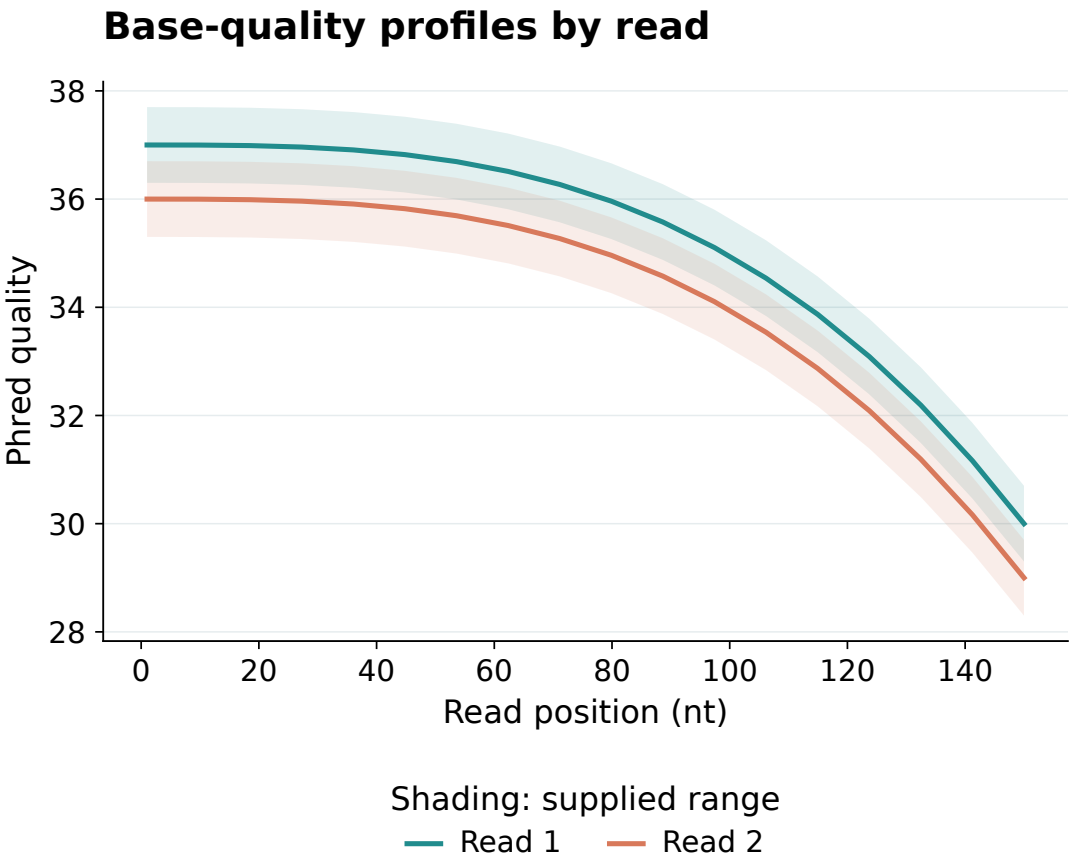
C

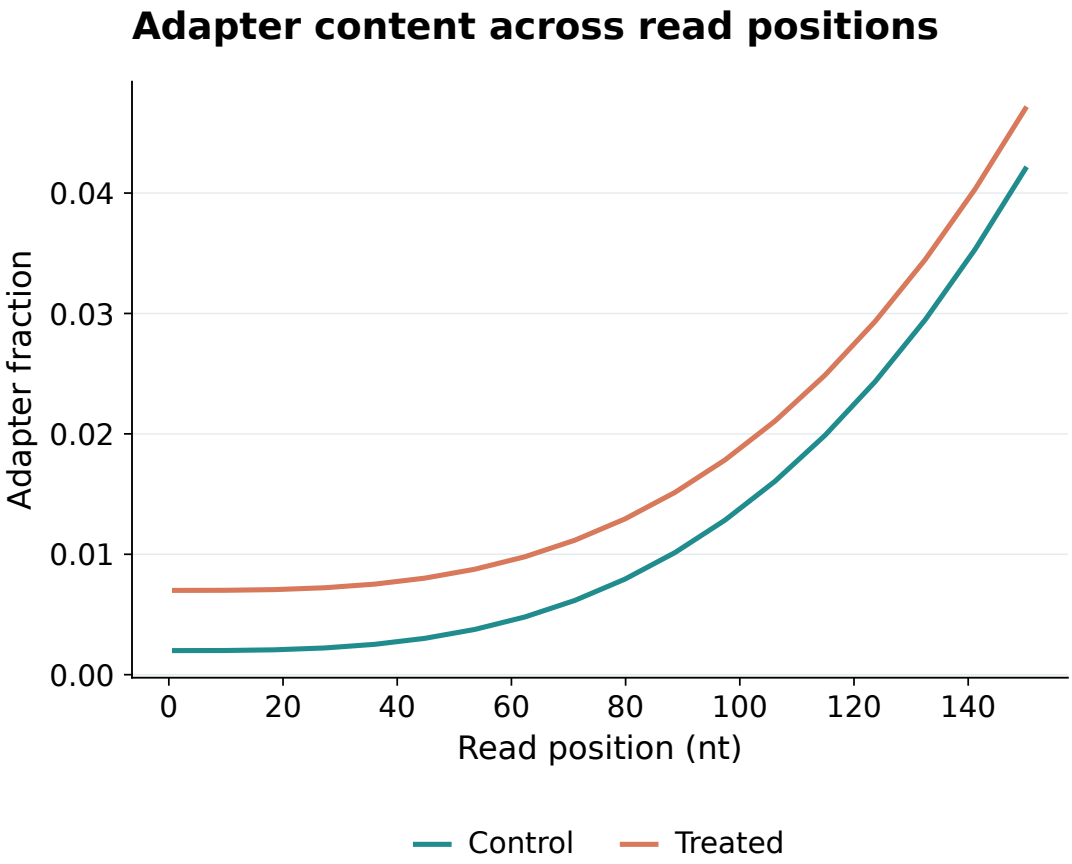
Prespecified design sensitivity



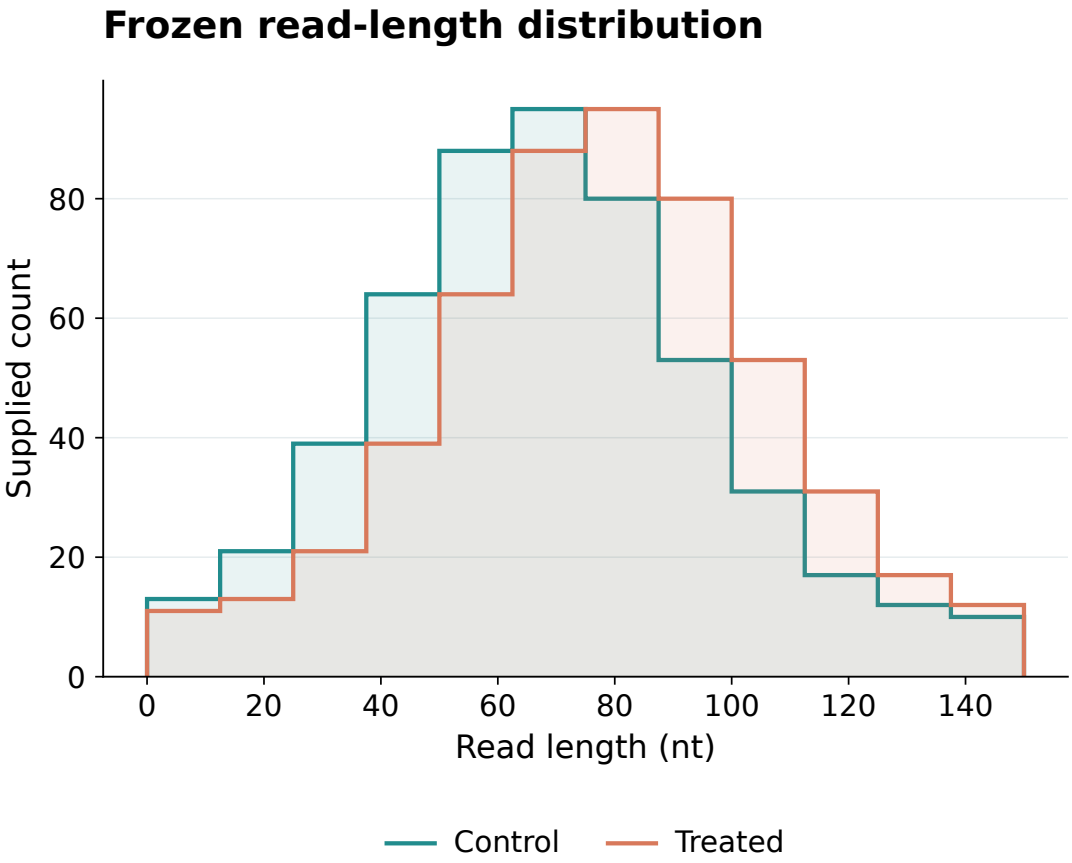
Estimate [interval] · q

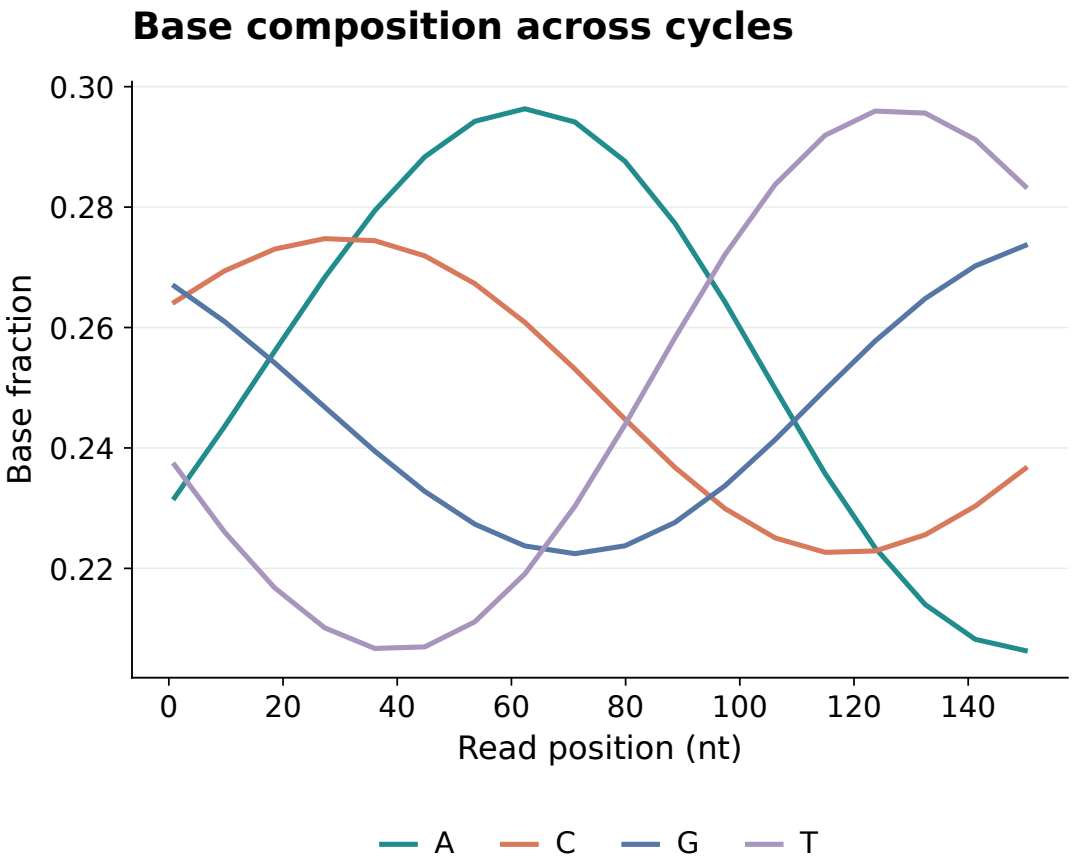
Primary design	-0.70 [-0.88, -0.52] 0.002 n=24
Batch adjusted	-0.17 [-0.35, 0.01] 0.65 n=24
Paired subset	0.37 [0.19, 0.55] 0.038 n=12
Complete cases	0.90 [0.72, 1.08] 0.056 n=22


Base-quality profiles by read
Phred quality
38
36
34
32
30
28
0
20
40
60
80
100
120
140
Read position (nt)
Shading: supplied range
Read 1
Read 2

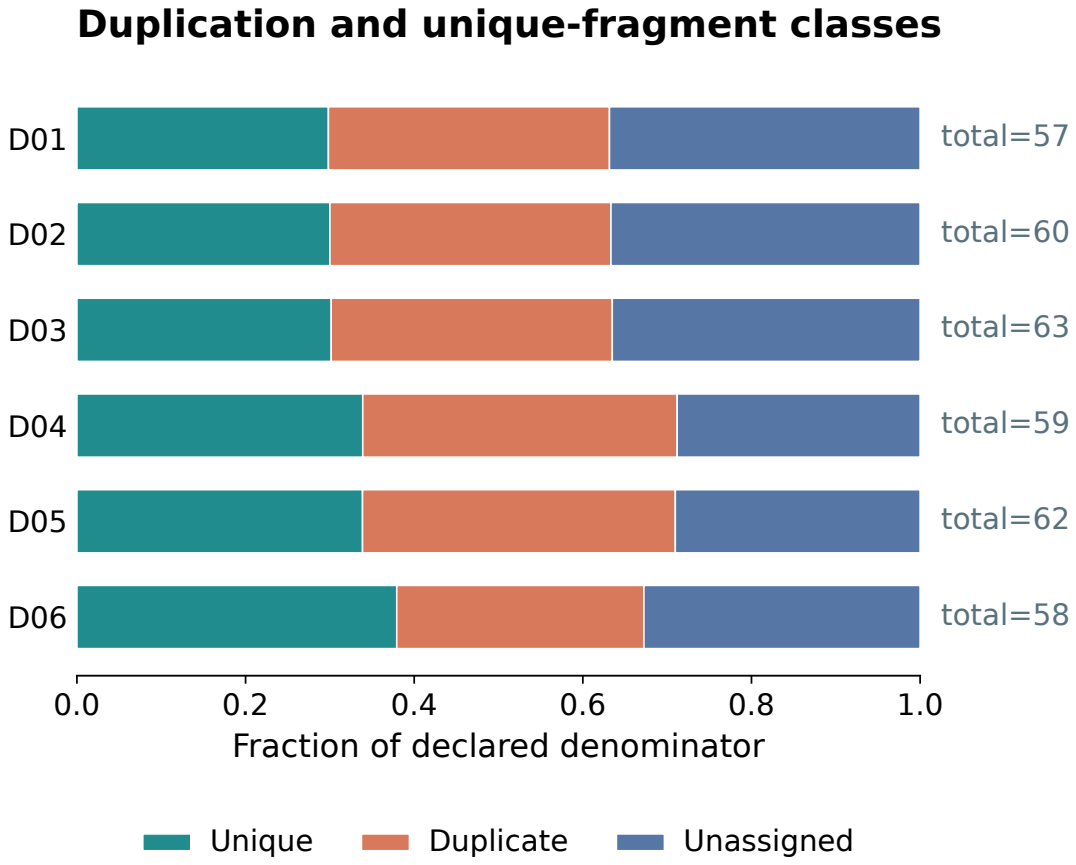


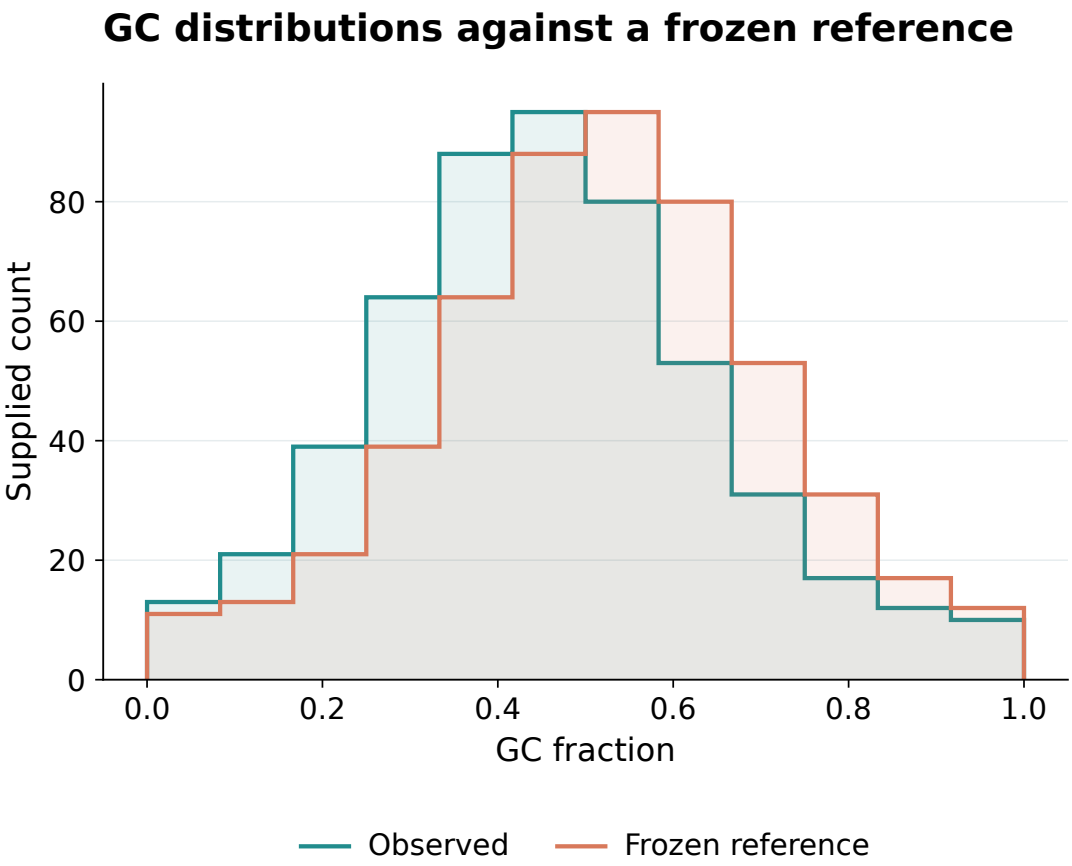
Software test fixture; not biological evidence


Frozen read-length distribution
Supplied count
0
20
40
60
80
Read length (nt)
0
20
40
60
80
100
120
140
Control
Treated

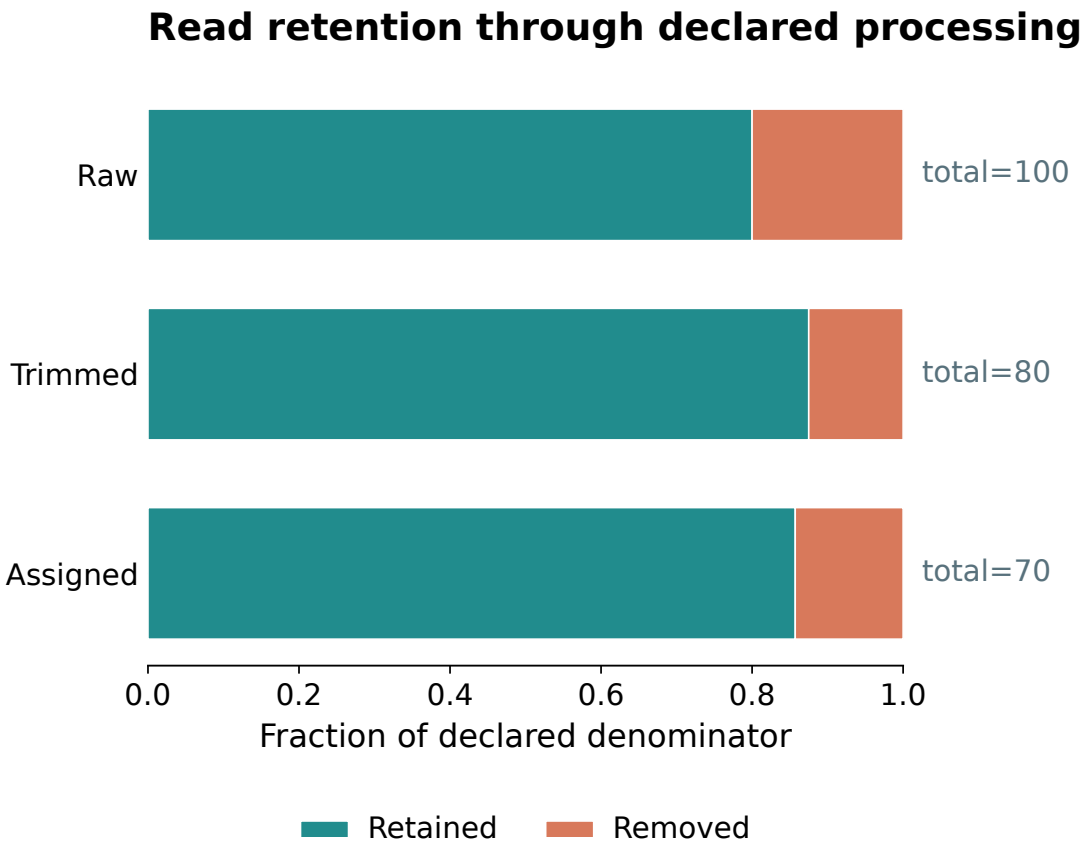


Software test fixture; not biological evidence

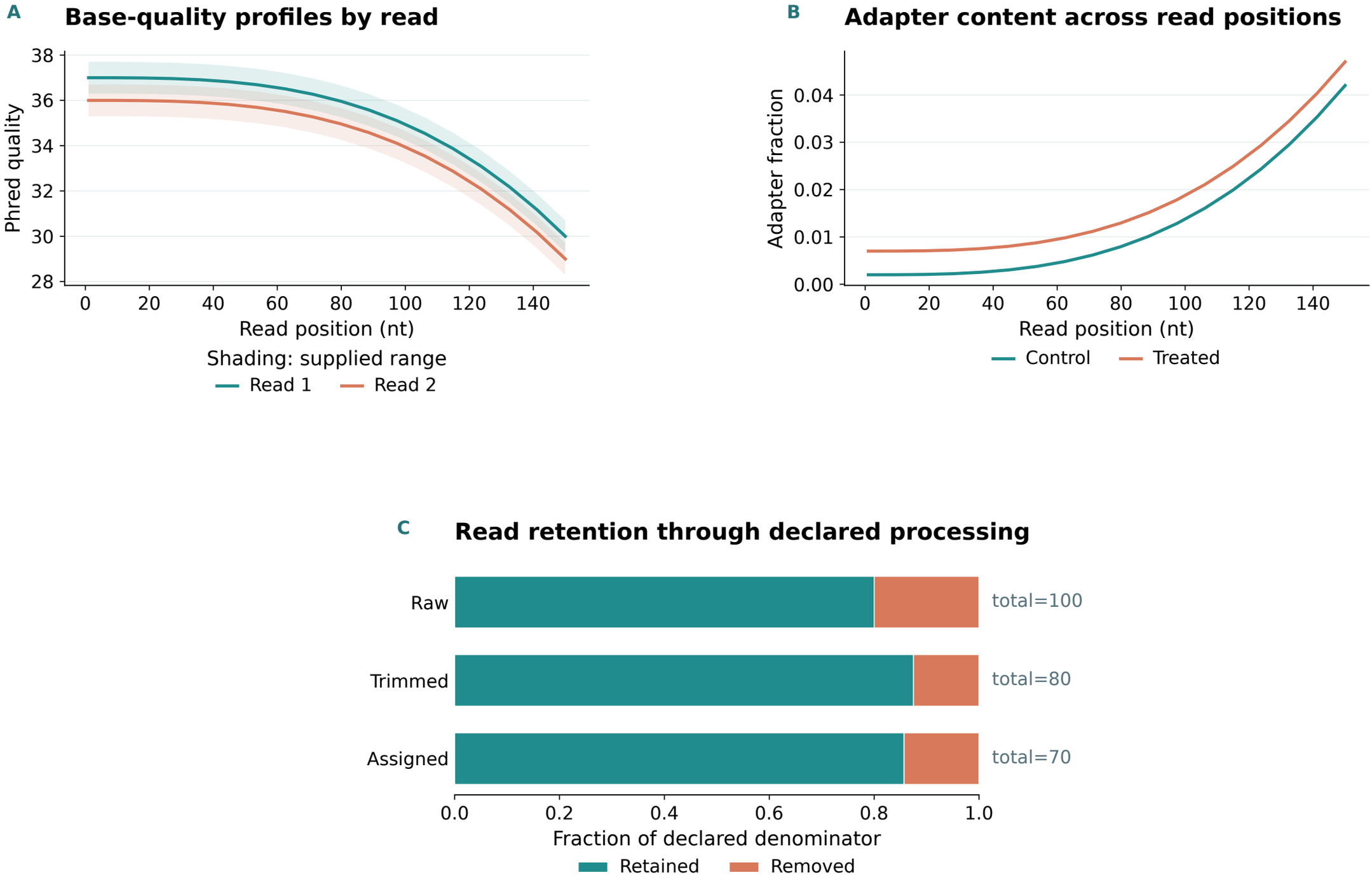

Duplication and unique-fragment classes
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Unique
Duplicate
Unassigned



Software test fixture; not biological evidence

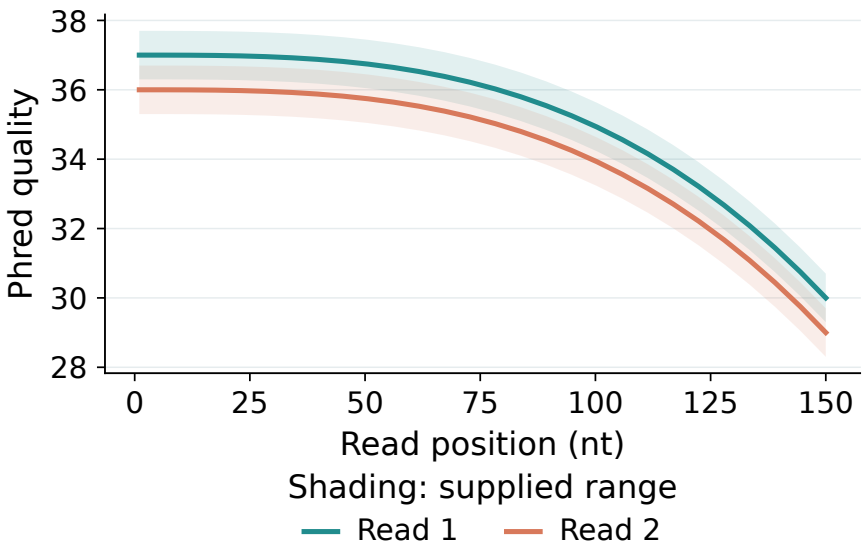

Read retention through declared processing
Raw
Trimmed
Assigned
total=100
total=80
total=70
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Retained
Removed

Bulk · reads — complementary evidence

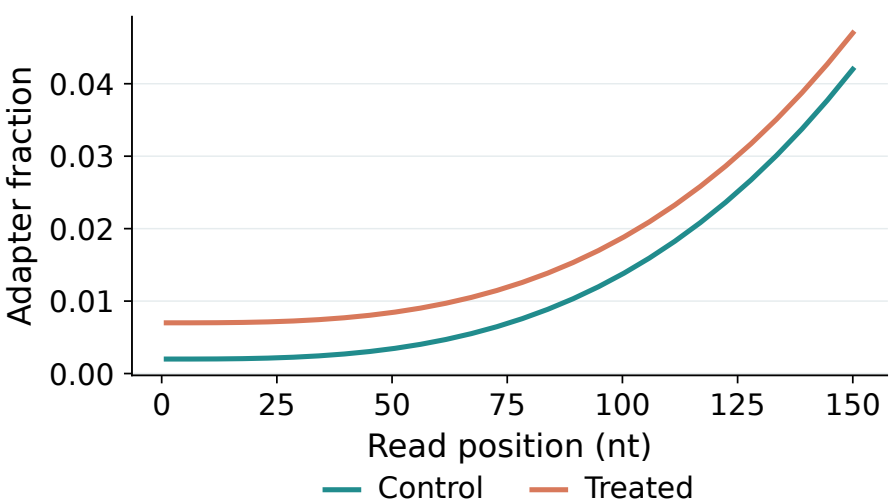


Bulk · reads — complementary evidence

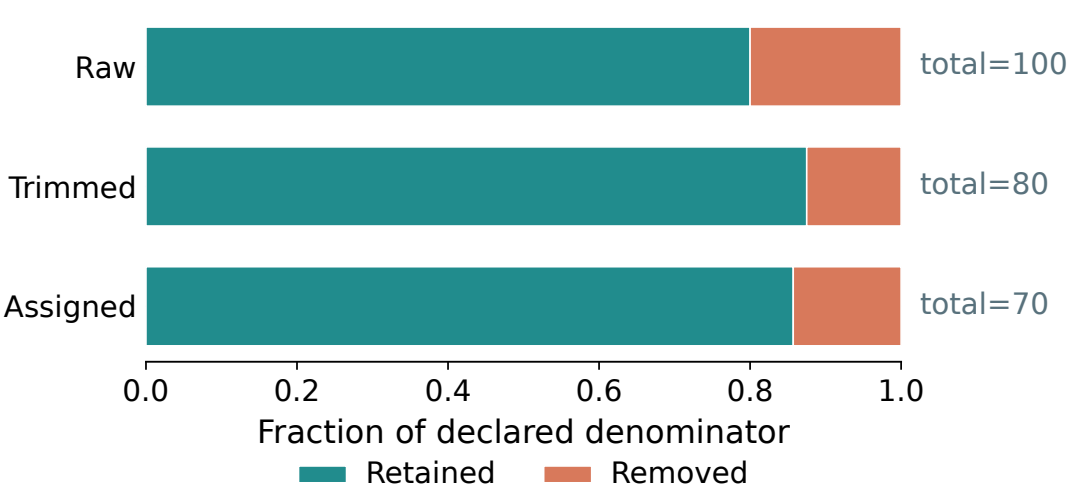
A Base-quality profiles by read

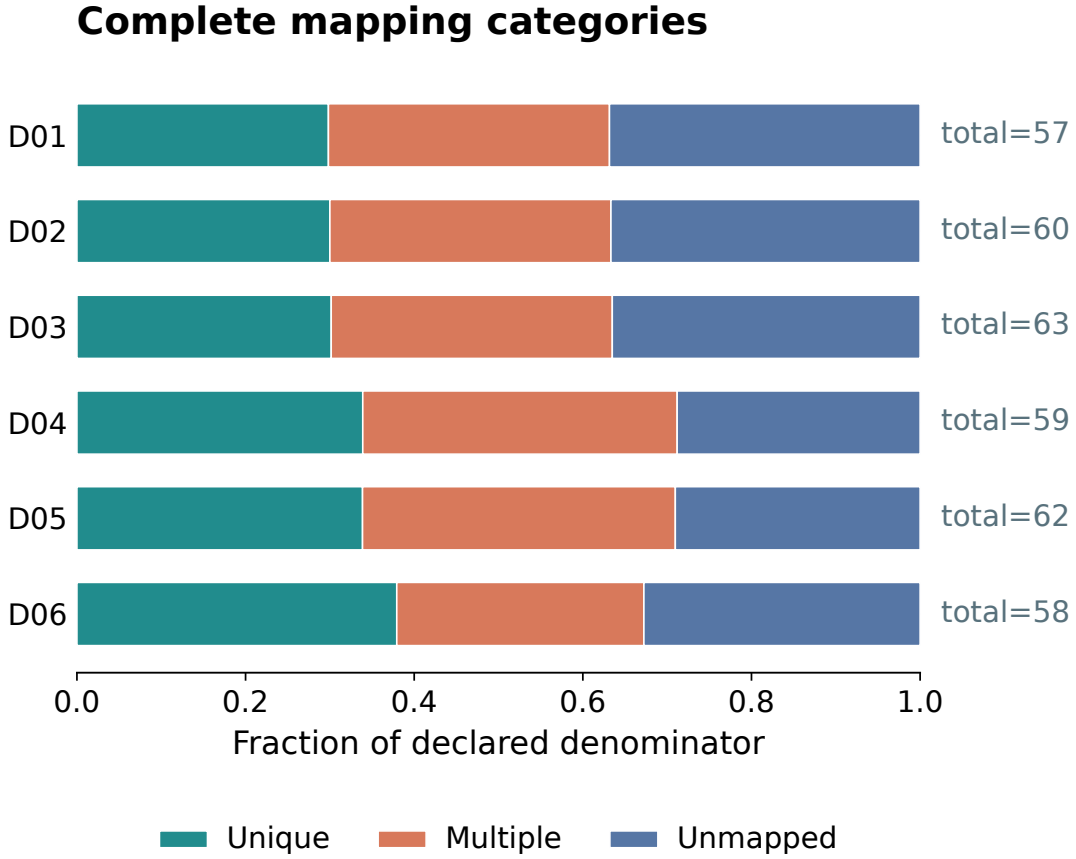


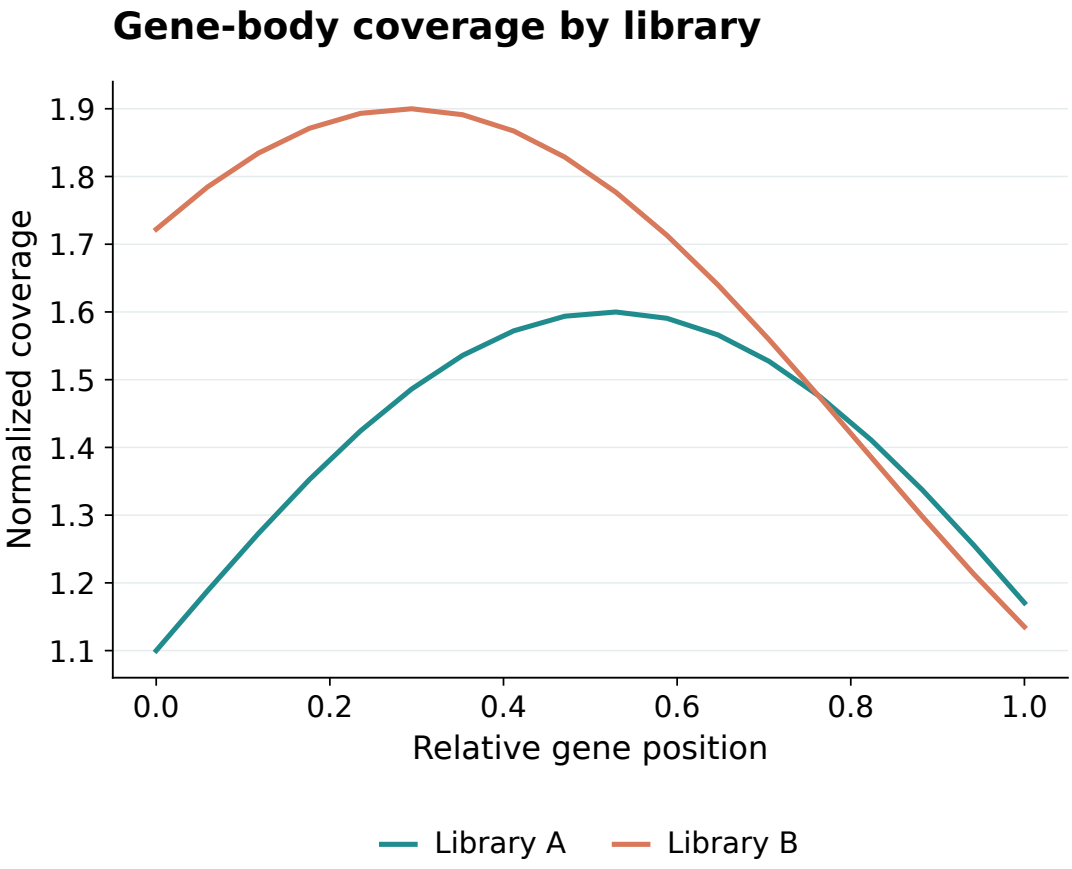
B Adapter content across read positions



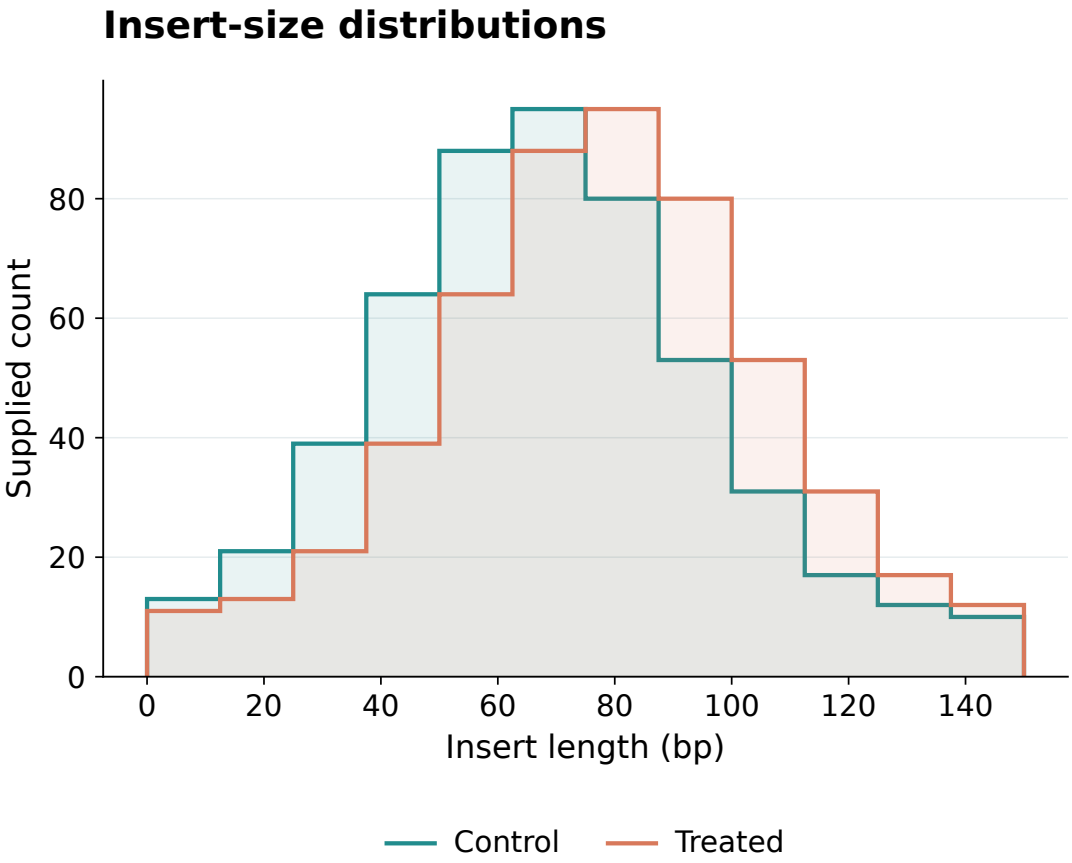
C Read retention through declared processing

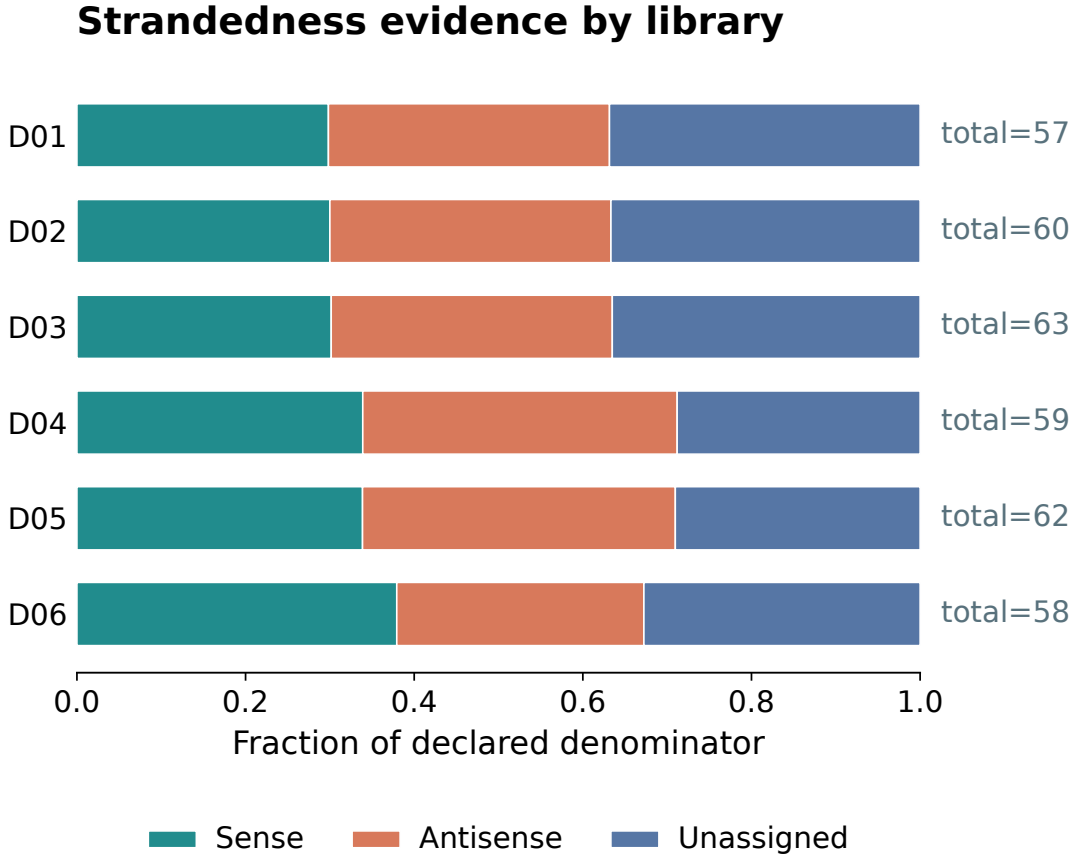


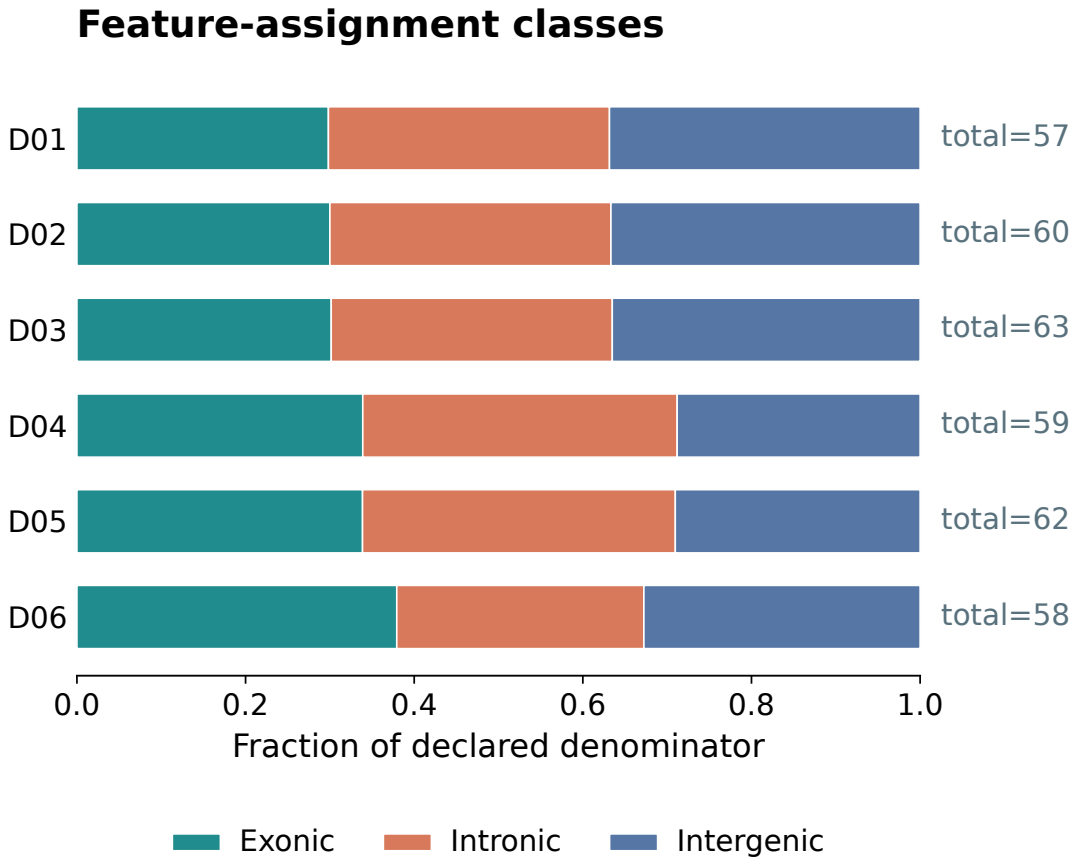

Complete mapping categories
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Unique
Multiple
Unmapped

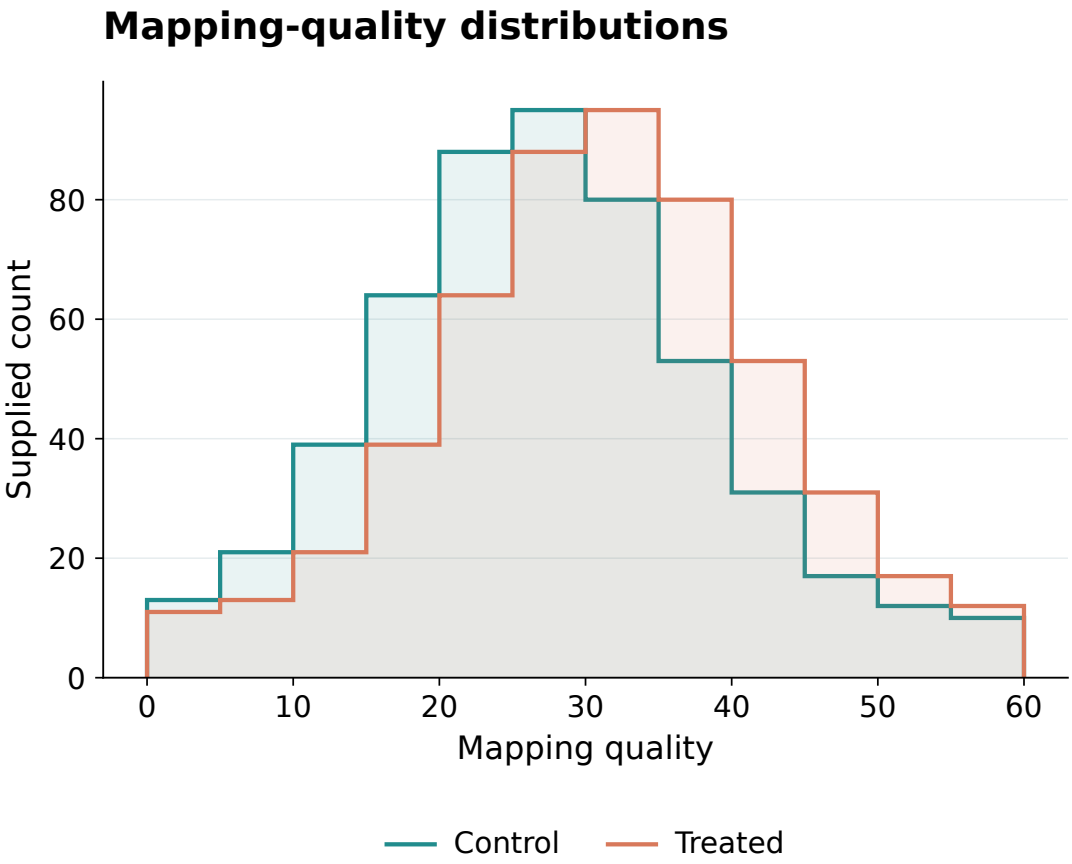


Software test fixture; not biological evidence

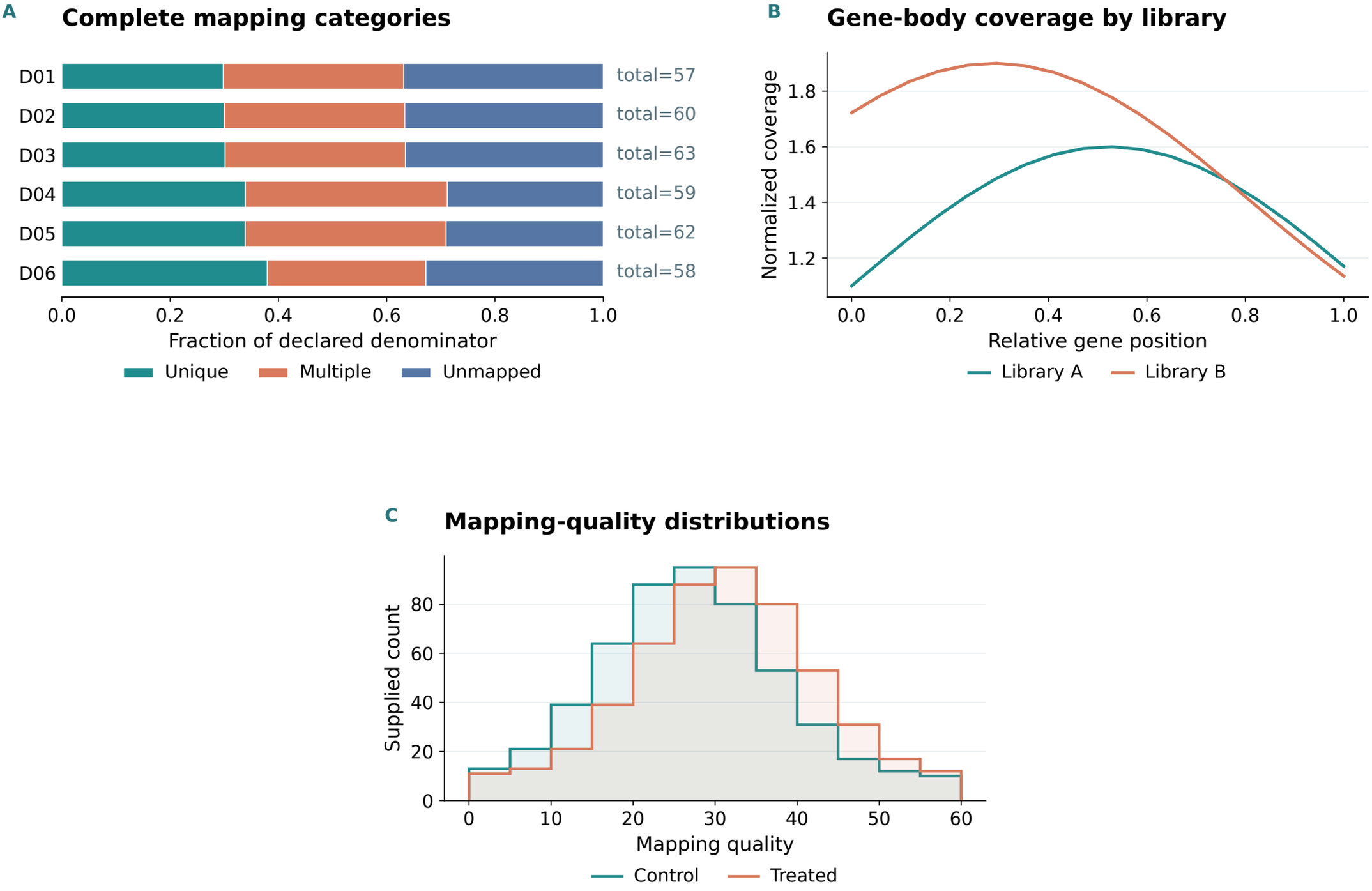

Insert-size distributions
Supplied count
0
20
40
60
80
0
20
40
60
80
100
120
140
Insert length (bp)
Control
Treated


Strandedness evidence by library
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Sense
Antisense
Unassigned


Feature-assignment classes
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Exonic
Intronic
Intergenic

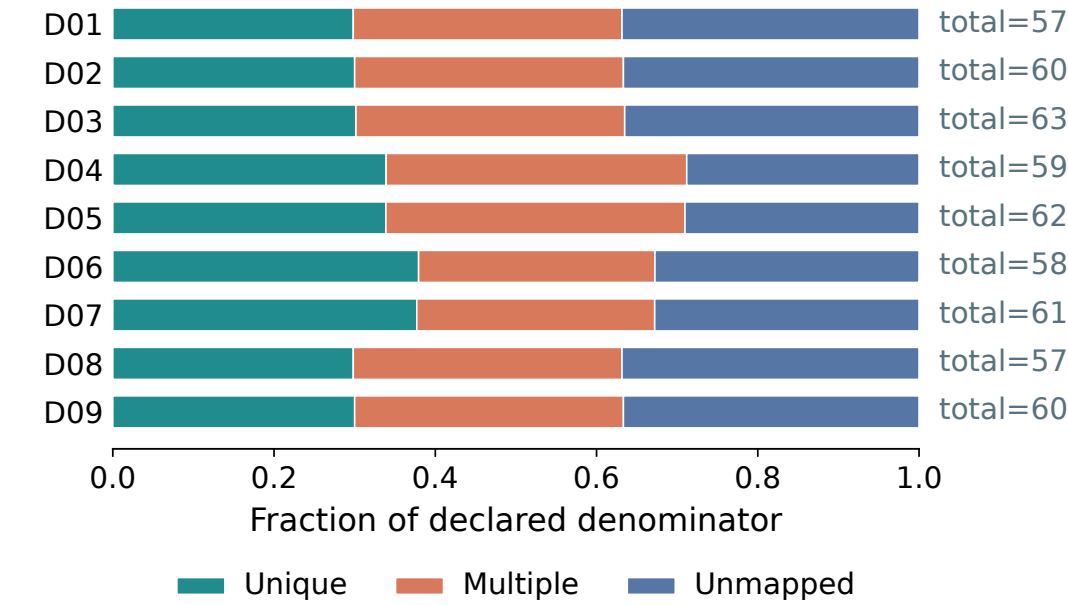

Mapping-quality distributions
Supplied count
0
20
40
60
80
0
10
20
30
40
50
60
Mapping quality
Control
Treated

Bulk · alignment — complementary evidence

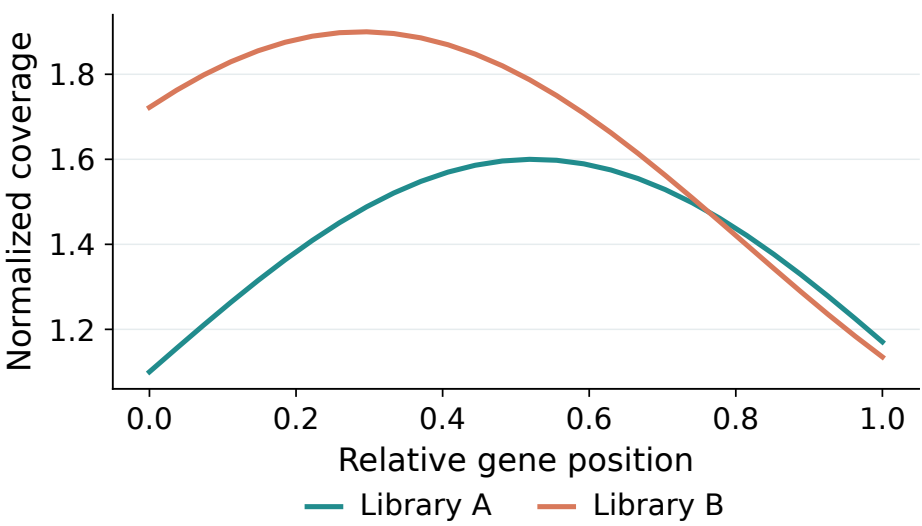


Bulk · alignment — complementary evidence

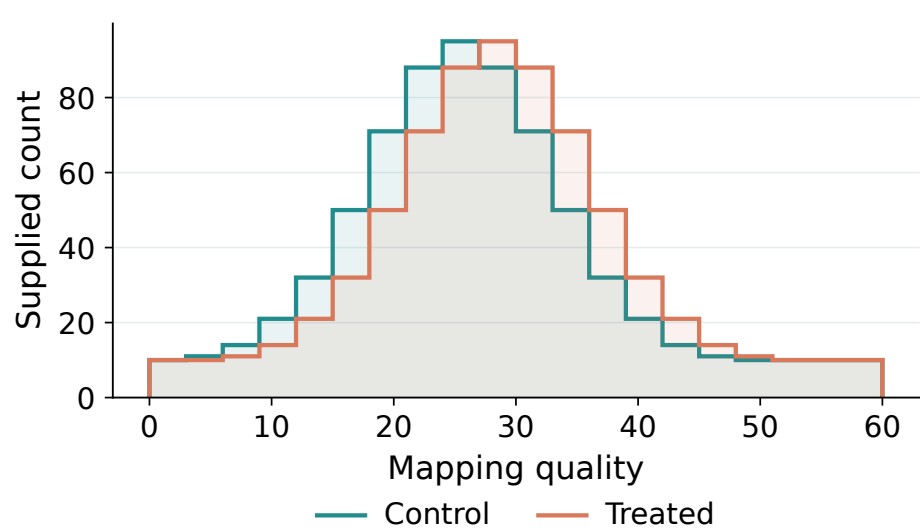
A Complete mapping categories

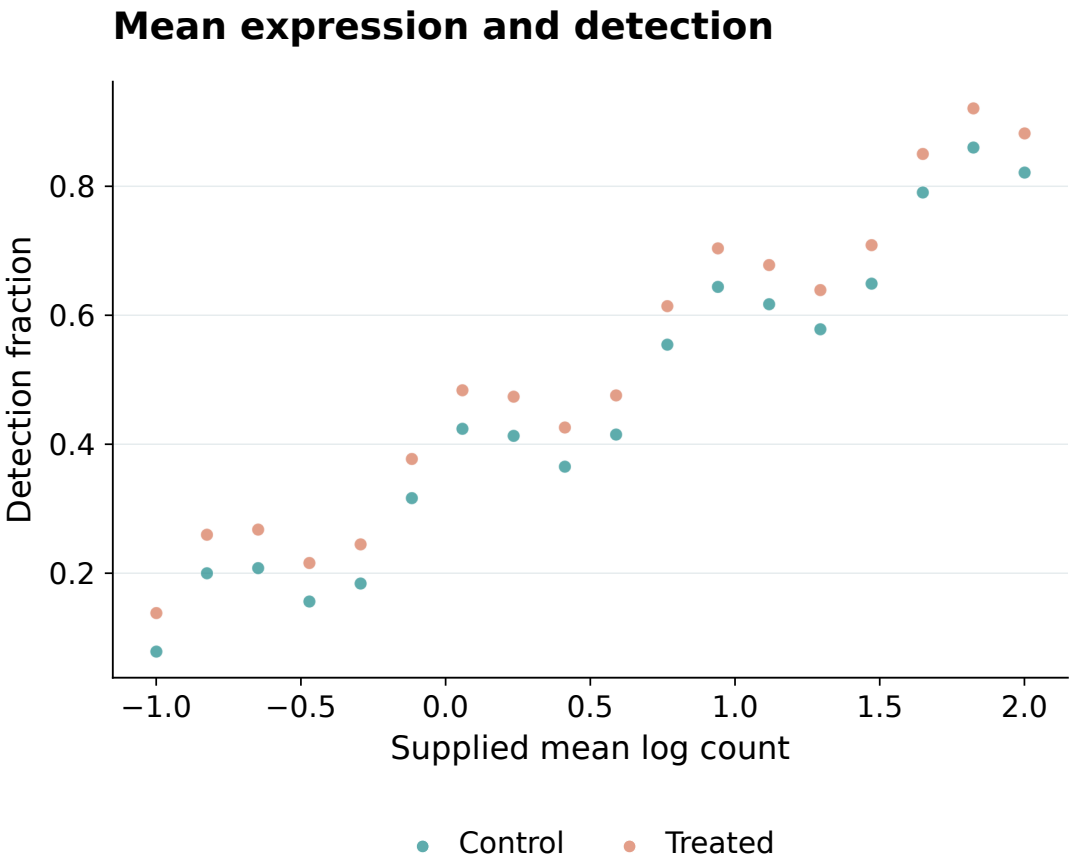


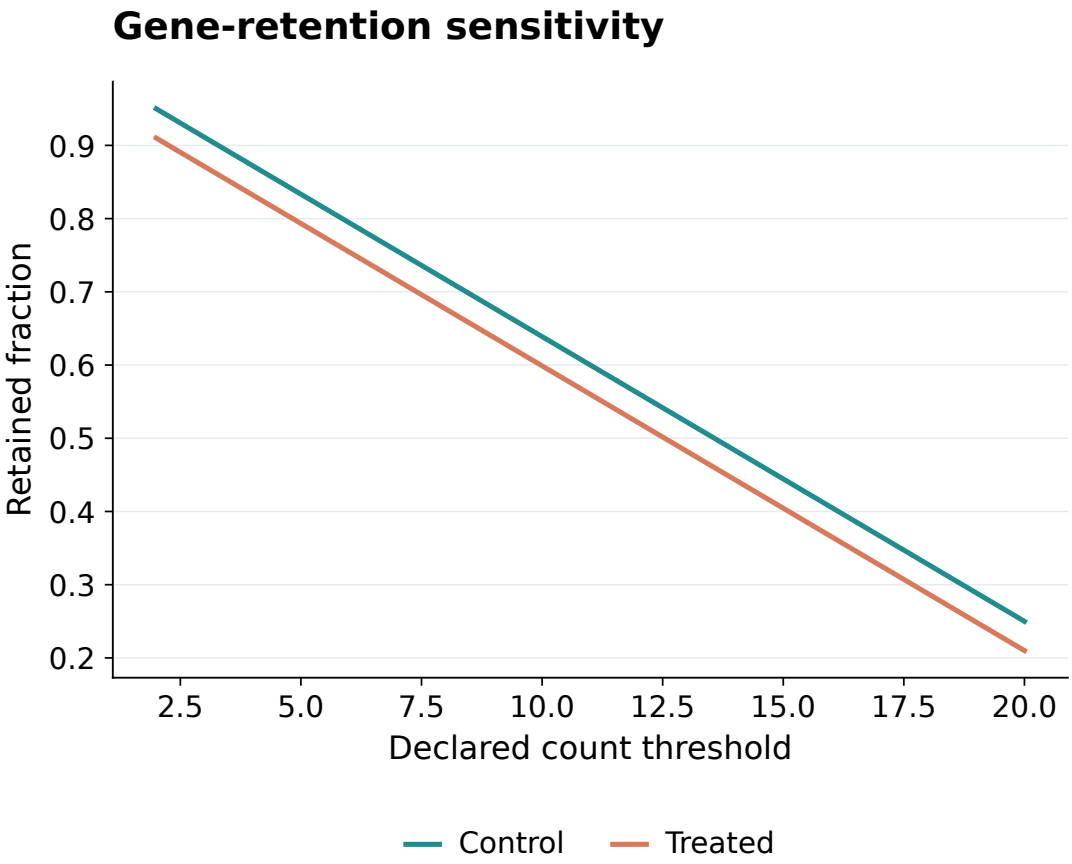
B Gene-body coverage by library



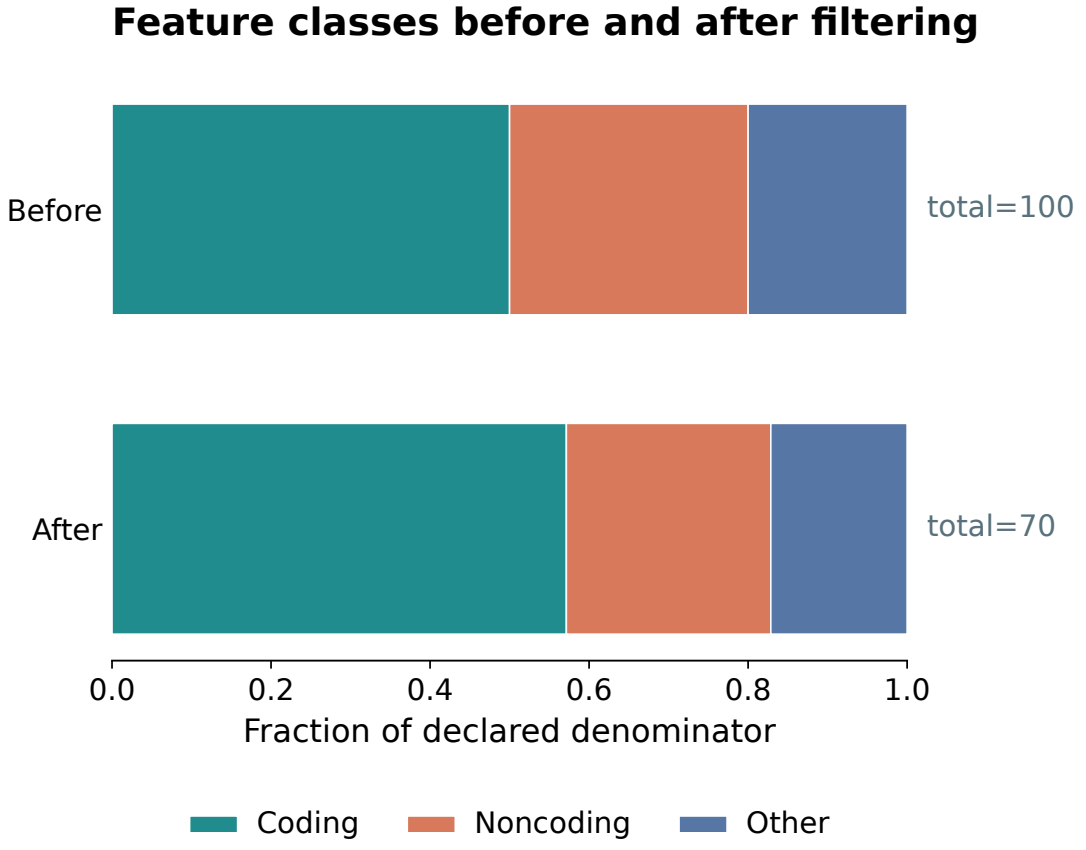
C Mapping-quality distributions



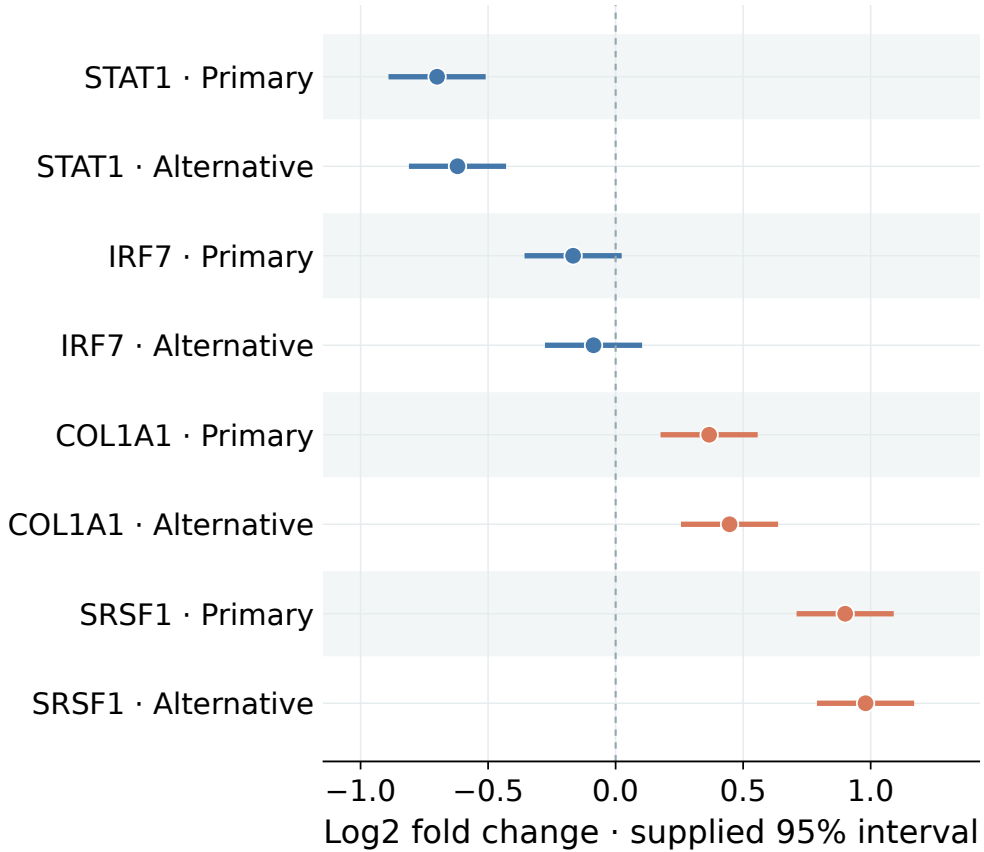

Mean expression and detection
Detection fraction
0.8
0.6
0.4
0.2
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Supplied mean log count
Control
Treated



Software test fixture; not biological evidence


Feature classes before and after filtering
Before
After
total=100
total=70
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Coding
Noncoding
Other

Effect stability across filter choices



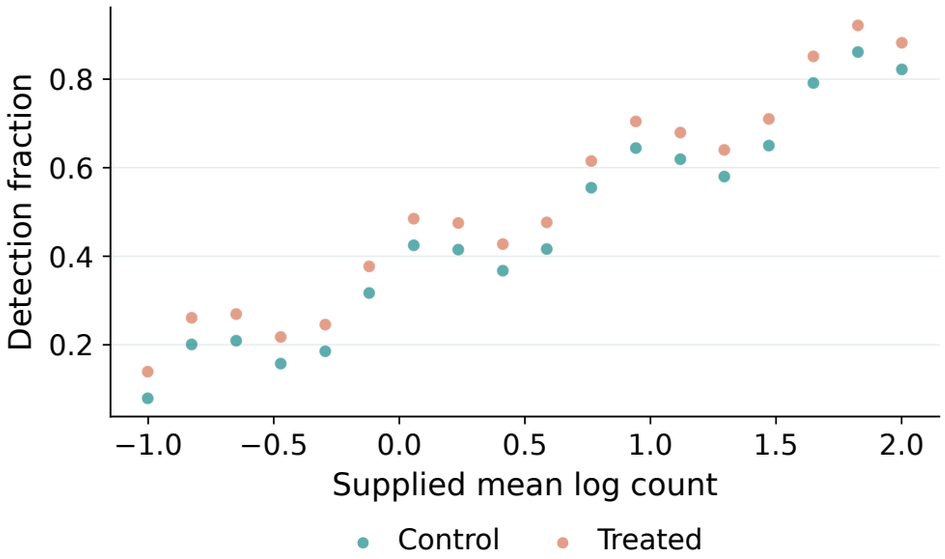
Estimate [interval] · q
Independent n = 12

STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.17 [-0.35, 0.01] 0.65
IRF7 · Alternative	-0.09 [-0.27, 0.09] 0.65
COL1A1 · Primary	0.37 [0.19, 0.55] 0.038
COL1A1 · Alternative	0.45 [0.27, 0.63] 0.041
SRSF1 · Primary	0.90 [0.72, 1.08] 0.056
SRSF1 · Alternative	0.98 [0.80, 1.16] 0.059

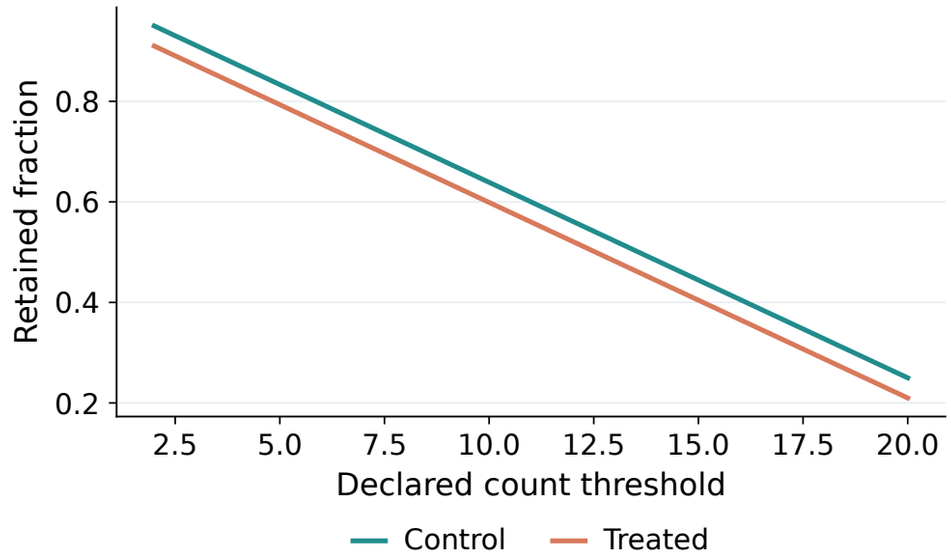

Detection support across groups
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0.15
0.31
0.31
0.47
0.47
0.64
0.64
0.8
0.8
0.15
0.15
0.31
Control
Treated
Detected donor fraction
1.0
0.8
0.6
0.4
0.2
0.0

Bulk · filtering — complementary evidence

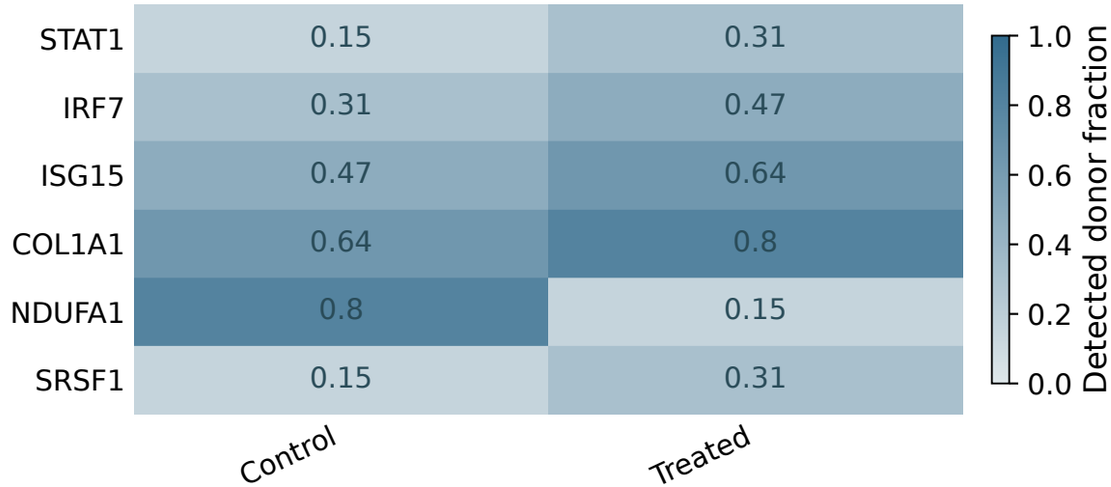
A Mean expression and detection



B Gene-retention sensitivity

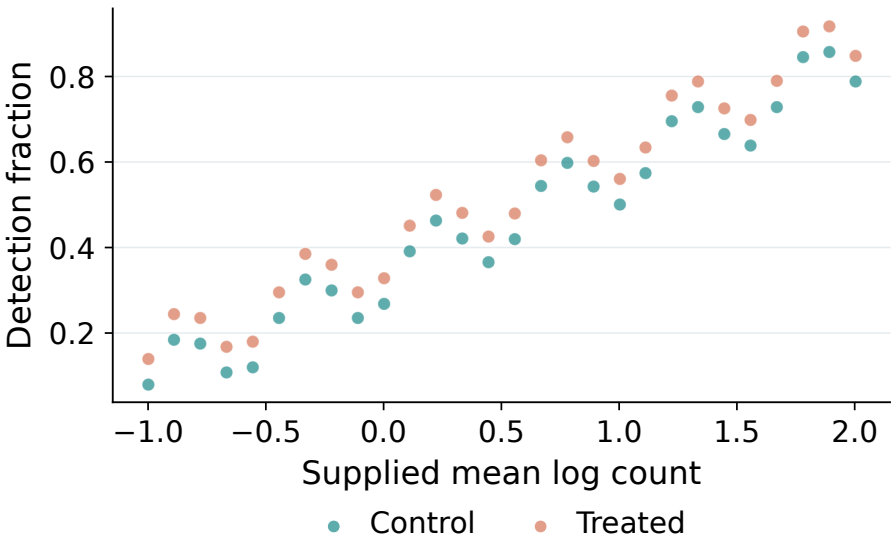


C Detection support across groups

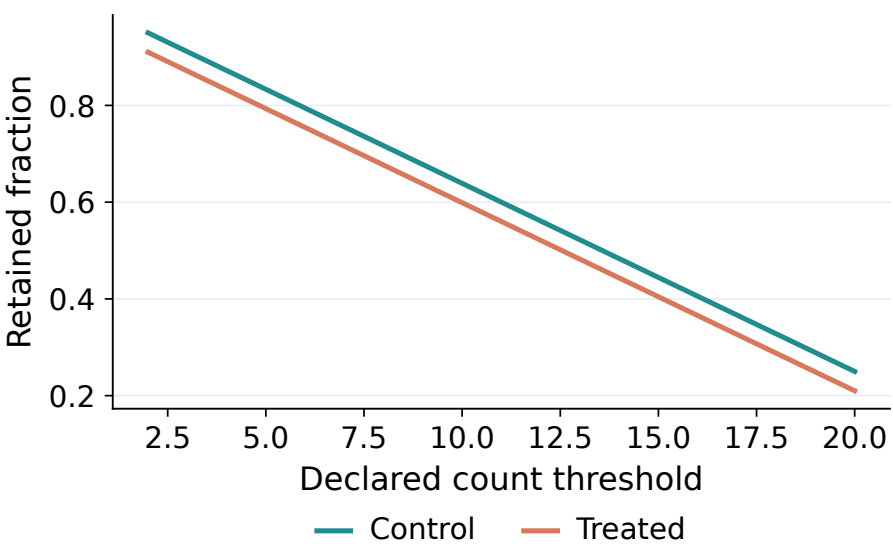


Bulk · filtering — complementary evidence

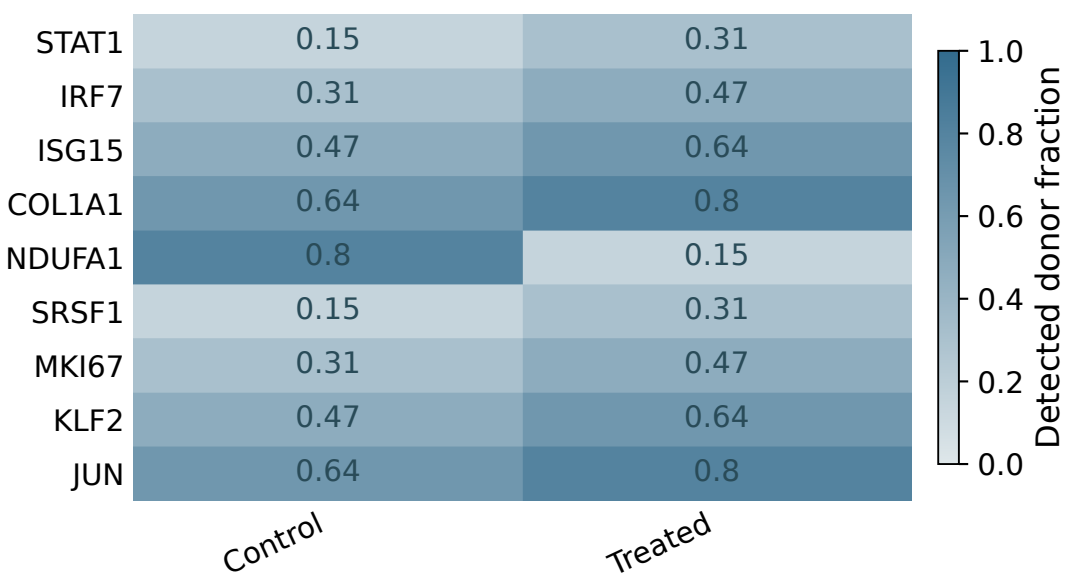
A Mean expression and detection

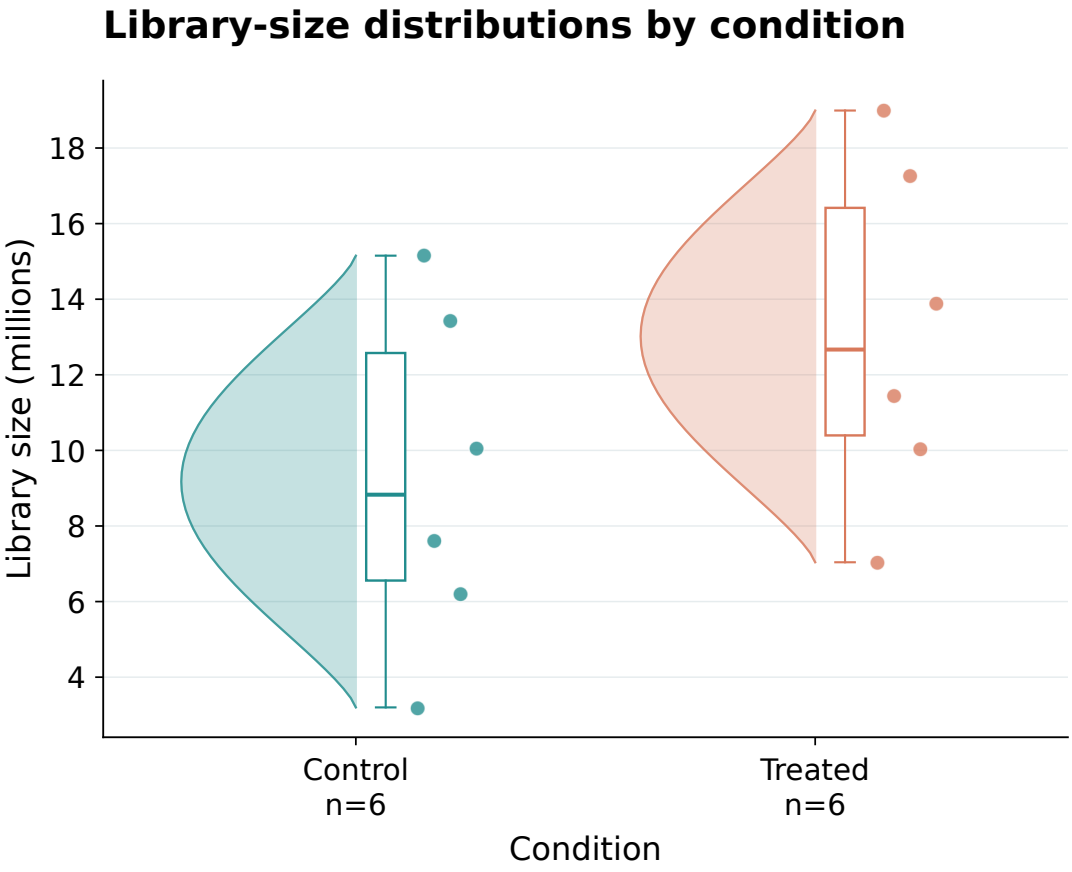


B Gene-retention sensitivity

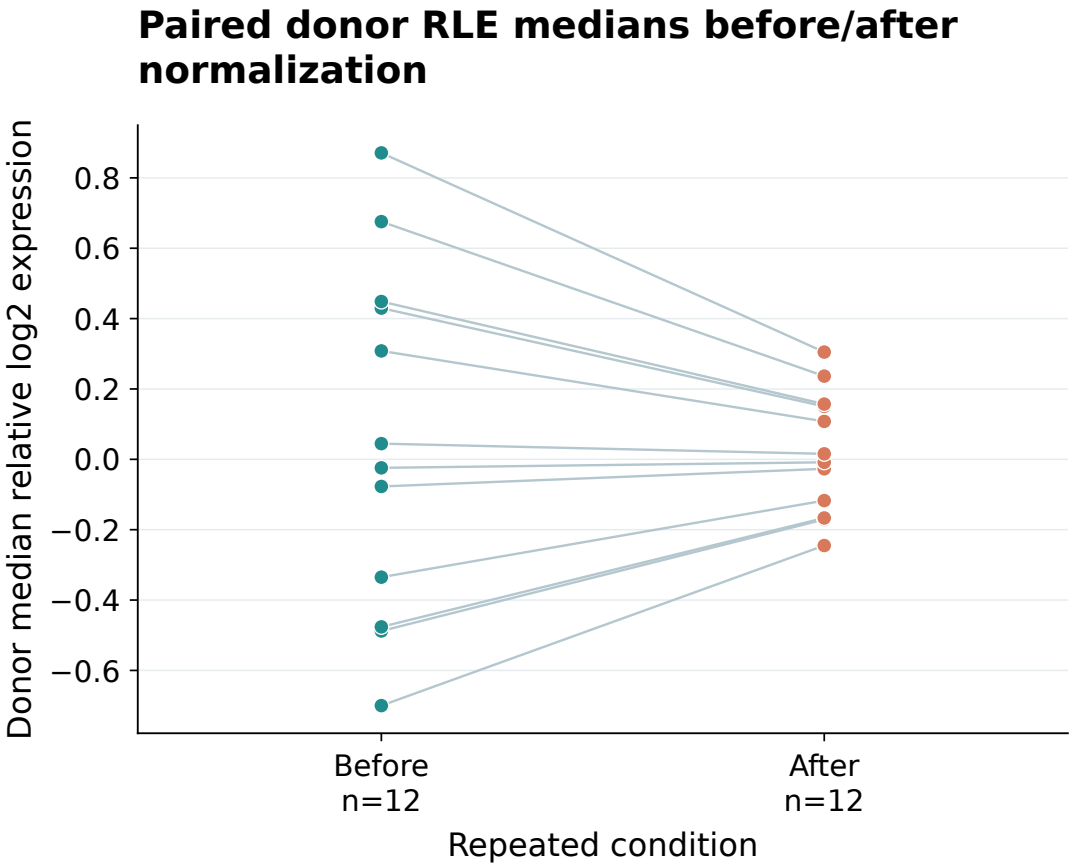


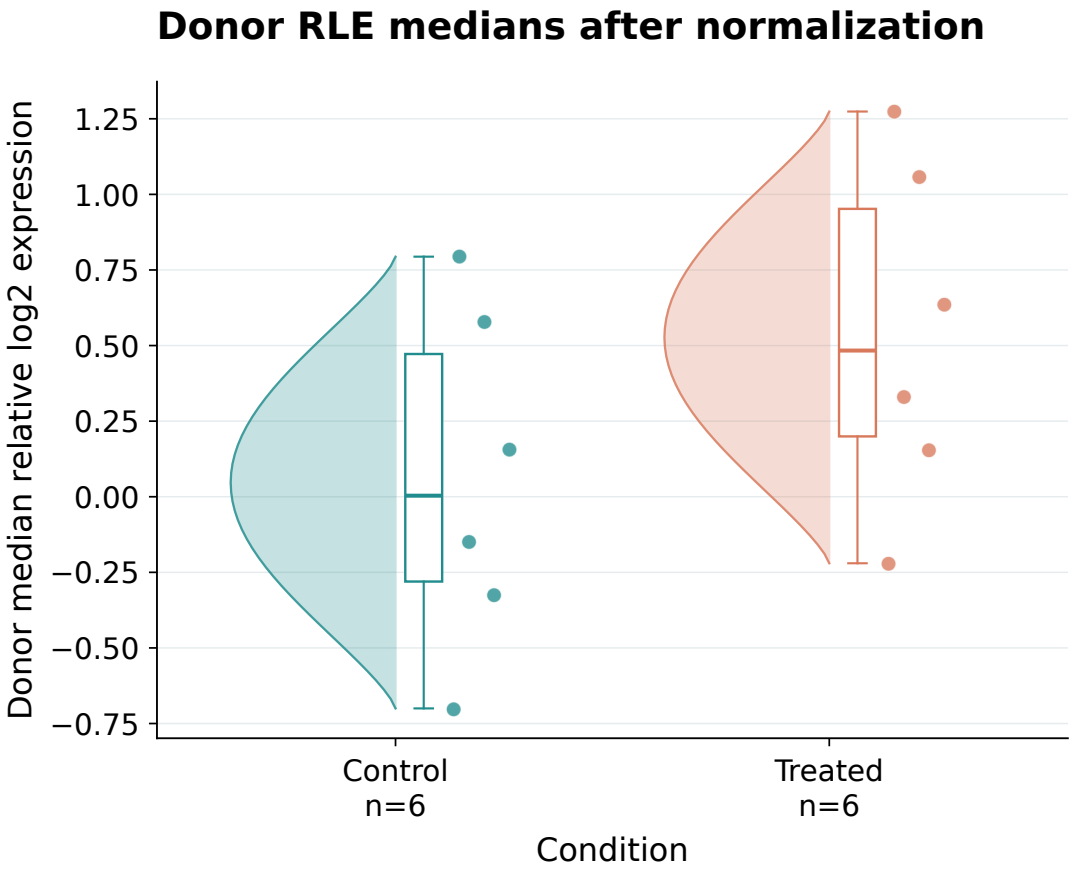
C Detection support across groups



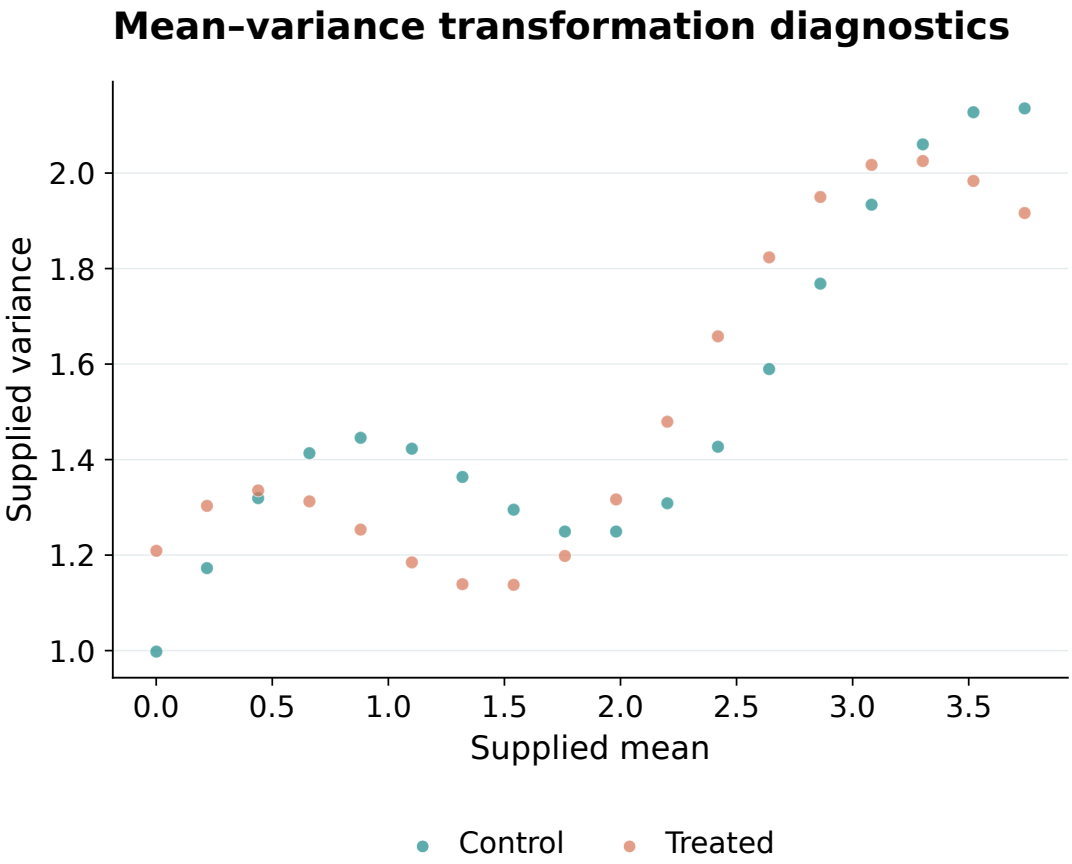


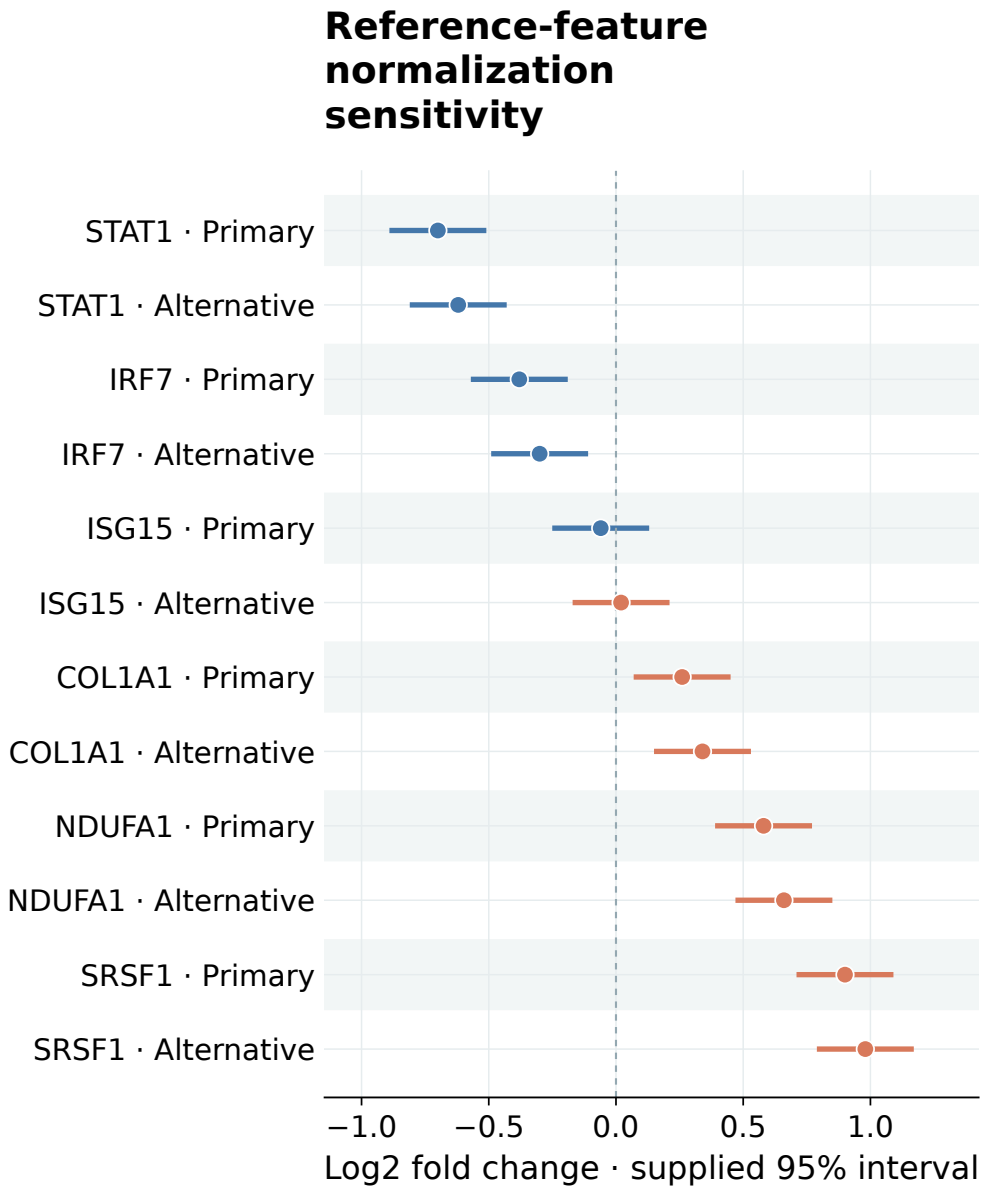
Software test fixture; not biological evidence


Paired donor RLE medians before/after normalization
Donor median relative log2 expression
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Before
n=12
After
n=12
Repeated condition



Software test fixture; not biological evidence


Mean–variance transformation diagnostics
Supplied variance
1.0
1.2
1.4
1.6
1.8
2.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Supplied mean
Control
Treated



Estimate [interval] · q
Independent n = 12

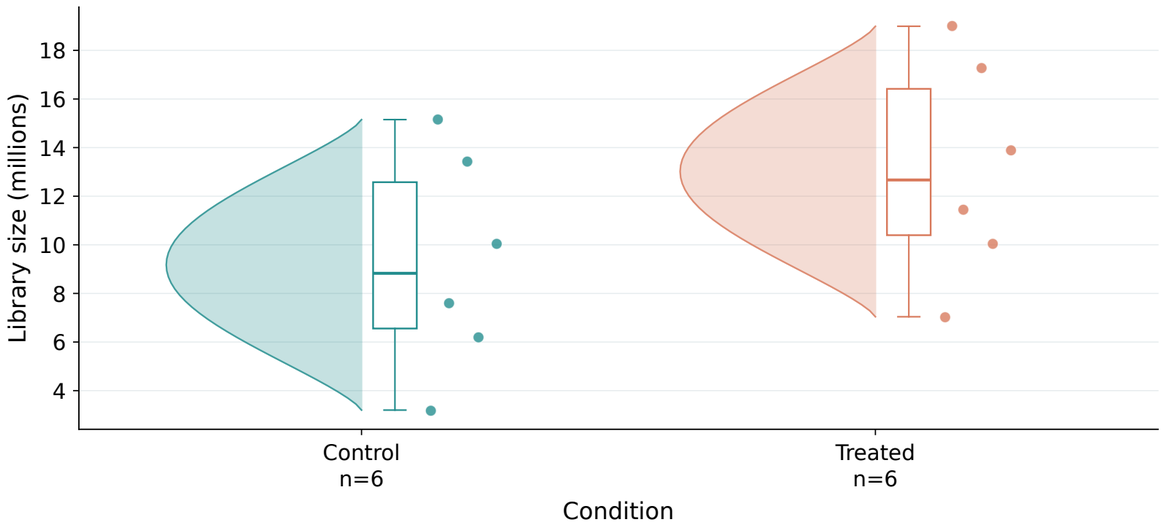
STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.38 [-0.56, -0.20] 0.02
IRF7 · Alternative	-0.30 [-0.48, -0.12] 0.023
ISG15 · Primary	-0.06 [-0.24, 0.12] 0.65
ISG15 · Alternative	0.02 [-0.16, 0.20] 0.65
COL1A1 · Primary	0.26 [0.08, 0.44] 0.056
COL1A1 · Alternative	0.34 [0.16, 0.52] 0.059
NDUFA1 · Primary	0.58 [0.40, 0.76] 0.074
NDUFA1 · Alternative	0.66 [0.48, 0.84] 0.077
SRSF1 · Primary	0.90 [0.72, 1.08] 0.092
SRSF1 · Alternative	0.98 [0.80, 1.16] 0.095

Software test fixture; not biological evidence

Bulk · normalization — complementary evidence

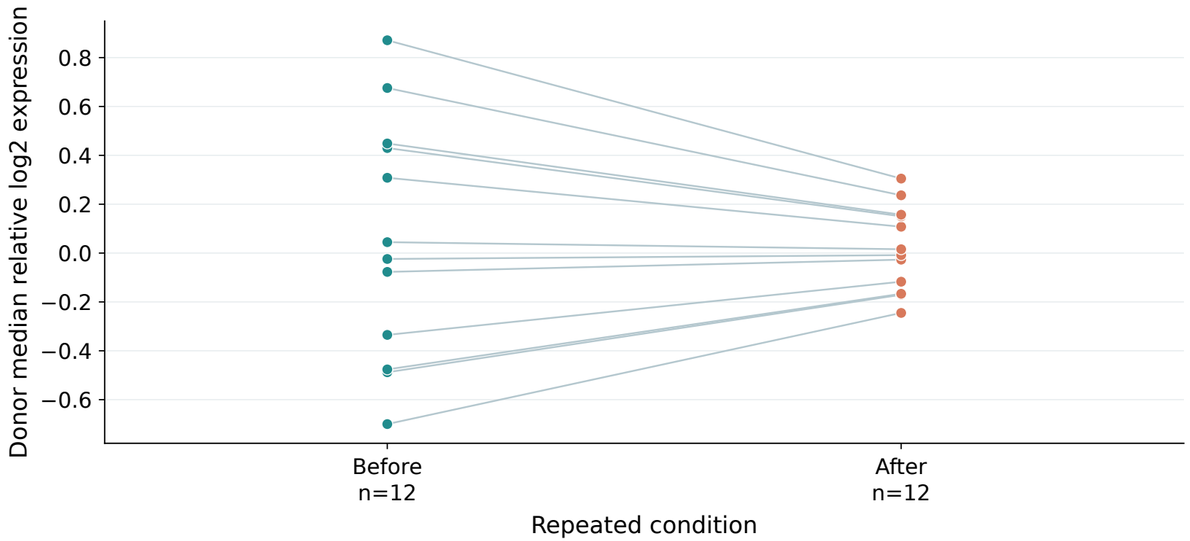
A

Library-size distributions by condition



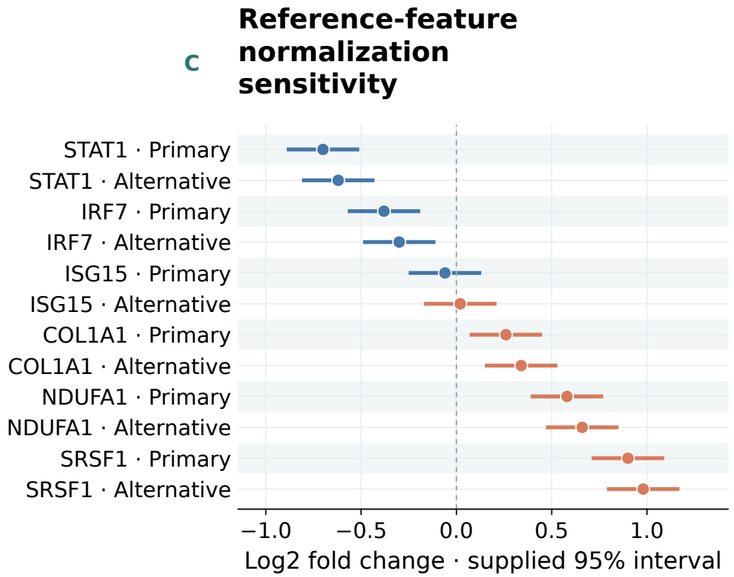
B

Paired donor RLE medians before/after normalization



C

Reference-feature normalization sensitivity



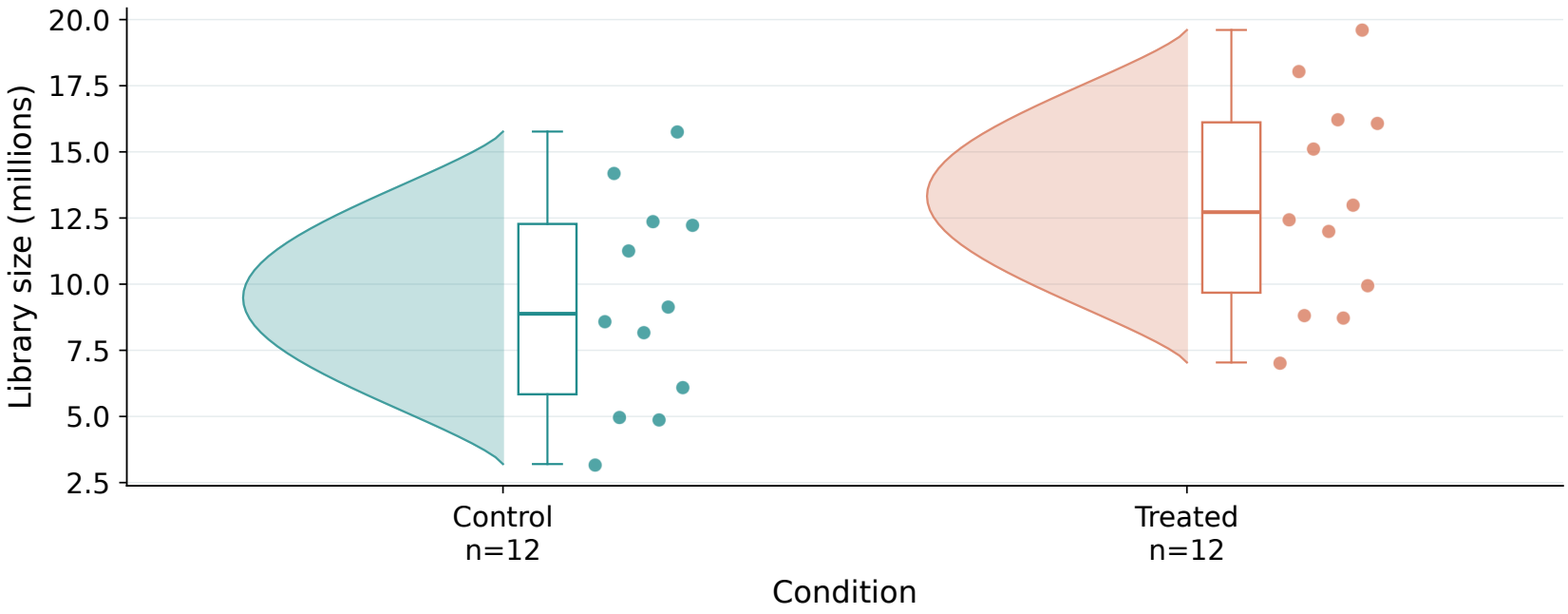
Estimate [interval] · q
Independent n = 12

STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.38 [-0.56, -0.20] 0.02
IRF7 · Alternative	-0.30 [-0.48, -0.12] 0.023
ISG15 · Primary	-0.06 [-0.24, 0.12] 0.65
ISG15 · Alternative	0.02 [-0.16, 0.20] 0.65
COL1A1 · Primary	0.26 [0.08, 0.44] 0.056
COL1A1 · Alternative	0.34 [0.16, 0.52] 0.059
NDUFA1 · Primary	0.58 [0.40, 0.76] 0.074
NDUFA1 · Alternative	0.66 [0.48, 0.84] 0.077
SRSF1 · Primary	0.90 [0.72, 1.08] 0.092
SRSF1 · Alternative	0.98 [0.80, 1.16] 0.095

Bulk · normalization — complementary evidence

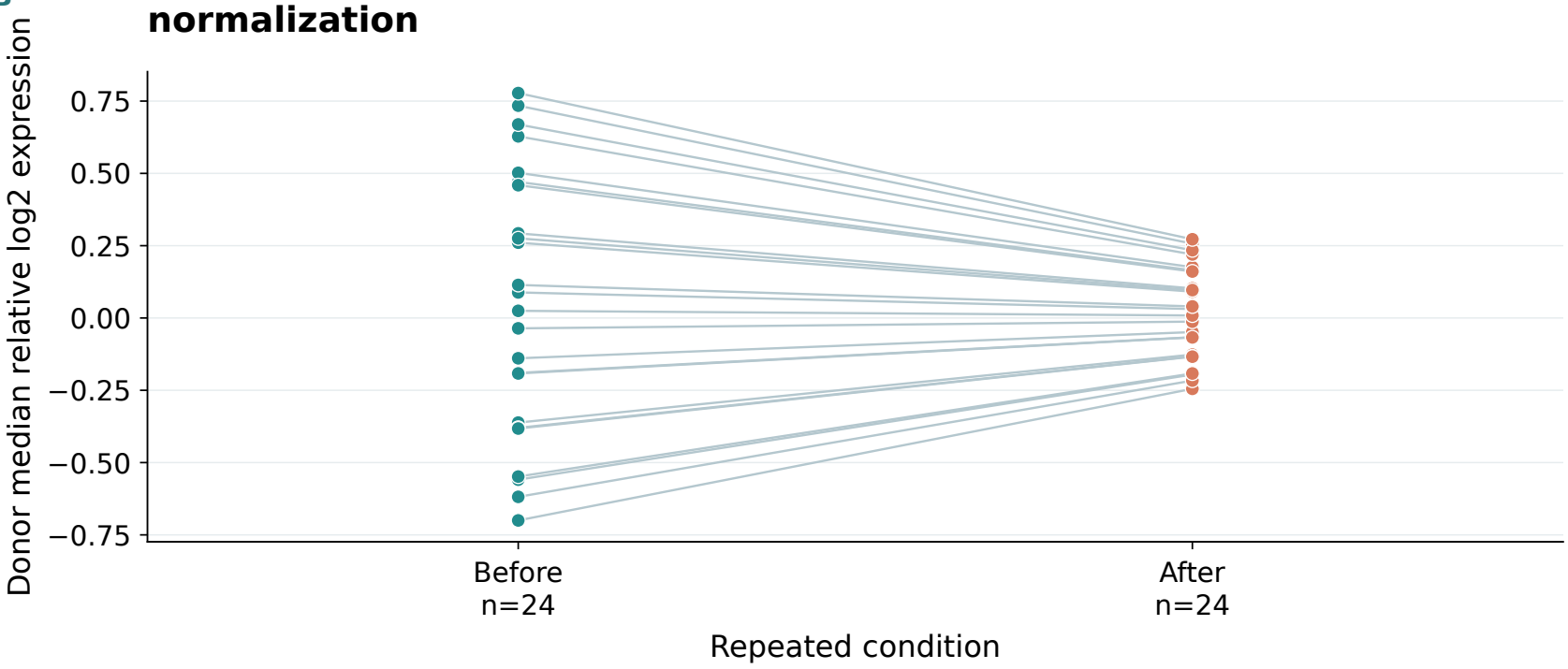
A

Library-size distributions by condition



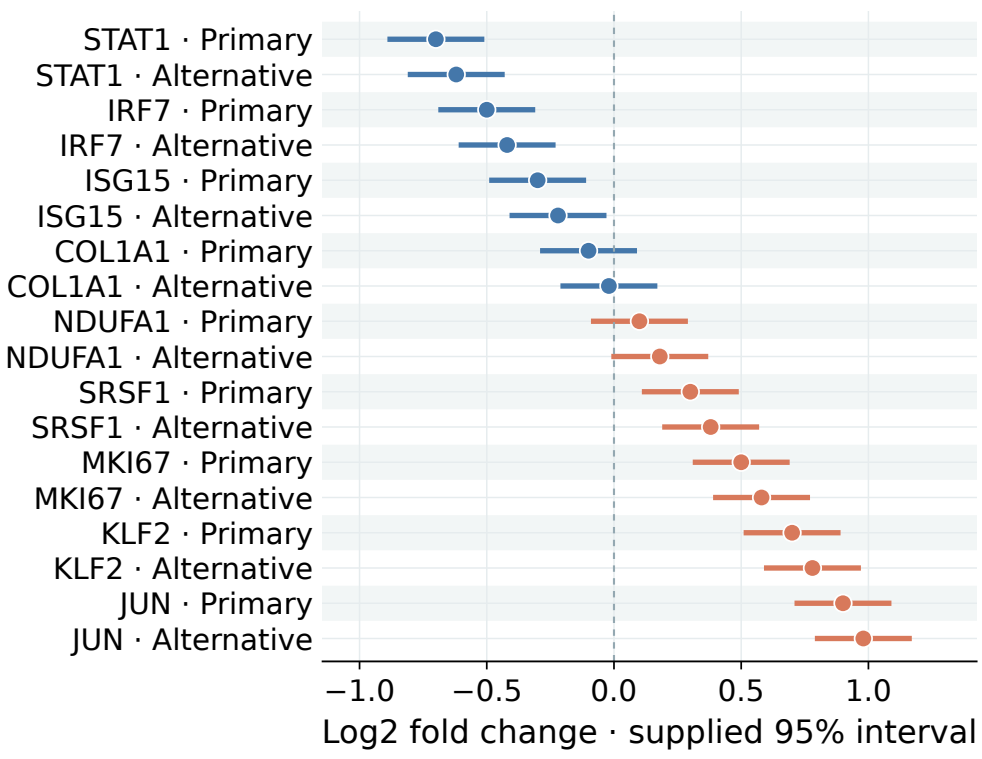
B

Paired donor RLE medians before/after normalization



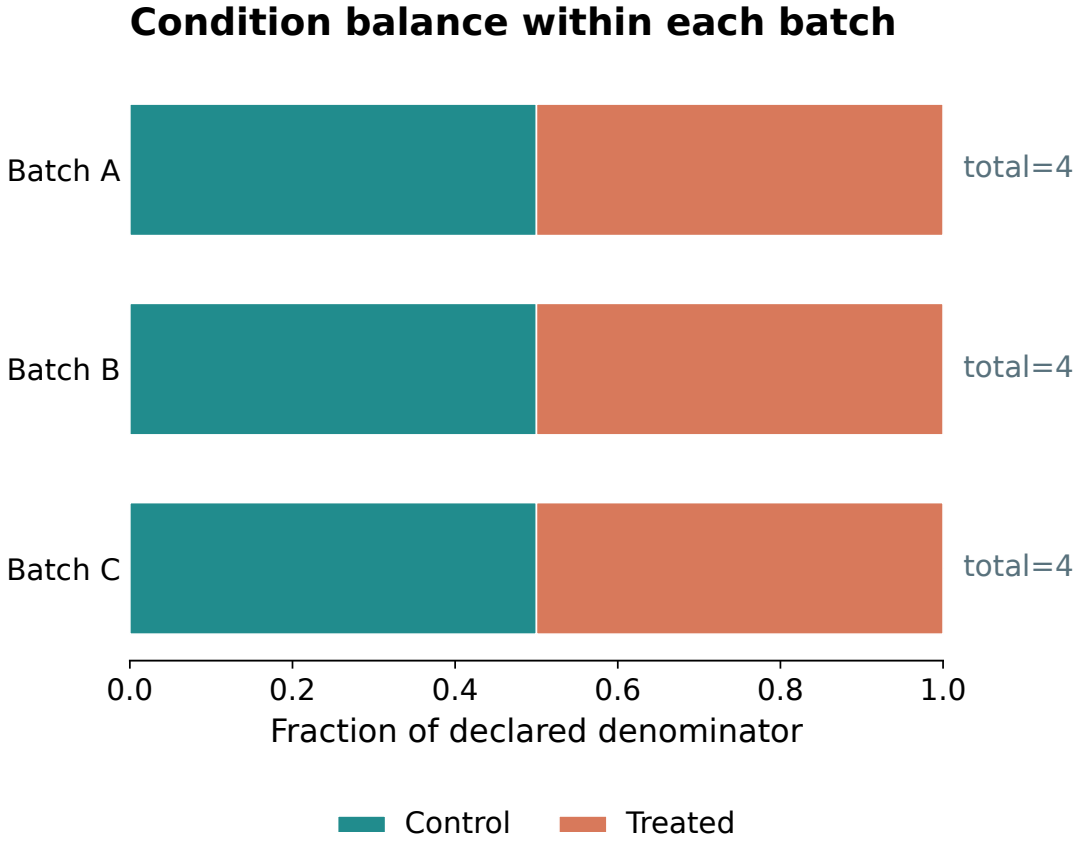
C

Reference-feature normalization sensitivity

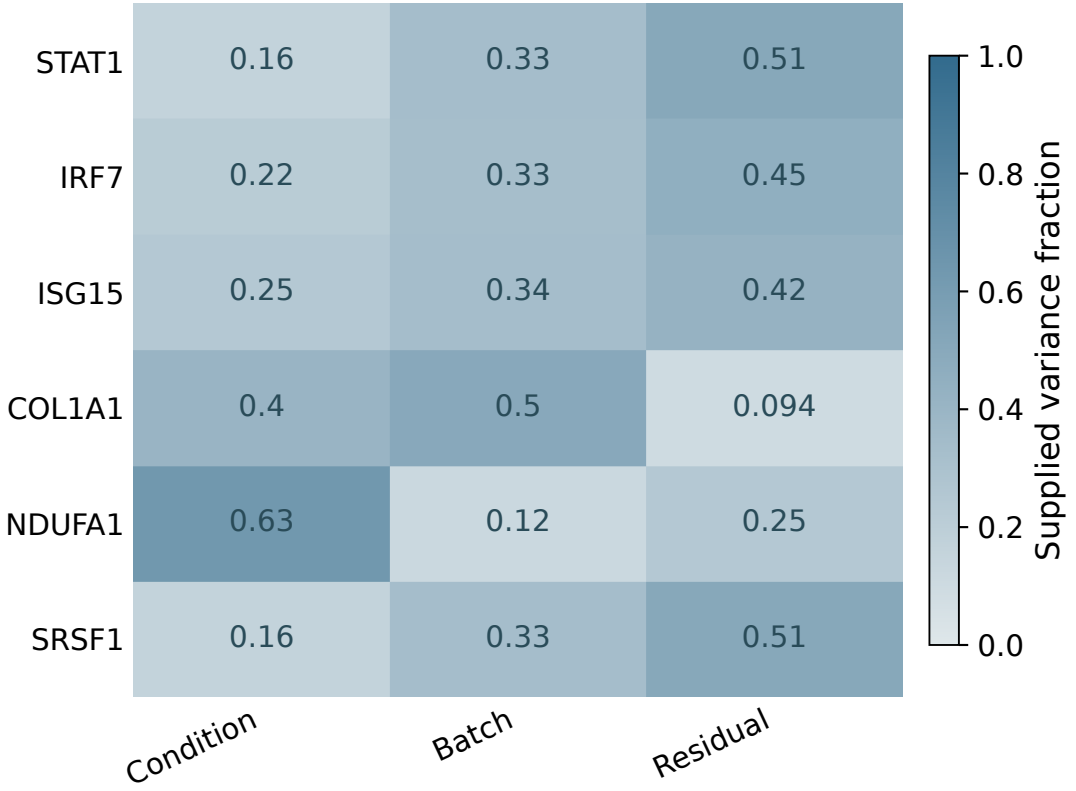


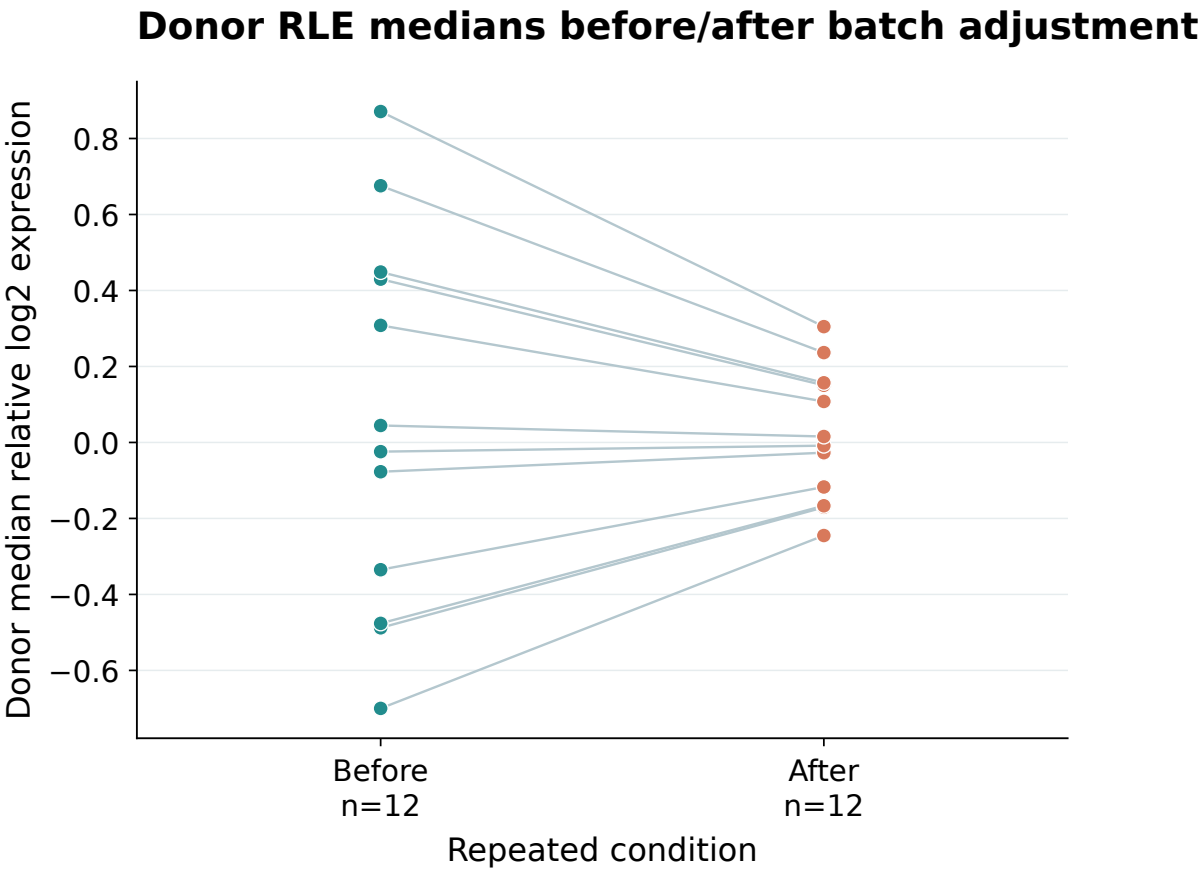
Estimate [interval] · q
Independent n = 24

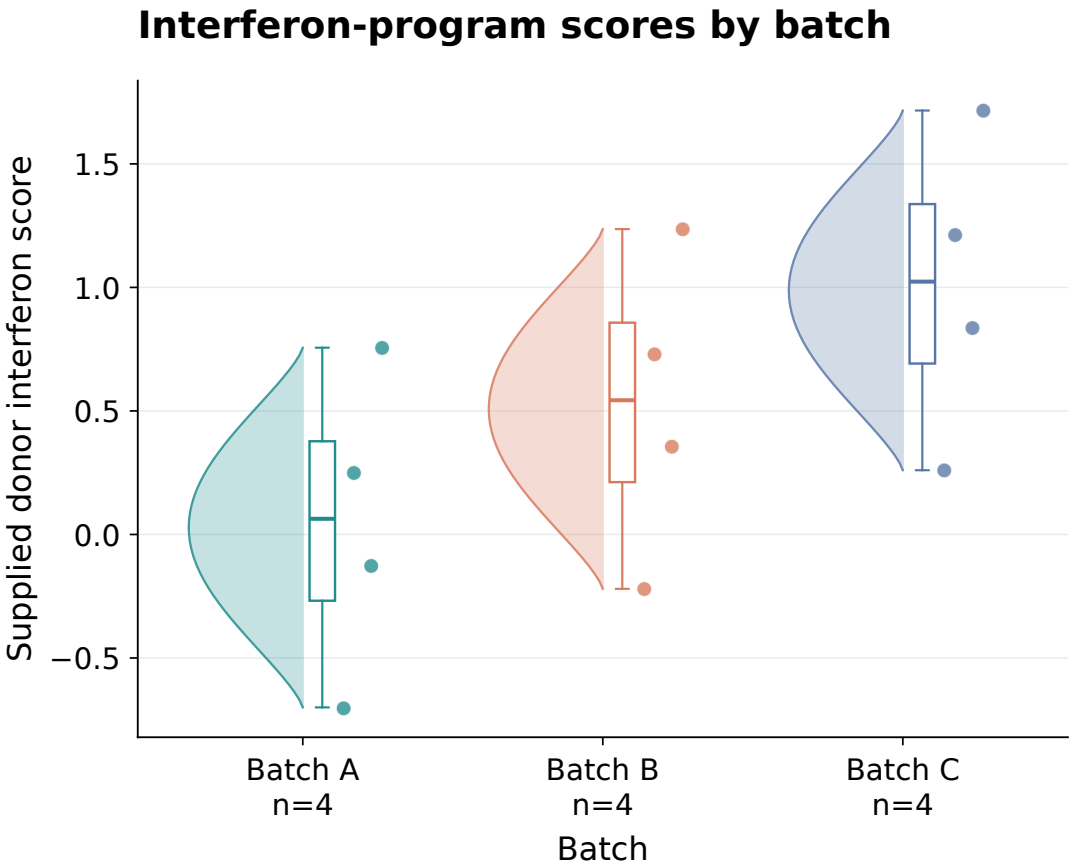
STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.50 [-0.68, -0.32] 0.02
IRF7 · Alternative	-0.42 [-0.60, -0.24] 0.023
ISG15 · Primary	-0.30 [-0.48, -0.12] 0.038
ISG15 · Alternative	-0.22 [-0.40, -0.04] 0.041
COL1A1 · Primary	-0.10 [-0.28, 0.08] 0.65
COL1A1 · Alternative	-0.02 [-0.20, 0.16] 0.65
NDUFA1 · Primary	0.10 [-0.08, 0.28] 0.65
NDUFA1 · Alternative	0.18 [0.00, 0.36] 0.077
SRSF1 · Primary	0.30 [0.12, 0.48] 0.092
SRSF1 · Alternative	0.38 [0.20, 0.56] 0.095
MKI67 · Primary	0.50 [0.32, 0.68] 0.11
MKI67 · Alternative	0.58 [0.40, 0.76] 0.11
KLF2 · Primary	0.70 [0.52, 0.88] 0.13
KLF2 · Alternative	0.78 [0.60, 0.96] 0.13
JUN · Primary	0.90 [0.72, 1.08] 0.15
JUN · Alternative	0.98 [0.80, 1.16] 0.15


Condition balance within each batch
Batch A
Batch B
Batch C
total=4
total=4
total=4
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Control
Treated

Complete covariate-associated variance

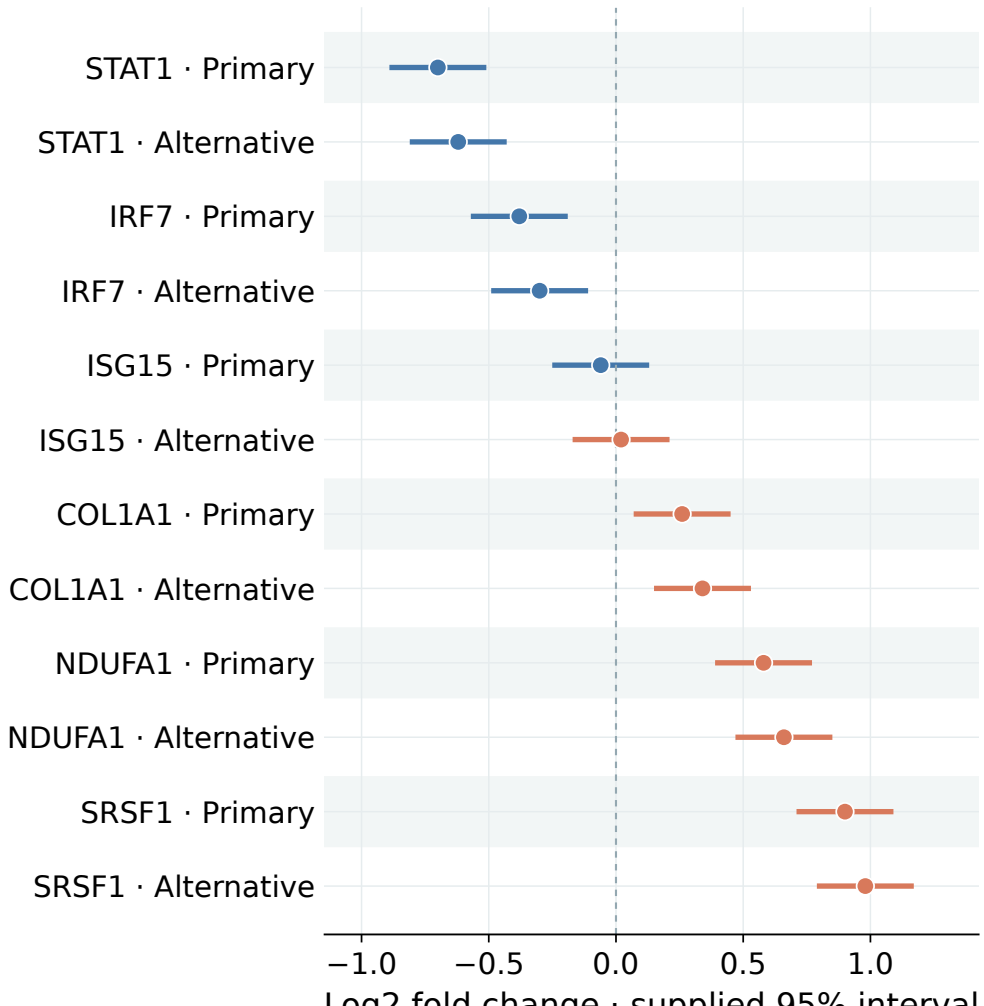



Donor RLE medians before/after batch adjustment
Donor median relative log2 expression
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Before
n=12
After
n=12
Repeated condition



Software test fixture; not biological evidence

Biological-effect preservation across choices

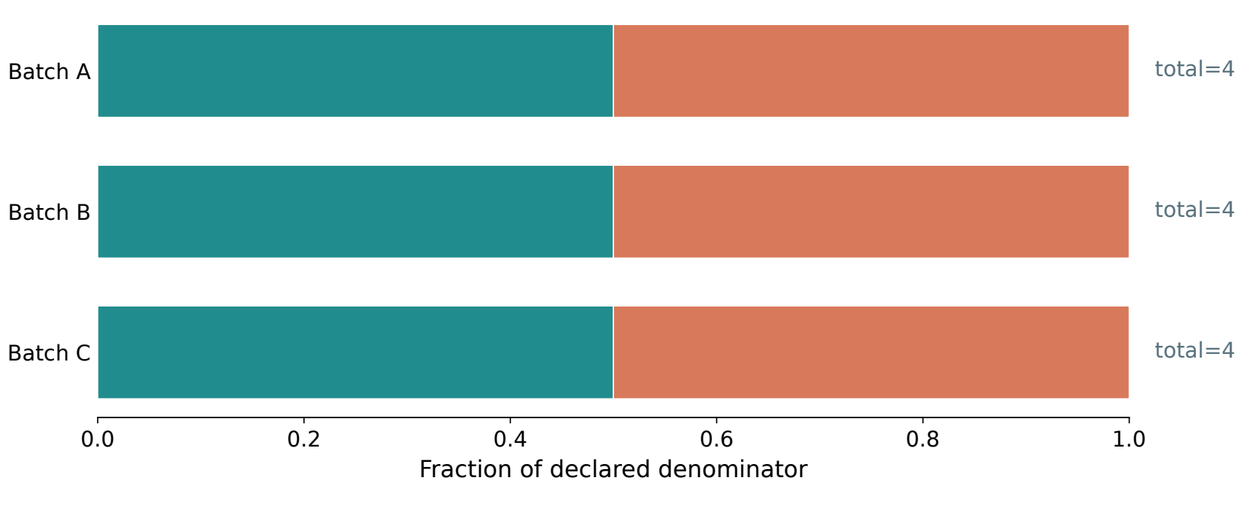


Estimate [interval] · q
Independent n = 12

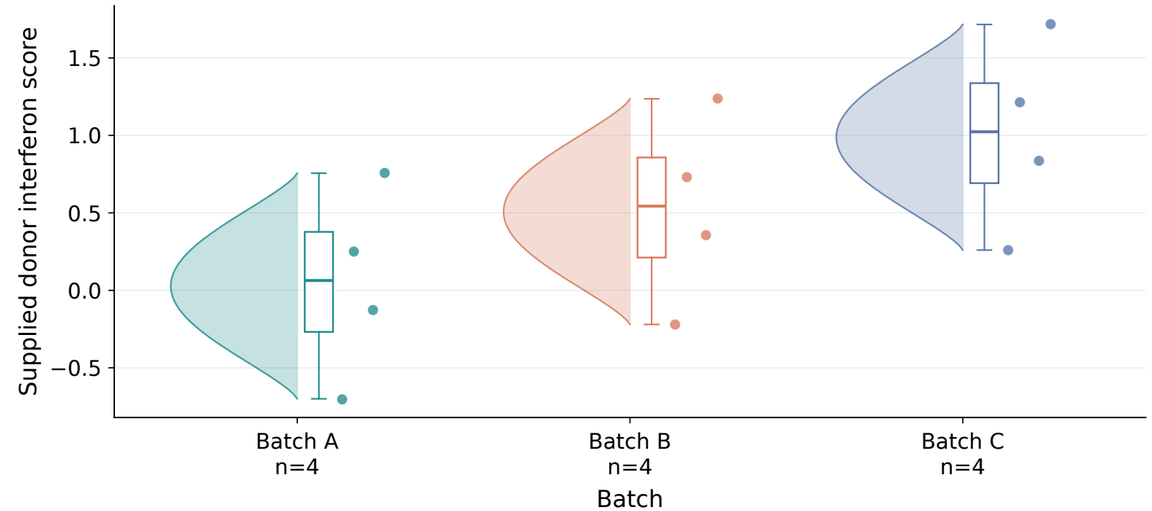
STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.38 [-0.56, -0.20] 0.02
IRF7 · Alternative	-0.30 [-0.48, -0.12] 0.023
ISG15 · Primary	-0.06 [-0.24, 0.12] 0.65
ISG15 · Alternative	0.02 [-0.16, 0.20] 0.65
COL1A1 · Primary	0.26 [0.08, 0.44] 0.056
COL1A1 · Alternative	0.34 [0.16, 0.52] 0.059
NDUFA1 · Primary	0.58 [0.40, 0.76] 0.074
NDUFA1 · Alternative	0.66 [0.48, 0.84] 0.077
SRSF1 · Primary	0.90 [0.72, 1.08] 0.092
SRSF1 · Alternative	0.98 [0.80, 1.16] 0.095

Bulk · batch — complementary evidence

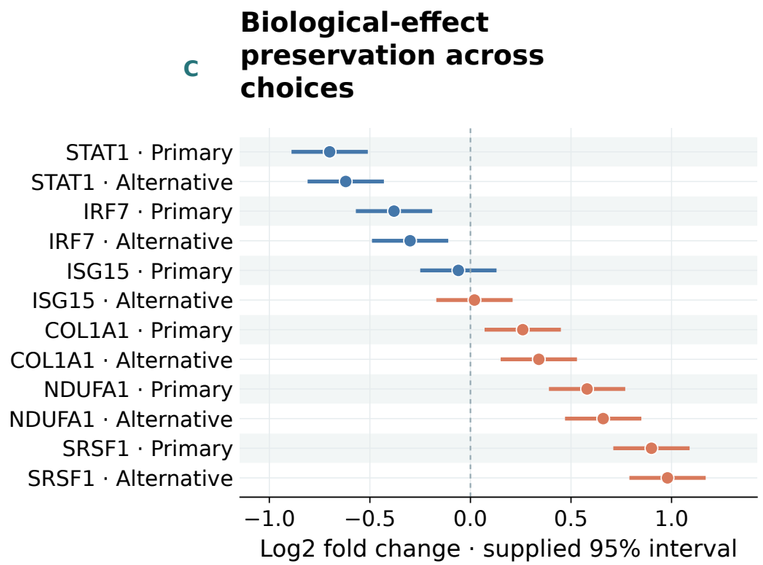
A



B



C



Estimate [interval] · q
Independent n = 12

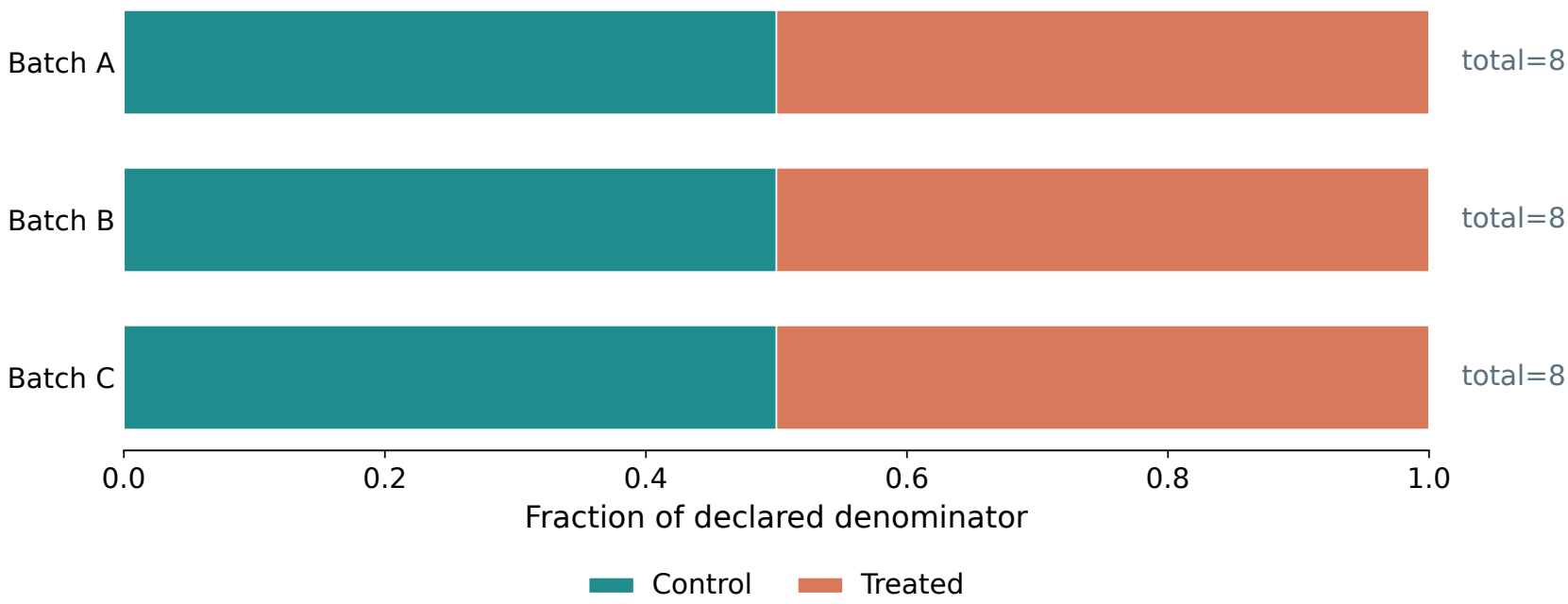
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095

Software test fixture; not biological evidence

Bulk · batch — complementary evidence

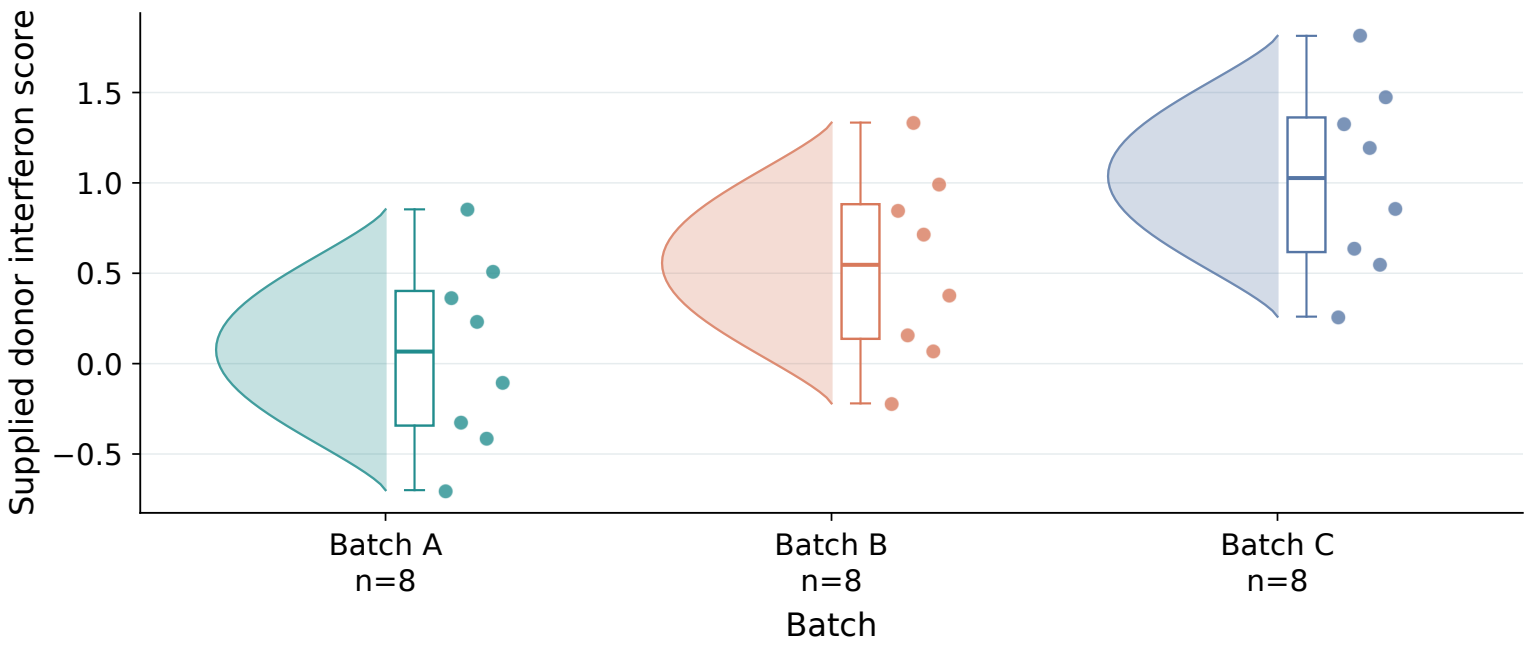
A

Condition balance within each batch



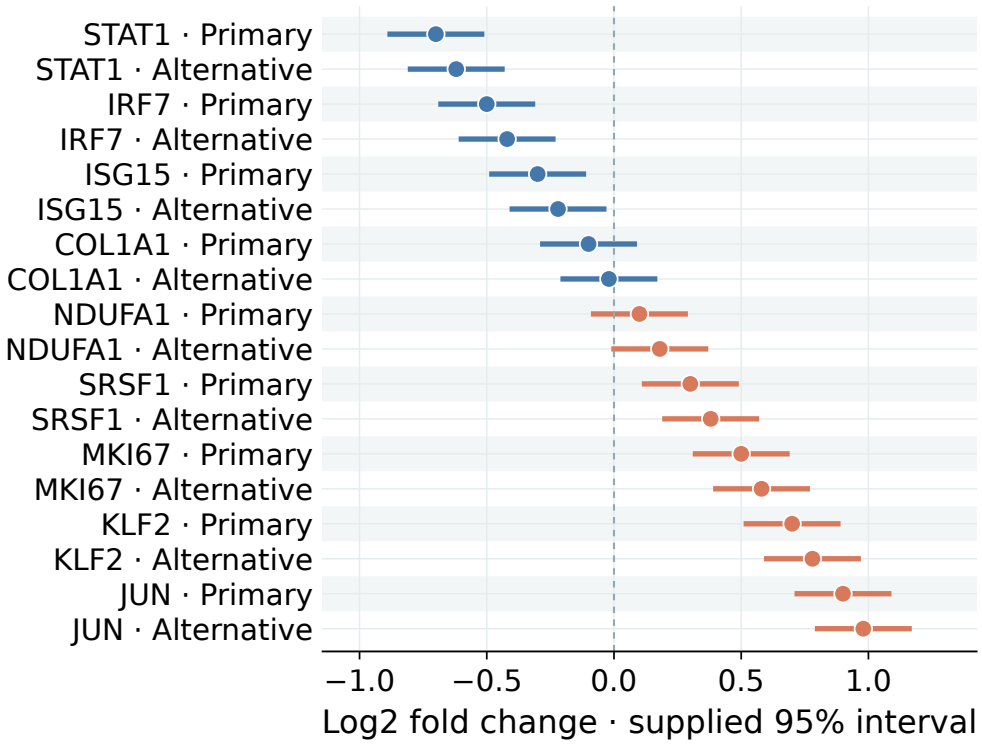
B

Interferon-program scores by batch



C

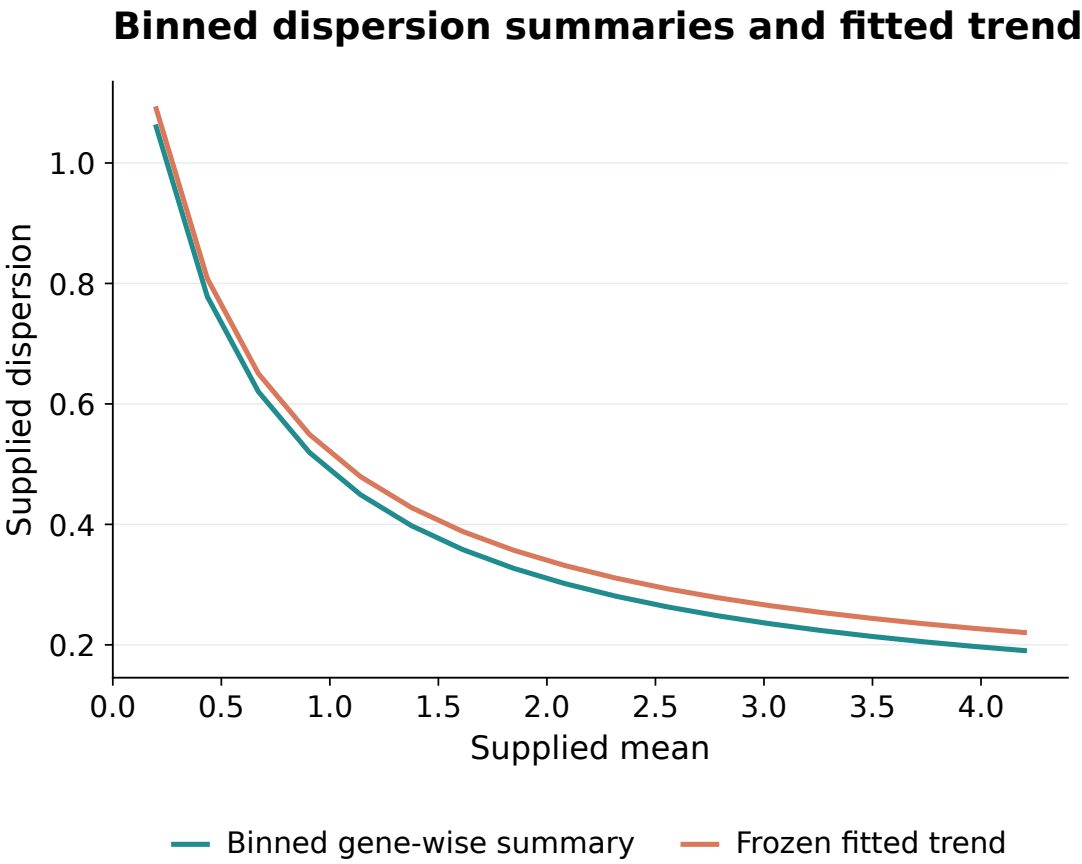
Biological-effect preservation across choices

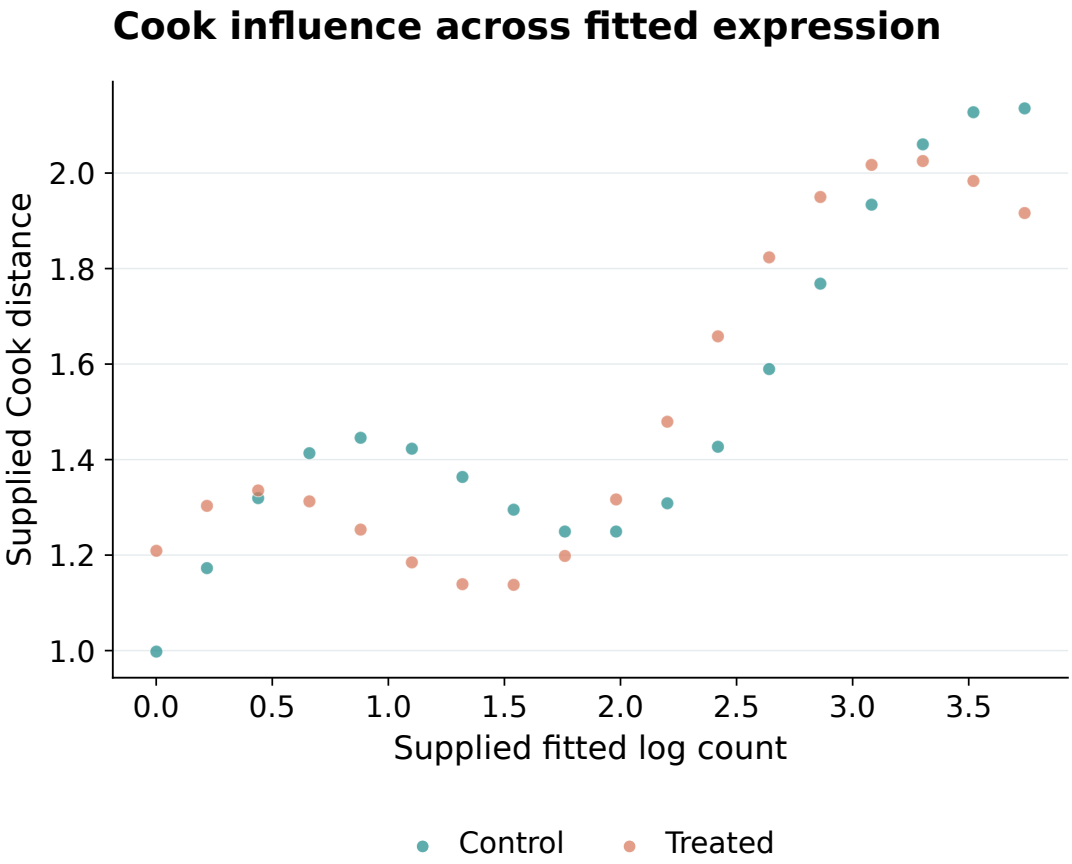


Estimate [interval] · q
Independent n = 24

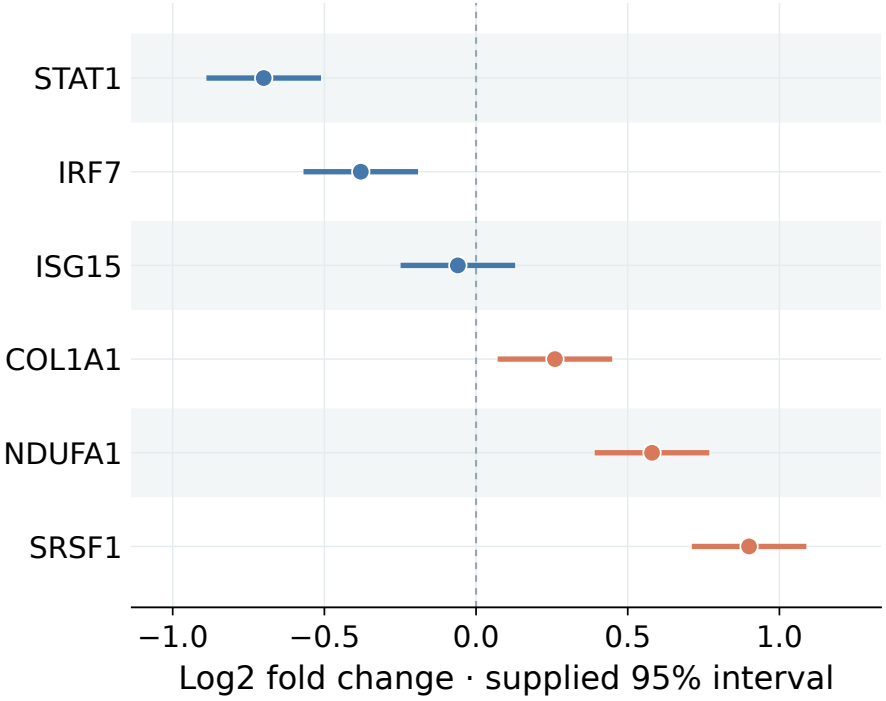
STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.50 [-0.68, -0.32] 0.02
IRF7 · Alternative	-0.42 [-0.60, -0.24] 0.023
ISG15 · Primary	-0.30 [-0.48, -0.12] 0.038
ISG15 · Alternative	-0.22 [-0.40, -0.04] 0.041
COL1A1 · Primary	-0.10 [-0.28, 0.08] 0.65
COL1A1 · Alternative	-0.02 [-0.20, 0.16] 0.65
NDUFA1 · Primary	0.10 [-0.08, 0.28] 0.65
NDUFA1 · Alternative	0.18 [0.00, 0.36] 0.077
SRSF1 · Primary	0.30 [0.12, 0.48] 0.092
SRSF1 · Alternative	0.38 [0.20, 0.56] 0.095
MKI67 · Primary	0.50 [0.32, 0.68] 0.11
MKI67 · Alternative	0.58 [0.40, 0.76] 0.11
KLF2 · Primary	0.70 [0.52, 0.88] 0.13
KLF2 · Alternative	0.78 [0.60, 0.96] 0.13
JUN · Primary	0.90 [0.72, 1.08] 0.15
JUN · Alternative	0.98 [0.80, 1.16] 0.15

Software test fixture; not biological evidence


Binned dispersion summaries and fitted trend
Supplied dispersion
1.0
0.8
0.6
0.4
0.2
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Supplied mean
Binned gene-wise summary
Frozen fitted trend

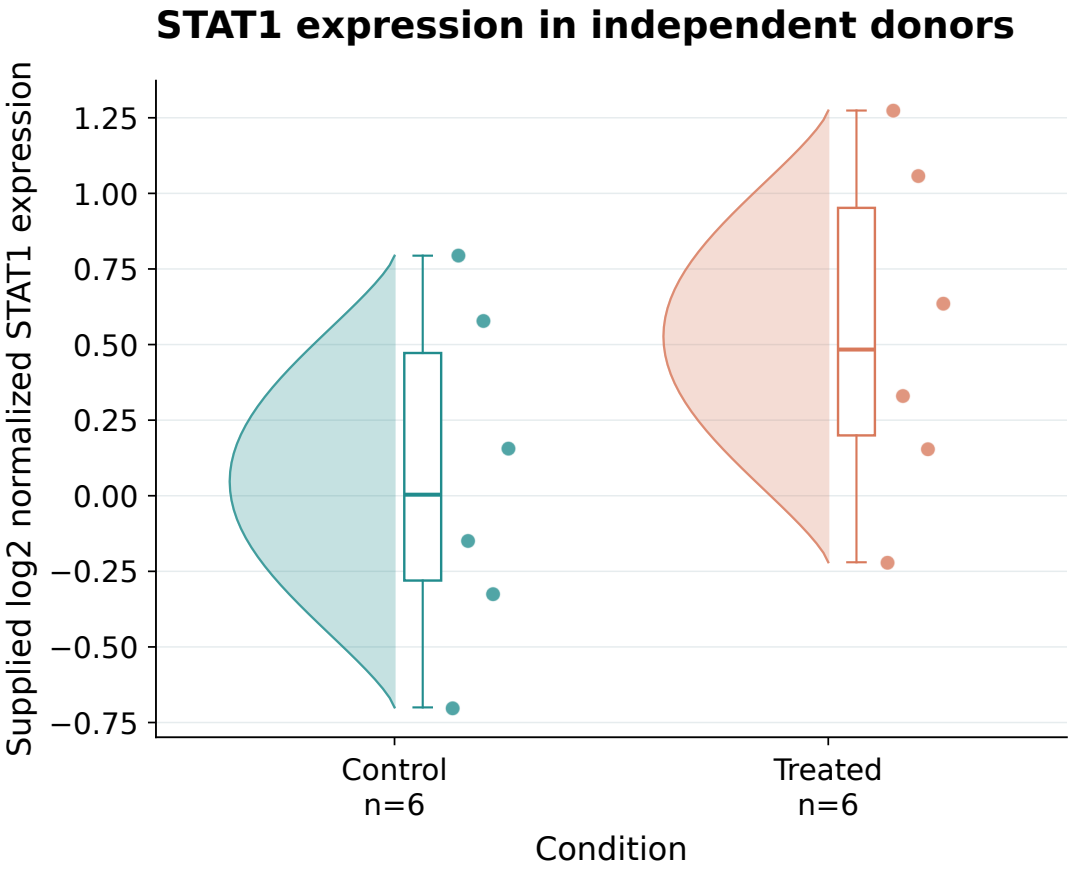

Cook influence across fitted expression
Supplied Cook distance
1.0
1.2
1.4
1.6
1.8
2.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Supplied fitted log count
Control
Treated

Prespecified gene effects with uncertainty

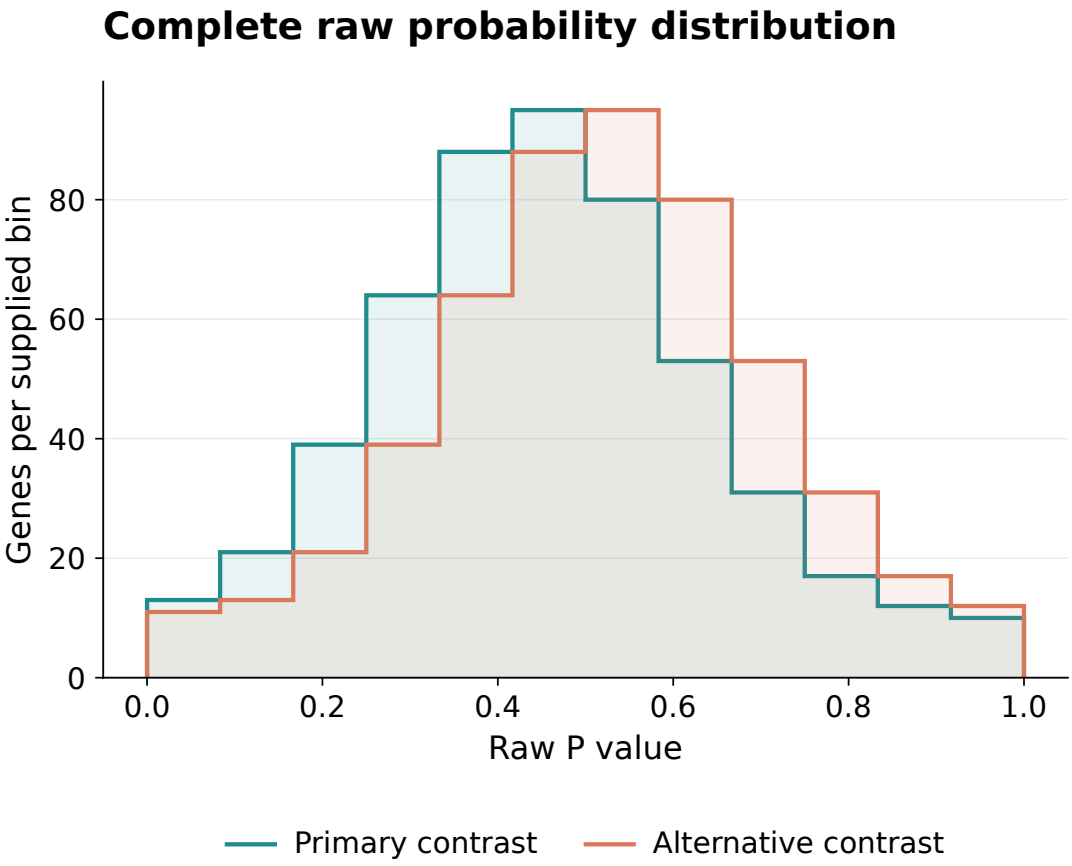


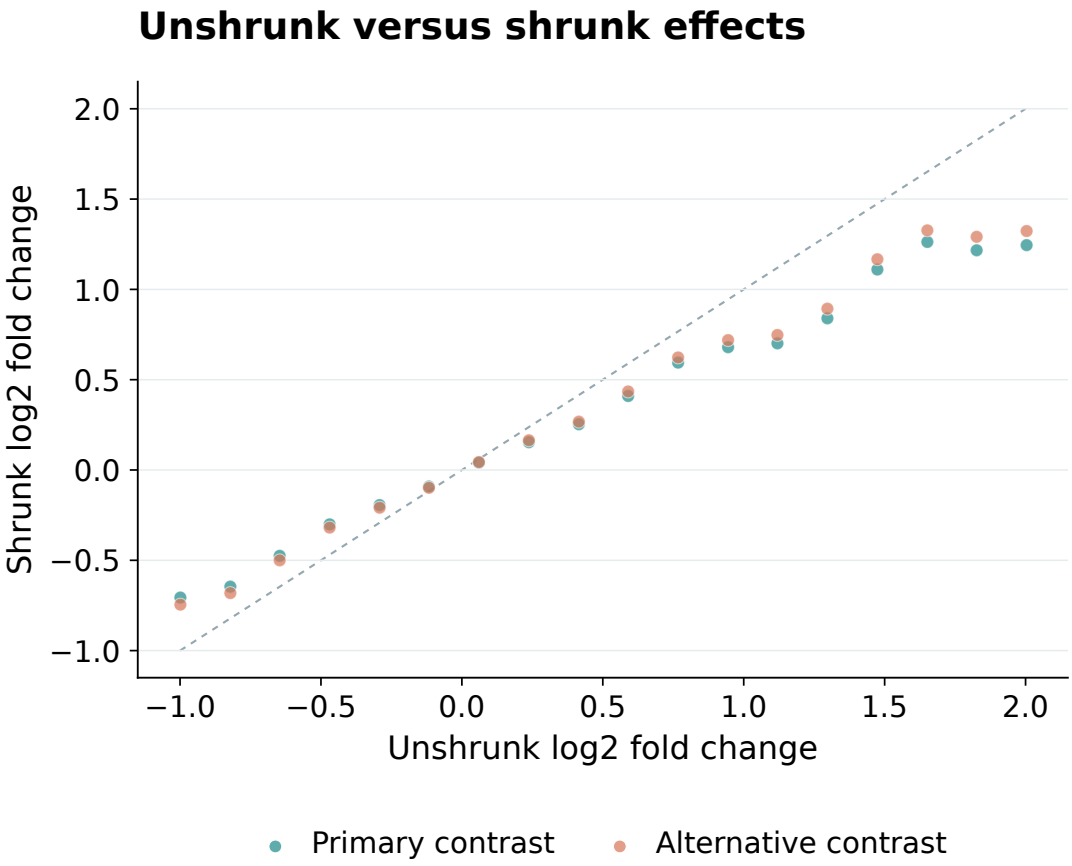
Estimate [interval] · q
Independent n = 12

STAT1	-0.70 [-0.88, -0.52] 0.002
IRF7	-0.38 [-0.56, -0.20] 0.02
ISG15	-0.06 [-0.24, 0.12] 0.65
COL1A1	0.26 [0.08, 0.44] 0.056
NDUFA1	0.58 [0.40, 0.76] 0.074
SRSF1	0.90 [0.72, 1.08] 0.092

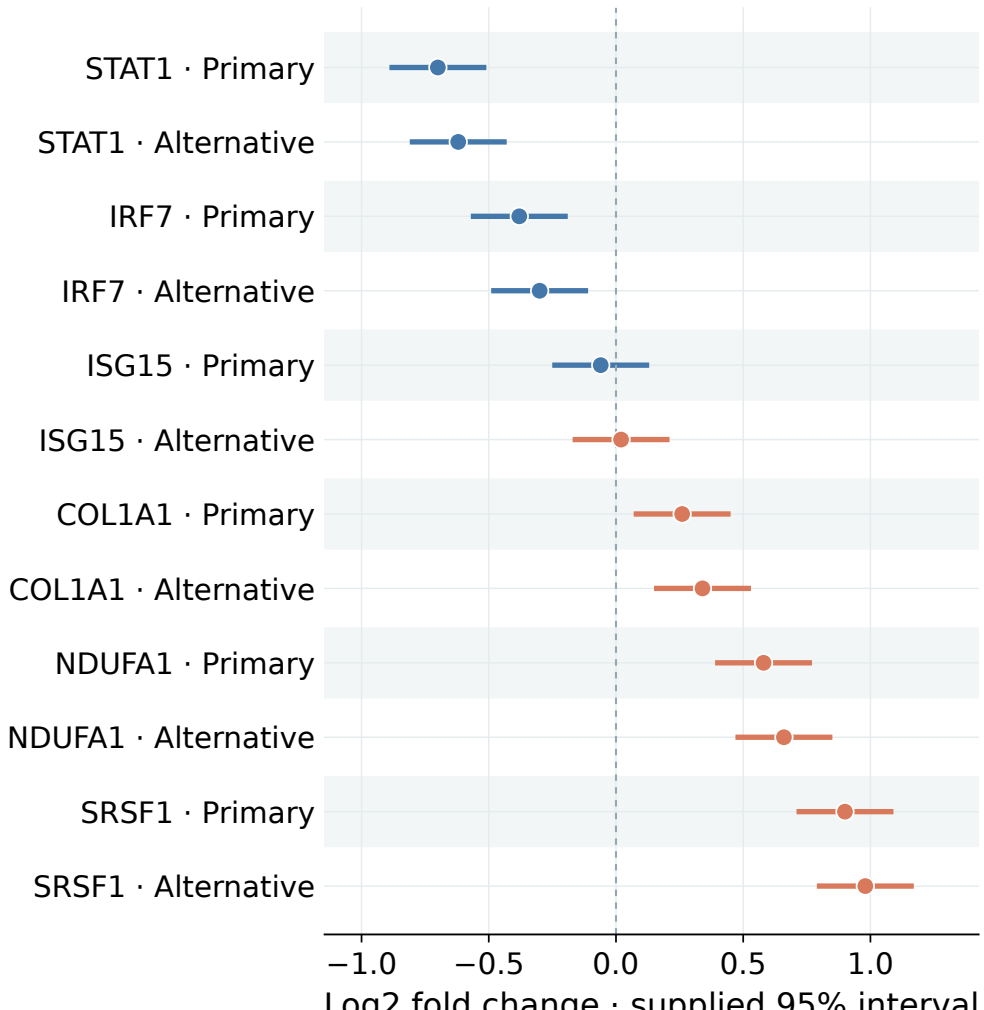


Software test fixture; not biological evidence


Complete raw probability distribution
Genes per supplied bin
0
20
40
60
80
0.0
0.2
0.4
0.6
0.8
1.0
Raw P value
Primary contrast
Alternative contrast


Unshrunk versus shrunk effects
Shrunk log2 fold change
Unshrunk log2 fold change
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Primary contrast
Alternative contrast

Effect stability across prespecified models



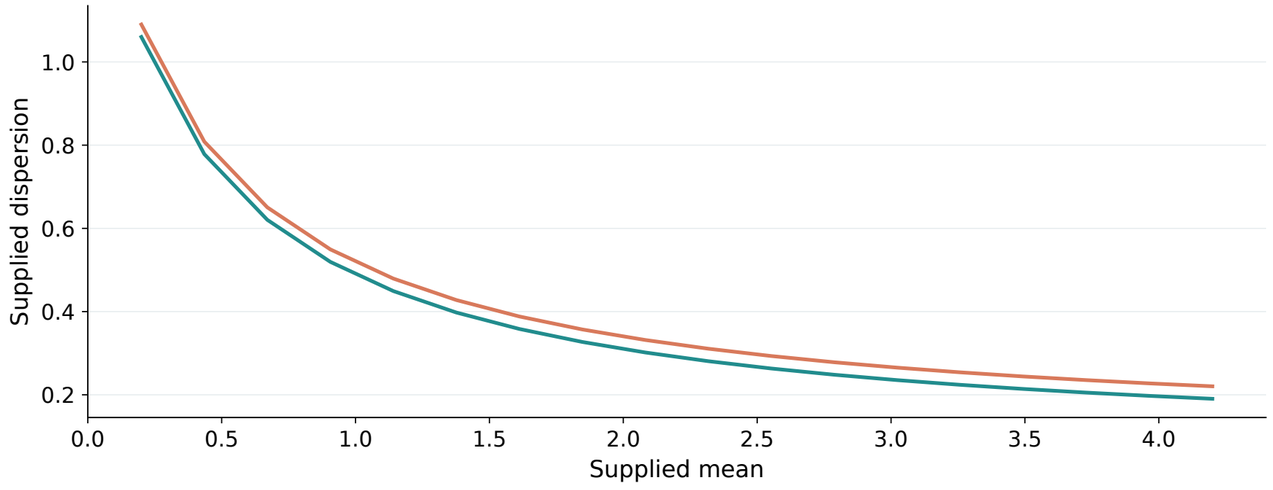
Estimate [interval] · q
Independent n = 12

STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.38 [-0.56, -0.20] 0.02
IRF7 · Alternative	-0.30 [-0.48, -0.12] 0.023
ISG15 · Primary	-0.06 [-0.24, 0.12] 0.65
ISG15 · Alternative	0.02 [-0.16, 0.20] 0.65
COL1A1 · Primary	0.26 [0.08, 0.44] 0.056
COL1A1 · Alternative	0.34 [0.16, 0.52] 0.059
NDUFA1 · Primary	0.58 [0.40, 0.76] 0.074
NDUFA1 · Alternative	0.66 [0.48, 0.84] 0.077
SRSF1 · Primary	0.90 [0.72, 1.08] 0.092
SRSF1 · Alternative	0.98 [0.80, 1.16] 0.095

Bulk · differential — complementary evidence

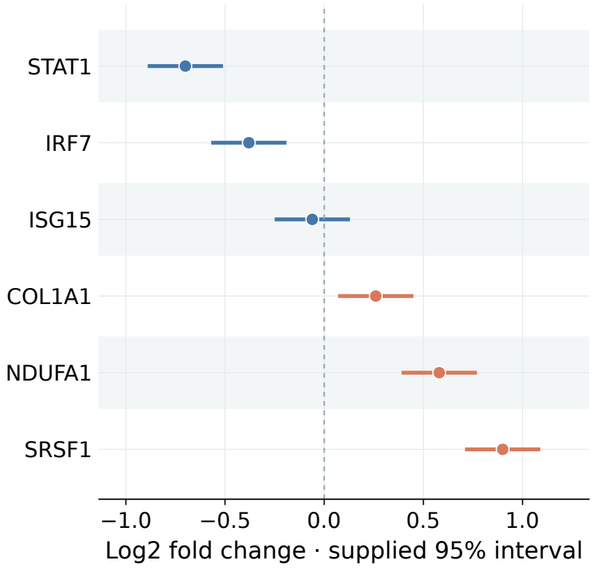
A

Binned dispersion summaries and fitted trend



B

Prespecified gene effects with uncertainty

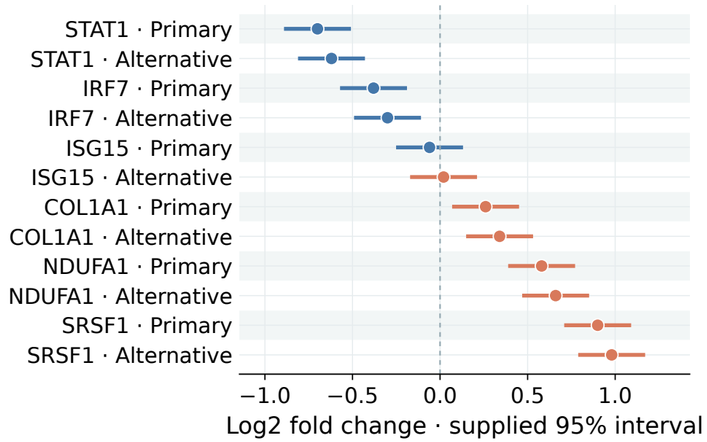


Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002
-0.38 [-0.56, -0.20] 0.02
-0.06 [-0.24, 0.12] 0.65
0.26 [0.08, 0.44] 0.056
0.58 [0.40, 0.76] 0.074
0.90 [0.72, 1.08] 0.092

C

Effect stability across prespecified models



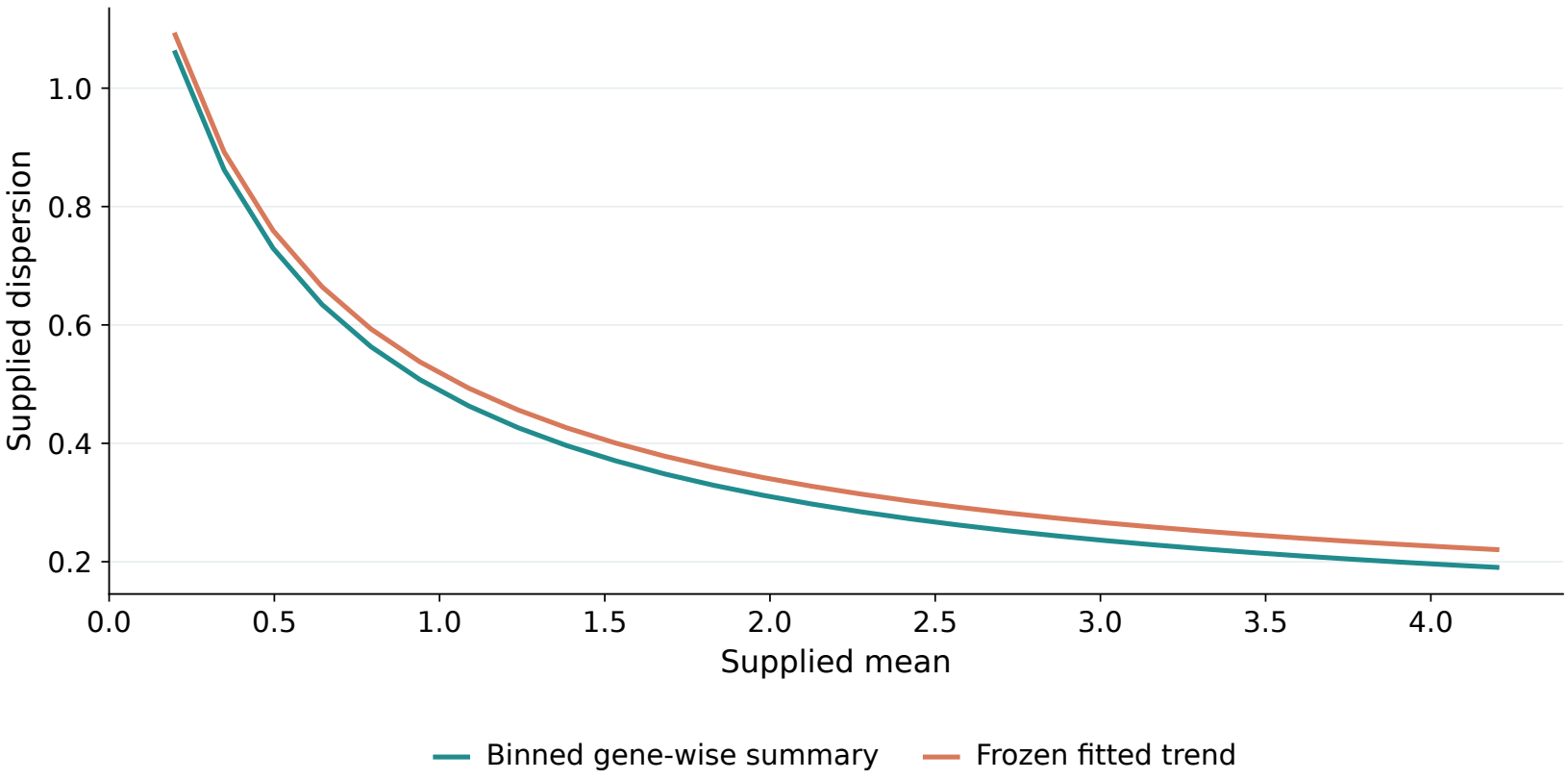
Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095

Bulk · differential — complementary evidence

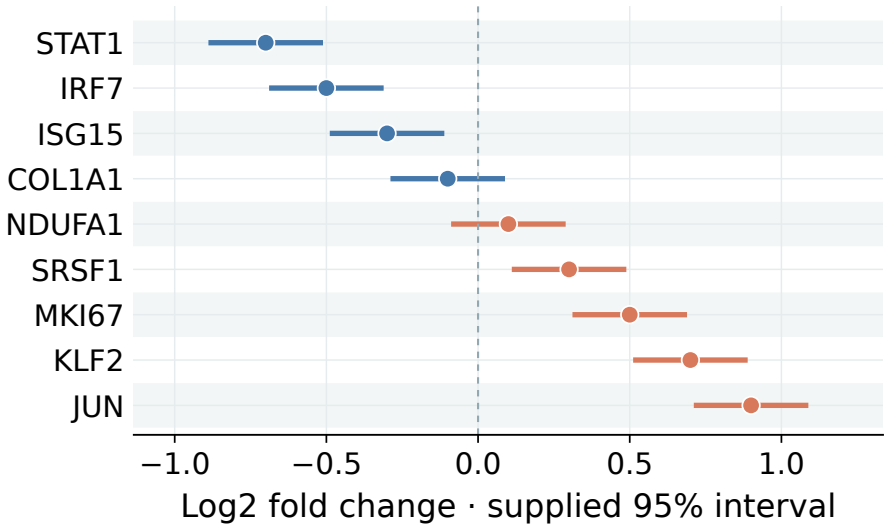
A

Binned dispersion summaries and fitted trend



B

Prespecified gene effects with uncertainty

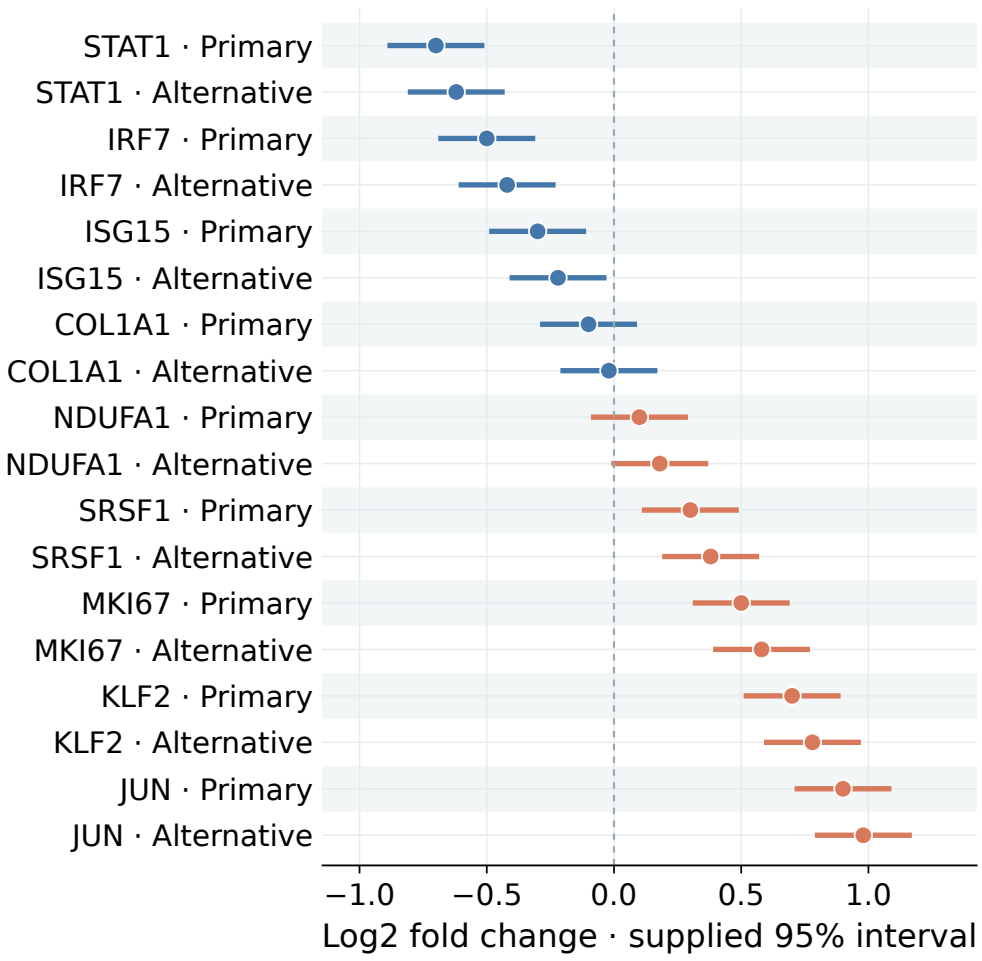


Estimate [interval] · q
Independent n = 24

-0.70	[-0.88, -0.52]	0.002
-0.50	[-0.68, -0.32]	0.02
-0.30	[-0.48, -0.12]	0.038
-0.10	[-0.28, 0.08]	0.65
0.10	[-0.08, 0.28]	0.65
0.30	[0.12, 0.48]	0.092
0.50	[0.32, 0.68]	0.11
0.70	[0.52, 0.88]	0.13
0.90	[0.72, 1.08]	0.15

C

Effect stability across prespecified models

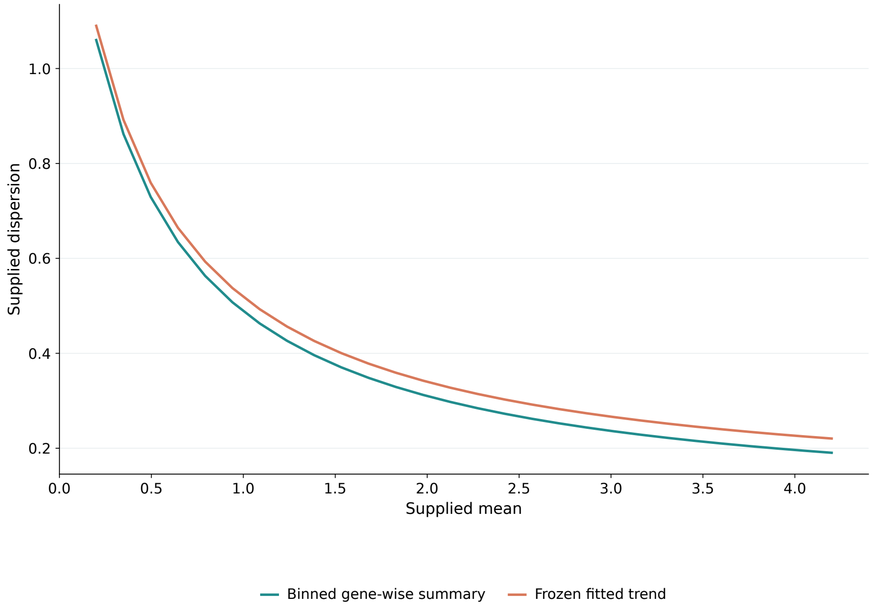


Estimate [interval] · q
Independent n = 24

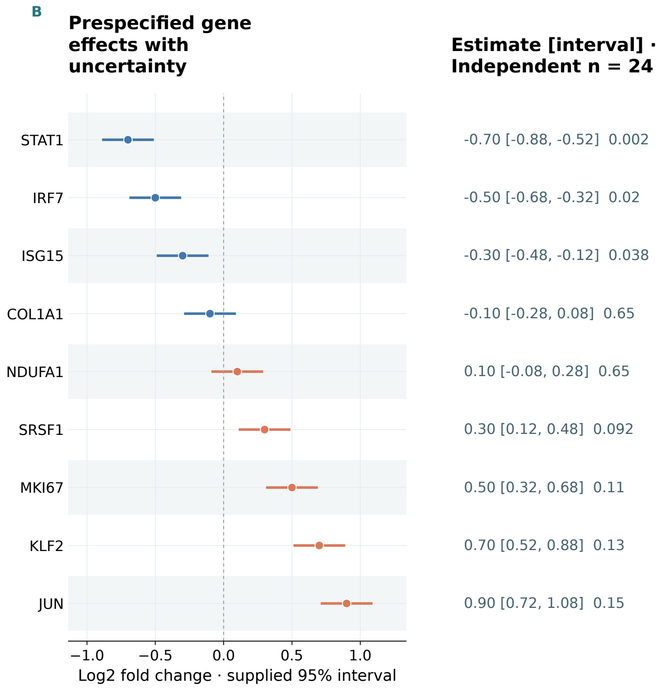
-0.70	[-0.88, -0.52]	0.002
-0.62	[-0.80, -0.44]	0.005
-0.50	[-0.68, -0.32]	0.02
-0.42	[-0.60, -0.24]	0.023
-0.30	[-0.48, -0.12]	0.038
-0.22	[-0.40, -0.04]	0.041
-0.10	[-0.28, 0.08]	0.65
-0.02	[-0.20, 0.16]	0.65
0.10	[-0.08, 0.28]	0.65
0.18	[0.00, 0.36]	0.077
0.30	[0.12, 0.48]	0.092
0.38	[0.20, 0.56]	0.095
0.50	[0.32, 0.68]	0.11
0.58	[0.40, 0.76]	0.11
0.70	[0.52, 0.88]	0.13
0.78	[0.60, 0.96]	0.13
0.90	[0.72, 1.08]	0.15
0.98	[0.80, 1.16]	0.15

Bulk · differential — complementary evidence

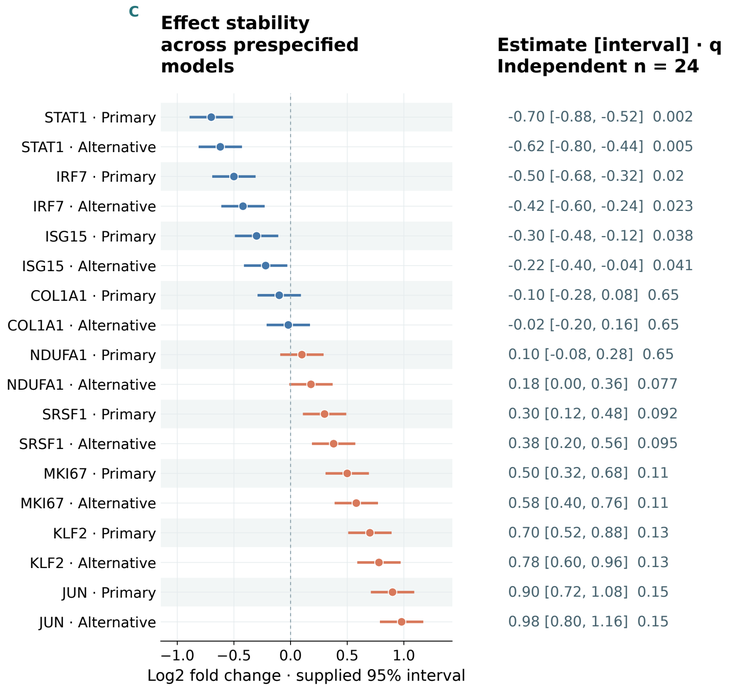
A

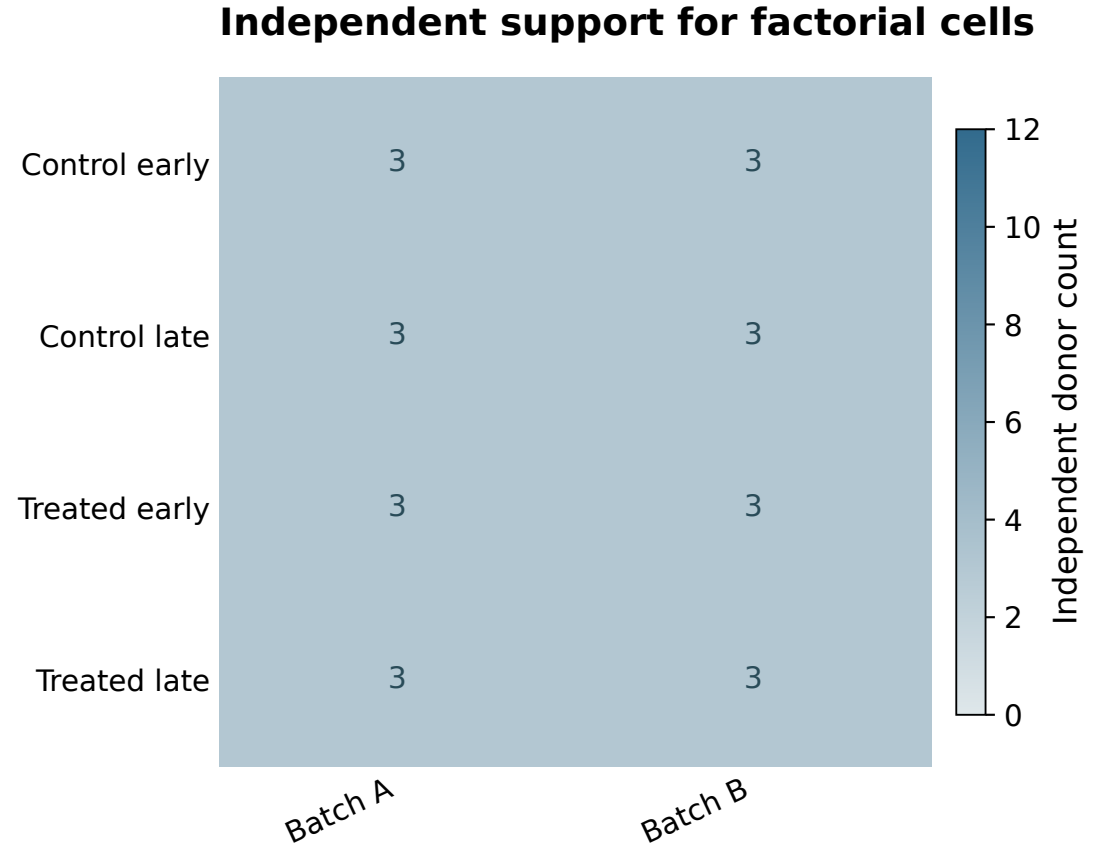


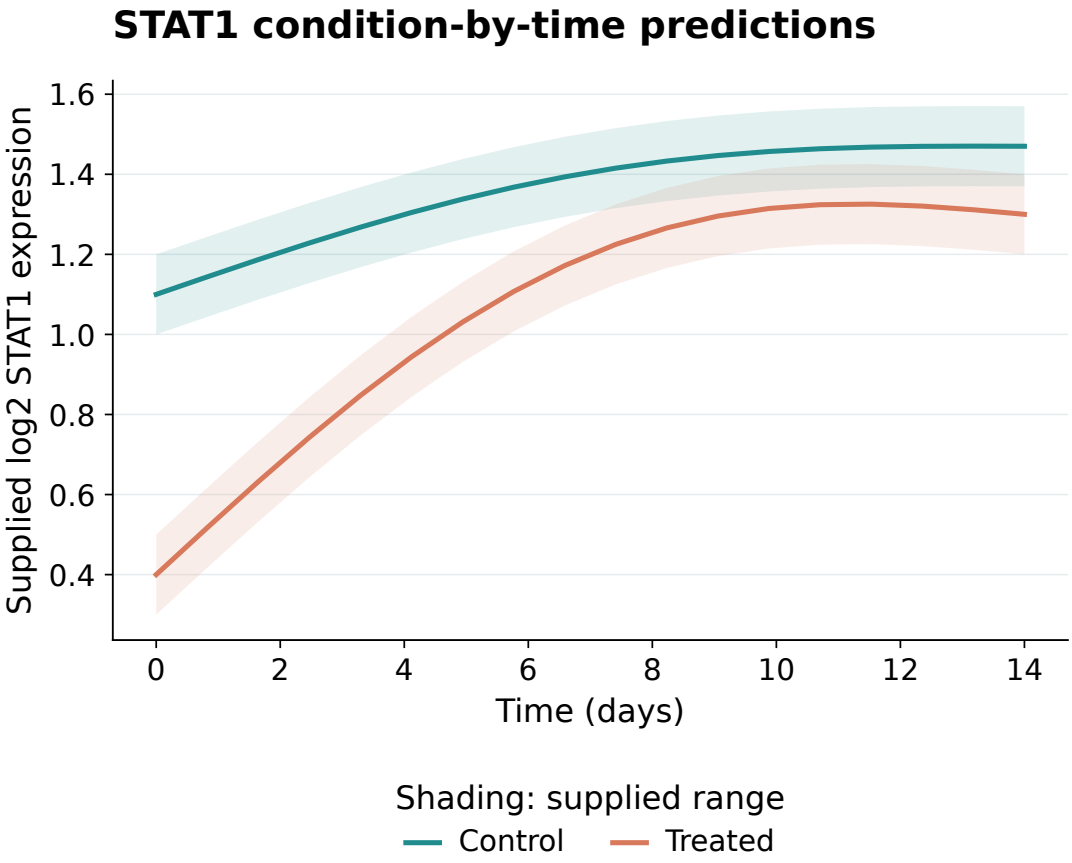
B



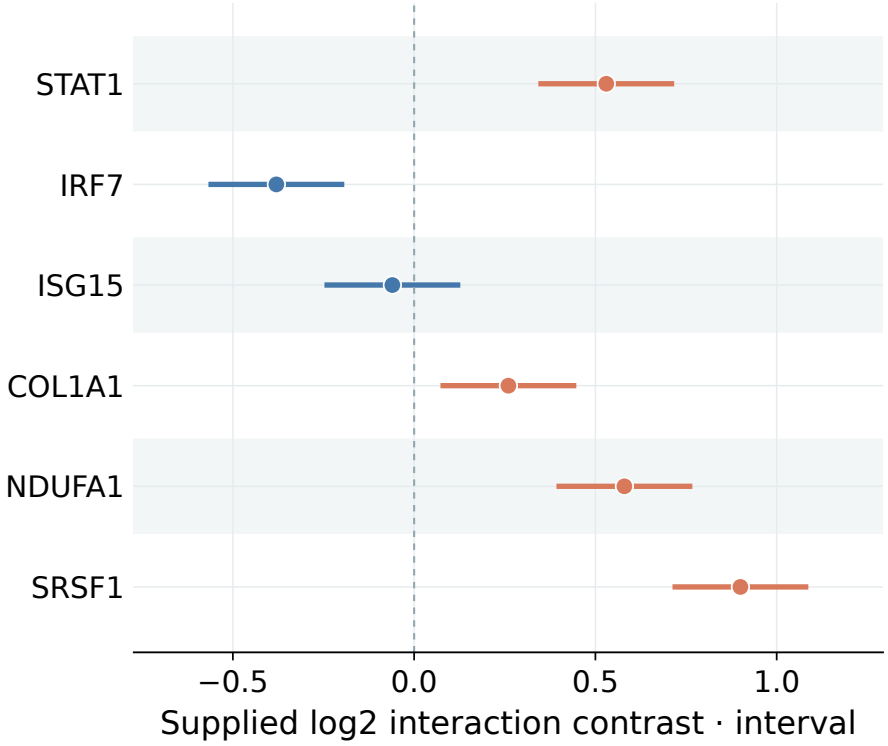
C




Independent support for factorial cells
Control early
Control late
Treated early
Treated late
3
3
3
3
3
3
3
3
Batch A
Batch B
Independent donor count
0
2
4
6
8
10
12


STAT1 condition-by-time predictions
Supplied log2 STAT1 expression
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0
2
4
6
8
10
12
14
Time (days)
Shading: supplied range
Control
Treated

Gene-wise interaction contrasts



Estimate [interval] · q
Independent n = 12

0.53 [0.35, 0.71] 0.002

-0.38 [-0.56, -0.20] 0.02

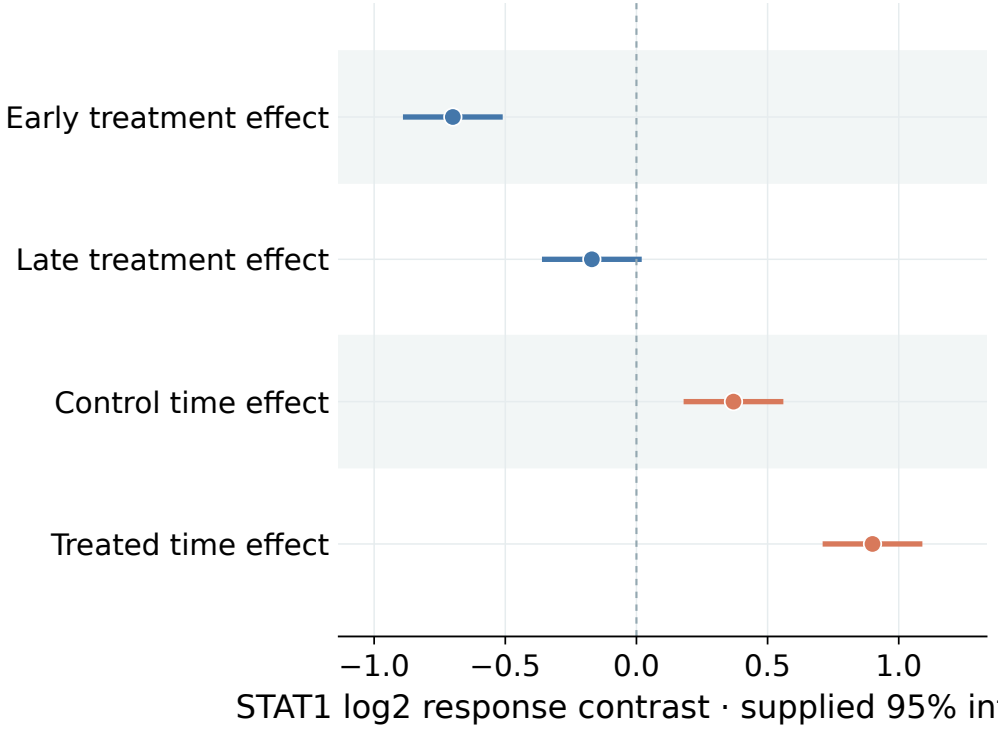
-0.06 [-0.24, 0.12] 0.65

0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092

STAT1 simple effects at prespecified times



Estimate [interval] · q
Independent n = 12

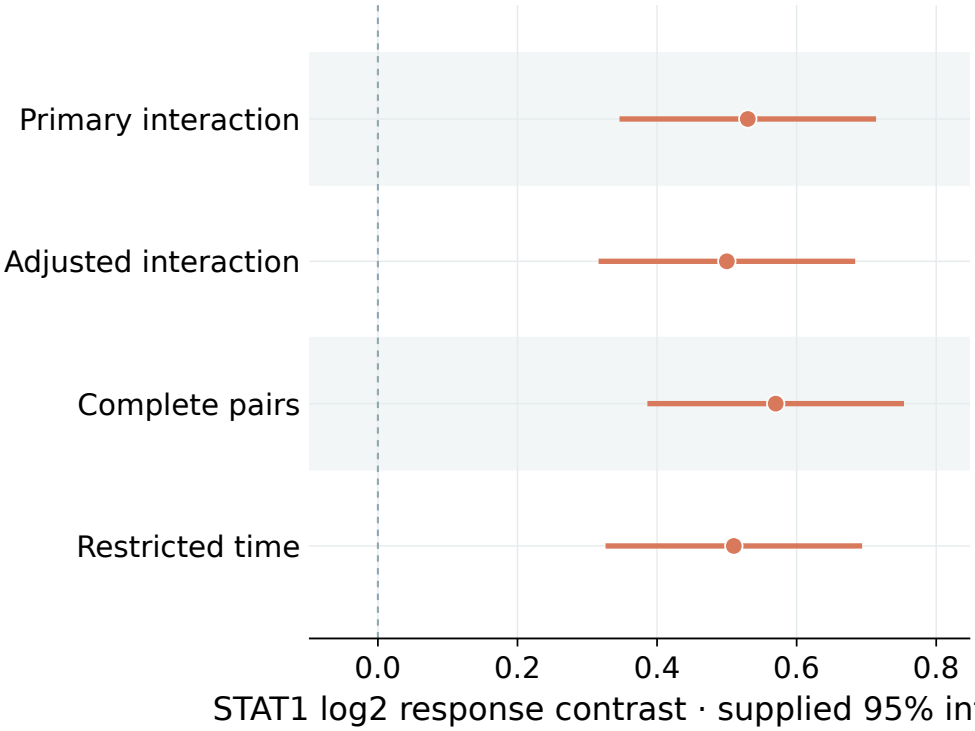
-0.70 [-0.88, -0.52] 0.002

-0.17 [-0.35, 0.01] 0.65

0.37 [0.19, 0.55] 0.038

0.90 [0.72, 1.08] 0.056

Interaction-model sensitivity



Estimate [interval] · q

Primary interaction	0.53 [0.35, 0.71] 0.002 n=12
Adjusted interaction	0.50 [0.32, 0.68] 0.02 n=12
Complete pairs	0.57 [0.39, 0.75] 0.038 n=10
Restricted time	0.51 [0.33, 0.69] 0.056 n=12

Bulk · interaction — complementary evidence

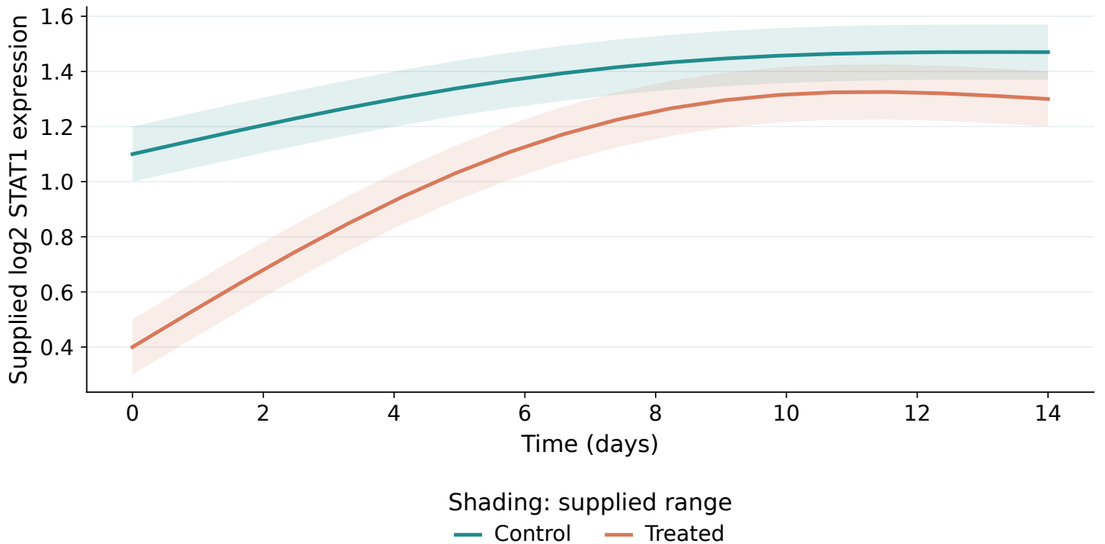
A

Independent support for factorial cells



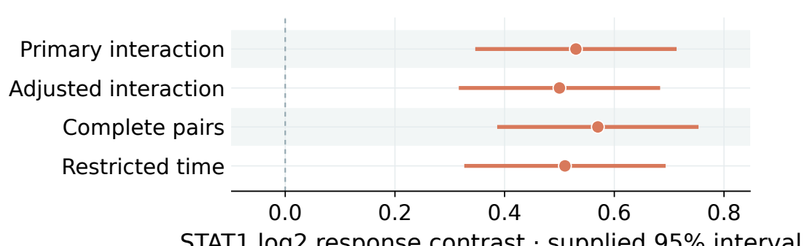
B

STAT1 condition-by-time predictions



C

Interaction-model sensitivity



Estimate [interval] · q

0.53 [0.35, 0.71] 0.002 n=12

0.50 [0.32, 0.68] 0.02 n=12

0.57 [0.39, 0.75] 0.038 n=10

0.51 [0.33, 0.69] 0.056 n=12

Bulk · interaction — complementary evidence

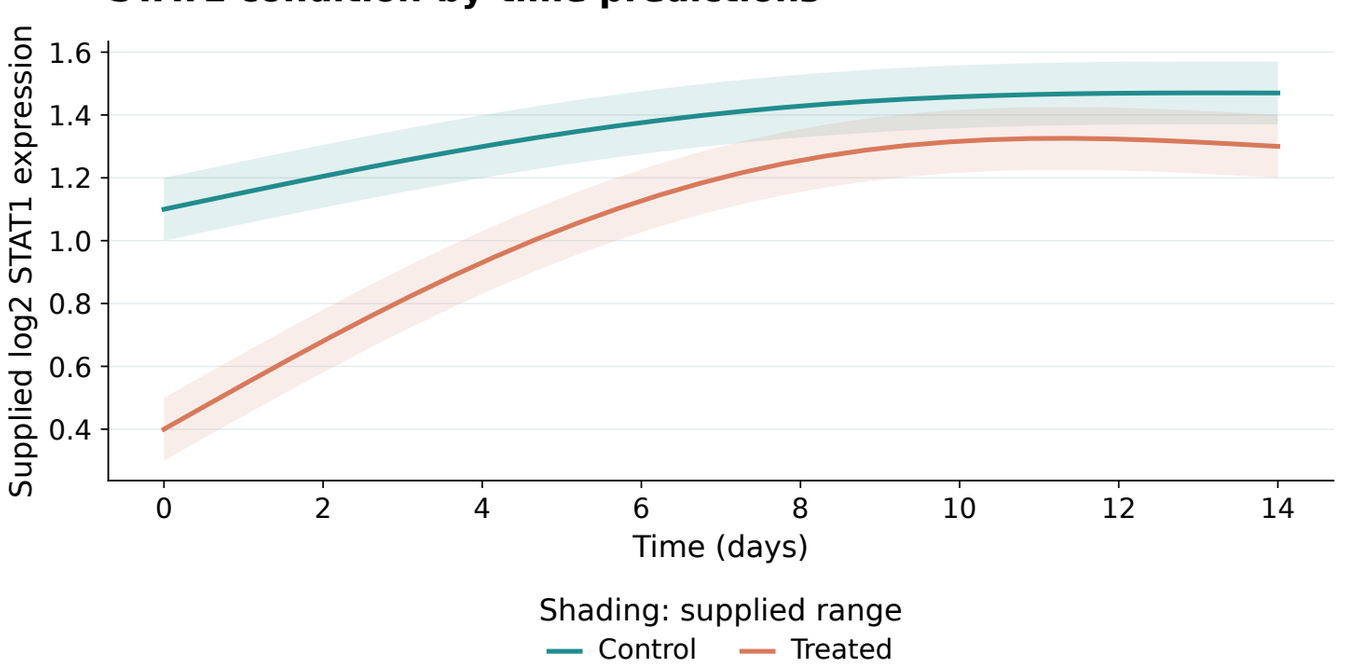
A

Independent support for factorial cells



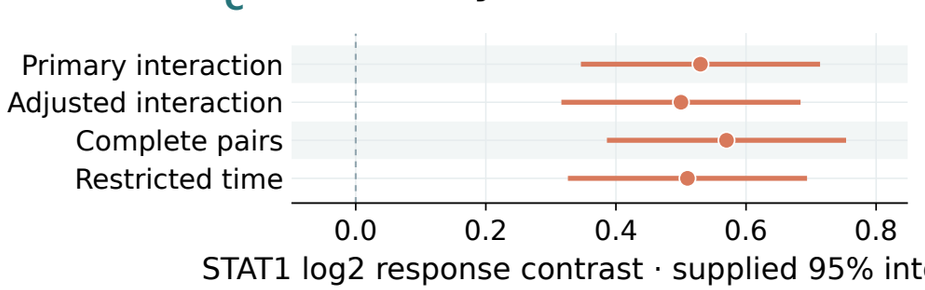
B

STAT1 condition-by-time predictions



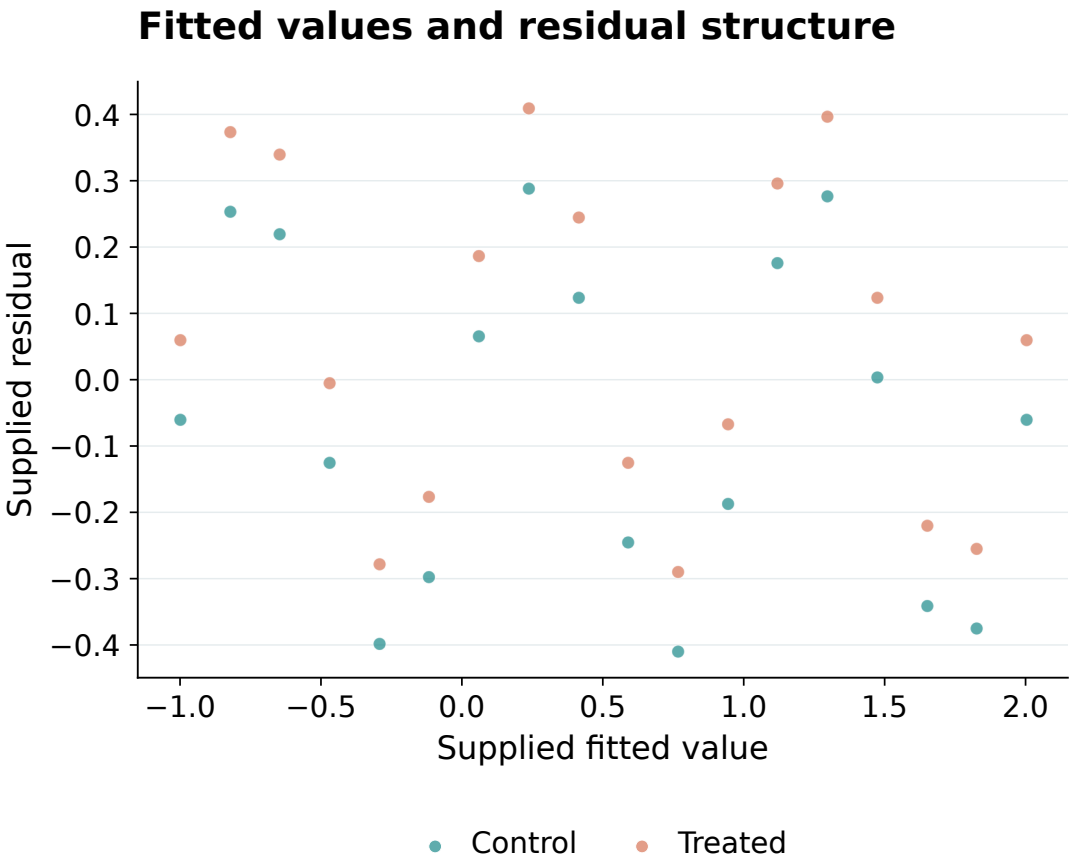
C

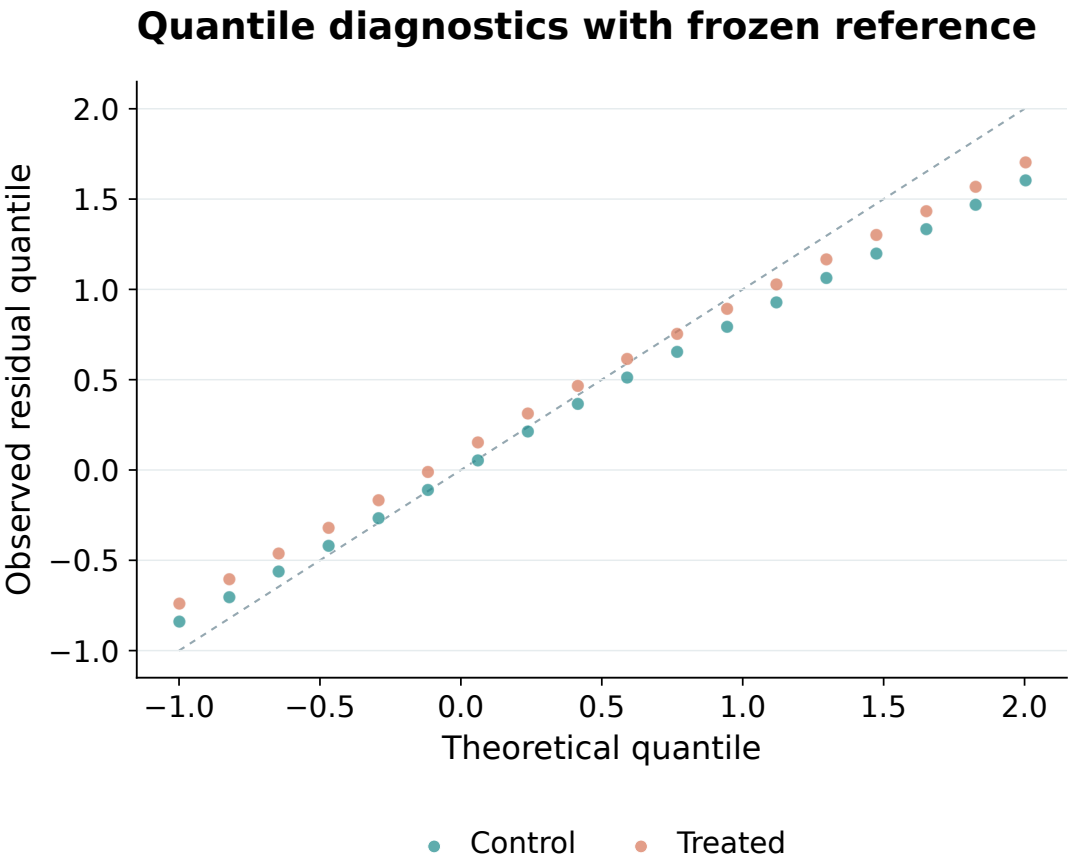
Interaction-model sensitivity

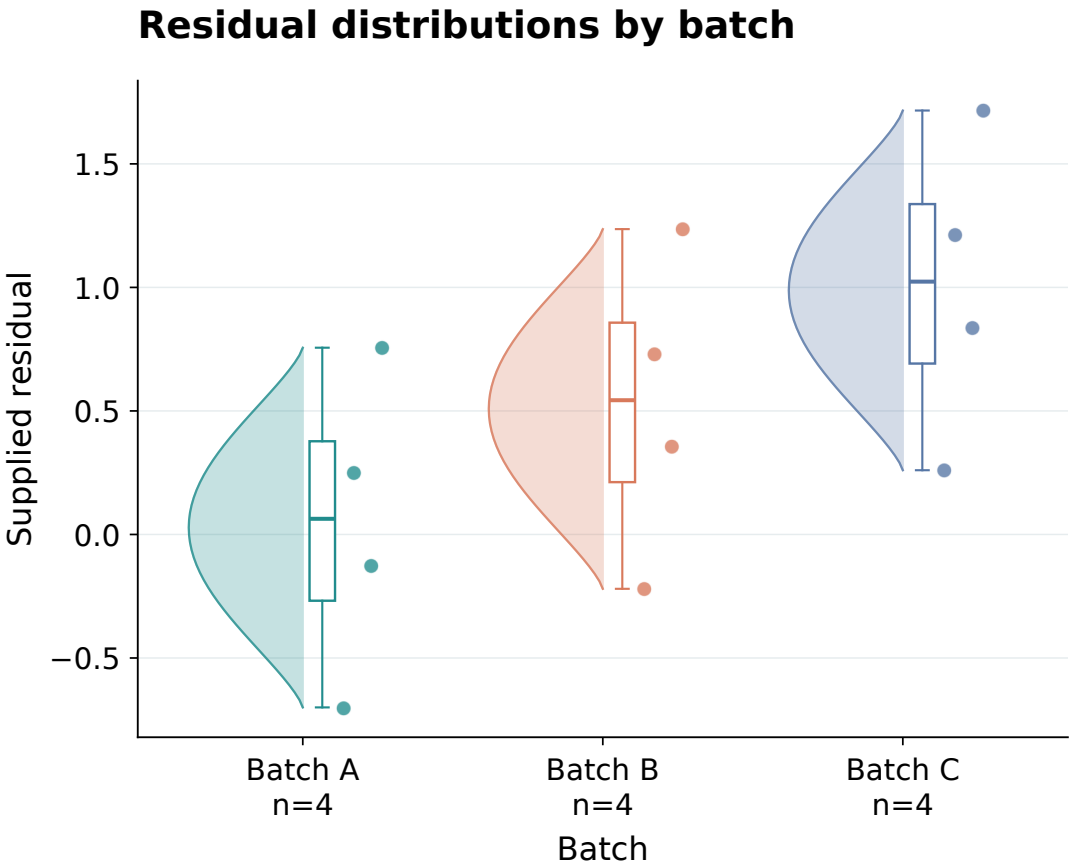


Estimate [interval] · q

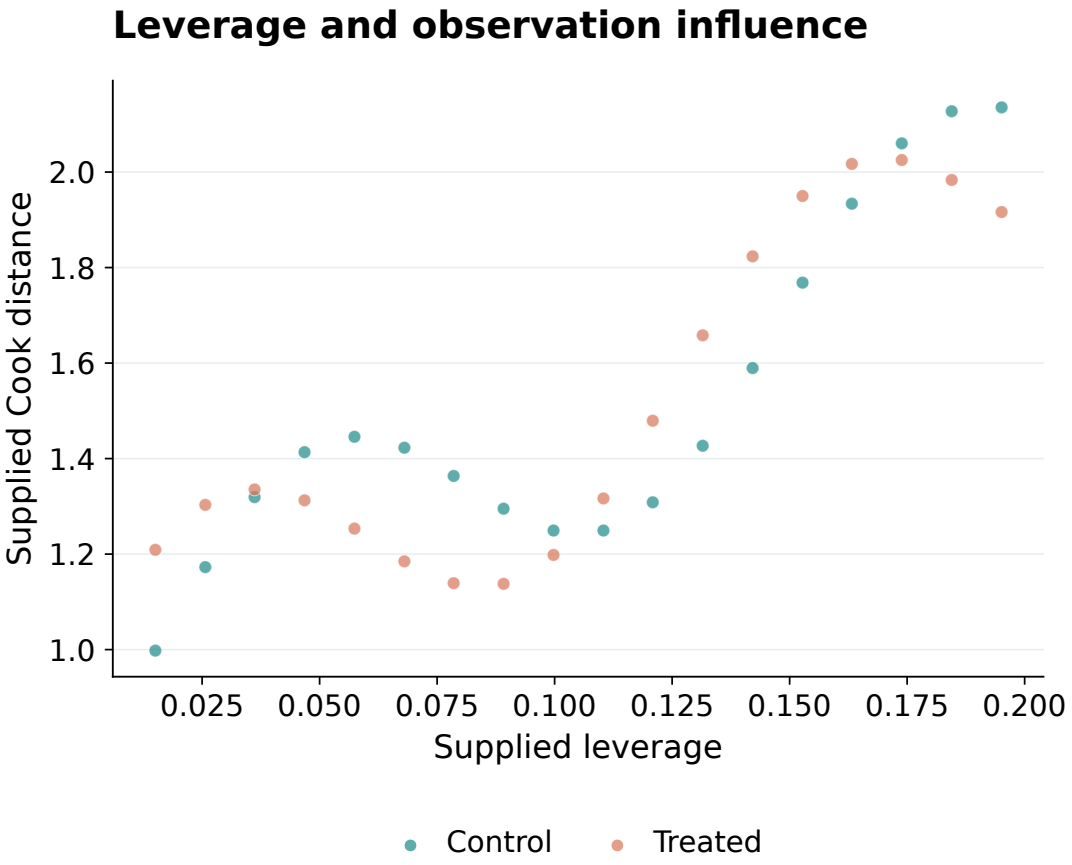
0.53 [0.35, 0.71] 0.002 n=24
0.50 [0.32, 0.68] 0.02 n=24
0.57 [0.39, 0.75] 0.038 n=22
0.51 [0.33, 0.69] 0.056 n=24


Fitted values and residual structure
Supplied residual
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Supplied fitted value
Control
Treated

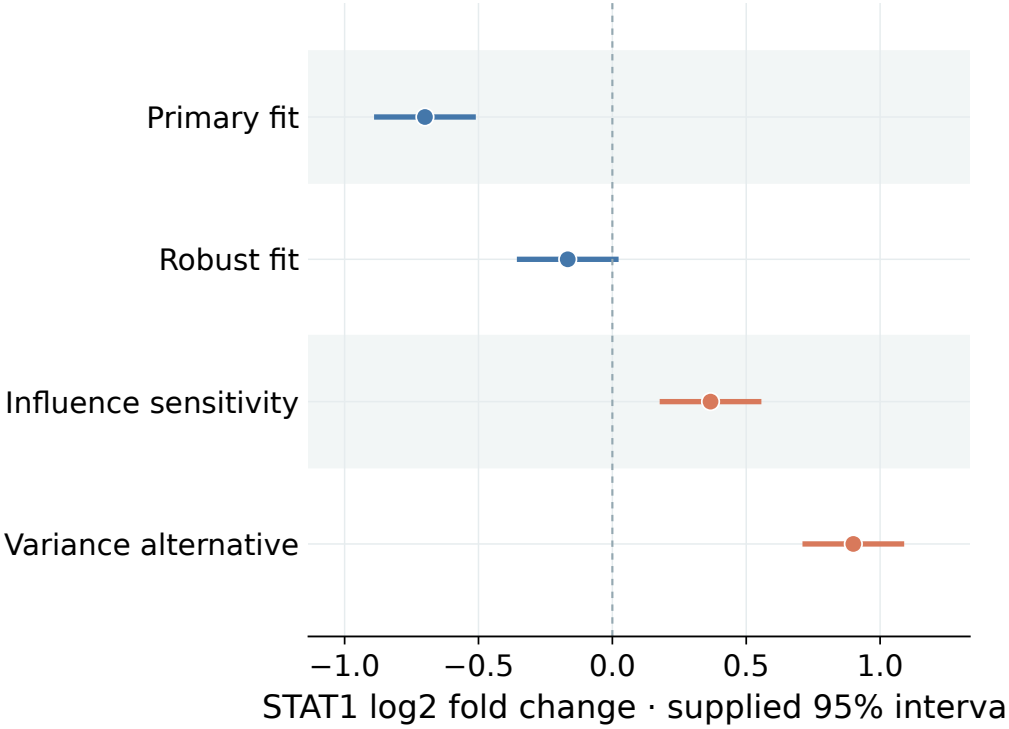

Quantile diagnostics with frozen reference
Observed residual quantile
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Theoretical quantile
Control
Treated



Software test fixture; not biological evidence


Leverage and observation influence
Supplied Cook distance
2.0
1.8
1.6
1.4
1.2
1.0
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Supplied leverage
Control
Treated

Prespecified diagnostic sensitivity



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

-0.17 [-0.35, 0.01] 0.65

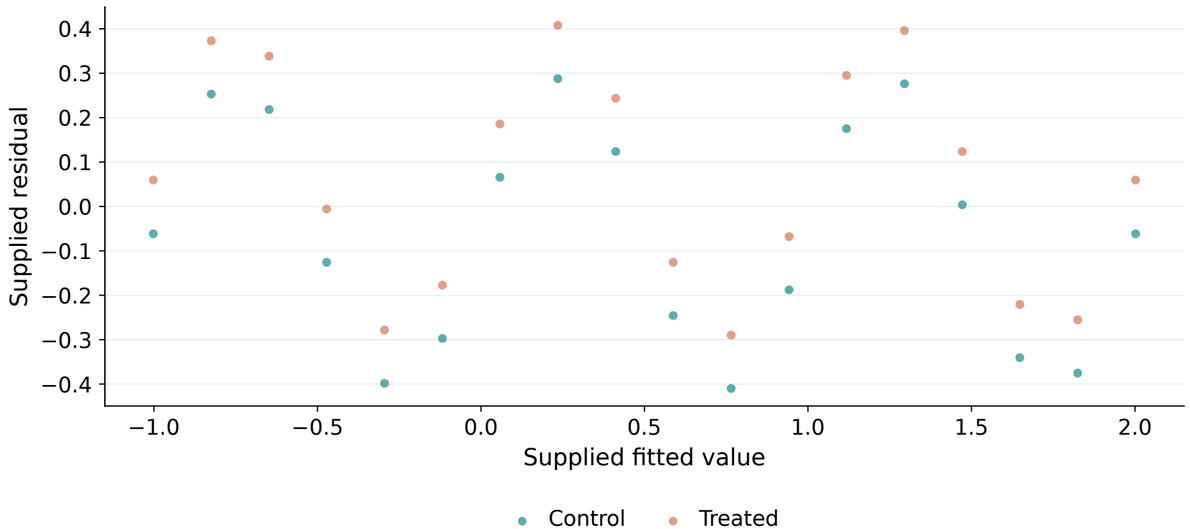
0.37 [0.19, 0.55] 0.038

0.90 [0.72, 1.08] 0.056

Bulk · model_qc — complementary evidence

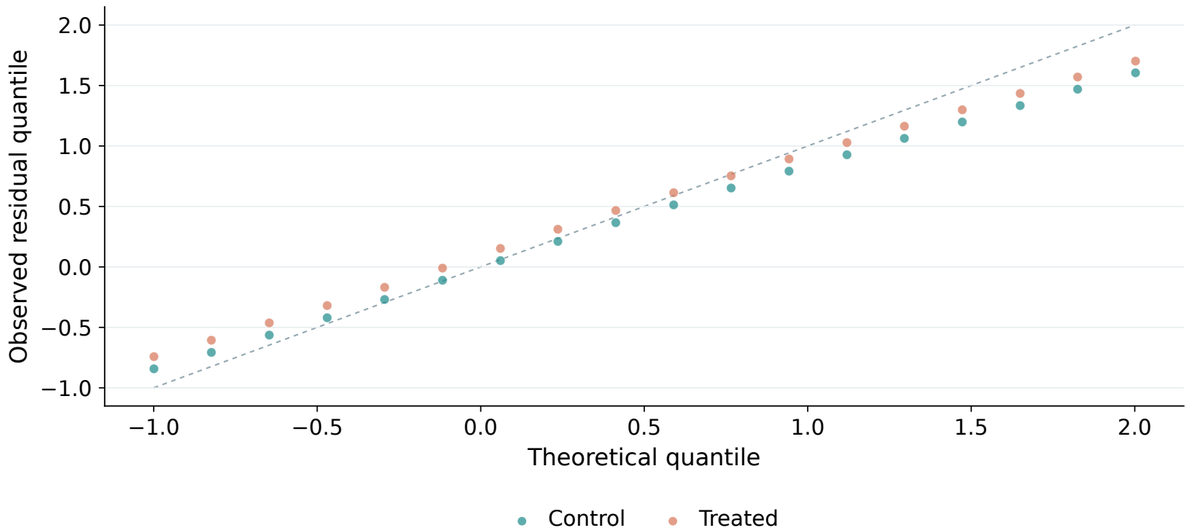
A

Fitted values and residual structure



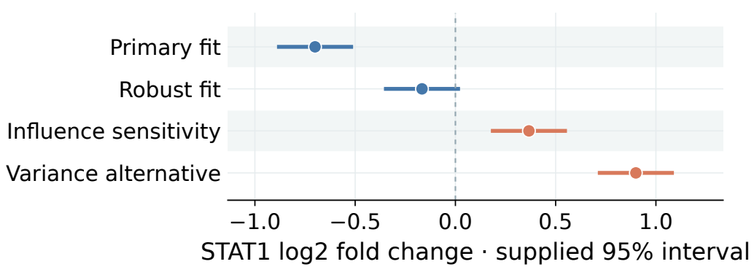
B

Quantile diagnostics with frozen reference



C

Prespecified diagnostic sensitivity



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

-0.17 [-0.35, 0.01] 0.65

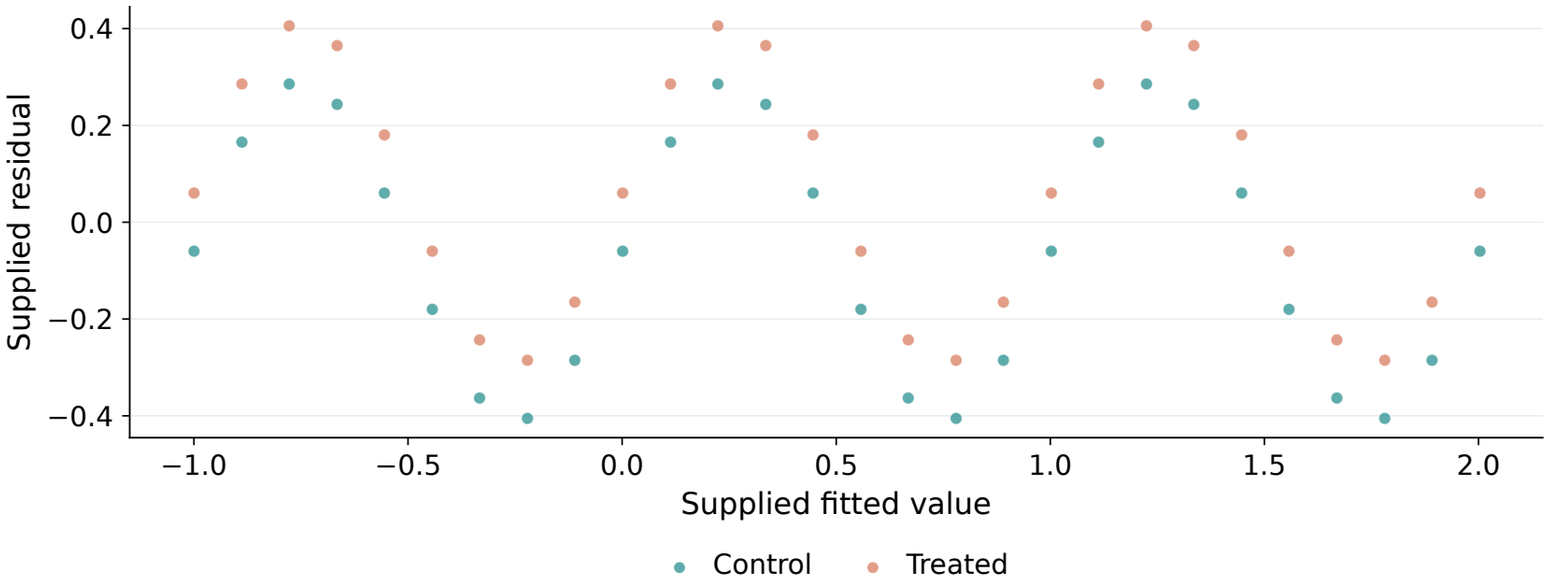
0.37 [0.19, 0.55] 0.038

0.90 [0.72, 1.08] 0.056

Bulk · model_qc — complementary evidence

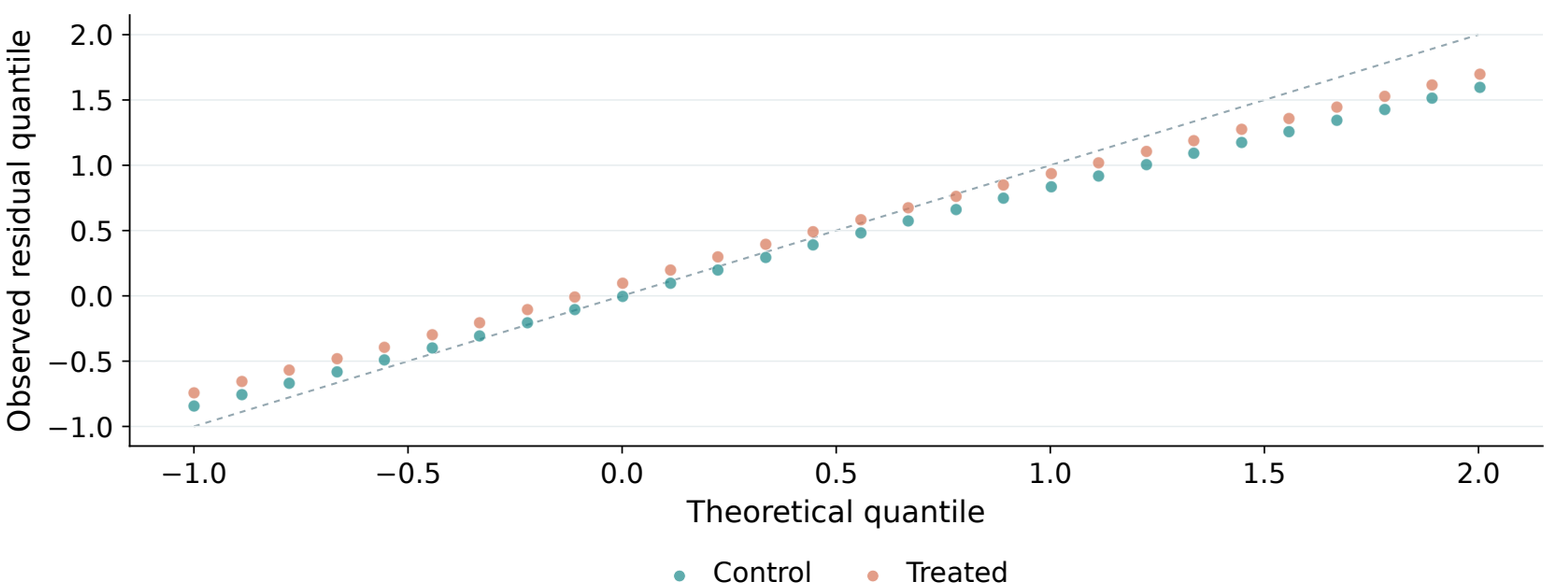
A

Fitted values and residual structure



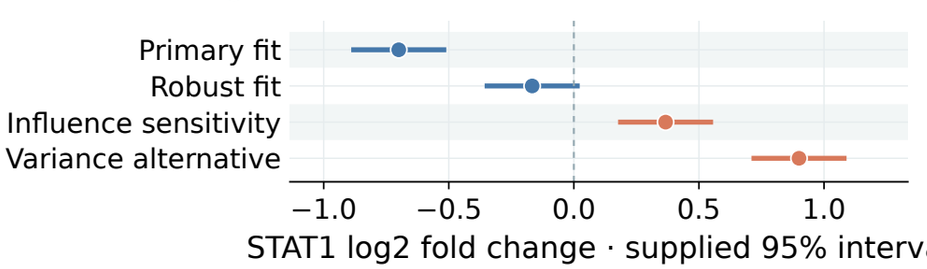
B

Quantile diagnostics with frozen reference



C

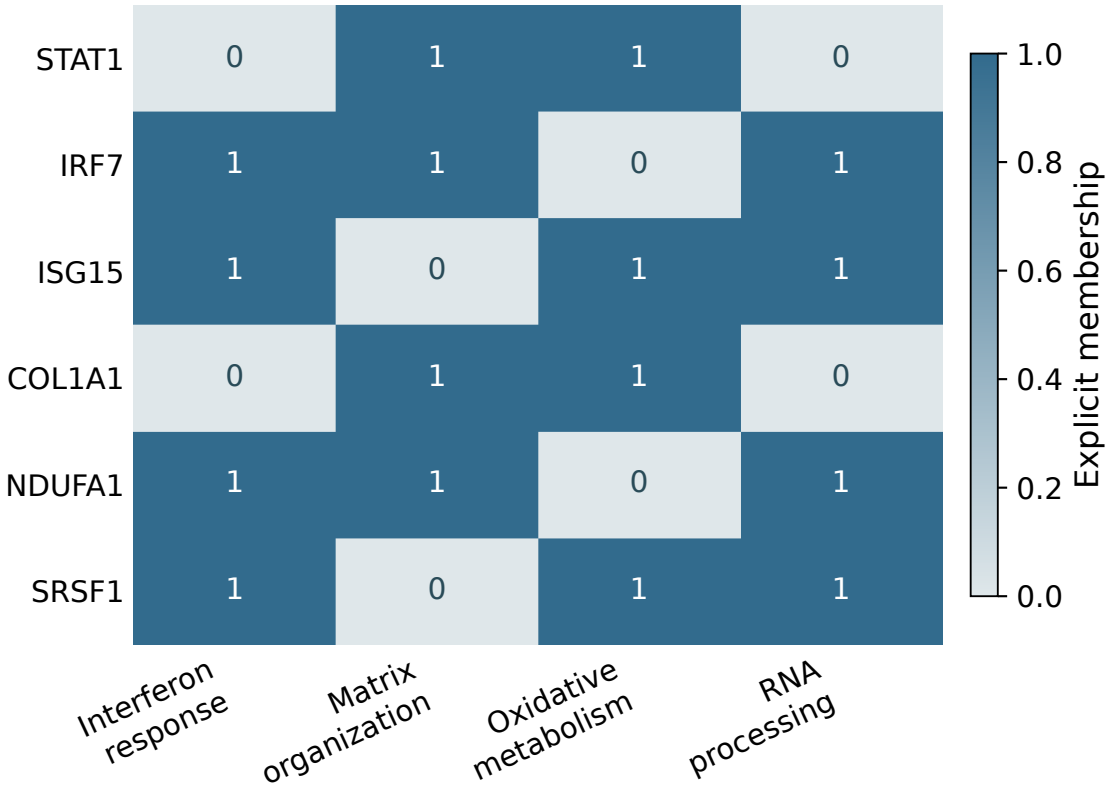
Prespecified diagnostic sensitivity



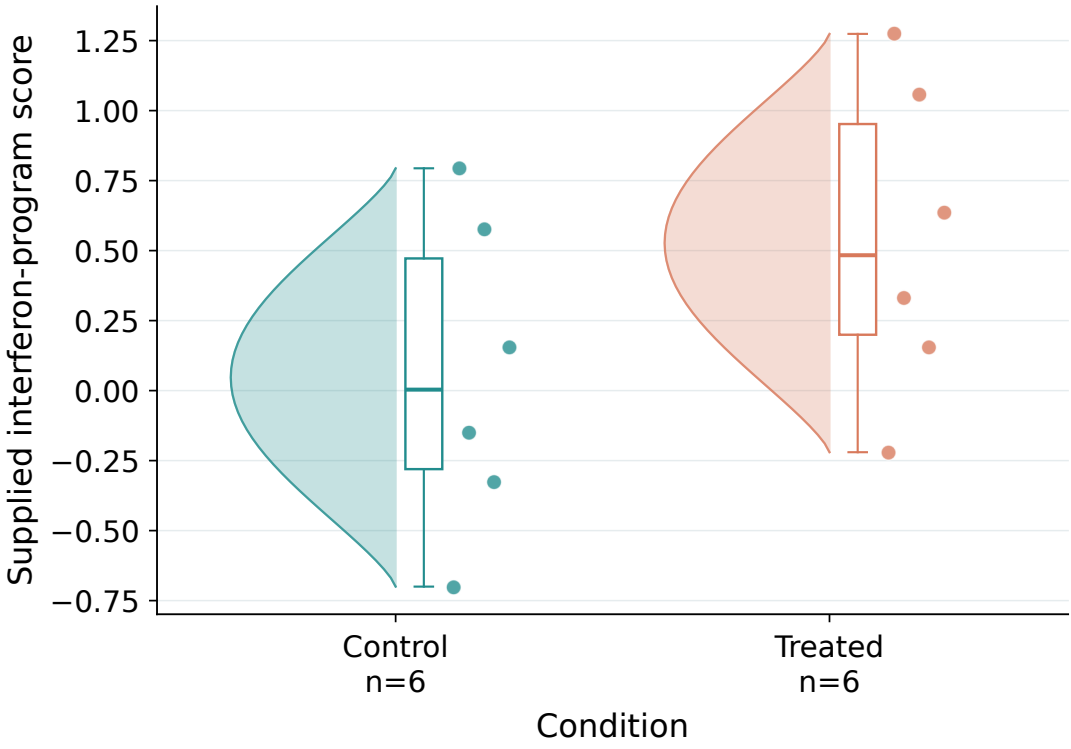
Estimate [interval] · q
Independent n = 24

-0.70 [-0.88, -0.52] 0.002
-0.17 [-0.35, 0.01] 0.65
0.37 [0.19, 0.55] 0.038
0.90 [0.72, 1.08] 0.056

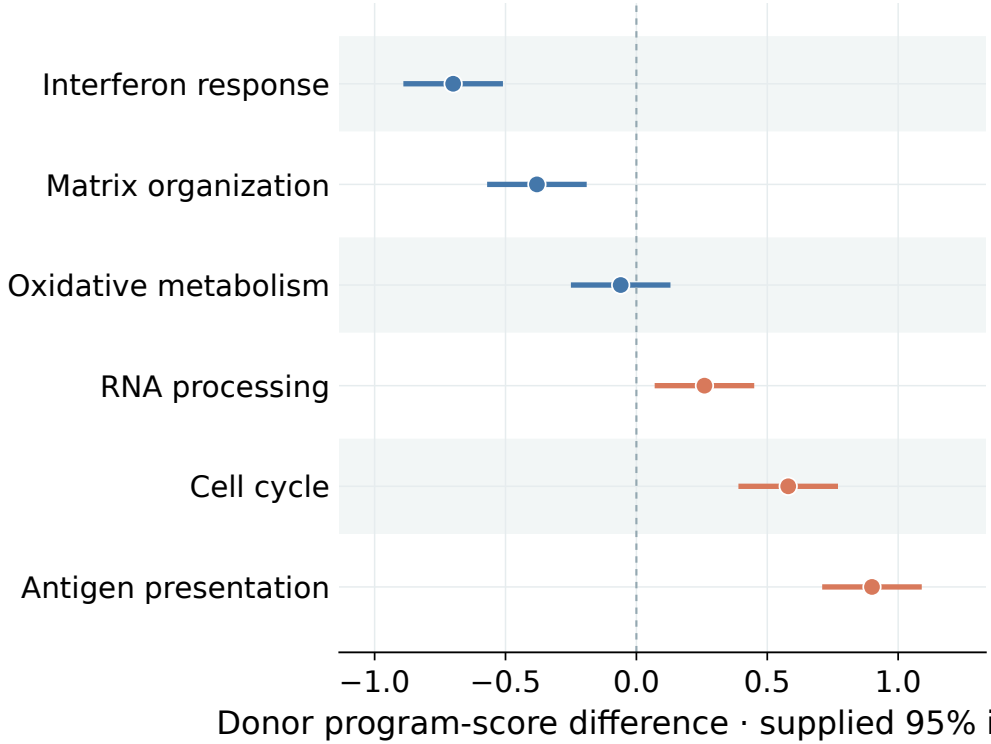
Program membership and complete gene support



Interferon-program distributions in independent donors



Program response estimates



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

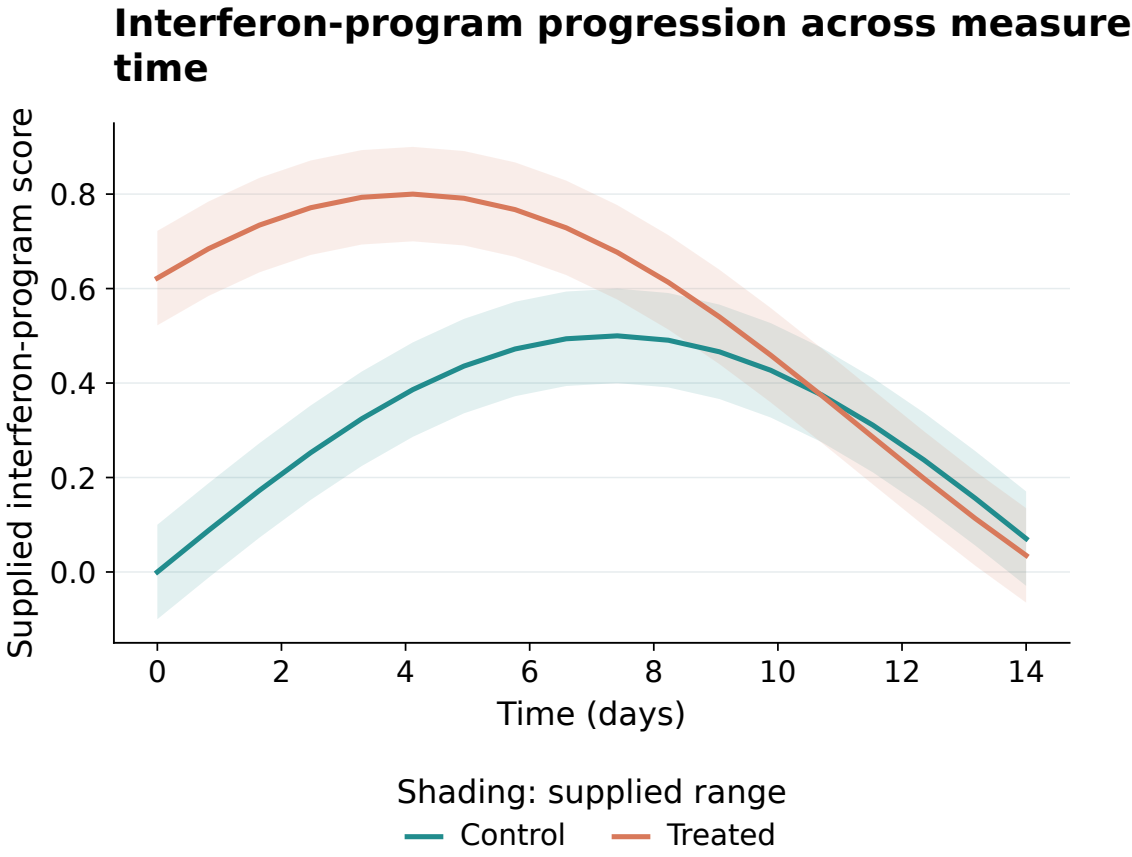
-0.38 [-0.56, -0.20] 0.02

-0.06 [-0.24, 0.12] 0.65

0.26 [0.08, 0.44] 0.056

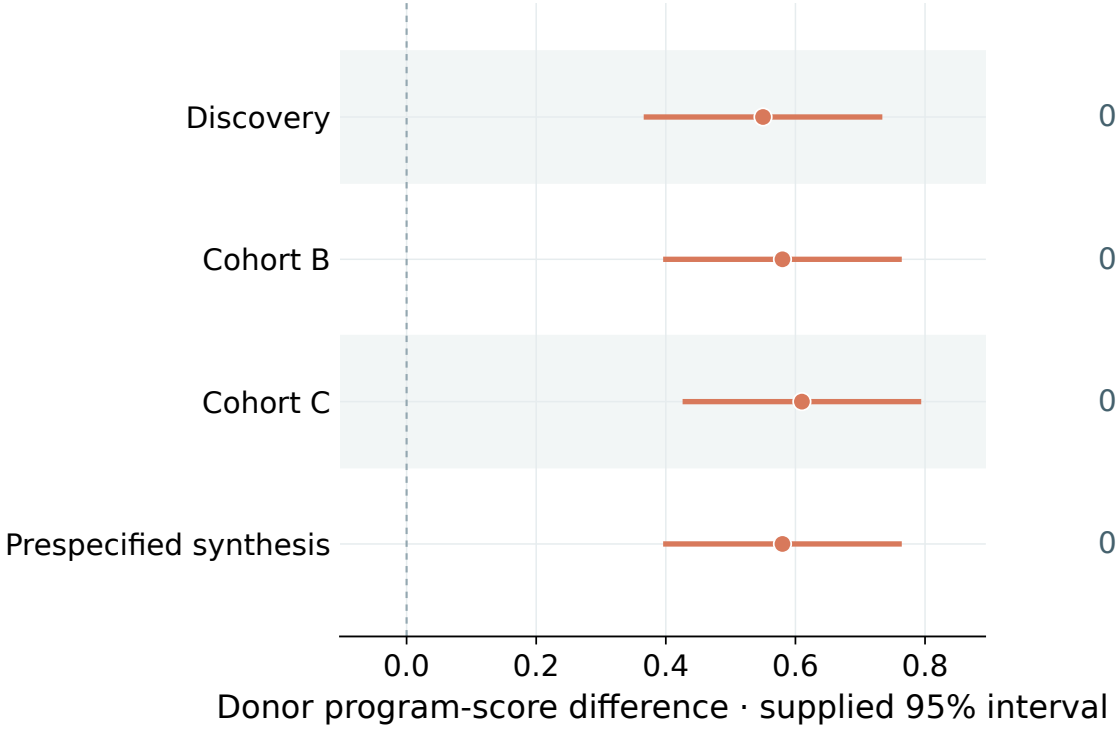
0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092



Software test fixture; not biological evidence

Interferon response in independent cohorts



Estimate [interval] · q

0.55 [0.37, 0.73] 0.002 n=4

0.58 [0.40, 0.76] 0.02 n=4

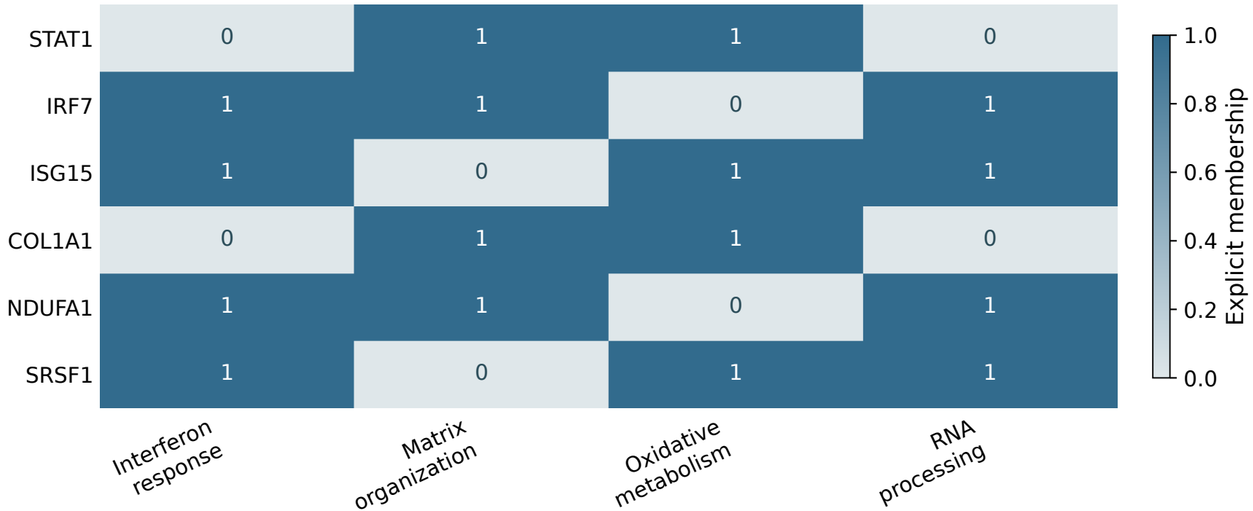
0.61 [0.43, 0.79] 0.038 n=4

0.58 [0.40, 0.76] 0.056 n=12

Bulk · programs — complementary evidence

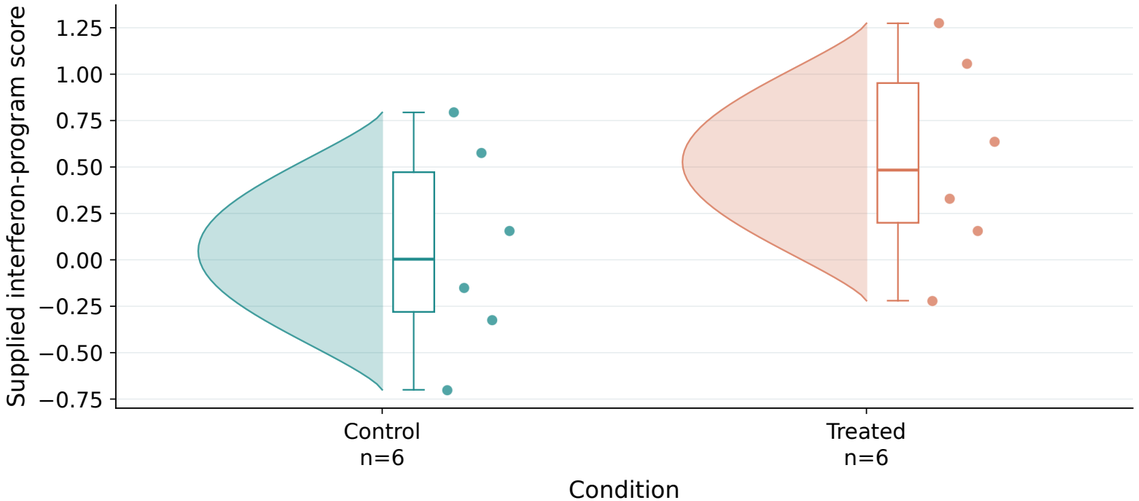
A

Program membership and complete gene support



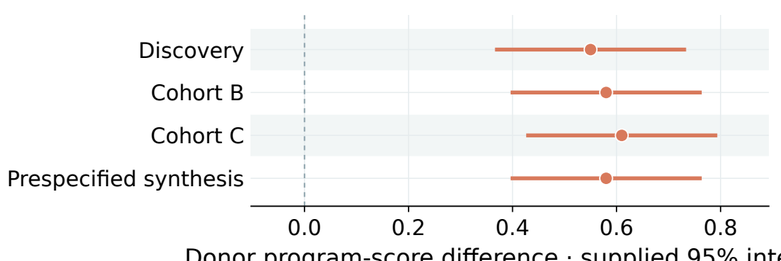
B

Interferon-program distributions in independent donors



C

Interferon response in independent cohorts



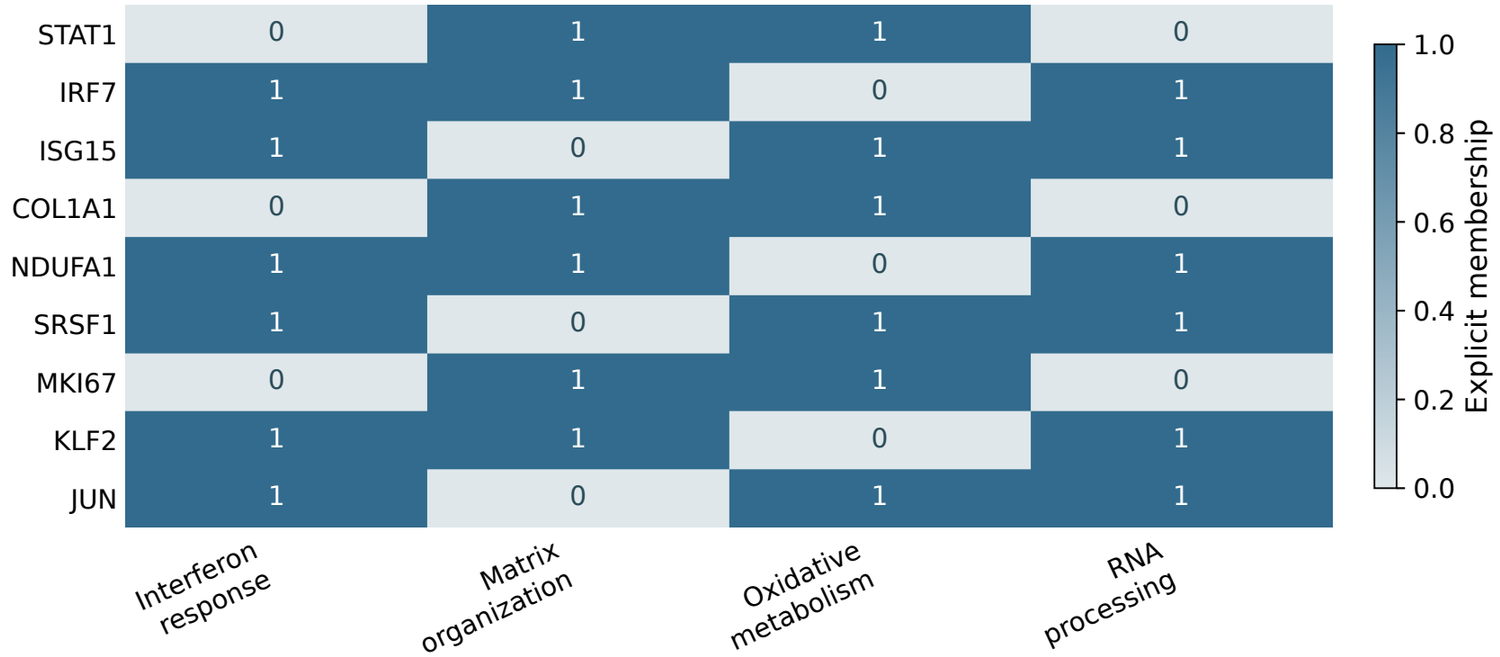
Estimate [interval] · q

Discovery	0.55 [0.37, 0.73] 0.002 n=4
Cohort B	0.58 [0.40, 0.76] 0.02 n=4
Cohort C	0.61 [0.43, 0.79] 0.038 n=4
Prespecified synthesis	0.58 [0.40, 0.76] 0.056 n=12

Bulk · programs — complementary evidence

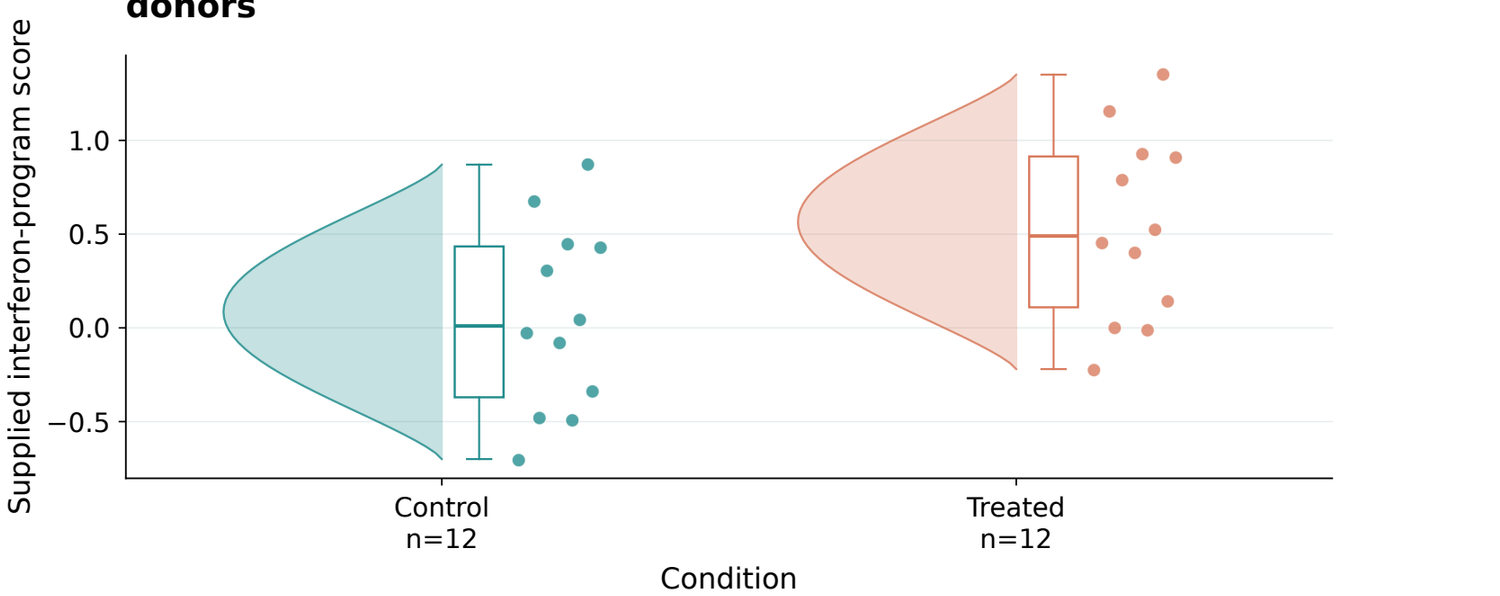
A

Program membership and complete gene support



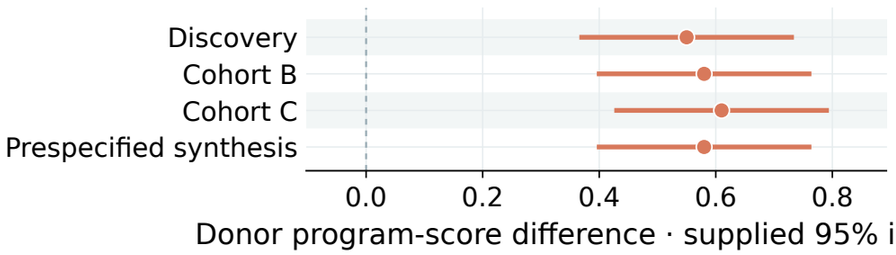
B

Interferon-program distributions in independent donors



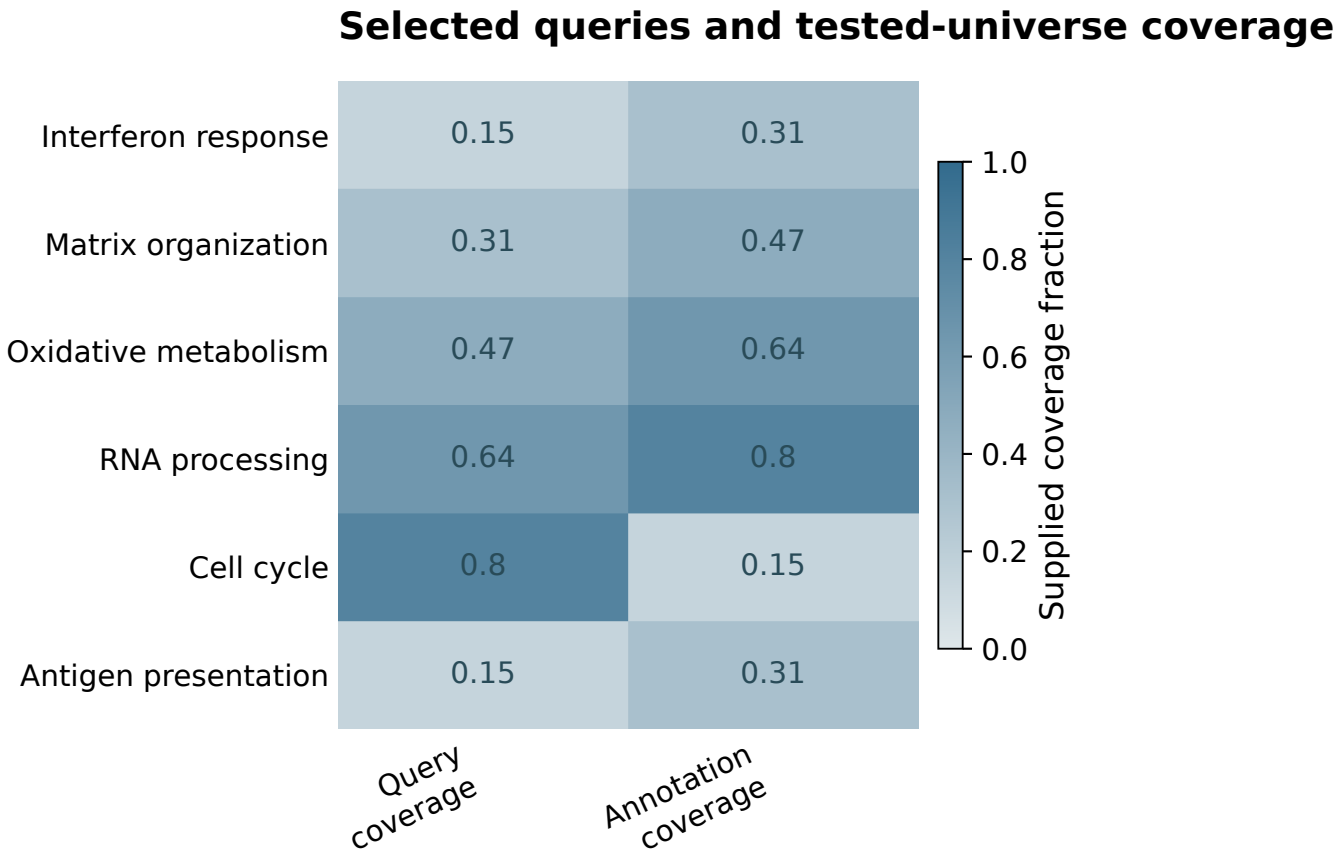
C

Interferon response in independent cohorts

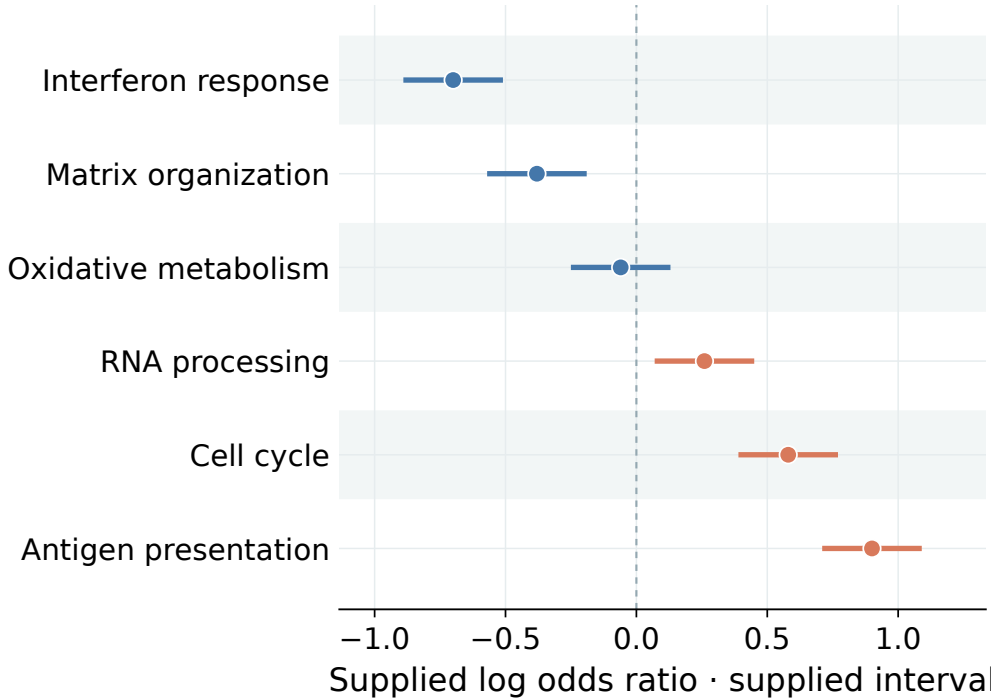


Estimate [interval] · q

0.55 [0.37, 0.73] 0.002 n=8
0.58 [0.40, 0.76] 0.02 n=8
0.61 [0.43, 0.79] 0.038 n=8
0.58 [0.40, 0.76] 0.056 n=24


Selected queries and tested-universe coverage
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
0.15
0.31
0.31
0.47
0.47
0.64
0.64
0.8
0.8
0.15
0.15
0.31
Query coverage
Annotation coverage
Supplied coverage fraction
1.0
0.8
0.6
0.4
0.2
0.0

Prespecified enrichment effect estimates



Estimate [interval] · q
Upstream donor n = 12

-0.70 [-0.88, -0.52] 0.002

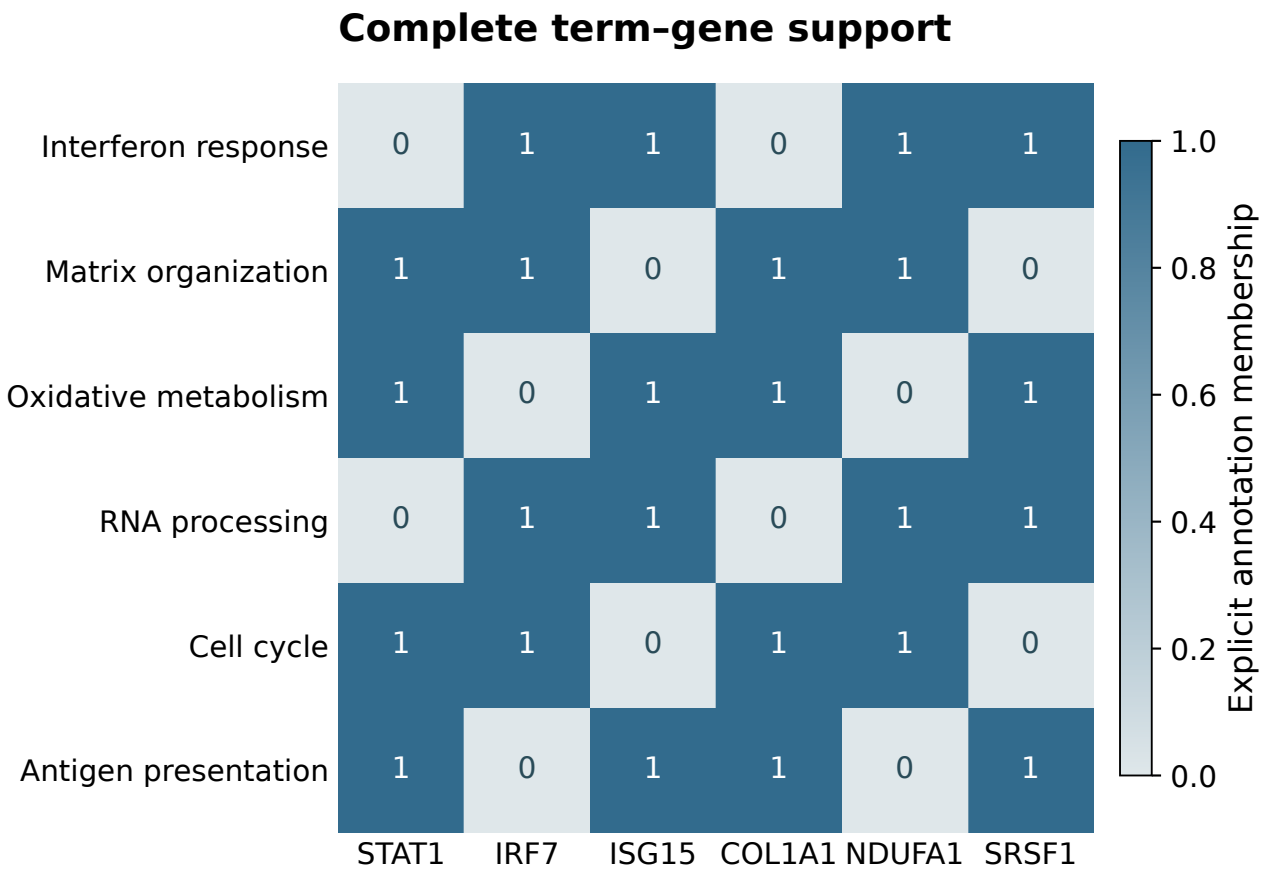
-0.38 [-0.56, -0.20] 0.02

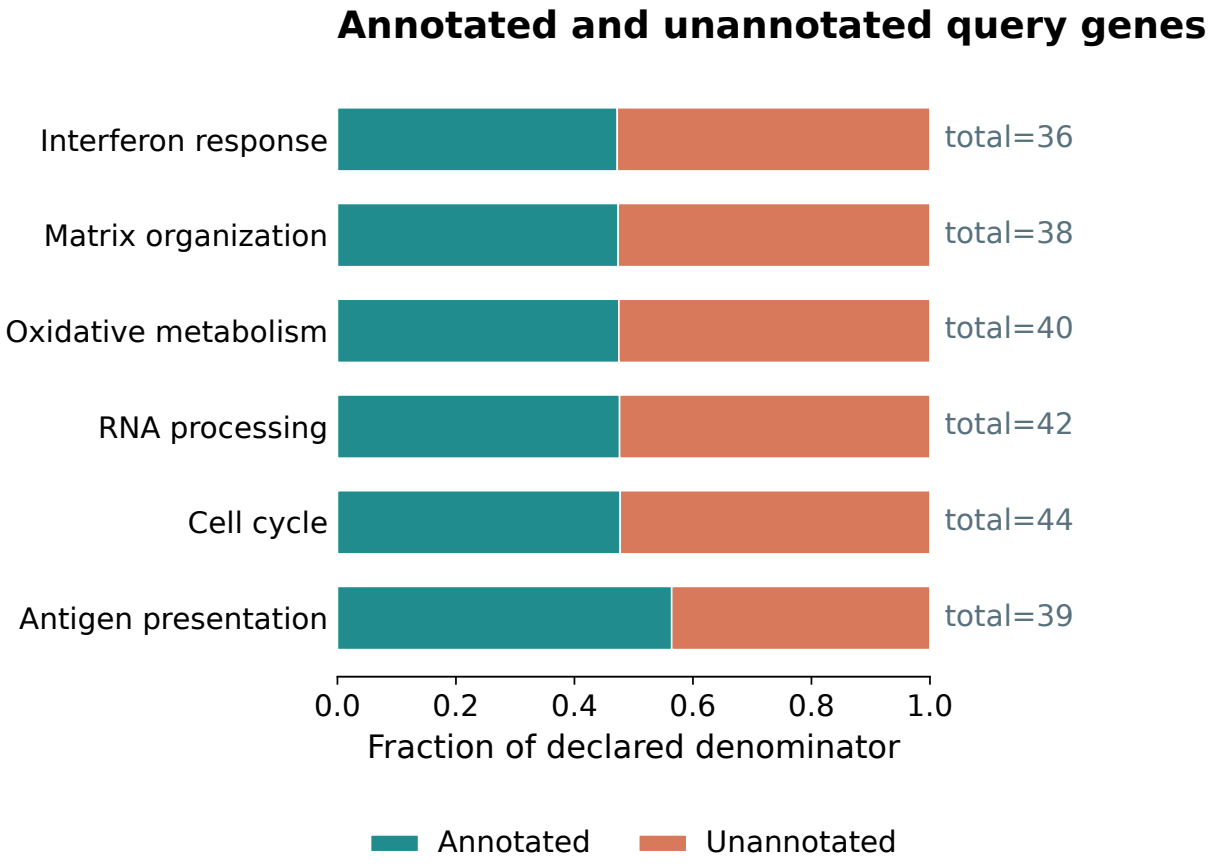
-0.06 [-0.24, 0.12] 0.65

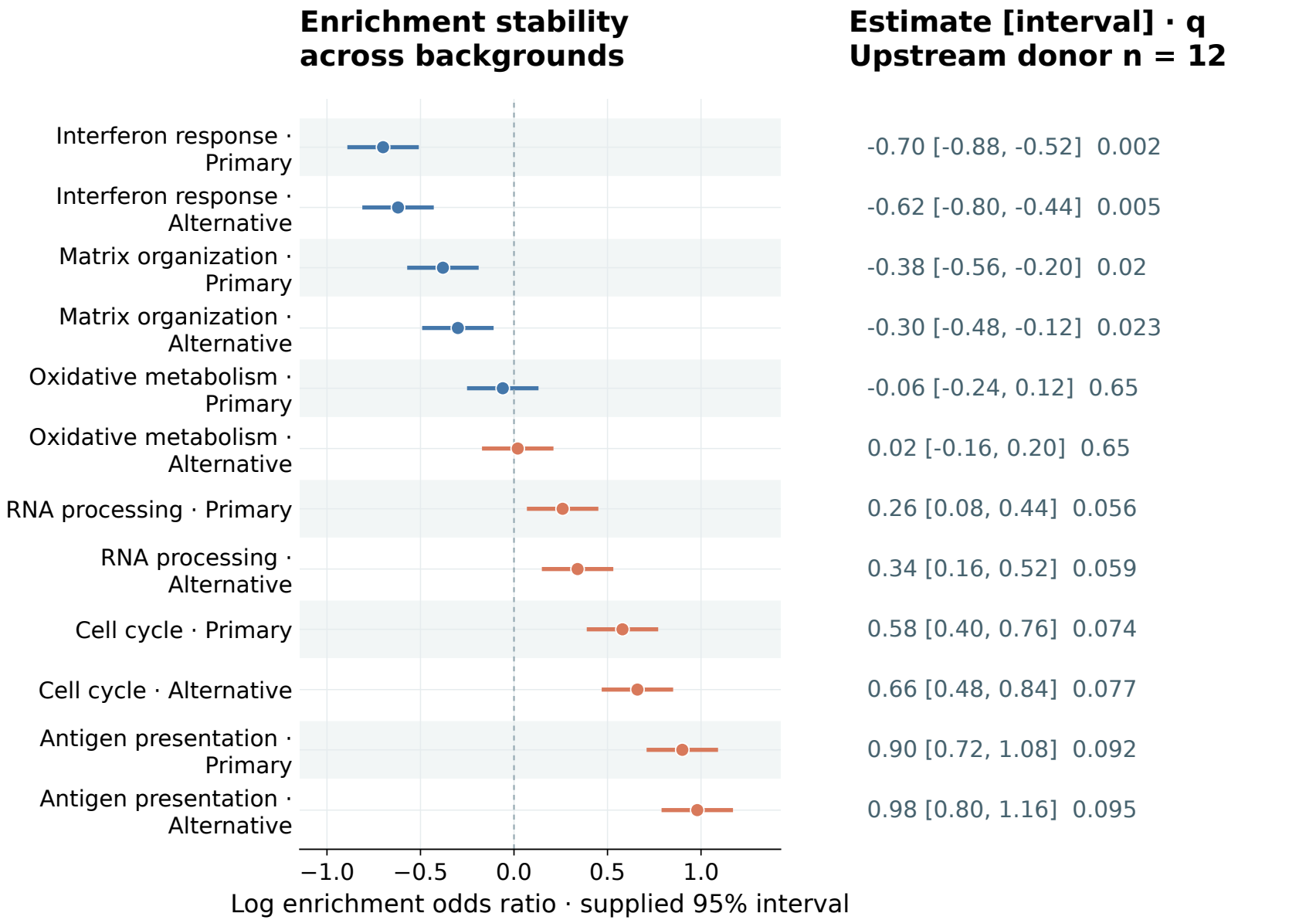
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

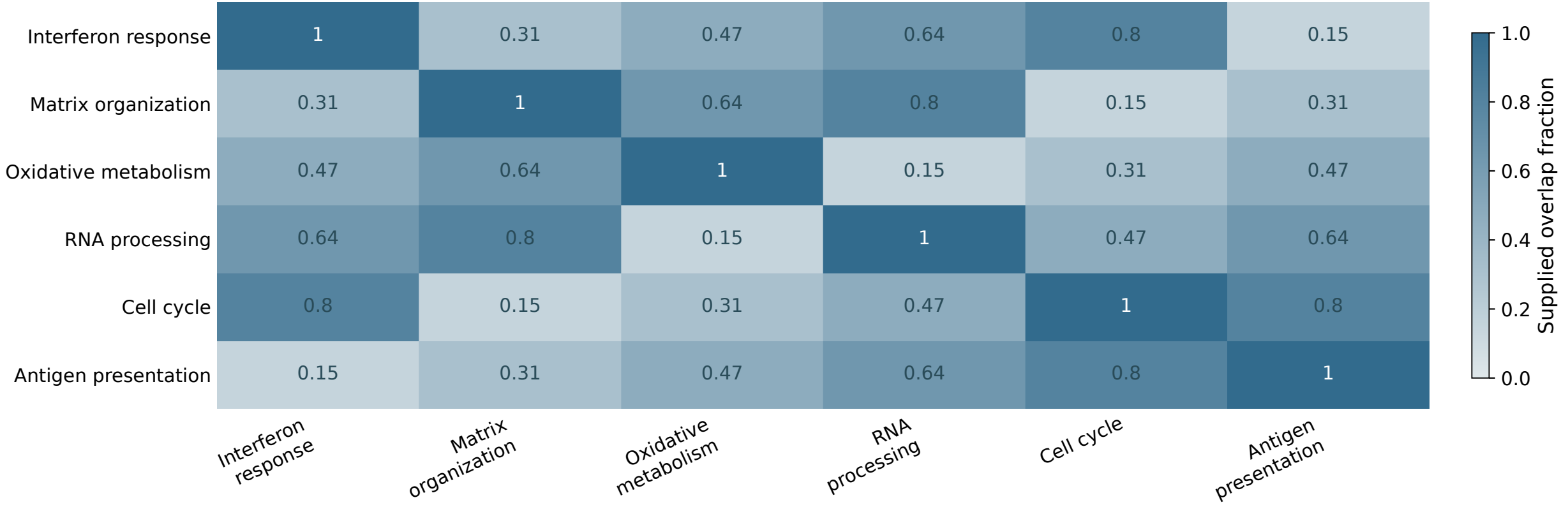
0.90 [0.72, 1.08] 0.092


Complete term–gene support
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
0 1 1 0 1 1
1 1 0 1 1 0
1 0 1 1 0 1
0 1 1 0 1 1
1 1 0 1 1 0
1 0 1 1 0 1
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
Explicit annotation membership
1.0
0.8
0.6
0.4
0.2
0.0

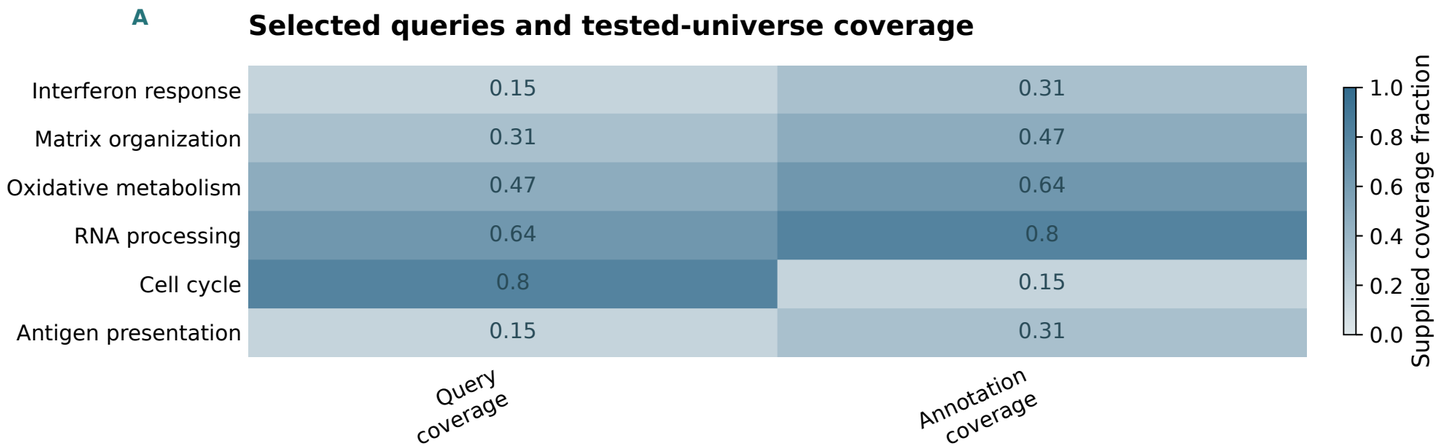

Annotated and unannotated query genes
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Annotated
Unannotated

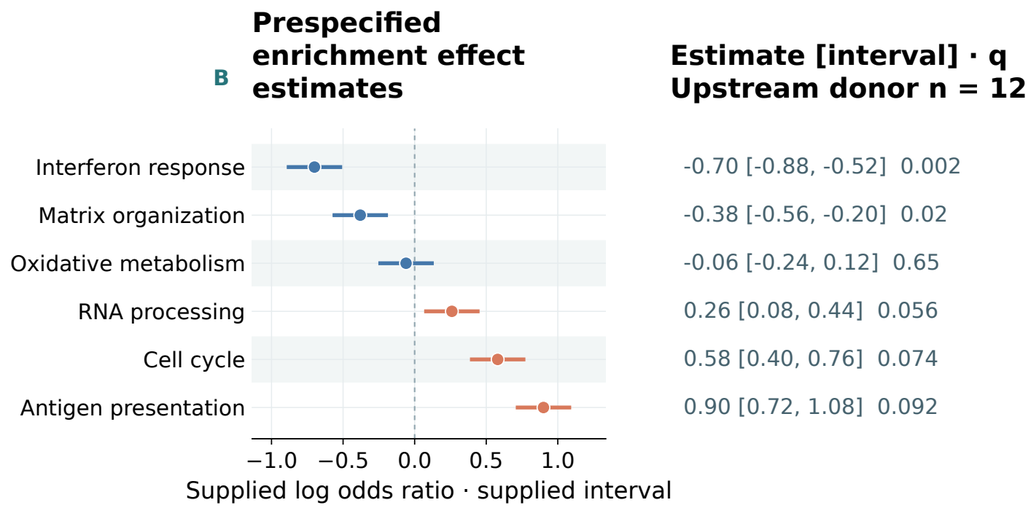

Enrichment stability
across backgrounds
Estimate [interval] · q
Upstream donor n = 12
Interferon response · Primary
Interferon response · Alternative
Matrix organization · Primary
Matrix organization · Alternative
Oxidative metabolism · Primary
Oxidative metabolism · Alternative
RNA processing · Primary
RNA processing · Alternative
Cell cycle · Primary
Cell cycle · Alternative
Antigen presentation · Primary
Antigen presentation · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Log enrichment odds ratio · supplied 95% interval

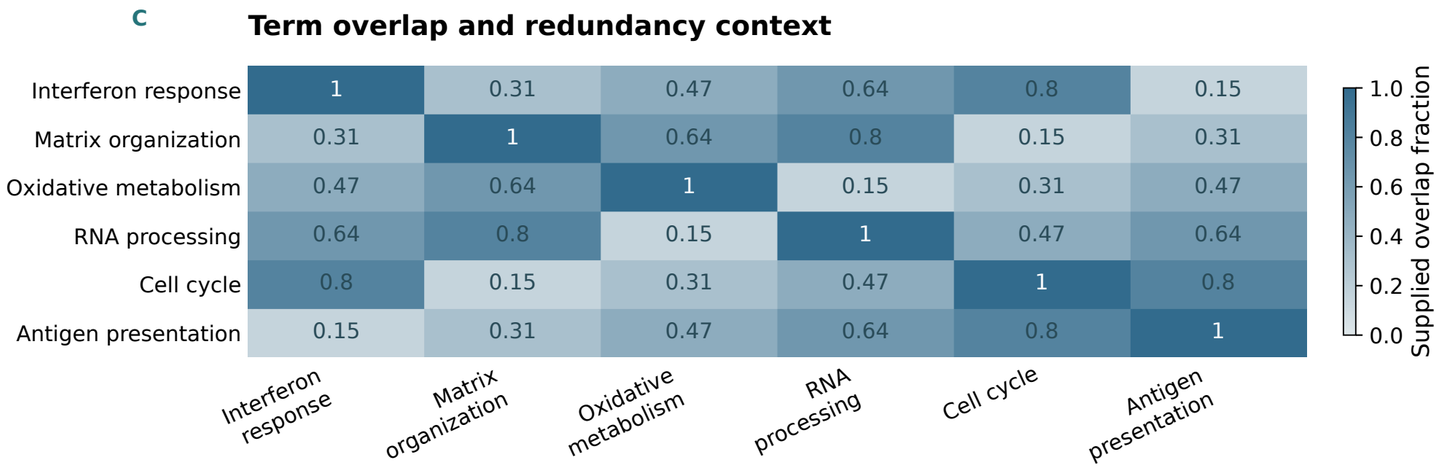
Term overlap and redundancy context



Bulk · ora — complementary evidence



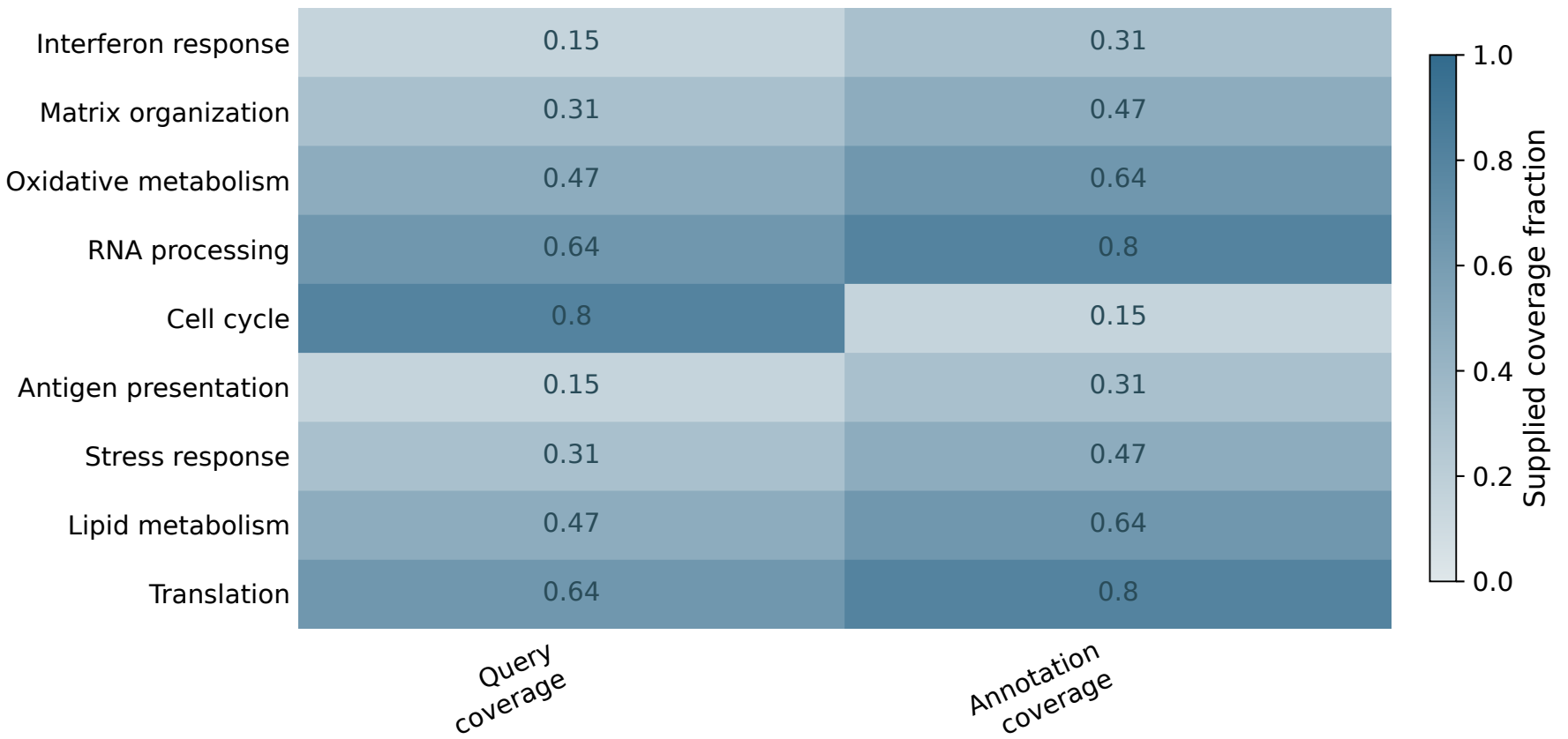




Bulk · ora — complementary evidence

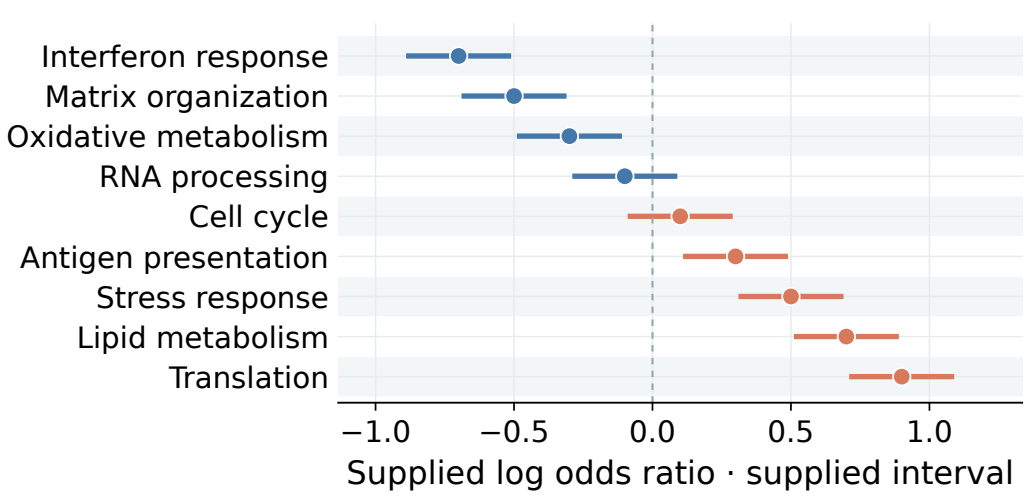
A

Selected queries and tested-universe coverage



B

Prespecified enrichment effect estimates

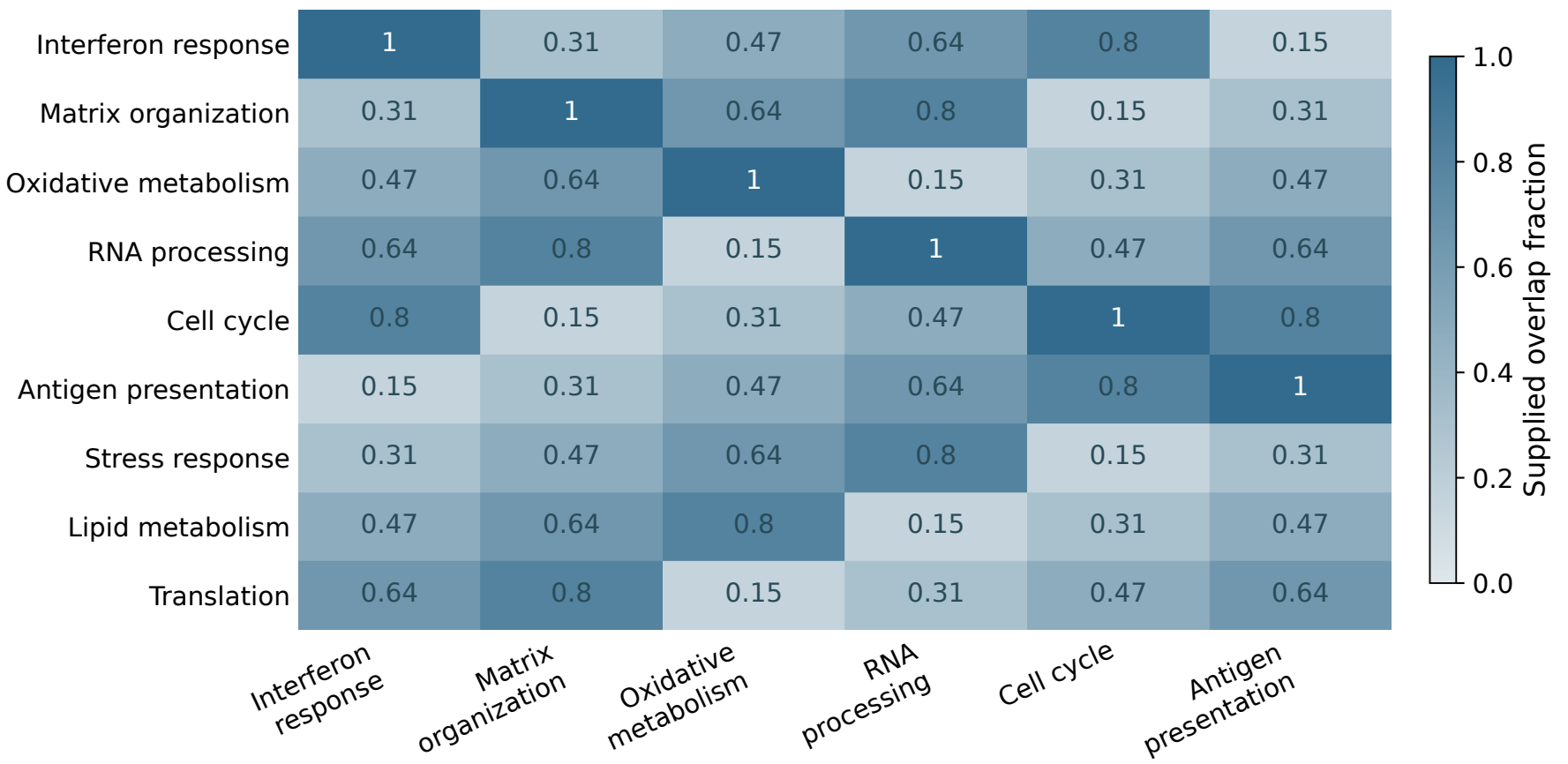


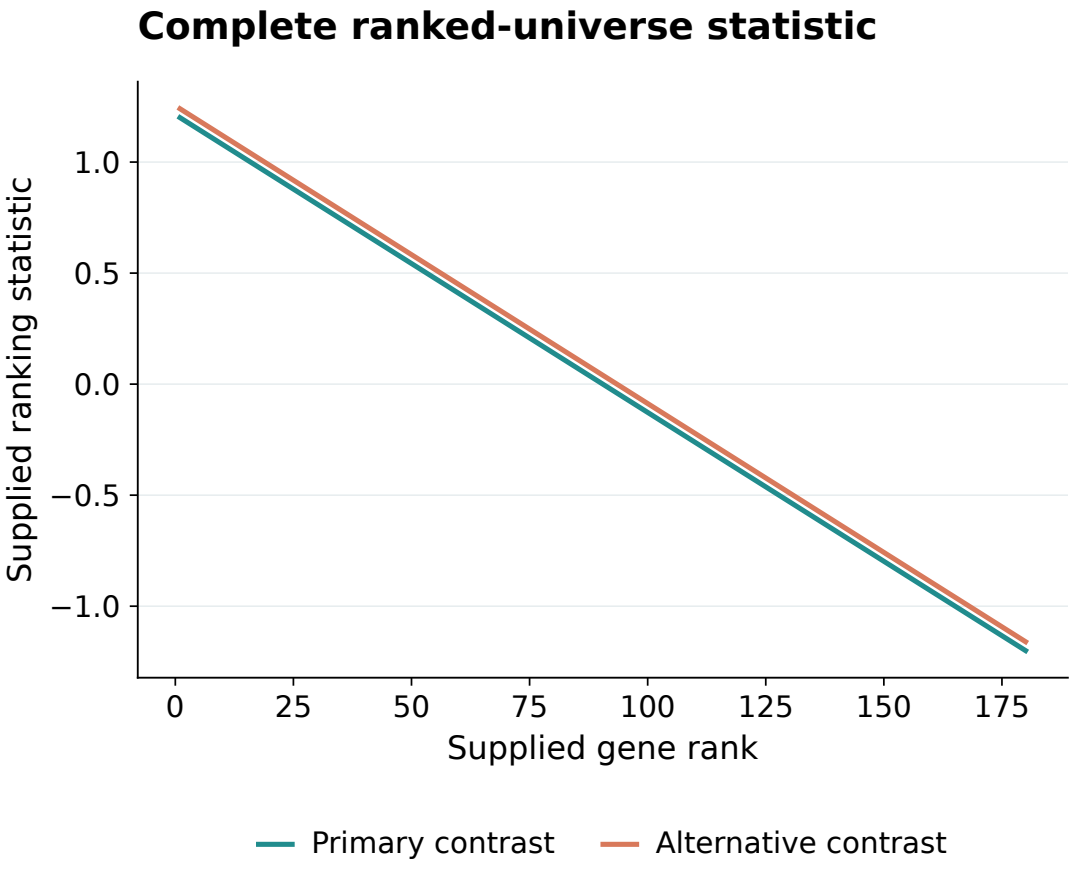
Estimate [interval] · q
Upstream donor n = 24

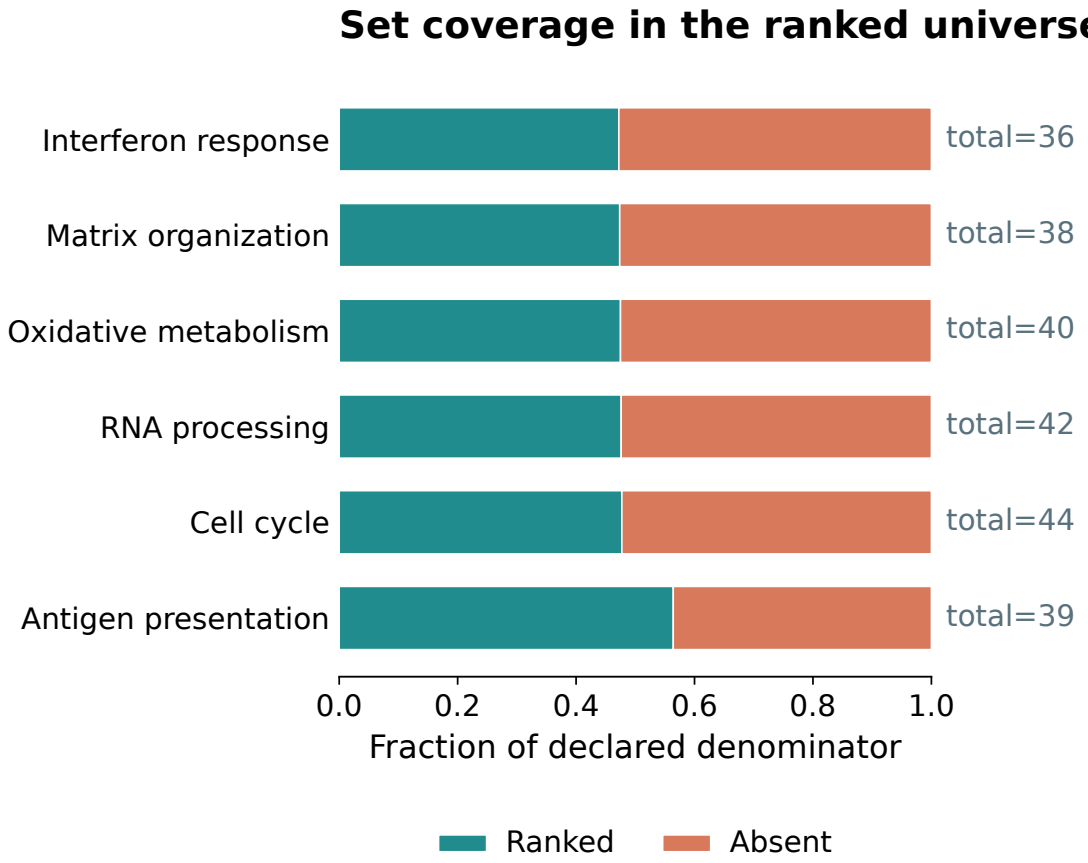
-0.70 [-0.88, -0.52] 0.002
-0.50 [-0.68, -0.32] 0.02
-0.30 [-0.48, -0.12] 0.038
-0.10 [-0.28, 0.08] 0.65
0.10 [-0.08, 0.28] 0.65
0.30 [0.12, 0.48] 0.092
0.50 [0.32, 0.68] 0.11
0.70 [0.52, 0.88] 0.13
0.90 [0.72, 1.08] 0.15

C

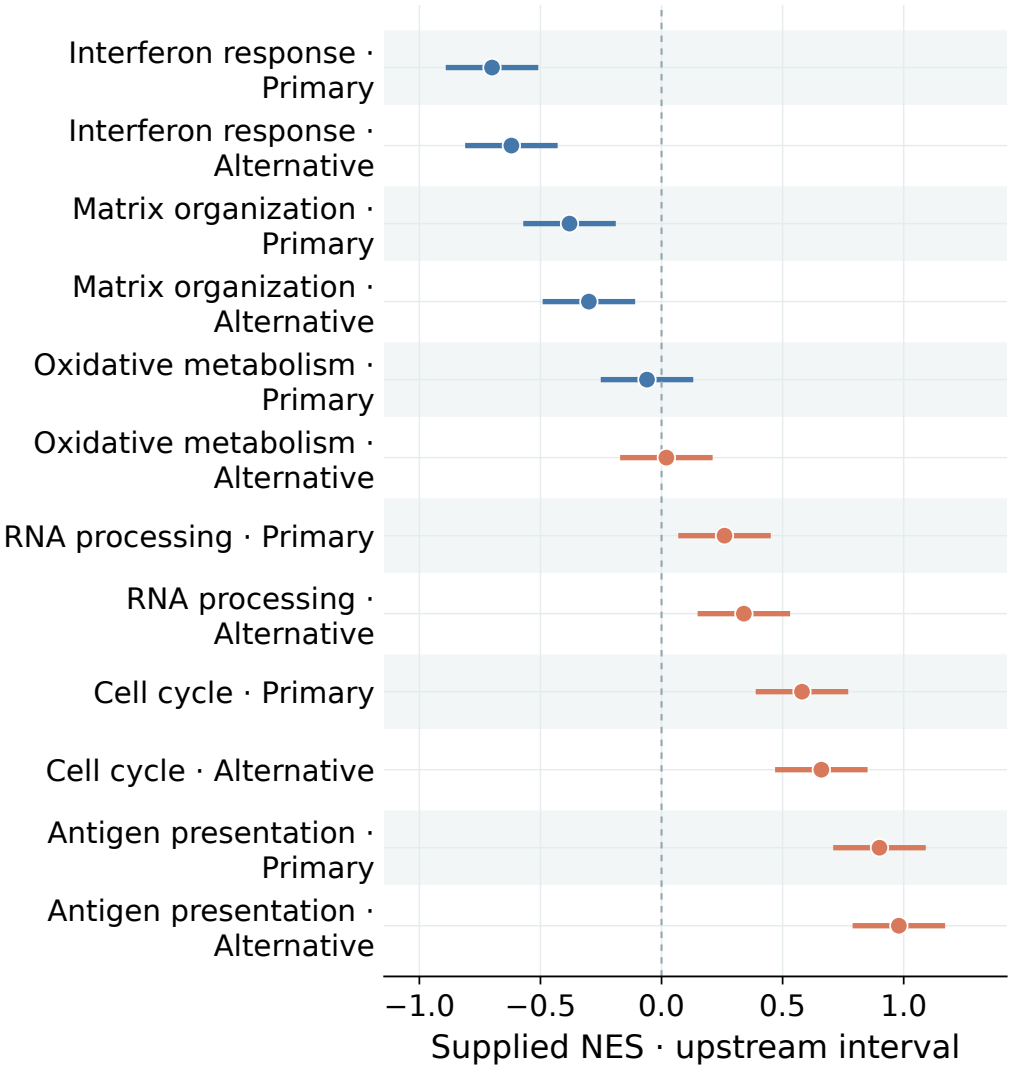
Term overlap and redundancy context




Complete ranked-universe statistic
Supplied ranking statistic
1.0
0.5
0.0
−0.5
−1.0
0
25
50
75
100
125
150
175
Supplied gene rank
Primary contrast
Alternative contrast

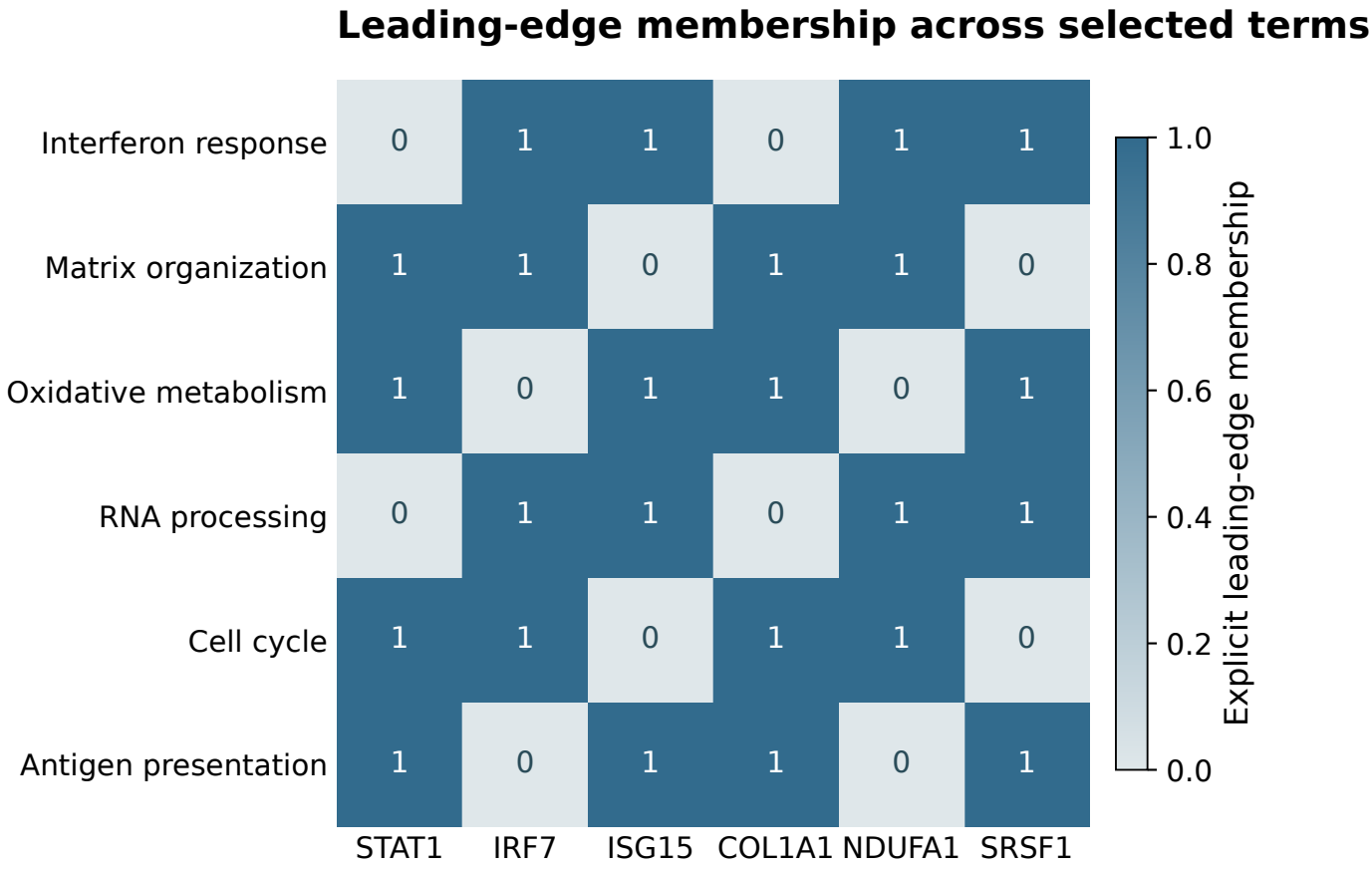

Set coverage in the ranked universe
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Ranked
Absent

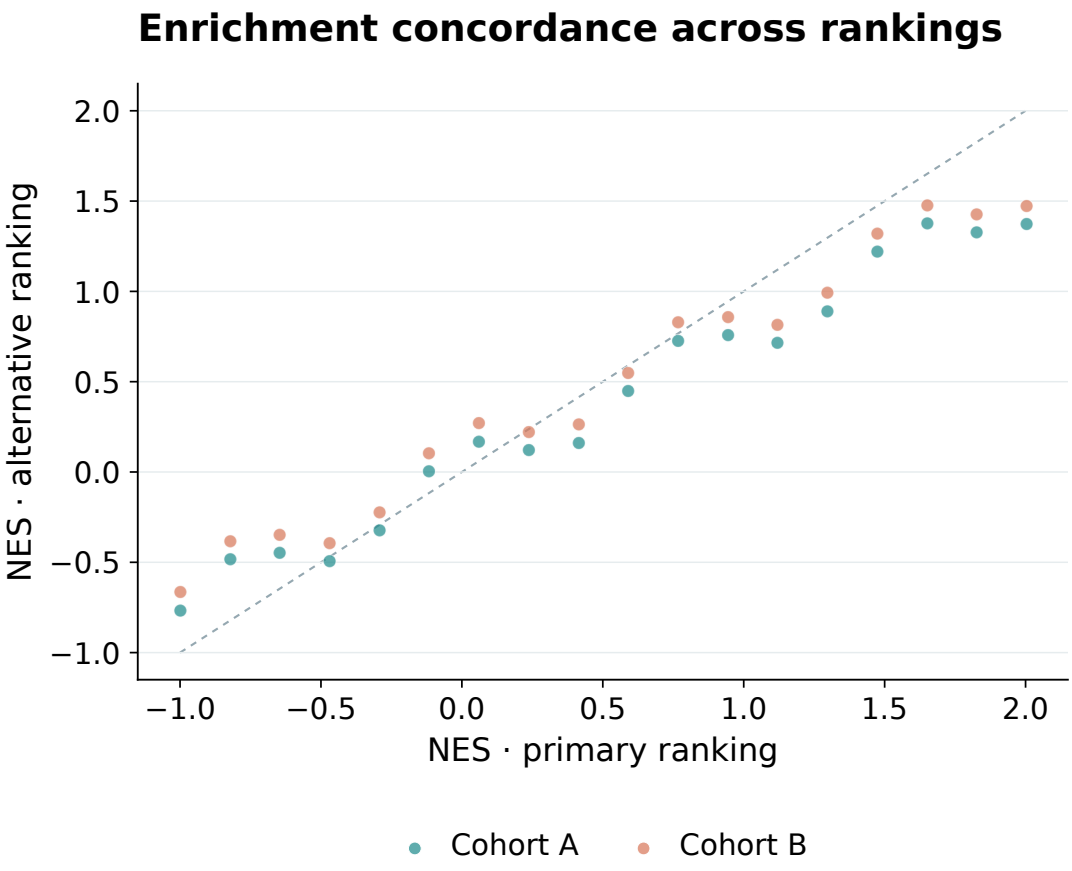
Normalized enrichment across contrasts



Estimate [interval] · q
Upstream donor n = 12

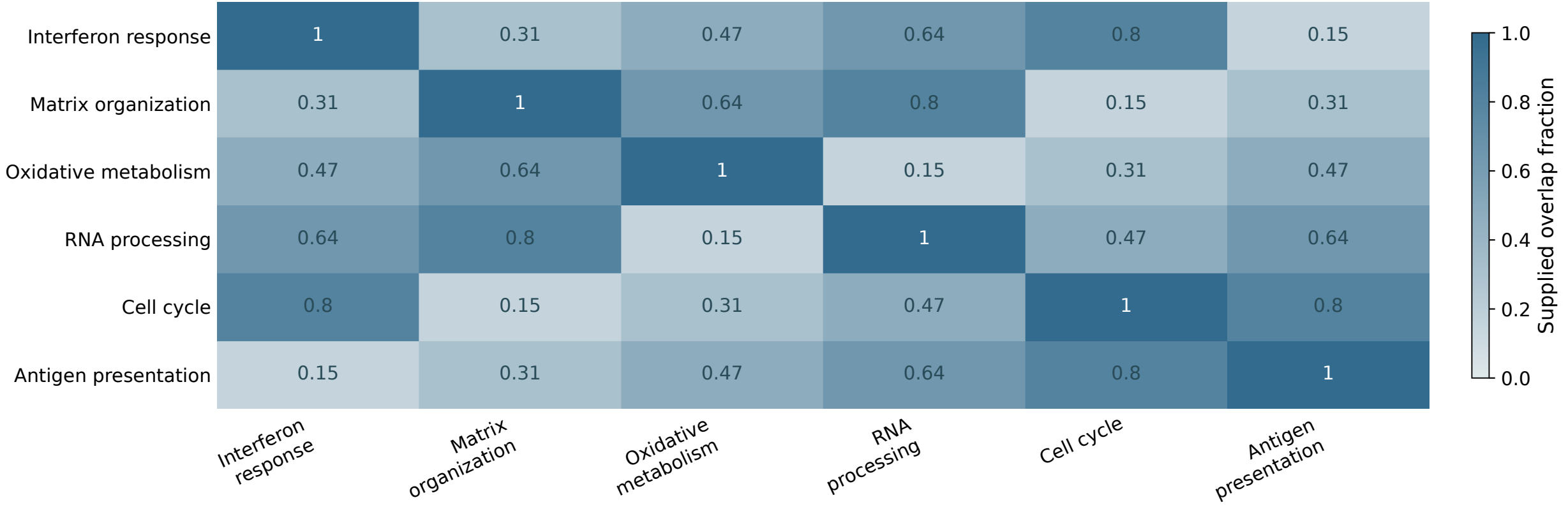
Interferon response · Primary	-0.70 [-0.88, -0.52] 0.002
Interferon response · Alternative	-0.62 [-0.80, -0.44] 0.005
Matrix organization · Primary	-0.38 [-0.56, -0.20] 0.02
Matrix organization · Alternative	-0.30 [-0.48, -0.12] 0.023
Oxidative metabolism · Primary	-0.06 [-0.24, 0.12] 0.65
Oxidative metabolism · Alternative	0.02 [-0.16, 0.20] 0.65
RNA processing · Primary	0.26 [0.08, 0.44] 0.056
RNA processing · Alternative	0.34 [0.16, 0.52] 0.059
Cell cycle · Primary	0.58 [0.40, 0.76] 0.074
Cell cycle · Alternative	0.66 [0.48, 0.84] 0.077
Antigen presentation · Primary	0.90 [0.72, 1.08] 0.092
Antigen presentation · Alternative	0.98 [0.80, 1.16] 0.095


Leading-edge membership across selected terms
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
0 1 1 0 1 1
1 1 0 1 1 0
1 0 1 1 0 1
0 1 1 0 1 1
1 1 0 1 1 0
1 0 1 1 0 1
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
Explicit leading-edge membership
1.0
0.8
0.6
0.4
0.2
0.0



Software test fixture; not biological evidence

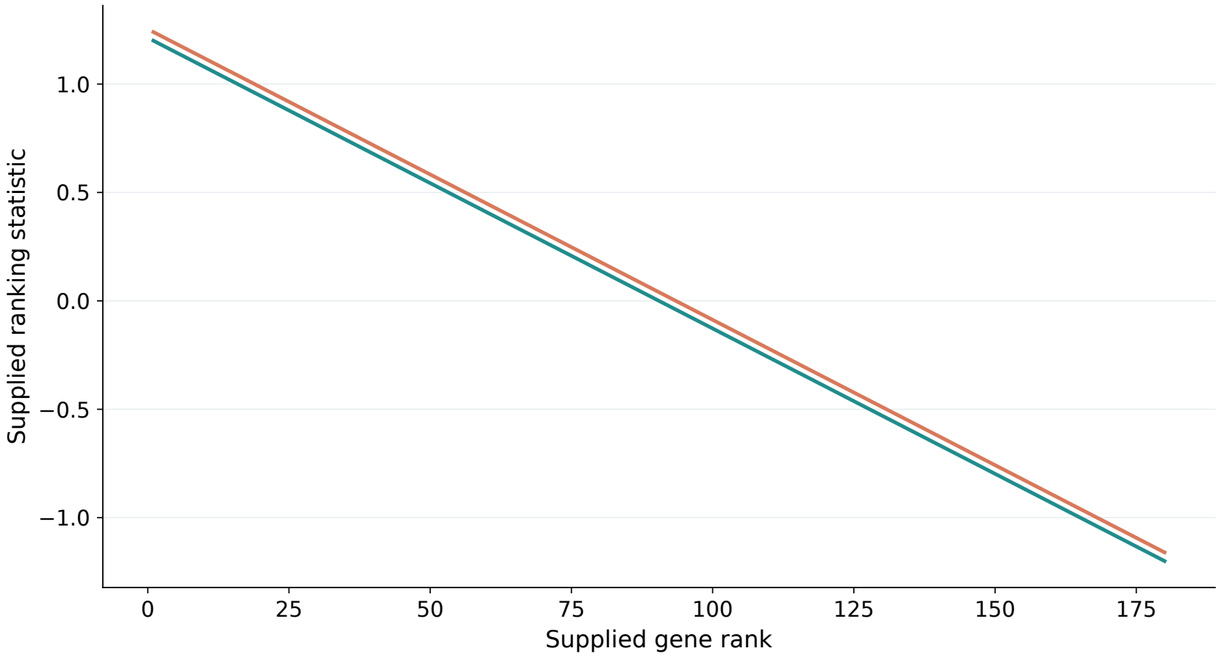
Leading-edge overlap across pathways



Bulk · gsea — complementary evidence

A

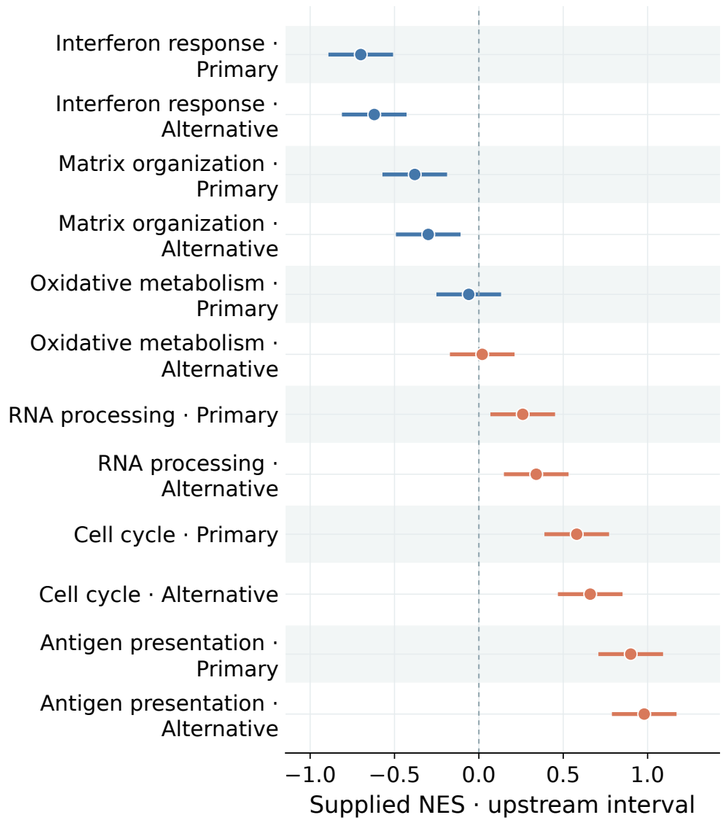
Complete ranked-universe statistic



B

Normalized enrichment across contrasts

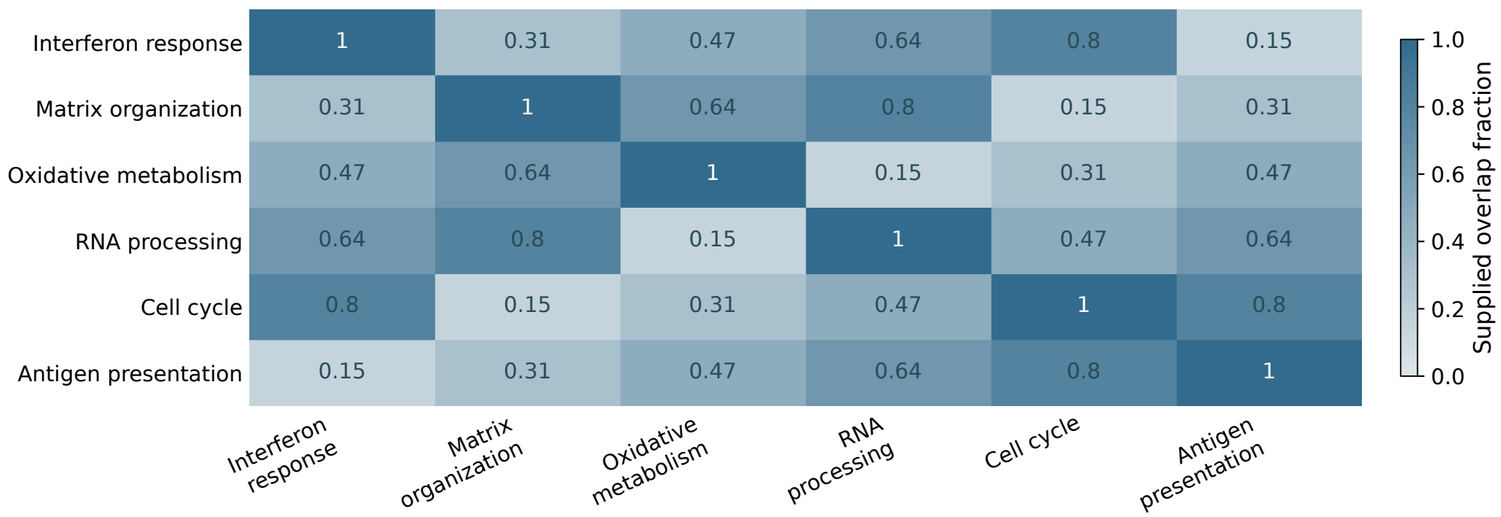
Estimate [interval] · q
Upstream donor n = 12



-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095

C

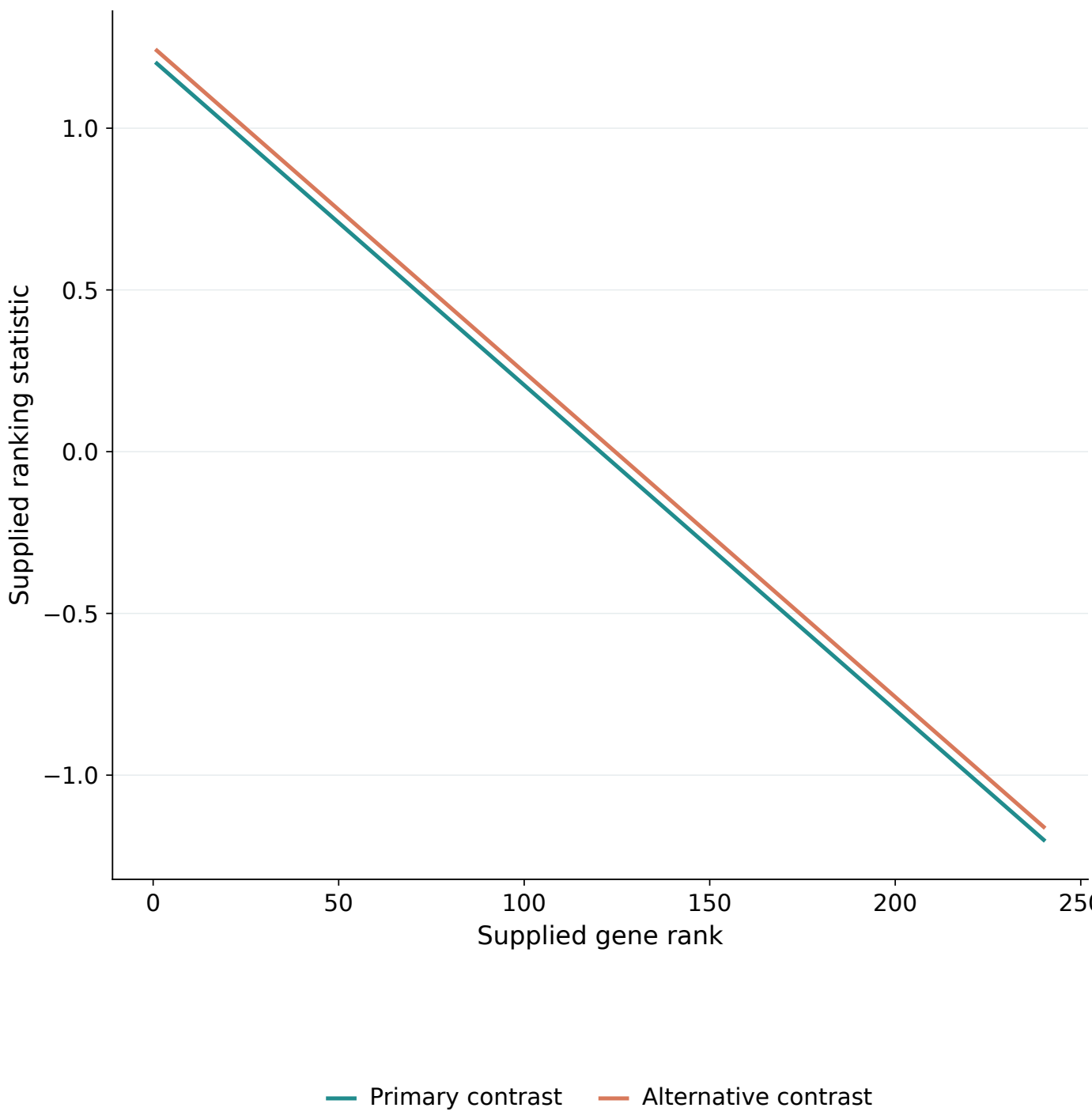
Leading-edge overlap across pathways



Bulk · gsea — complementary evidence

A

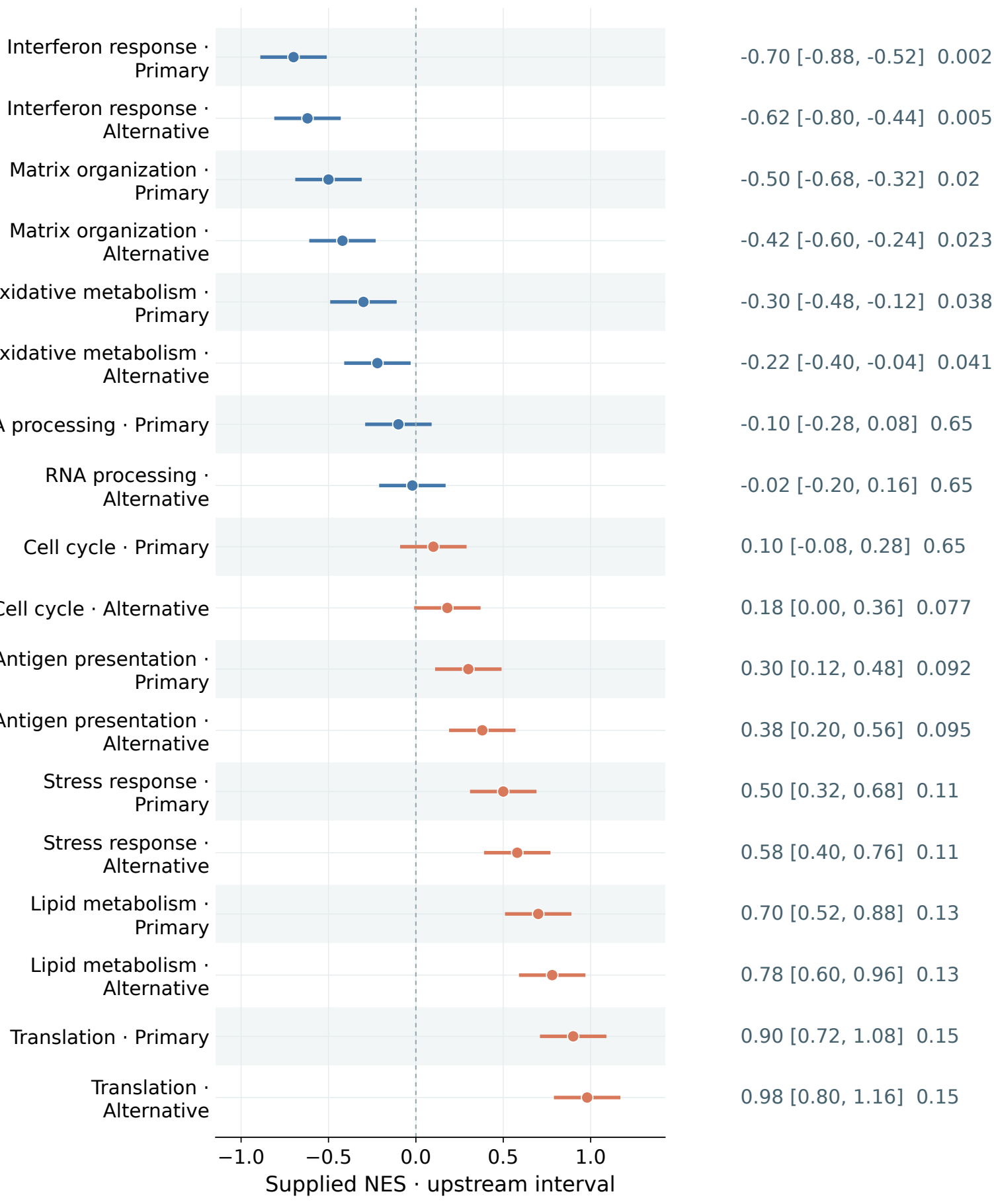
Complete ranked-universe statistic



B

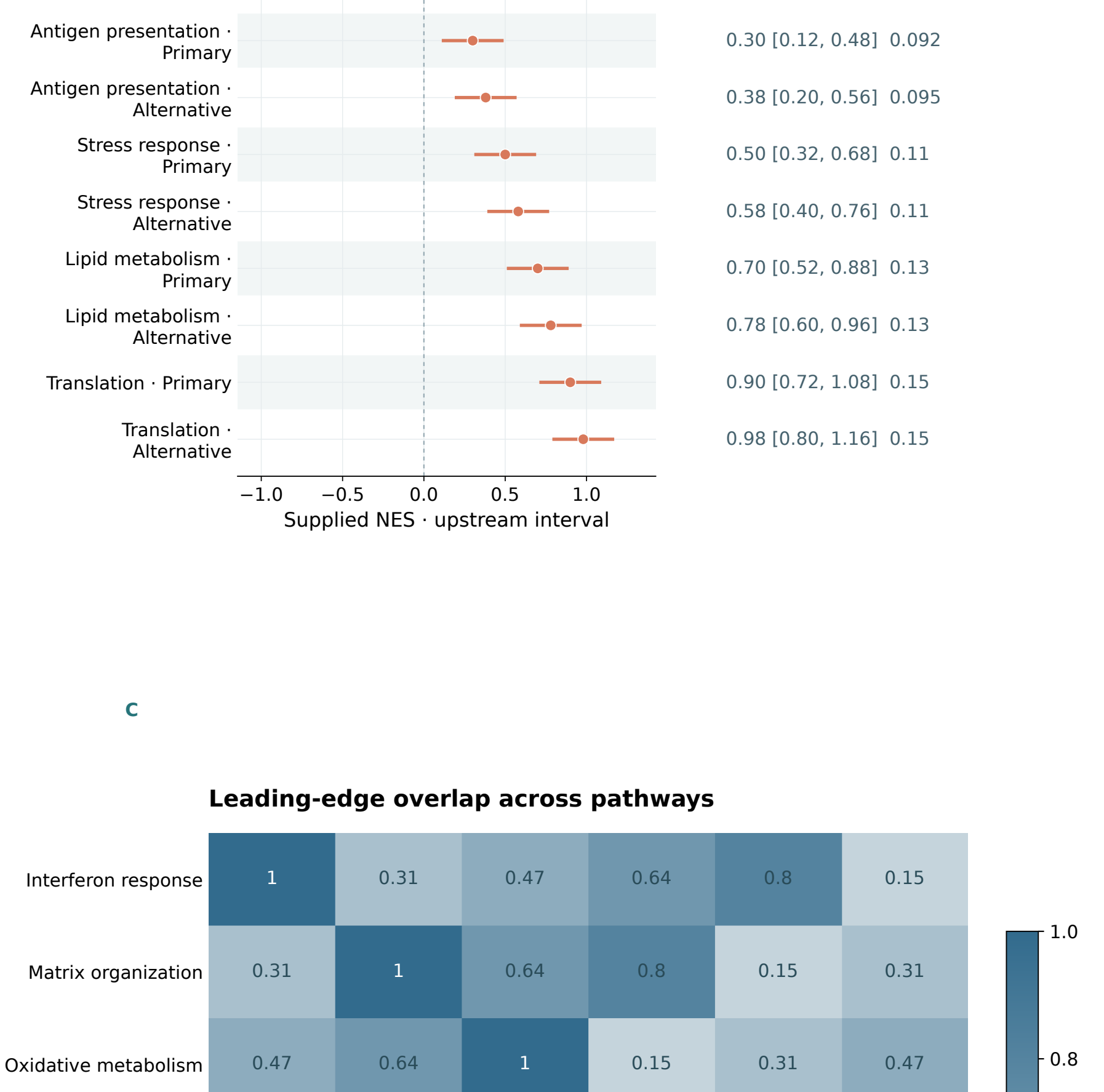
Normalized enrichment across contrasts

Estimate [interval] · q
Upstream donor n = 24



C

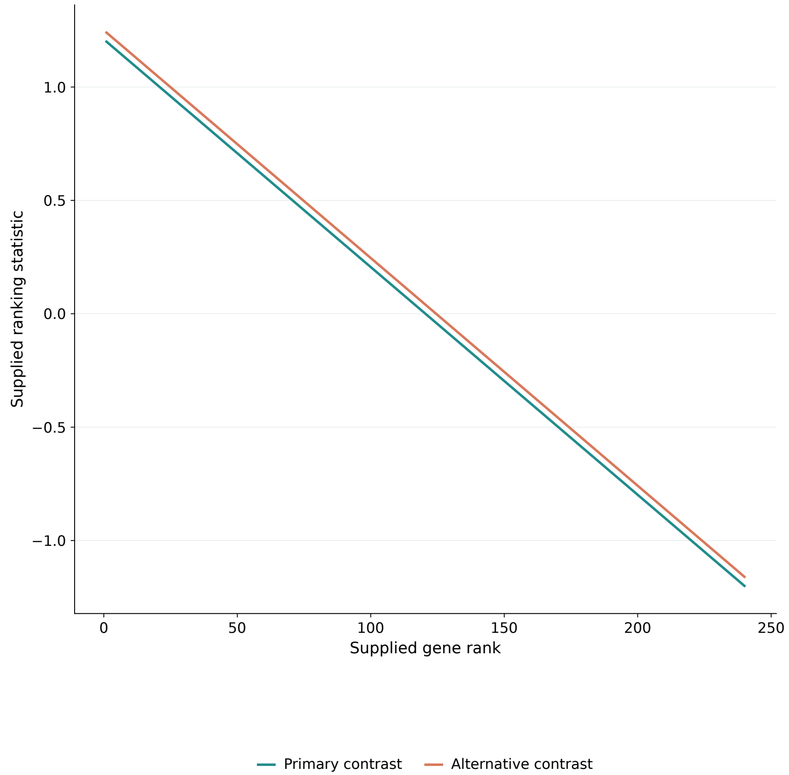
Leading-edge overlap across pathways



Bulk · gsea — complementary evidence

A

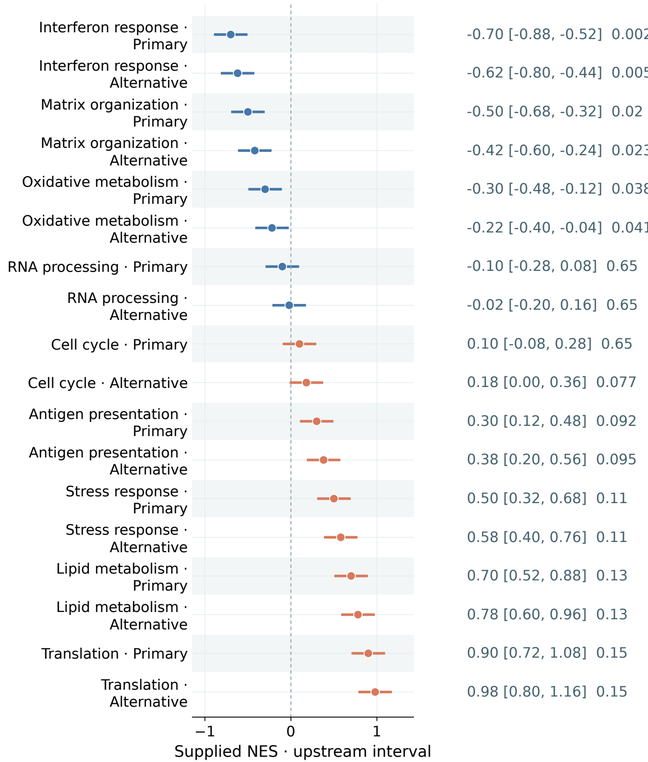
Complete ranked-universe statistic



B

Normalized enrichment across contrasts

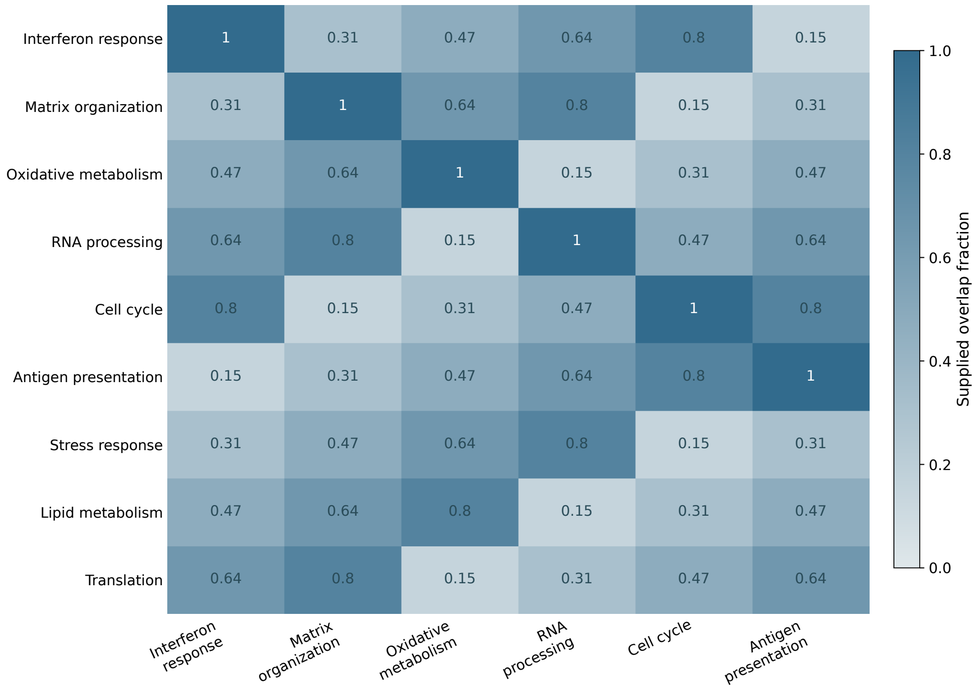
Estimate [interval] · q
Upstream donor n = 24

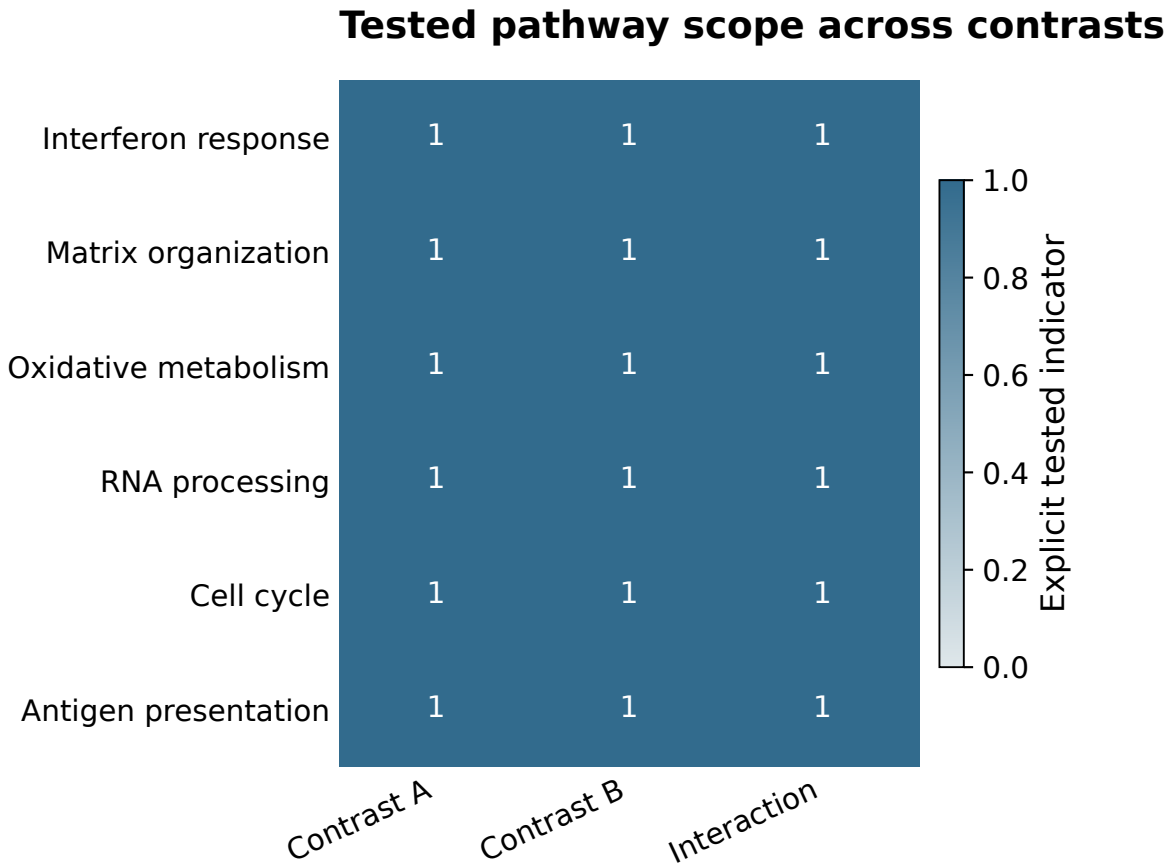


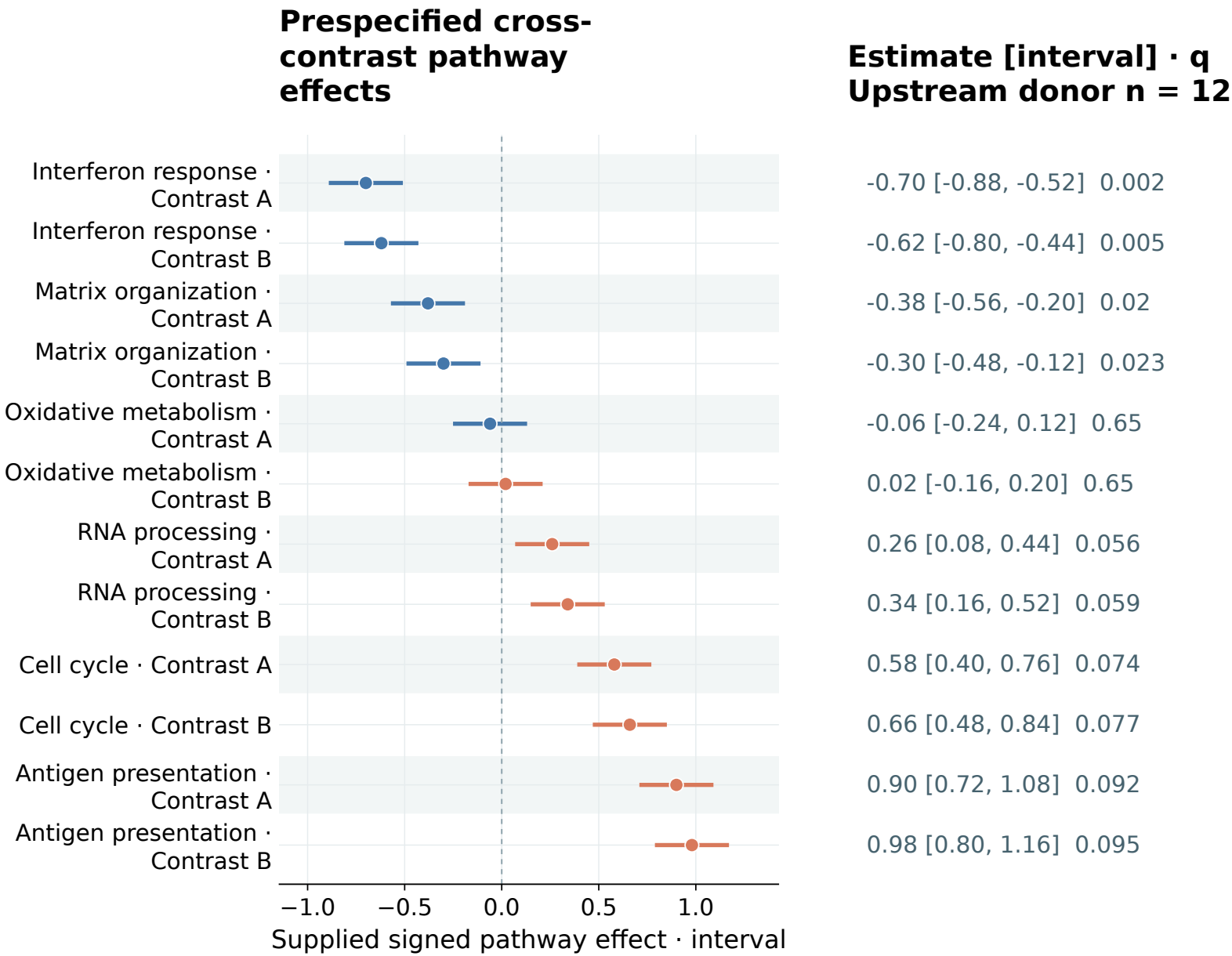
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.50 [-0.68, -0.32] 0.02
-0.42 [-0.60, -0.24] 0.023
-0.30 [-0.48, -0.12] 0.038
-0.22 [-0.40, -0.04] 0.041
-0.10 [-0.28, 0.08] 0.65
-0.02 [-0.20, 0.16] 0.65
0.10 [-0.08, 0.28] 0.65
0.18 [0.00, 0.36] 0.077
0.30 [0.12, 0.48] 0.092
0.38 [0.20, 0.56] 0.095
0.50 [0.32, 0.68] 0.11
0.58 [0.40, 0.76] 0.11
0.70 [0.52, 0.88] 0.13
0.78 [0.60, 0.96] 0.13
0.90 [0.72, 1.08] 0.15
0.98 [0.80, 1.16] 0.15

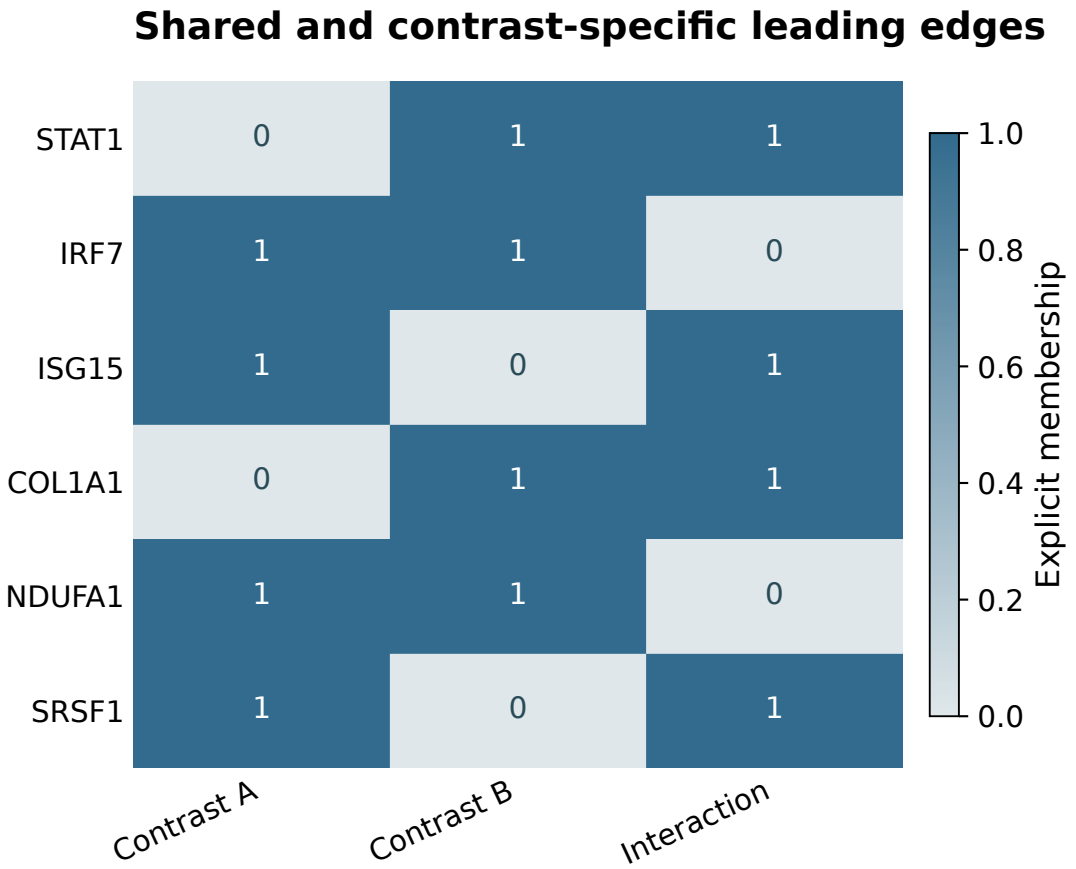
C

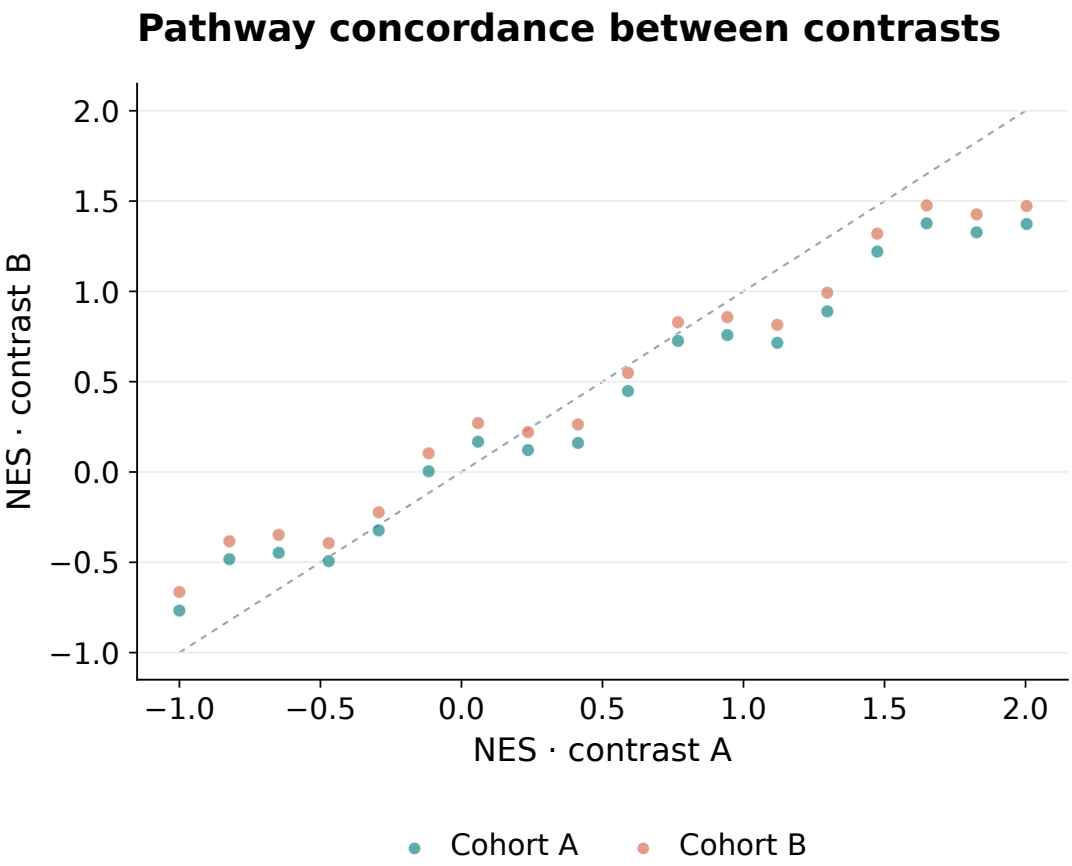
Leading-edge overlap across pathways

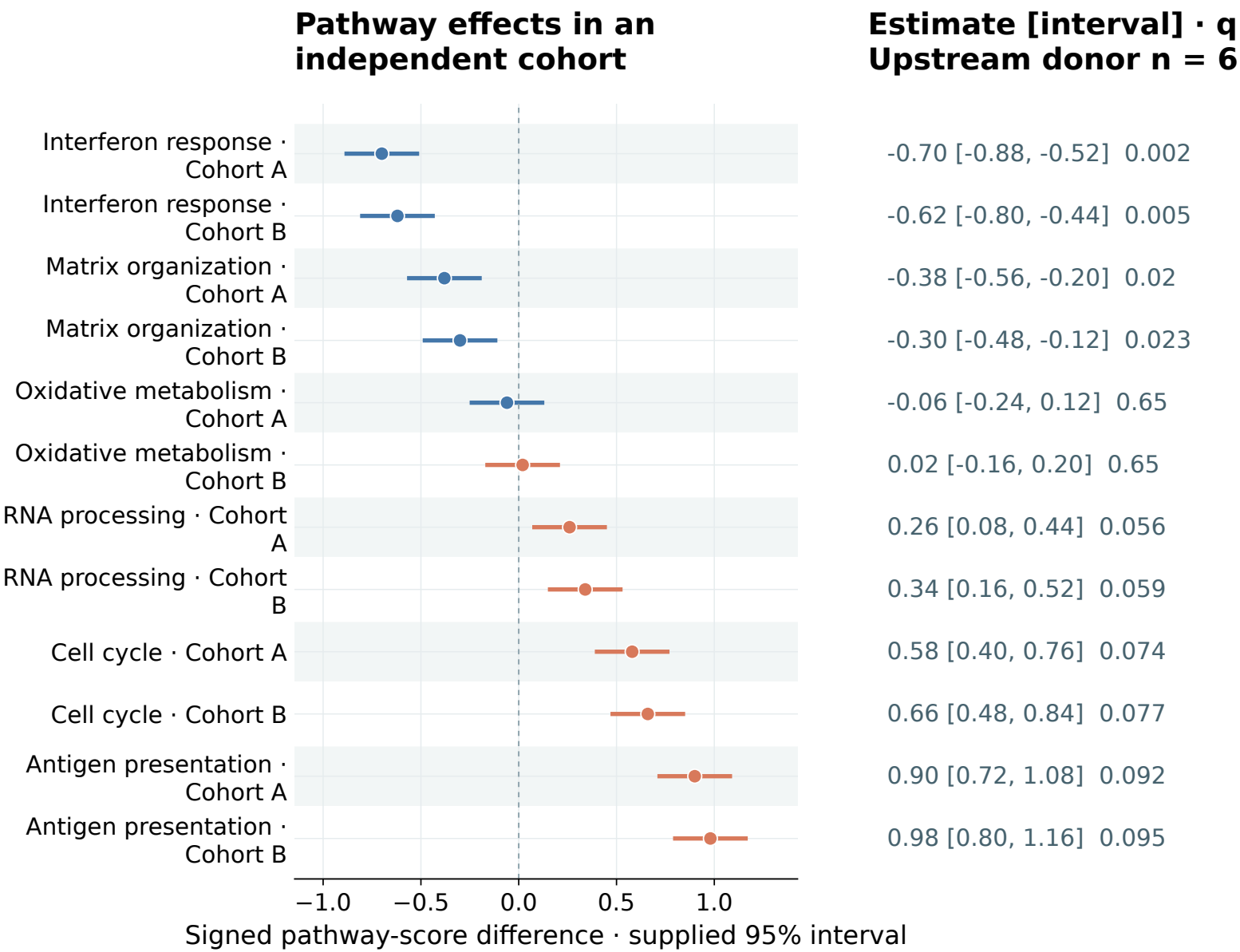



Tested pathway scope across contrasts
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
1
Contrast A
Contrast B
Interaction
Explicit tested indicator
1.0
0.8
0.6
0.4
0.2
0.0


Prespecified cross-contrast pathway effects
Estimate [interval] · q
Upstream donor n = 12
Interferon response · Contrast A
Interferon response · Contrast B
Matrix organization · Contrast A
Matrix organization · Contrast B
Oxidative metabolism · Contrast A
Oxidative metabolism · Contrast B
RNA processing · Contrast A
RNA processing · Contrast B
Cell cycle · Contrast A
Cell cycle · Contrast B
Antigen presentation · Contrast A
Antigen presentation · Contrast B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Supplied signed pathway effect · interval


Shared and contrast-specific leading edges
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0
1
1
1
1
0
1
0
1
0
1
1
1
1
0
1
0
1
Contrast A
Contrast B
Interaction
Explicit membership
1.0
0.8
0.6
0.4
0.2
0.0

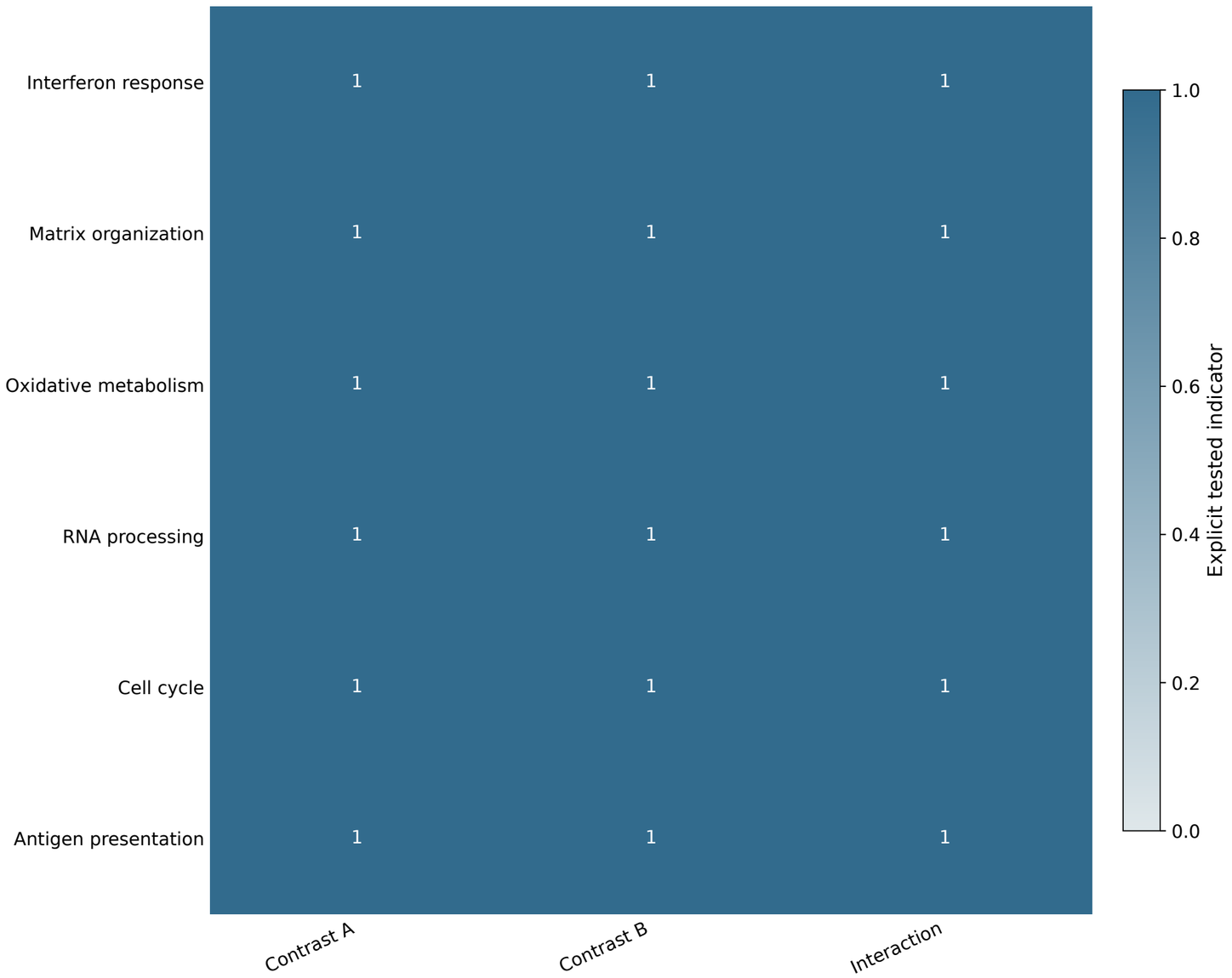

Pathway concordance between contrasts
NES · contrast B
NES · contrast A
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Cohort A
Cohort B


Pathway effects in an independent cohort
Estimate [interval] · q
Upstream donor n = 6
Interferon response · Cohort A
Interferon response · Cohort B
Matrix organization · Cohort A
Matrix organization · Cohort B
Oxidative metabolism · Cohort A
Oxidative metabolism · Cohort B
RNA processing · Cohort A
RNA processing · Cohort B
Cell cycle · Cohort A
Cell cycle · Cohort B
Antigen presentation · Cohort A
Antigen presentation · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Signed pathway-score difference · supplied 95% interval

Bulk · pathway_contrasts — complementary evidence

A

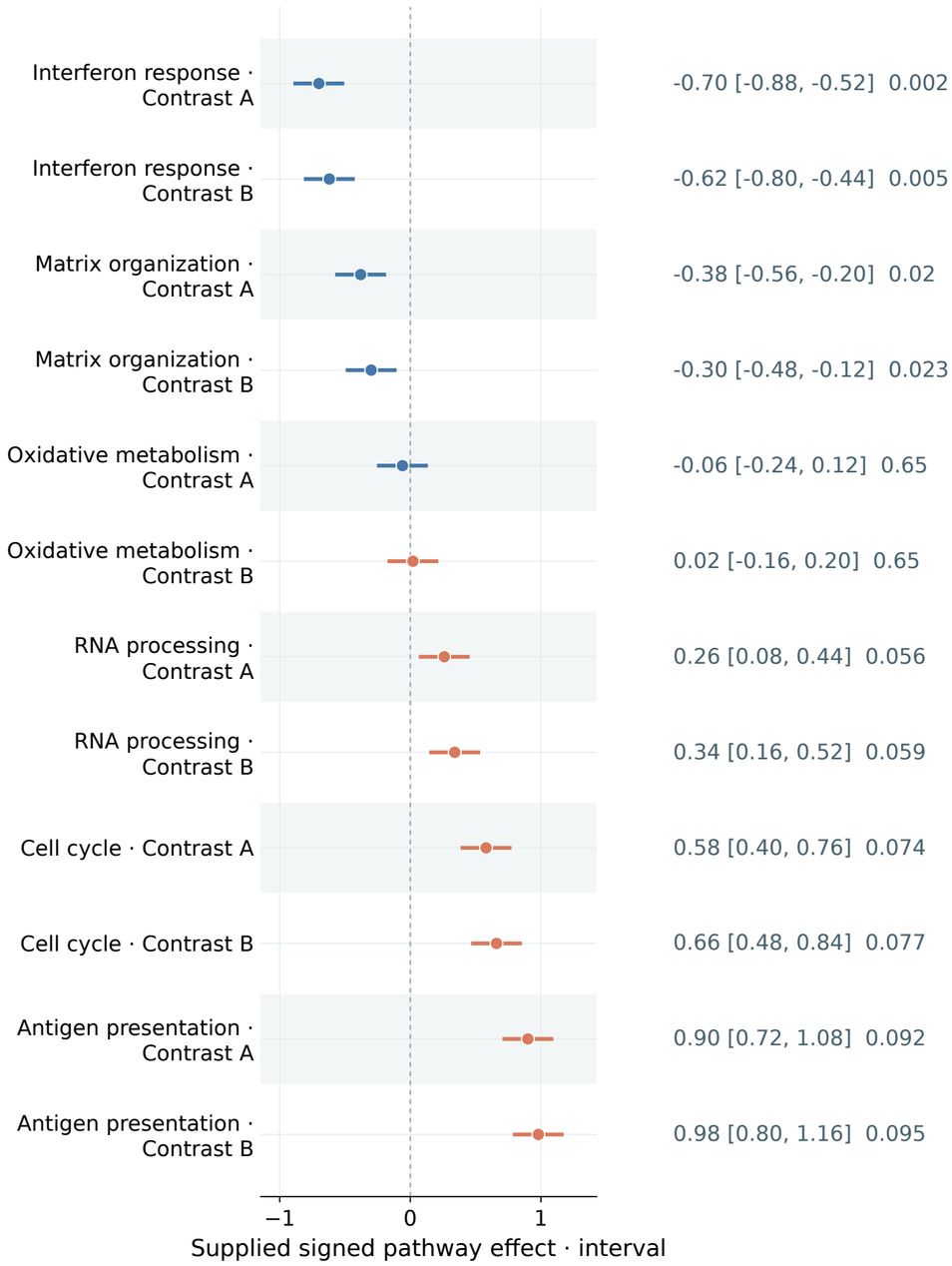
Tested pathway scope across contrasts



B

Prespecified cross-contrast pathway effects

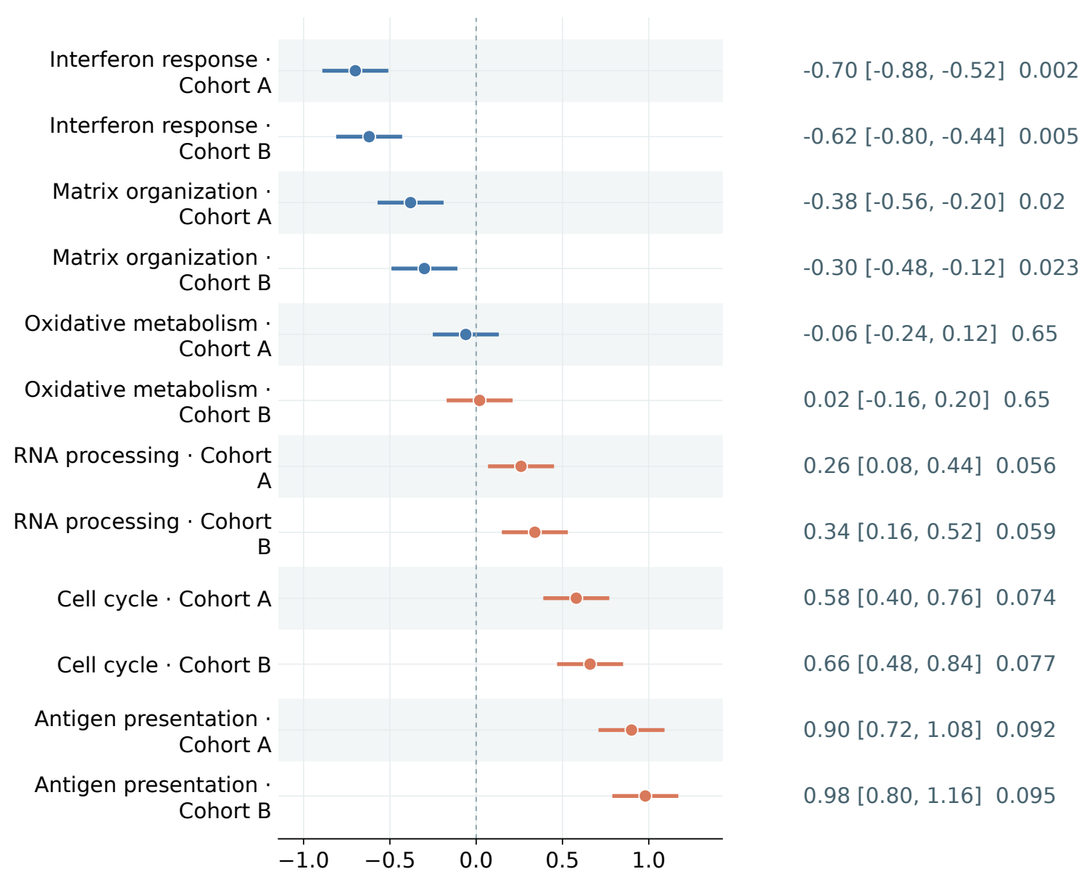
Estimate [interval] · q
Upstream donor n = 12



C

Pathway effects in an independent cohort

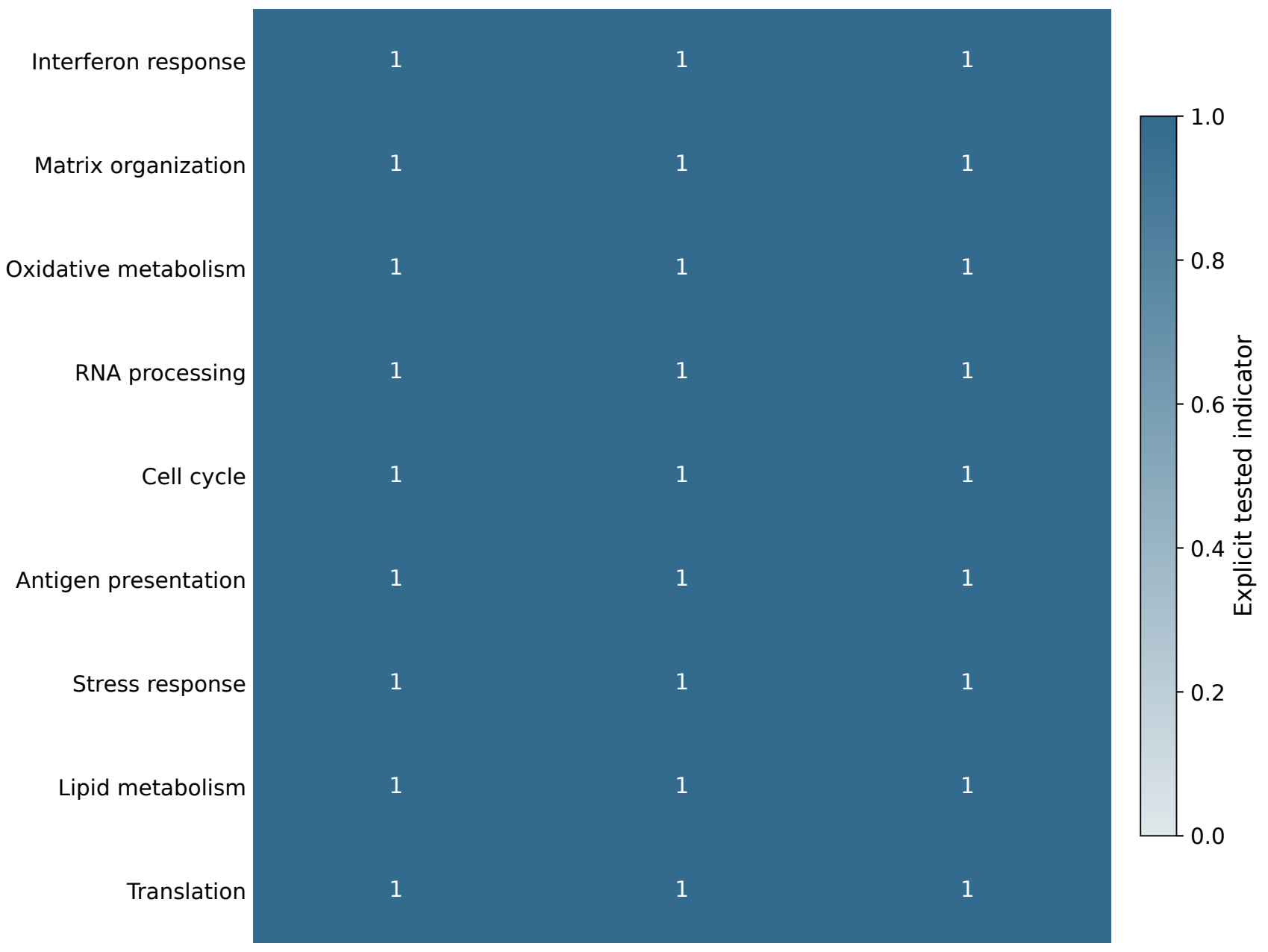
Estimate [interval] · q
Upstream donor n = 6



Bulk · pathway_contrasts — complementary evidence

A

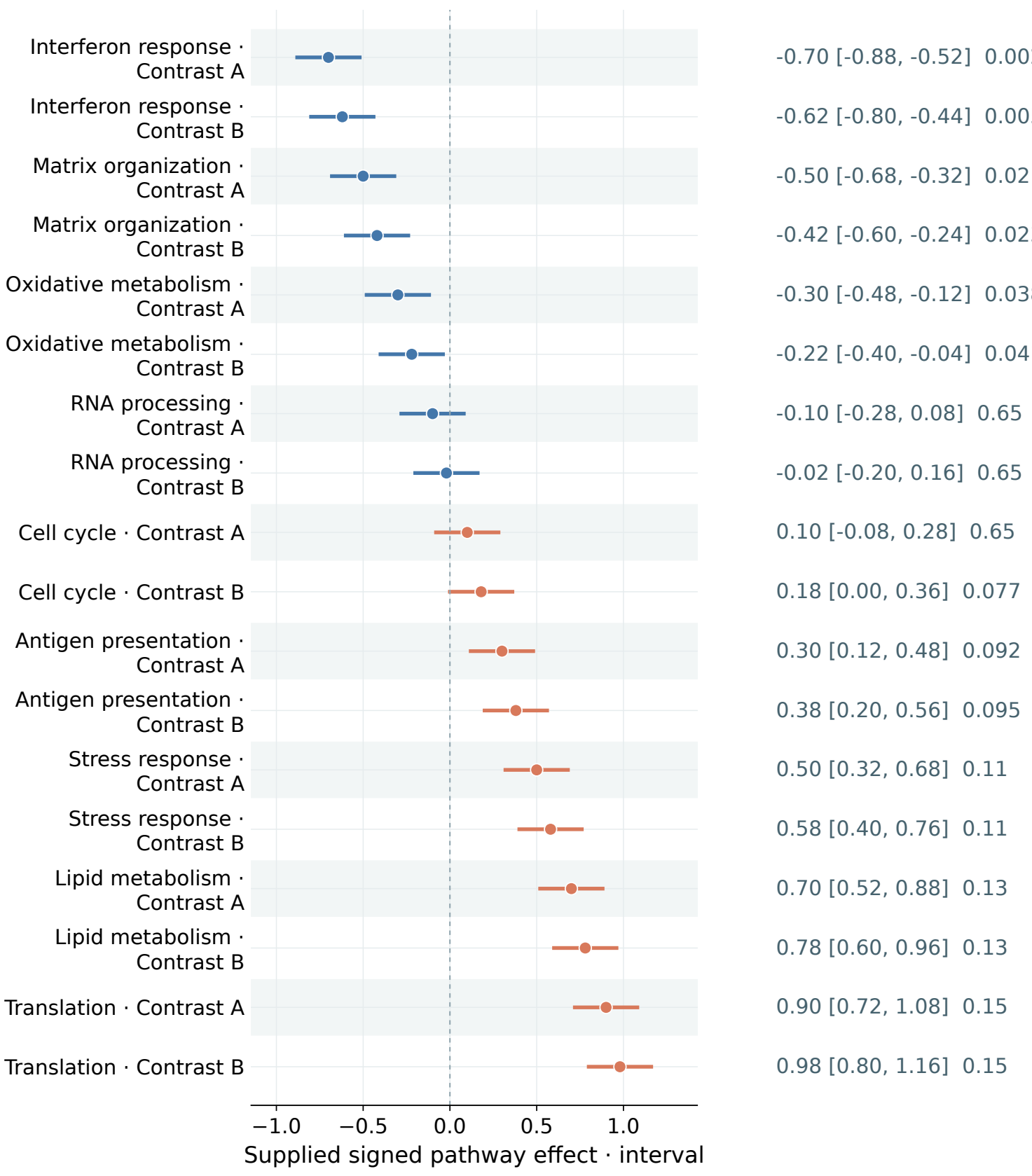
Tested pathway scope across contrasts



B

Prespecified cross-contrast pathway effects

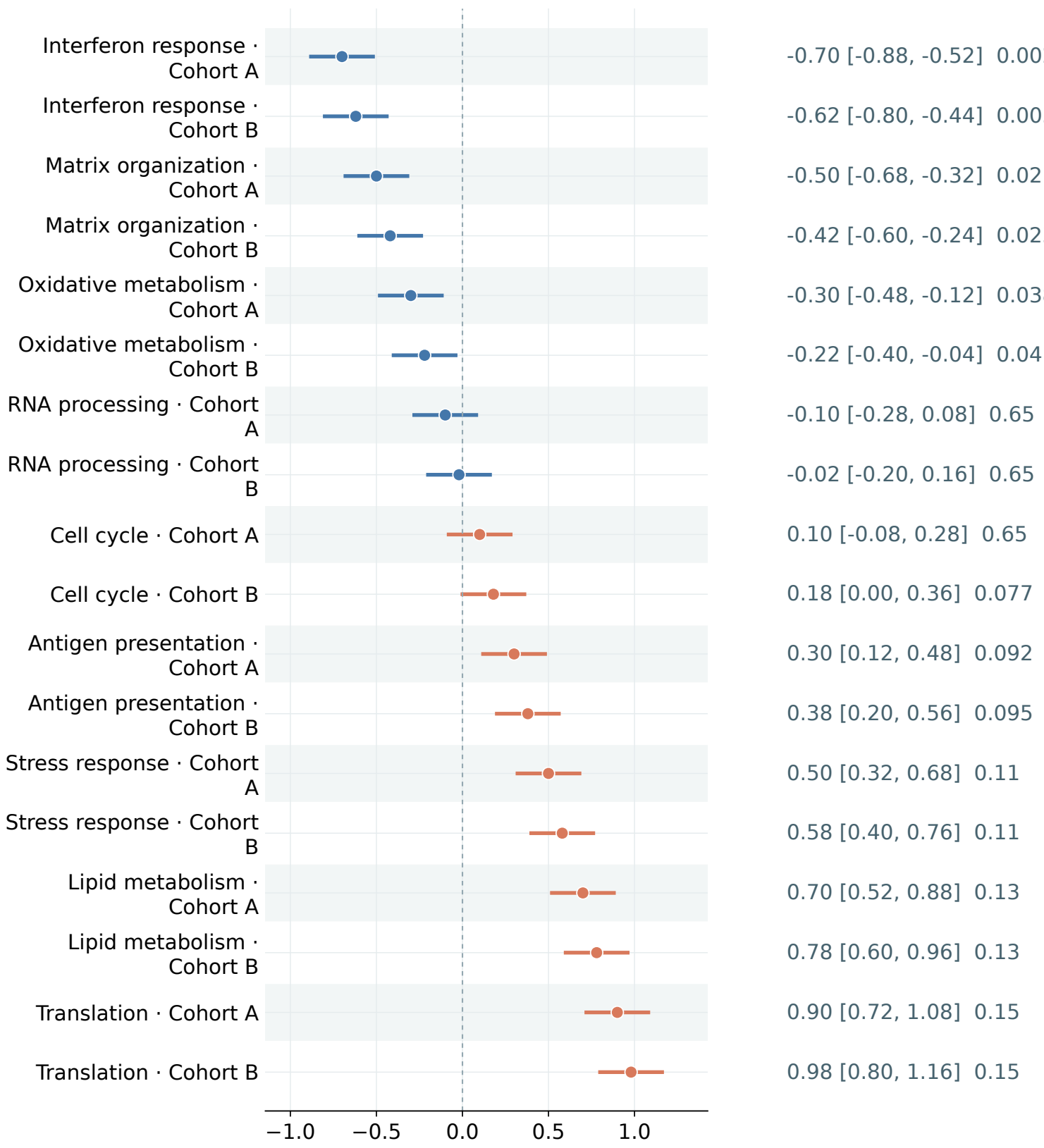
Estimate [interval] · q
Upstream donor n = 24



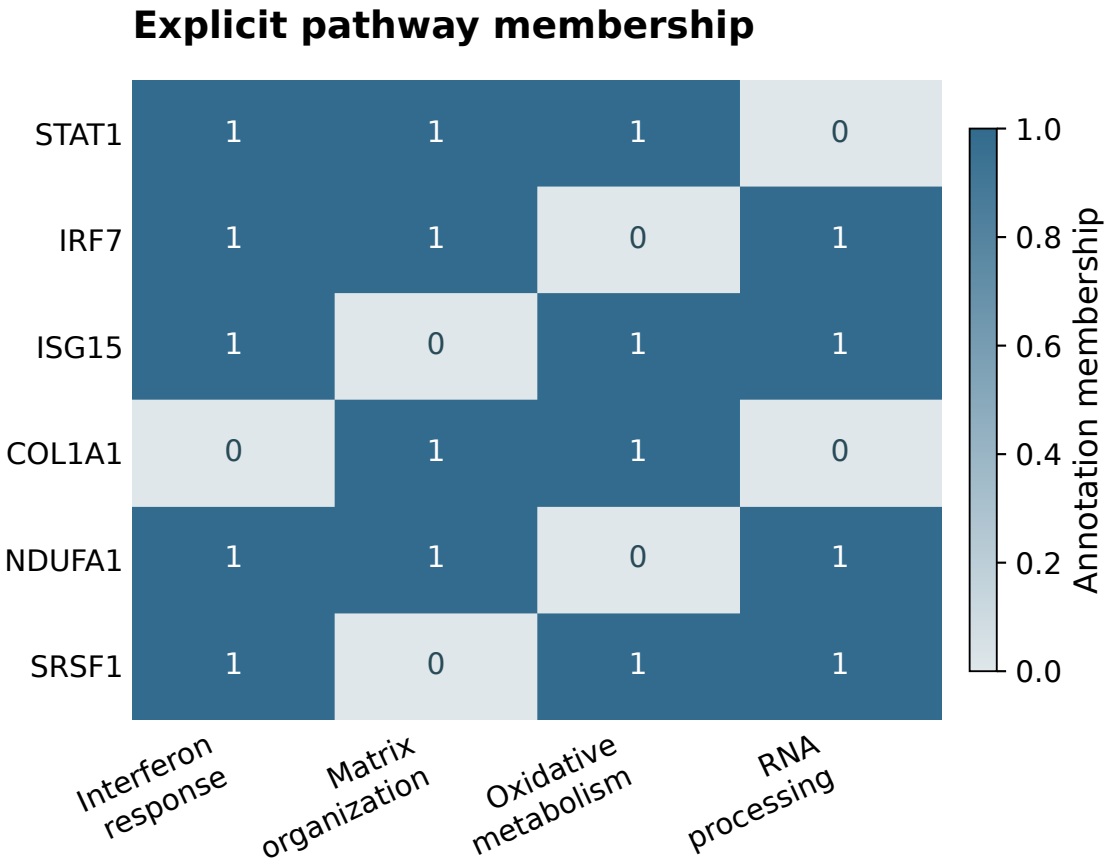
C

Pathway effects in an independent cohort

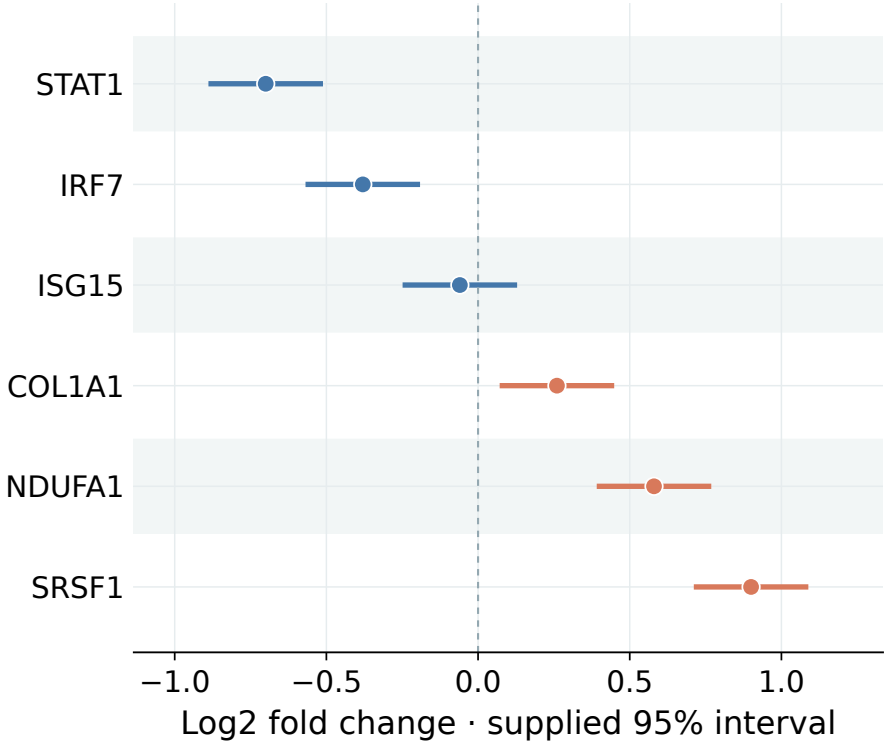
Estimate [interval] · q
Upstream donor n = 12



Software test fixture; not biological evidence


Explicit pathway membership
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
1 1 1 0
1 1 0 1
1 0 1 1
0 1 1 0
1 1 0 1
1 0 1 1
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Annotation membership
1.0
0.8
0.6
0.4
0.2
0.0

Supporting-gene effect estimates



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

-0.38 [-0.56, -0.20] 0.02

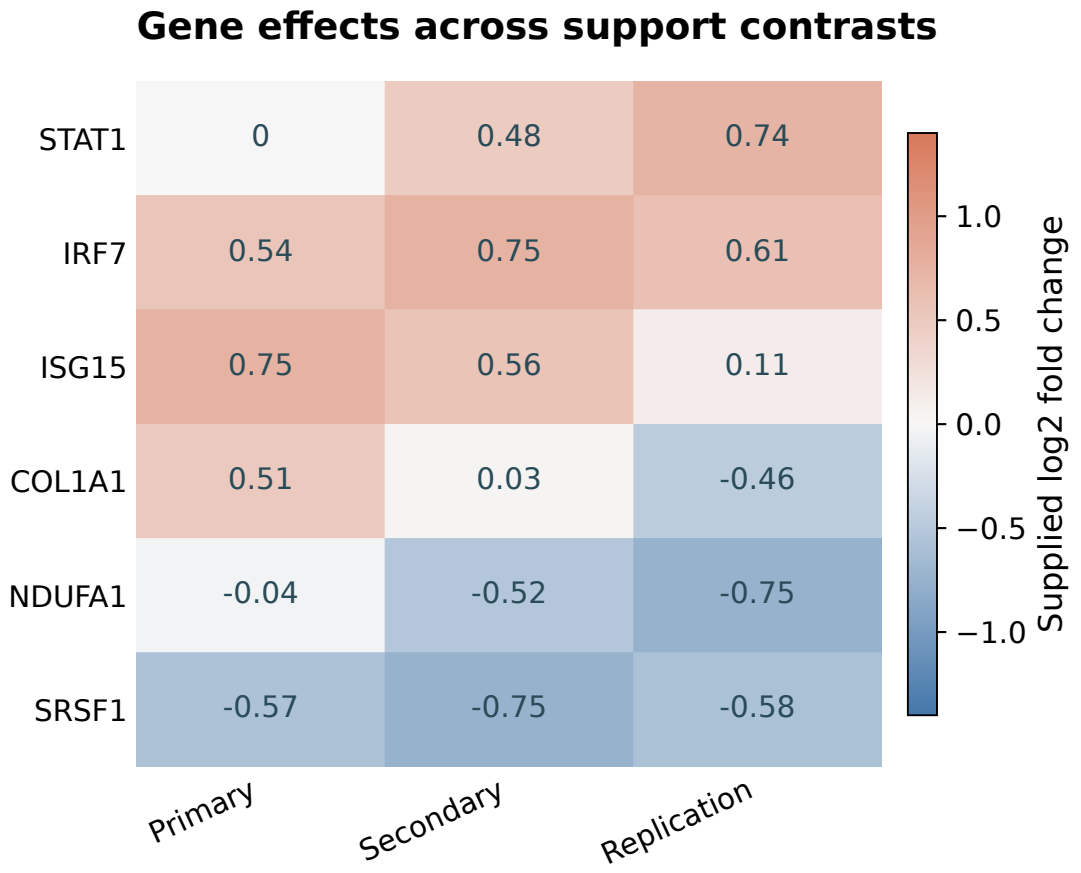
-0.06 [-0.24, 0.12] 0.65

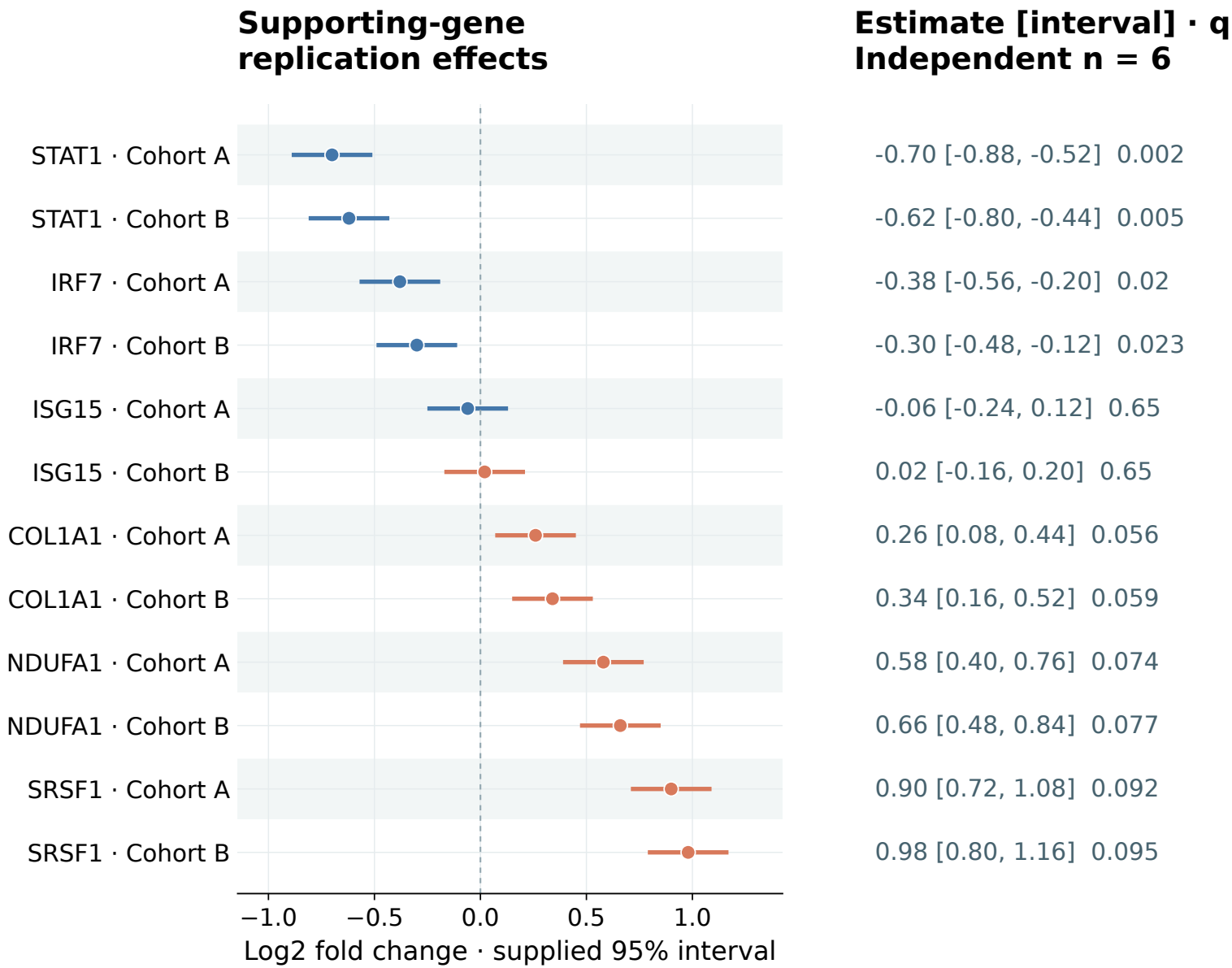
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

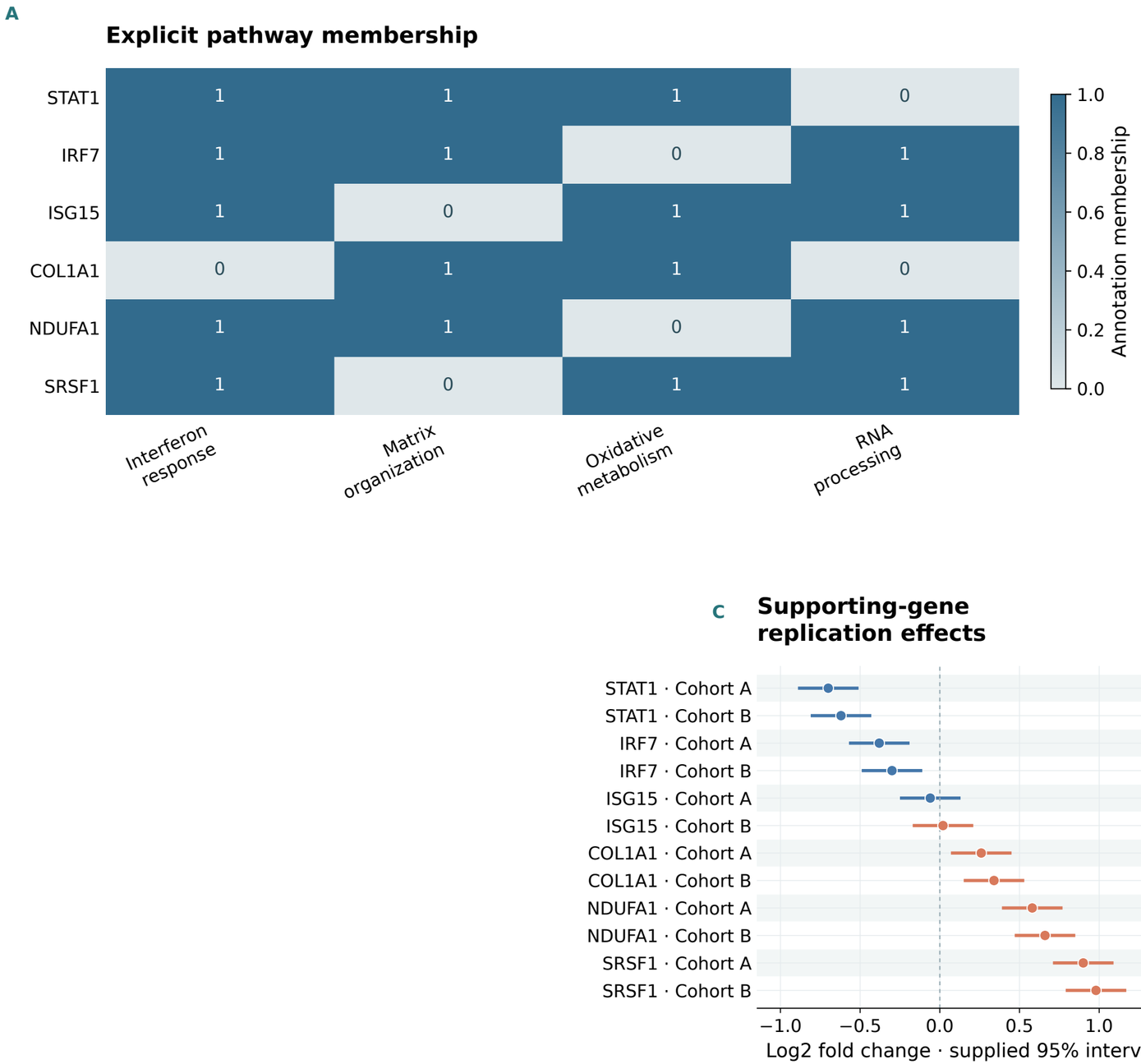
0.90 [0.72, 1.08] 0.092

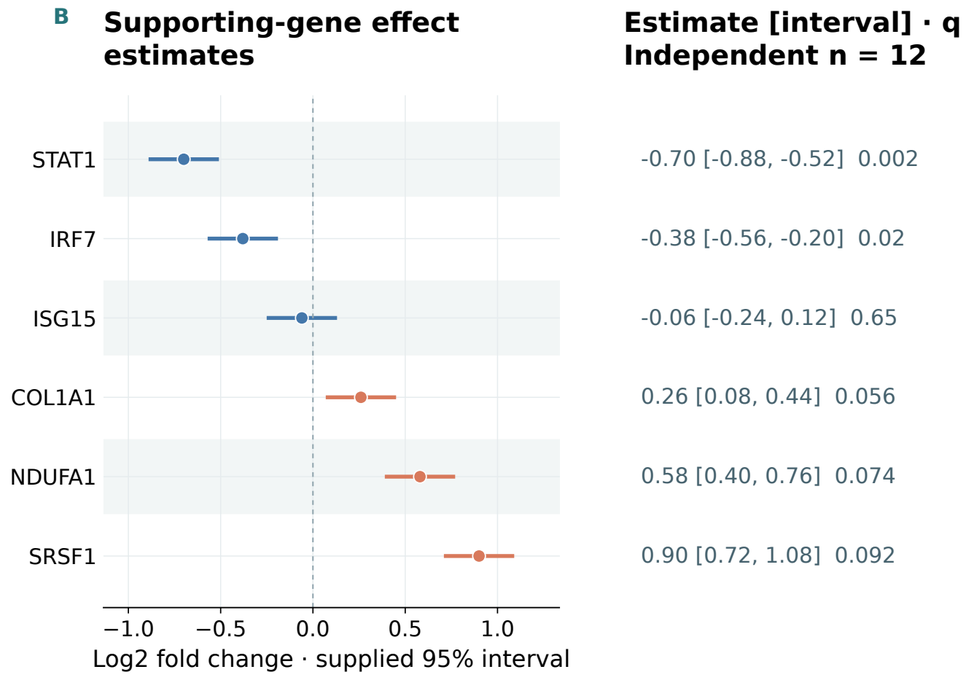

STAT1 donor expression in the selected pathway
Supplied log2 STAT1 expression
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Control
n=6
Treated
n=6
Condition

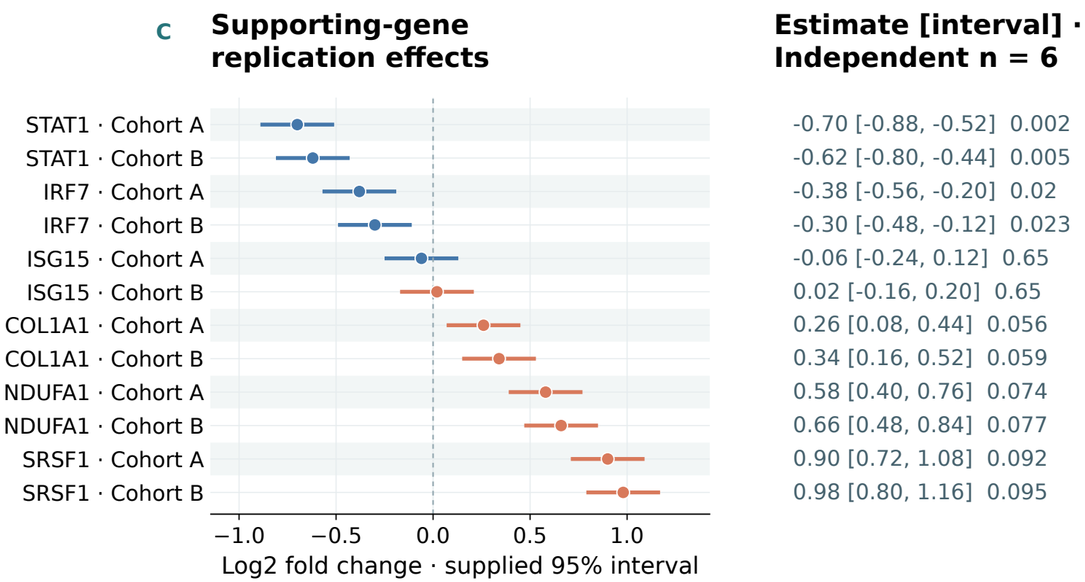

Gene effects across support contrasts
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0
0.48
0.74
0.54
0.75
0.61
0.75
0.56
0.11
0.51
0.03
-0.46
-0.04
-0.52
-0.75
-0.57
-0.75
-0.58
Primary
Secondary
Replication
1.0
0.5
0.0
−0.5
−1.0
Supplied log2 fold change


Supporting-gene replication effects
Estimate [interval] · q
Independent n = 6
STAT1 · Cohort A
STAT1 · Cohort B
IRF7 · Cohort A
IRF7 · Cohort B
ISG15 · Cohort A
ISG15 · Cohort B
COL1A1 · Cohort A
COL1A1 · Cohort B
NDUFA1 · Cohort A
NDUFA1 · Cohort B
SRSF1 · Cohort A
SRSF1 · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Log2 fold change · supplied 95% interval

Bulk · gene_support — complementary evidence



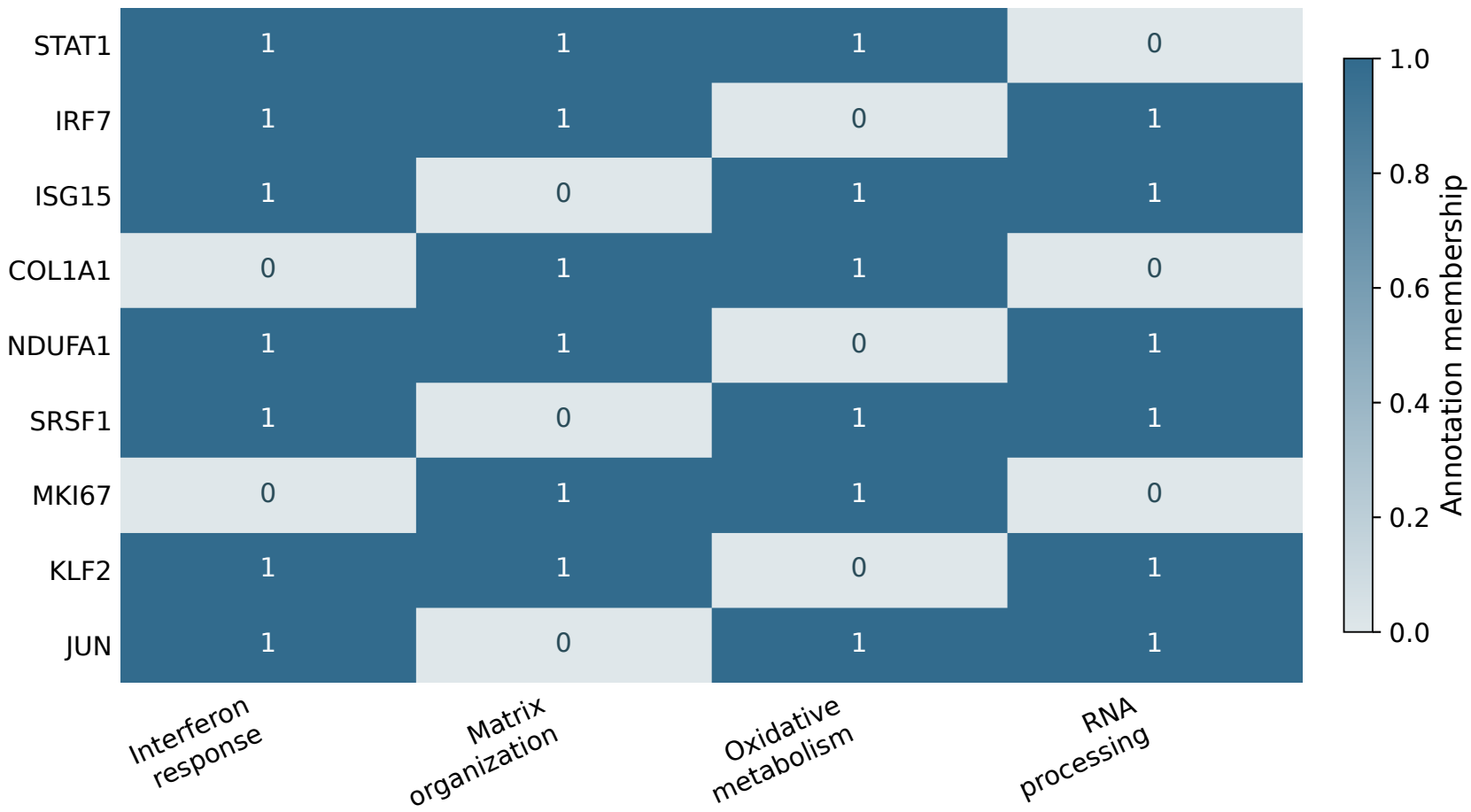




Bulk · gene_support — complementary evidence

A

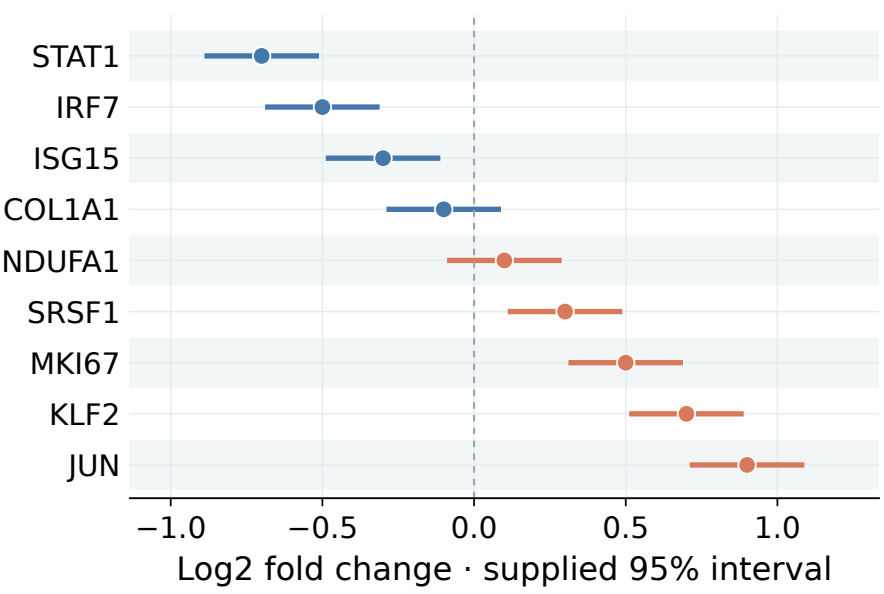
Explicit pathway membership



B

Supporting-gene effect estimates

Estimate [interval] · q Independent n = 24

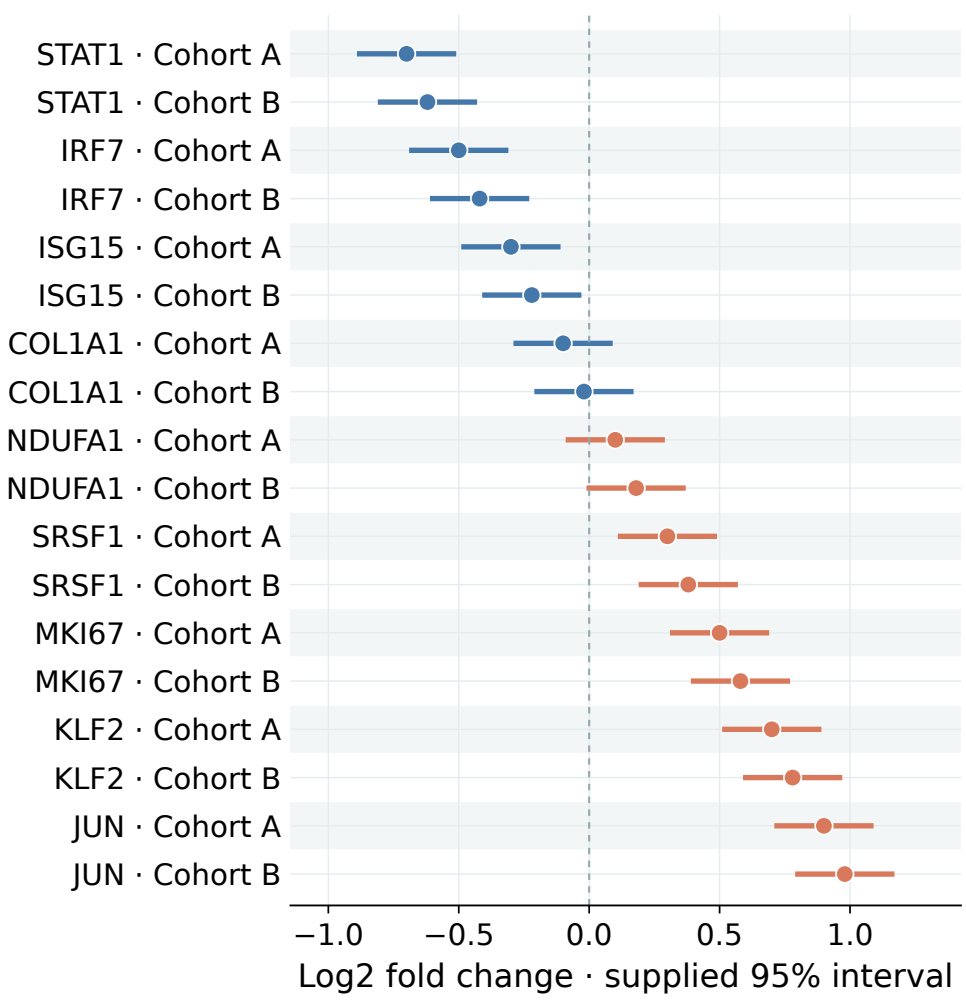


STAT1	-0.70 [-0.88, -0.52] 0.002
IRF7	-0.50 [-0.68, -0.32] 0.02
ISG15	-0.30 [-0.48, -0.12] 0.038
COL1A1	-0.10 [-0.28, 0.08] 0.65
NDUFA1	0.10 [-0.08, 0.28] 0.65
SRSF1	0.30 [0.12, 0.48] 0.092
MKI67	0.50 [0.32, 0.68] 0.11
KLF2	0.70 [0.52, 0.88] 0.13
JUN	0.90 [0.72, 1.08] 0.15

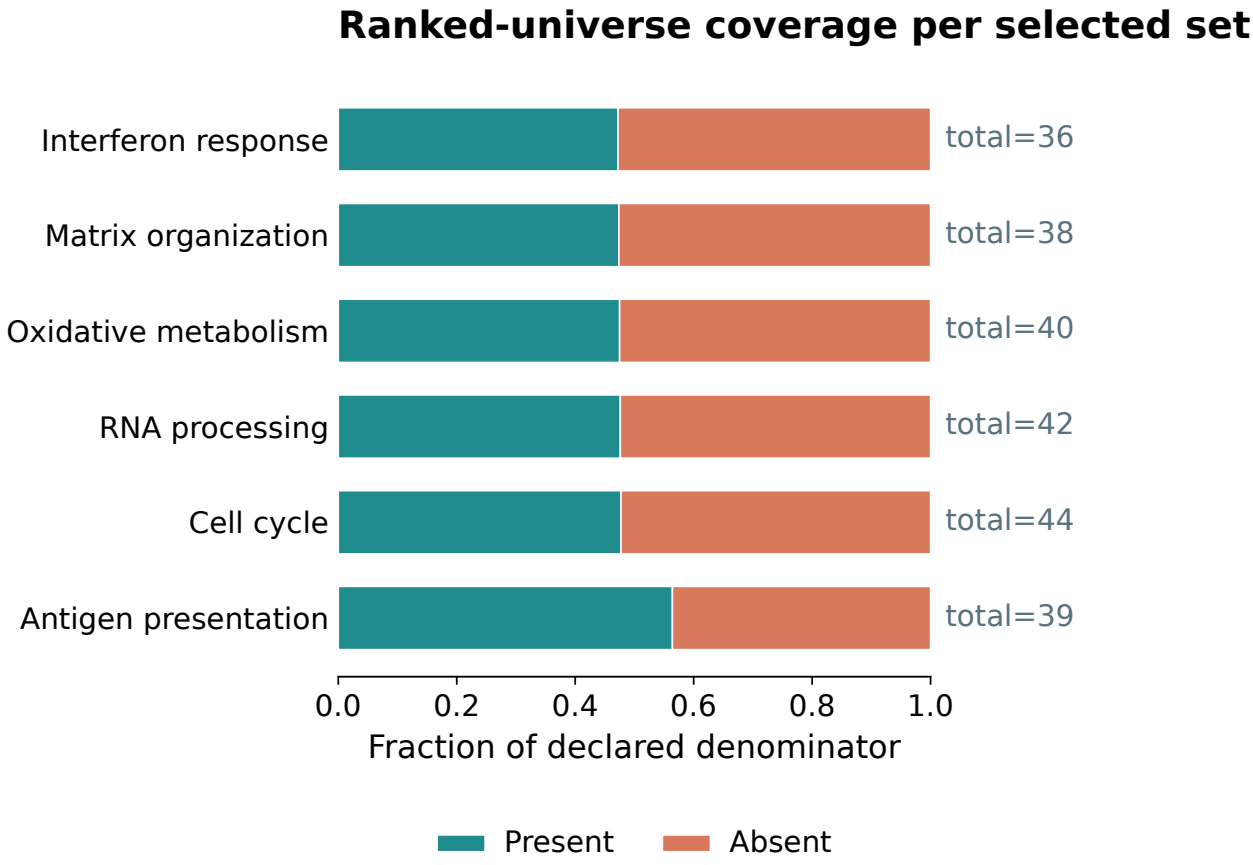
C

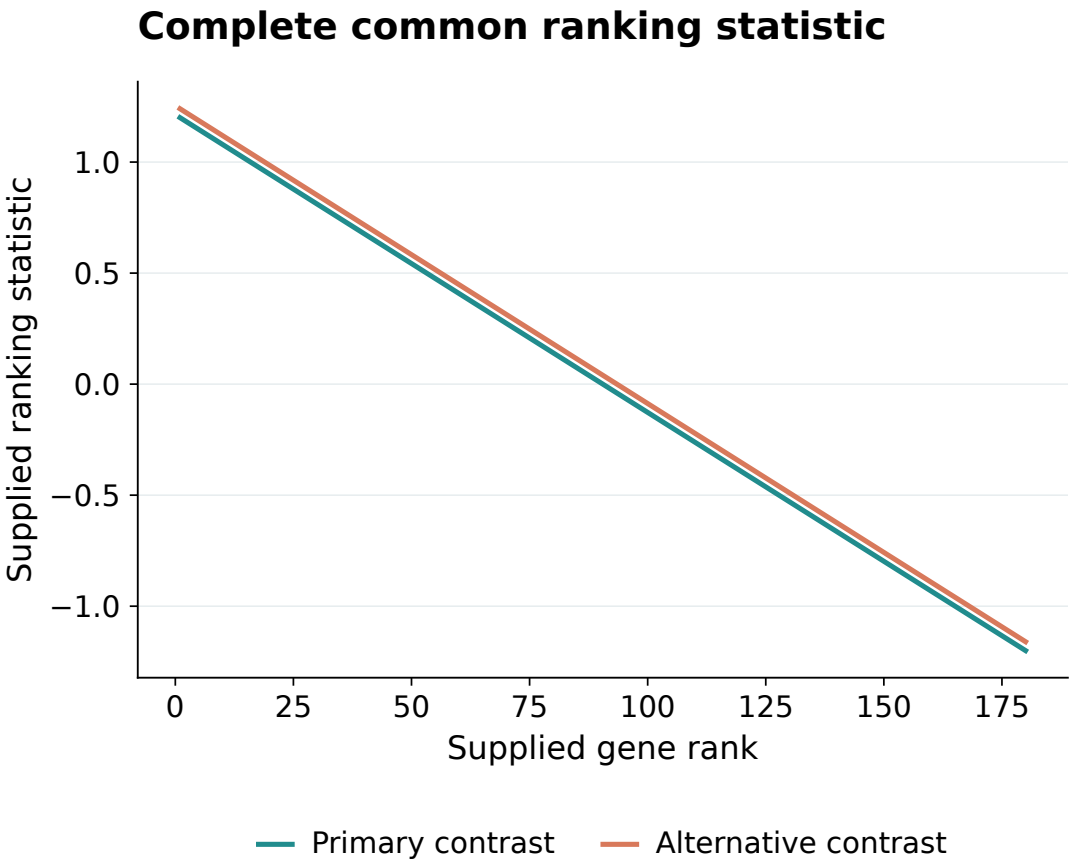
Supporting-gene replication effects

Estimate [interval] · q Independent n = 12

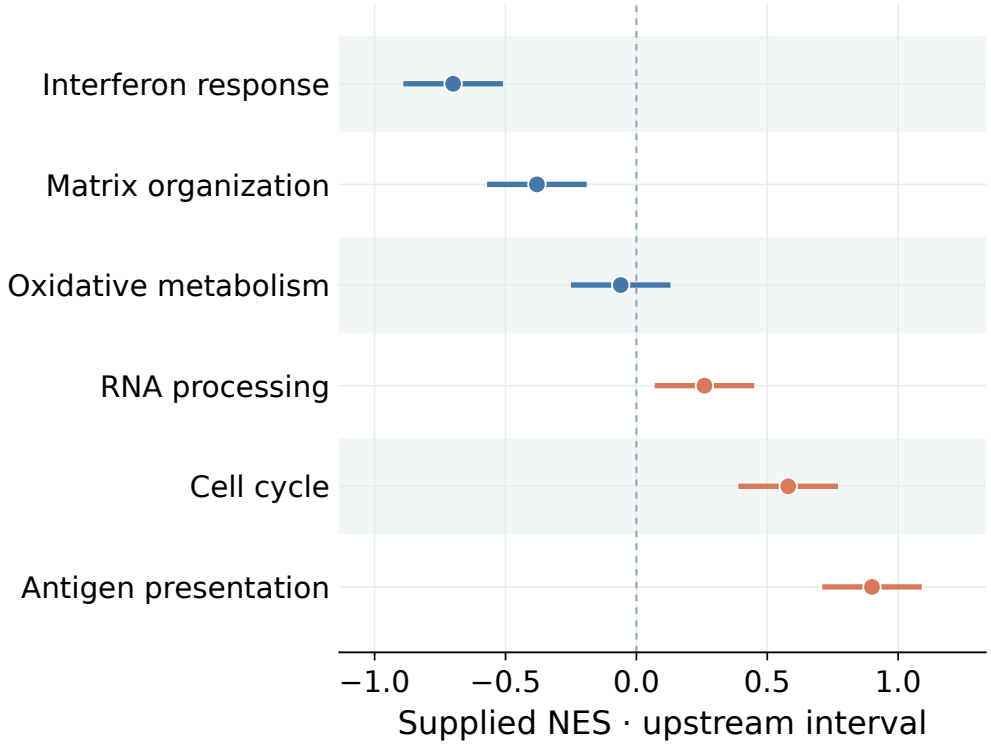


STAT1 · Cohort A	-0.70 [-0.88, -0.52] 0.002
STAT1 · Cohort B	-0.62 [-0.80, -0.44] 0.005
IRF7 · Cohort A	-0.50 [-0.68, -0.32] 0.02
IRF7 · Cohort B	-0.42 [-0.60, -0.24] 0.023
ISG15 · Cohort A	-0.30 [-0.48, -0.12] 0.038
ISG15 · Cohort B	-0.22 [-0.40, -0.04] 0.041
COL1A1 · Cohort A	-0.10 [-0.28, 0.08] 0.65
COL1A1 · Cohort B	-0.02 [-0.20, 0.16] 0.65
NDUFA1 · Cohort A	0.10 [-0.08, 0.28] 0.65
NDUFA1 · Cohort B	0.18 [0.00, 0.36] 0.077
SRSF1 · Cohort A	0.30 [0.12, 0.48] 0.092
SRSF1 · Cohort B	0.38 [0.20, 0.56] 0.095
MKI67 · Cohort A	0.50 [0.32, 0.68] 0.11
MKI67 · Cohort B	0.58 [0.40, 0.76] 0.11
KLF2 · Cohort A	0.70 [0.52, 0.88] 0.13
KLF2 · Cohort B	0.78 [0.60, 0.96] 0.13
JUN · Cohort A	0.90 [0.72, 1.08] 0.15
JUN · Cohort B	0.98 [0.80, 1.16] 0.15


Ranked-universe coverage per selected set
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Present
Absent


Complete common ranking statistic
Supplied ranking statistic
1.0
0.5
0.0
−0.5
−1.0
0
25
50
75
100
125
150
175
Supplied gene rank
Primary contrast
Alternative contrast

Selected pathways on a common NES scale



Estimate [interval] · q
Upstream donor n = 12

-0.70 [-0.88, -0.52] 0.002

-0.38 [-0.56, -0.20] 0.02

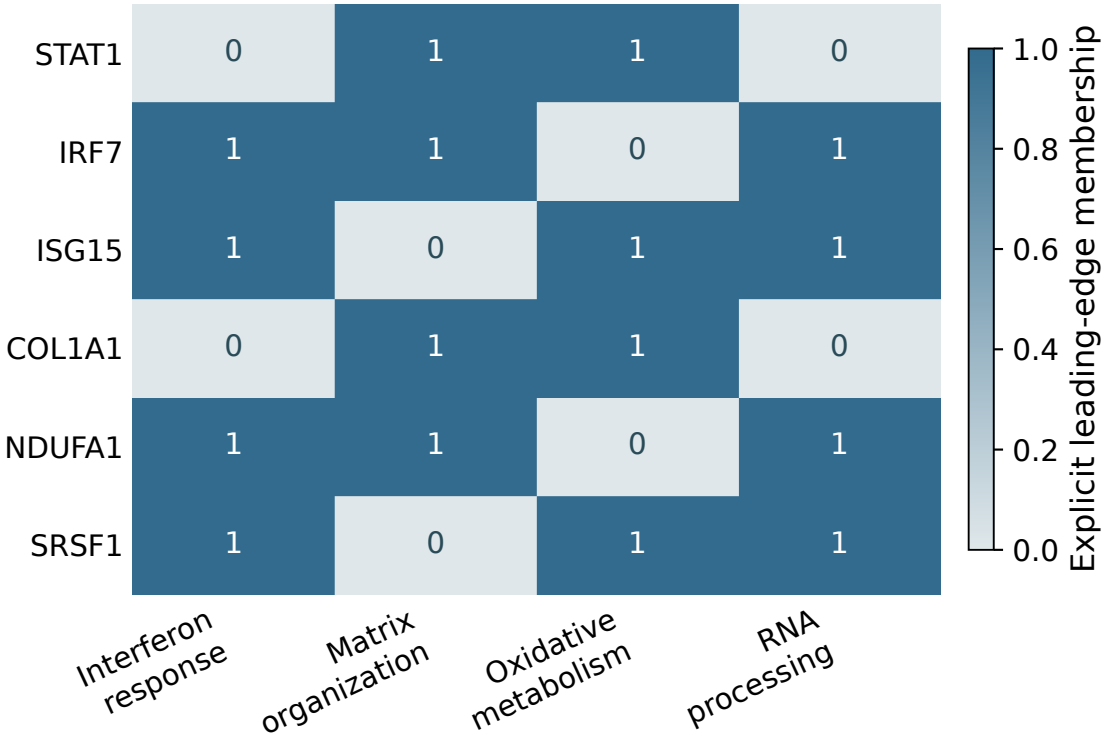
-0.06 [-0.24, 0.12] 0.65

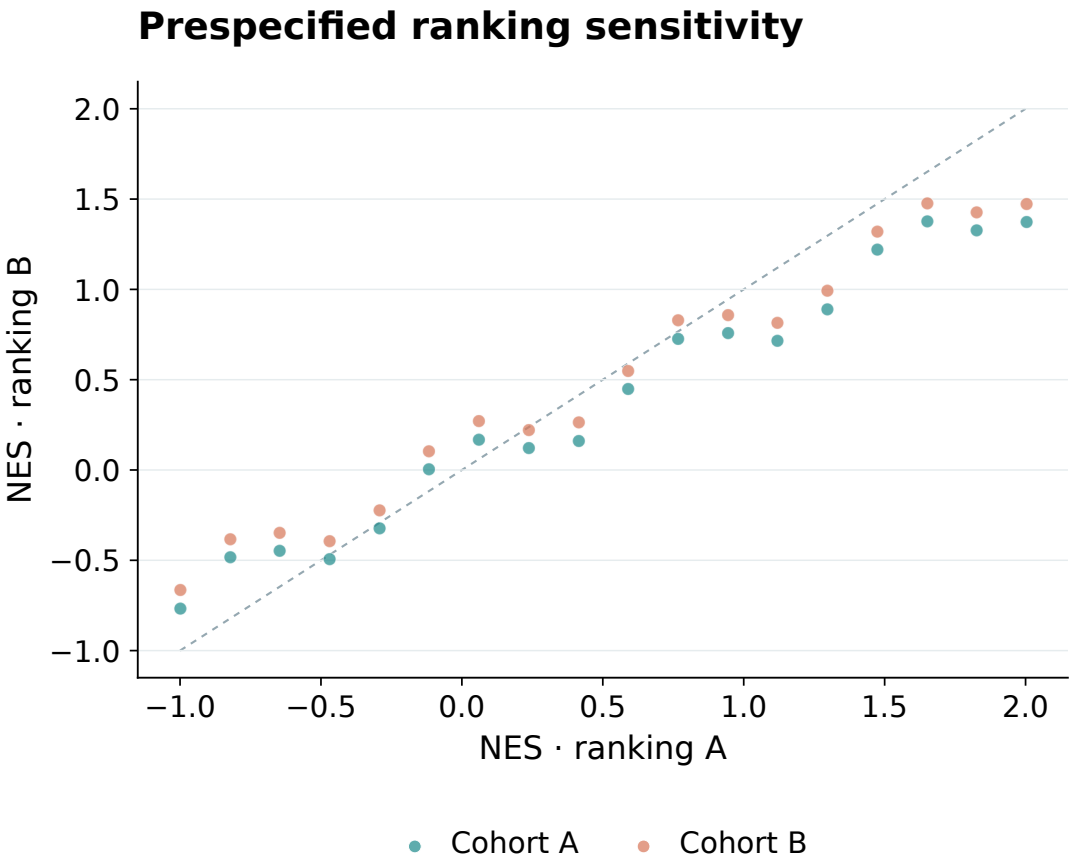
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

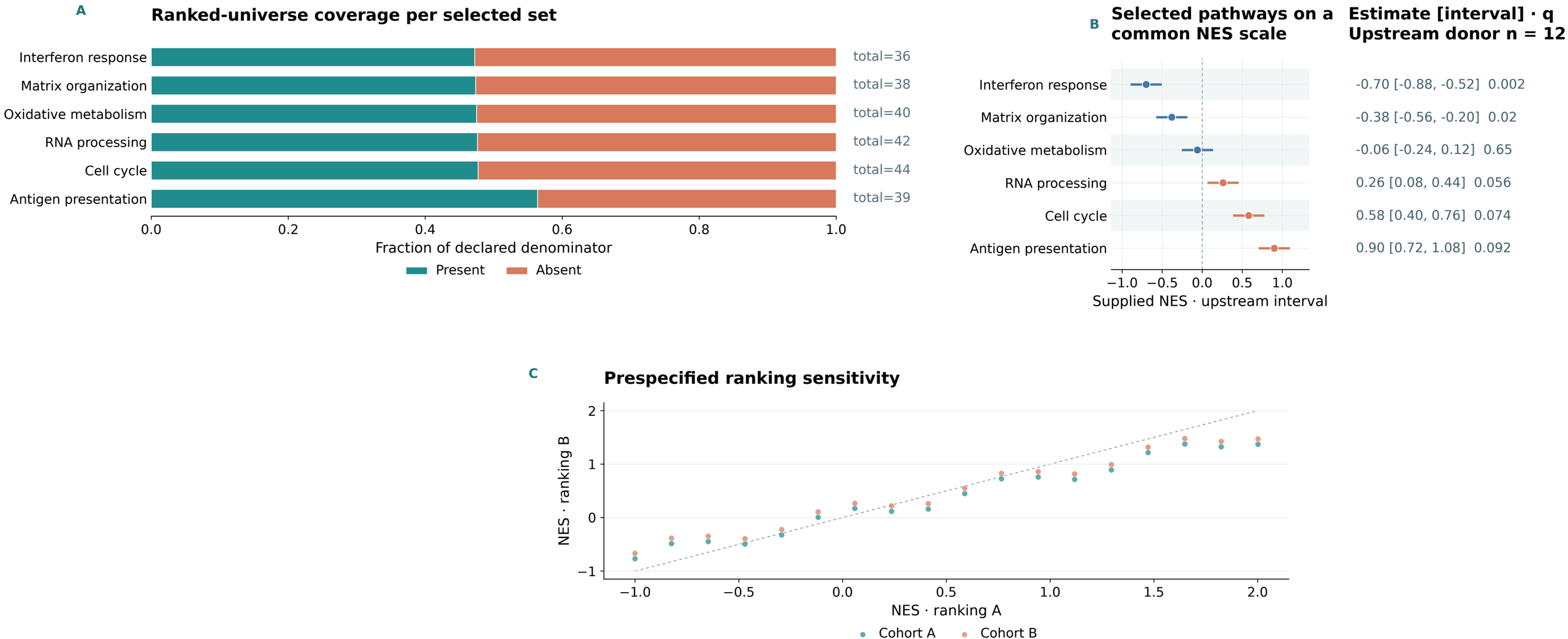
0.90 [0.72, 1.08] 0.092

Leading-edge support across the common gene scope




Prespecified ranking sensitivity
NES · ranking B
NES · ranking A
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Cohort A
Cohort B

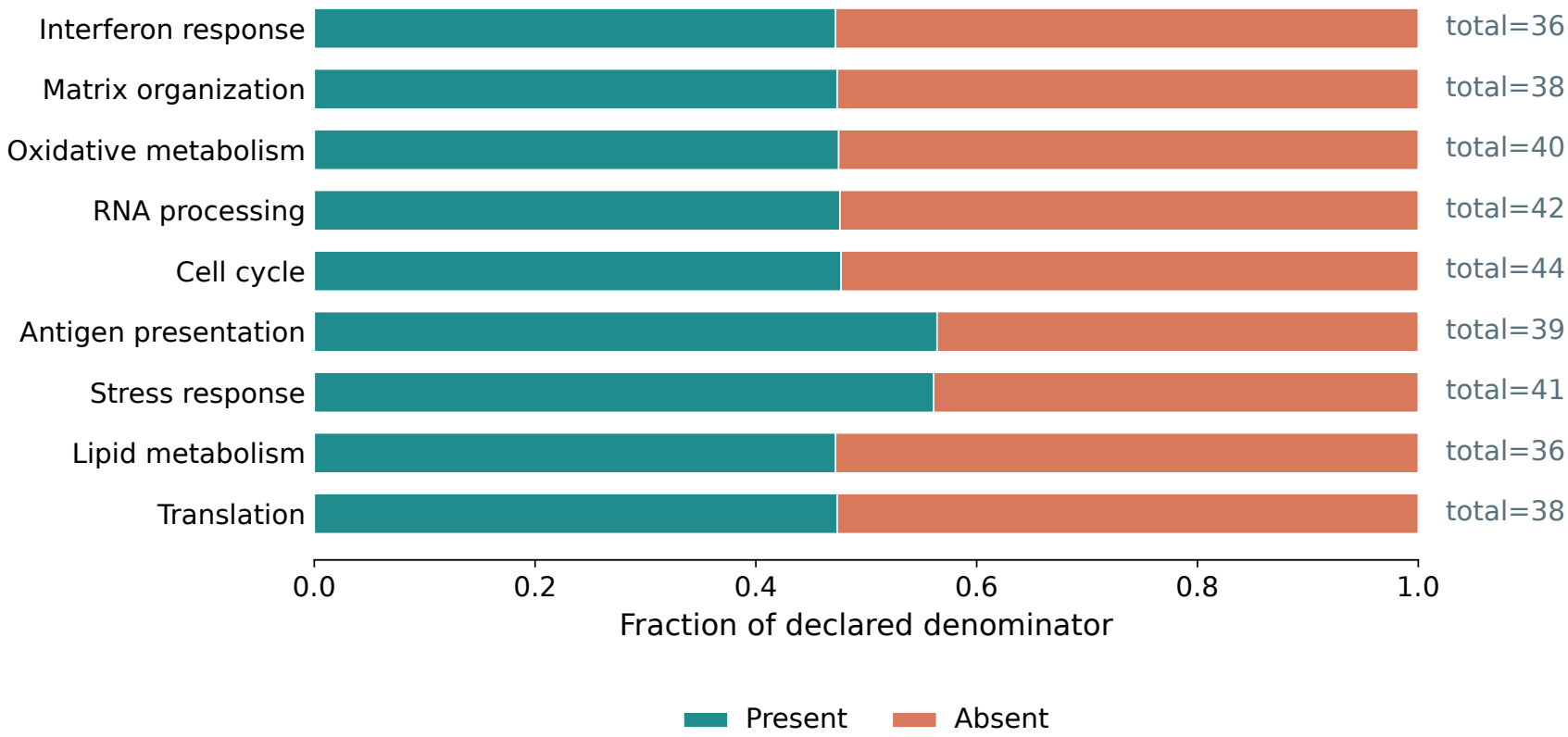
Bulk · gsea_multitrack — complementary evidence



Bulk · gsea_multitrack — complementary evidence

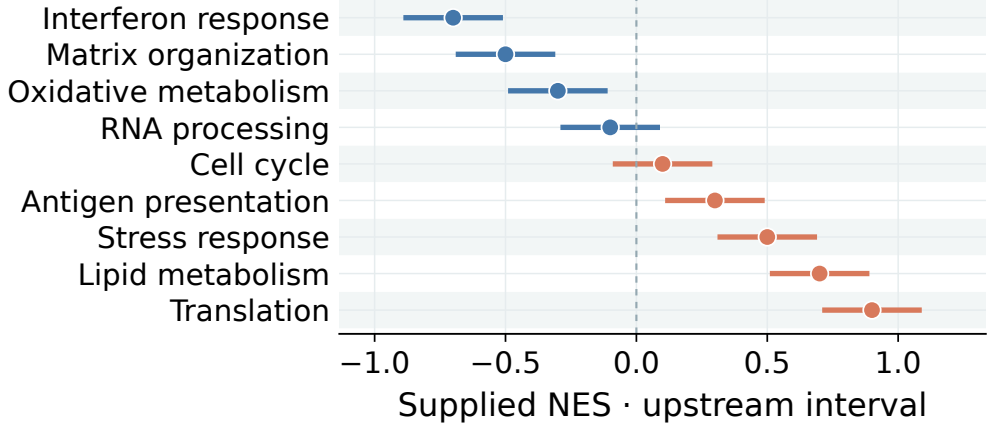
A

Ranked-universe coverage per selected set



B

Selected pathways on a common NES scale

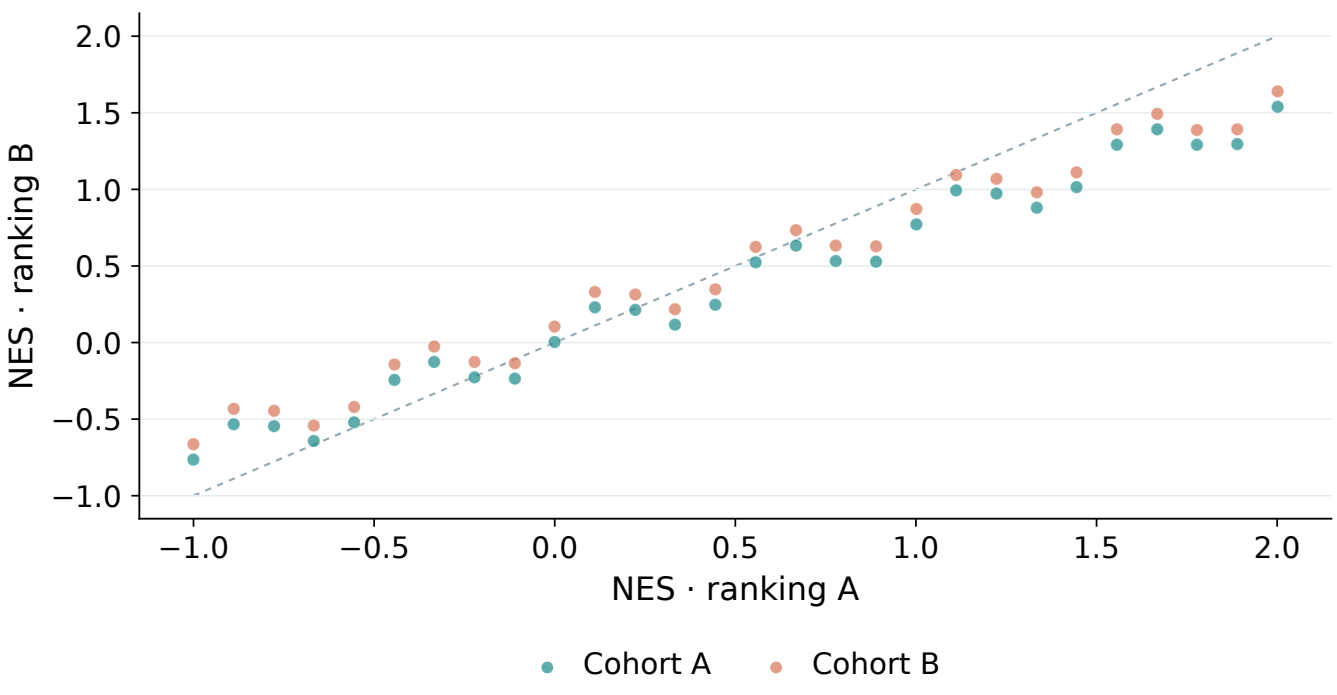


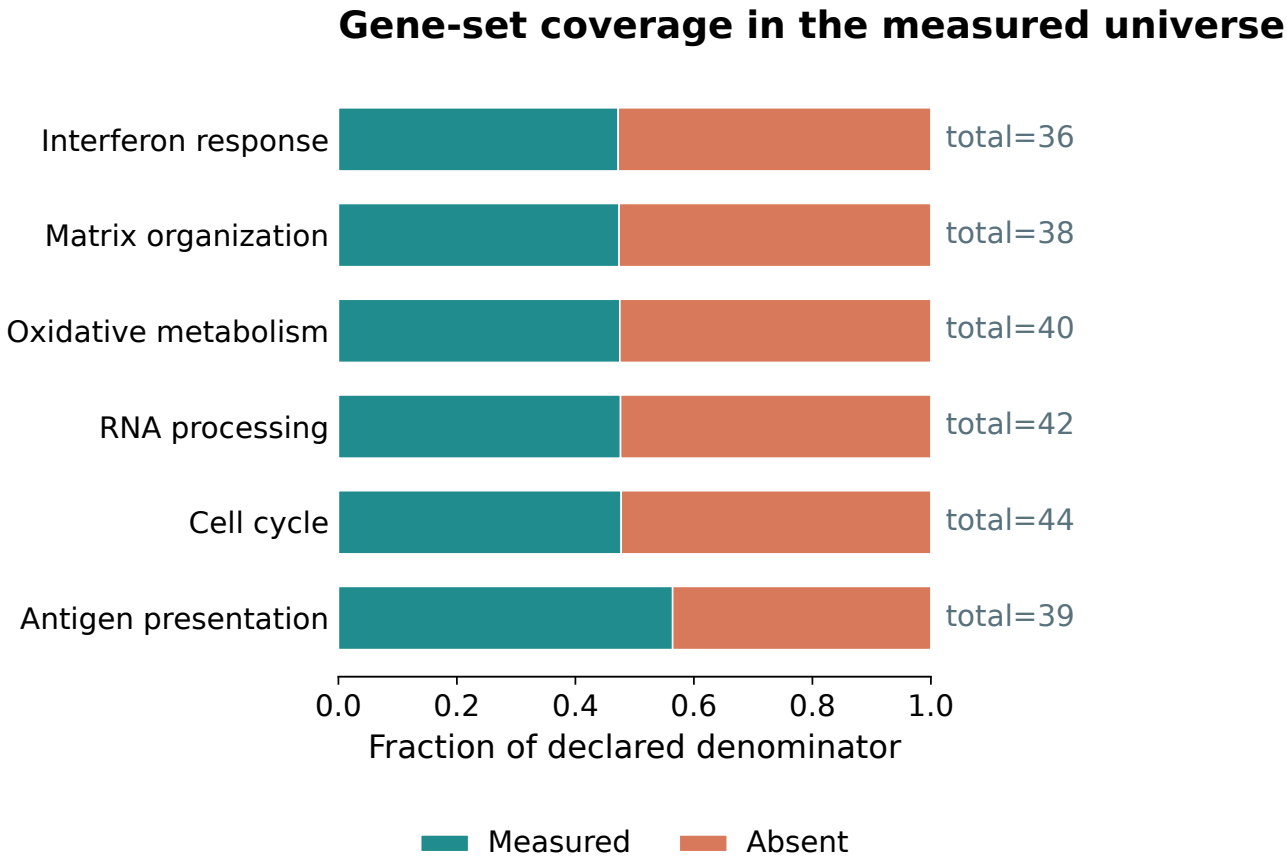
Estimate [interval] · q
Upstream donor n = 24

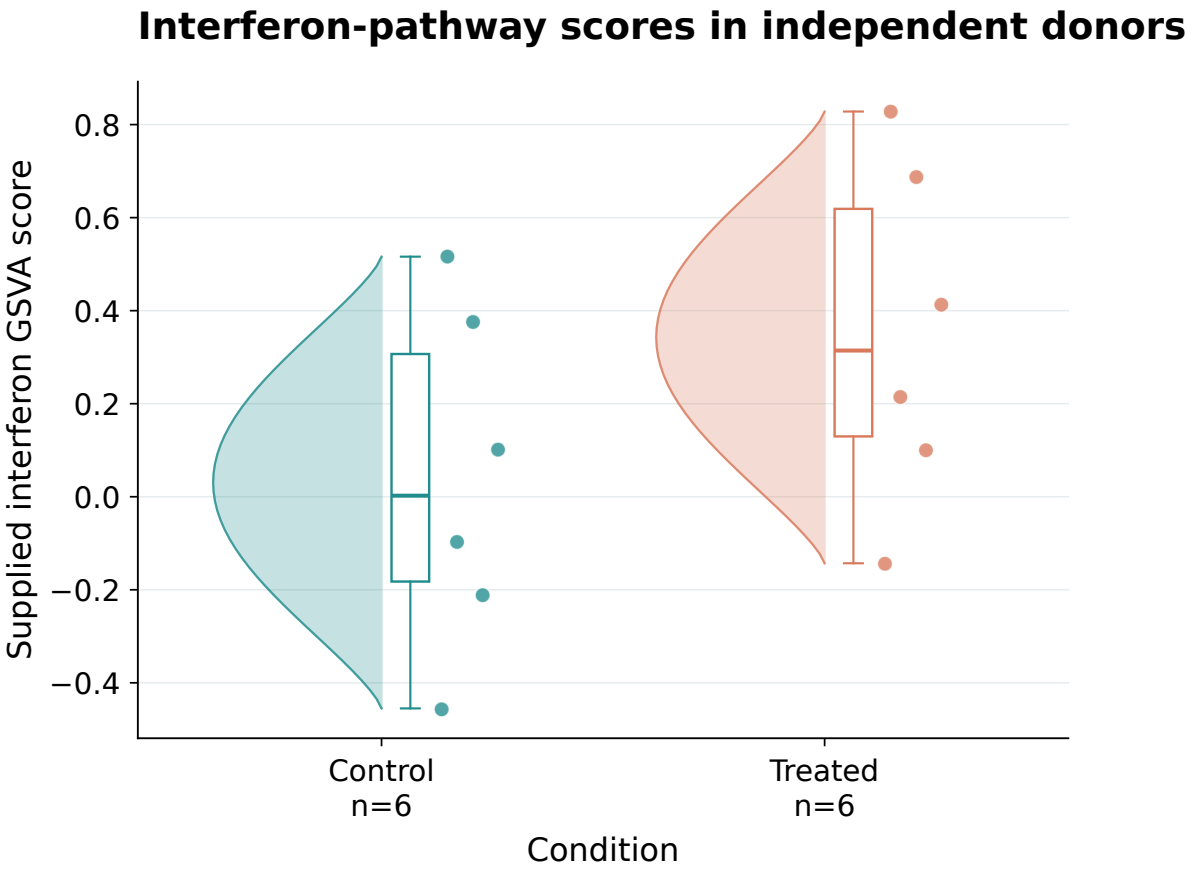
-0.70 [-0.88, -0.52] 0.002
-0.50 [-0.68, -0.32] 0.02
-0.30 [-0.48, -0.12] 0.038
-0.10 [-0.28, 0.08] 0.65
0.10 [-0.08, 0.28] 0.65
0.30 [0.12, 0.48] 0.092
0.50 [0.32, 0.68] 0.11
0.70 [0.52, 0.88] 0.13
0.90 [0.72, 1.08] 0.15

C

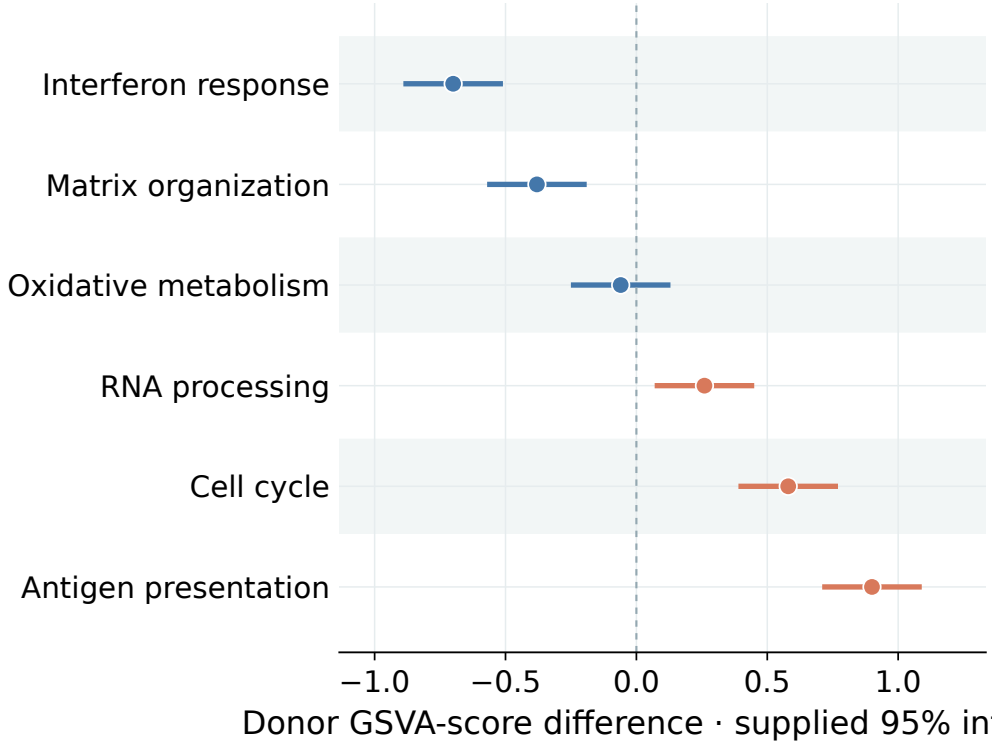
Prespecified ranking sensitivity




Gene-set coverage in the measured universe
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Measured
Absent


Interferon-pathway scores in independent donors
Supplied interferon GSVA score
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
Control
n=6
Treated
n=6
Condition

Adjusted donor-level pathway effects



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

-0.38 [-0.56, -0.20] 0.02

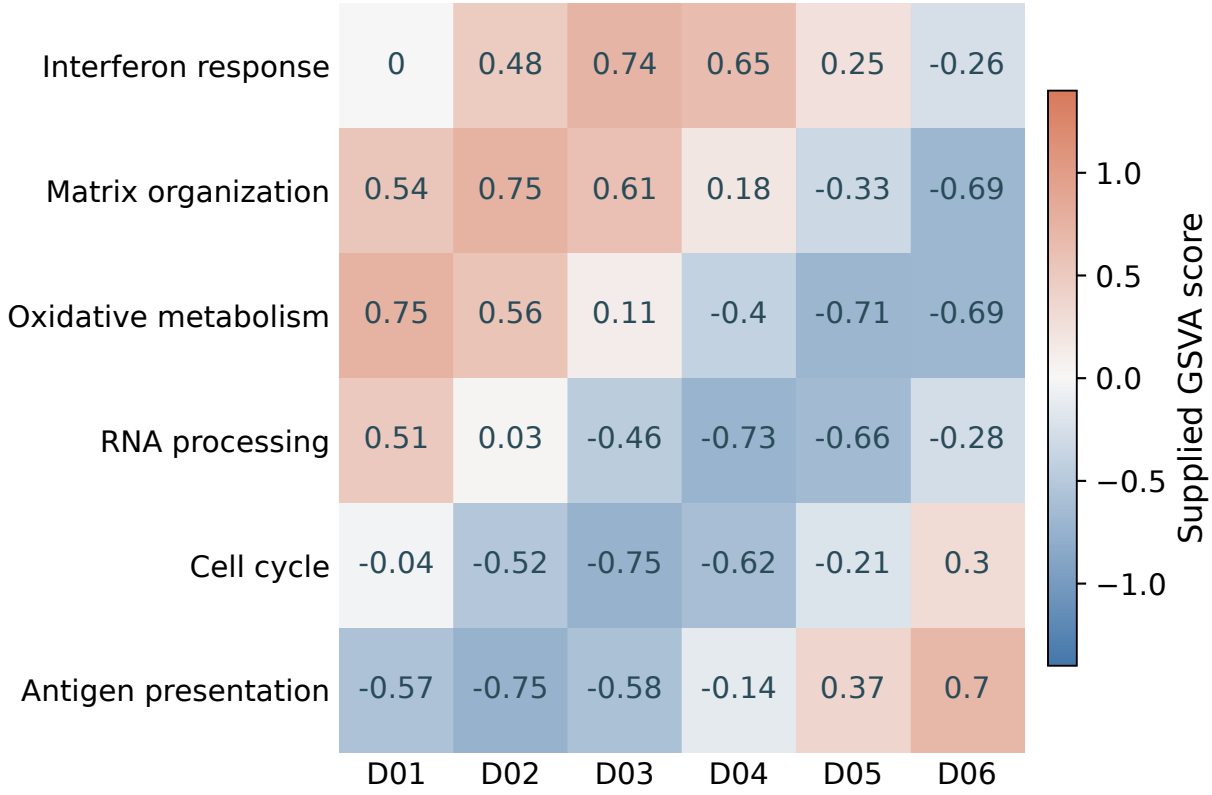
-0.06 [-0.24, 0.12] 0.65

0.26 [0.08, 0.44] 0.056

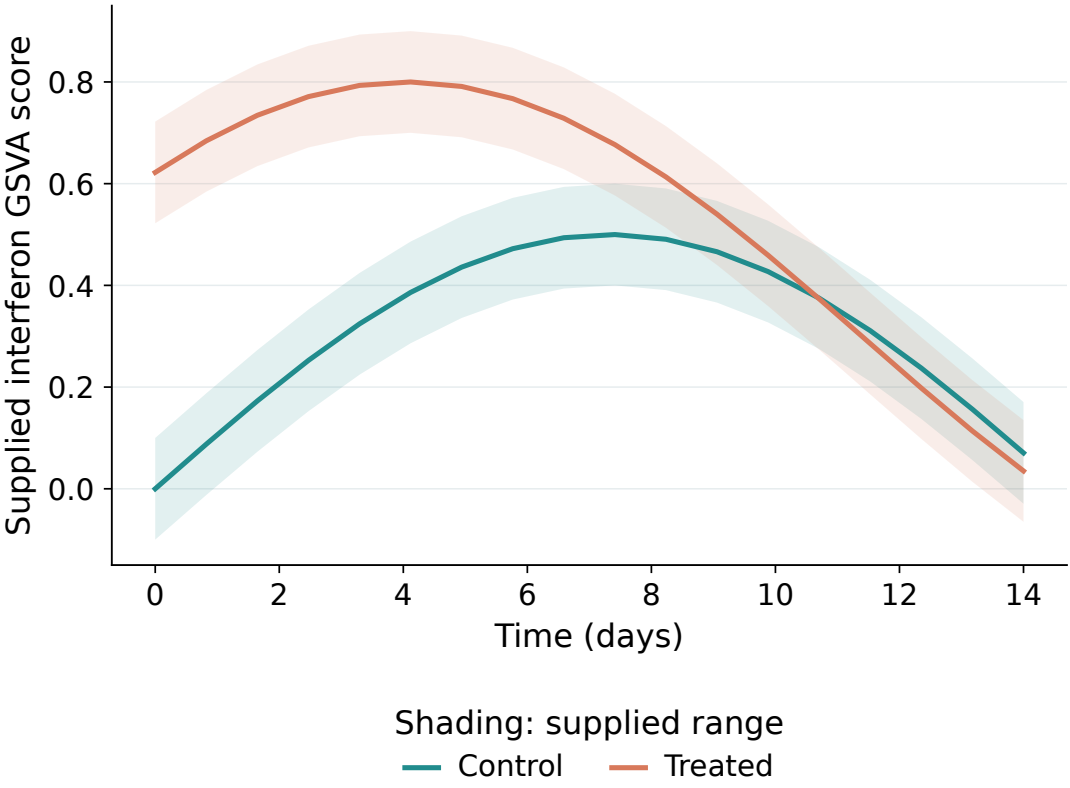
0.58 [0.40, 0.76] 0.074

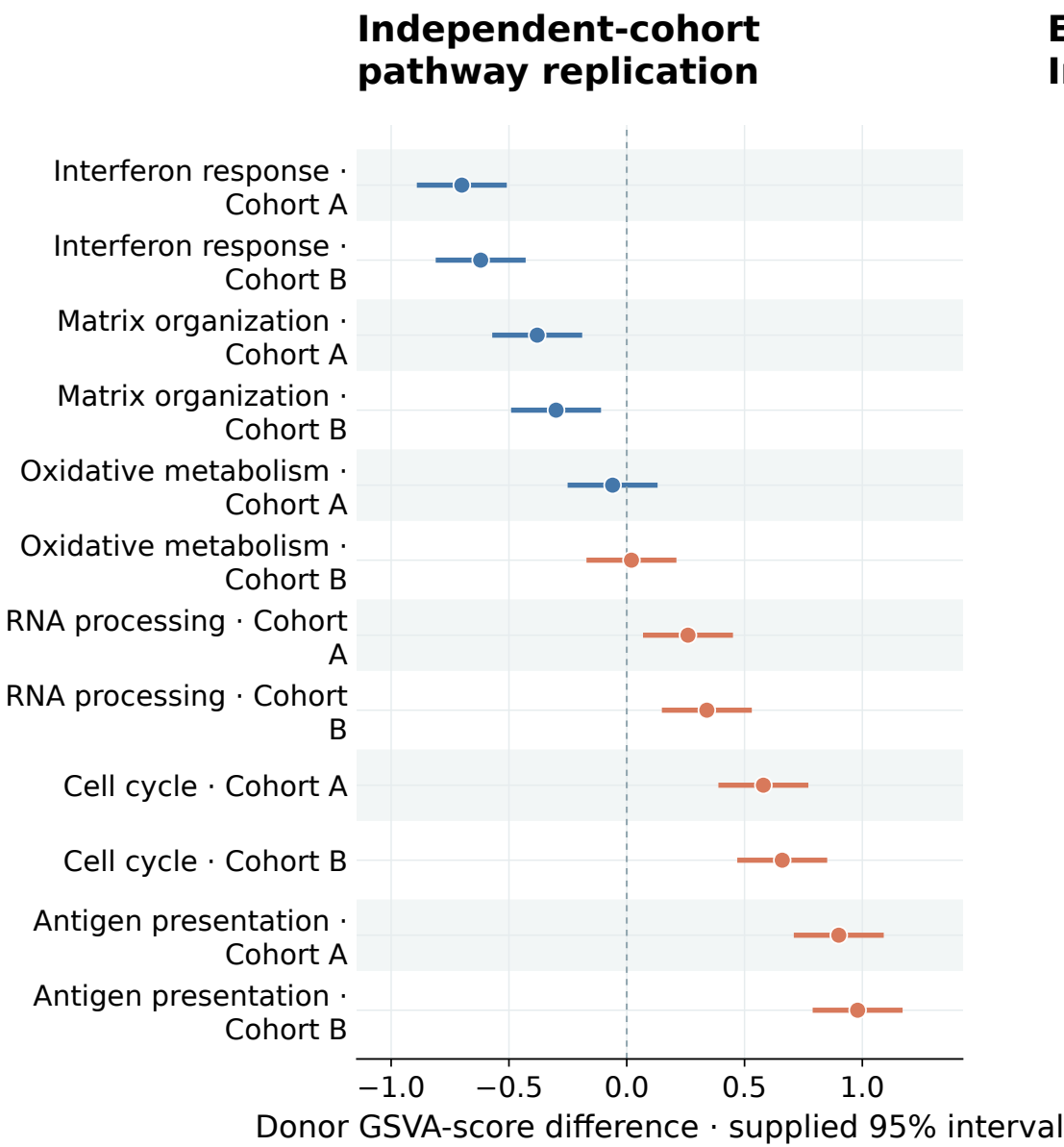
0.90 [0.72, 1.08] 0.092

GSVA scores across independent donors



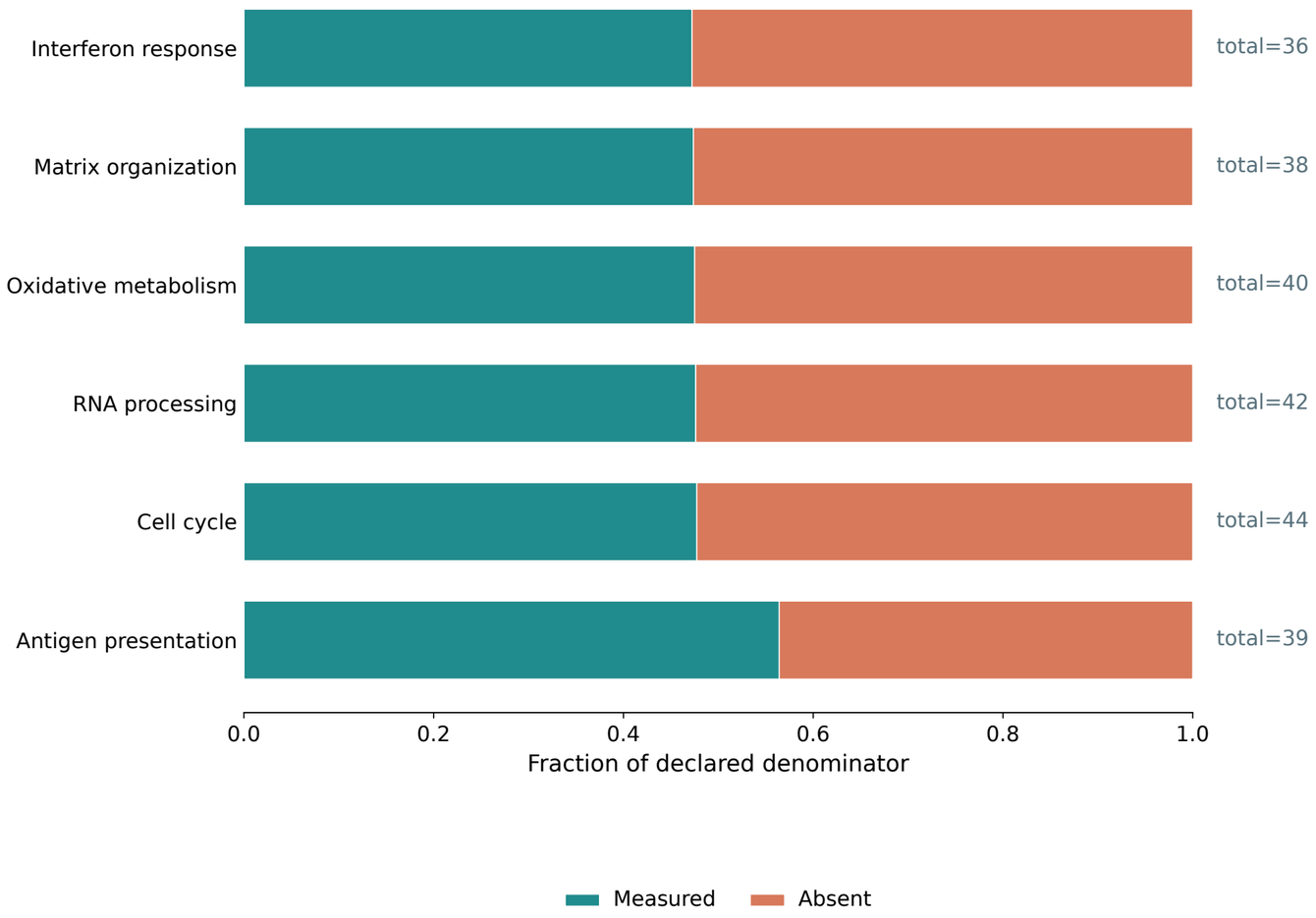
Interferon-pathway response across measured time



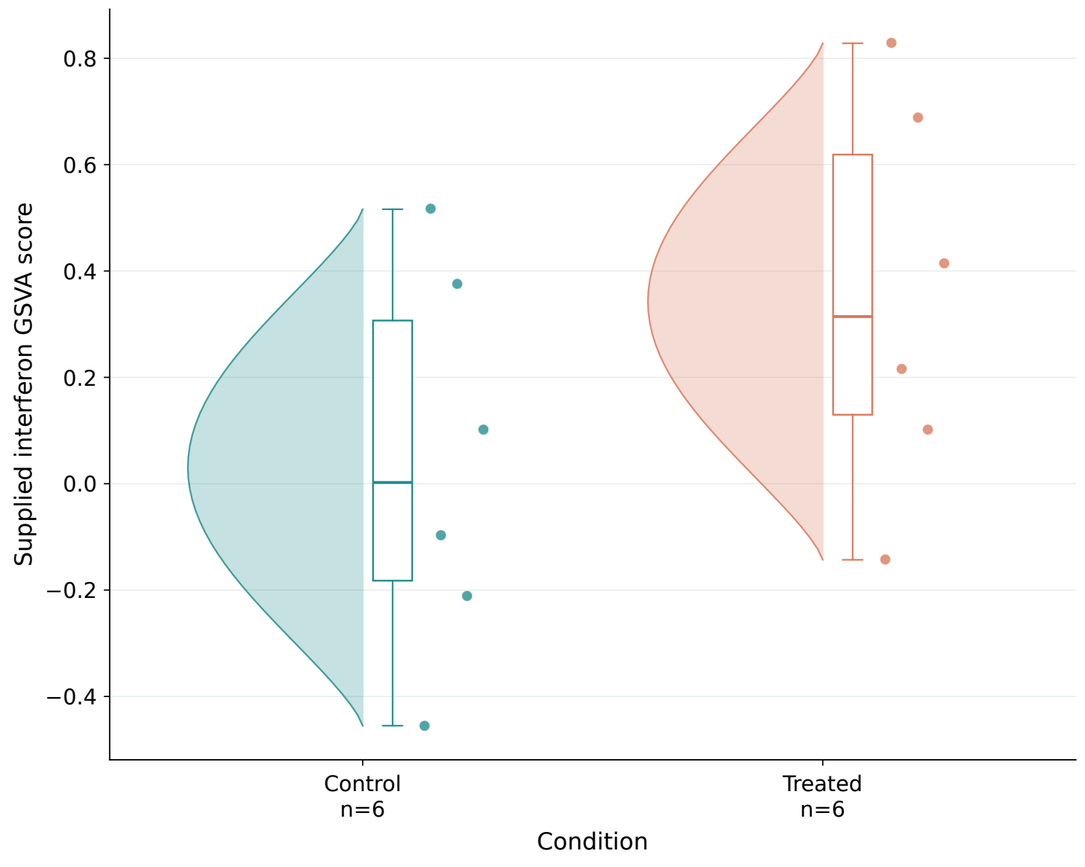

Independent-cohort pathway replication
Estimate [interval] · q
Independent n = 6
Interferon response · Cohort A
Interferon response · Cohort B
Matrix organization · Cohort A
Matrix organization · Cohort B
Oxidative metabolism · Cohort A
Oxidative metabolism · Cohort B
RNA processing · Cohort A
RNA processing · Cohort B
Cell cycle · Cohort A
Cell cycle · Cohort B
Antigen presentation · Cohort A
Antigen presentation · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Donor GSVA-score difference · supplied 95% interval

Bulk · gsva — complementary evidence

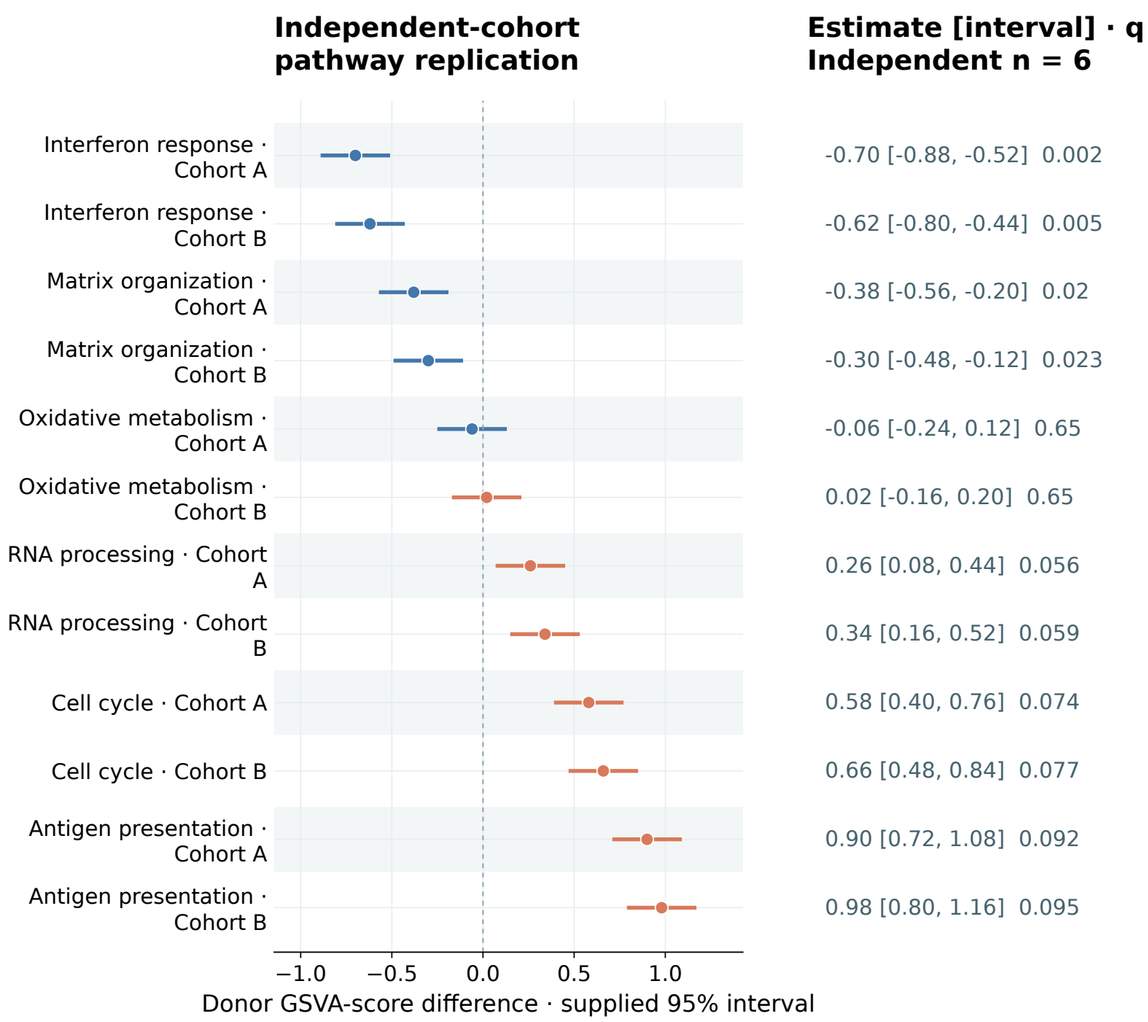
A



B



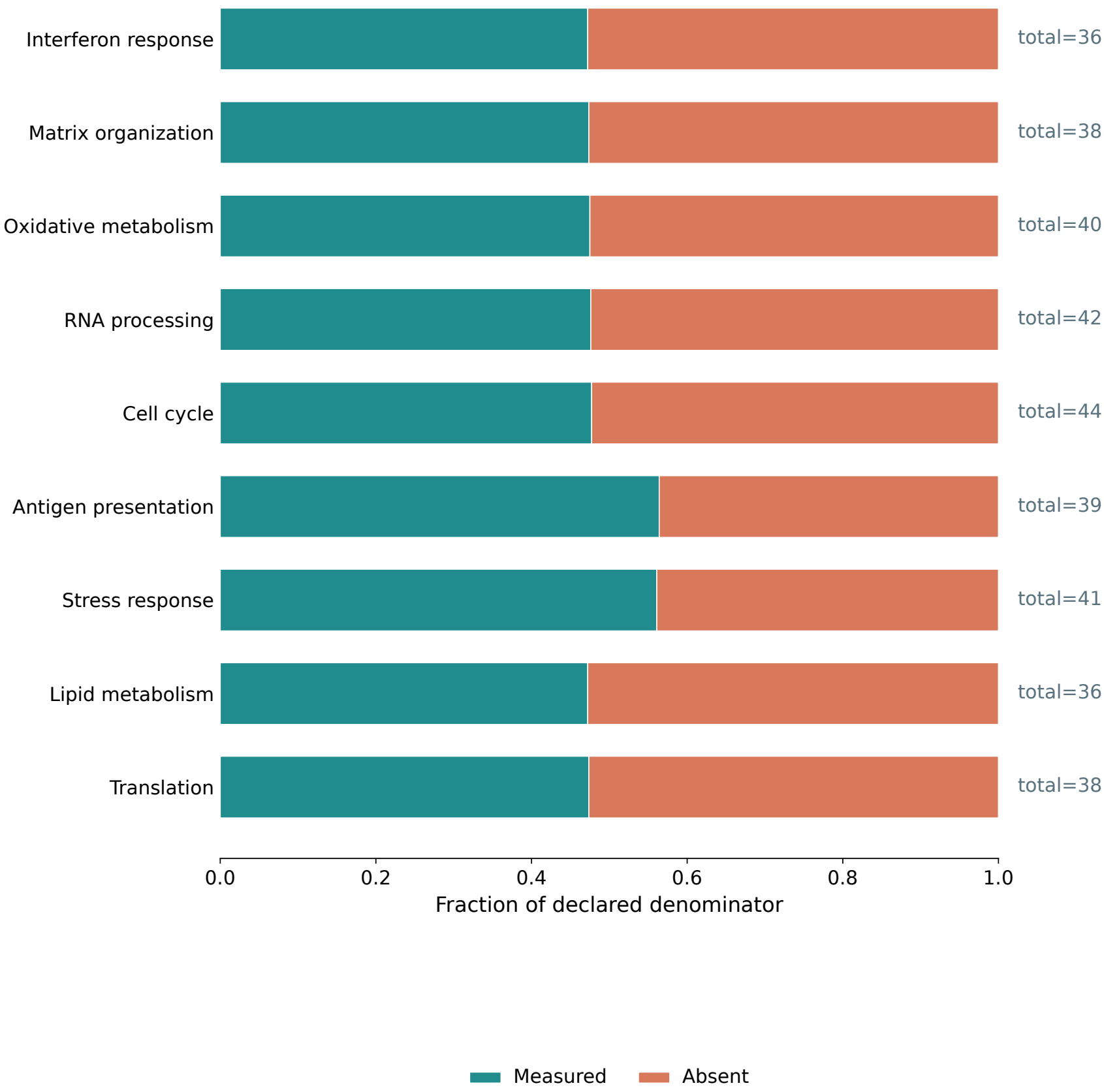
C



Bulk · gsva — complementary evidence

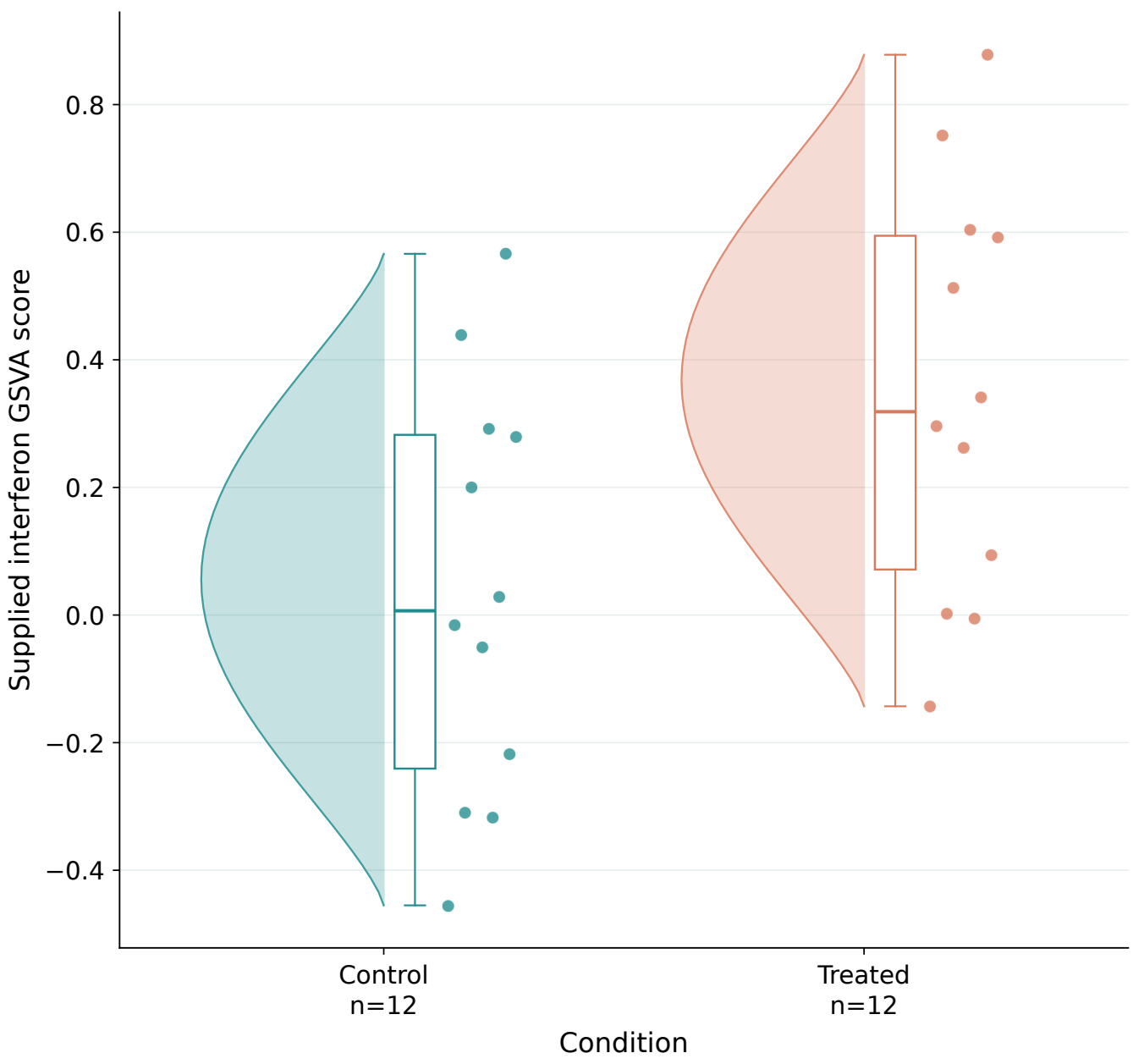
A

Gene-set coverage in the measured universe



B

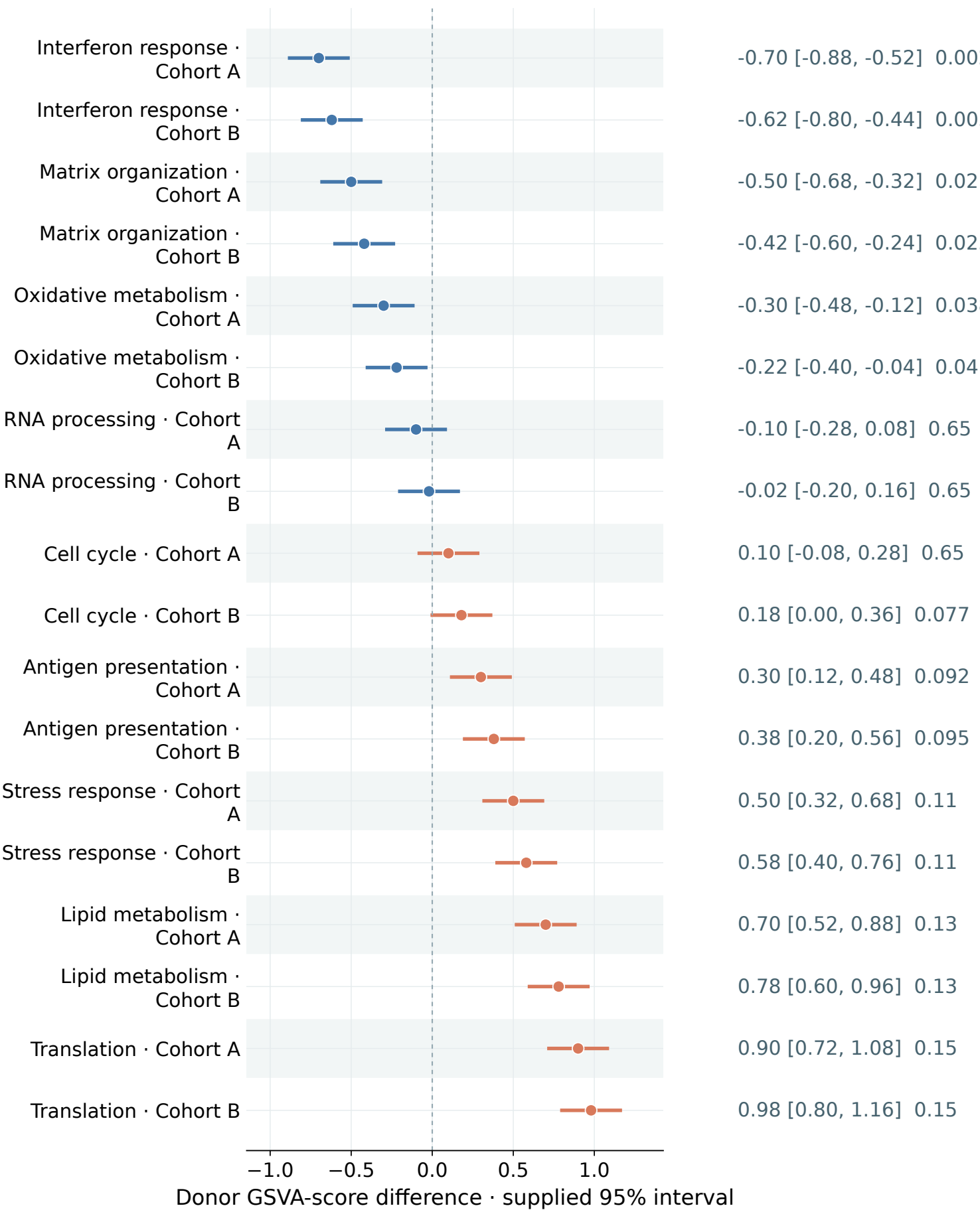
Interferon-pathway scores in independent donors

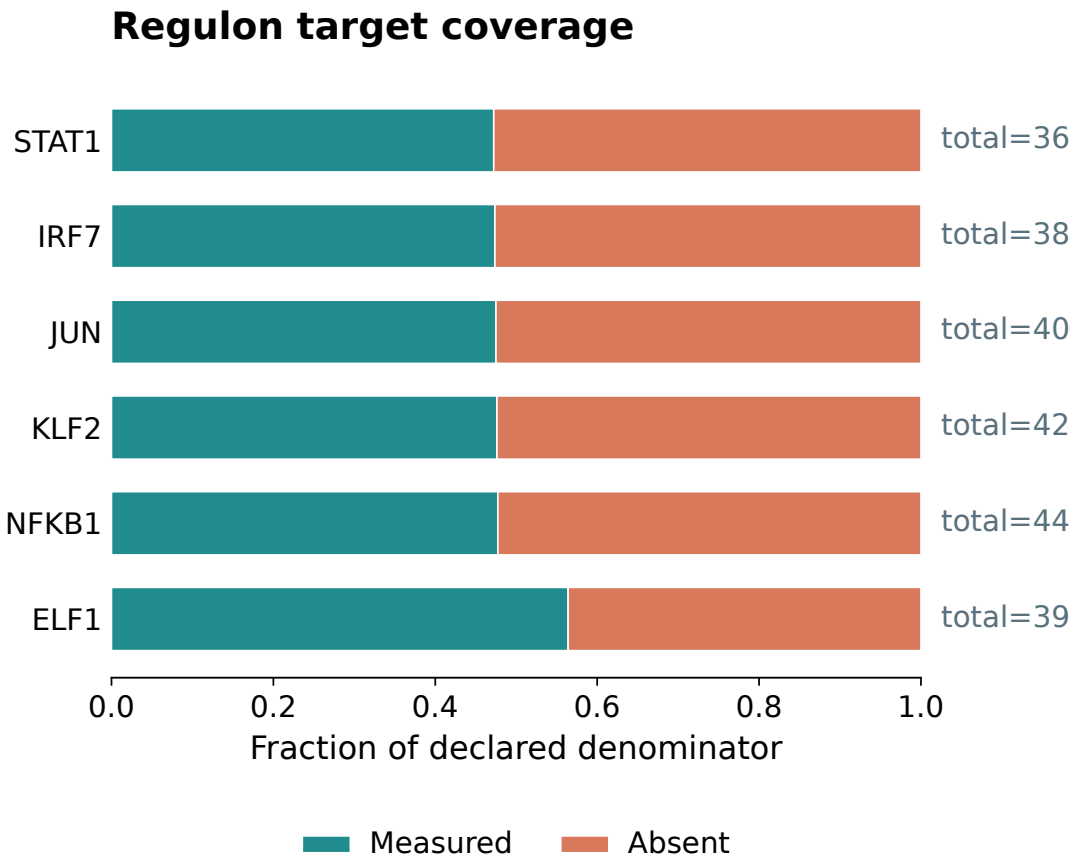


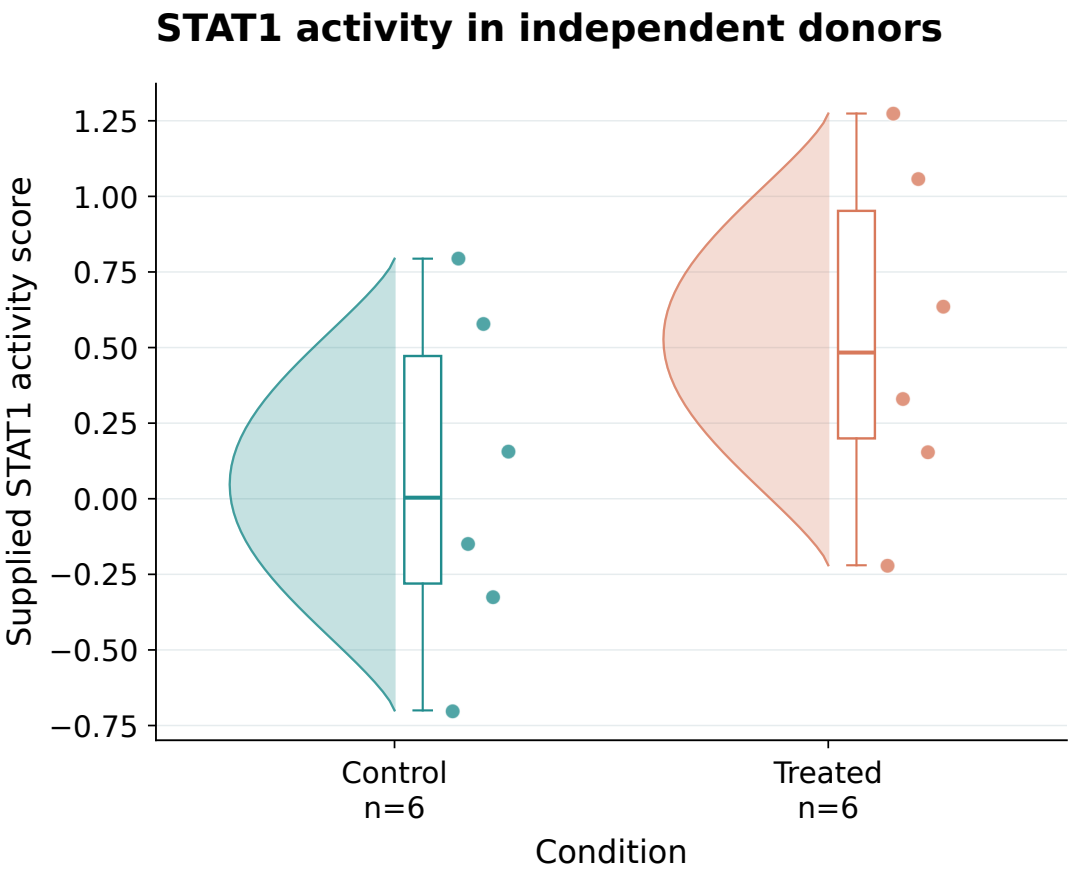
C

Independent-cohort pathway replication

Estimate [interval] · q
Independent n = 12

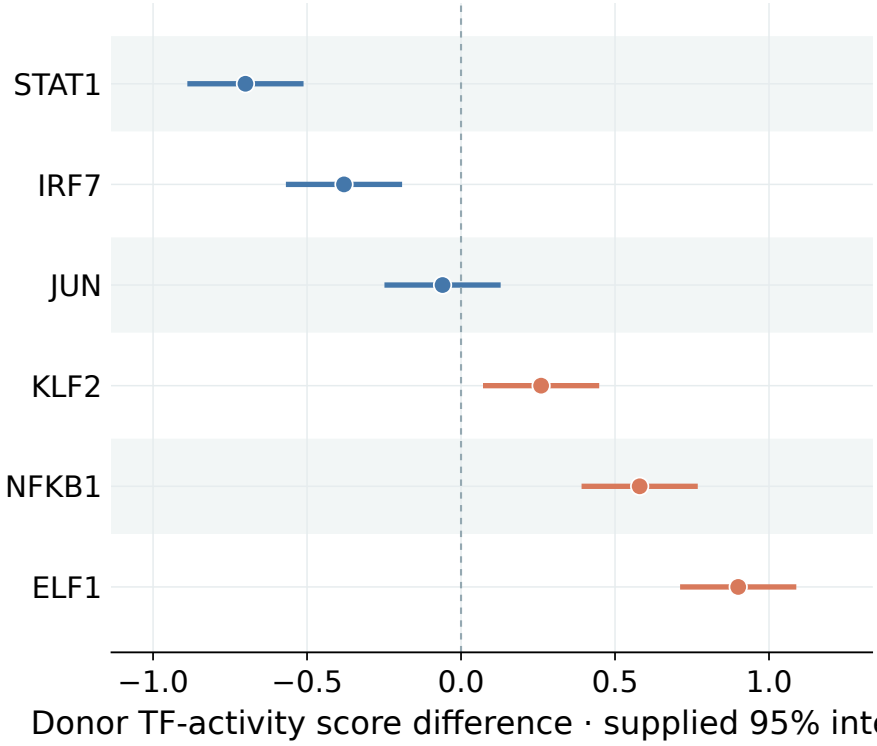



Regulon target coverage
STAT1
IRF7
JUN
KLF2
NFKB1
ELF1
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Measured
Absent



Software test fixture; not biological evidence

Adjusted TF activity effects



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

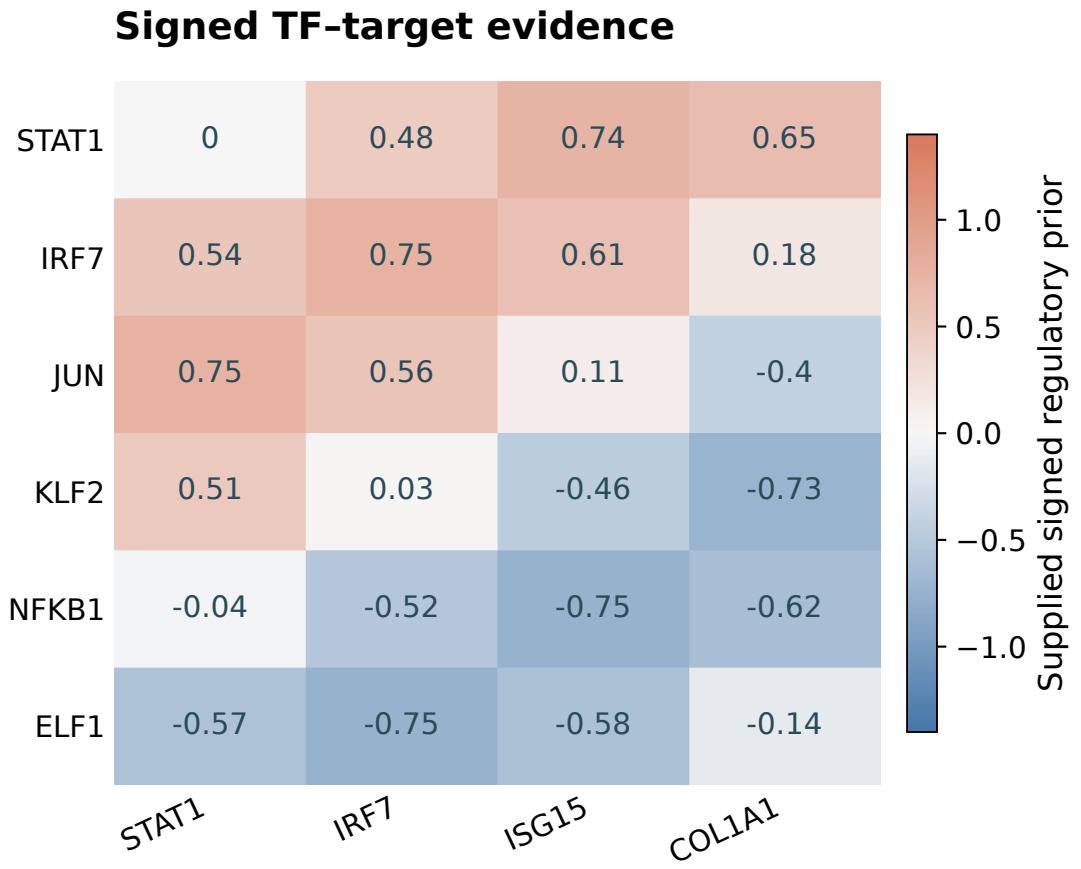
-0.38 [-0.56, -0.20] 0.02

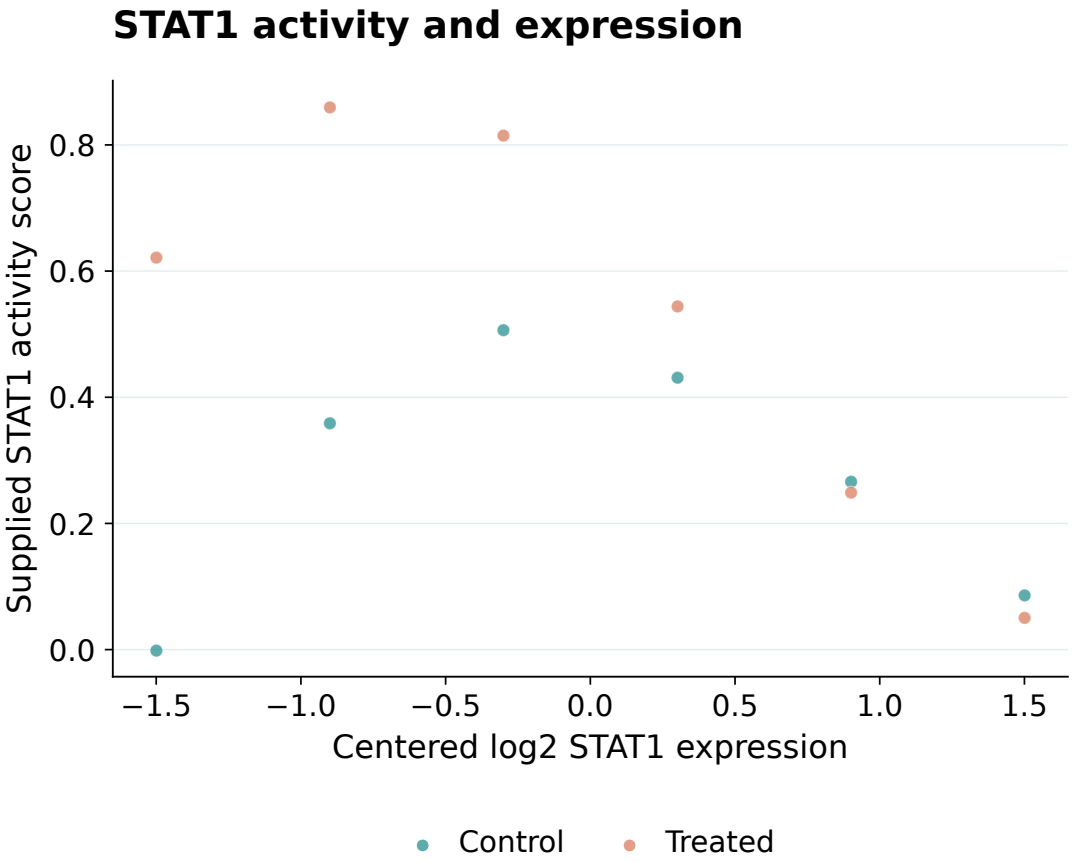
-0.06 [-0.24, 0.12] 0.65

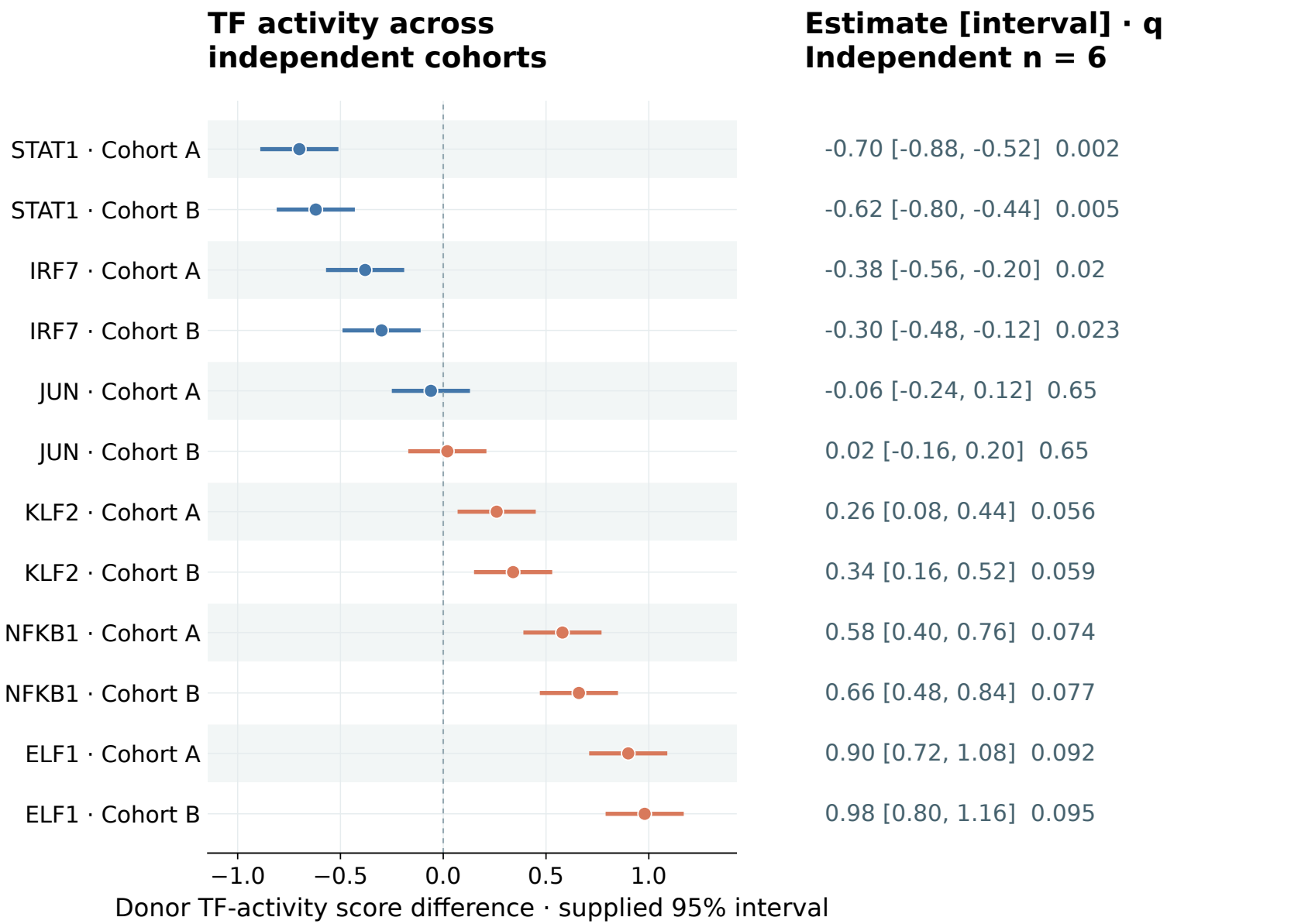
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092


Signed TF–target evidence
STAT1
IRF7
JUN
KLF2
NFKB1
ELF1
0
0.48
0.74
0.65
0.54
0.75
0.61
0.18
0.75
0.56
0.11
-0.4
0.51
0.03
-0.46
-0.73
-0.04
-0.52
-0.75
-0.62
-0.57
-0.75
-0.58
-0.14
STAT1
IRF7
ISG15
COL1A1
1.0
0.5
0.0
−0.5
−1.0
Supplied signed regulatory prior

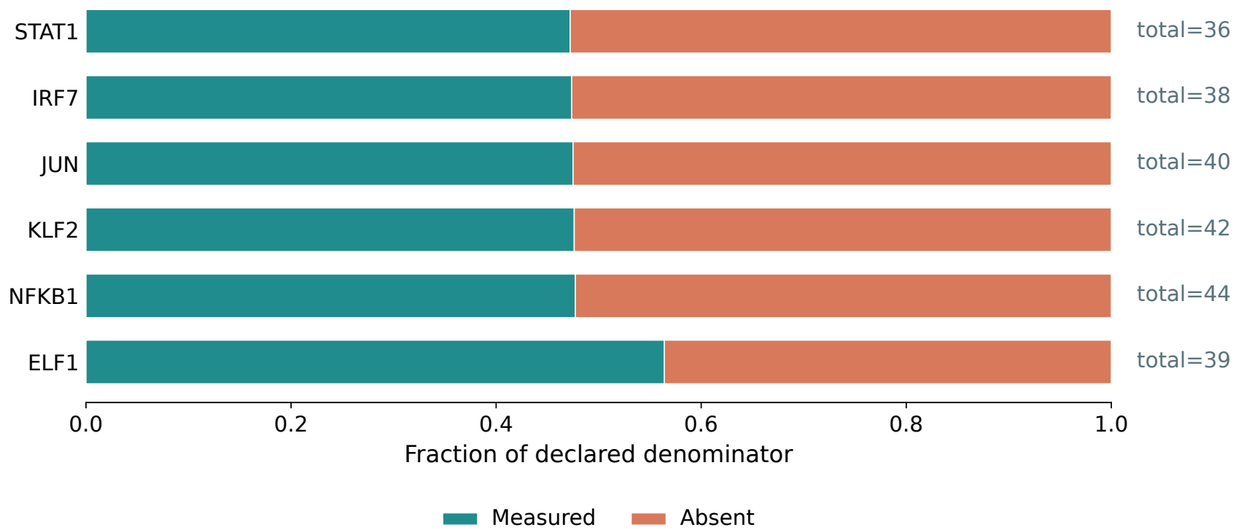

STAT1 activity and expression
Supplied STAT1 activity score
0.0
0.2
0.4
0.6
0.8
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Centered log2 STAT1 expression
Control
Treated


TF activity across independent cohorts
Estimate [interval] · q
Independent n = 6
STAT1 · Cohort A
STAT1 · Cohort B
IRF7 · Cohort A
IRF7 · Cohort B
JUN · Cohort A
JUN · Cohort B
KLF2 · Cohort A
KLF2 · Cohort B
NFKB1 · Cohort A
NFKB1 · Cohort B
ELF1 · Cohort A
ELF1 · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Donor TF-activity score difference · supplied 95% interval

Bulk · tf_activity — complementary evidence

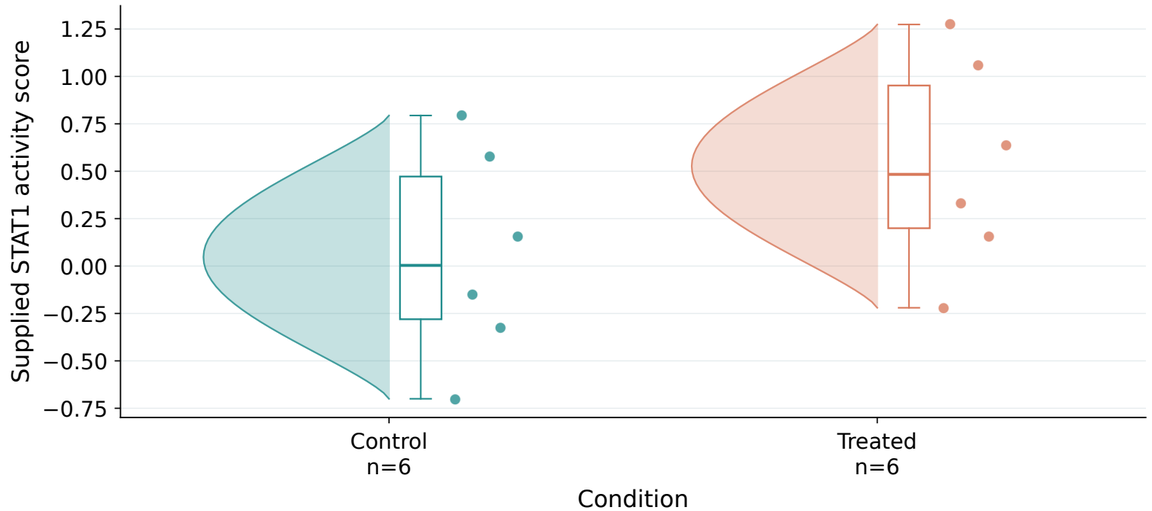
A

Regulon target coverage



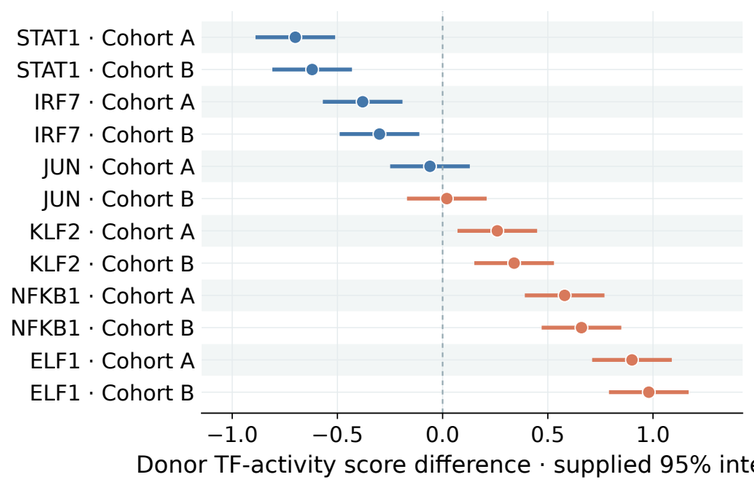
B

STAT1 activity in independent donors



C

TF activity across independent cohorts



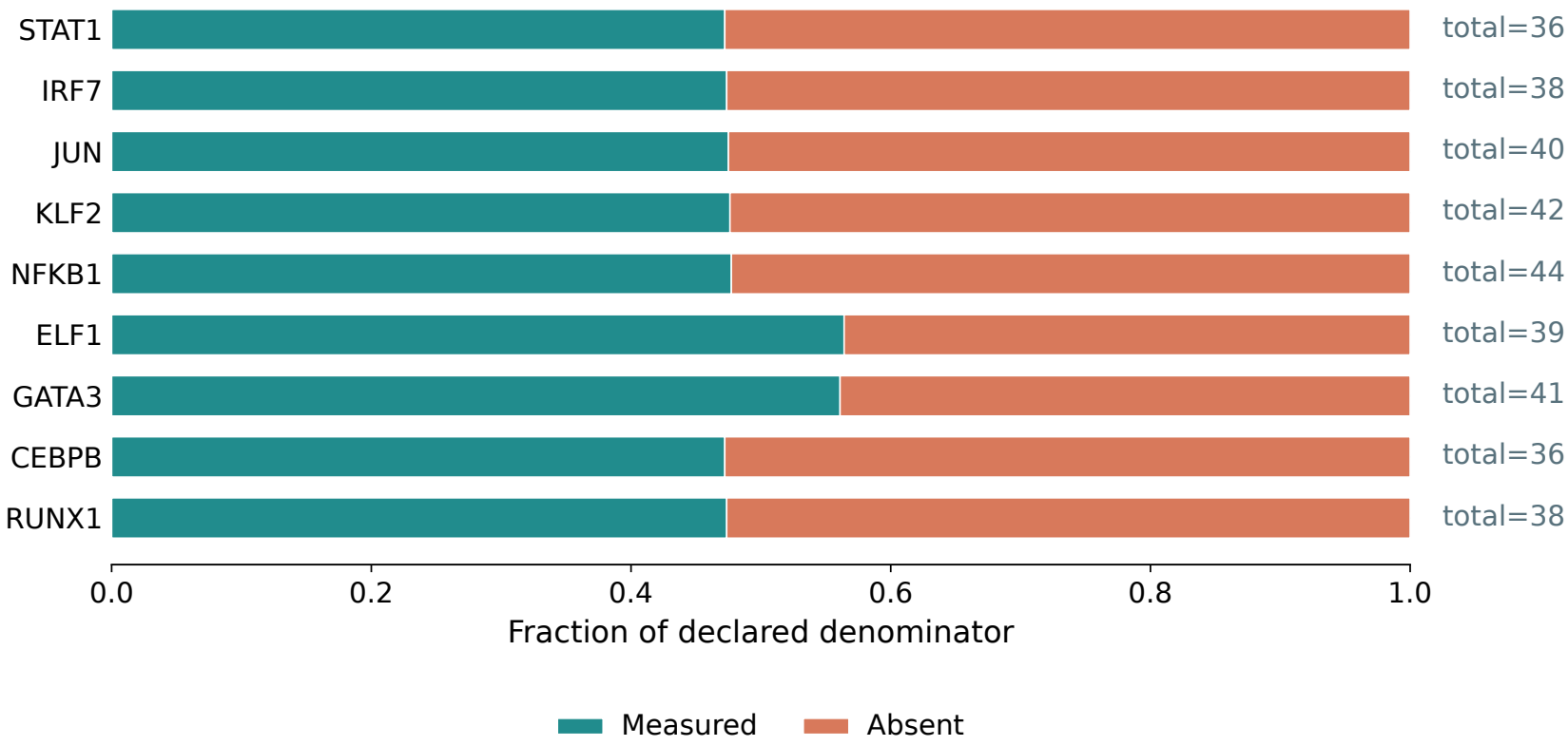
Estimate [interval] · q
Independent n = 6

STAT1 · Cohort A	-0.70 [-0.88, -0.52] 0.002
STAT1 · Cohort B	-0.62 [-0.80, -0.44] 0.005
IRF7 · Cohort A	-0.38 [-0.56, -0.20] 0.02
IRF7 · Cohort B	-0.30 [-0.48, -0.12] 0.023
JUN · Cohort A	-0.06 [-0.24, 0.12] 0.65
JUN · Cohort B	0.02 [-0.16, 0.20] 0.65
KLF2 · Cohort A	0.26 [0.08, 0.44] 0.056
KLF2 · Cohort B	0.34 [0.16, 0.52] 0.059
NFKB1 · Cohort A	0.58 [0.40, 0.76] 0.074
NFKB1 · Cohort B	0.66 [0.48, 0.84] 0.077
ELF1 · Cohort A	0.90 [0.72, 1.08] 0.092
ELF1 · Cohort B	0.98 [0.80, 1.16] 0.095

Bulk · tf_activity — complementary evidence

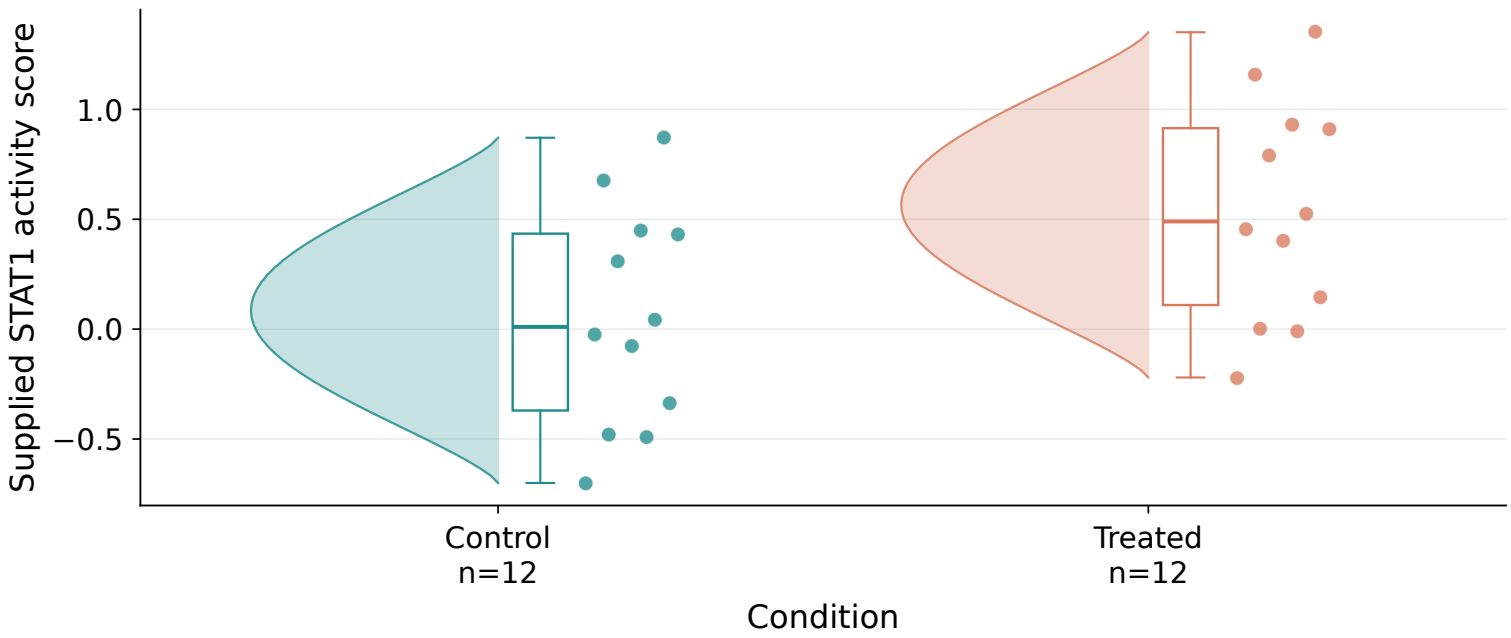
A

Regulon target coverage



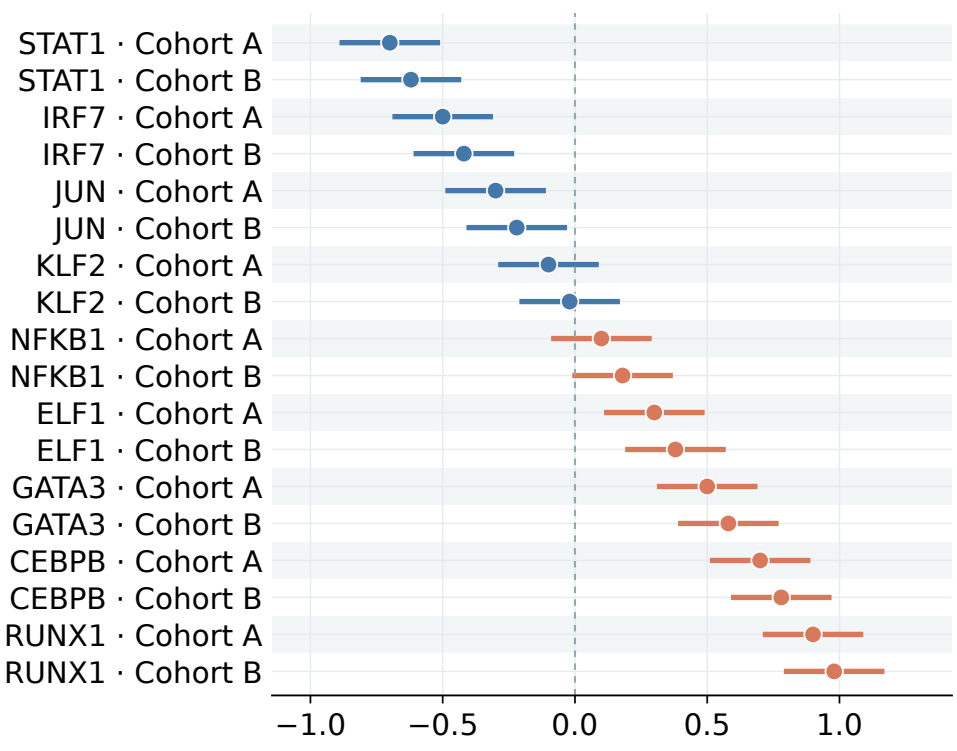
B

STAT1 activity in independent donors



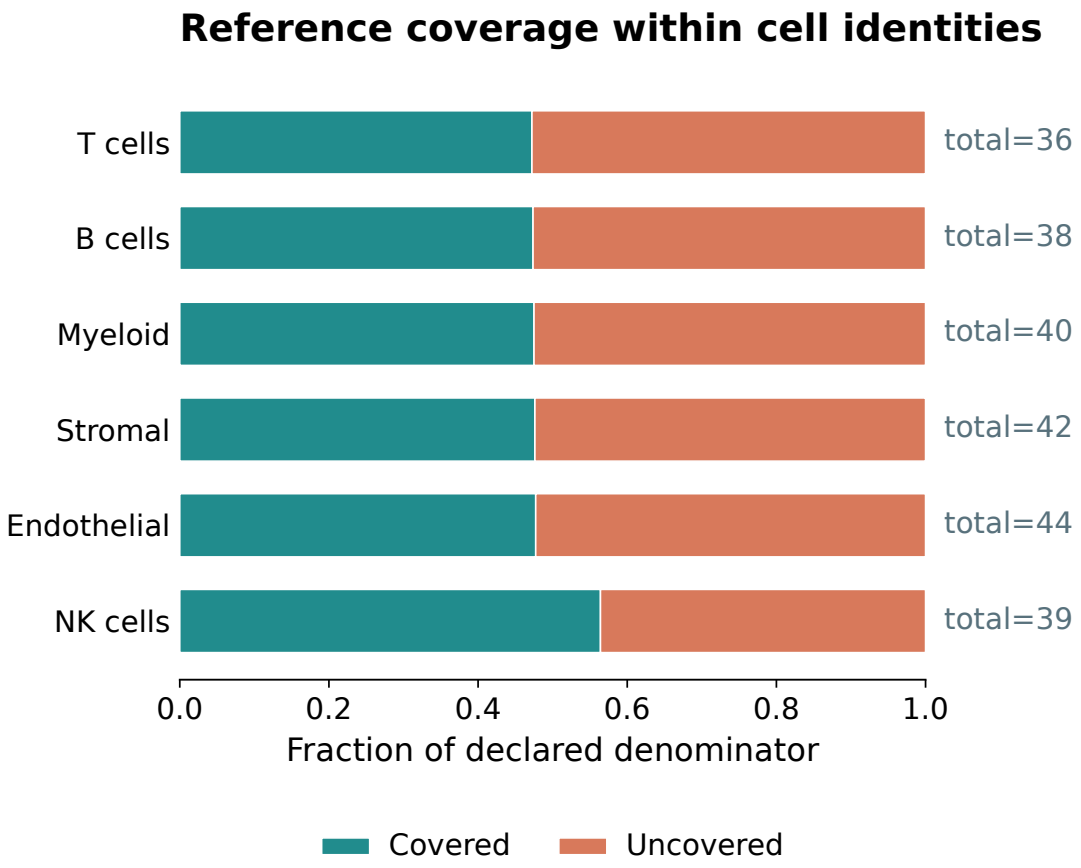
C

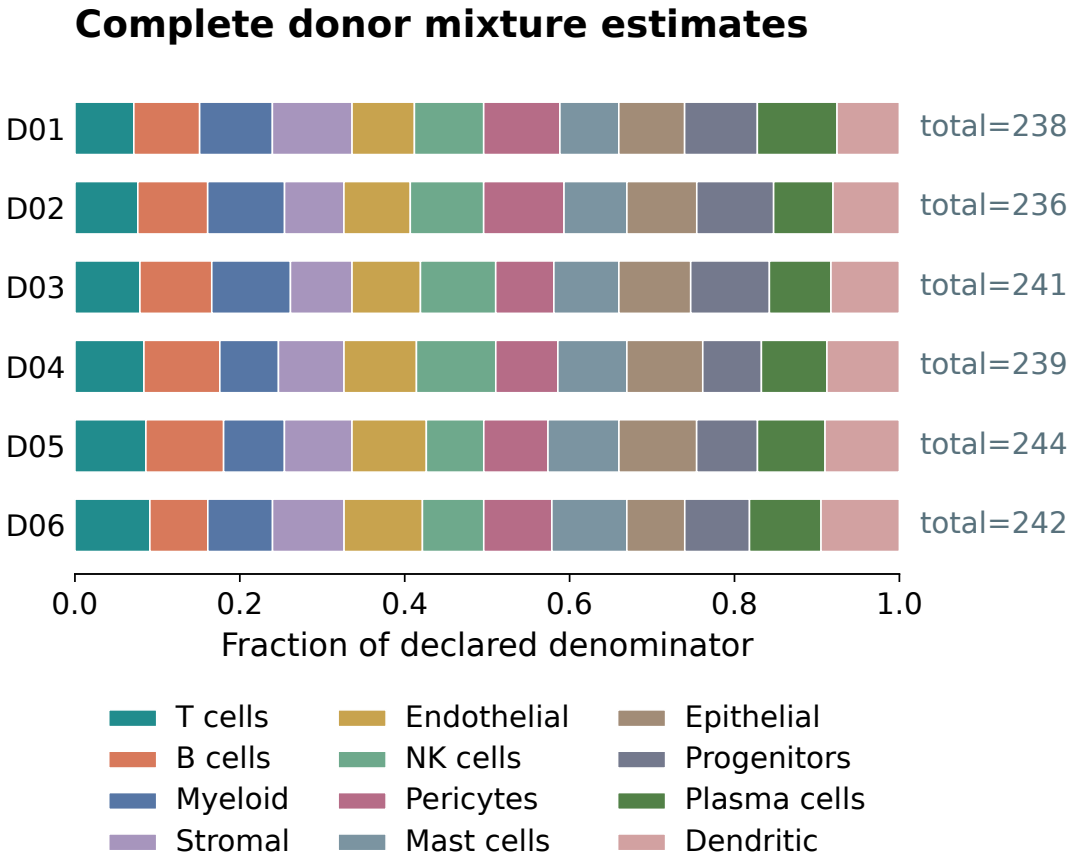
TF activity across independent cohorts



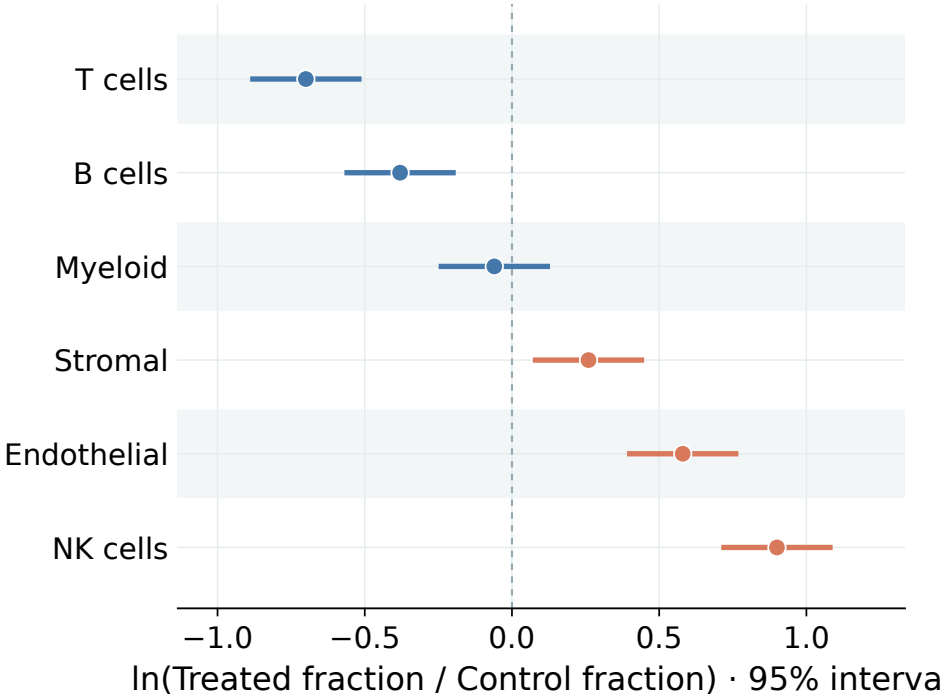
Estimate [interval] · q
Independent n = 12

STAT1 · Cohort A	-0.70 [-0.88, -0.52] 0.002
STAT1 · Cohort B	-0.62 [-0.80, -0.44] 0.005
IRF7 · Cohort A	-0.50 [-0.68, -0.32] 0.02
IRF7 · Cohort B	-0.42 [-0.60, -0.24] 0.023
JUN · Cohort A	-0.30 [-0.48, -0.12] 0.038
JUN · Cohort B	-0.22 [-0.40, -0.04] 0.041
KLF2 · Cohort A	-0.10 [-0.28, 0.08] 0.65
KLF2 · Cohort B	-0.02 [-0.20, 0.16] 0.65
NFKB1 · Cohort A	0.10 [-0.08, 0.28] 0.65
NFKB1 · Cohort B	0.18 [0.00, 0.36] 0.077
ELF1 · Cohort A	0.30 [0.12, 0.48] 0.092
ELF1 · Cohort B	0.38 [0.20, 0.56] 0.095
GATA3 · Cohort A	0.50 [0.32, 0.68] 0.11
GATA3 · Cohort B	0.58 [0.40, 0.76] 0.11
CEBPB · Cohort A	0.70 [0.52, 0.88] 0.13
CEBPB · Cohort B	0.78 [0.60, 0.96] 0.13
RUNX1 · Cohort A	0.90 [0.72, 1.08] 0.15
RUNX1 · Cohort B	0.98 [0.80, 1.16] 0.15


Reference coverage within cell identities
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Covered
Uncovered


Complete donor mixture estimates
D01
D02
D03
D04
D05
D06
total=238
total=236
total=241
total=239
total=244
total=242
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
Pericytes
Mast cells
Epithelial
Progenitors
Plasma cells
Dendritic

Donor-level compositional contrasts



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

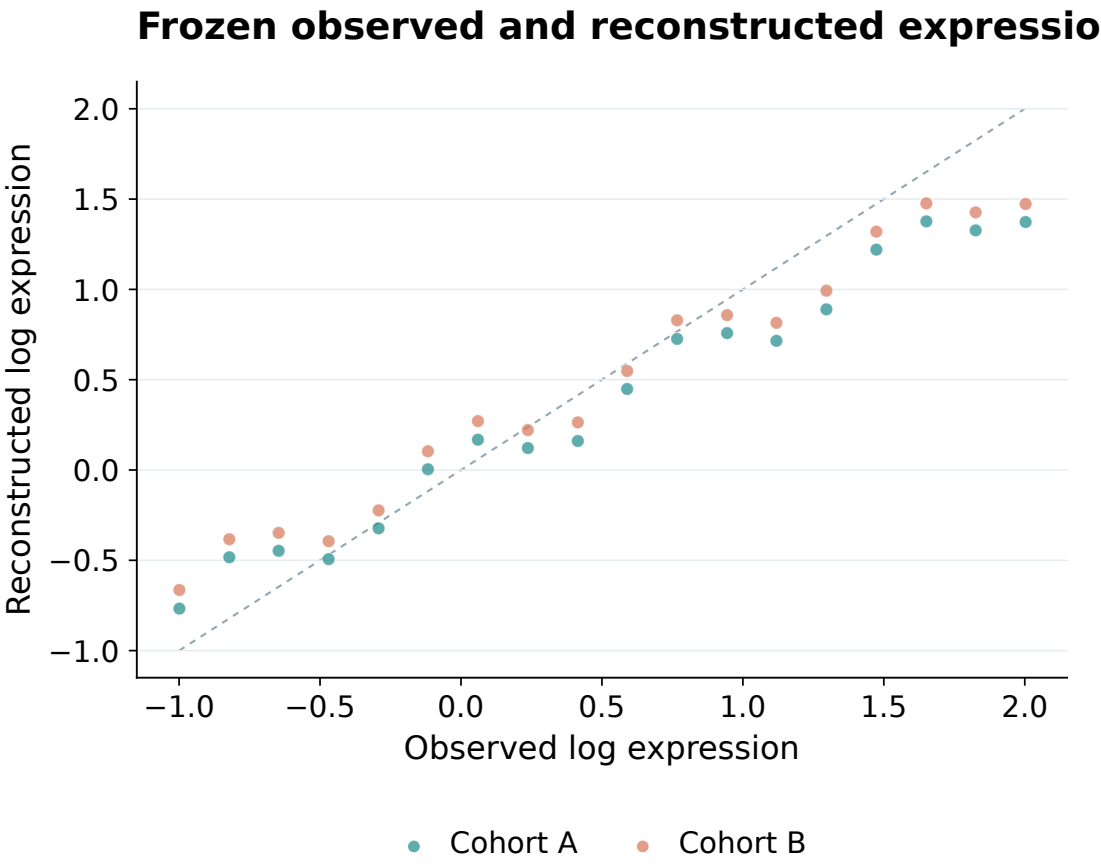
-0.38 [-0.56, -0.20] 0.02

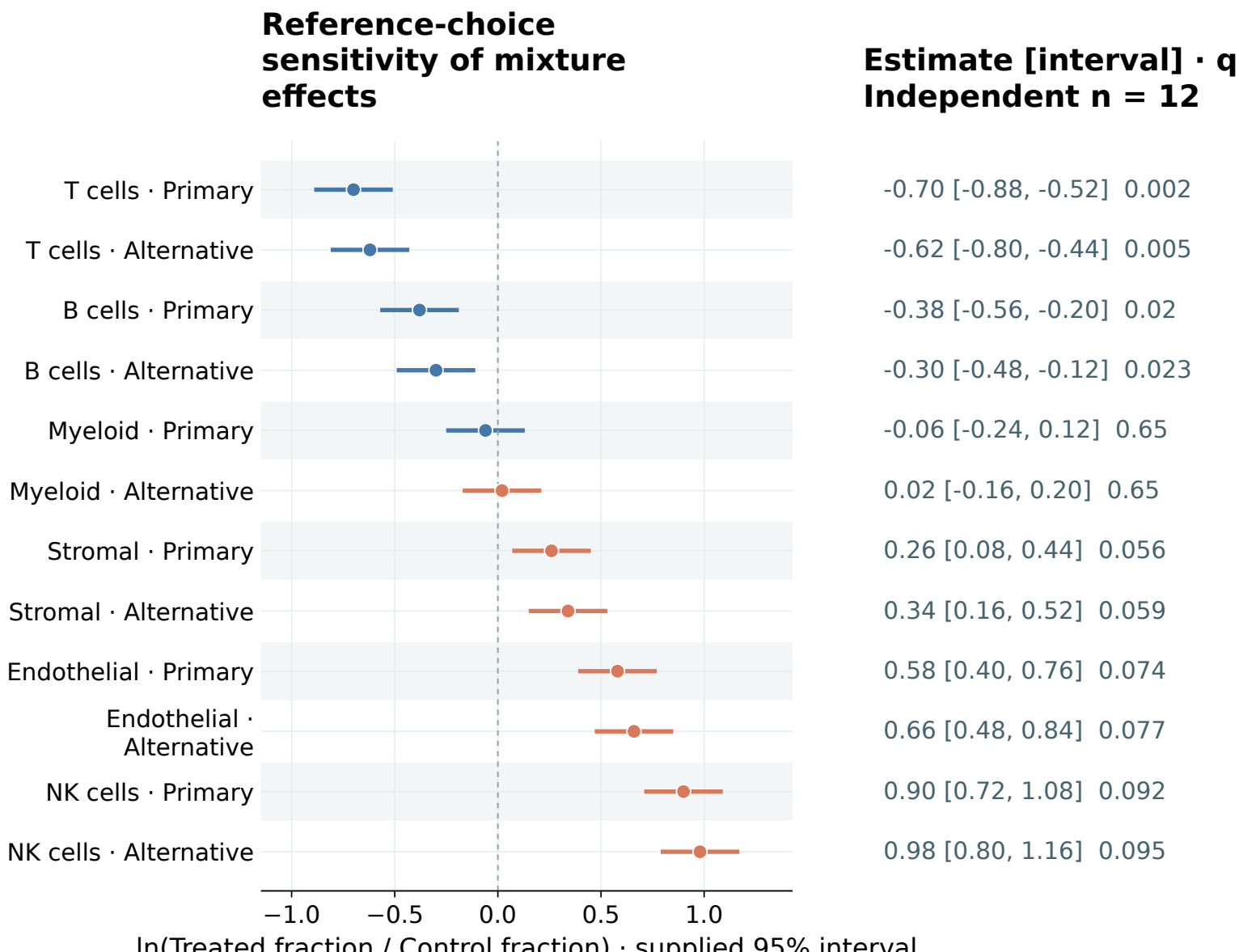
-0.06 [-0.24, 0.12] 0.65

0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092

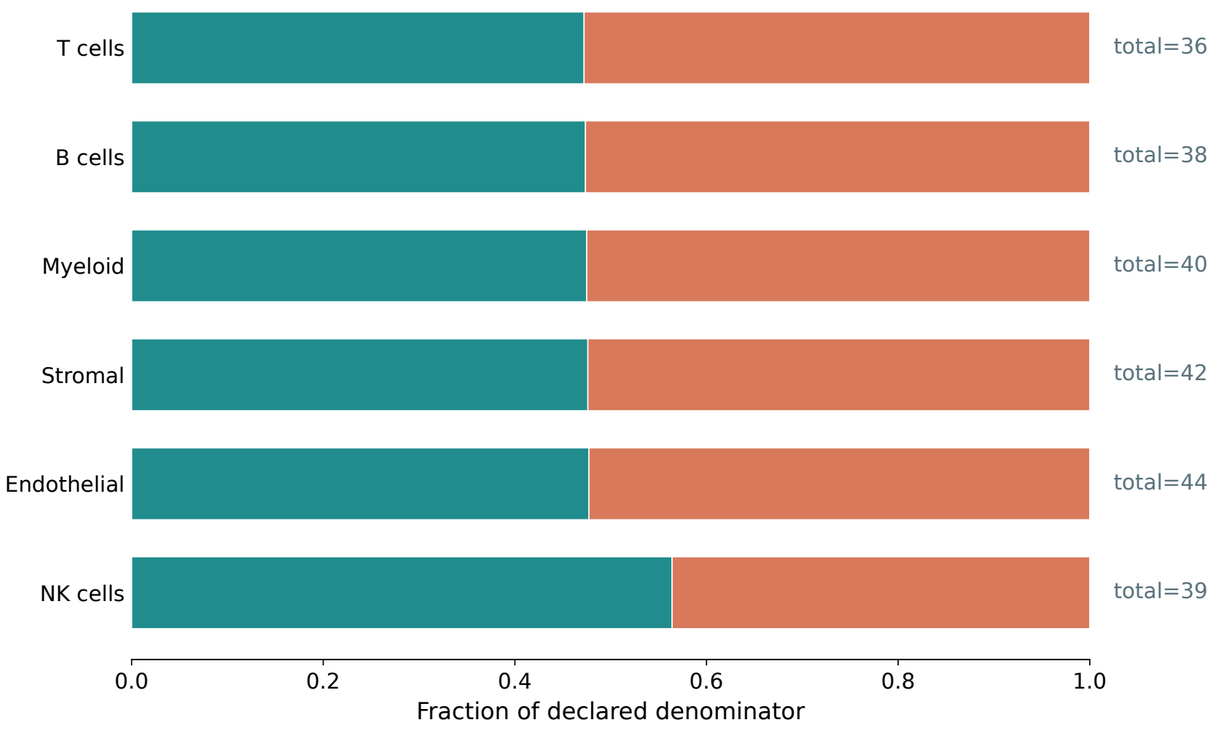

Frozen observed and reconstructed expression
Reconstructed log expression
Observed log expression
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
Cohort A
Cohort B


Reference-choice sensitivity of mixture effects
Estimate [interval] · q
Independent n = 12
T cells · Primary
T cells · Alternative
B cells · Primary
B cells · Alternative
Myeloid · Primary
Myeloid · Alternative
Stromal · Primary
Stromal · Alternative
Endothelial · Primary
Endothelial · Alternative
NK cells · Primary
NK cells · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
ln(Treated fraction / Control fraction) · supplied 95% interval

Bulk · deconvolution — complementary evidence

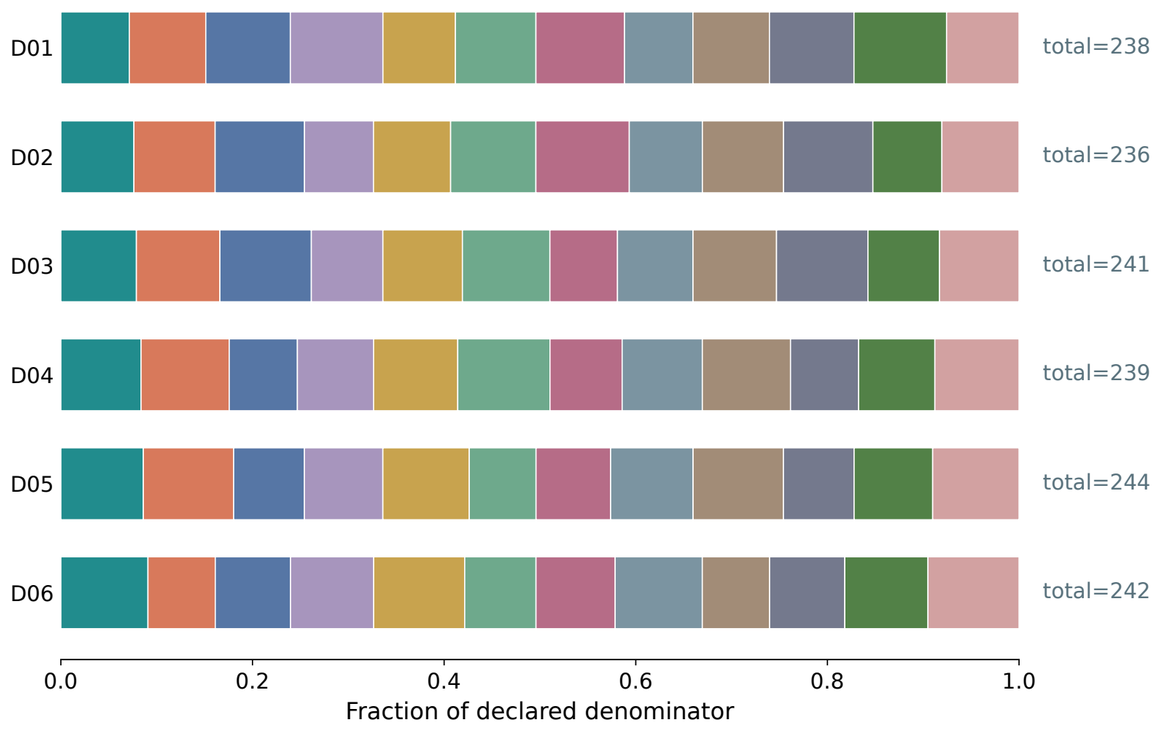
A

Reference coverage within cell identities



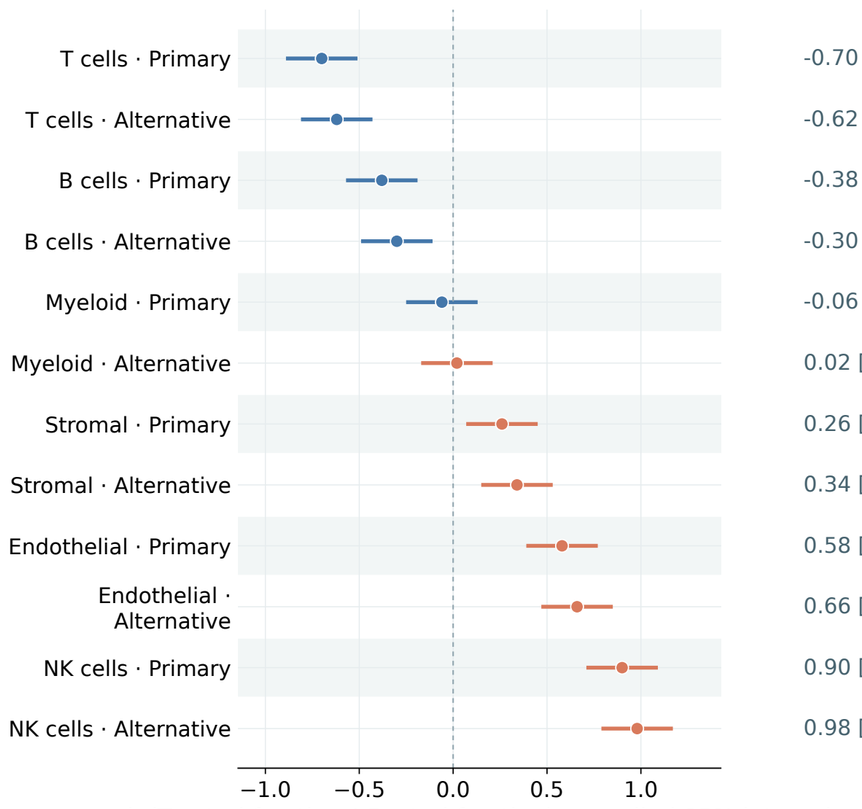
B

Complete donor mixture estimates



C

Reference-choice sensitivity of mixture effects

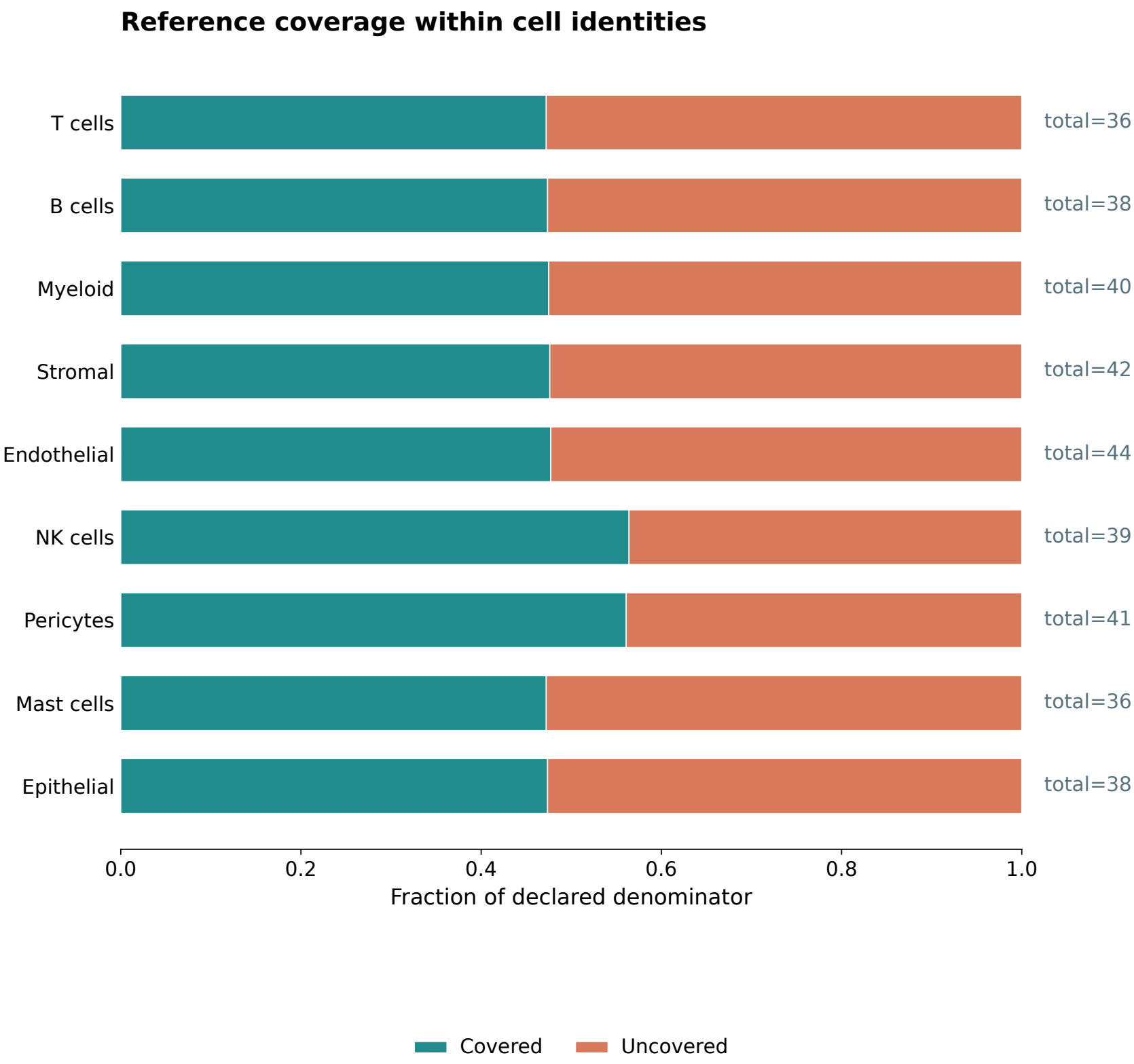


Estimate [interval] · q
Independent n = 12

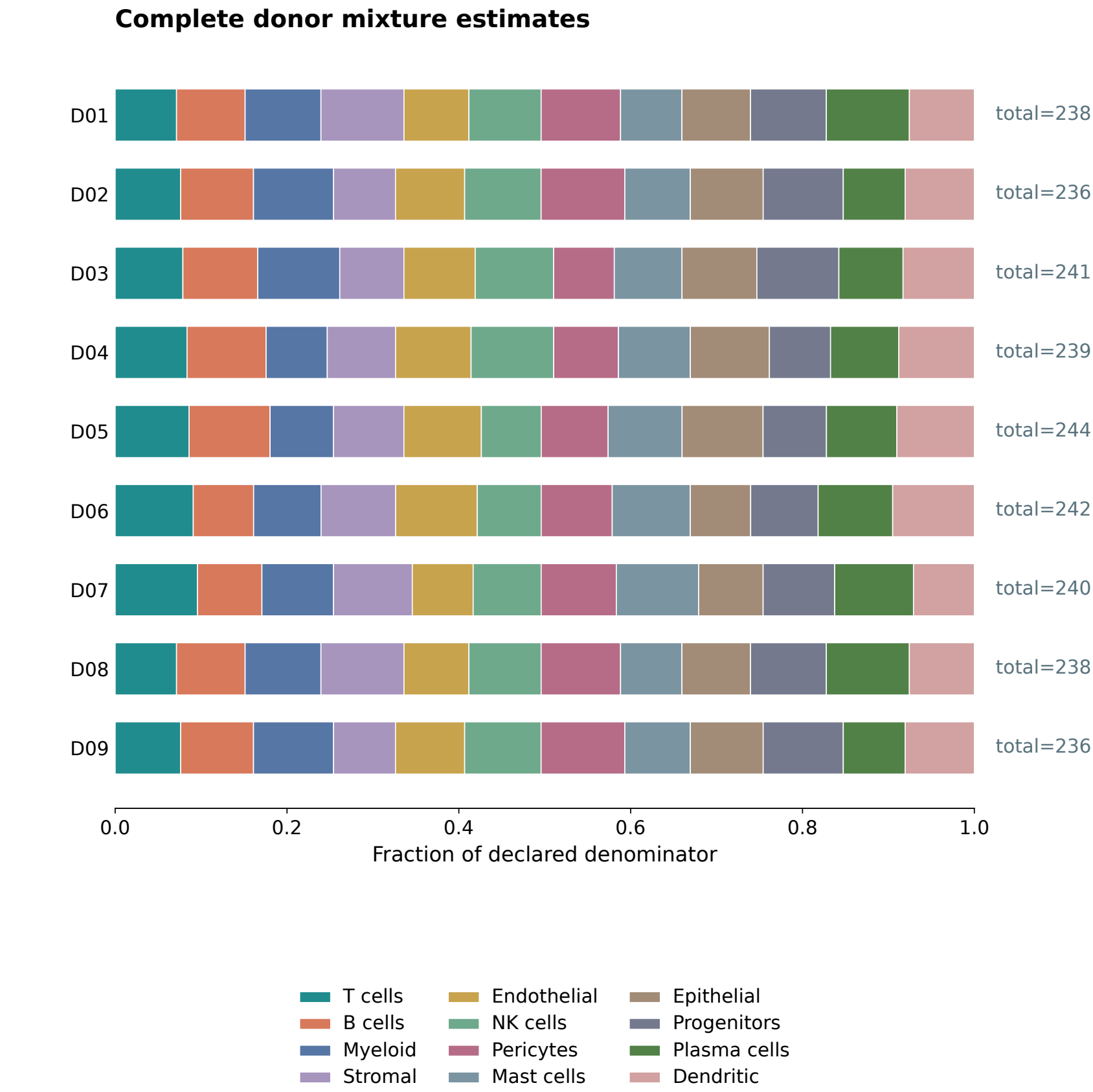
T cells · Primary	-0.70 [-0.88, -0.52] 0.002
T cells · Alternative	-0.62 [-0.80, -0.44] 0.005
B cells · Primary	-0.38 [-0.56, -0.20] 0.02
B cells · Alternative	-0.30 [-0.48, -0.12] 0.023
Myeloid · Primary	-0.06 [-0.24, 0.12] 0.65
Myeloid · Alternative	0.02 [-0.16, 0.20] 0.65
Stromal · Primary	0.26 [0.08, 0.44] 0.056
Stromal · Alternative	0.34 [0.16, 0.52] 0.059
Endothelial · Primary	0.58 [0.40, 0.76] 0.074
Endothelial · Alternative	0.66 [0.48, 0.84] 0.077
NK cells · Primary	0.90 [0.72, 1.08] 0.092
NK cells · Alternative	0.98 [0.80, 1.16] 0.095

Bulk · deconvolution — complementary evidence

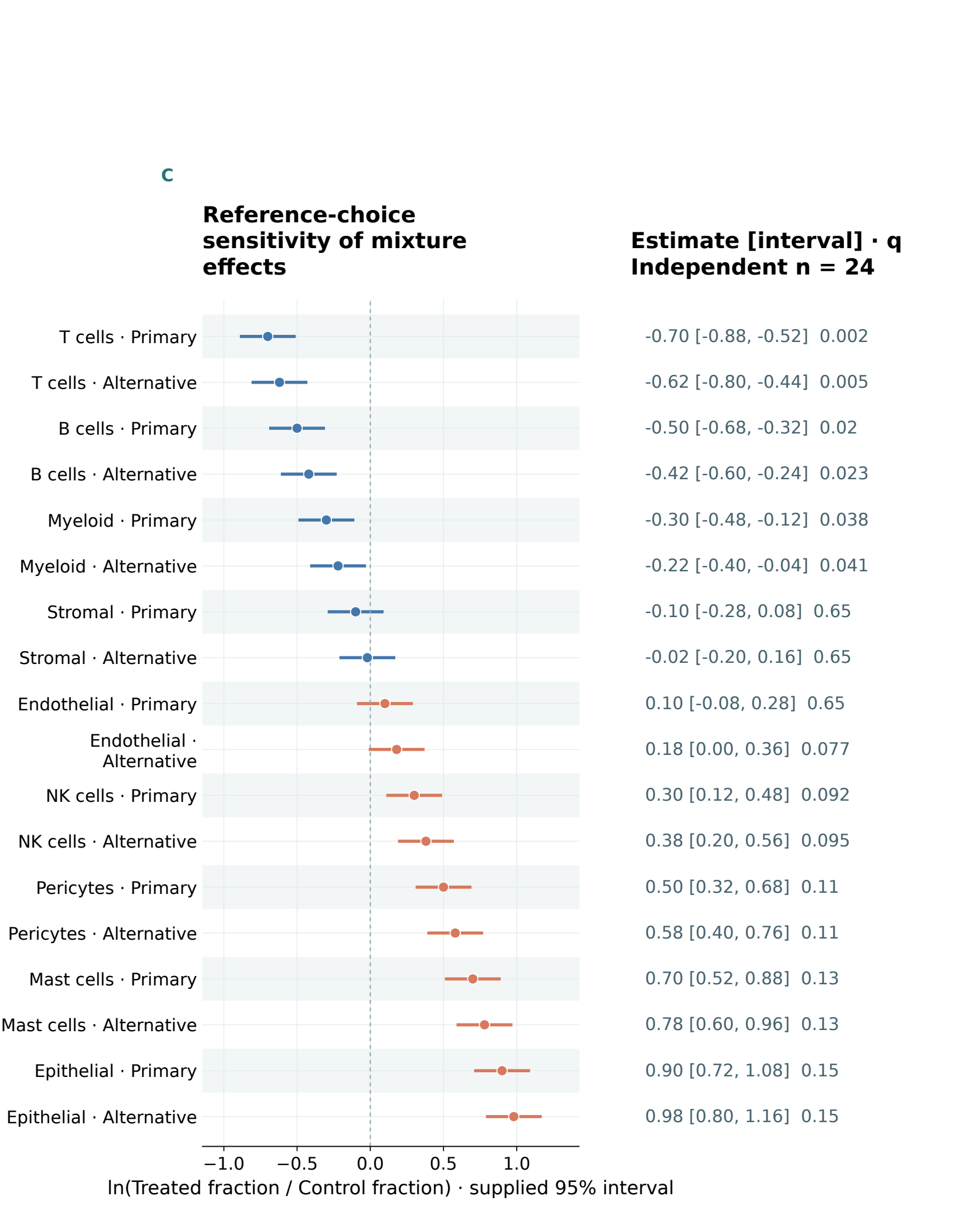
A



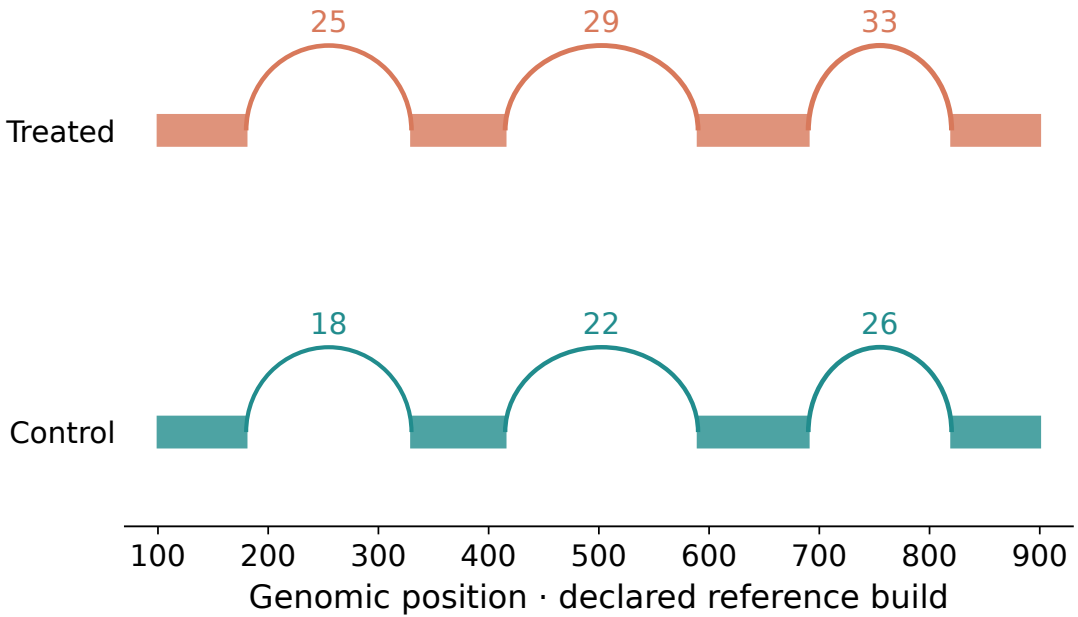
B

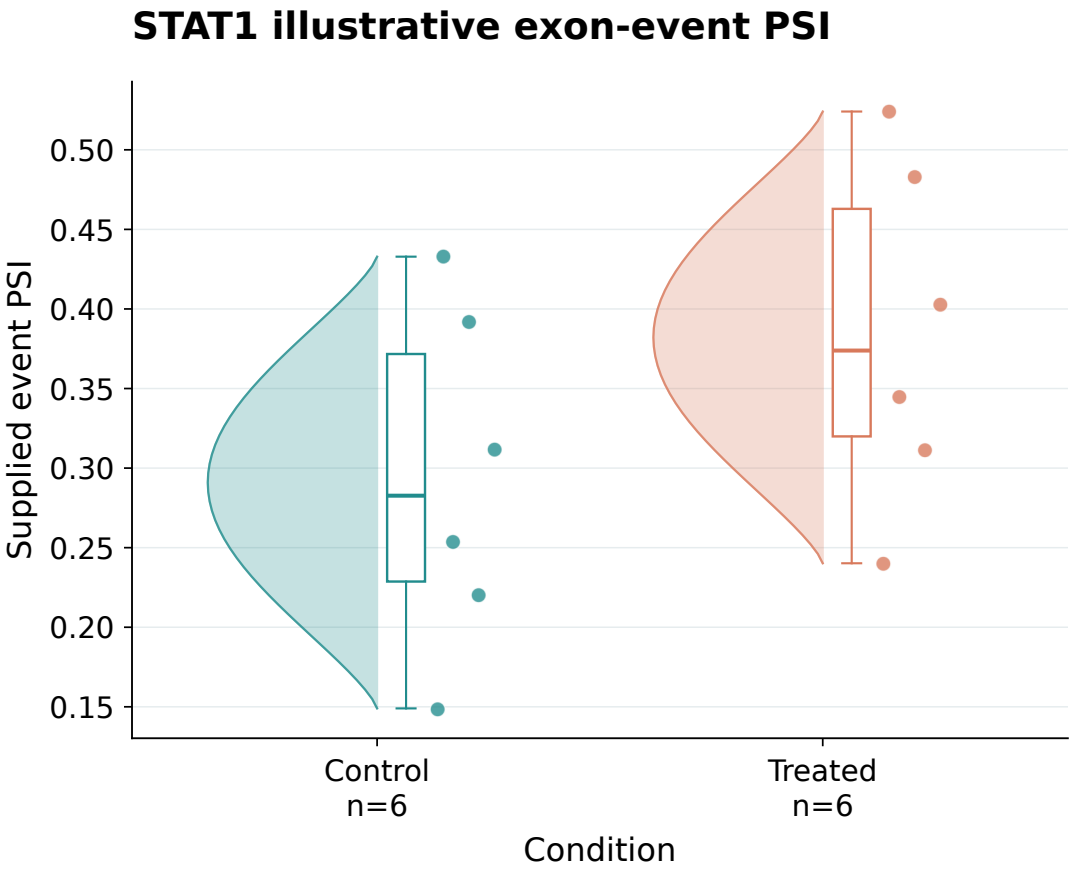


C



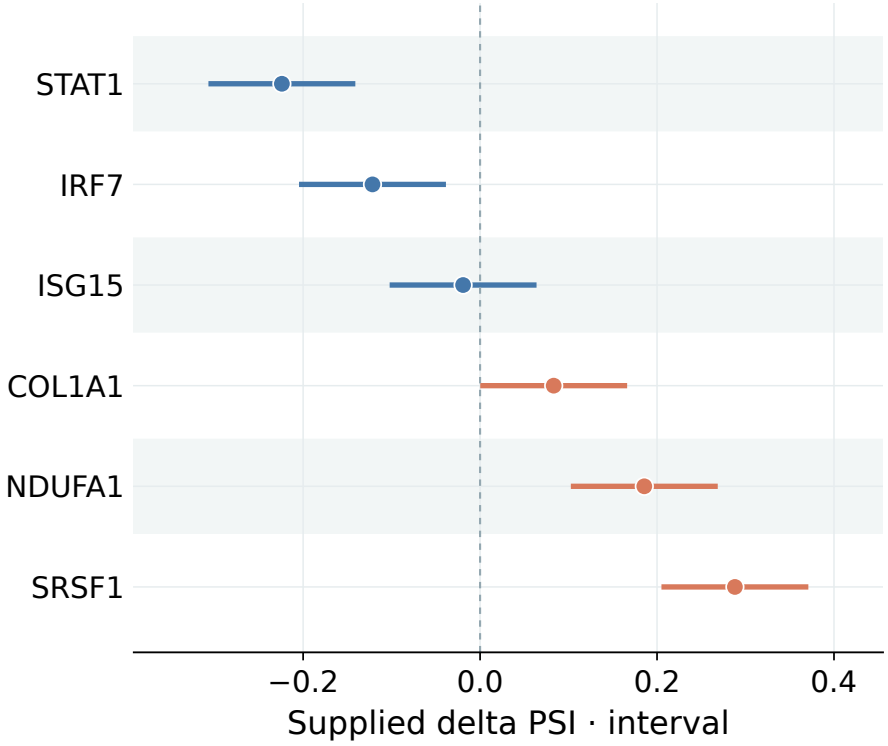
Physical exon and junction support





Software test fixture; not biological evidence

Event-level differential splicing



Estimate [interval] · q
Independent n = 12

-0.22 [-0.30, -0.14] 0.002

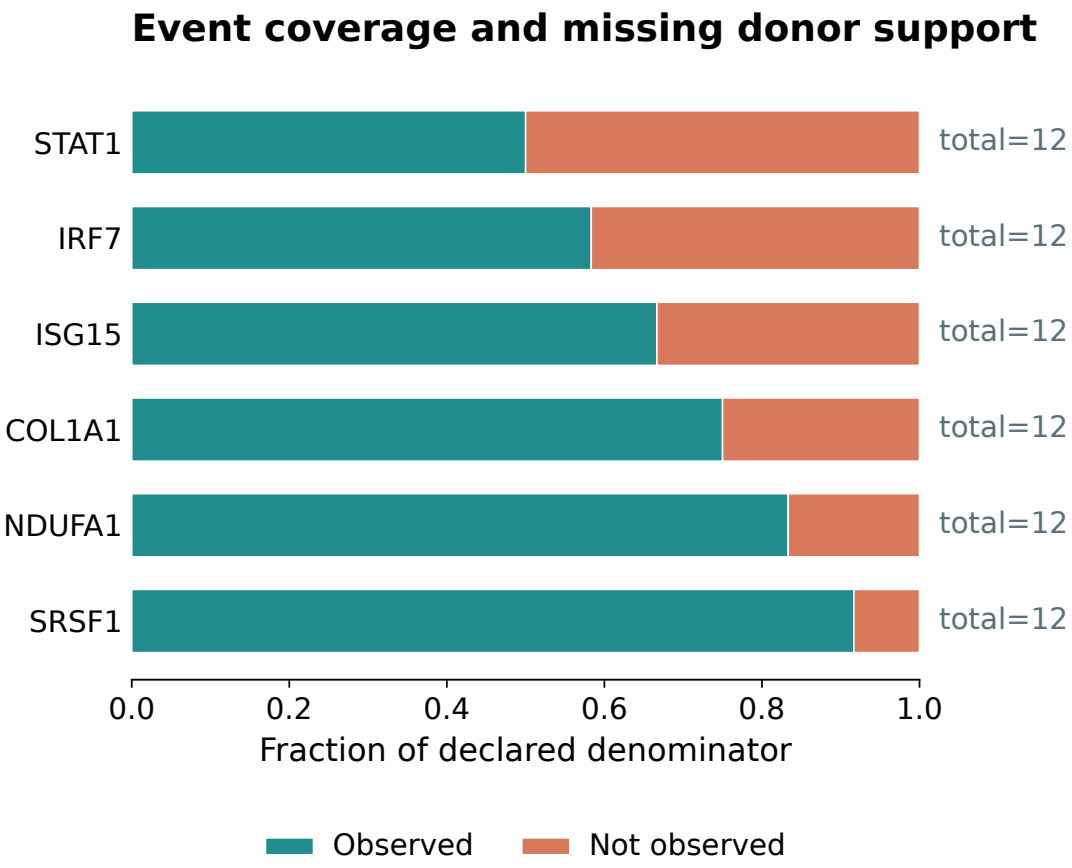
-0.12 [-0.20, -0.04] 0.02

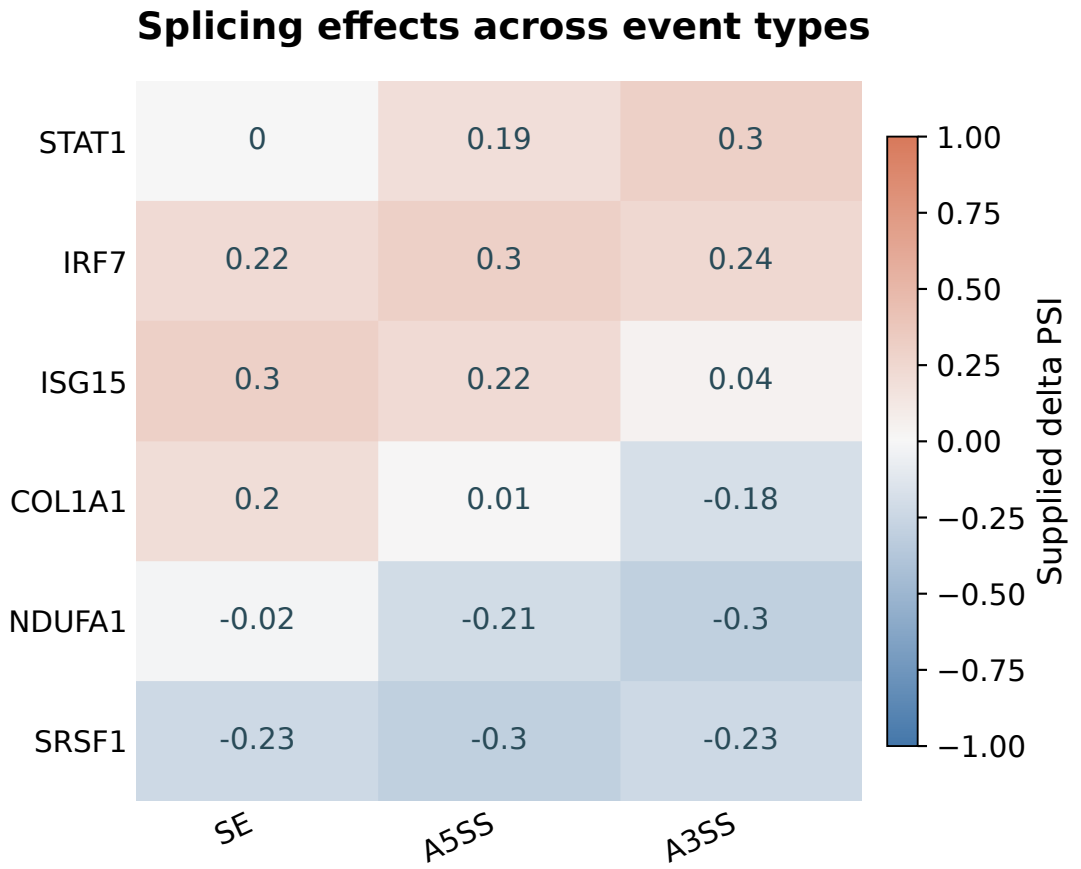
-0.02 [-0.10, 0.06] 0.65

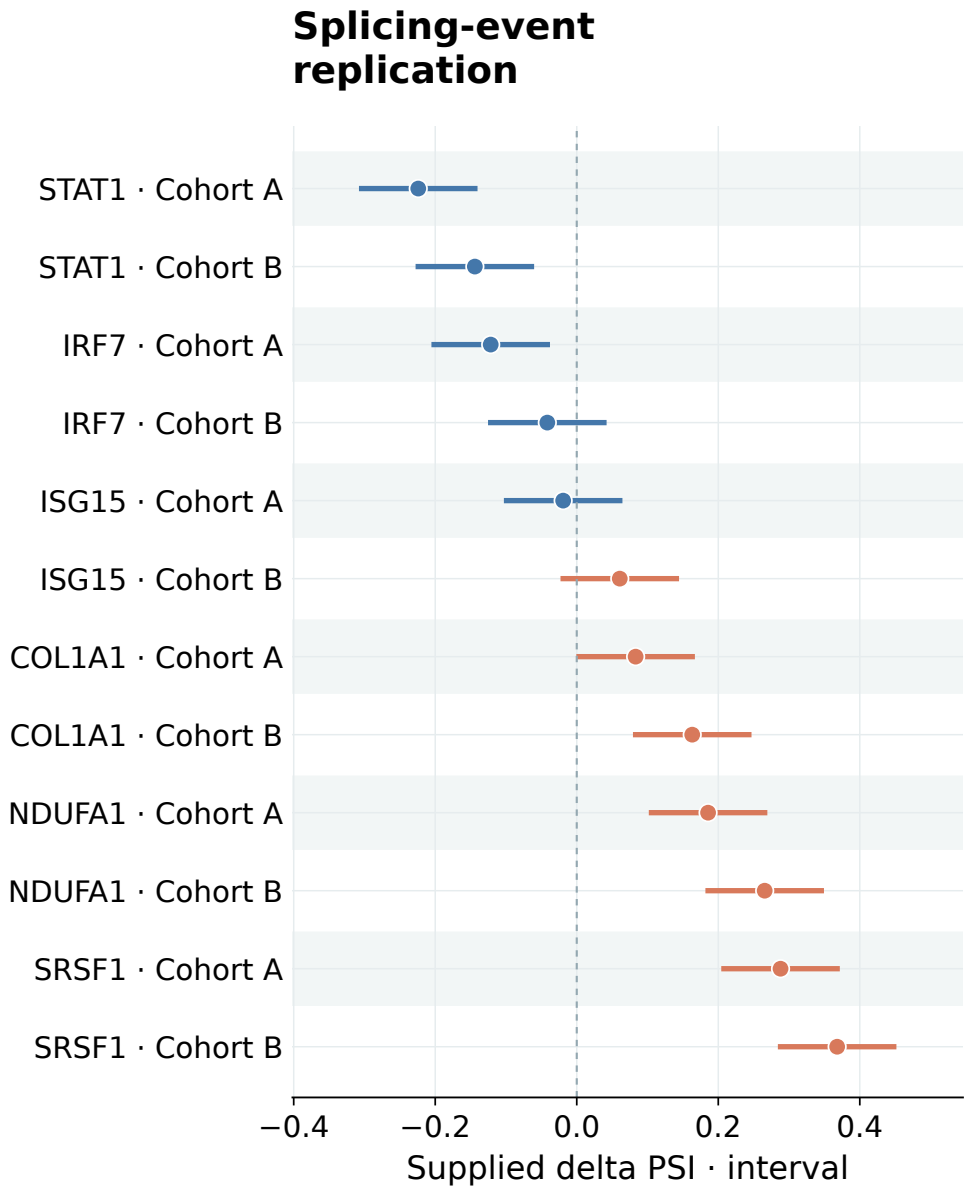
0.08 [0.00, 0.16] 0.056

0.19 [0.11, 0.27] 0.074

0.29 [0.21, 0.37] 0.092


Event coverage and missing donor support
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
total=12
total=12
total=12
total=12
total=12
total=12
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Observed
Not observed

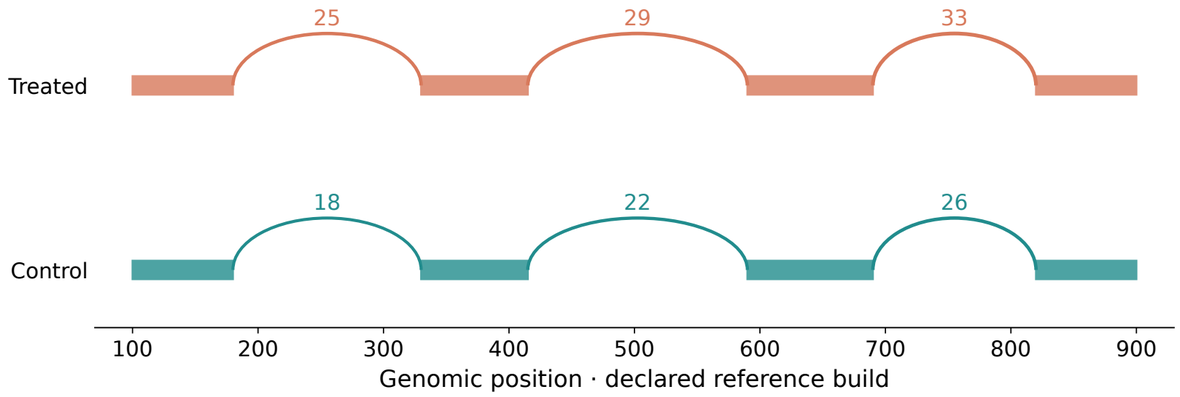

Splicing effects across event types
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0
0.19
0.3
0.22
0.3
0.24
0.3
0.22
0.04
0.2
0.01
-0.18
-0.02
-0.21
-0.3
-0.23
-0.3
-0.23
SE
A5SS
A3SS
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Supplied delta PSI


Splicing-event replication
Estimate [interval] · q
Independent n = 6
STAT1 · Cohort A
STAT1 · Cohort B
IRF7 · Cohort A
IRF7 · Cohort B
ISG15 · Cohort A
ISG15 · Cohort B
COL1A1 · Cohort A
COL1A1 · Cohort B
NDUFA1 · Cohort A
NDUFA1 · Cohort B
SRSF1 · Cohort A
SRSF1 · Cohort B
-0.22 [-0.30, -0.14] 0.002
-0.14 [-0.22, -0.06] 0.005
-0.12 [-0.20, -0.04] 0.02
-0.04 [-0.12, 0.04] 0.65
-0.02 [-0.10, 0.06] 0.65
0.06 [-0.02, 0.14] 0.65
0.08 [0.00, 0.16] 0.056
0.16 [0.08, 0.24] 0.059
0.19 [0.11, 0.27] 0.074
0.27 [0.19, 0.35] 0.077
0.29 [0.21, 0.37] 0.092
0.37 [0.29, 0.45] 0.095
−0.4
−0.2
0.0
0.2
0.4
Supplied delta PSI · interval

Bulk · splicing — complementary evidence

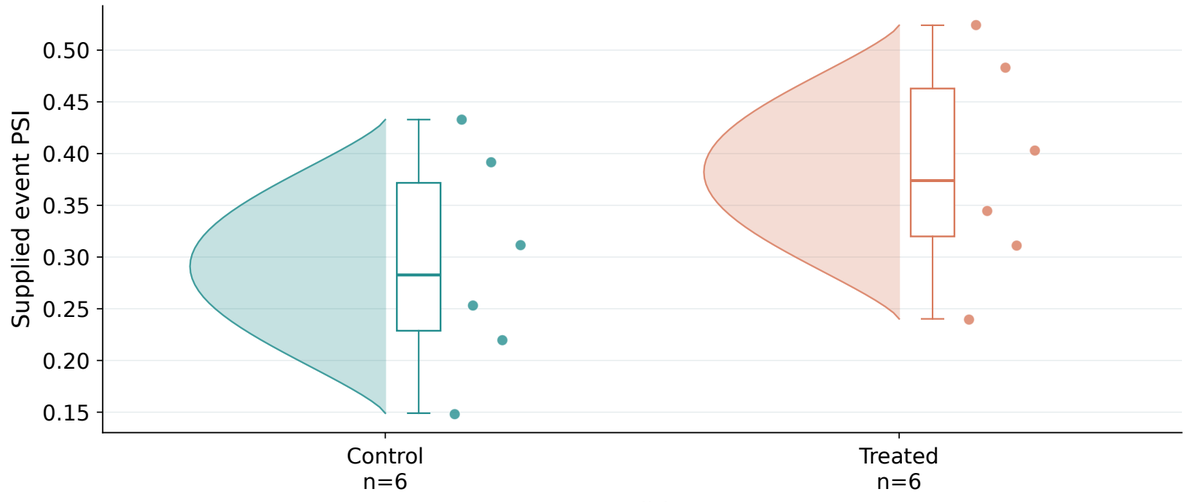
A

Physical exon and junction support



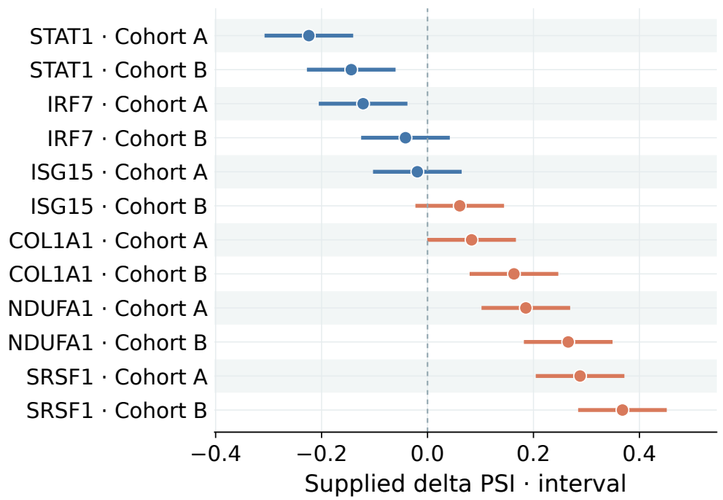
B

STAT1 illustrative exon-event PSI



C

Splicing-event replication



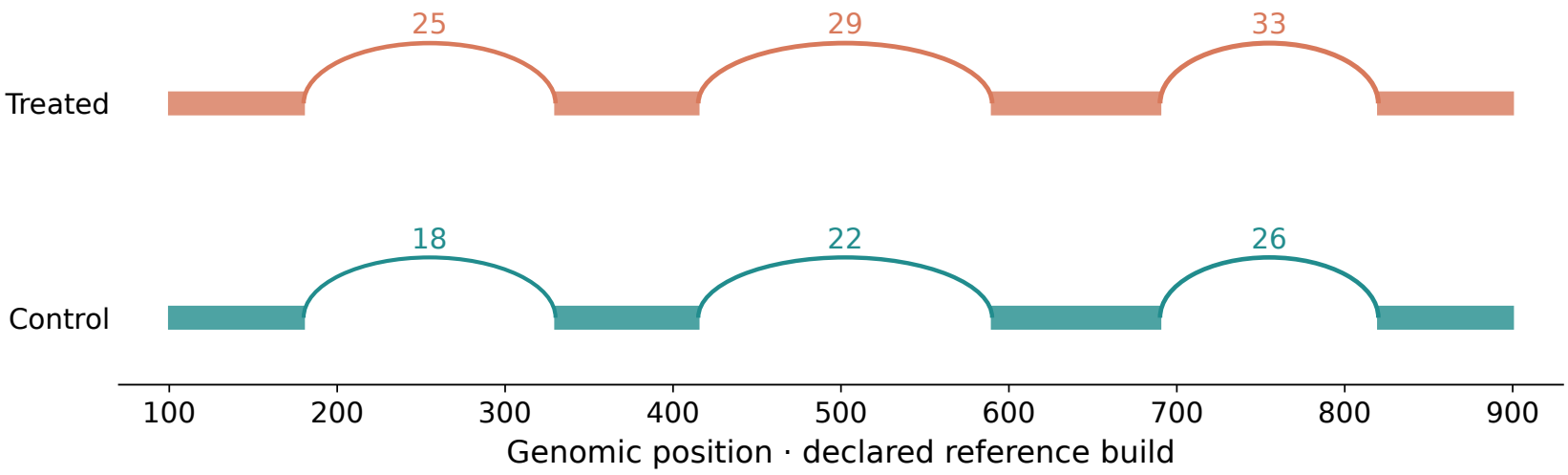
Estimate [interval] · q
Independent n = 6

STAT1 · Cohort A	-0.22 [-0.30, -0.14] 0.002
STAT1 · Cohort B	-0.14 [-0.22, -0.06] 0.005
IRF7 · Cohort A	-0.12 [-0.20, -0.04] 0.02
IRF7 · Cohort B	-0.04 [-0.12, 0.04] 0.65
ISG15 · Cohort A	-0.02 [-0.10, 0.06] 0.65
ISG15 · Cohort B	0.06 [-0.02, 0.14] 0.65
COL1A1 · Cohort A	0.08 [0.00, 0.16] 0.056
COL1A1 · Cohort B	0.16 [0.08, 0.24] 0.059
NDUFA1 · Cohort A	0.19 [0.11, 0.27] 0.074
NDUFA1 · Cohort B	0.27 [0.19, 0.35] 0.077
SRSF1 · Cohort A	0.29 [0.21, 0.37] 0.092
SRSF1 · Cohort B	0.37 [0.29, 0.45] 0.095

Bulk · splicing — complementary evidence

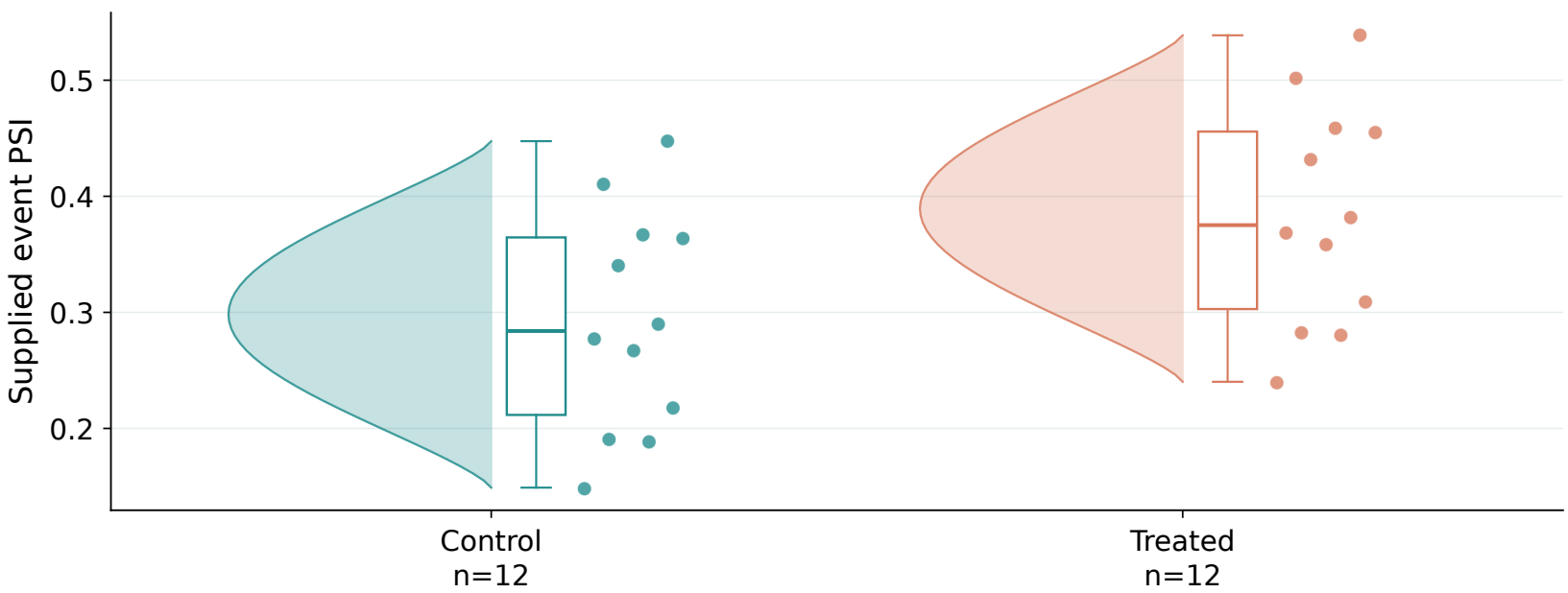
A

Physical exon and junction support



B

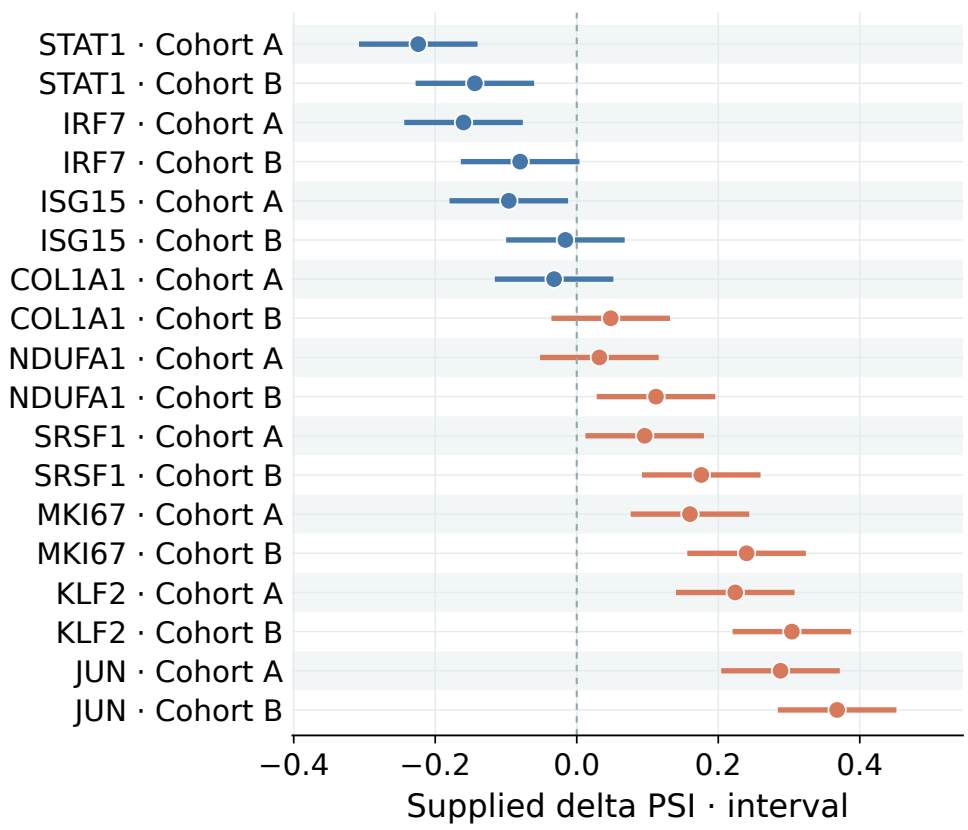
STAT1 illustrative exon-event PSI



C

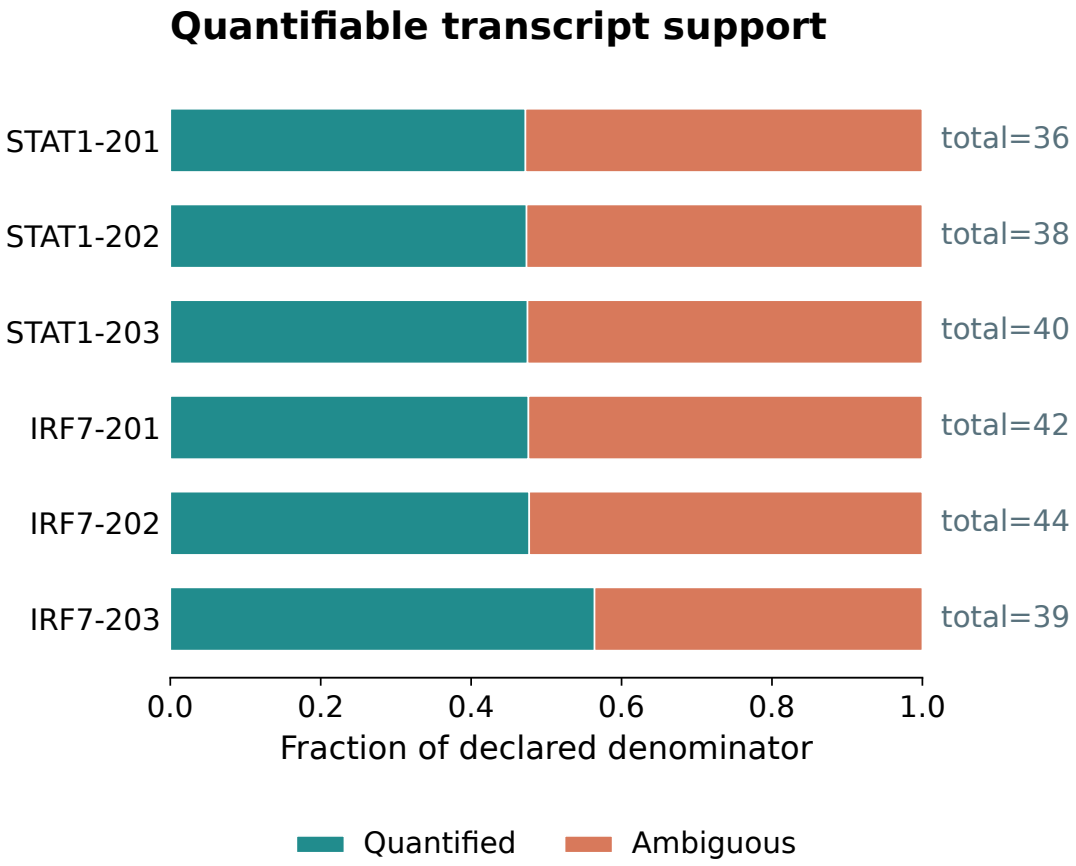
Splicing-event replication

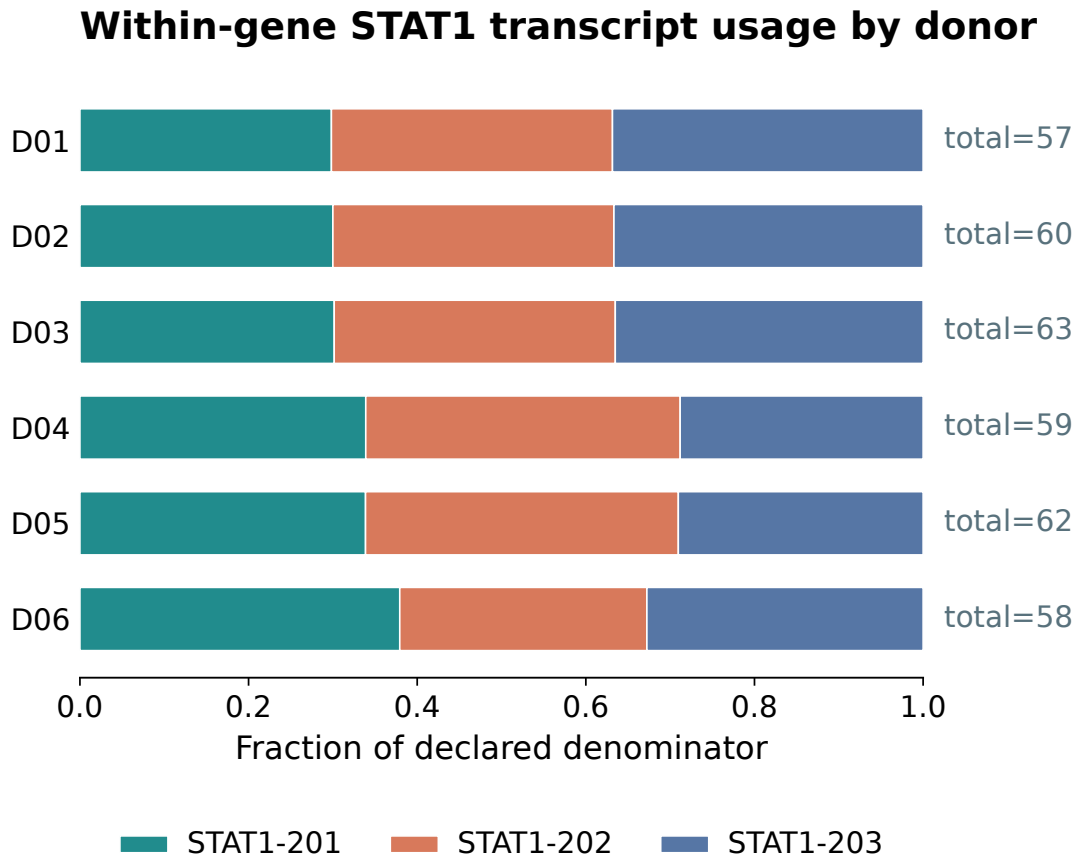
Estimate [interval] · q
Independent n = 12



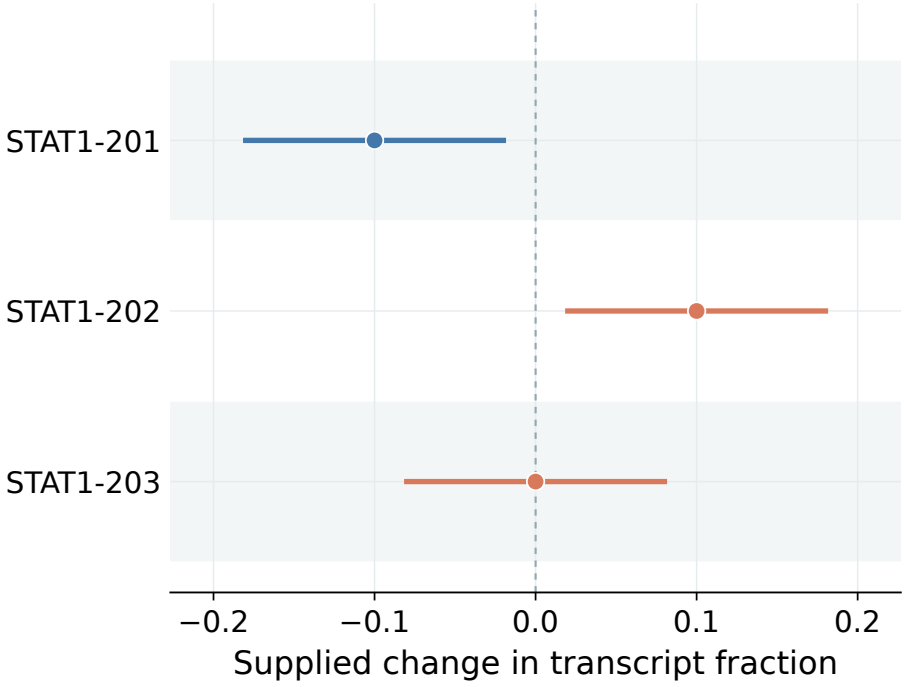
STAT1 · Cohort A	-0.22 [-0.30, -0.14] 0.002
STAT1 · Cohort B	-0.14 [-0.22, -0.06] 0.005
IRF7 · Cohort A	-0.16 [-0.24, -0.08] 0.02
IRF7 · Cohort B	-0.08 [-0.16, 0.00] 0.65
ISG15 · Cohort A	-0.10 [-0.18, -0.02] 0.038
ISG15 · Cohort B	-0.02 [-0.10, 0.06] 0.65
COL1A1 · Cohort A	-0.03 [-0.11, 0.05] 0.65
COL1A1 · Cohort B	0.05 [-0.03, 0.13] 0.65
NDUFA1 · Cohort A	0.03 [-0.05, 0.11] 0.65
NDUFA1 · Cohort B	0.11 [0.03, 0.19] 0.077
SRSF1 · Cohort A	0.10 [0.02, 0.18] 0.092
SRSF1 · Cohort B	0.18 [0.10, 0.26] 0.095
MKI67 · Cohort A	0.16 [0.08, 0.24] 0.11
MKI67 · Cohort B	0.24 [0.16, 0.32] 0.11
KLF2 · Cohort A	0.22 [0.14, 0.30] 0.13
KLF2 · Cohort B	0.30 [0.22, 0.38] 0.13
JUN · Cohort A	0.29 [0.21, 0.37] 0.15
JUN · Cohort B	0.37 [0.29, 0.45] 0.15

Software test fixture; not biological evidence


Quantifiable transcript support
STAT1-201
STAT1-202
STAT1-203
IRF7-201
IRF7-202
IRF7-203
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Quantified
Ambiguous


Within-gene STAT1 transcript usage by donor
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
STAT1-201
STAT1-202
STAT1-203

Prespecified STAT1 transcript-usage changes



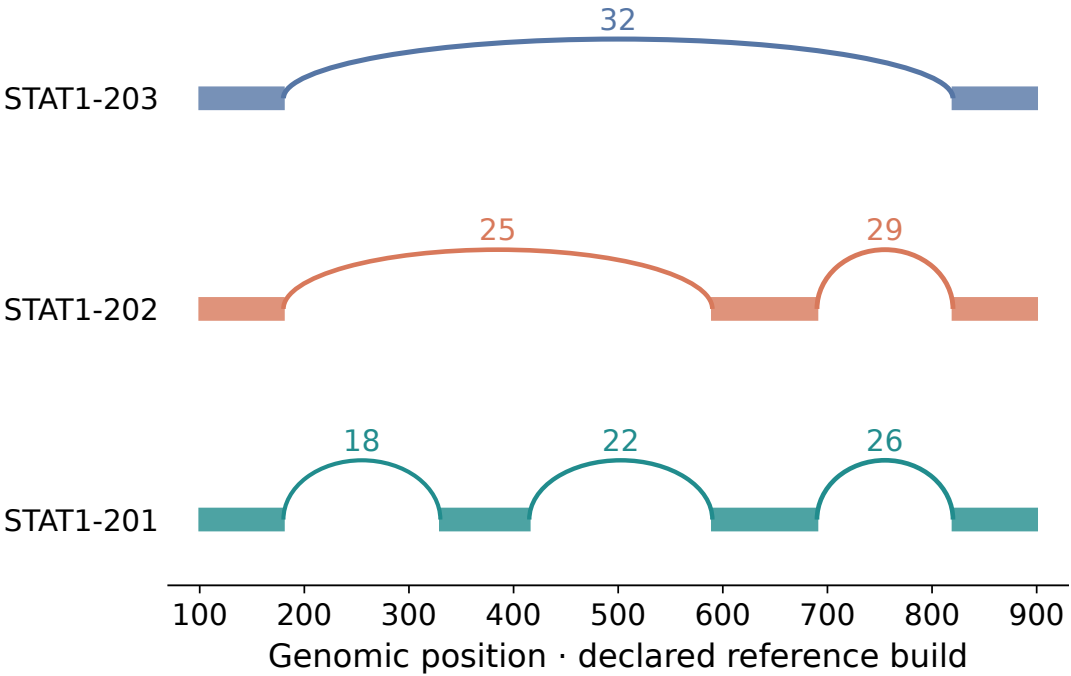
Estimate [interval] · q
Independent n = 12

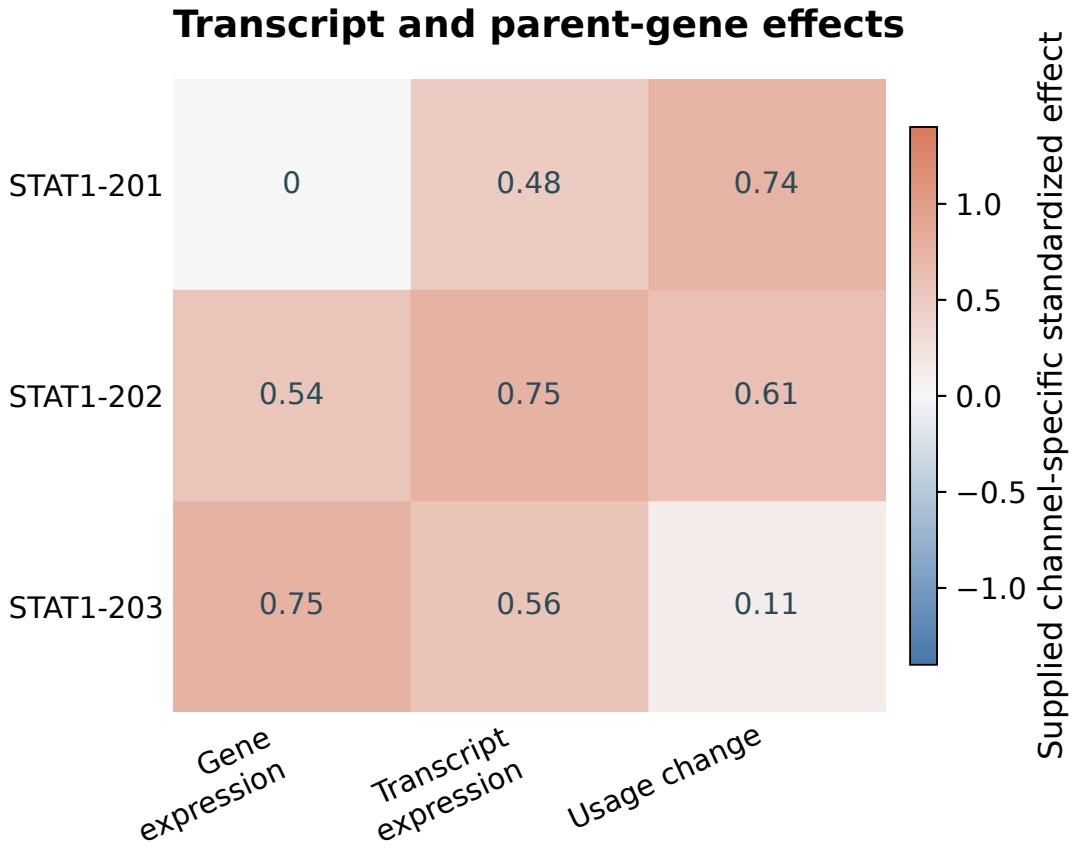
-0.10 [-0.18, -0.02] 0.002

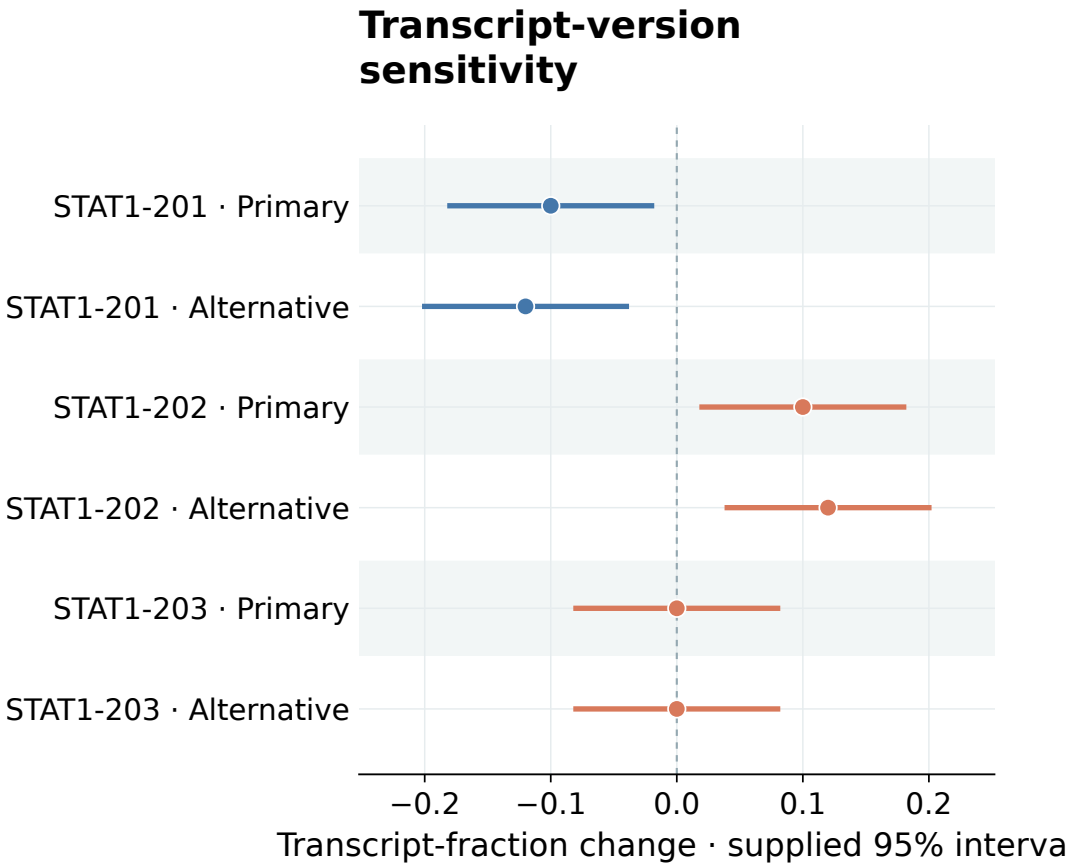
0.10 [0.02, 0.18] 0.02

0.00 [-0.08, 0.08] 0.65

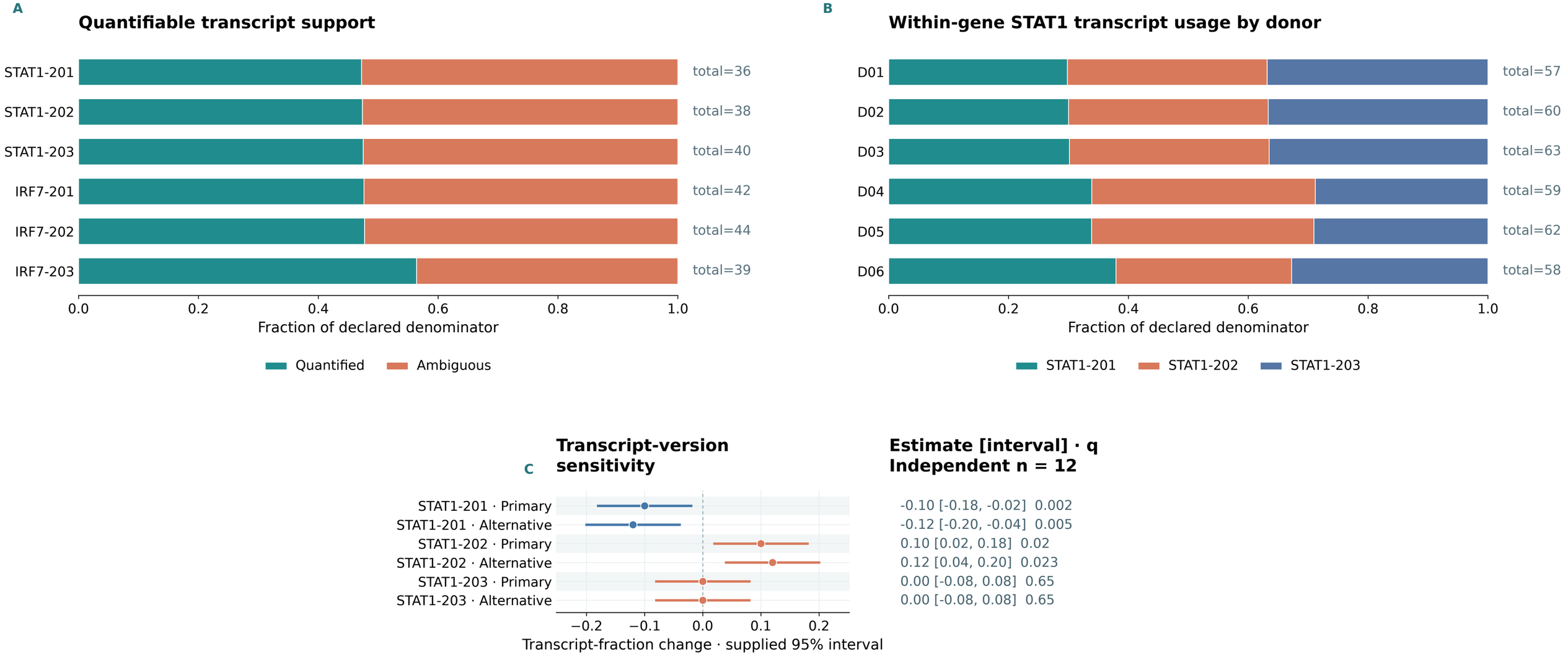
STAT1 transcript structure and junction support




Transcript and parent-gene effects
STAT1-201
STAT1-202
STAT1-203
0
0.48
0.74
0.54
0.75
0.61
0.75
0.56
0.11
Gene expression
Transcript expression
Usage change
1.0
0.5
0.0
−0.5
−1.0
Supplied channel-specific standardized effect


Transcript-version sensitivity
Estimate [interval] · q
Independent n = 12
STAT1-201 · Primary
STAT1-201 · Alternative
STAT1-202 · Primary
STAT1-202 · Alternative
STAT1-203 · Primary
STAT1-203 · Alternative
-0.10 [-0.18, -0.02] 0.002
-0.12 [-0.20, -0.04] 0.005
0.10 [0.02, 0.18] 0.02
0.12 [0.04, 0.20] 0.023
0.00 [-0.08, 0.08] 0.65
0.00 [-0.08, 0.08] 0.65
−0.2
−0.1
0.0
0.1
0.2
Transcript-fraction change · supplied 95% interval

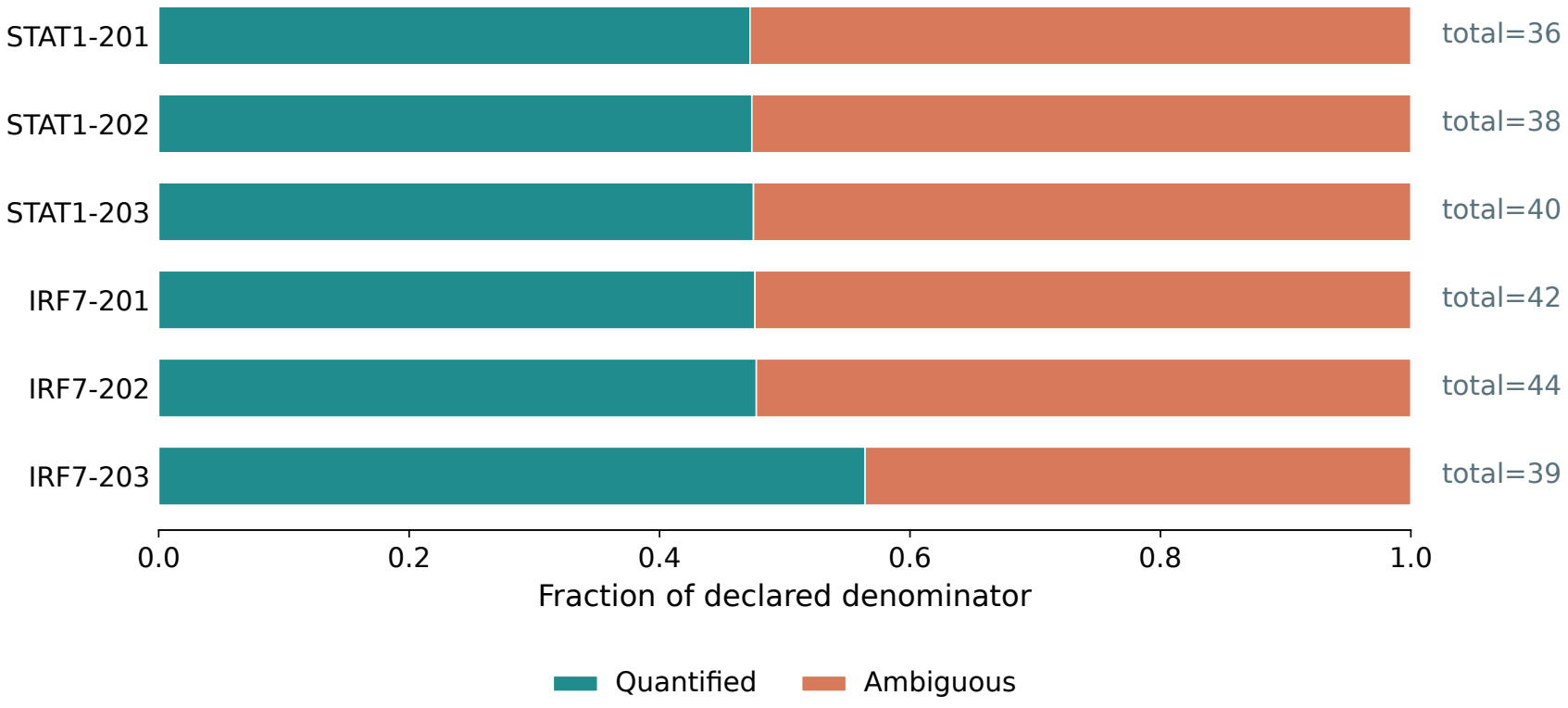
Bulk · dtu — complementary evidence



Bulk · dtu — complementary evidence

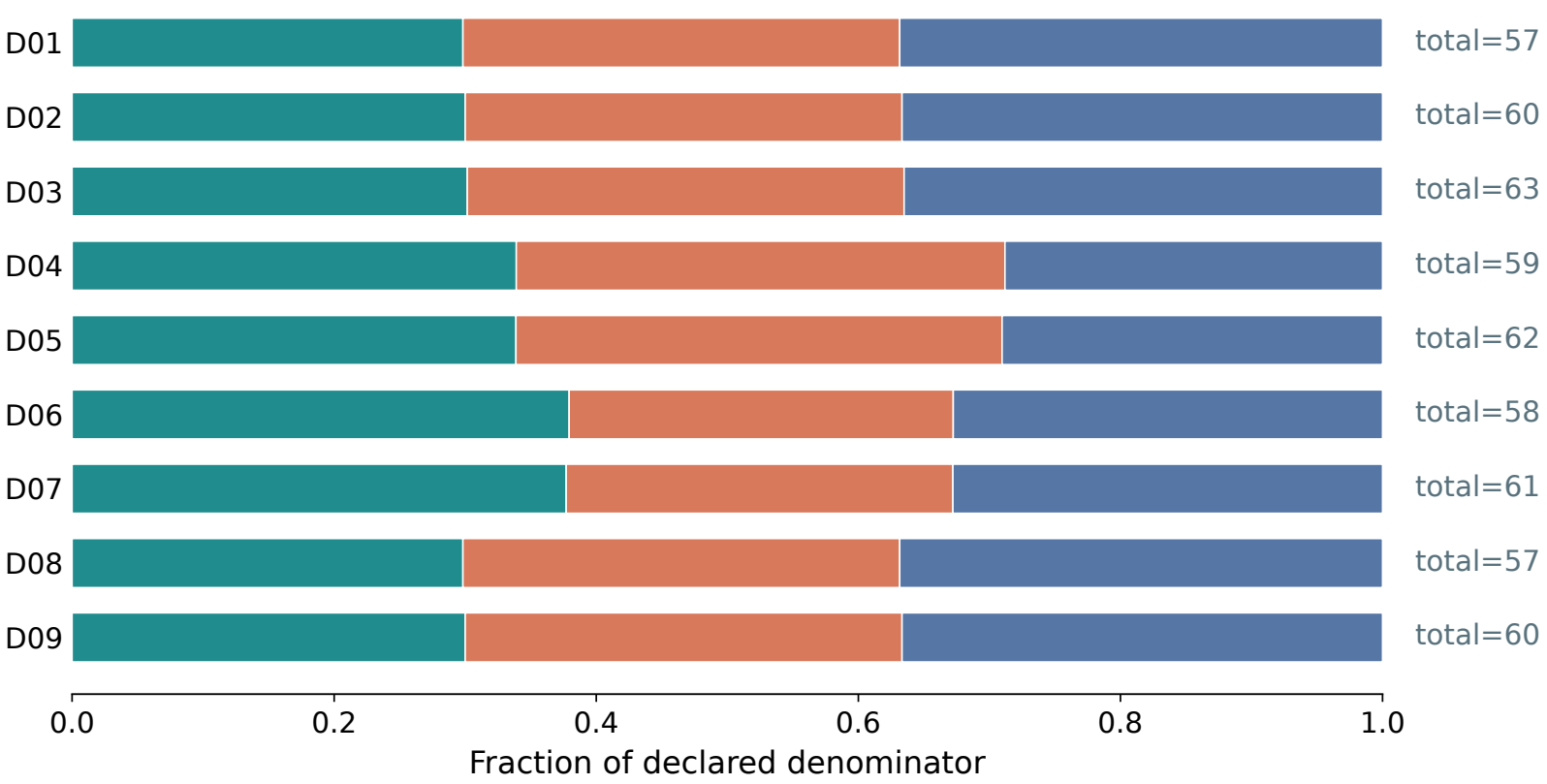
A

Quantifiable transcript support



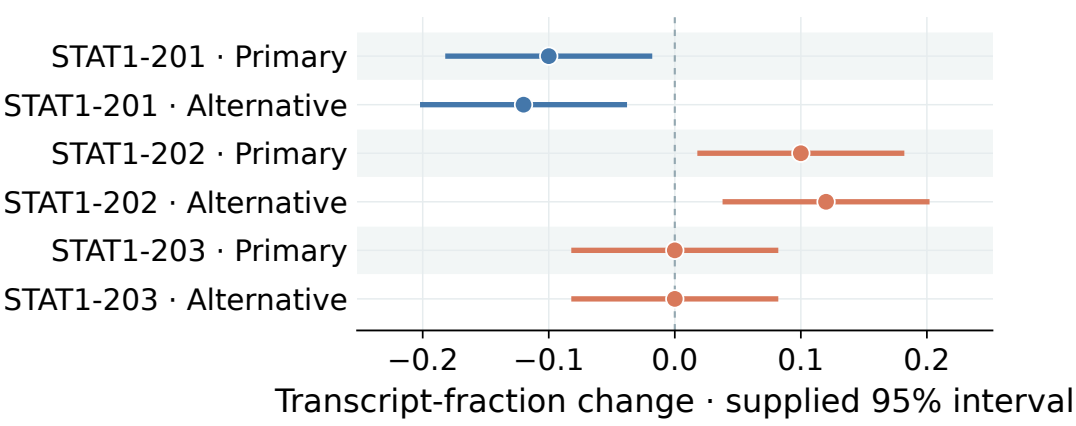
B

Within-gene STAT1 transcript usage by donor



C

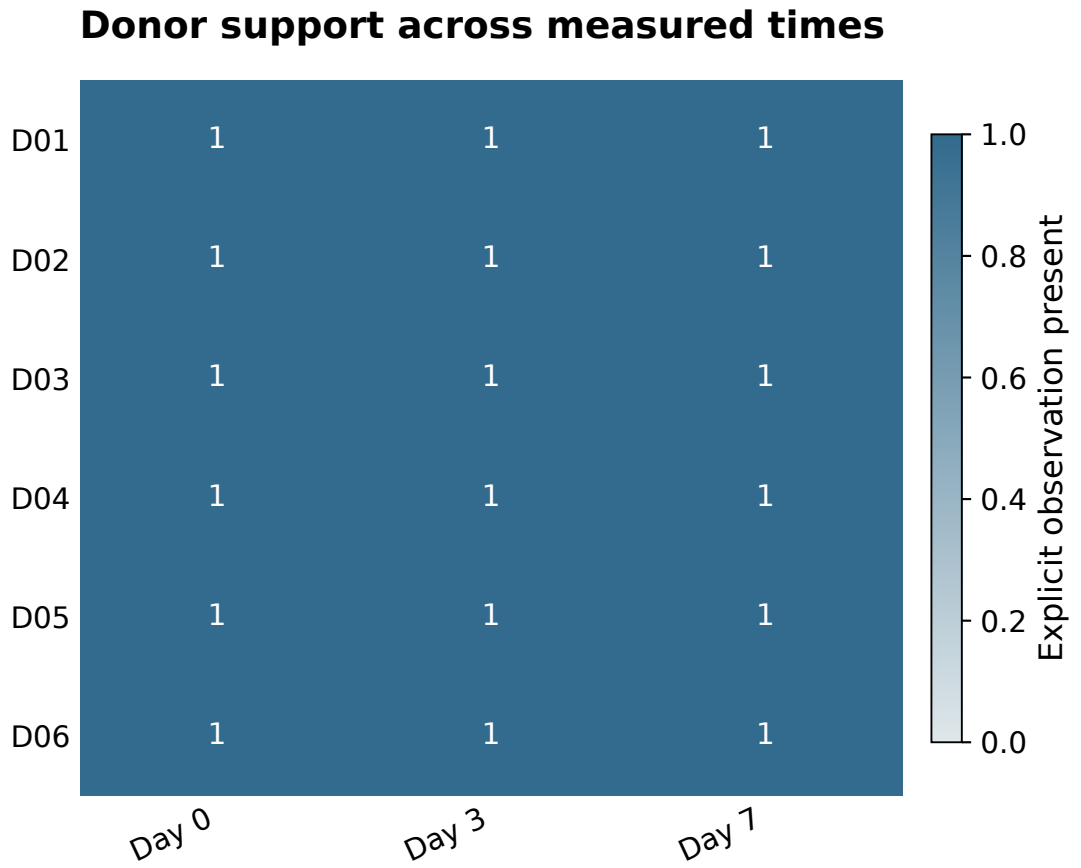
Transcript-version sensitivity

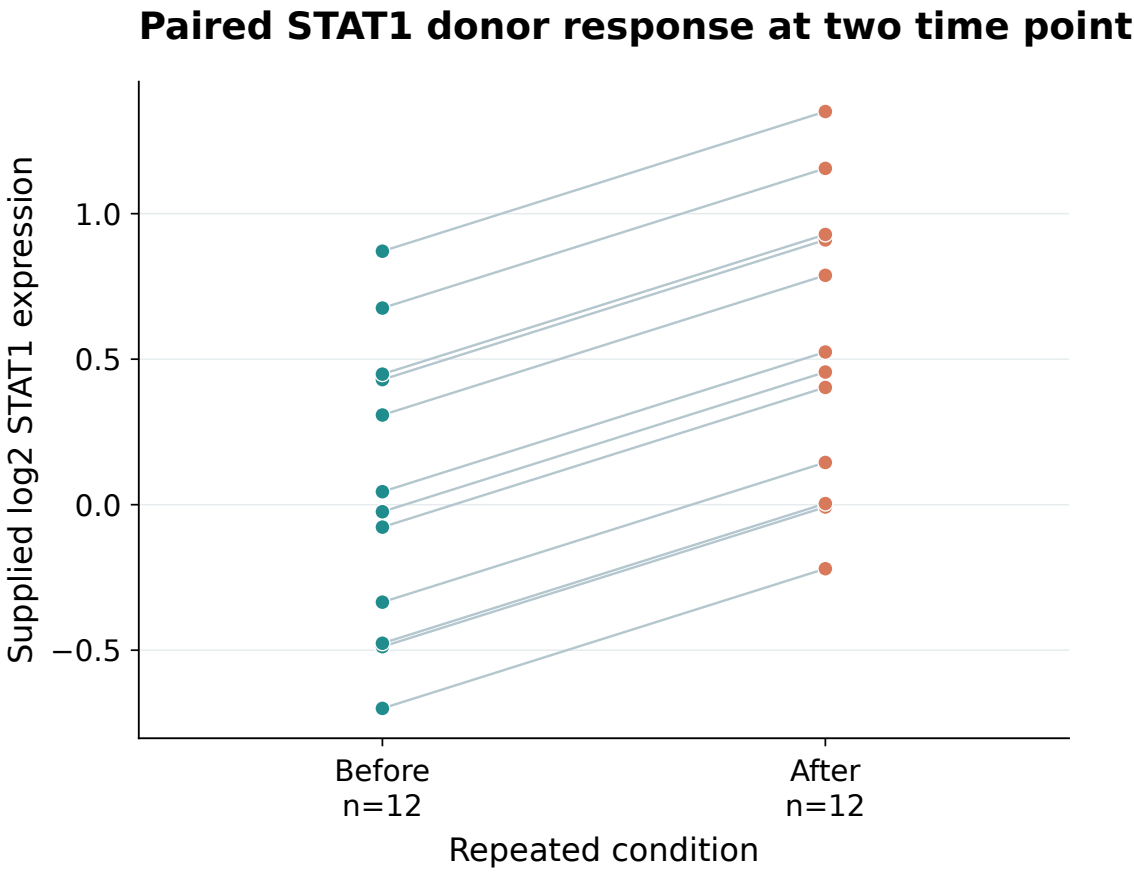


Estimate [interval] · q
Independent n = 24

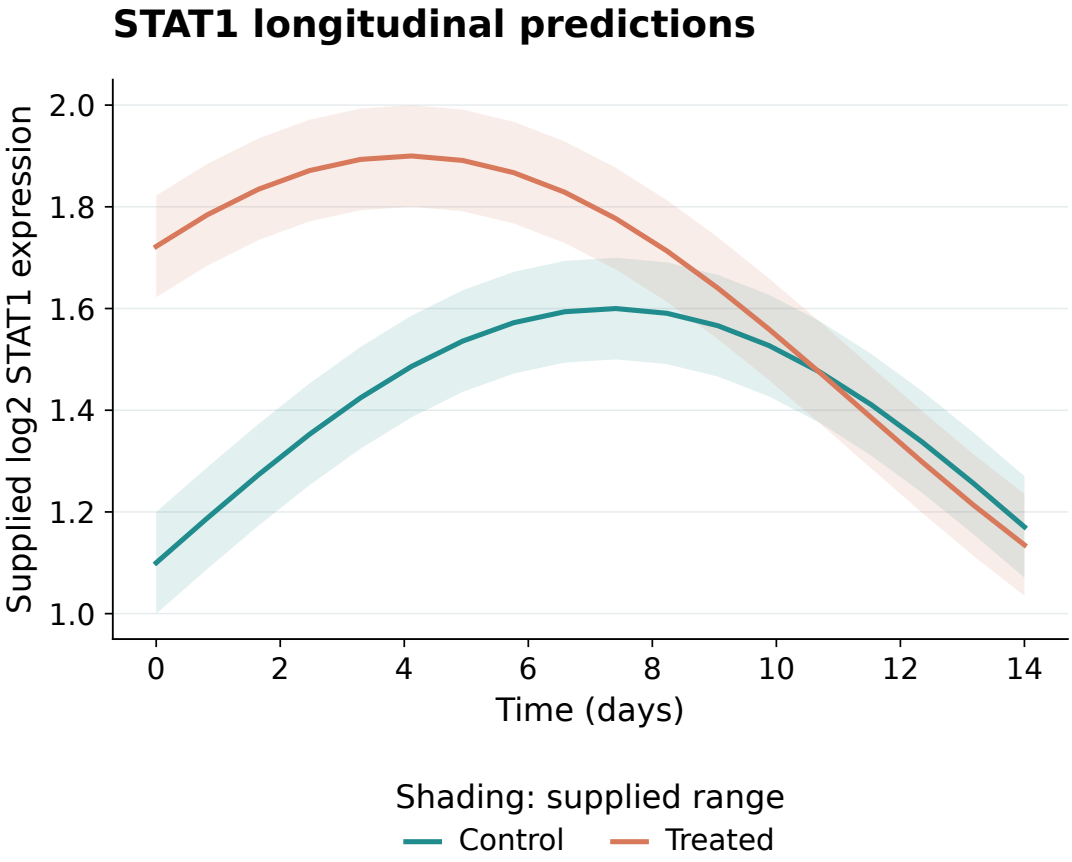
-0.10 [-0.18, -0.02] 0.002
-0.12 [-0.20, -0.04] 0.005
0.10 [0.02, 0.18] 0.02
0.12 [0.04, 0.20] 0.023
0.00 [-0.08, 0.08] 0.65
0.00 [-0.08, 0.08] 0.65

Software test fixture; not biological evidence


Donor support across measured times
D01
D02
D03
D04
D05
D06
1
Day 0
Day 3
Day 7
Explicit observation present
1.0
0.8
0.6
0.4
0.2
0.0

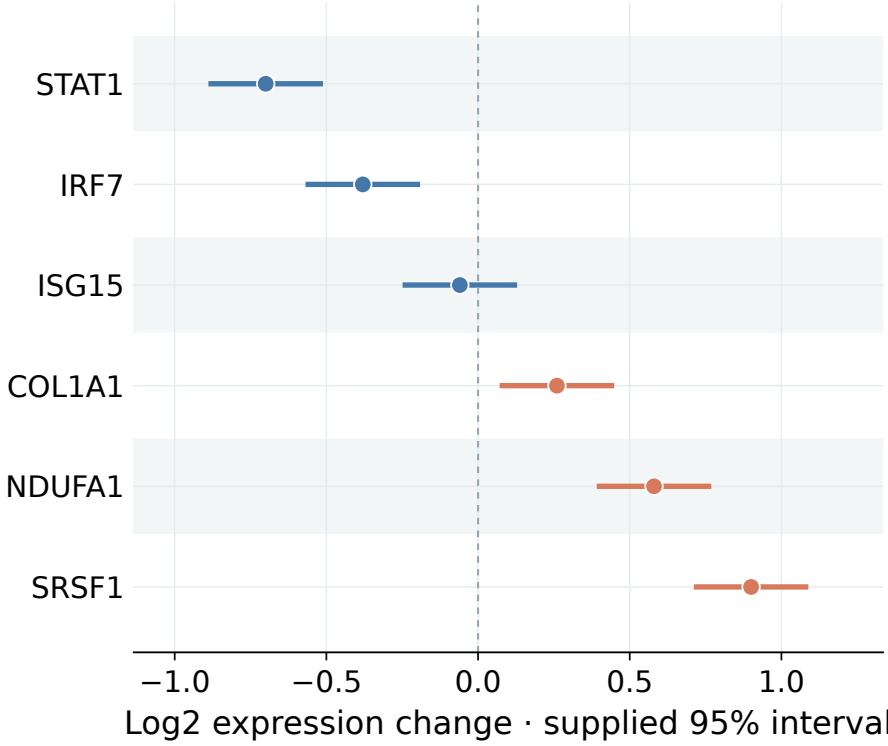


Software test fixture; not biological evidence



Software test fixture; not biological evidence

Prespecified time-by-condition effects



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

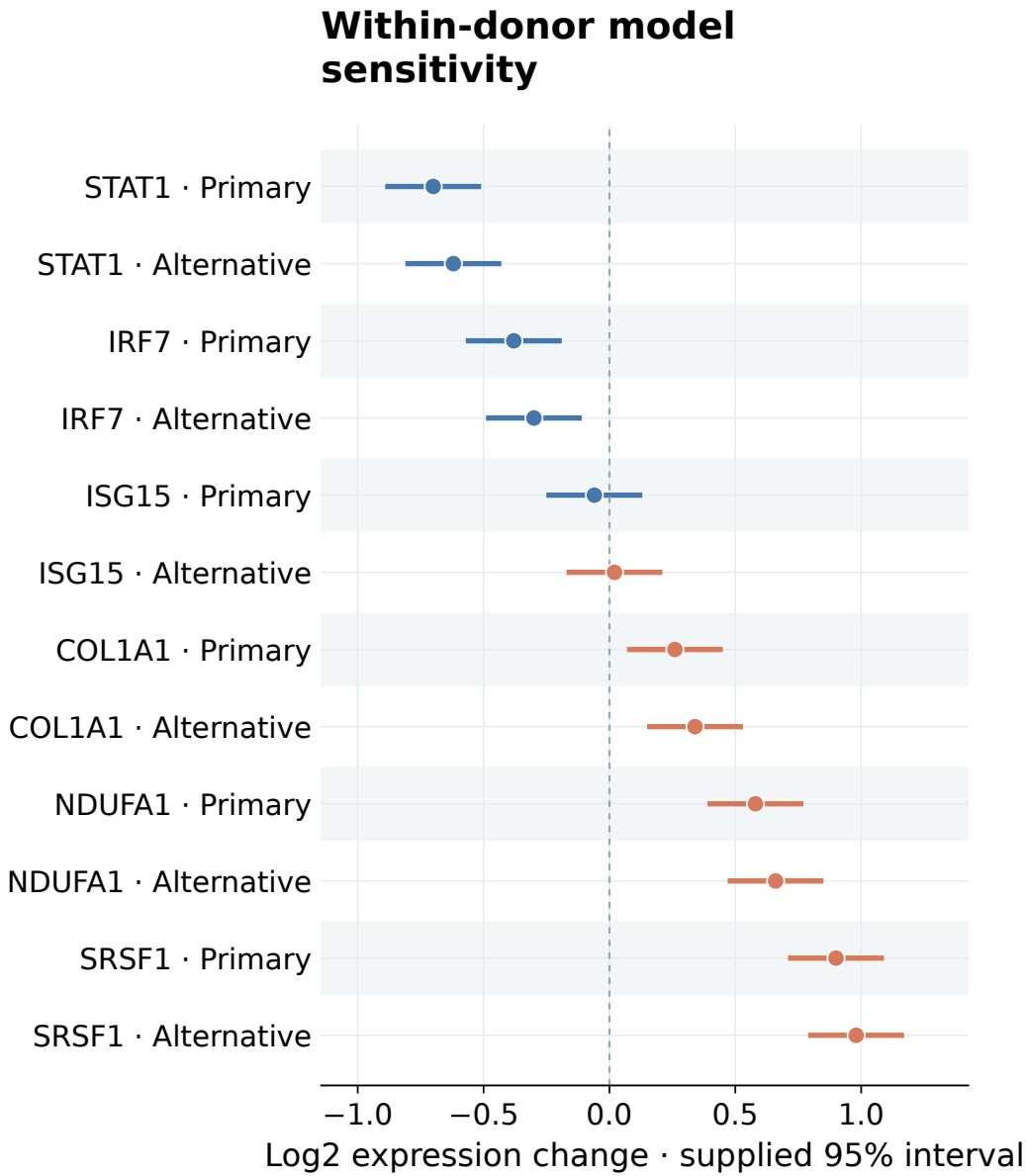
-0.38 [-0.56, -0.20] 0.02

-0.06 [-0.24, 0.12] 0.65

0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

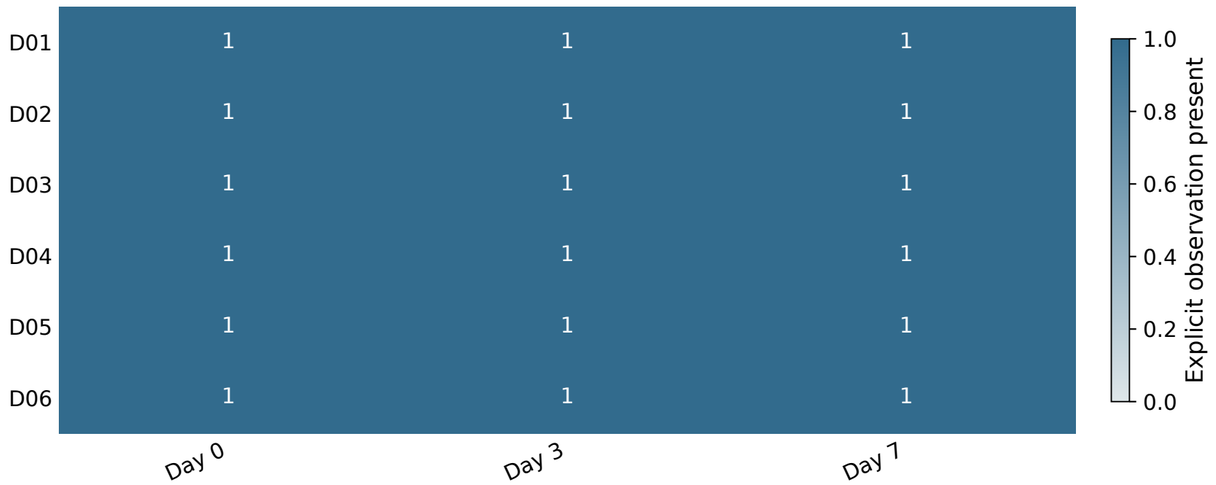
0.90 [0.72, 1.08] 0.092


Within-donor model sensitivity
Estimate [interval] · q
Independent n = 12
STAT1 · Primary
STAT1 · Alternative
IRF7 · Primary
IRF7 · Alternative
ISG15 · Primary
ISG15 · Alternative
COL1A1 · Primary
COL1A1 · Alternative
NDUFA1 · Primary
NDUFA1 · Alternative
SRSF1 · Primary
SRSF1 · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Log2 expression change · supplied 95% interval

Bulk · timecourse — complementary evidence

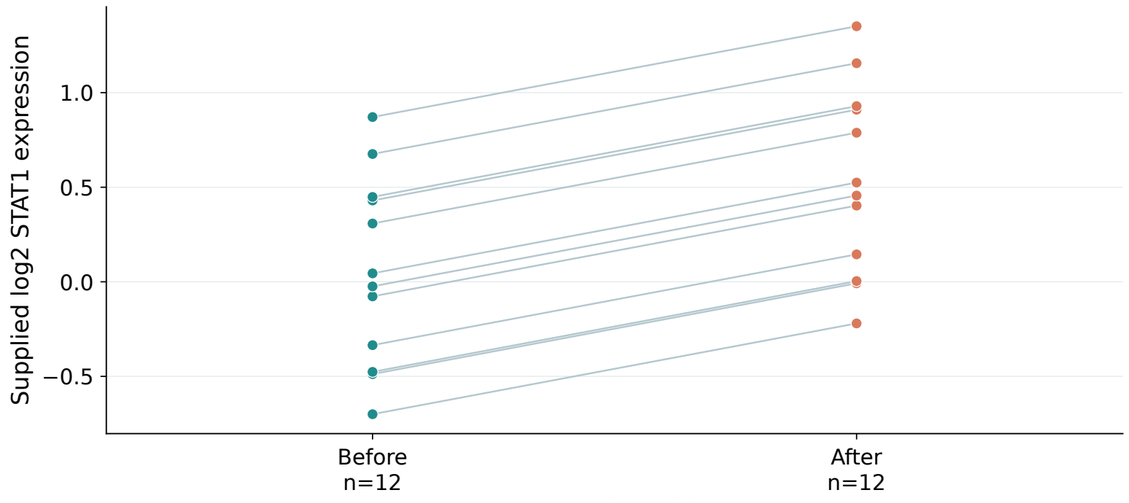
A

Donor support across measured times



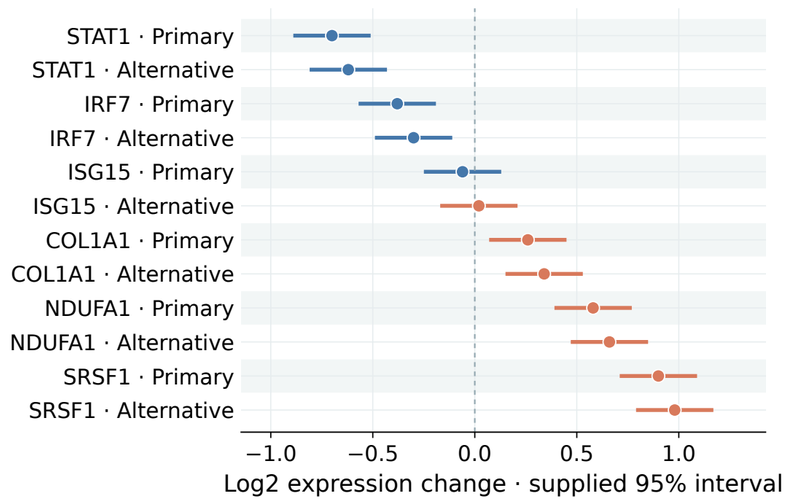
B

Paired STAT1 donor response at two time points



C

Within-donor model sensitivity



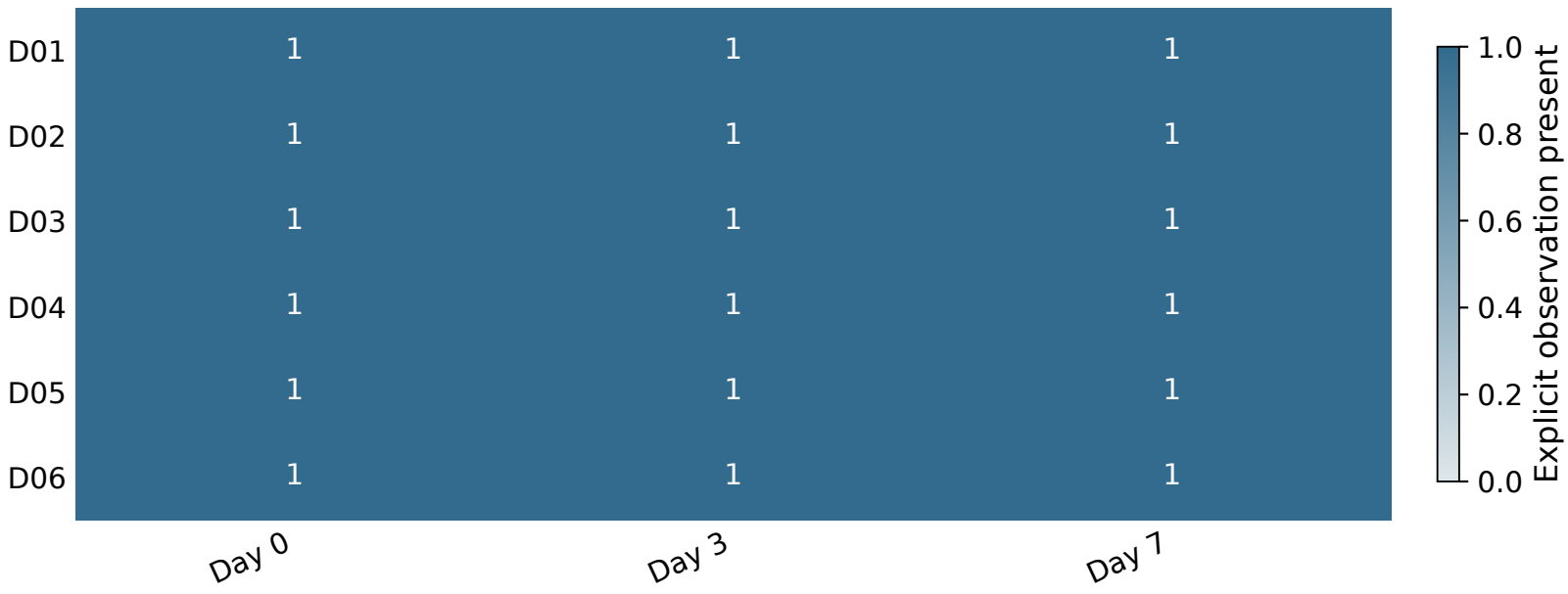
Estimate [interval] · q
Independent n = 12

STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.38 [-0.56, -0.20] 0.02
IRF7 · Alternative	-0.30 [-0.48, -0.12] 0.023
ISG15 · Primary	-0.06 [-0.24, 0.12] 0.65
ISG15 · Alternative	0.02 [-0.16, 0.20] 0.65
COL1A1 · Primary	0.26 [0.08, 0.44] 0.056
COL1A1 · Alternative	0.34 [0.16, 0.52] 0.059
NDUFA1 · Primary	0.58 [0.40, 0.76] 0.074
NDUFA1 · Alternative	0.66 [0.48, 0.84] 0.077
SRSF1 · Primary	0.90 [0.72, 1.08] 0.092
SRSF1 · Alternative	0.98 [0.80, 1.16] 0.095

Bulk · timecourse — complementary evidence

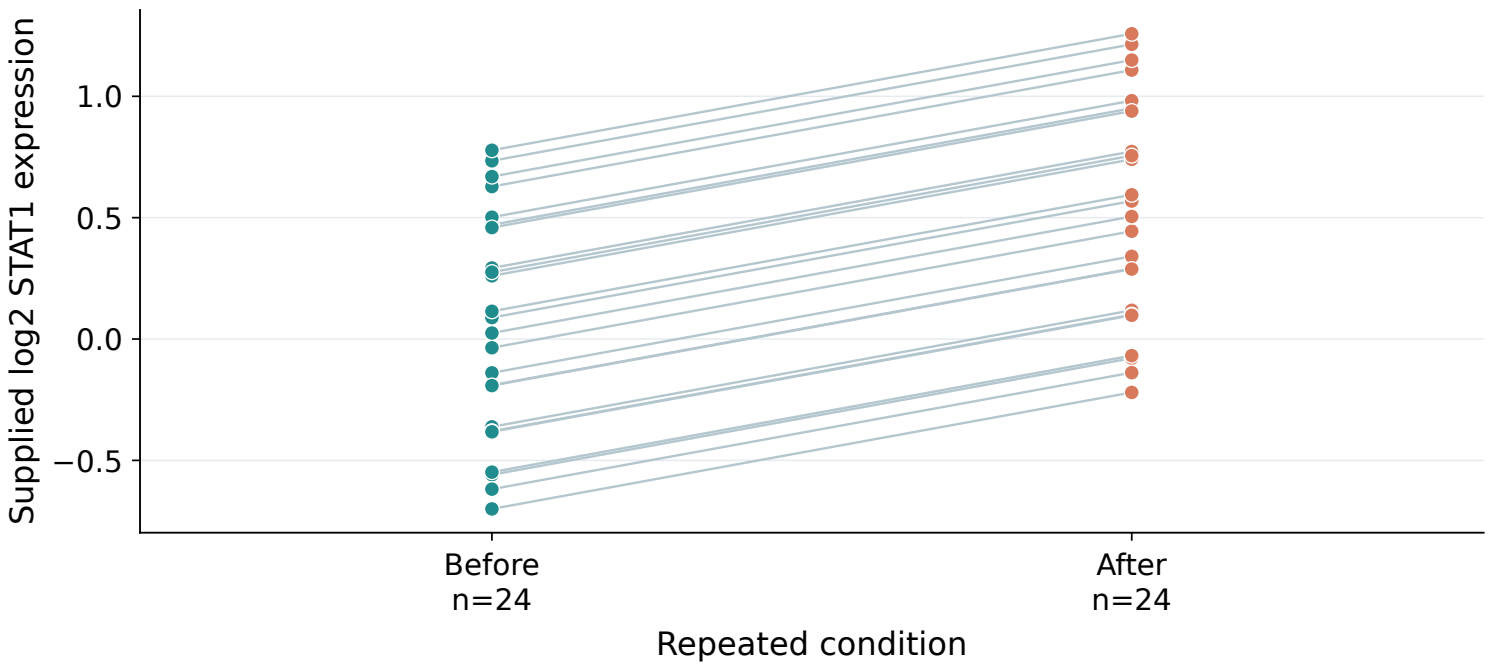
A

Donor support across measured times



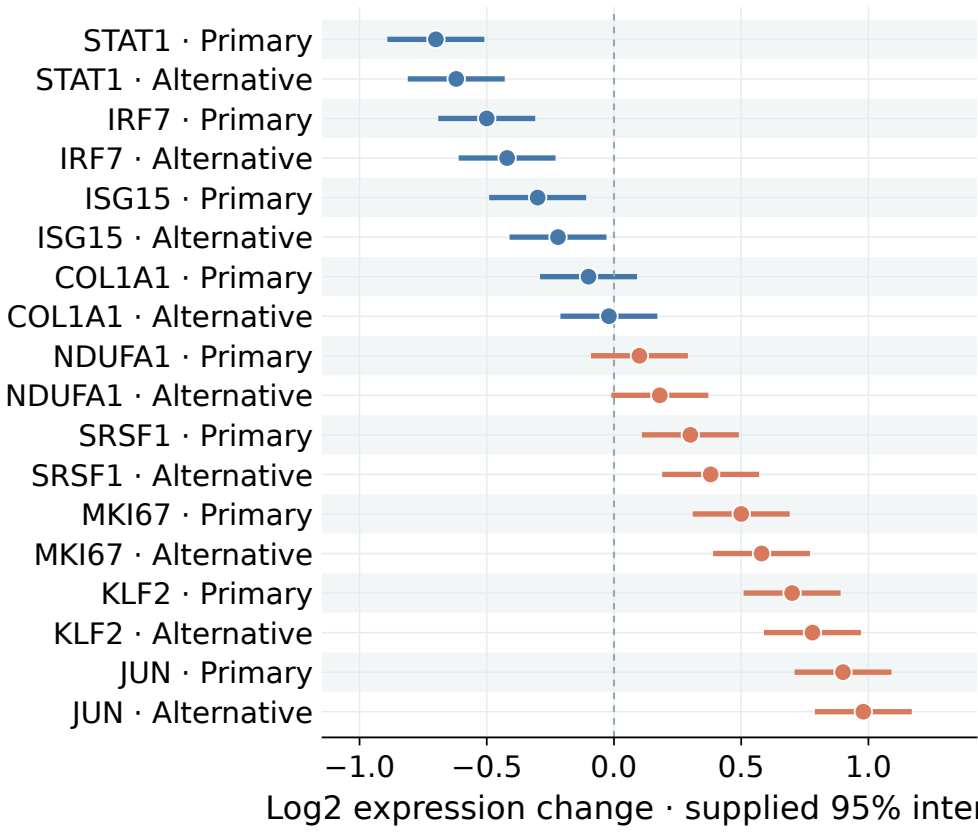
B

Paired STAT1 donor response at two time points



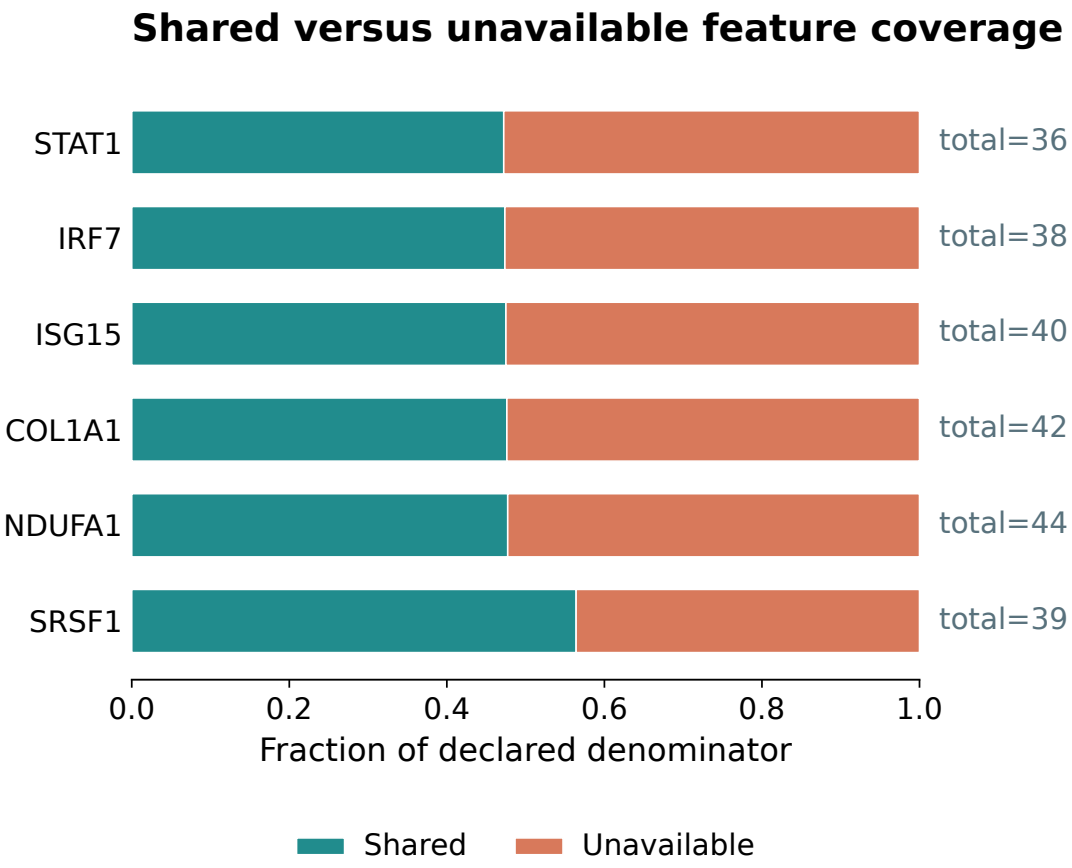
C

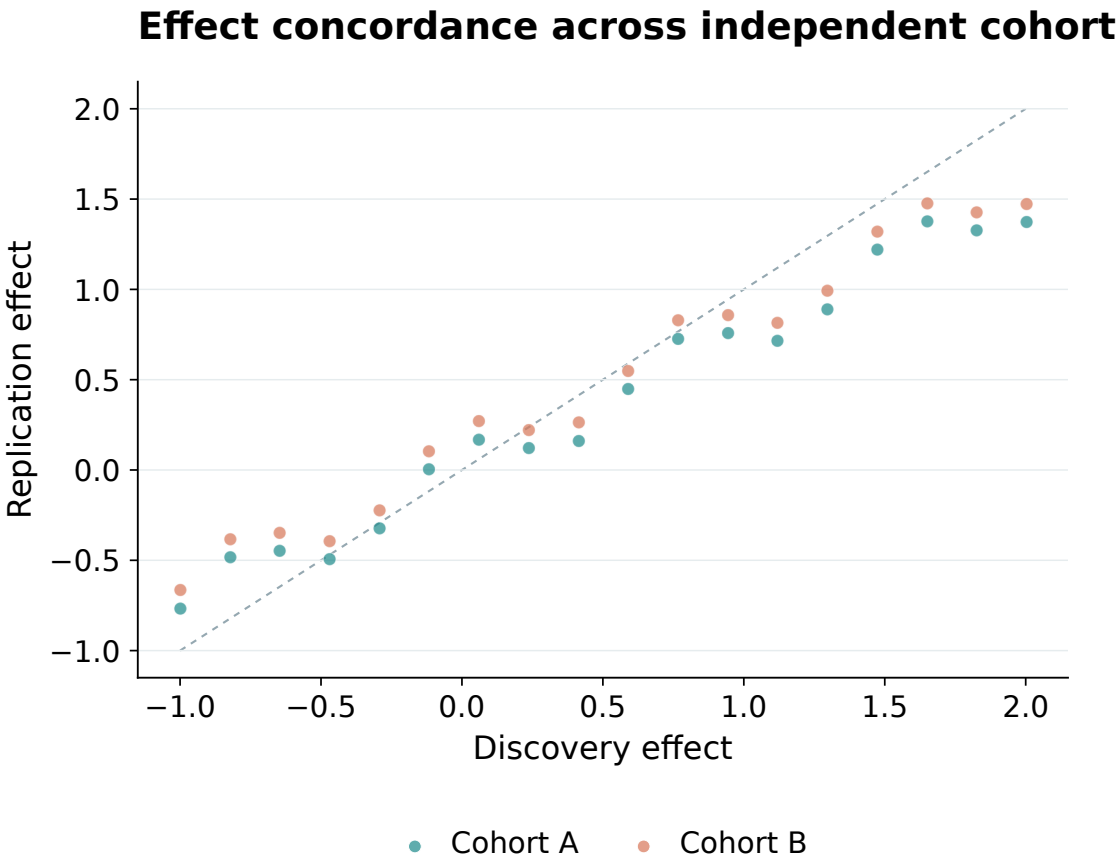
Within-donor model sensitivity



Estimate [interval] · q
Independent n = 24

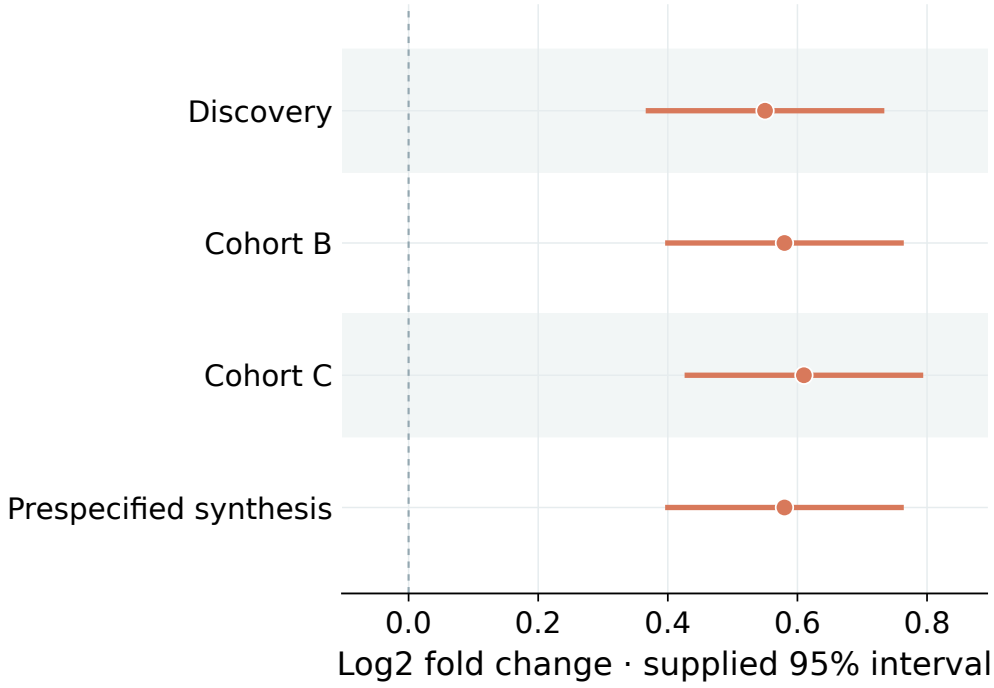
STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.50 [-0.68, -0.32] 0.02
IRF7 · Alternative	-0.42 [-0.60, -0.24] 0.023
ISG15 · Primary	-0.30 [-0.48, -0.12] 0.038
ISG15 · Alternative	-0.22 [-0.40, -0.04] 0.041
COL1A1 · Primary	-0.10 [-0.28, 0.08] 0.65
COL1A1 · Alternative	-0.02 [-0.20, 0.16] 0.65
NDUFA1 · Primary	0.10 [-0.08, 0.28] 0.65
NDUFA1 · Alternative	0.18 [0.00, 0.36] 0.077
SRSF1 · Primary	0.30 [0.12, 0.48] 0.092
SRSF1 · Alternative	0.38 [0.20, 0.56] 0.095
MKI67 · Primary	0.50 [0.32, 0.68] 0.11
MKI67 · Alternative	0.58 [0.40, 0.76] 0.11
KLF2 · Primary	0.70 [0.52, 0.88] 0.13
KLF2 · Alternative	0.78 [0.60, 0.96] 0.13
JUN · Primary	0.90 [0.72, 1.08] 0.15
JUN · Alternative	0.98 [0.80, 1.16] 0.15


Shared versus unavailable feature coverage
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Shared
Unavailable



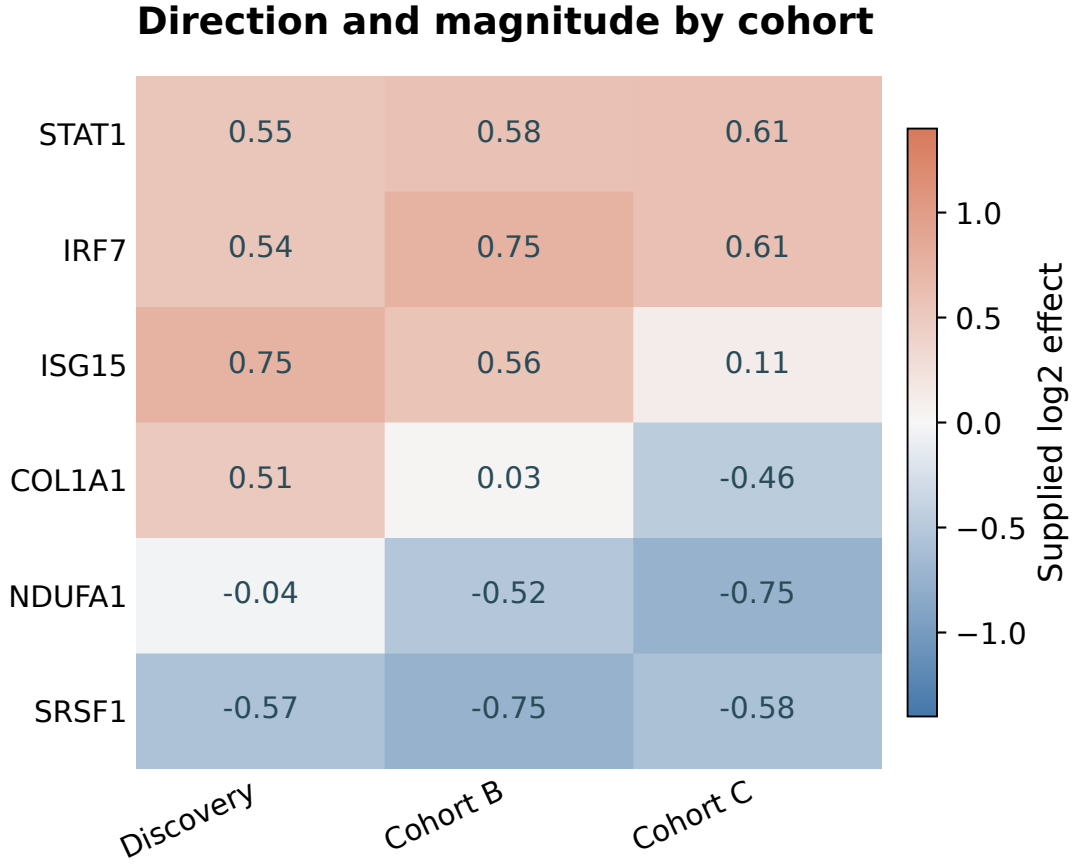
Software test fixture; not biological evidence

STAT1 independent-cohort effects and uncertainty

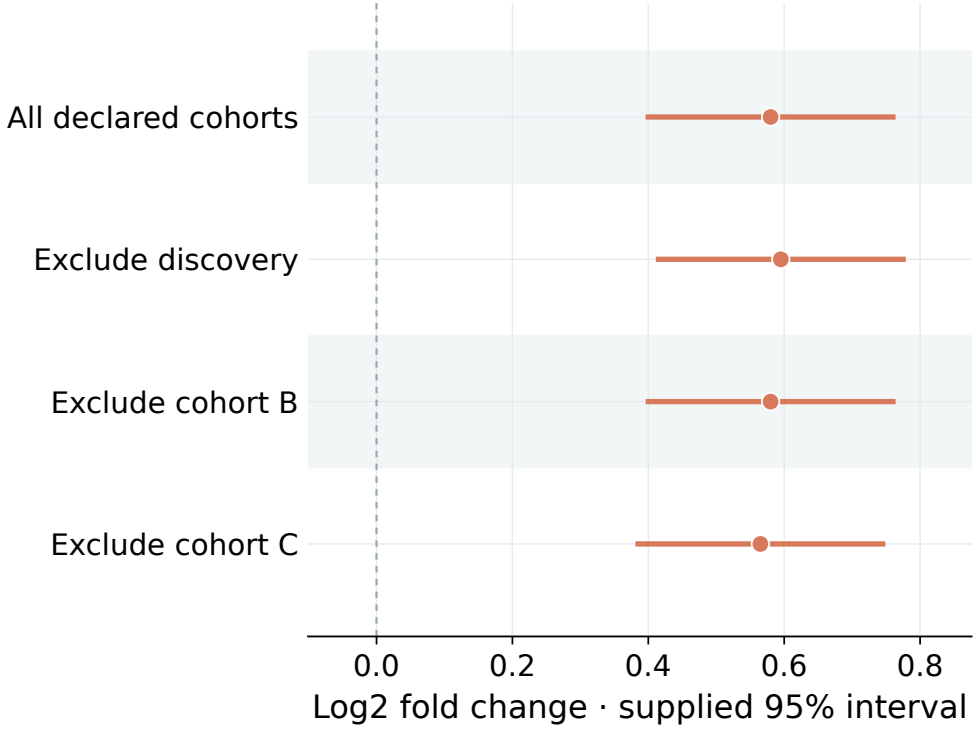


Estimate [interval] · q

Discovery	0.55 [0.37, 0.73] 0.002 n=4
Cohort B	0.58 [0.40, 0.76] 0.02 n=4
Cohort C	0.61 [0.43, 0.79] 0.038 n=4
Prespecified synthesis	0.58 [0.40, 0.76] 0.056 n=12


Direction and magnitude by cohort
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0.55
0.58
0.61
0.54
0.75
0.61
0.75
0.56
0.11
0.51
0.03
-0.46
-0.04
-0.52
-0.75
-0.57
-0.75
-0.58
Discovery
Cohort B
Cohort C
1.0
0.5
0.0
−0.5
−1.0
Supplied log2 effect

Leave-one-cohort sensitivity



Estimate [interval] · q

0.58 [0.40, 0.76] 0.002 n=12

0.59 [0.41, 0.77] 0.02 n=8

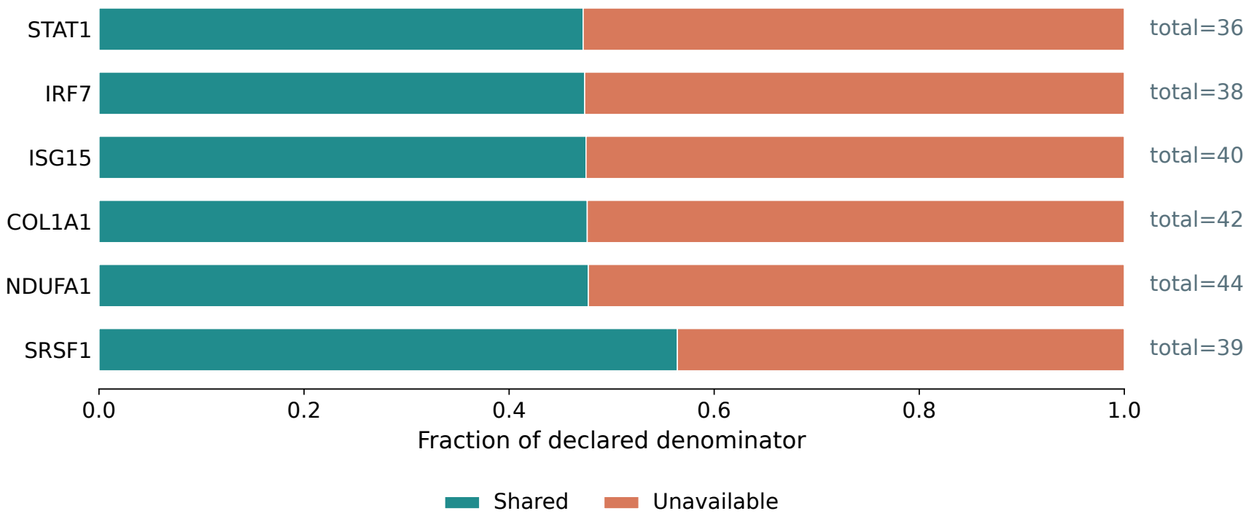
0.58 [0.40, 0.76] 0.038 n=8

0.56 [0.38, 0.74] 0.056 n=8

Bulk · replication — complementary evidence

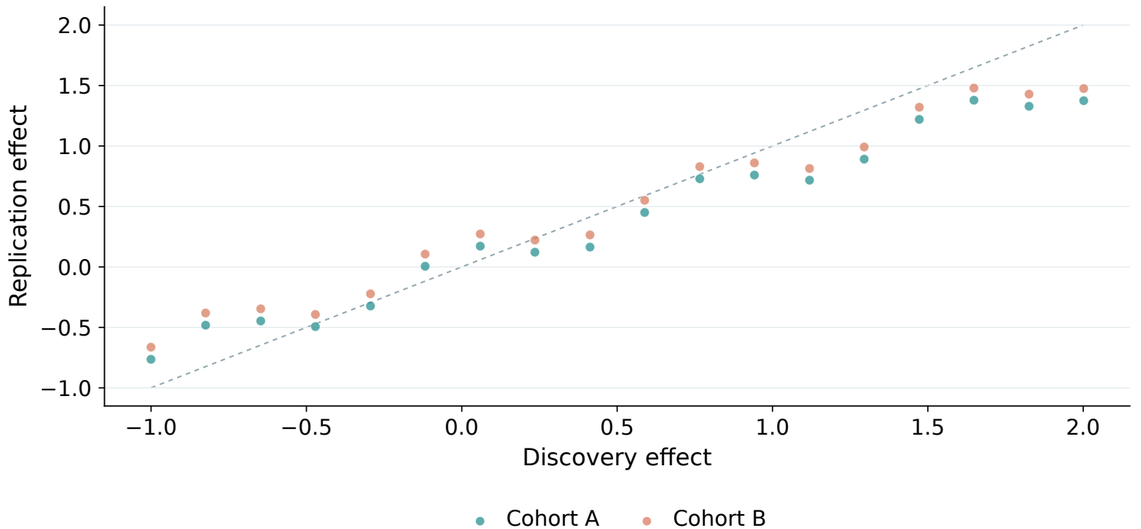
A

Shared versus unavailable feature coverage



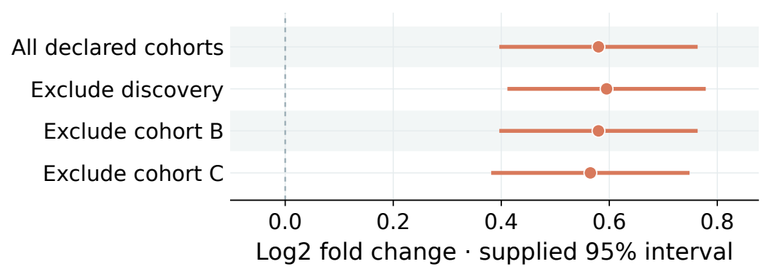
B

Effect concordance across independent cohorts



C

Leave-one-cohort sensitivity



Estimate [interval] · q

0.58 [0.40, 0.76] 0.002 n=12

0.59 [0.41, 0.77] 0.02 n=8

0.58 [0.40, 0.76] 0.038 n=8

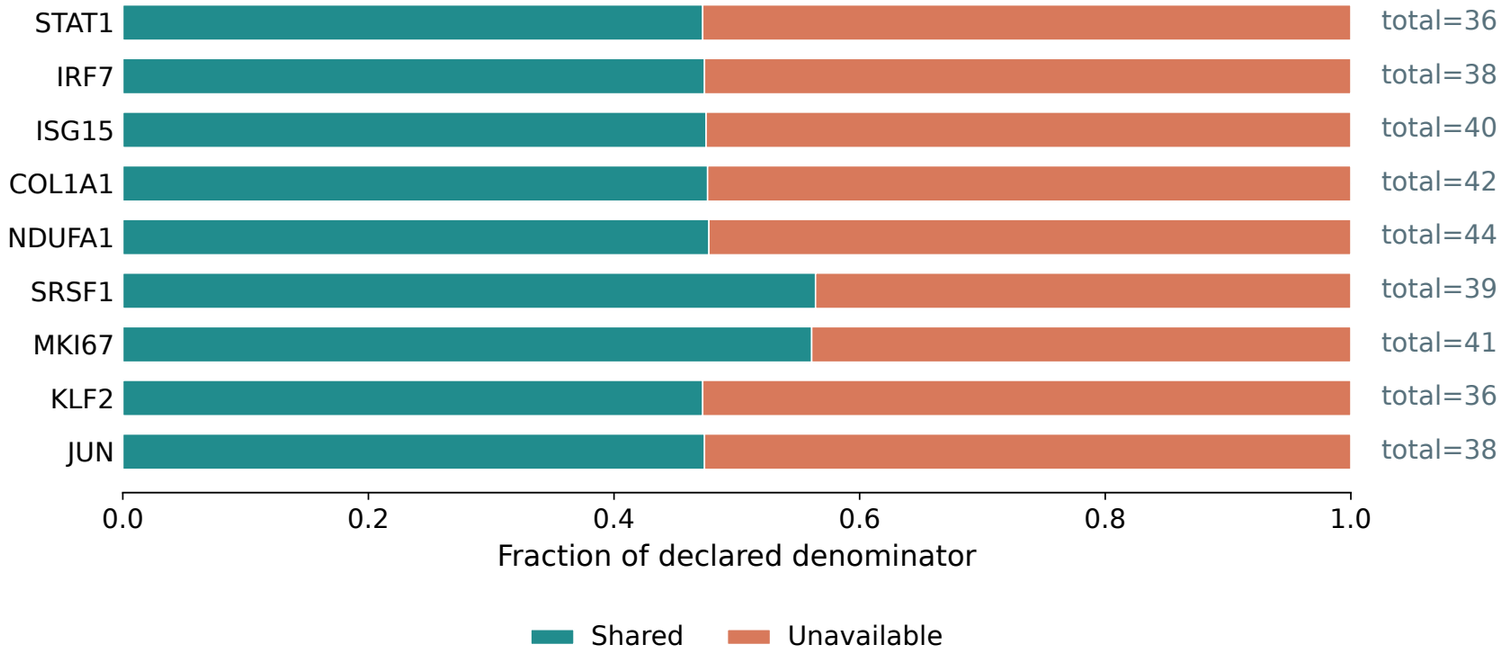
0.56 [0.38, 0.74] 0.056 n=8

Software test fixture; not biological evidence

Bulk · replication — complementary evidence

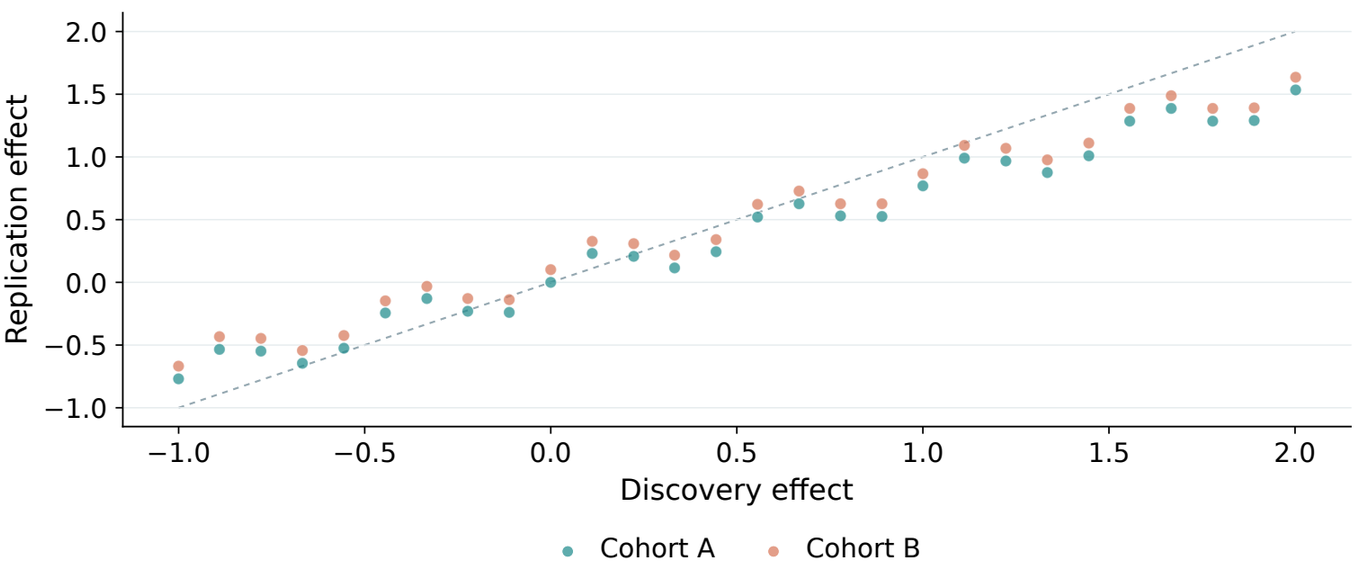
A

Shared versus unavailable feature coverage



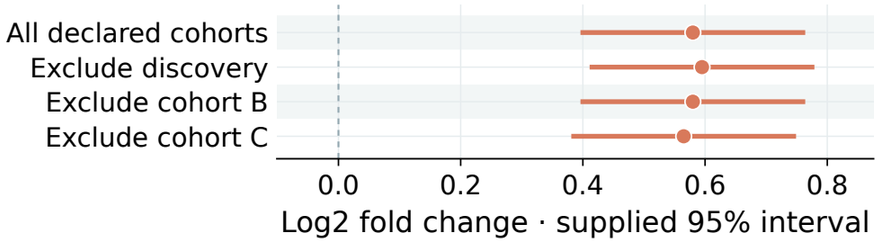
B

Effect concordance across independent cohorts



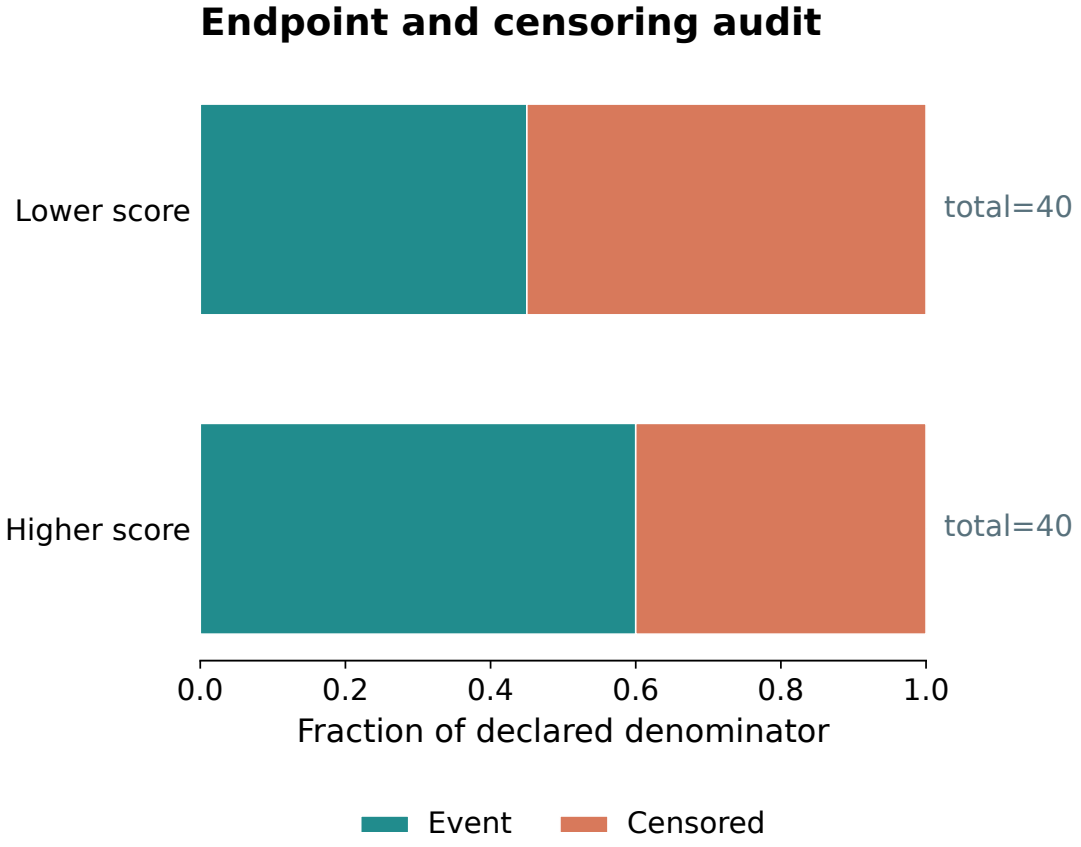
C

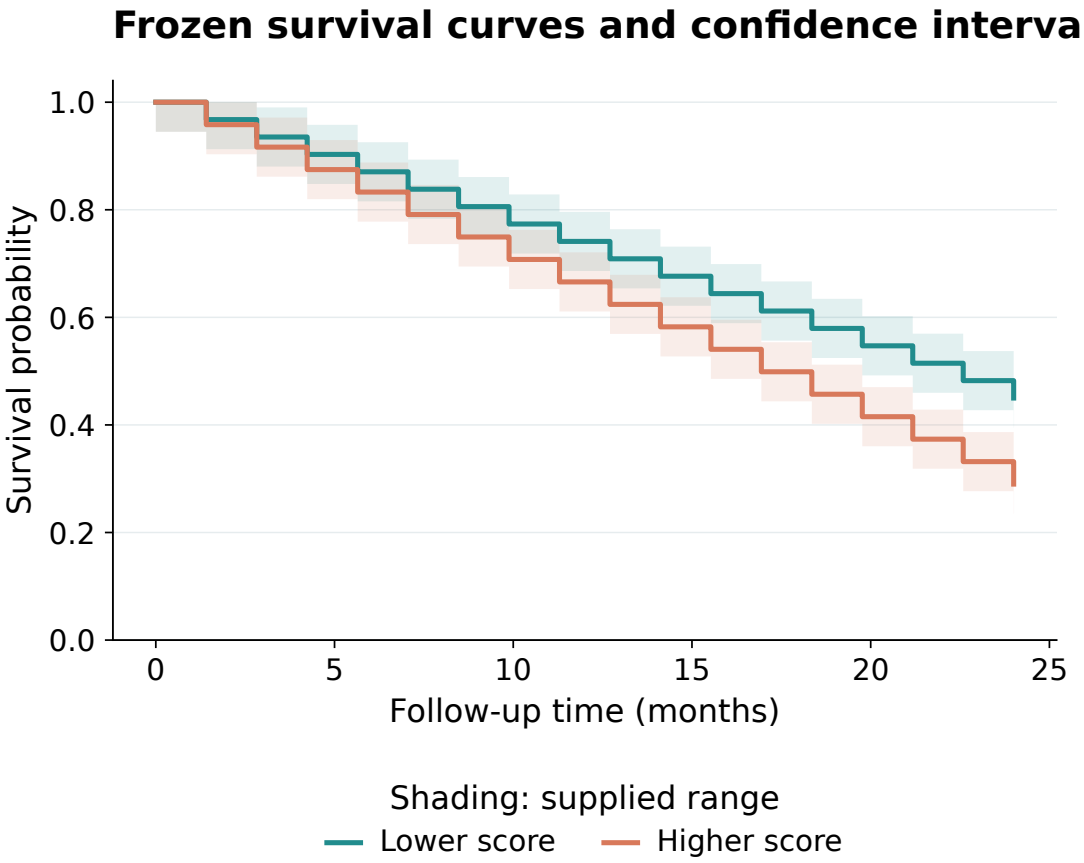
Leave-one-cohort sensitivity

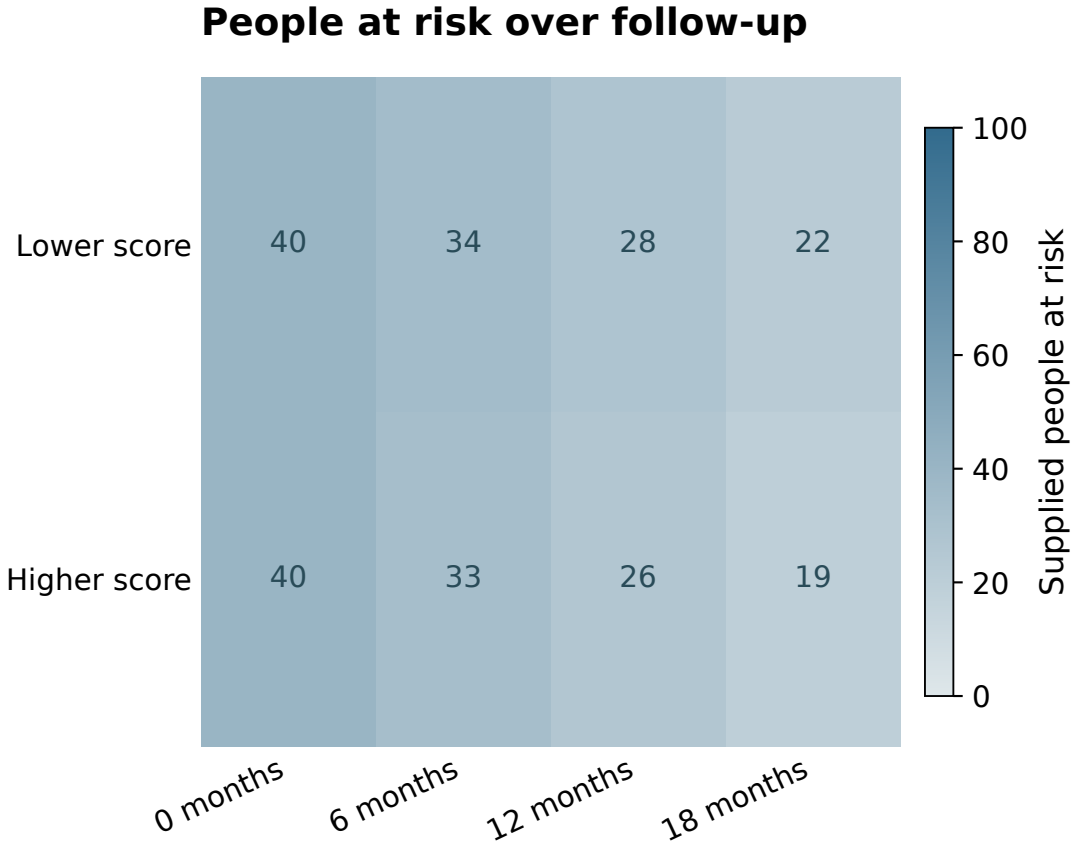


Estimate [interval] · q

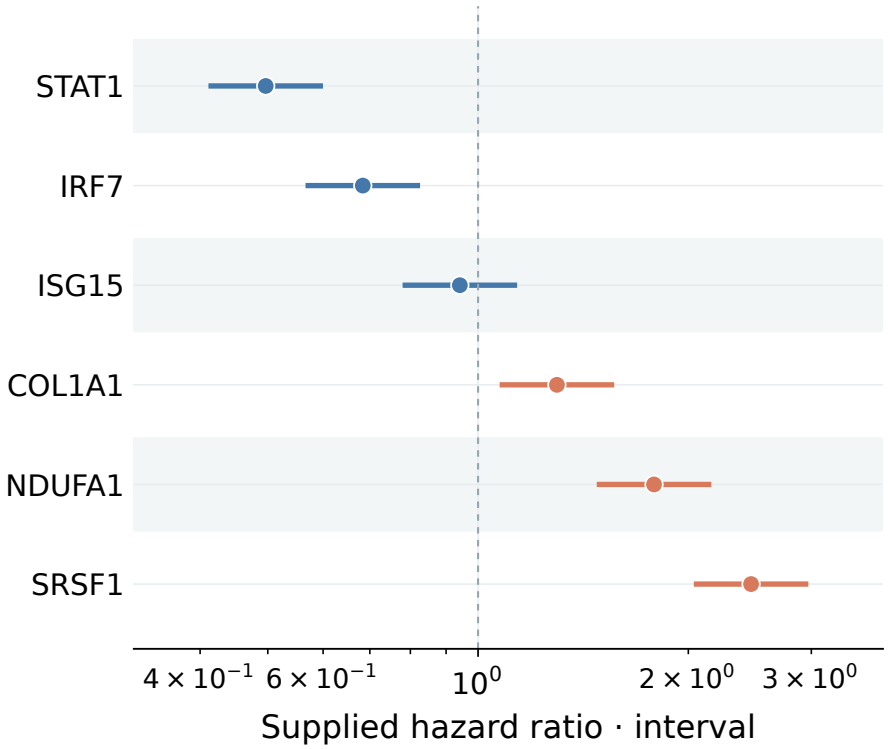
0.58 [0.40, 0.76] 0.002 n=24
0.59 [0.41, 0.77] 0.02 n=16
0.58 [0.40, 0.76] 0.038 n=16
0.56 [0.38, 0.74] 0.056 n=16


Endpoint and censoring audit
Lower score
total=40
Higher score
total=40
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Event
Censored


Frozen survival curves and confidence intervals
Survival probability
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
Follow-up time (months)
Shading: supplied range
Lower score
Higher score


People at risk over follow-up
Lower score
Higher score
40
34
28
22
40
33
26
19
0 months
6 months
12 months
18 months
Supplied people at risk
100
80
60
40
20
0

Adjusted outcome associations



Estimate [interval] · q
Independent n = 80

0.50 [0.41, 0.59] 0.002

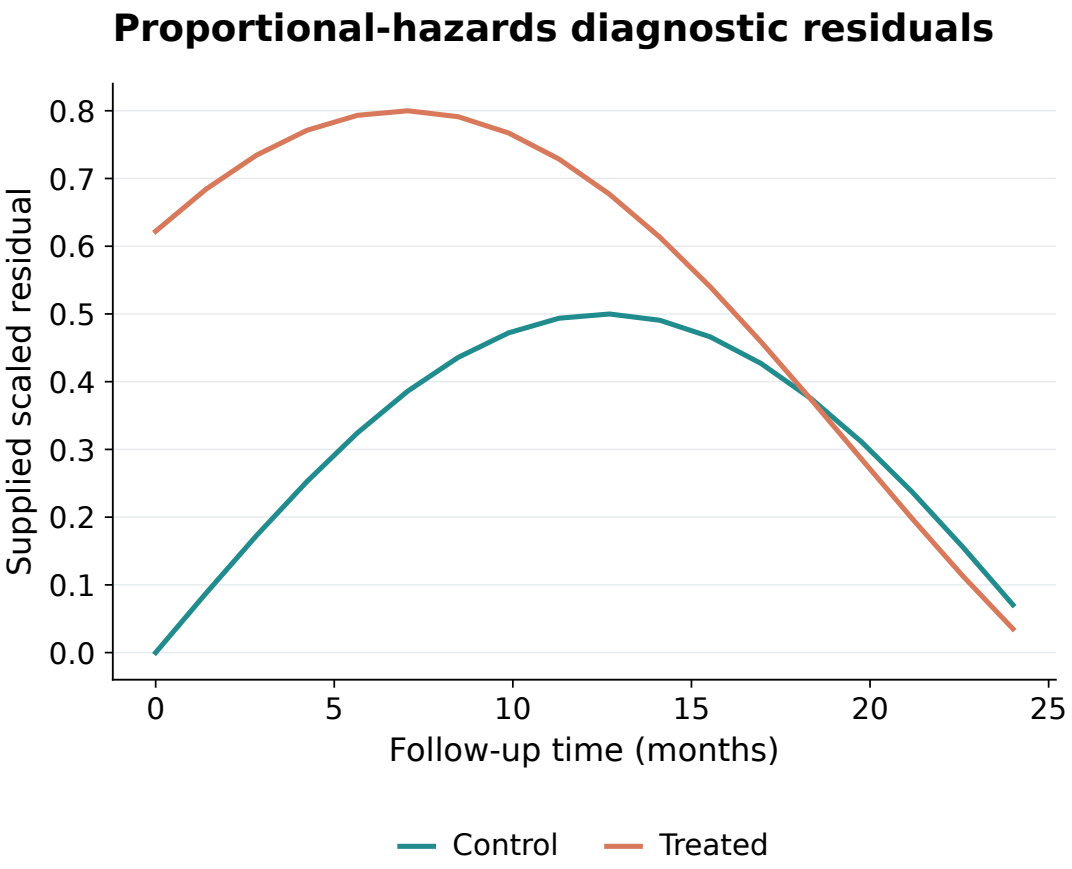
0.68 [0.57, 0.82] 0.02

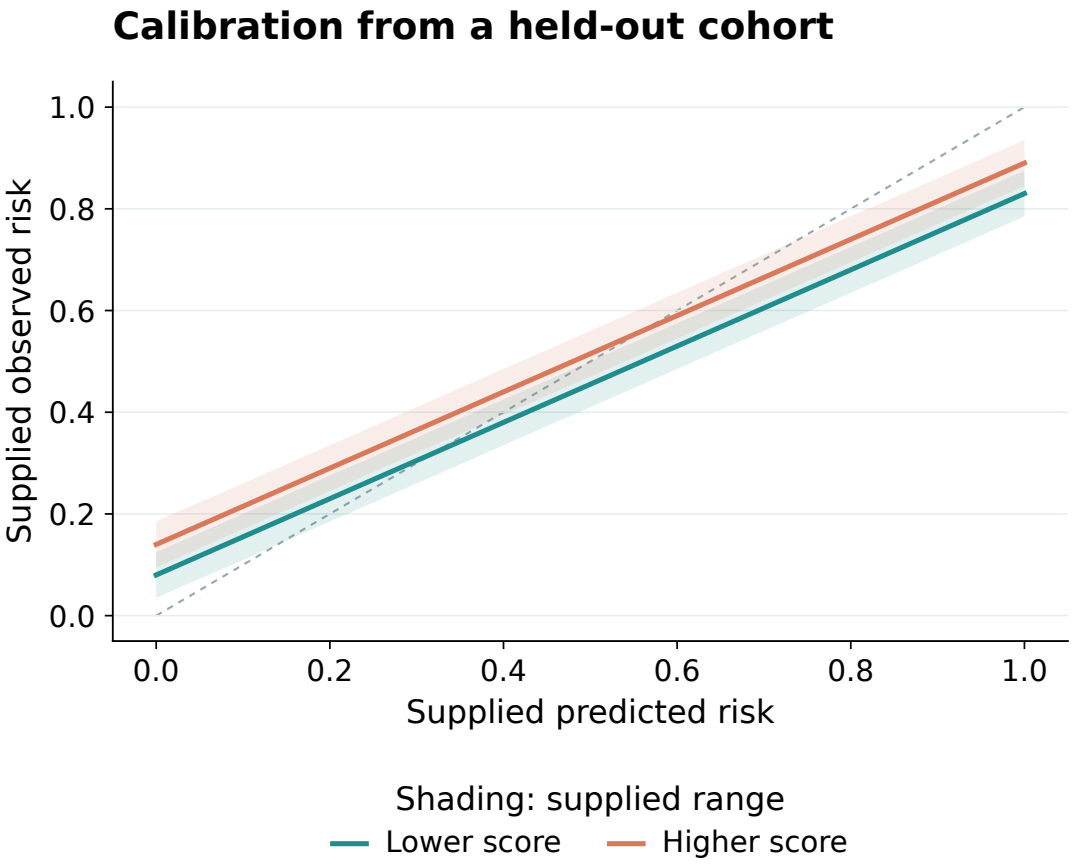
0.94 [0.79, 1.13] 0.65

1.30 [1.08, 1.55] 0.056

1.79 [1.49, 2.14] 0.074

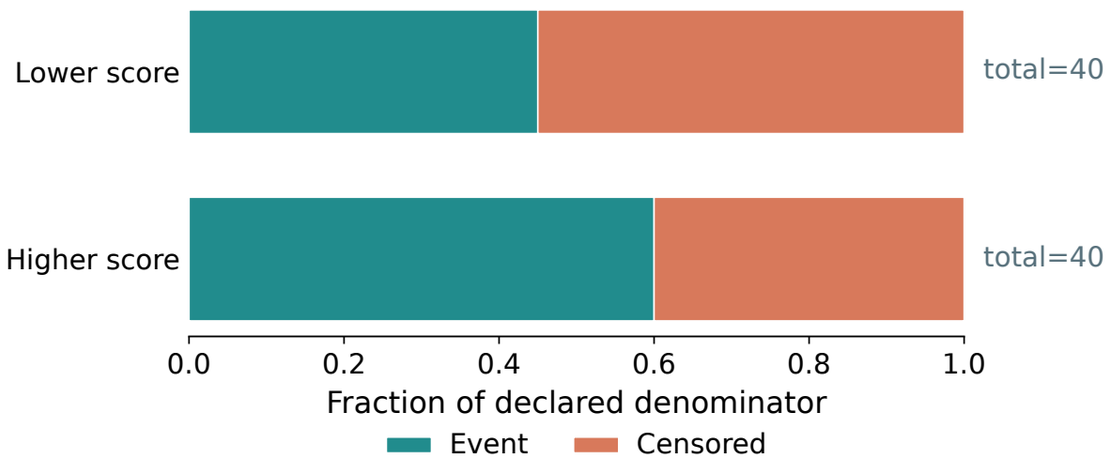
2.46 [2.05, 2.94] 0.092


Proportional-hazards diagnostic residuals
Supplied scaled residual
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
5
10
15
20
25
Follow-up time (months)
Control
Treated

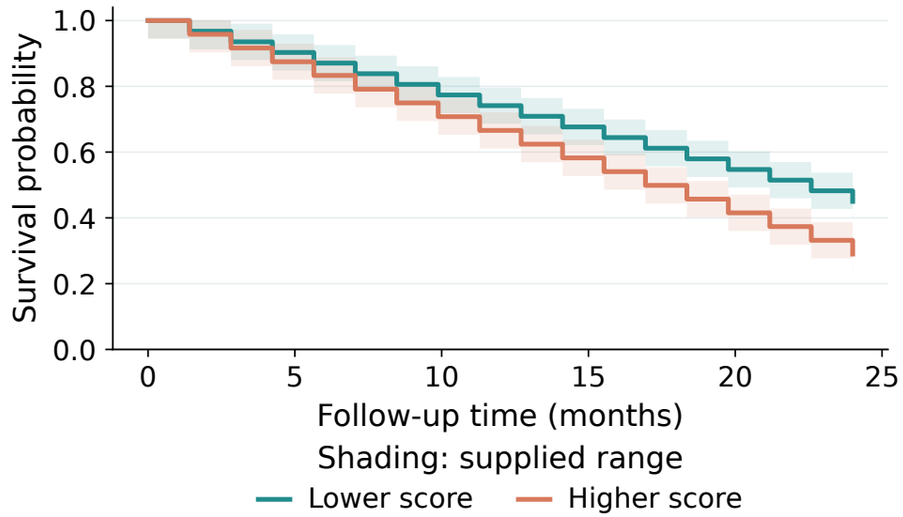

Calibration from a held-out cohort
Supplied observed risk
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Supplied predicted risk
Shading: supplied range
Lower score
Higher score

Bulk · survival — complementary evidence

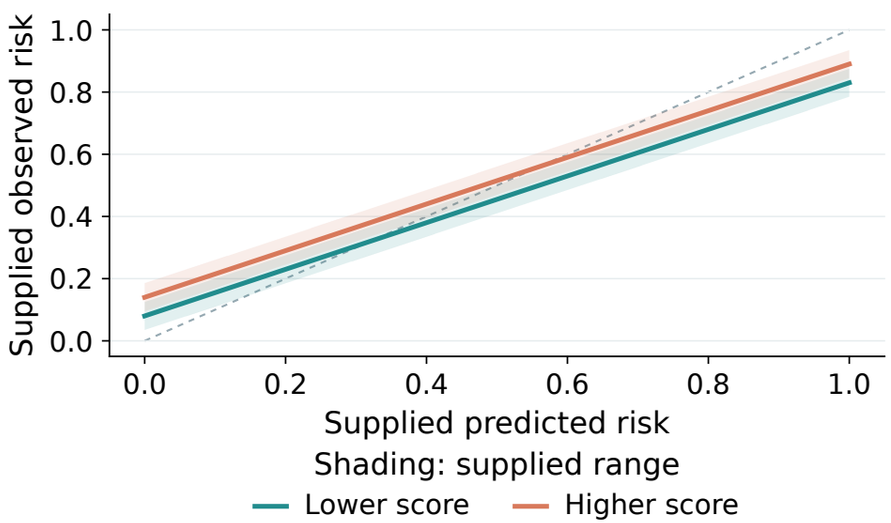
A Endpoint and censoring audit



B Frozen survival curves and confidence intervals

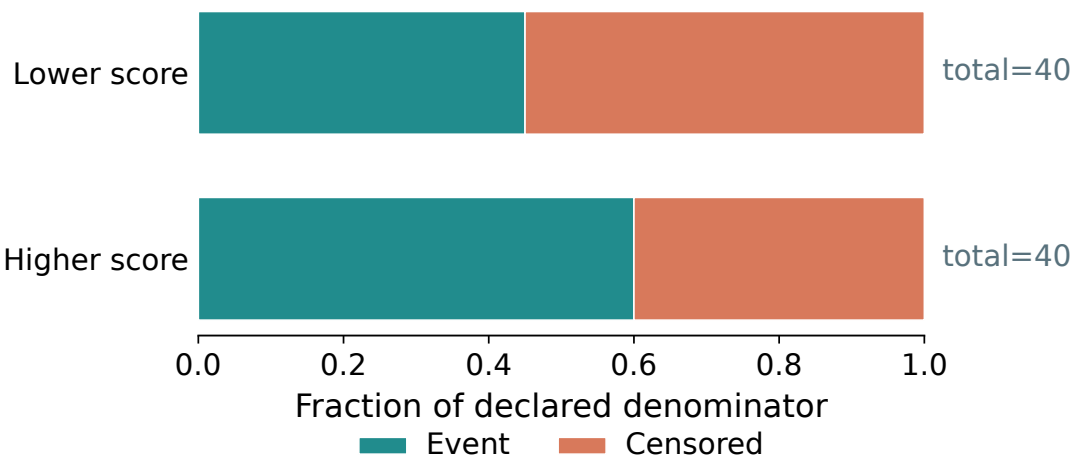


C Calibration from a held-out cohort

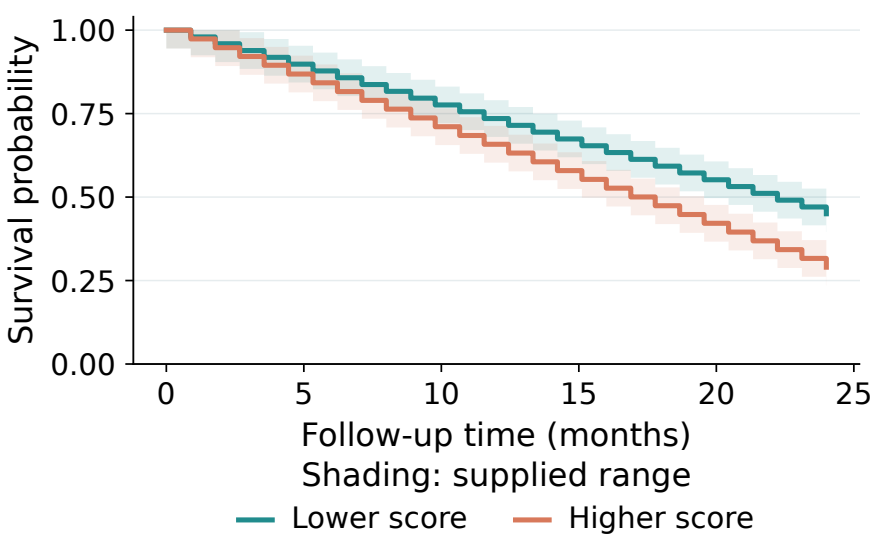


Bulk · survival — complementary evidence

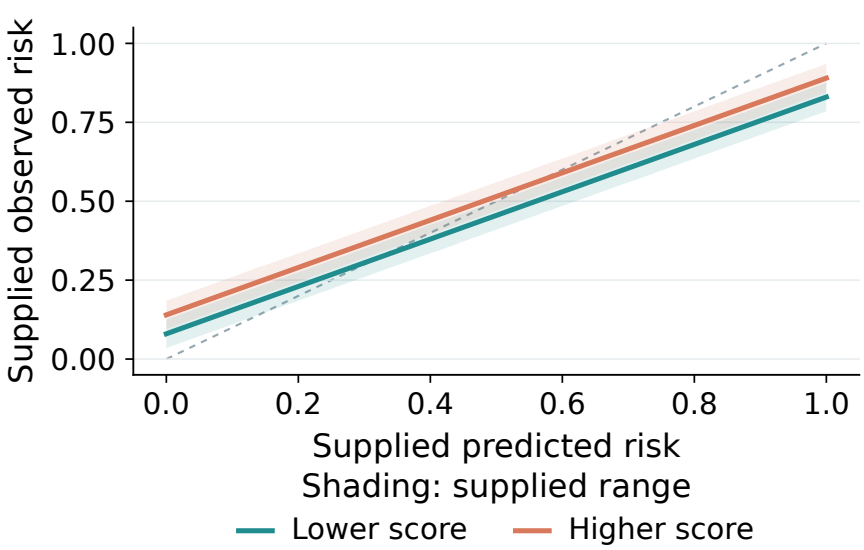
A Endpoint and censoring audit

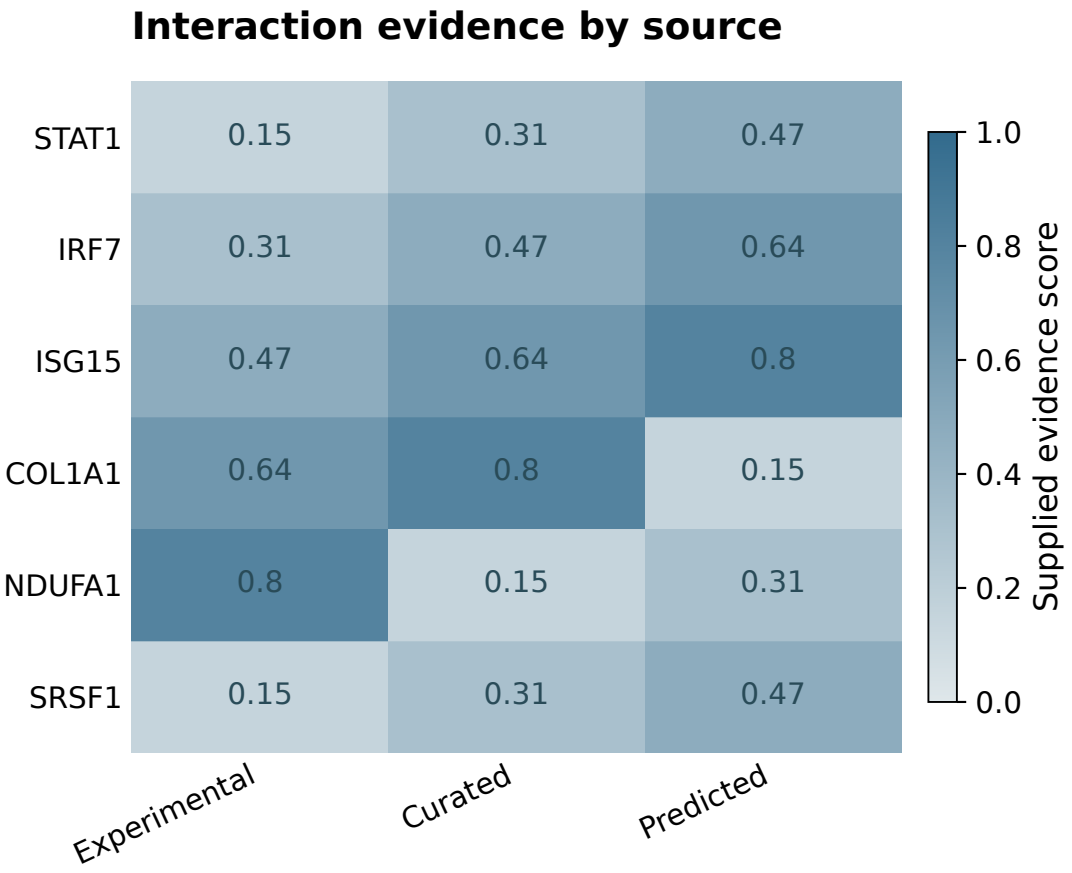


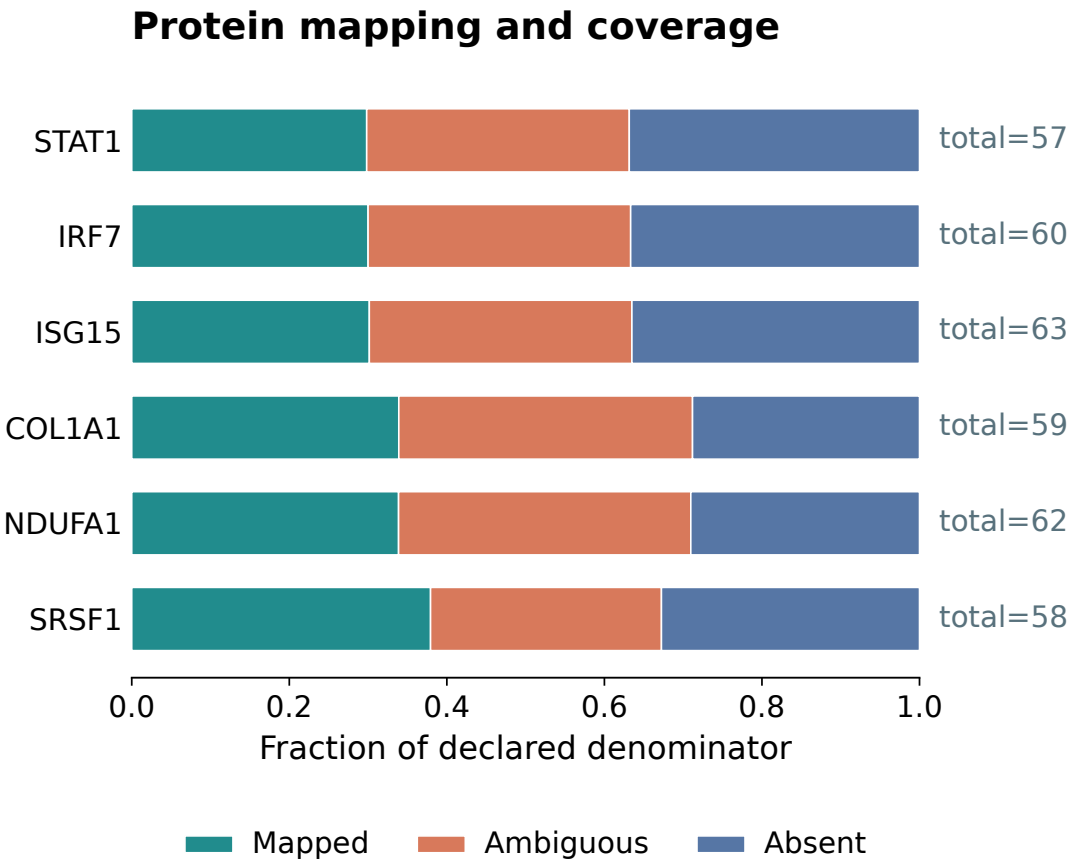
B Frozen survival curves and confidence intervals



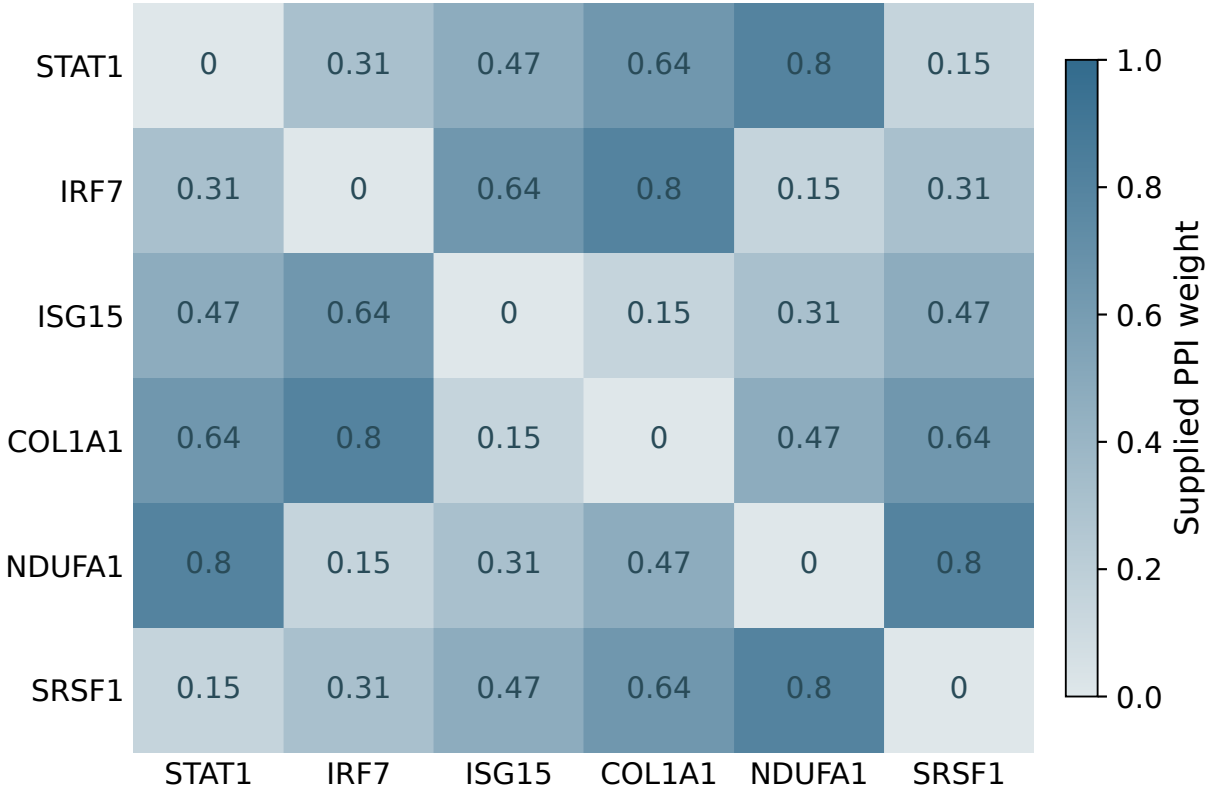
C Calibration from a held-out cohort



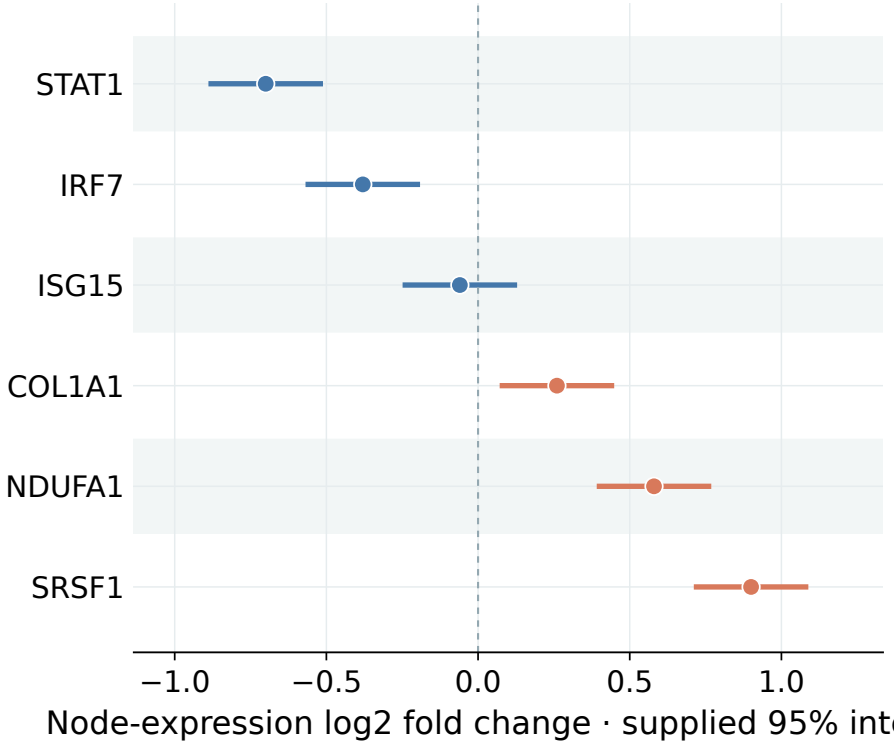

Interaction evidence by source
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0.15
0.31
0.47
0.31
0.47
0.64
0.47
0.64
0.8
0.64
0.8
0.15
0.8
0.15
0.31
0.15
0.31
0.47
Experimental
Curated
Predicted
Supplied evidence score
1.0
0.8
0.6
0.4
0.2
0.0


Protein mapping and coverage
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Mapped
Ambiguous
Absent

Complete selected interaction weights



Prespecified node-level response



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

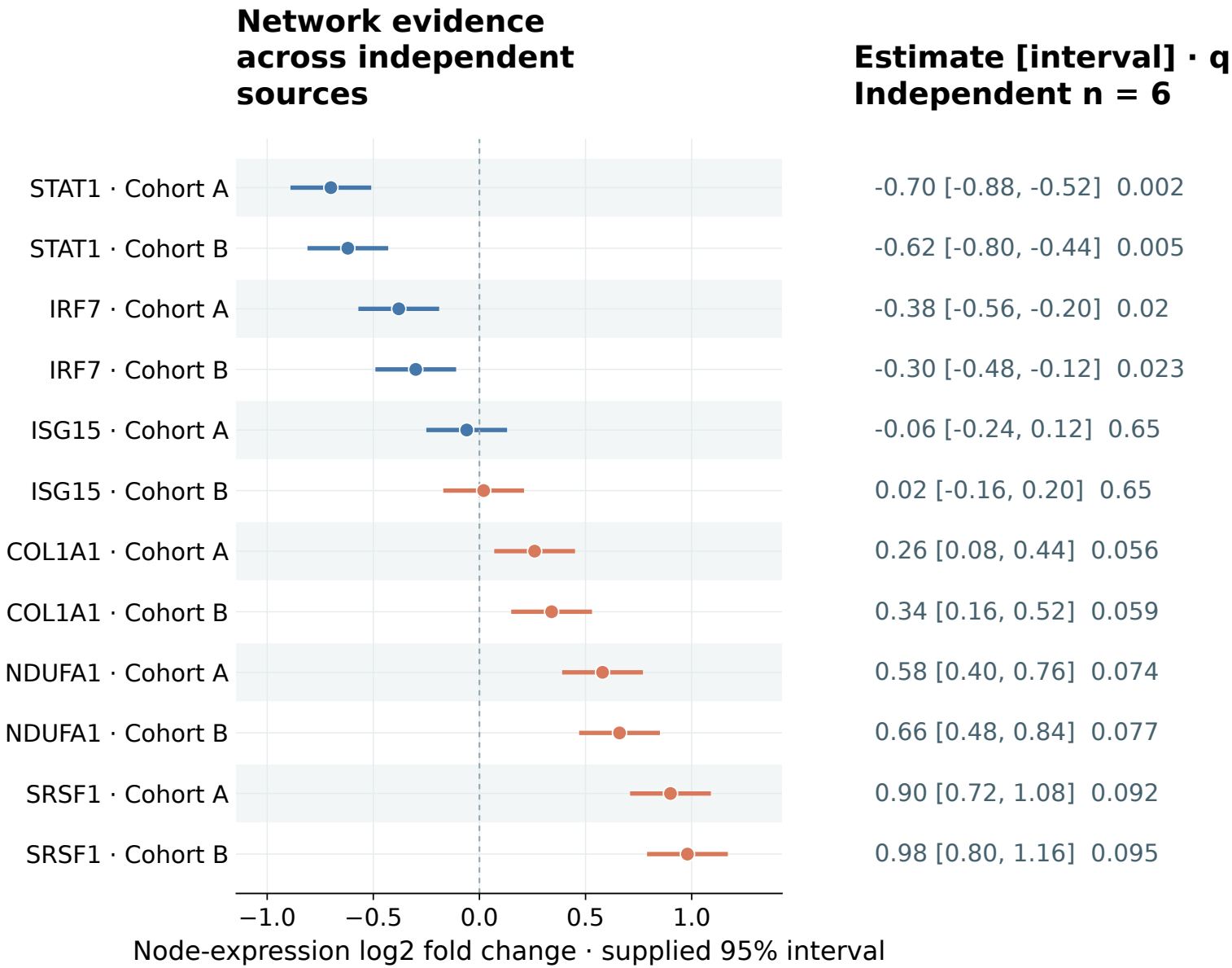
-0.38 [-0.56, -0.20] 0.02

-0.06 [-0.24, 0.12] 0.65

0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

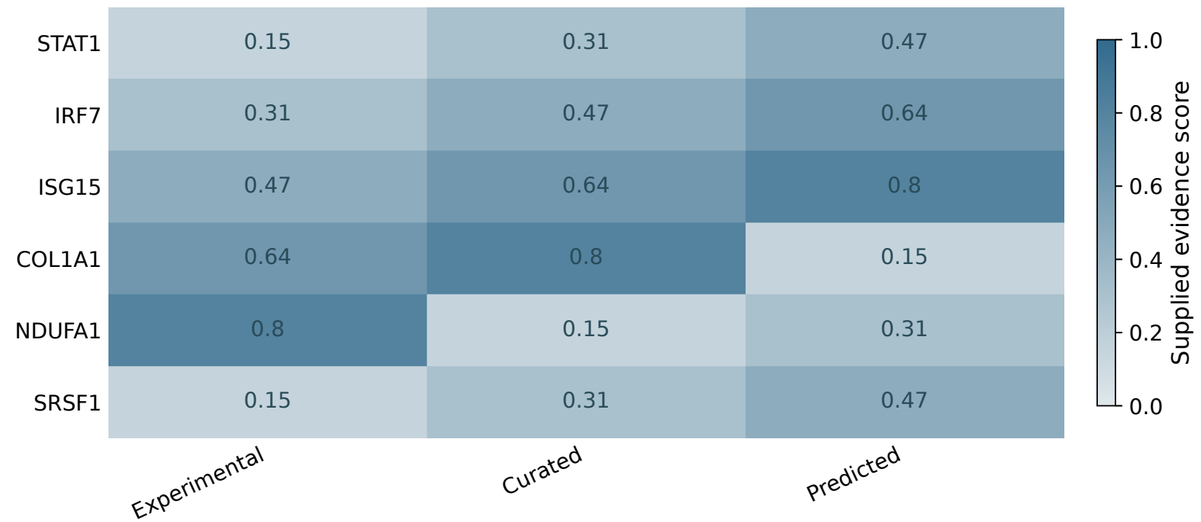
0.90 [0.72, 1.08] 0.092


Network evidence across independent sources
Estimate [interval] · q
Independent n = 6
STAT1 · Cohort A
STAT1 · Cohort B
IRF7 · Cohort A
IRF7 · Cohort B
ISG15 · Cohort A
ISG15 · Cohort B
COL1A1 · Cohort A
COL1A1 · Cohort B
NDUFA1 · Cohort A
NDUFA1 · Cohort B
SRSF1 · Cohort A
SRSF1 · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Node-expression log2 fold change · supplied 95% interval

Bulk · ppi — complementary evidence

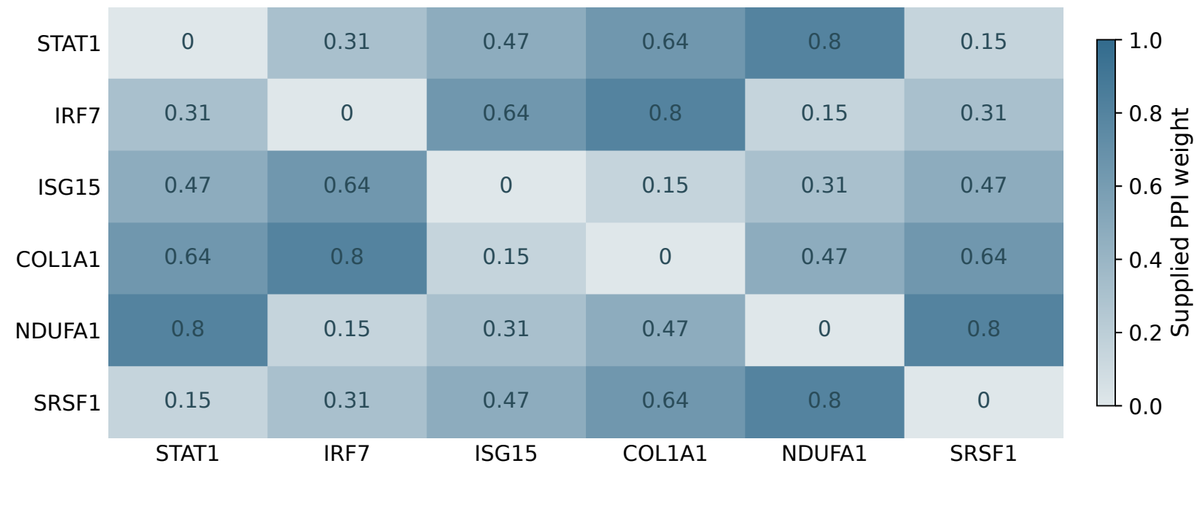
A

Interaction evidence by source



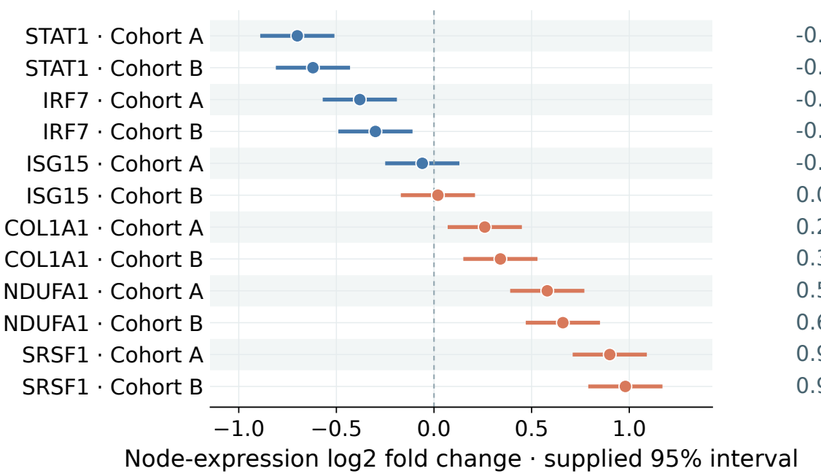
B

Complete selected interaction weights



C

Network evidence across independent sources



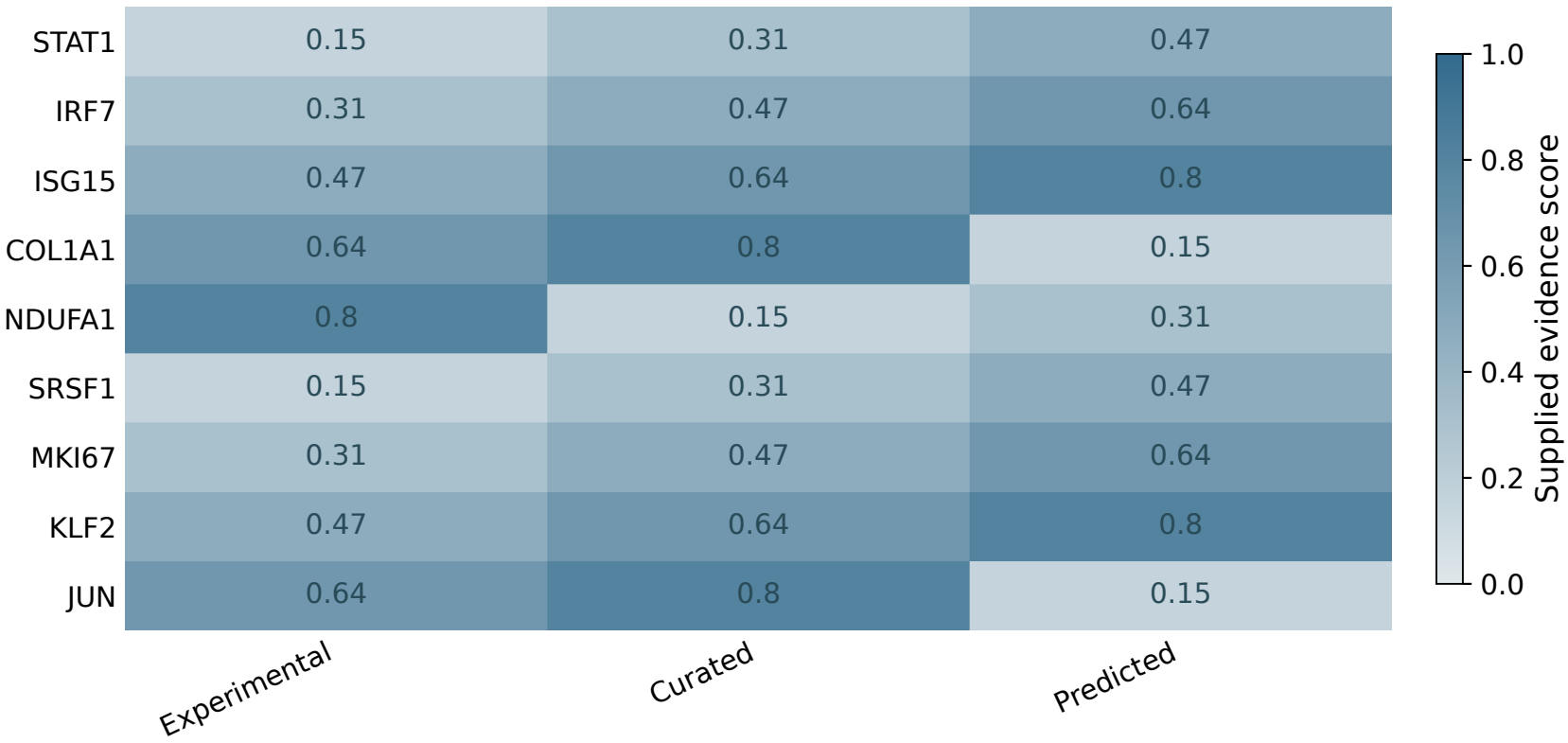
Estimate [interval] · q
Independent n = 6

-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095

Bulk · ppi — complementary evidence

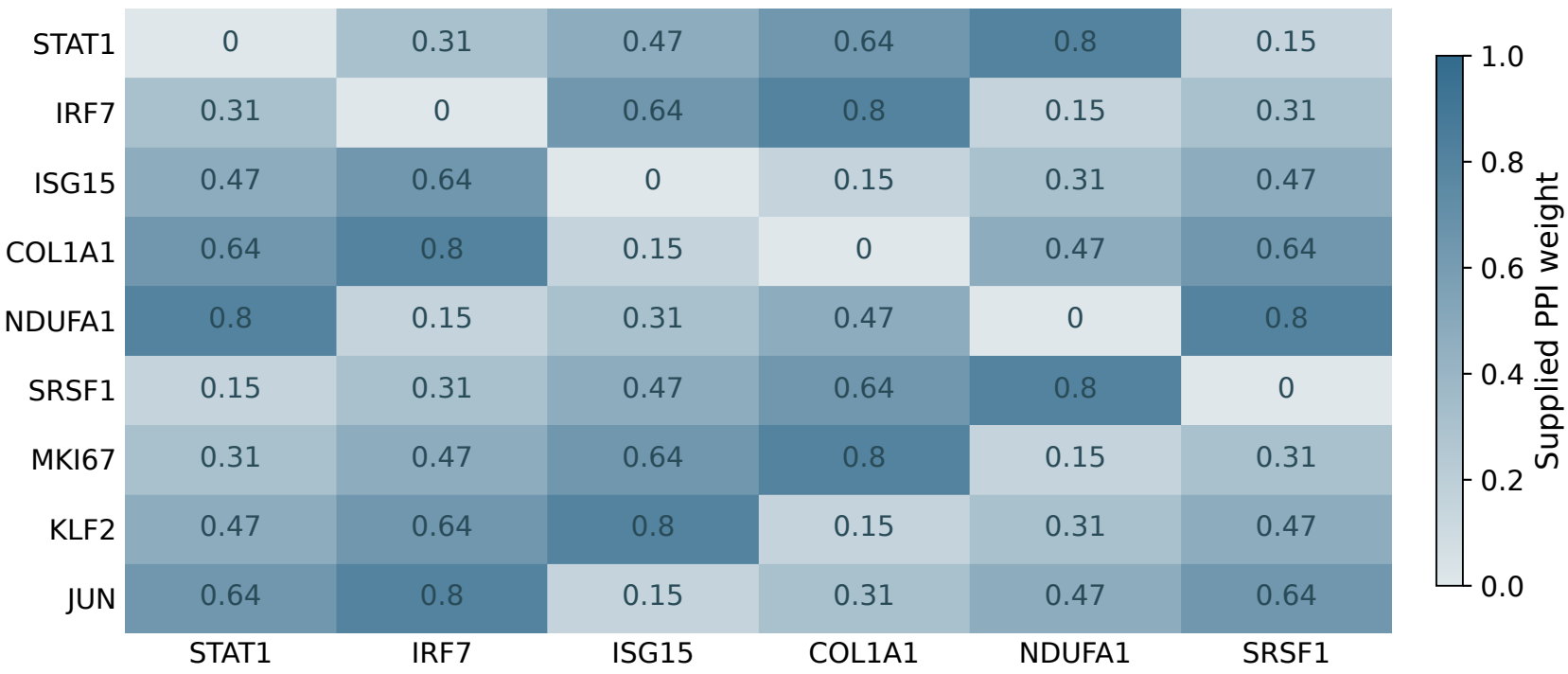
A

Interaction evidence by source



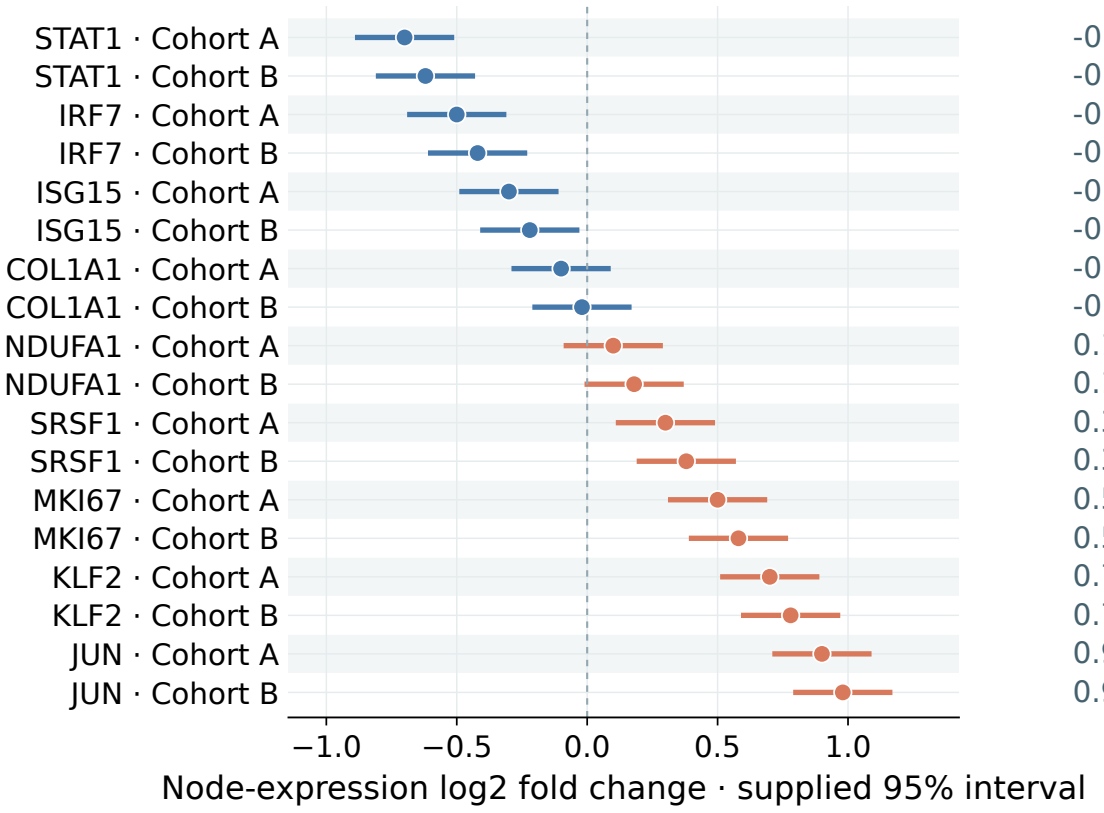
B

Complete selected interaction weights



C

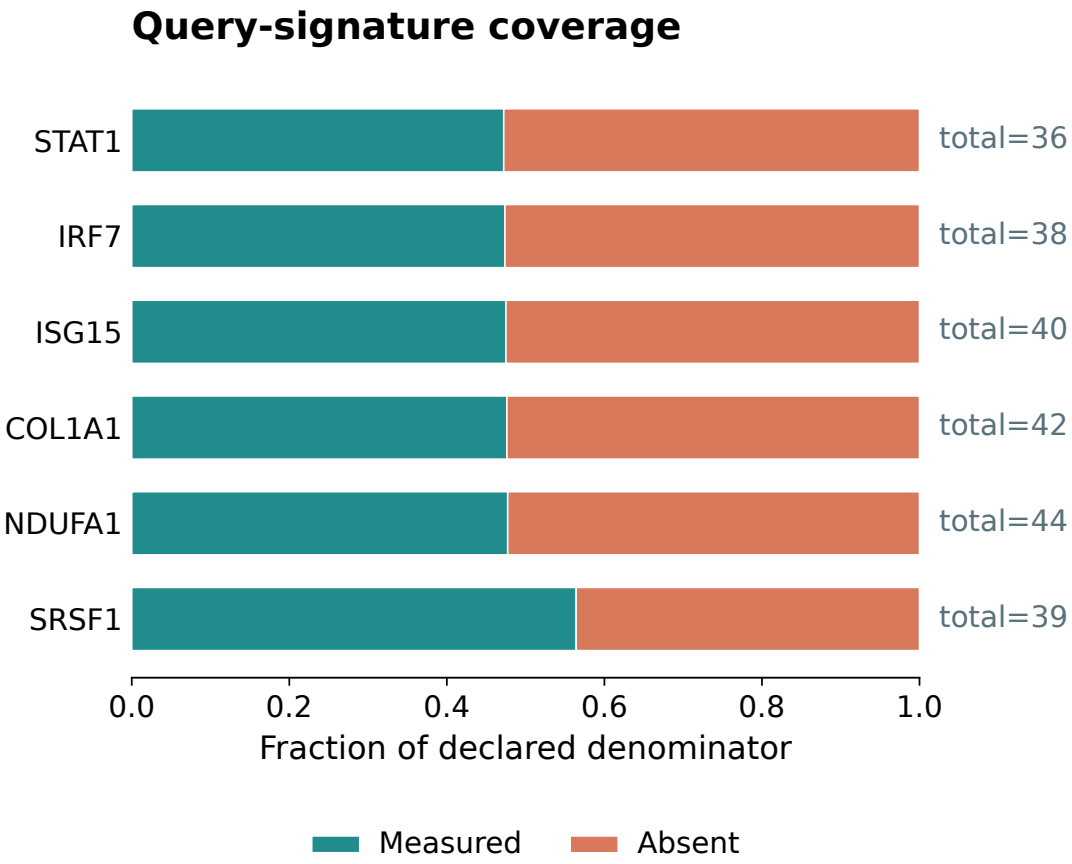
Network evidence across independent sources



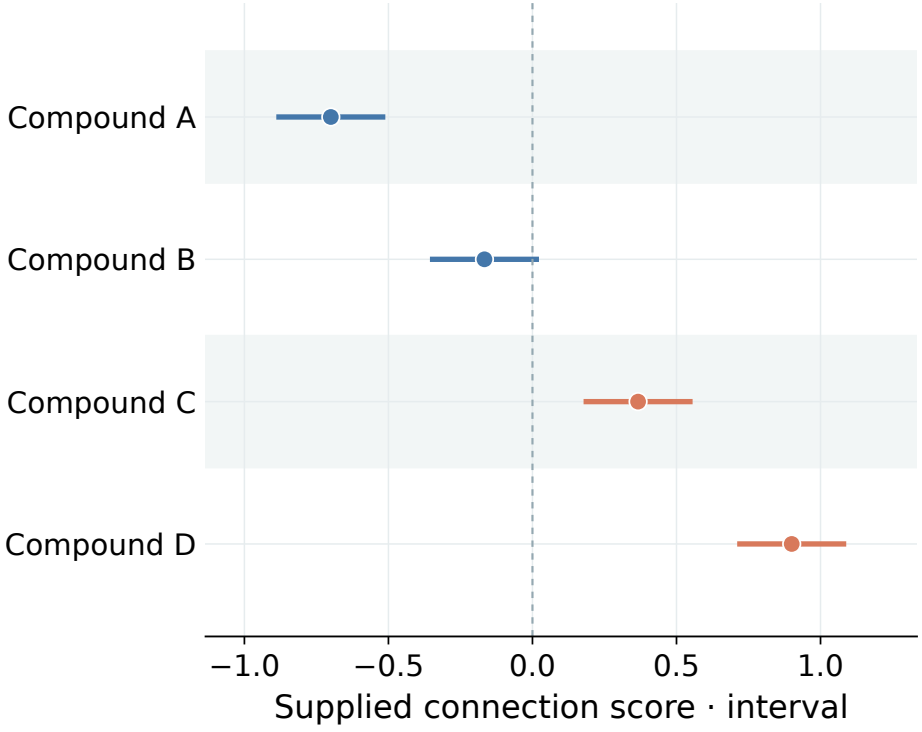
Estimate [interval] · q
Independent n = 12

STAT1 · Cohort A	-0.70 [-0.88, -0.52] 0.002
STAT1 · Cohort B	-0.62 [-0.80, -0.44] 0.005
IRF7 · Cohort A	-0.50 [-0.68, -0.32] 0.02
IRF7 · Cohort B	-0.42 [-0.60, -0.24] 0.023
ISG15 · Cohort A	-0.30 [-0.48, -0.12] 0.038
ISG15 · Cohort B	-0.22 [-0.40, -0.04] 0.041
COL1A1 · Cohort A	-0.10 [-0.28, 0.08] 0.65
COL1A1 · Cohort B	-0.02 [-0.20, 0.16] 0.65
NDUFA1 · Cohort A	0.10 [-0.08, 0.28] 0.65
NDUFA1 · Cohort B	0.18 [0.00, 0.36] 0.077
SRSF1 · Cohort A	0.30 [0.12, 0.48] 0.092
SRSF1 · Cohort B	0.38 [0.20, 0.56] 0.095
MKI67 · Cohort A	0.50 [0.32, 0.68] 0.11
MKI67 · Cohort B	0.58 [0.40, 0.76] 0.11
KLF2 · Cohort A	0.70 [0.52, 0.88] 0.13
KLF2 · Cohort B	0.78 [0.60, 0.96] 0.13
JUN · Cohort A	0.90 [0.72, 1.08] 0.15
JUN · Cohort B	0.98 [0.80, 1.16] 0.15

Software test fixture; not biological evidence


Query-signature coverage
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Measured
Absent

Connection scores with supplied uncertainty



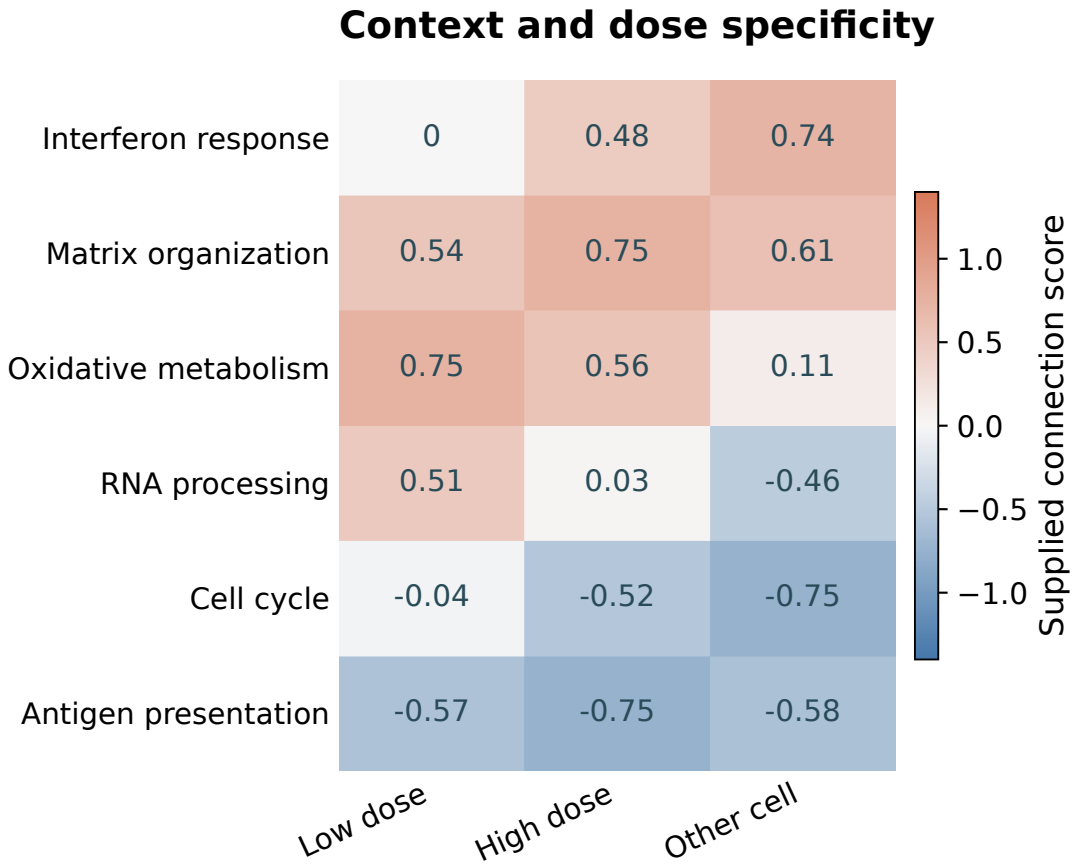
Estimate [interval] · q
Independent n = 12

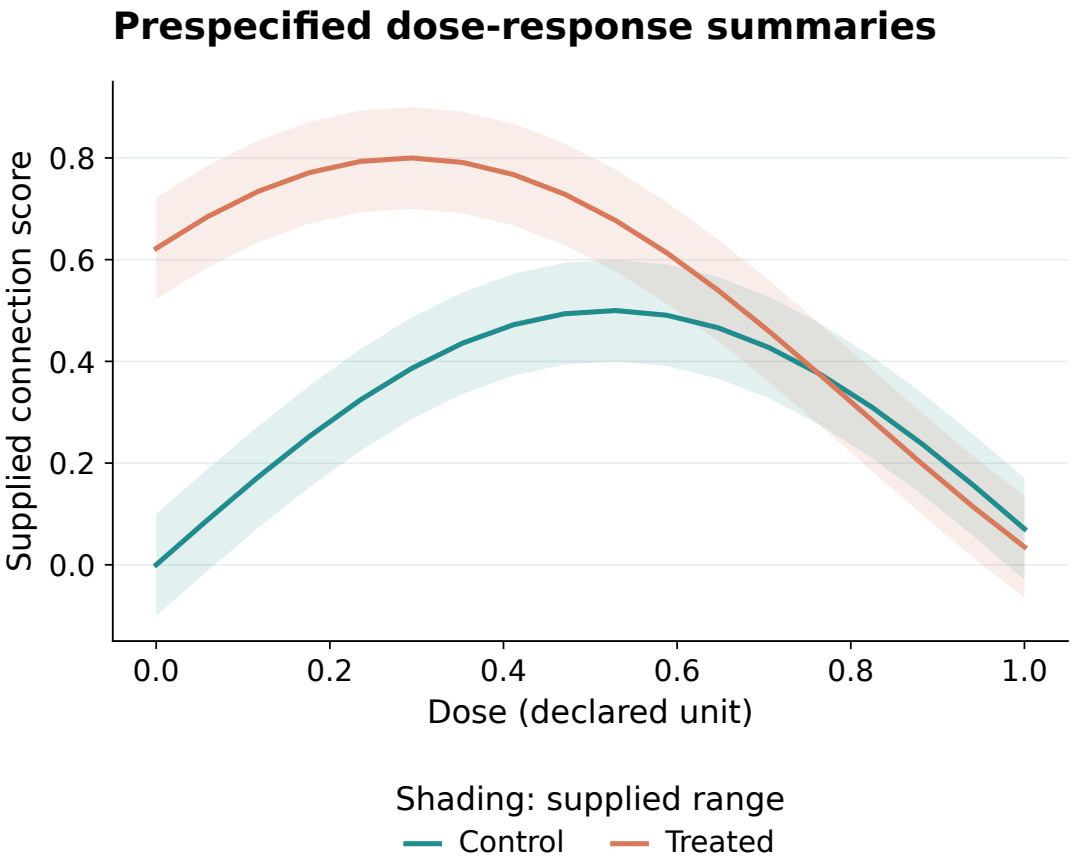
-0.70 [-0.88, -0.52] 0.002

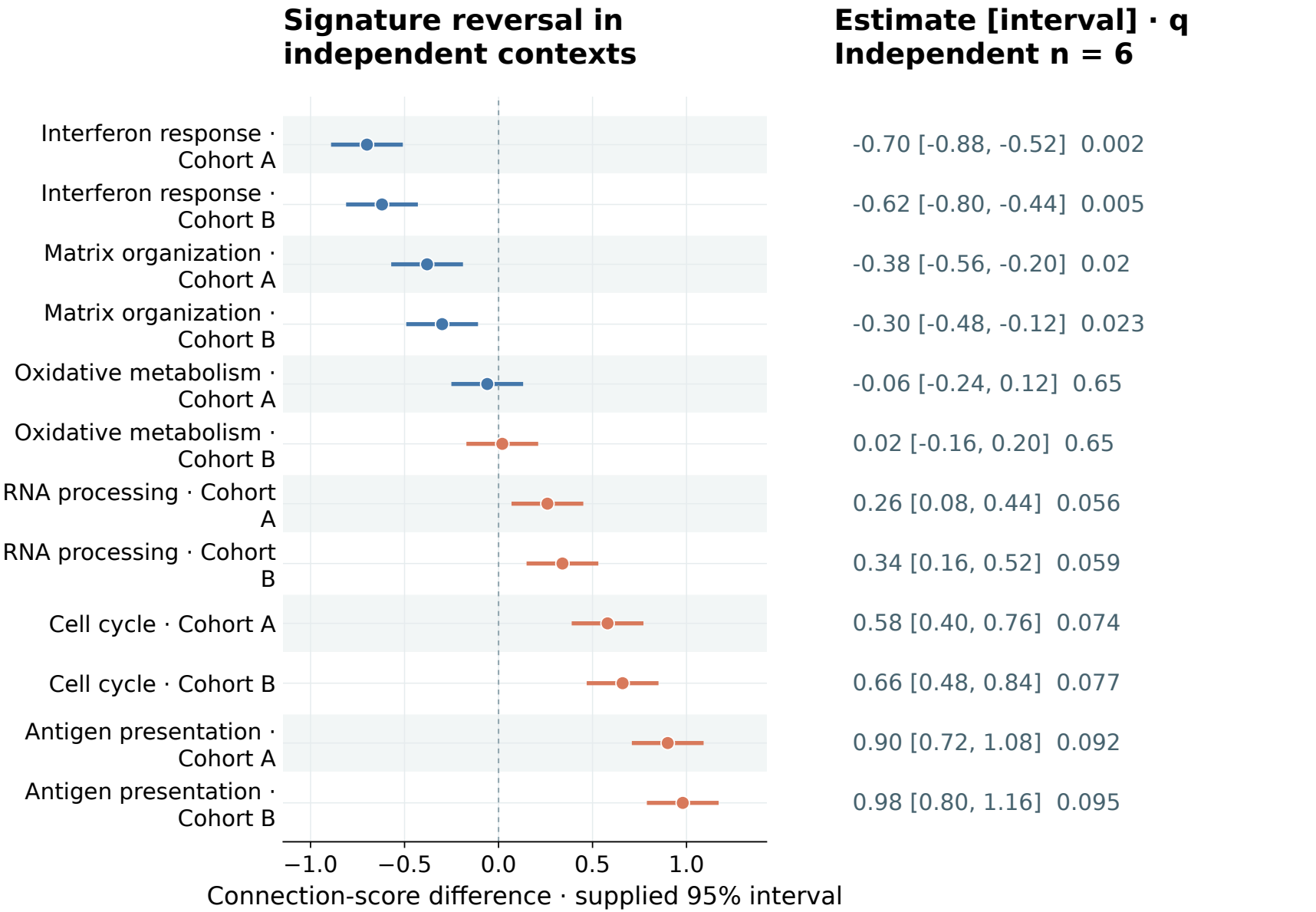
-0.17 [-0.35, 0.01] 0.65

0.37 [0.19, 0.55] 0.038

0.90 [0.72, 1.08] 0.056

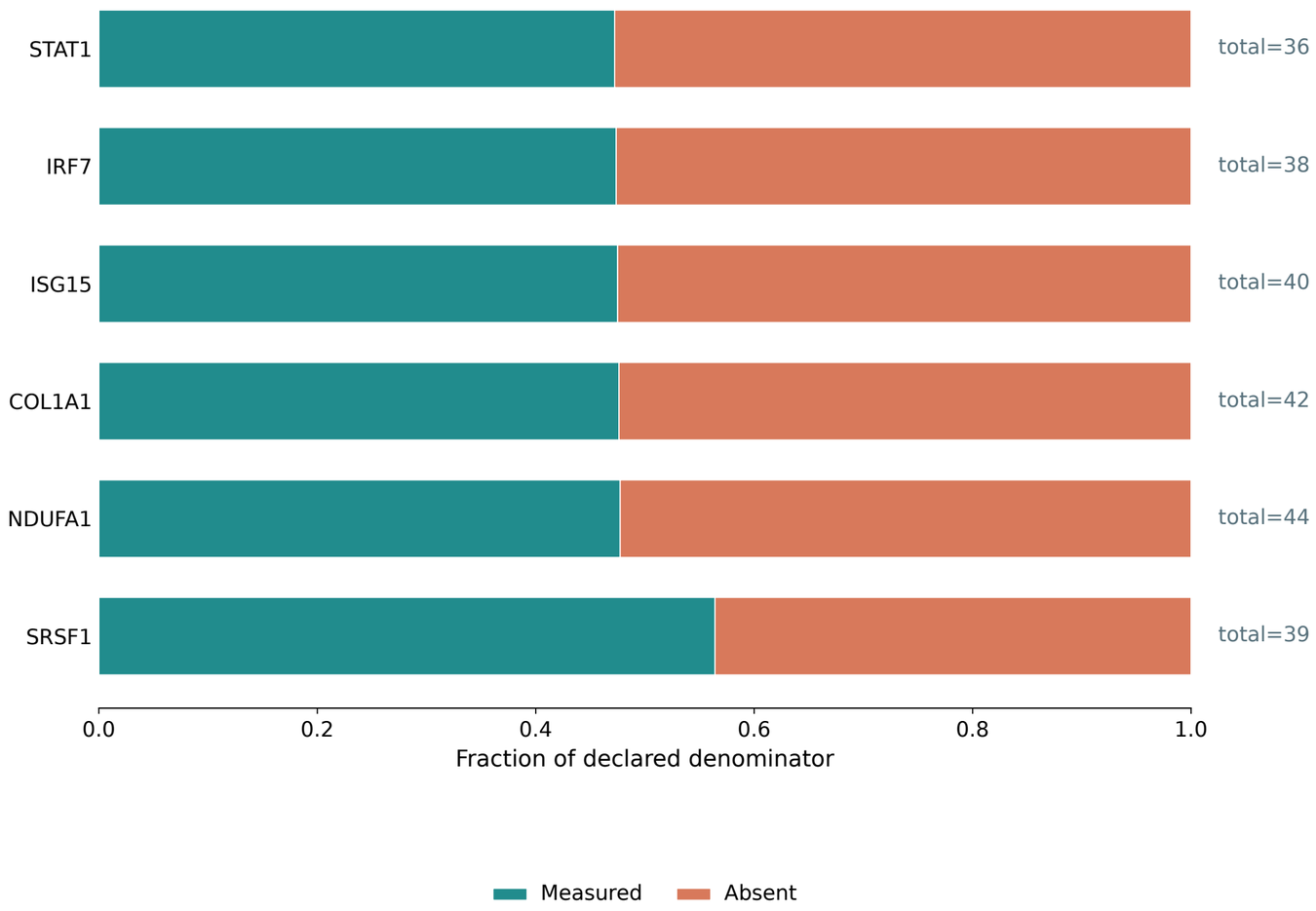

Context and dose specificity
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
0
0.48
0.74
0.54
0.75
0.61
0.75
0.56
0.11
0.51
0.03
-0.46
-0.04
-0.52
-0.75
-0.57
-0.75
-0.58
Low dose
High dose
Other cell
1.0
0.5
0.0
−0.5
−1.0
Supplied connection score


Prespecified dose-response summaries
Supplied connection score
0.0
0.2
0.4
0.6
0.8
0.0
0.2
0.4
0.6
0.8
1.0
Dose (declared unit)
Shading: supplied range
Control
Treated

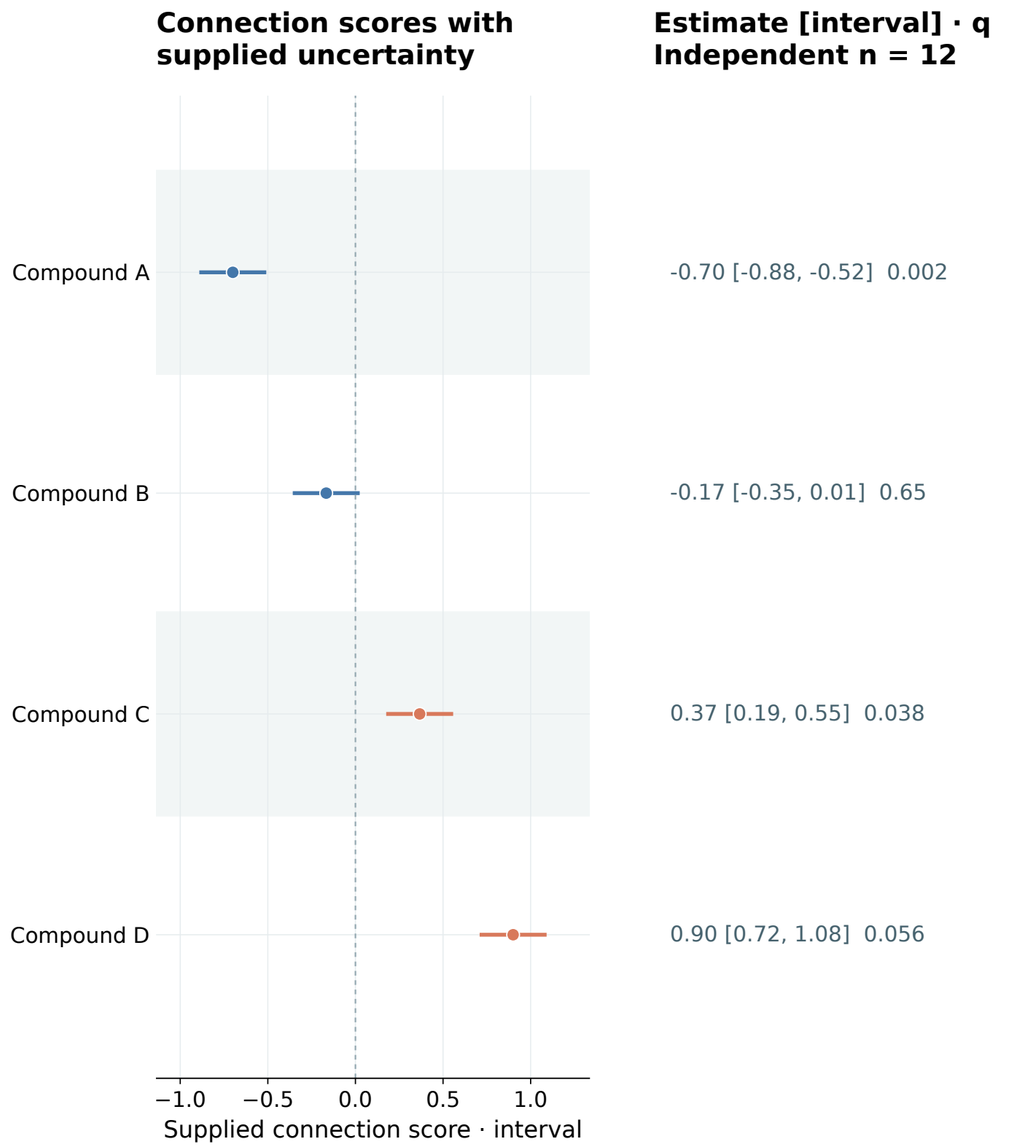

Signature reversal in independent contexts
Estimate [interval] · q
Independent n = 6
Interferon response · Cohort A
Interferon response · Cohort B
Matrix organization · Cohort A
Matrix organization · Cohort B
Oxidative metabolism · Cohort A
Oxidative metabolism · Cohort B
RNA processing · Cohort A
RNA processing · Cohort B
Cell cycle · Cohort A
Cell cycle · Cohort B
Antigen presentation · Cohort A
Antigen presentation · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Connection-score difference · supplied 95% interval

Bulk · connectivity — complementary evidence

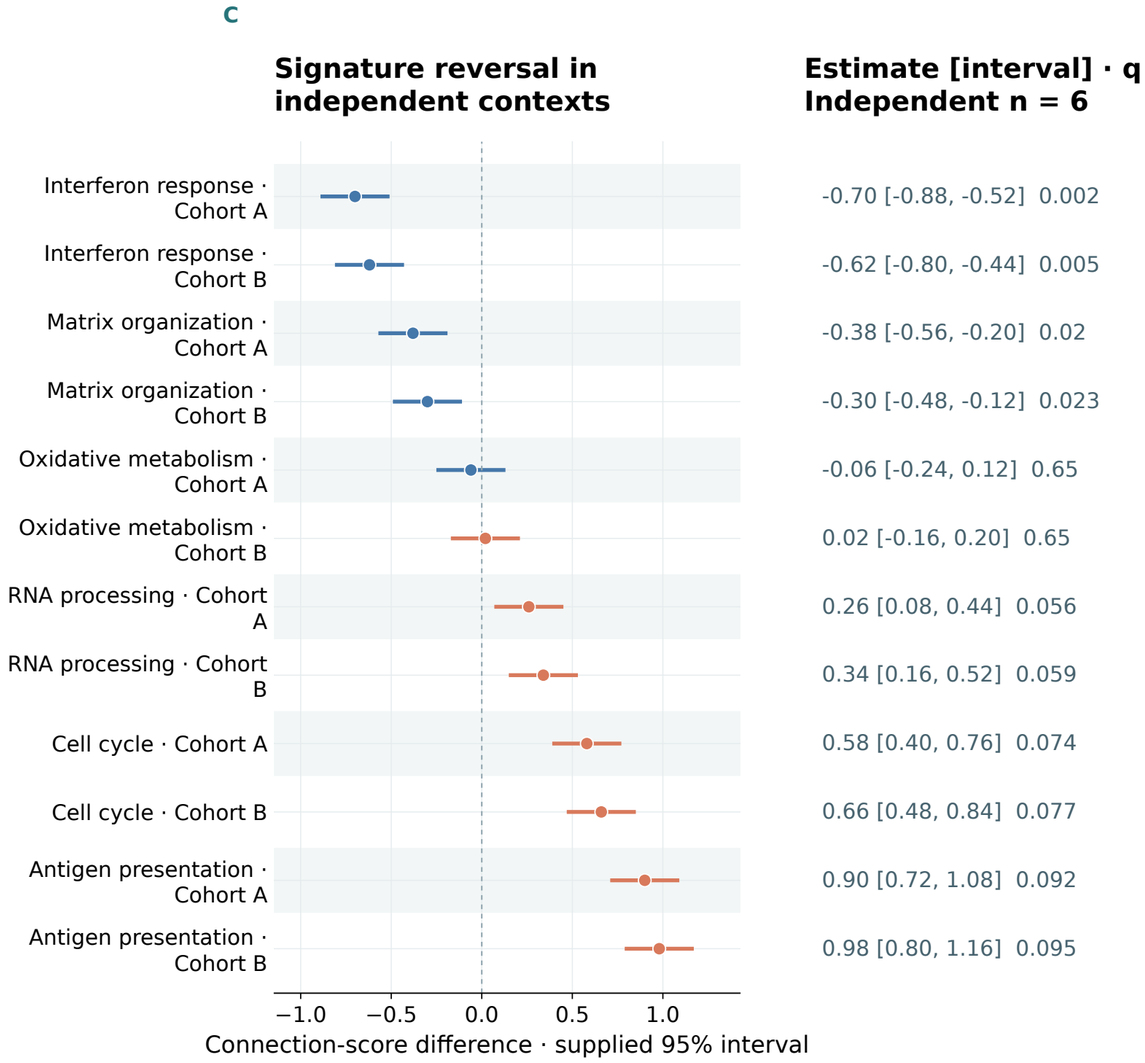
A



B



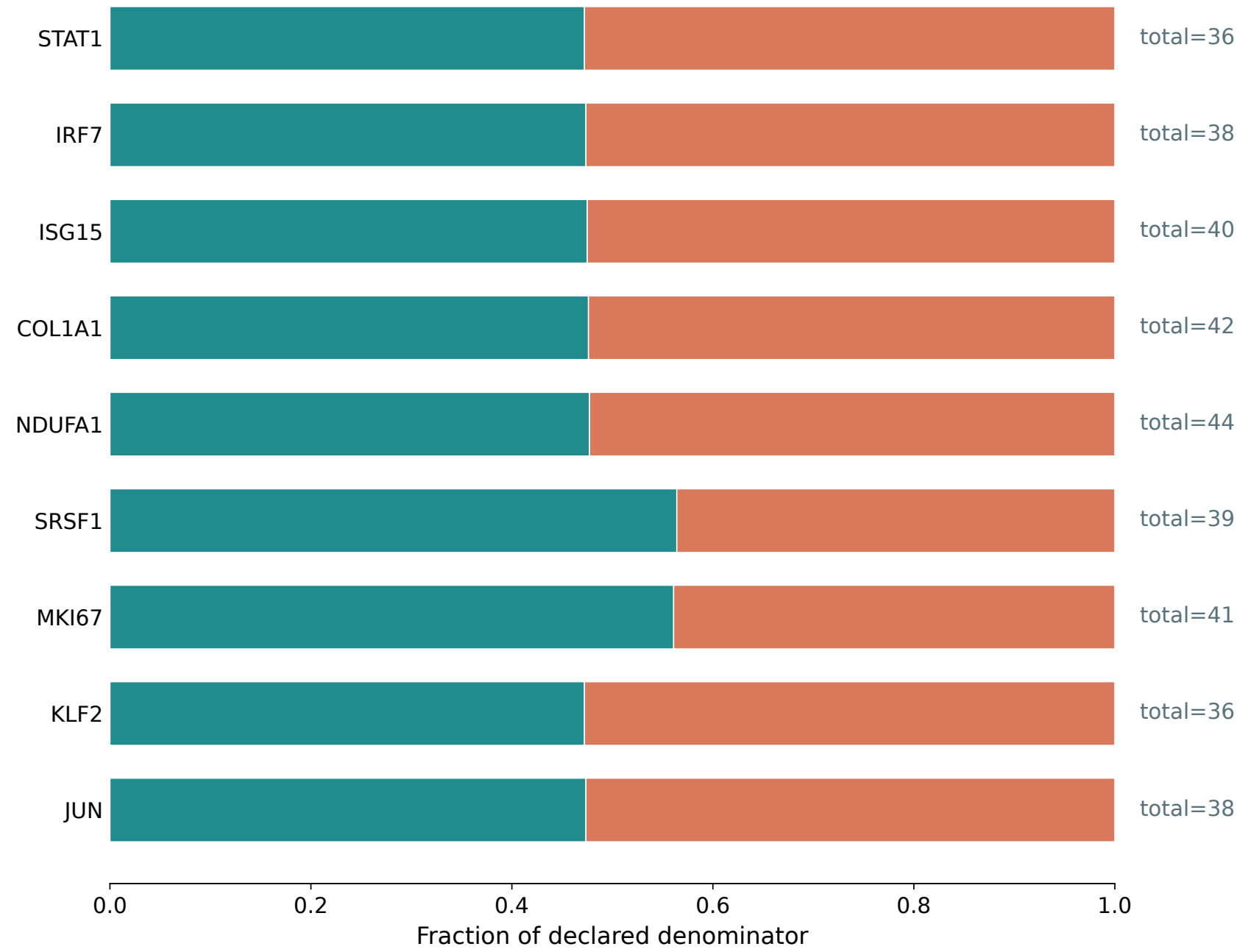
C



Bulk · connectivity — complementary evidence

A

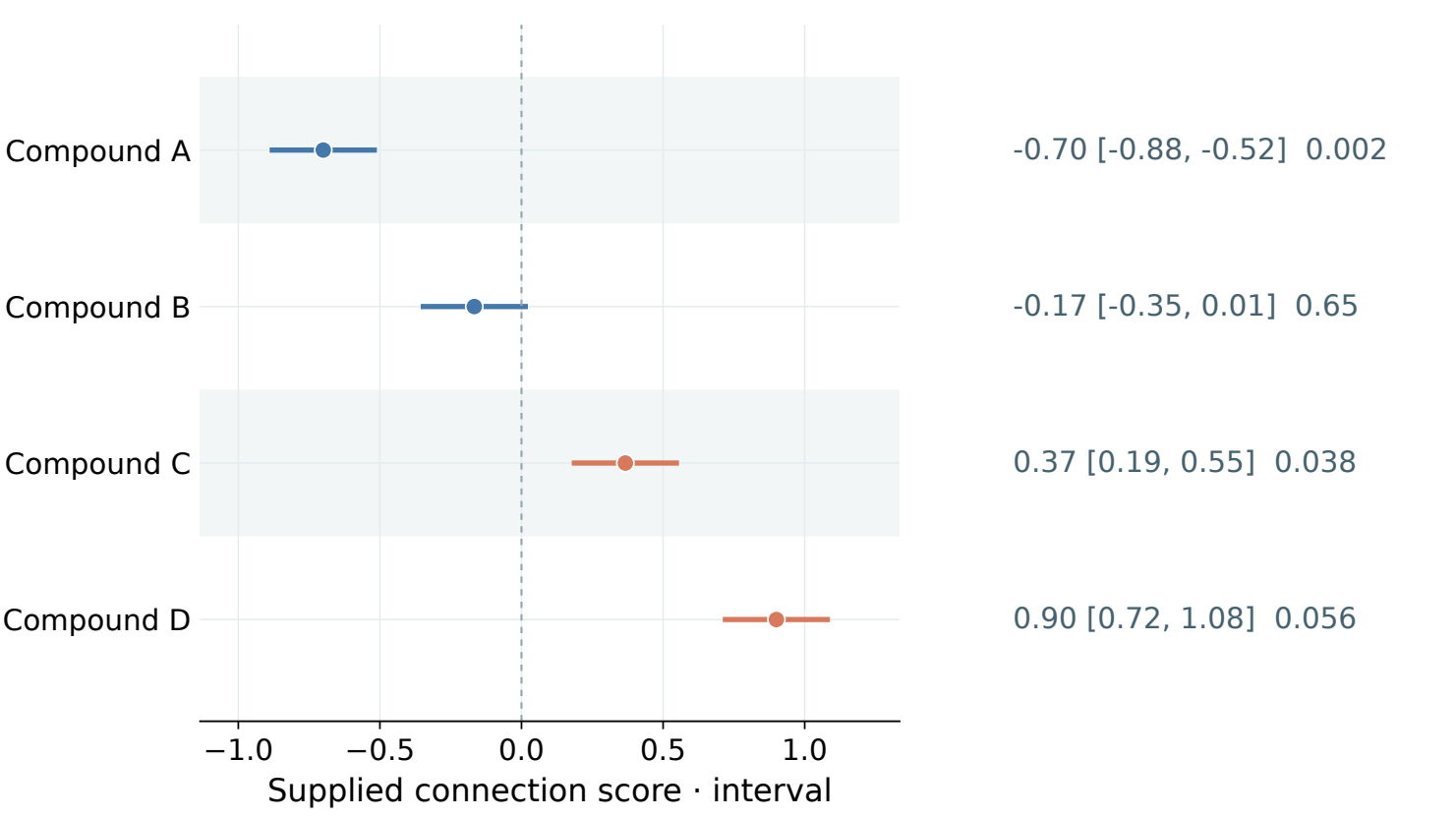
Query-signature coverage



B

Connection scores with supplied uncertainty

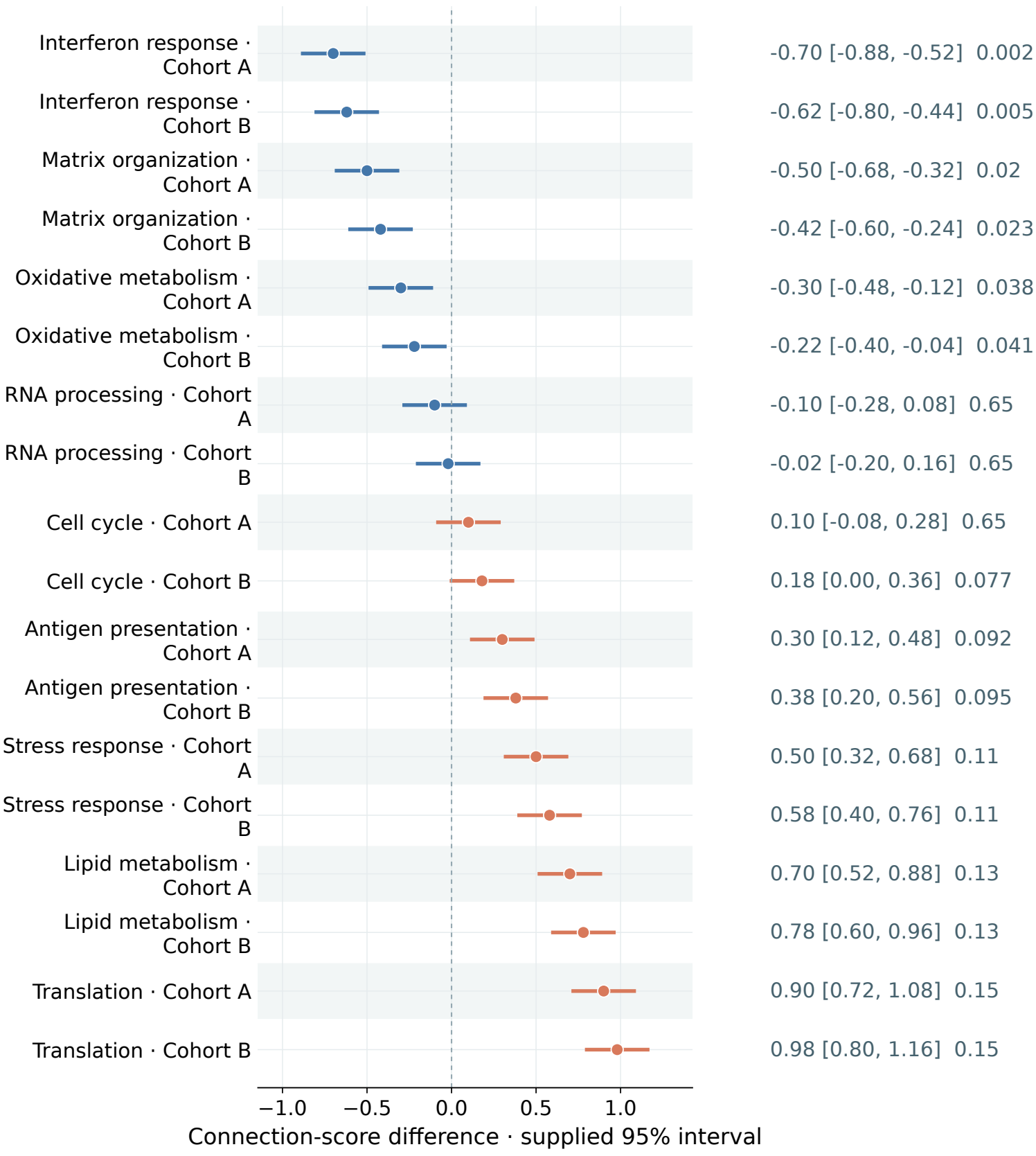
Estimate [interval] · q
Independent n = 24

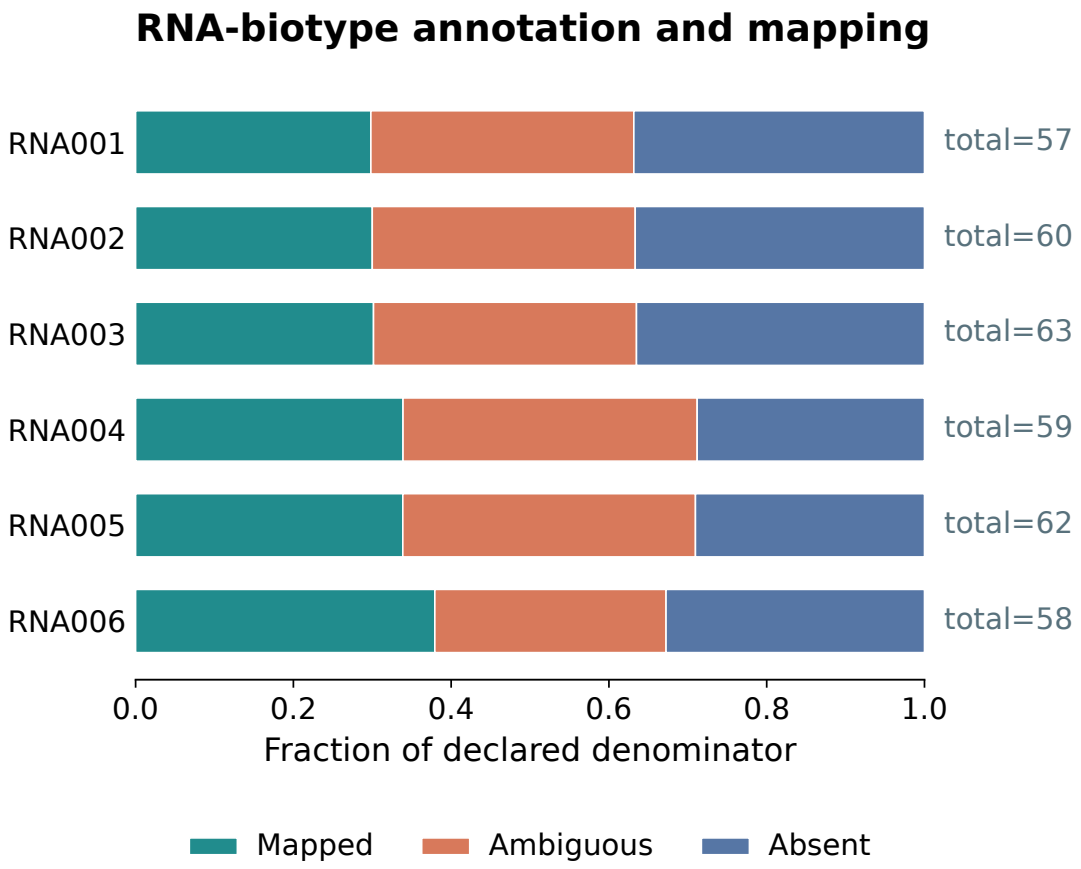


C

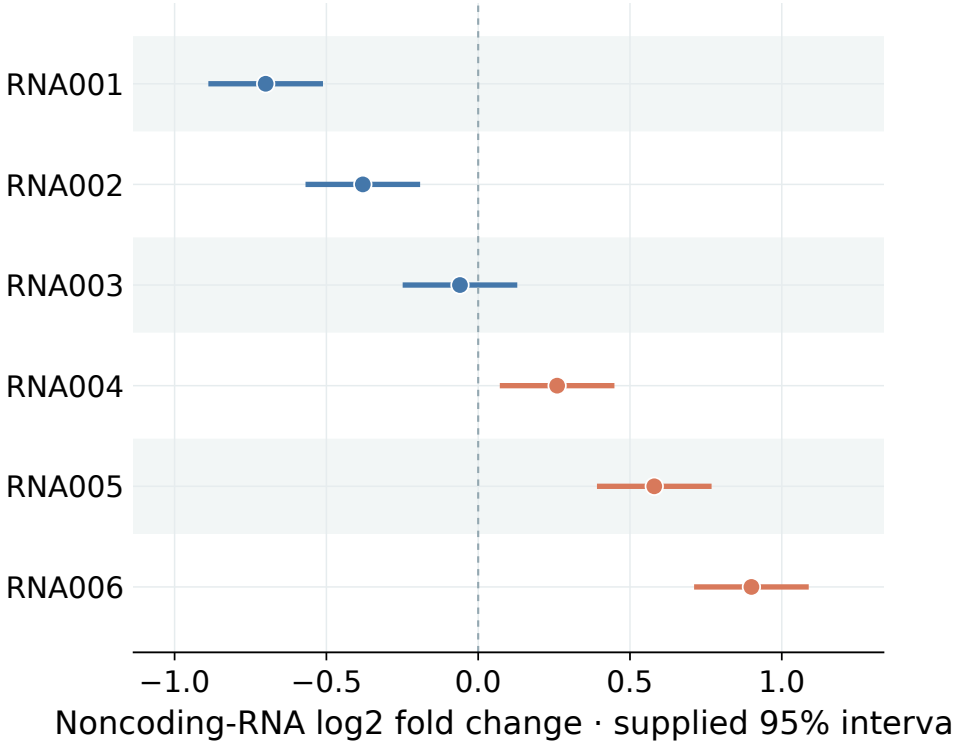
Signature reversal in independent contexts

Estimate [interval] · q
Independent n = 12




RNA-biotype annotation and mapping
RNA001
RNA002
RNA003
RNA004
RNA005
RNA006
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Mapped
Ambiguous
Absent

Prespecified noncoding RNA effects



Estimate [interval] · q Independent n = 12

-0.70 [-0.88, -0.52] 0.002

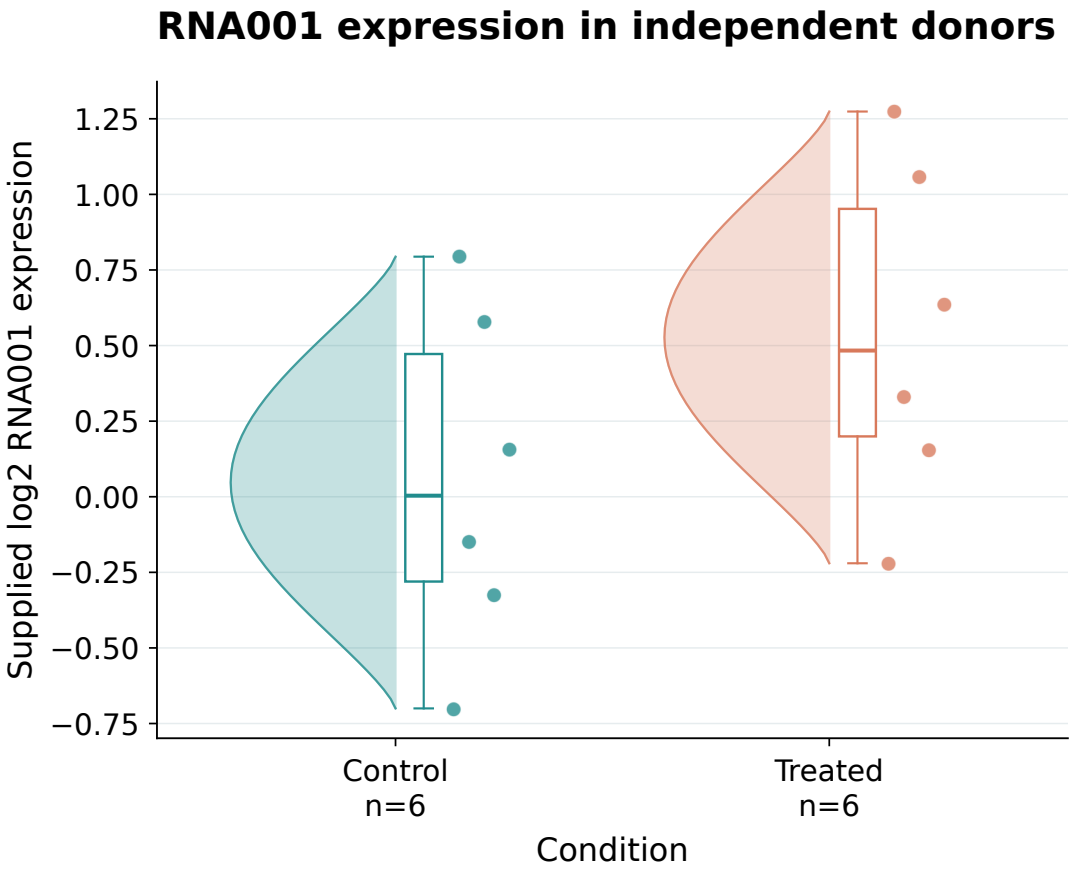
-0.38 [-0.56, -0.20] 0.02

-0.06 [-0.24, 0.12] 0.65

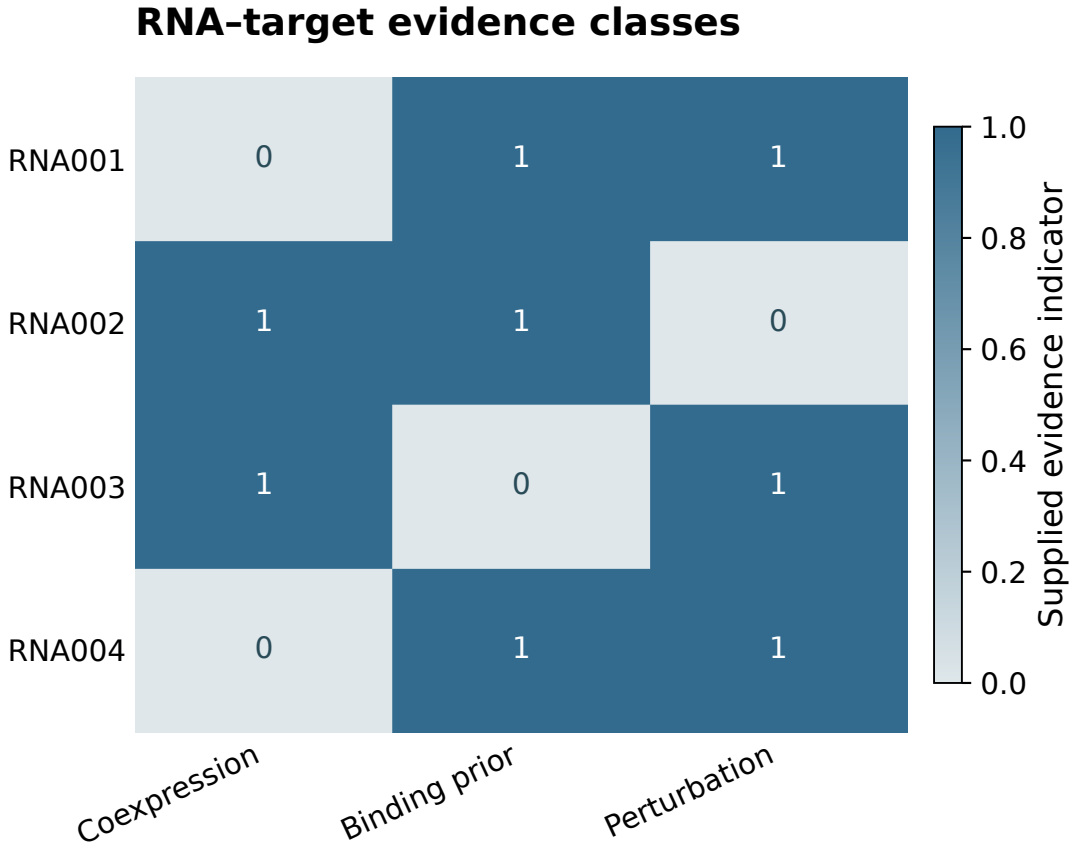
0.26 [0.08, 0.44] 0.056

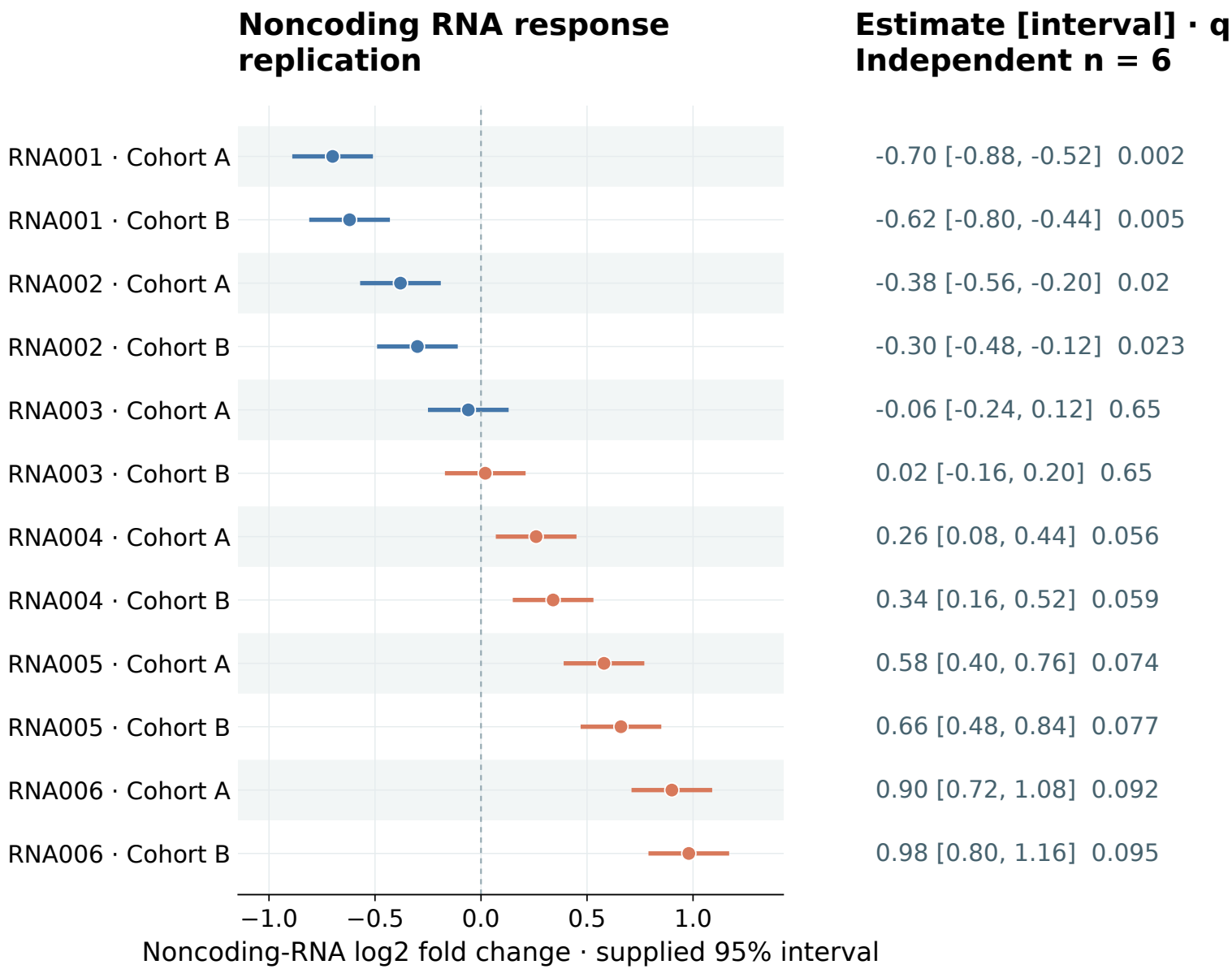
0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092



Software test fixture; not biological evidence

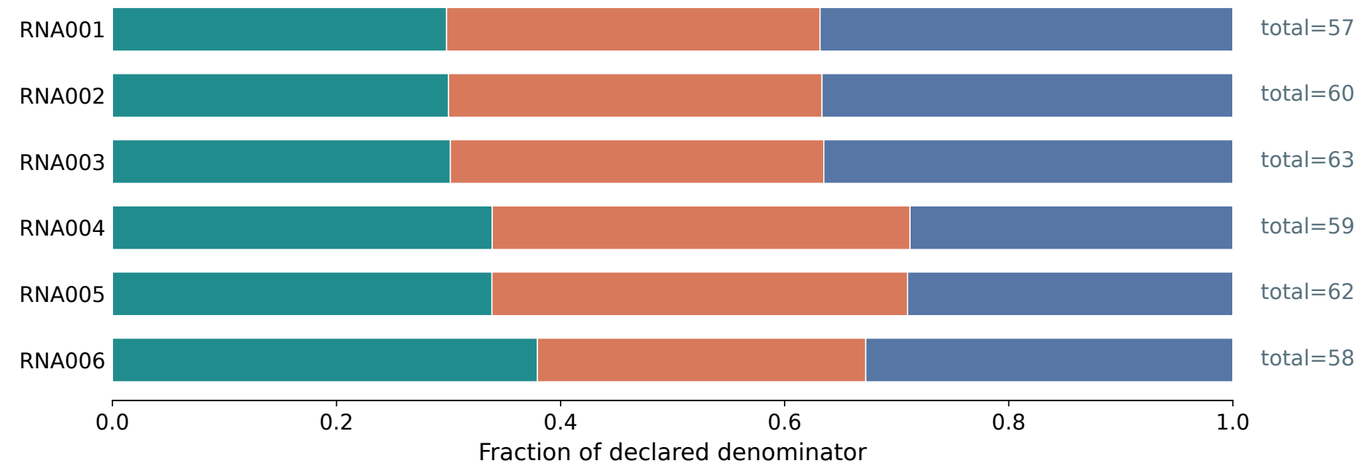

RNA–target evidence classes
RNA001
RNA002
RNA003
RNA004
0
1
1
1
1
0
1
0
1
0
1
1
Coexpression
Binding prior
Perturbation
Supplied evidence indicator
1.0
0.8
0.6
0.4
0.2
0.0


Noncoding RNA response replication
Estimate [interval] · q
Independent n = 6
RNA001 · Cohort A
RNA001 · Cohort B
RNA002 · Cohort A
RNA002 · Cohort B
RNA003 · Cohort A
RNA003 · Cohort B
RNA004 · Cohort A
RNA004 · Cohort B
RNA005 · Cohort A
RNA005 · Cohort B
RNA006 · Cohort A
RNA006 · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Noncoding-RNA log2 fold change · supplied 95% interval

Bulk · noncoding — complementary evidence

A

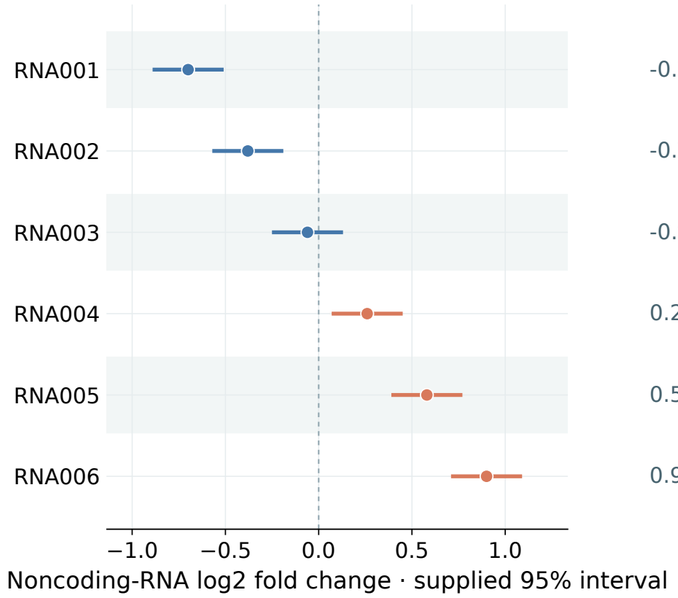
RNA-biotype annotation and mapping



B

Prespecified noncoding RNA effects

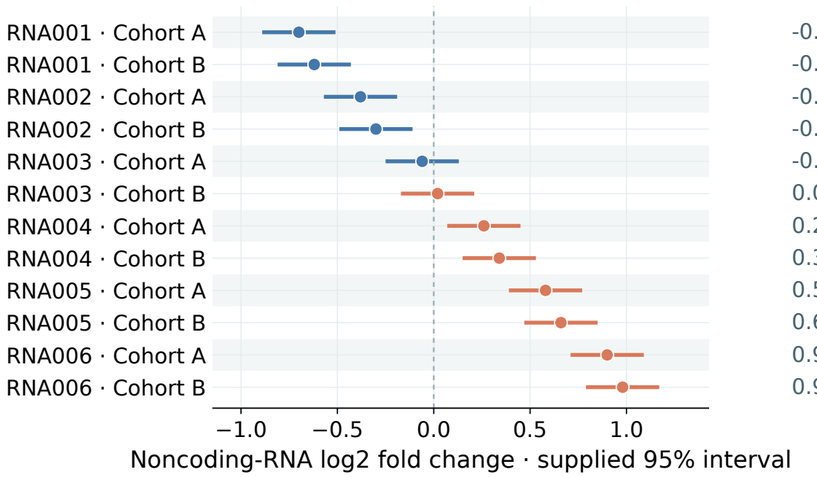
Estimate [interval] · q
Independent n = 12



C

Noncoding RNA response replication

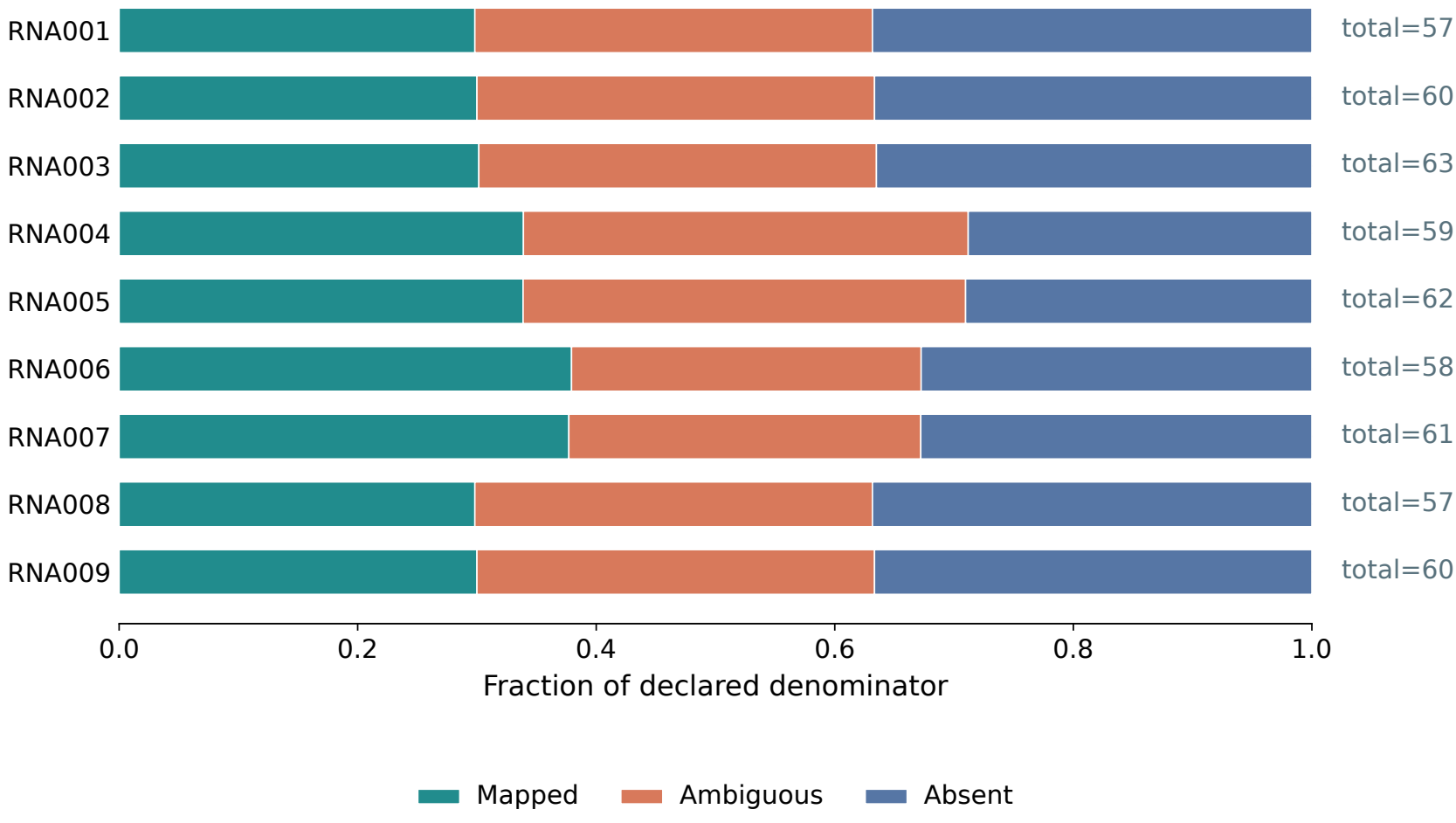
Estimate [interval] · q
Independent n = 6



Bulk · noncoding — complementary evidence

A

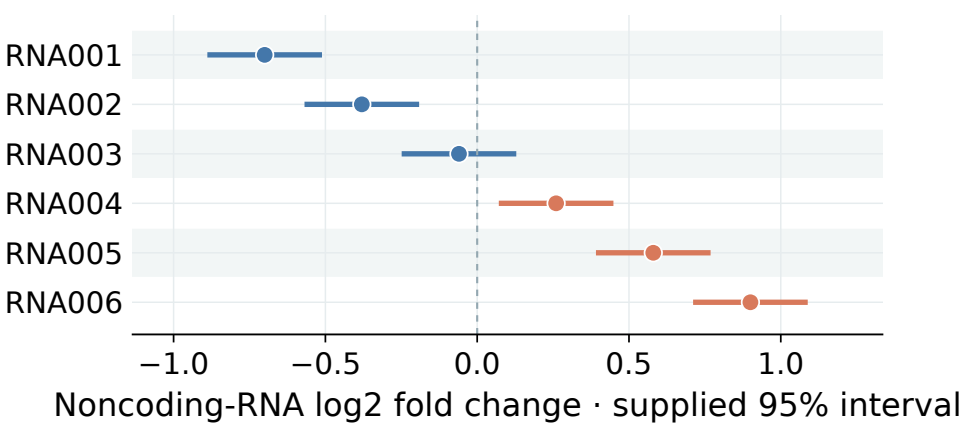
RNA-biotype annotation and mapping



B

Prespecified noncoding RNA effects

Estimate [interval] · q
Independent n = 24

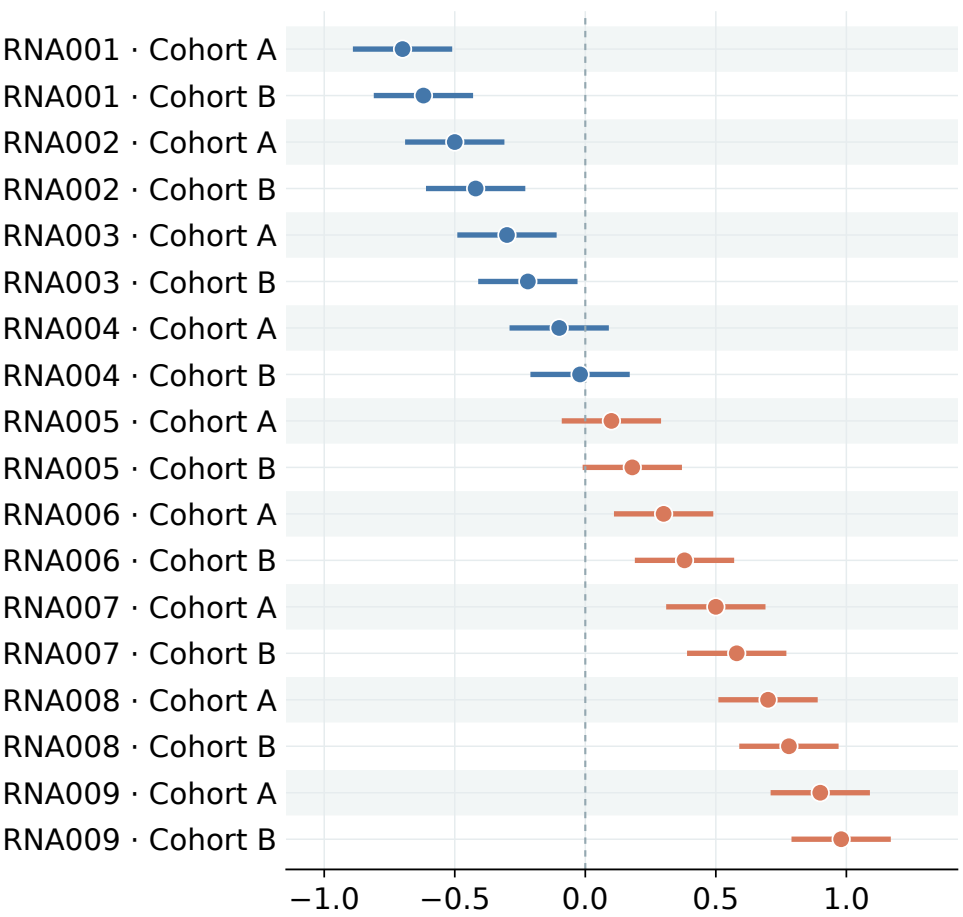


-0.70 [-0.88, -0.52] 0.002
-0.38 [-0.56, -0.20] 0.02
-0.06 [-0.24, 0.12] 0.65
0.26 [0.08, 0.44] 0.056
0.58 [0.40, 0.76] 0.074
0.90 [0.72, 1.08] 0.092

C

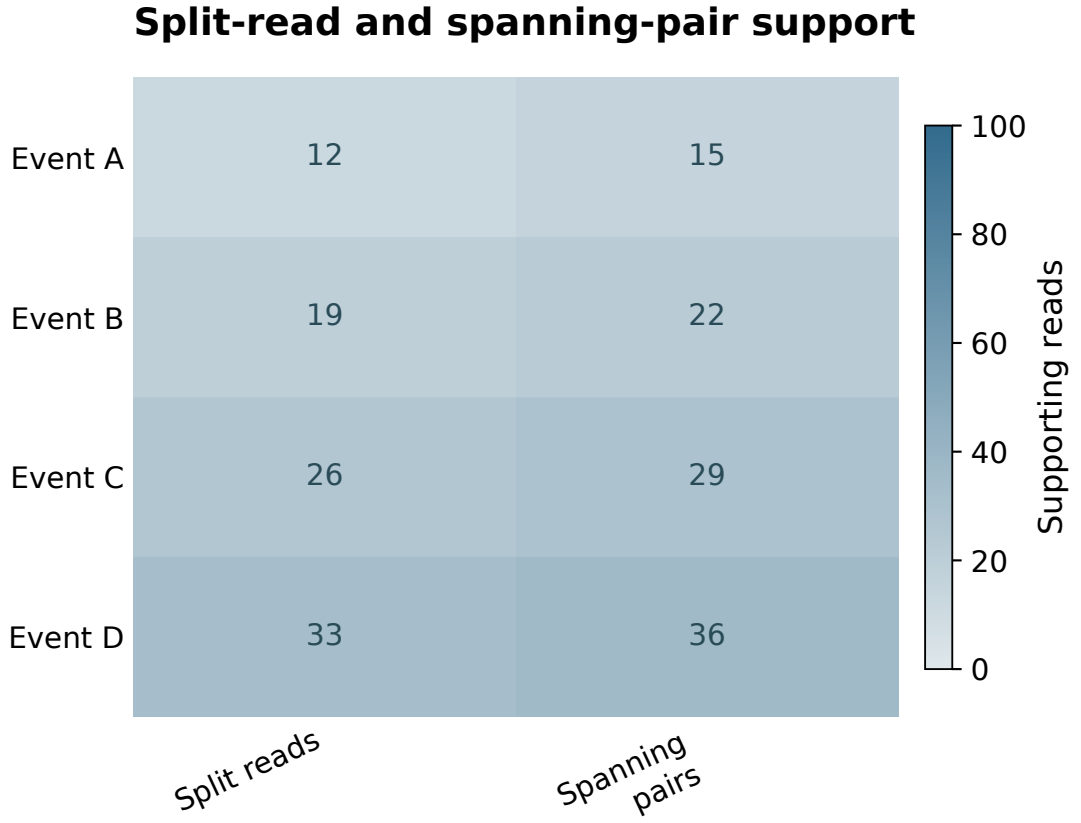
Noncoding RNA response replication

Estimate [interval] · q
Independent n = 12

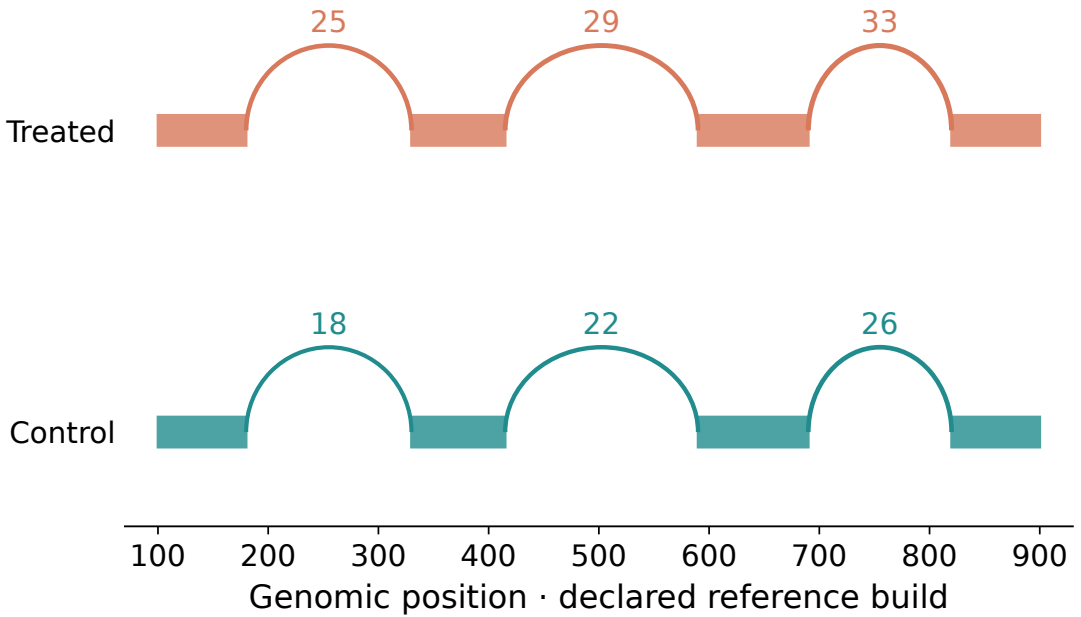


-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.50 [-0.68, -0.32] 0.02
-0.42 [-0.60, -0.24] 0.023
-0.30 [-0.48, -0.12] 0.038
-0.22 [-0.40, -0.04] 0.041
-0.10 [-0.28, 0.08] 0.65
-0.02 [-0.20, 0.16] 0.65
0.10 [-0.08, 0.28] 0.65
0.18 [0.00, 0.36] 0.077
0.30 [0.12, 0.48] 0.092
0.38 [0.20, 0.56] 0.095
0.50 [0.32, 0.68] 0.11
0.58 [0.40, 0.76] 0.11
0.70 [0.52, 0.88] 0.13
0.78 [0.60, 0.96] 0.13
0.90 [0.72, 1.08] 0.15
0.98 [0.80, 1.16] 0.15

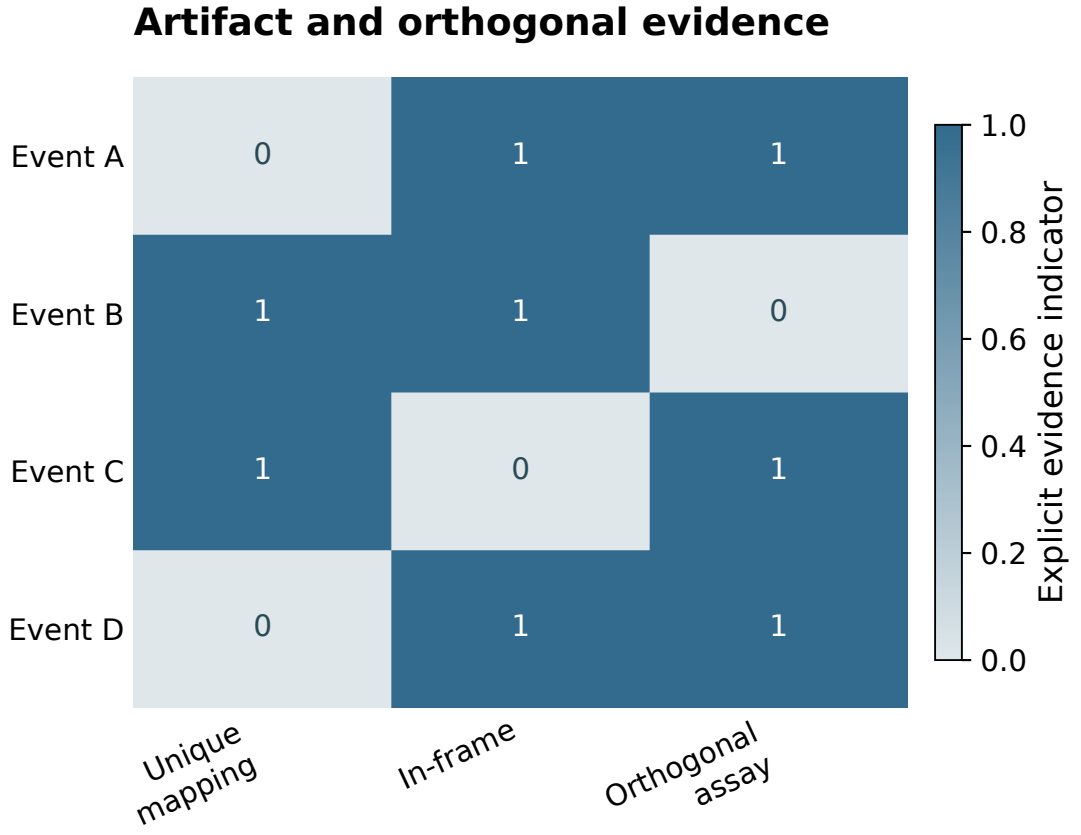
Software test fixture; not biological evidence


Split-read and spanning-pair support
Event A
Event B
Event C
Event D
12
15
19
22
26
29
33
36
Split reads
Spanning pairs
Supporting reads
100
80
60
40
20
0

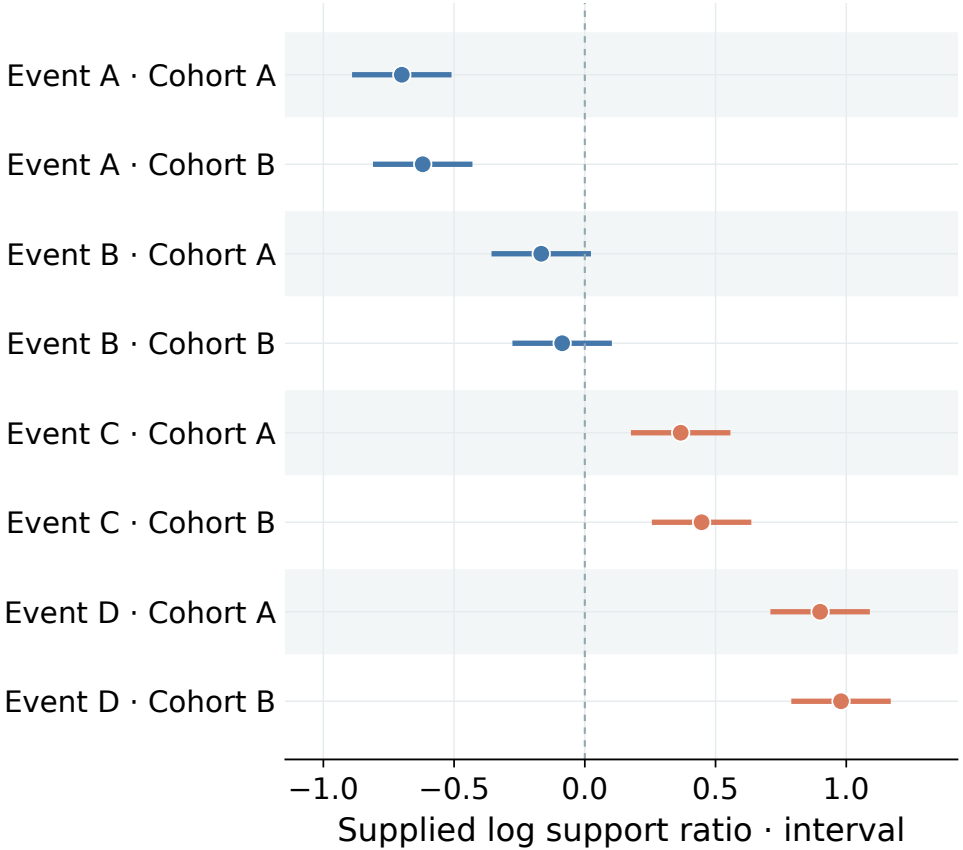
Breakpoint-flanking transcript support




Breakpoint read classes
Event A
Event B
Event C
Event D
total=57
total=60
total=63
total=59
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Unique
Multi-map
Low quality


Artifact and orthogonal evidence
Event A
Event B
Event C
Event D
0
1
1
1
1
0
1
0
1
0
1
1
Unique mapping
In-frame
Orthogonal assay
Explicit evidence indicator
1.0
0.8
0.6
0.4
0.2
0.0

Event support in independent libraries



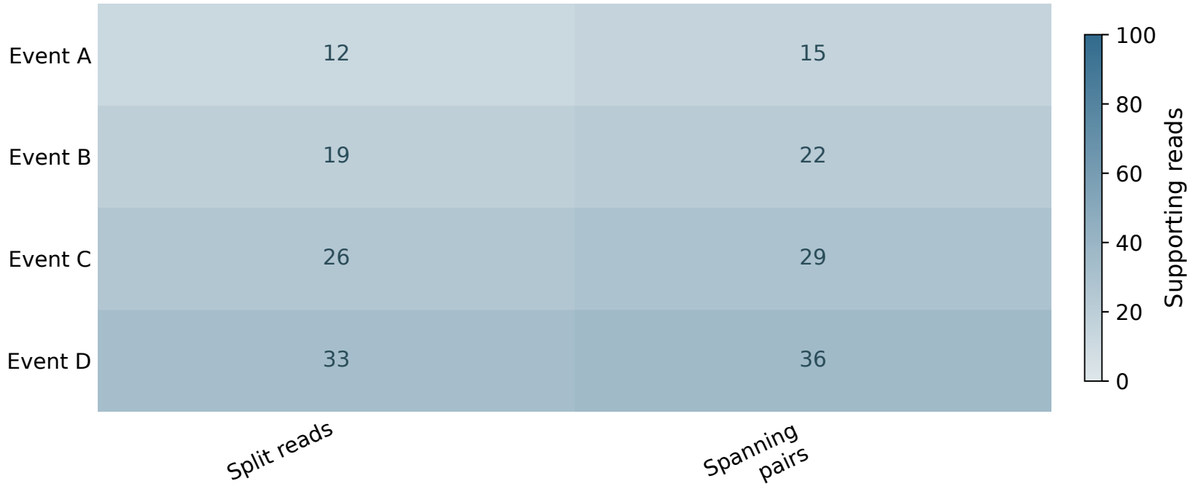
Estimate [interval] · q
Independent n = 6

Event A · Cohort A	-0.70 [-0.88, -0.52] 0.002
Event A · Cohort B	-0.62 [-0.80, -0.44] 0.005
Event B · Cohort A	-0.17 [-0.35, 0.01] 0.65
Event B · Cohort B	-0.09 [-0.27, 0.09] 0.65
Event C · Cohort A	0.37 [0.19, 0.55] 0.038
Event C · Cohort B	0.45 [0.27, 0.63] 0.041
Event D · Cohort A	0.90 [0.72, 1.08] 0.056
Event D · Cohort B	0.98 [0.80, 1.16] 0.059

Bulk · fusions — complementary evidence

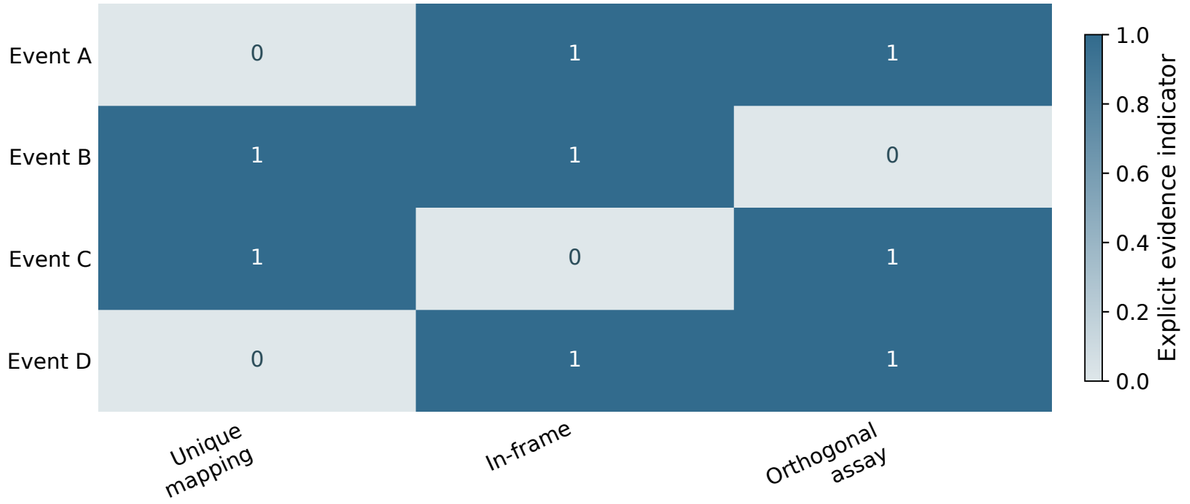
A

Split-read and spanning-pair support



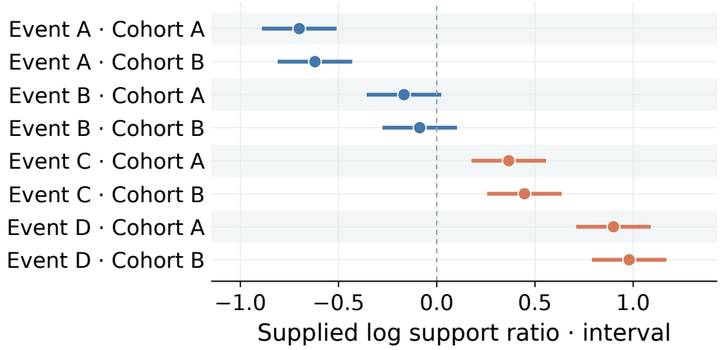
B

Artifact and orthogonal evidence



C

Event support in independent libraries



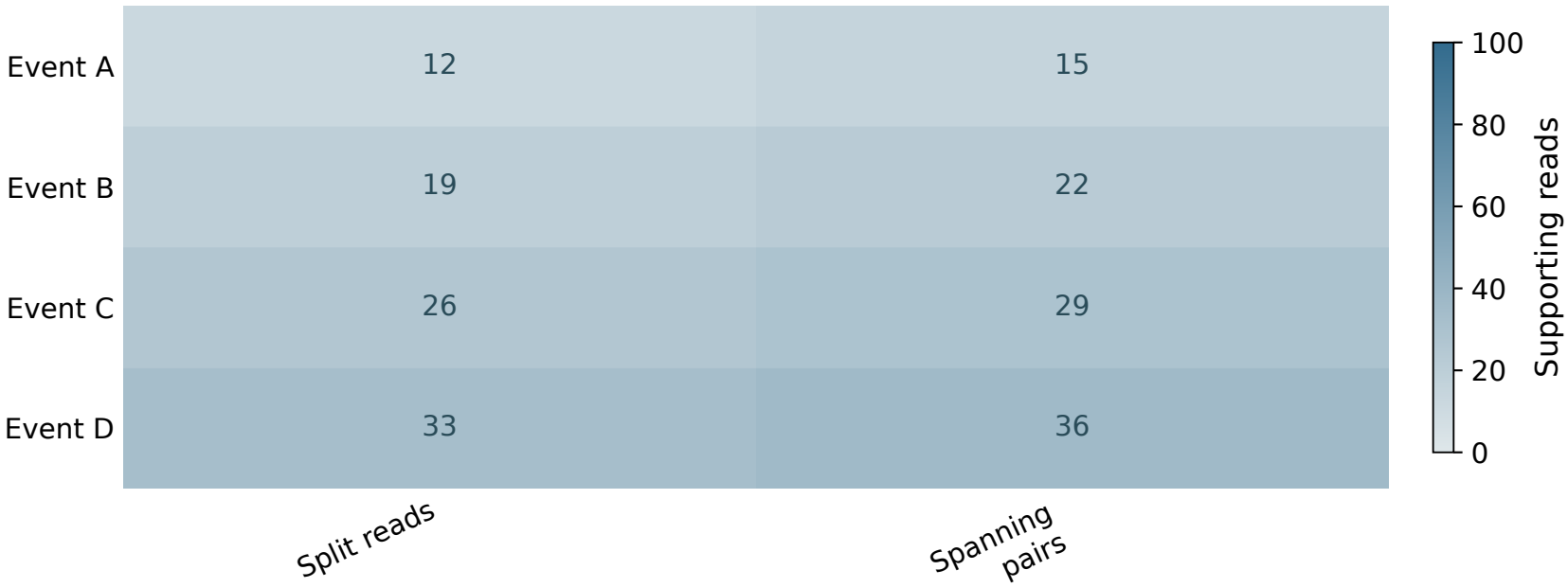
Estimate [interval] · q
Independent n = 6

-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.17 [-0.35, 0.01] 0.65
-0.09 [-0.27, 0.09] 0.65
0.37 [0.19, 0.55] 0.038
0.45 [0.27, 0.63] 0.041
0.90 [0.72, 1.08] 0.056
0.98 [0.80, 1.16] 0.059

Bulk · fusions — complementary evidence

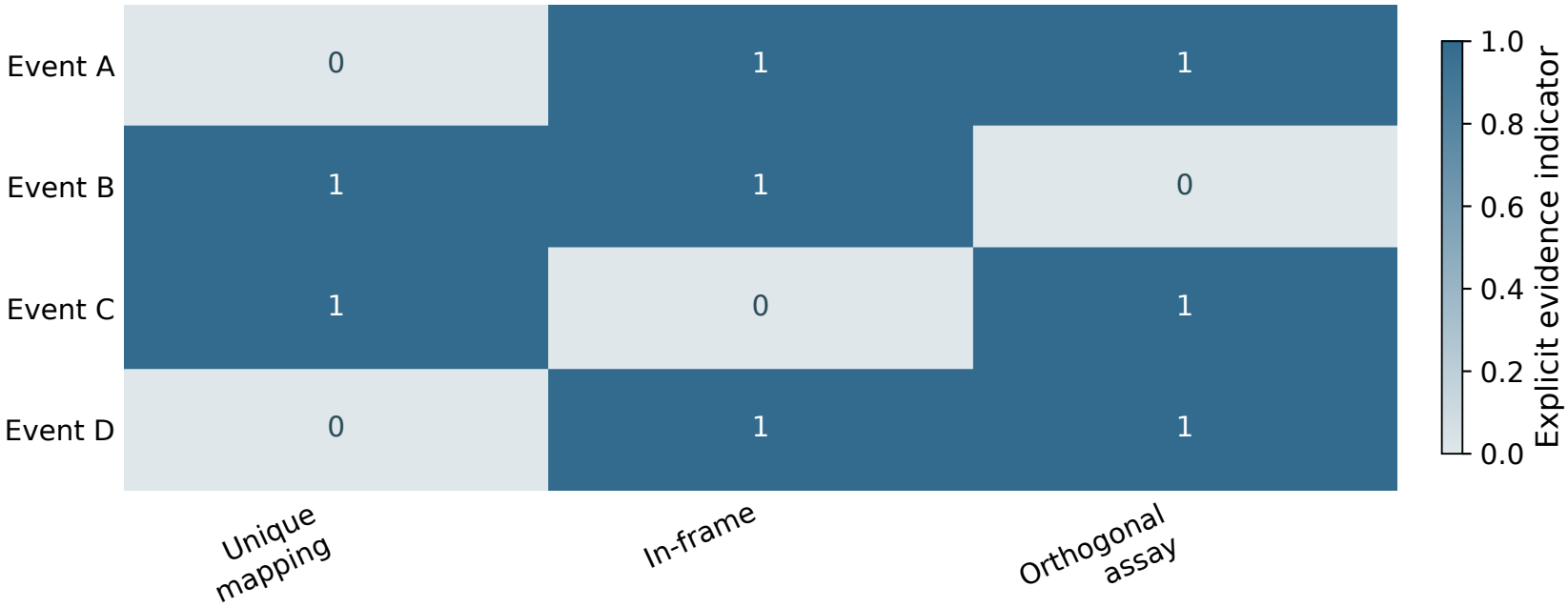
A

Split-read and spanning-pair support



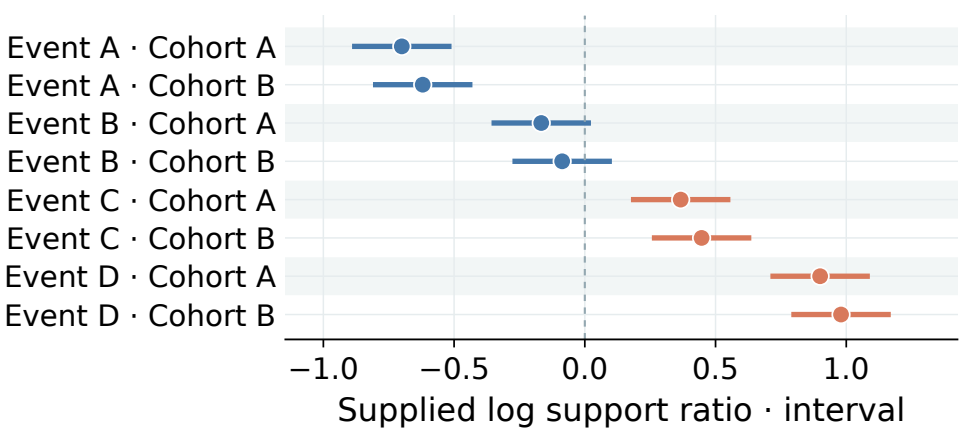
B

Artifact and orthogonal evidence



C

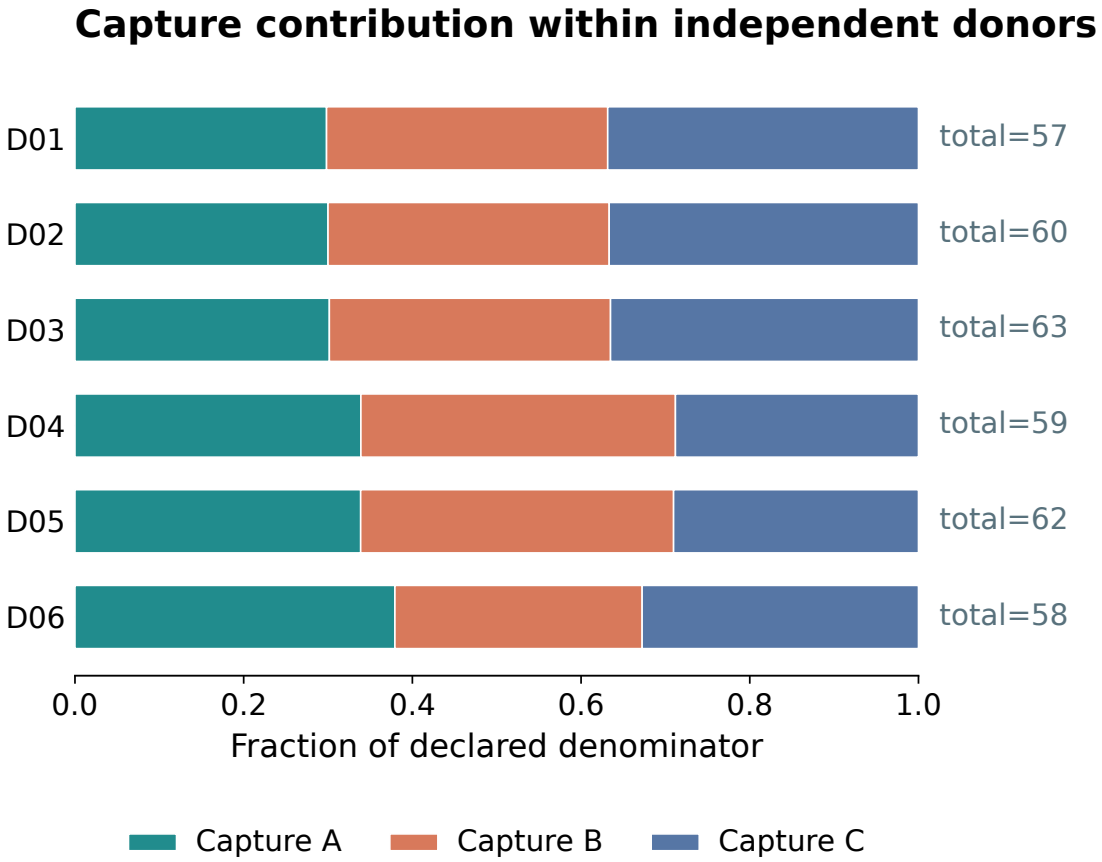
Event support in independent libraries

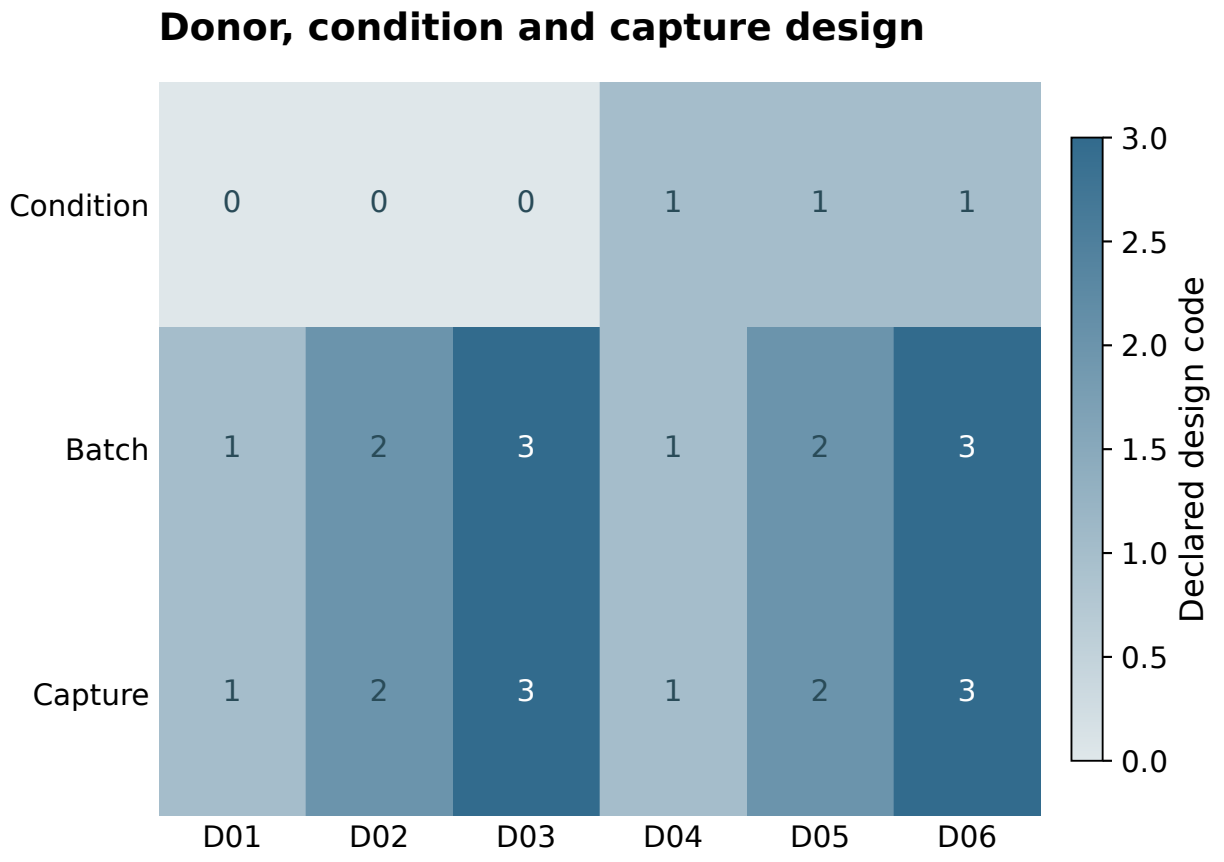


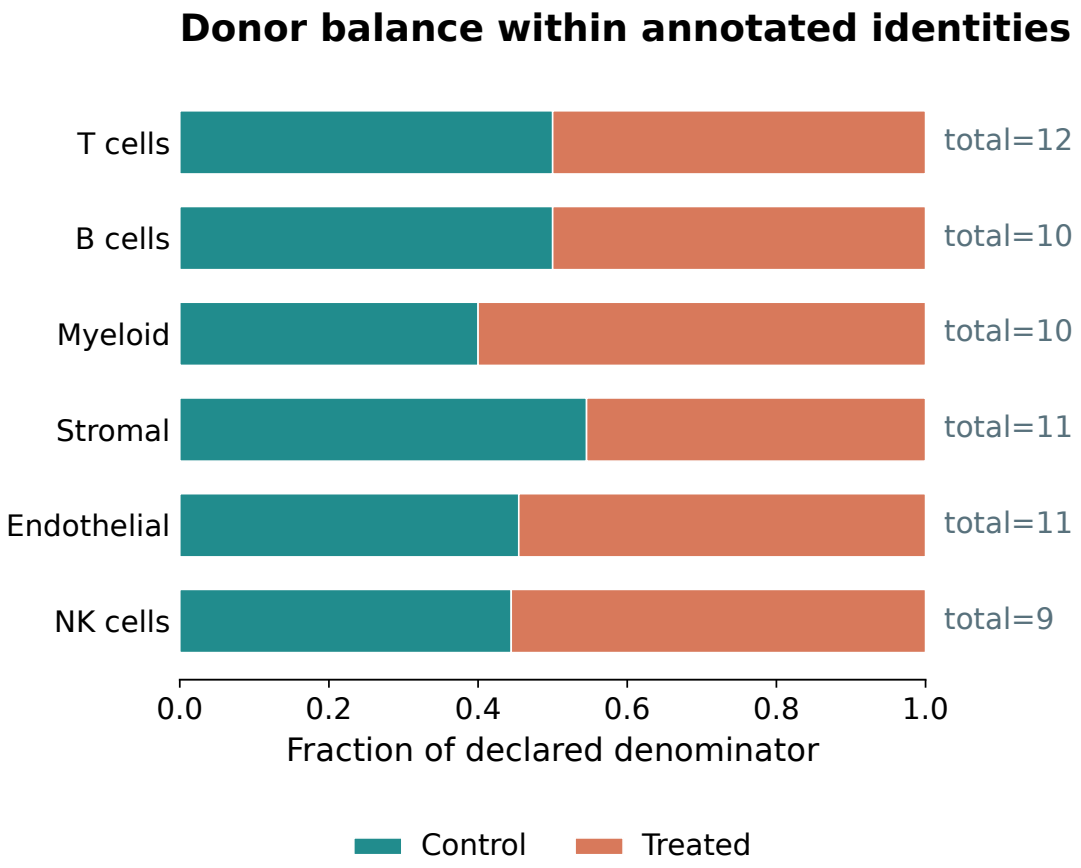
Estimate [interval] · q
Independent n = 12

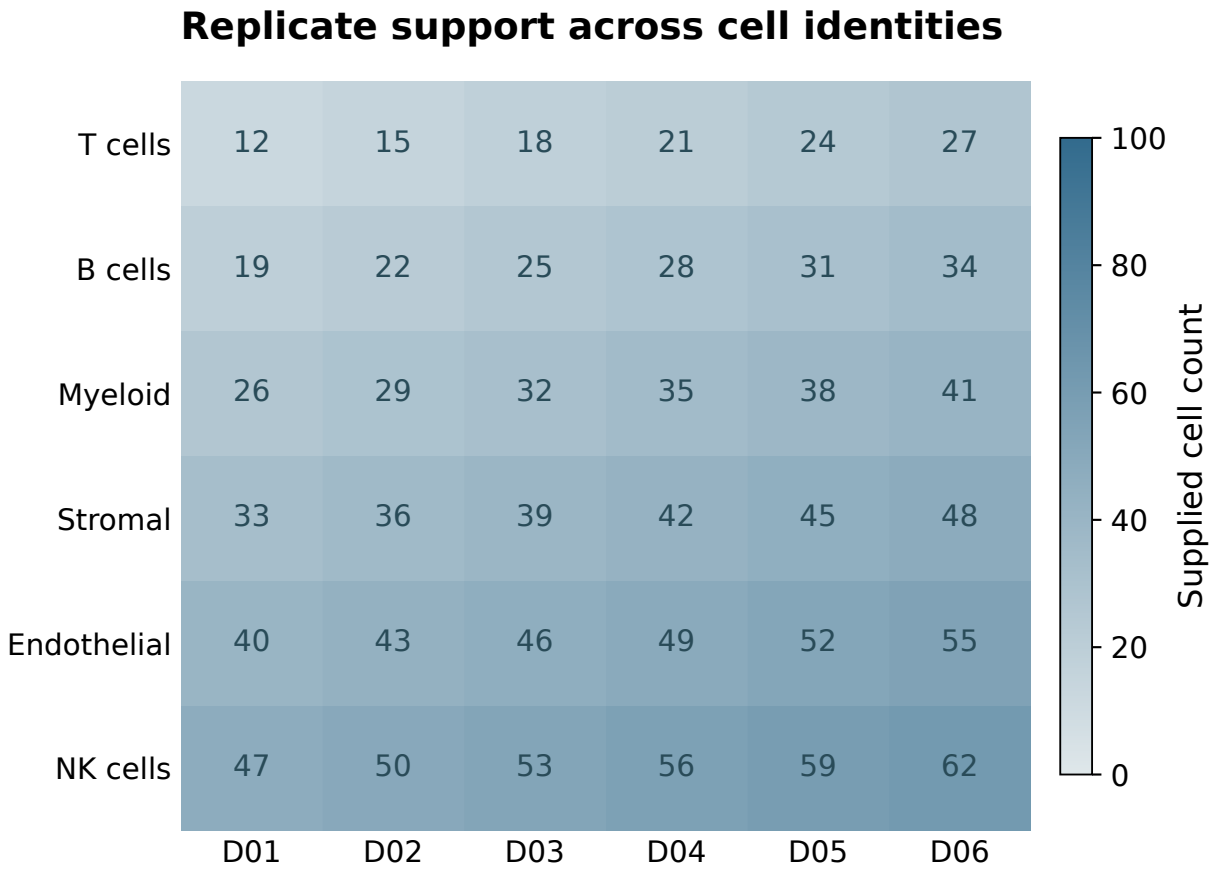
-0.70 [-0.88, -0.52]	0.002
-0.62 [-0.80, -0.44]	0.005
-0.17 [-0.35, 0.01]	0.65
-0.09 [-0.27, 0.09]	0.65
0.37 [0.19, 0.55]	0.038
0.45 [0.27, 0.63]	0.041
0.90 [0.72, 1.08]	0.056
0.98 [0.80, 1.16]	0.059

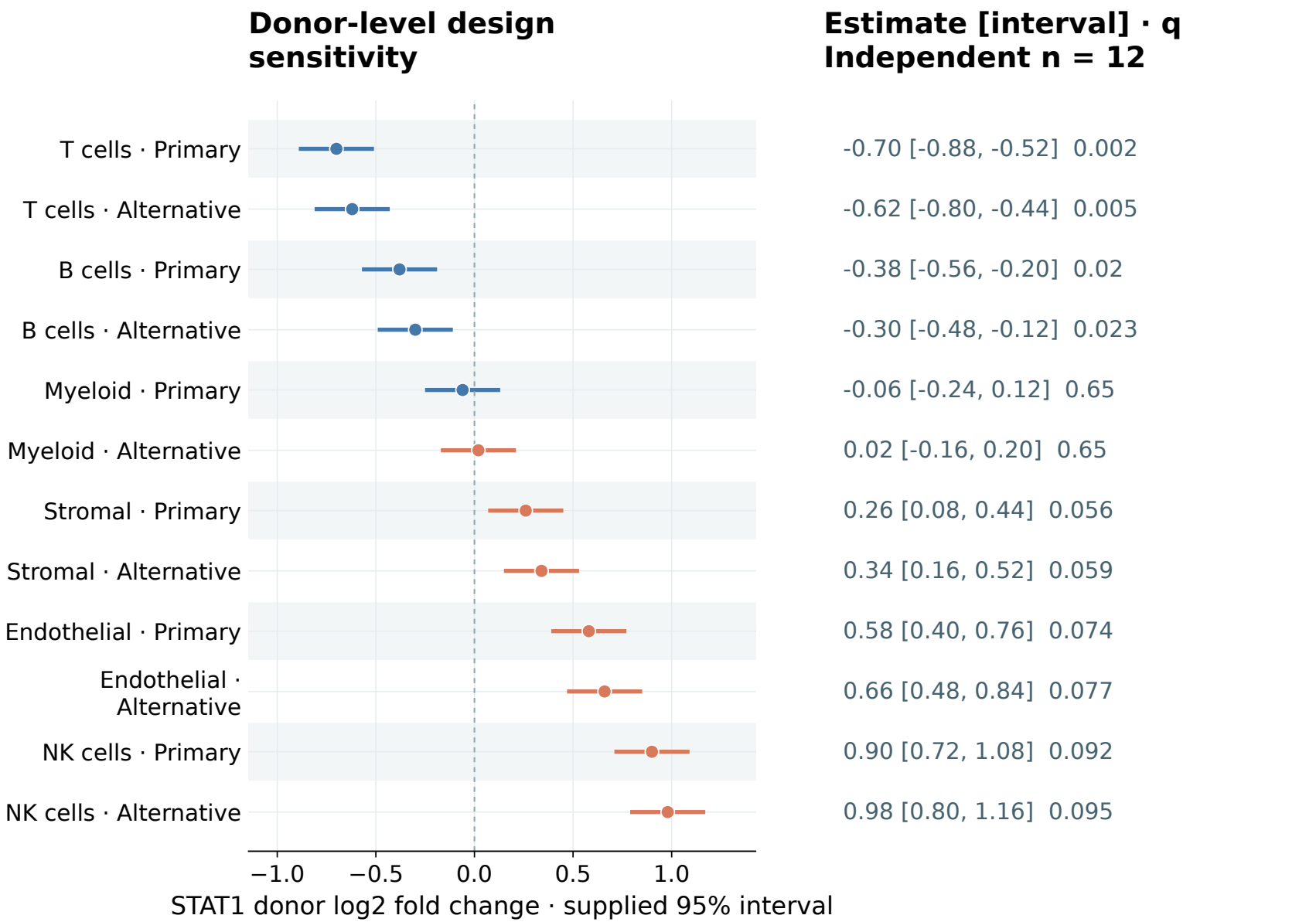
Software test fixture; not biological evidence


Capture contribution within independent donors
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Capture A
Capture B
Capture C


Donor, condition and capture design
Condition
Batch
Capture
0
0
0
1
1
1
1
2
3
1
2
3
1
2
3
1
2
3
D01
D02
D03
D04
D05
D06
Declared design code
3.0
2.5
2.0
1.5
1.0
0.5
0.0


Donor balance within annotated identities
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
total=12
total=10
total=10
total=11
total=11
total=9
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Control
Treated

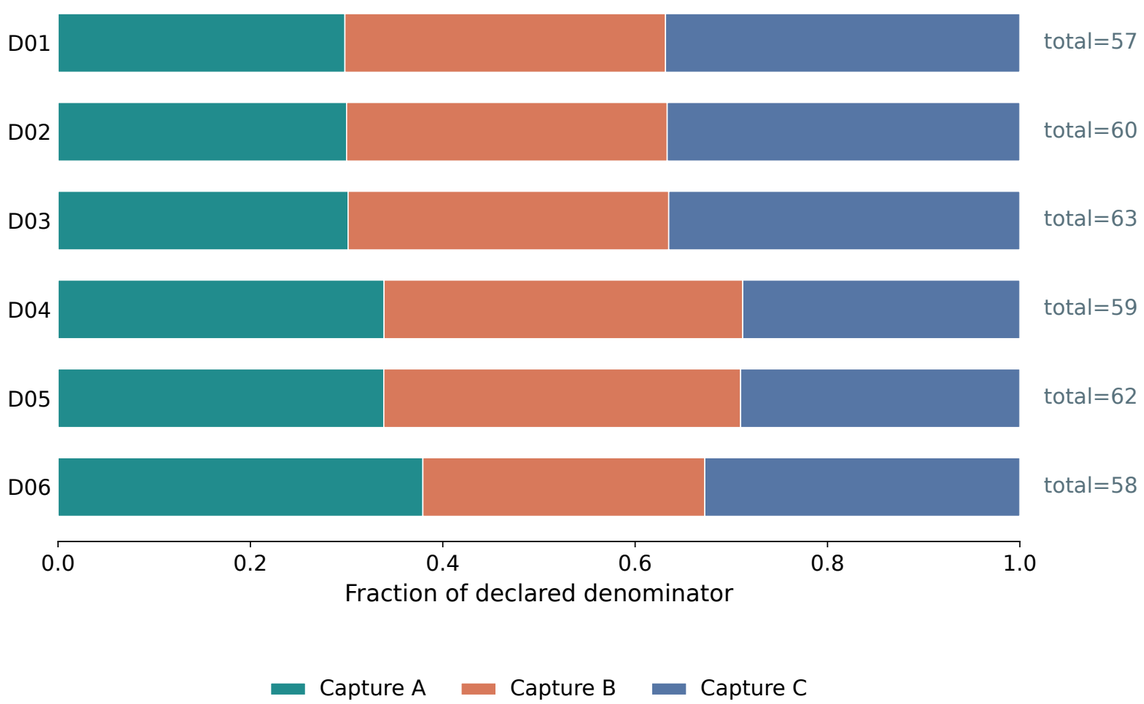

Replicate support across cell identities
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
12 15 18 21 24 27
19 22 25 28 31 34
26 29 32 35 38 41
33 36 39 42 45 48
40 43 46 49 52 55
47 50 53 56 59 62
D01 D02 D03 D04 D05 D06
Supplied cell count
100
80
60
40
20
0


Donor-level design sensitivity
Estimate [interval] · q
Independent n = 12
T cells · Primary
T cells · Alternative
B cells · Primary
B cells · Alternative
Myeloid · Primary
Myeloid · Alternative
Stromal · Primary
Stromal · Alternative
Endothelial · Primary
Endothelial · Alternative
NK cells · Primary
NK cells · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
STAT1 donor log2 fold change · supplied 95% interval

Single cell · design — complementary evidence

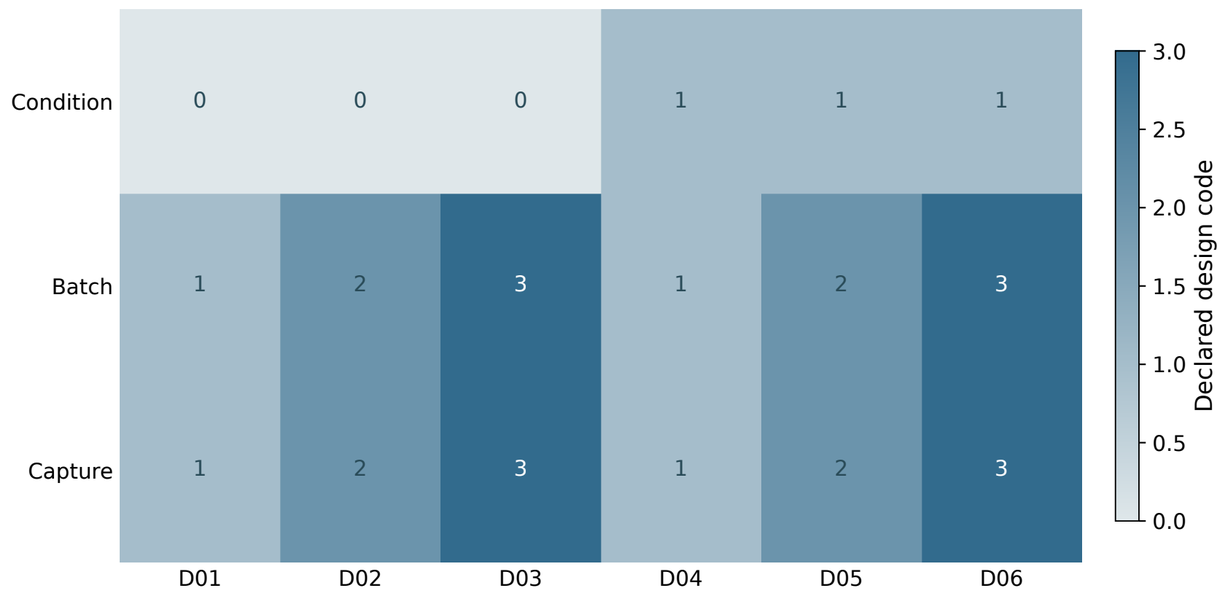
A

Capture contribution within independent donors



B

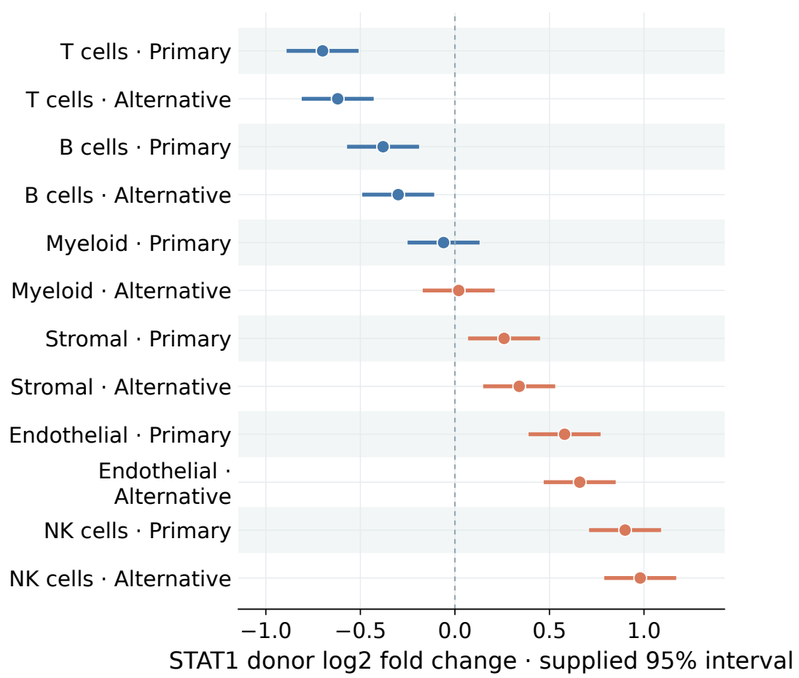
Donor, condition and capture design



C

Donor-level design sensitivity

Estimate [interval] · q
Independent n = 12

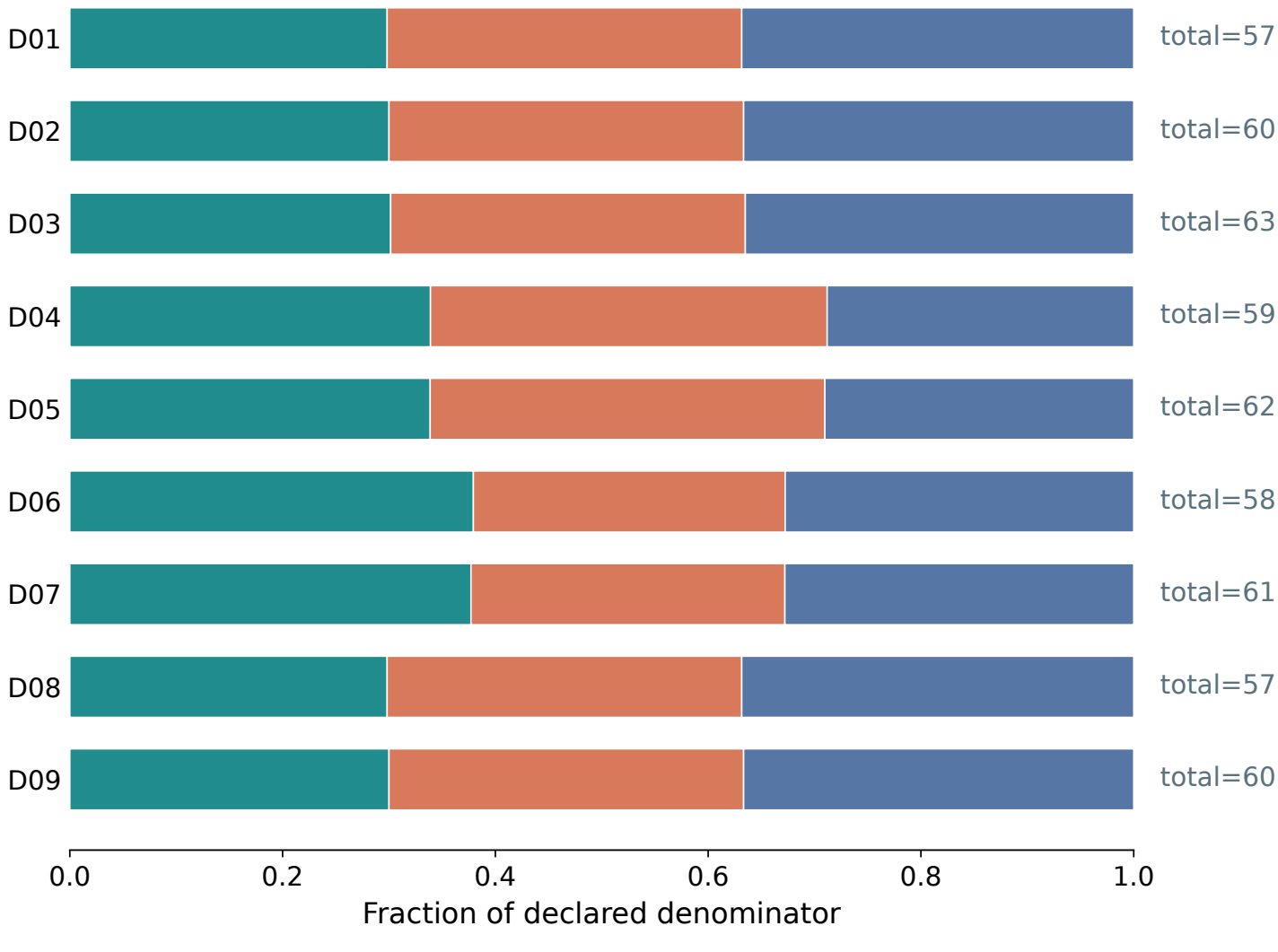


-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095

Single cell · design — complementary evidence

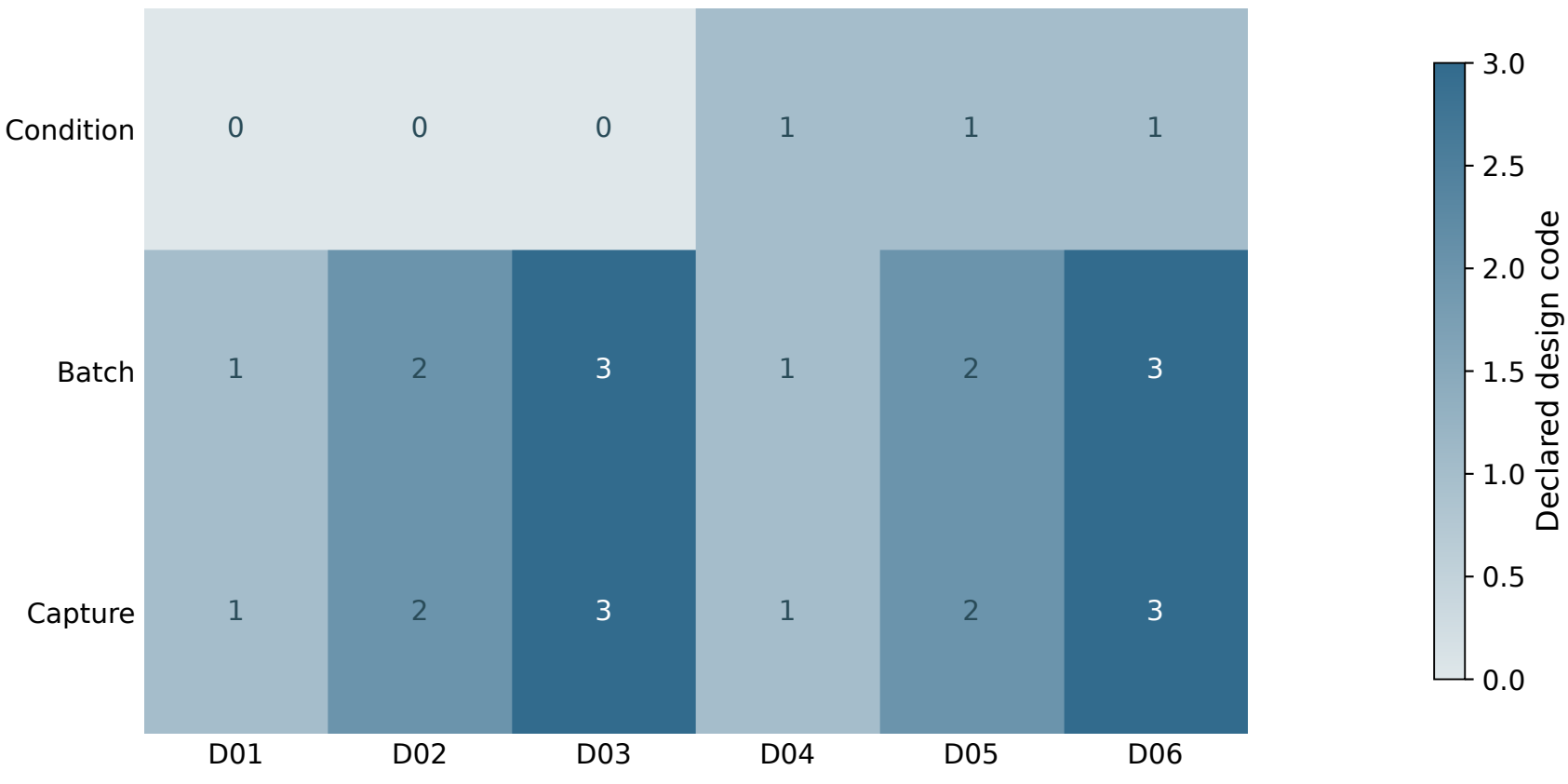
A

Capture contribution within independent donors



B

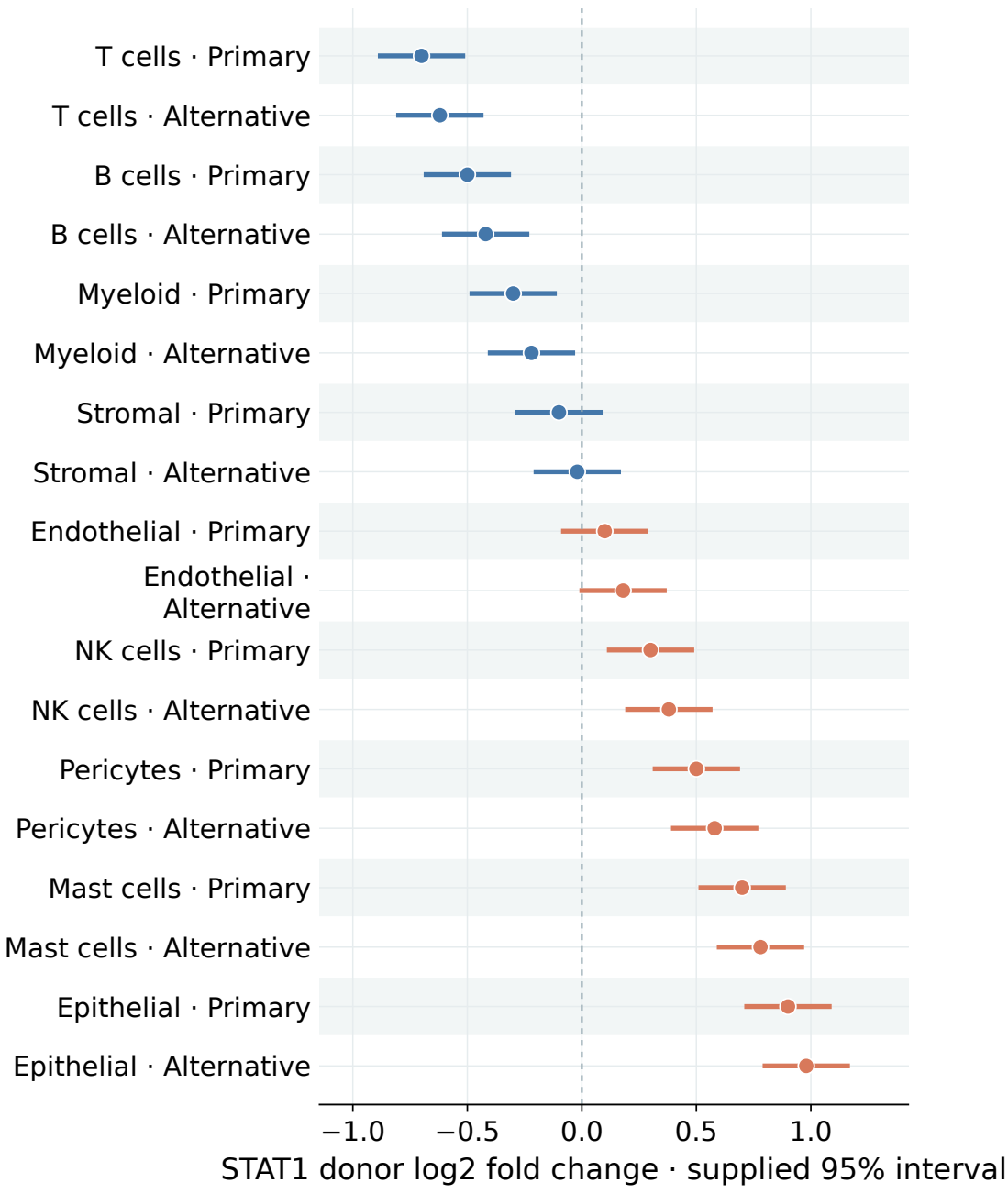
Donor, condition and capture design



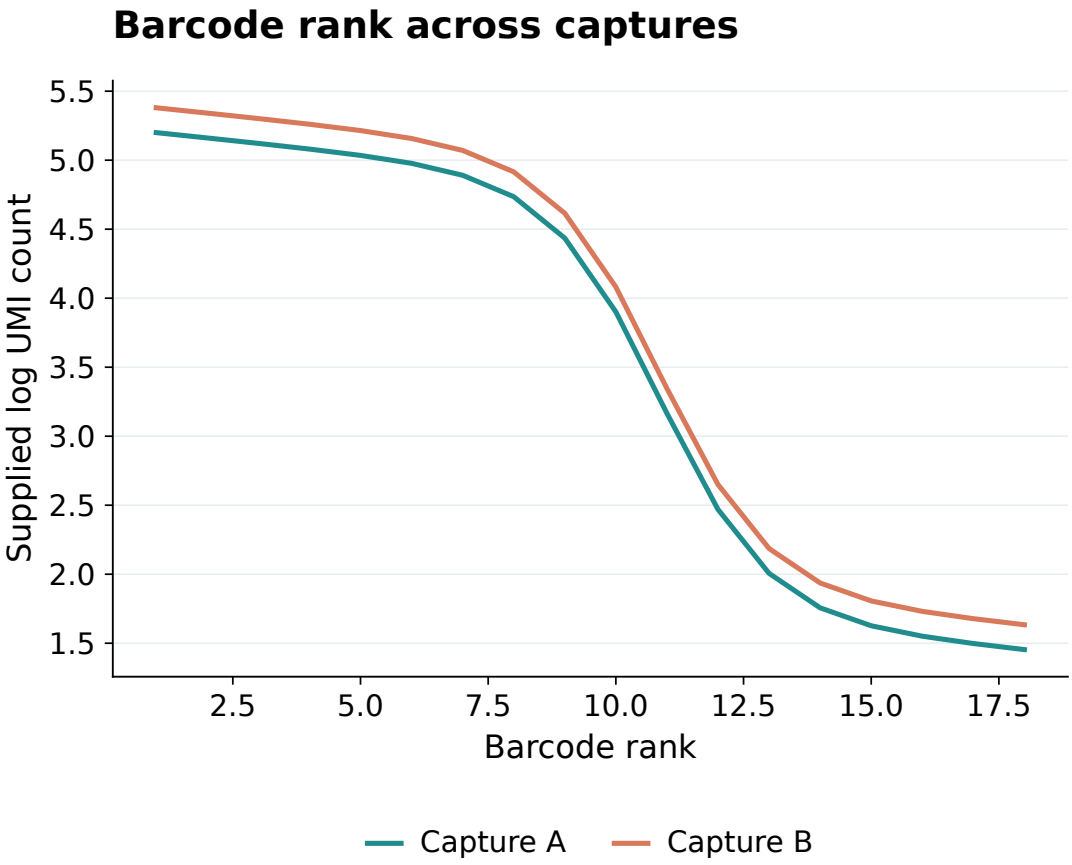
C

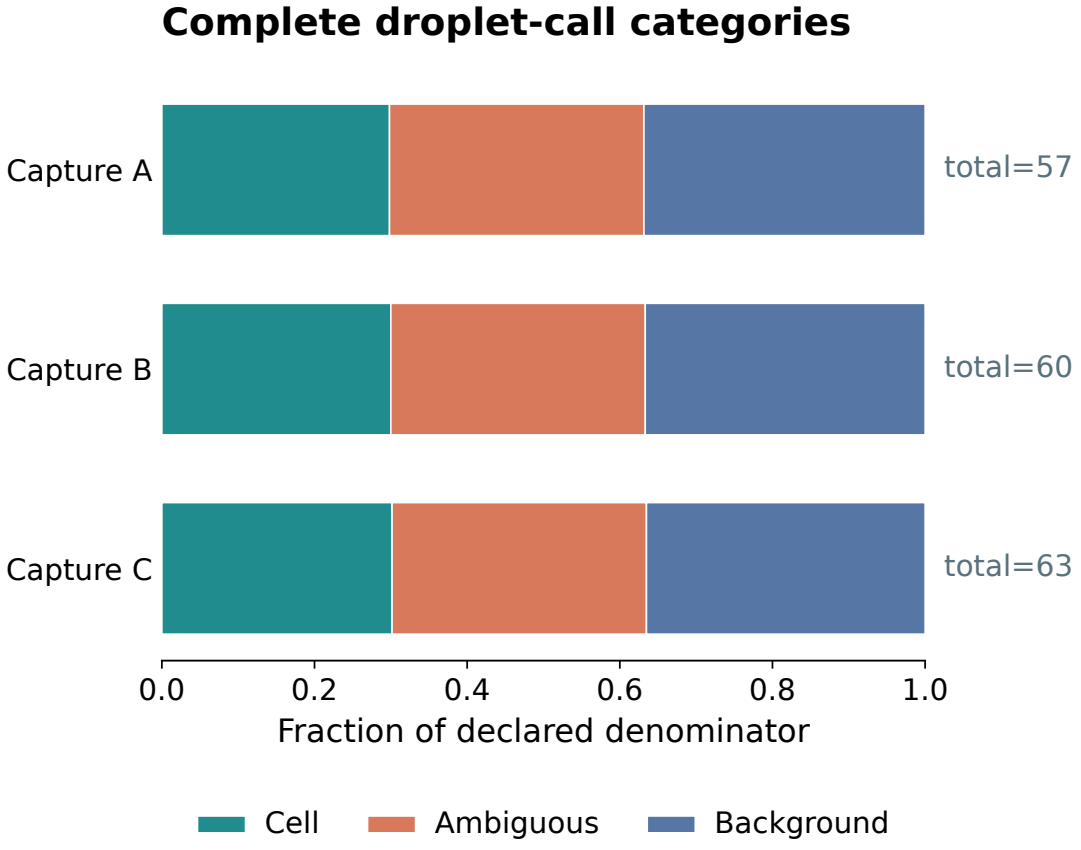
Donor-level design sensitivity

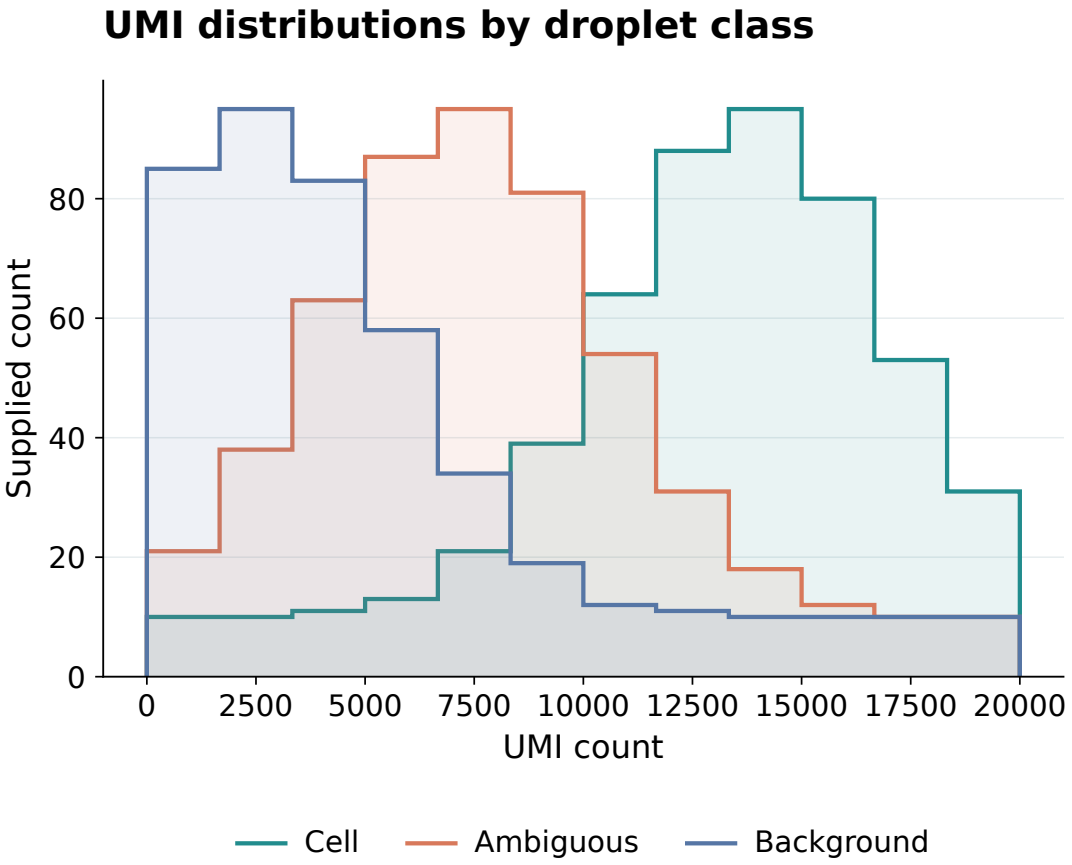
Estimate [interval] · q
Independent n = 24



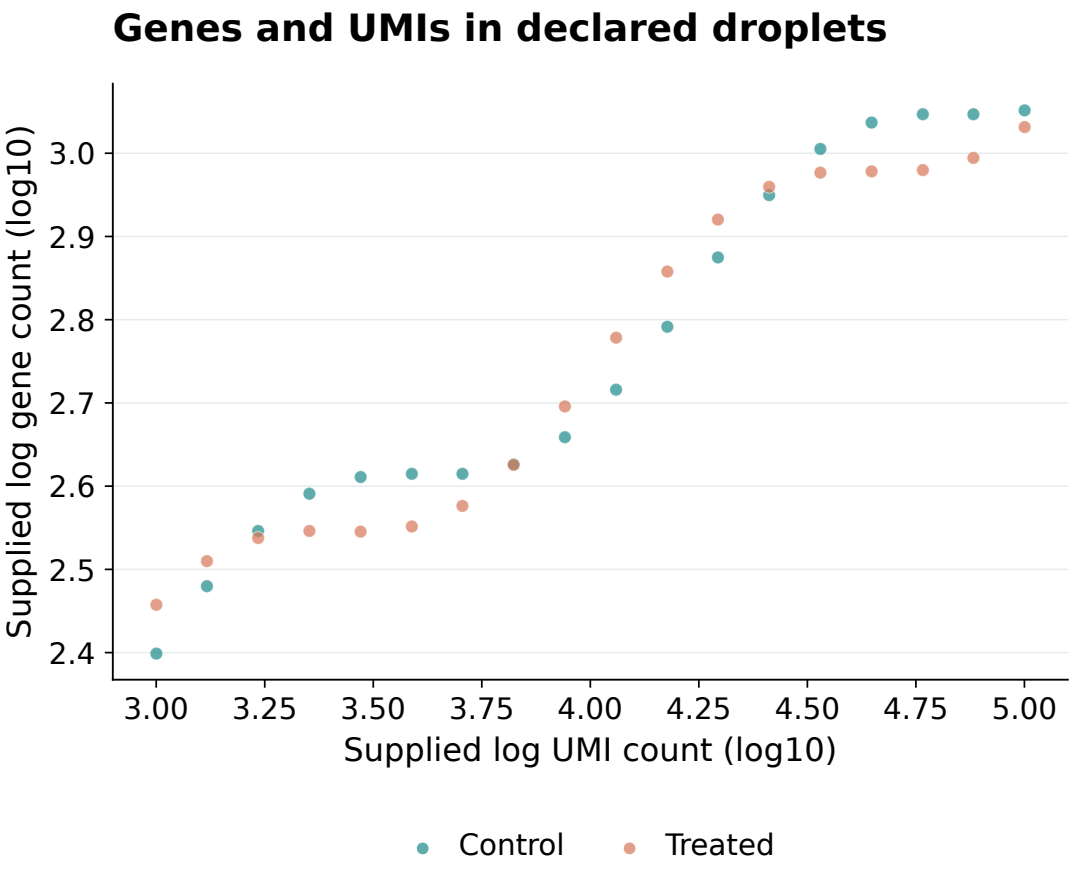
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.50 [-0.68, -0.32] 0.02
-0.42 [-0.60, -0.24] 0.023
-0.30 [-0.48, -0.12] 0.038
-0.22 [-0.40, -0.04] 0.041
-0.10 [-0.28, 0.08] 0.65
-0.02 [-0.20, 0.16] 0.65
0.10 [-0.08, 0.28] 0.65
0.18 [0.00, 0.36] 0.077
0.30 [0.12, 0.48] 0.092
0.38 [0.20, 0.56] 0.095
0.50 [0.32, 0.68] 0.11
0.58 [0.40, 0.76] 0.11
0.70 [0.52, 0.88] 0.13
0.78 [0.60, 0.96] 0.13
0.90 [0.72, 1.08] 0.15
0.98 [0.80, 1.16] 0.15

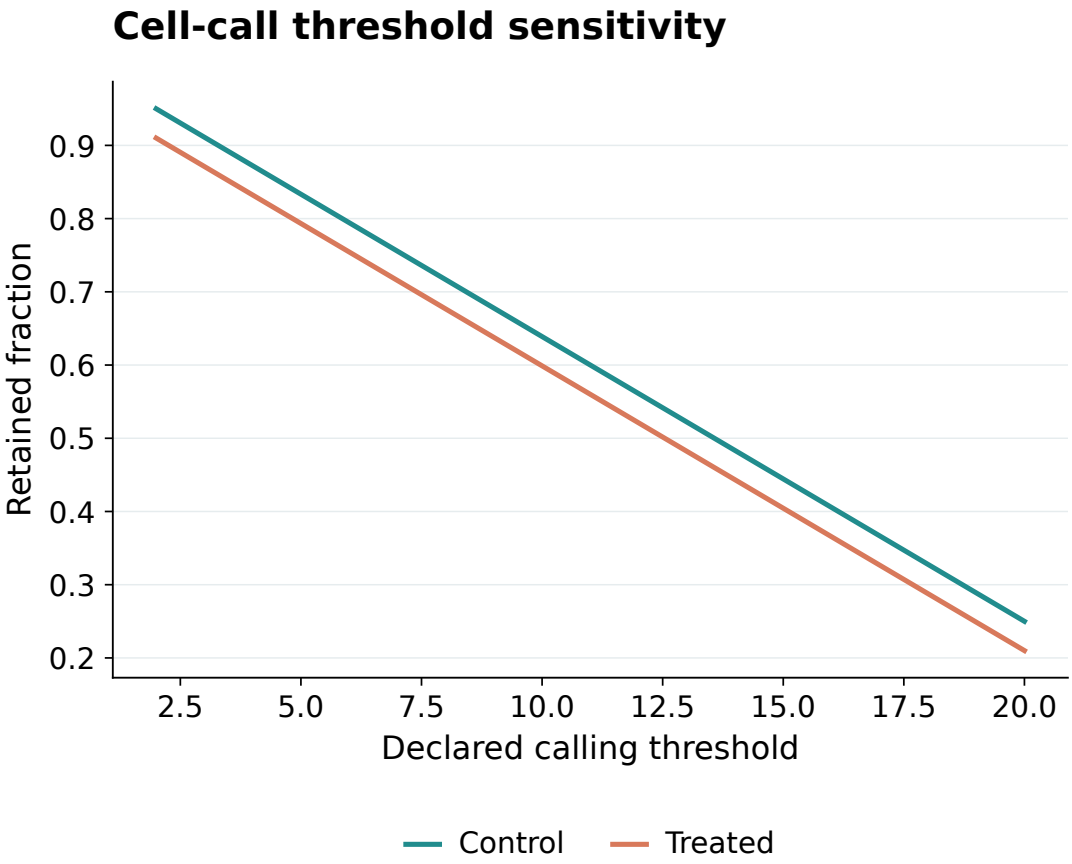

Barcode rank across captures
Supplied log UMI count
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Barcode rank
Capture A
Capture B


Complete droplet-call categories
Capture A
Capture B
Capture C
total=57
total=60
total=63
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Cell
Ambiguous
Background

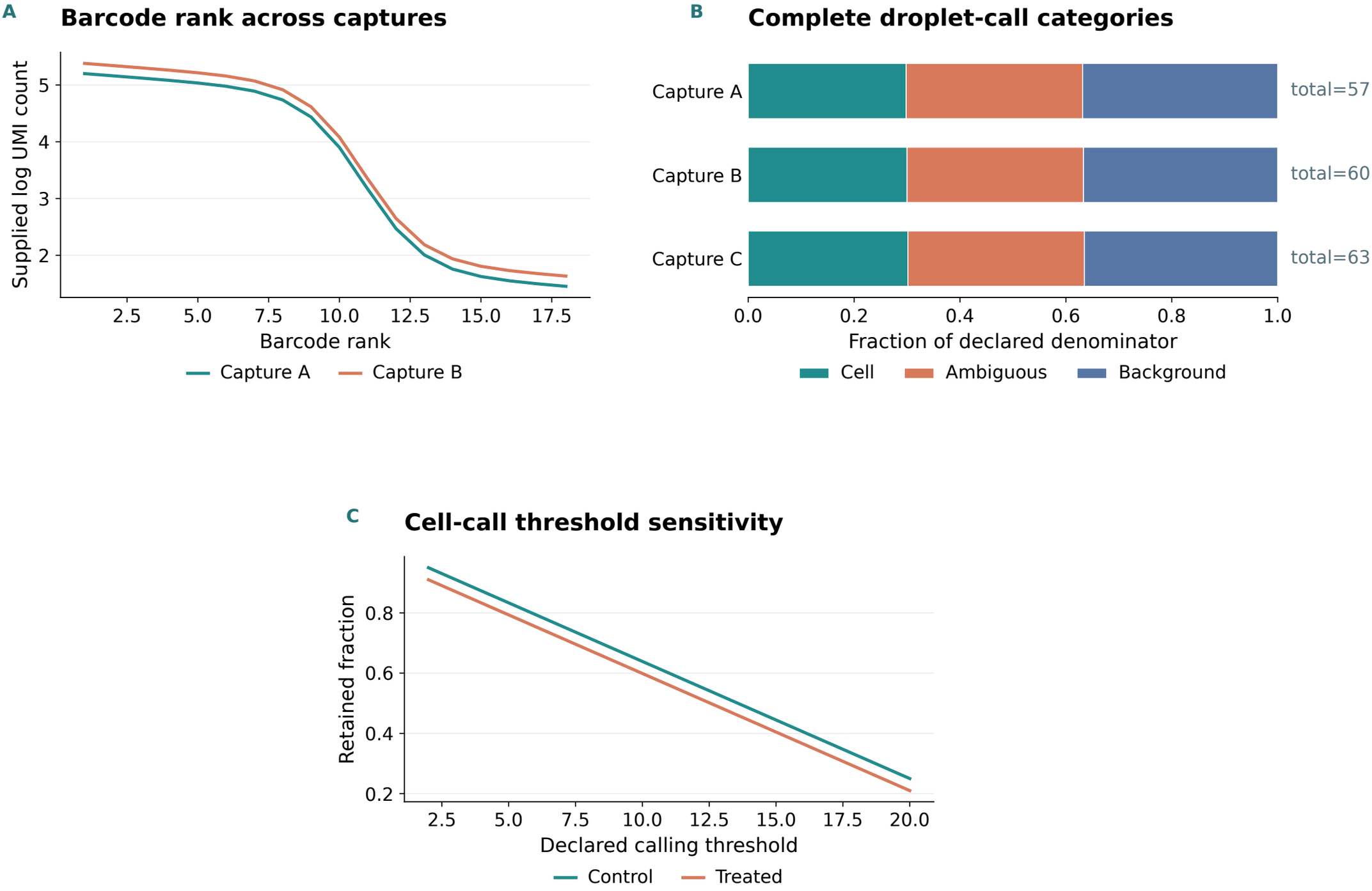


Software test fixture; not biological evidence


Genes and UMIs in declared droplets
Supplied log gene count (log10)
3.0
2.9
2.8
2.7
2.6
2.5
2.4
3.00
3.25
3.50
3.75
4.00
4.25
4.50
4.75
5.00
Supplied log UMI count (log10)
Control
Treated

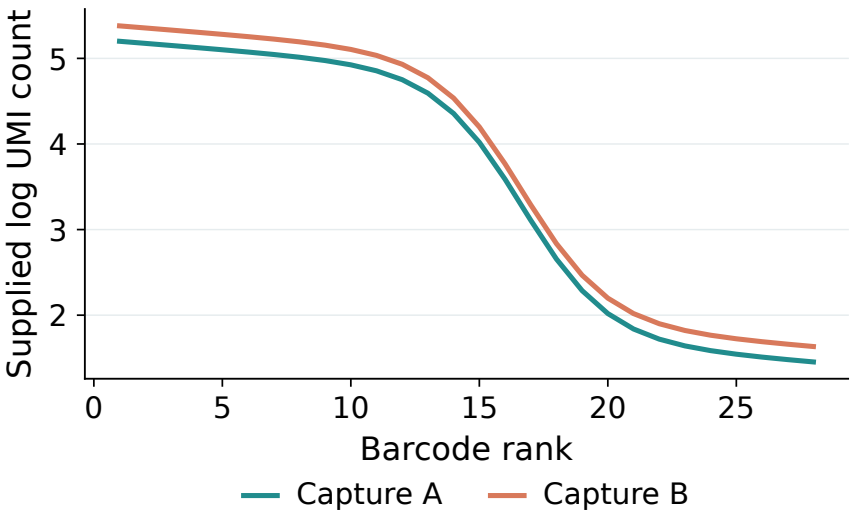

Cell-call threshold sensitivity
Retained fraction
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Declared calling threshold
Control
Treated

Single cell · barcode — complementary evidence

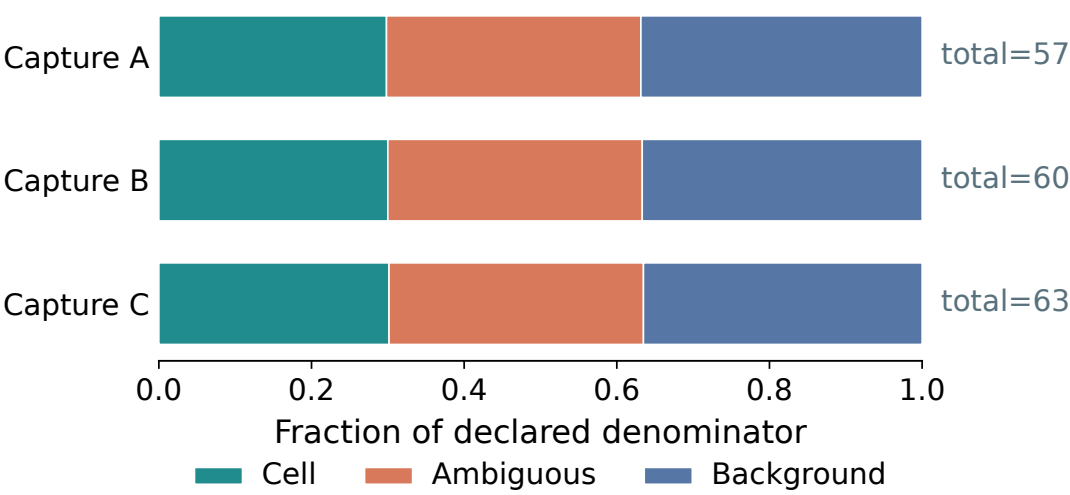


Single cell · barcode — complementary evidence

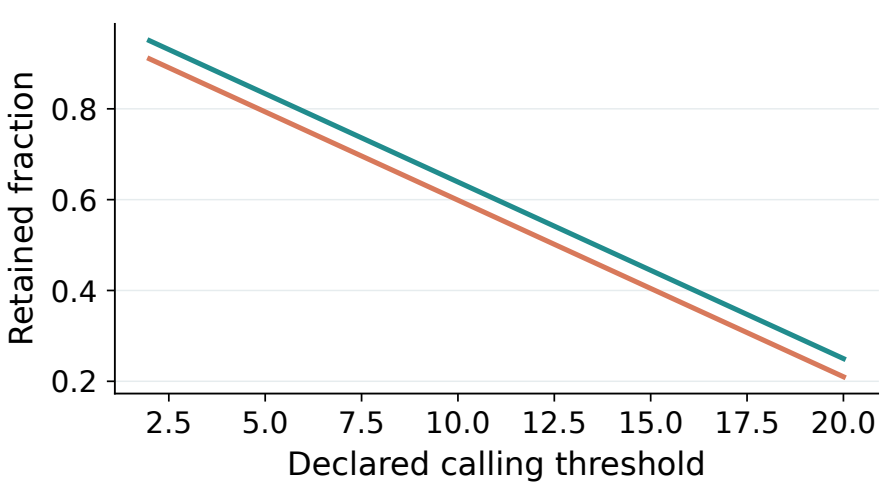
A Barcode rank across captures

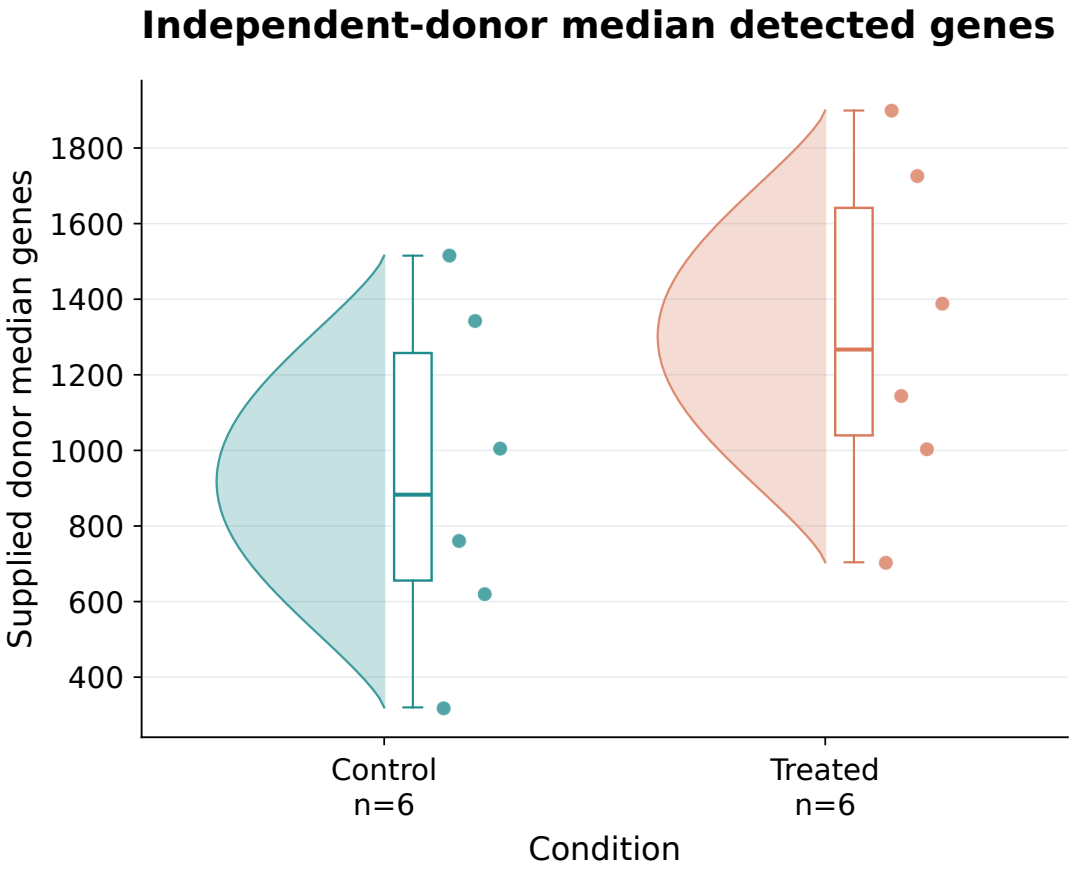


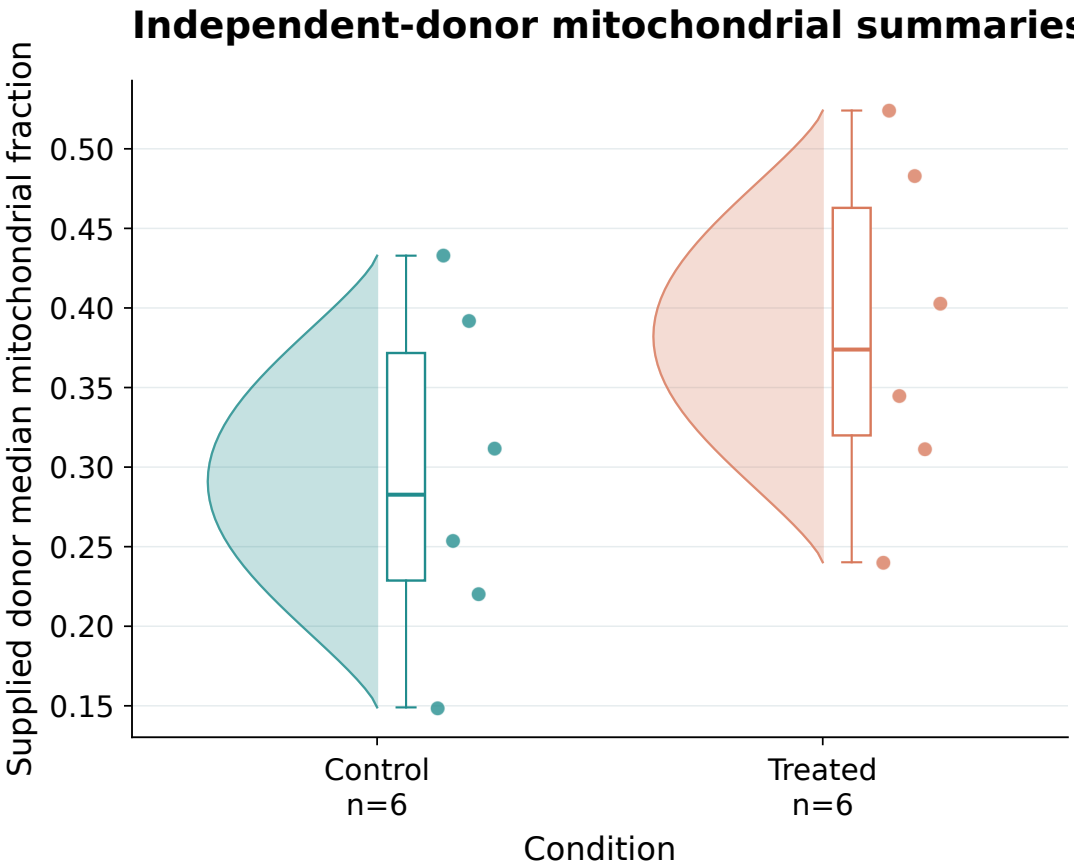
B Complete droplet-call categories

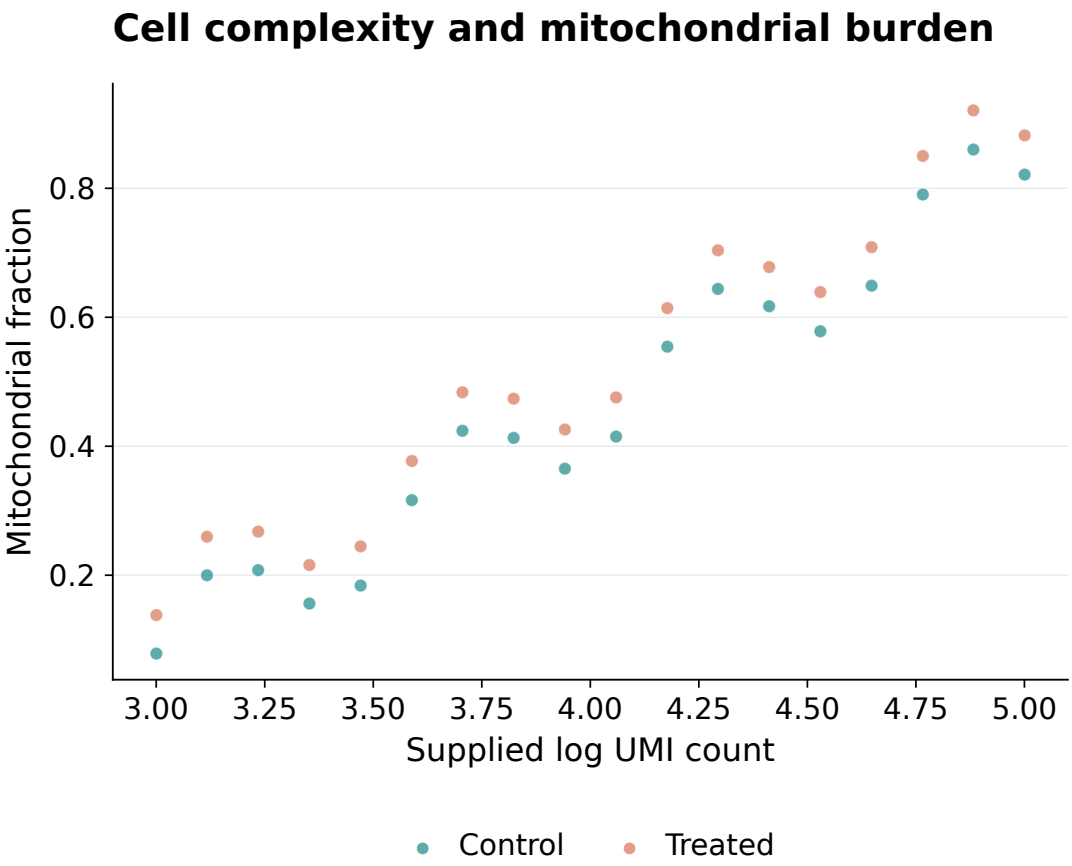


C Cell-call threshold sensitivity

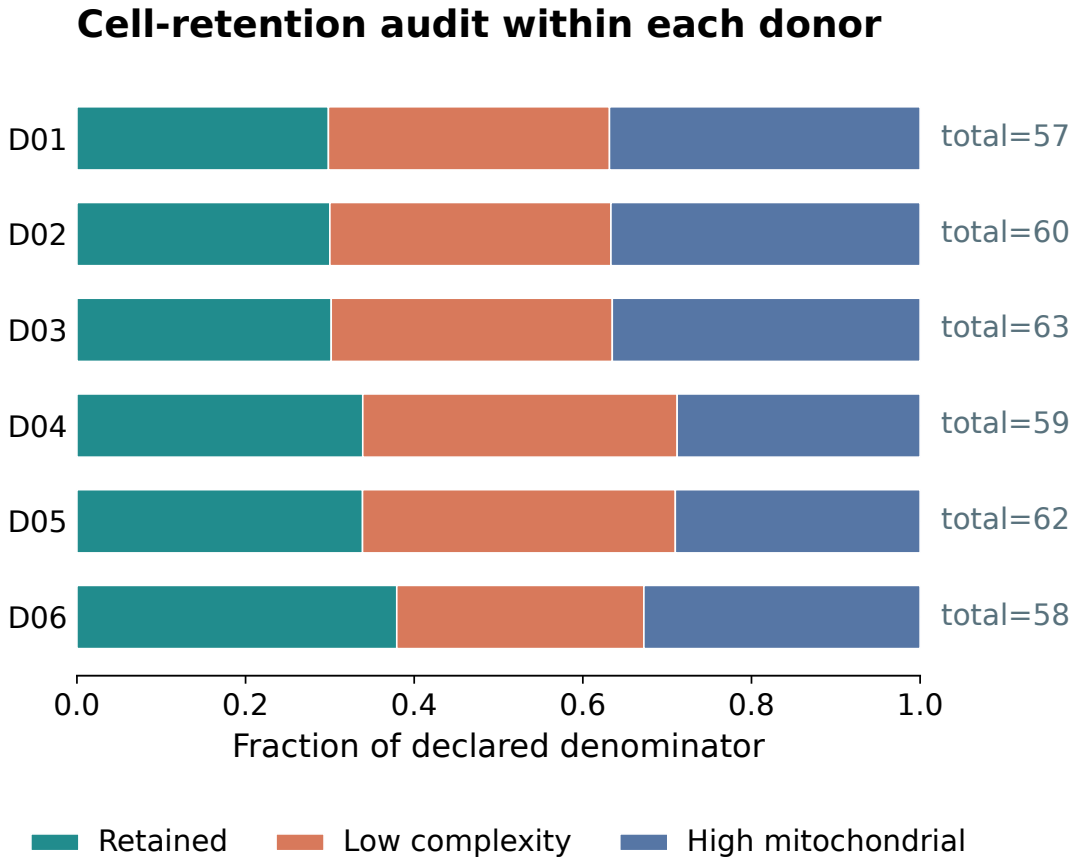


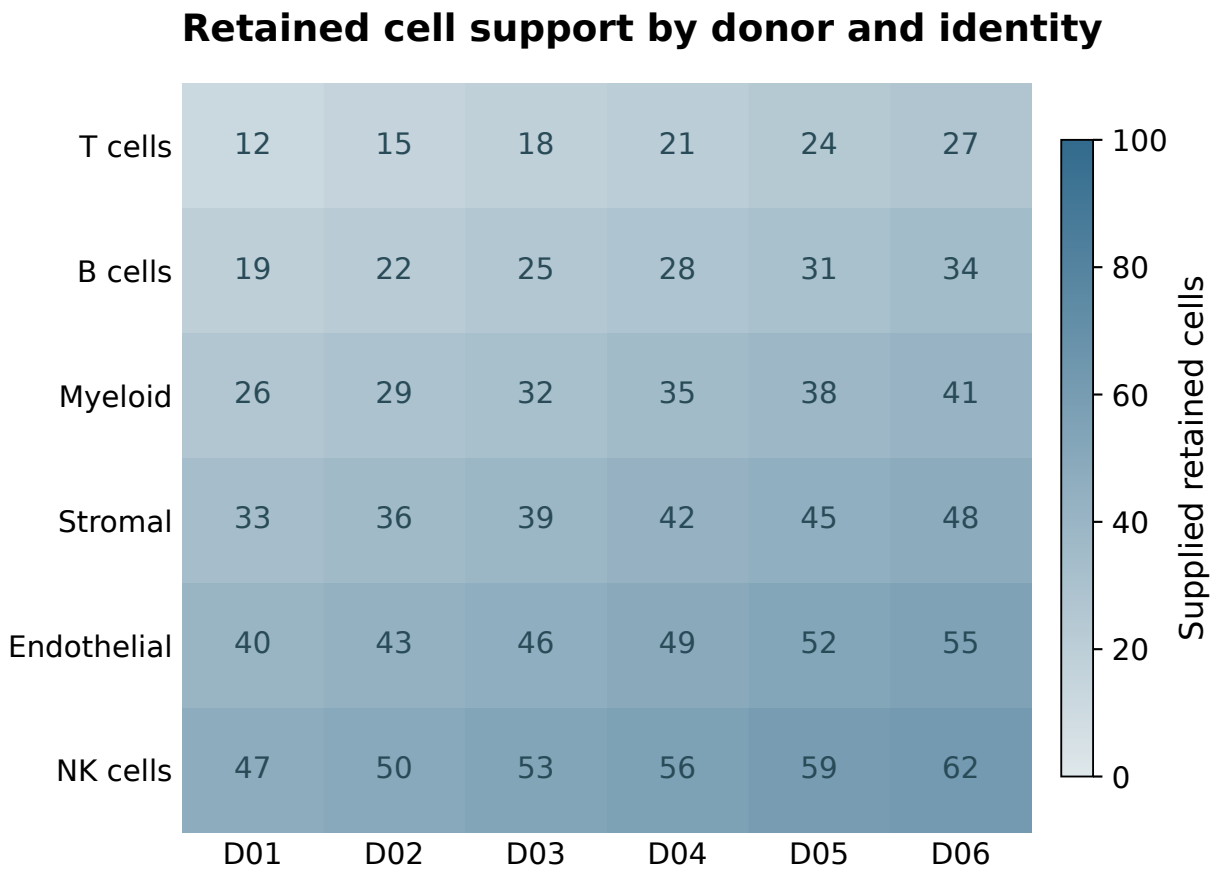

Independent-donor median detected genes
Supplied donor median genes
1800
1600
1400
1200
1000
800
600
400
Control
n=6
Treated
n=6
Condition

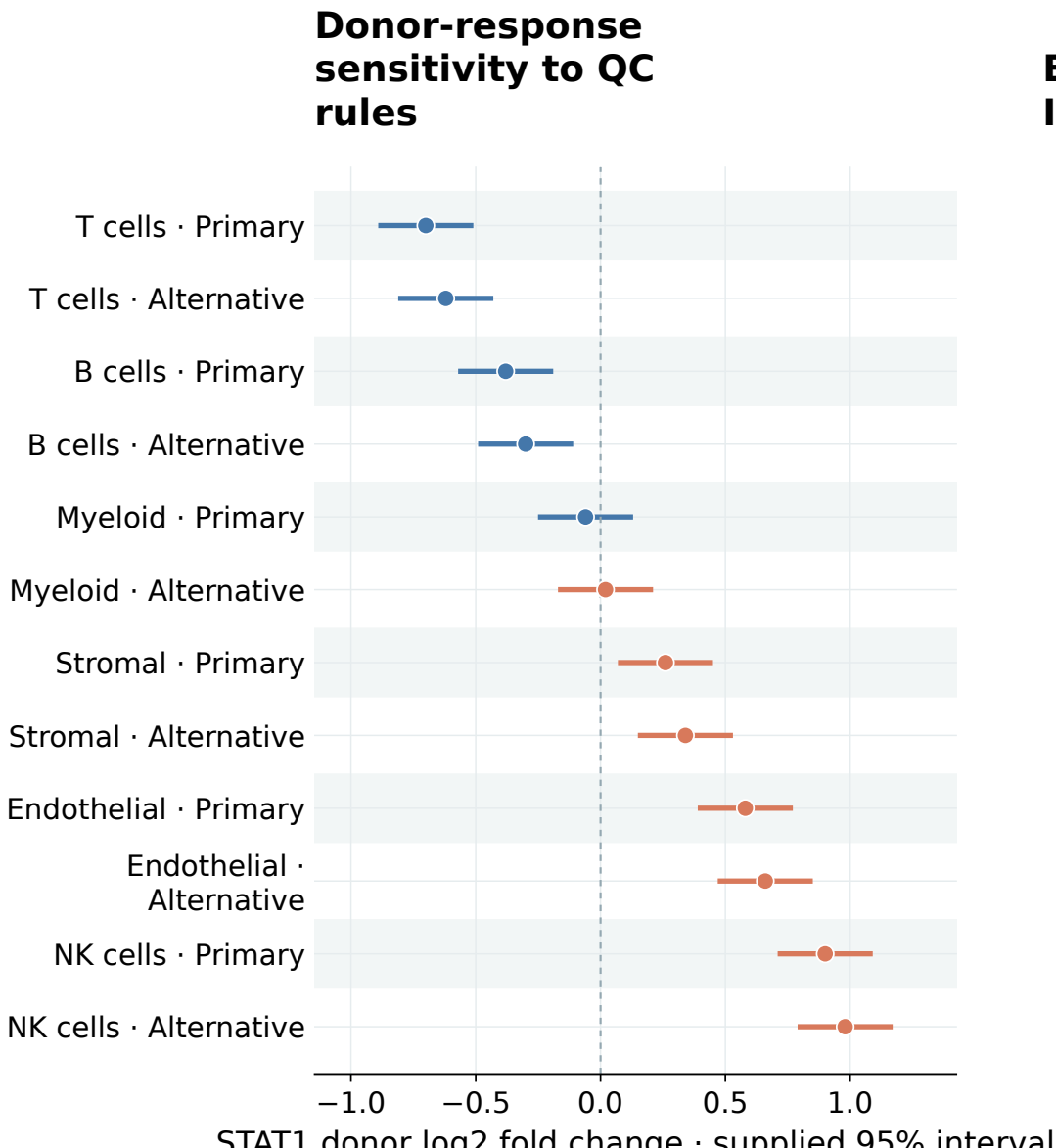

Independent-donor mitochondrial summaries
Supplied donor median mitochondrial fraction
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
Control
n=6
Treated
n=6
Condition



Software test fixture; not biological evidence

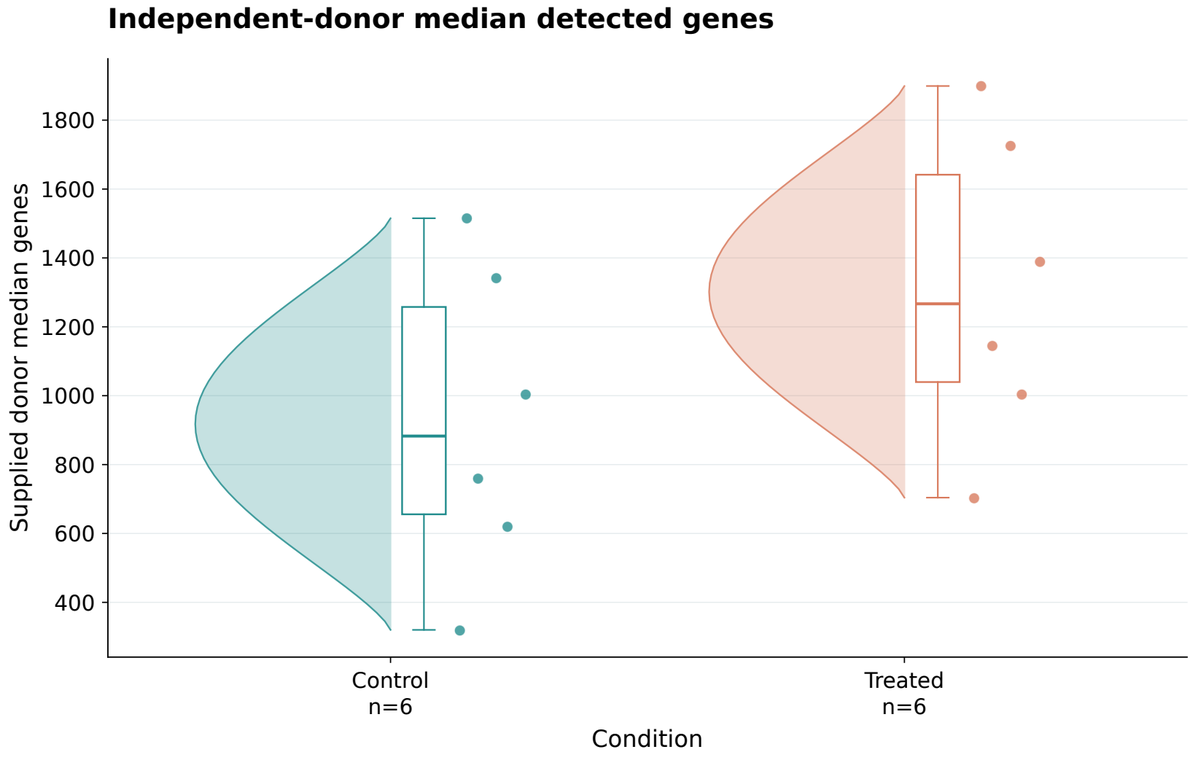

Cell-retention audit within each donor
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Retained
Low complexity
High mitochondrial


Retained cell support by donor and identity
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
12 15 18 21 24 27
19 22 25 28 31 34
26 29 32 35 38 41
33 36 39 42 45 48
40 43 46 49 52 55
47 50 53 56 59 62
D01 D02 D03 D04 D05 D06
Supplied retained cells
100
80
60
40
20
0

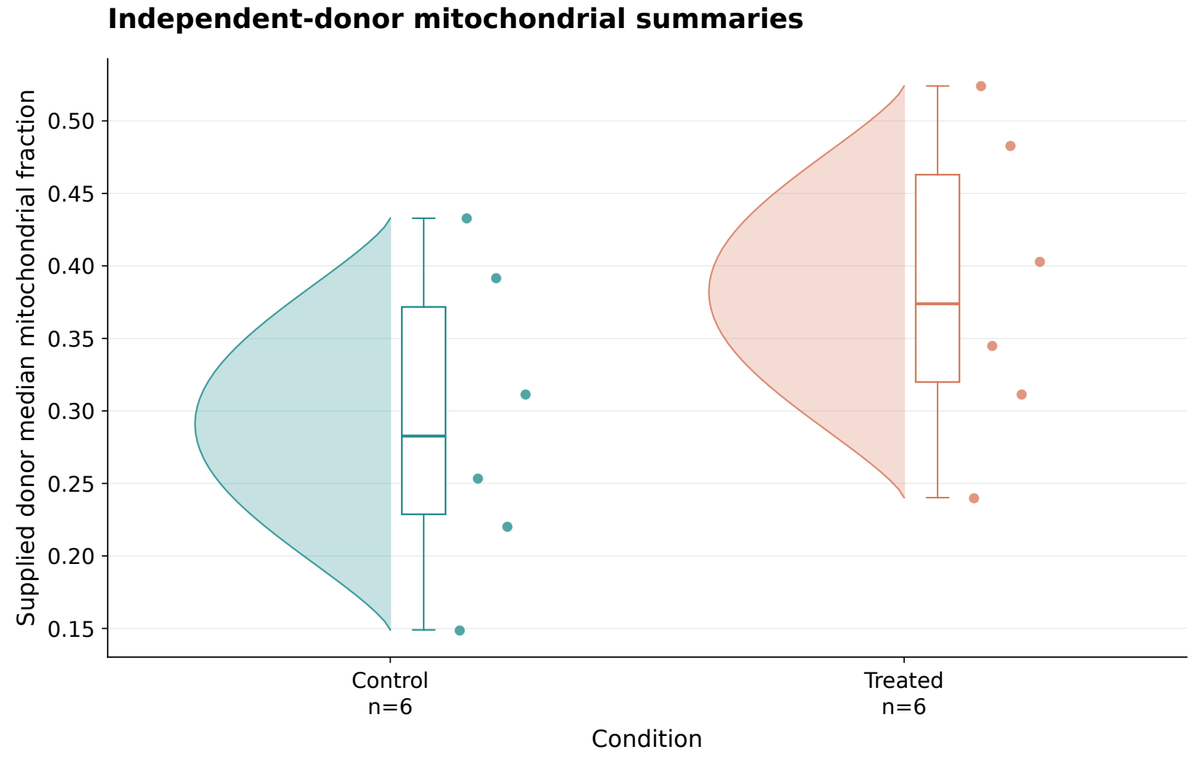

Donor-response sensitivity to QC rules
Estimate [interval] · q
Independent n = 12
T cells · Primary
T cells · Alternative
B cells · Primary
B cells · Alternative
Myeloid · Primary
Myeloid · Alternative
Stromal · Primary
Stromal · Alternative
Endothelial · Primary
Endothelial · Alternative
NK cells · Primary
NK cells · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
STAT1 donor log2 fold change · supplied 95% interval

Single cell · qc — complementary evidence

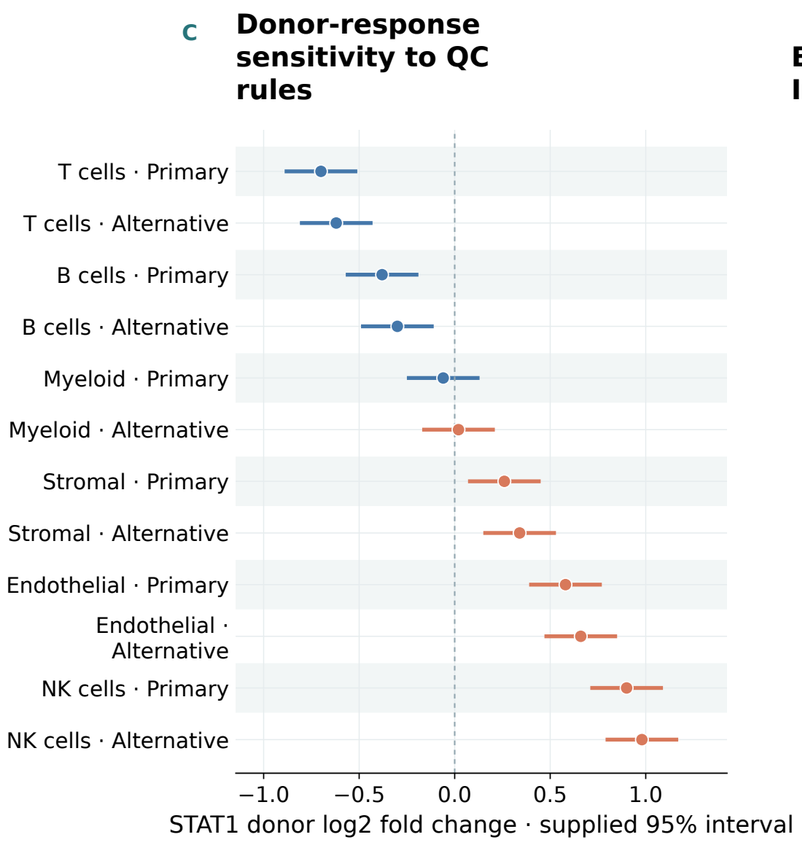
A

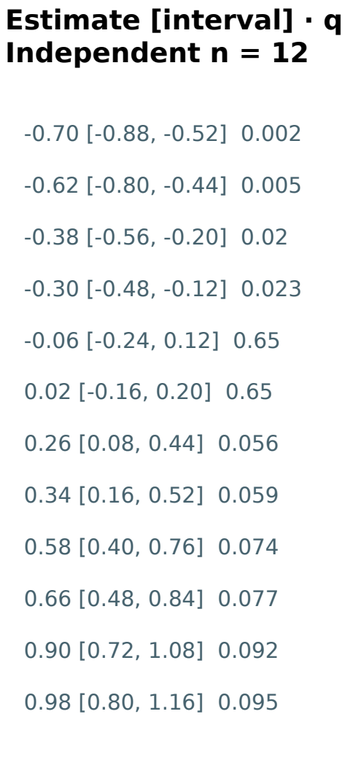


B



C

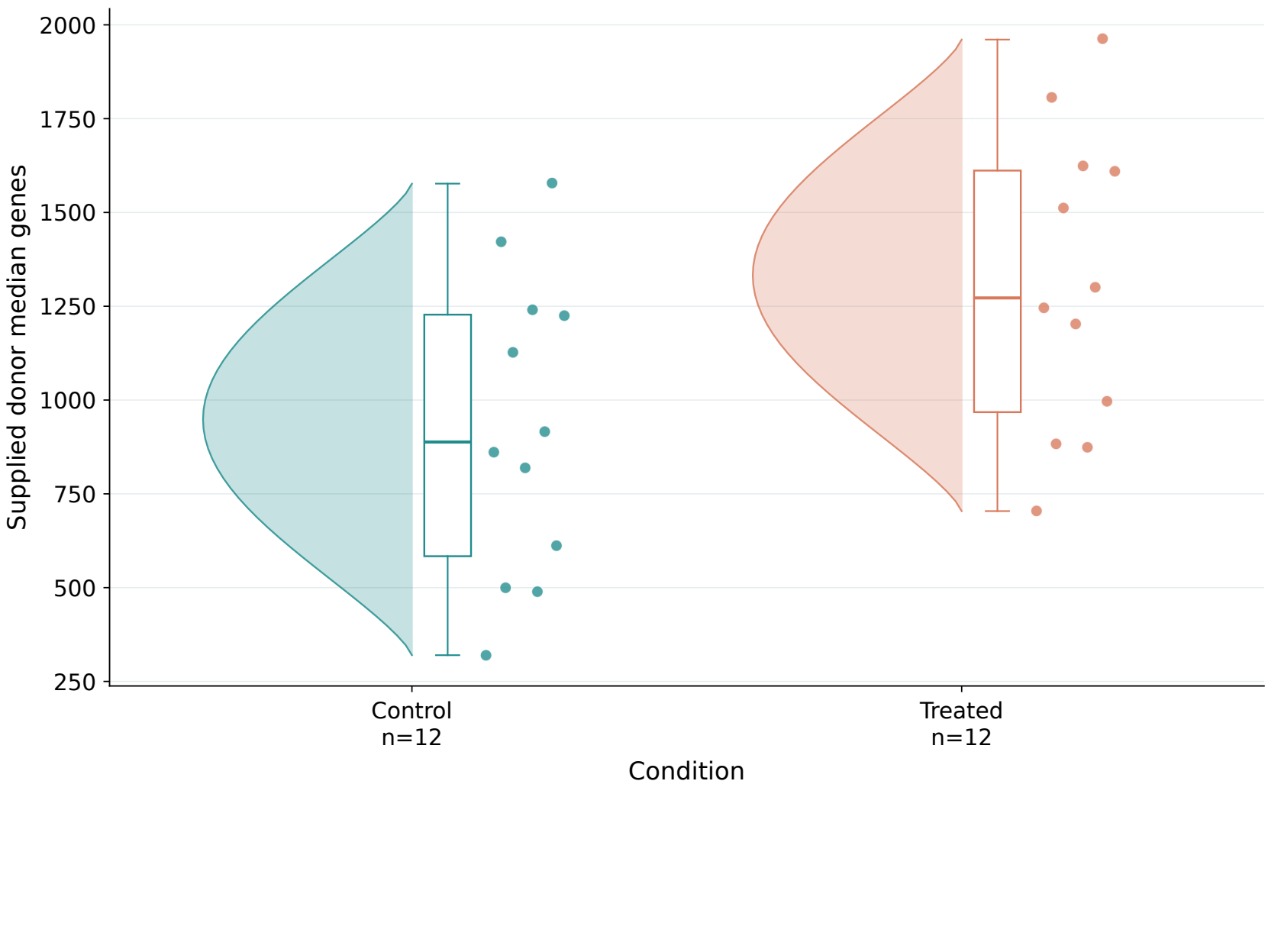




Single cell · qc — complementary evidence

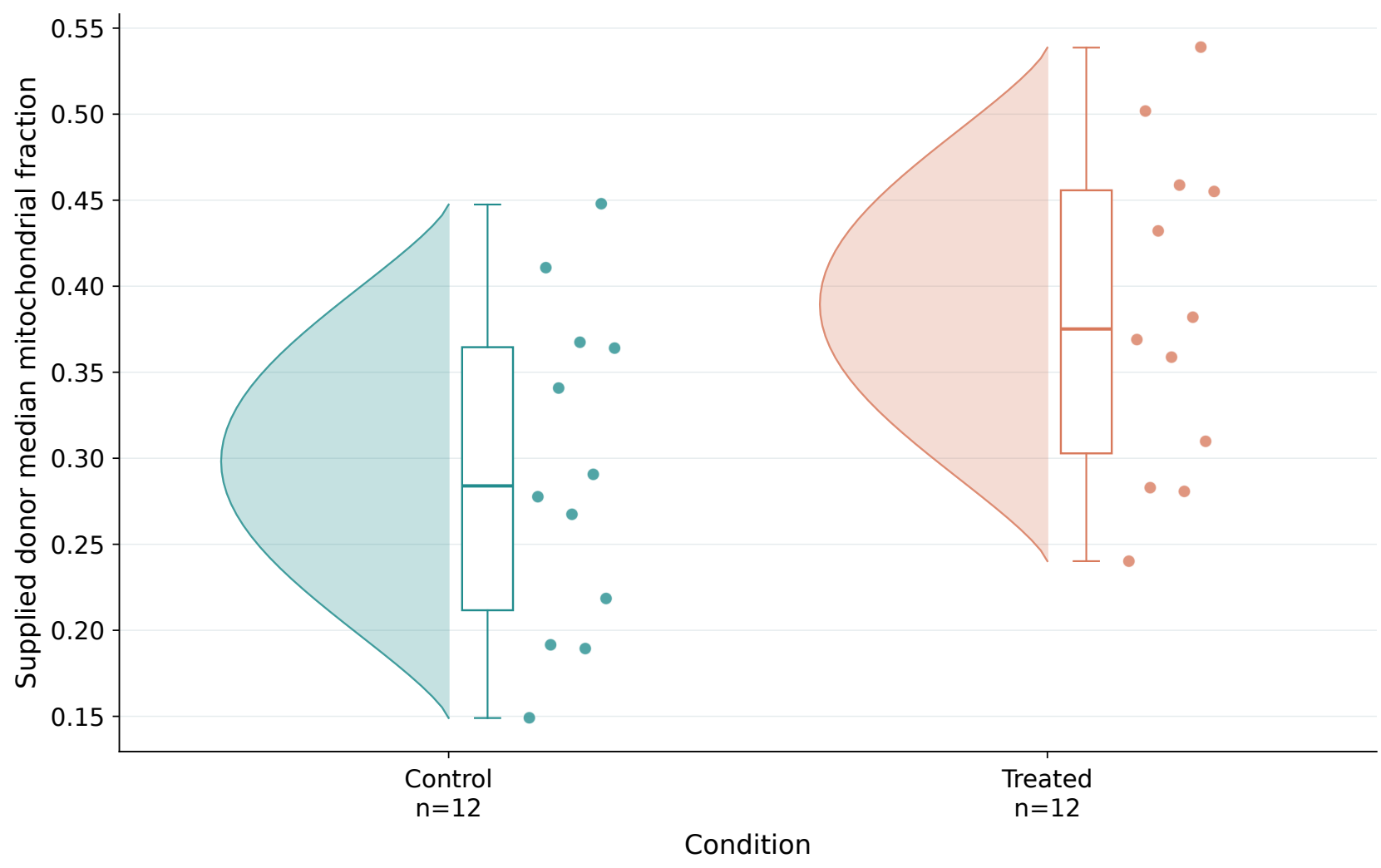
A

Independent-donor median detected genes



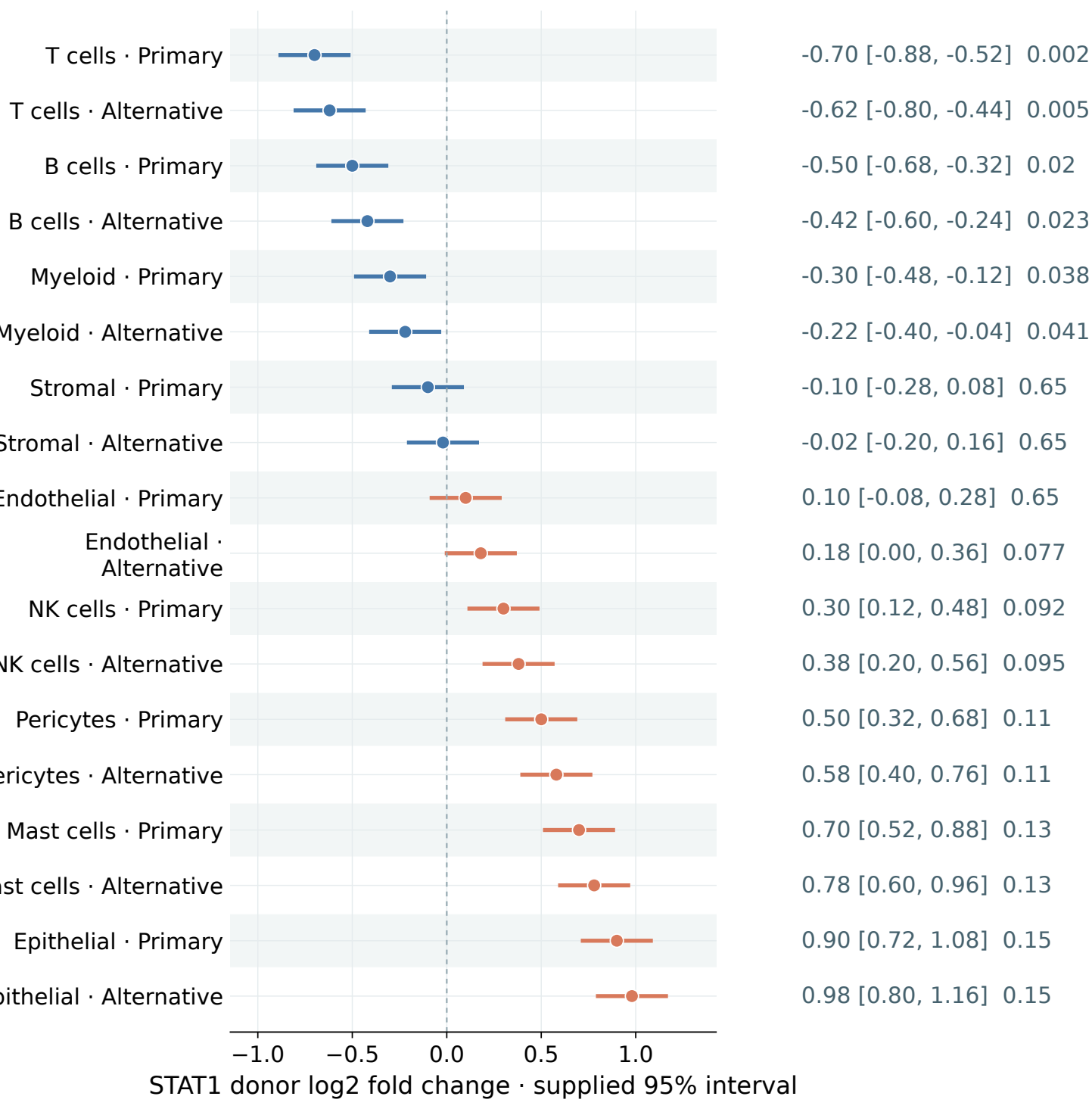
B

Independent-donor mitochondrial summaries



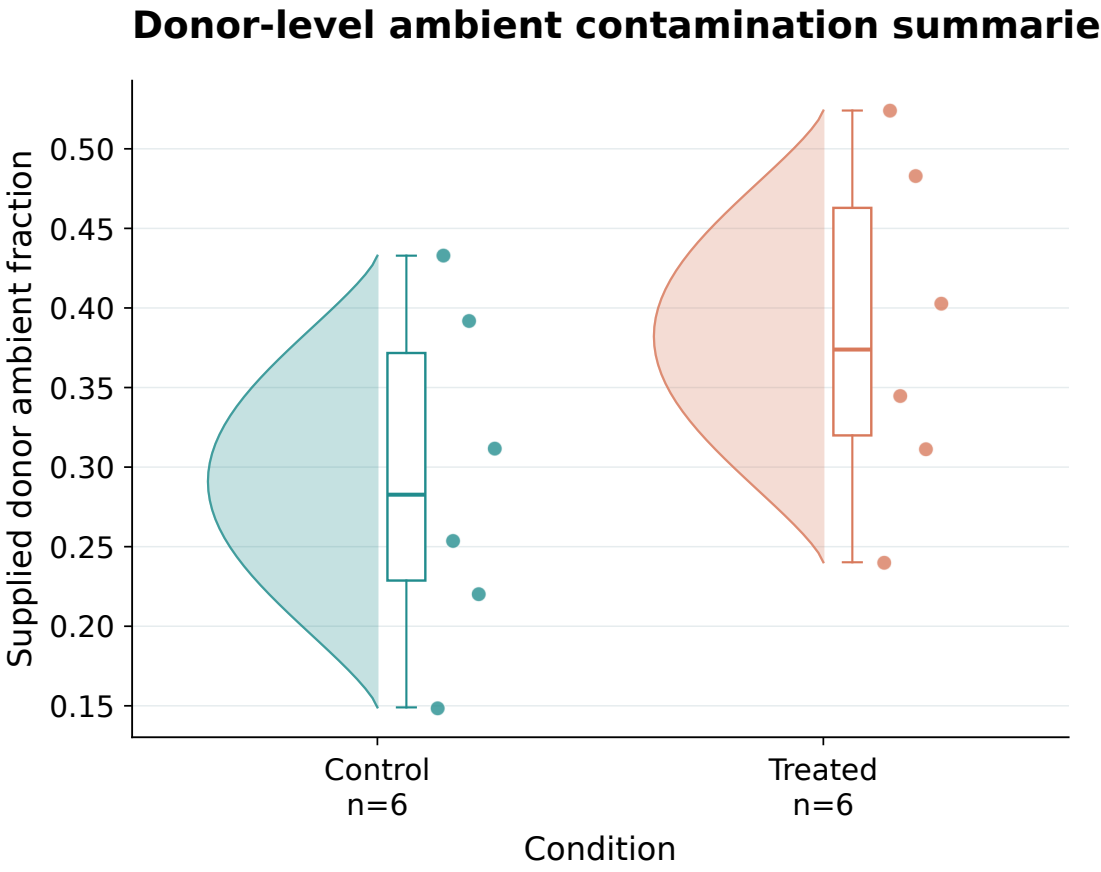
C

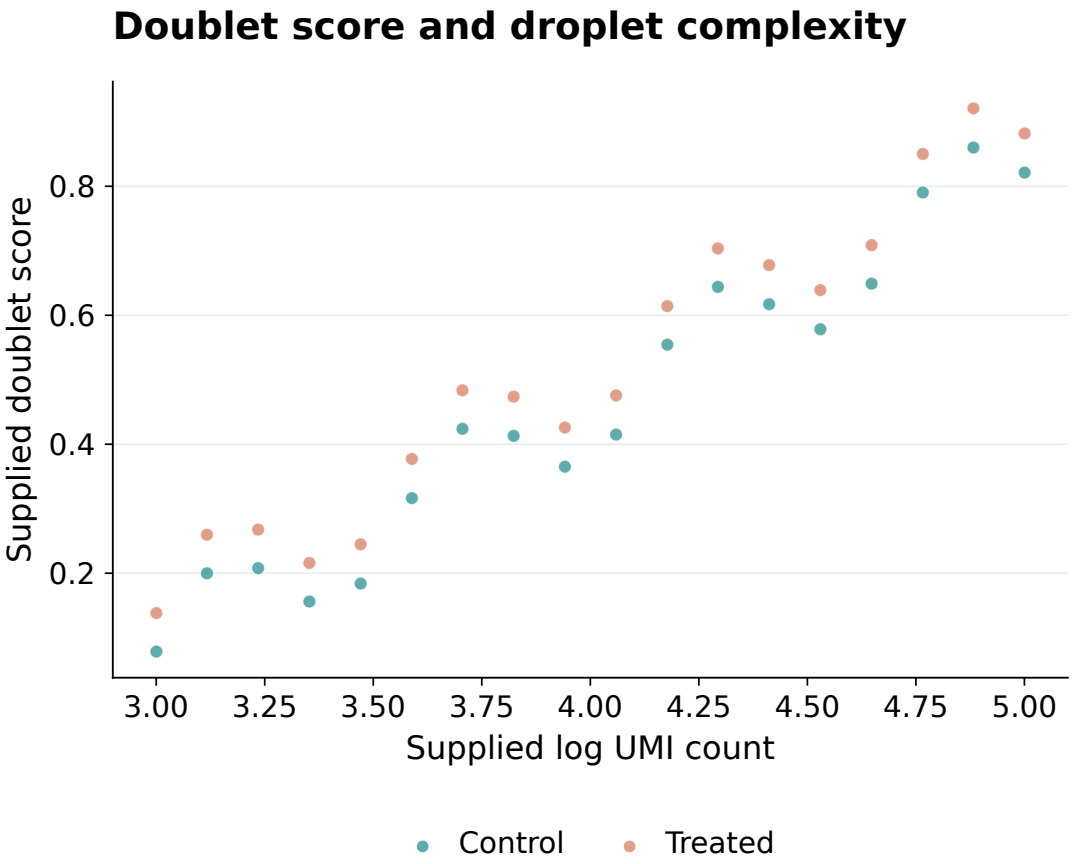
Donor-response sensitivity to QC rules

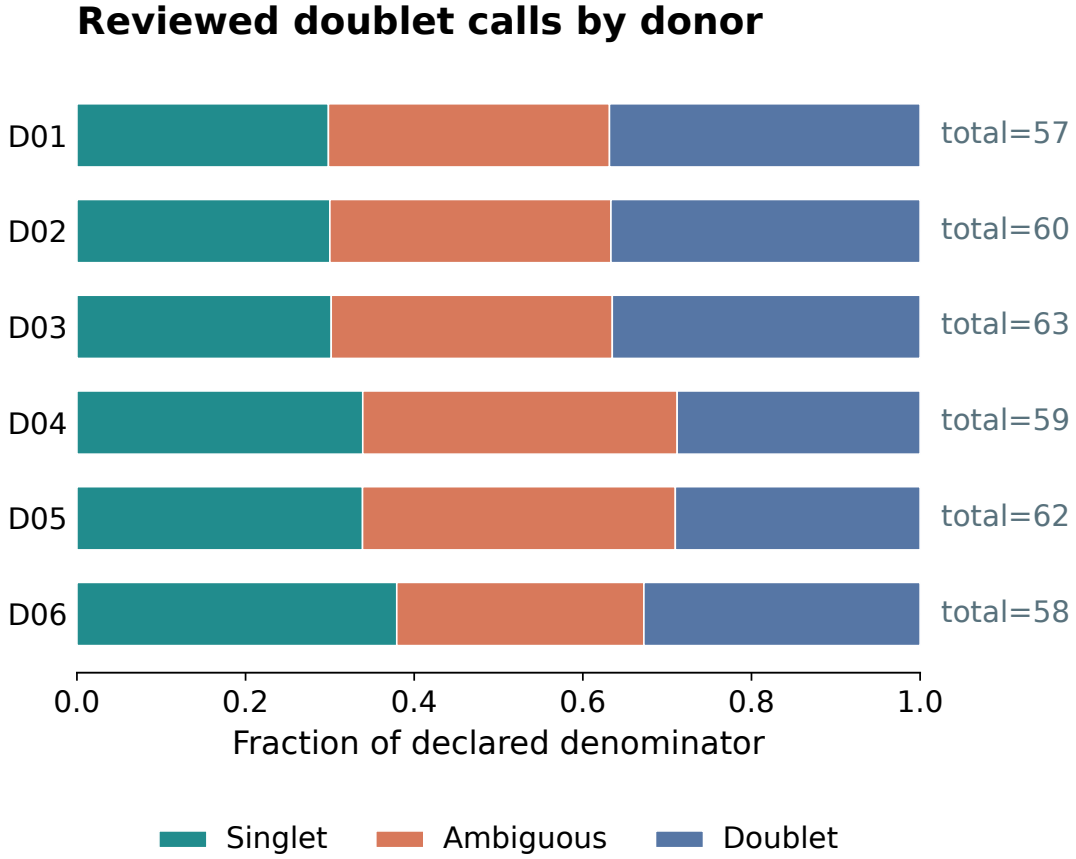


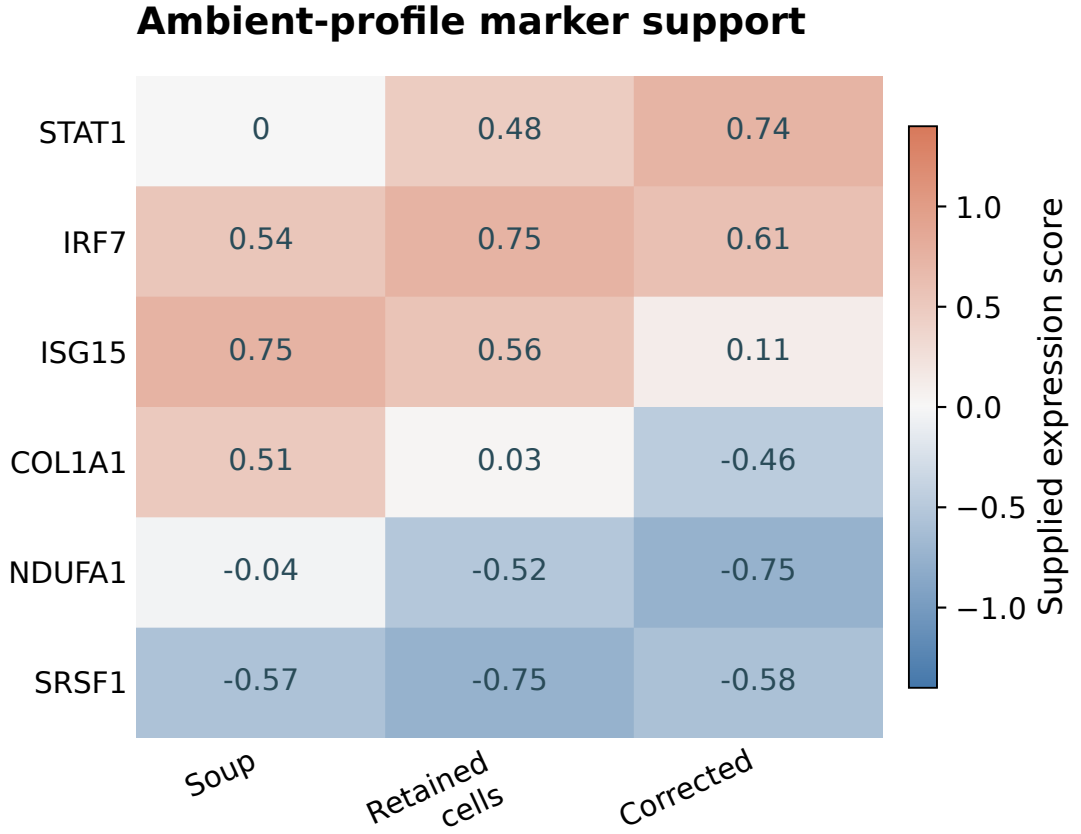
Estimate [interval] · q Independent n = 24

T cells · Primary	-0.70 [-0.88, -0.52]	0.002
T cells · Alternative	-0.62 [-0.80, -0.44]	0.005
B cells · Primary	-0.50 [-0.68, -0.32]	0.02
B cells · Alternative	-0.42 [-0.60, -0.24]	0.023
Myeloid · Primary	-0.30 [-0.48, -0.12]	0.038
Myeloid · Alternative	-0.22 [-0.40, -0.04]	0.041
Stromal · Primary	-0.10 [-0.28, 0.08]	0.65
Stromal · Alternative	-0.02 [-0.20, 0.16]	0.65
Endothelial · Primary	0.10 [-0.08, 0.28]	0.65
Endothelial · Alternative	0.18 [0.00, 0.36]	0.077
NK cells · Primary	0.30 [0.12, 0.48]	0.092
NK cells · Alternative	0.38 [0.20, 0.56]	0.095
Pericytes · Primary	0.50 [0.32, 0.68]	0.11
Pericytes · Alternative	0.58 [0.40, 0.76]	0.11
Mast cells · Primary	0.70 [0.52, 0.88]	0.13
Mast cells · Alternative	0.78 [0.60, 0.96]	0.13
Epithelial · Primary	0.90 [0.72, 1.08]	0.15
Epithelial · Alternative	0.98 [0.80, 1.16]	0.15

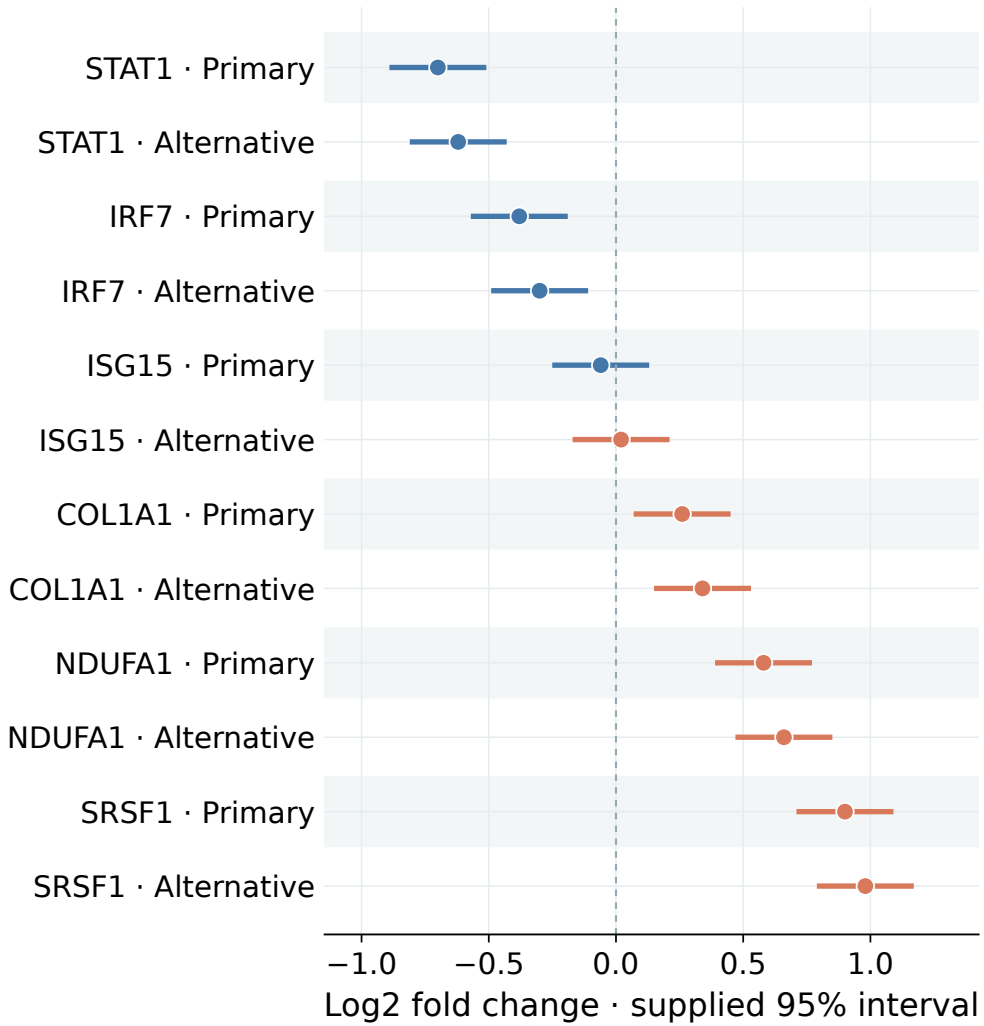

Donor-level ambient contamination summaries
Supplied donor ambient fraction
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
Control
n=6
Treated
n=6
Condition


Doublet score and droplet complexity
Supplied doublet score
0.2
0.4
0.6
0.8
3.00
3.25
3.50
3.75
4.00
4.25
4.50
4.75
5.00
Supplied log UMI count
Control
Treated


Reviewed doublet calls by donor
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Singlet
Ambiguous
Doublet


Ambient-profile marker support
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0
0.48
0.74
0.54
0.75
0.61
0.75
0.56
0.11
0.51
0.03
-0.46
-0.04
-0.52
-0.75
-0.57
-0.75
-0.58
Soup
Retained cells
Corrected
1.0
0.5
0.0
−0.5
−1.0
Supplied expression score

Biological-response sensitivity to correction



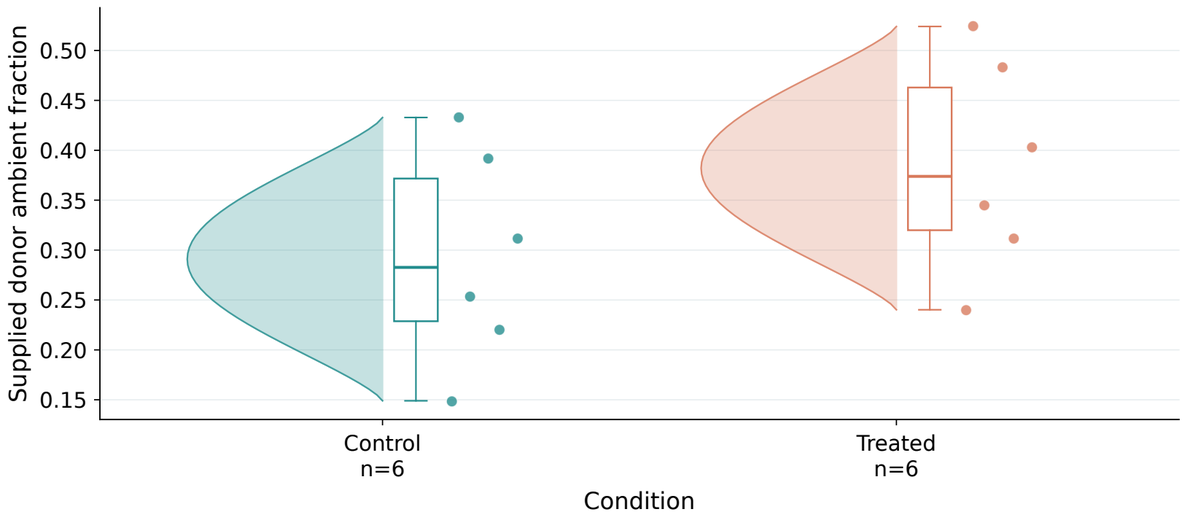
Estimate [interval] · q Independent n = 12

STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.38 [-0.56, -0.20] 0.02
IRF7 · Alternative	-0.30 [-0.48, -0.12] 0.023
ISG15 · Primary	-0.06 [-0.24, 0.12] 0.65
ISG15 · Alternative	0.02 [-0.16, 0.20] 0.65
COL1A1 · Primary	0.26 [0.08, 0.44] 0.056
COL1A1 · Alternative	0.34 [0.16, 0.52] 0.059
NDUFA1 · Primary	0.58 [0.40, 0.76] 0.074
NDUFA1 · Alternative	0.66 [0.48, 0.84] 0.077
SRSF1 · Primary	0.90 [0.72, 1.08] 0.092
SRSF1 · Alternative	0.98 [0.80, 1.16] 0.095

Single cell · ambient_doublet — complementary evidence

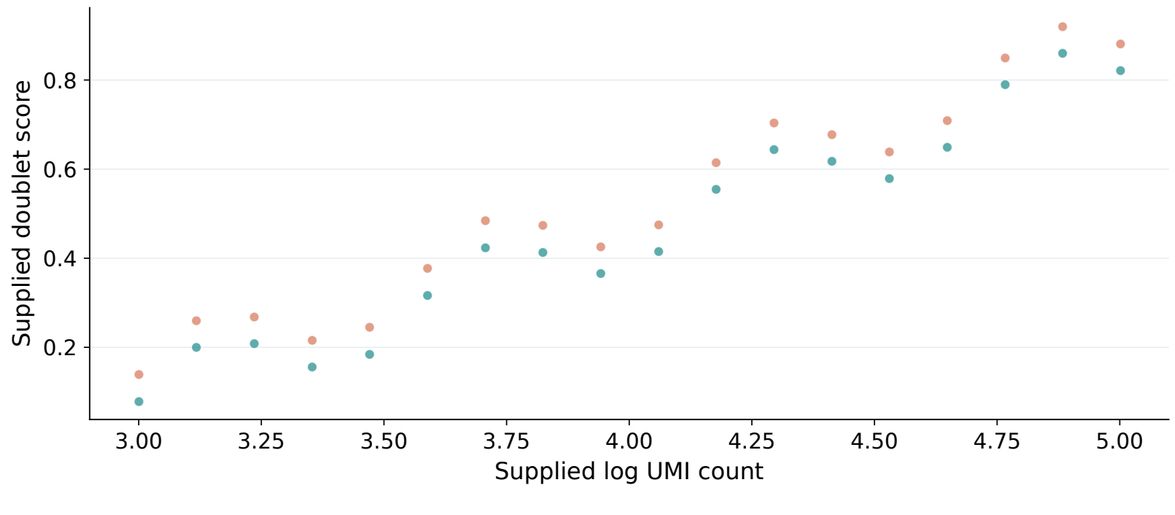
A

Donor-level ambient contamination summaries



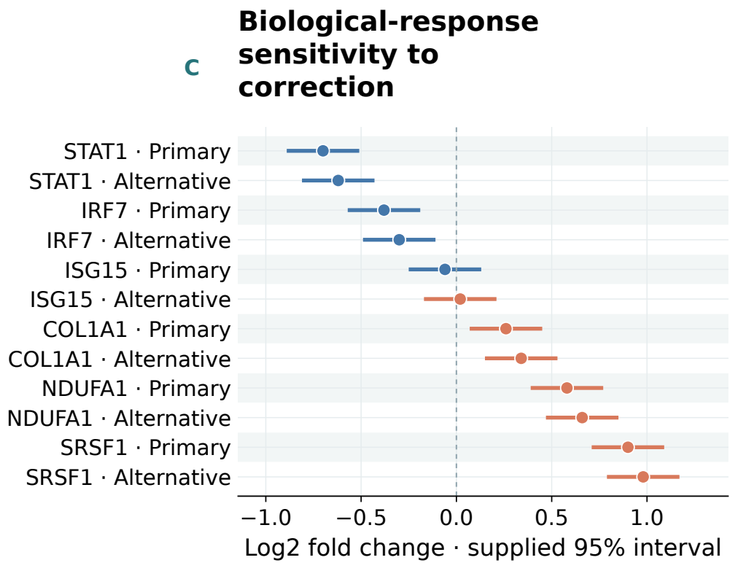
B

Doublet score and droplet complexity



C

Biological-response sensitivity to correction



Estimate [interval] · q
Independent n = 12

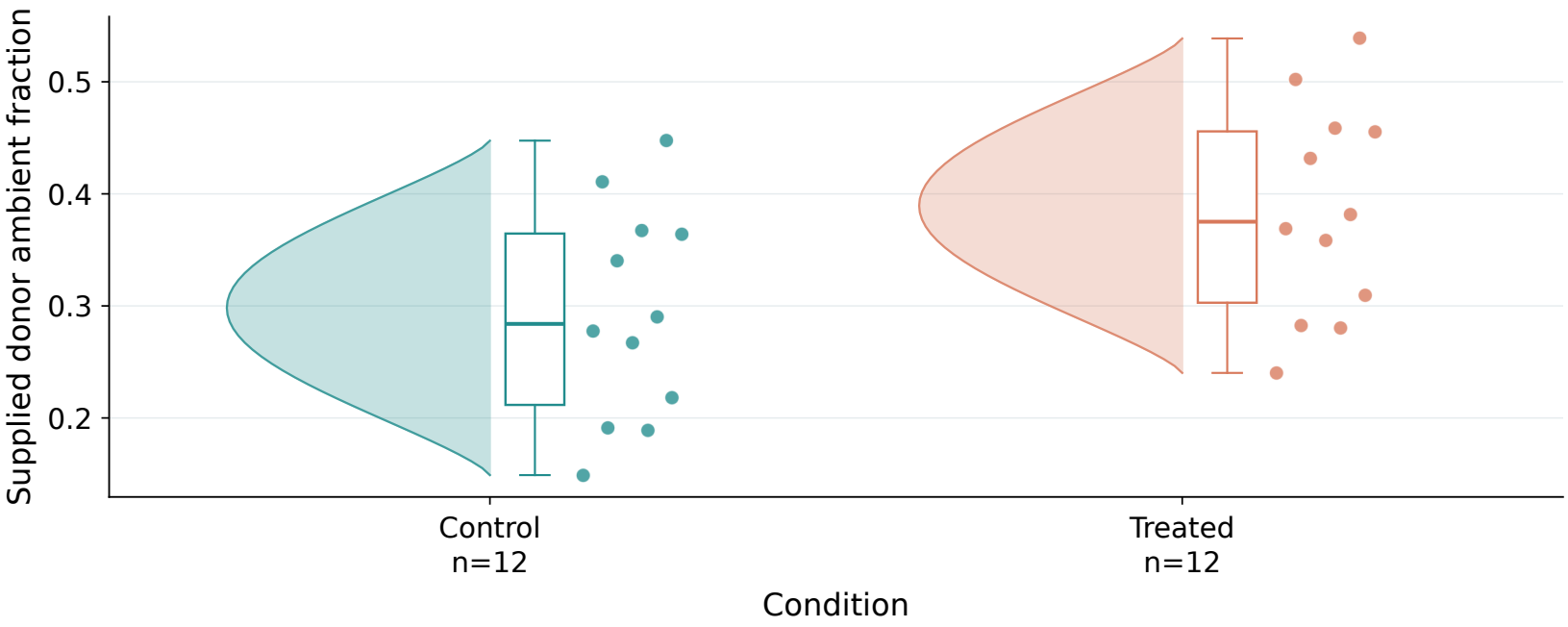
STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.38 [-0.56, -0.20] 0.02
IRF7 · Alternative	-0.30 [-0.48, -0.12] 0.023
ISG15 · Primary	-0.06 [-0.24, 0.12] 0.65
ISG15 · Alternative	0.02 [-0.16, 0.20] 0.65
COL1A1 · Primary	0.26 [0.08, 0.44] 0.056
COL1A1 · Alternative	0.34 [0.16, 0.52] 0.059
NDUFA1 · Primary	0.58 [0.40, 0.76] 0.074
NDUFA1 · Alternative	0.66 [0.48, 0.84] 0.077
SRSF1 · Primary	0.90 [0.72, 1.08] 0.092
SRSF1 · Alternative	0.98 [0.80, 1.16] 0.095

Software test fixture; not biological evidence

Single cell · ambient_doublet — complementary evidence

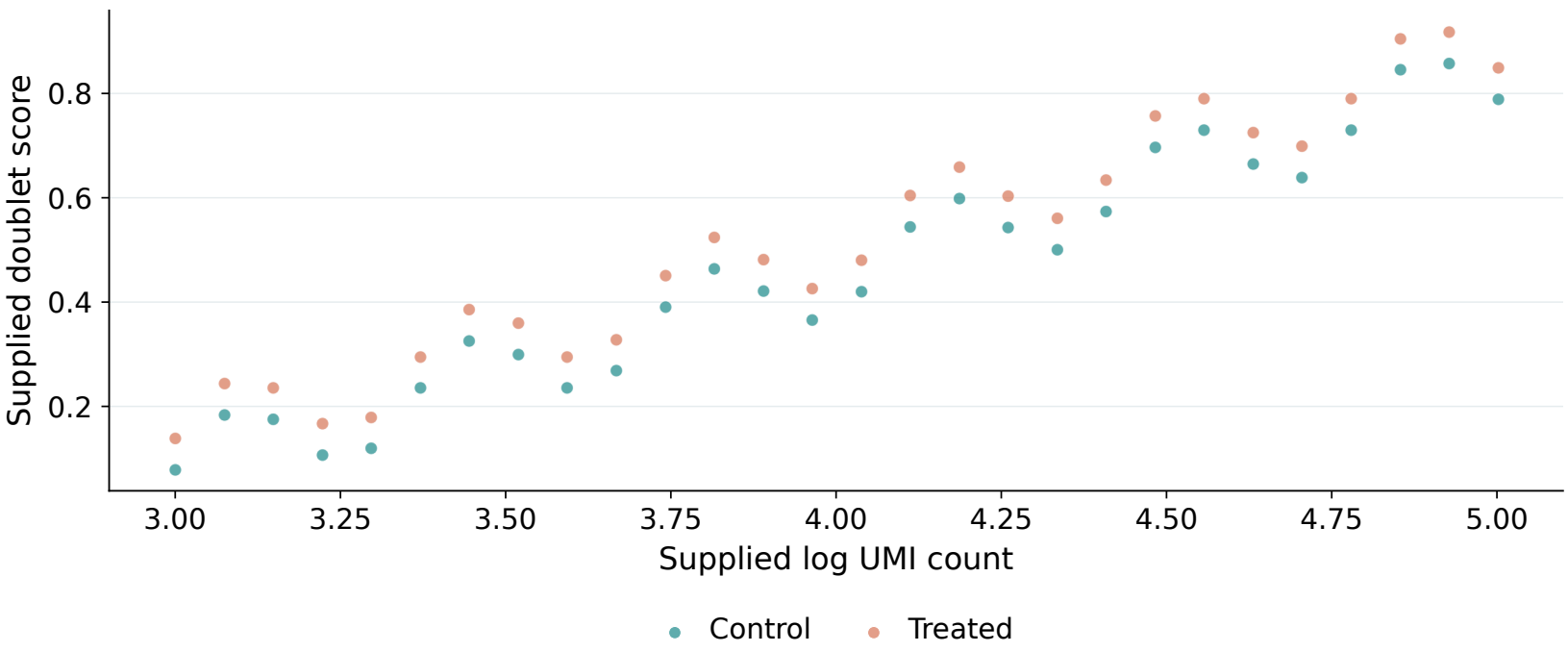
A

Donor-level ambient contamination summaries



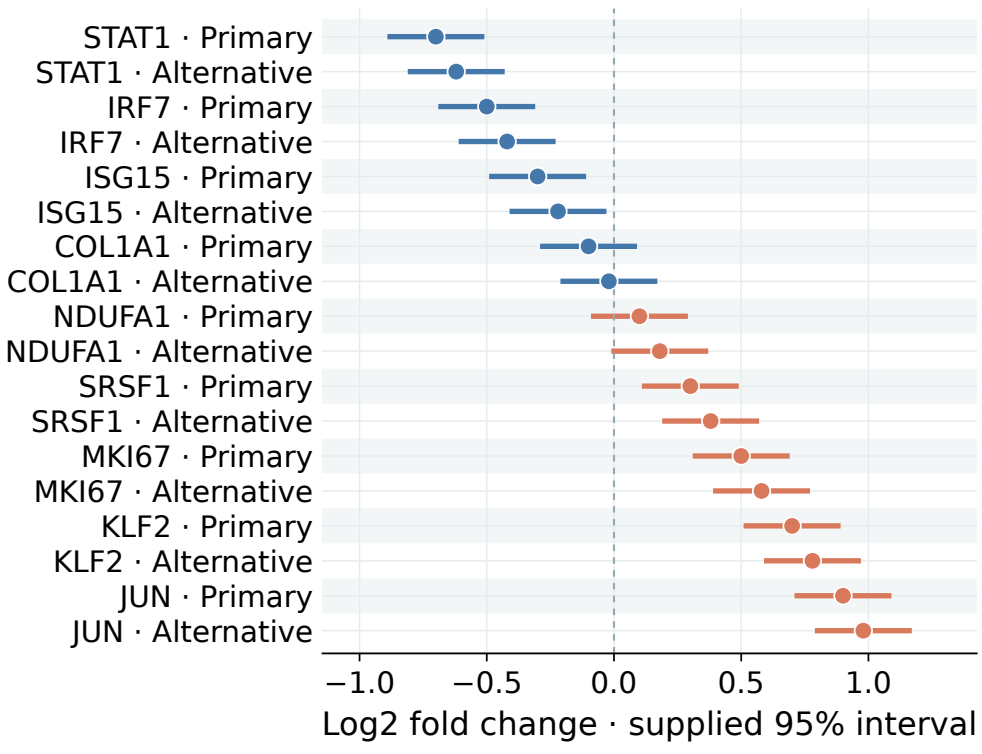
B

Doublet score and droplet complexity



C

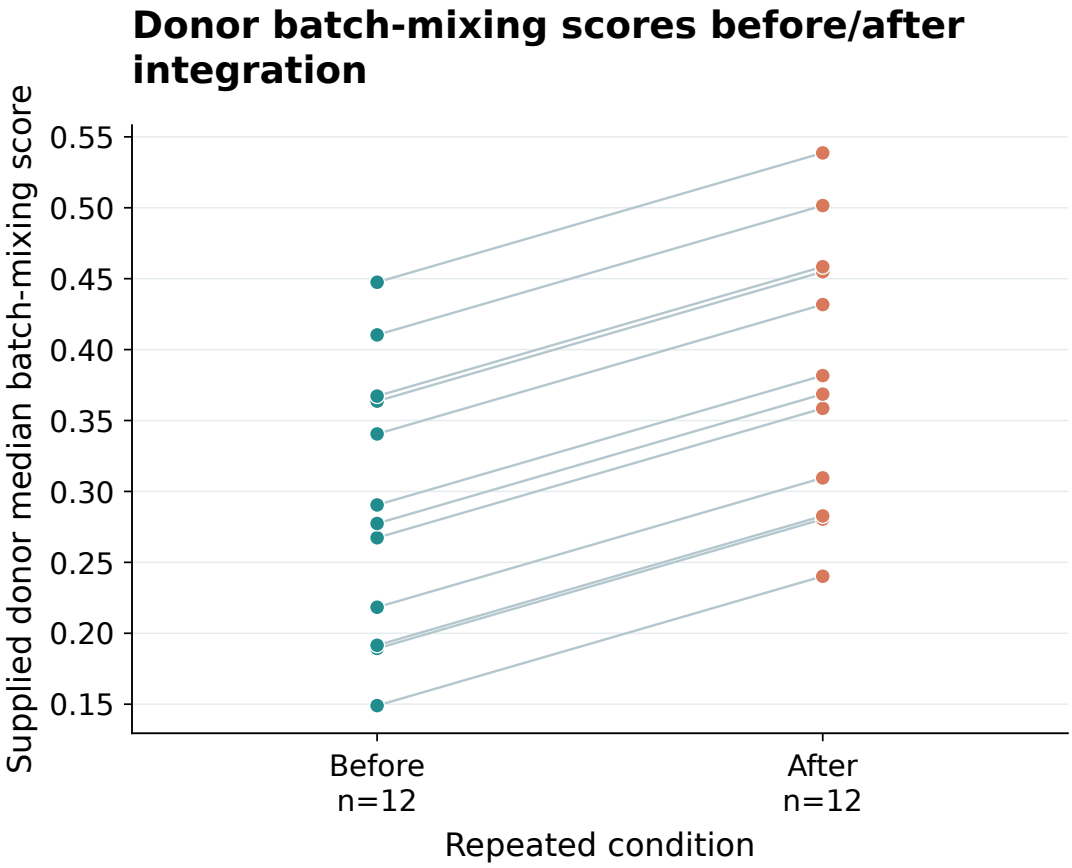
Biological-response sensitivity to correction

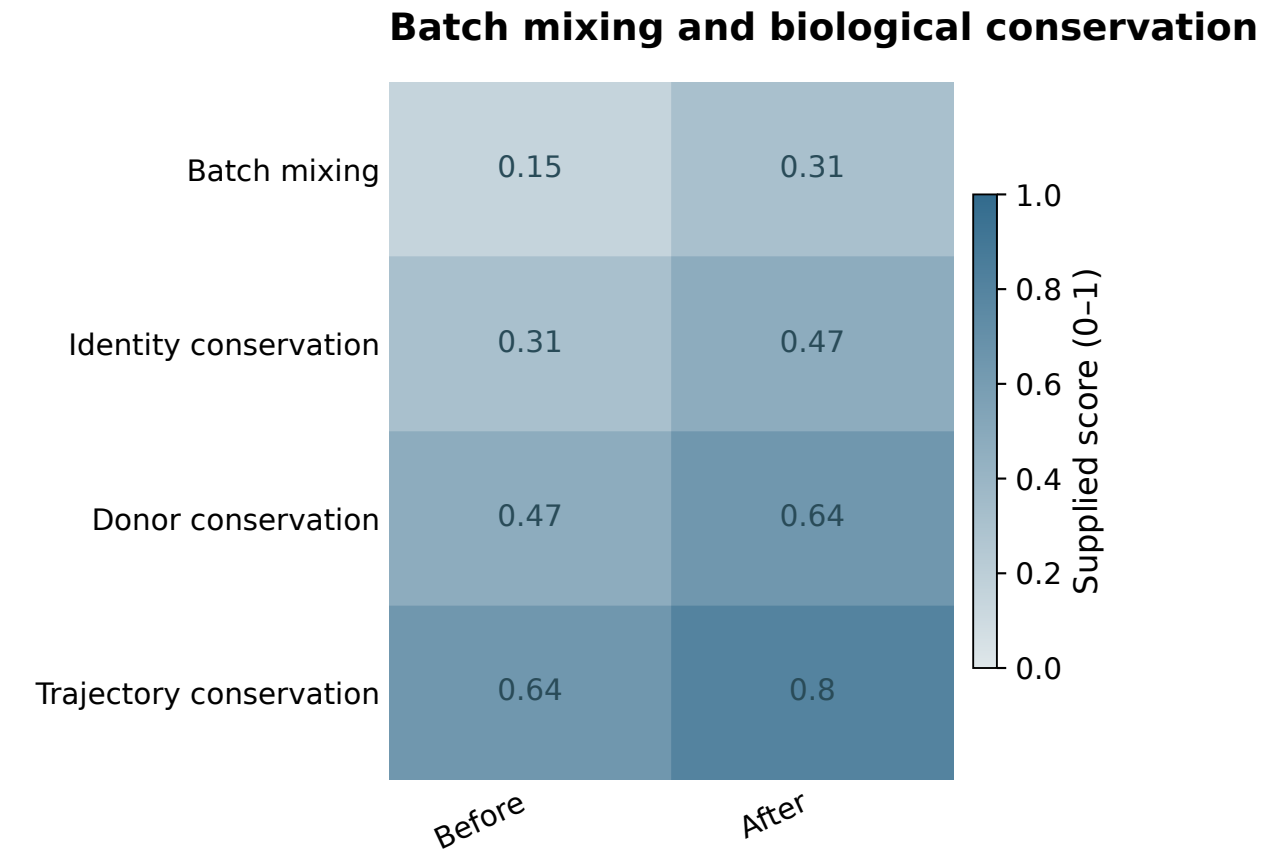


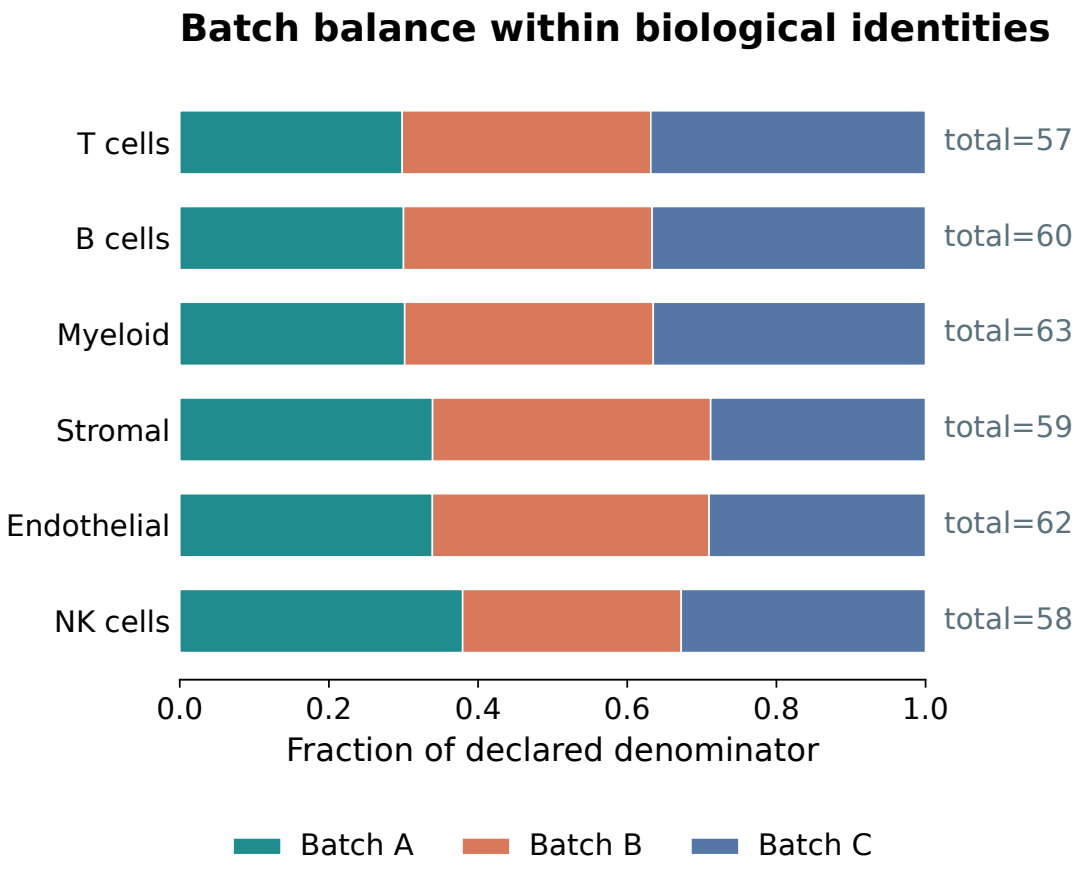
Estimate [interval] · q
Independent n = 24

STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.50 [-0.68, -0.32] 0.02
IRF7 · Alternative	-0.42 [-0.60, -0.24] 0.023
ISG15 · Primary	-0.30 [-0.48, -0.12] 0.038
ISG15 · Alternative	-0.22 [-0.40, -0.04] 0.041
COL1A1 · Primary	-0.10 [-0.28, 0.08] 0.65
COL1A1 · Alternative	-0.02 [-0.20, 0.16] 0.65
NDUFA1 · Primary	0.10 [-0.08, 0.28] 0.65
NDUFA1 · Alternative	0.18 [0.00, 0.36] 0.077
SRSF1 · Primary	0.30 [0.12, 0.48] 0.092
SRSF1 · Alternative	0.38 [0.20, 0.56] 0.095
MKI67 · Primary	0.50 [0.32, 0.68] 0.11
MKI67 · Alternative	0.58 [0.40, 0.76] 0.11
KLF2 · Primary	0.70 [0.52, 0.88] 0.13
KLF2 · Alternative	0.78 [0.60, 0.96] 0.13
JUN · Primary	0.90 [0.72, 1.08] 0.15
JUN · Alternative	0.98 [0.80, 1.16] 0.15

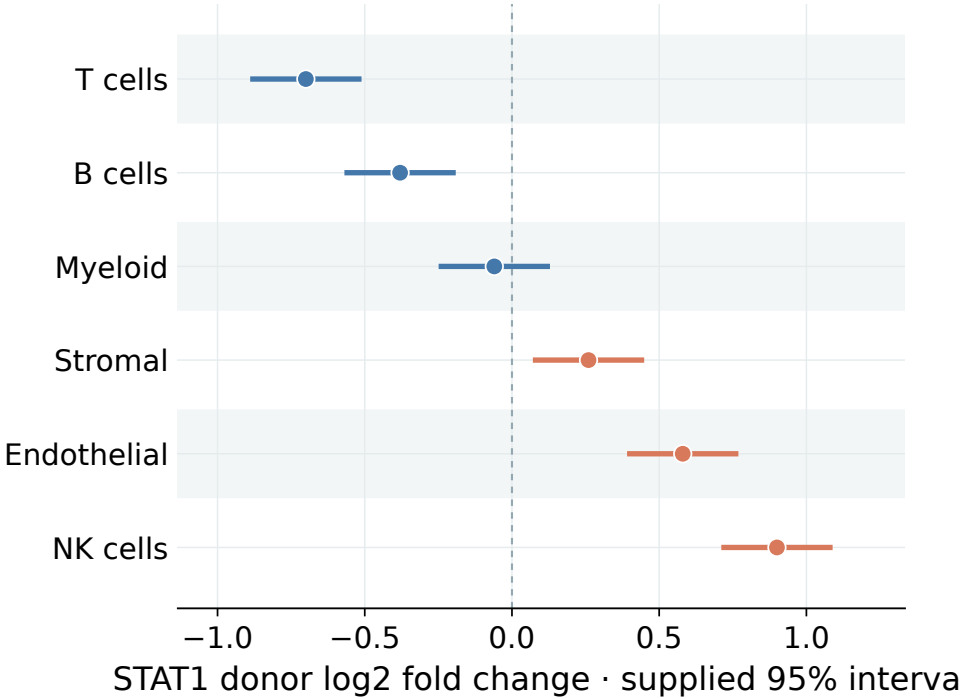
Software test fixture; not biological evidence


Donor batch-mixing scores before/after integration
Supplied donor median batch-mixing score
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
0.15
Before
n=12
After
n=12
Repeated condition


Batch mixing and biological conservation
Batch mixing
Identity conservation
Donor conservation
Trajectory conservation
0.15
0.31
0.31
0.47
0.47
0.64
0.64
0.8
Before
After
Supplied score (0–1)
1.0
0.8
0.6
0.4
0.2
0.0


Batch balance within biological identities
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Batch A
Batch B
Batch C

Donor-level biological-effect preservation



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

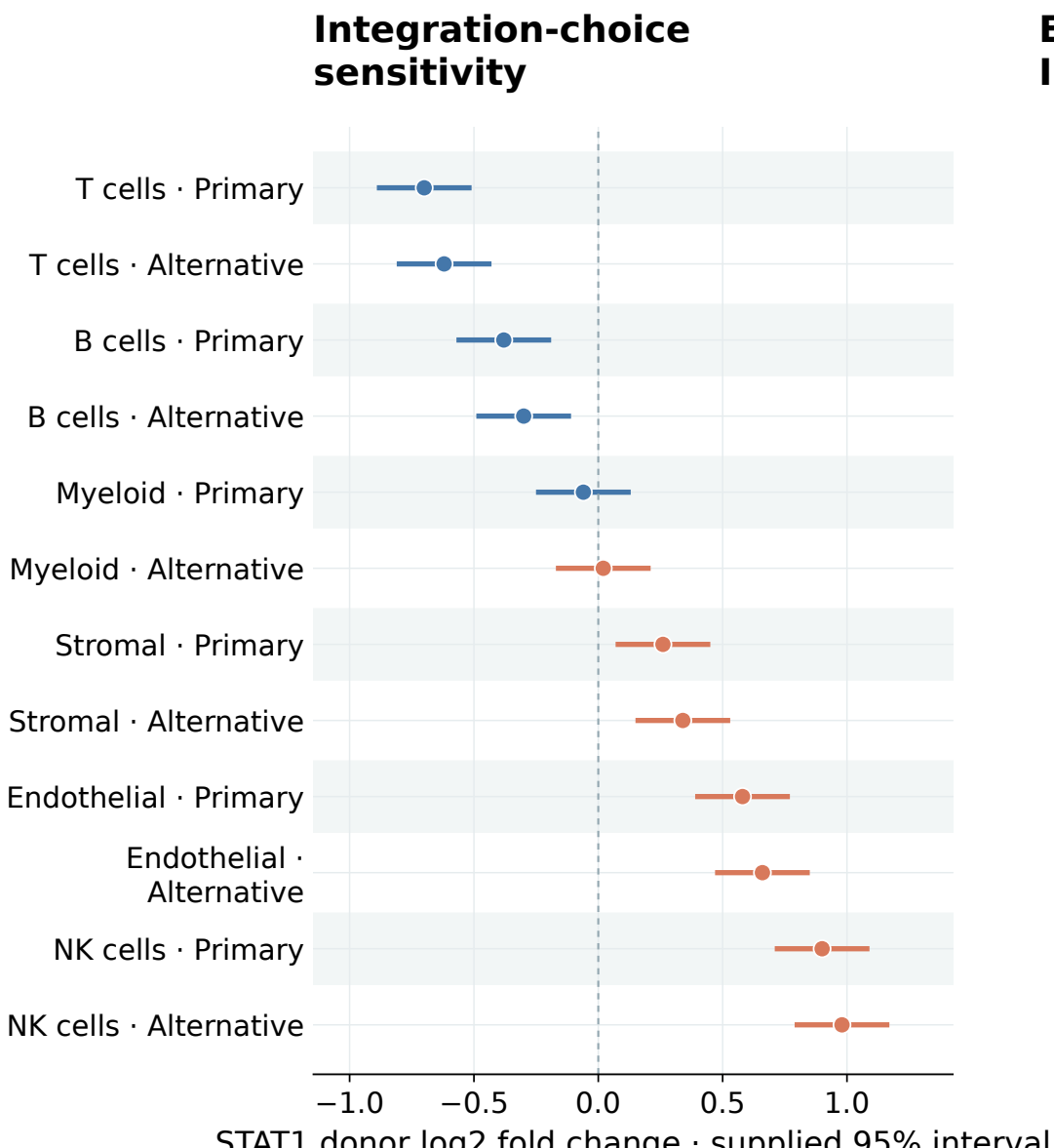
-0.38 [-0.56, -0.20] 0.02

-0.06 [-0.24, 0.12] 0.65

0.26 [0.08, 0.44] 0.056

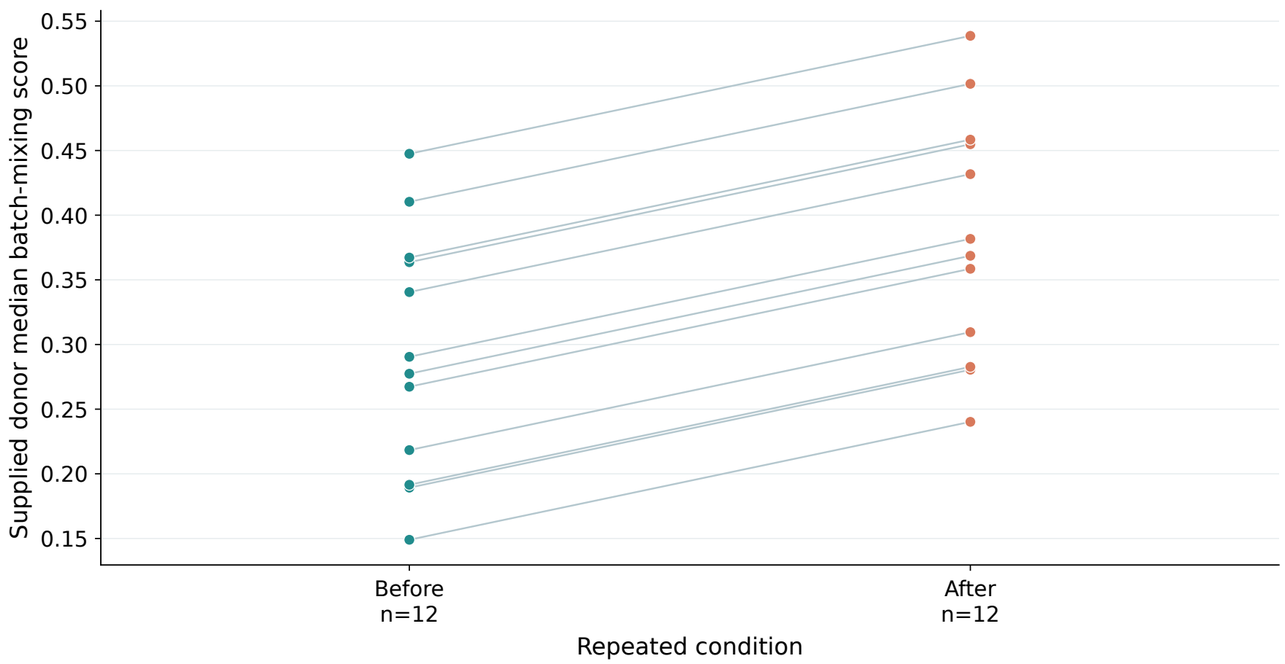
0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092

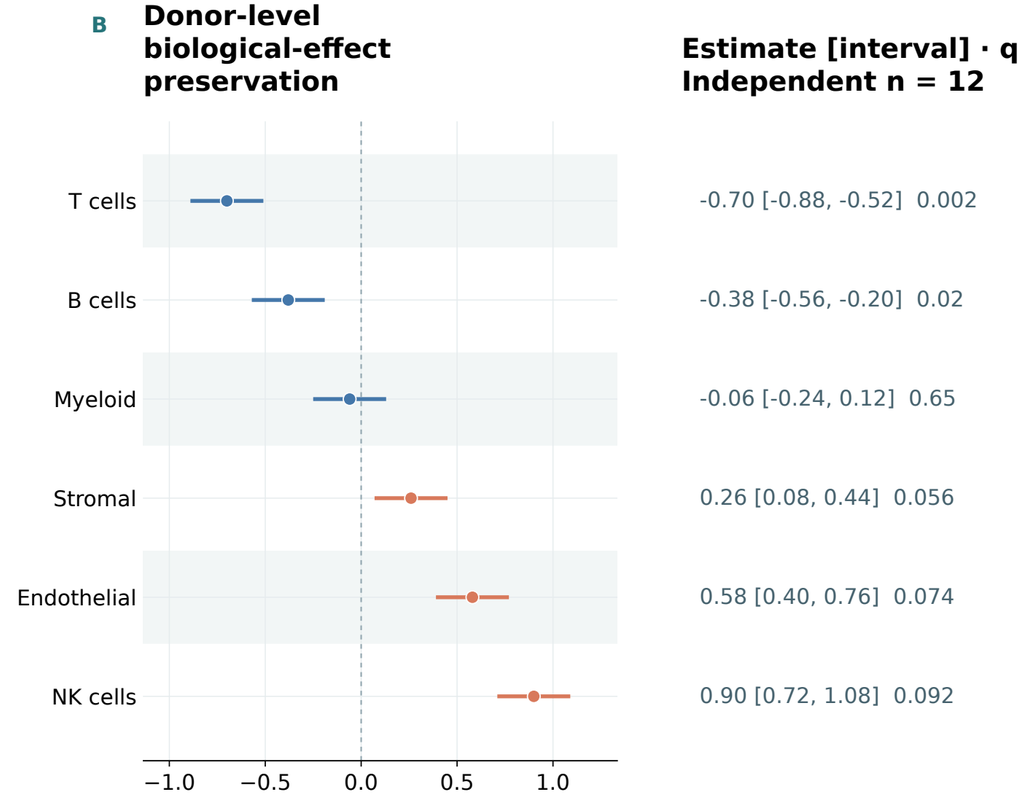

Integration-choice sensitivity
Estimate [interval] · q
Independent n = 12
T cells · Primary
T cells · Alternative
B cells · Primary
B cells · Alternative
Myeloid · Primary
Myeloid · Alternative
Stromal · Primary
Stromal · Alternative
Endothelial · Primary
Endothelial · Alternative
NK cells · Primary
NK cells · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
STAT1 donor log2 fold change · supplied 95% interval

Single cell · integration — complementary evidence

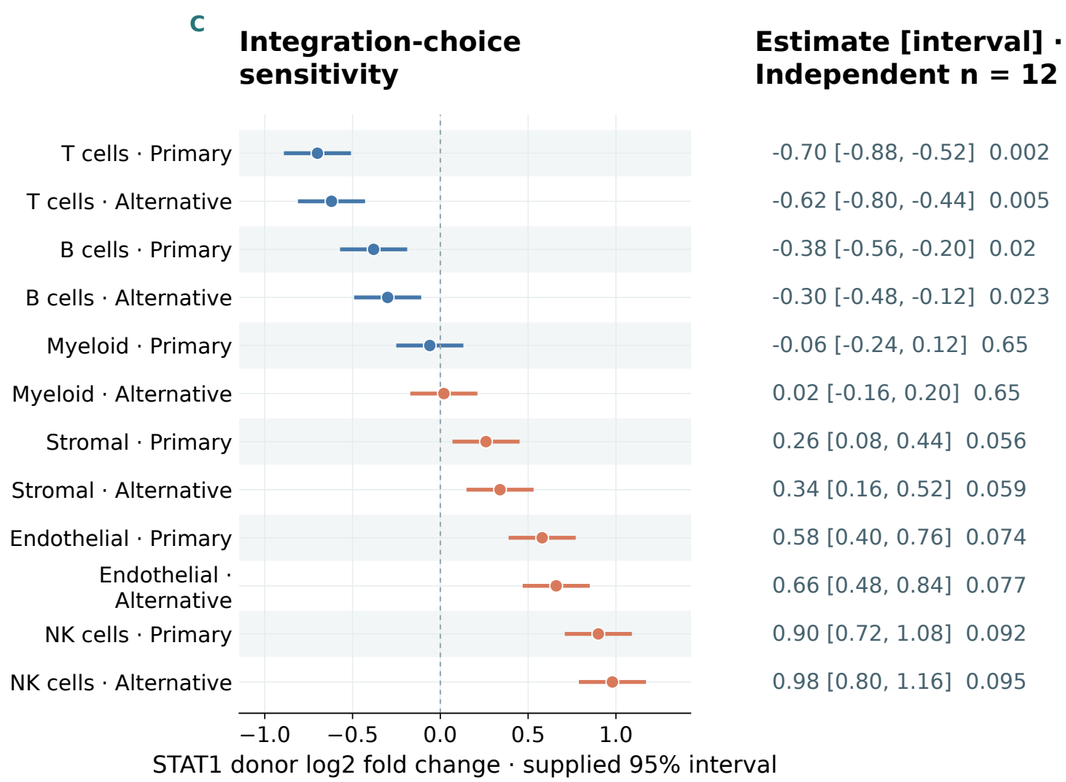
A



B



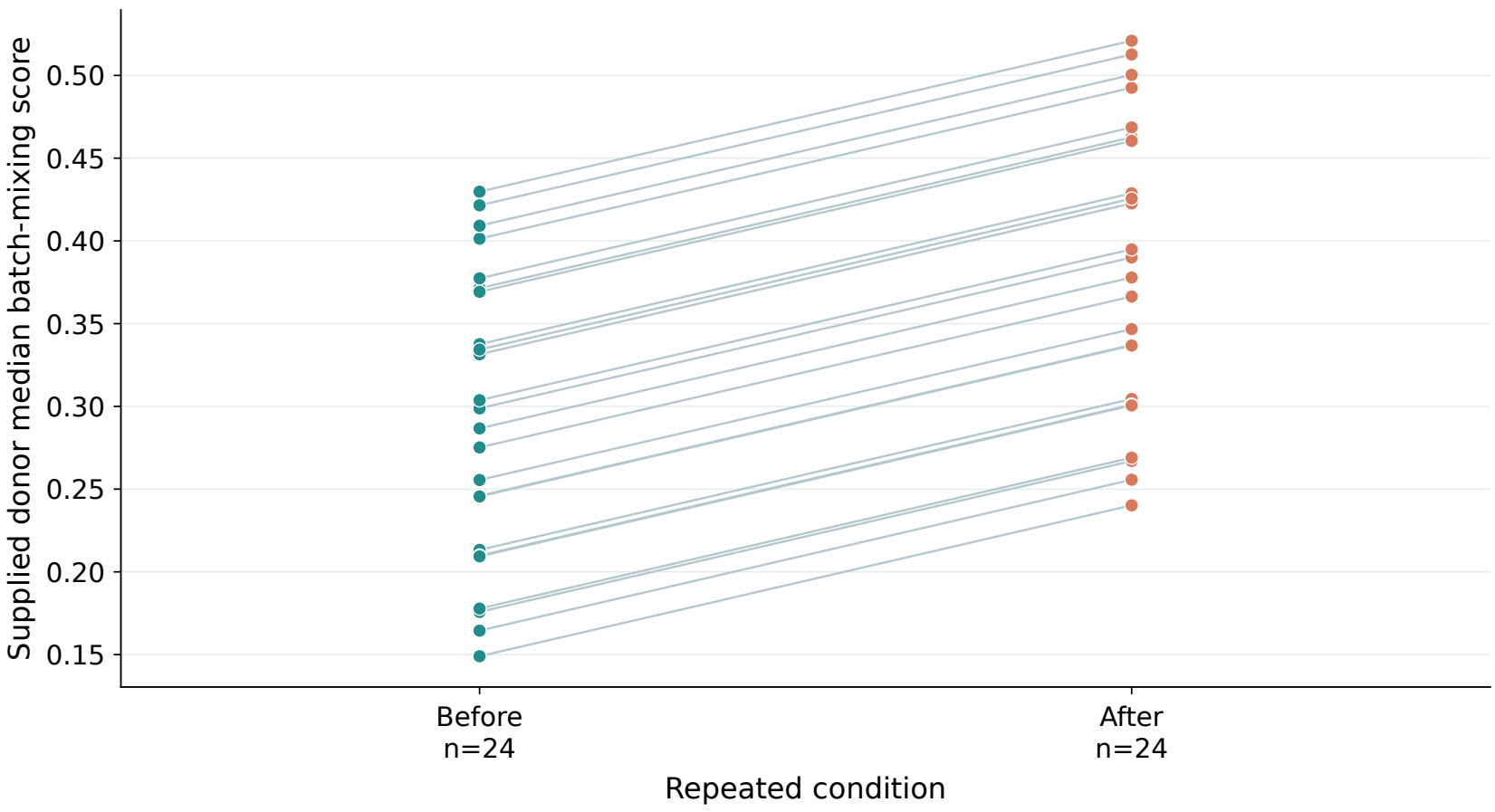
C



Single cell · integration — complementary evidence

A

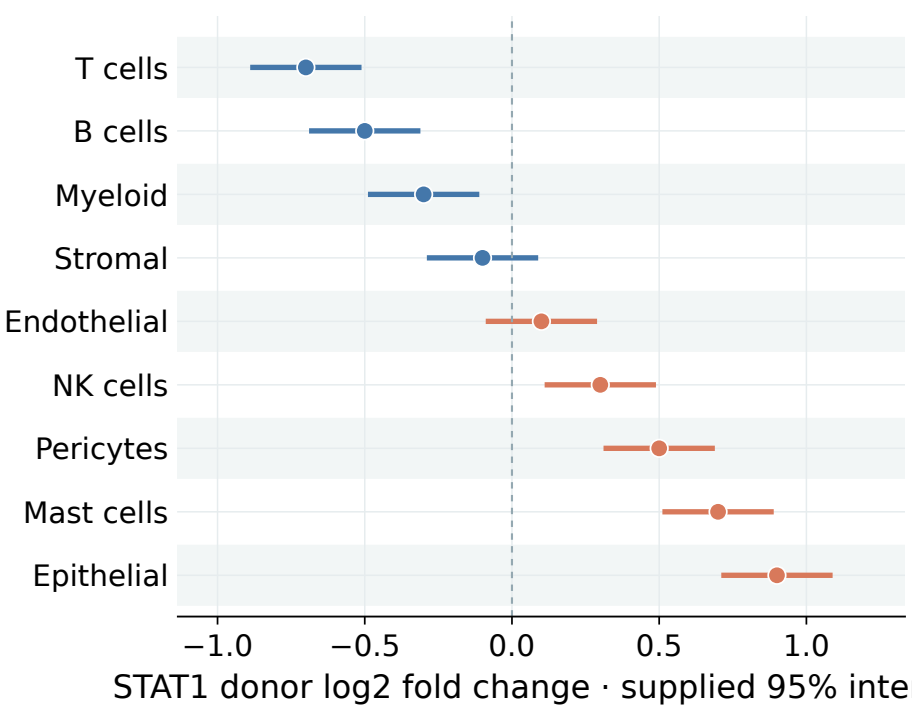
Donor batch-mixing scores before/after integration



B

Donor-level biological-effect preservation

Estimate [interval] · q
Independent n = 24

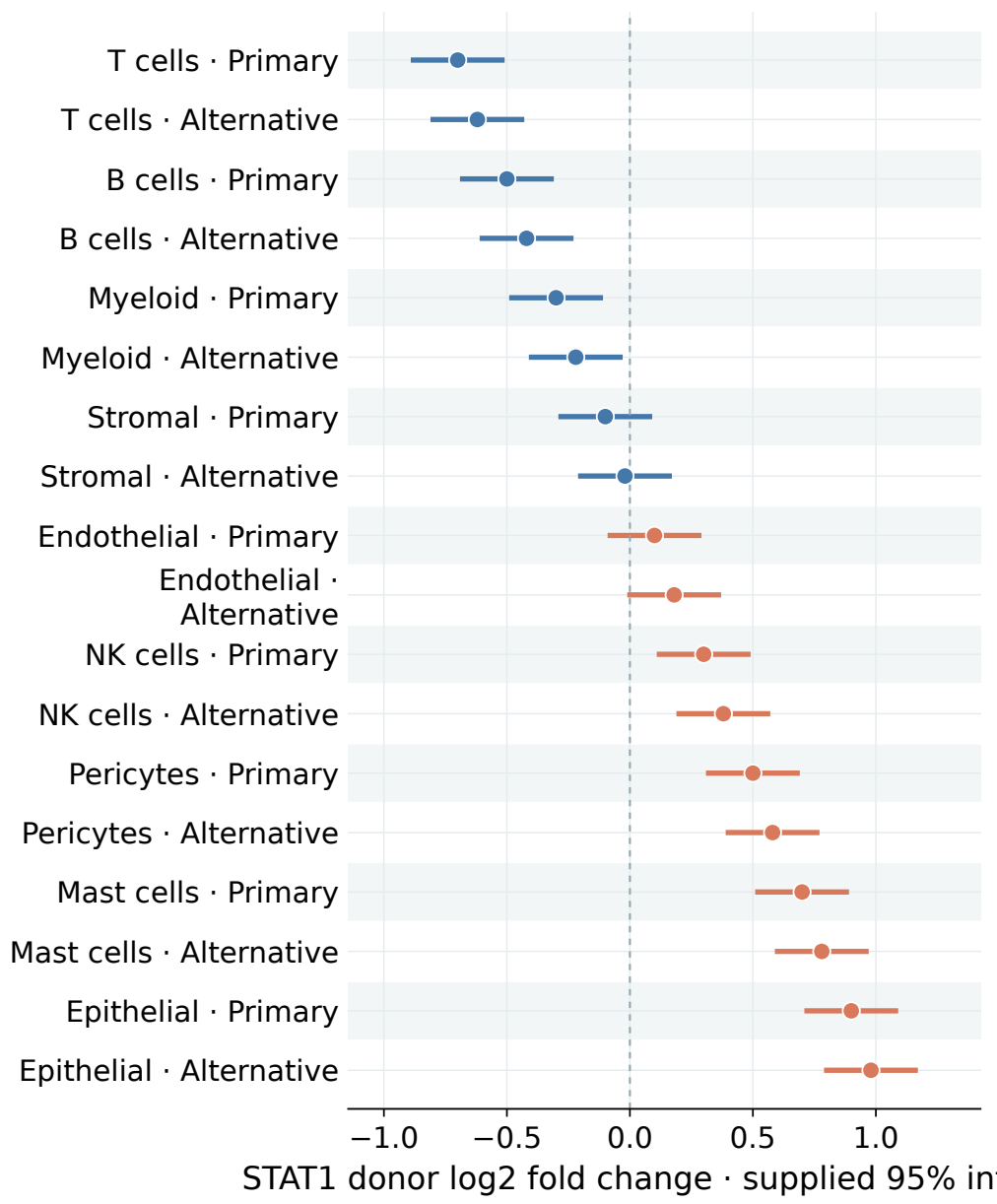


T cells	-0.70 [-0.88, -0.52] 0.002
B cells	-0.50 [-0.68, -0.32] 0.02
Myeloid	-0.30 [-0.48, -0.12] 0.038
Stromal	-0.10 [-0.28, 0.08] 0.65
Endothelial	0.10 [-0.08, 0.28] 0.65
NK cells	0.30 [0.12, 0.48] 0.092
Pericytes	0.50 [0.32, 0.68] 0.11
Mast cells	0.70 [0.52, 0.88] 0.13
Epithelial	0.90 [0.72, 1.08] 0.15

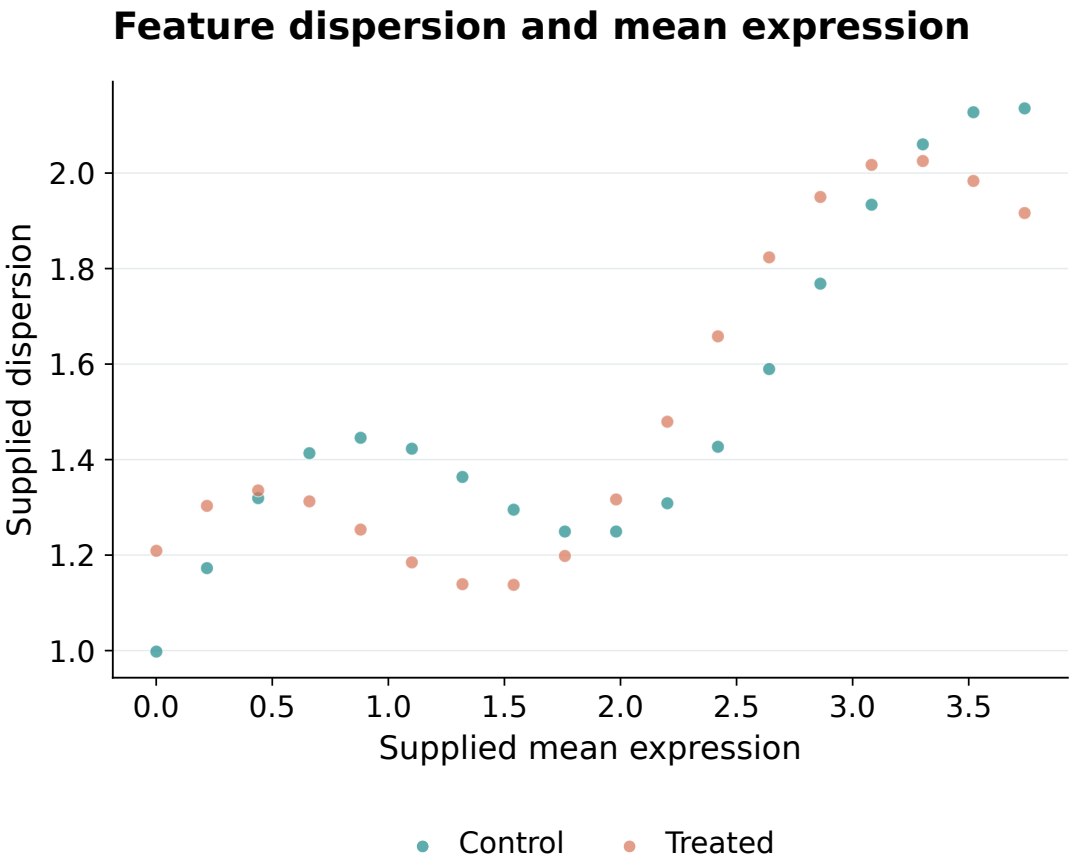
C

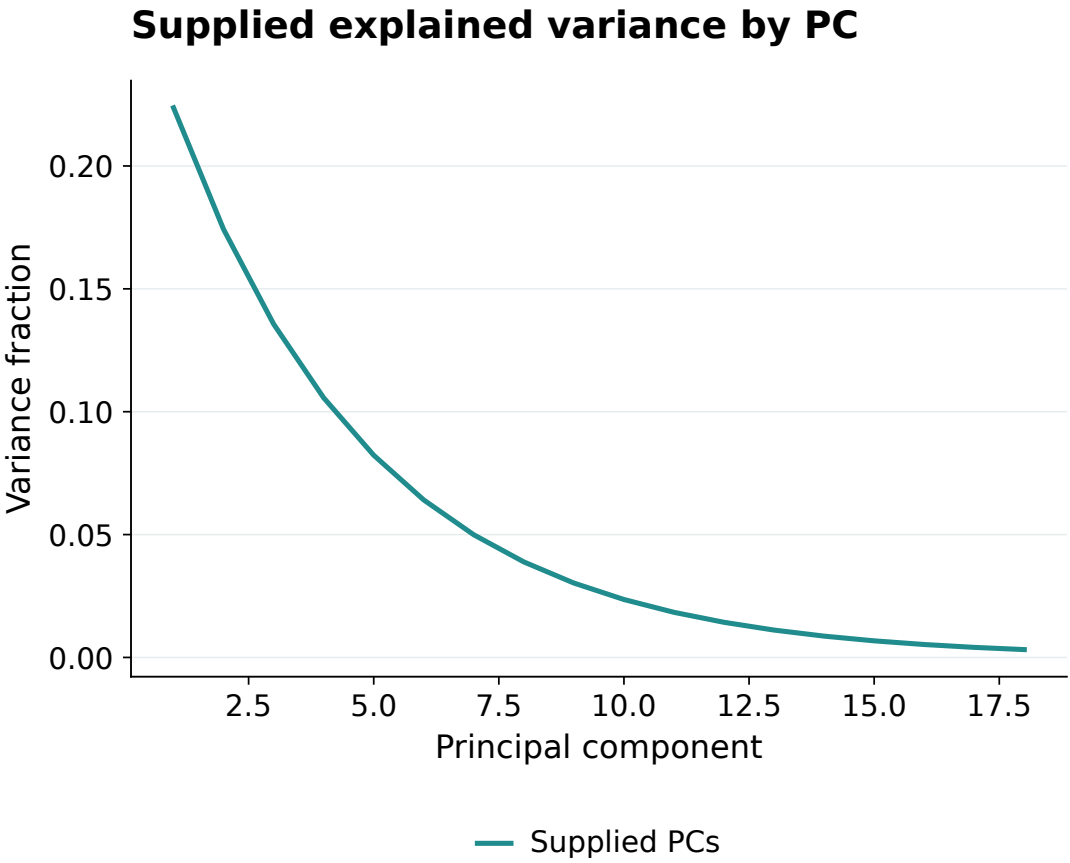
Integration-choice sensitivity

Estimate [interval] · q
Independent n = 24

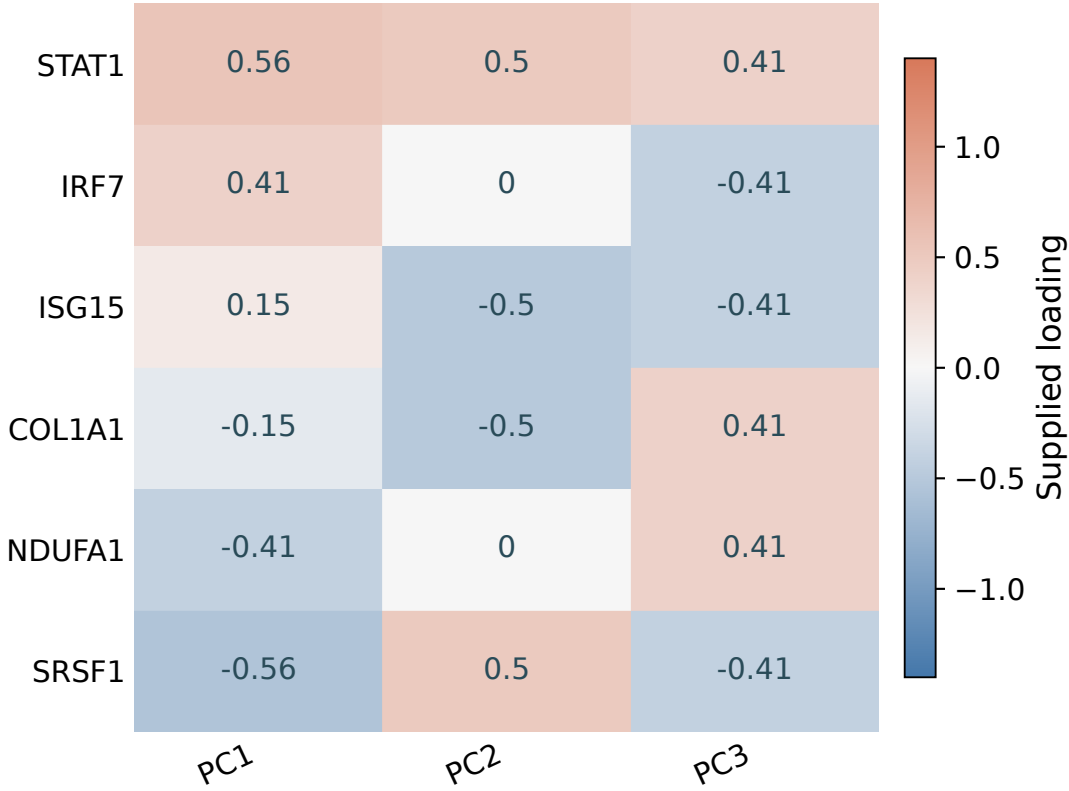


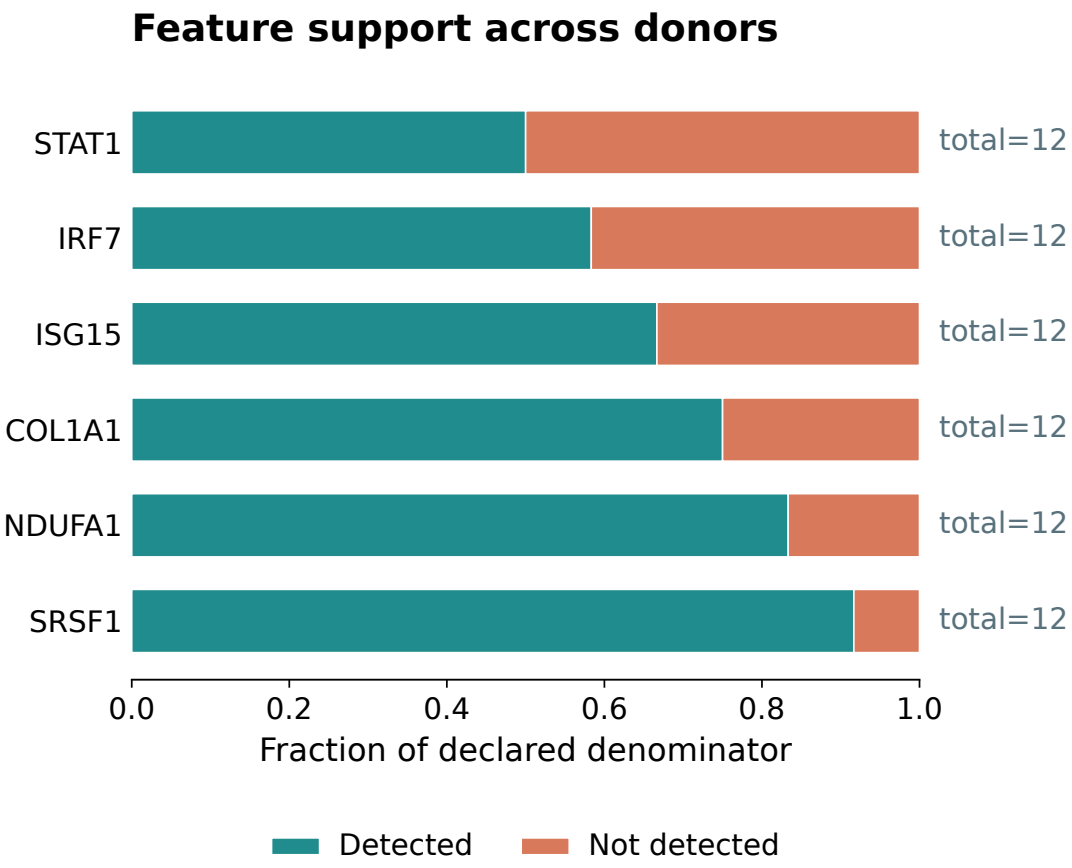
T cells · Primary	-0.70 [-0.88, -0.52] 0.002
T cells · Alternative	-0.62 [-0.80, -0.44] 0.005
B cells · Primary	-0.50 [-0.68, -0.32] 0.02
B cells · Alternative	-0.42 [-0.60, -0.24] 0.023
Myeloid · Primary	-0.30 [-0.48, -0.12] 0.038
Myeloid · Alternative	-0.22 [-0.40, -0.04] 0.041
Stromal · Primary	-0.10 [-0.28, 0.08] 0.65
Stromal · Alternative	-0.02 [-0.20, 0.16] 0.65
Endothelial · Primary	0.10 [-0.08, 0.28] 0.65
Endothelial · Alternative	0.18 [0.00, 0.36] 0.077
NK cells · Primary	0.30 [0.12, 0.48] 0.092
NK cells · Alternative	0.38 [0.20, 0.56] 0.095
Pericytes · Primary	0.50 [0.32, 0.68] 0.11
Pericytes · Alternative	0.58 [0.40, 0.76] 0.11
Mast cells · Primary	0.70 [0.52, 0.88] 0.13
Mast cells · Alternative	0.78 [0.60, 0.96] 0.13
Epithelial · Primary	0.90 [0.72, 1.08] 0.15
Epithelial · Alternative	0.98 [0.80, 1.16] 0.15

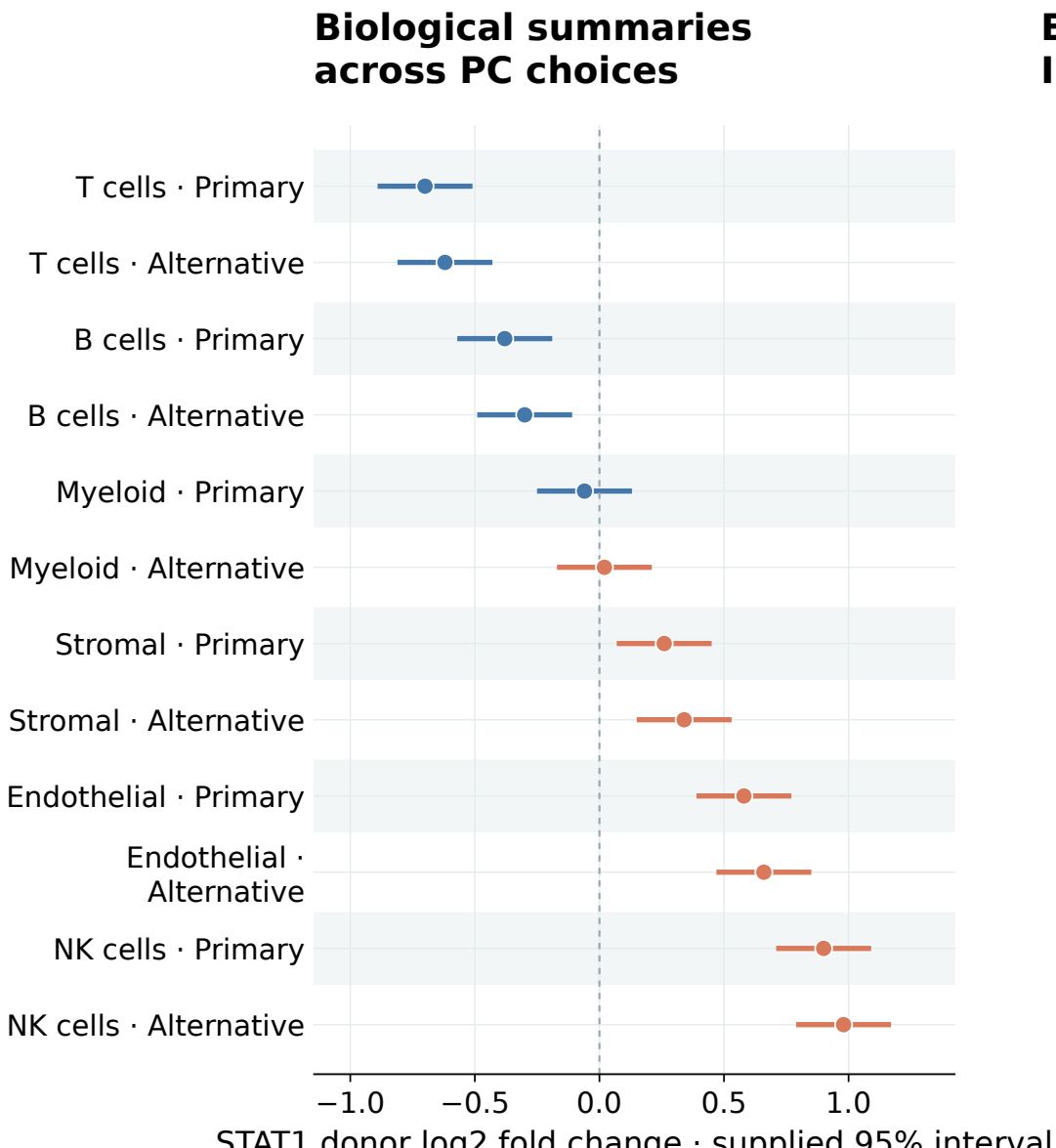

Feature dispersion and mean expression
Supplied dispersion
1.0
1.2
1.4
1.6
1.8
2.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Supplied mean expression
Control
Treated


Supplied explained variance by PC
Variance fraction
0.20
0.15
0.10
0.05
0.00
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Principal component
Supplied PCs

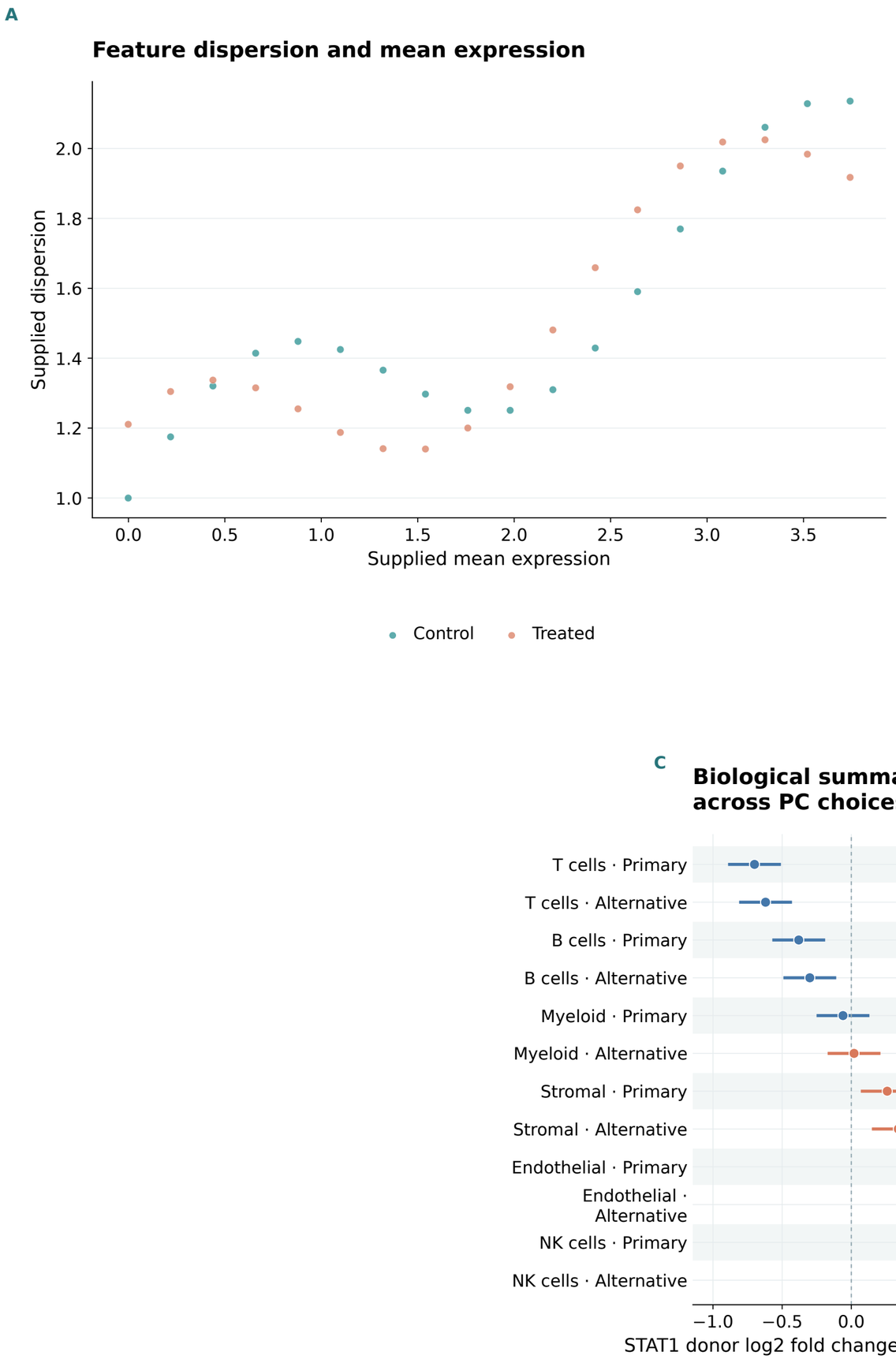
Selected genes and principal-component loadings

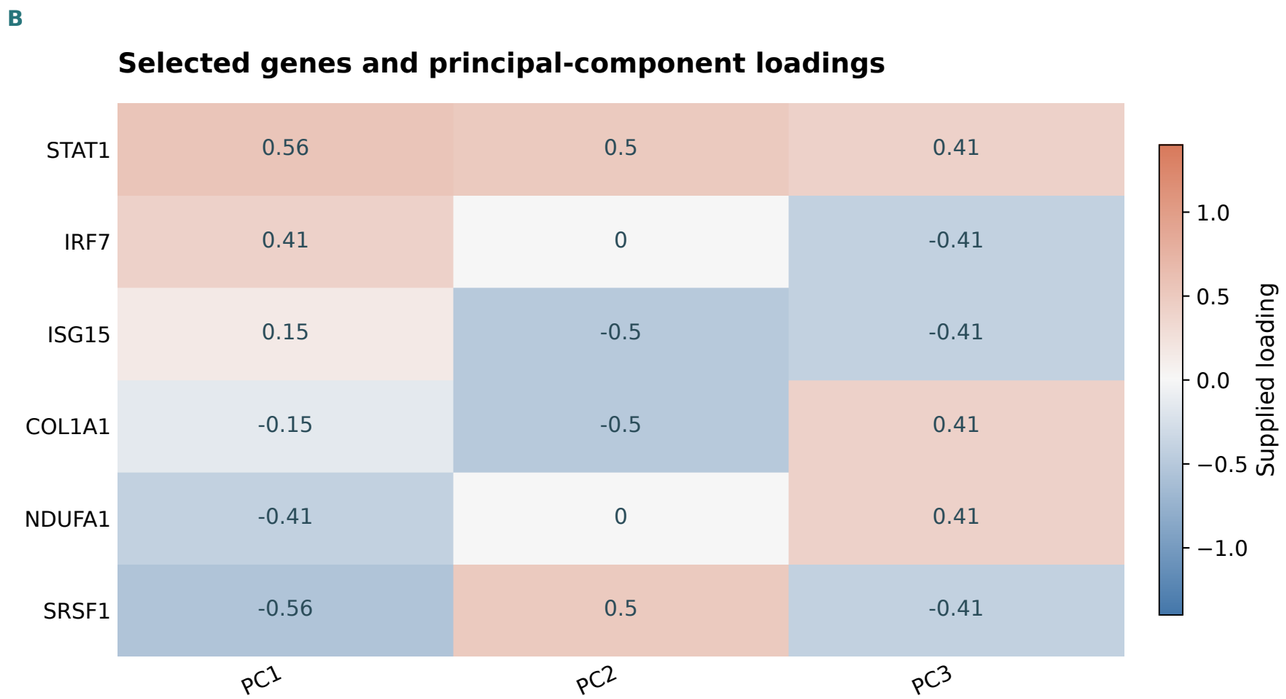


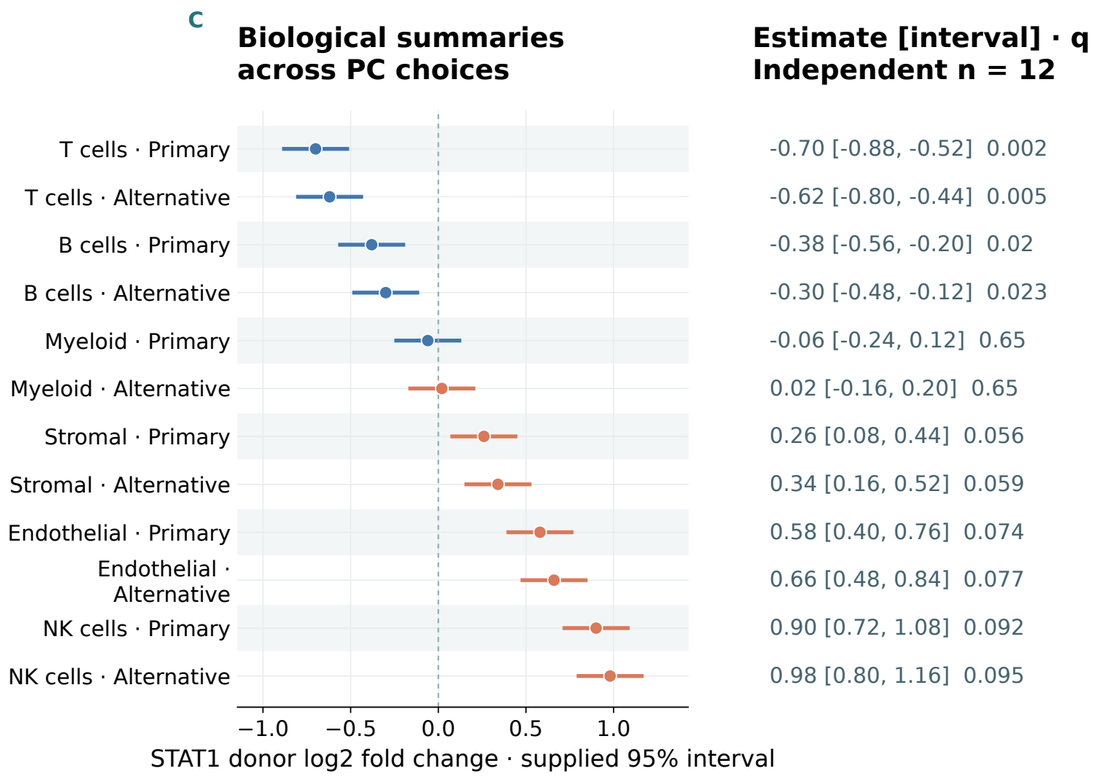

Feature support across donors
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
total=12
total=12
total=12
total=12
total=12
total=12
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Detected
Not detected


Biological summaries
across PC choices
Estimate [interval] · q
Independent n = 12
T cells · Primary
T cells · Alternative
B cells · Primary
B cells · Alternative
Myeloid · Primary
Myeloid · Alternative
Stromal · Primary
Stromal · Alternative
Endothelial · Primary
Endothelial ·
Alternative
NK cells · Primary
NK cells · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
STAT1 donor log2 fold change · supplied 95% interval

Single cell · hvg_pca — complementary evidence



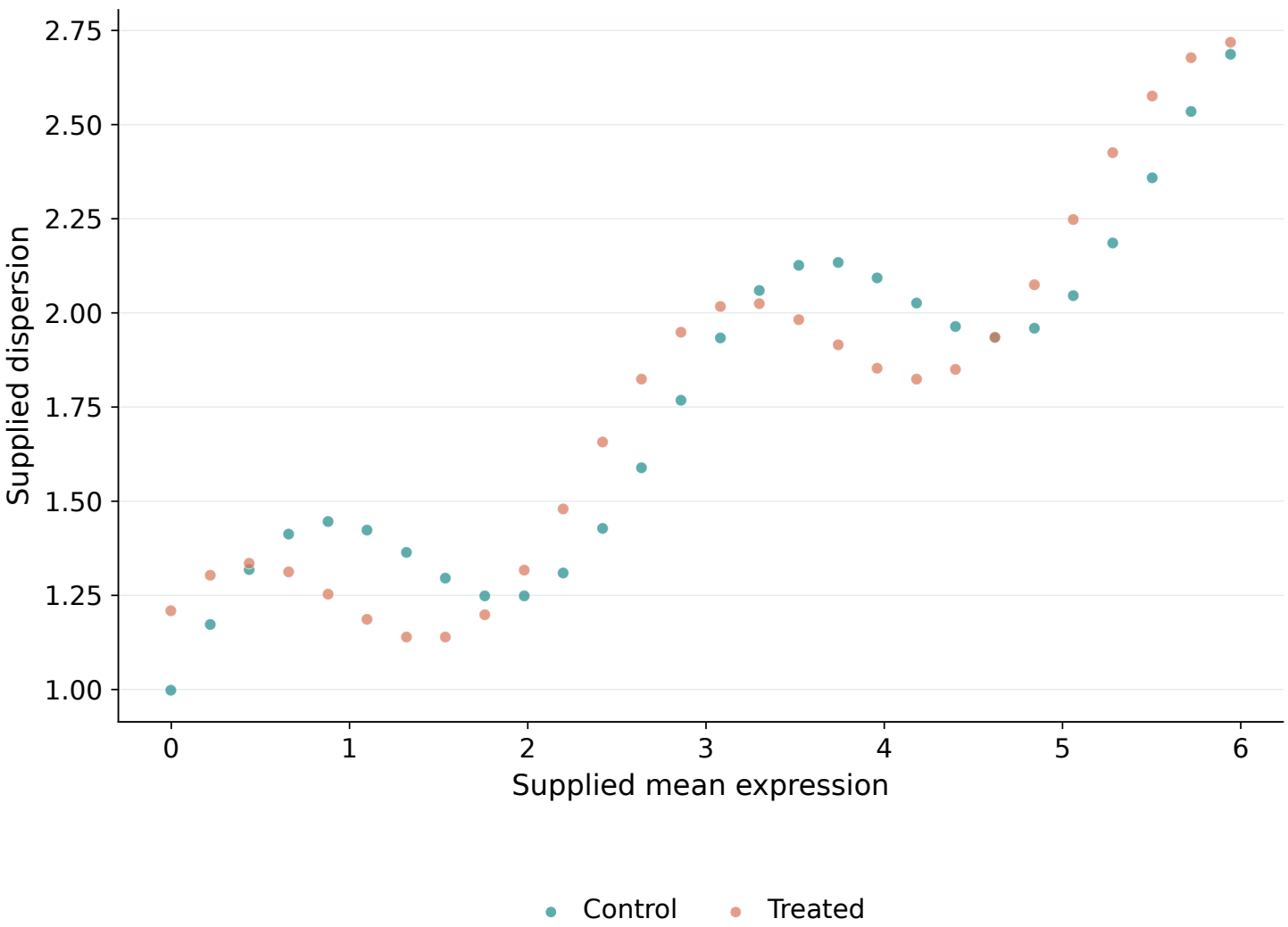




Single cell · hvg_pca — complementary evidence

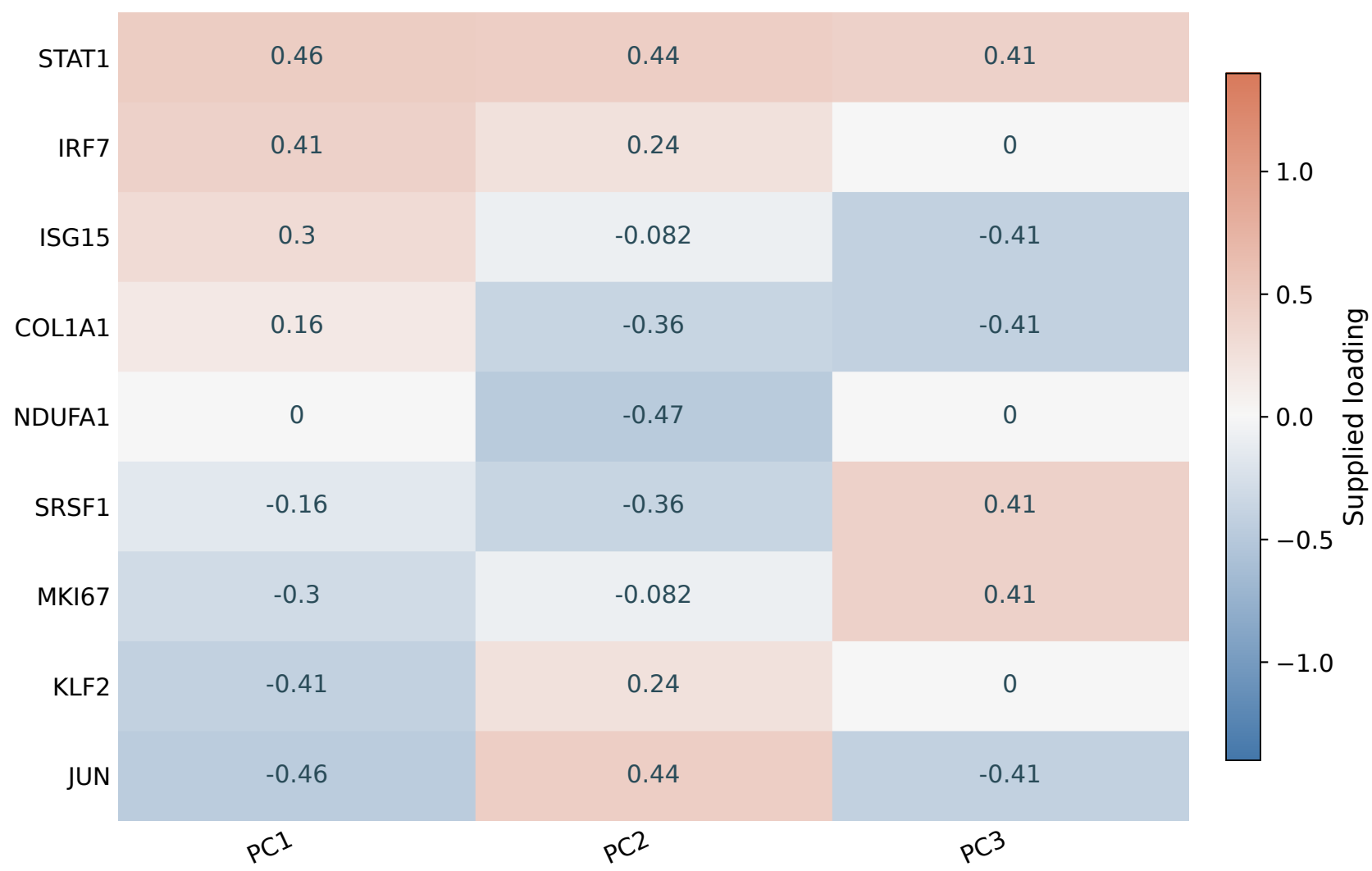
A

Feature dispersion and mean expression



B

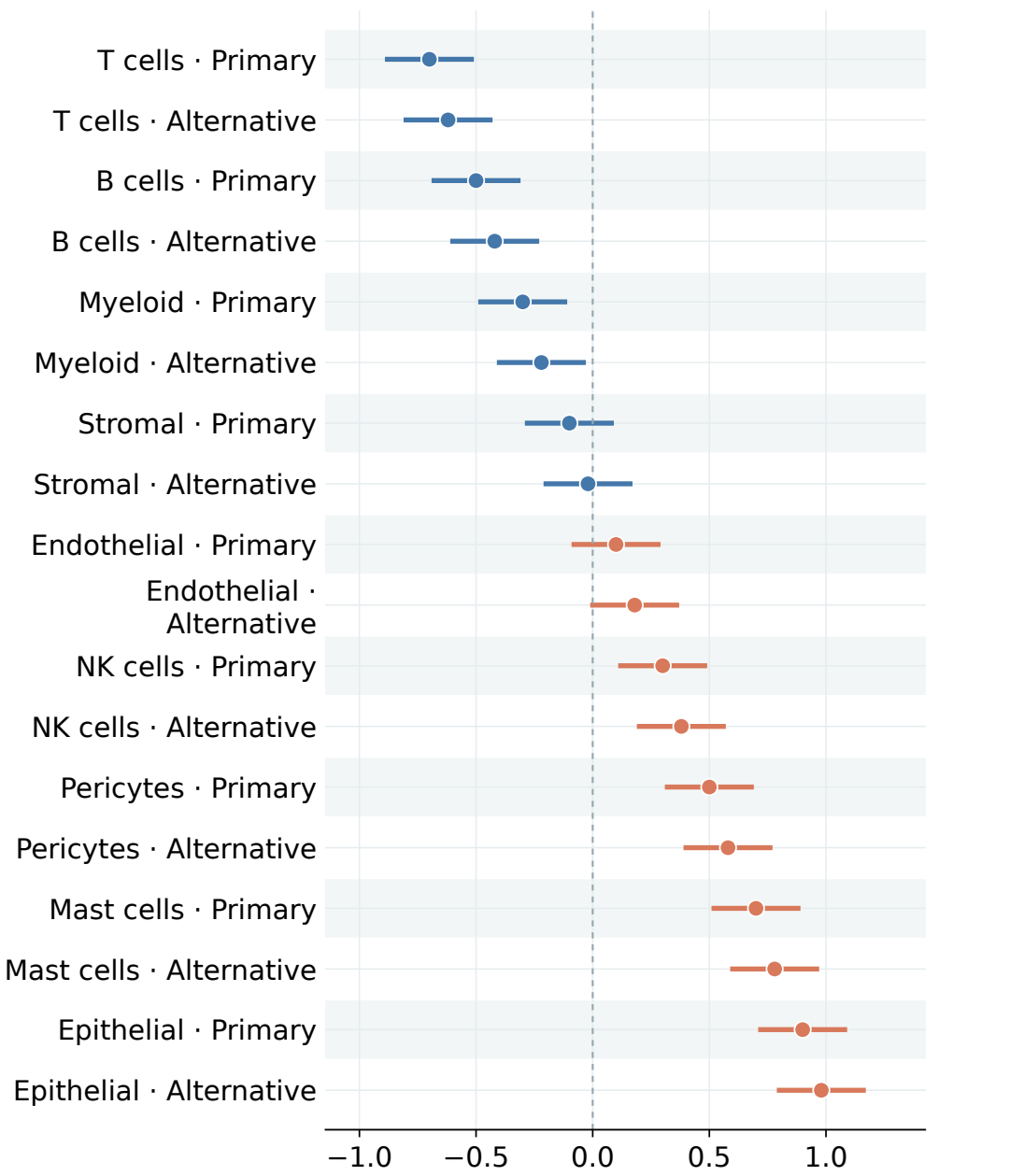
Selected genes and principal-component loadings



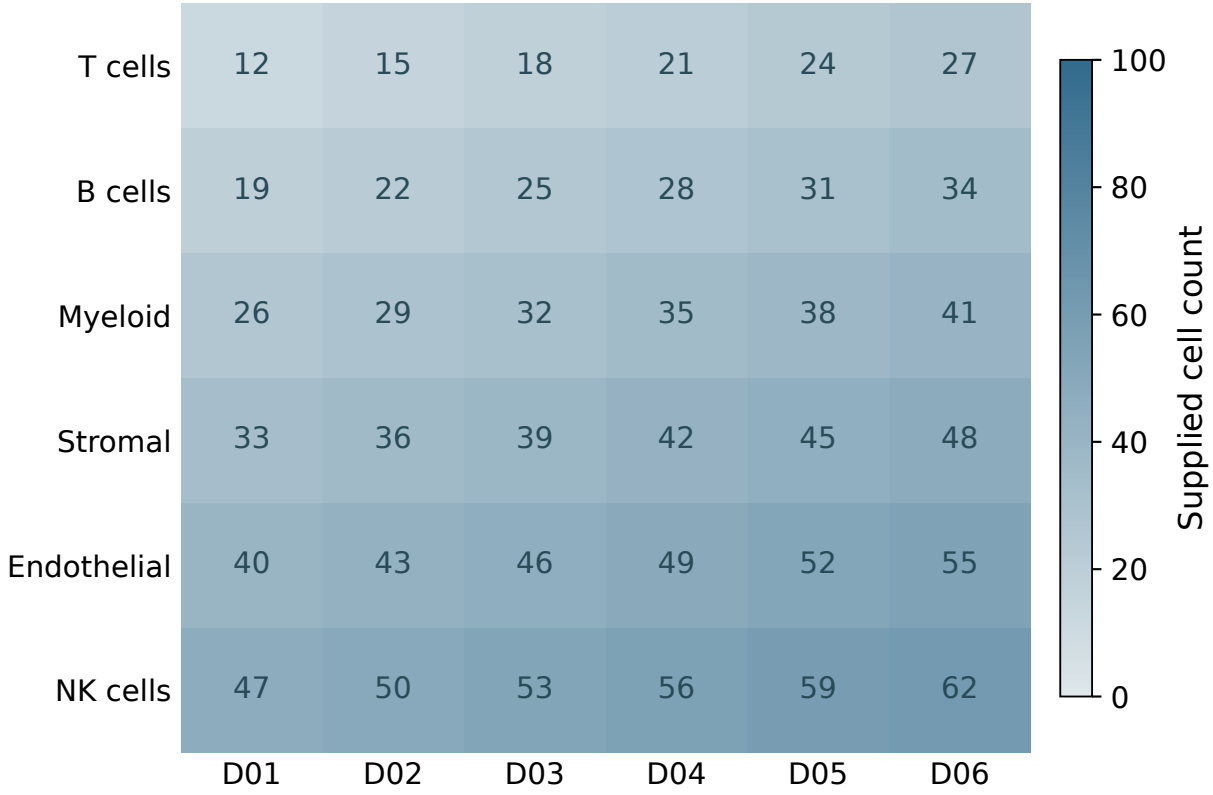
C

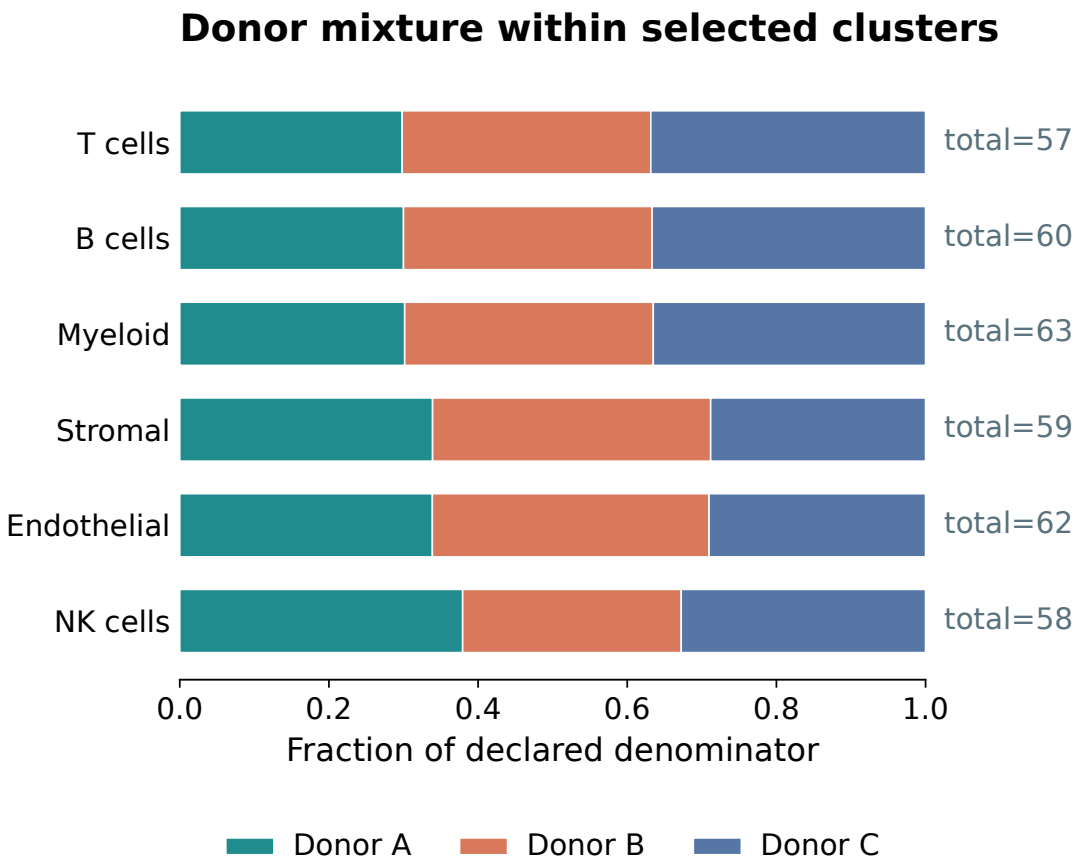
Biological summaries across PC choices

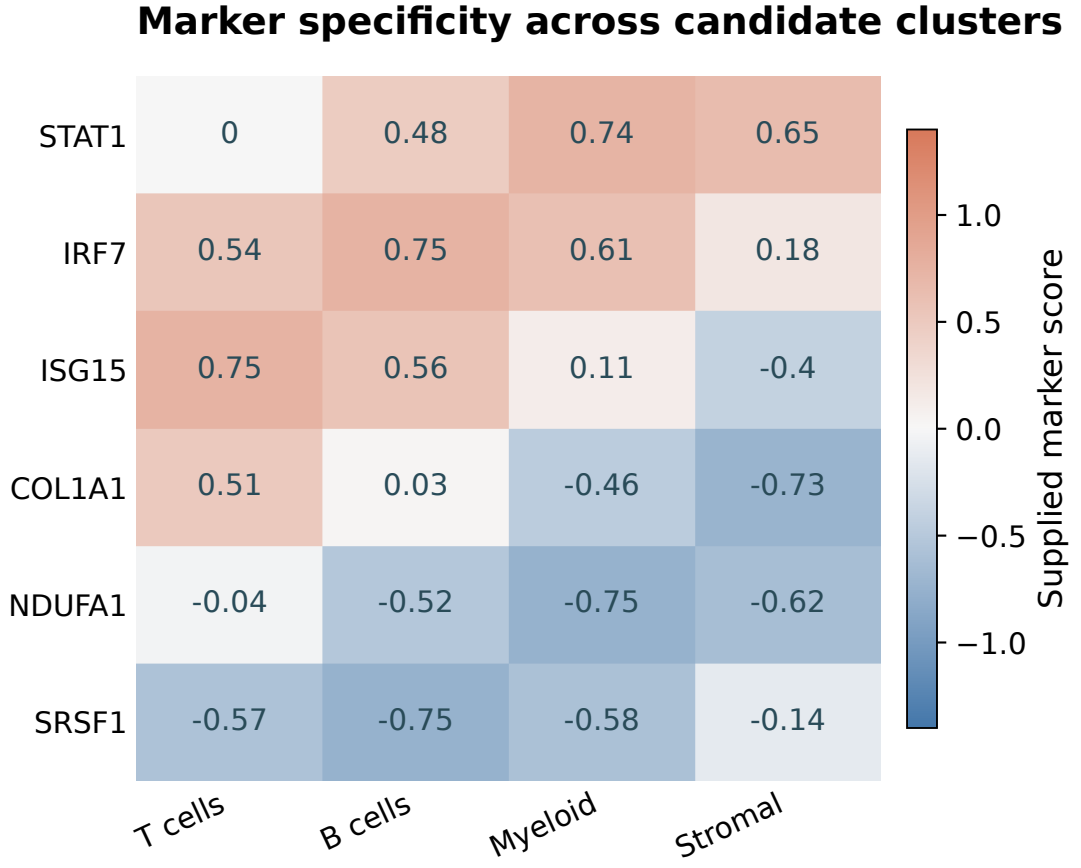
Estimate [interval] · q Independent n = 24

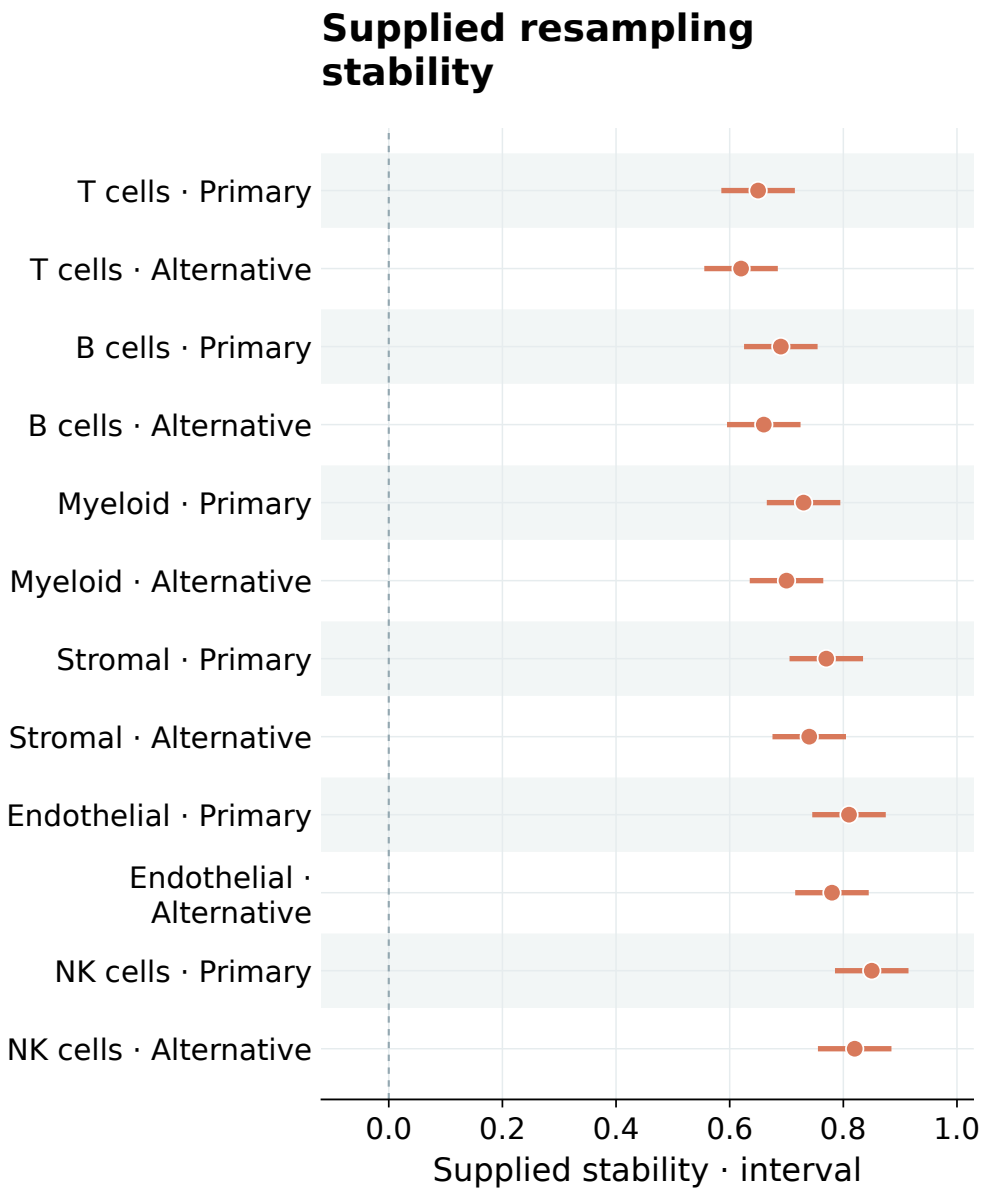


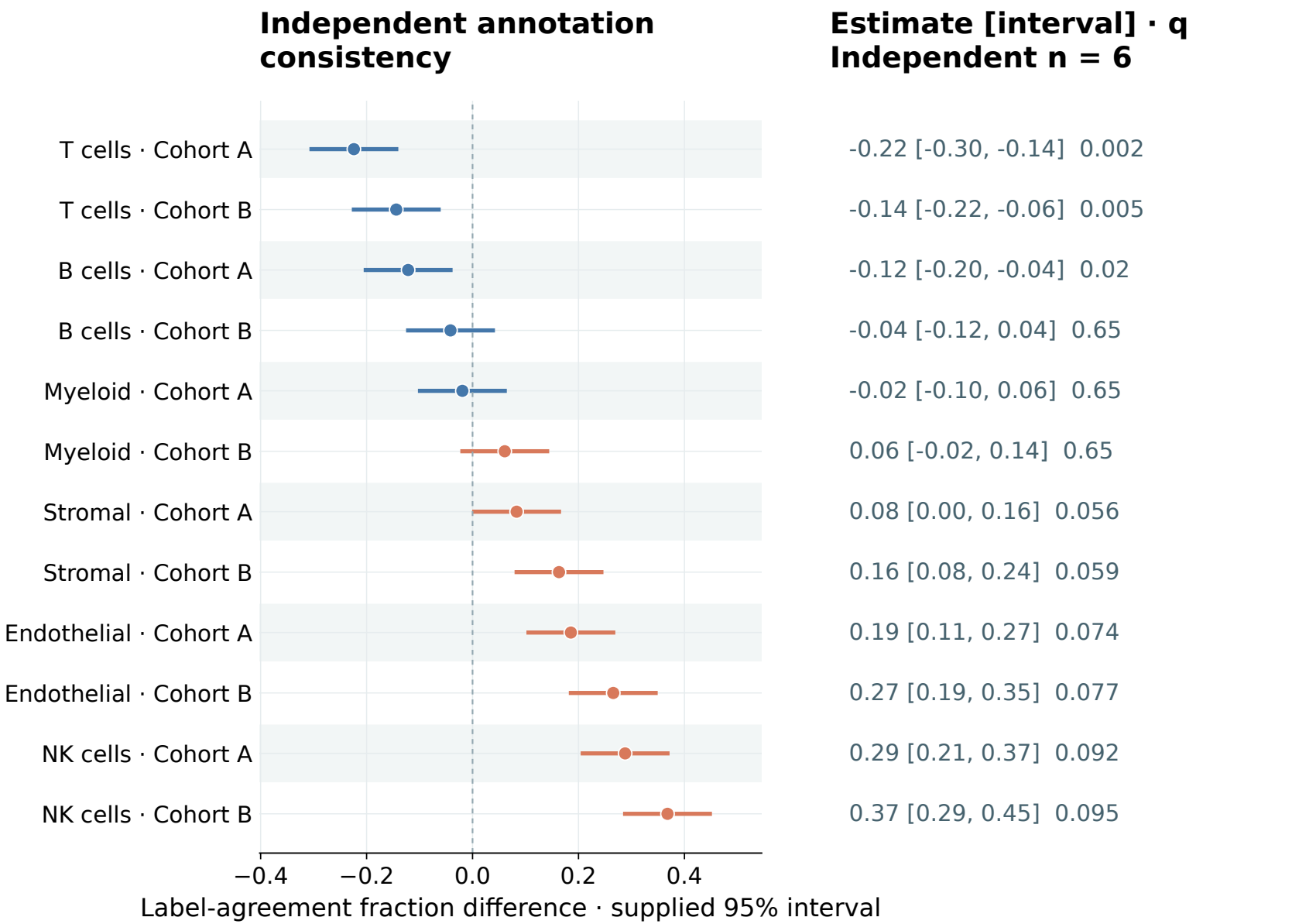
Donor support at declared resolutions




Donor mixture within selected clusters
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Donor A
Donor B
Donor C


Marker specificity across candidate clusters
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0
0.48
0.74
0.65
0.54
0.75
0.61
0.18
0.75
0.56
0.11
-0.4
0.51
0.03
-0.46
-0.73
-0.04
-0.52
-0.75
-0.62
-0.57
-0.75
-0.58
-0.14
T cells
B cells
Myeloid
Stromal
1.0
0.5
0.0
−0.5
−1.0
Supplied marker score

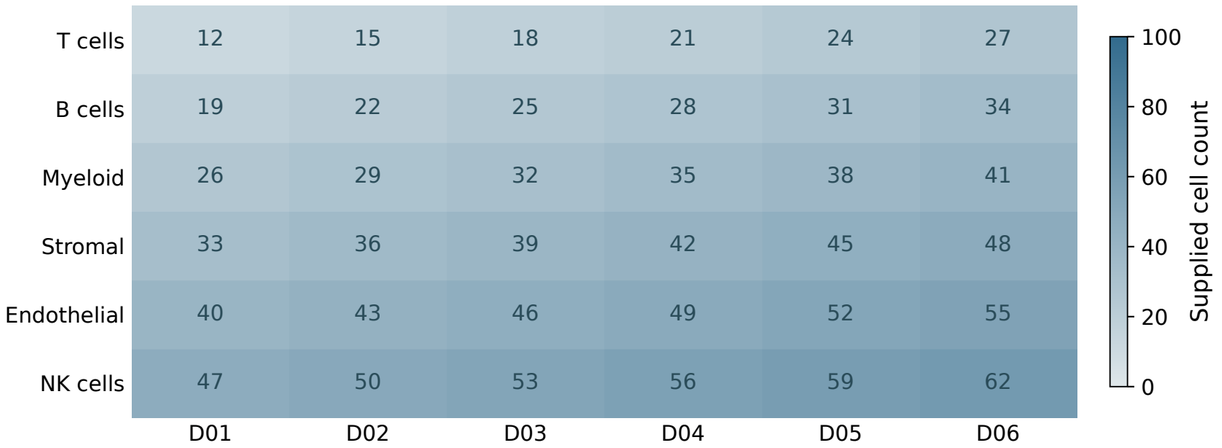

Supplied resampling stability
Estimate [interval] · q
Independent n = 12
T cells · Primary
T cells · Alternative
B cells · Primary
B cells · Alternative
Myeloid · Primary
Myeloid · Alternative
Stromal · Primary
Stromal · Alternative
Endothelial · Primary
Endothelial · Alternative
NK cells · Primary
NK cells · Alternative
0.65 [0.59, 0.71] 0.002
0.62 [0.56, 0.68] 0.005
0.69 [0.63, 0.75] 0.02
0.66 [0.60, 0.72] 0.023
0.73 [0.67, 0.79] 0.038
0.70 [0.64, 0.76] 0.041
0.77 [0.71, 0.83] 0.056
0.74 [0.68, 0.80] 0.059
0.81 [0.75, 0.87] 0.074
0.78 [0.72, 0.84] 0.077
0.85 [0.79, 0.91] 0.092
0.82 [0.76, 0.88] 0.095
0.0
0.2
0.4
0.6
0.8
1.0
Supplied stability · interval


Independent annotation consistency
Estimate [interval] · q
Independent n = 6
T cells · Cohort A
T cells · Cohort B
B cells · Cohort A
B cells · Cohort B
Myeloid · Cohort A
Myeloid · Cohort B
Stromal · Cohort A
Stromal · Cohort B
Endothelial · Cohort A
Endothelial · Cohort B
NK cells · Cohort A
NK cells · Cohort B
-0.22 [-0.30, -0.14] 0.002
-0.14 [-0.22, -0.06] 0.005
-0.12 [-0.20, -0.04] 0.02
-0.04 [-0.12, 0.04] 0.65
-0.02 [-0.10, 0.06] 0.65
0.06 [-0.02, 0.14] 0.65
0.08 [0.00, 0.16] 0.056
0.16 [0.08, 0.24] 0.059
0.19 [0.11, 0.27] 0.074
0.27 [0.19, 0.35] 0.077
0.29 [0.21, 0.37] 0.092
0.37 [0.29, 0.45] 0.095
−0.4
−0.2
0.0
0.2
0.4
Label-agreement fraction difference · supplied 95% interval

Single cell · resolution — complementary evidence

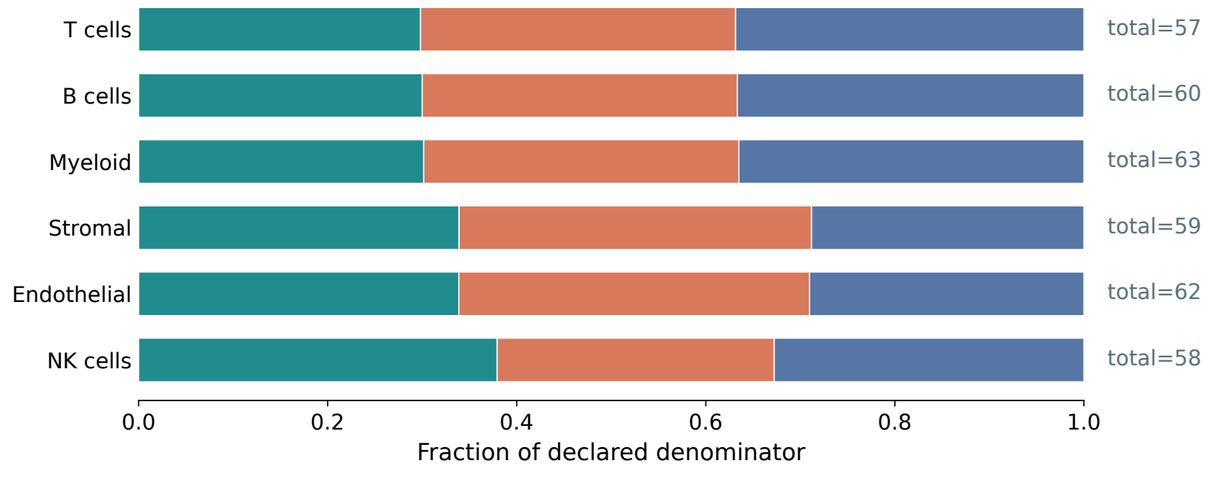
A

Donor support at declared resolutions



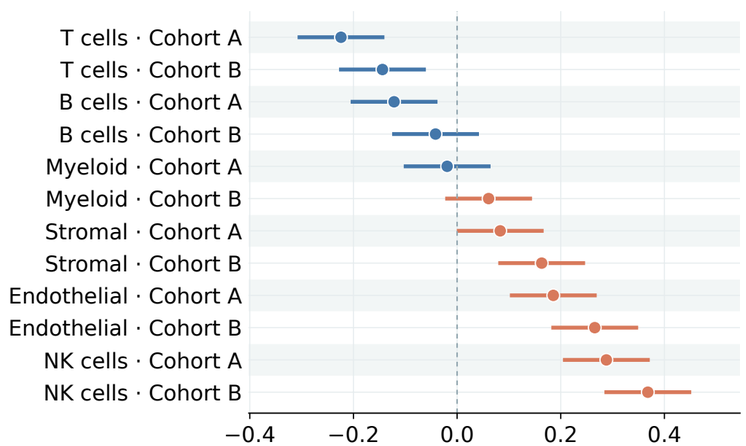
B

Donor mixture within selected clusters



C

Independent annotation consistency



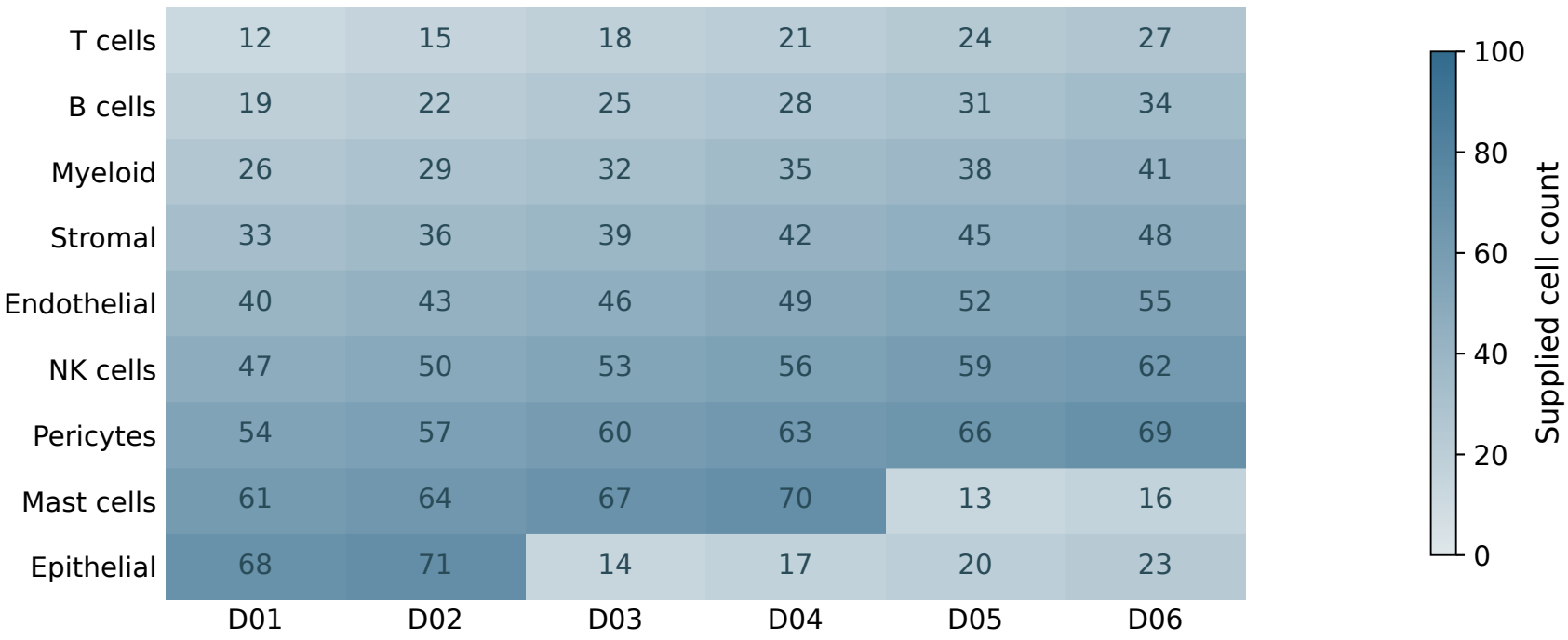
Estimate [interval] · q
Independent n = 6

T cells · Cohort A	-0.22 [-0.30, -0.14] 0.002
T cells · Cohort B	-0.14 [-0.22, -0.06] 0.005
B cells · Cohort A	-0.12 [-0.20, -0.04] 0.02
B cells · Cohort B	-0.04 [-0.12, 0.04] 0.65
Myeloid · Cohort A	-0.02 [-0.10, 0.06] 0.65
Myeloid · Cohort B	0.06 [-0.02, 0.14] 0.65
Stromal · Cohort A	0.08 [0.00, 0.16] 0.056
Stromal · Cohort B	0.16 [0.08, 0.24] 0.059
Endothelial · Cohort A	0.19 [0.11, 0.27] 0.074
Endothelial · Cohort B	0.27 [0.19, 0.35] 0.077
NK cells · Cohort A	0.29 [0.21, 0.37] 0.092
NK cells · Cohort B	0.37 [0.29, 0.45] 0.095

Single cell · resolution — complementary evidence

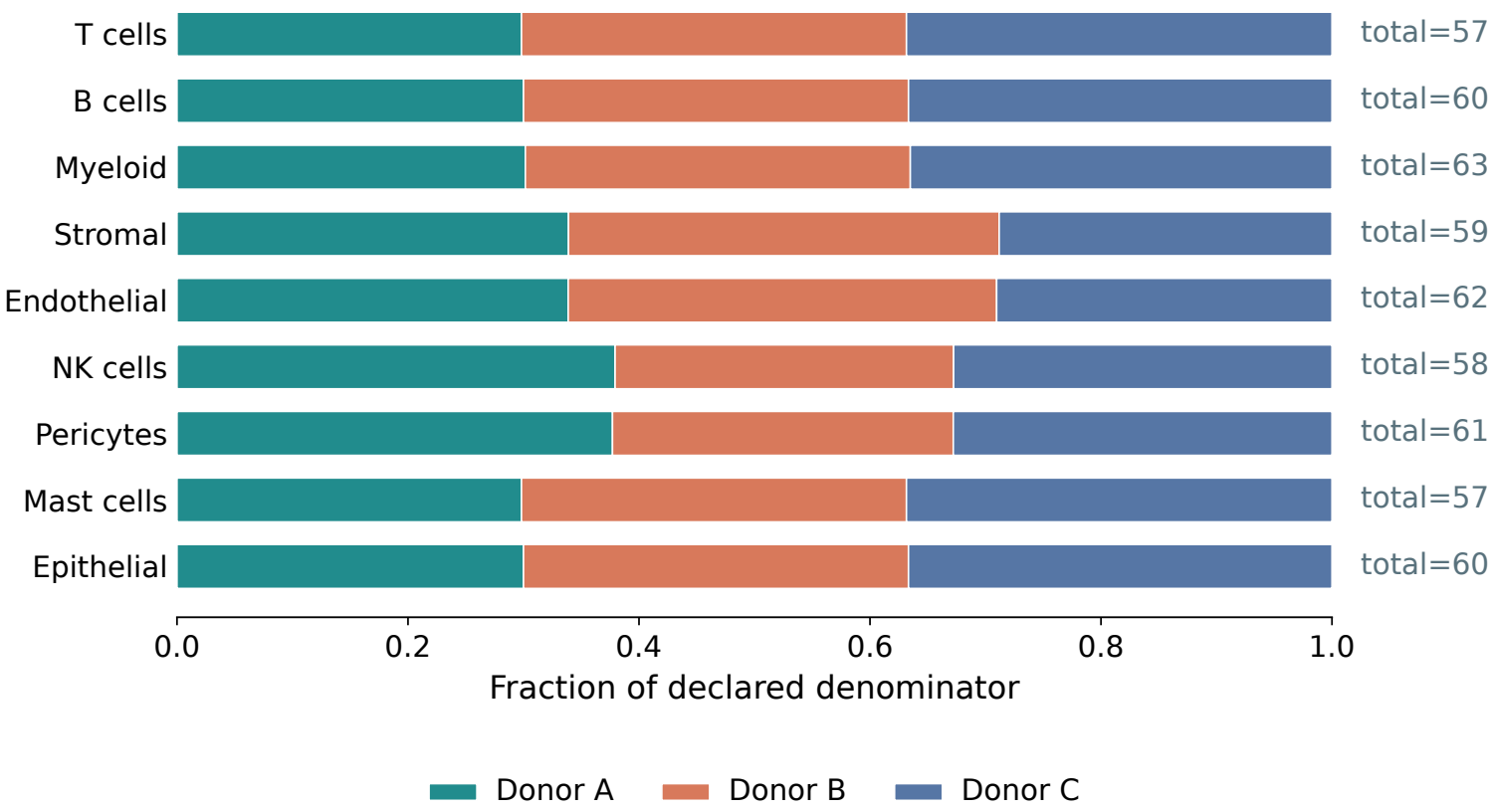
A

Donor support at declared resolutions



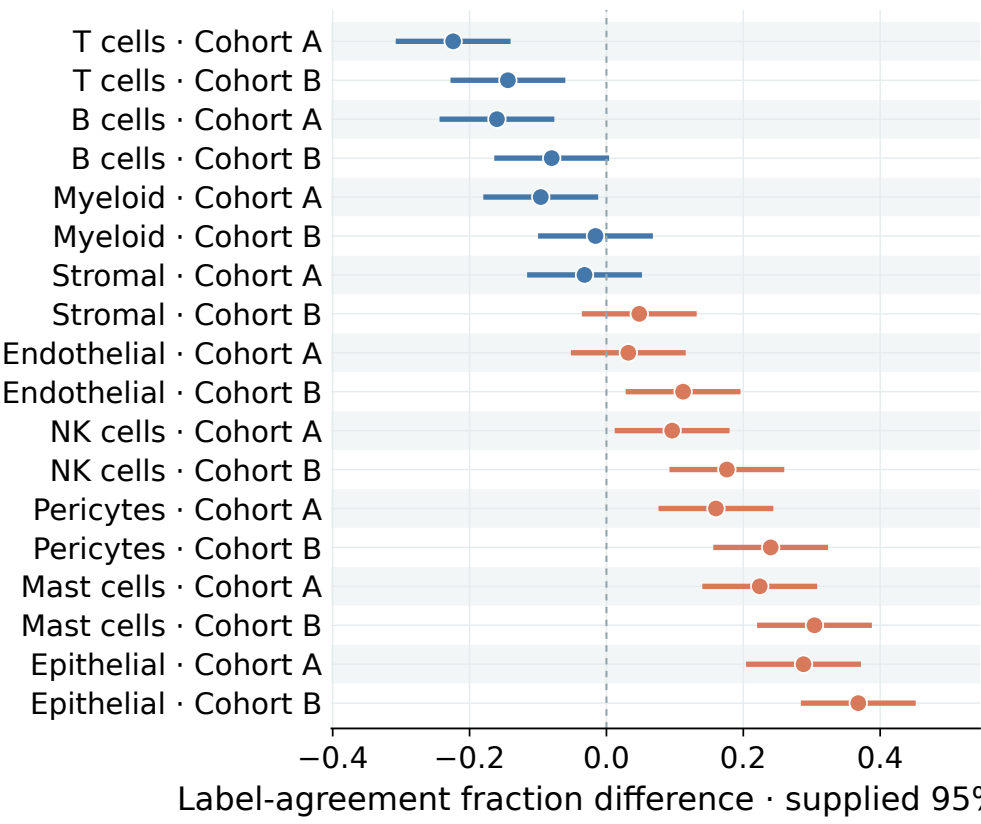
B

Donor mixture within selected clusters



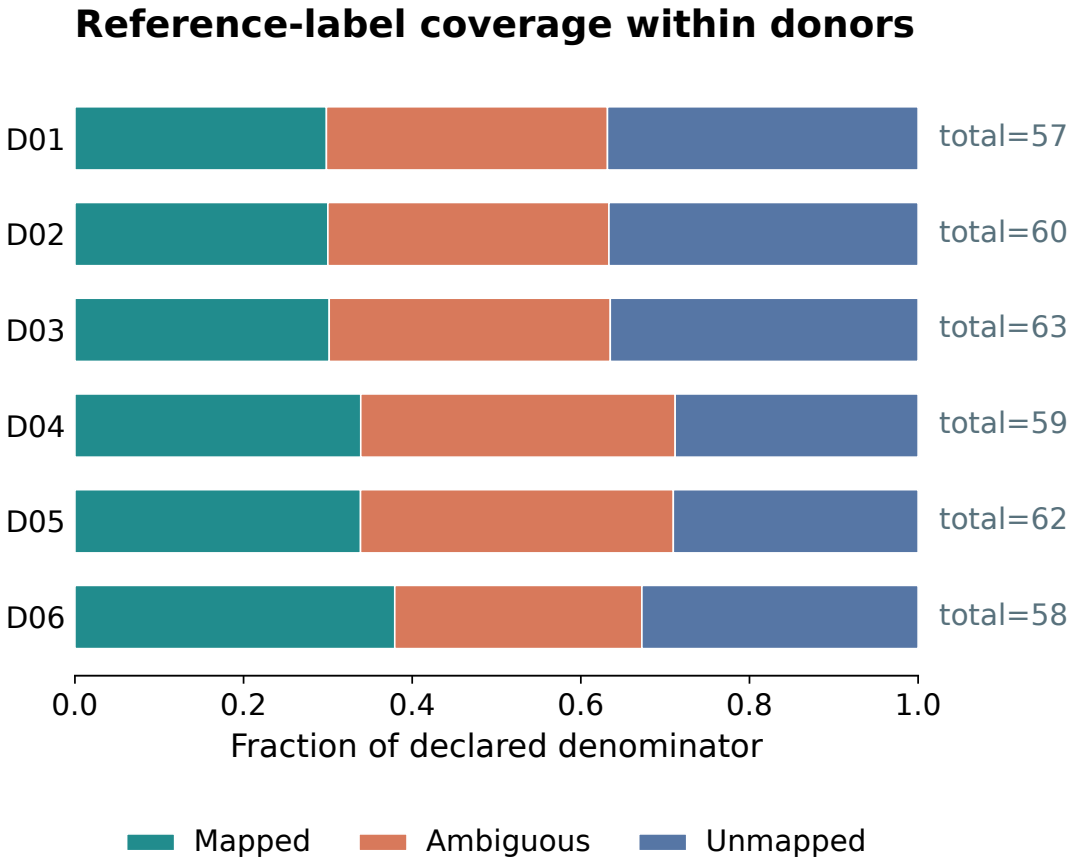
C

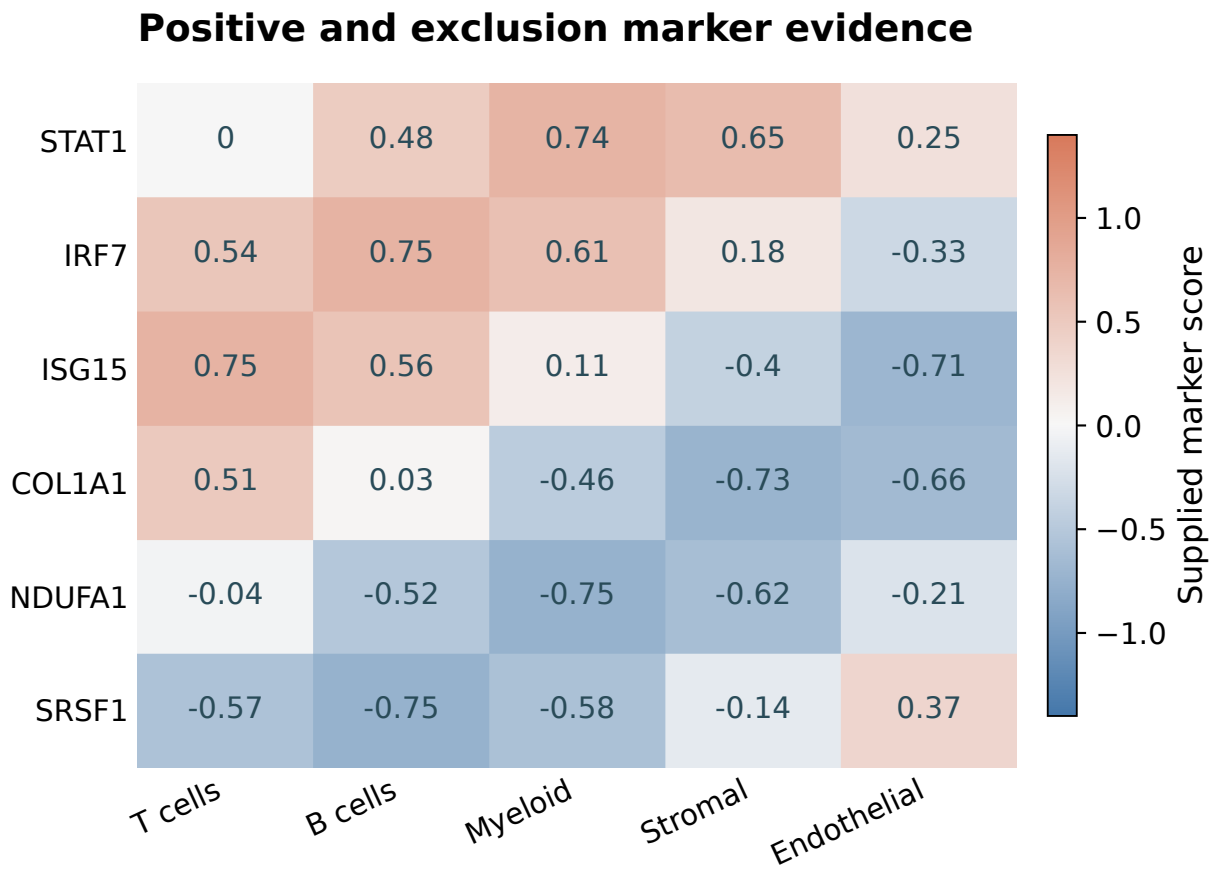
Independent annotation consistency

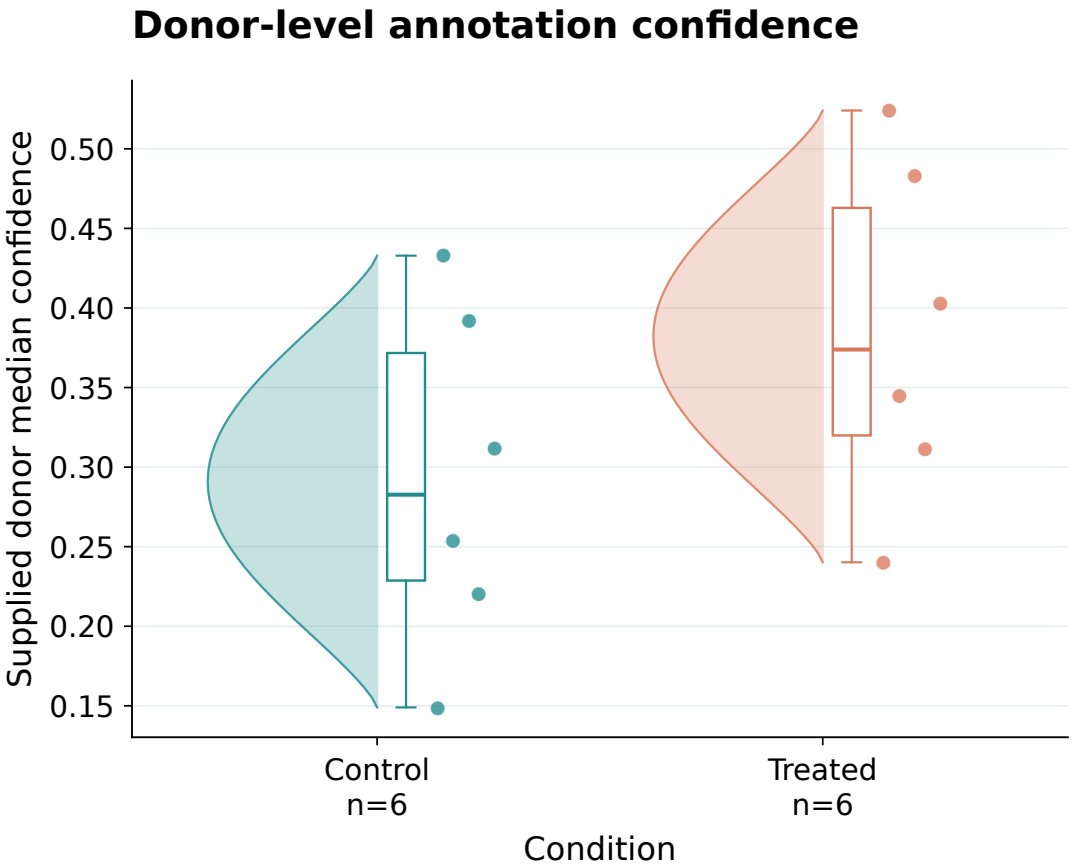


Estimate [interval] · q
Independent n = 12

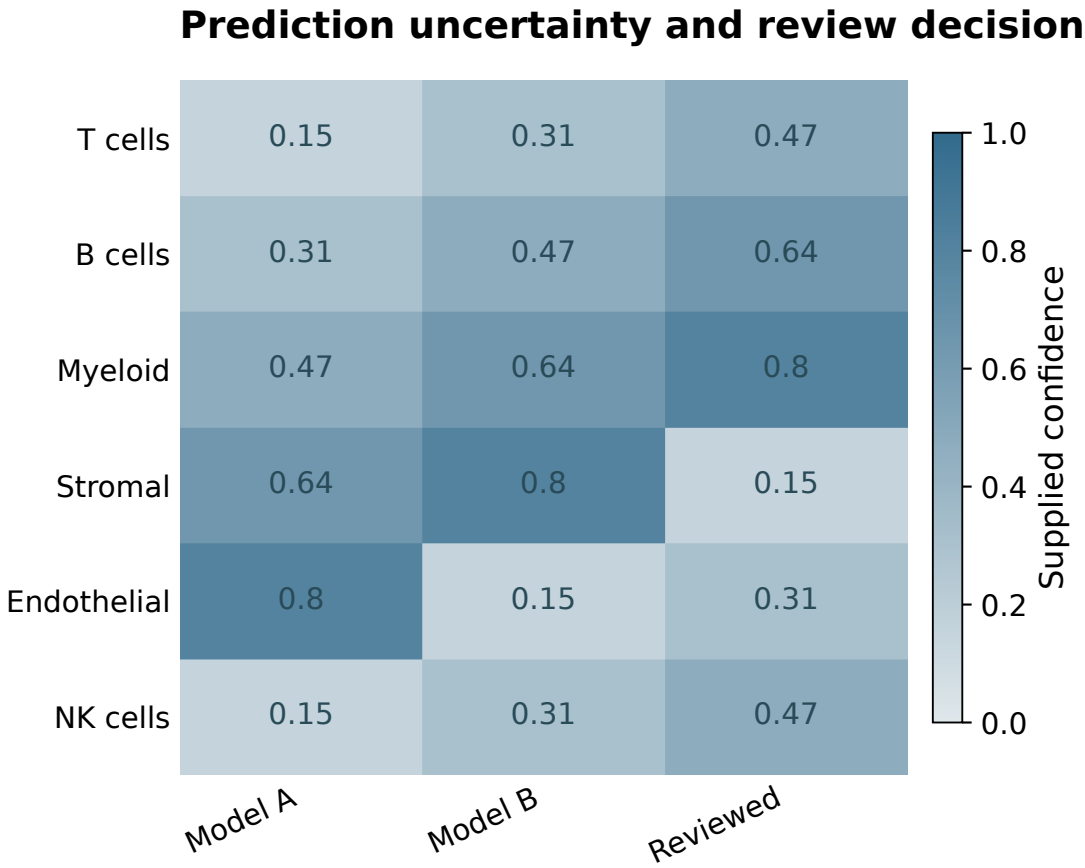
T cells · Cohort A	-0.22 [-0.30, -0.14] 0.002
T cells · Cohort B	-0.14 [-0.22, -0.06] 0.005
B cells · Cohort A	-0.16 [-0.24, -0.08] 0.02
B cells · Cohort B	-0.08 [-0.16, 0.00] 0.65
Myeloid · Cohort A	-0.10 [-0.18, -0.02] 0.038
Myeloid · Cohort B	-0.02 [-0.10, 0.06] 0.65
Stromal · Cohort A	-0.03 [-0.11, 0.05] 0.65
Stromal · Cohort B	0.05 [-0.03, 0.13] 0.65
Endothelial · Cohort A	0.03 [-0.05, 0.11] 0.65
Endothelial · Cohort B	0.11 [0.03, 0.19] 0.077
NK cells · Cohort A	0.10 [0.02, 0.18] 0.092
NK cells · Cohort B	0.18 [0.10, 0.26] 0.095
Pericytes · Cohort A	0.16 [0.08, 0.24] 0.11
Pericytes · Cohort B	0.24 [0.16, 0.32] 0.11
Mast cells · Cohort A	0.22 [0.14, 0.30] 0.13
Mast cells · Cohort B	0.30 [0.22, 0.38] 0.13
Epithelial · Cohort A	0.29 [0.21, 0.37] 0.15
Epithelial · Cohort B	0.37 [0.29, 0.45] 0.15

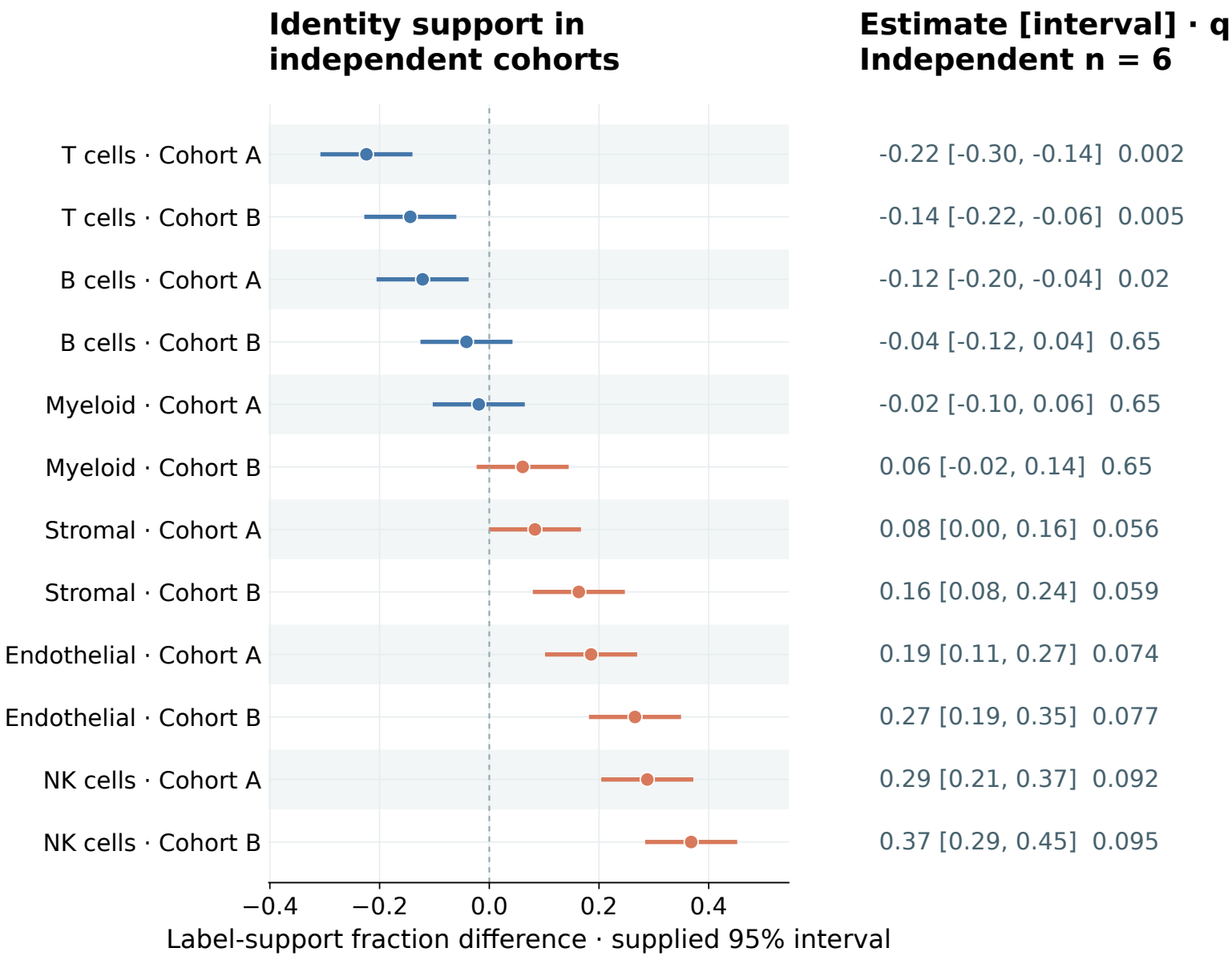

Reference-label coverage within donors
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Mapped
Ambiguous
Unmapped

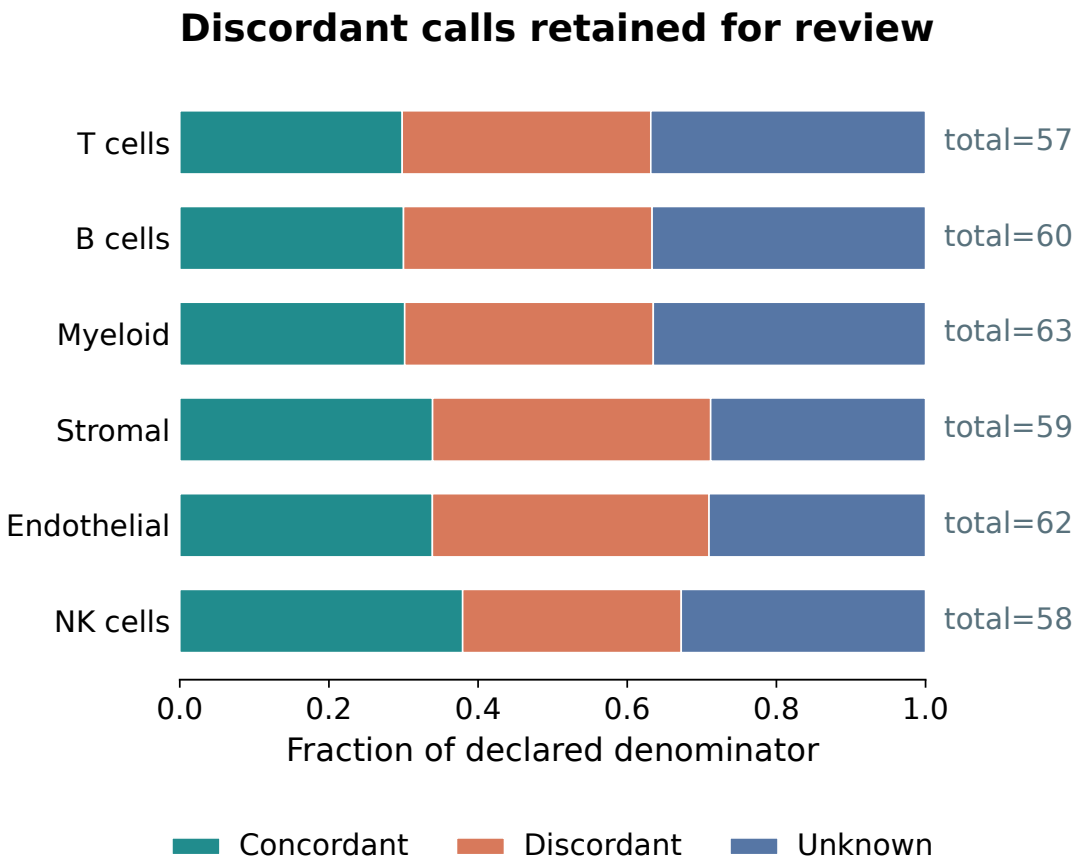

Positive and exclusion marker evidence
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0
0.48
0.74
0.65
0.25
0.54
0.75
0.61
0.18
-0.33
0.75
0.56
0.11
-0.4
-0.71
0.51
0.03
-0.46
-0.73
-0.66
-0.04
-0.52
-0.75
-0.62
-0.21
-0.57
-0.75
-0.58
-0.14
0.37
T cells
B cells
Myeloid
Stromal
Endothelial
1.0
0.5
0.0
−0.5
−1.0
Supplied marker score



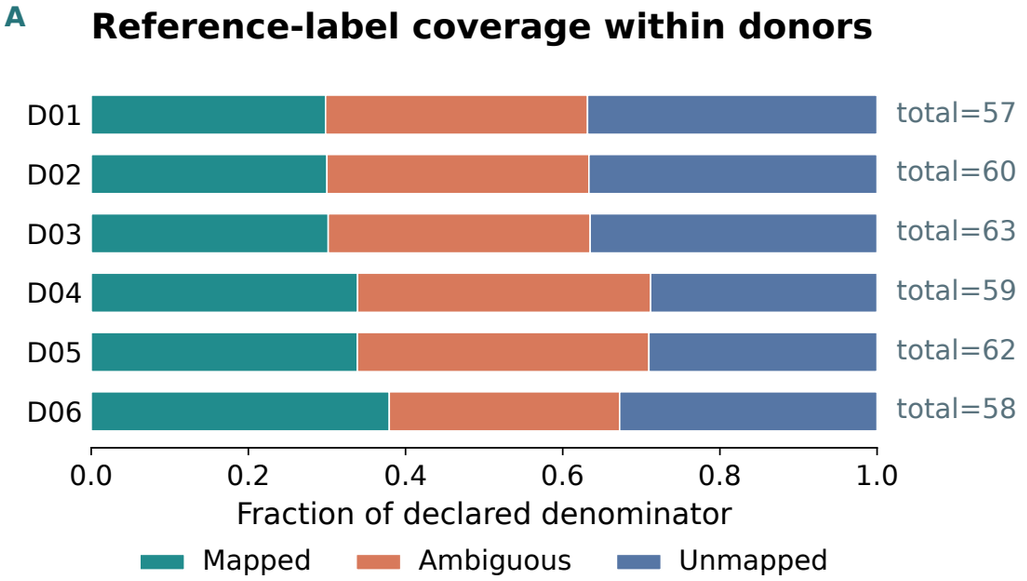
Software test fixture; not biological evidence

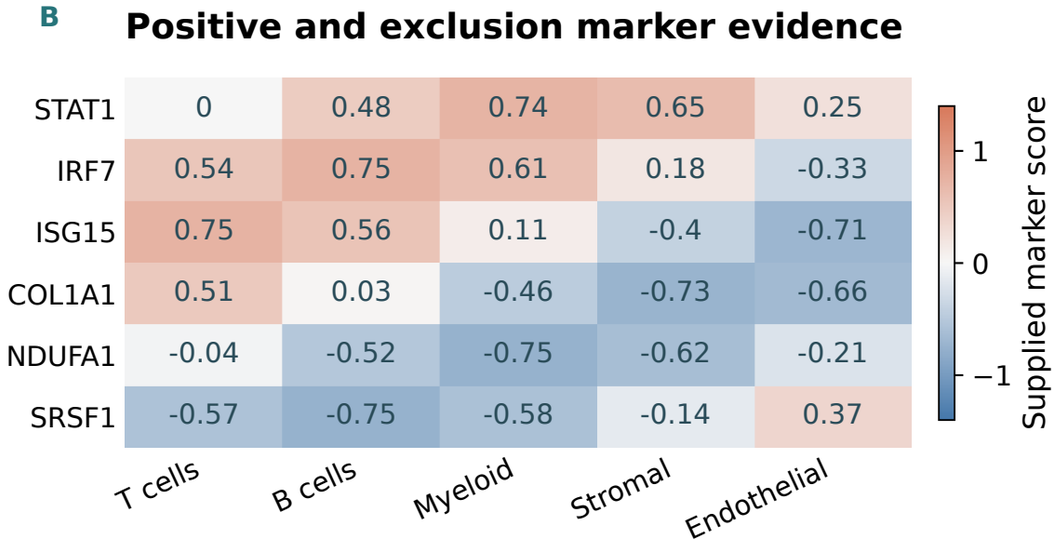

Prediction uncertainty and review decisions
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
0.15
0.31
0.47
0.31
0.47
0.64
0.47
0.64
0.8
0.64
0.8
0.15
0.8
0.15
0.31
0.15
0.31
0.47
Model A
Model B
Reviewed
Supplied confidence
1.0
0.8
0.6
0.4
0.2
0.0

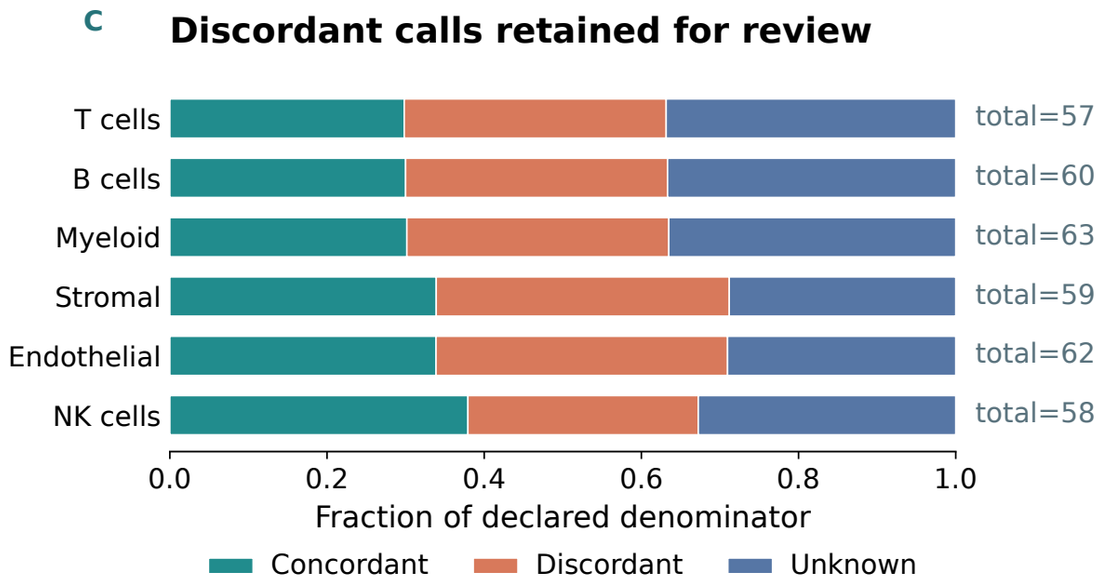

Identity support in independent cohorts
Estimate [interval] · q
Independent n = 6
T cells · Cohort A
T cells · Cohort B
B cells · Cohort A
B cells · Cohort B
Myeloid · Cohort A
Myeloid · Cohort B
Stromal · Cohort A
Stromal · Cohort B
Endothelial · Cohort A
Endothelial · Cohort B
NK cells · Cohort A
NK cells · Cohort B
-0.22 [-0.30, -0.14] 0.002
-0.14 [-0.22, -0.06] 0.005
-0.12 [-0.20, -0.04] 0.02
-0.04 [-0.12, 0.04] 0.65
-0.02 [-0.10, 0.06] 0.65
0.06 [-0.02, 0.14] 0.65
0.08 [0.00, 0.16] 0.056
0.16 [0.08, 0.24] 0.059
0.19 [0.11, 0.27] 0.074
0.27 [0.19, 0.35] 0.077
0.29 [0.21, 0.37] 0.092
0.37 [0.29, 0.45] 0.095
−0.4
−0.2
0.0
0.2
0.4
Label-support fraction difference · supplied 95% interval


Discordant calls retained for review
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Concordant
Discordant
Unknown

Single cell · annotation — complementary evidence



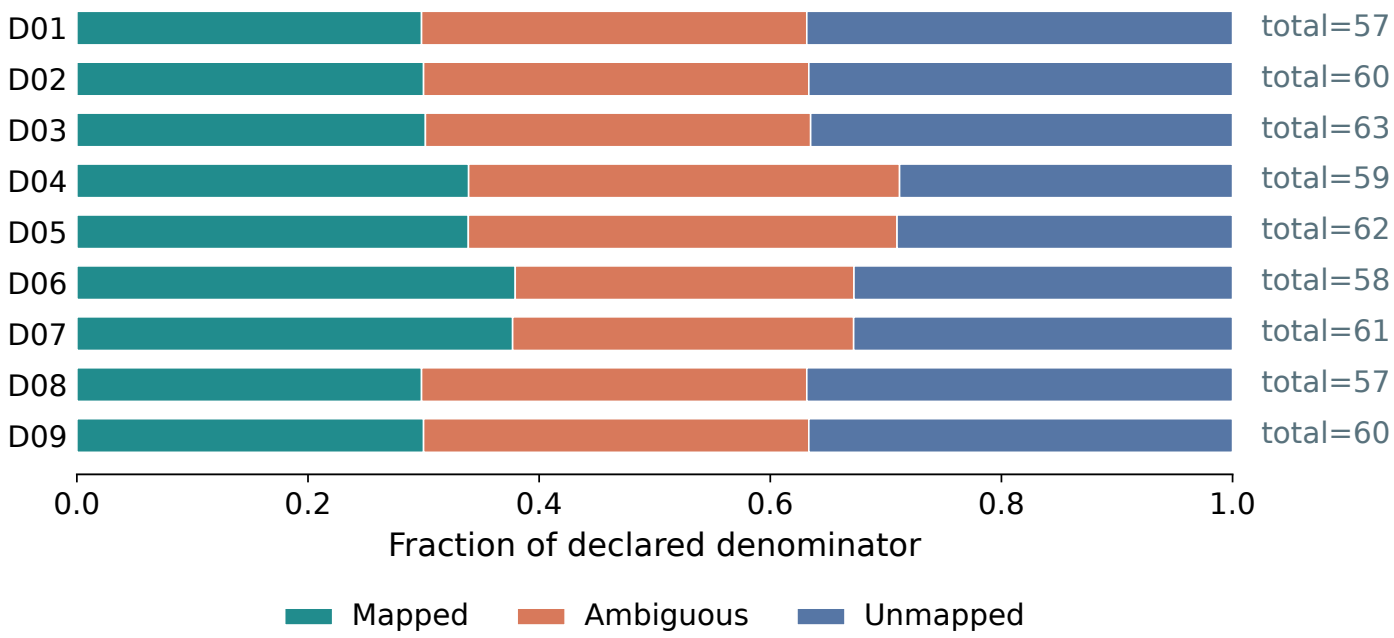




Single cell · annotation — complementary evidence

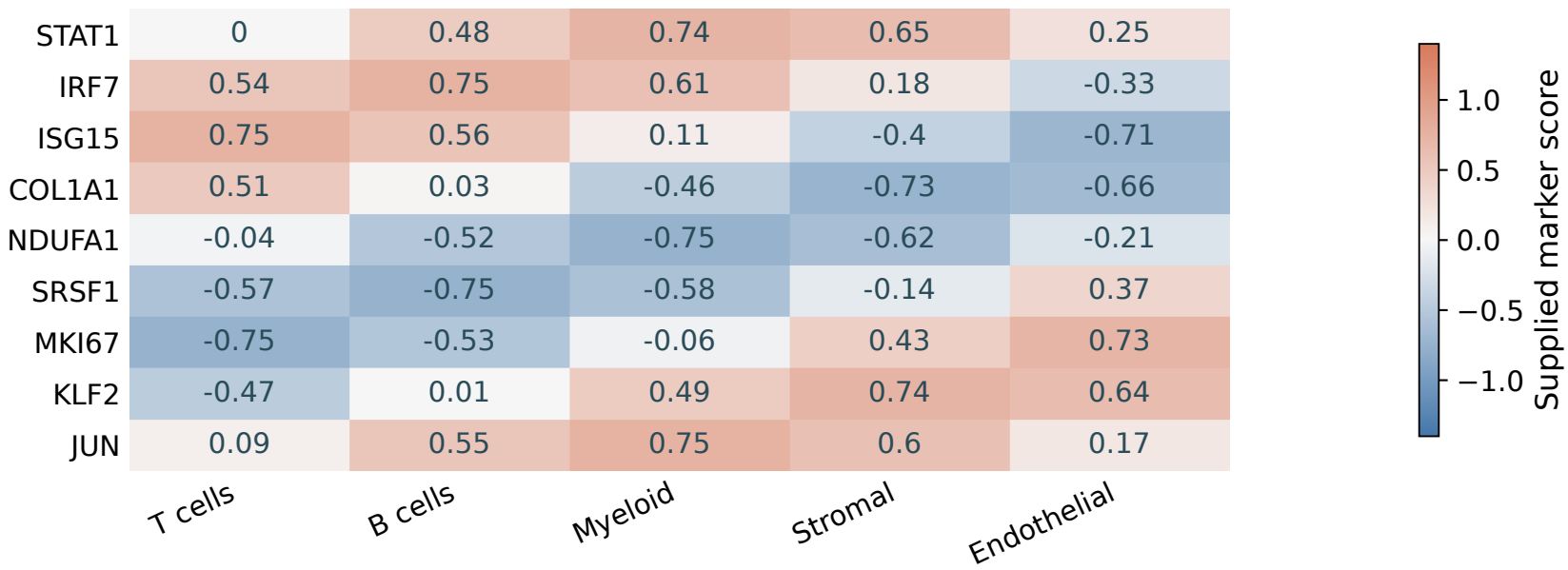
A

Reference-label coverage within donors



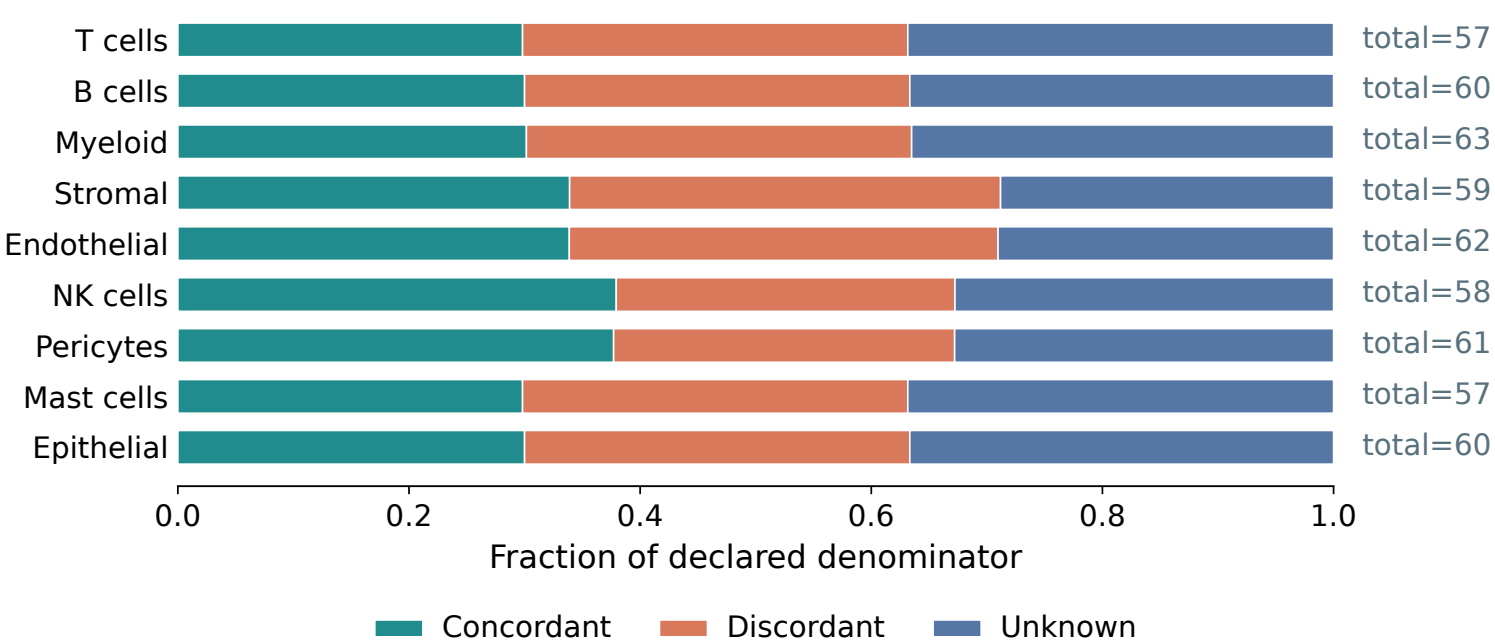
B

Positive and exclusion marker evidence

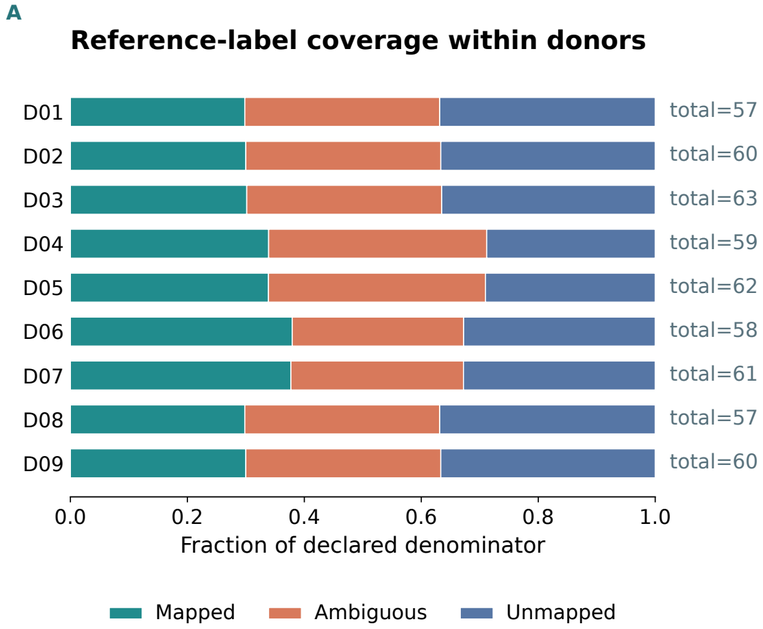


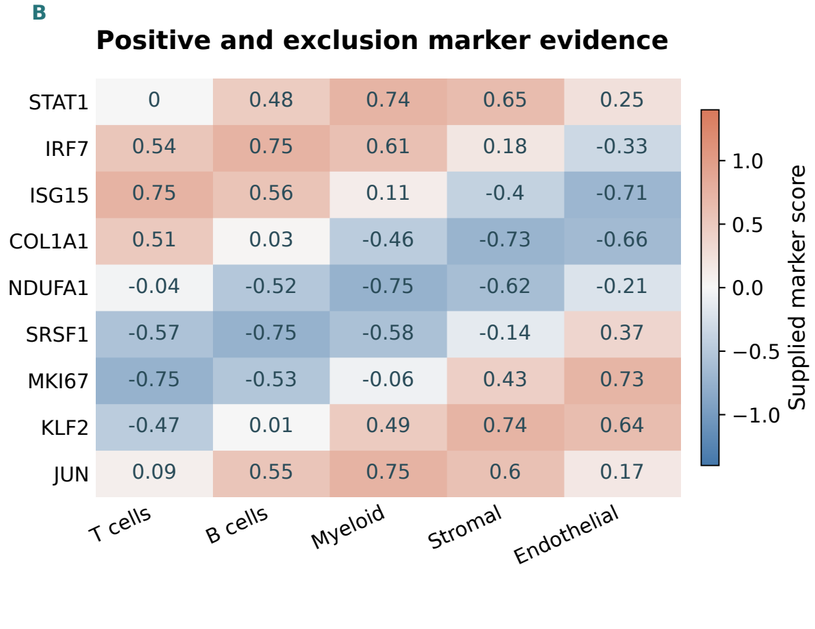
C

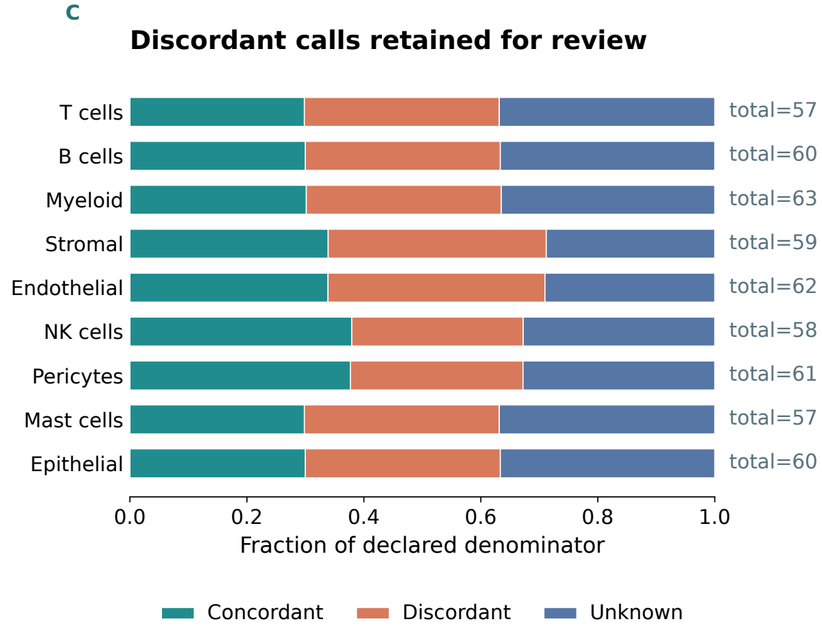
Discordant calls retained for review

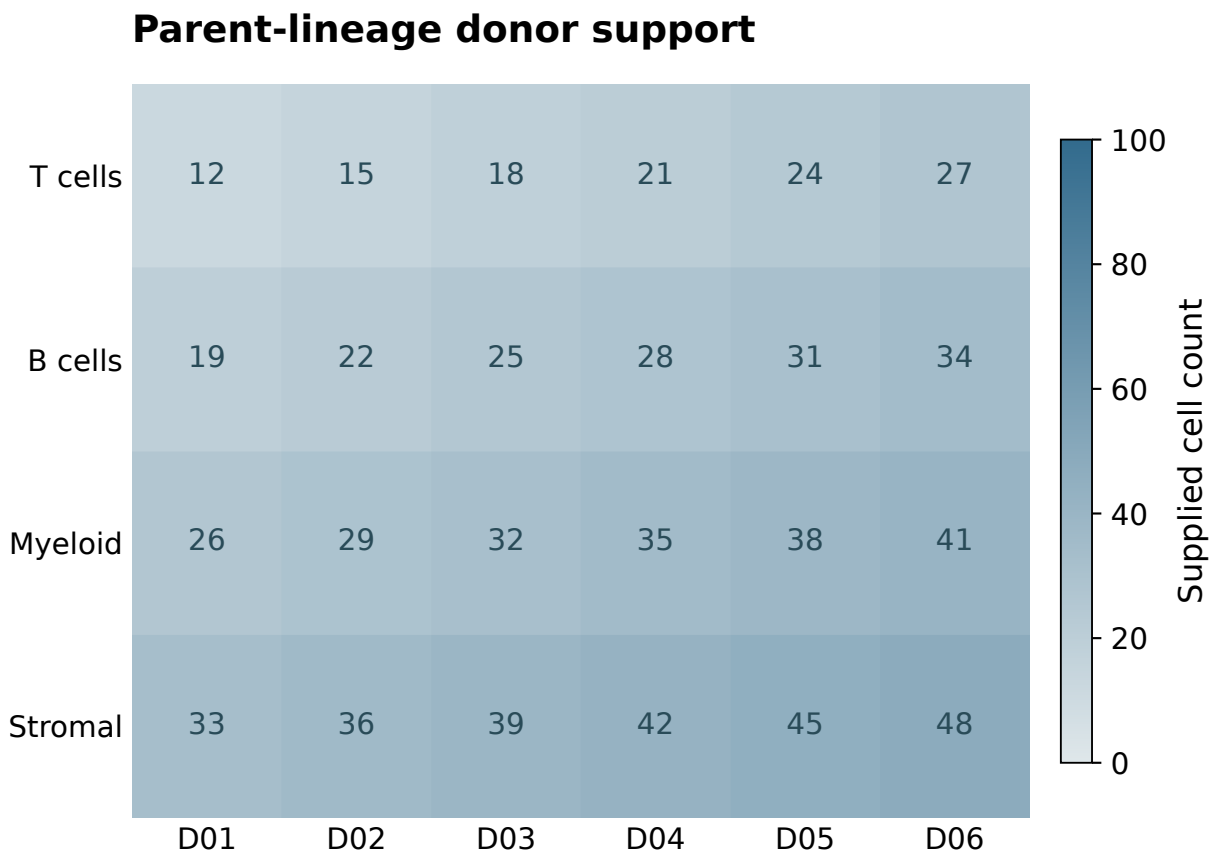


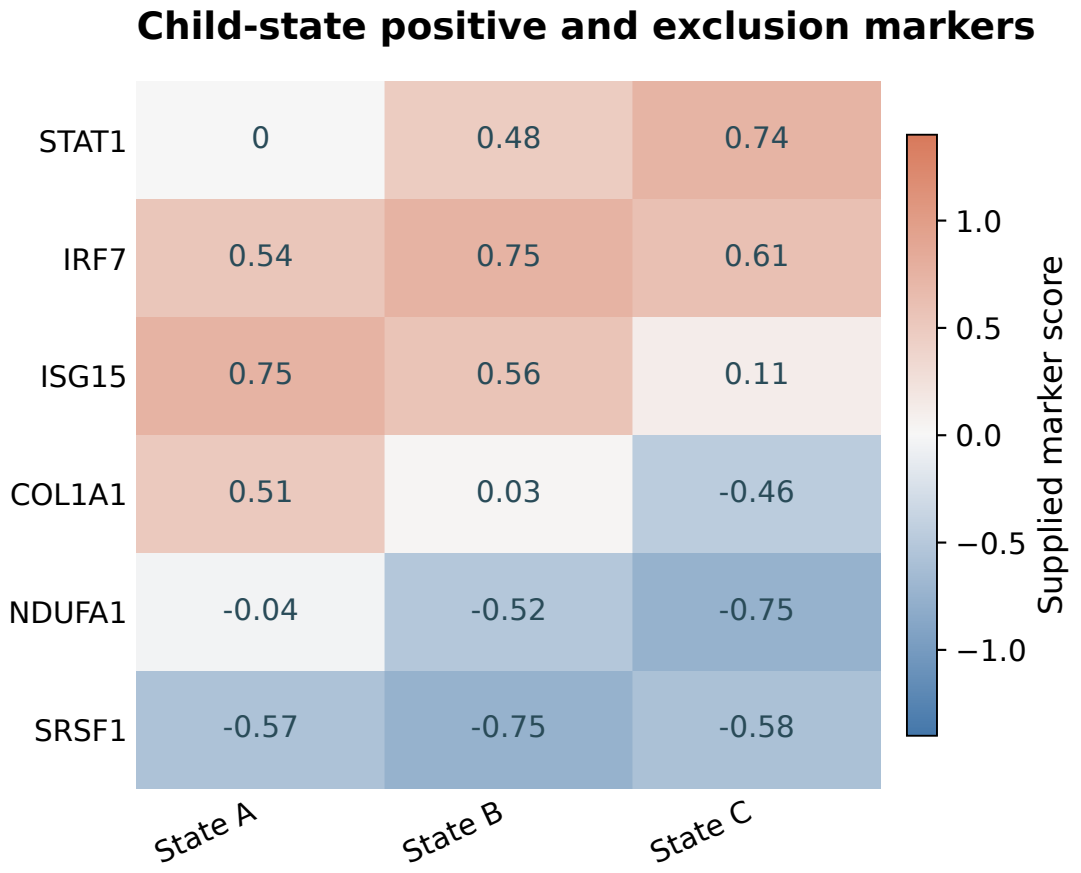
Single cell · annotation — complementary evidence

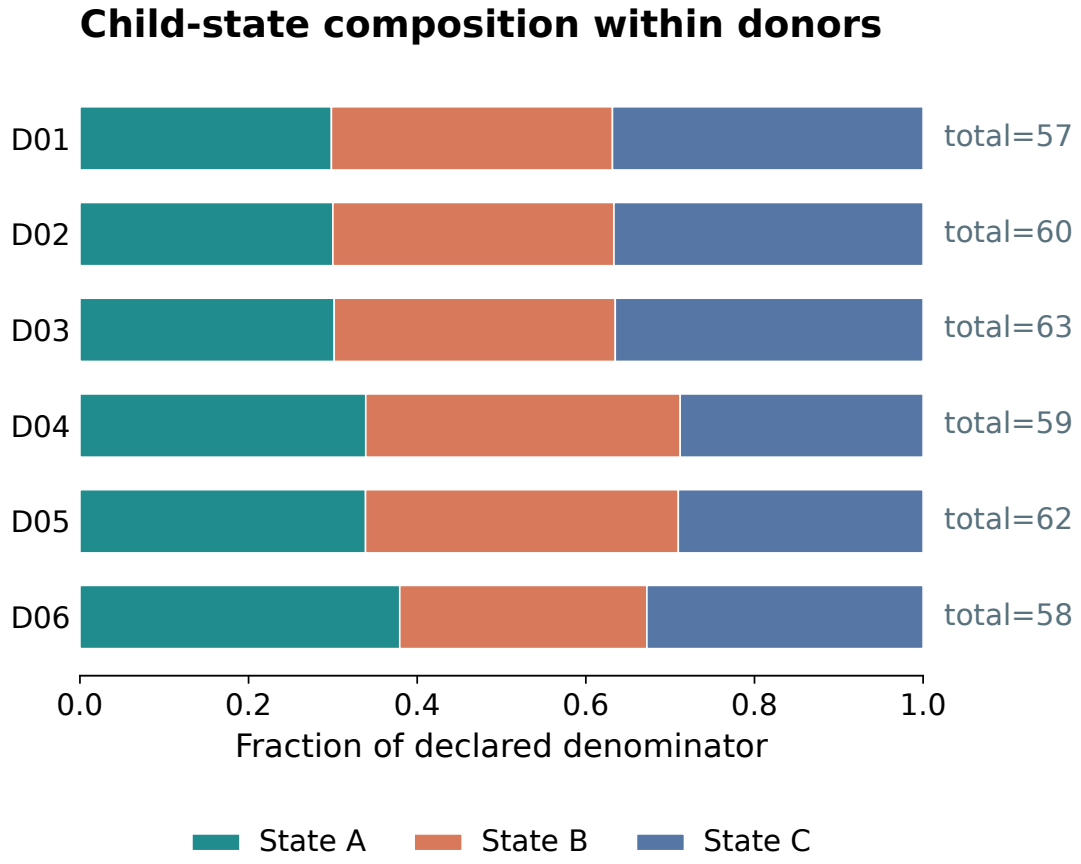




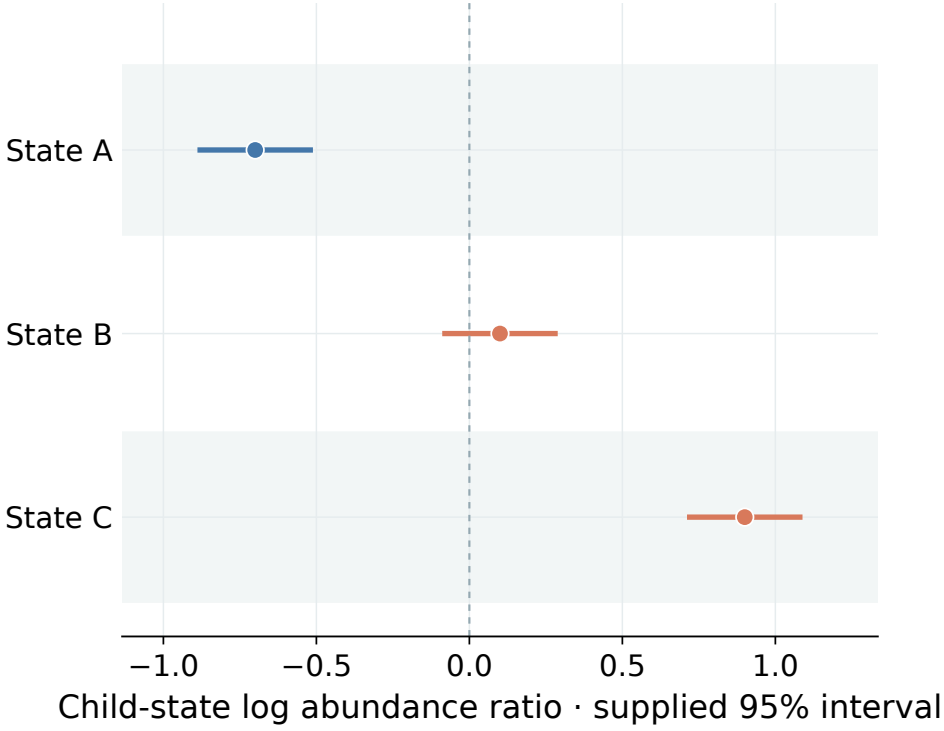



Parent-lineage donor support
T cells
B cells
Myeloid
Stromal
12
15
18
21
24
27
19
22
25
28
31
34
26
29
32
35
38
41
33
36
39
42
45
48
D01
D02
D03
D04
D05
D06
100
80
60
40
20
0
Supplied cell count


Child-state positive and exclusion markers
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0
0.48
0.74
0.54
0.75
0.61
0.75
0.56
0.11
0.51
0.03
-0.46
-0.04
-0.52
-0.75
-0.57
-0.75
-0.58
State A
State B
State C
1.0
0.5
0.0
−0.5
−1.0
Supplied marker score


Child-state composition within donors
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
State A
State B
State C

Donor-level child-state response



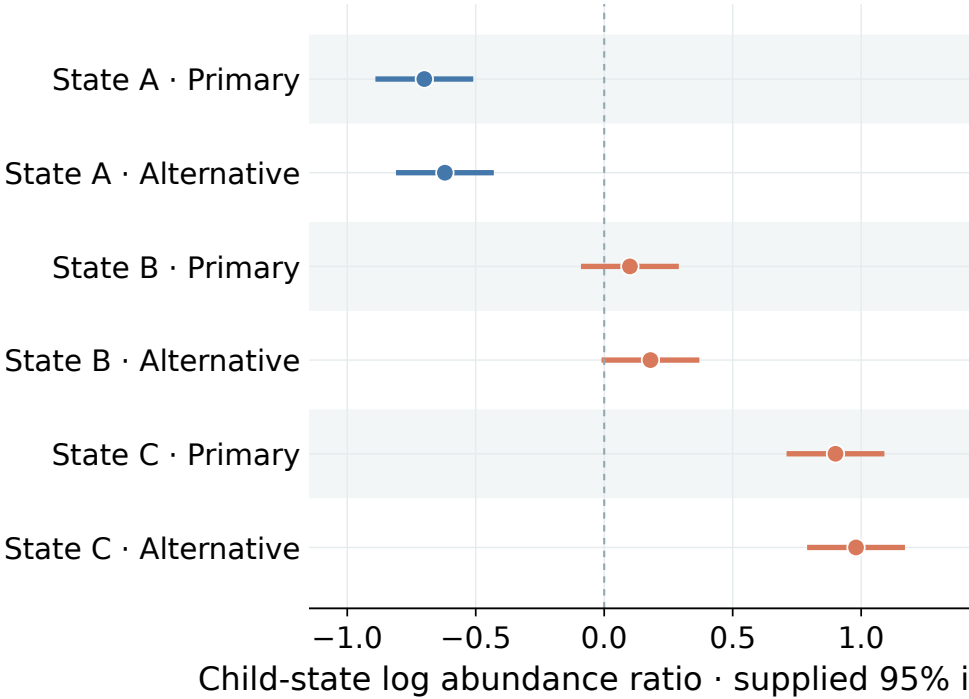
Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

0.10 [-0.08, 0.28] 0.65

0.90 [0.72, 1.08] 0.038

Child-state reproducibility across choices



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

-0.62 [-0.80, -0.44] 0.005

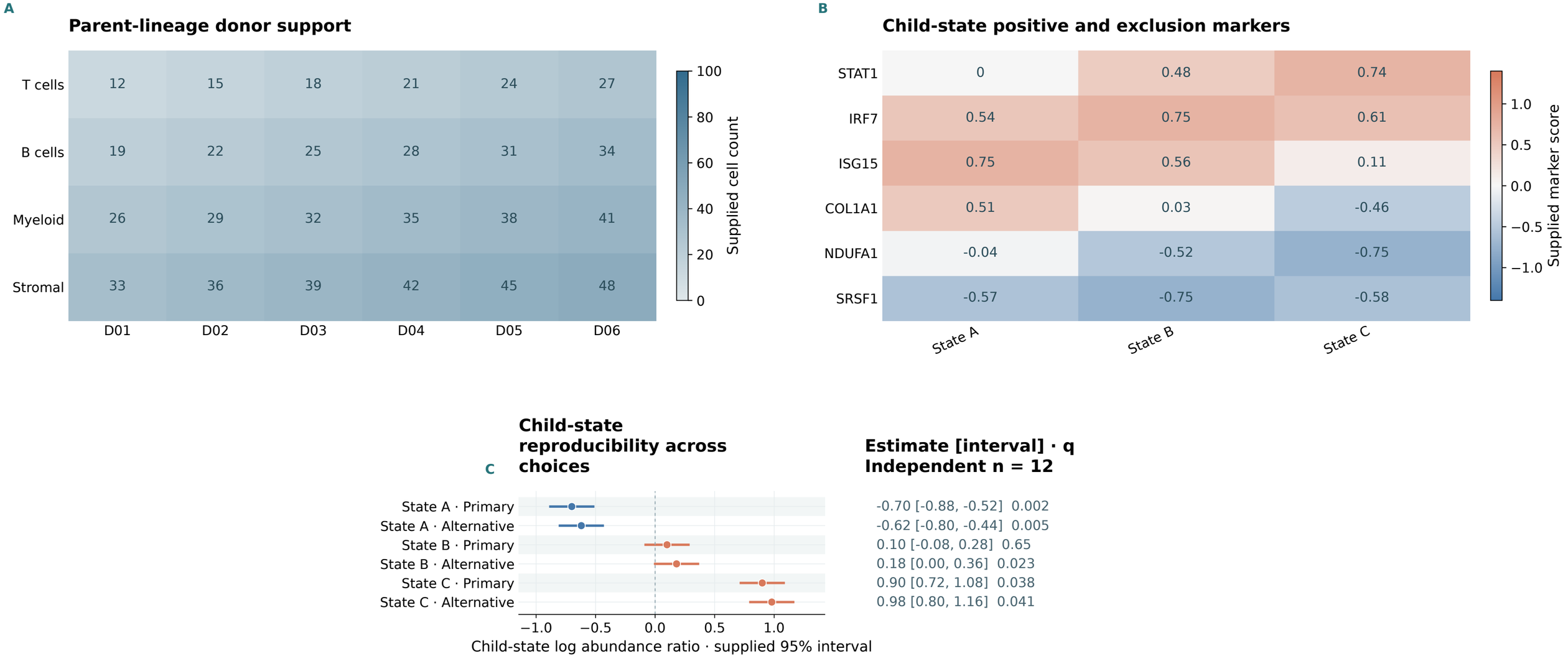
0.10 [-0.08, 0.28] 0.65

0.18 [0.00, 0.36] 0.023

0.90 [0.72, 1.08] 0.038

0.98 [0.80, 1.16] 0.041

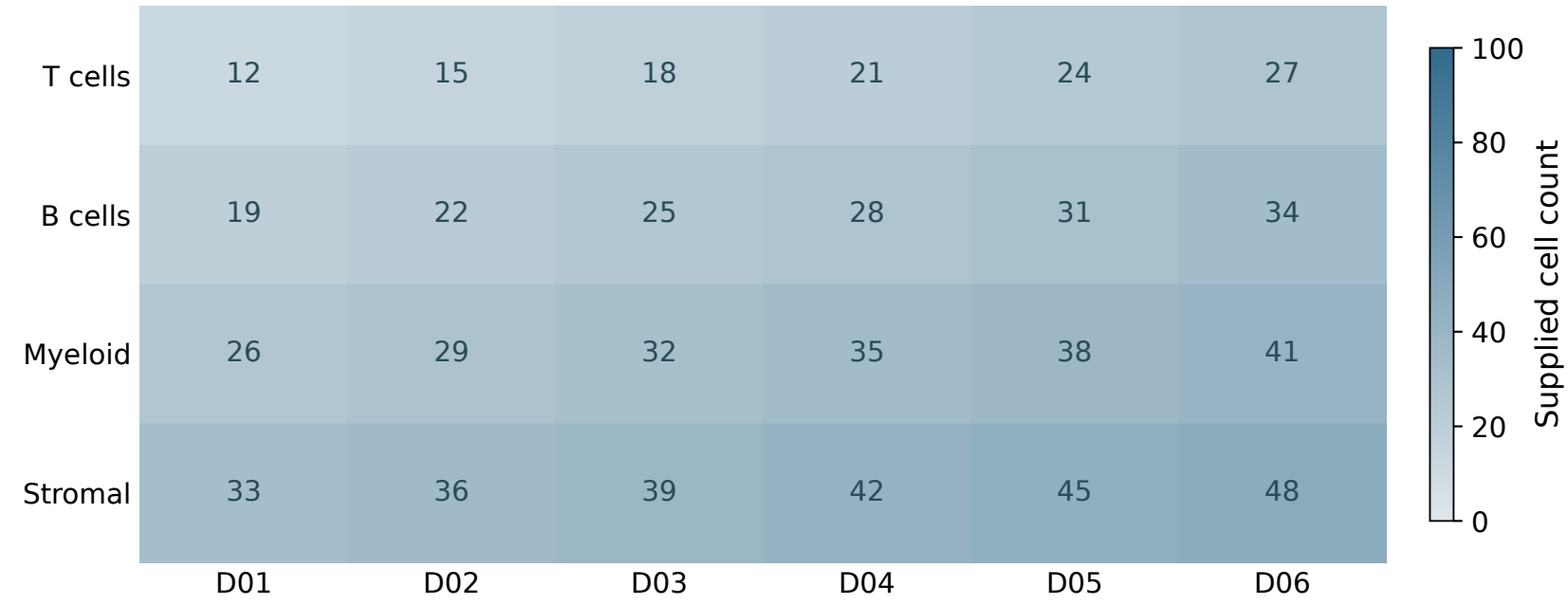
Single cell · subcluster — complementary evidence



Single cell · subcluster — complementary evidence

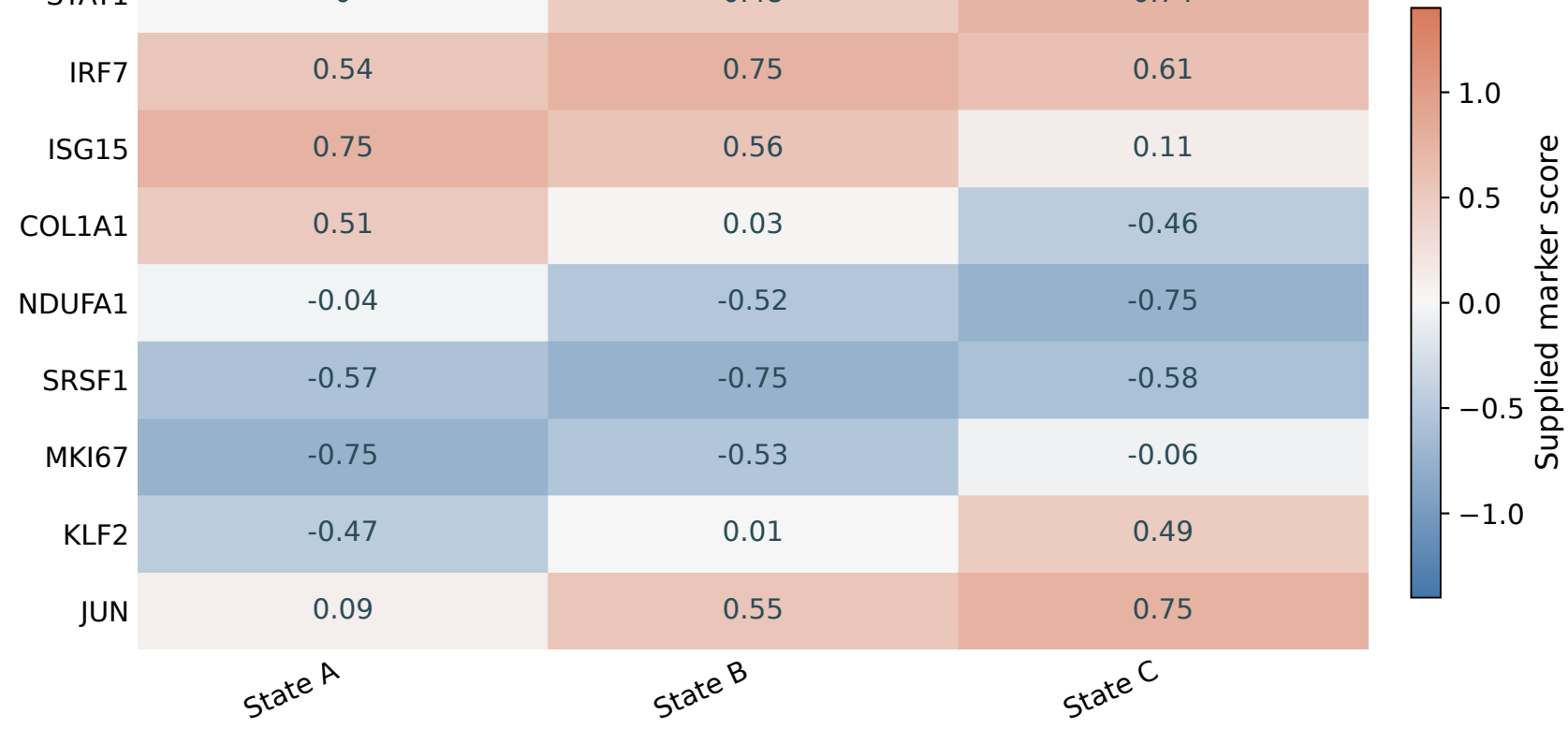
A

Parent-lineage donor support



B

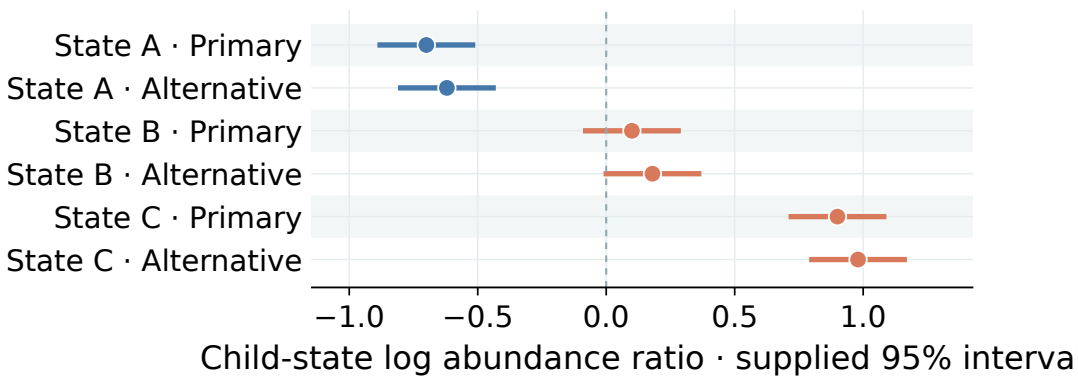
Child-state positive and exclusion markers



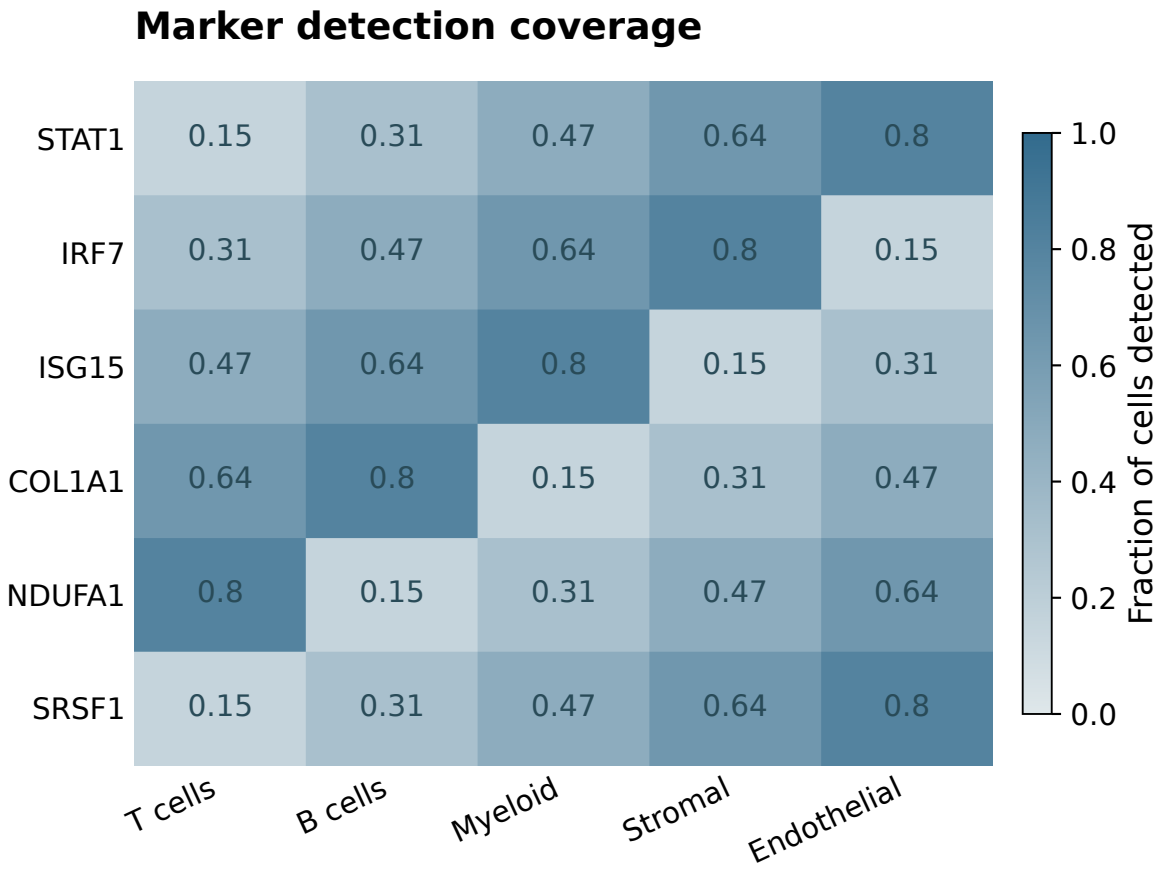
C

Child-state reproducibility across choices

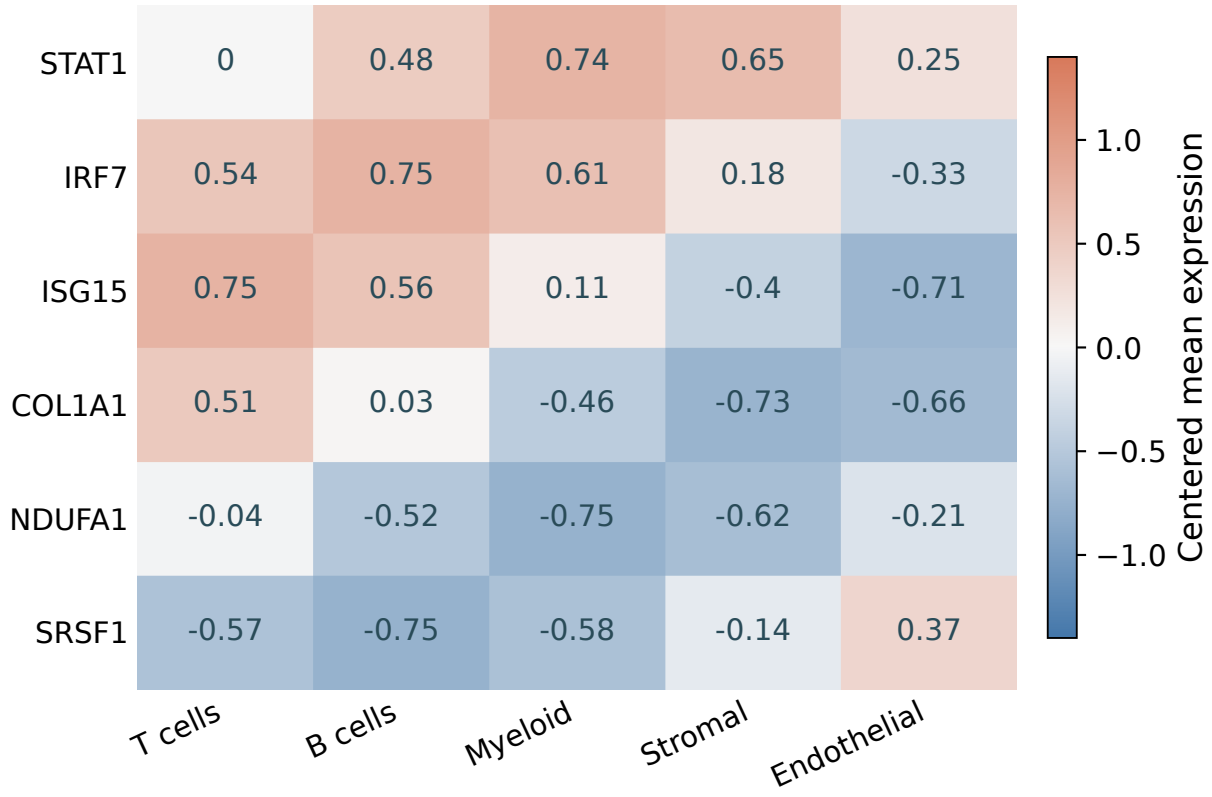
Estimate [interval] · q Independent n = 24



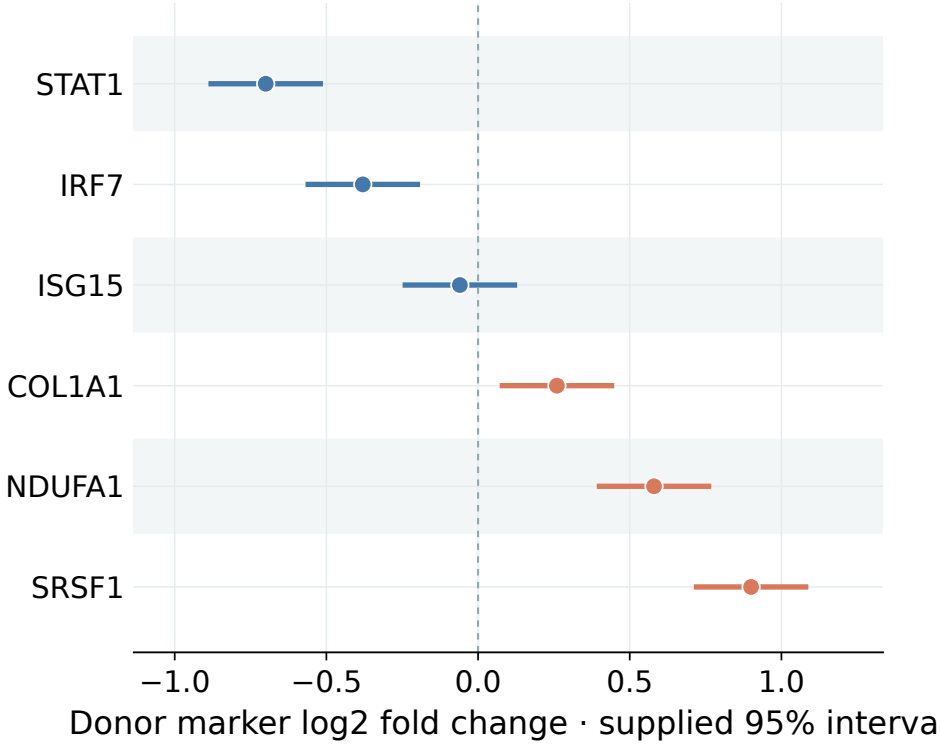
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
0.10 [-0.08, 0.28] 0.65
0.18 [0.00, 0.36] 0.023
0.90 [0.72, 1.08] 0.038
0.98 [0.80, 1.16] 0.041


Marker detection coverage
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0.15
0.31
0.47
0.64
0.8
T cells
B cells
Myeloid
Stromal
Endothelial
Fraction of cells detected
1.0
0.8
0.6
0.4
0.2
0.0

Marker mean-expression specificity



Donor-level marker effect estimates



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

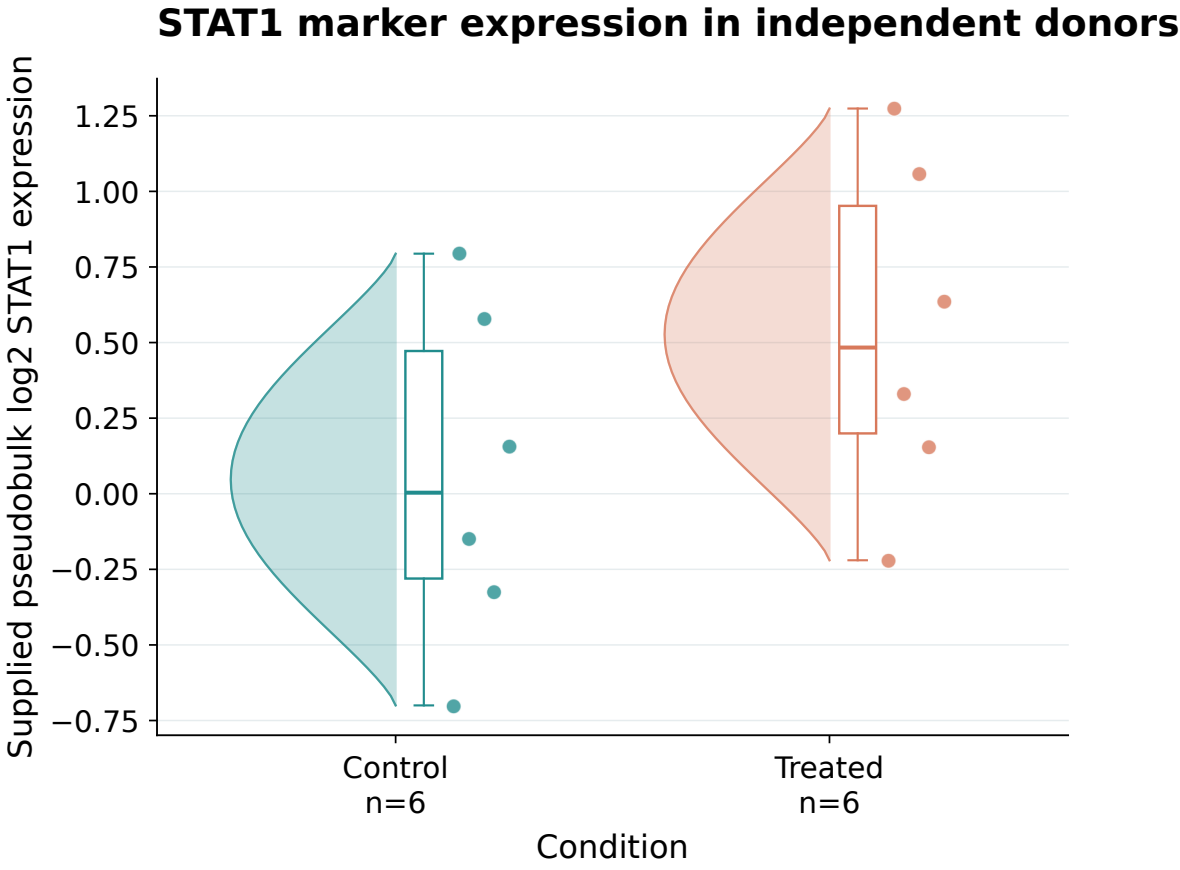
-0.38 [-0.56, -0.20] 0.02

-0.06 [-0.24, 0.12] 0.65

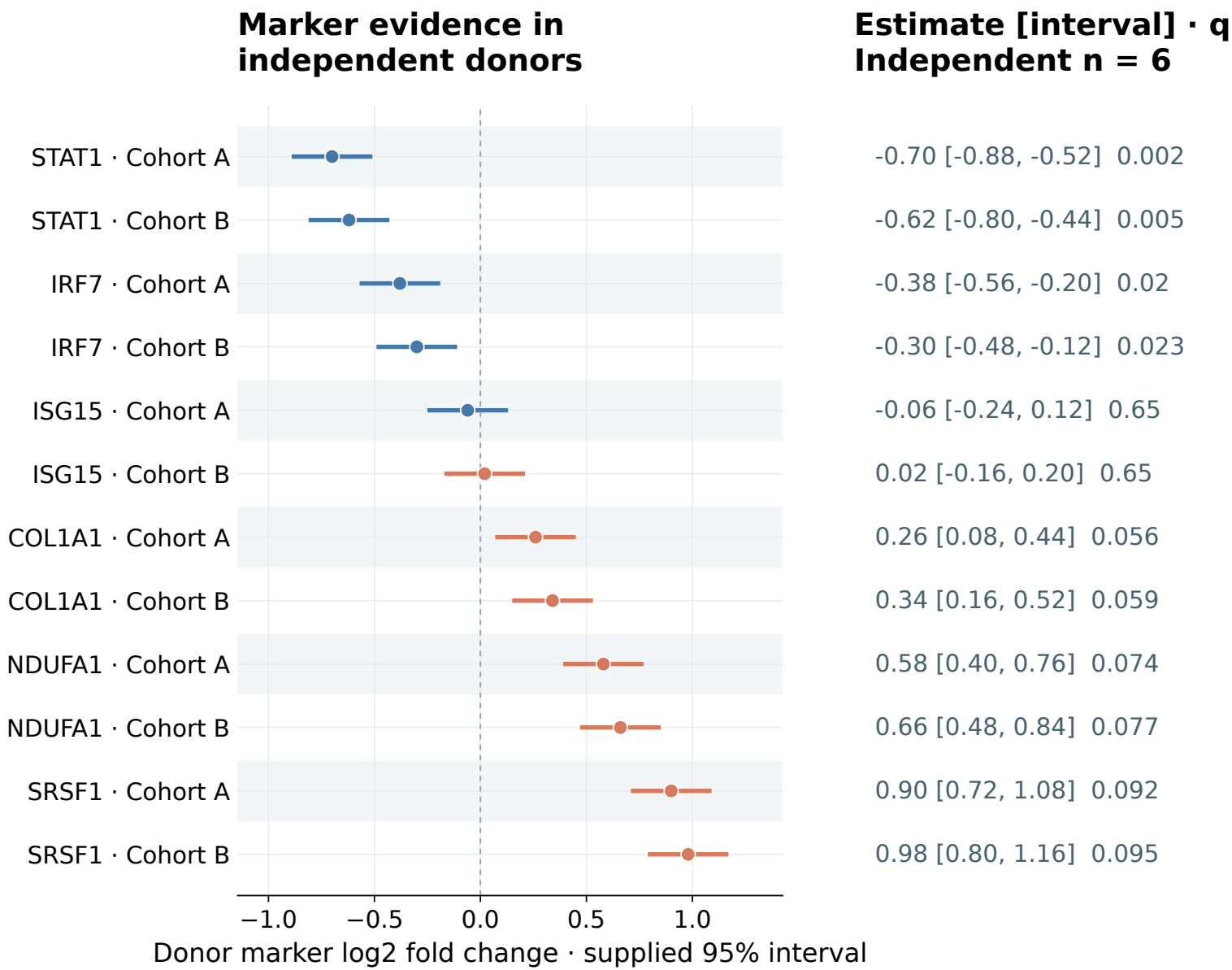
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092



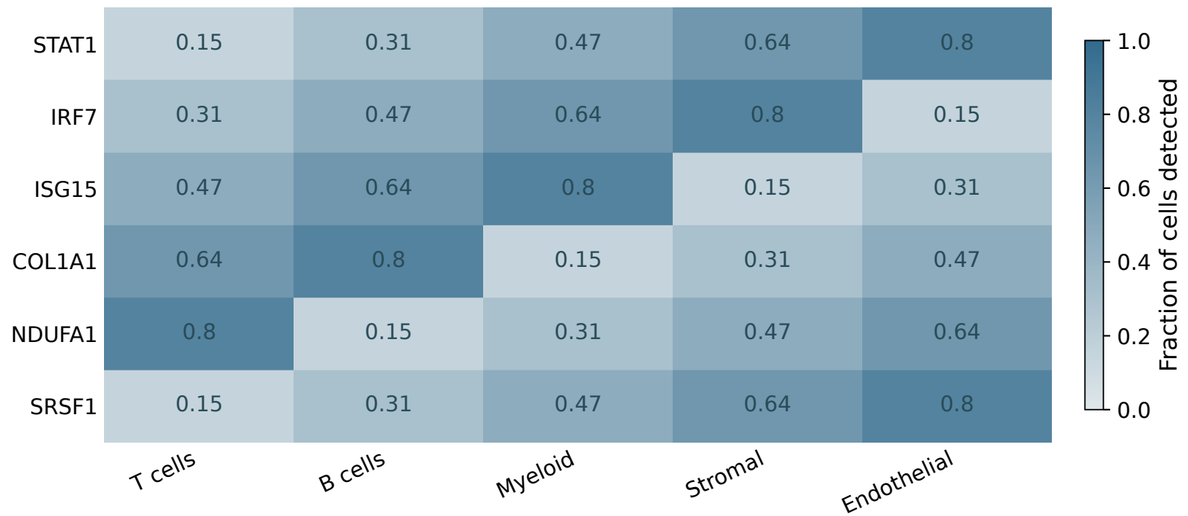
Software test fixture; not biological evidence


Marker evidence in independent donors
Estimate [interval] · q
Independent n = 6
STAT1 · Cohort A
STAT1 · Cohort B
IRF7 · Cohort A
IRF7 · Cohort B
ISG15 · Cohort A
ISG15 · Cohort B
COL1A1 · Cohort A
COL1A1 · Cohort B
NDUFA1 · Cohort A
NDUFA1 · Cohort B
SRSF1 · Cohort A
SRSF1 · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Donor marker log2 fold change · supplied 95% interval

Single cell · markers — complementary evidence

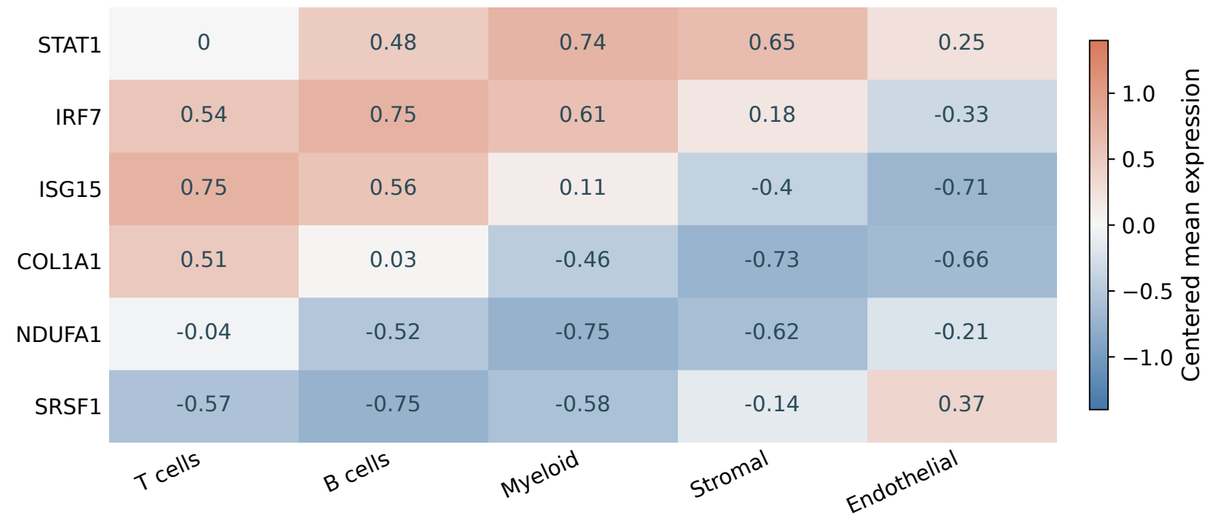
A

Marker detection coverage



B

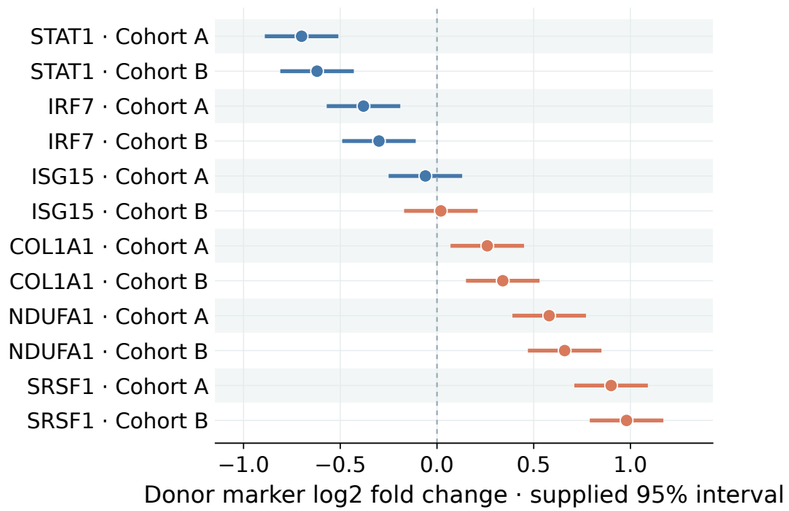
Marker mean-expression specificity



C

Marker evidence in independent donors

Estimate [interval] · q
Independent n = 6

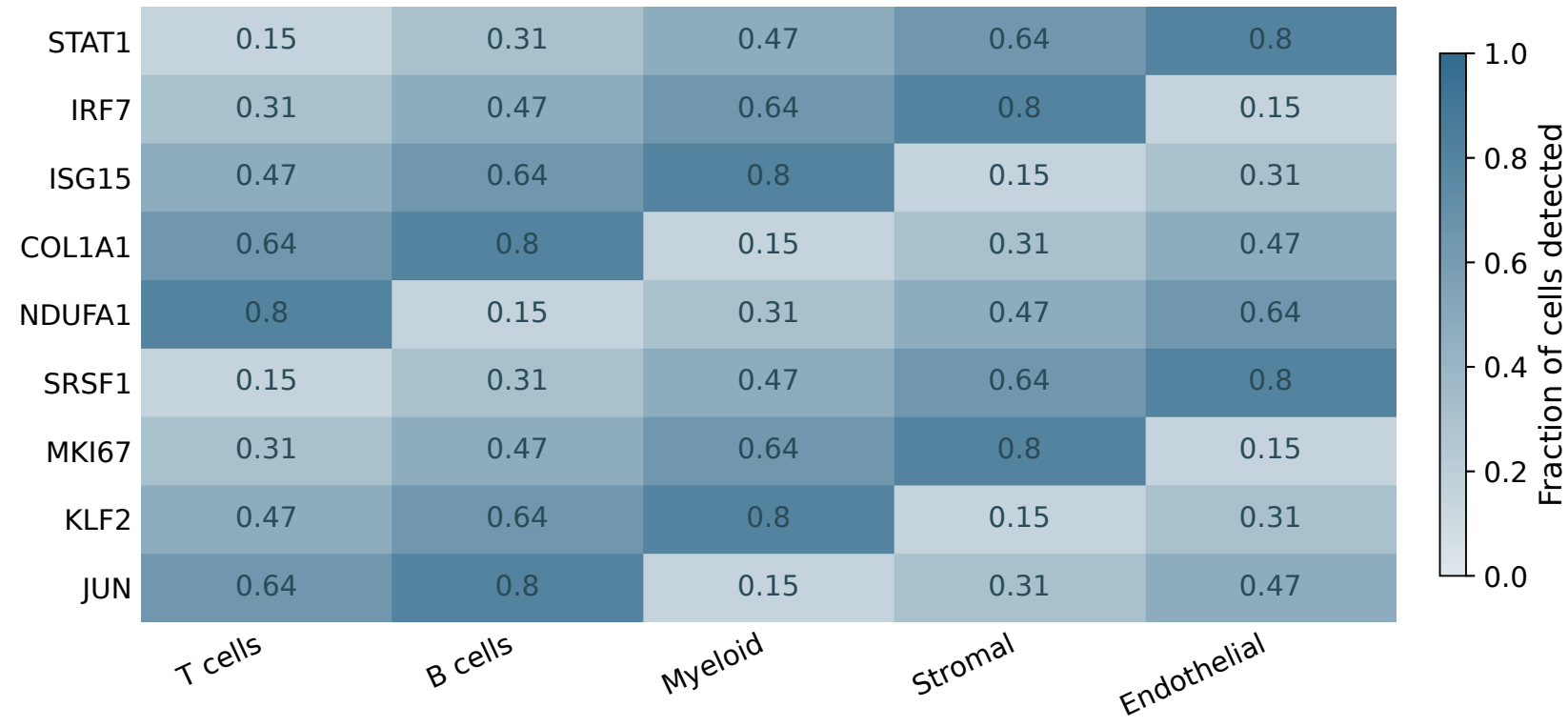


-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095

Single cell · markers — complementary evidence

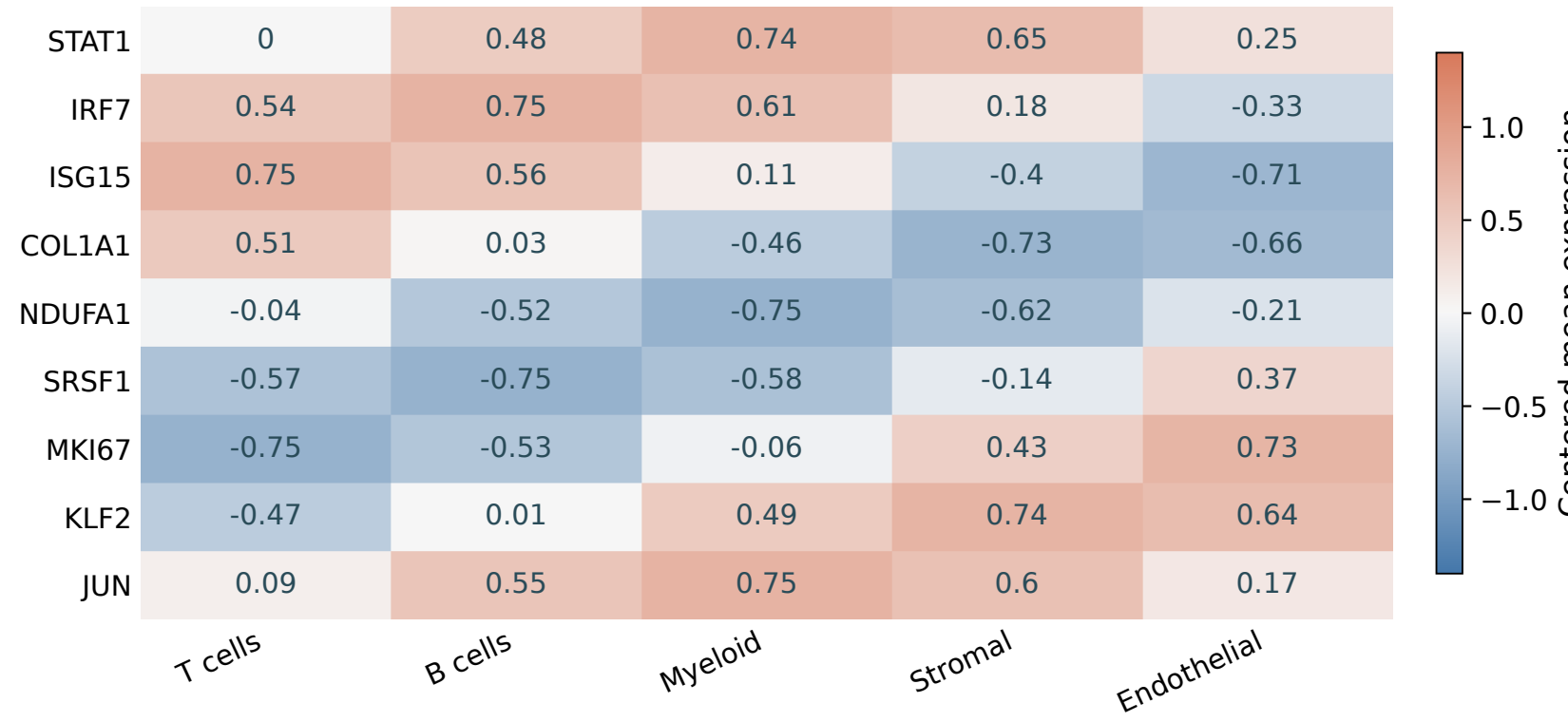
A

Marker detection coverage



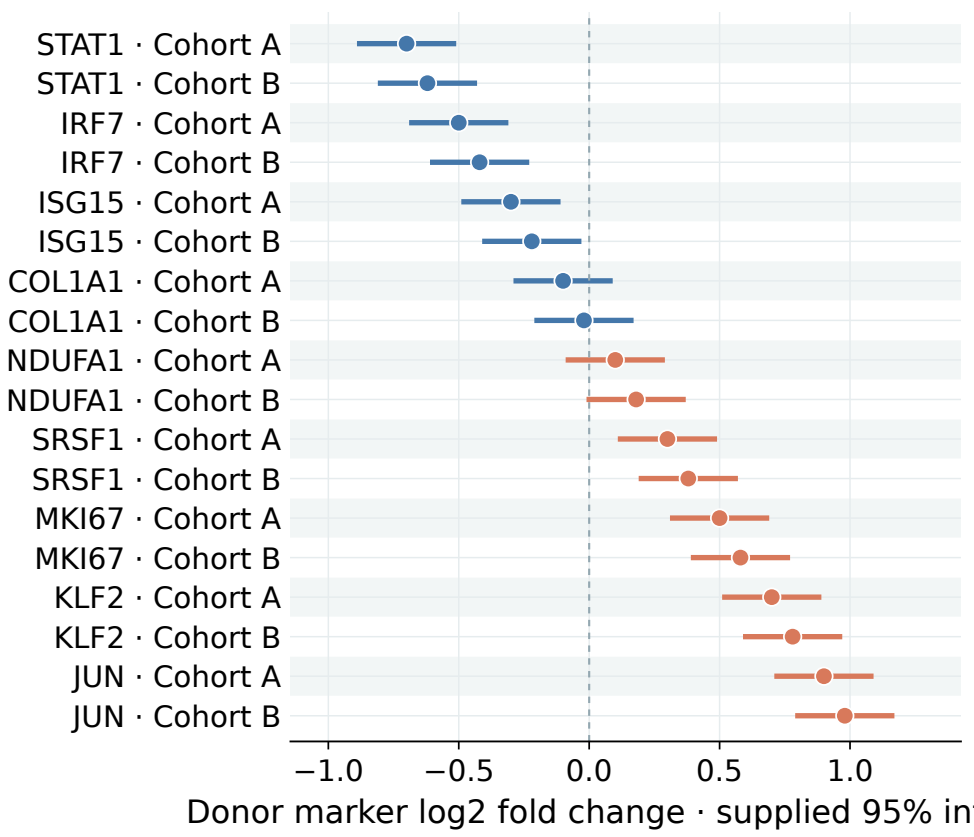
B

Marker mean-expression specificity



C

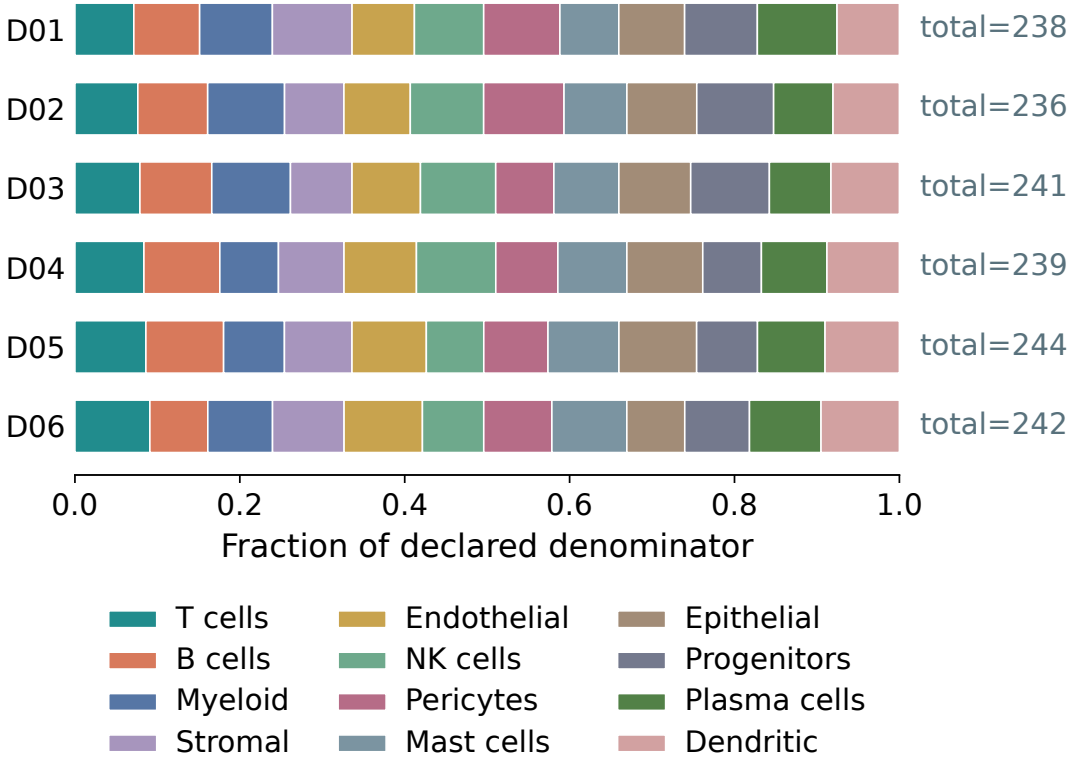
Marker evidence in independent donors



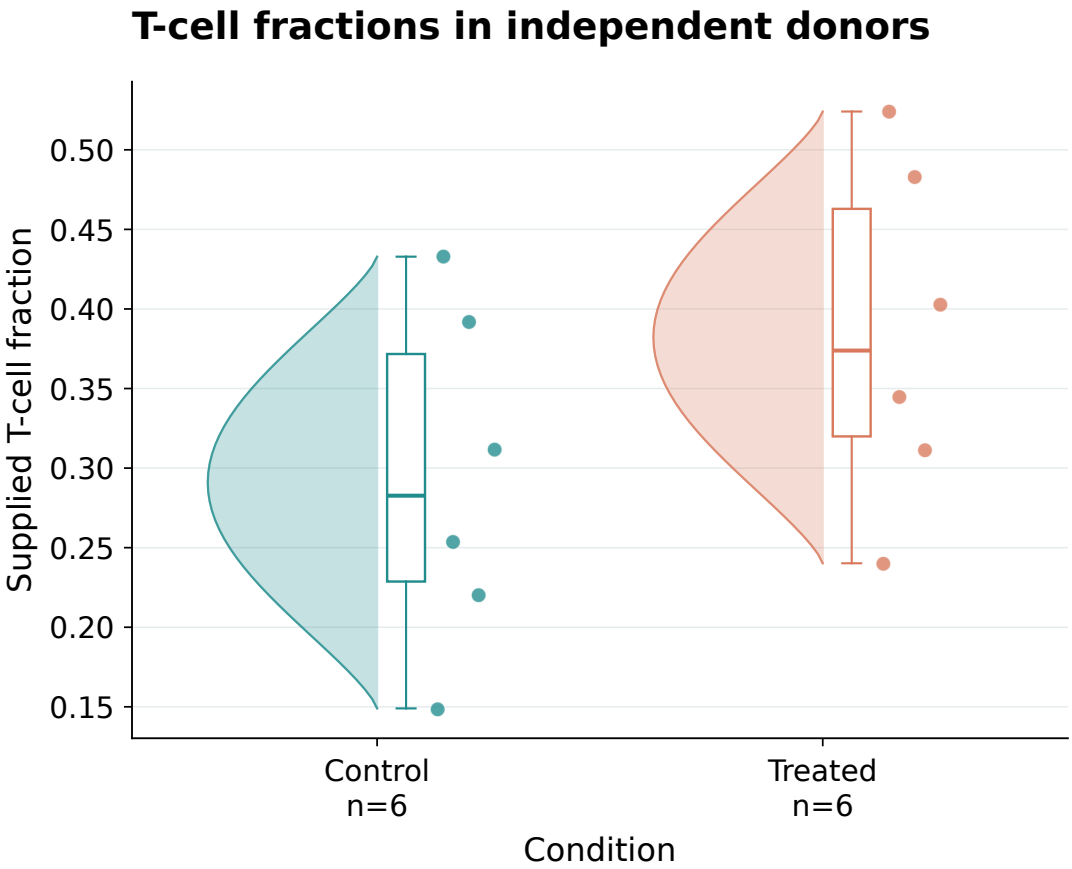
Estimate [interval] · q
Independent n = 12

STAT1 · Cohort A	-0.70 [-0.88, -0.52] 0.002
STAT1 · Cohort B	-0.62 [-0.80, -0.44] 0.005
IRF7 · Cohort A	-0.50 [-0.68, -0.32] 0.02
IRF7 · Cohort B	-0.42 [-0.60, -0.24] 0.023
ISG15 · Cohort A	-0.30 [-0.48, -0.12] 0.038
ISG15 · Cohort B	-0.22 [-0.40, -0.04] 0.041
COL1A1 · Cohort A	-0.10 [-0.28, 0.08] 0.65
COL1A1 · Cohort B	-0.02 [-0.20, 0.16] 0.65
NDUFA1 · Cohort A	0.10 [-0.08, 0.28] 0.65
NDUFA1 · Cohort B	0.18 [0.00, 0.36] 0.077
SRSF1 · Cohort A	0.30 [0.12, 0.48] 0.092
SRSF1 · Cohort B	0.38 [0.20, 0.56] 0.095
MKI67 · Cohort A	0.50 [0.32, 0.68] 0.11
MKI67 · Cohort B	0.58 [0.40, 0.76] 0.11
KLF2 · Cohort A	0.70 [0.52, 0.88] 0.13
KLF2 · Cohort B	0.78 [0.60, 0.96] 0.13
JUN · Cohort A	0.90 [0.72, 1.08] 0.15
JUN · Cohort B	0.98 [0.80, 1.16] 0.15

Complete donor cell-identity composition

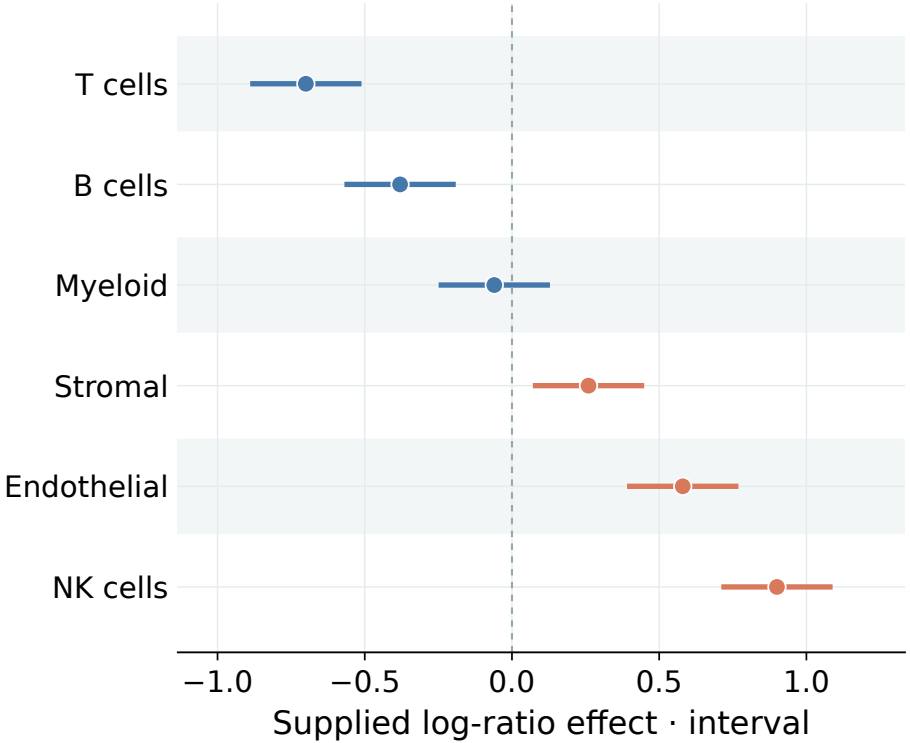


Software test fixture; not biological evidence



Software test fixture; not biological evidence

Prespecified compositional effects



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

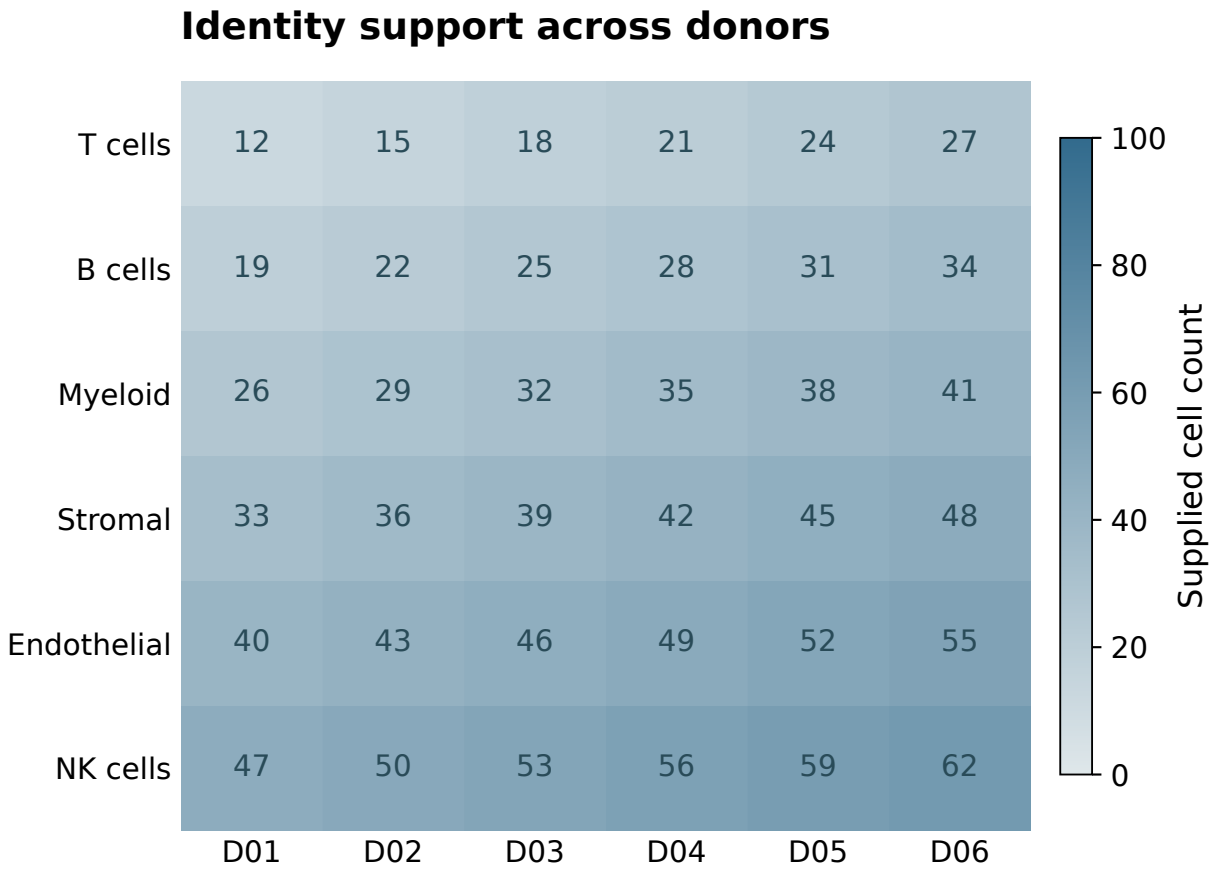
-0.38 [-0.56, -0.20] 0.02

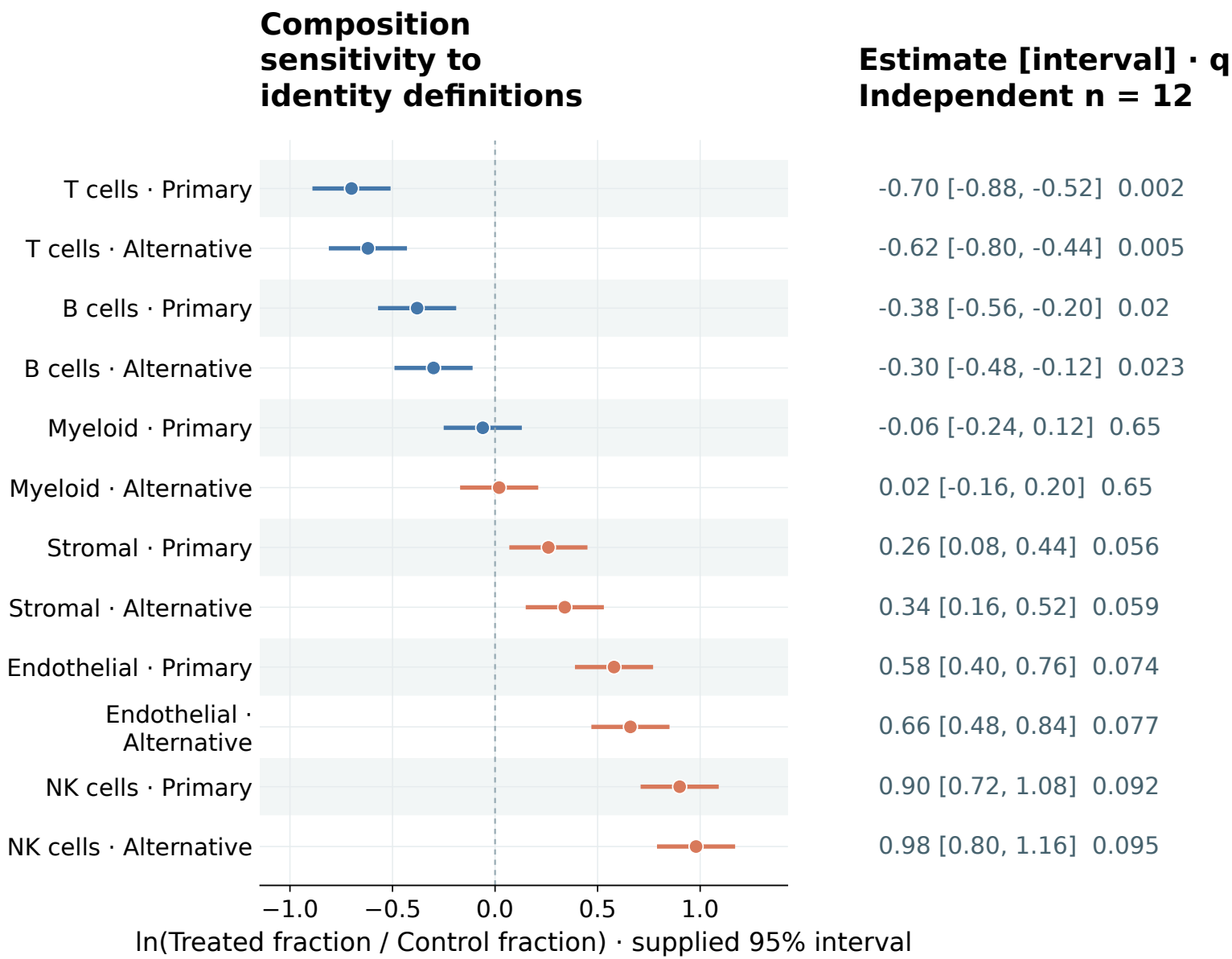
-0.06 [-0.24, 0.12] 0.65

0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092

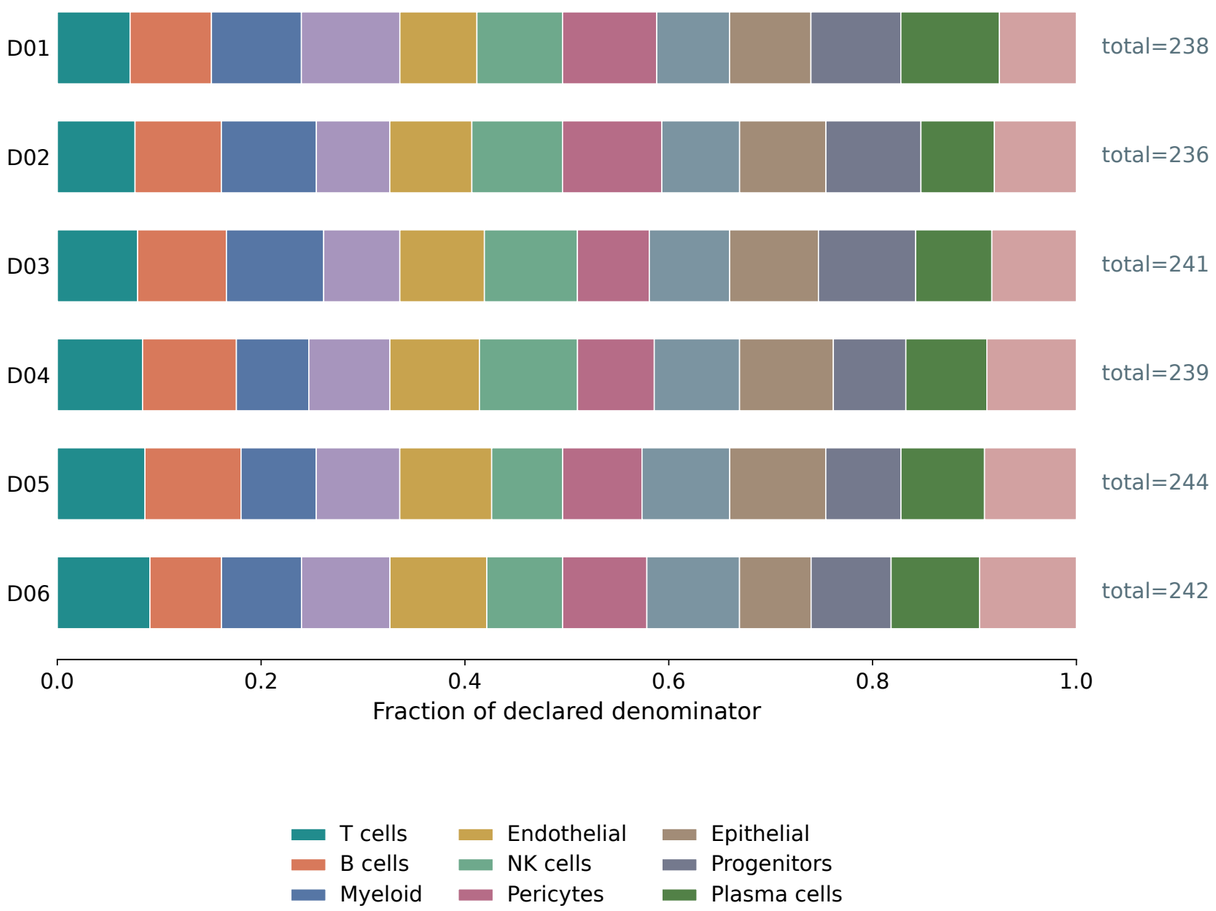

Identity support across donors
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
12 15 18 21 24 27
19 22 25 28 31 34
26 29 32 35 38 41
33 36 39 42 45 48
40 43 46 49 52 55
47 50 53 56 59 62
D01 D02 D03 D04 D05 D06
Supplied cell count
100
80
60
40
20
0


Composition sensitivity to identity definitions
Estimate [interval] · q
Independent n = 12
T cells · Primary
T cells · Alternative
B cells · Primary
B cells · Alternative
Myeloid · Primary
Myeloid · Alternative
Stromal · Primary
Stromal · Alternative
Endothelial · Primary
Endothelial · Alternative
NK cells · Primary
NK cells · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
ln(Treated fraction / Control fraction) · supplied 95% interval

Single cell · composition — complementary evidence

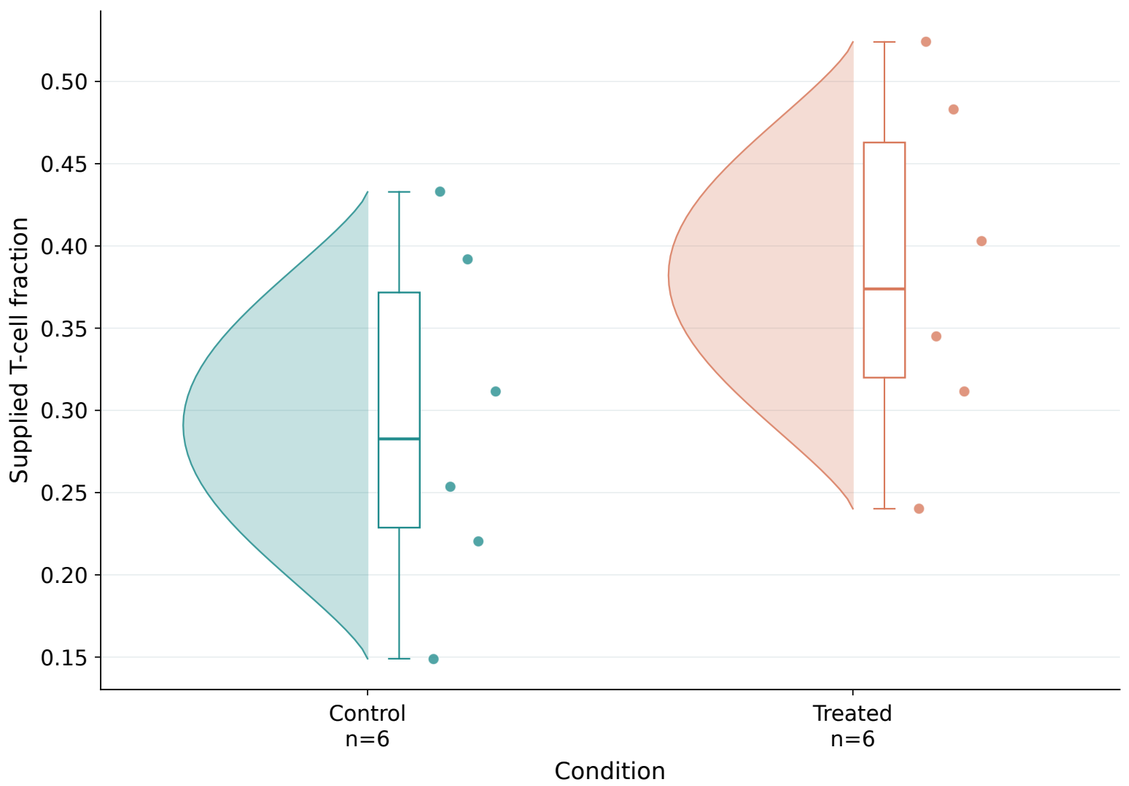
A

Complete donor cell-identity composition



B

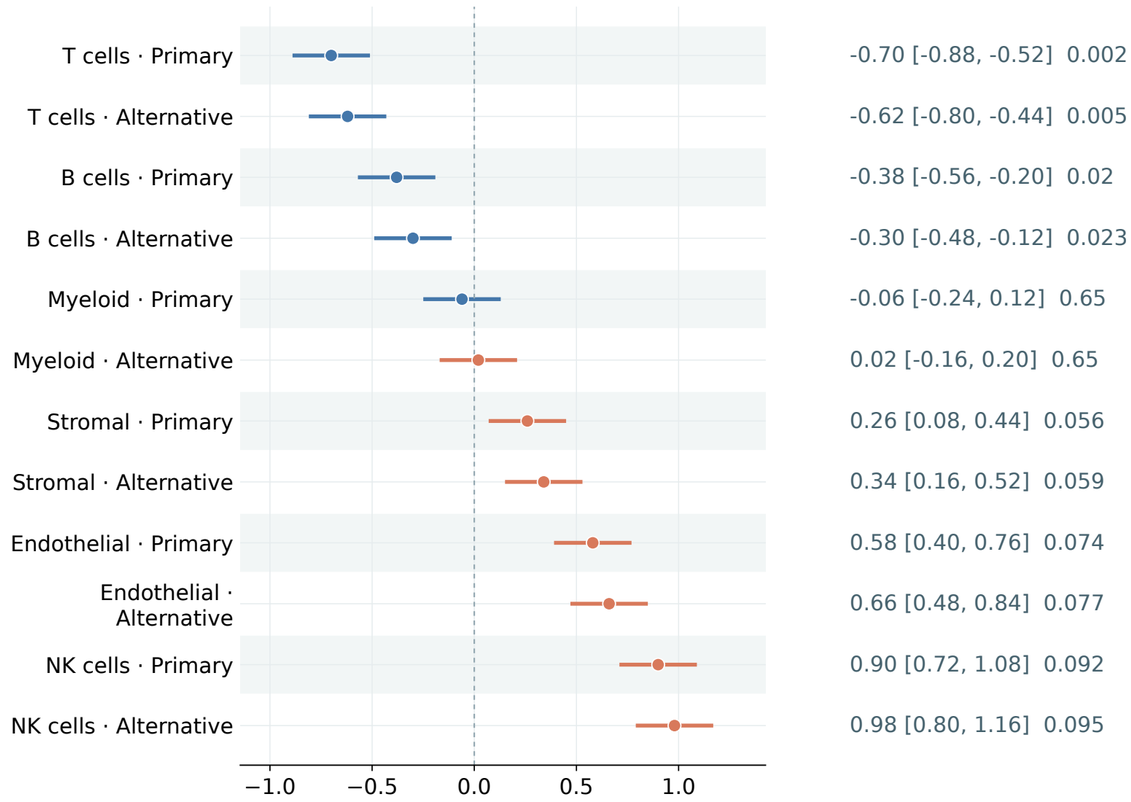
T-cell fractions in independent donors



C

Composition sensitivity to identity definitions

Estimate [interval] · q
Independent n = 12

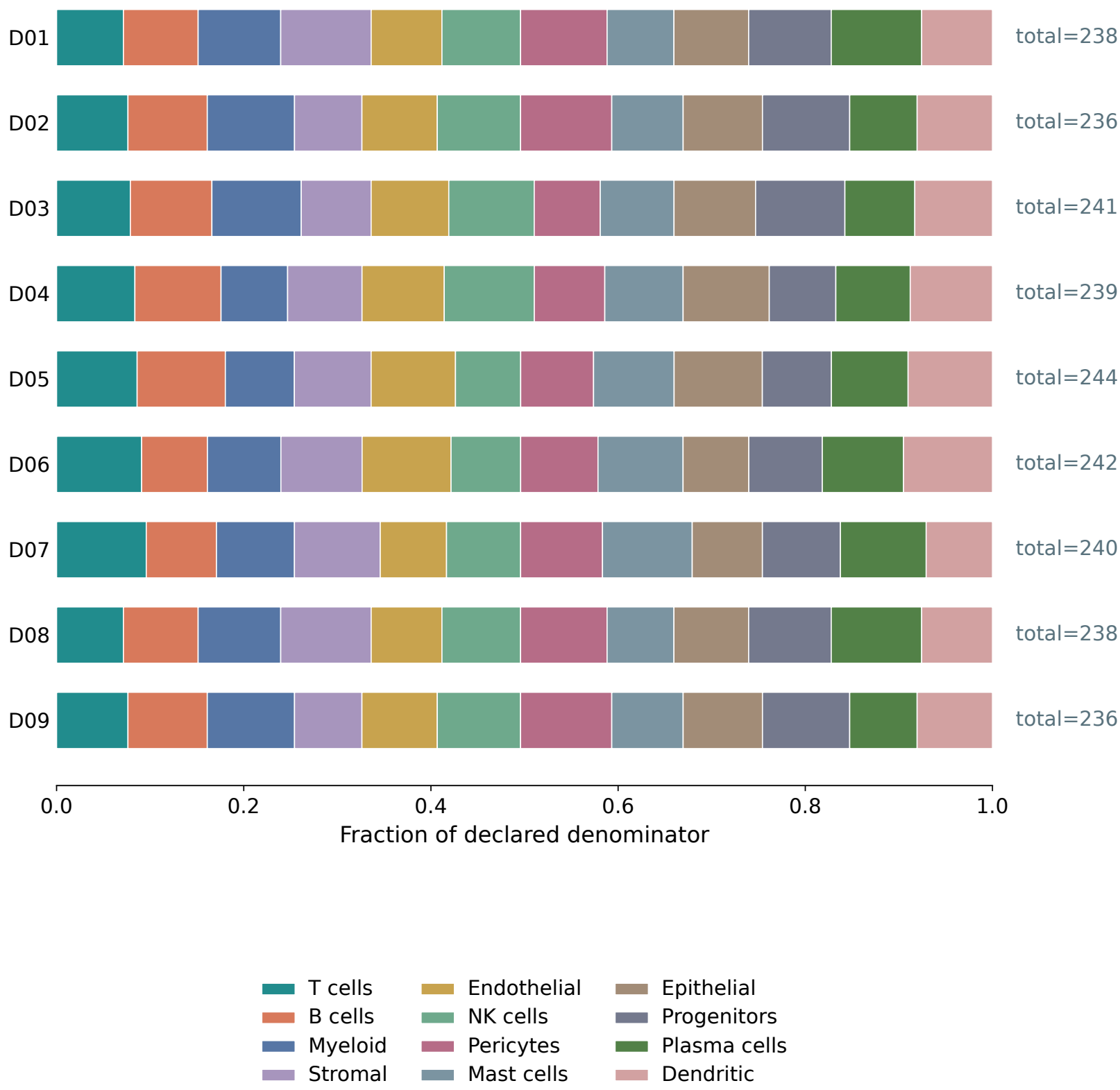


-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095

Single cell · composition — complementary evidence

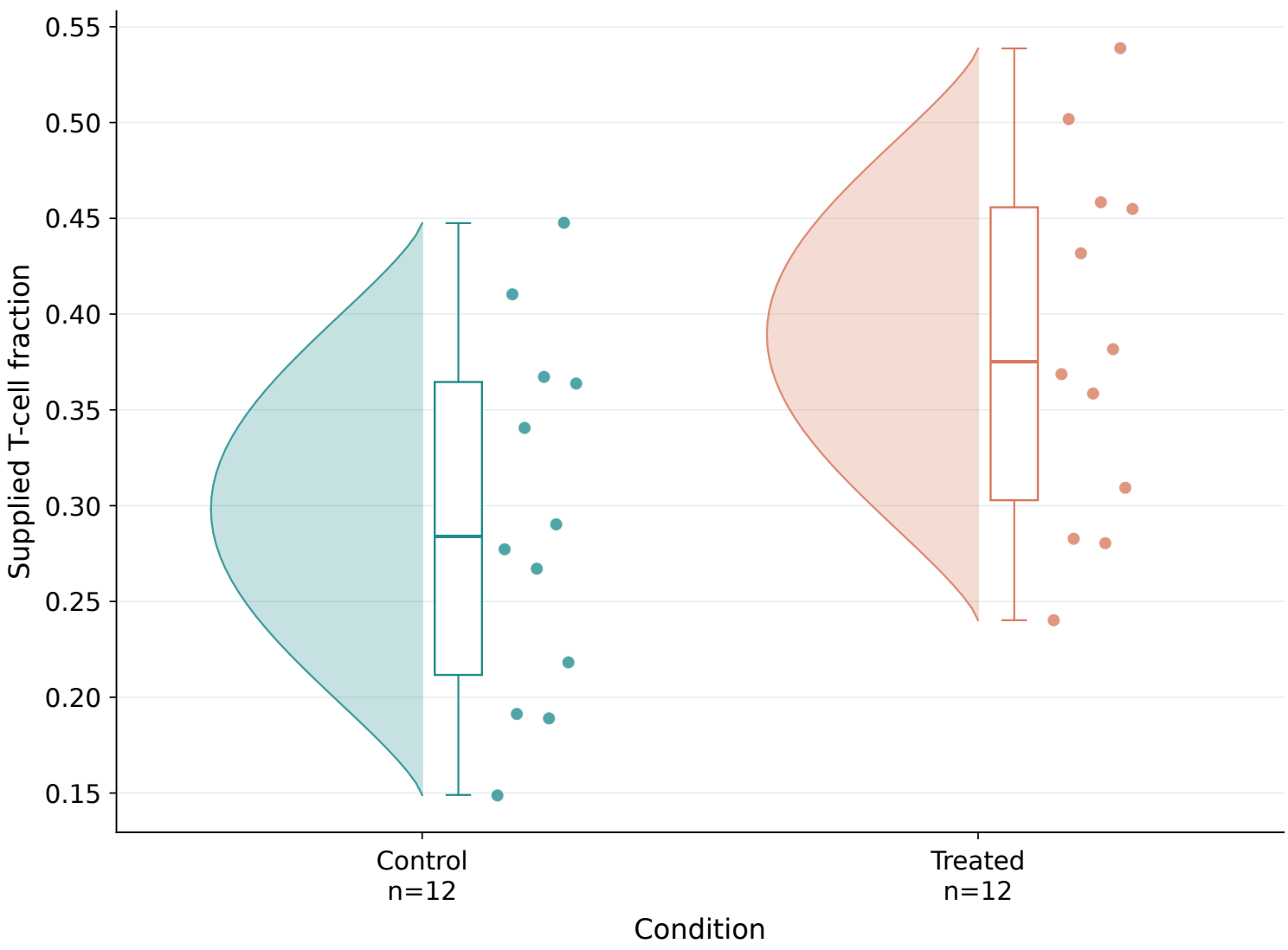
A

Complete donor cell-identity composition

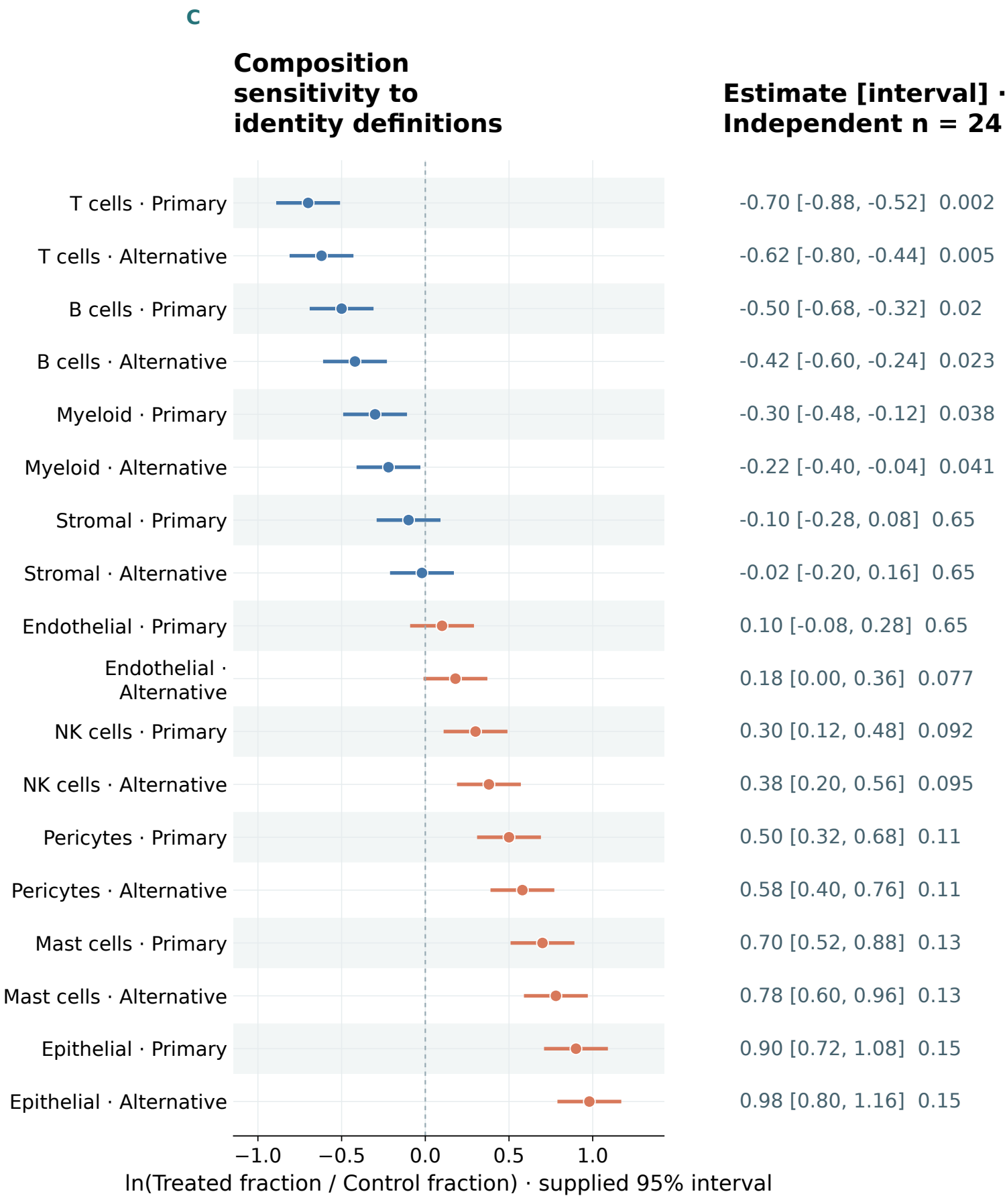


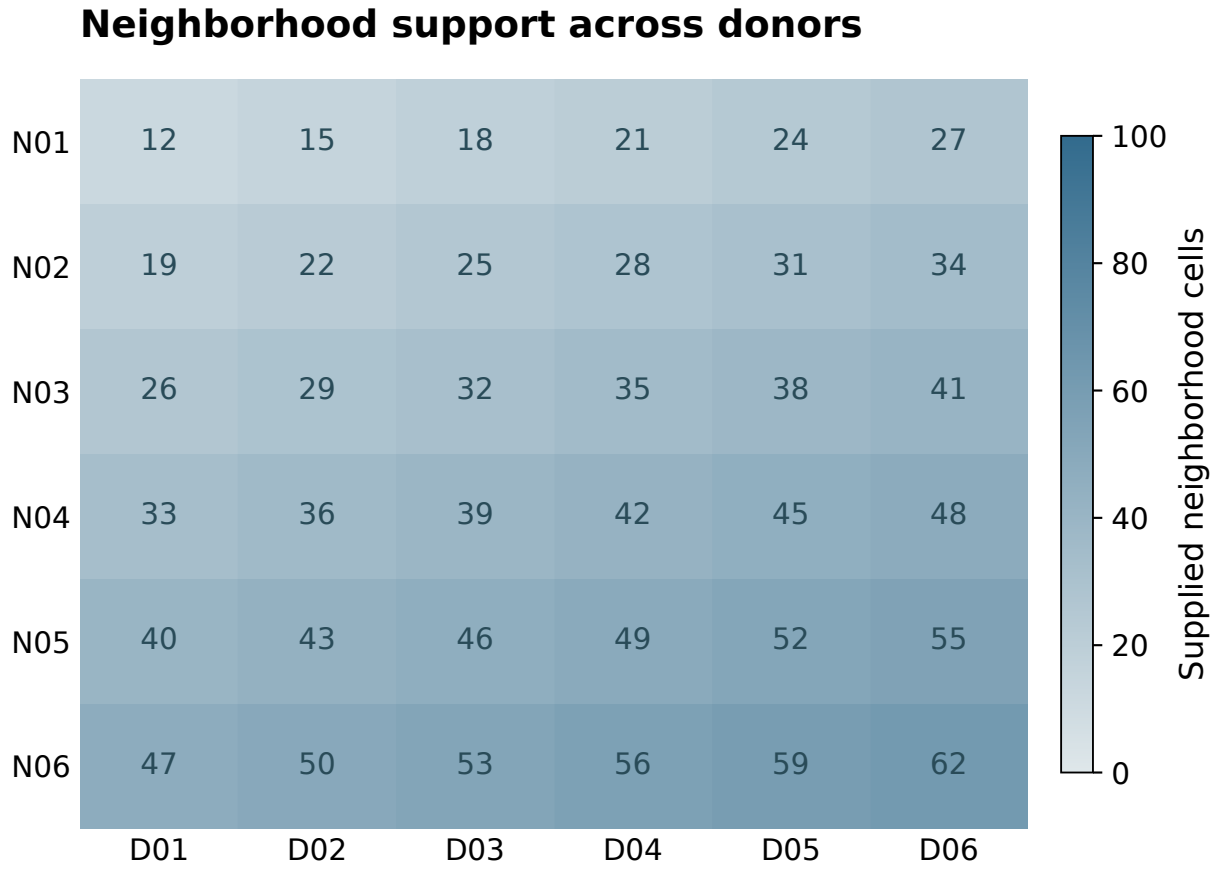
B

T-cell fractions in independent donors

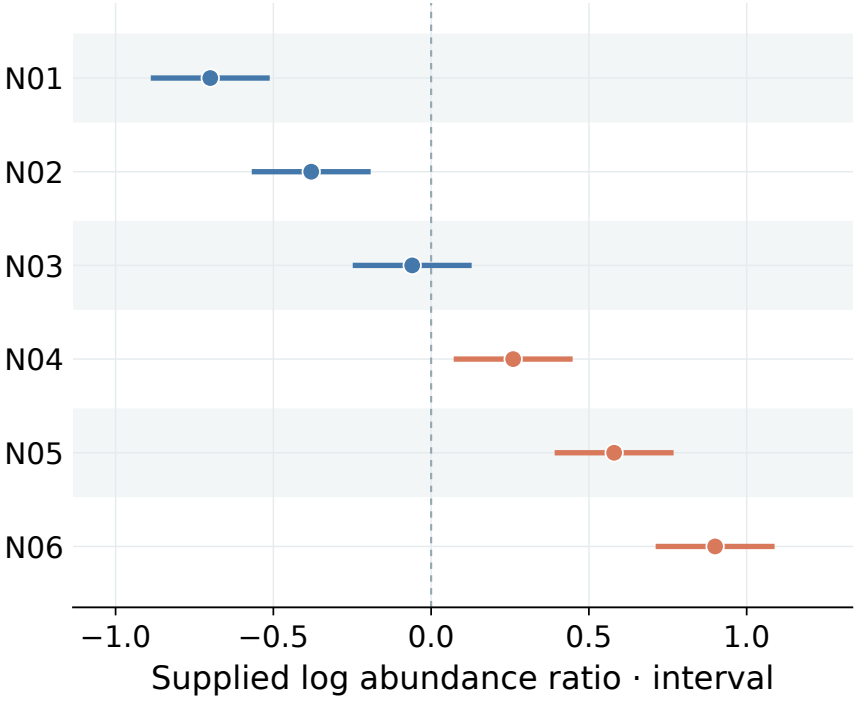


C




Neighborhood support across donors
N01
N02
N03
N04
N05
N06
12
15
18
21
24
27
19
22
25
28
31
34
26
29
32
35
38
41
33
36
39
42
45
48
40
43
46
49
52
55
47
50
53
56
59
62
D01
D02
D03
D04
D05
D06
100
80
60
40
20
0
Supplied neighborhood cells

Donor-aware neighborhood abundance effects



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

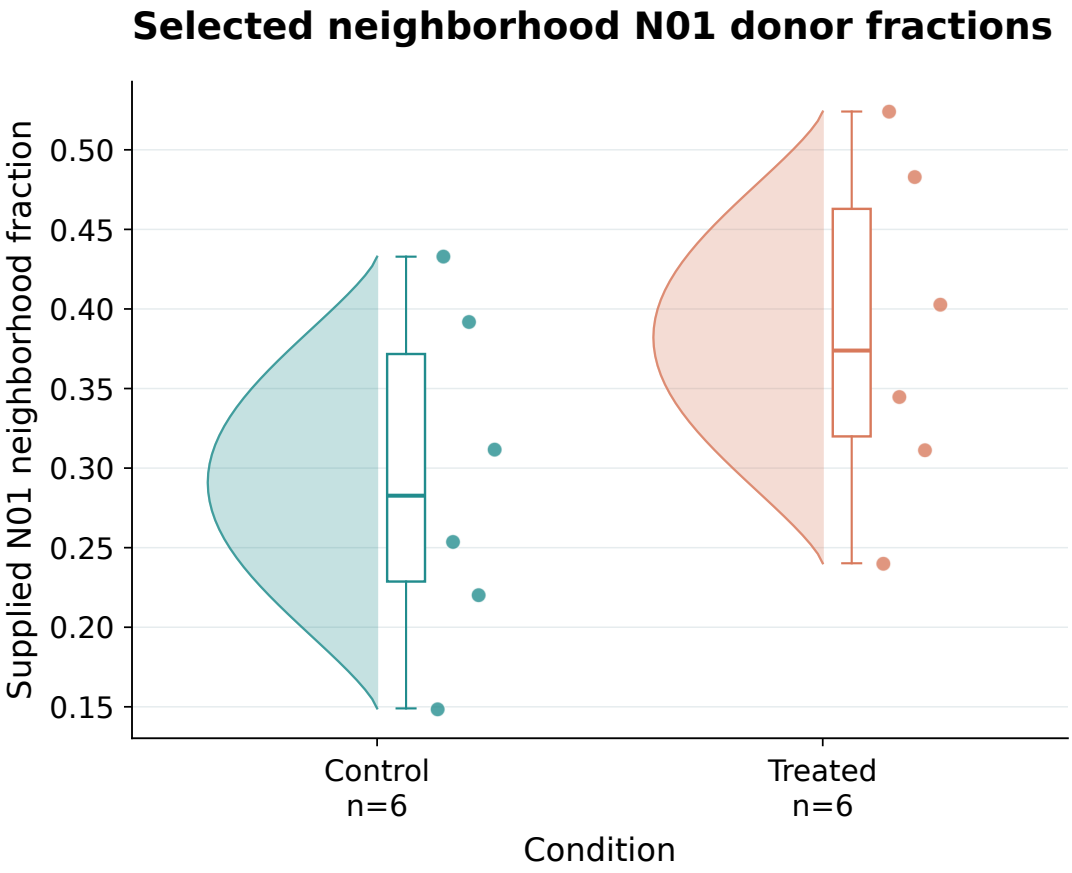
-0.38 [-0.56, -0.20] 0.02

-0.06 [-0.24, 0.12] 0.65

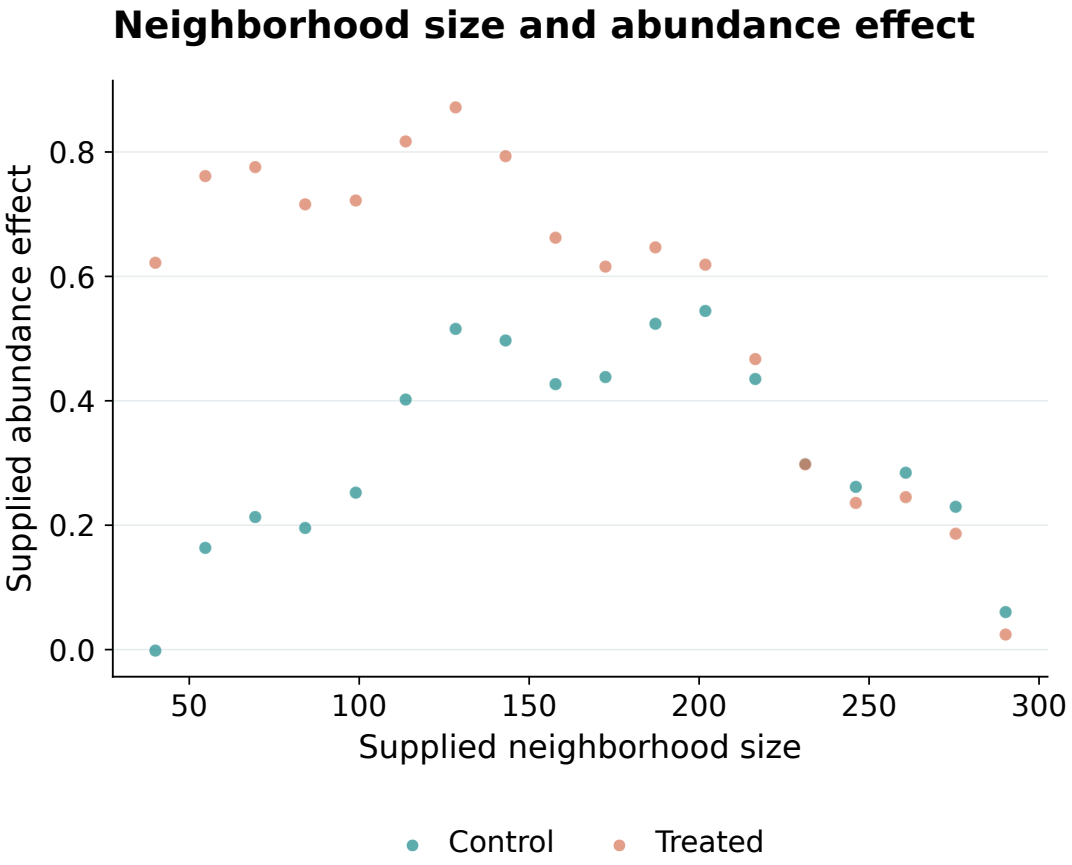
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

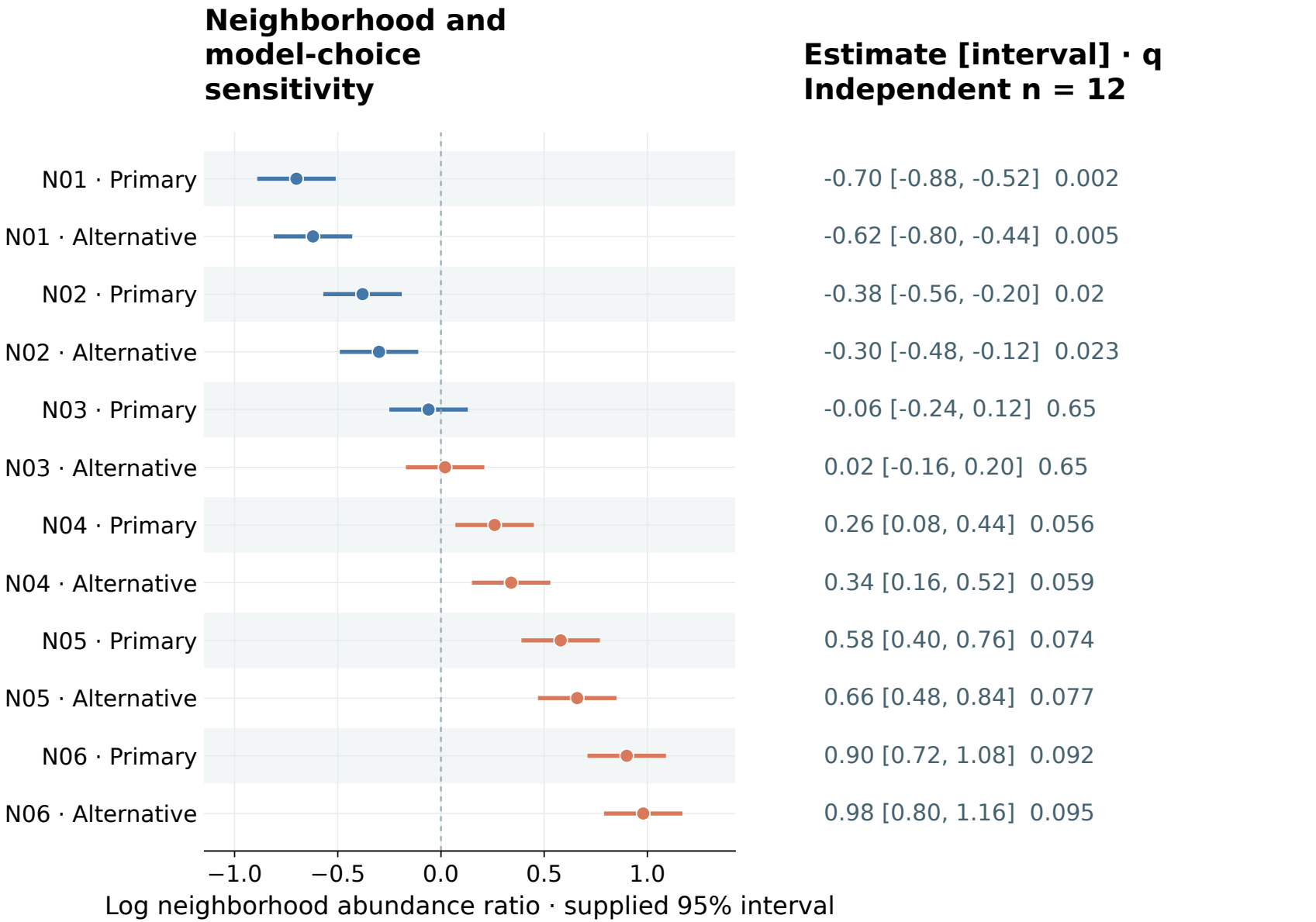
0.90 [0.72, 1.08] 0.092



Software test fixture; not biological evidence



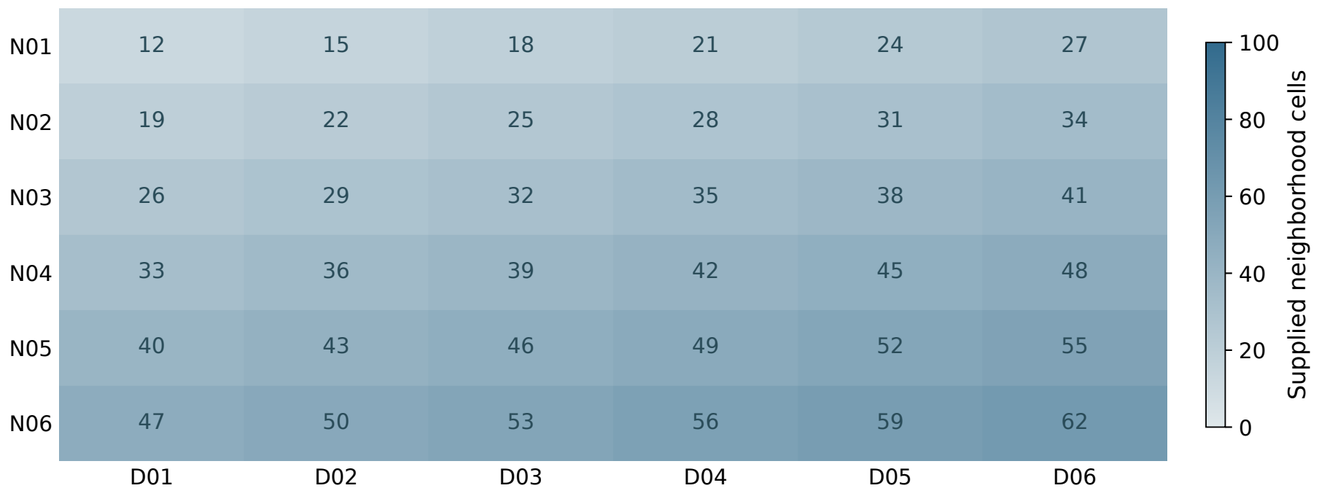
Software test fixture; not biological evidence


Neighborhood and model-choice sensitivity
Estimate [interval] · q
Independent n = 12
N01 · Primary
N01 · Alternative
N02 · Primary
N02 · Alternative
N03 · Primary
N03 · Alternative
N04 · Primary
N04 · Alternative
N05 · Primary
N05 · Alternative
N06 · Primary
N06 · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Log neighborhood abundance ratio · supplied 95% interval

Single cell · abundance — complementary evidence

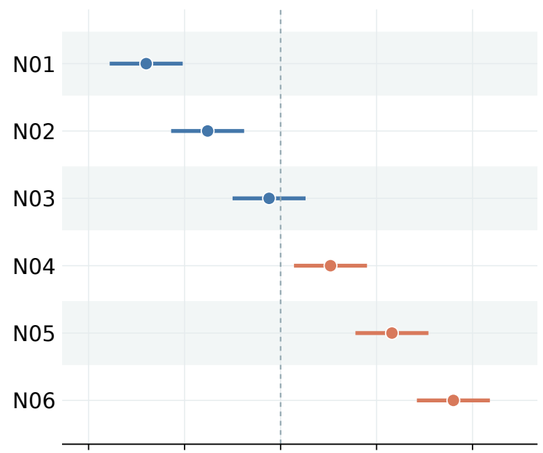
A

Neighborhood support across donors



B

Donor-aware neighborhood abundance effects

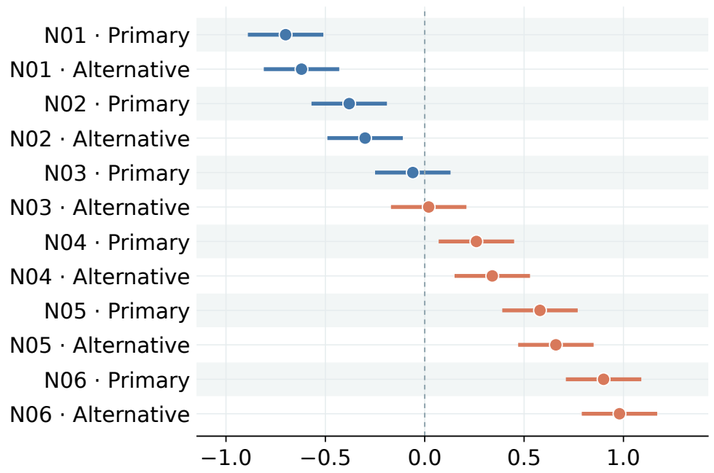


Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002
-0.38 [-0.56, -0.20] 0.02
-0.06 [-0.24, 0.12] 0.65
0.26 [0.08, 0.44] 0.056
0.58 [0.40, 0.76] 0.074
0.90 [0.72, 1.08] 0.092

C

Neighborhood and model-choice sensitivity



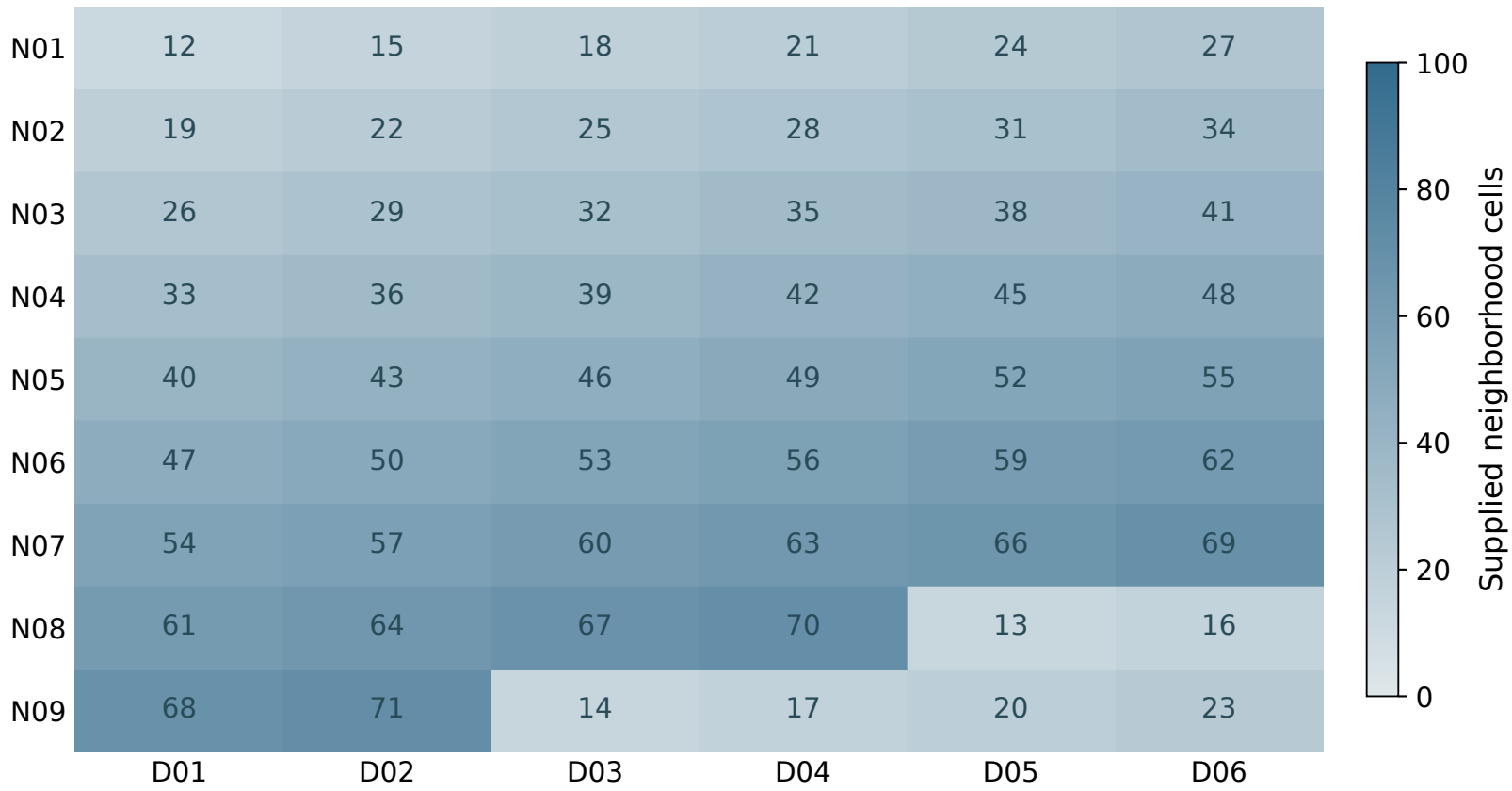
Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095

Single cell · abundance — complementary evidence

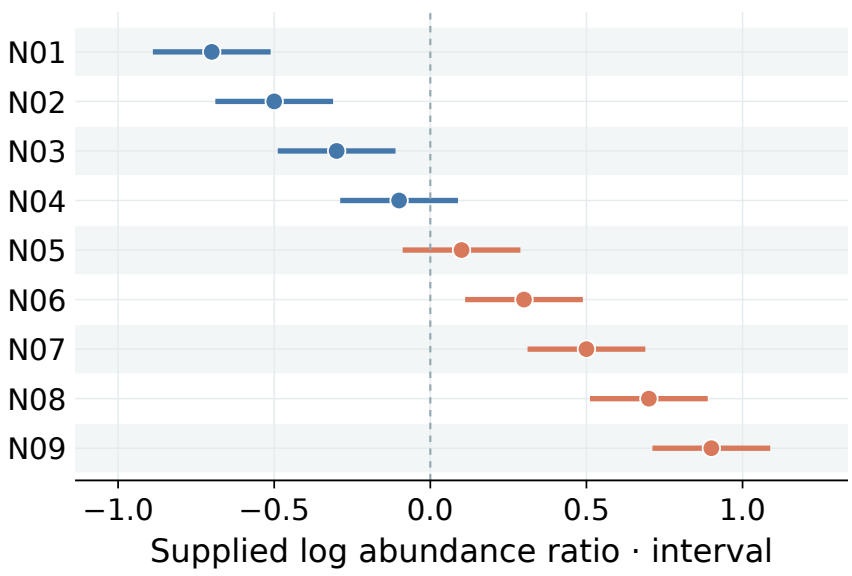
A

Neighborhood support across donors



B

Donor-aware neighborhood abundance effects

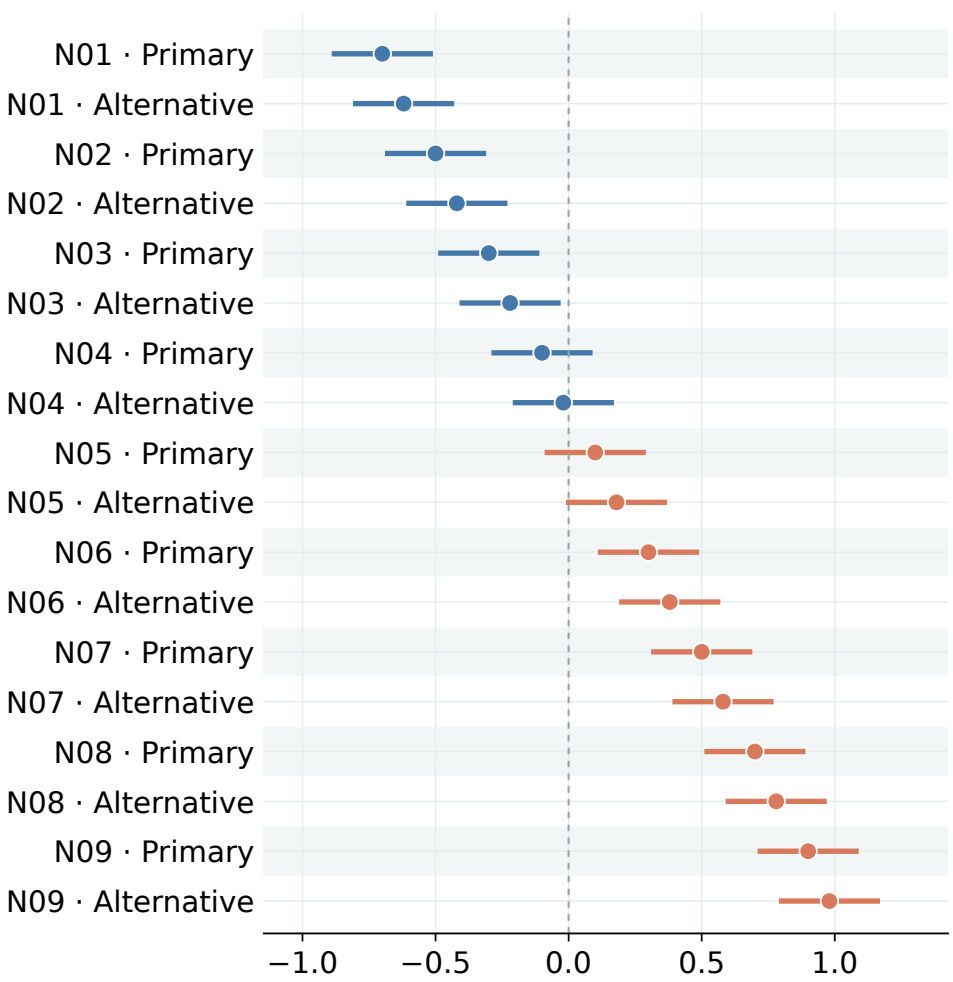


Estimate [interval] · q
Independent n = 24

N01	-0.70 [-0.88, -0.52] 0.002
N02	-0.50 [-0.68, -0.32] 0.02
N03	-0.30 [-0.48, -0.12] 0.038
N04	-0.10 [-0.28, 0.08] 0.65
N05	0.10 [-0.08, 0.28] 0.65
N06	0.30 [0.12, 0.48] 0.092
N07	0.50 [0.32, 0.68] 0.11
N08	0.70 [0.52, 0.88] 0.13
N09	0.90 [0.72, 1.08] 0.15

C

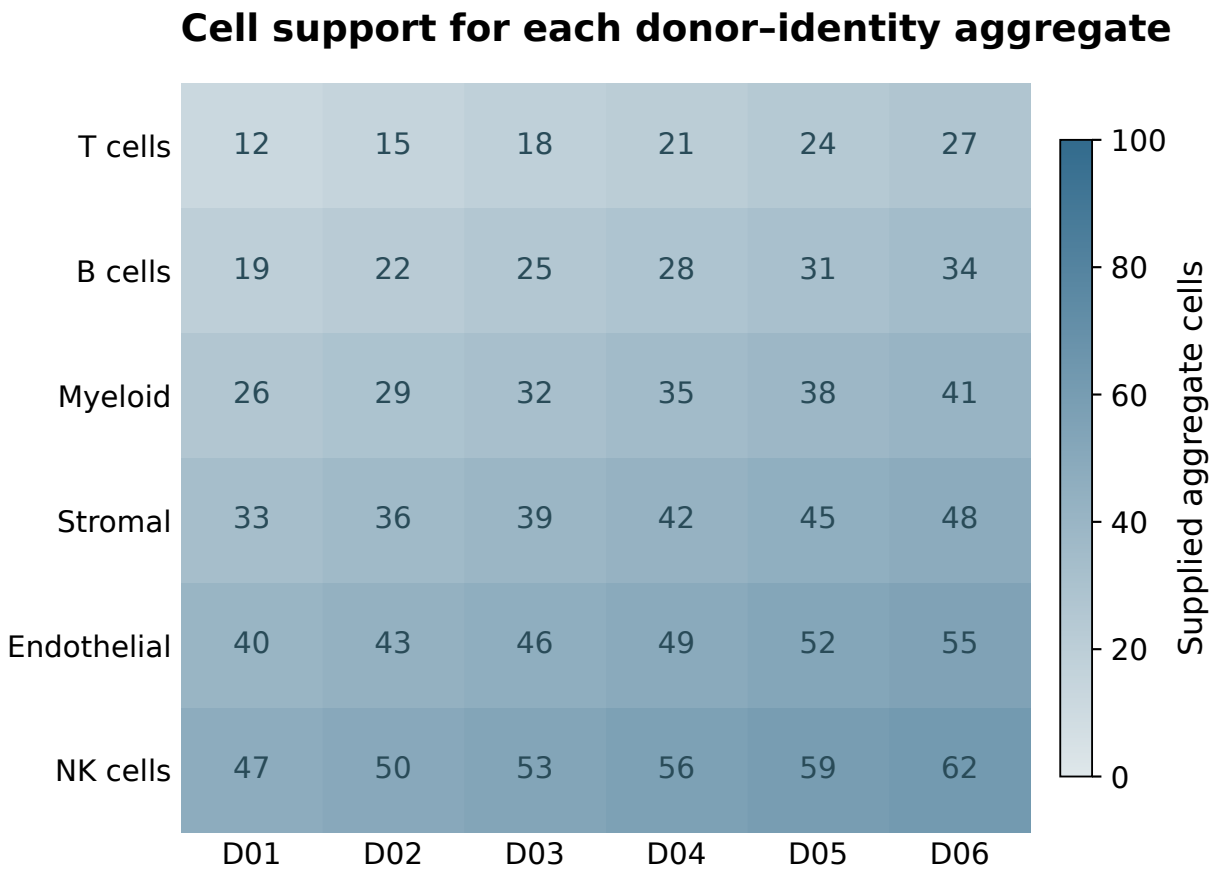
Neighborhood and model-choice sensitivity

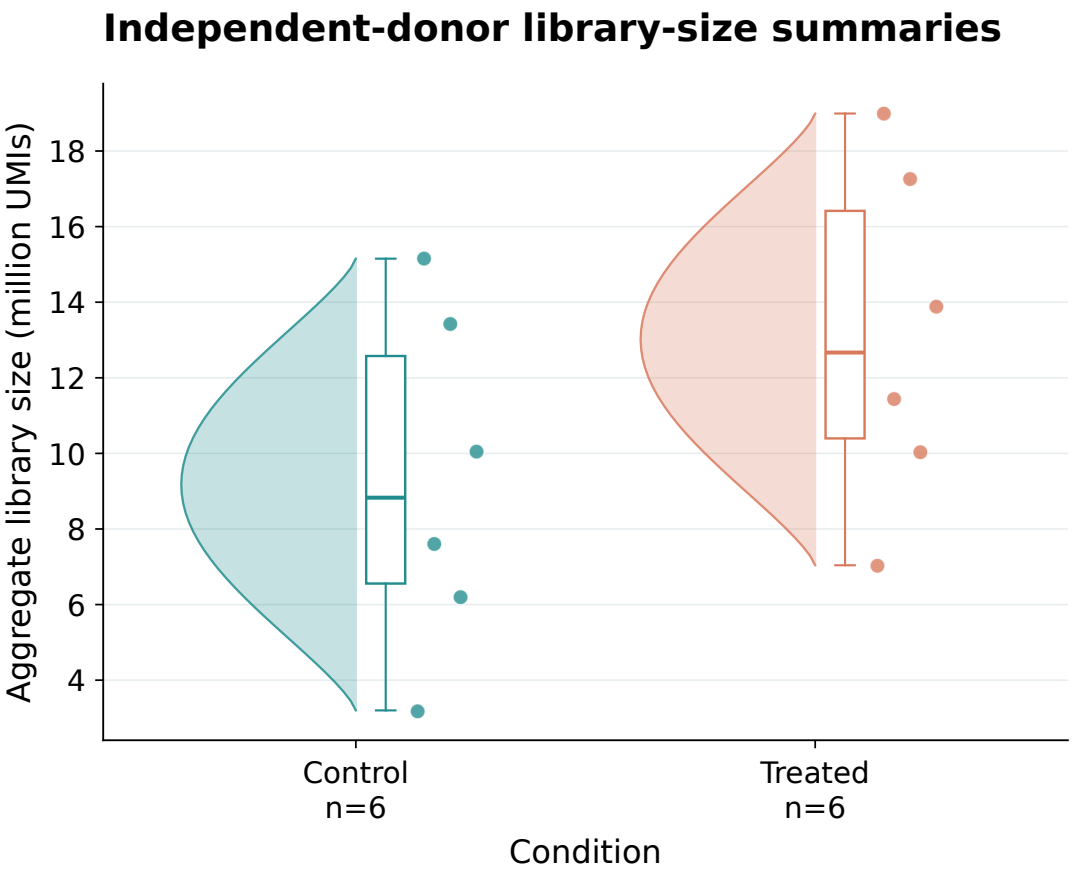


Estimate [interval] · q
Independent n = 24

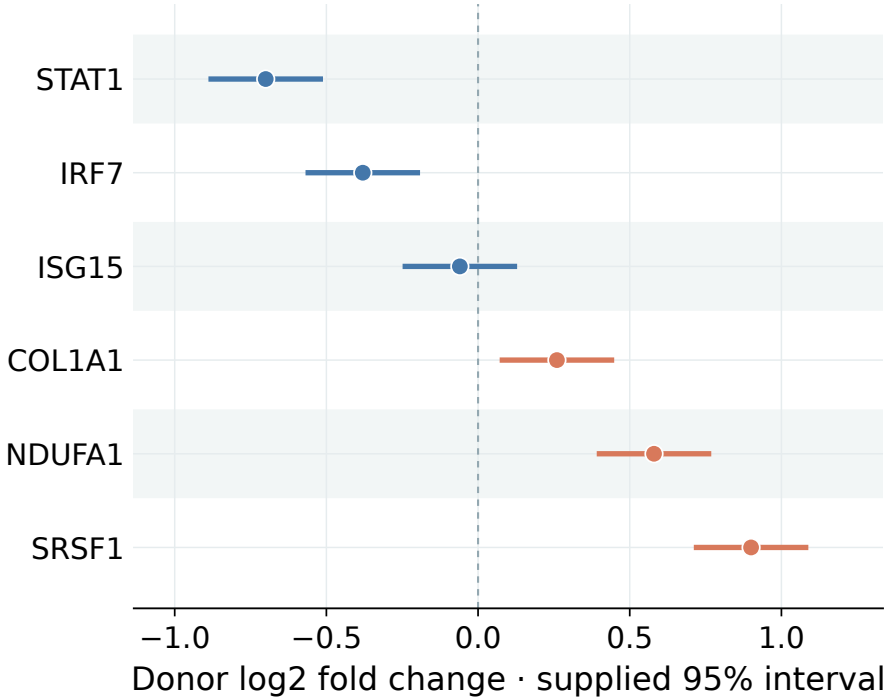
N01 · Primary	-0.70 [-0.88, -0.52] 0.002
N01 · Alternative	-0.62 [-0.80, -0.44] 0.005
N02 · Primary	-0.50 [-0.68, -0.32] 0.02
N02 · Alternative	-0.42 [-0.60, -0.24] 0.023
N03 · Primary	-0.30 [-0.48, -0.12] 0.038
N03 · Alternative	-0.22 [-0.40, -0.04] 0.041
N04 · Primary	-0.10 [-0.28, 0.08] 0.65
N04 · Alternative	-0.02 [-0.20, 0.16] 0.65
N05 · Primary	0.10 [-0.08, 0.28] 0.65
N05 · Alternative	0.18 [0.00, 0.36] 0.077
N06 · Primary	0.30 [0.12, 0.48] 0.092
N06 · Alternative	0.38 [0.20, 0.56] 0.095
N07 · Primary	0.50 [0.32, 0.68] 0.11
N07 · Alternative	0.58 [0.40, 0.76] 0.11
N08 · Primary	0.70 [0.52, 0.88] 0.13
N08 · Alternative	0.78 [0.60, 0.96] 0.13
N09 · Primary	0.90 [0.72, 1.08] 0.15
N09 · Alternative	0.98 [0.80, 1.16] 0.15

Software test fixture; not biological evidence


Cell support for each donor–identity aggregate
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
12 15 18 21 24 27
19 22 25 28 31 34
26 29 32 35 38 41
33 36 39 42 45 48
40 43 46 49 52 55
47 50 53 56 59 62
D01 D02 D03 D04 D05 D06
Supplied aggregate cells
100
80
60
40
20
0

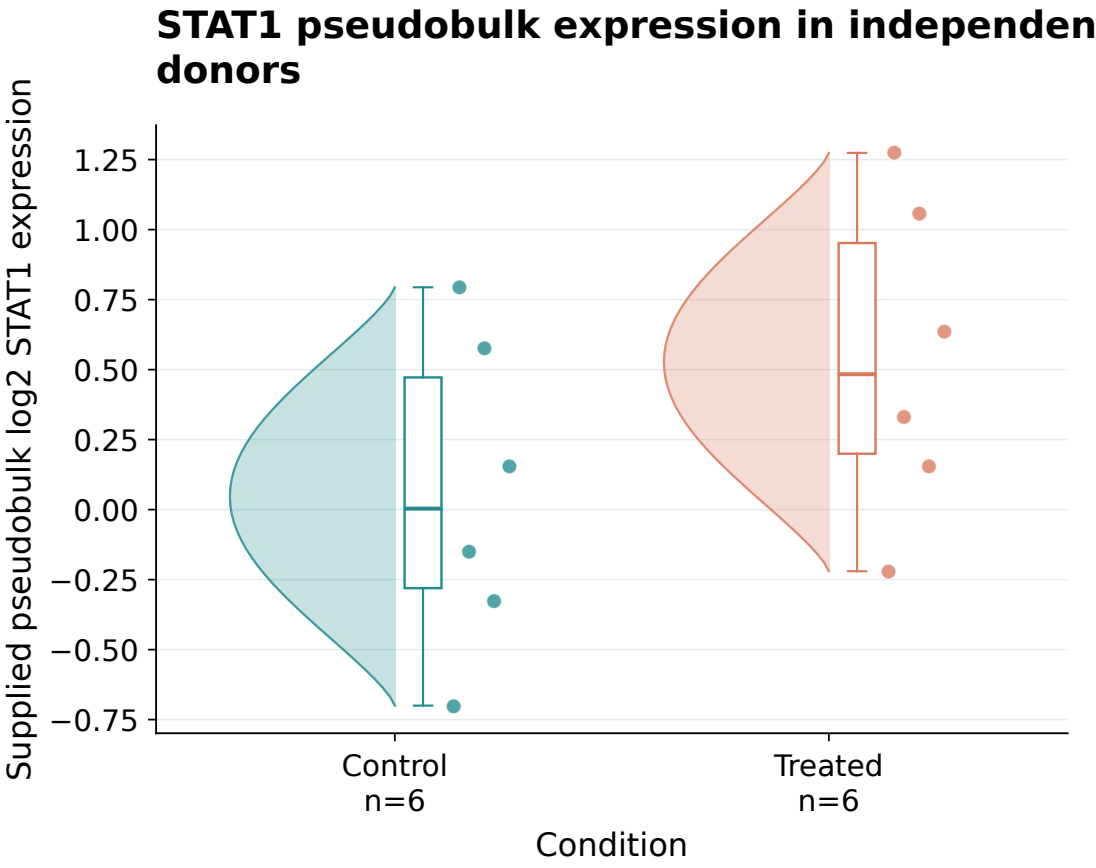

Independent-donor library-size summaries
Aggregate library size (million UMIs)
18
16
14
12
10
8
6
4
Control
n=6
Treated
n=6
Condition

Donor-level differential-expression effects



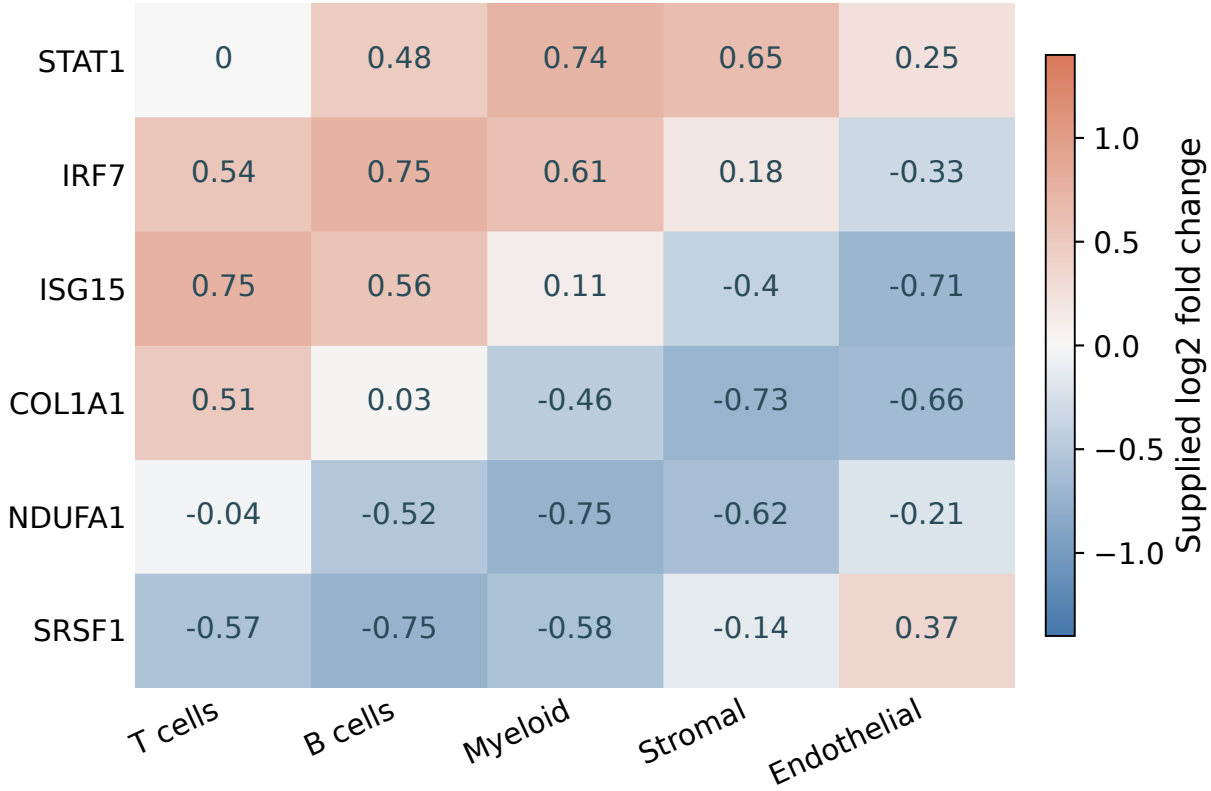
Estimate [interval] · q
Independent n = 12

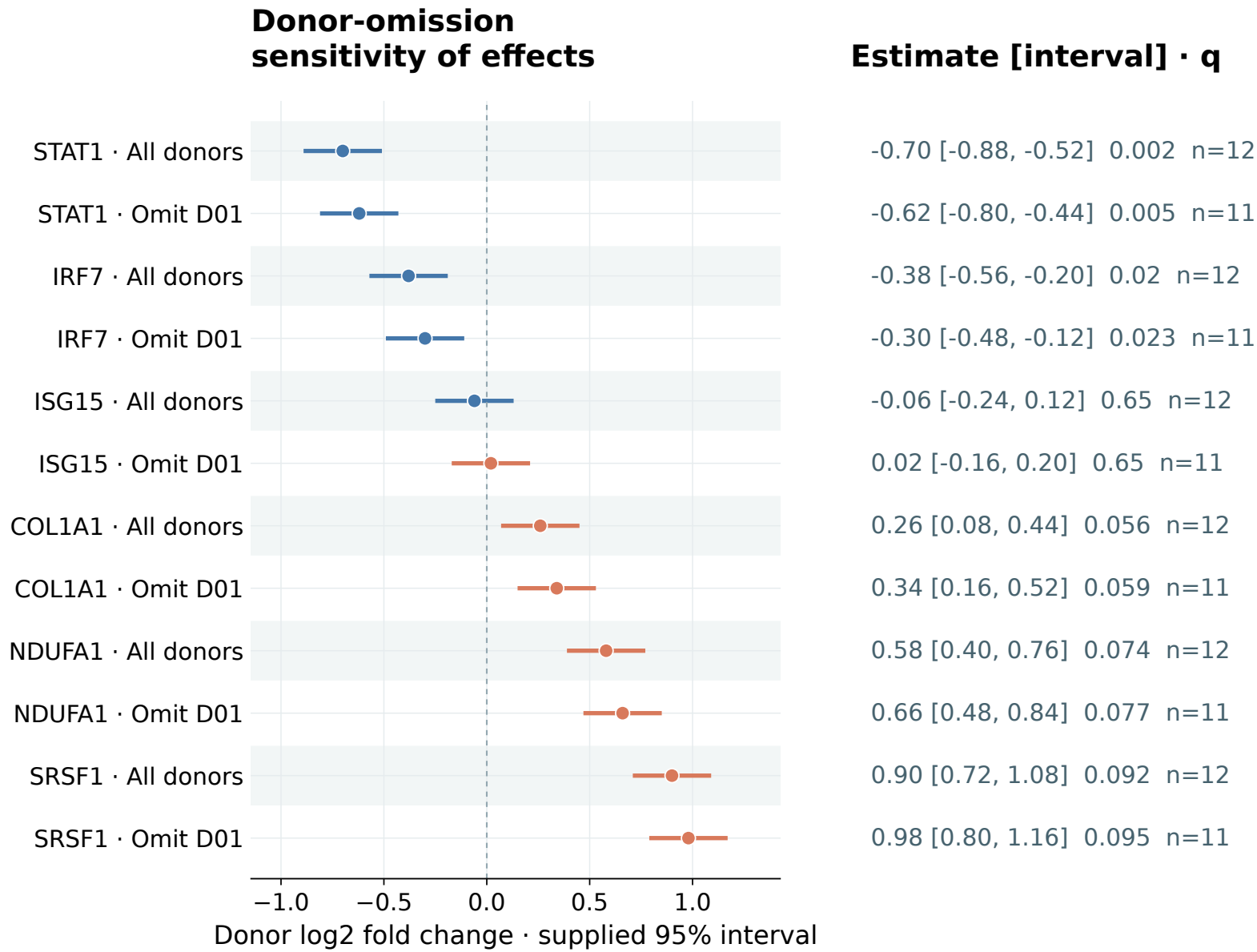
STAT1	-0.70 [-0.88, -0.52] 0.002
IRF7	-0.38 [-0.56, -0.20] 0.02
ISG15	-0.06 [-0.24, 0.12] 0.65
COL1A1	0.26 [0.08, 0.44] 0.056
NDUFA1	0.58 [0.40, 0.76] 0.074
SRSF1	0.90 [0.72, 1.08] 0.092



Software test fixture; not biological evidence

Celltype-specific differential responses

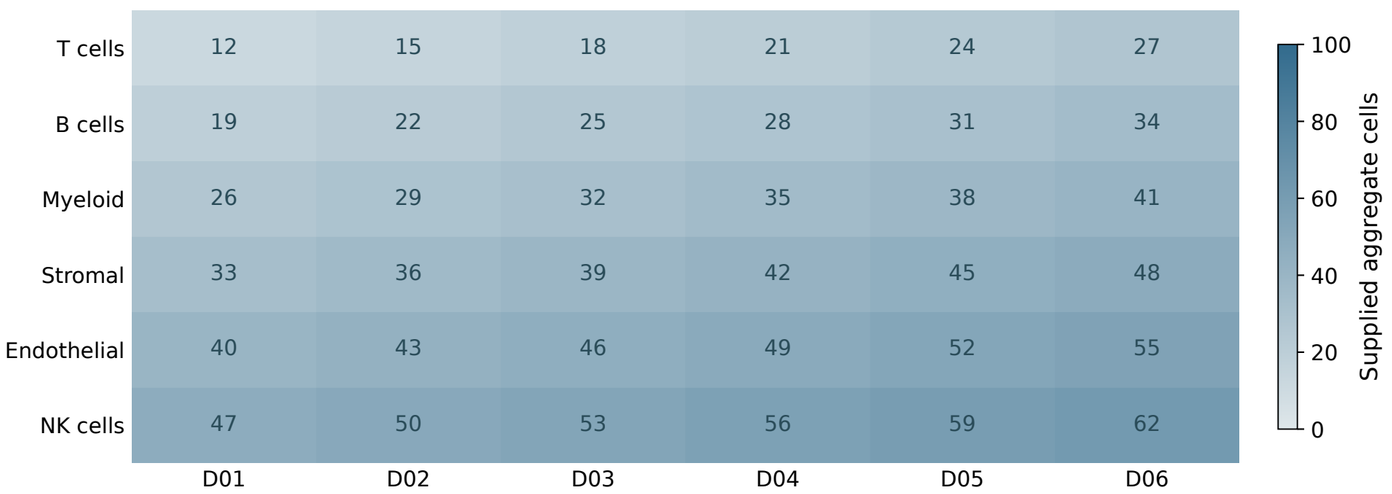



Donor-omission sensitivity of effects
Estimate [interval] · q
STAT1 · All donors
STAT1 · Omit D01
IRF7 · All donors
IRF7 · Omit D01
ISG15 · All donors
ISG15 · Omit D01
COL1A1 · All donors
COL1A1 · Omit D01
NDUFA1 · All donors
NDUFA1 · Omit D01
SRSF1 · All donors
SRSF1 · Omit D01
-0.70 [-0.88, -0.52] 0.002 n=12
-0.62 [-0.80, -0.44] 0.005 n=11
-0.38 [-0.56, -0.20] 0.02 n=12
-0.30 [-0.48, -0.12] 0.023 n=11
-0.06 [-0.24, 0.12] 0.65 n=12
0.02 [-0.16, 0.20] 0.65 n=11
0.26 [0.08, 0.44] 0.056 n=12
0.34 [0.16, 0.52] 0.059 n=11
0.58 [0.40, 0.76] 0.074 n=12
0.66 [0.48, 0.84] 0.077 n=11
0.90 [0.72, 1.08] 0.092 n=12
0.98 [0.80, 1.16] 0.095 n=11
−1.0
−0.5
0.0
0.5
1.0
Donor log2 fold change · supplied 95% interval

Single cell · pseudobulk — complementary evidence

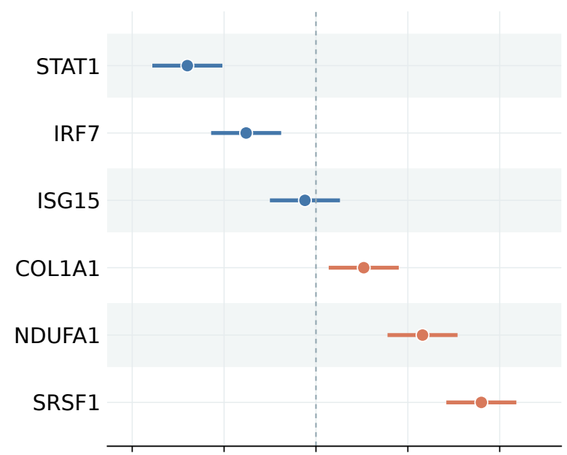
A

Cell support for each donor–identity aggregate



B

Donor-level differential-expression effects

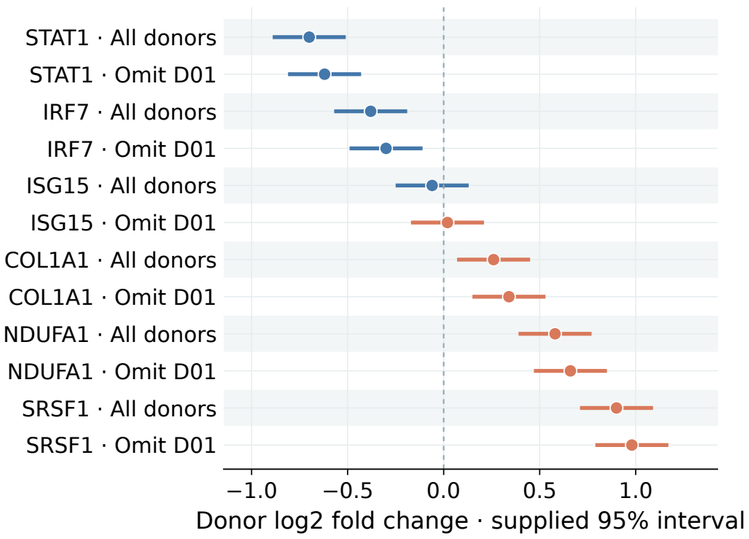


Estimate [interval] · q
Independent n = 12

- -0.70 [-0.88, -0.52] 0.002
- -0.38 [-0.56, -0.20] 0.02
- -0.06 [-0.24, 0.12] 0.65
- 0.26 [0.08, 0.44] 0.056
- 0.58 [0.40, 0.76] 0.074
- 0.90 [0.72, 1.08] 0.092

C

Donor-omission sensitivity of effects



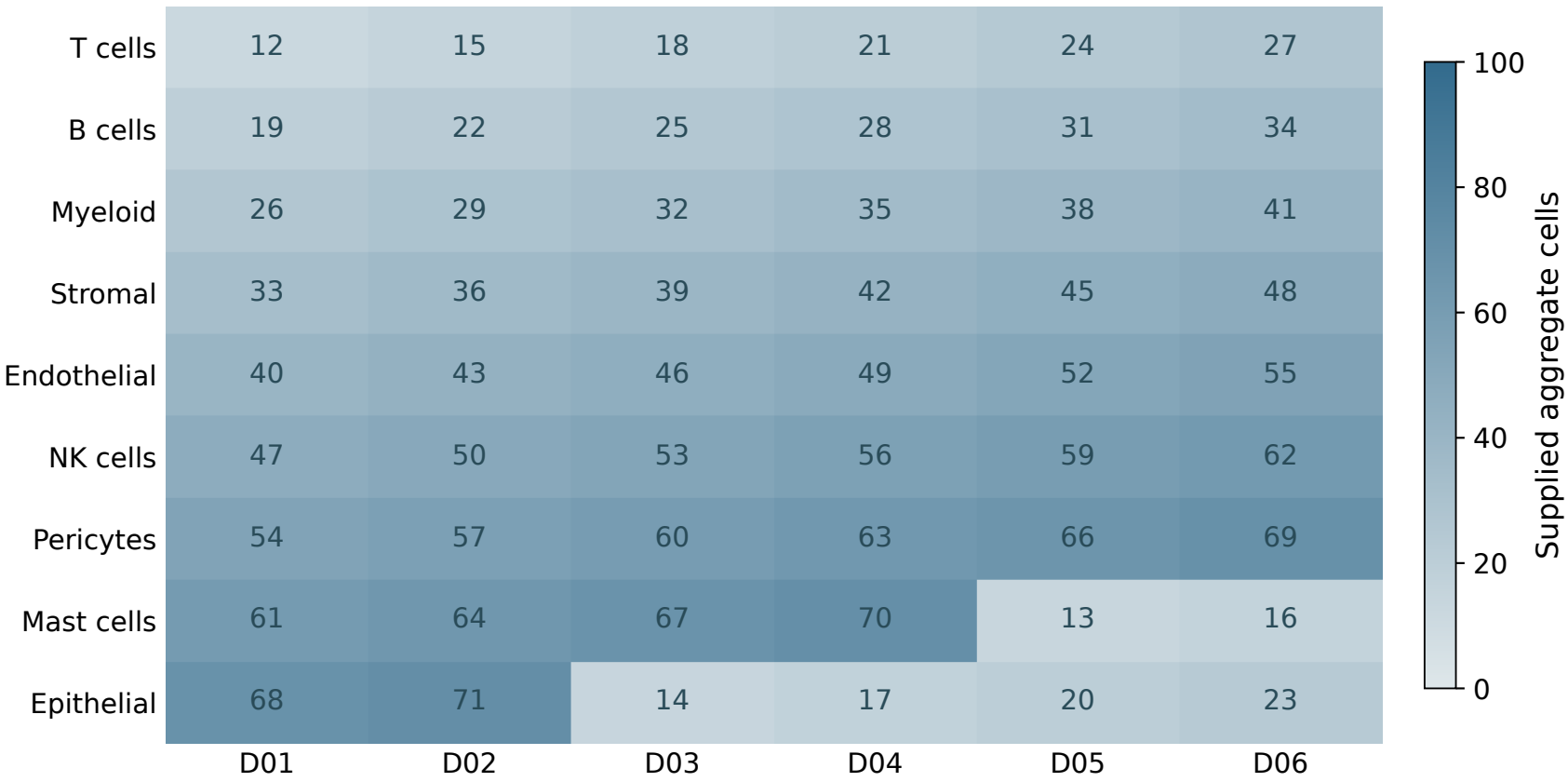
Estimate [interval] · q

- -0.70 [-0.88, -0.52] 0.002 n=12
- -0.62 [-0.80, -0.44] 0.005 n=11
- -0.38 [-0.56, -0.20] 0.02 n=12
- -0.30 [-0.48, -0.12] 0.023 n=11
- -0.06 [-0.24, 0.12] 0.65 n=12
- 0.02 [-0.16, 0.20] 0.65 n=11
- 0.26 [0.08, 0.44] 0.056 n=12
- 0.34 [0.16, 0.52] 0.059 n=11
- 0.58 [0.40, 0.76] 0.074 n=12
- 0.66 [0.48, 0.84] 0.077 n=11
- 0.90 [0.72, 1.08] 0.092 n=12
- 0.98 [0.80, 1.16] 0.095 n=11

Single cell · pseudobulk — complementary evidence

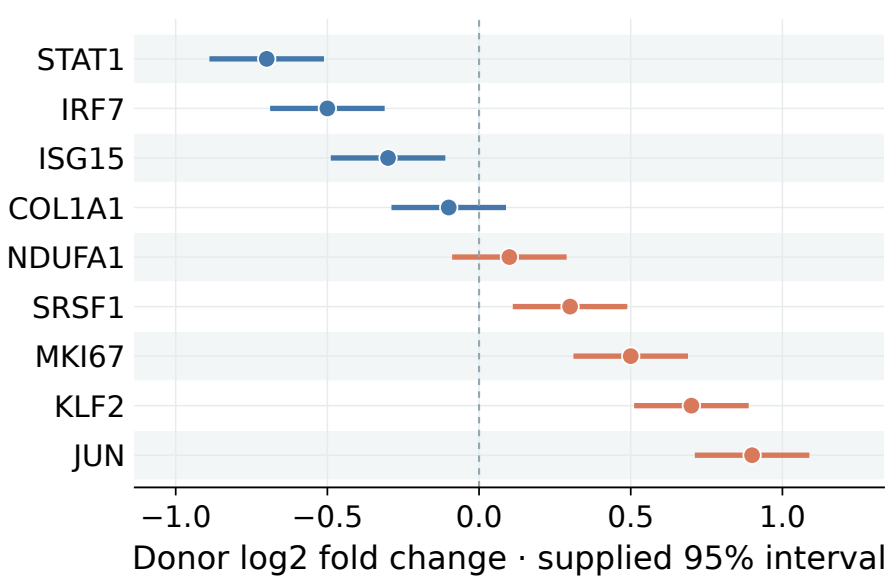
A

Cell support for each donor–identity aggregate



B

Donor-level differential-expression effects

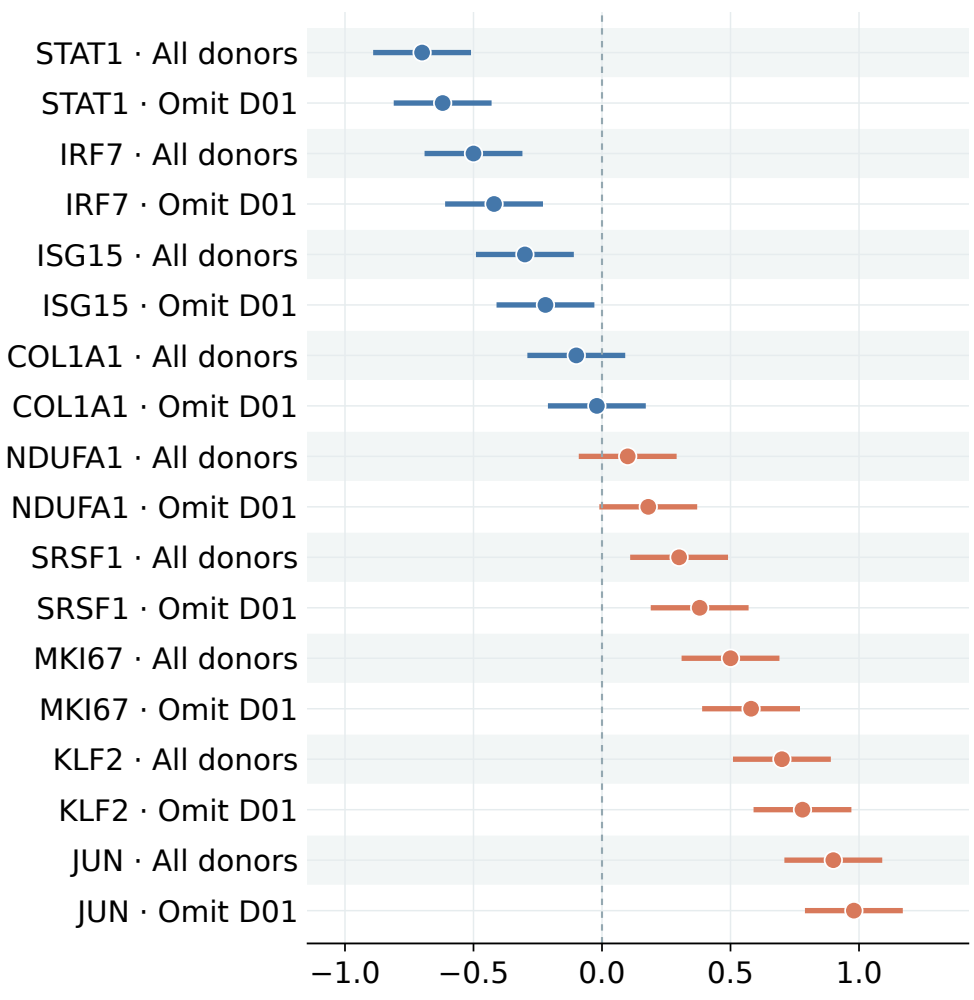


Estimate [interval] · q
Independent n = 24

STAT1	-0.70 [-0.88, -0.52] 0.002
IRF7	-0.50 [-0.68, -0.32] 0.02
ISG15	-0.30 [-0.48, -0.12] 0.038
COL1A1	-0.10 [-0.28, 0.08] 0.65
NDUFA1	0.10 [-0.08, 0.28] 0.65
SRSF1	0.30 [0.12, 0.48] 0.092
MKI67	0.50 [0.32, 0.68] 0.11
KLF2	0.70 [0.52, 0.88] 0.13
JUN	0.90 [0.72, 1.08] 0.15

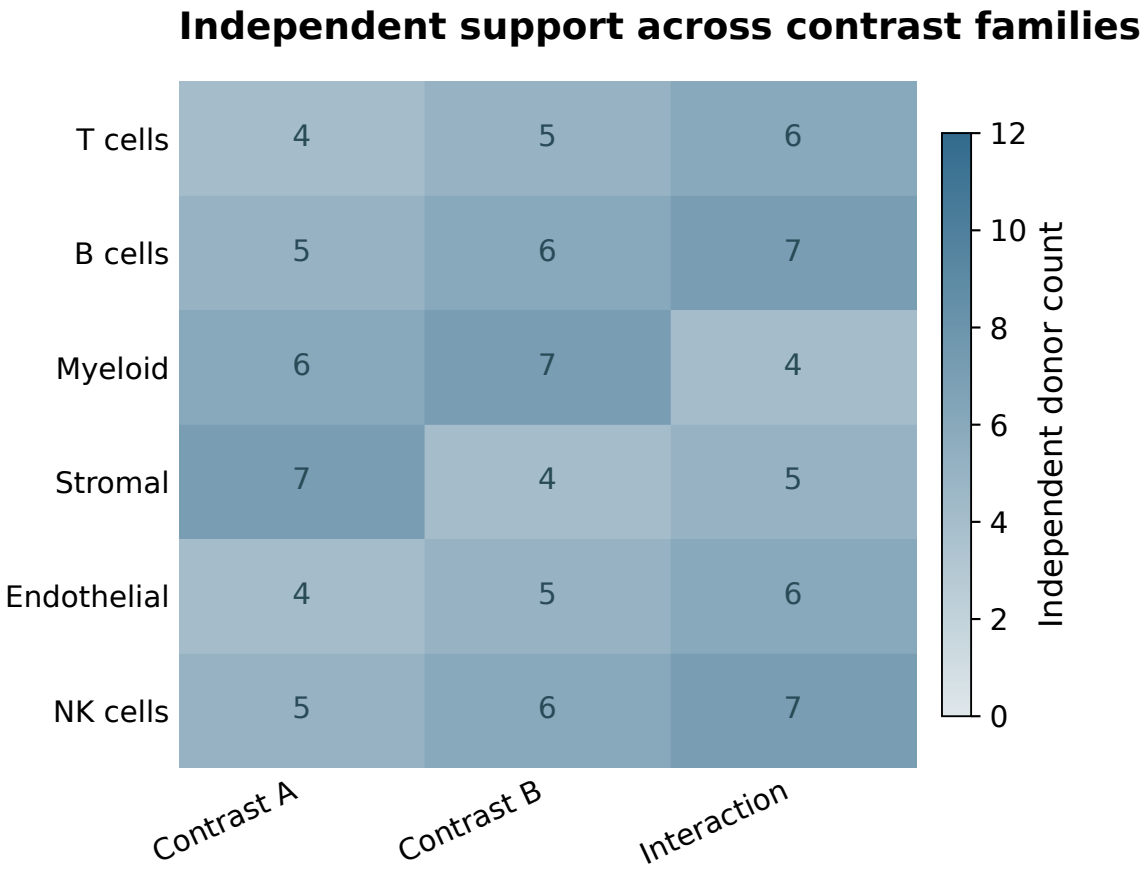
C

Donor-omission sensitivity of effects

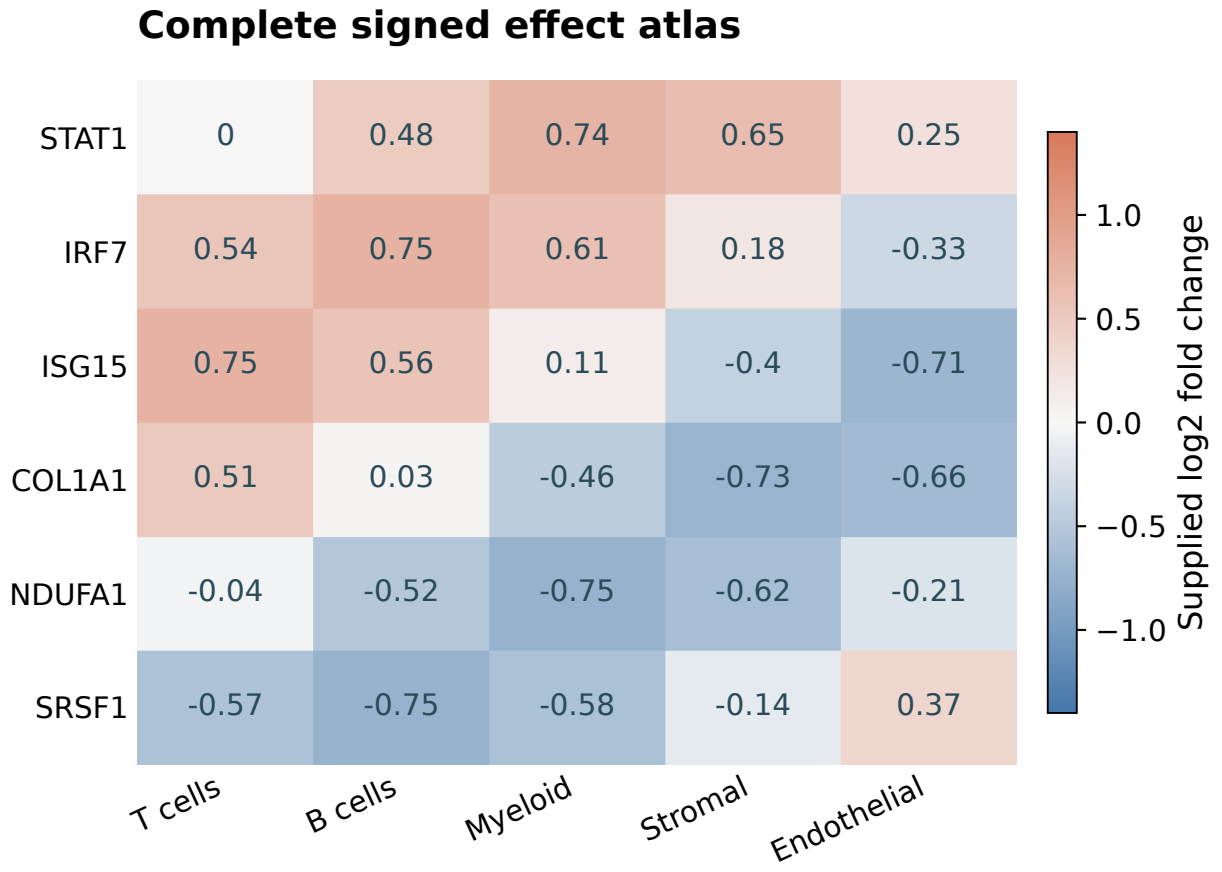


Estimate [interval] · q

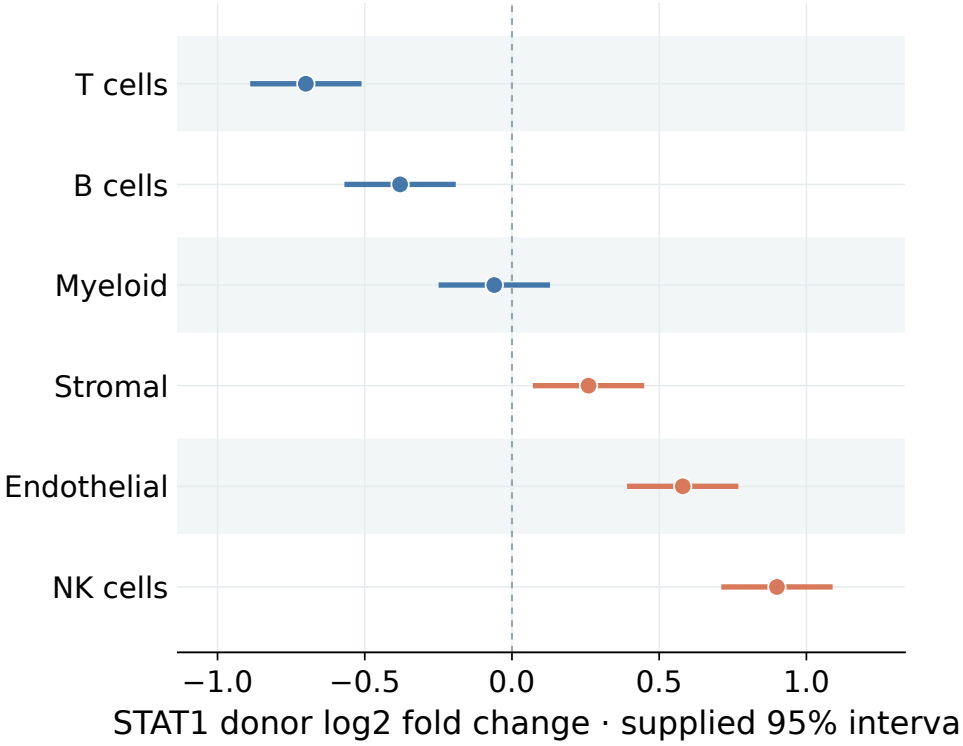
STAT1 · All donors	-0.70 [-0.88, -0.52] 0.002 n=24
STAT1 · Omit D01	-0.62 [-0.80, -0.44] 0.005 n=23
IRF7 · All donors	-0.50 [-0.68, -0.32] 0.02 n=24
IRF7 · Omit D01	-0.42 [-0.60, -0.24] 0.023 n=23
ISG15 · All donors	-0.30 [-0.48, -0.12] 0.038 n=24
ISG15 · Omit D01	-0.22 [-0.40, -0.04] 0.041 n=23
COL1A1 · All donors	-0.10 [-0.28, 0.08] 0.65 n=24
COL1A1 · Omit D01	-0.02 [-0.20, 0.16] 0.65 n=23
NDUFA1 · All donors	0.10 [-0.08, 0.28] 0.65 n=24
NDUFA1 · Omit D01	0.18 [0.00, 0.36] 0.077 n=23
SRSF1 · All donors	0.30 [0.12, 0.48] 0.092 n=24
SRSF1 · Omit D01	0.38 [0.20, 0.56] 0.095 n=23
MKI67 · All donors	0.50 [0.32, 0.68] 0.11 n=24
MKI67 · Omit D01	0.58 [0.40, 0.76] 0.11 n=23
KLF2 · All donors	0.70 [0.52, 0.88] 0.13 n=24
KLF2 · Omit D01	0.78 [0.60, 0.96] 0.13 n=23
JUN · All donors	0.90 [0.72, 1.08] 0.15 n=24
JUN · Omit D01	0.98 [0.80, 1.16] 0.15 n=23



Software test fixture; not biological evidence


Complete signed effect atlas
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0
0.48
0.74
0.65
0.25
0.54
0.75
0.61
0.18
-0.33
0.75
0.56
0.11
-0.4
-0.71
0.51
0.03
-0.46
-0.73
-0.66
-0.04
-0.52
-0.75
-0.62
-0.21
-0.57
-0.75
-0.58
-0.14
0.37
T cells
B cells
Myeloid
Stromal
Endothelial
1.0
0.5
0.0
−0.5
−1.0
Supplied log2 fold change

Prespecified identity-by-condition effects



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

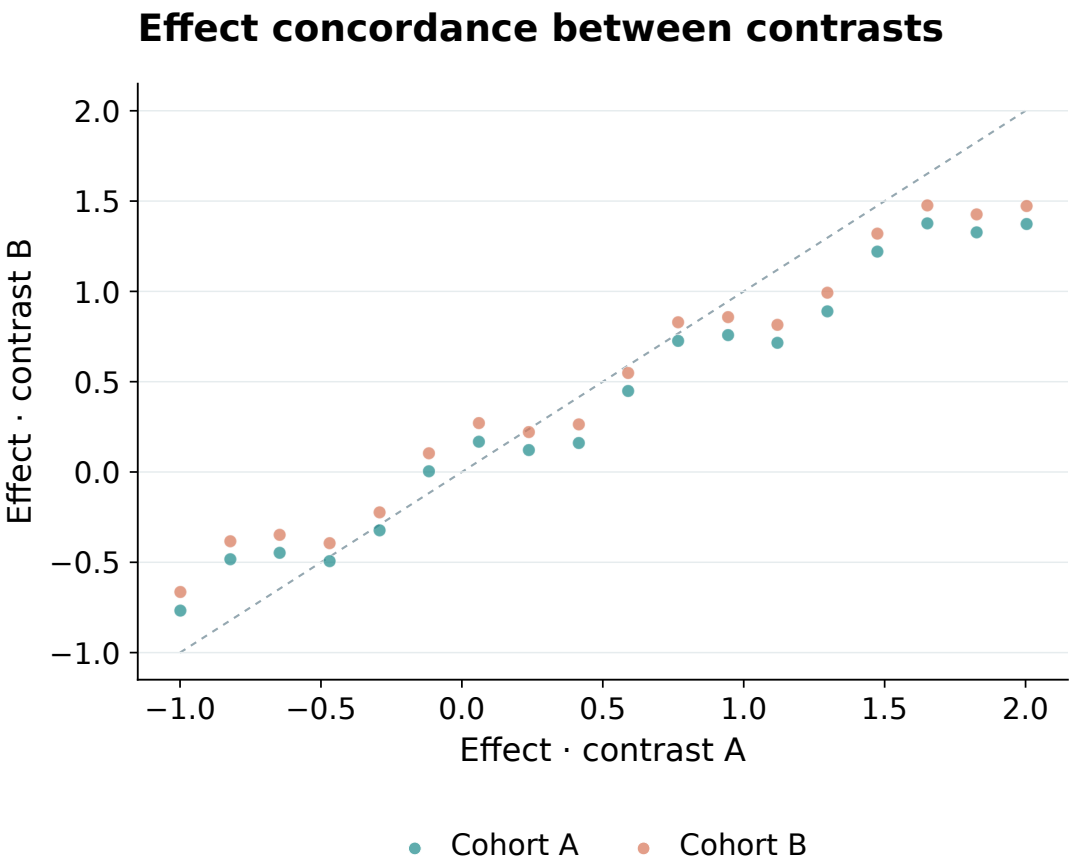
-0.38 [-0.56, -0.20] 0.02

-0.06 [-0.24, 0.12] 0.65

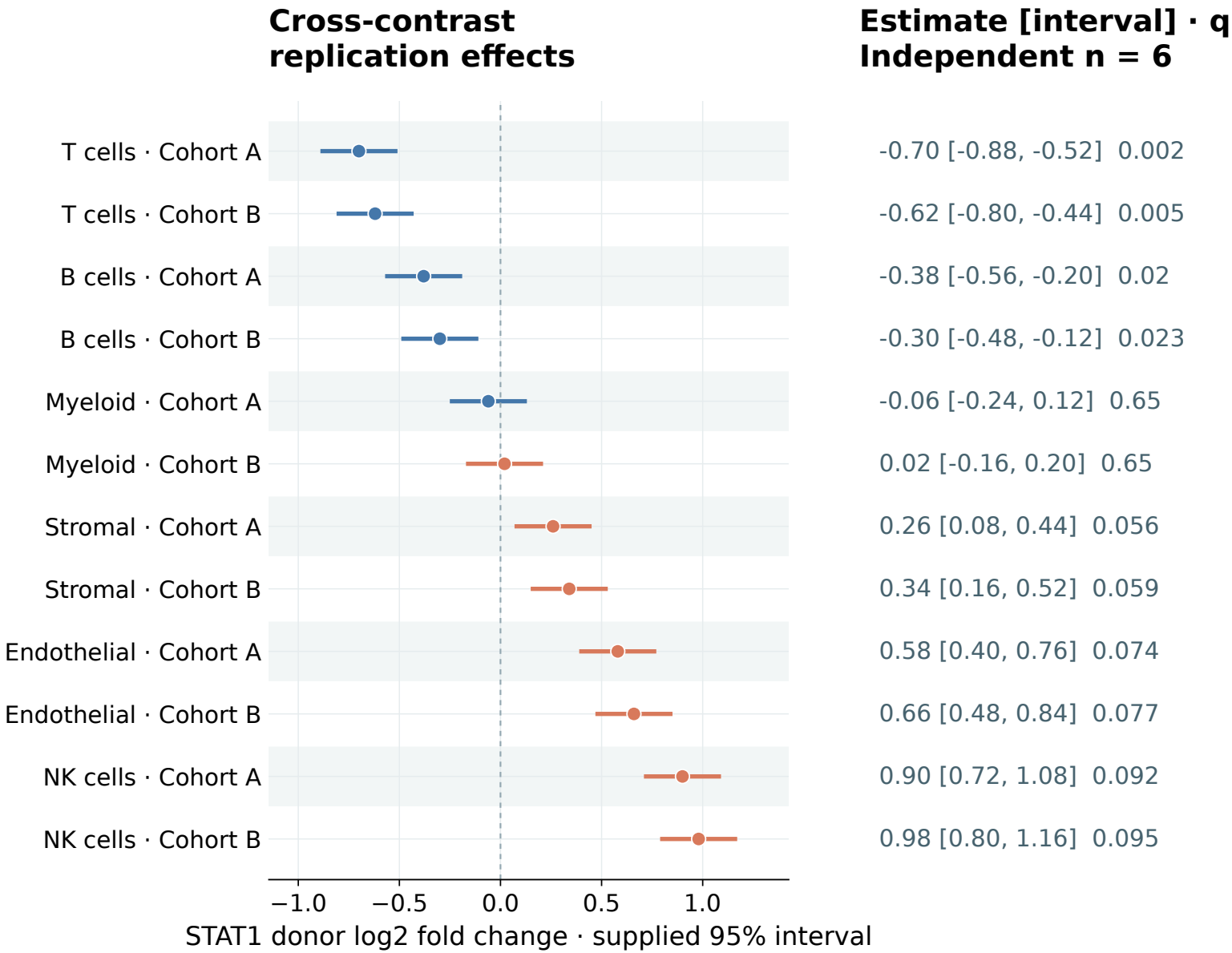
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092



Software test fixture; not biological evidence

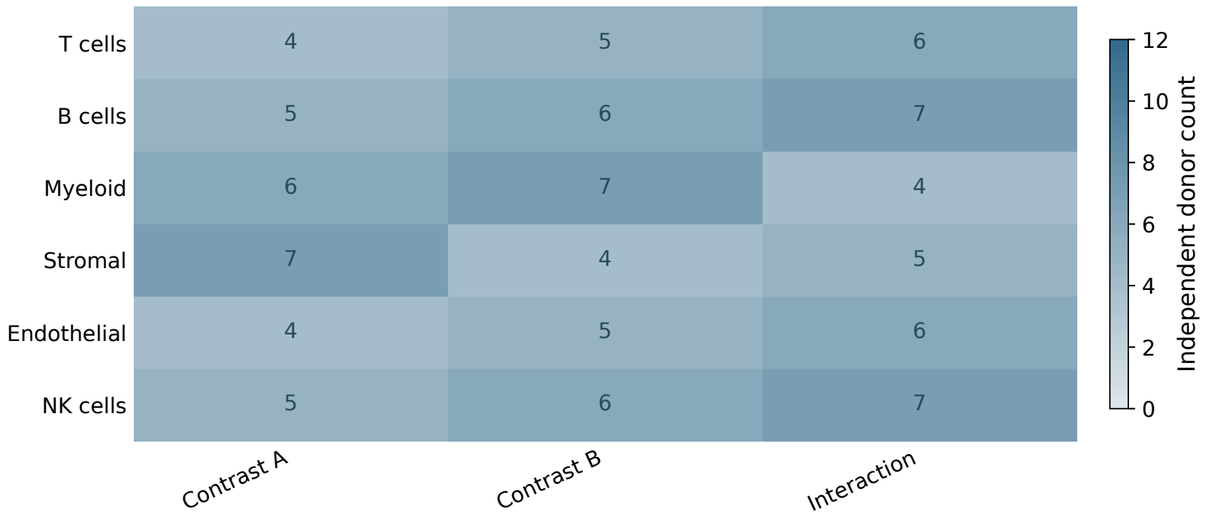


Software test fixture; not biological evidence

Single cell · multi_contrast — complementary evidence

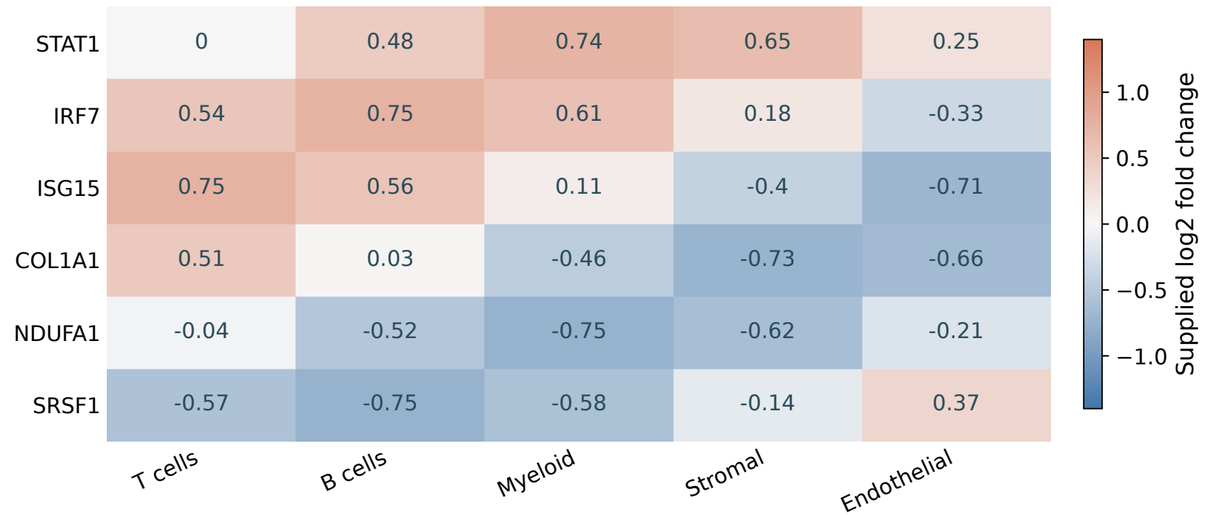
A

Independent support across contrast families



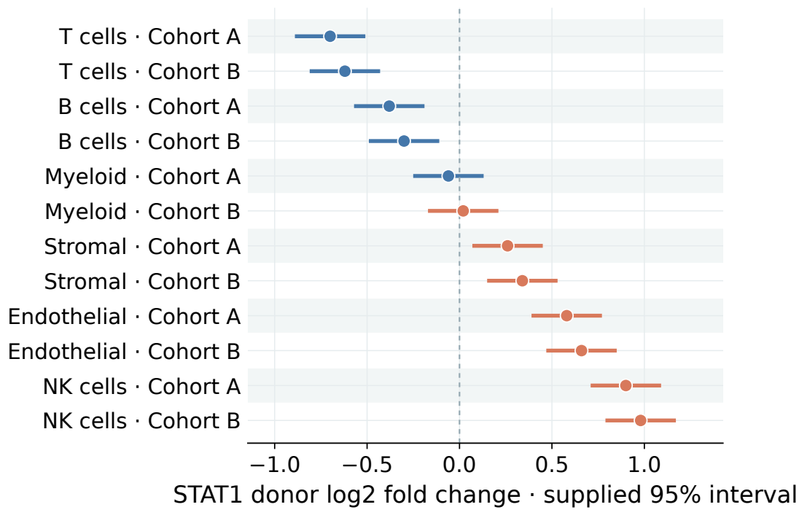
B

Complete signed effect atlas



C

Cross-contrast replication effects



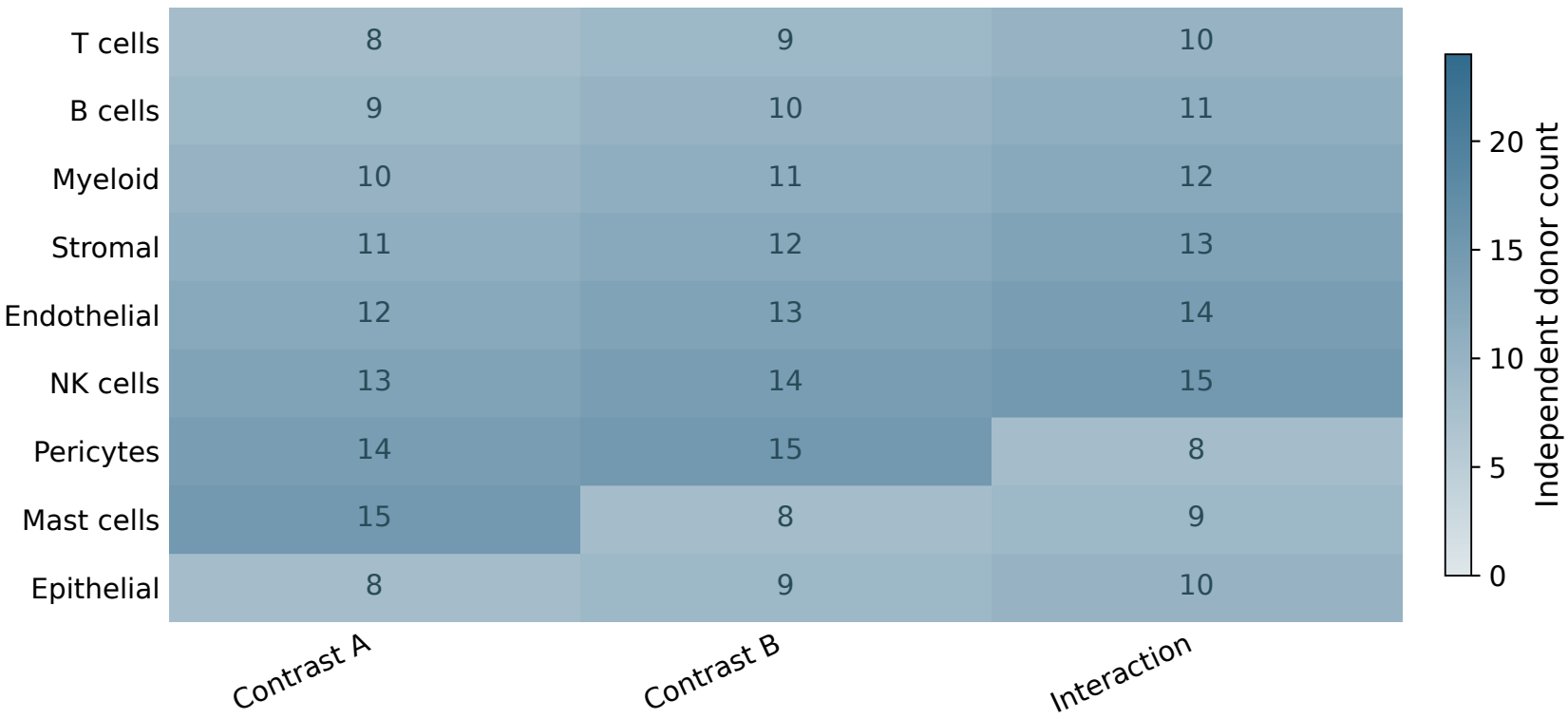
Estimate [interval] · q
Independent n = 6

T cells · Cohort A	-0.70 [-0.88, -0.52] 0.002
T cells · Cohort B	-0.62 [-0.80, -0.44] 0.005
B cells · Cohort A	-0.38 [-0.56, -0.20] 0.02
B cells · Cohort B	-0.30 [-0.48, -0.12] 0.023
Myeloid · Cohort A	-0.06 [-0.24, 0.12] 0.65
Myeloid · Cohort B	0.02 [-0.16, 0.20] 0.65
Stromal · Cohort A	0.26 [0.08, 0.44] 0.056
Stromal · Cohort B	0.34 [0.16, 0.52] 0.059
Endothelial · Cohort A	0.58 [0.40, 0.76] 0.074
Endothelial · Cohort B	0.66 [0.48, 0.84] 0.077
NK cells · Cohort A	0.90 [0.72, 1.08] 0.092
NK cells · Cohort B	0.98 [0.80, 1.16] 0.095

Single cell · multi_contrast — complementary evidence

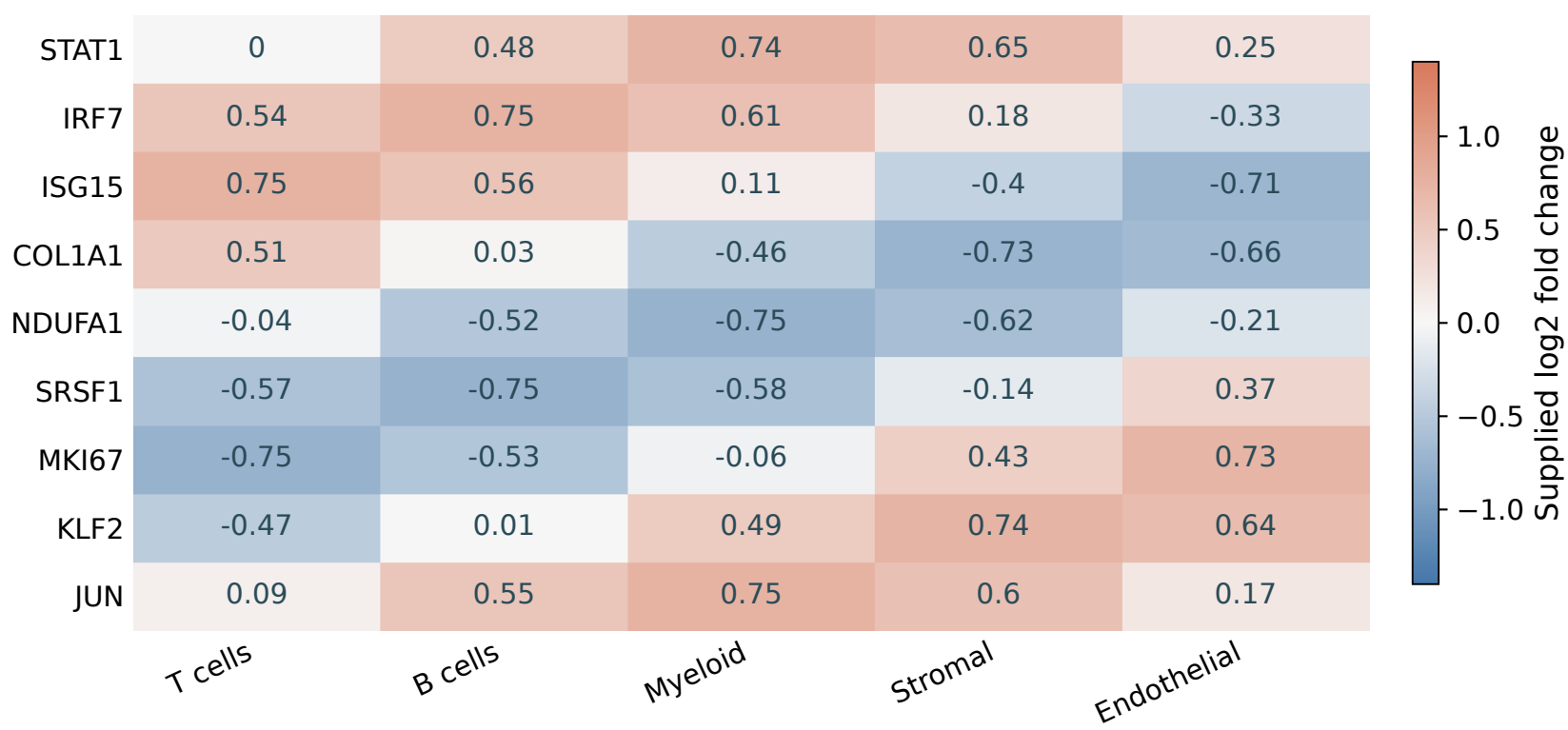
A

Independent support across contrast families



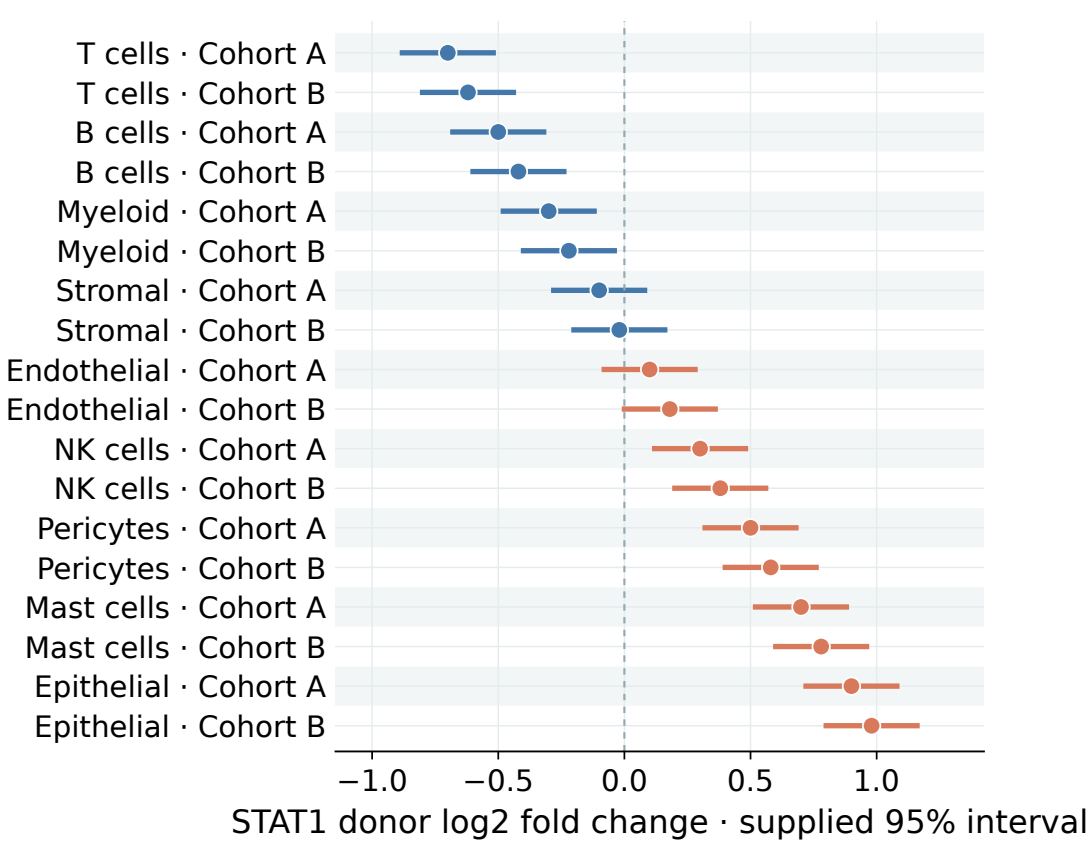
B

Complete signed effect atlas



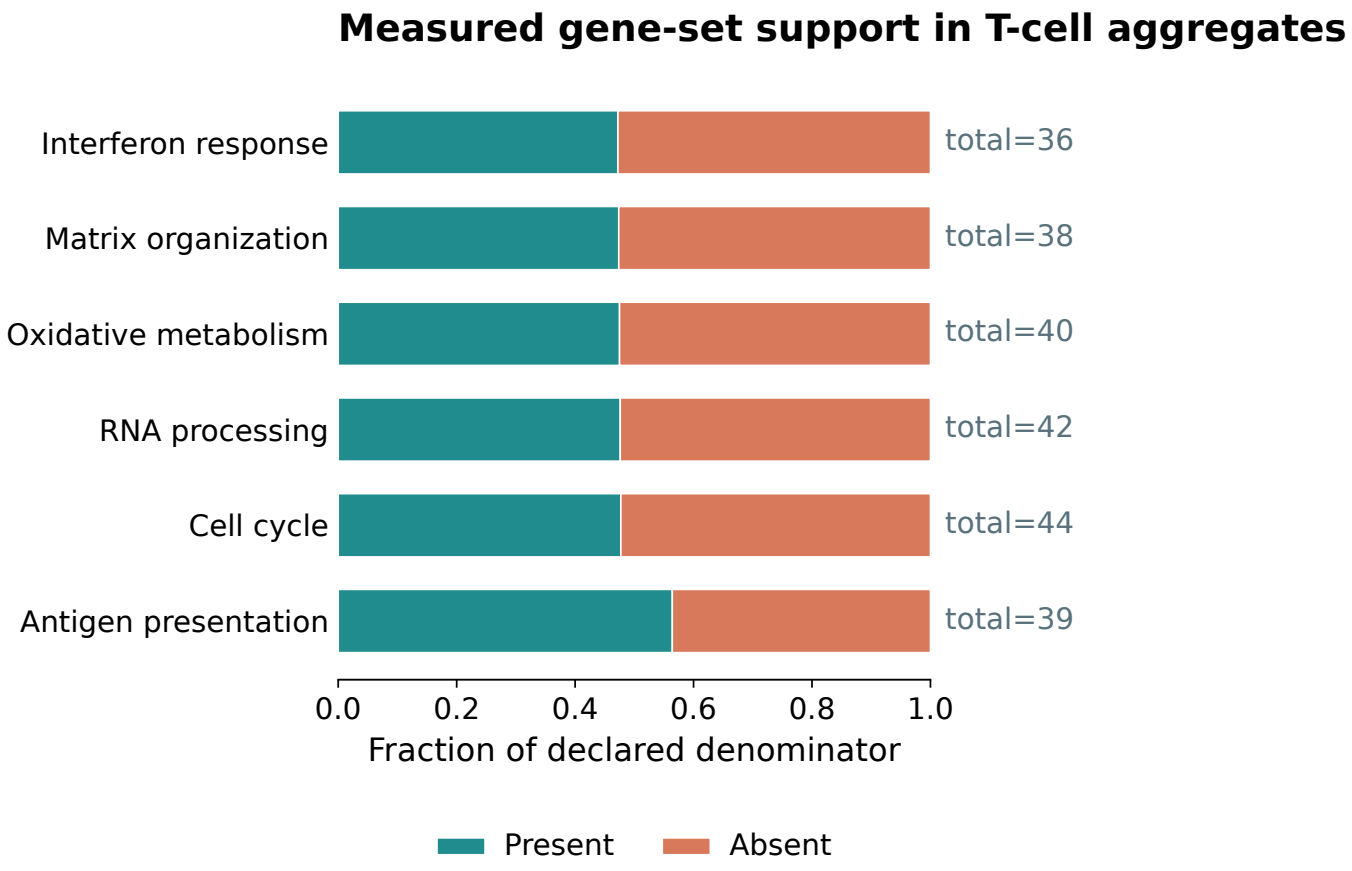
C

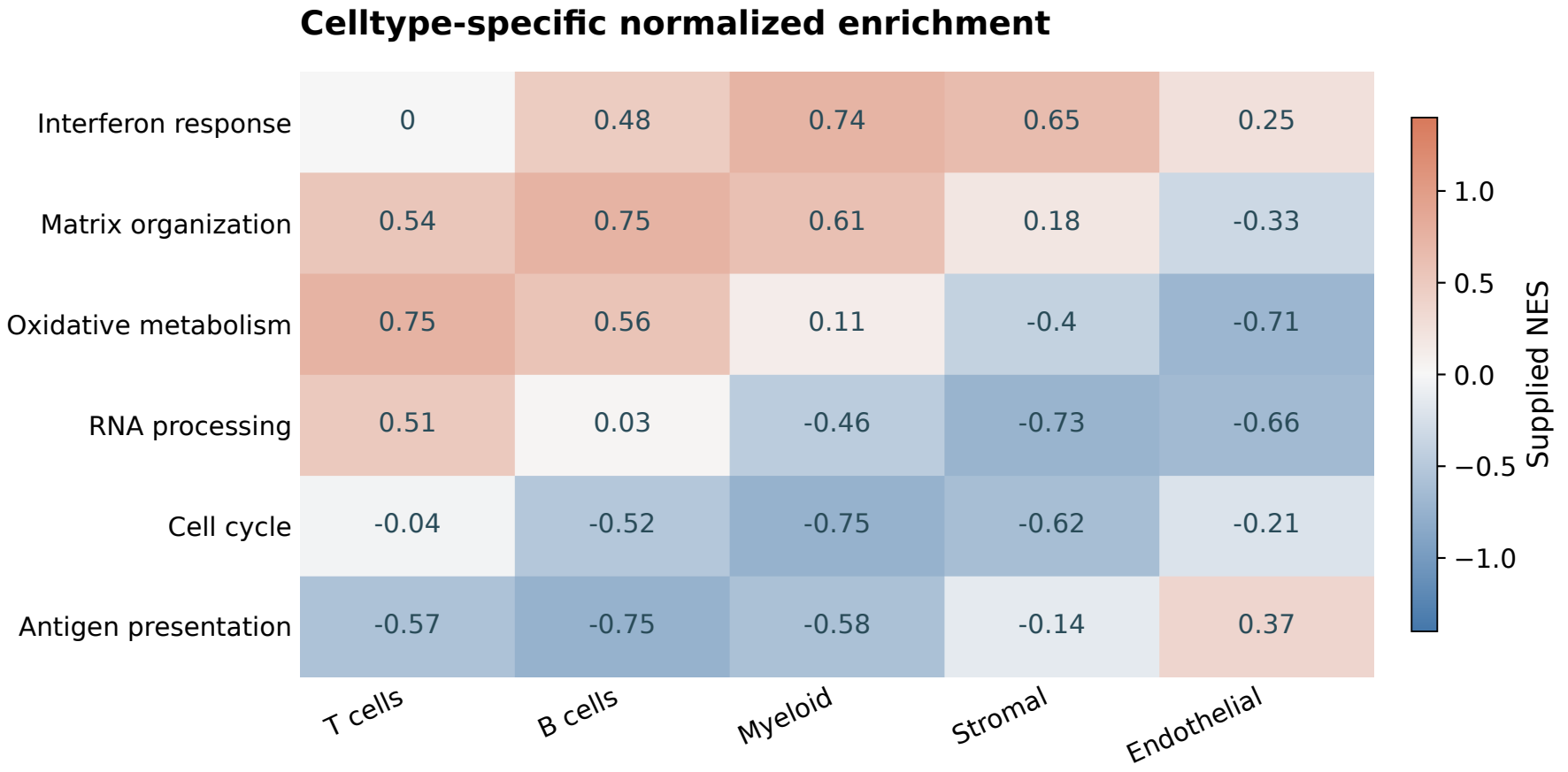
Cross-contrast replication effects

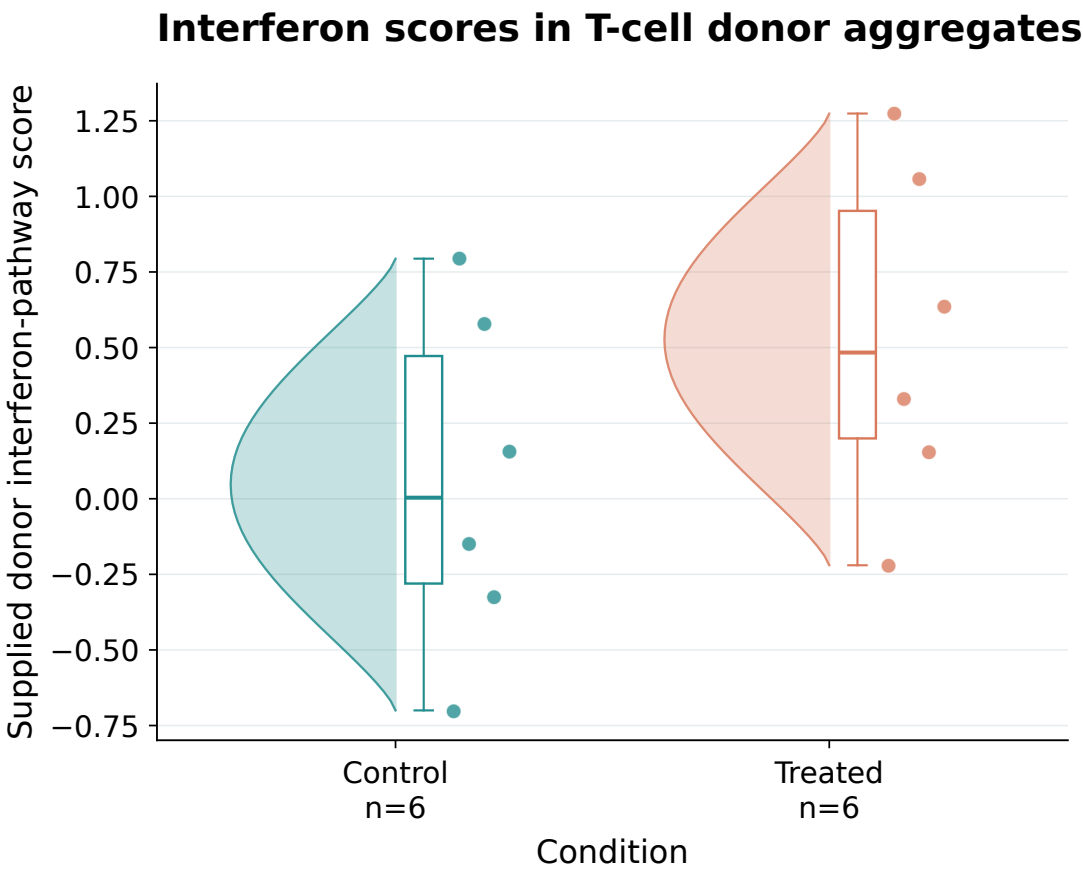


Estimate [interval] · q
Independent n = 12

T cells · Cohort A	-0.70 [-0.88, -0.52] 0.002
T cells · Cohort B	-0.62 [-0.80, -0.44] 0.005
B cells · Cohort A	-0.50 [-0.68, -0.32] 0.02
B cells · Cohort B	-0.42 [-0.60, -0.24] 0.023
Myeloid · Cohort A	-0.30 [-0.48, -0.12] 0.038
Myeloid · Cohort B	-0.22 [-0.40, -0.04] 0.041
Stromal · Cohort A	-0.10 [-0.28, 0.08] 0.65
Stromal · Cohort B	-0.02 [-0.20, 0.16] 0.65
Endothelial · Cohort A	0.10 [-0.08, 0.28] 0.65
Endothelial · Cohort B	0.18 [0.00, 0.36] 0.077
NK cells · Cohort A	0.30 [0.12, 0.48] 0.092
NK cells · Cohort B	0.38 [0.20, 0.56] 0.095
Pericytes · Cohort A	0.50 [0.32, 0.68] 0.11
Pericytes · Cohort B	0.58 [0.40, 0.76] 0.11
Mast cells · Cohort A	0.70 [0.52, 0.88] 0.13
Mast cells · Cohort B	0.78 [0.60, 0.96] 0.13
Epithelial · Cohort A	0.90 [0.72, 1.08] 0.15
Epithelial · Cohort B	0.98 [0.80, 1.16] 0.15

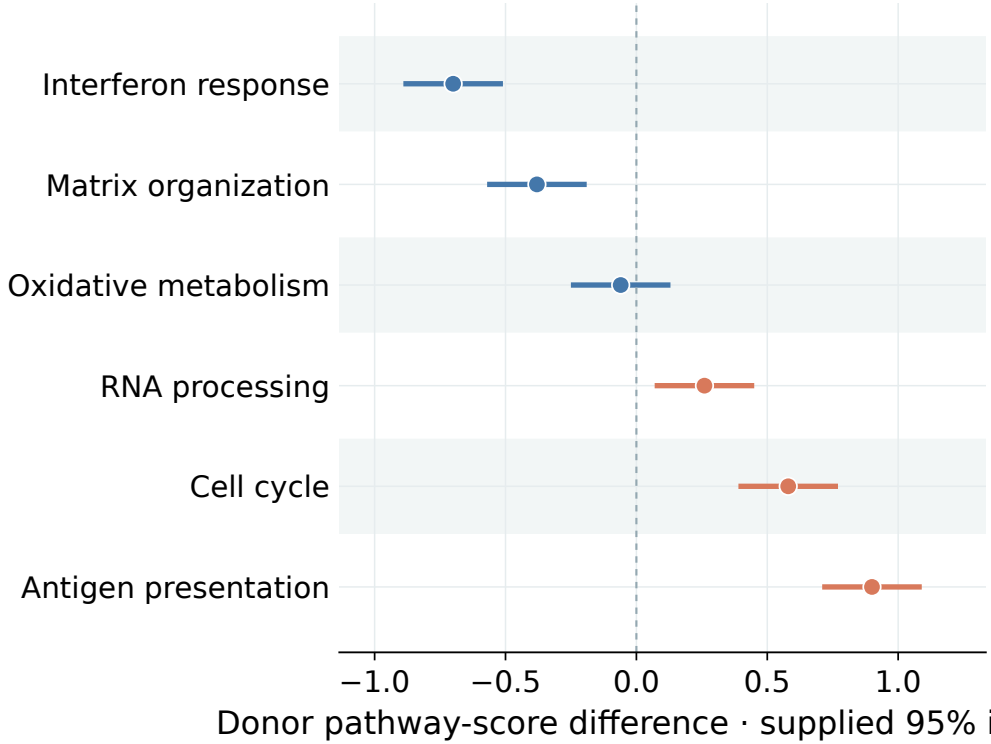

Measured gene-set support in T-cell aggregates
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Present
Absent


Celltype-specific normalized enrichment
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
0
0.48
0.74
0.65
0.25
0.54
0.75
0.61
0.18
-0.33
0.75
0.56
0.11
-0.4
-0.71
0.51
0.03
-0.46
-0.73
-0.66
-0.04
-0.52
-0.75
-0.62
-0.21
-0.57
-0.75
-0.58
-0.14
0.37
T cells
B cells
Myeloid
Stromal
Endothelial
1.0
0.5
0.0
−0.5
−1.0
Supplied NES



Software test fixture; not biological evidence

Donor-level pathway responses



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

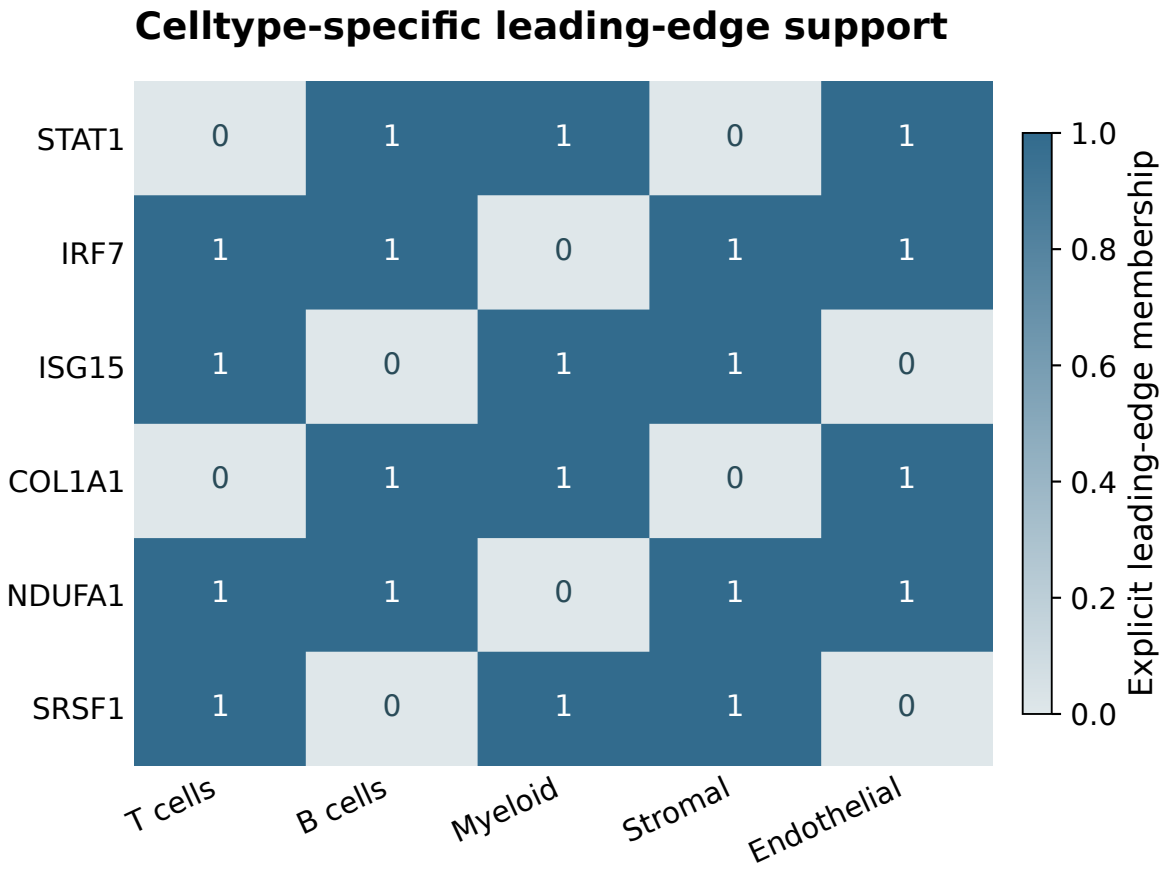
-0.38 [-0.56, -0.20] 0.02

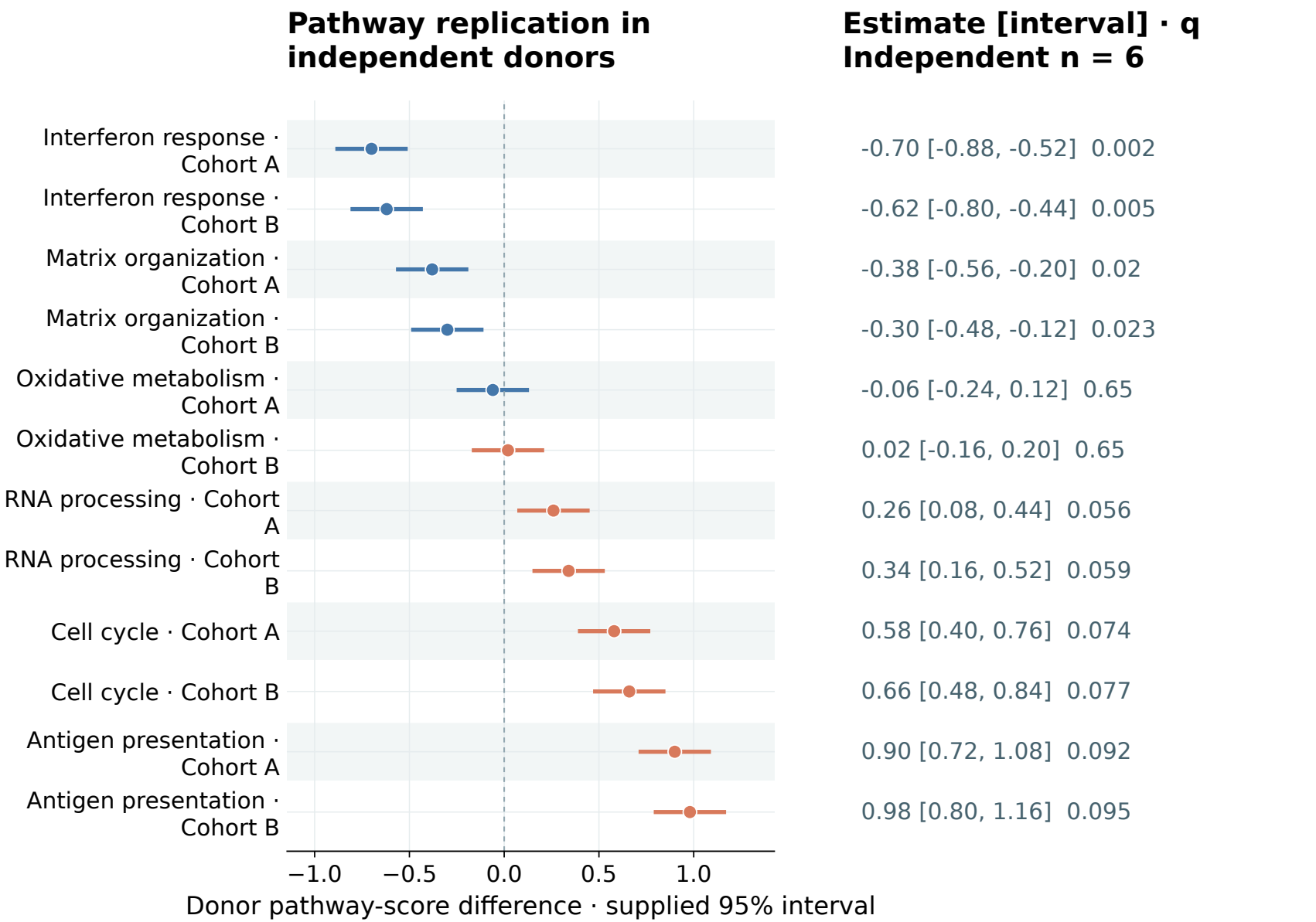
-0.06 [-0.24, 0.12] 0.65

0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092

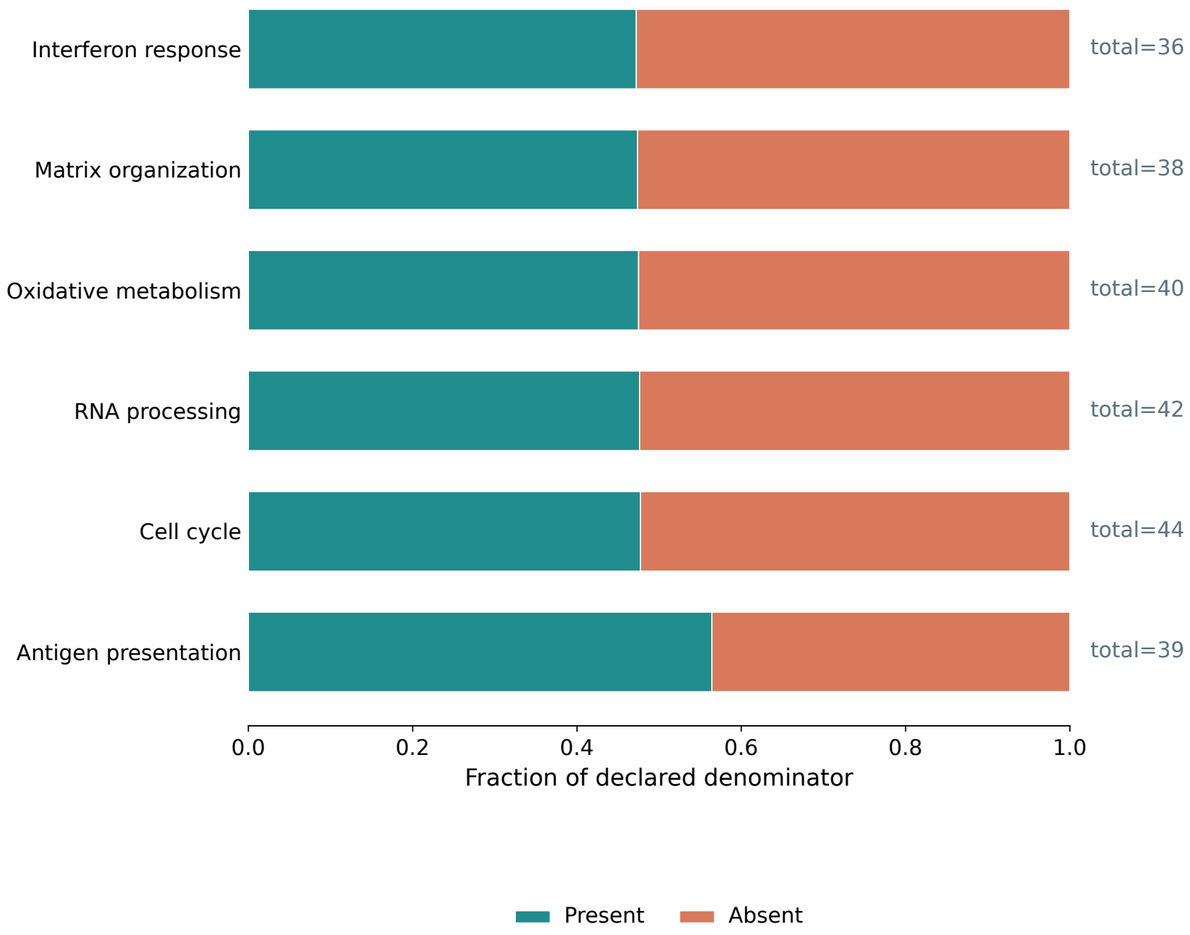

Celltype-specific leading-edge support
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0 1 1 0 1
1 1 0 1 1
1 0 1 1 0
0 1 1 0 1
1 1 0 1 1
1 0 1 1 0
T cells
B cells
Myeloid
Stromal
Endothelial
Explicit leading-edge membership
1.0
0.8
0.6
0.4
0.2
0.0


Pathway replication in independent donors
Estimate [interval] · q
Independent n = 6
Interferon response · Cohort A
Interferon response · Cohort B
Matrix organization · Cohort A
Matrix organization · Cohort B
Oxidative metabolism · Cohort A
Oxidative metabolism · Cohort B
RNA processing · Cohort A
RNA processing · Cohort B
Cell cycle · Cohort A
Cell cycle · Cohort B
Antigen presentation · Cohort A
Antigen presentation · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Donor pathway-score difference · supplied 95% interval

Single cell · pathways — complementary evidence

A

Measured gene-set support in T-cell aggregates

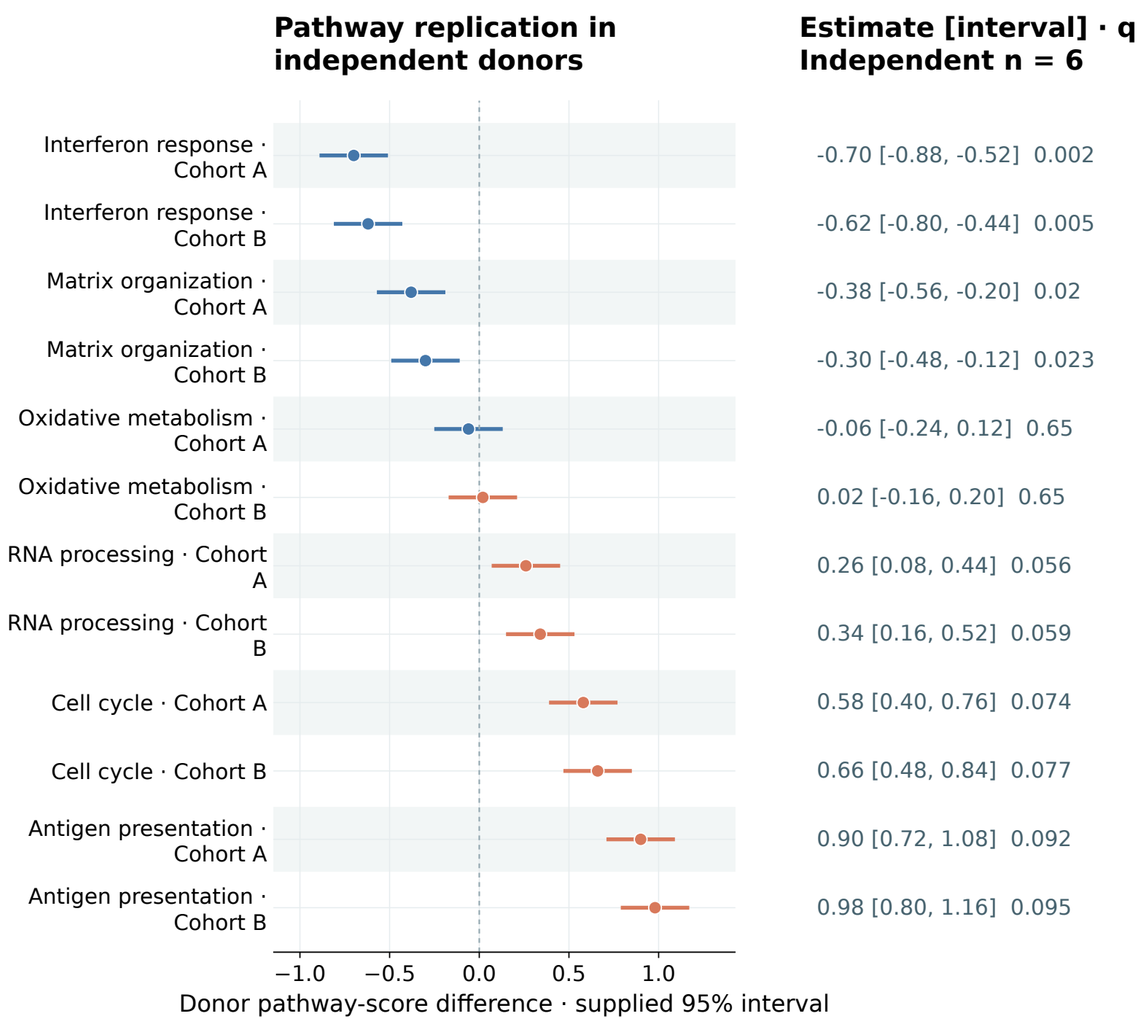


B

Celltype-specific normalized enrichment



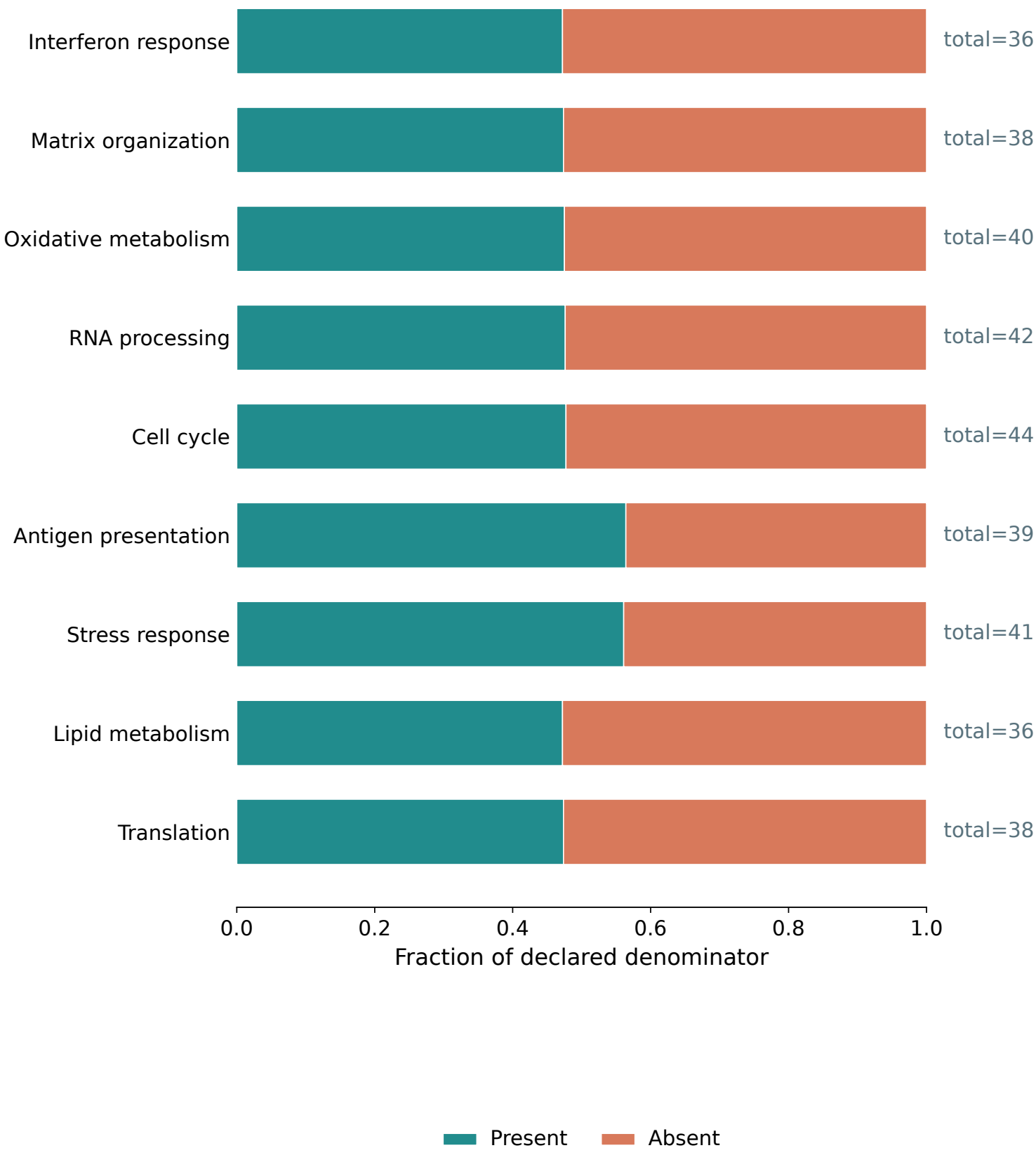
C



Single cell · pathways — complementary evidence

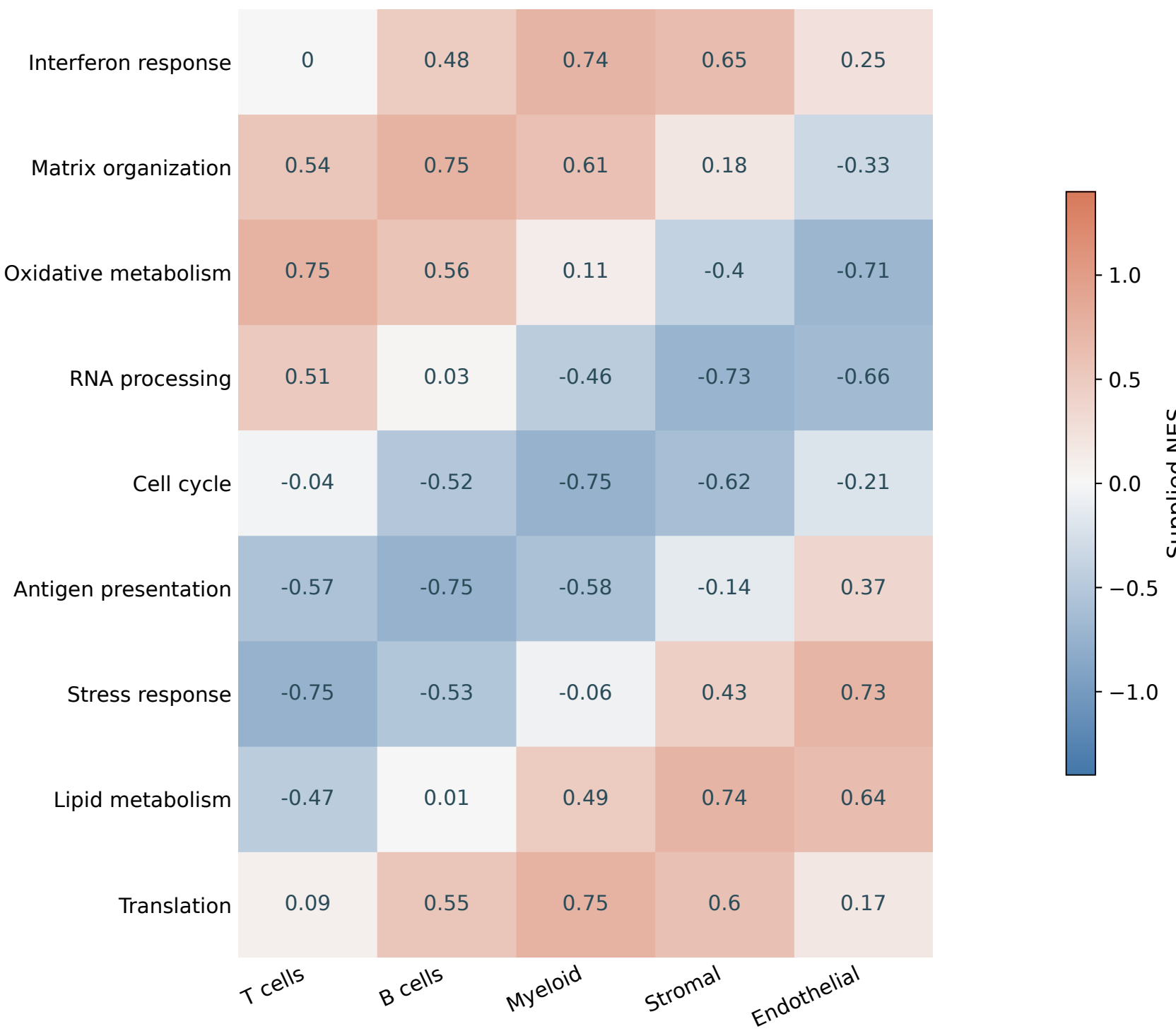
A

Measured gene-set support in T-cell aggregates

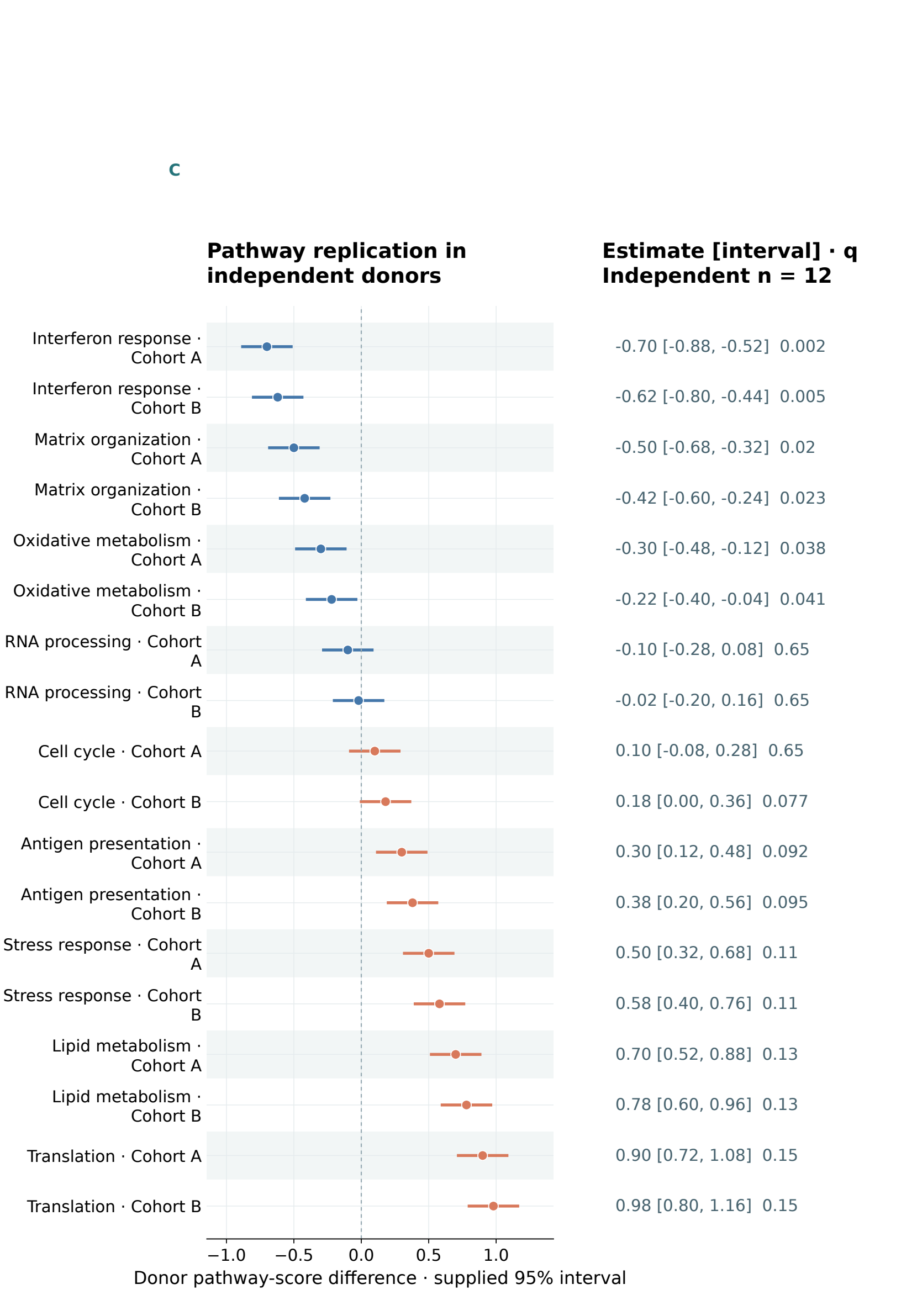


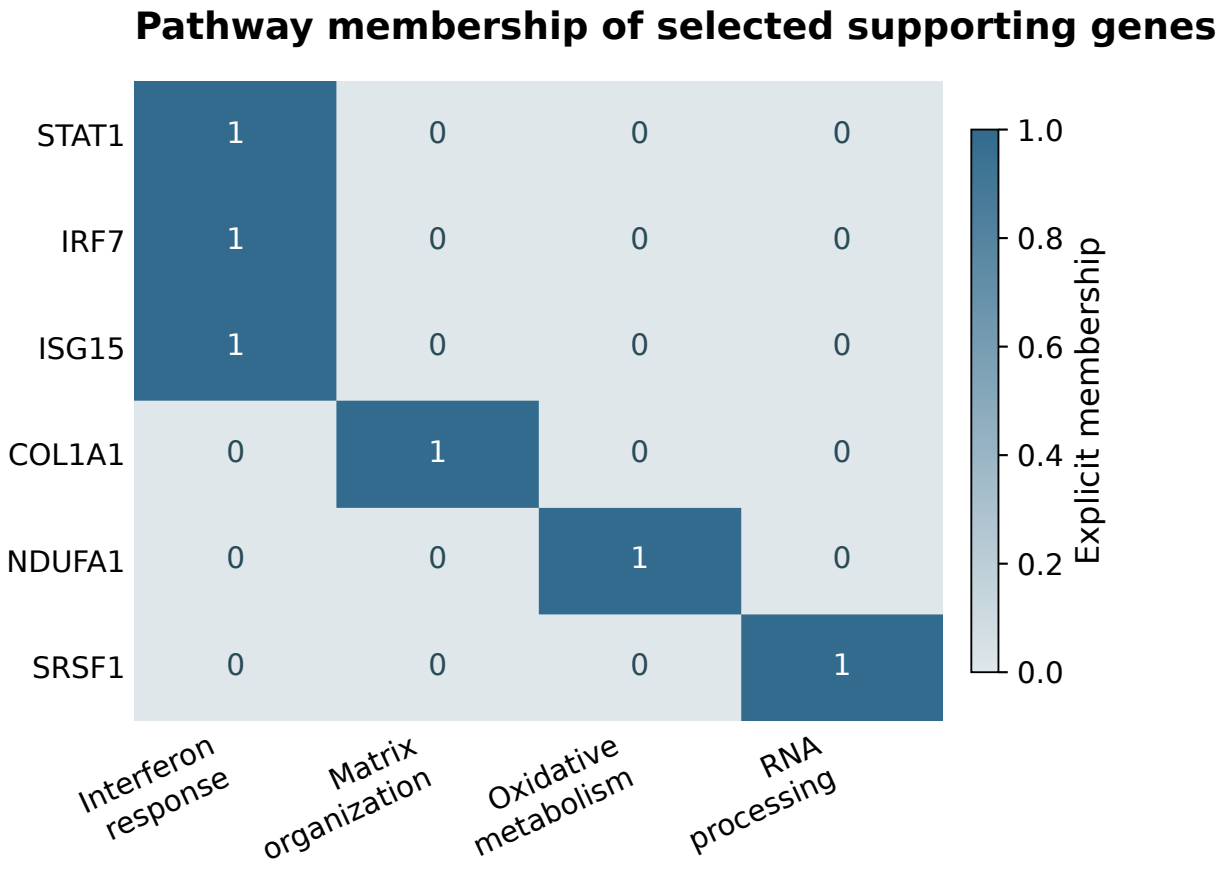
B

Celltype-specific normalized enrichment

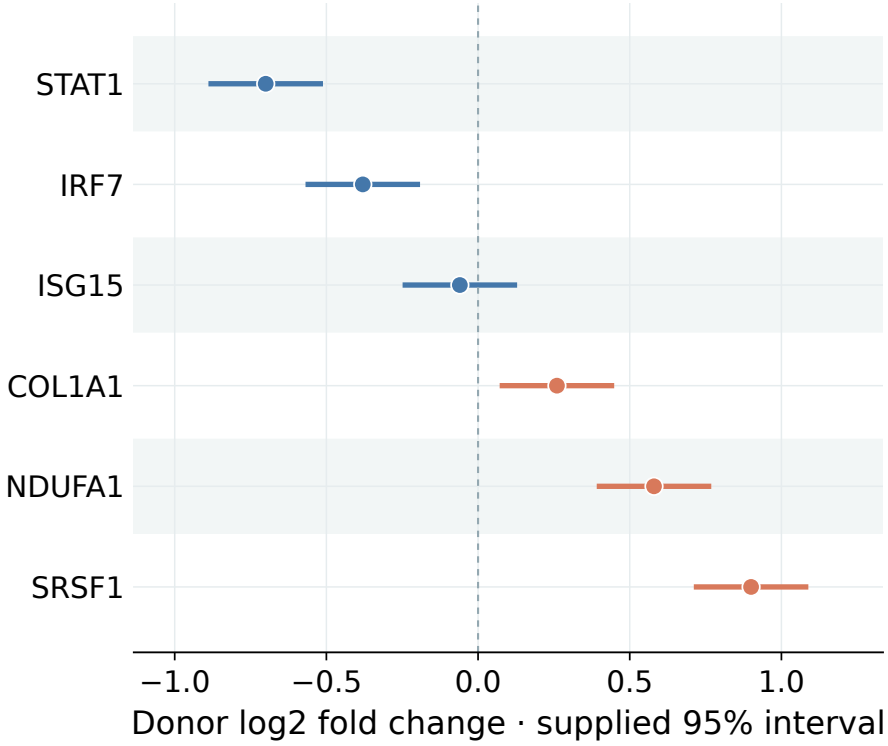


C




Pathway membership of selected supporting genes
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
1
0
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Explicit membership
1.0
0.8
0.6
0.4
0.2
0.0

Supporting-gene donor-level effects



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

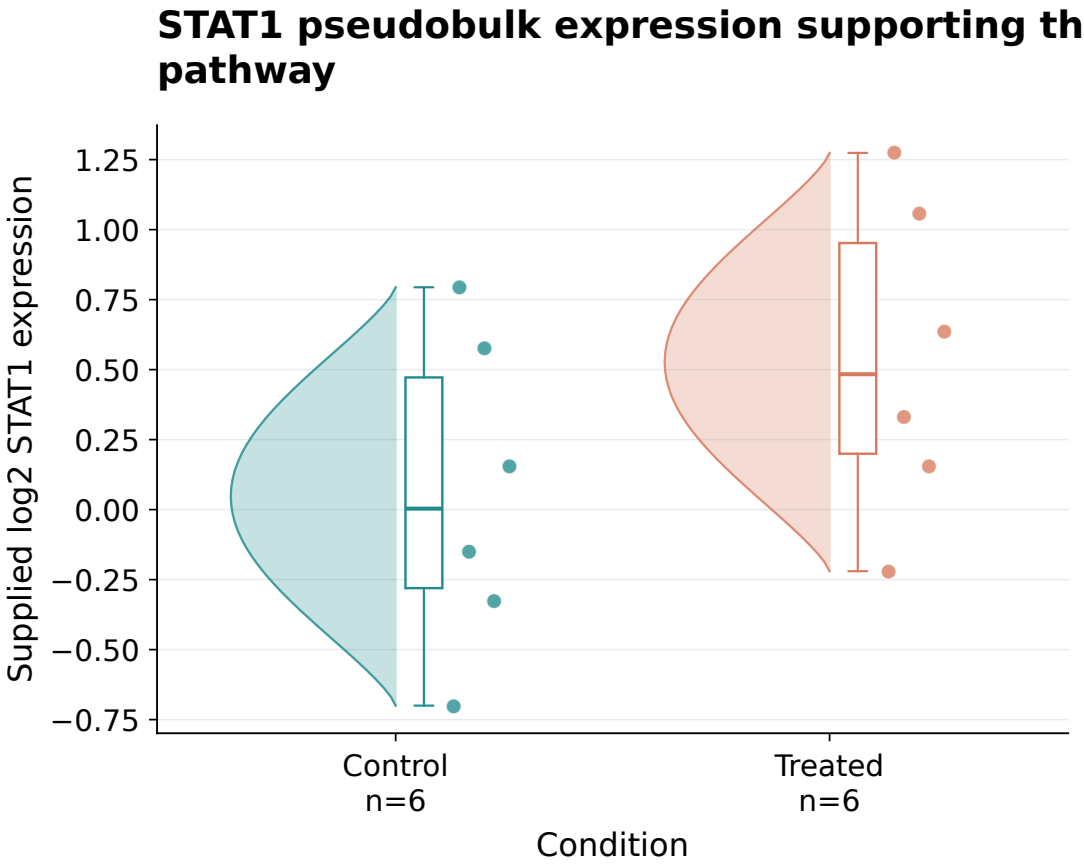
-0.38 [-0.56, -0.20] 0.02

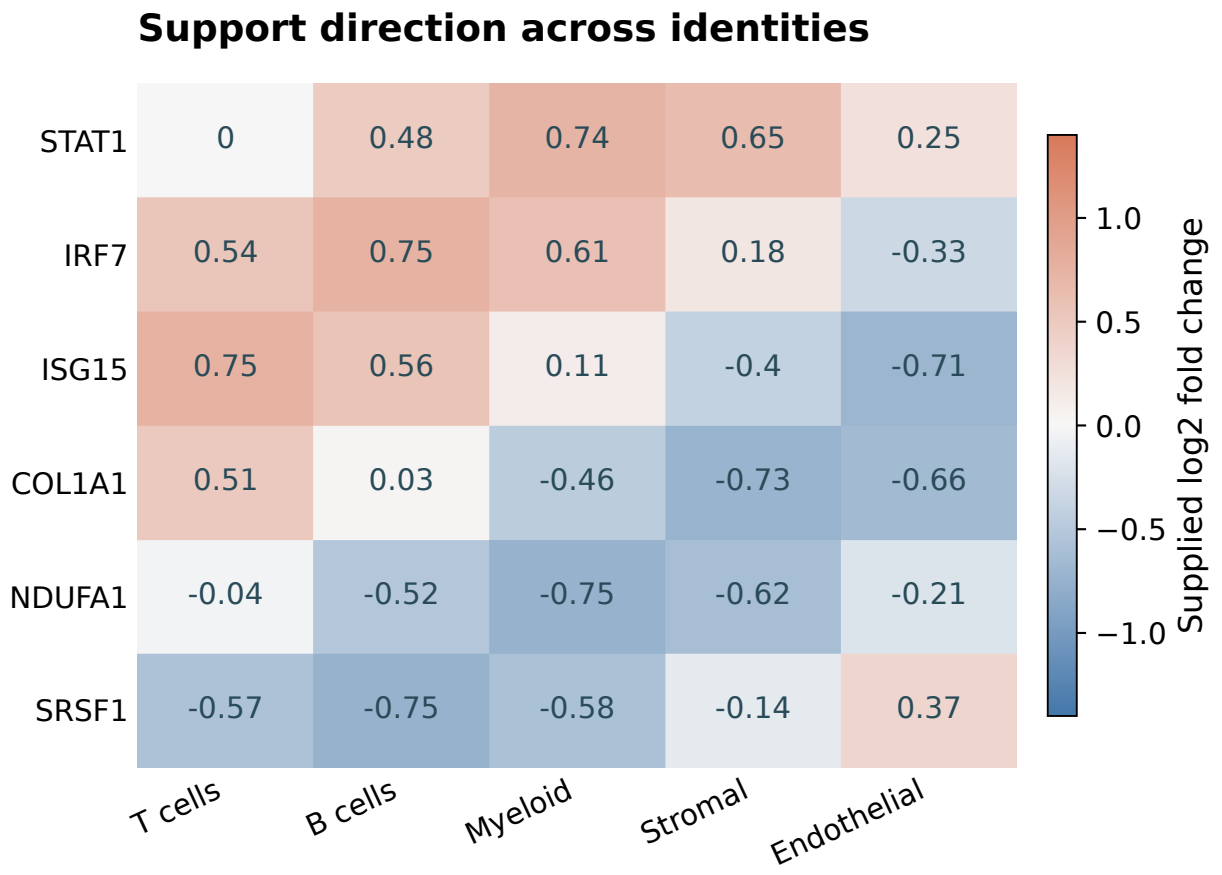
-0.06 [-0.24, 0.12] 0.65

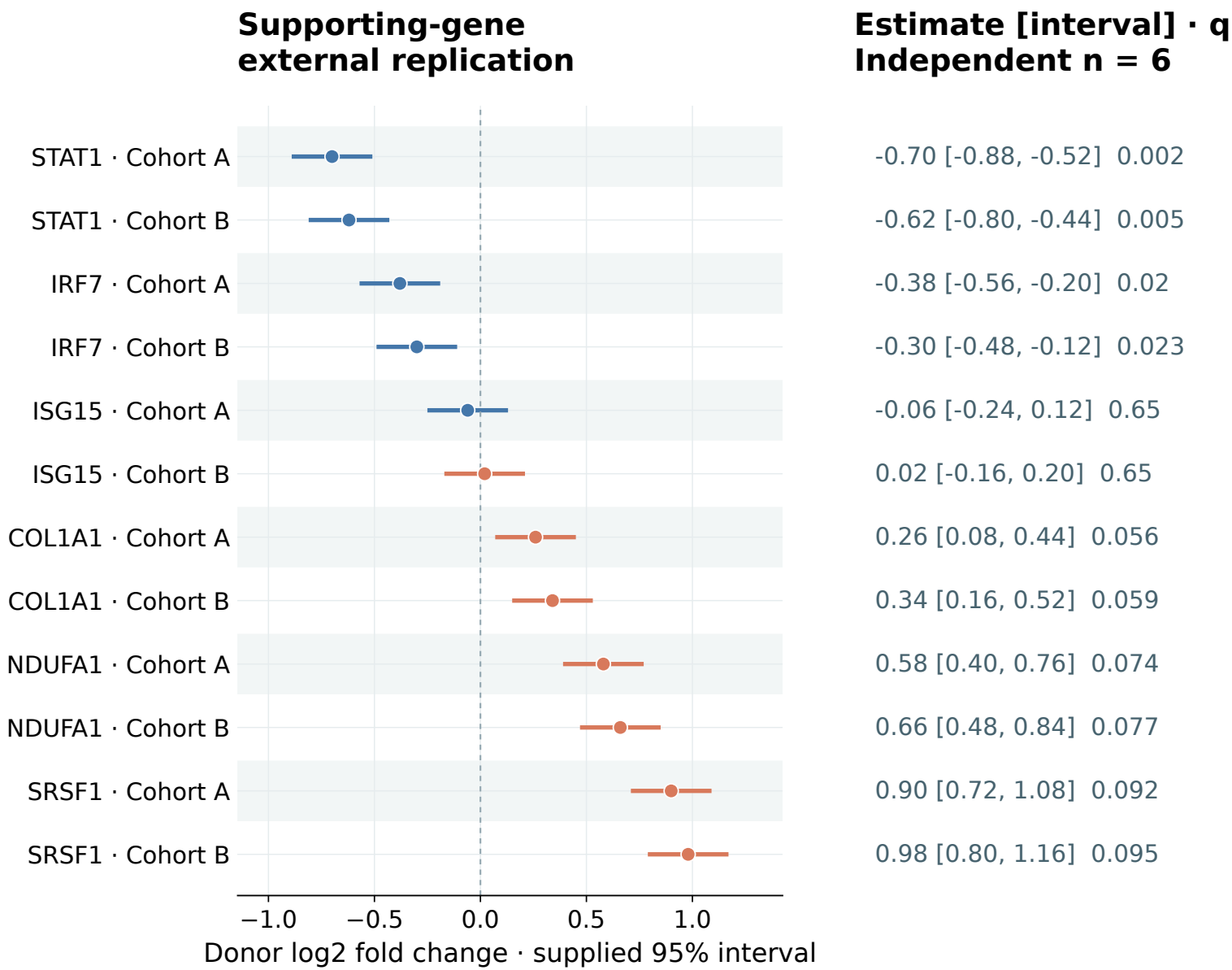
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092


STAT1 pseudobulk expression supporting the pathway
Supplied log2 STAT1 expression
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Control
n=6
Treated
n=6
Condition

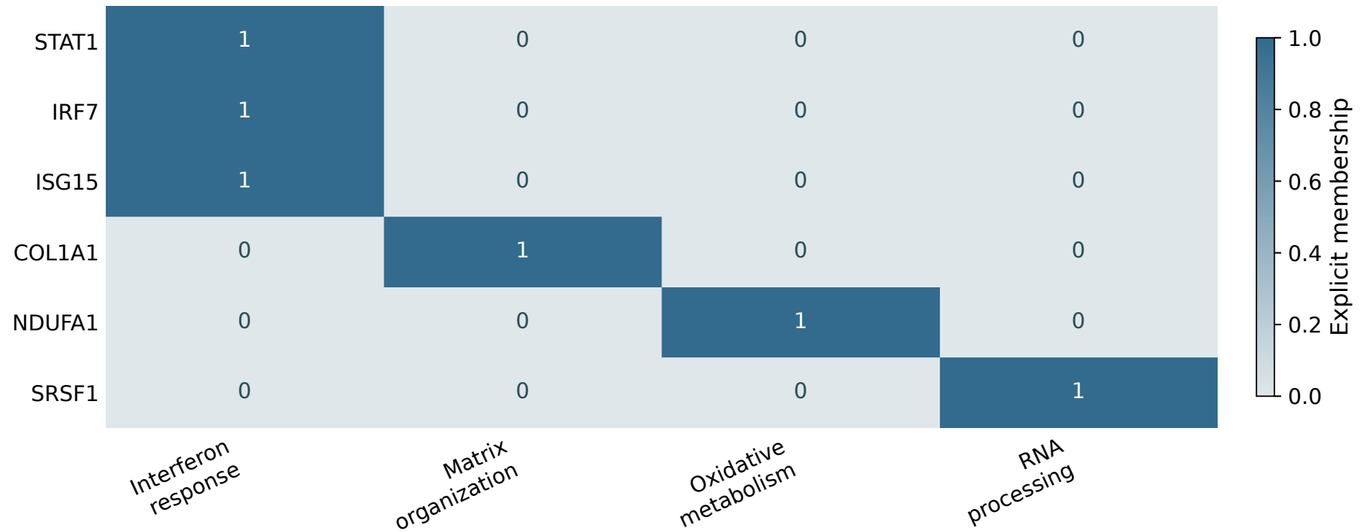

Support direction across identities
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0
0.48
0.74
0.65
0.25
0.54
0.75
0.61
0.18
-0.33
0.75
0.56
0.11
-0.4
-0.71
0.51
0.03
-0.46
-0.73
-0.66
-0.04
-0.52
-0.75
-0.62
-0.21
-0.57
-0.75
-0.58
-0.14
0.37
T cells
B cells
Myeloid
Stromal
Endothelial
1.0
0.5
0.0
−0.5
−1.0
Supplied log2 fold change


Supporting-gene external replication
Estimate [interval] · q
Independent n = 6
STAT1 · Cohort A
STAT1 · Cohort B
IRF7 · Cohort A
IRF7 · Cohort B
ISG15 · Cohort A
ISG15 · Cohort B
COL1A1 · Cohort A
COL1A1 · Cohort B
NDUFA1 · Cohort A
NDUFA1 · Cohort B
SRSF1 · Cohort A
SRSF1 · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Donor log2 fold change · supplied 95% interval

Single cell · gene_support — complementary evidence

A

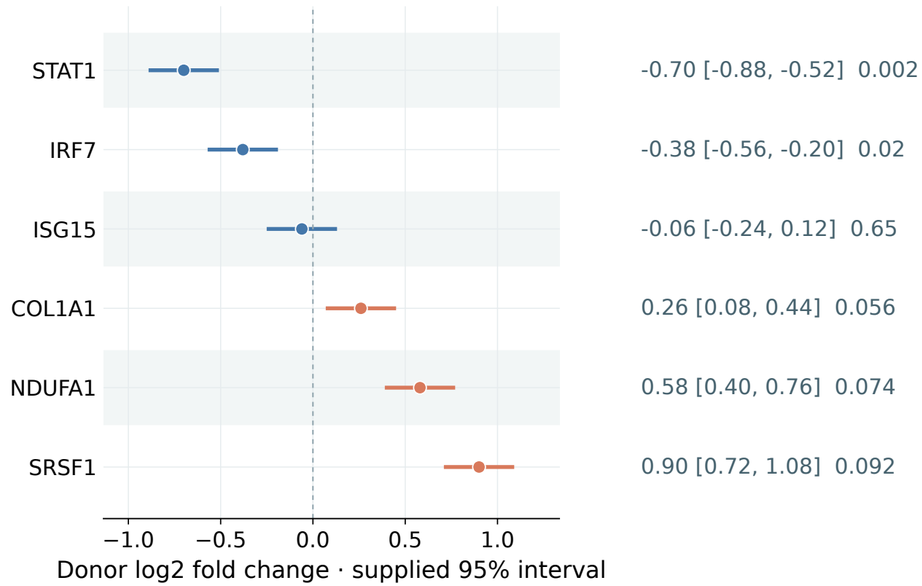
Pathway membership of selected supporting genes



B

Supporting-gene donor-level effects

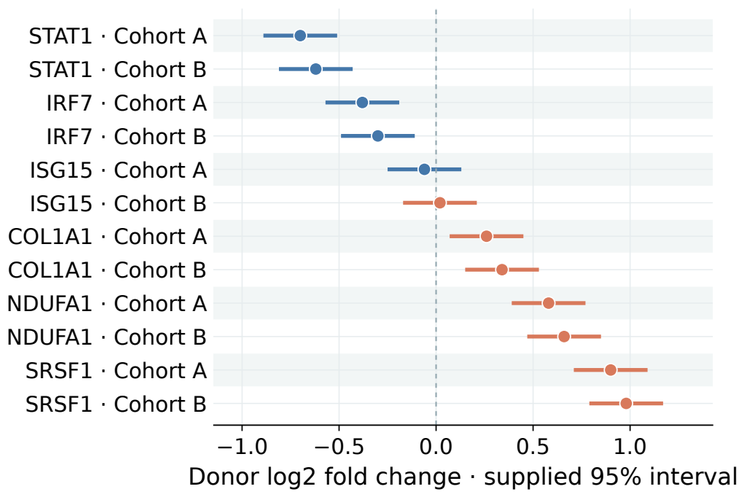
Estimate [interval] · q
Independent n = 12



C

Supporting-gene external replication

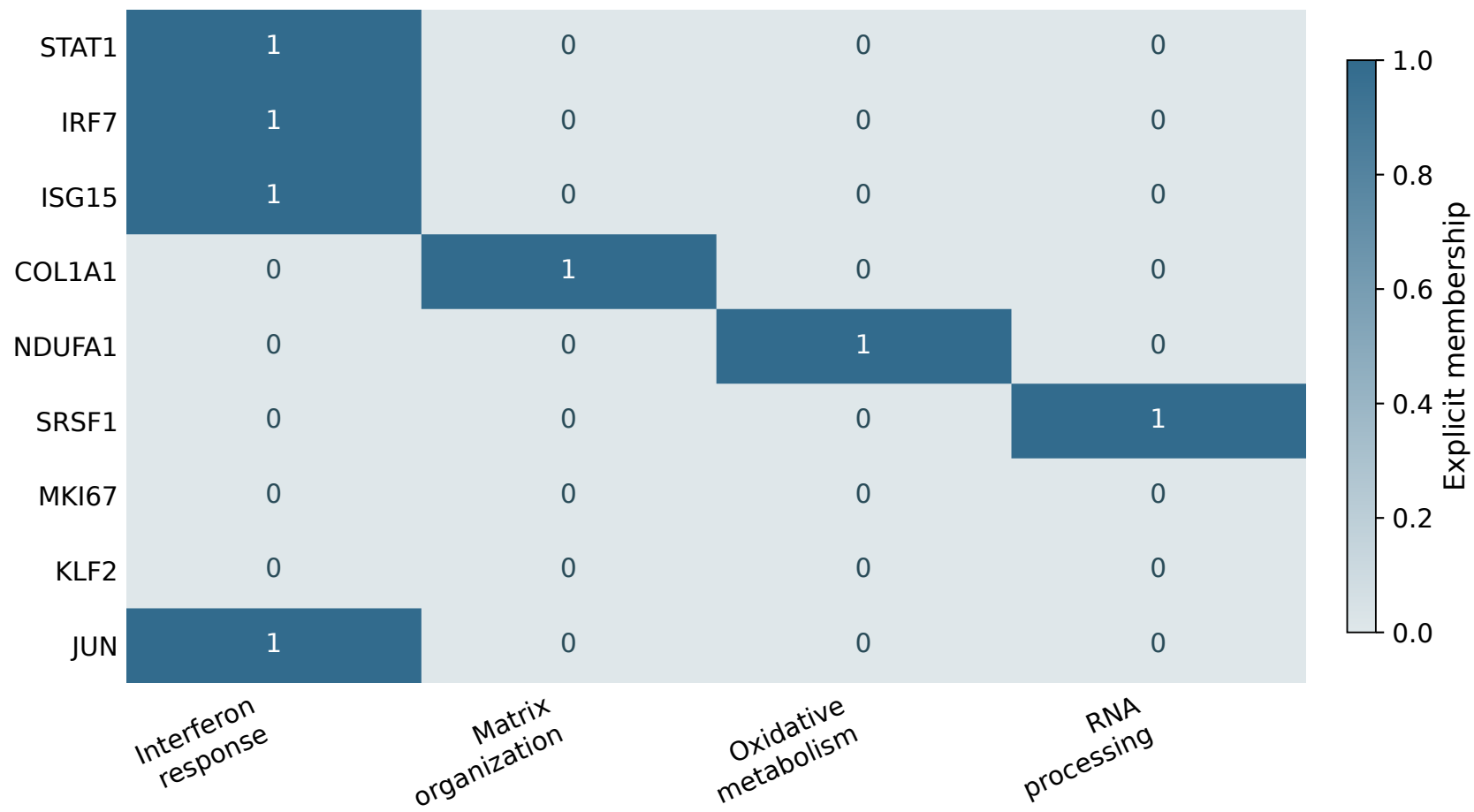
Estimate [interval] · q
Independent n = 6



Single cell · gene_support — complementary evidence

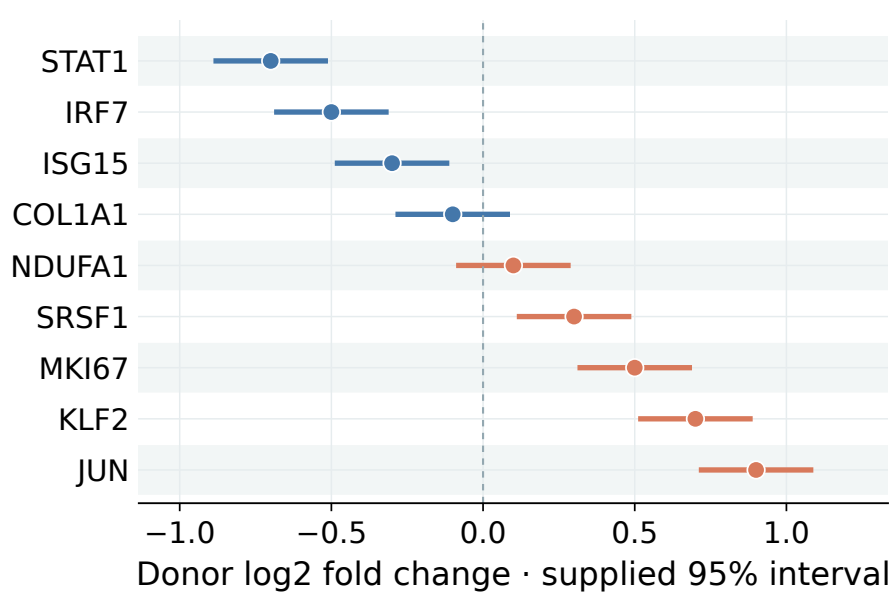
A

Pathway membership of selected supporting genes



B

Supporting-gene donor-level effects

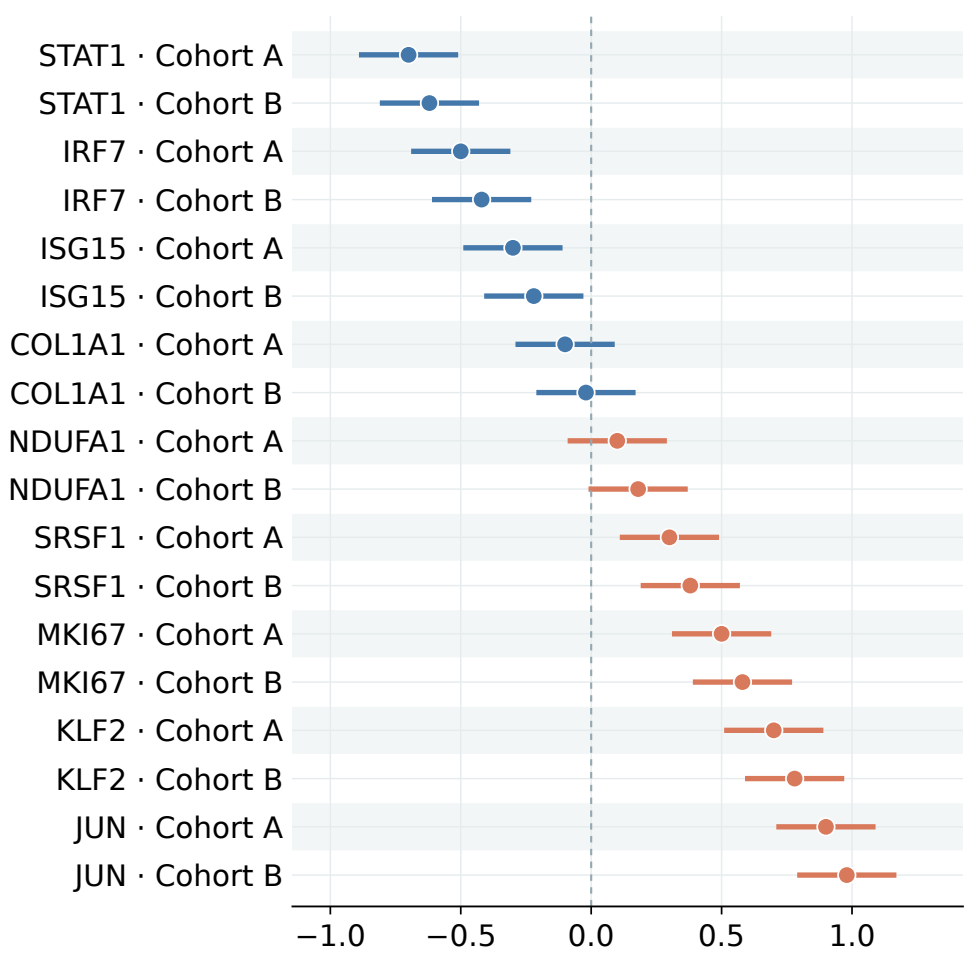


Estimate [interval] · q
Independent n = 24

-0.70	[-0.88, -0.52]	0.002
-0.50	[-0.68, -0.32]	0.02
-0.30	[-0.48, -0.12]	0.038
-0.10	[-0.28, 0.08]	0.65
0.10	[-0.08, 0.28]	0.65
0.30	[0.12, 0.48]	0.092
0.50	[0.32, 0.68]	0.11
0.70	[0.52, 0.88]	0.13
0.90	[0.72, 1.08]	0.15

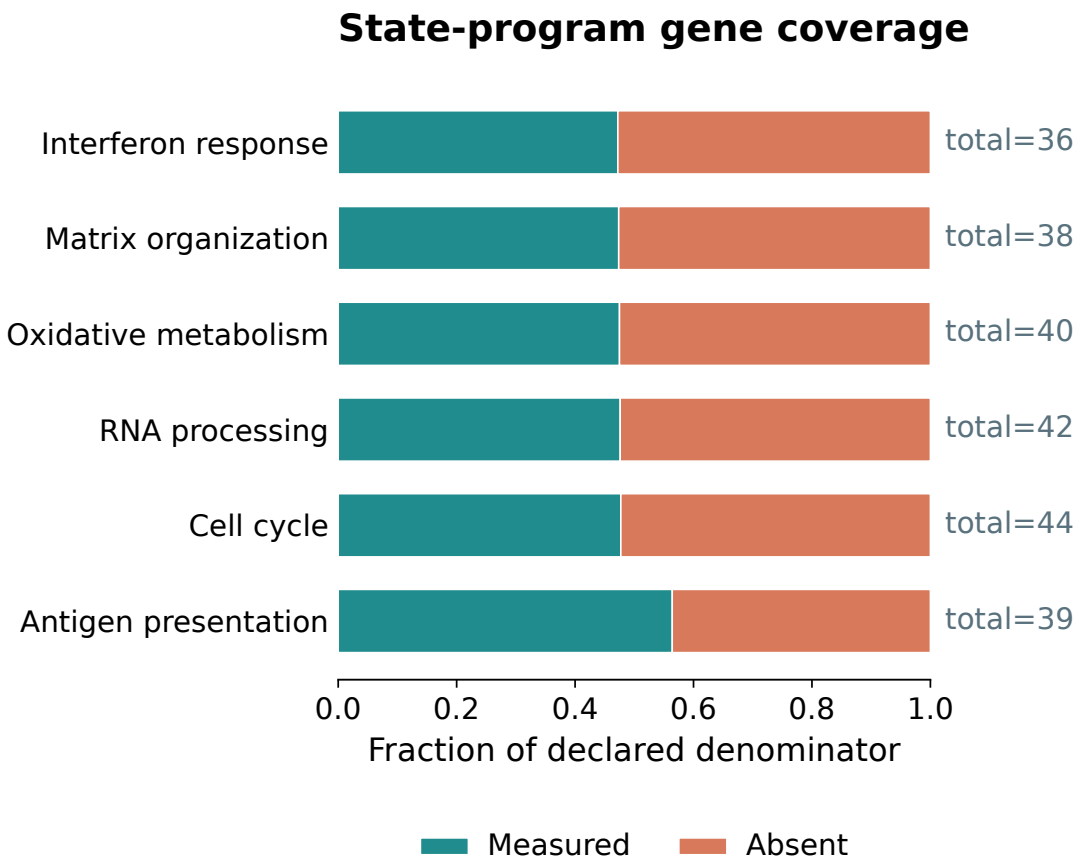
C

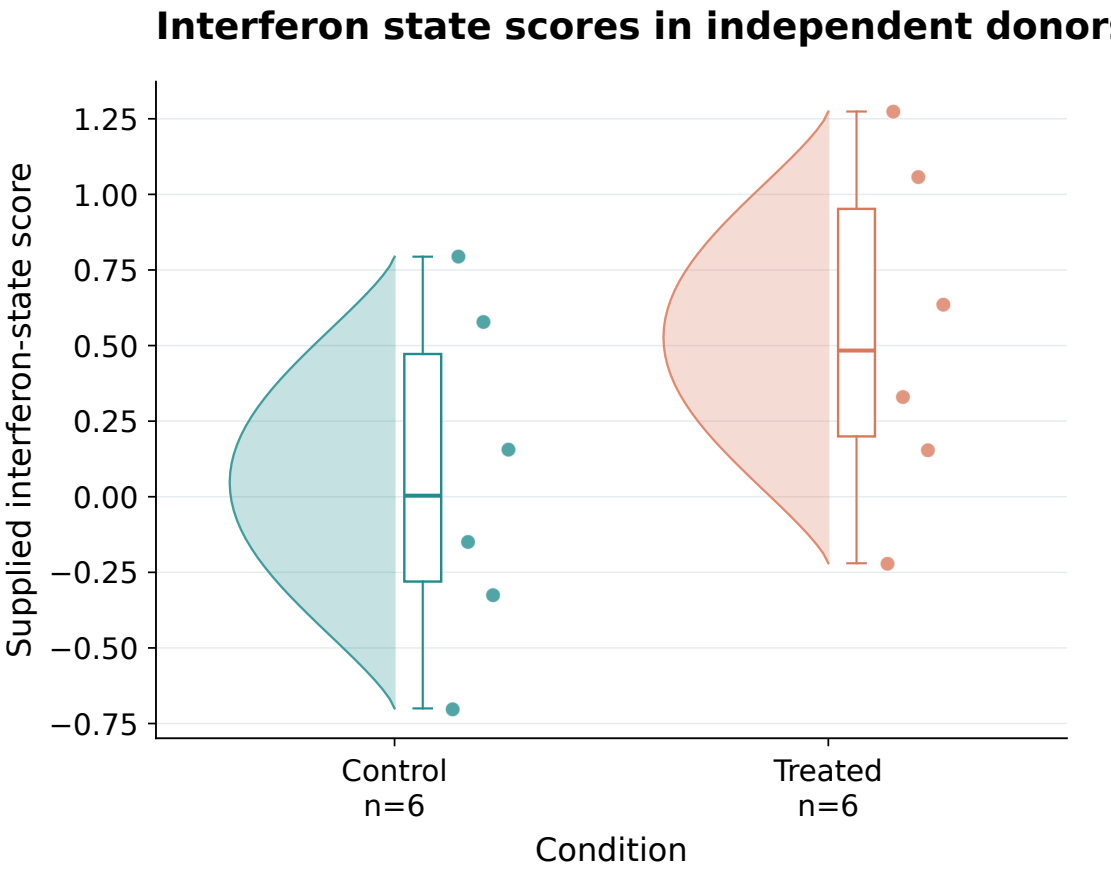
Supporting-gene external replication



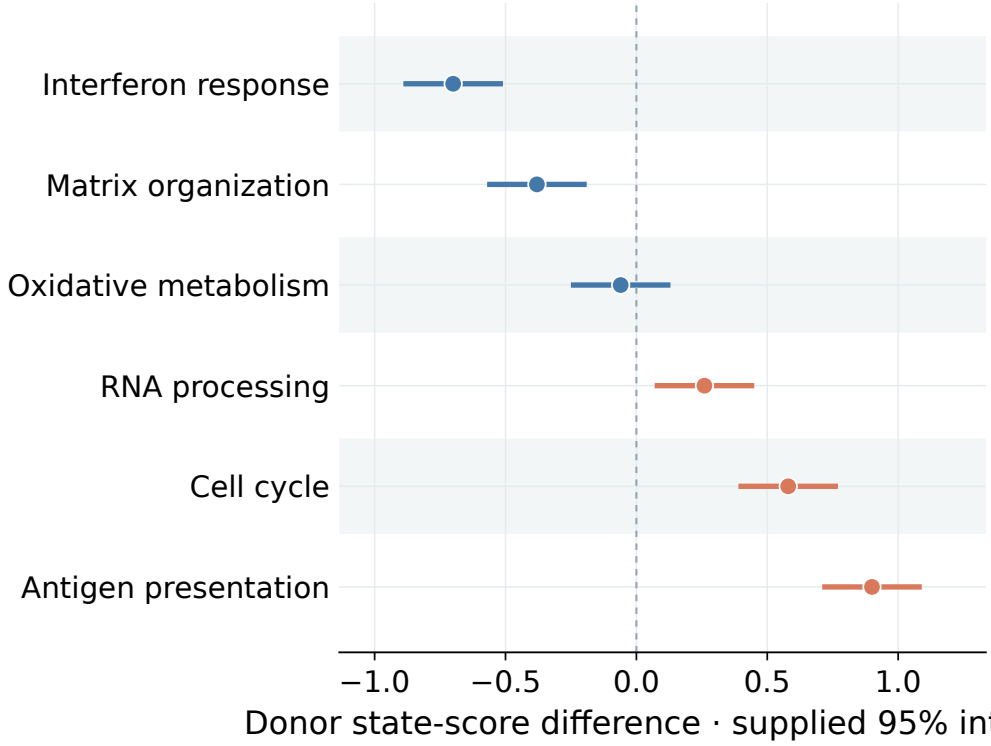
Estimate [interval] · q
Independent n = 12

-0.70	[-0.88, -0.52]	0.002
-0.62	[-0.80, -0.44]	0.005
-0.50	[-0.68, -0.32]	0.02
-0.42	[-0.60, -0.24]	0.023
-0.30	[-0.48, -0.12]	0.038
-0.22	[-0.40, -0.04]	0.041
-0.10	[-0.28, 0.08]	0.65
-0.02	[-0.20, 0.16]	0.65
0.10	[-0.08, 0.28]	0.65
0.18	[0.00, 0.36]	0.077
0.30	[0.12, 0.48]	0.092
0.38	[0.20, 0.56]	0.095
0.50	[0.32, 0.68]	0.11
0.58	[0.40, 0.76]	0.11
0.70	[0.52, 0.88]	0.13
0.78	[0.60, 0.96]	0.13
0.90	[0.72, 1.08]	0.15
0.98	[0.80, 1.16]	0.15


State-program gene coverage
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Measured
Absent


Interferon state scores in independent donors
Supplied interferon-state score
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Control
n=6
Treated
n=6
Condition

Donor-level state-program effects



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

-0.38 [-0.56, -0.20] 0.02

-0.06 [-0.24, 0.12] 0.65

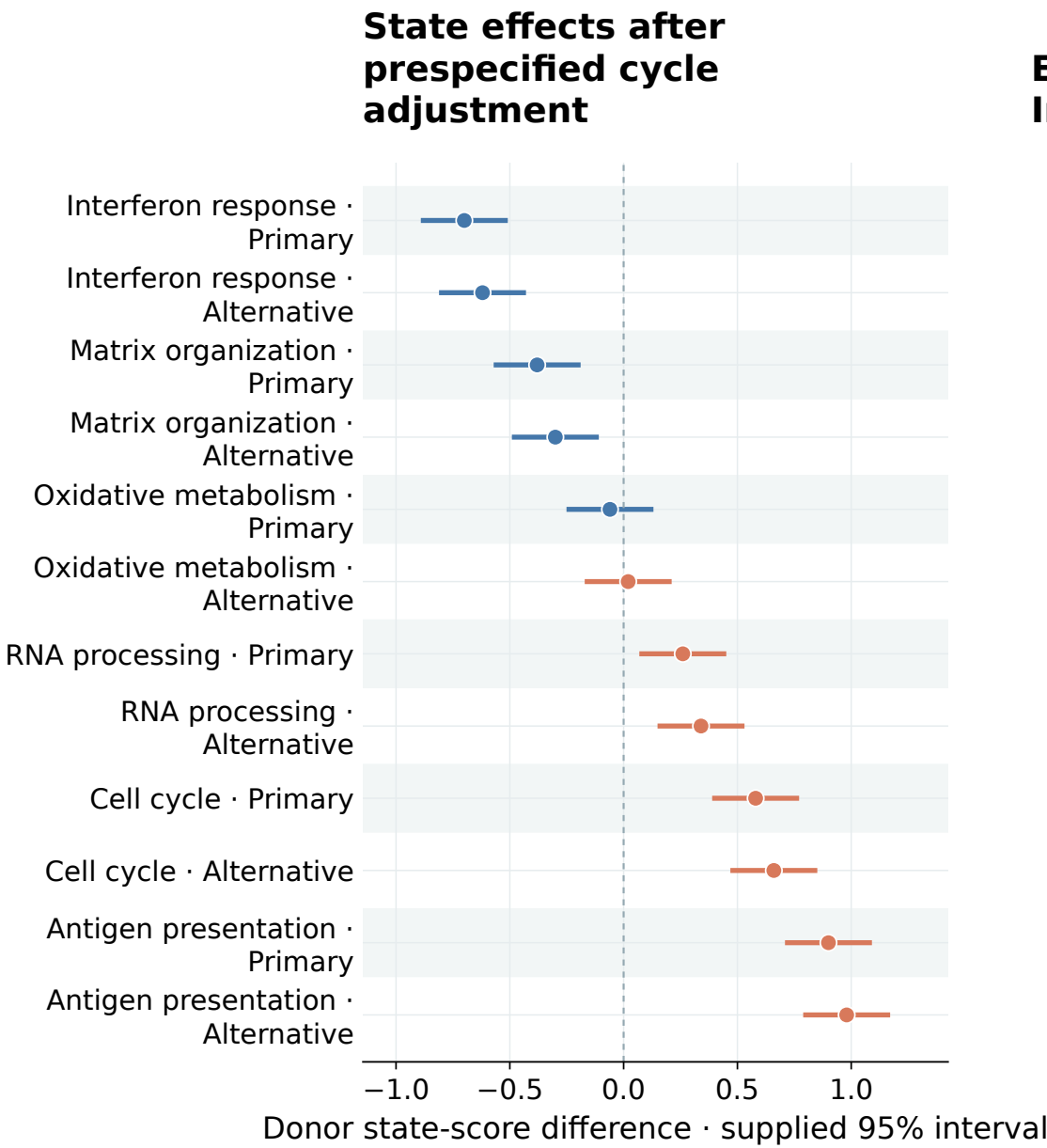
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092

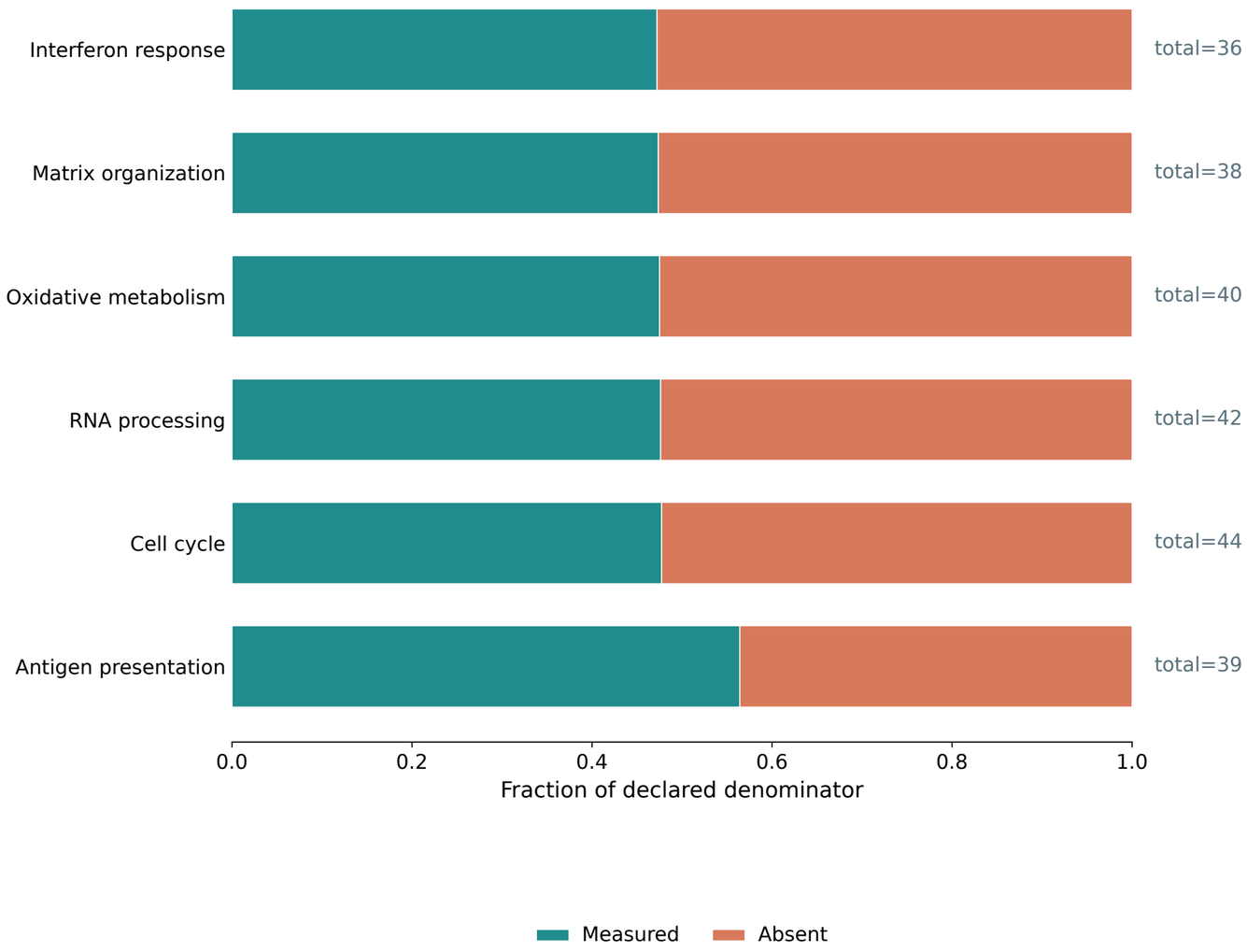
State, stress and cycle contexts



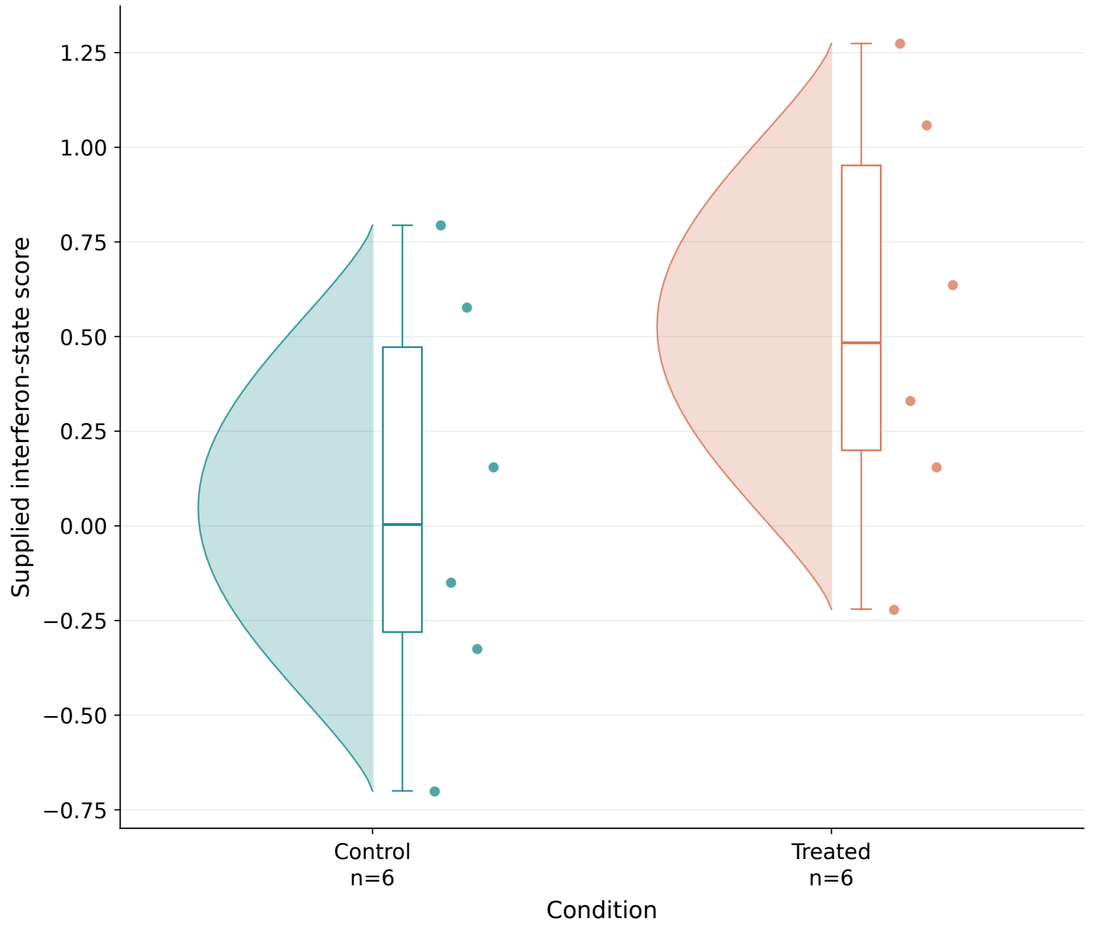

State effects after prespecified cycle adjustment
Estimate [interval] · q
Independent n = 12
Interferon response · Primary
Interferon response · Alternative
Matrix organization · Primary
Matrix organization · Alternative
Oxidative metabolism · Primary
Oxidative metabolism · Alternative
RNA processing · Primary
RNA processing · Alternative
Cell cycle · Primary
Cell cycle · Alternative
Antigen presentation · Primary
Antigen presentation · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Donor state-score difference · supplied 95% interval

Single cell · states — complementary evidence

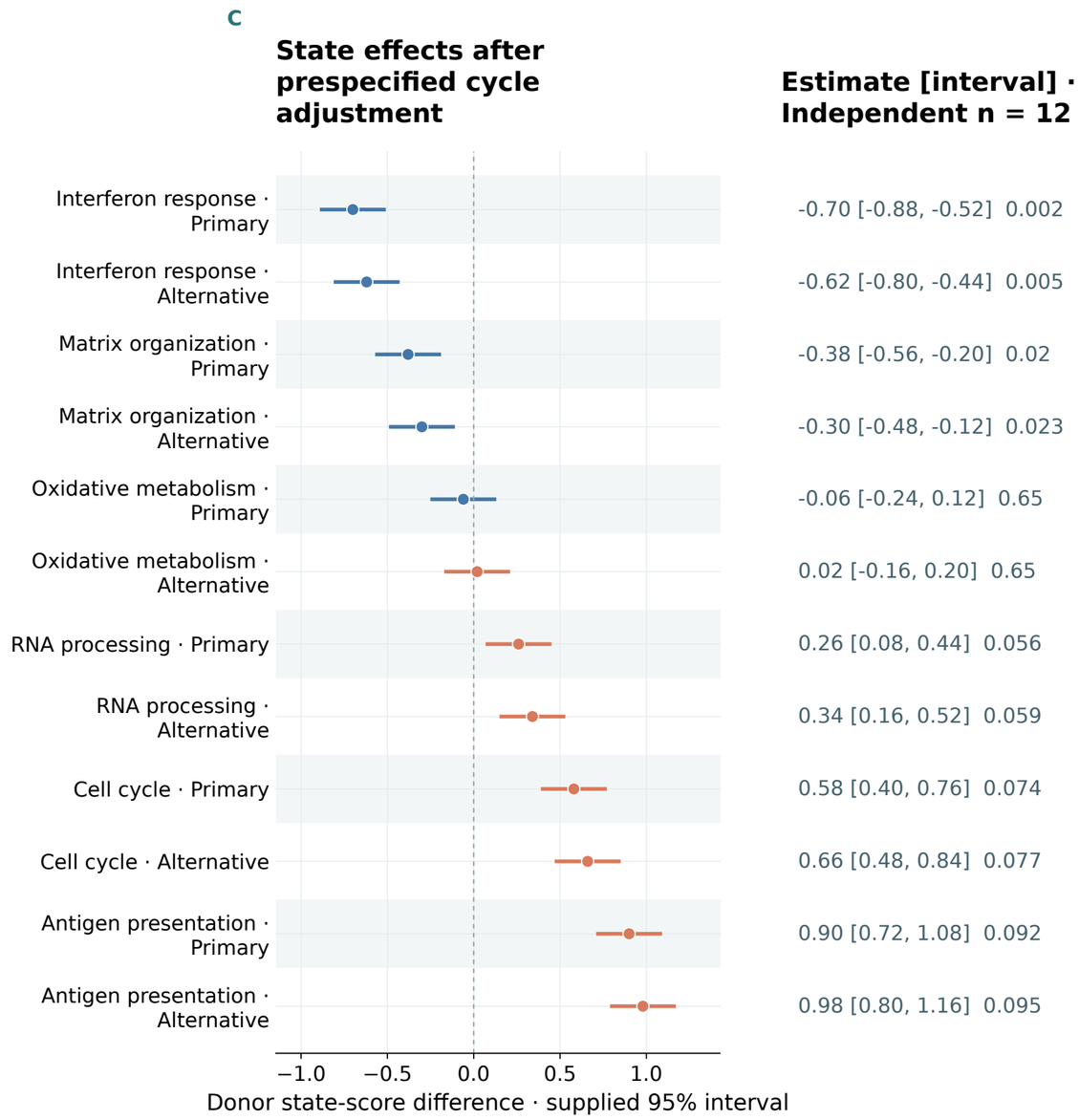
A



B



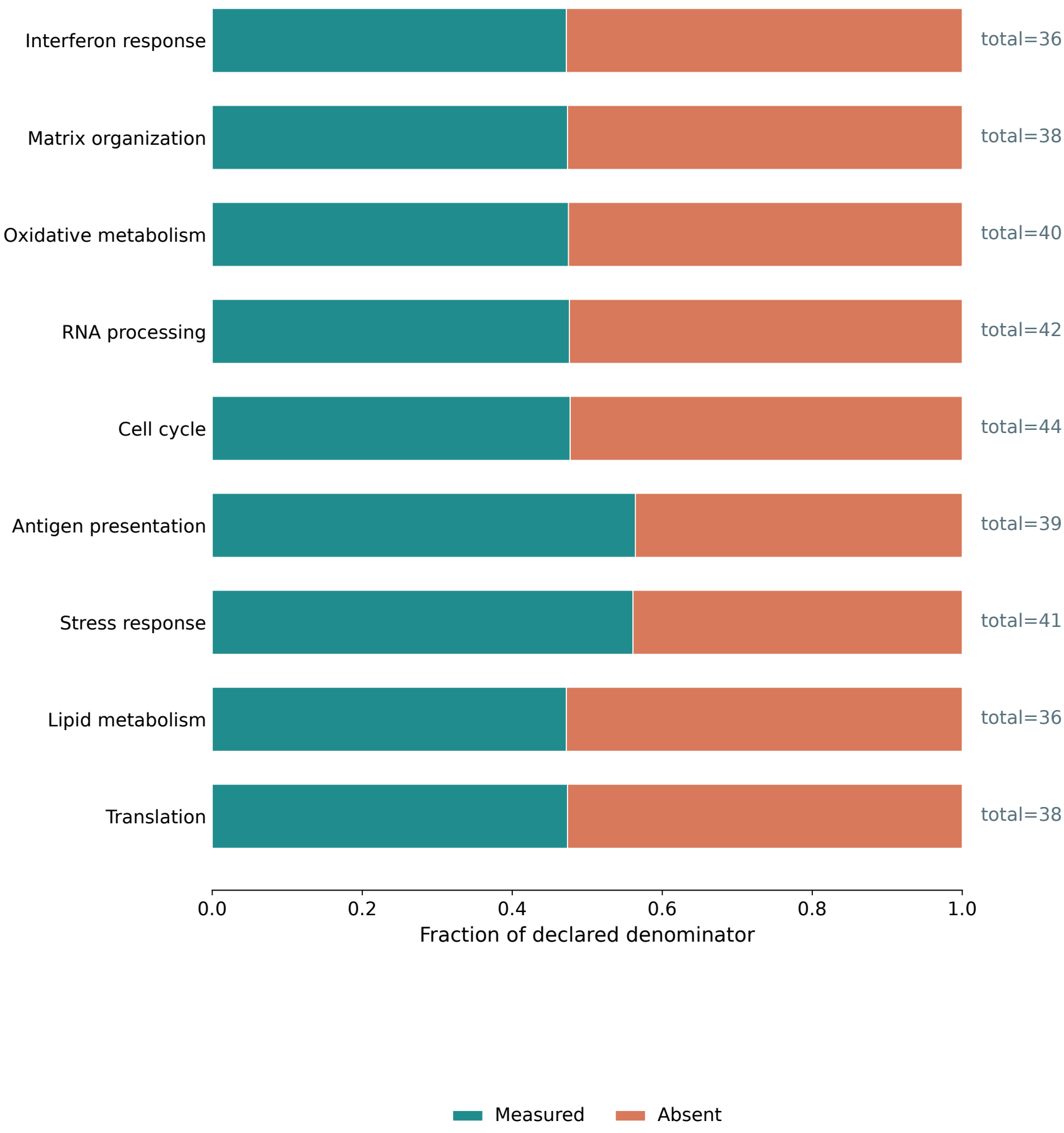
C



Single cell · states — complementary evidence

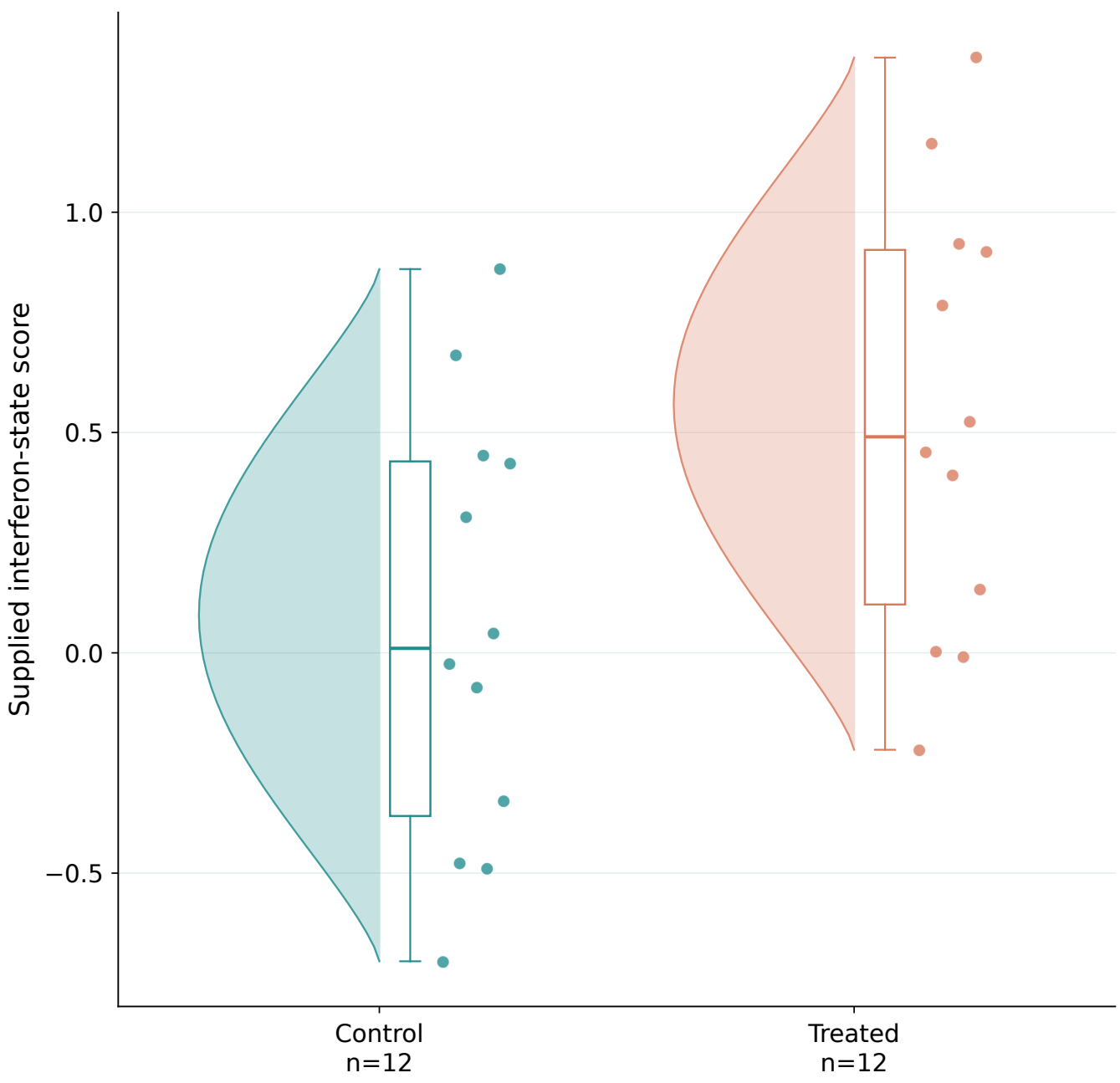
A

State-program gene coverage



B

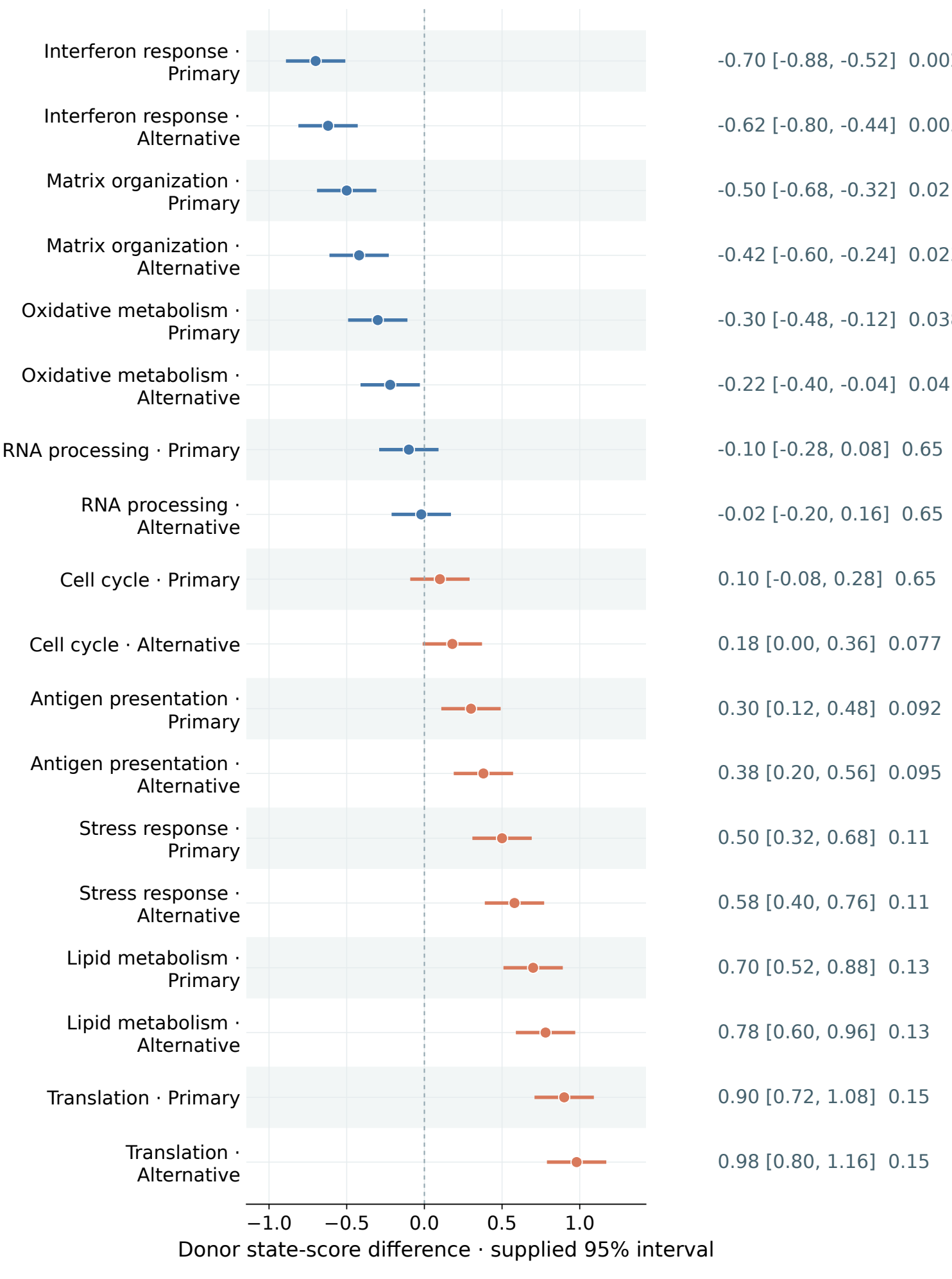
Interferon state scores in independent donors

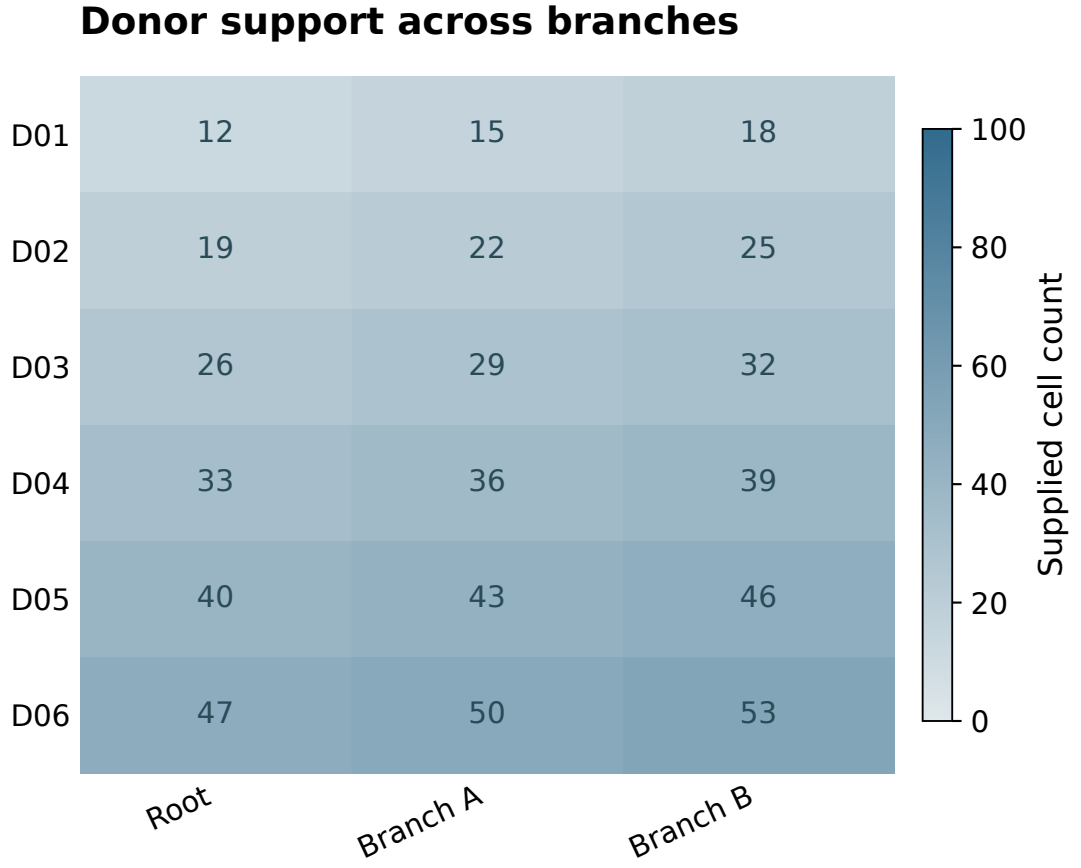


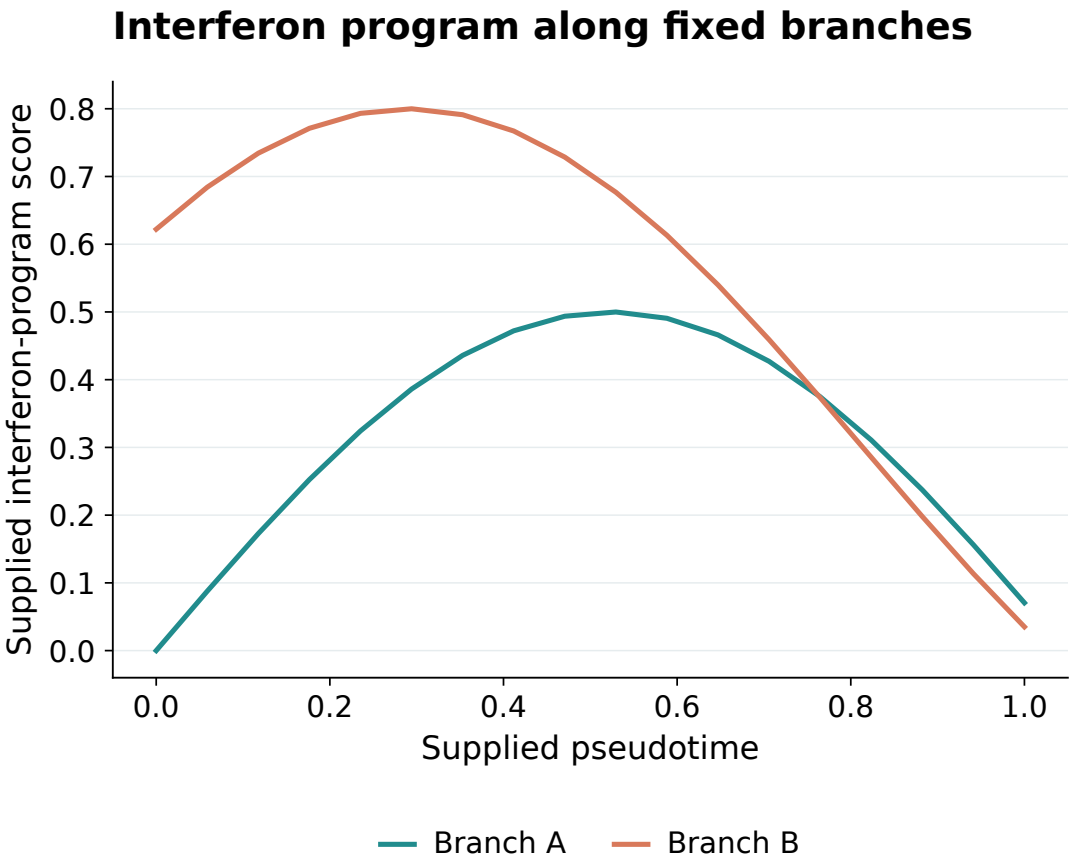
C

State effects after prespecified cycle adjustment

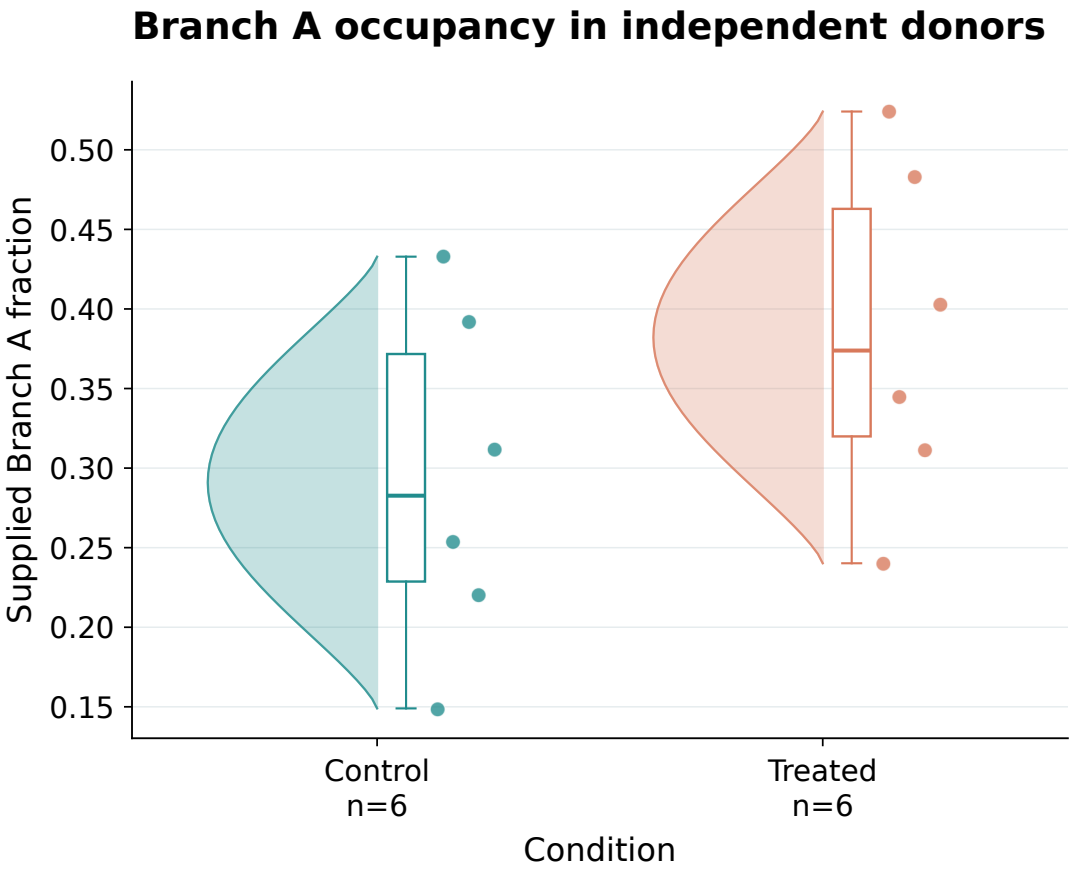
Estimate [interval] · q
Independent n = 24




Donor support across branches
D01
D02
D03
D04
D05
D06
12
15
18
19
22
25
26
29
32
33
36
39
40
43
46
47
50
53
Root
Branch A
Branch B
Supplied cell count
100
80
60
40
20
0

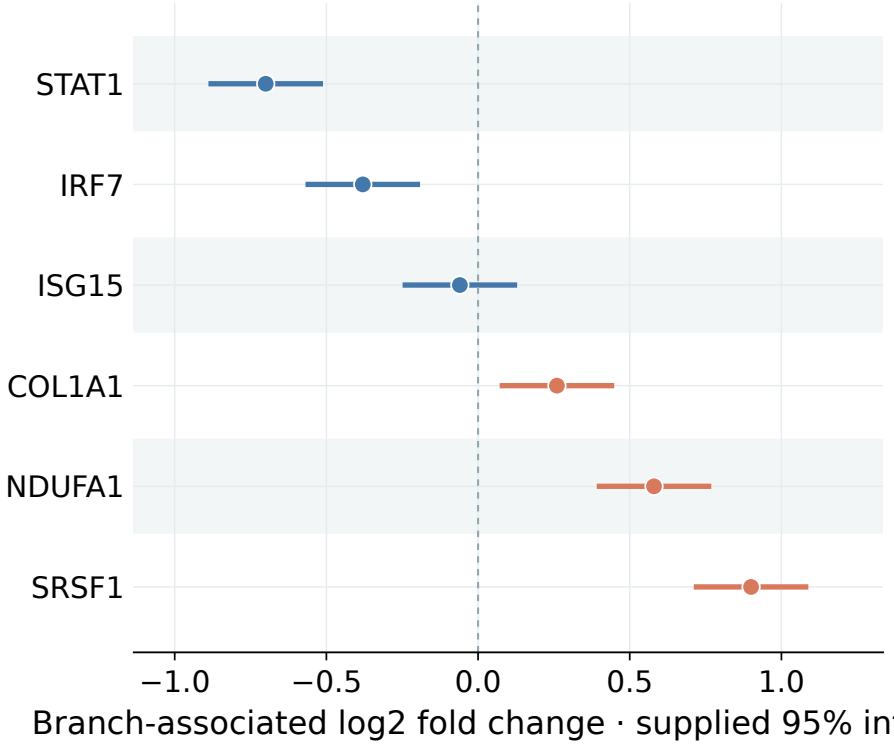


Software test fixture; not biological evidence



Software test fixture; not biological evidence

Branch-specific donor-level effects



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

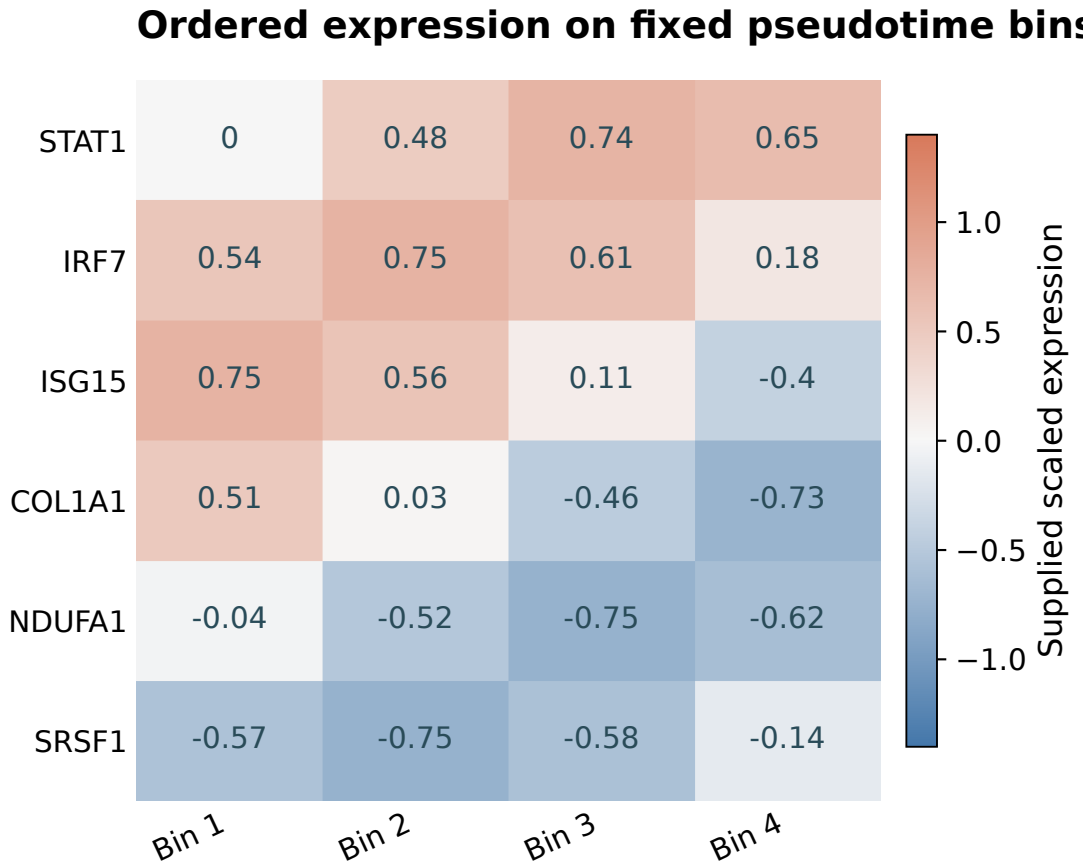
-0.38 [-0.56, -0.20] 0.02

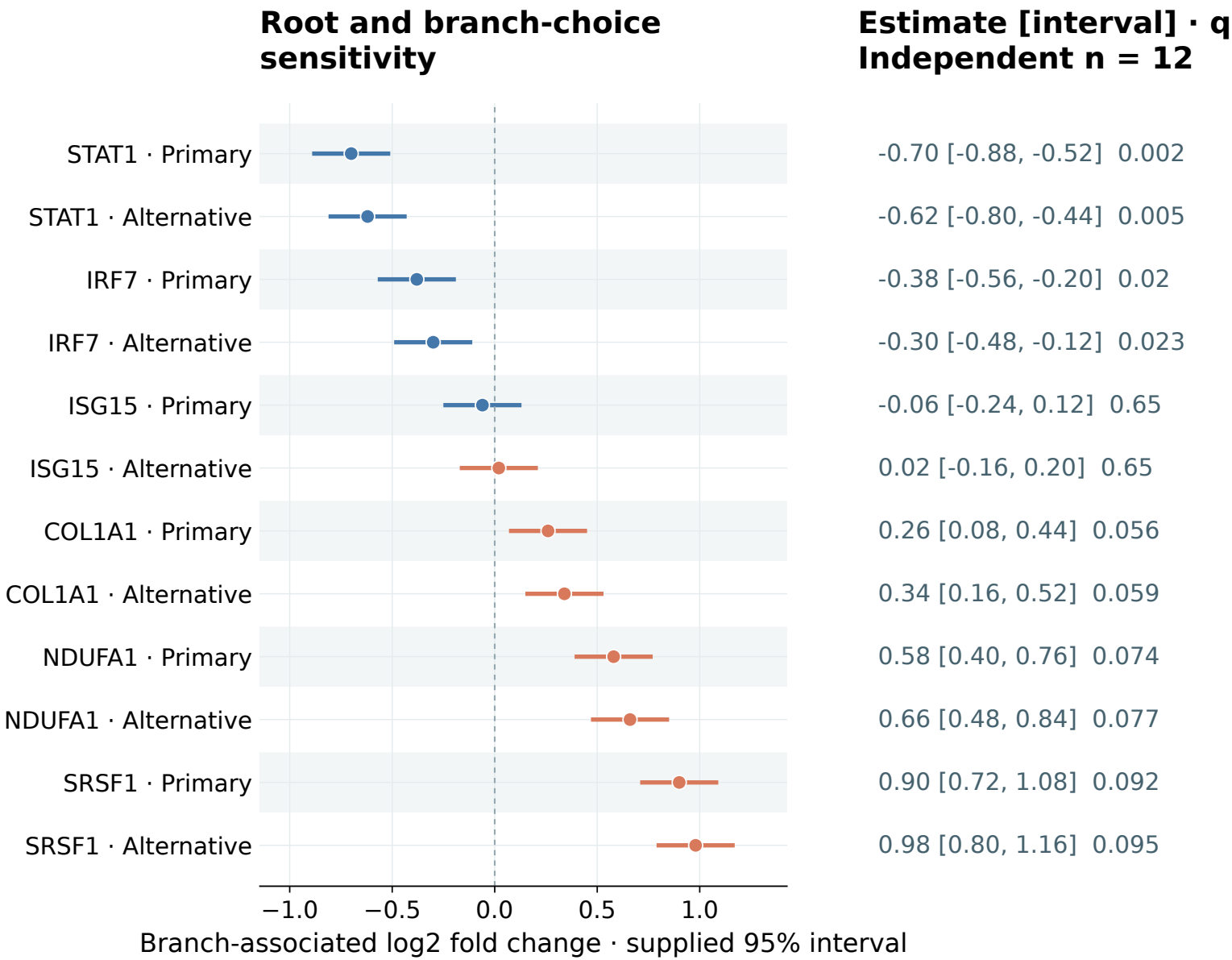
-0.06 [-0.24, 0.12] 0.65

0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

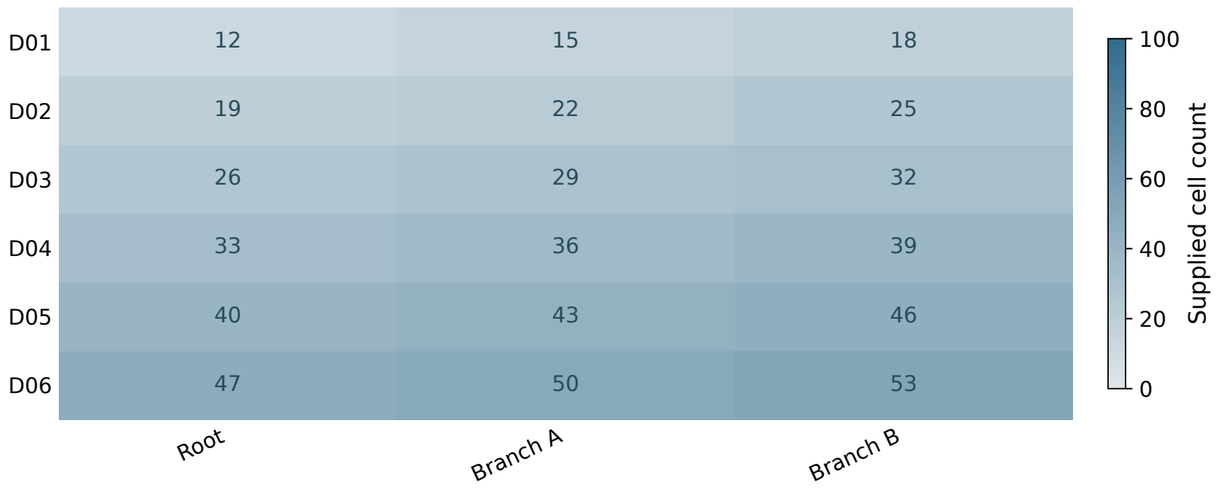
0.90 [0.72, 1.08] 0.092


Ordered expression on fixed pseudotime bins
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0
0.48
0.74
0.65
0.54
0.75
0.61
0.18
0.75
0.56
0.11
-0.4
0.51
0.03
-0.46
-0.73
-0.04
-0.52
-0.75
-0.62
-0.57
-0.75
-0.58
-0.14
Bin 1
Bin 2
Bin 3
Bin 4
1.0
0.5
0.0
−0.5
−1.0
Supplied scaled expression

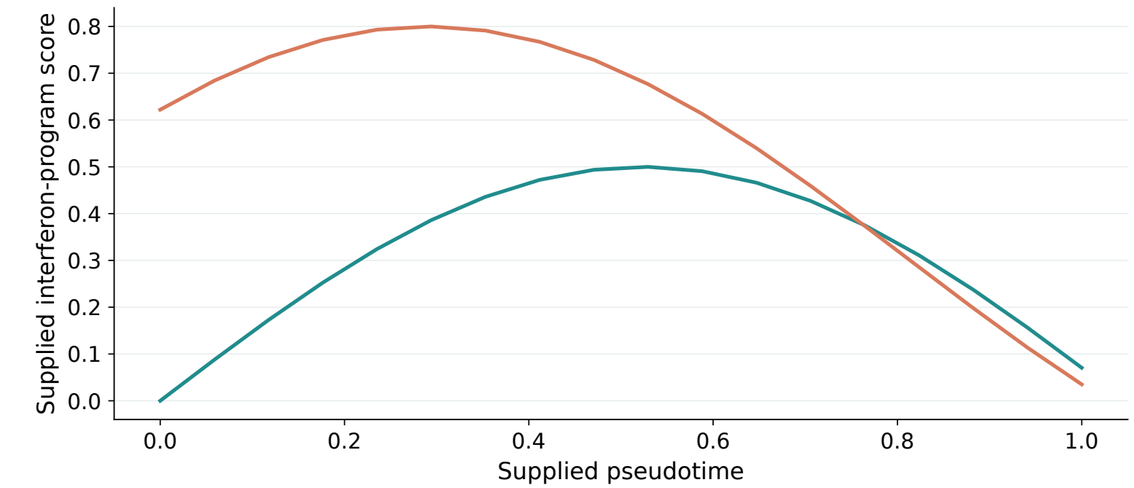

Root and branch-choice sensitivity
Estimate [interval] · q
Independent n = 12
STAT1 · Primary
STAT1 · Alternative
IRF7 · Primary
IRF7 · Alternative
ISG15 · Primary
ISG15 · Alternative
COL1A1 · Primary
COL1A1 · Alternative
NDUFA1 · Primary
NDUFA1 · Alternative
SRSF1 · Primary
SRSF1 · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Branch-associated log2 fold change · supplied 95% interval

Single cell · trajectory — complementary evidence

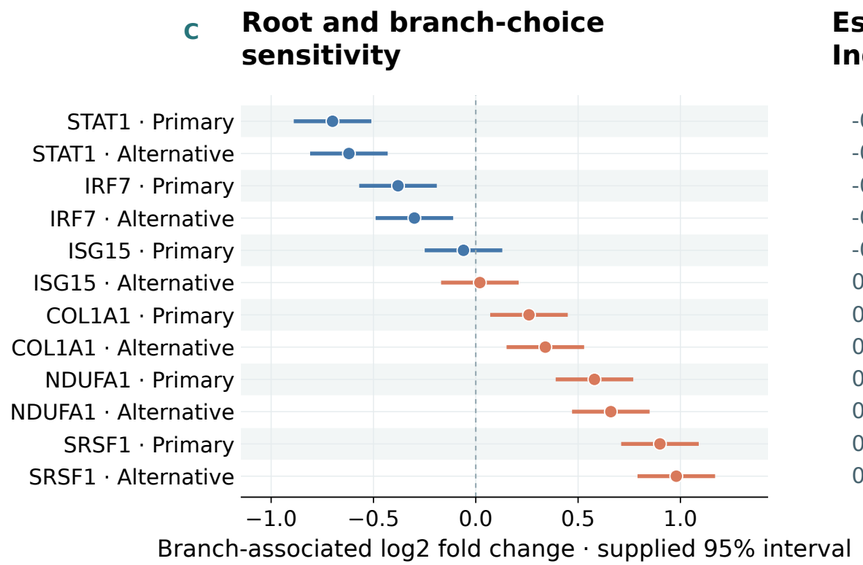
A



B



C



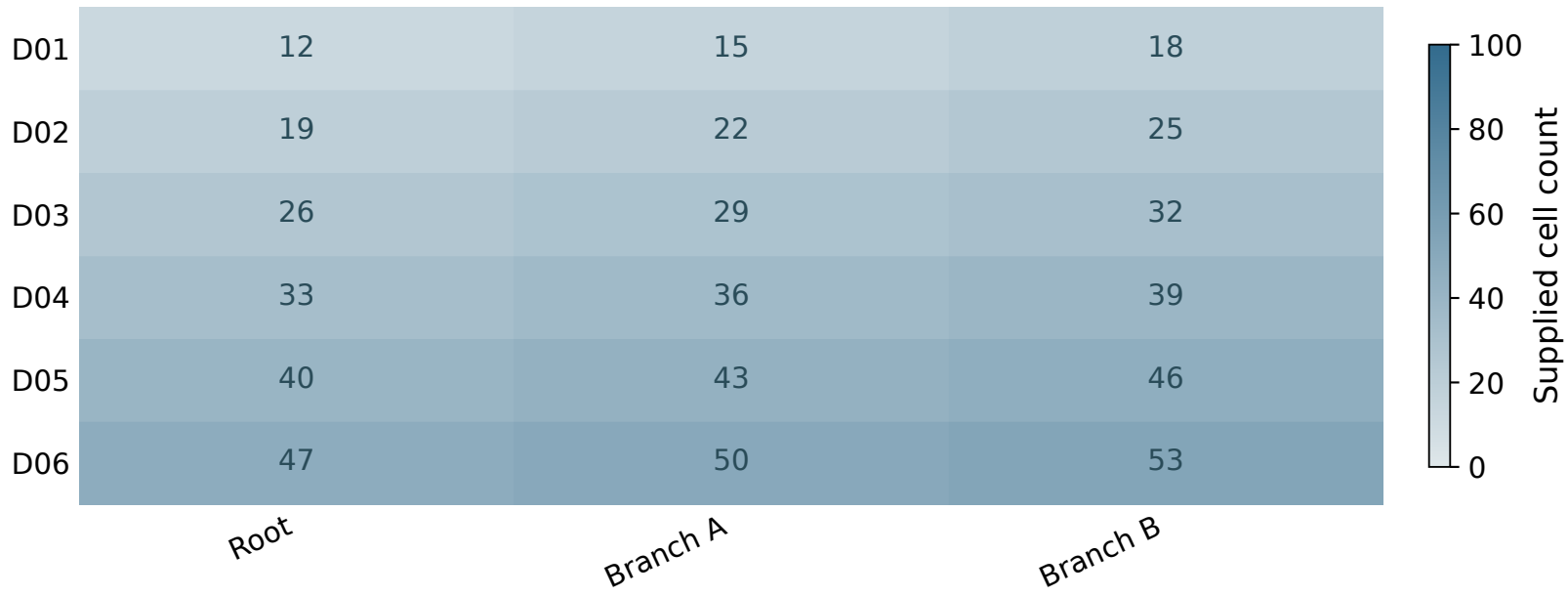
Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095

Single cell · trajectory — complementary evidence

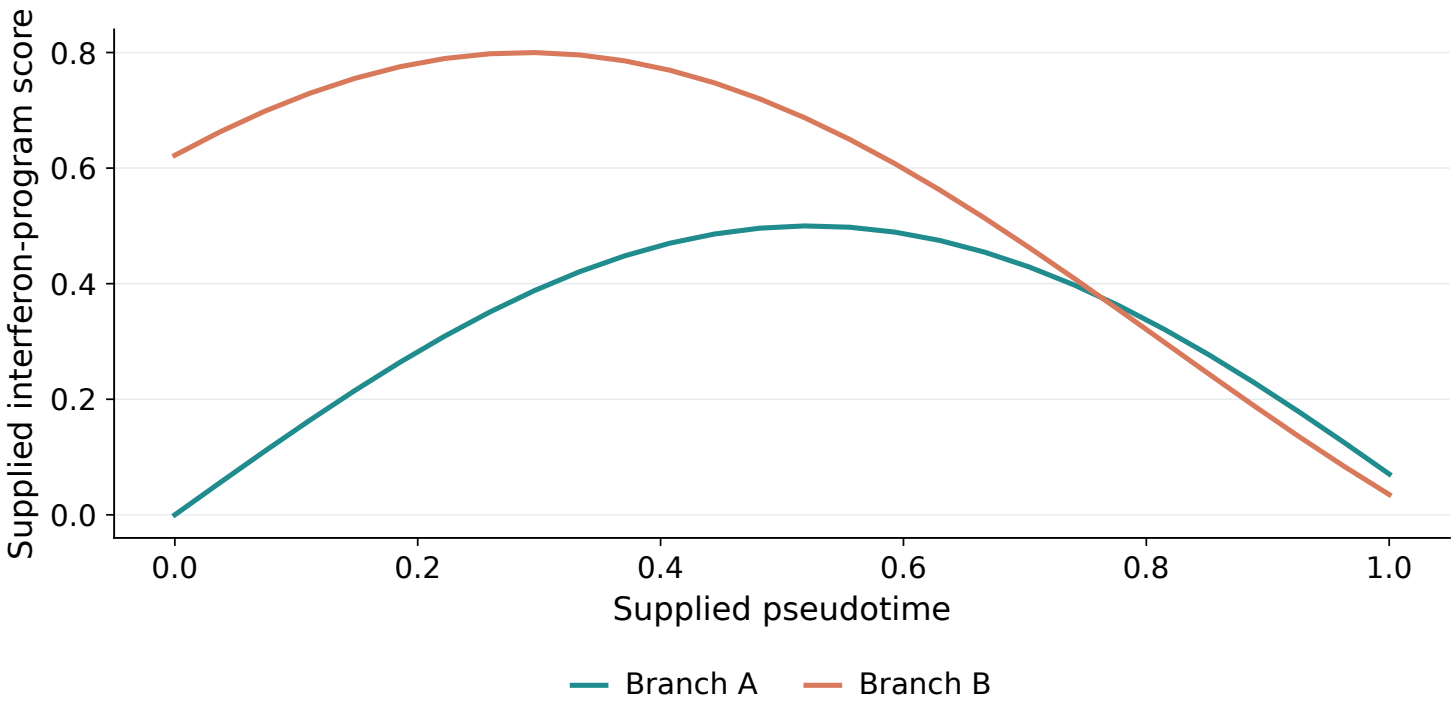
A

Donor support across branches



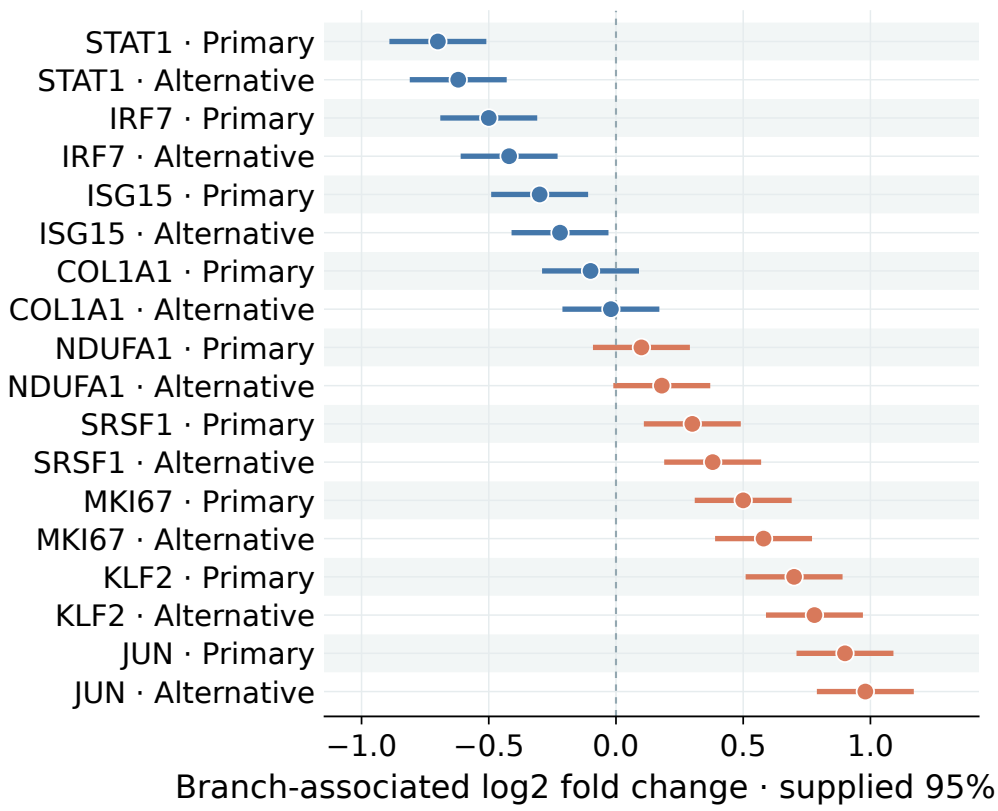
B

Interferon program along fixed branches



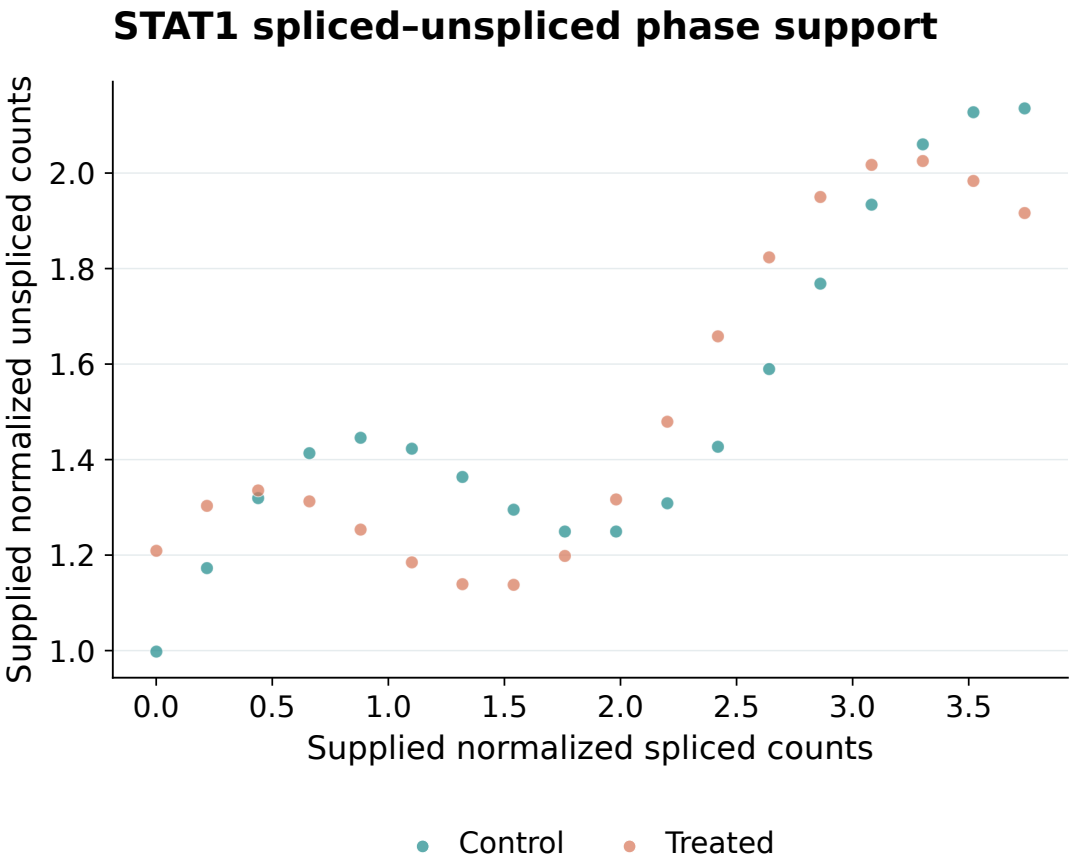
C

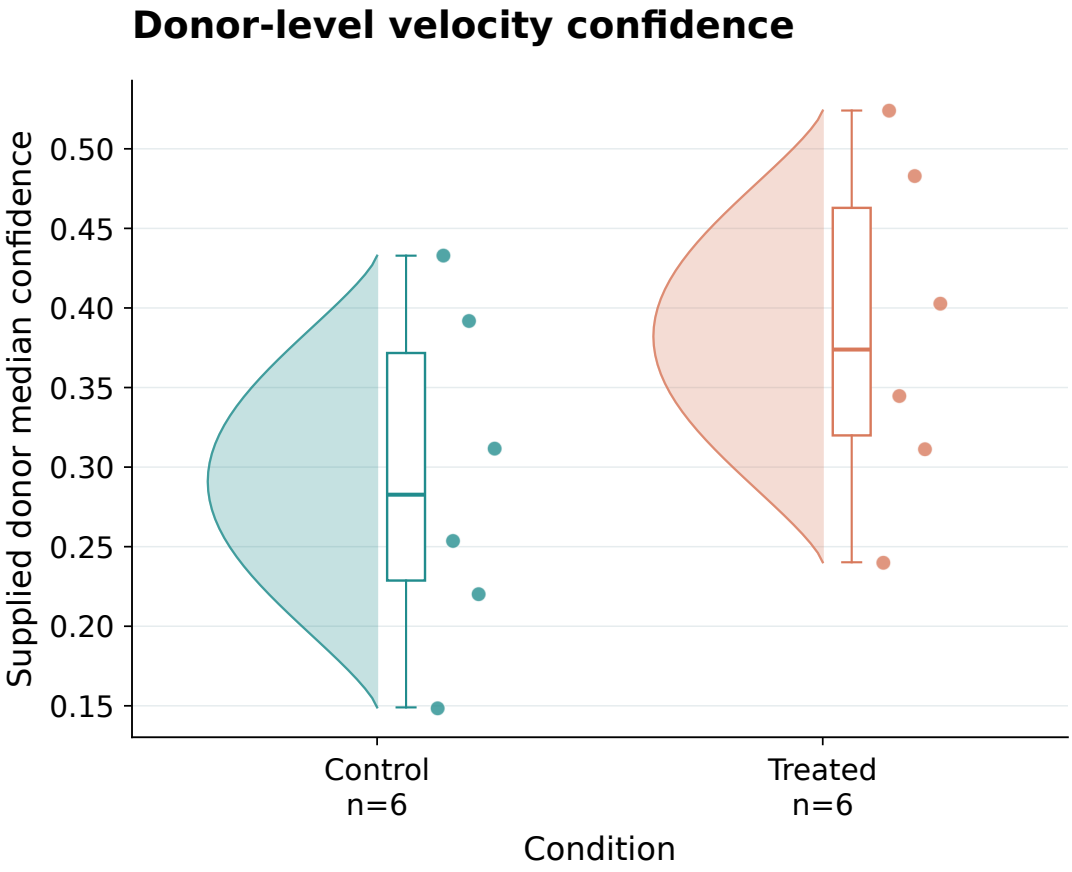
Root and branch-choice sensitivity



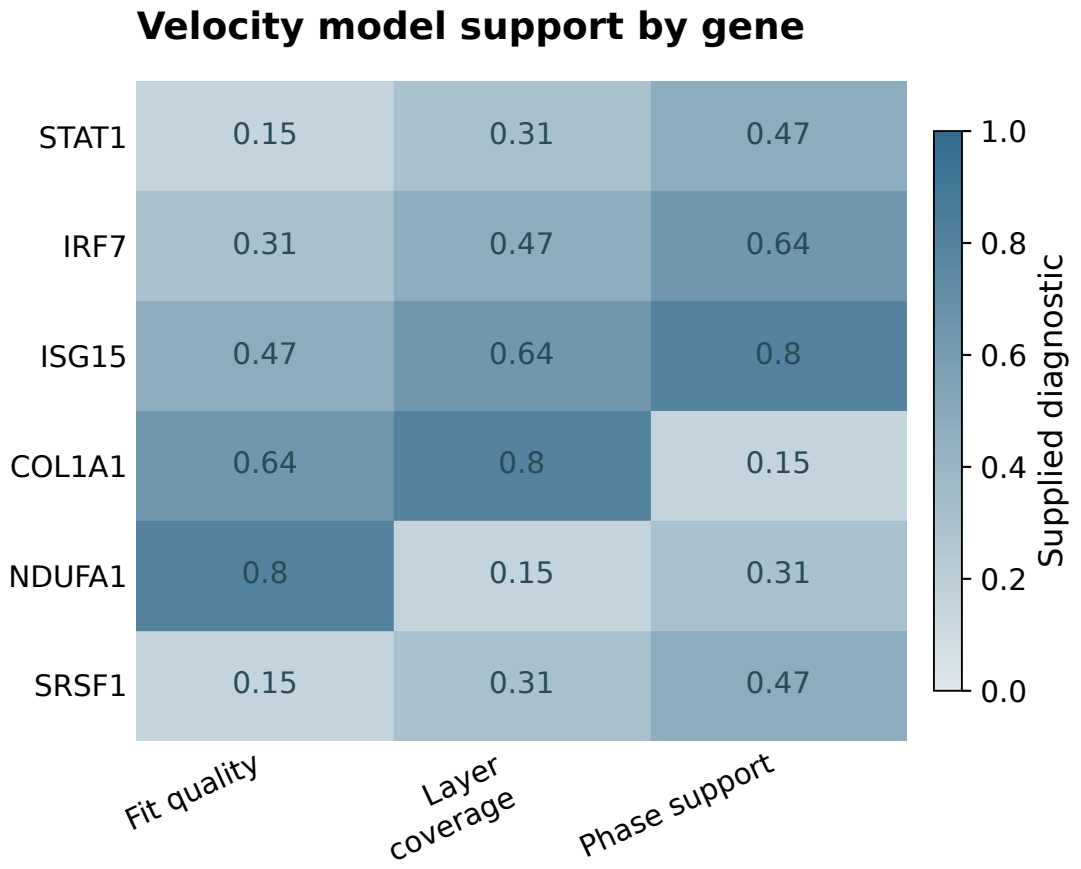
Estimate [interval] · q
Independent n = 24

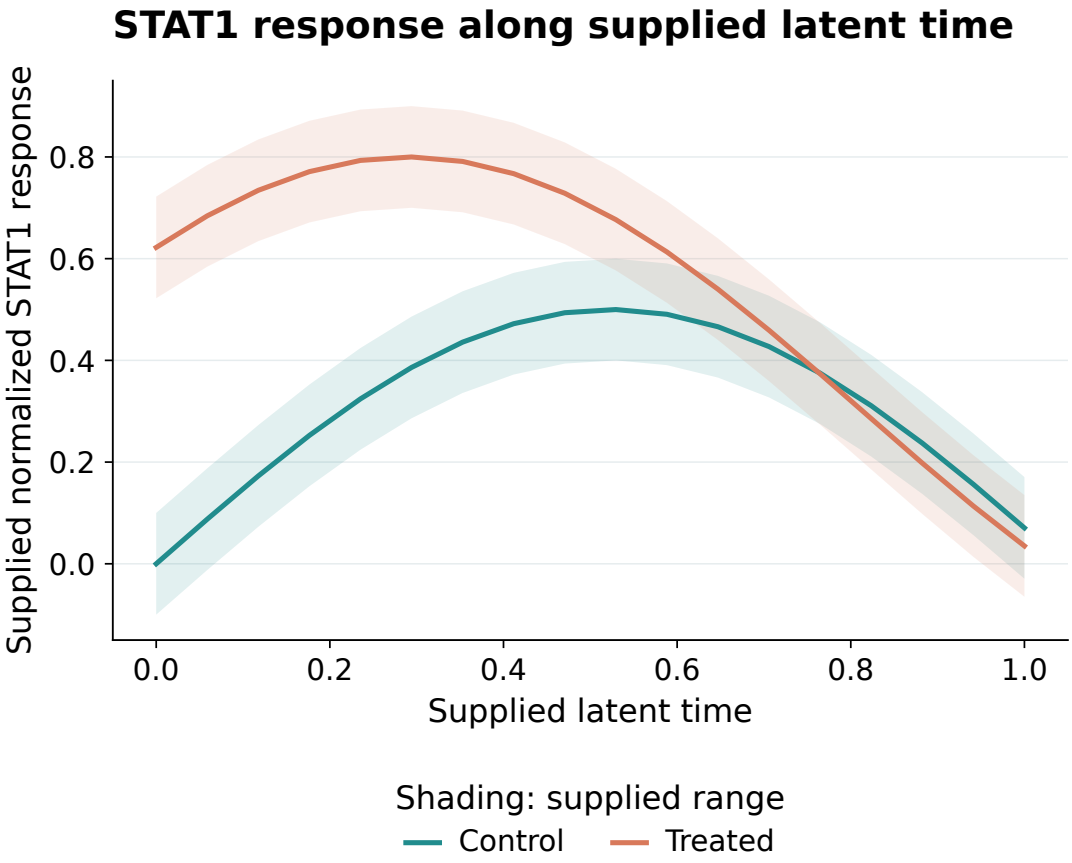
STAT1 · Primary	-0.70 [-0.88, -0.52] 0.002
STAT1 · Alternative	-0.62 [-0.80, -0.44] 0.005
IRF7 · Primary	-0.50 [-0.68, -0.32] 0.02
IRF7 · Alternative	-0.42 [-0.60, -0.24] 0.023
ISG15 · Primary	-0.30 [-0.48, -0.12] 0.038
ISG15 · Alternative	-0.22 [-0.40, -0.04] 0.041
COL1A1 · Primary	-0.10 [-0.28, 0.08] 0.65
COL1A1 · Alternative	-0.02 [-0.20, 0.16] 0.65
NDUFA1 · Primary	0.10 [-0.08, 0.28] 0.65
NDUFA1 · Alternative	0.18 [0.00, 0.36] 0.077
SRSF1 · Primary	0.30 [0.12, 0.48] 0.092
SRSF1 · Alternative	0.38 [0.20, 0.56] 0.095
MKI67 · Primary	0.50 [0.32, 0.68] 0.11
MKI67 · Alternative	0.58 [0.40, 0.76] 0.11
KLF2 · Primary	0.70 [0.52, 0.88] 0.13
KLF2 · Alternative	0.78 [0.60, 0.96] 0.13
JUN · Primary	0.90 [0.72, 1.08] 0.15
JUN · Alternative	0.98 [0.80, 1.16] 0.15

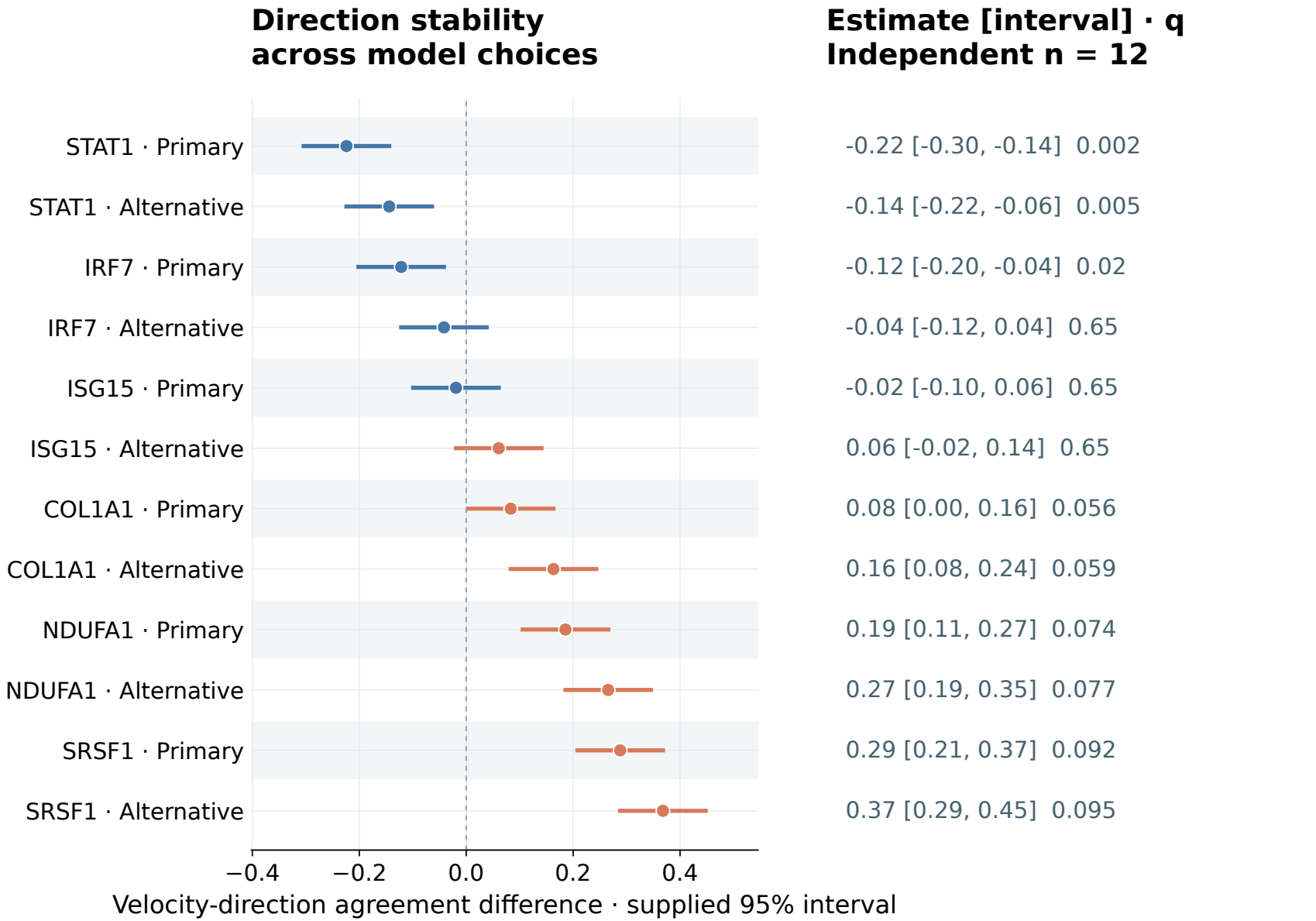

STAT1 spliced–unspliced phase support
Supplied normalized unspliced counts
1.0
1.2
1.4
1.6
1.8
2.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Supplied normalized spliced counts
Control
Treated



Software test fixture; not biological evidence

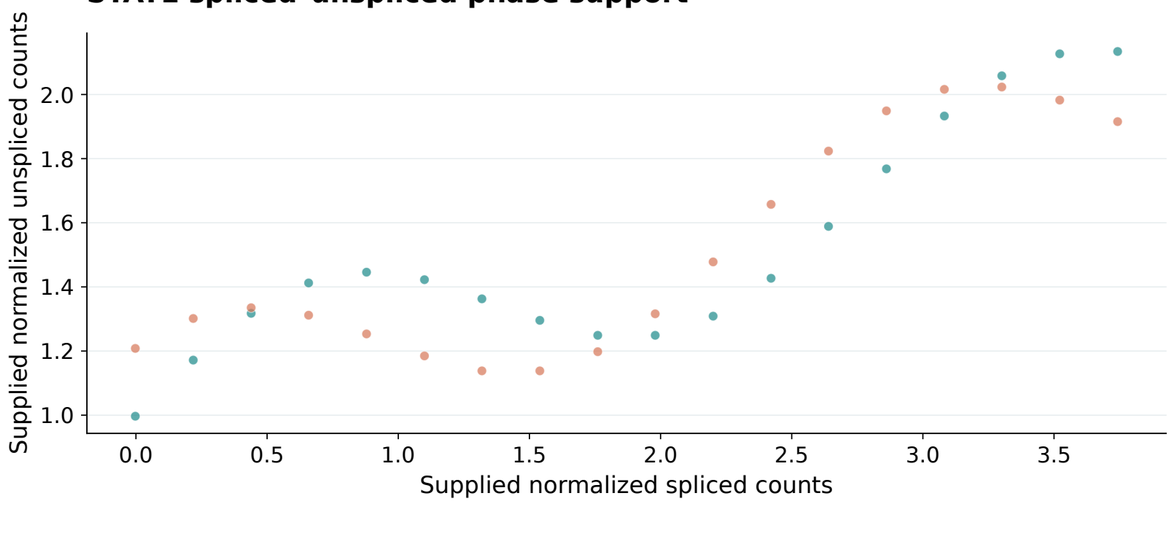

Velocity model support by gene
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0.15
0.31
0.47
0.31
0.47
0.64
0.47
0.64
0.8
0.64
0.8
0.15
0.8
0.15
0.31
0.15
0.31
0.47
Fit quality
Layer coverage
Phase support
Supplied diagnostic
1.0
0.8
0.6
0.4
0.2
0.0


STAT1 response along supplied latent time
Supplied normalized STAT1 response
0.0
0.2
0.4
0.6
0.8
0.0
0.2
0.4
0.6
0.8
1.0
Supplied latent time
Shading: supplied range
Control
Treated

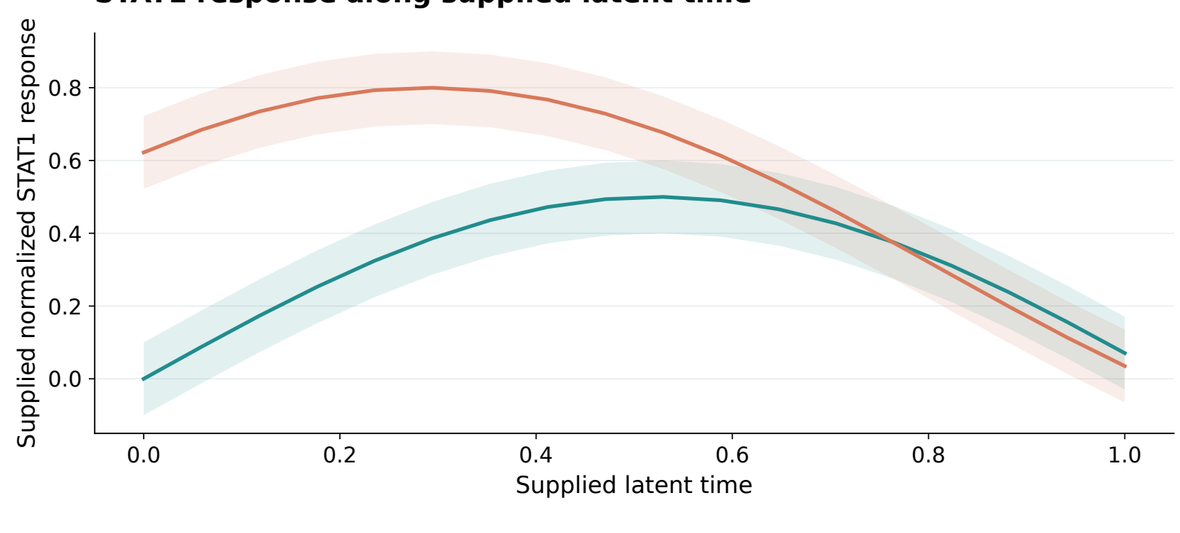

Direction stability
across model choices
Estimate [interval] · q
Independent n = 12
STAT1 · Primary
STAT1 · Alternative
IRF7 · Primary
IRF7 · Alternative
ISG15 · Primary
ISG15 · Alternative
COL1A1 · Primary
COL1A1 · Alternative
NDUFA1 · Primary
NDUFA1 · Alternative
SRSF1 · Primary
SRSF1 · Alternative
-0.22 [-0.30, -0.14] 0.002
-0.14 [-0.22, -0.06] 0.005
-0.12 [-0.20, -0.04] 0.02
-0.04 [-0.12, 0.04] 0.65
-0.02 [-0.10, 0.06] 0.65
0.06 [-0.02, 0.14] 0.65
0.08 [0.00, 0.16] 0.056
0.16 [0.08, 0.24] 0.059
0.19 [0.11, 0.27] 0.074
0.27 [0.19, 0.35] 0.077
0.29 [0.21, 0.37] 0.092
0.37 [0.29, 0.45] 0.095
−0.4
−0.2
0.0
0.2
0.4
Velocity-direction agreement difference · supplied 95% interval

Single cell · velocity — complementary evidence

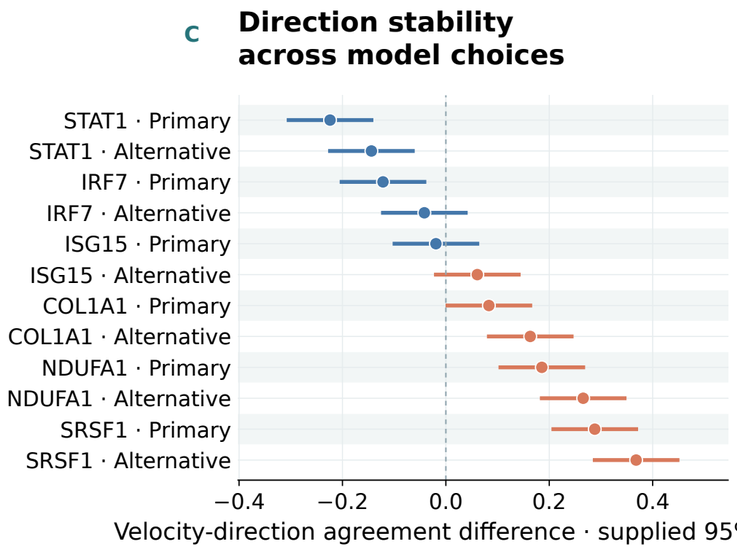
A



B



C



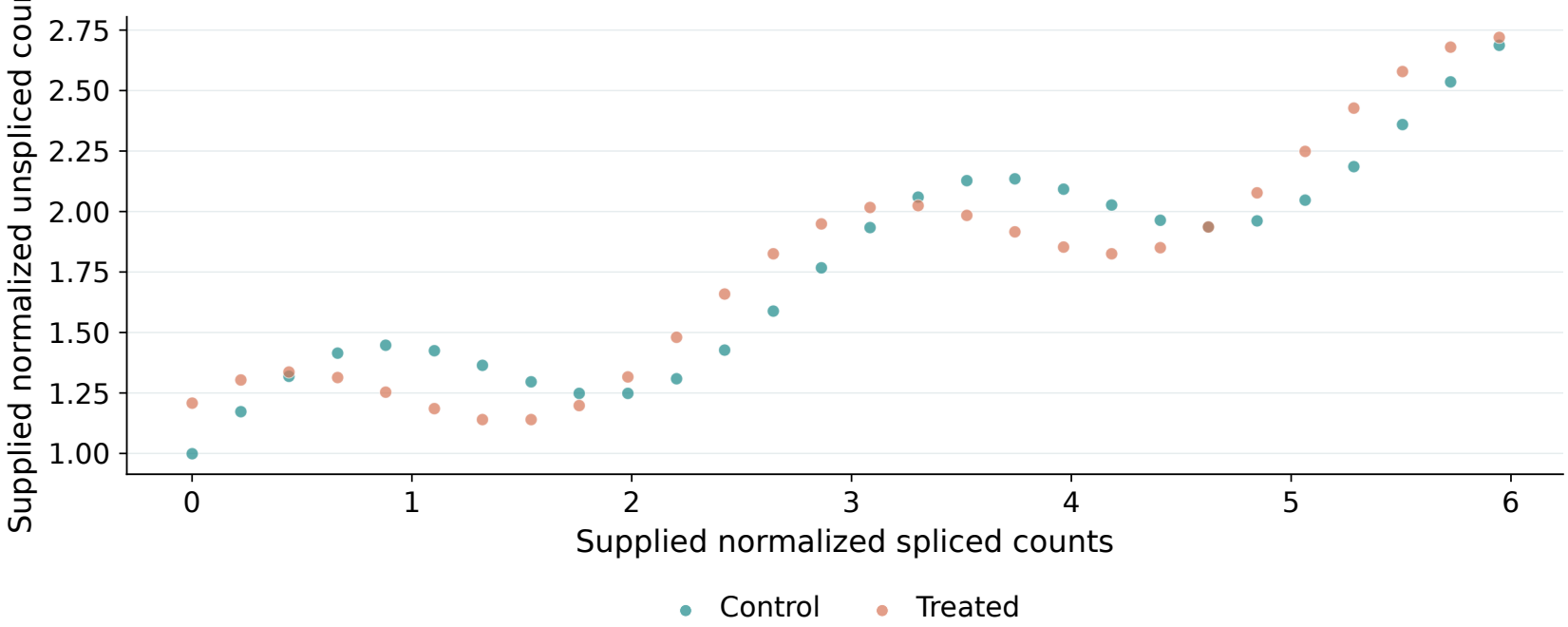
Estimate [interval] · q
Independent n = 12

STAT1 · Primary	-0.22 [-0.30, -0.14] 0.002
STAT1 · Alternative	-0.14 [-0.22, -0.06] 0.005
IRF7 · Primary	-0.12 [-0.20, -0.04] 0.02
IRF7 · Alternative	-0.04 [-0.12, 0.04] 0.65
ISG15 · Primary	-0.02 [-0.10, 0.06] 0.65
ISG15 · Alternative	0.06 [-0.02, 0.14] 0.65
COL1A1 · Primary	0.08 [0.00, 0.16] 0.056
COL1A1 · Alternative	0.16 [0.08, 0.24] 0.059
NDUFA1 · Primary	0.19 [0.11, 0.27] 0.074
NDUFA1 · Alternative	0.27 [0.19, 0.35] 0.077
SRSF1 · Primary	0.29 [0.21, 0.37] 0.092
SRSF1 · Alternative	0.37 [0.29, 0.45] 0.095

Single cell · velocity — complementary evidence

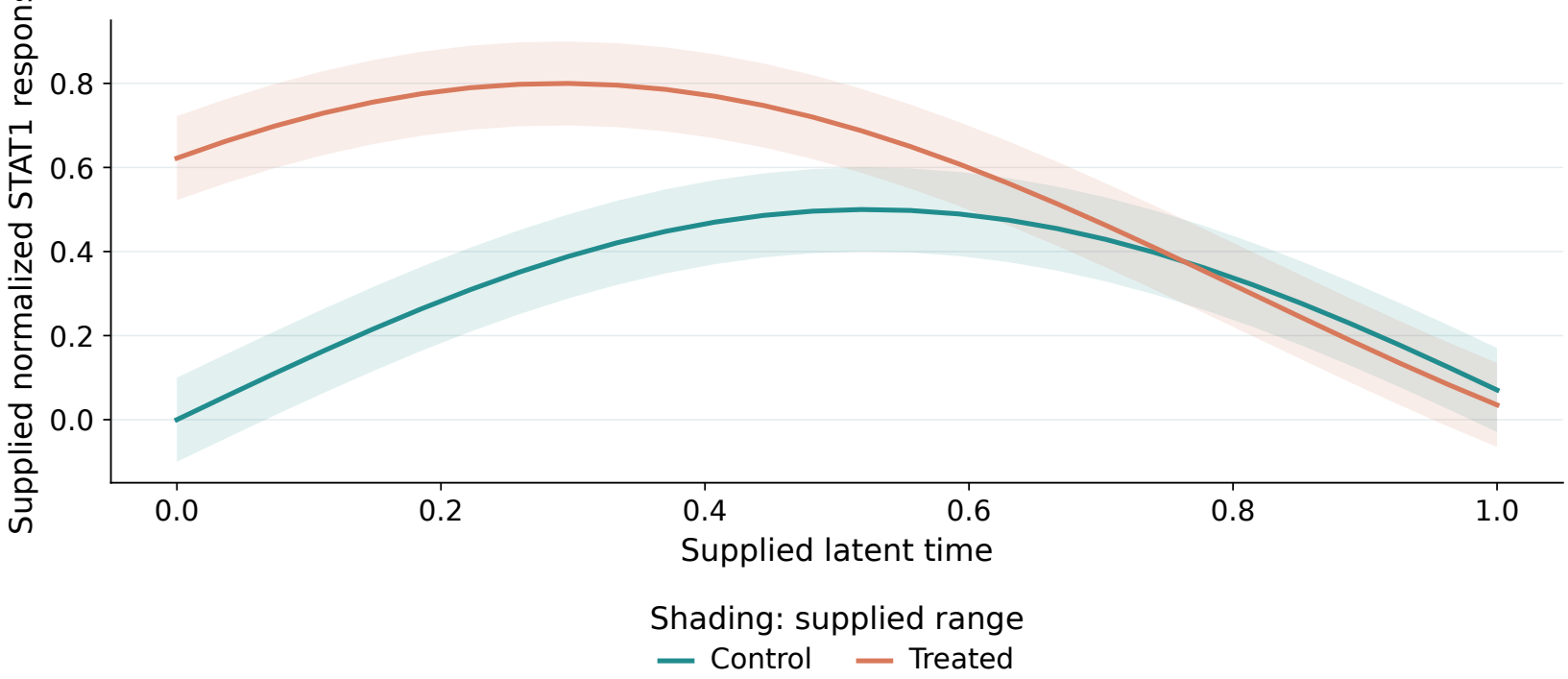
A

STAT1 spliced–unspliced phase support



B

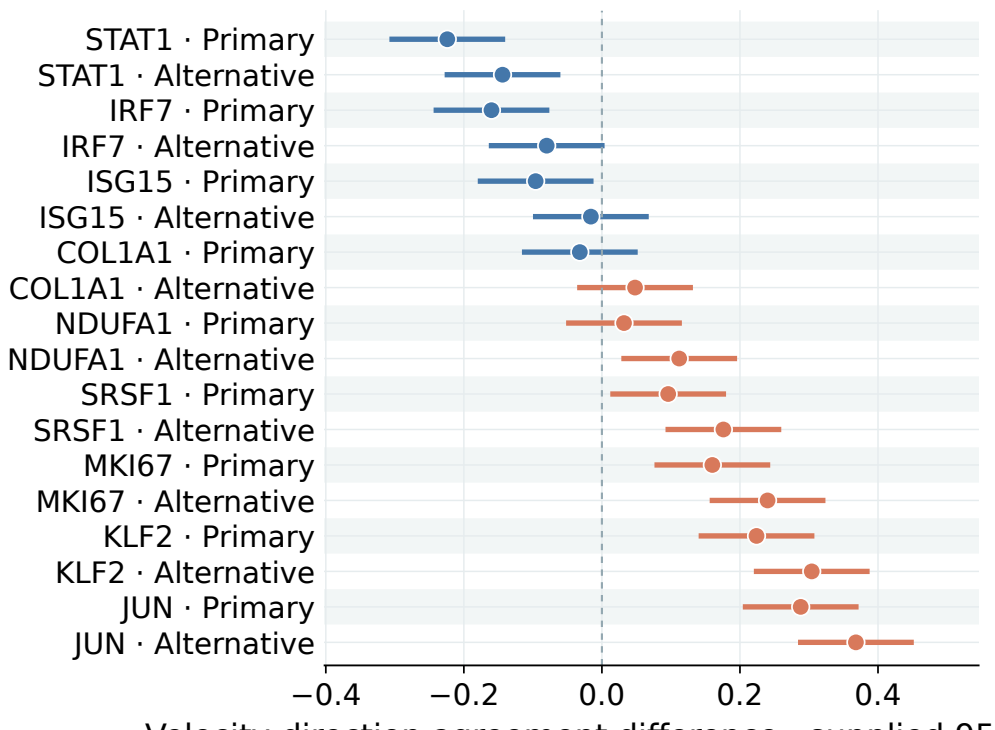
STAT1 response along supplied latent time



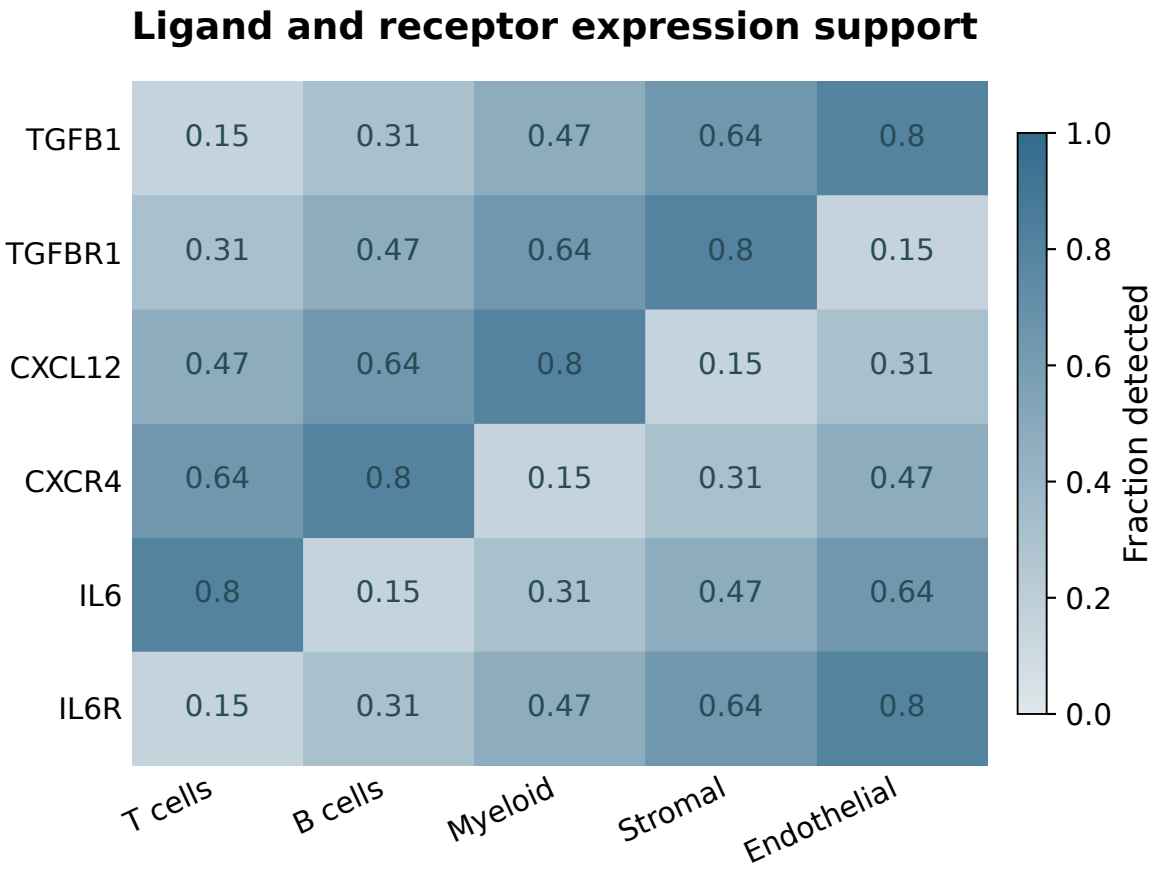
C

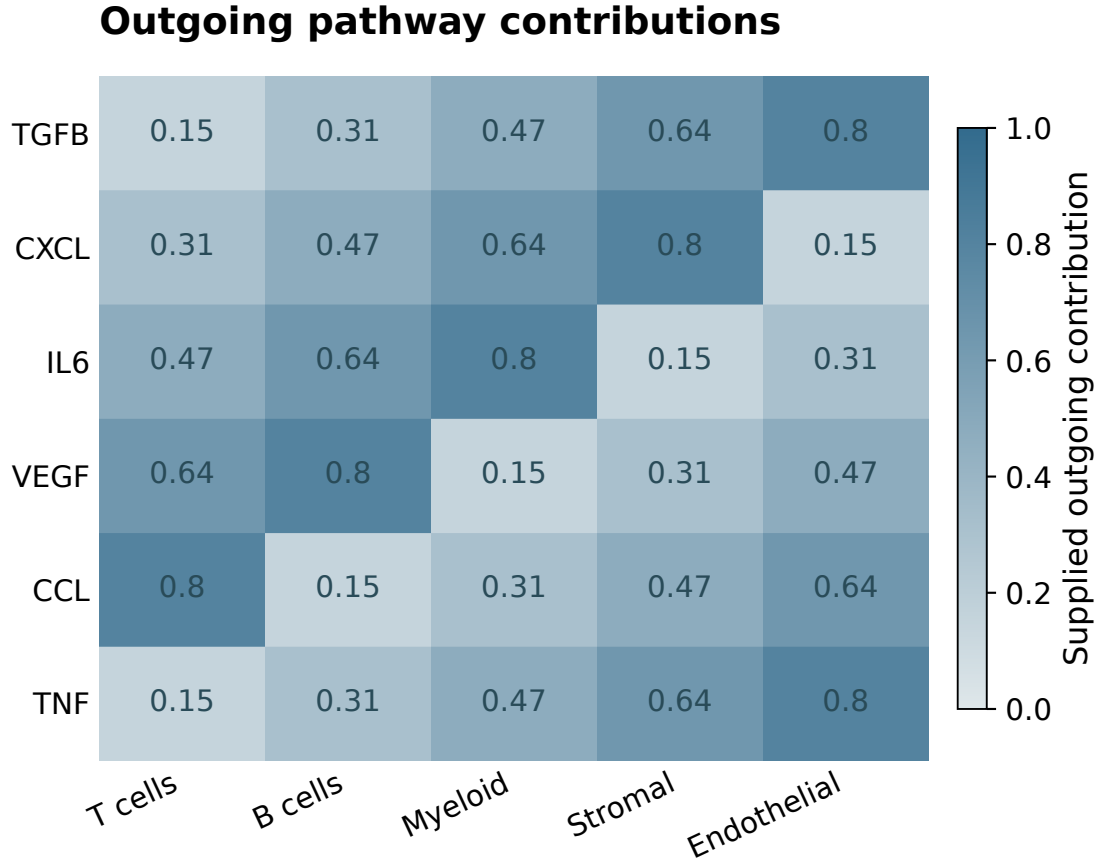
Direction stability across model choices

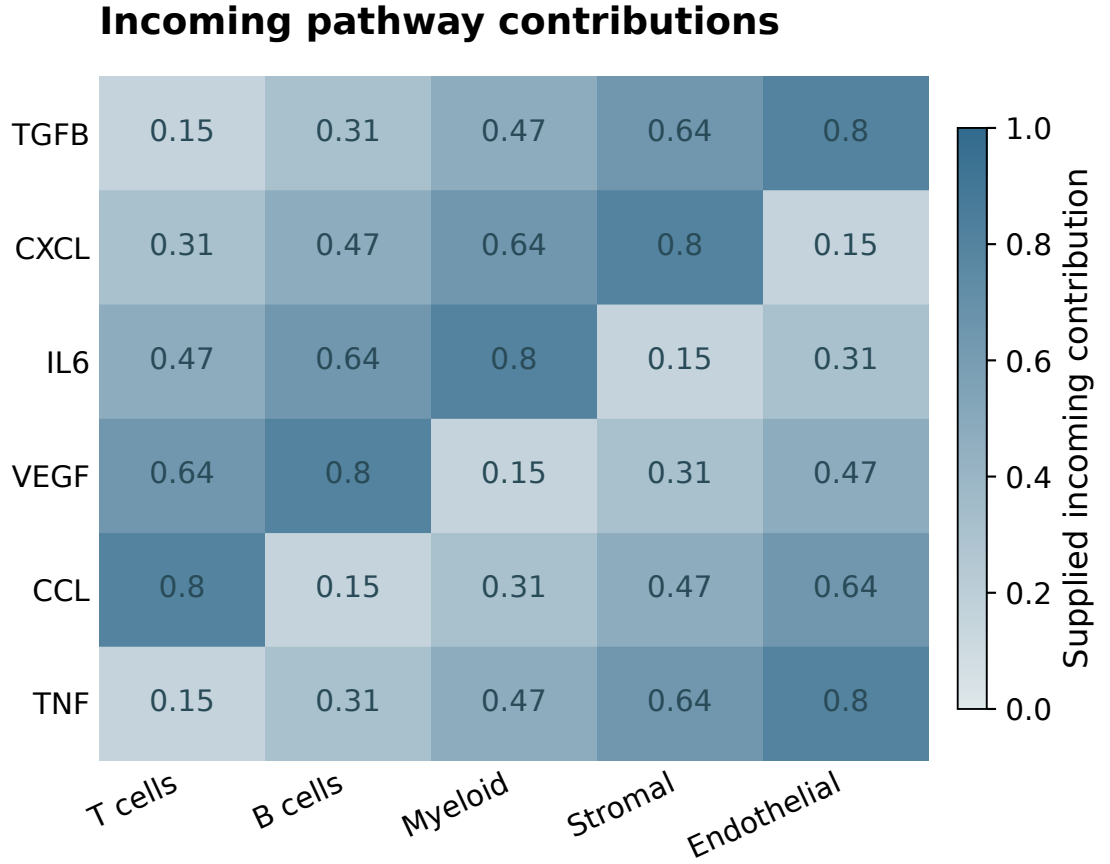
Estimate [interval] · q Independent n = 24



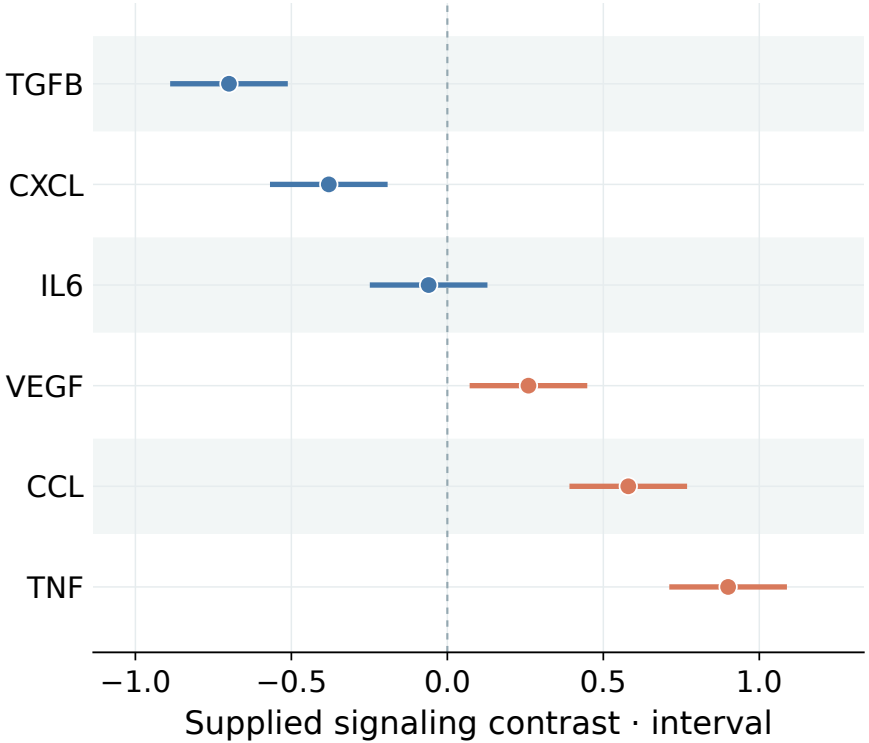
STAT1 · Primary	-0.22 [-0.30, -0.14] 0.002
STAT1 · Alternative	-0.14 [-0.22, -0.06] 0.005
IRF7 · Primary	-0.16 [-0.24, -0.08] 0.02
IRF7 · Alternative	-0.08 [-0.16, 0.00] 0.65
ISG15 · Primary	-0.10 [-0.18, -0.02] 0.038
ISG15 · Alternative	-0.02 [-0.10, 0.06] 0.65
COL1A1 · Primary	-0.03 [-0.11, 0.05] 0.65
COL1A1 · Alternative	0.05 [-0.03, 0.13] 0.65
NDUFA1 · Primary	0.03 [-0.05, 0.11] 0.65
NDUFA1 · Alternative	0.11 [0.03, 0.19] 0.077
SRSF1 · Primary	0.10 [0.02, 0.18] 0.092
SRSF1 · Alternative	0.18 [0.10, 0.26] 0.095
MKI67 · Primary	0.16 [0.08, 0.24] 0.11
MKI67 · Alternative	0.24 [0.16, 0.32] 0.11
KLF2 · Primary	0.22 [0.14, 0.30] 0.13
KLF2 · Alternative	0.30 [0.22, 0.38] 0.13
JUN · Primary	0.29 [0.21, 0.37] 0.15
JUN · Alternative	0.37 [0.29, 0.45] 0.15


Ligand and receptor expression support
TGFB1
TGFBR1
CXCL12
CXCR4
IL6
IL6R
0.15 0.31 0.47 0.64 0.8
0.31 0.47 0.64 0.8 0.15
0.47 0.64 0.8 0.15 0.31
0.64 0.8 0.15 0.31 0.47
0.8 0.15 0.31 0.47 0.64
0.15 0.31 0.47 0.64 0.8
T cells
B cells
Myeloid
Stromal
Endothelial
Fraction detected
1.0
0.8
0.6
0.4
0.2
0.0


Outgoing pathway contributions
TGFB
CXCL
IL6
VEGF
CCL
TNF
0.15
0.31
0.47
0.64
0.8
0.31
0.47
0.64
0.8
0.15
0.47
0.64
0.8
0.15
0.31
0.64
0.8
0.15
0.31
0.47
0.8
0.15
0.31
0.47
0.64
0.15
0.31
0.47
0.64
0.8
T cells
B cells
Myeloid
Stromal
Endothelial
Supplied outgoing contribution
1.0
0.8
0.6
0.4
0.2
0.0


Incoming pathway contributions
TGFB
CXCL
IL6
VEGF
CCL
TNF
0.15
0.31
0.47
0.64
0.8
0.31
0.47
0.64
0.8
0.15
0.47
0.64
0.8
0.15
0.31
0.64
0.8
0.15
0.31
0.47
0.8
0.15
0.31
0.47
0.64
0.15
0.31
0.47
0.64
0.8
T cells
B cells
Myeloid
Stromal
Endothelial
Supplied incoming contribution
1.0
0.8
0.6
0.4
0.2
0.0

Donor-aware signaling contrasts



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

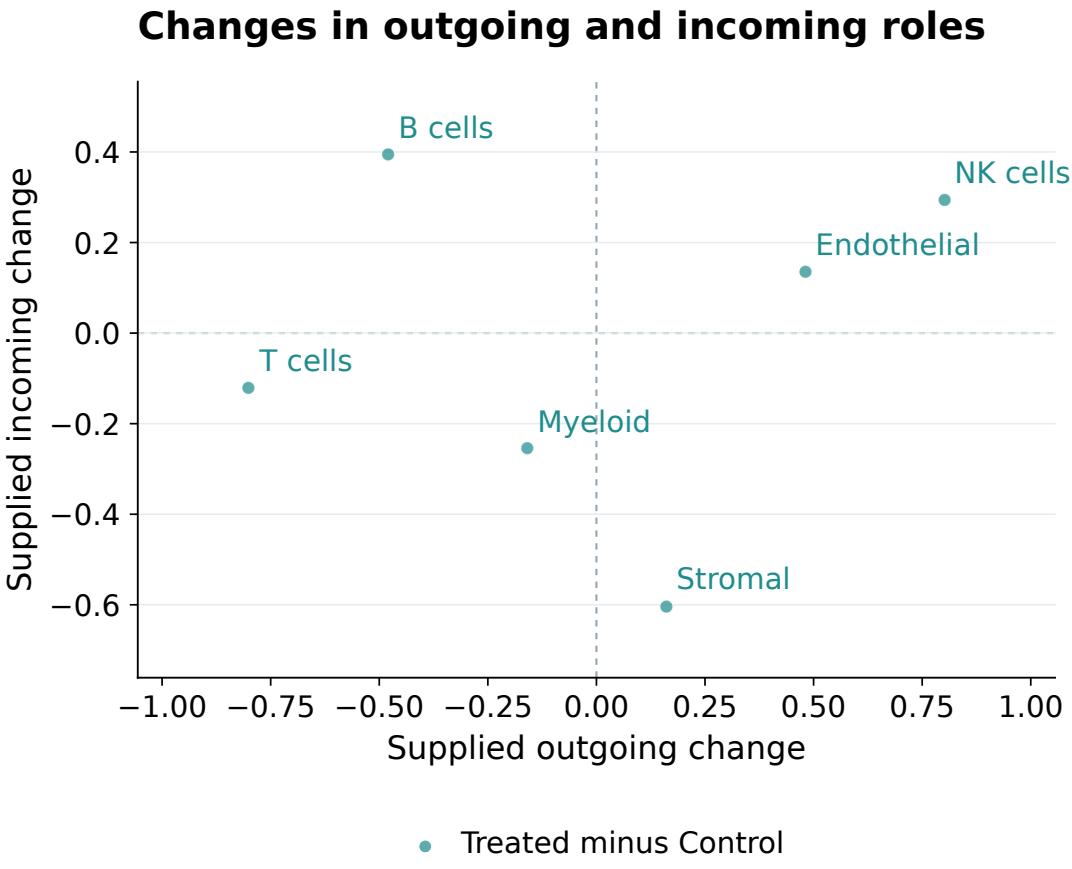
-0.38 [-0.56, -0.20] 0.02

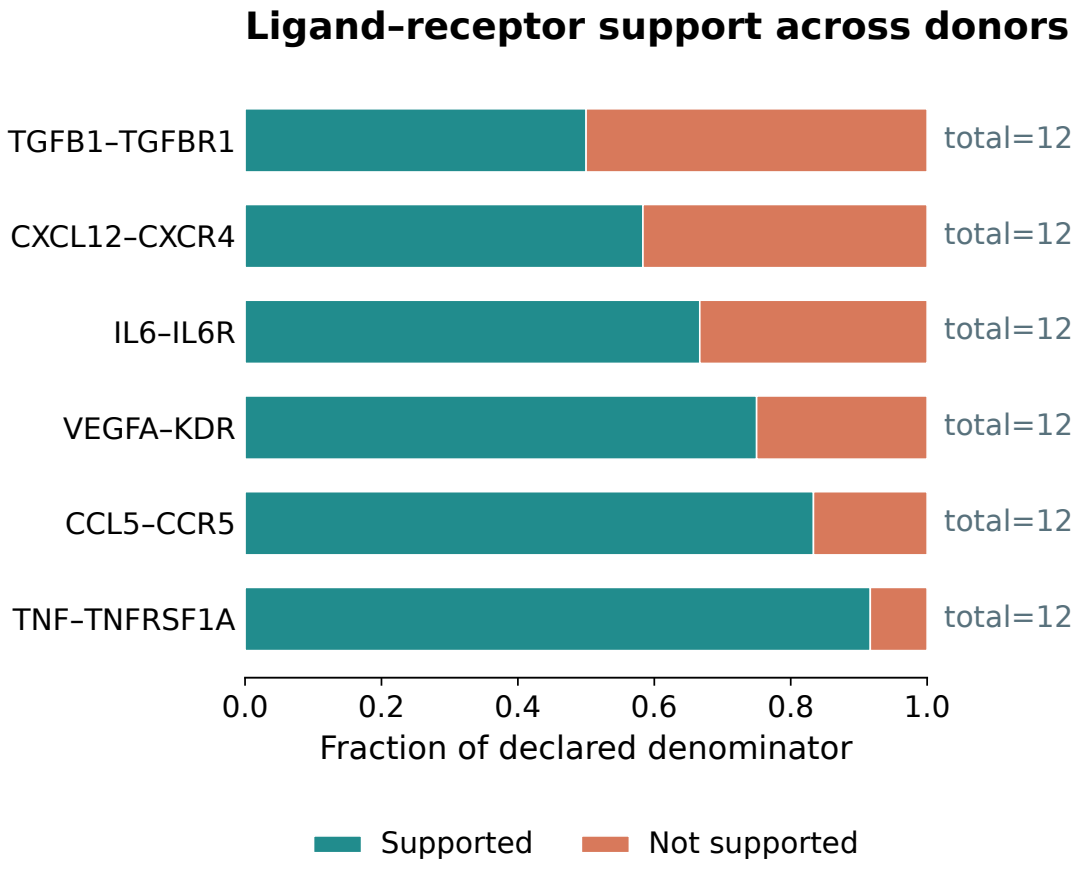
-0.06 [-0.24, 0.12] 0.65

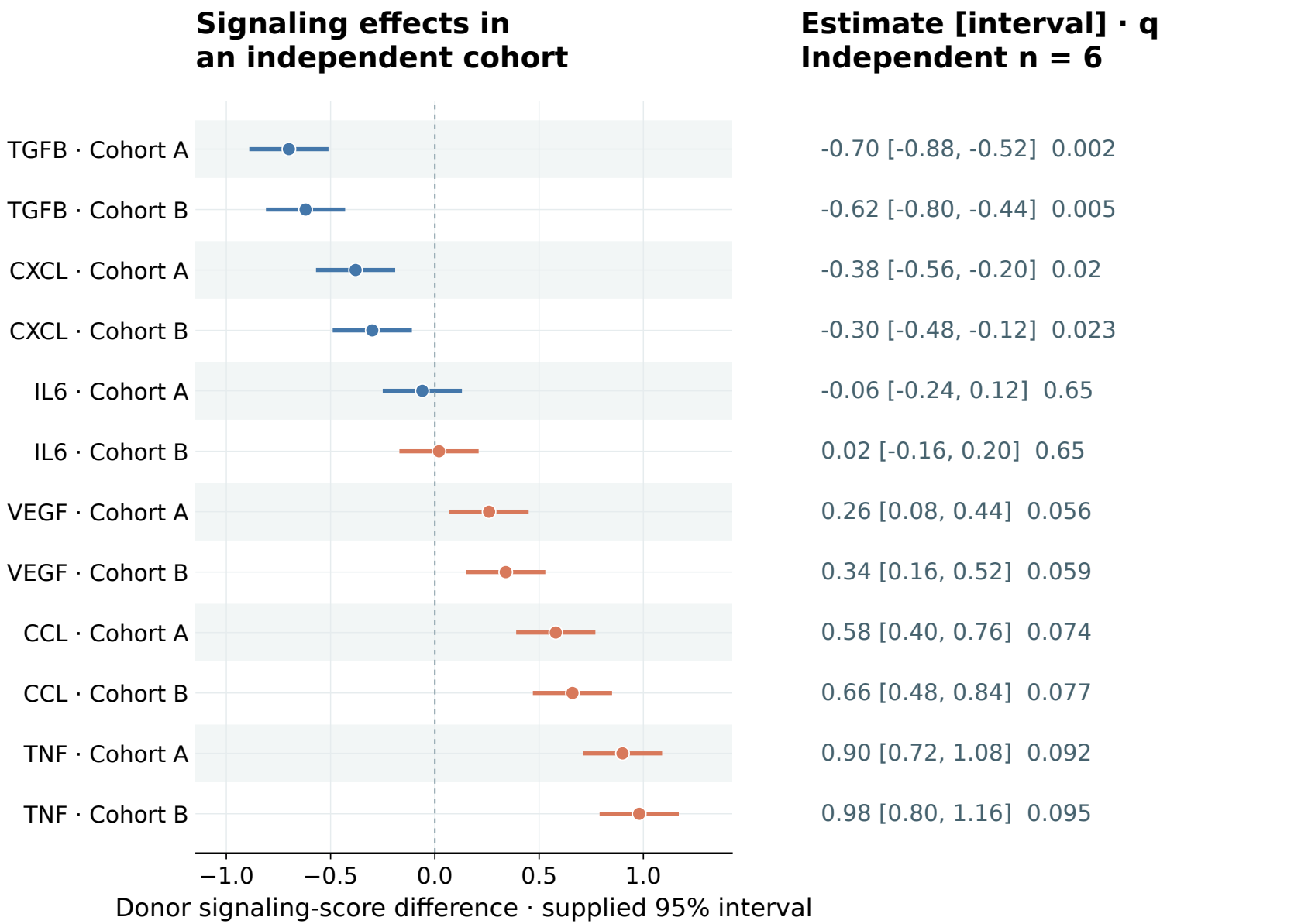
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092


Changes in outgoing and incoming roles
B cells
NK cells
Endothelial
T cells
Myeloid
Stromal
Supplied incoming change
0.4
0.2
0.0
−0.2
−0.4
−0.6
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Supplied outgoing change
Treated minus Control

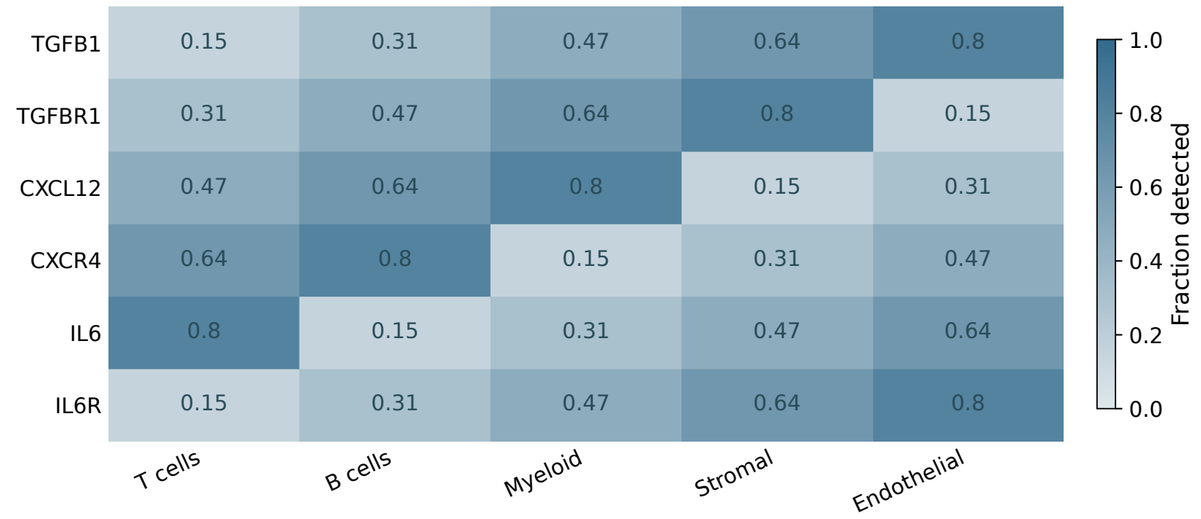

Ligand–receptor support across donors
TGFB1–TGFBR1
CXCL12–CXCR4
IL6–IL6R
VEGFA–KDR
CCL5–CCR5
TNF–TNFRSF1A
total=12
total=12
total=12
total=12
total=12
total=12
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Supported
Not supported


Signaling effects in an independent cohort
Estimate [interval] · q
Independent n = 6
TGFB · Cohort A
TGFB · Cohort B
CXCL · Cohort A
CXCL · Cohort B
IL6 · Cohort A
IL6 · Cohort B
VEGF · Cohort A
VEGF · Cohort B
CCL · Cohort A
CCL · Cohort B
TNF · Cohort A
TNF · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Donor signaling-score difference · supplied 95% interval

Single cell · communication — complementary evidence

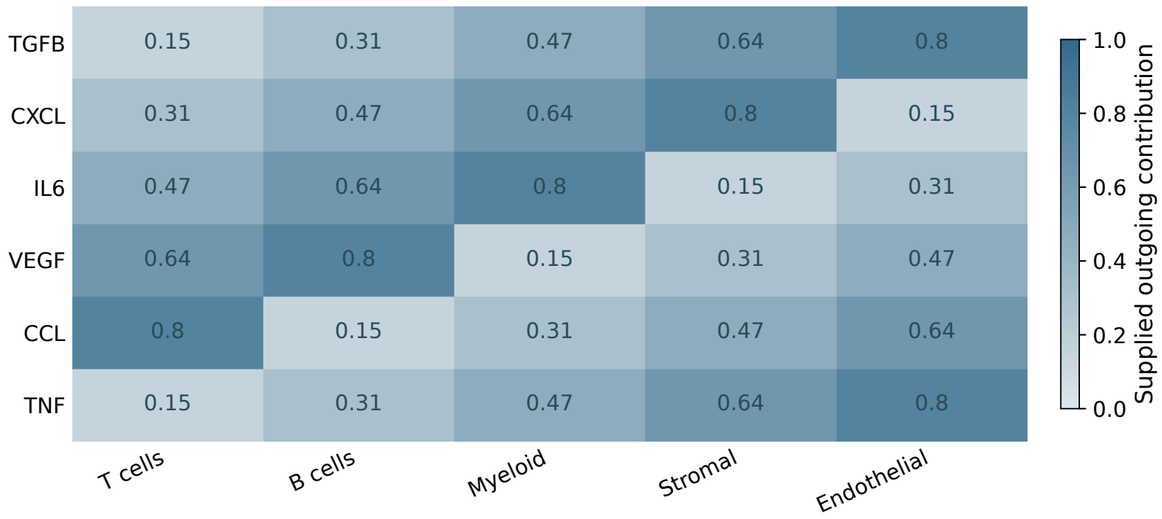
A

Ligand and receptor expression support



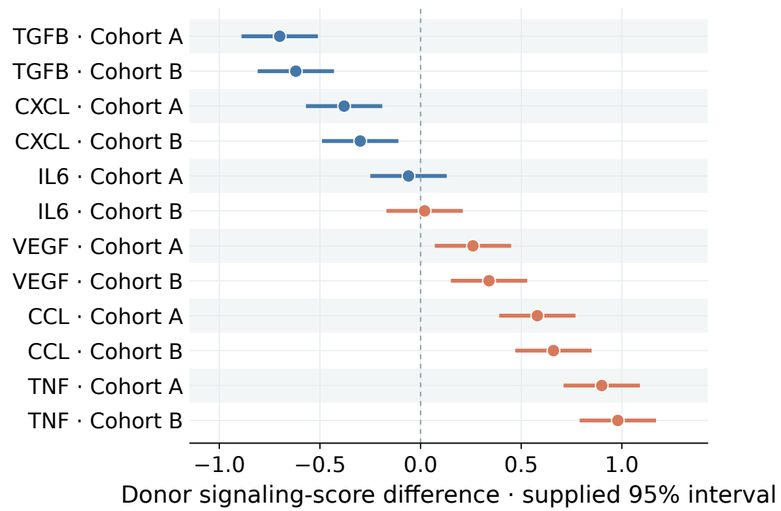
B

Outgoing pathway contributions



C

Signaling effects in an independent cohort



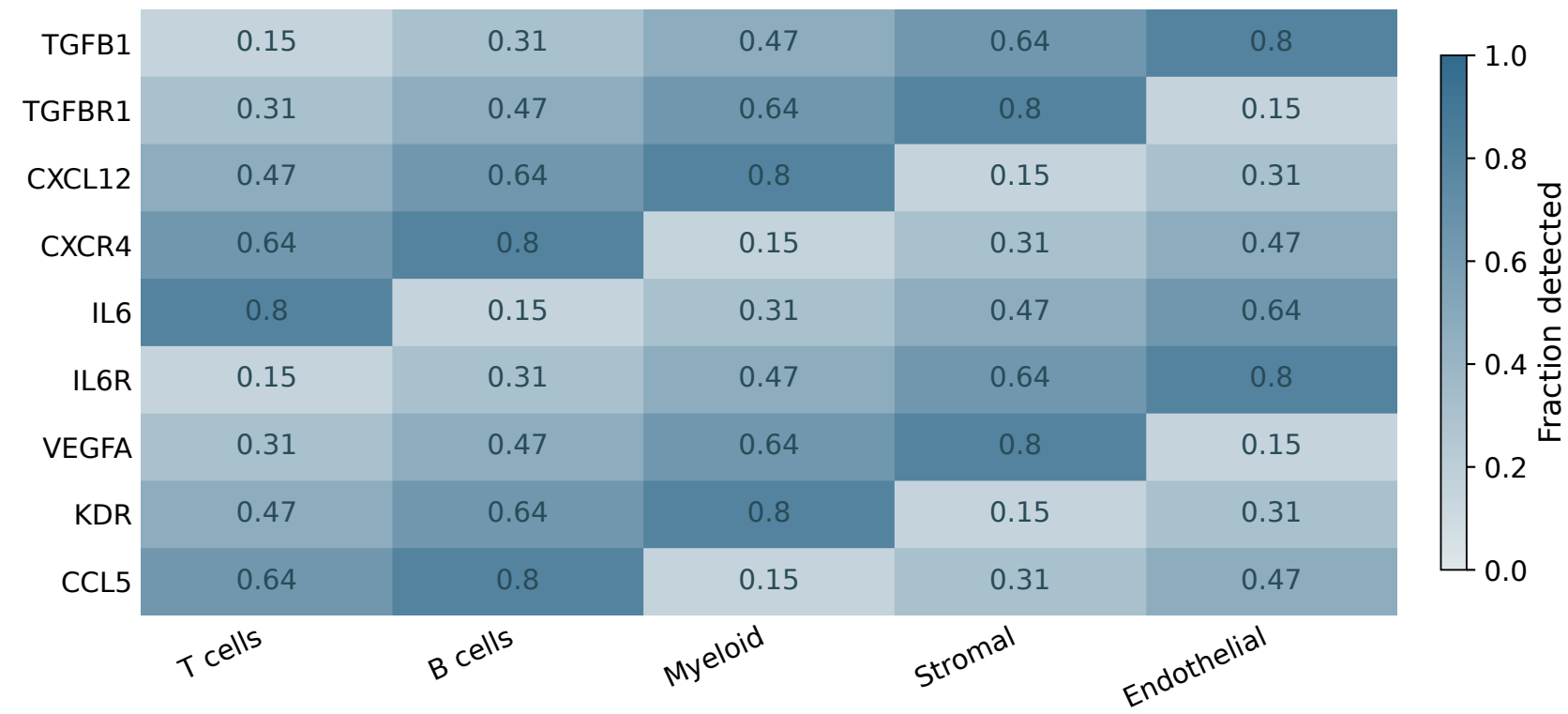
Estimate [interval] · q
Independent n = 6

TGFB · Cohort A	-0.70 [-0.88, -0.52] 0.002
TGFB · Cohort B	-0.62 [-0.80, -0.44] 0.005
CXCL · Cohort A	-0.38 [-0.56, -0.20] 0.02
CXCL · Cohort B	-0.30 [-0.48, -0.12] 0.023
IL6 · Cohort A	-0.06 [-0.24, 0.12] 0.65
IL6 · Cohort B	0.02 [-0.16, 0.20] 0.65
VEGF · Cohort A	0.26 [0.08, 0.44] 0.056
VEGF · Cohort B	0.34 [0.16, 0.52] 0.059
CCL · Cohort A	0.58 [0.40, 0.76] 0.074
CCL · Cohort B	0.66 [0.48, 0.84] 0.077
TNF · Cohort A	0.90 [0.72, 1.08] 0.092
TNF · Cohort B	0.98 [0.80, 1.16] 0.095

Single cell · communication — complementary evidence

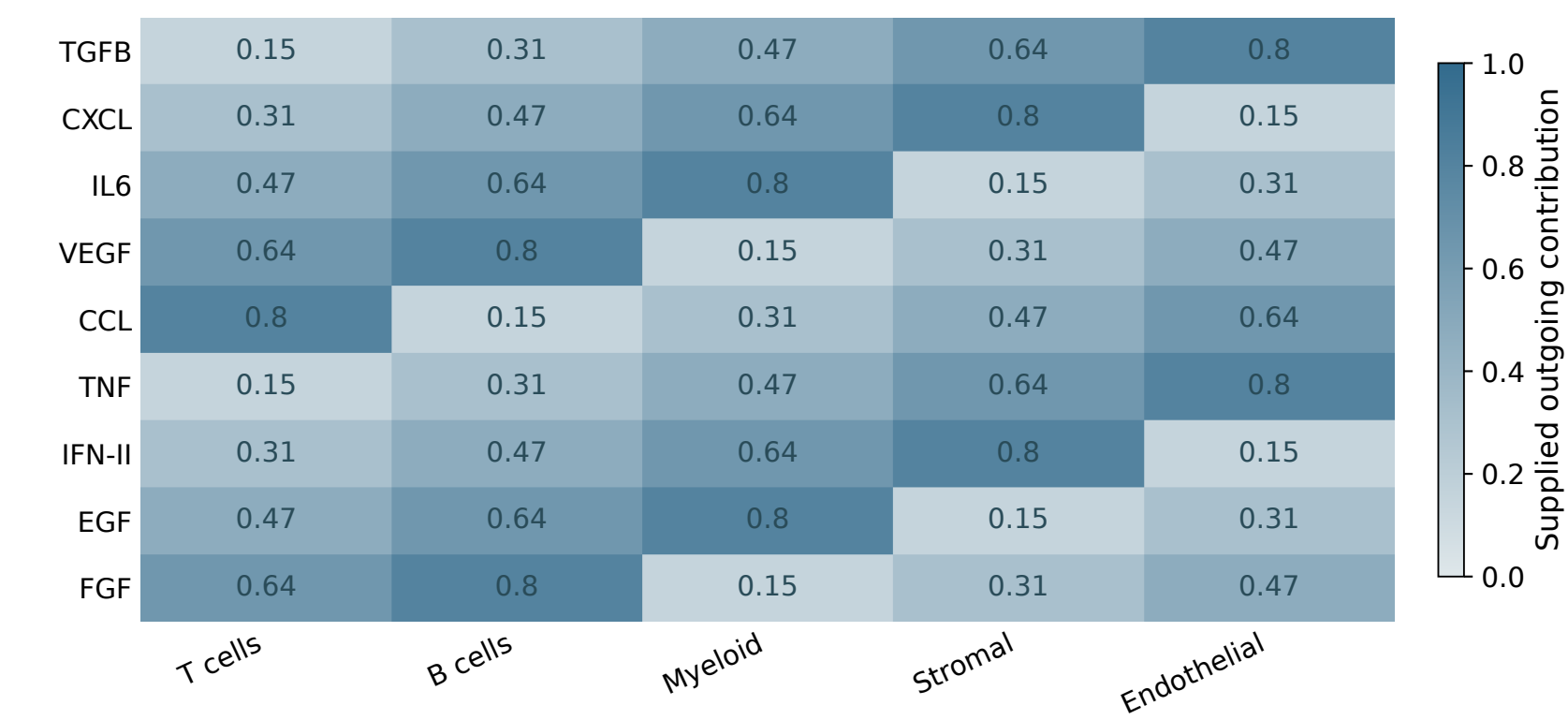
A

Ligand and receptor expression support



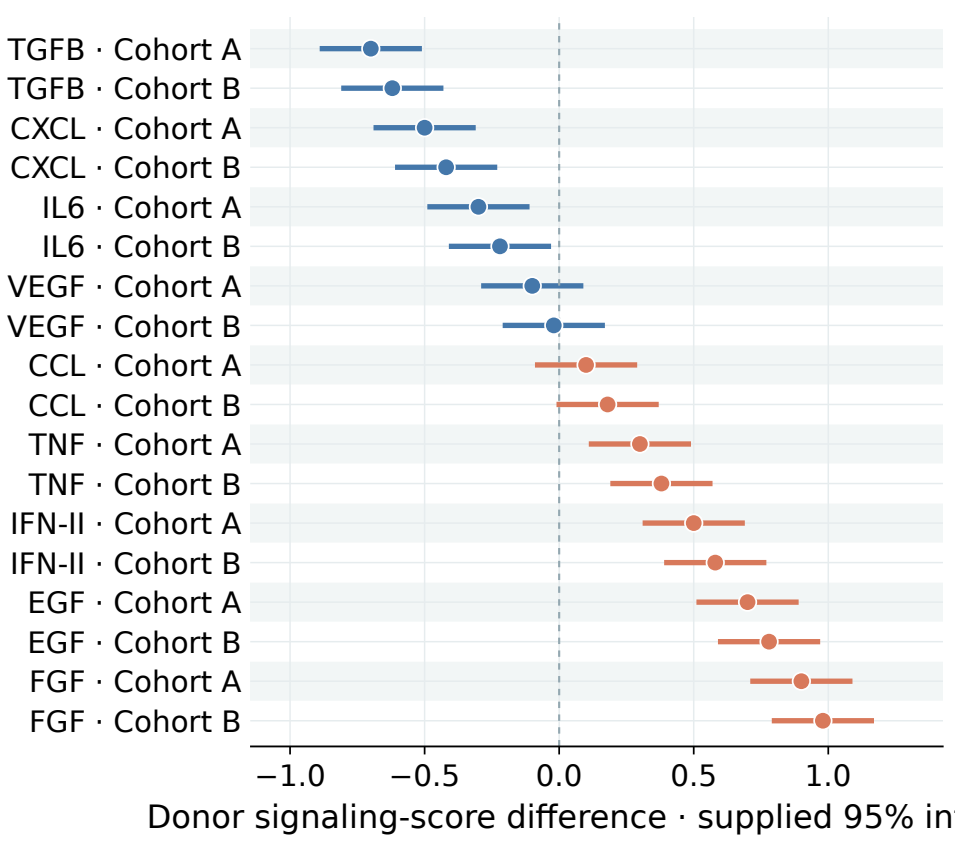
B

Outgoing pathway contributions



C

Signaling effects in an independent cohort



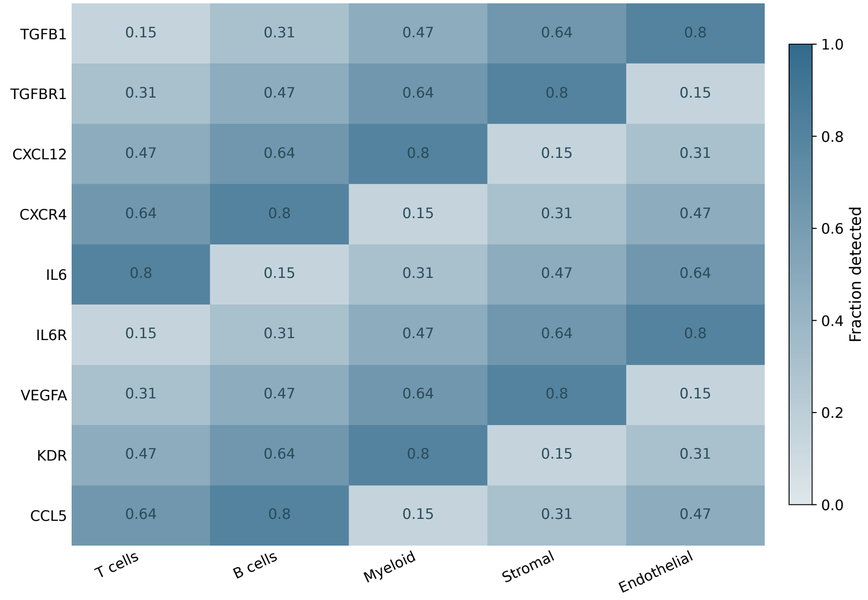
Estimate [interval] · q
Independent n = 12

TGFB · Cohort A	-0.70 [-0.88, -0.52] 0.002
TGFB · Cohort B	-0.62 [-0.80, -0.44] 0.005
CXCL · Cohort A	-0.50 [-0.68, -0.32] 0.02
CXCL · Cohort B	-0.42 [-0.60, -0.24] 0.023
IL6 · Cohort A	-0.30 [-0.48, -0.12] 0.038
IL6 · Cohort B	-0.22 [-0.40, -0.04] 0.041
VEGF · Cohort A	-0.10 [-0.28, 0.08] 0.65
VEGF · Cohort B	-0.02 [-0.20, 0.16] 0.65
CCL · Cohort A	0.10 [-0.08, 0.28] 0.65
CCL · Cohort B	0.18 [0.00, 0.36] 0.077
TNF · Cohort A	0.30 [0.12, 0.48] 0.092
TNF · Cohort B	0.38 [0.20, 0.56] 0.095
IFN-II · Cohort A	0.50 [0.32, 0.68] 0.11
IFN-II · Cohort B	0.58 [0.40, 0.76] 0.11
EGF · Cohort A	0.70 [0.52, 0.88] 0.13
EGF · Cohort B	0.78 [0.60, 0.96] 0.13
FGF · Cohort A	0.90 [0.72, 1.08] 0.15
FGF · Cohort B	0.98 [0.80, 1.16] 0.15

Single cell · communication — complementary evidence

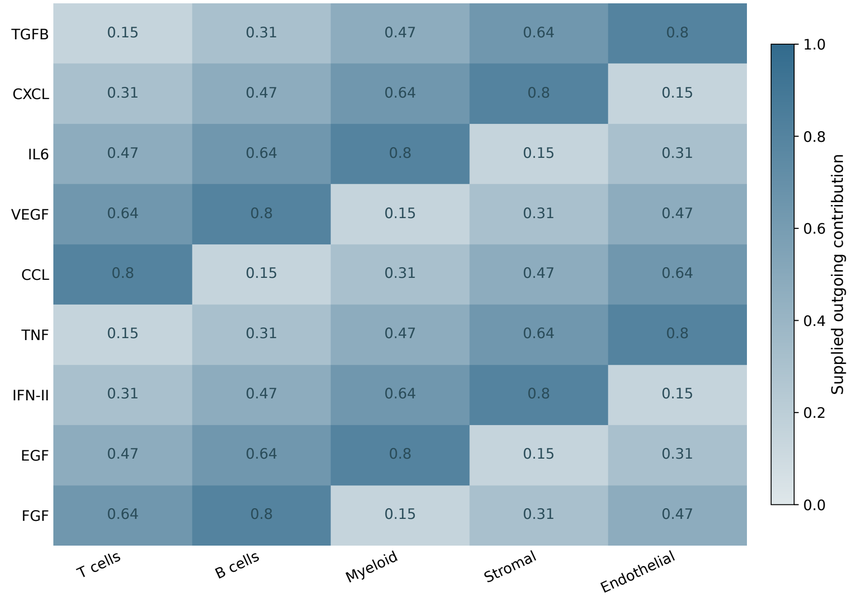
A

Ligand and receptor expression support



B

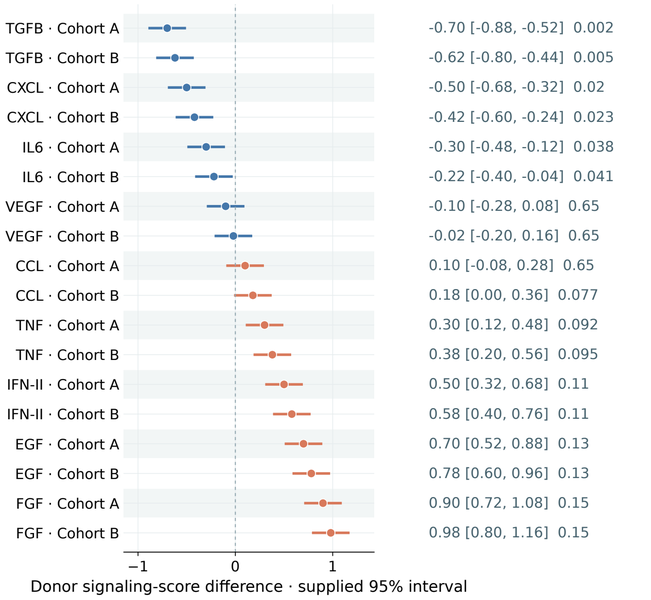
Outgoing pathway contributions

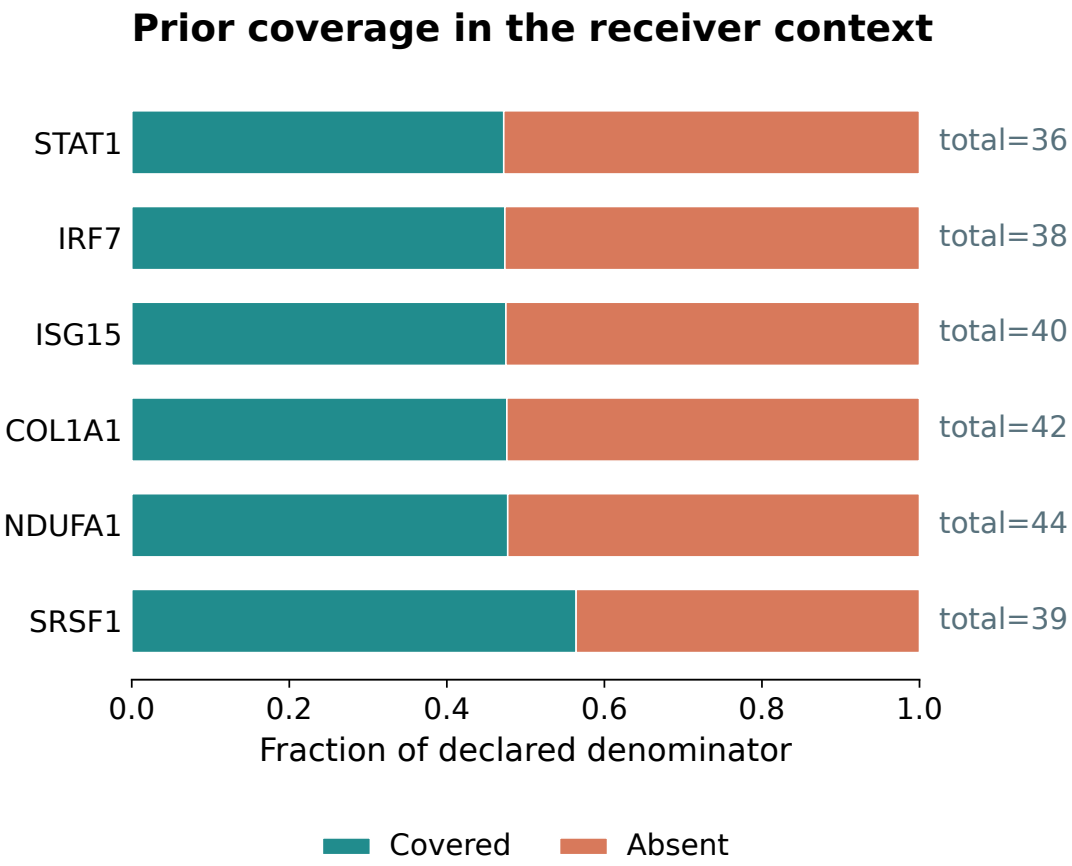


C

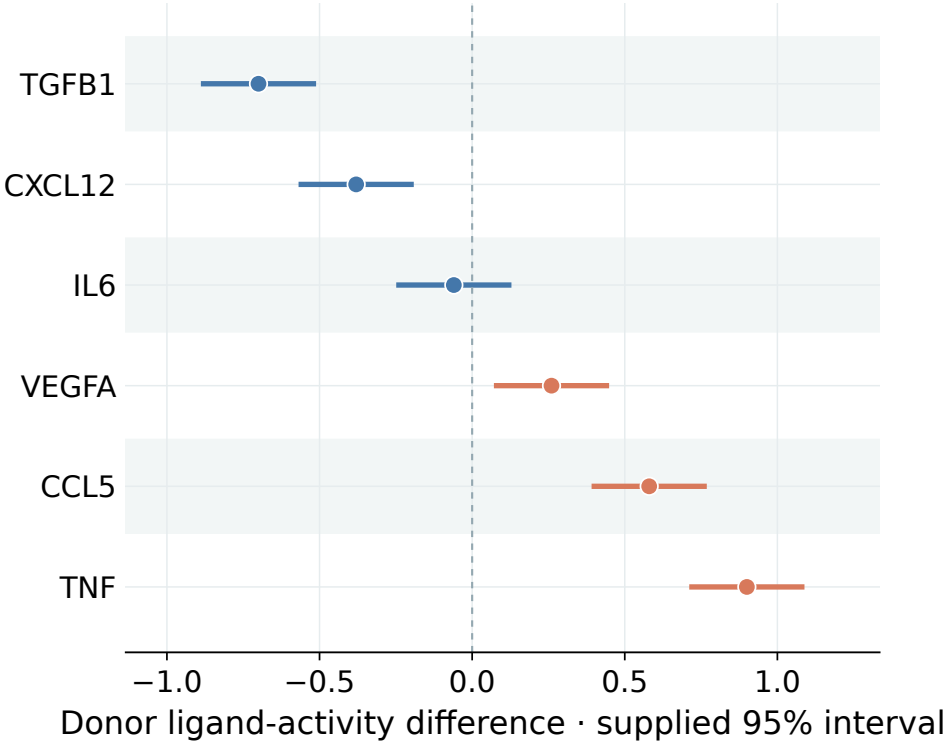
Signaling effects in an independent cohort

Estimate [interval] · q
Independent n = 12




Prior coverage in the receiver context
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Covered
Absent

Prespecified ligand-activity contrasts



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

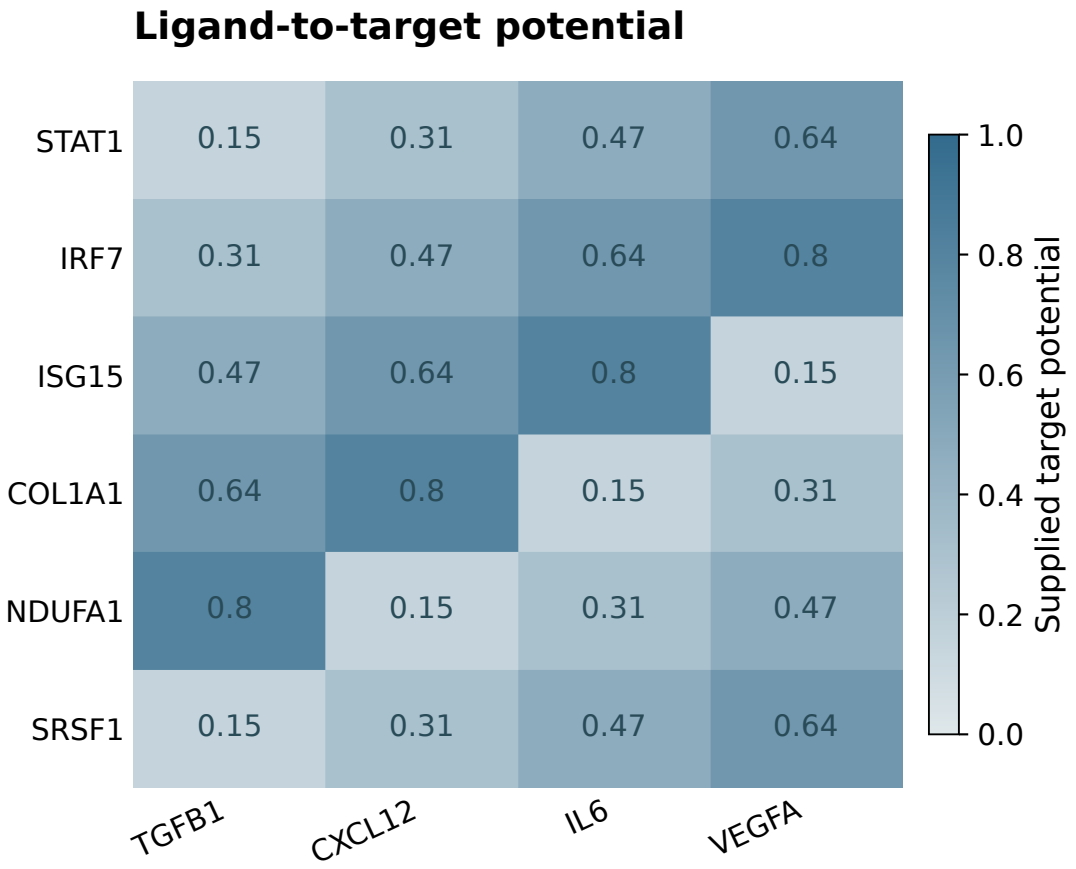
-0.38 [-0.56, -0.20] 0.02

-0.06 [-0.24, 0.12] 0.65

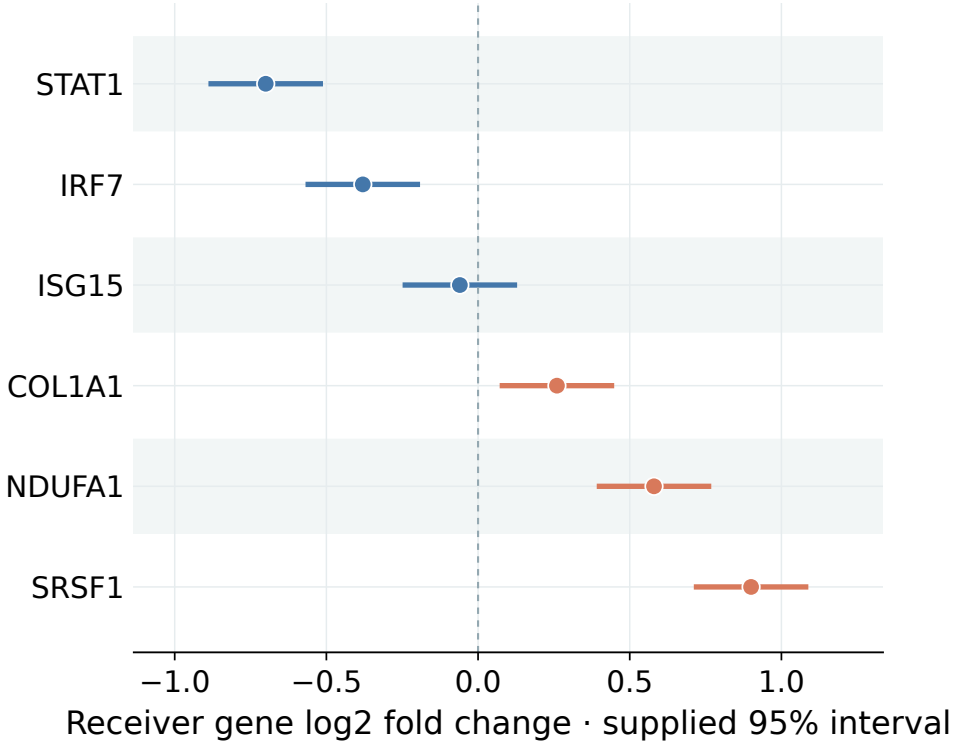
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092


Ligand-to-target potential
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0.15
0.31
0.47
0.64
0.31
0.47
0.64
0.8
0.47
0.64
0.8
0.15
0.64
0.8
0.15
0.31
0.8
0.15
0.31
0.47
0.15
0.31
0.47
0.64
TGFB1
CXCL12
IL6
VEGFA
1.0
0.8
0.6
0.4
0.2
0.0
Supplied target potential

Receiver target-gene differential response



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

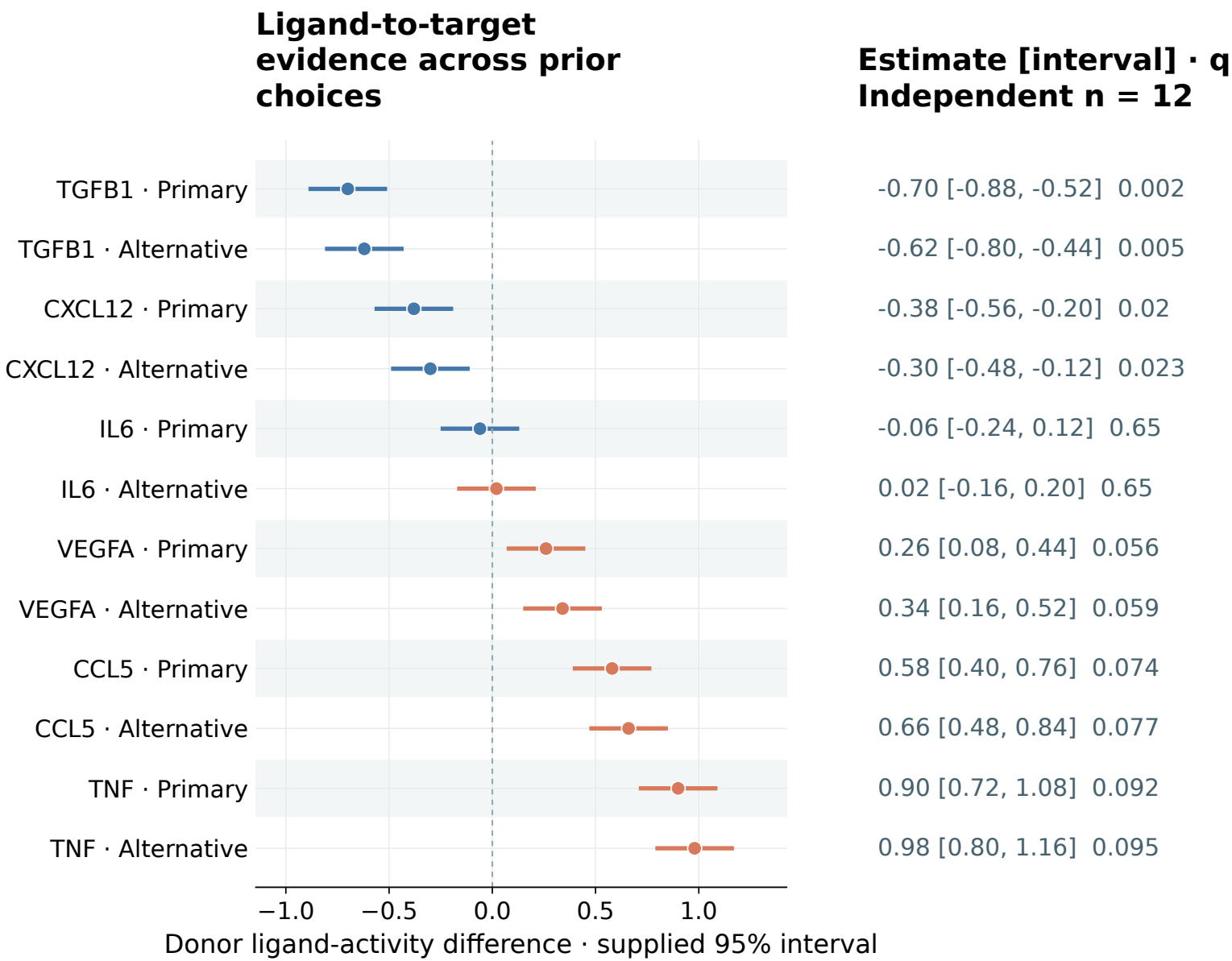
-0.38 [-0.56, -0.20] 0.02

-0.06 [-0.24, 0.12] 0.65

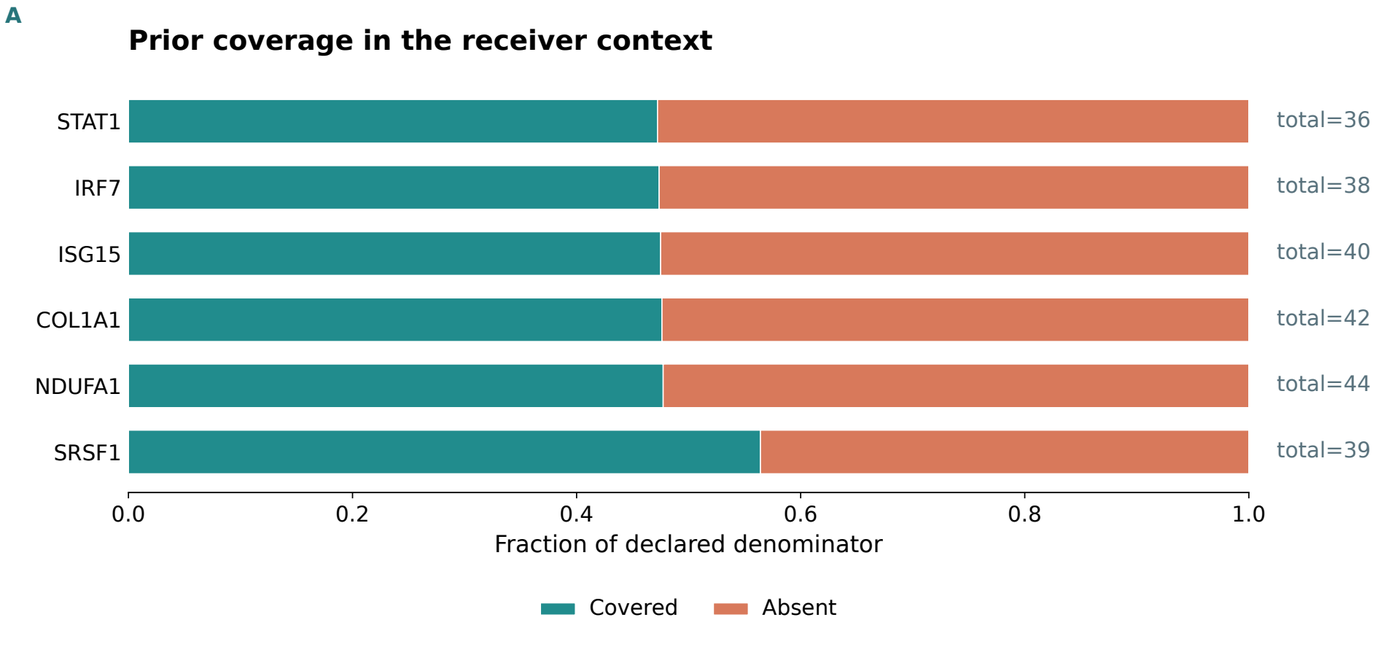
0.26 [0.08, 0.44] 0.056

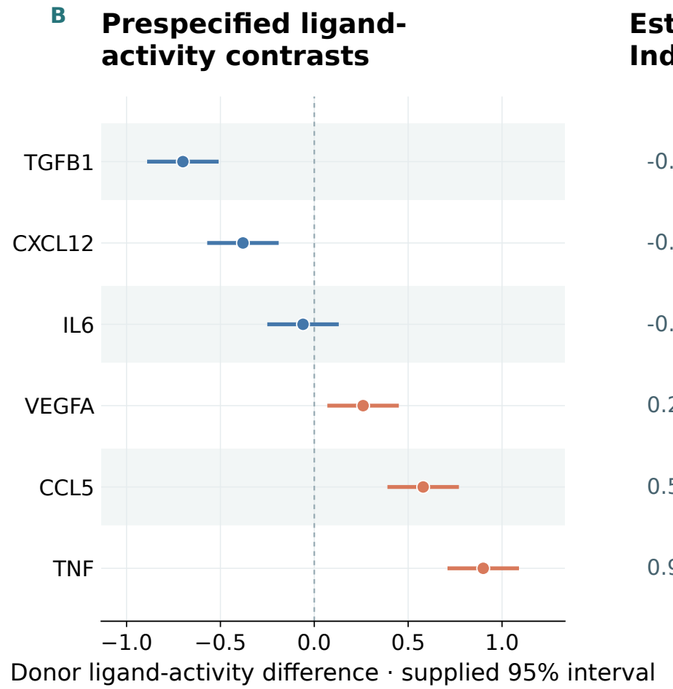
0.58 [0.40, 0.76] 0.074

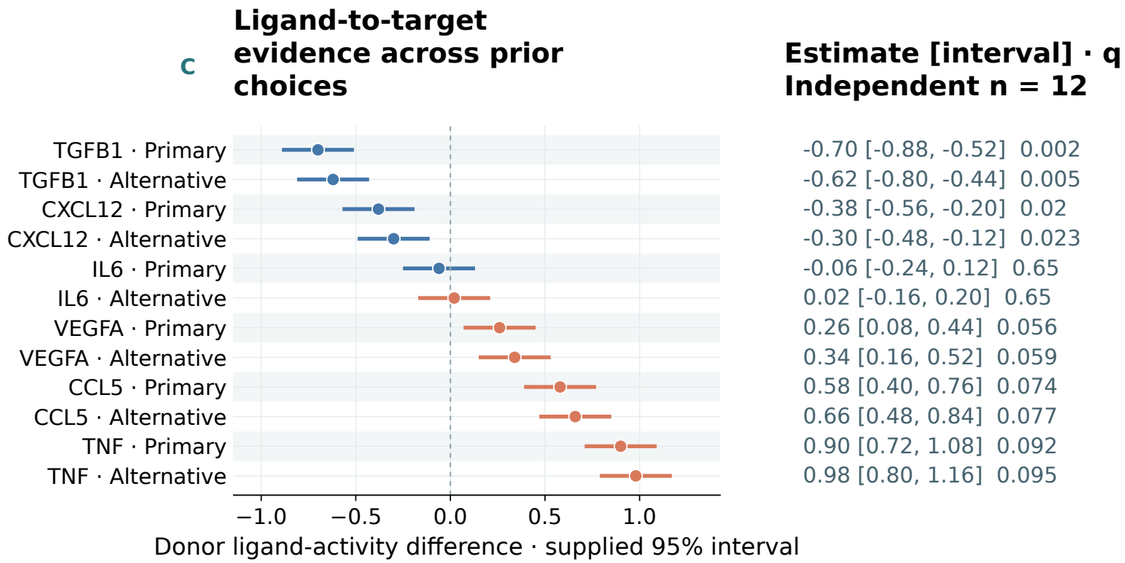
0.90 [0.72, 1.08] 0.092


Ligand-to-target evidence across prior choices
Estimate [interval] · q
Independent n = 12
TGFB1 · Primary
TGFB1 · Alternative
CXCL12 · Primary
CXCL12 · Alternative
IL6 · Primary
IL6 · Alternative
VEGFA · Primary
VEGFA · Alternative
CCL5 · Primary
CCL5 · Alternative
TNF · Primary
TNF · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Donor ligand-activity difference · supplied 95% interval

Single cell · ligand_target — complementary evidence



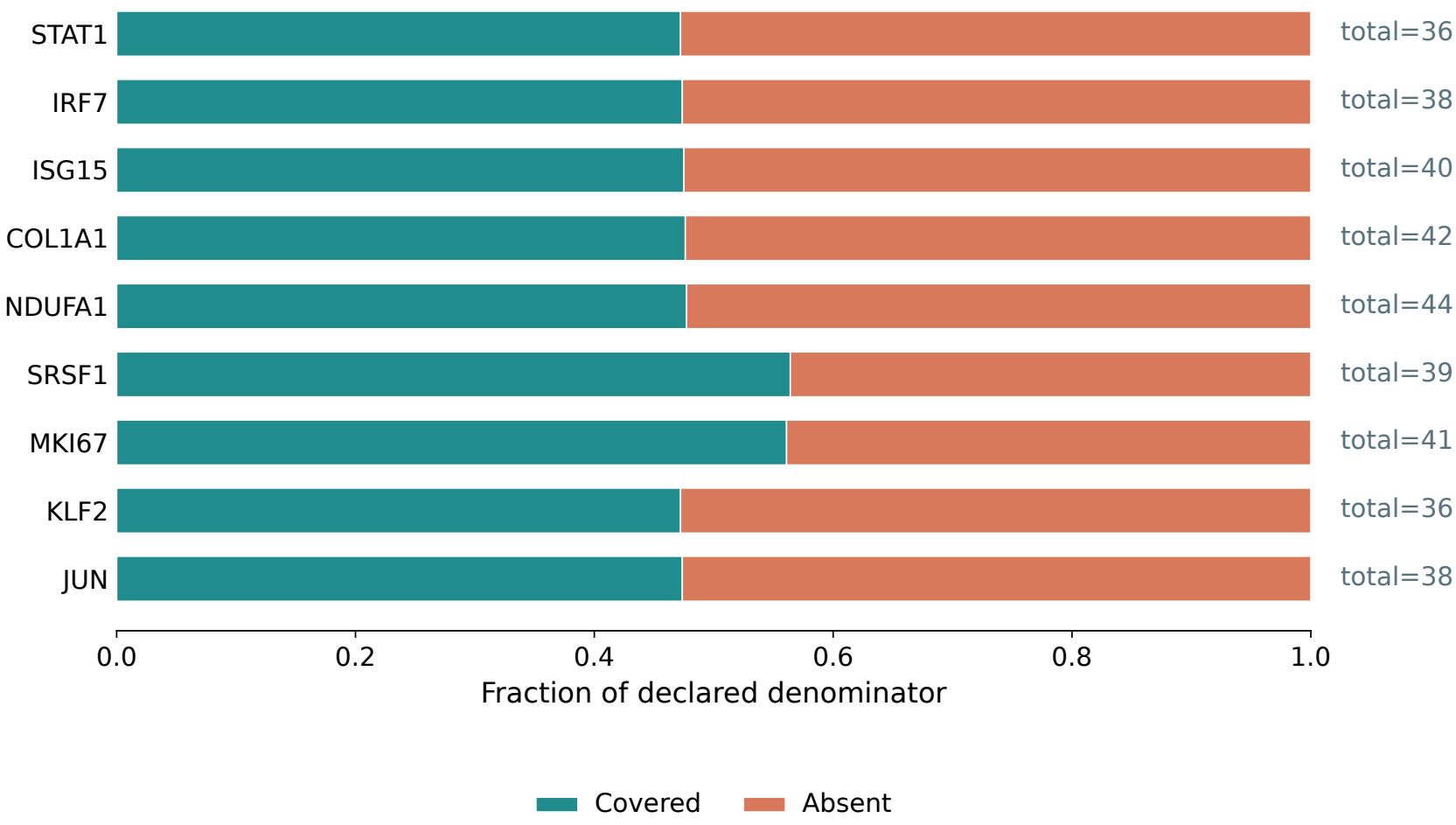




Single cell · ligand_target — complementary evidence

A

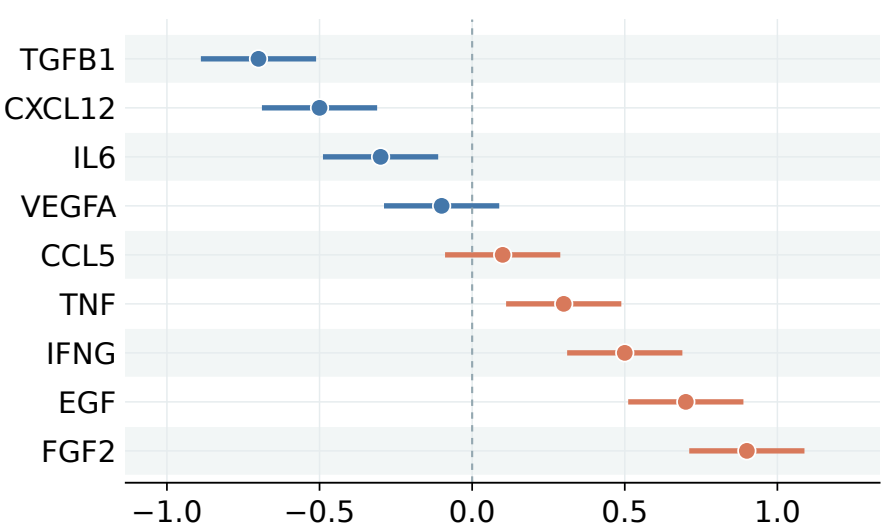
Prior coverage in the receiver context



B

Prespecified ligand-activity contrasts

Estimate [interval] · q
Independent n = 24

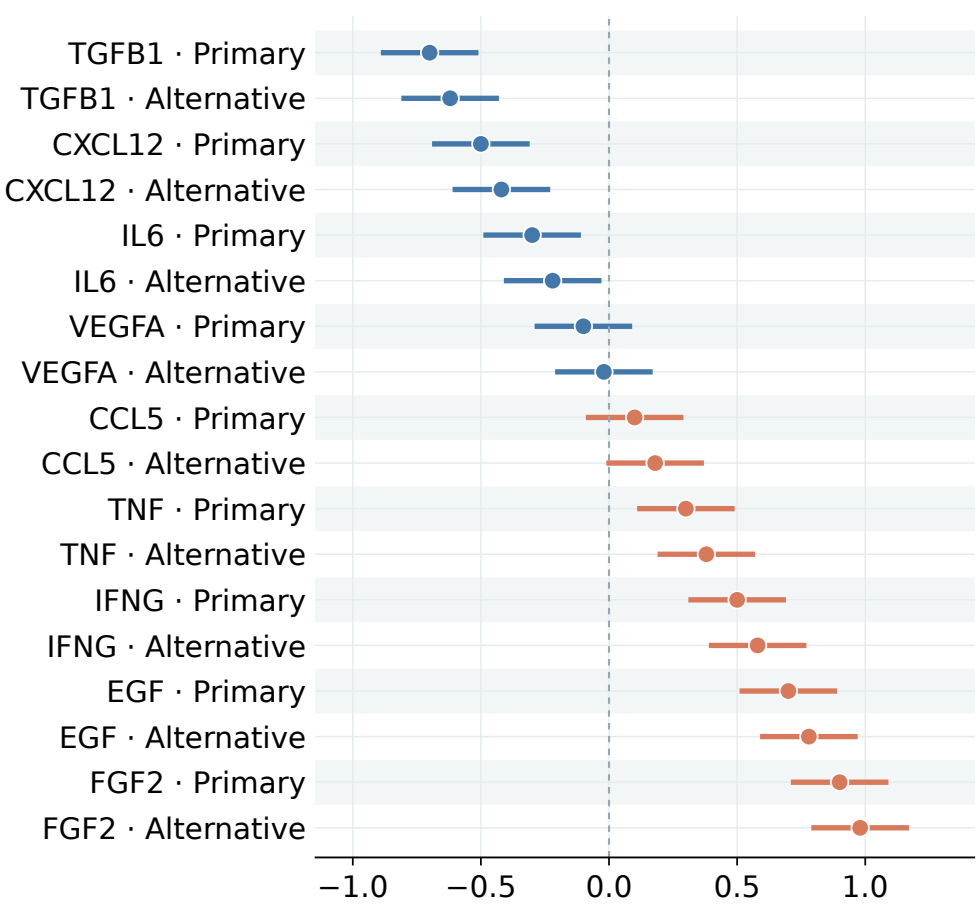


-0.70 [-0.88, -0.52] 0.002
-0.50 [-0.68, -0.32] 0.02
-0.30 [-0.48, -0.12] 0.038
-0.10 [-0.28, 0.08] 0.65
0.10 [-0.08, 0.28] 0.65
0.30 [0.12, 0.48] 0.092
0.50 [0.32, 0.68] 0.11
0.70 [0.52, 0.88] 0.13
0.90 [0.72, 1.08] 0.15

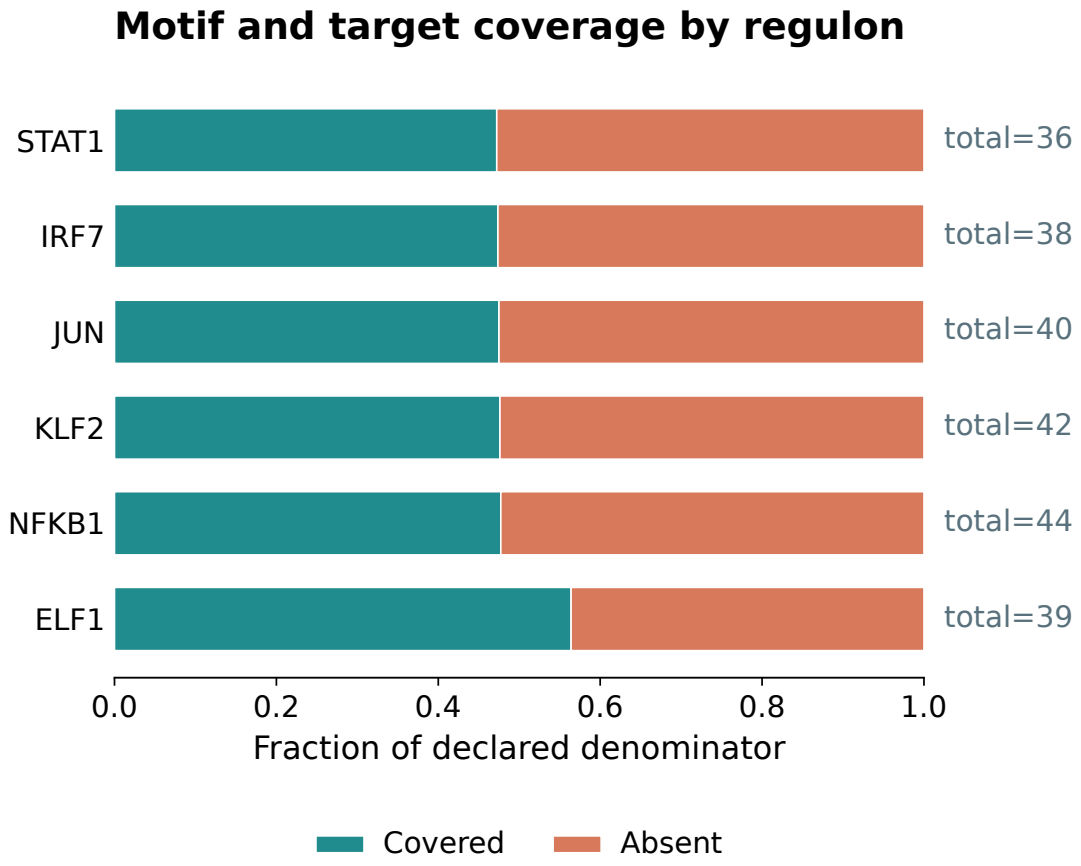
C

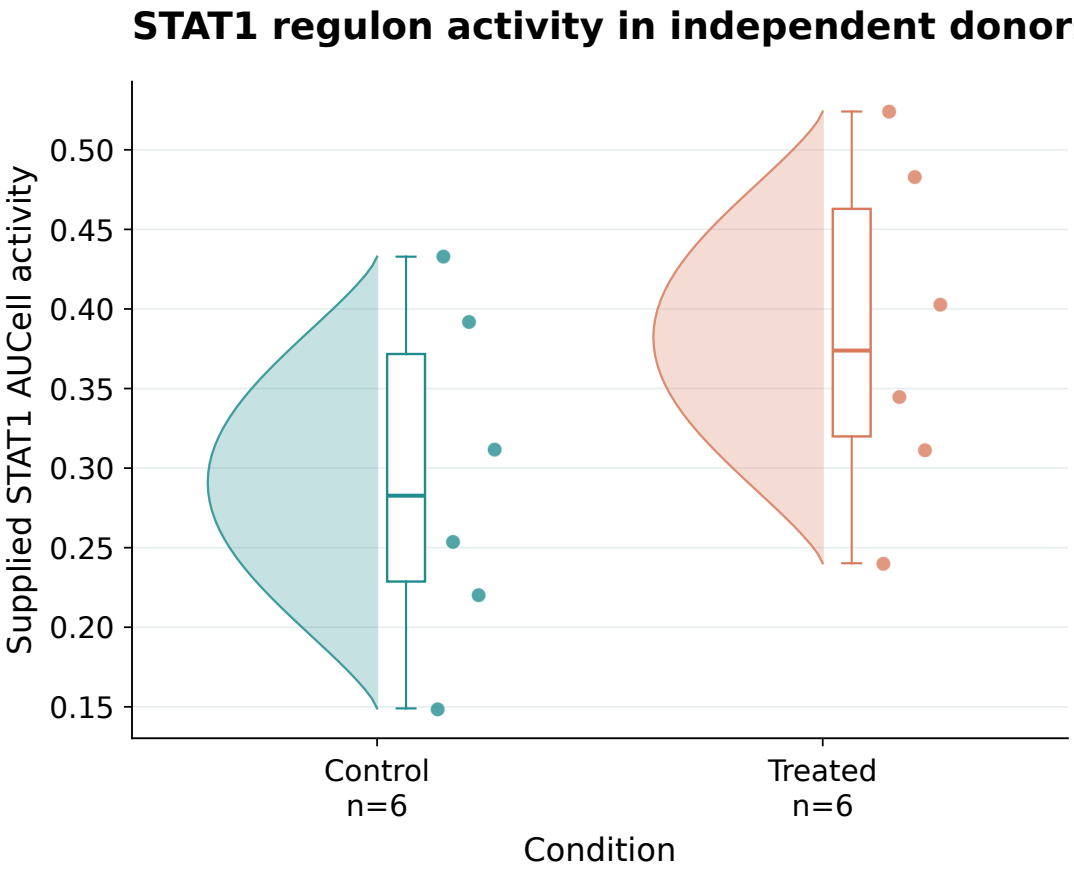
Ligand-to-target evidence across prior choices

Estimate [interval] · q
Independent n = 24



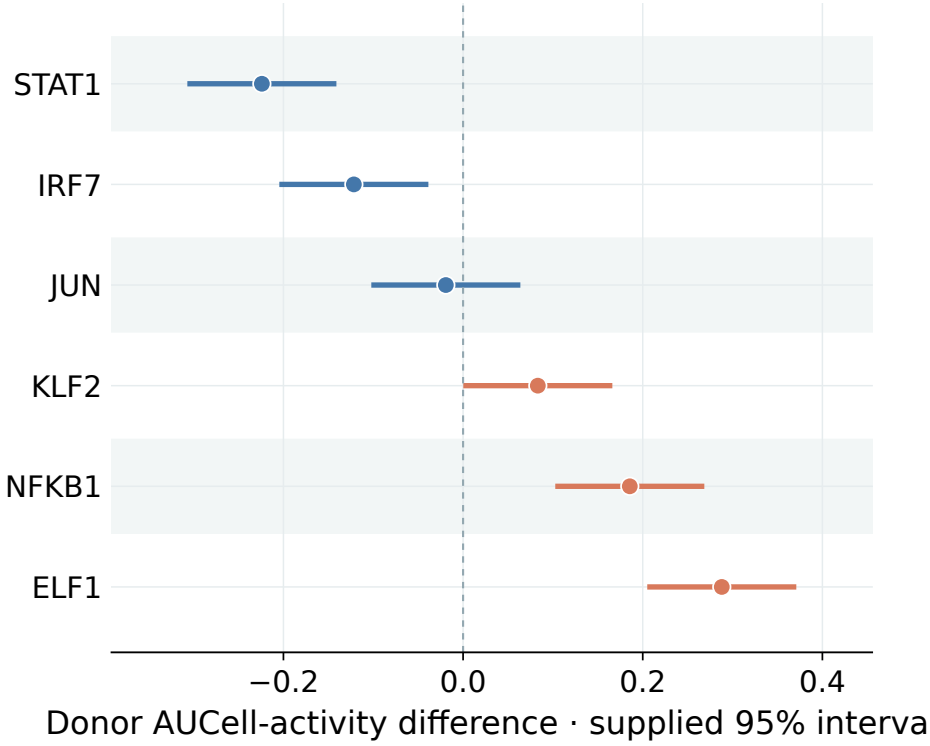
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.50 [-0.68, -0.32] 0.02
-0.42 [-0.60, -0.24] 0.023
-0.30 [-0.48, -0.12] 0.038
-0.22 [-0.40, -0.04] 0.041
-0.10 [-0.28, 0.08] 0.65
-0.02 [-0.20, 0.16] 0.65
0.10 [-0.08, 0.28] 0.65
0.18 [0.00, 0.36] 0.077
0.30 [0.12, 0.48] 0.092
0.38 [0.20, 0.56] 0.095
0.50 [0.32, 0.68] 0.11
0.58 [0.40, 0.76] 0.11
0.70 [0.52, 0.88] 0.13
0.78 [0.60, 0.96] 0.13
0.90 [0.72, 1.08] 0.15
0.98 [0.80, 1.16] 0.15


Motif and target coverage by regulon
STAT1
IRF7
JUN
KLF2
NFKB1
ELF1
total=36
total=38
total=40
total=42
total=44
total=39
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Covered
Absent



Software test fixture; not biological evidence

Donor-level regulon activity effects



Estimate [interval] · q
Independent n = 12

-0.22 [-0.30, -0.14] 0.002

-0.12 [-0.20, -0.04] 0.02

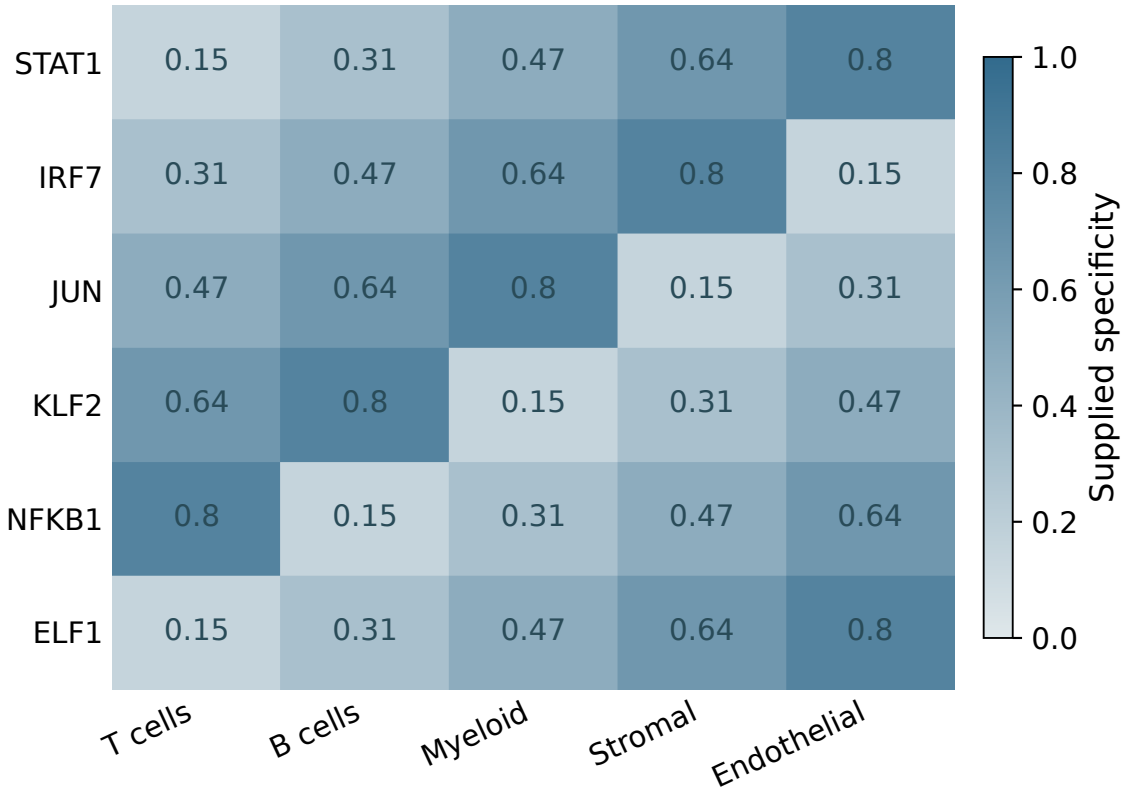
-0.02 [-0.10, 0.06] 0.65

0.08 [0.00, 0.16] 0.056

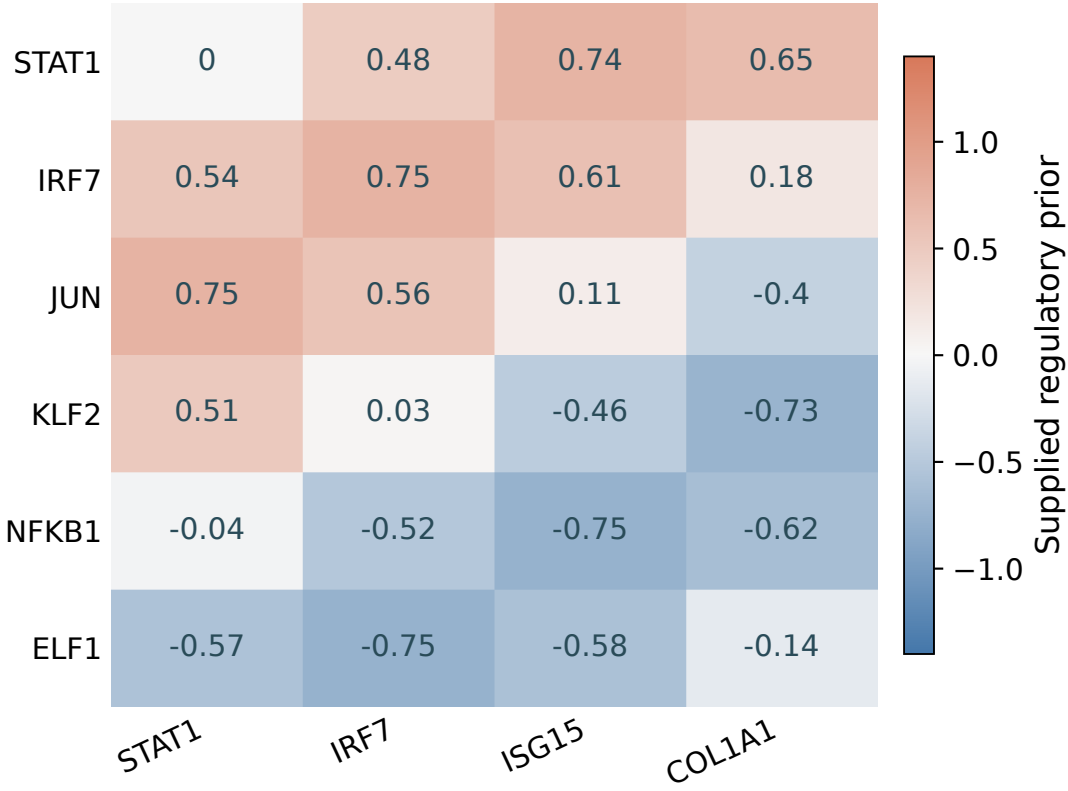
0.19 [0.11, 0.27] 0.074

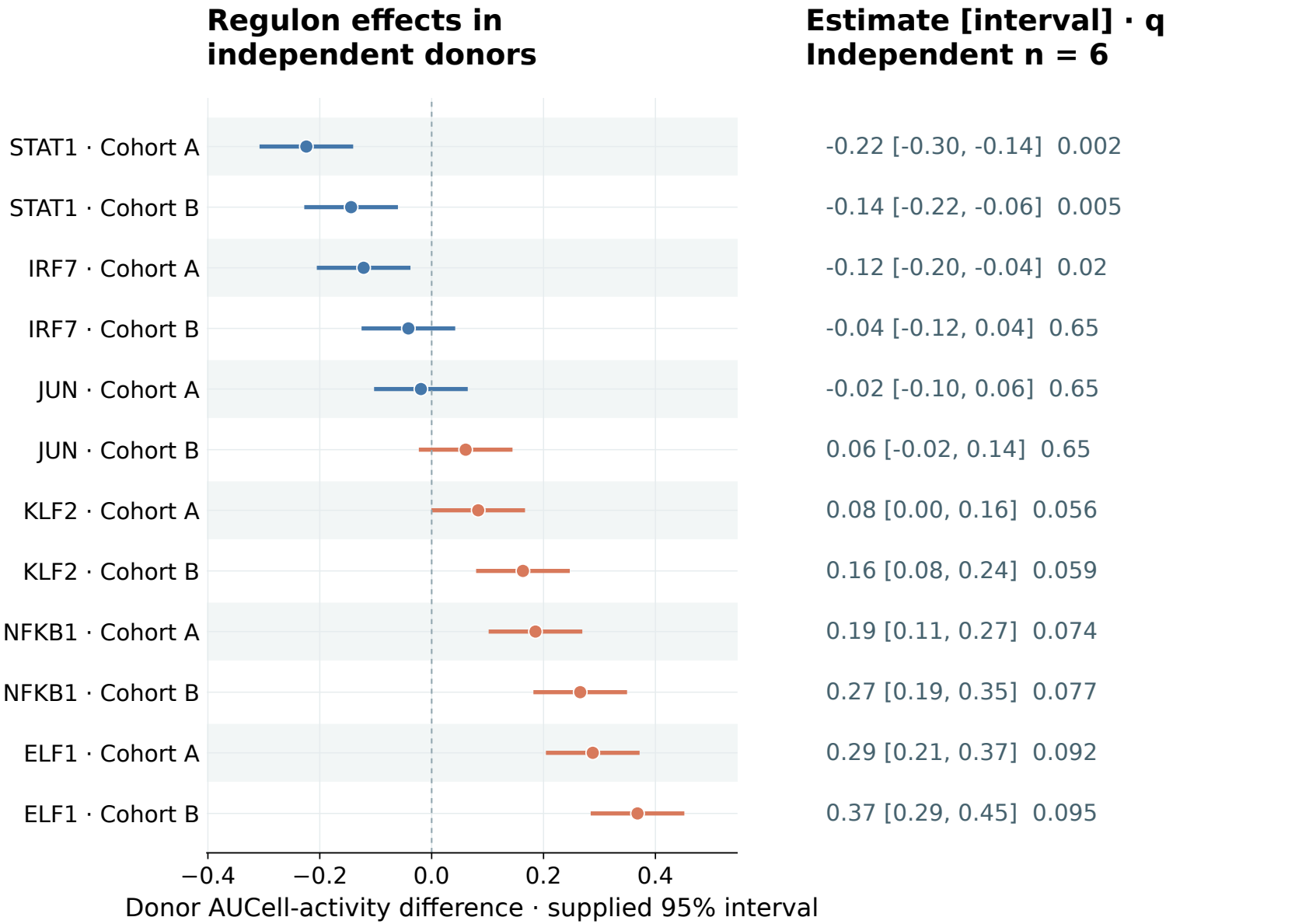
0.29 [0.21, 0.37] 0.092

Regulon specificity across identities



Motif support and signed target priors

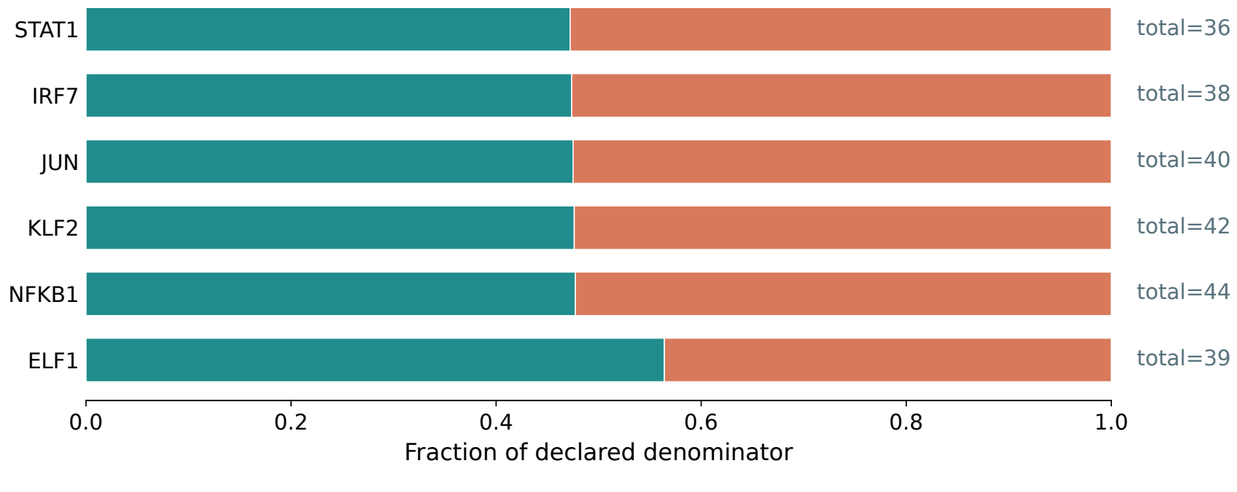



Regulon effects in independent donors
Estimate [interval] · q
Independent n = 6
STAT1 · Cohort A
STAT1 · Cohort B
IRF7 · Cohort A
IRF7 · Cohort B
JUN · Cohort A
JUN · Cohort B
KLF2 · Cohort A
KLF2 · Cohort B
NFKB1 · Cohort A
NFKB1 · Cohort B
ELF1 · Cohort A
ELF1 · Cohort B
-0.22 [-0.30, -0.14] 0.002
-0.14 [-0.22, -0.06] 0.005
-0.12 [-0.20, -0.04] 0.02
-0.04 [-0.12, 0.04] 0.65
-0.02 [-0.10, 0.06] 0.65
0.06 [-0.02, 0.14] 0.65
0.08 [0.00, 0.16] 0.056
0.16 [0.08, 0.24] 0.059
0.19 [0.11, 0.27] 0.074
0.27 [0.19, 0.35] 0.077
0.29 [0.21, 0.37] 0.092
0.37 [0.29, 0.45] 0.095
−0.4
−0.2
0.0
0.2
0.4
Donor AUCell-activity difference · supplied 95% interval

Single cell · regulons — complementary evidence

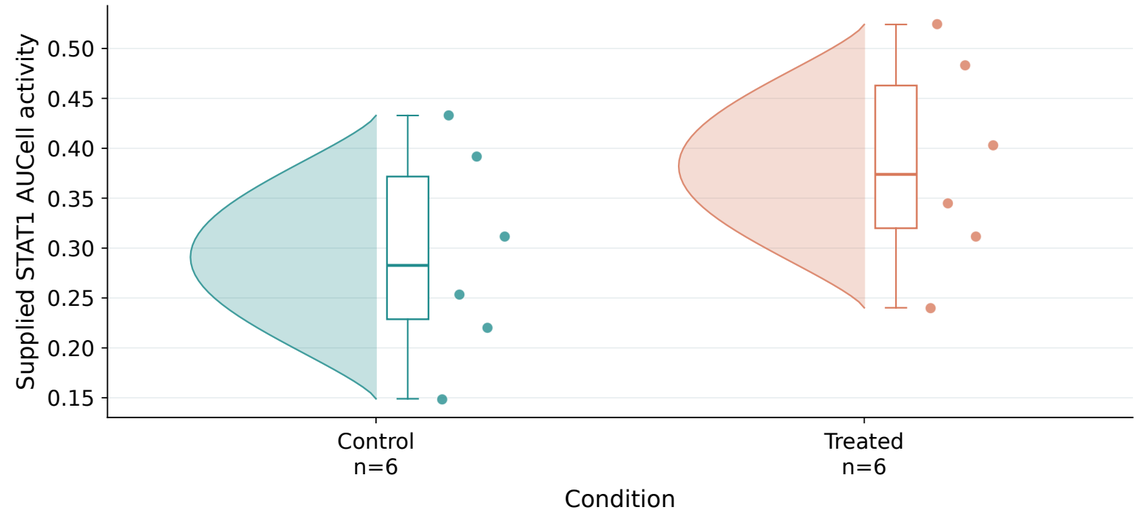
A

Motif and target coverage by regulon



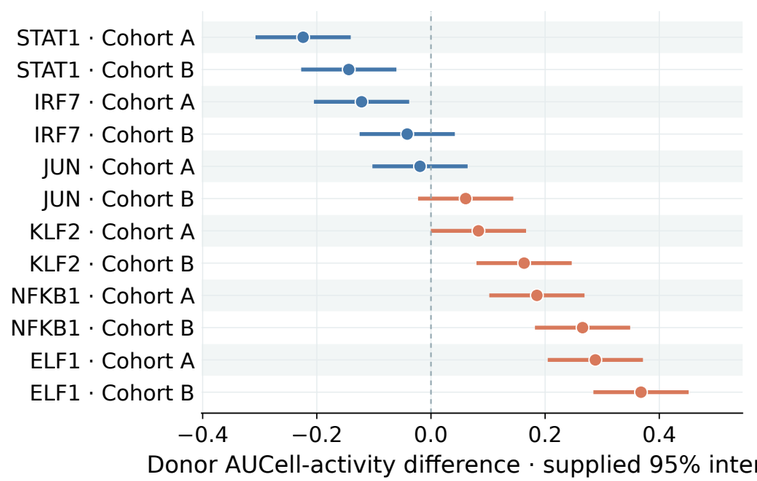
B

STAT1 regulon activity in independent donors



C

Regulon effects in independent donors



Estimate [interval] · q
Independent n = 6

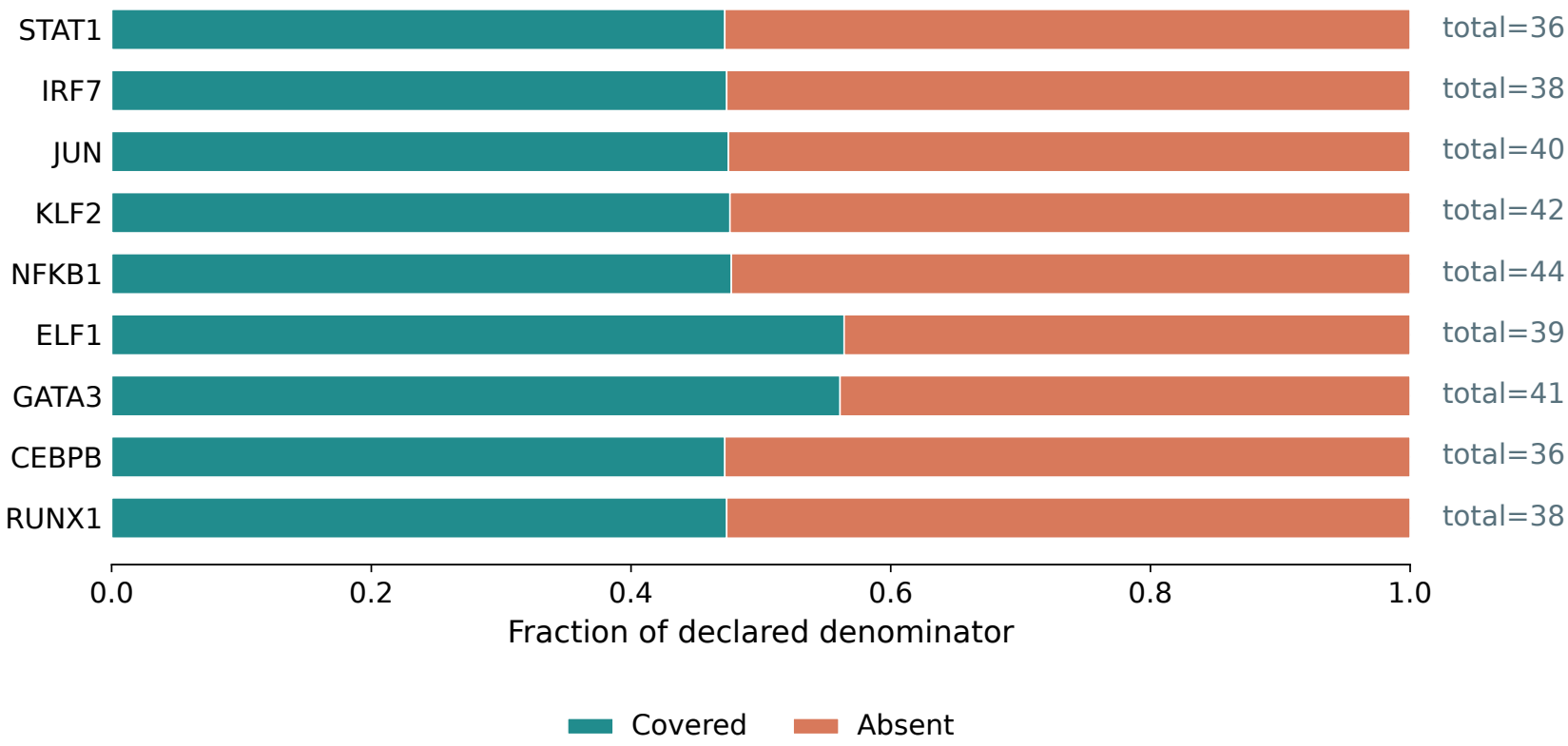
STAT1 · Cohort A	-0.22 [-0.30, -0.14] 0.002
STAT1 · Cohort B	-0.14 [-0.22, -0.06] 0.005
IRF7 · Cohort A	-0.12 [-0.20, -0.04] 0.02
IRF7 · Cohort B	-0.04 [-0.12, 0.04] 0.65
JUN · Cohort A	-0.02 [-0.10, 0.06] 0.65
JUN · Cohort B	0.06 [-0.02, 0.14] 0.65
KLF2 · Cohort A	0.08 [0.00, 0.16] 0.056
KLF2 · Cohort B	0.16 [0.08, 0.24] 0.059
NFKB1 · Cohort A	0.19 [0.11, 0.27] 0.074
NFKB1 · Cohort B	0.27 [0.19, 0.35] 0.077
ELF1 · Cohort A	0.29 [0.21, 0.37] 0.092
ELF1 · Cohort B	0.37 [0.29, 0.45] 0.095

Software test fixture; not biological evidence

Single cell · regulons — complementary evidence

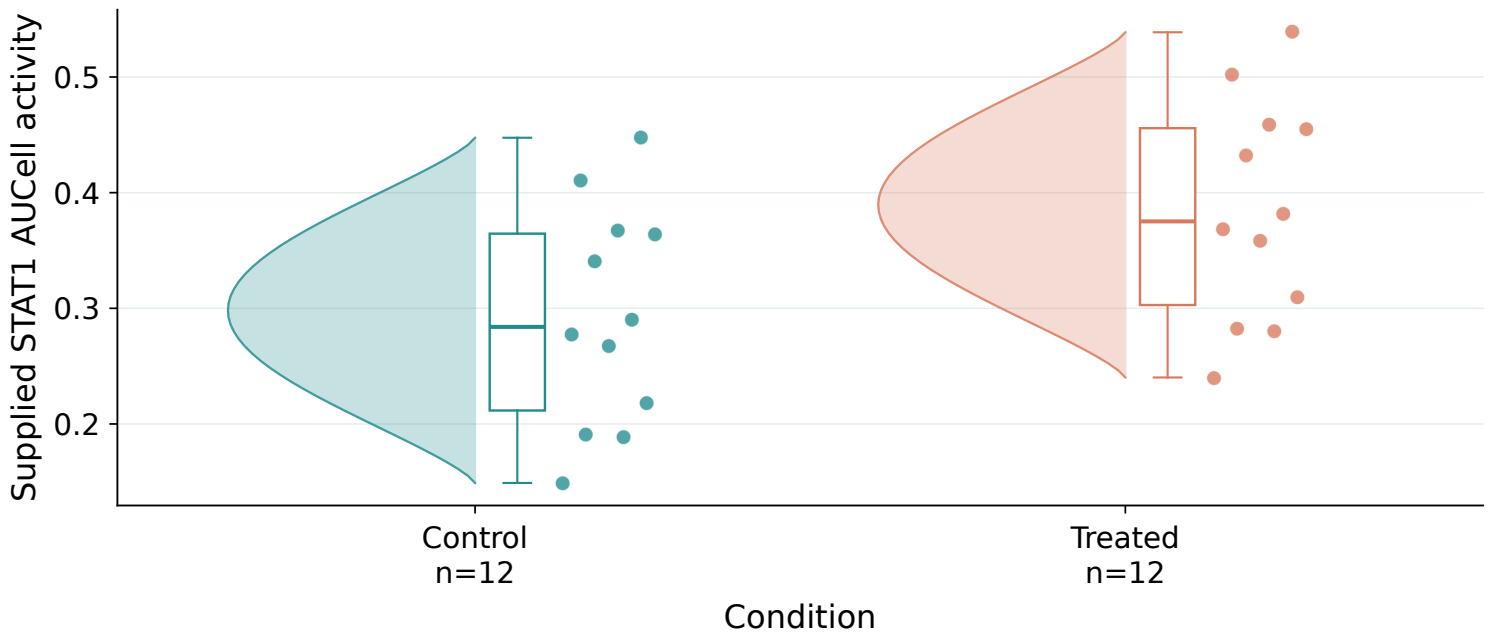
A

Motif and target coverage by regulon



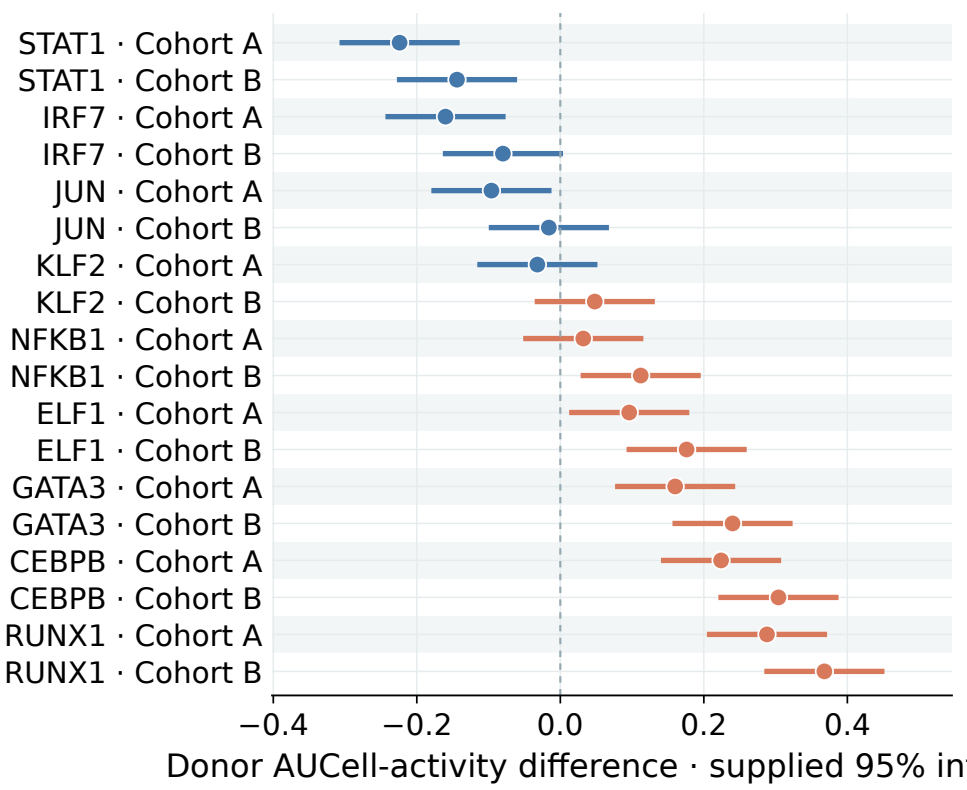
B

STAT1 regulon activity in independent donors



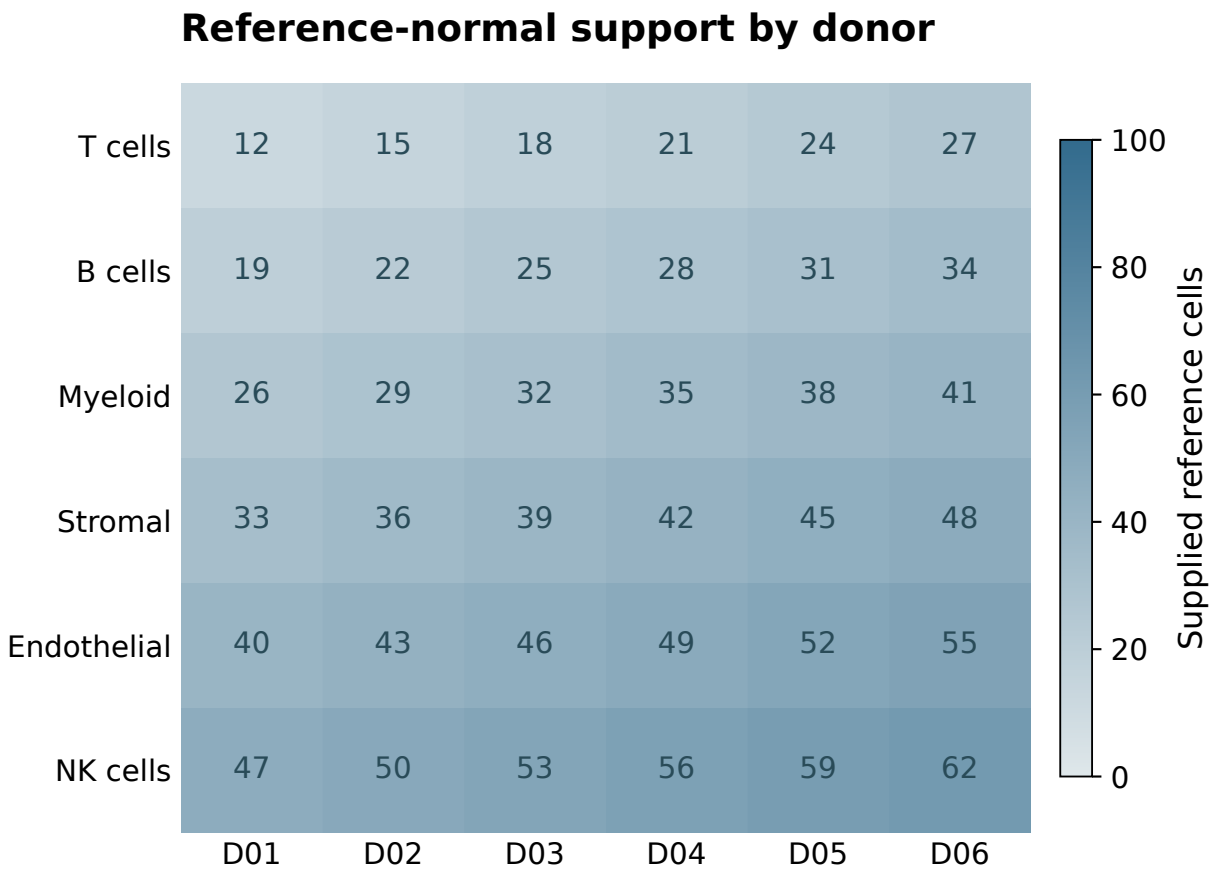
C

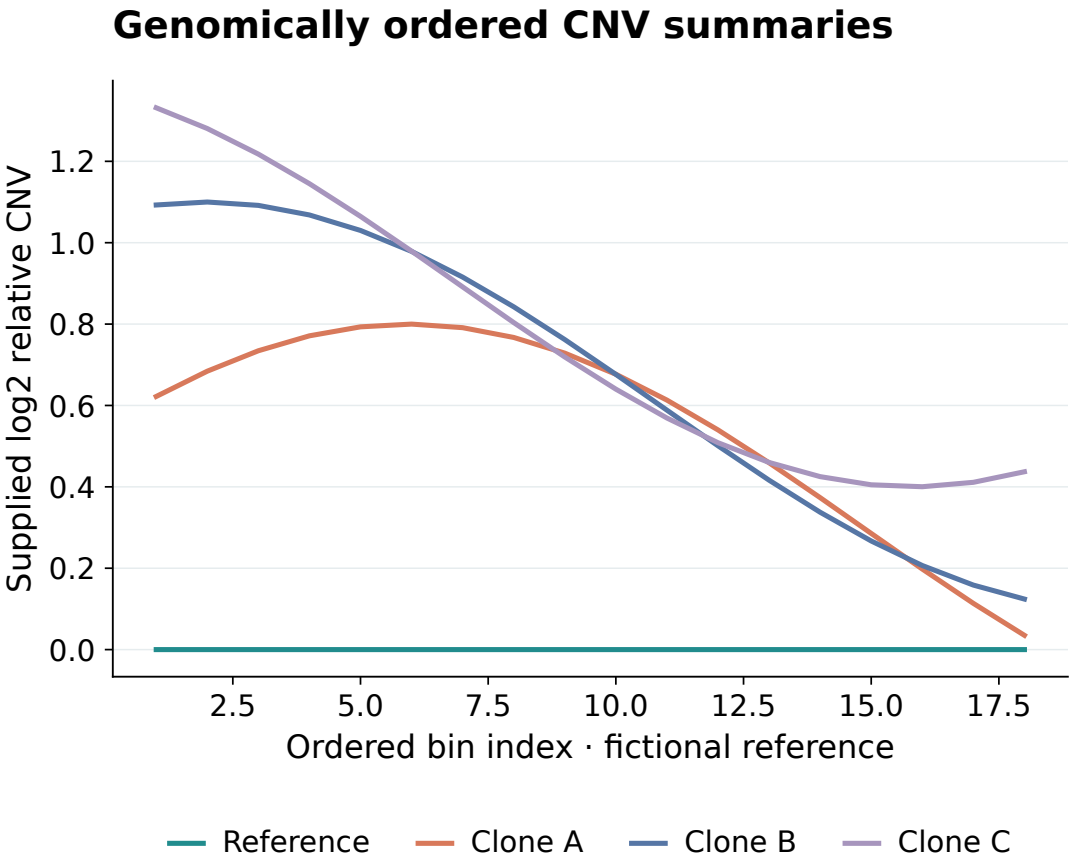
Regulon effects in independent donors

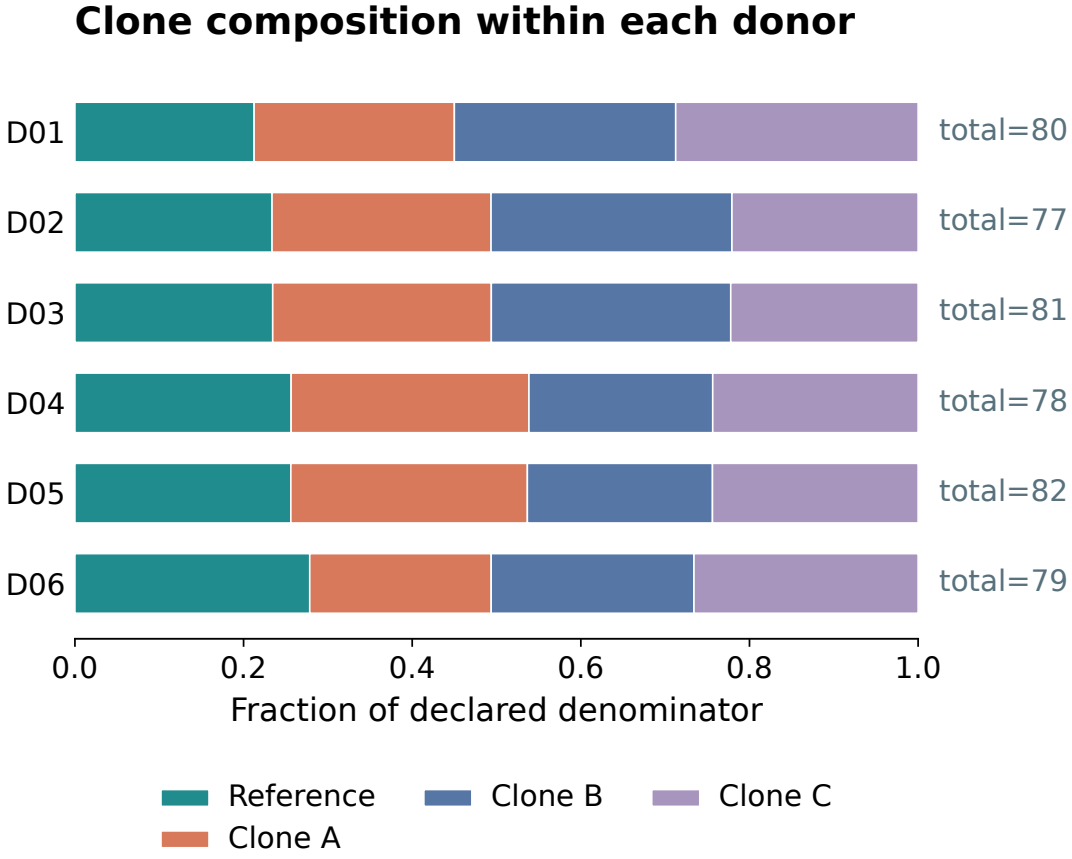


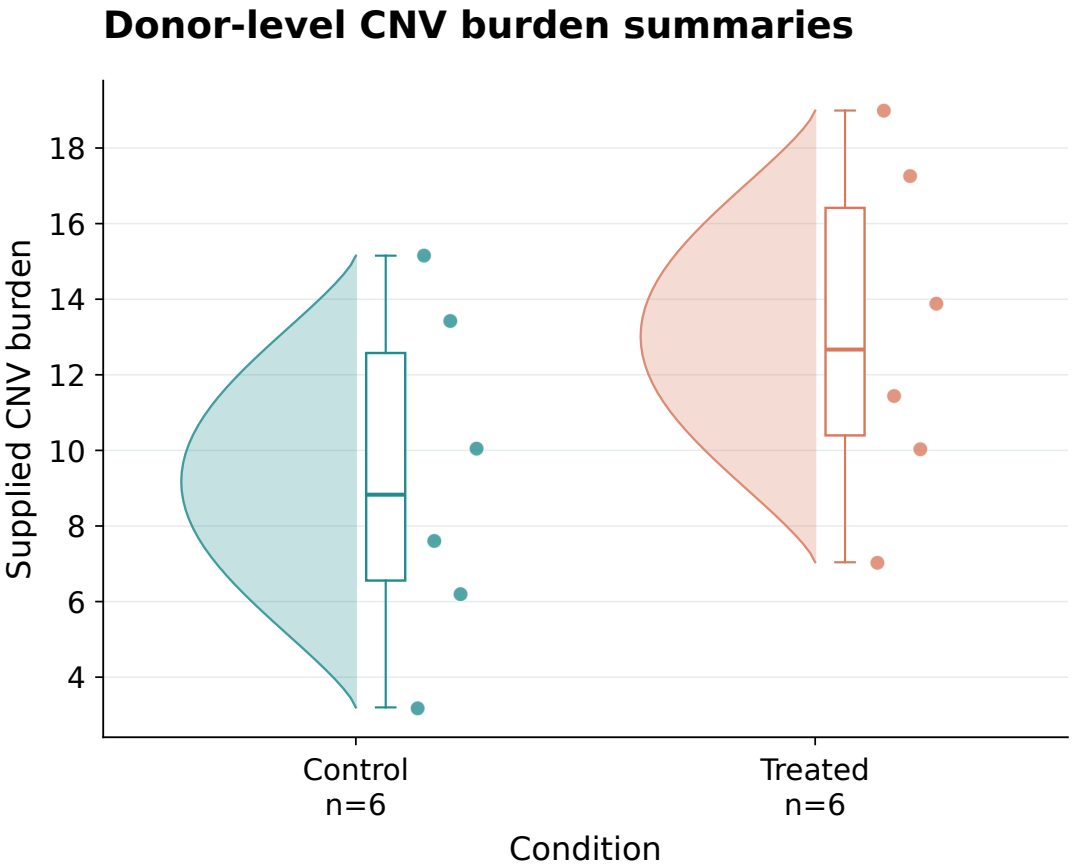
Estimate [interval] · q
Independent n = 12

STAT1 · Cohort A	-0.22 [-0.30, -0.14] 0.002
STAT1 · Cohort B	-0.14 [-0.22, -0.06] 0.005
IRF7 · Cohort A	-0.16 [-0.24, -0.08] 0.02
IRF7 · Cohort B	-0.08 [-0.16, 0.00] 0.65
JUN · Cohort A	-0.10 [-0.18, -0.02] 0.038
JUN · Cohort B	-0.02 [-0.10, 0.06] 0.65
KLF2 · Cohort A	-0.03 [-0.11, 0.05] 0.65
KLF2 · Cohort B	0.05 [-0.03, 0.13] 0.65
NFKB1 · Cohort A	0.03 [-0.05, 0.11] 0.65
NFKB1 · Cohort B	0.11 [0.03, 0.19] 0.077
ELF1 · Cohort A	0.10 [0.02, 0.18] 0.092
ELF1 · Cohort B	0.18 [0.10, 0.26] 0.095
GATA3 · Cohort A	0.16 [0.08, 0.24] 0.11
GATA3 · Cohort B	0.24 [0.16, 0.32] 0.11
CEBPB · Cohort A	0.22 [0.14, 0.30] 0.13
CEBPB · Cohort B	0.30 [0.22, 0.38] 0.13
RUNX1 · Cohort A	0.29 [0.21, 0.37] 0.15
RUNX1 · Cohort B	0.37 [0.29, 0.45] 0.15


Reference-normal support by donor
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
12 15 18 21 24 27
19 22 25 28 31 34
26 29 32 35 38 41
33 36 39 42 45 48
40 43 46 49 52 55
47 50 53 56 59 62
D01 D02 D03 D04 D05 D06
Supplied reference cells
100
80
60
40
20
0

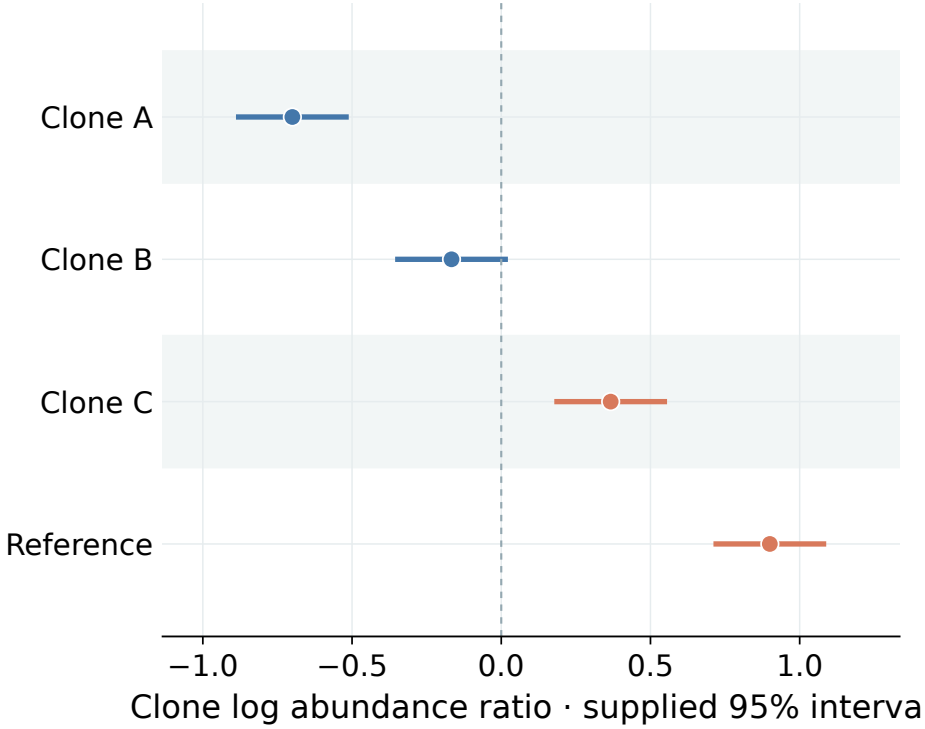

Genomically ordered CNV summaries
Supplied log2 relative CNV
1.2
1.0
0.8
0.6
0.4
0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Ordered bin index · fictional reference
Reference
Clone A
Clone B
Clone C


Clone composition within each donor
D01
D02
D03
D04
D05
D06
total=80
total=77
total=81
total=78
total=82
total=79
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Reference
Clone A
Clone B
Clone C



Software test fixture; not biological evidence

Clone-associated donor effects



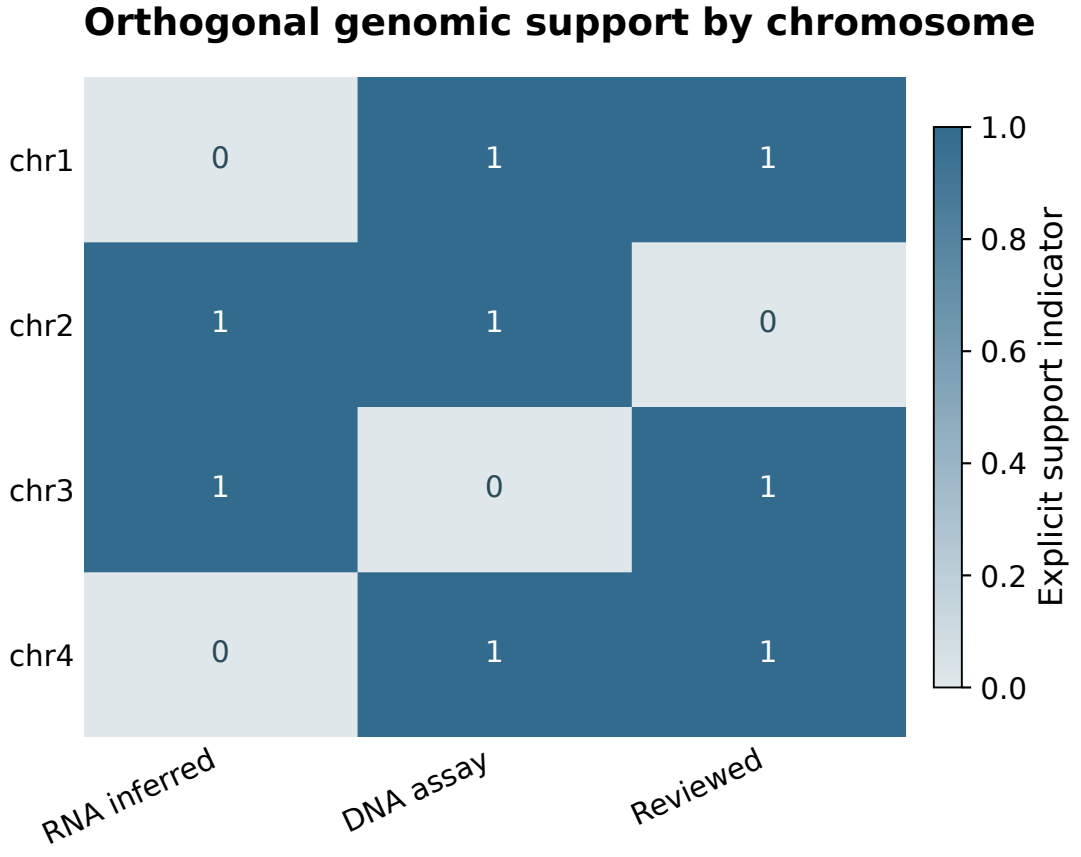
Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

-0.17 [-0.35, 0.01] 0.65

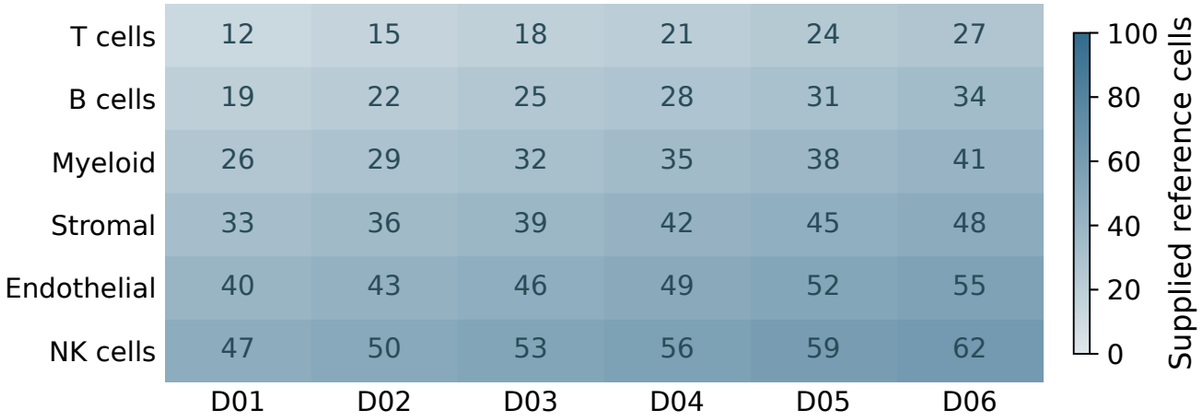
0.37 [0.19, 0.55] 0.038

0.90 [0.72, 1.08] 0.056

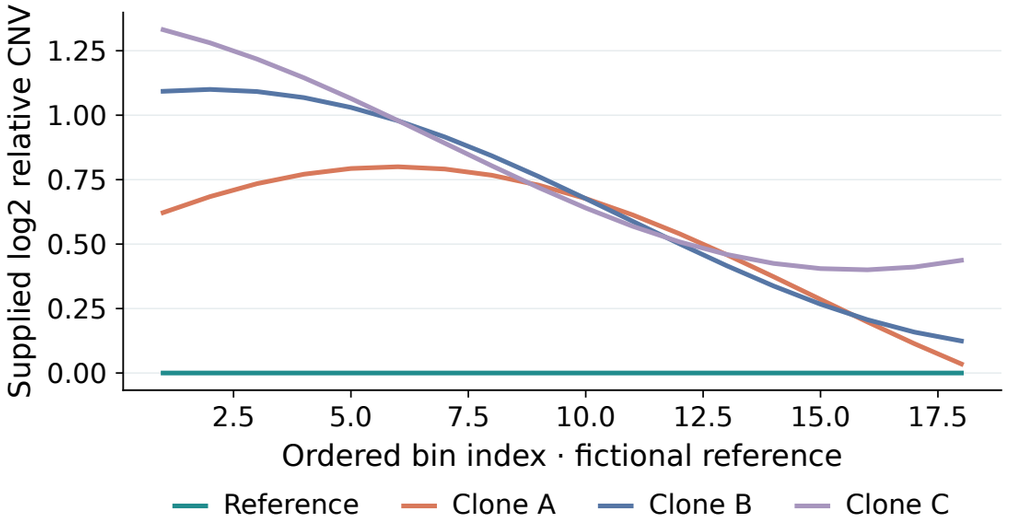

Orthogonal genomic support by chromosome
chr1
chr2
chr3
chr4
0
1
1
1
1
0
1
0
1
0
1
1
RNA inferred
DNA assay
Reviewed
Explicit support indicator
1.0
0.8
0.6
0.4
0.2
0.0

Single cell · cnv — complementary evidence

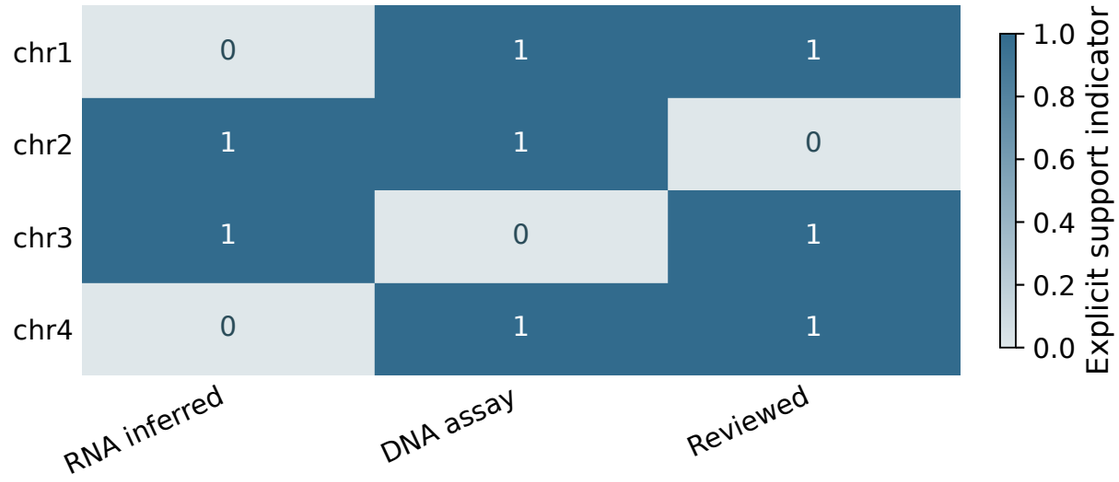
A **Reference-normal support by donor**



B **Genomically ordered CNV summaries**



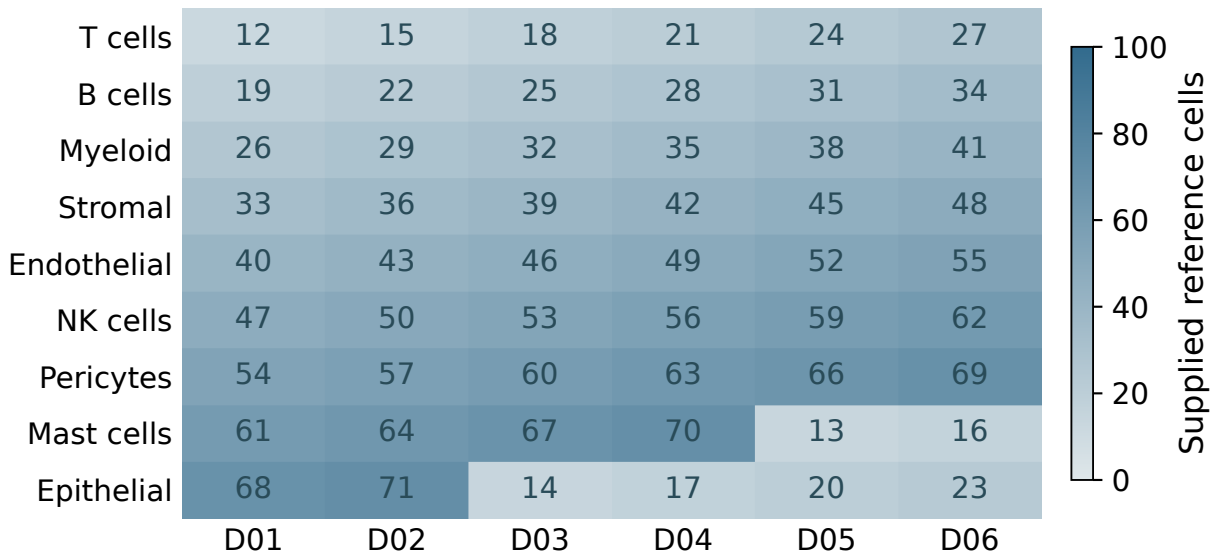
C **Orthogonal genomic support by chromosome**



Single cell · cnv — complementary evidence

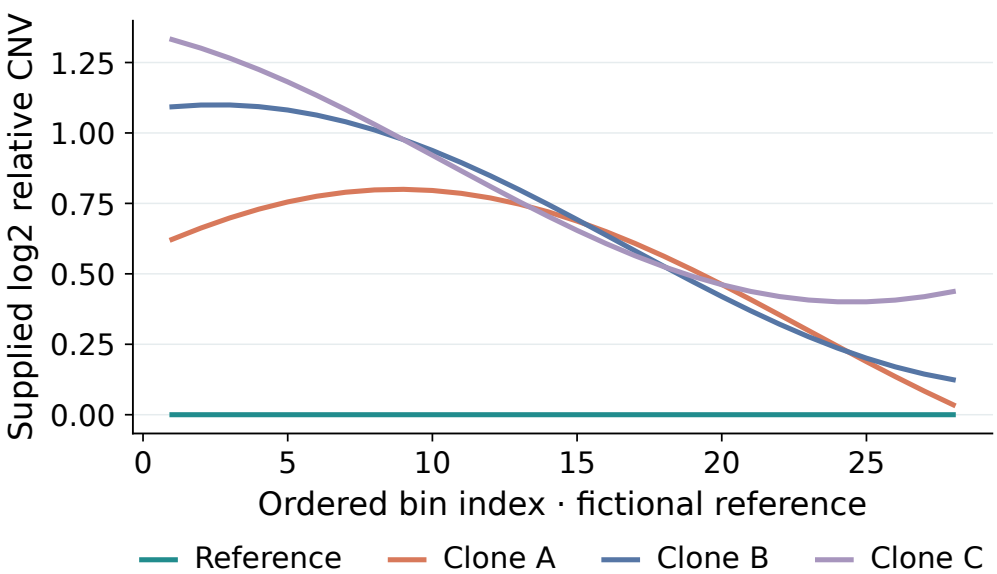
A

Reference-normal support by donor



B

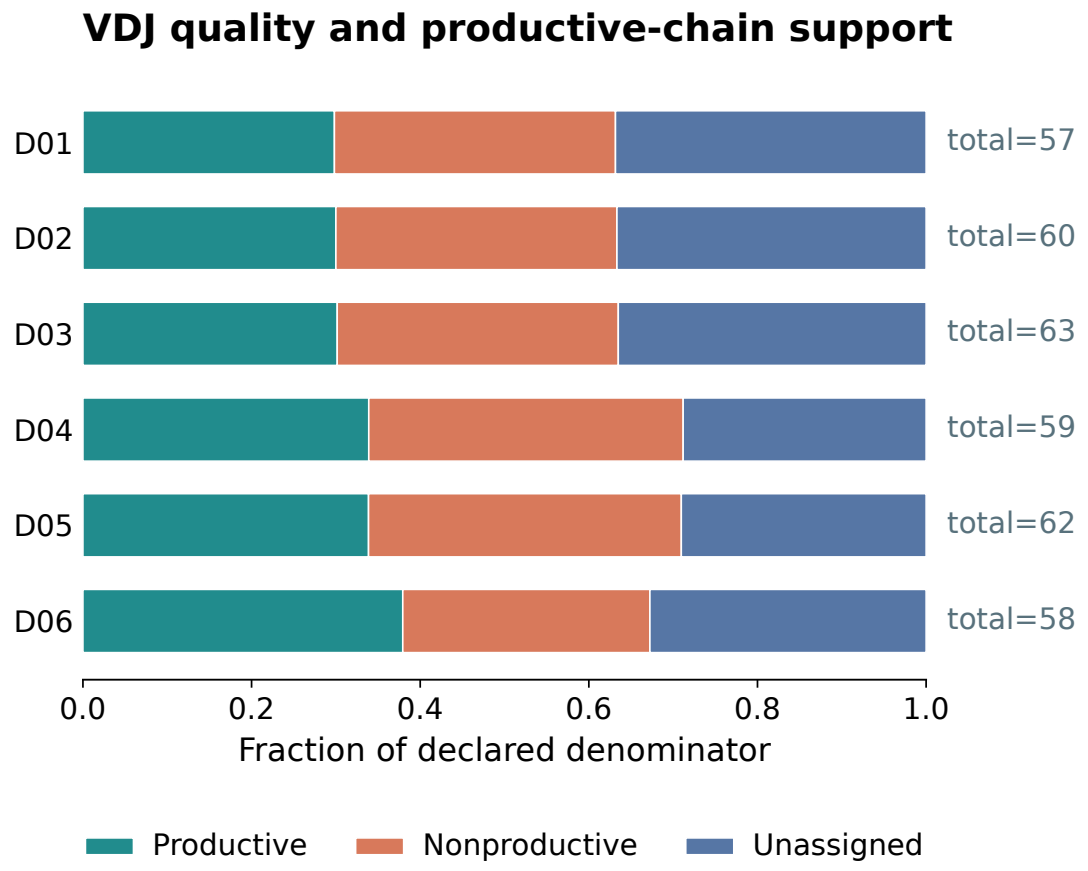
Genomically ordered CNV summaries

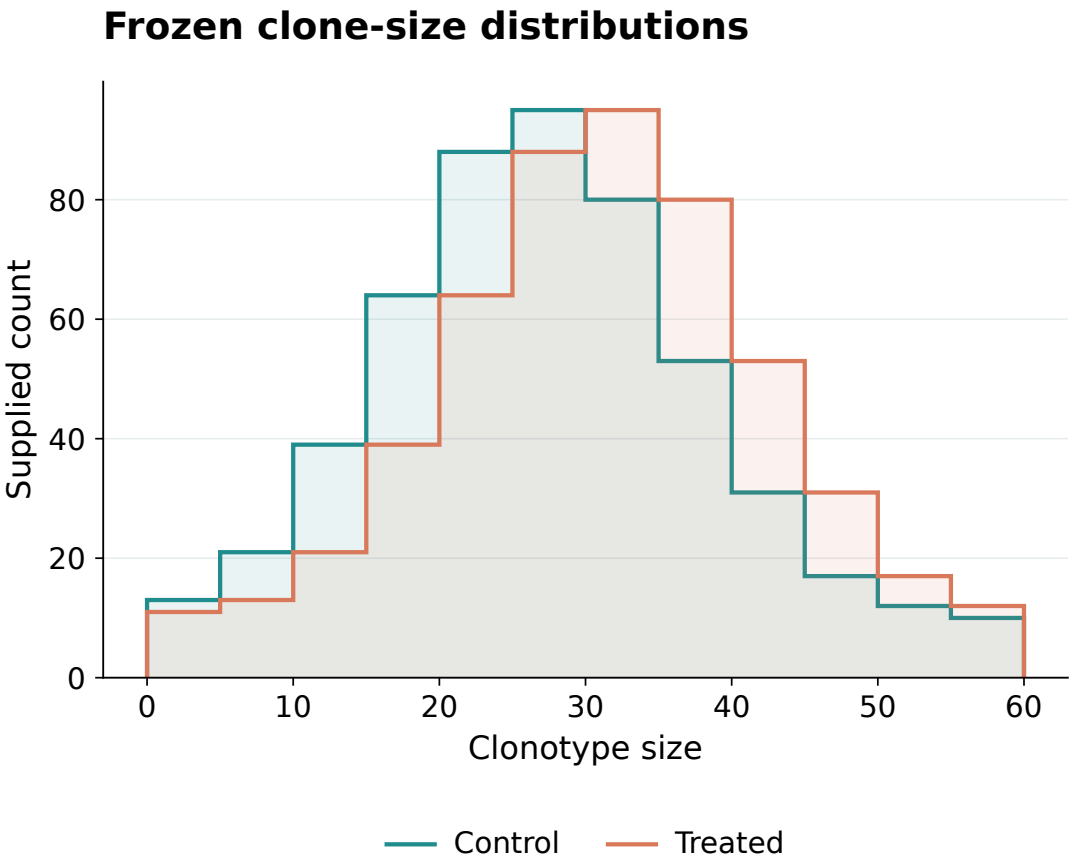


C

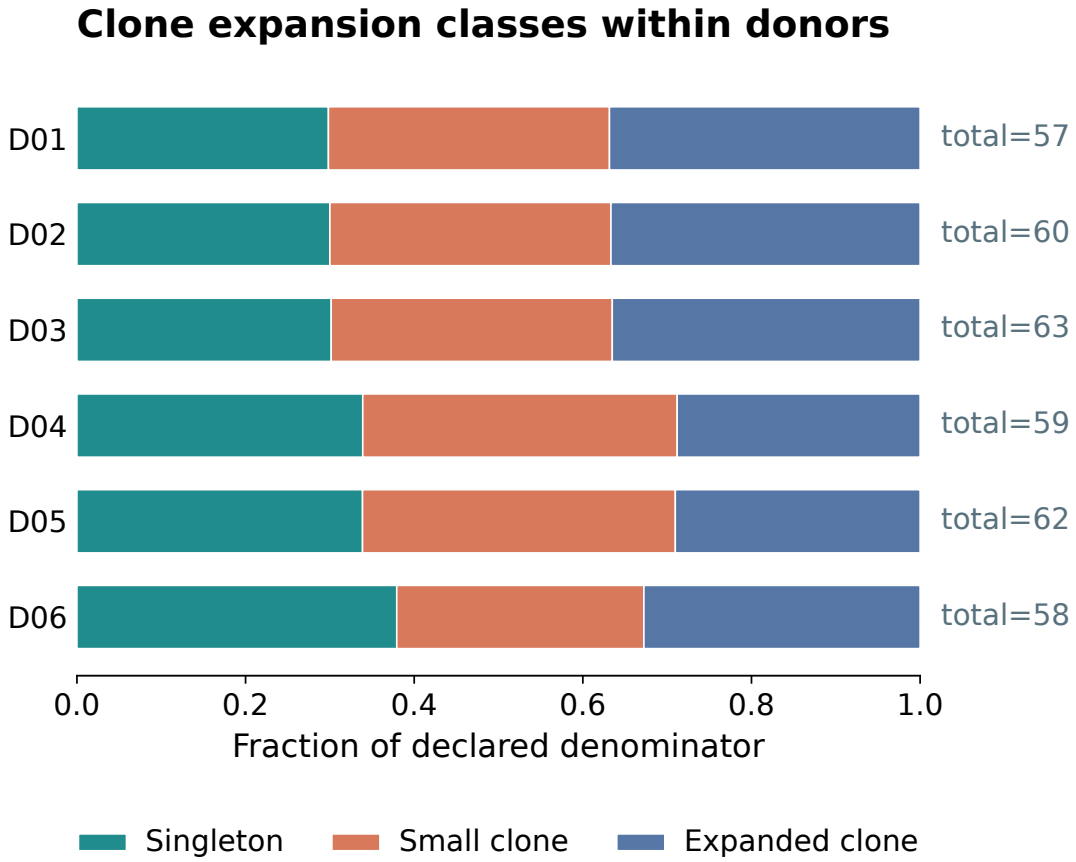
Orthogonal genomic support by chromosome



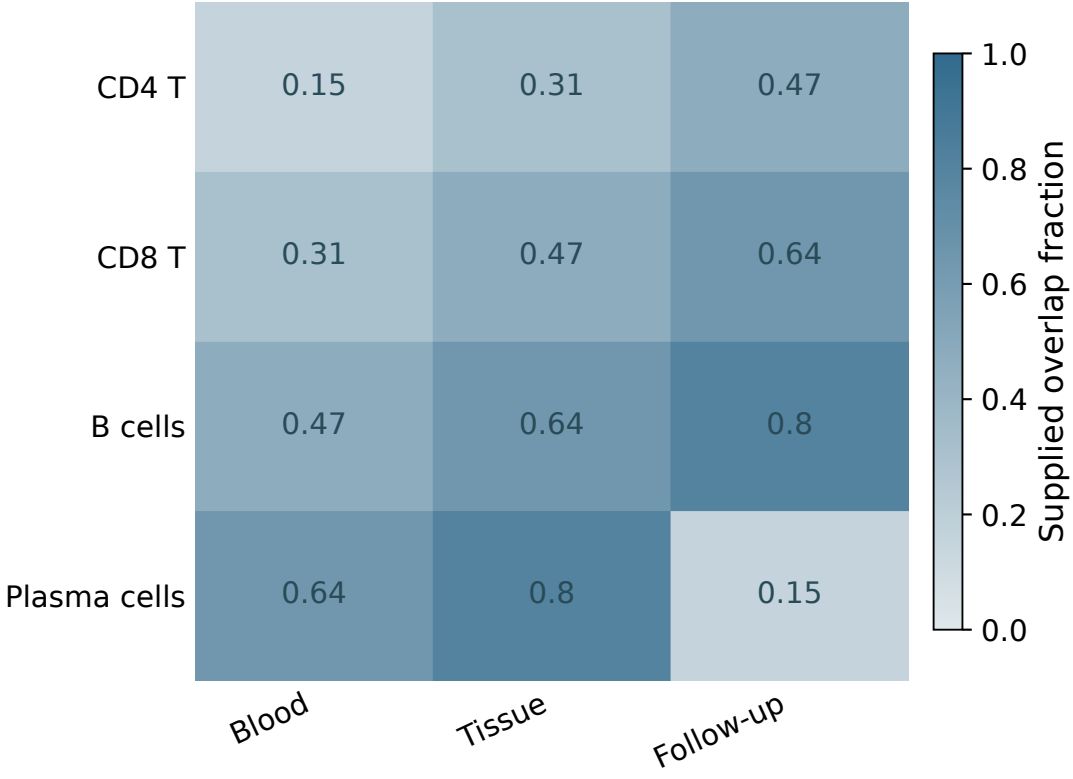

VDJ quality and productive-chain support
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Productive
Nonproductive
Unassigned



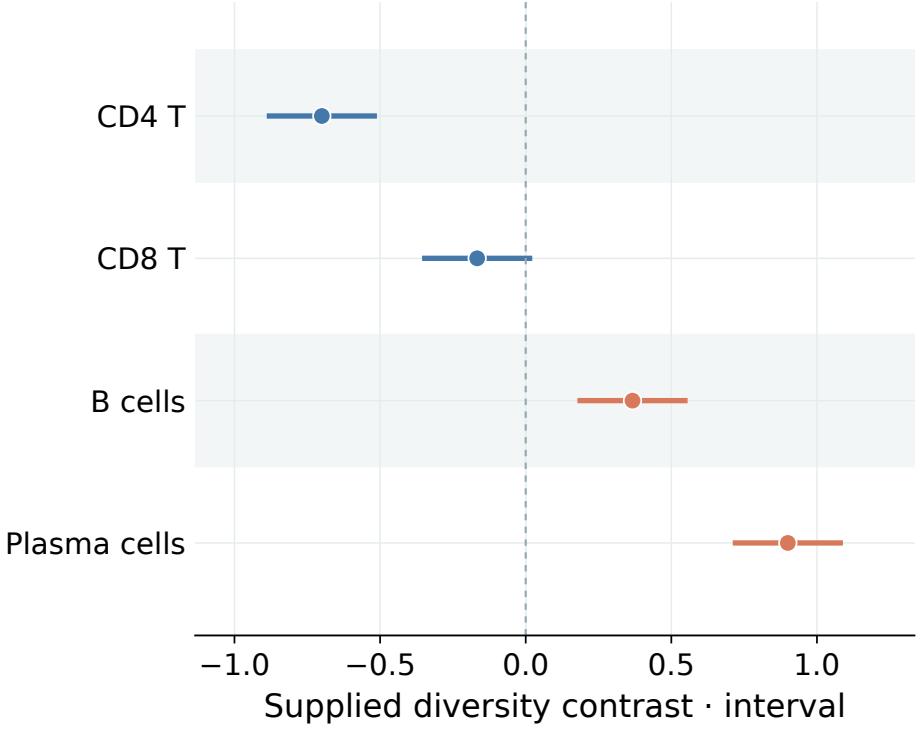
Software test fixture; not biological evidence


Clone expansion classes within donors
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Singleton
Small clone
Expanded clone

Clonotype sharing across sampled compartments



Donor-level diversity contrasts



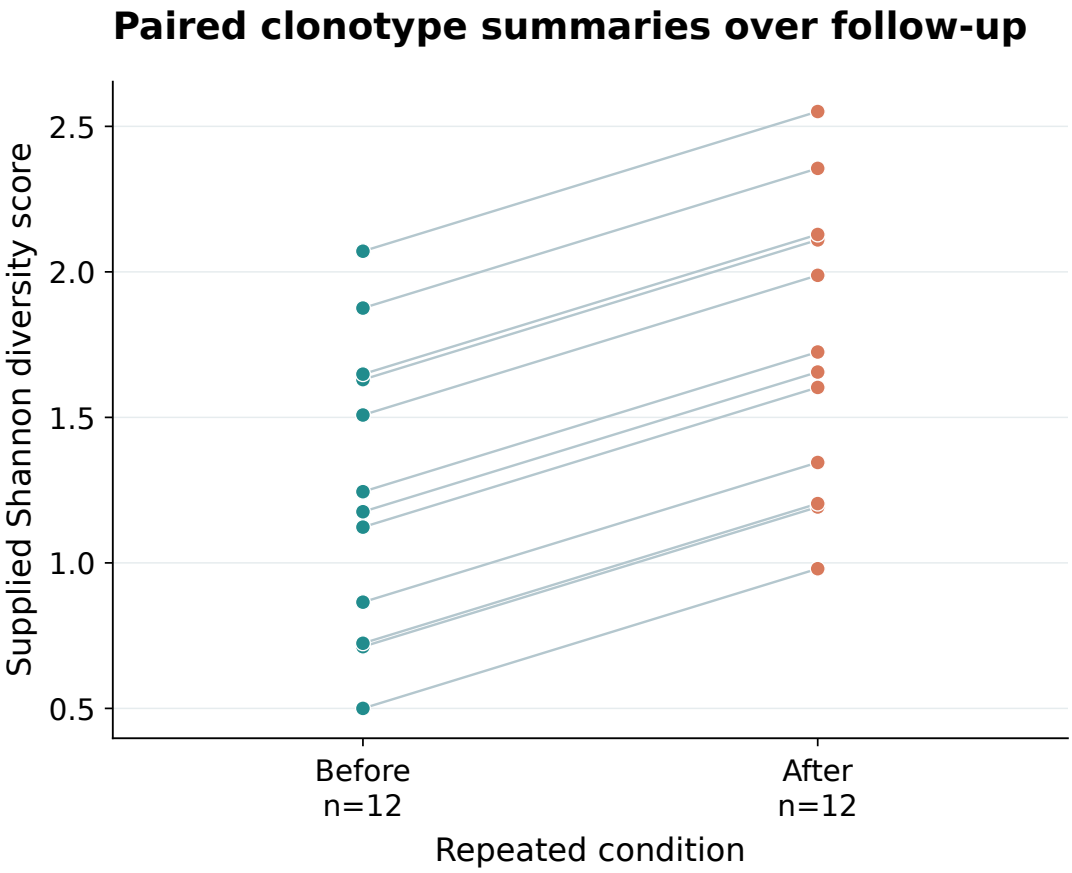
Estimate [interval] · q
Independent n = 12

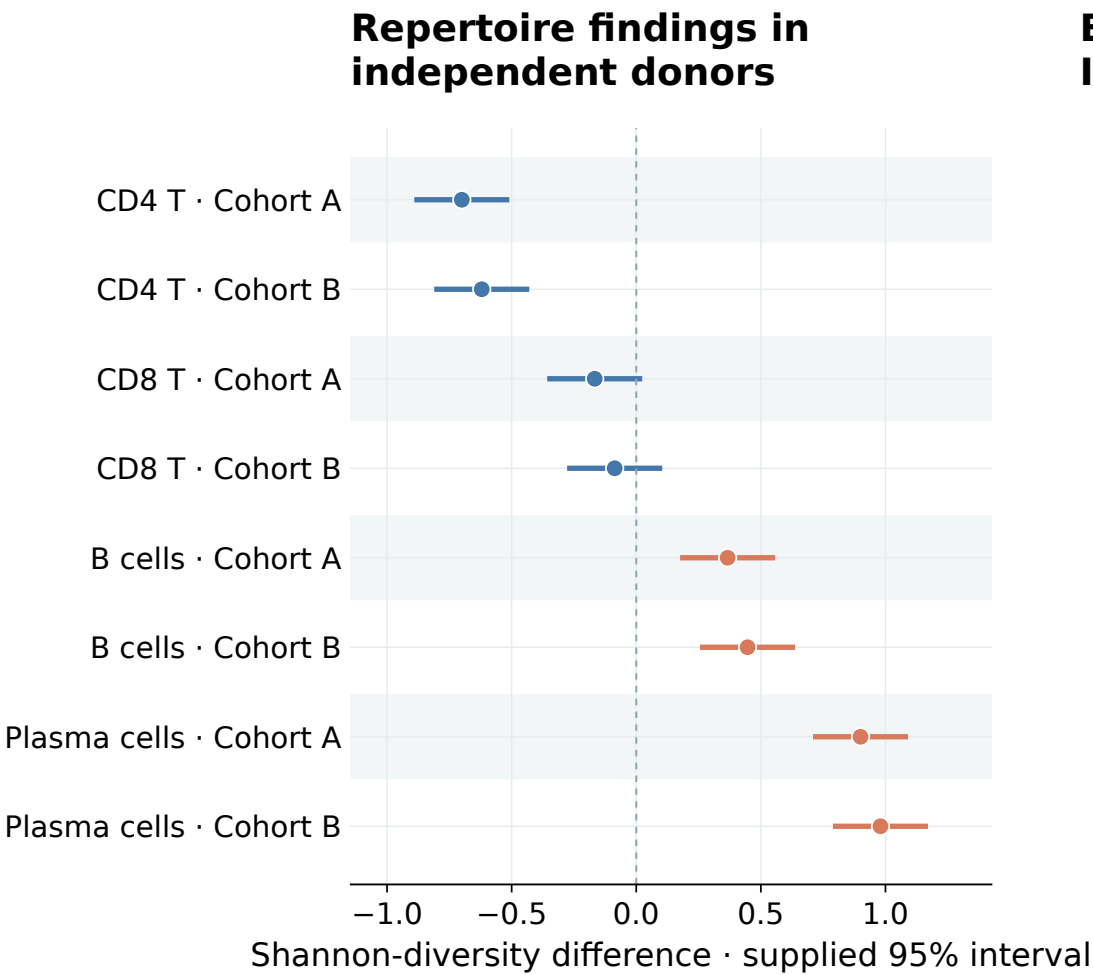
-0.70 [-0.88, -0.52] 0.002

-0.17 [-0.35, 0.01] 0.65

0.37 [0.19, 0.55] 0.038

0.90 [0.72, 1.08] 0.056

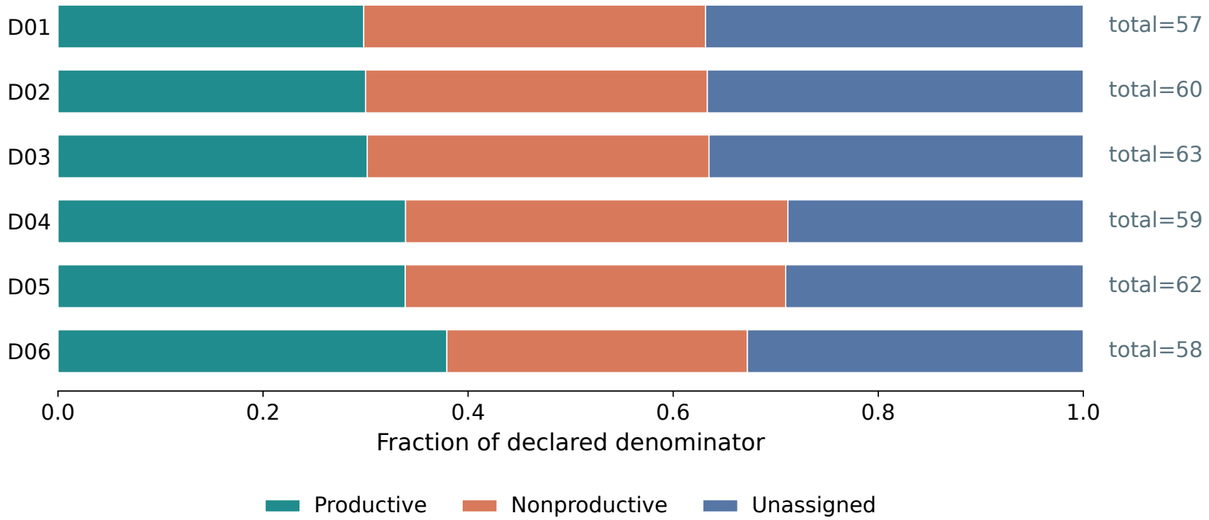

Paired clonotype summaries over follow-up
Supplied Shannon diversity score
2.5
2.0
1.5
1.0
0.5
Before
n=12
After
n=12
Repeated condition


Repertoire findings in independent donors
Estimate [interval] · q
Independent n = 6
CD4 T · Cohort A
CD4 T · Cohort B
CD8 T · Cohort A
CD8 T · Cohort B
B cells · Cohort A
B cells · Cohort B
Plasma cells · Cohort A
Plasma cells · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.17 [-0.35, 0.01] 0.65
-0.09 [-0.27, 0.09] 0.65
0.37 [0.19, 0.55] 0.038
0.45 [0.27, 0.63] 0.041
0.90 [0.72, 1.08] 0.056
0.98 [0.80, 1.16] 0.059
−1.0
−0.5
0.0
0.5
1.0
Shannon-diversity difference · supplied 95% interval

Single cell · vdj — complementary evidence

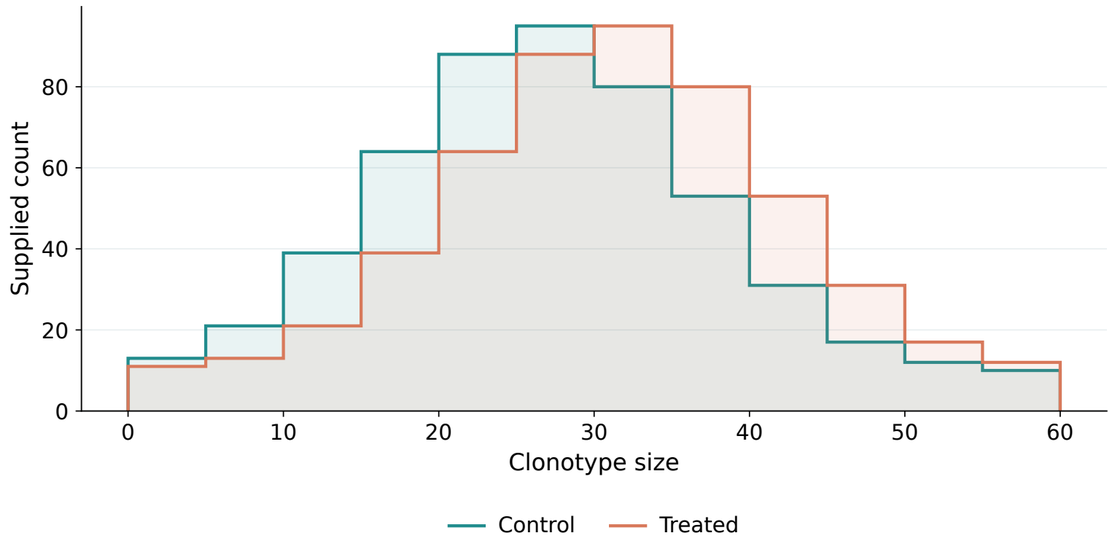
A

VDJ quality and productive-chain support



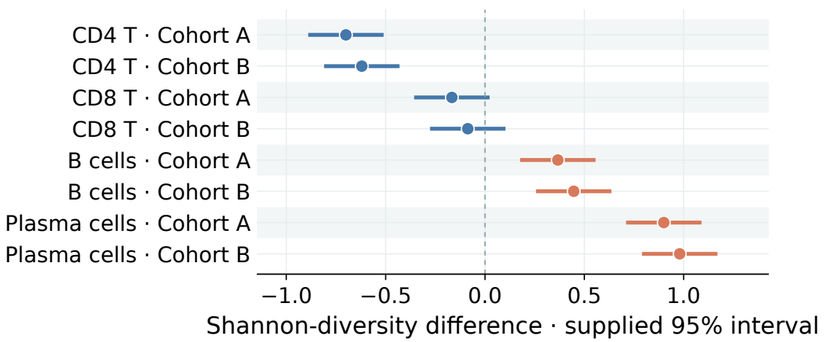
B

Frozen clone-size distributions



C

Repertoire findings in independent donors



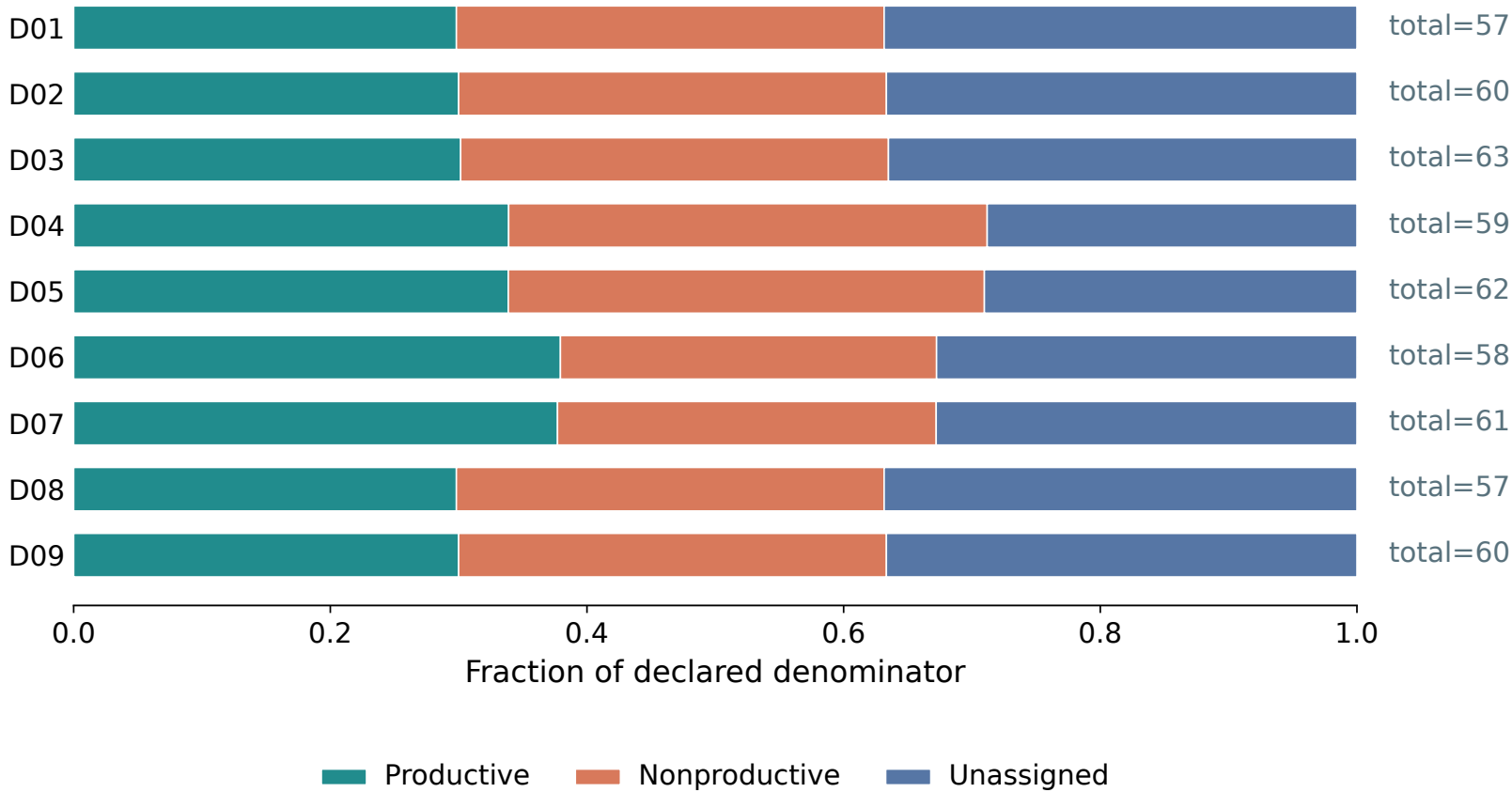
Estimate [interval] · q
Independent n = 6

-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.17 [-0.35, 0.01] 0.65
-0.09 [-0.27, 0.09] 0.65
0.37 [0.19, 0.55] 0.038
0.45 [0.27, 0.63] 0.041
0.90 [0.72, 1.08] 0.056
0.98 [0.80, 1.16] 0.059

Single cell · vdj — complementary evidence

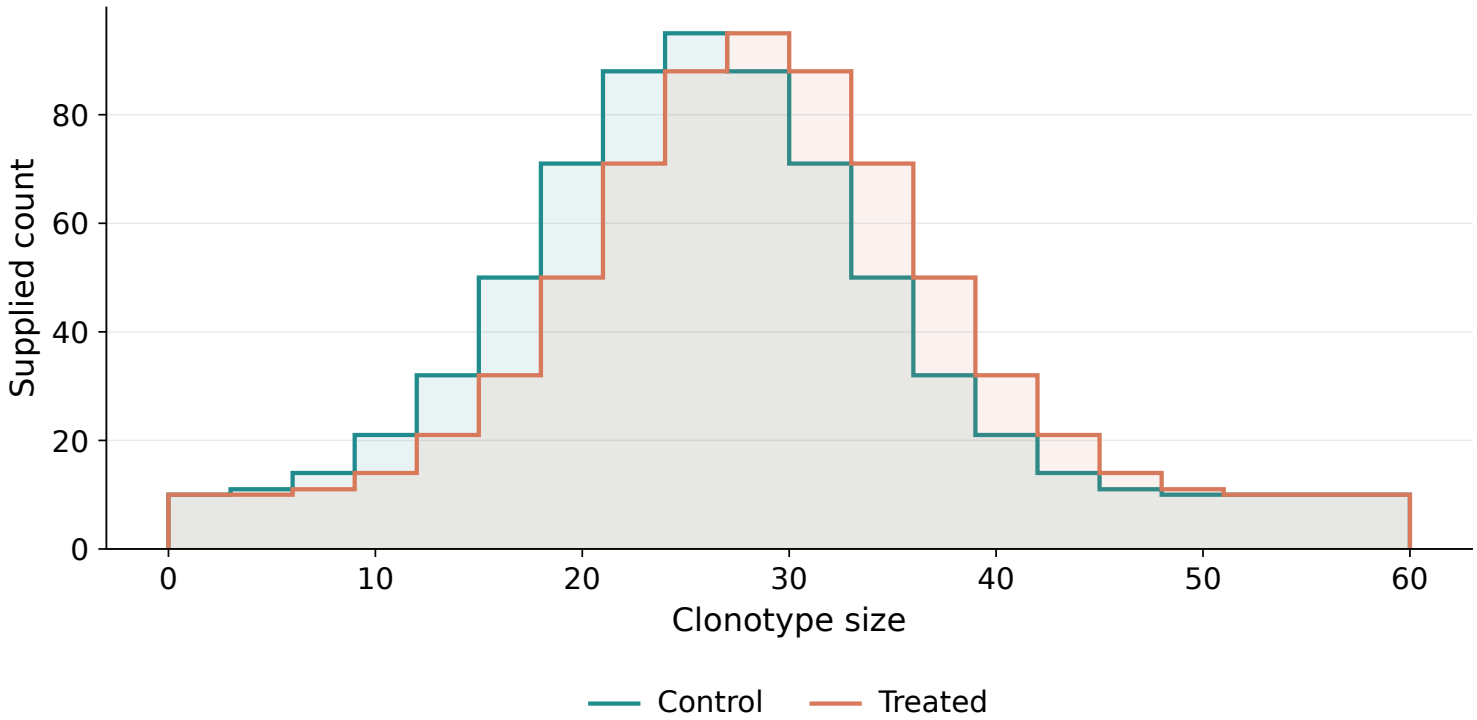
A

VDJ quality and productive-chain support



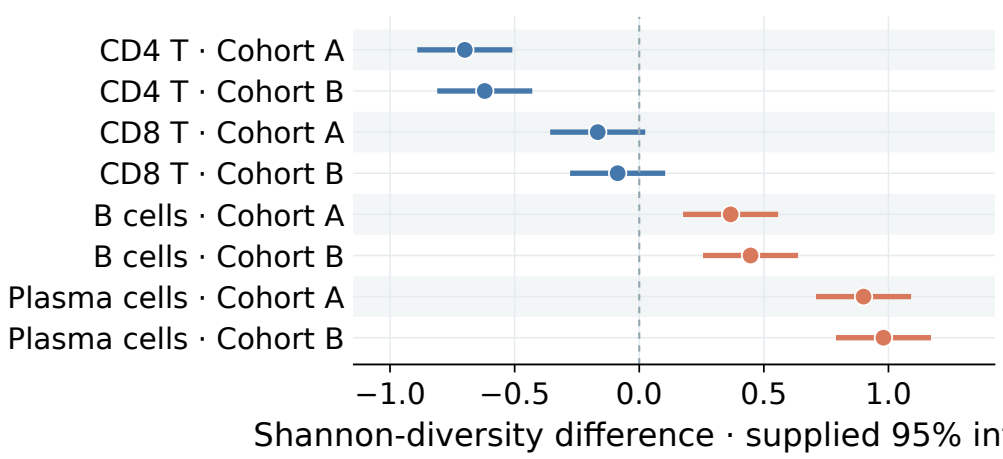
B

Frozen clone-size distributions



C

Repertoire findings in independent donors

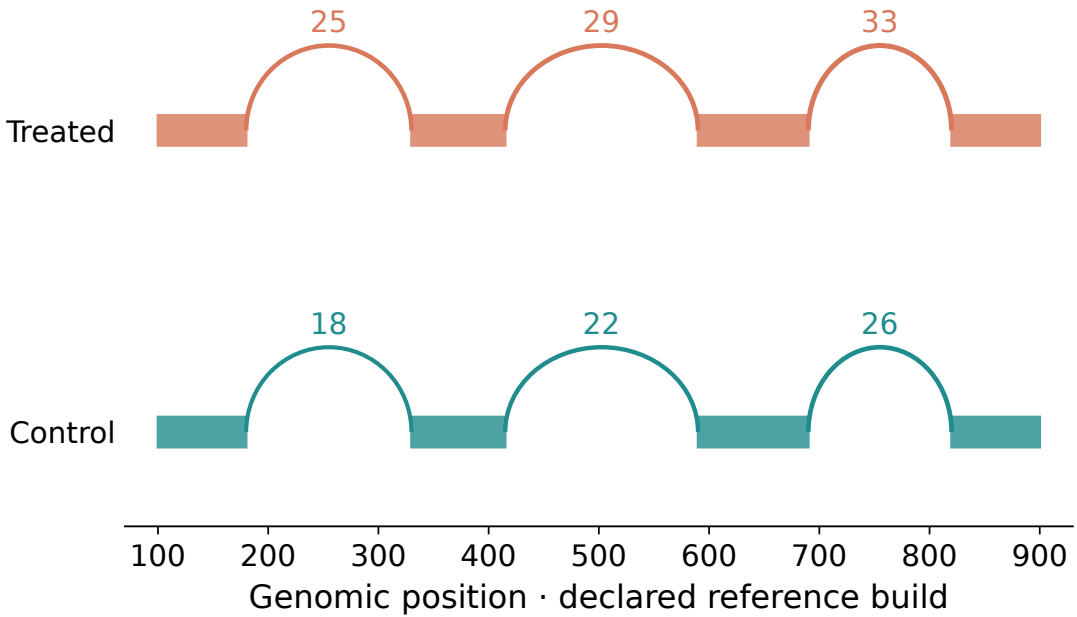


Estimate [interval] · q
Independent n = 12

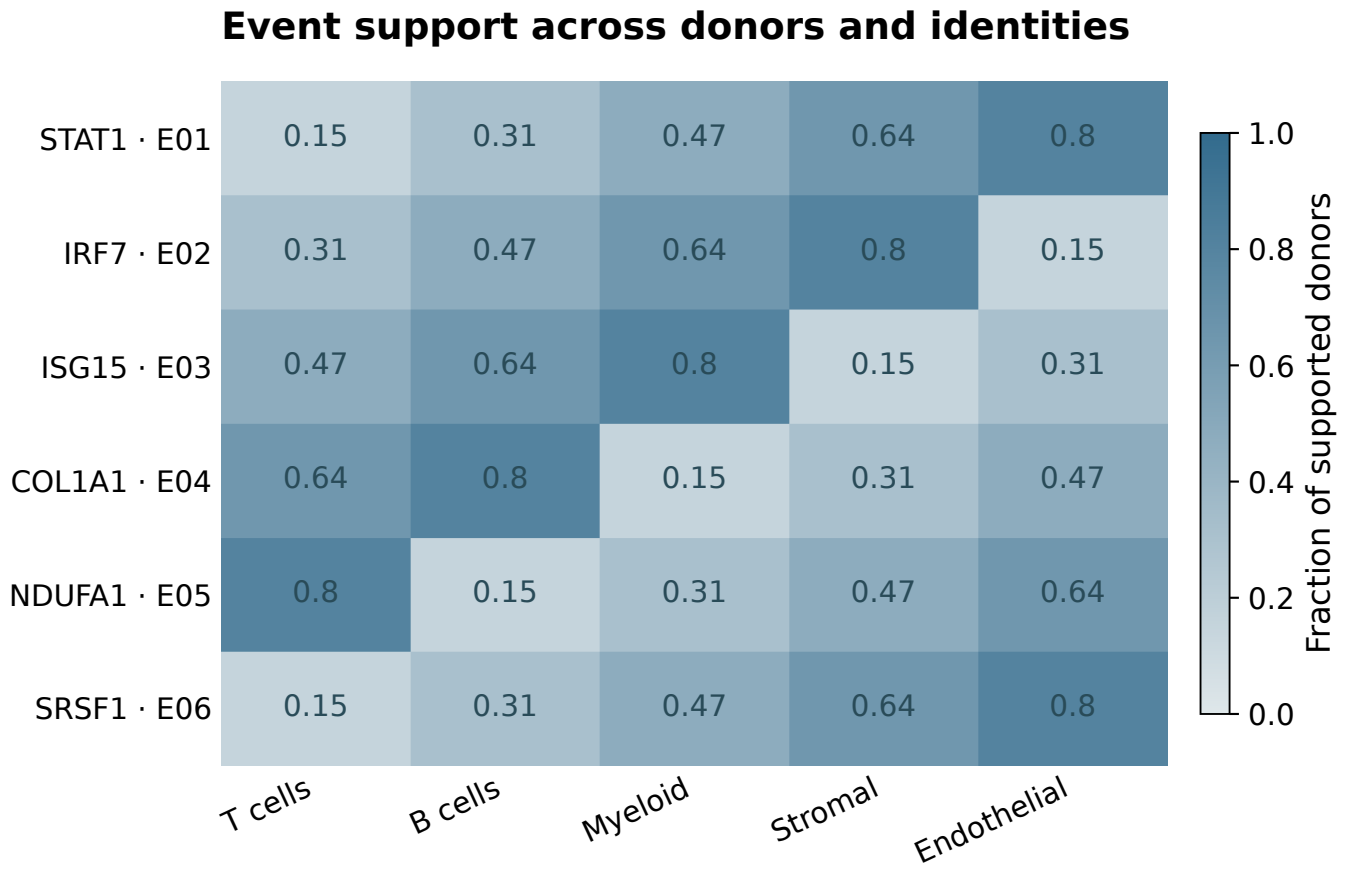
CD4 T · Cohort A	-0.70 [-0.88, -0.52] 0.002
CD4 T · Cohort B	-0.62 [-0.80, -0.44] 0.005
CD8 T · Cohort A	-0.17 [-0.35, 0.01] 0.65
CD8 T · Cohort B	-0.09 [-0.27, 0.09] 0.65
B cells · Cohort A	0.37 [0.19, 0.55] 0.038
B cells · Cohort B	0.45 [0.27, 0.63] 0.041
Plasma cells · Cohort A	0.90 [0.72, 1.08] 0.056
Plasma cells · Cohort B	0.98 [0.80, 1.16] 0.059

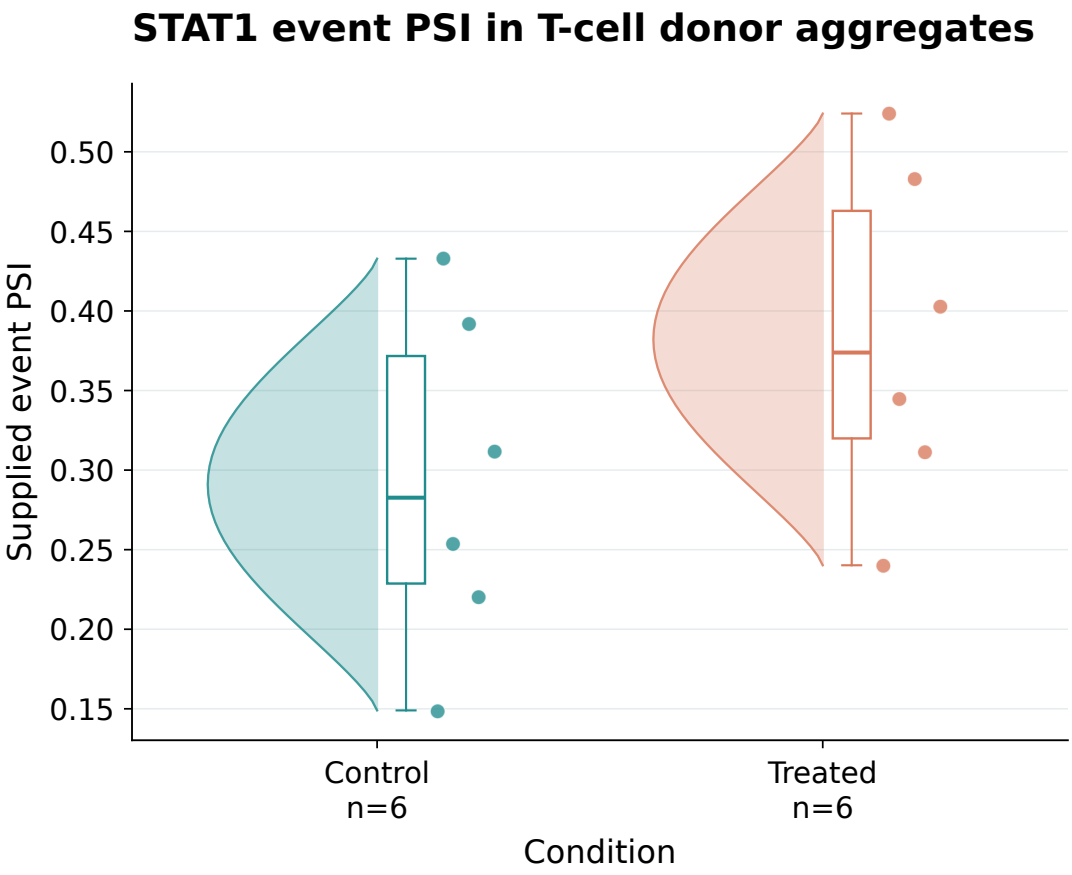
Software test fixture; not biological evidence

T-cell STAT1 E01 physical junction support



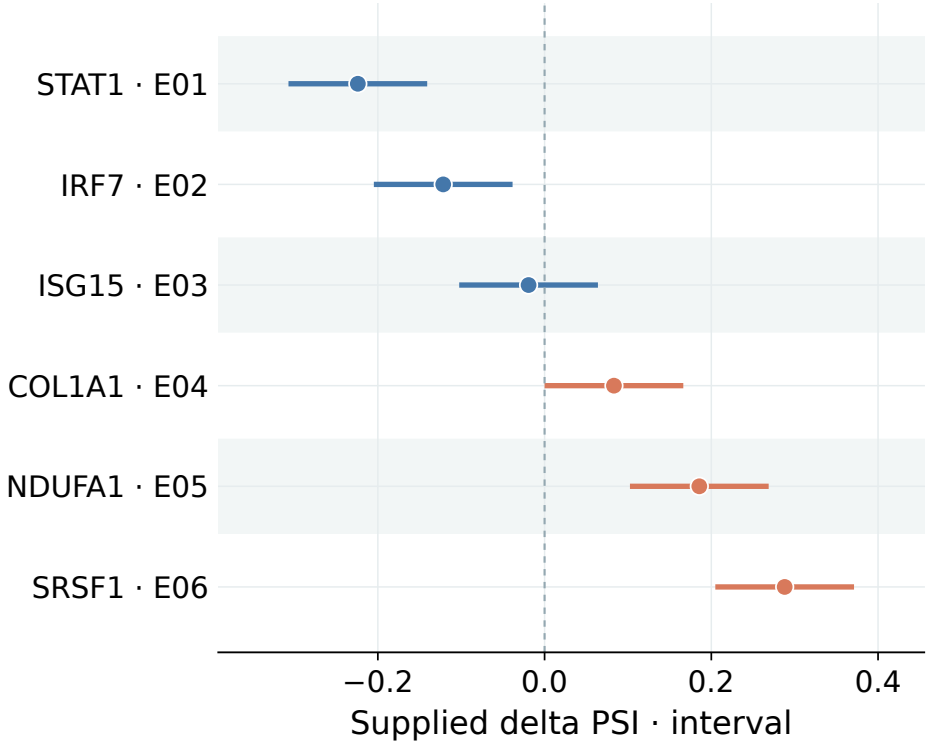
Software test fixture; not biological evidence


Event support across donors and identities
STAT1 · E01
IRF7 · E02
ISG15 · E03
COL1A1 · E04
NDUFA1 · E05
SRSF1 · E06
0.15 0.31 0.47 0.64 0.8
0.31 0.47 0.64 0.8 0.15
0.47 0.64 0.8 0.15 0.31
0.64 0.8 0.15 0.31 0.47
0.8 0.15 0.31 0.47 0.64
0.15 0.31 0.47 0.64 0.8
T cells
B cells
Myeloid
Stromal
Endothelial
Fraction of supported donors
1.0
0.8
0.6
0.4
0.2
0.0



Software test fixture; not biological evidence

T-cell donor differential splicing



Estimate [interval] · q
Independent n = 12

-0.22 [-0.30, -0.14] 0.002

-0.12 [-0.20, -0.04] 0.02

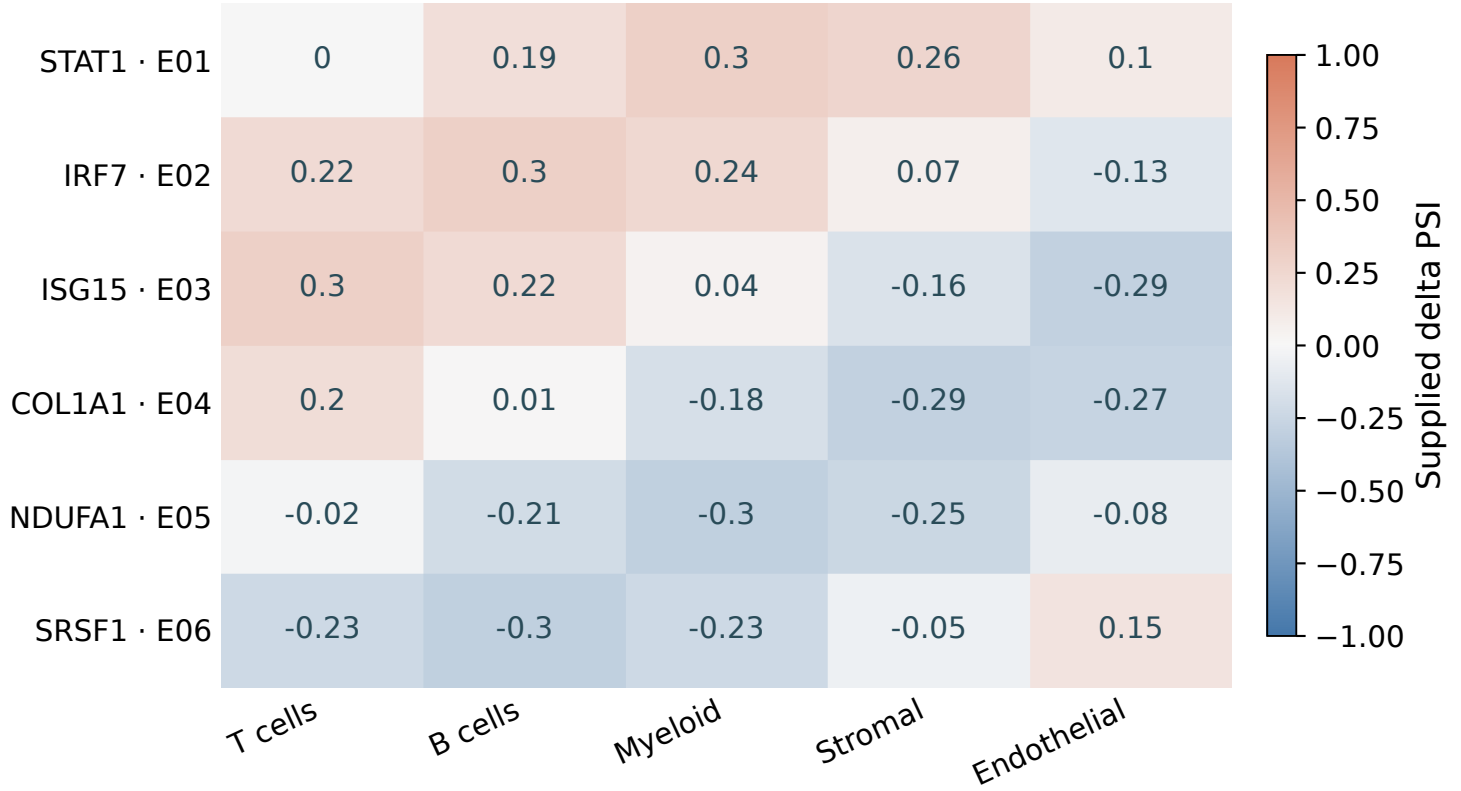
-0.02 [-0.10, 0.06] 0.65

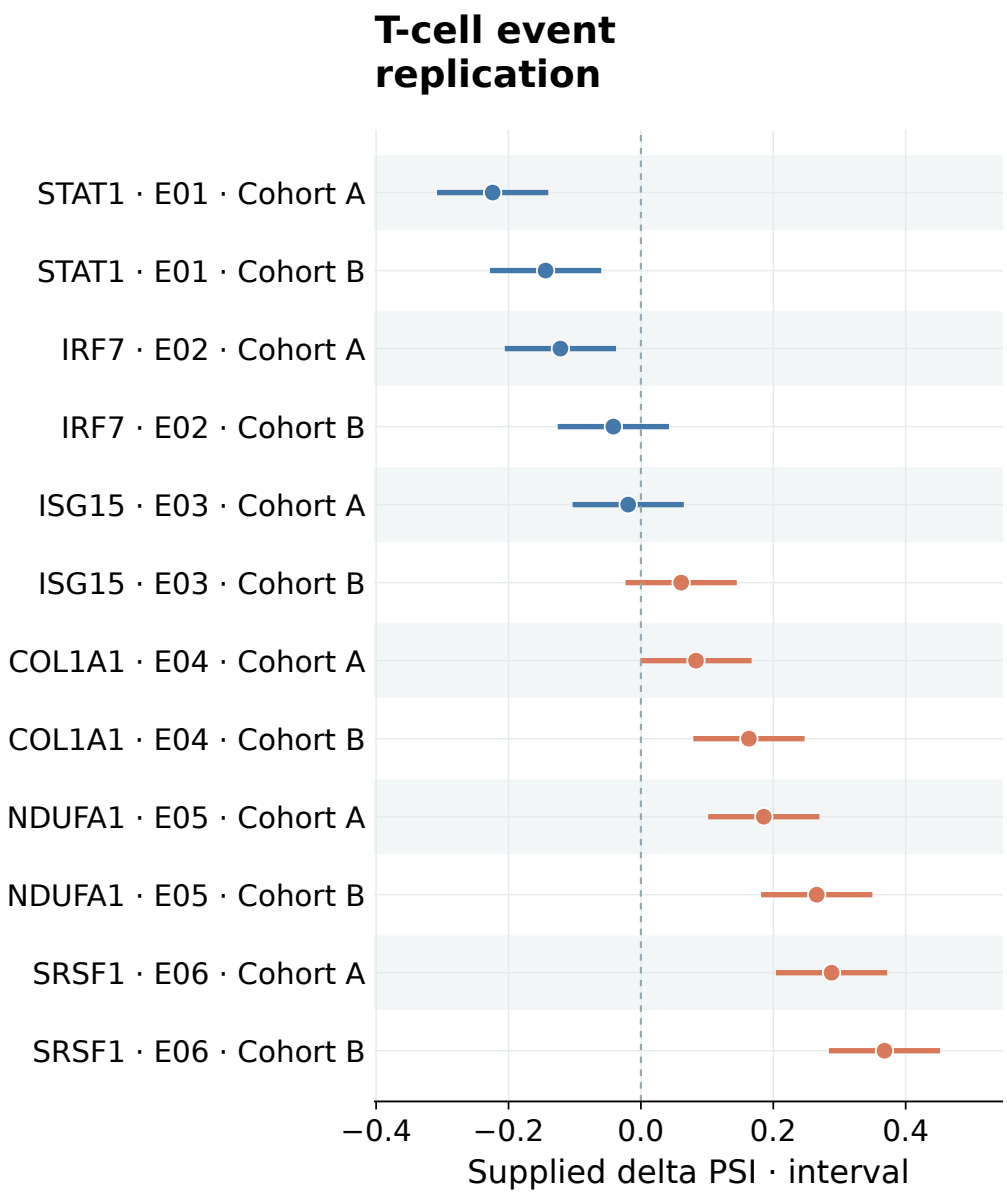
0.08 [0.00, 0.16] 0.056

0.19 [0.11, 0.27] 0.074

0.29 [0.21, 0.37] 0.092

Splicing-event effects across identities





Estimate [interval] · q
Independent n = 6

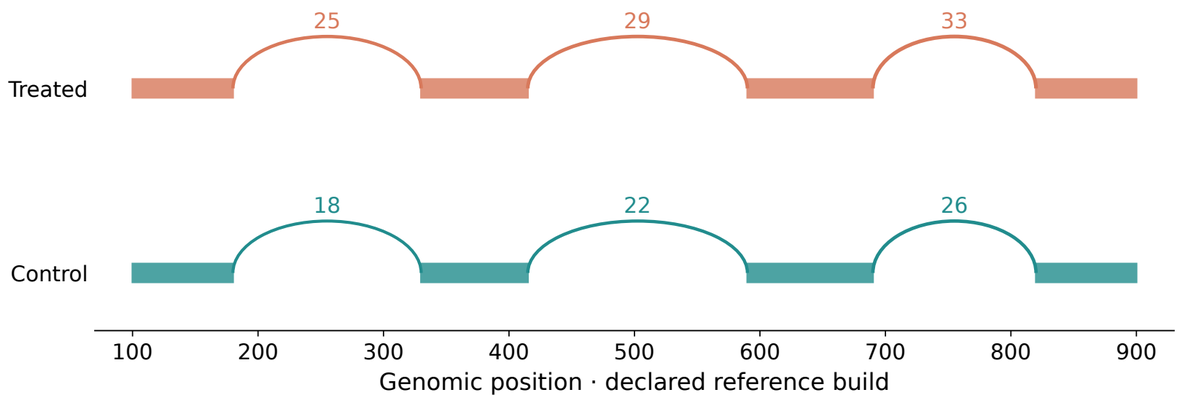
-0.22 [-0.30, -0.14] 0.002
-0.14 [-0.22, -0.06] 0.005
-0.12 [-0.20, -0.04] 0.02
-0.04 [-0.12, 0.04] 0.65
-0.02 [-0.10, 0.06] 0.65
0.06 [-0.02, 0.14] 0.65
0.08 [0.00, 0.16] 0.056
0.16 [0.08, 0.24] 0.059
0.19 [0.11, 0.27] 0.074
0.27 [0.19, 0.35] 0.077
0.29 [0.21, 0.37] 0.092
0.37 [0.29, 0.45] 0.095

Software test fixture; not biological evidence

Single cell · splicing — complementary evidence

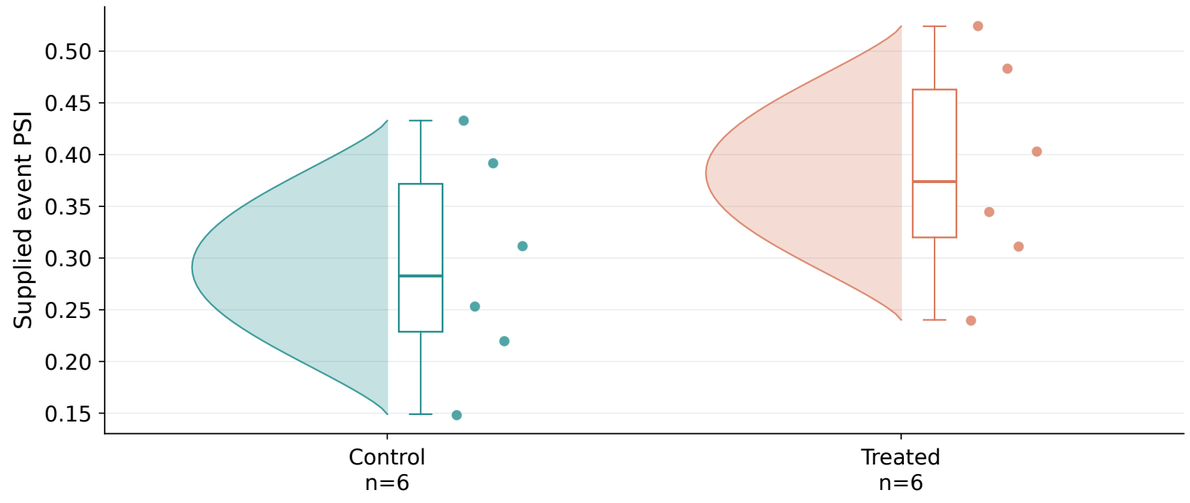
A

T-cell STAT1 E01 physical junction support



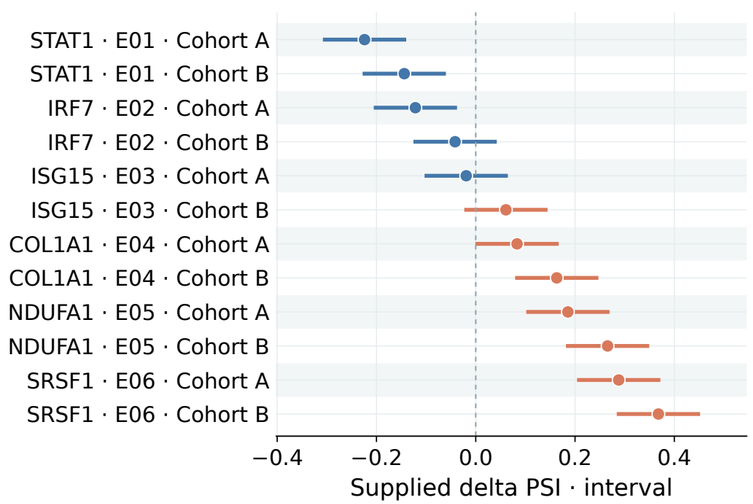
B

STAT1 event PSI in T-cell donor aggregates



C

T-cell event replication



Estimate [interval] · q
Independent n = 6

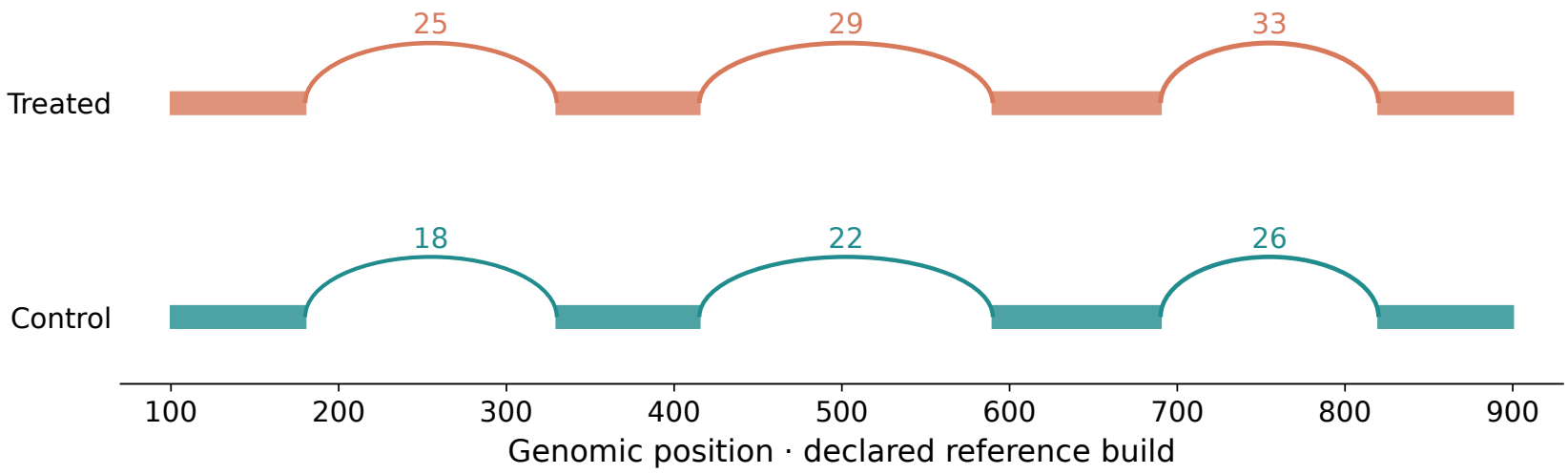
STAT1 · E01 · Cohort A	-0.22 [-0.30, -0.14] 0.002
STAT1 · E01 · Cohort B	-0.14 [-0.22, -0.06] 0.005
IRF7 · E02 · Cohort A	-0.12 [-0.20, -0.04] 0.02
IRF7 · E02 · Cohort B	-0.04 [-0.12, 0.04] 0.65
ISG15 · E03 · Cohort A	-0.02 [-0.10, 0.06] 0.65
ISG15 · E03 · Cohort B	0.06 [-0.02, 0.14] 0.65
COL1A1 · E04 · Cohort A	0.08 [0.00, 0.16] 0.056
COL1A1 · E04 · Cohort B	0.16 [0.08, 0.24] 0.059
NDUFA1 · E05 · Cohort A	0.19 [0.11, 0.27] 0.074
NDUFA1 · E05 · Cohort B	0.27 [0.19, 0.35] 0.077
SRSF1 · E06 · Cohort A	0.29 [0.21, 0.37] 0.092
SRSF1 · E06 · Cohort B	0.37 [0.29, 0.45] 0.095

Software test fixture; not biological evidence

Single cell · splicing — complementary evidence

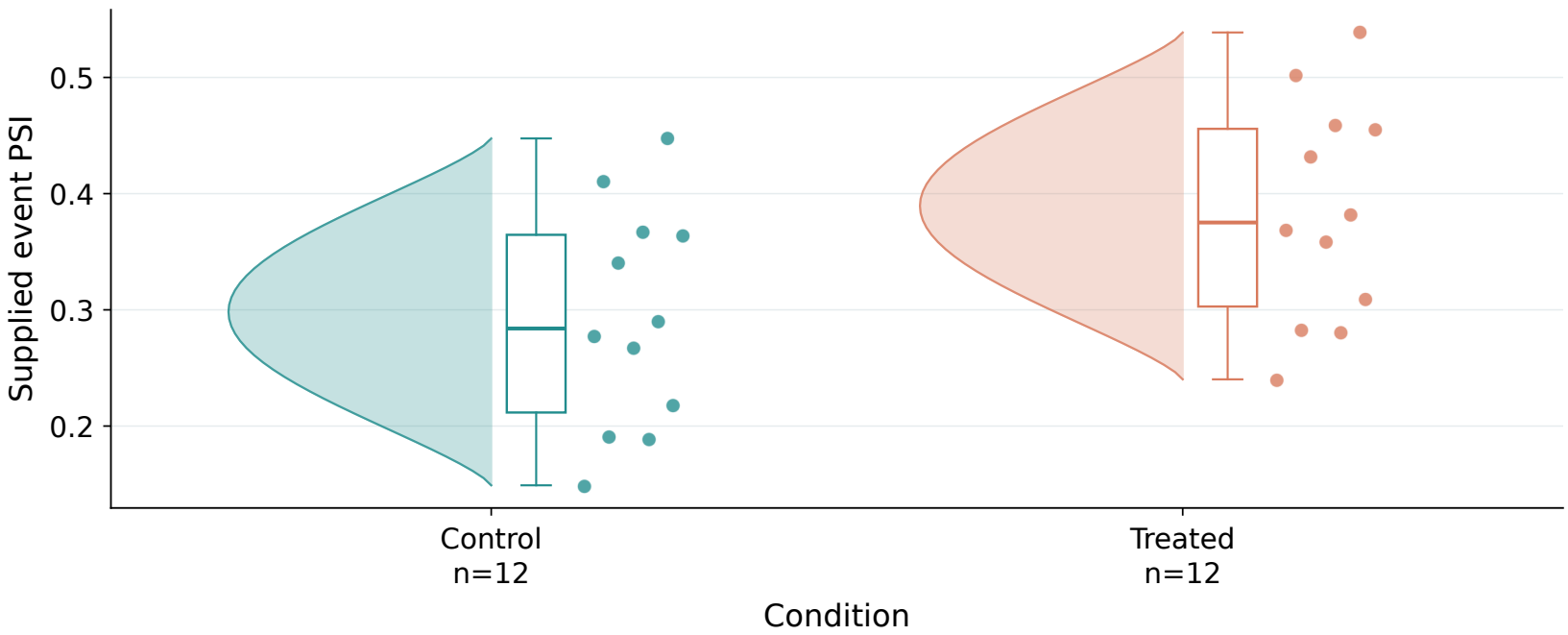
A

T-cell STAT1 E01 physical junction support



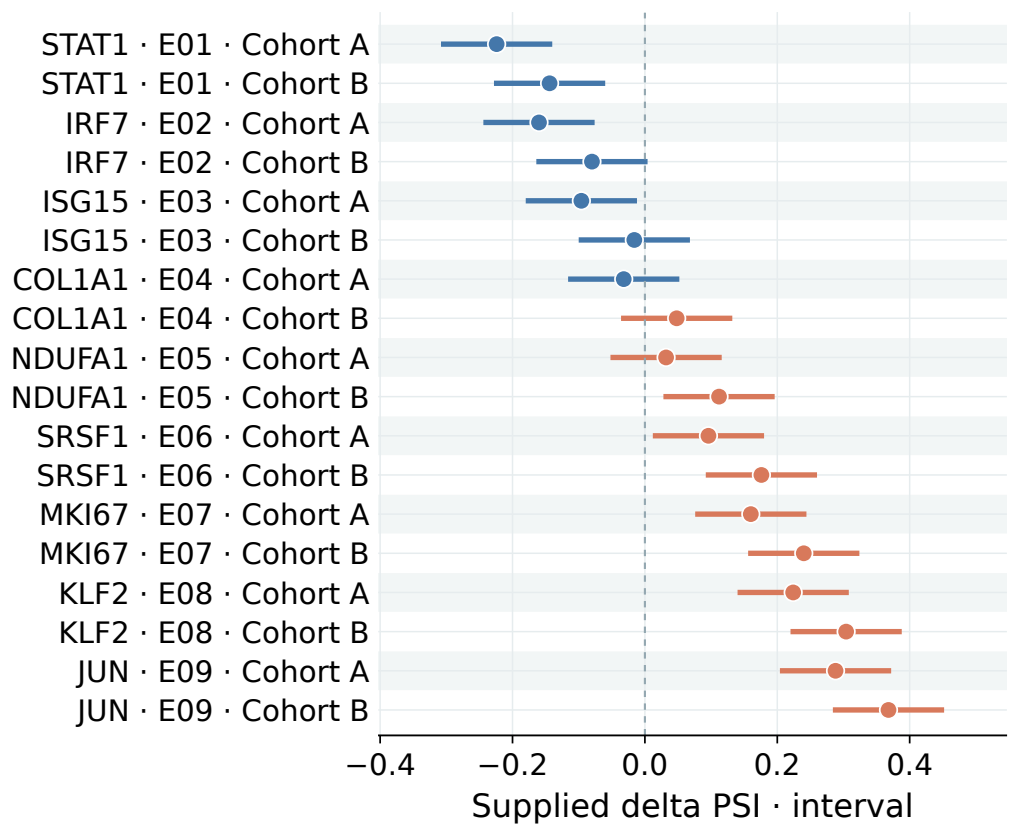
B

STAT1 event PSI in T-cell donor aggregates



C

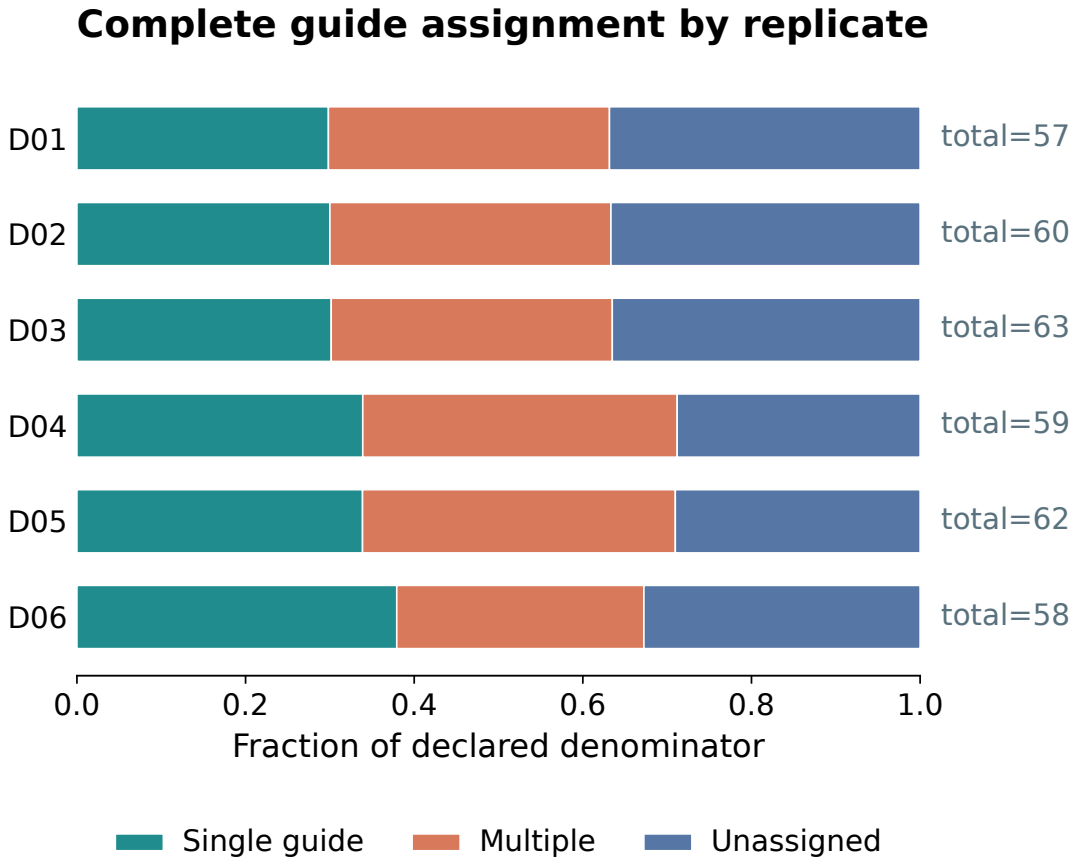
T-cell event replication



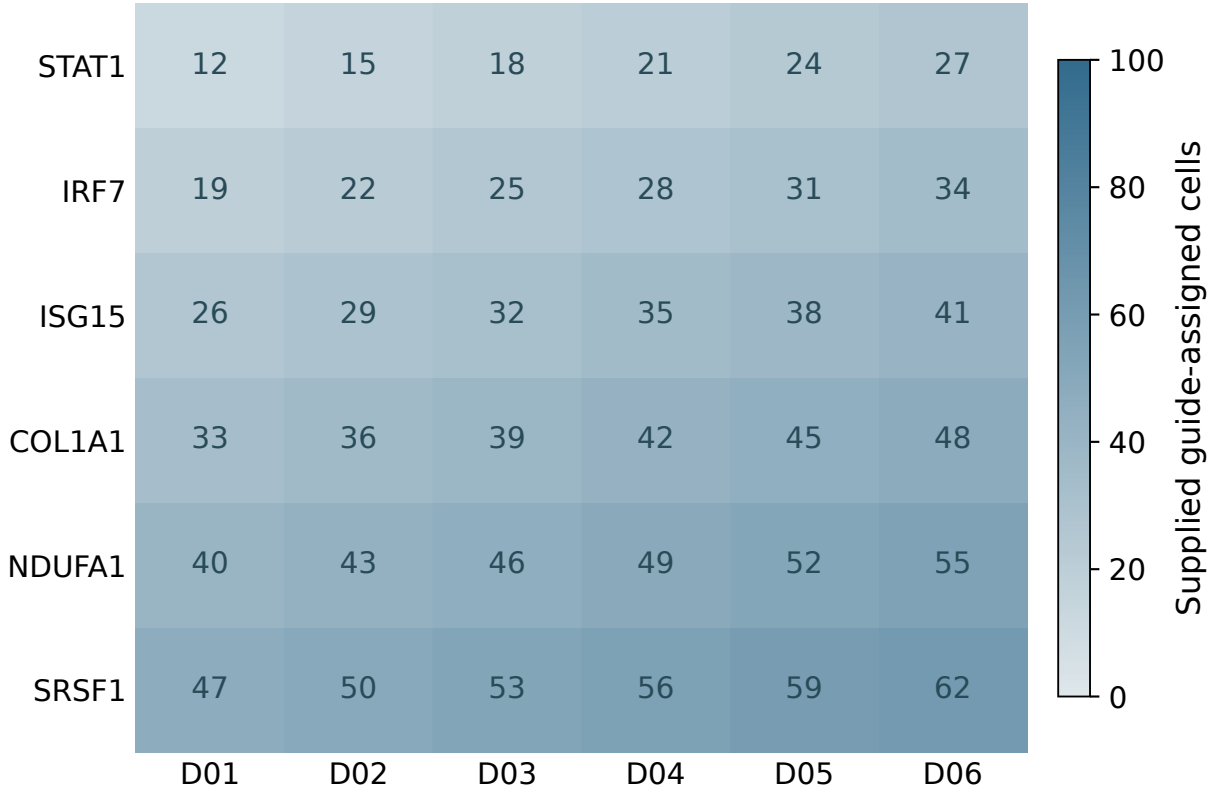
Estimate [interval] · q
Independent n = 12

STAT1 · E01 · Cohort A	-0.22 [-0.30, -0.14] 0.002
STAT1 · E01 · Cohort B	-0.14 [-0.22, -0.06] 0.005
IRF7 · E02 · Cohort A	-0.16 [-0.24, -0.08] 0.02
IRF7 · E02 · Cohort B	-0.08 [-0.16, 0.00] 0.65
ISG15 · E03 · Cohort A	-0.10 [-0.18, -0.02] 0.038
ISG15 · E03 · Cohort B	-0.02 [-0.10, 0.06] 0.65
COL1A1 · E04 · Cohort A	-0.03 [-0.11, 0.05] 0.65
COL1A1 · E04 · Cohort B	0.05 [-0.03, 0.13] 0.65
NDUFA1 · E05 · Cohort A	0.03 [-0.05, 0.11] 0.65
NDUFA1 · E05 · Cohort B	0.11 [0.03, 0.19] 0.077
SRSF1 · E06 · Cohort A	0.10 [0.02, 0.18] 0.092
SRSF1 · E06 · Cohort B	0.18 [0.10, 0.26] 0.095
MKI67 · E07 · Cohort A	0.16 [0.08, 0.24] 0.11
MKI67 · E07 · Cohort B	0.24 [0.16, 0.32] 0.11
KLF2 · E08 · Cohort A	0.22 [0.14, 0.30] 0.13
KLF2 · E08 · Cohort B	0.30 [0.22, 0.38] 0.13
JUN · E09 · Cohort A	0.29 [0.21, 0.37] 0.15
JUN · E09 · Cohort B	0.37 [0.29, 0.45] 0.15

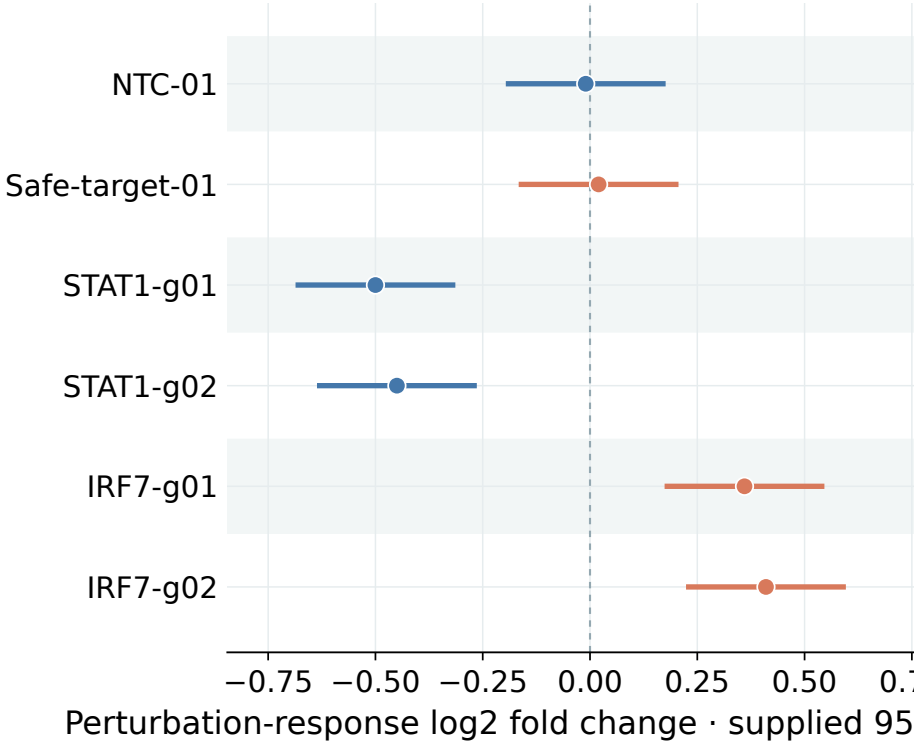
Software test fixture; not biological evidence


Complete guide assignment by replicate
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Single guide
Multiple
Unassigned

Guide support across donors



STAT1 response by guide and control



Estimate [interval] · q
Independent n = 12

-0.01 [-0.19, 0.17] 0.65

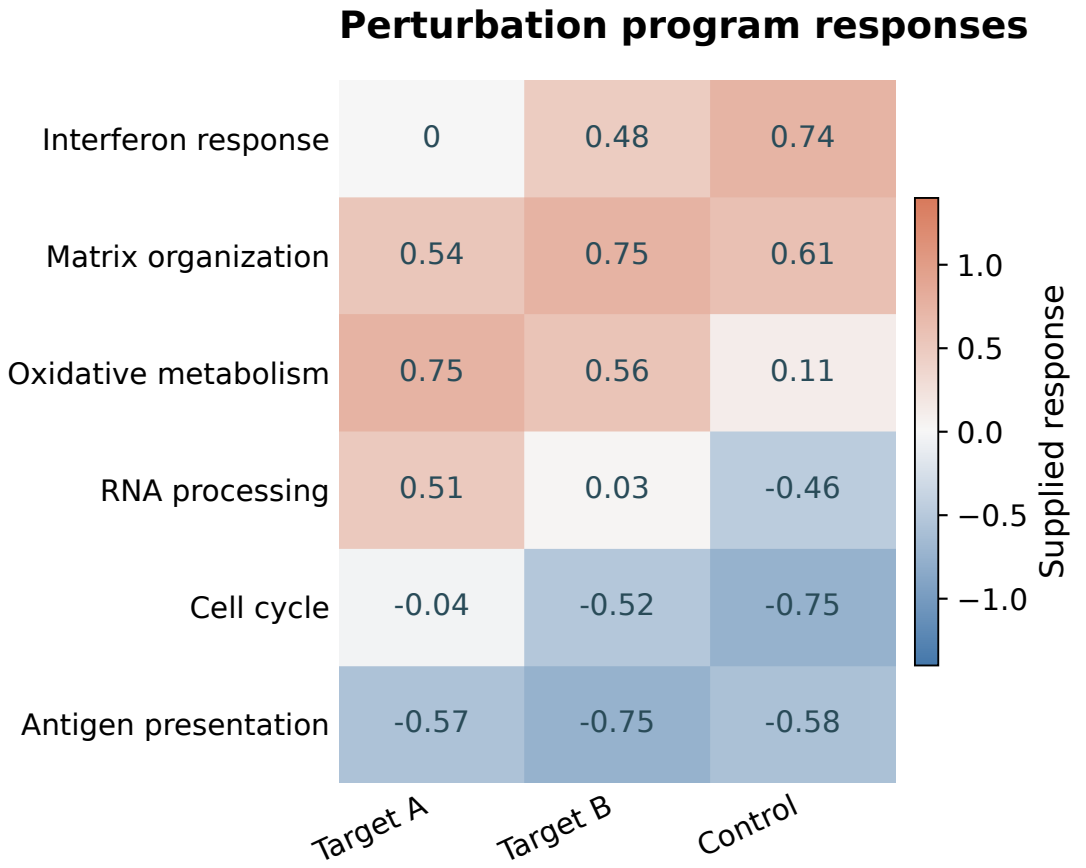
0.02 [-0.16, 0.20] 0.65

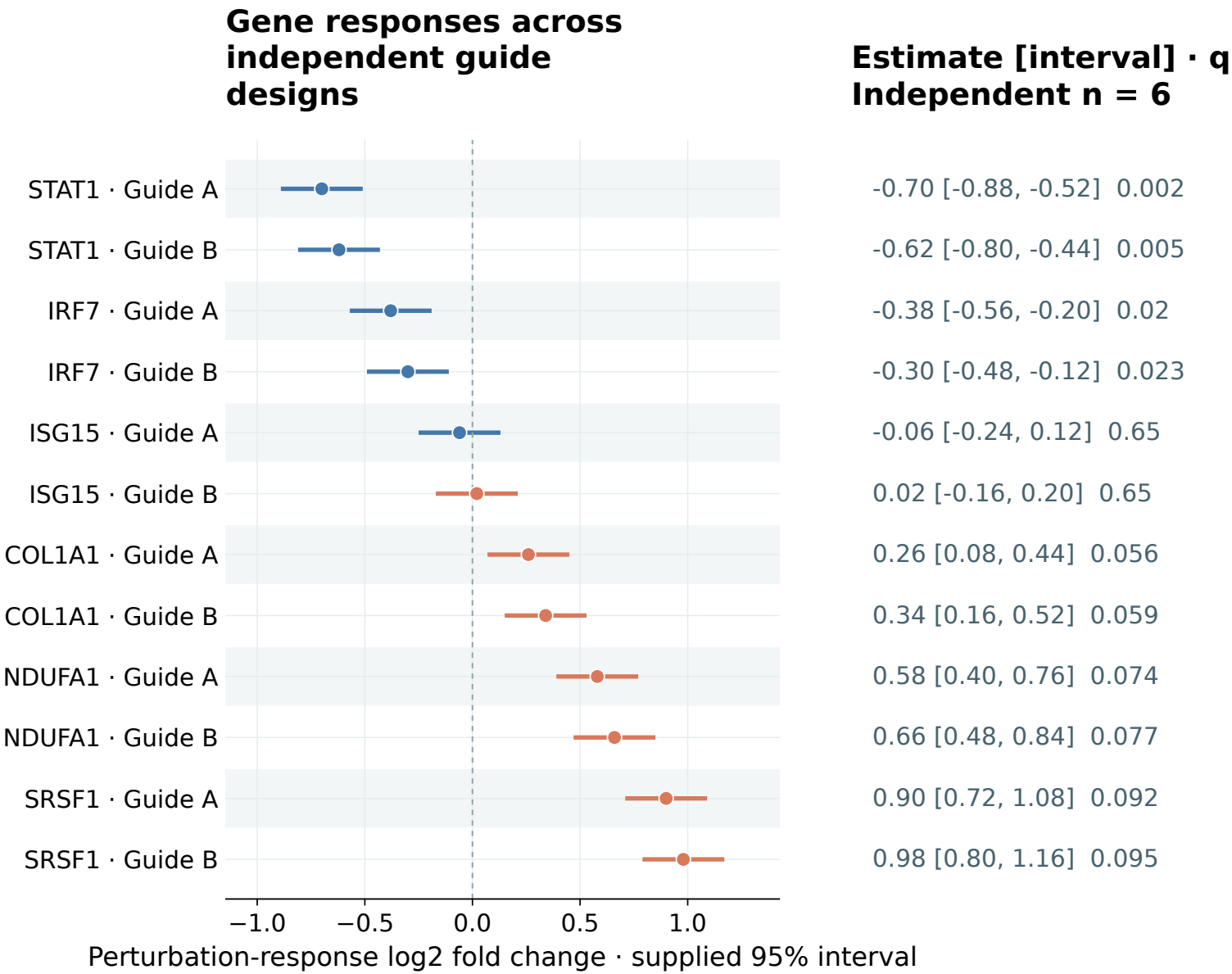
-0.50 [-0.68, -0.32] 0.038

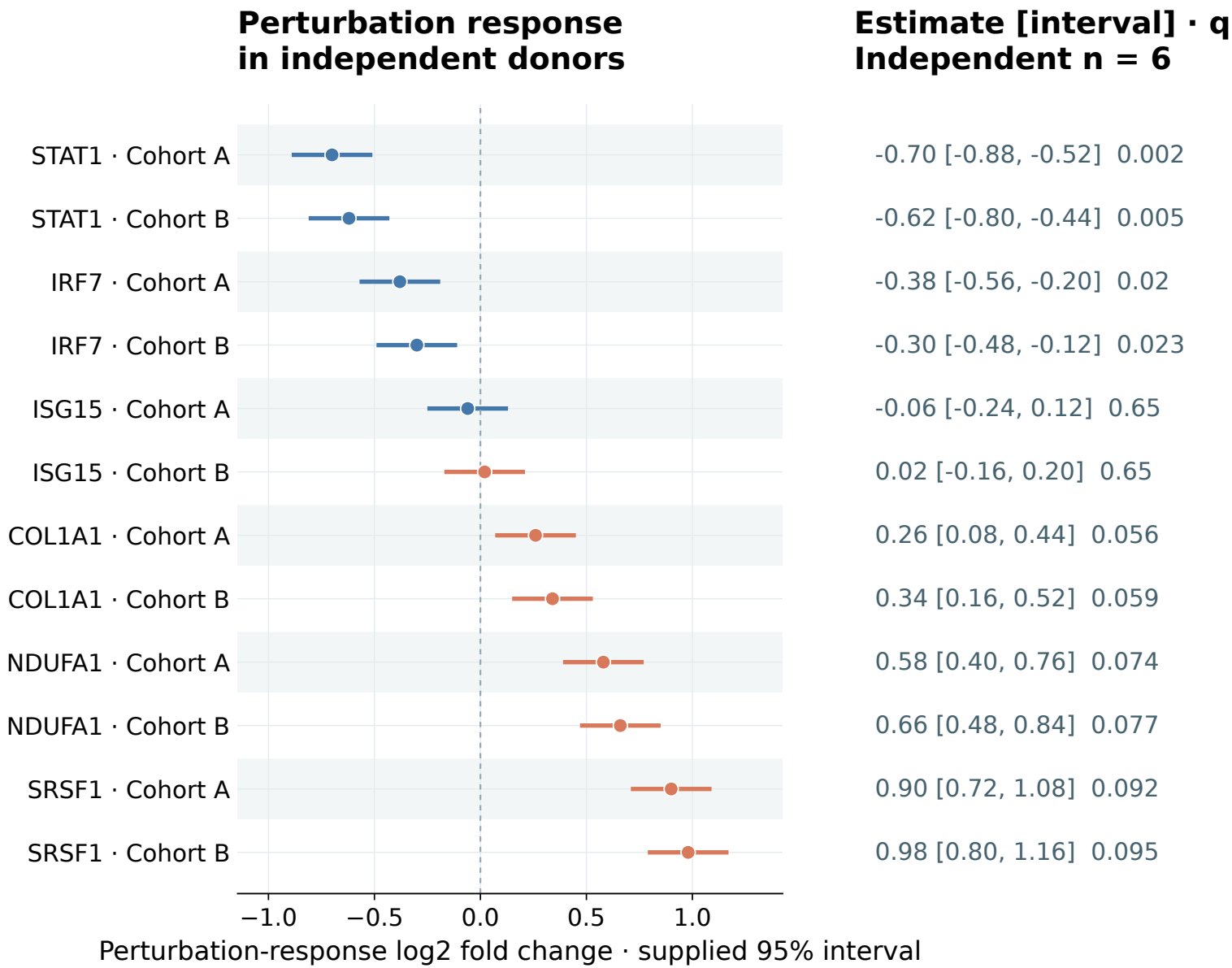
-0.45 [-0.63, -0.27] 0.056

0.36 [0.18, 0.54] 0.074

0.41 [0.23, 0.59] 0.092


Perturbation program responses
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
0
0.48
0.74
0.54
0.75
0.61
0.75
0.56
0.11
0.51
0.03
-0.46
-0.04
-0.52
-0.75
-0.57
-0.75
-0.58
Target A
Target B
Control
1.0
0.5
0.0
−0.5
−1.0
Supplied response


Gene responses across independent guide designs
Estimate [interval] · q
Independent n = 6
STAT1 · Guide A
STAT1 · Guide B
IRF7 · Guide A
IRF7 · Guide B
ISG15 · Guide A
ISG15 · Guide B
COL1A1 · Guide A
COL1A1 · Guide B
NDUFA1 · Guide A
NDUFA1 · Guide B
SRSF1 · Guide A
SRSF1 · Guide B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Perturbation-response log2 fold change · supplied 95% interval

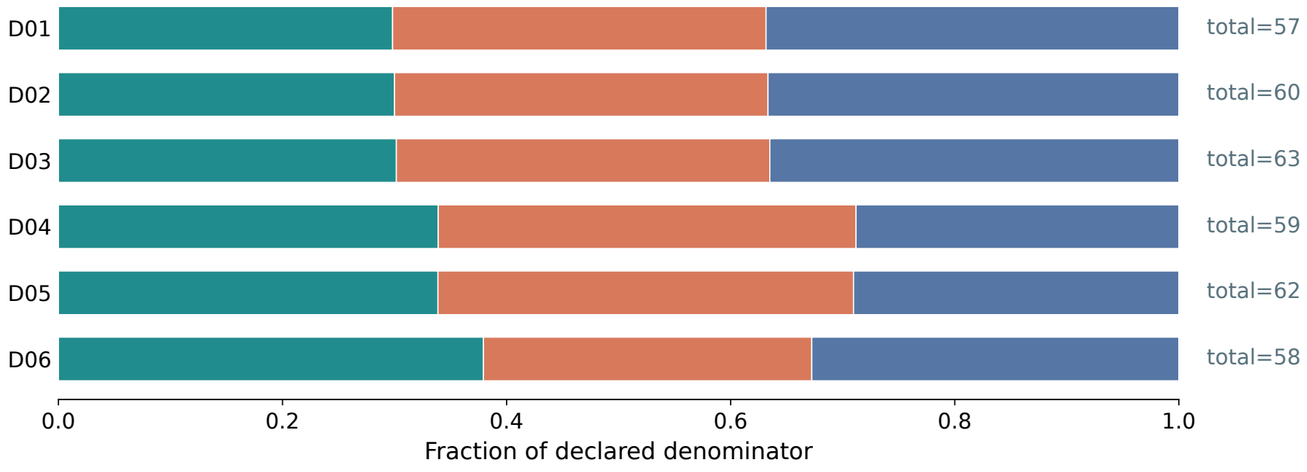


Software test fixture; not biological evidence

Single cell · perturbseq — complementary evidence

A

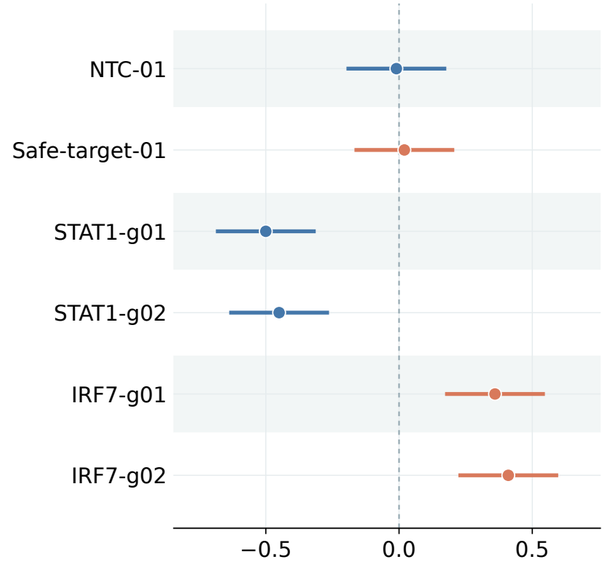
Complete guide assignment by replicate



B

STAT1 response by guide and control

Estimate [interval] · q
Independent n = 12

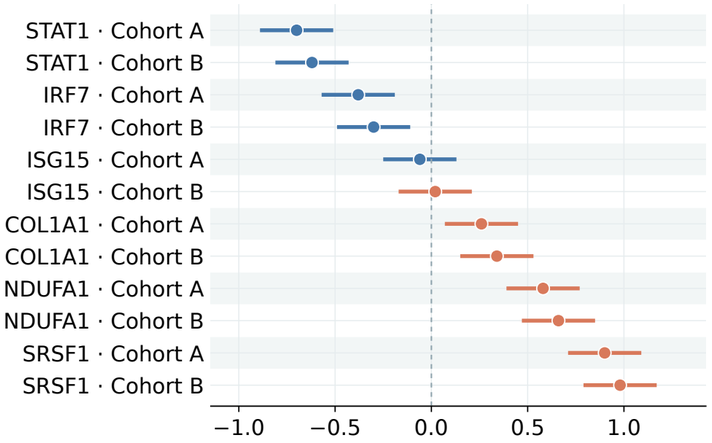


-0.01 [-0.19, 0.17] 0.65
0.02 [-0.16, 0.20] 0.65
-0.50 [-0.68, -0.32] 0.038
-0.45 [-0.63, -0.27] 0.056
0.36 [0.18, 0.54] 0.074
0.41 [0.23, 0.59] 0.092

C

Perturbation response in independent donors

Estimate [interval] · q
Independent n = 6

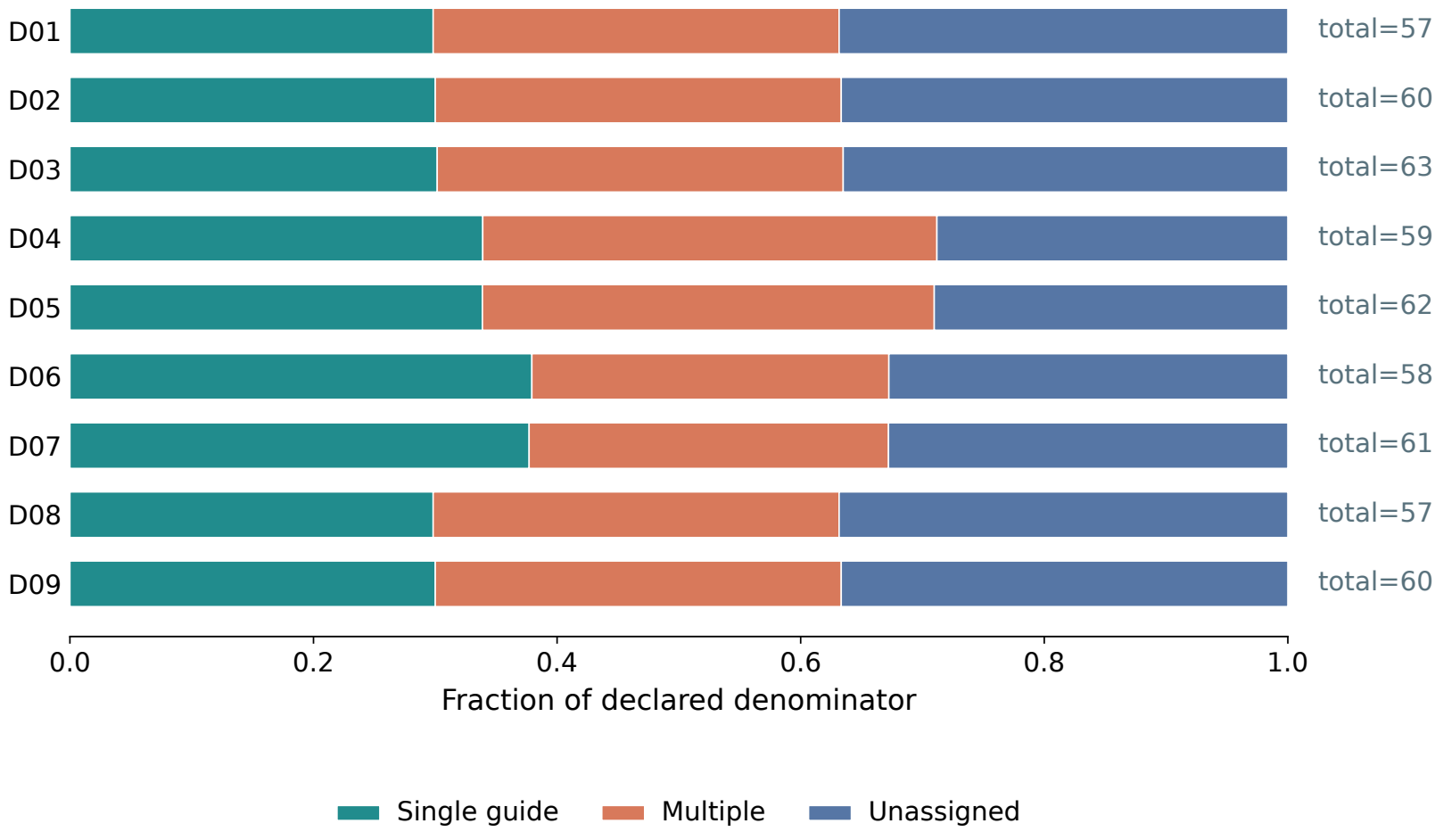


-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095

Single cell · perturbseq — complementary evidence

A

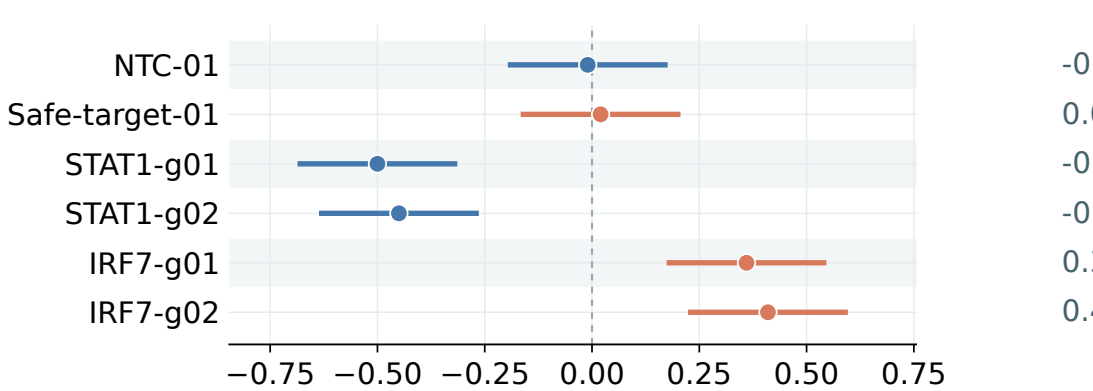
Complete guide assignment by replicate



B

STAT1 response by guide and control

Estimate [interval] · q Independent n = 24

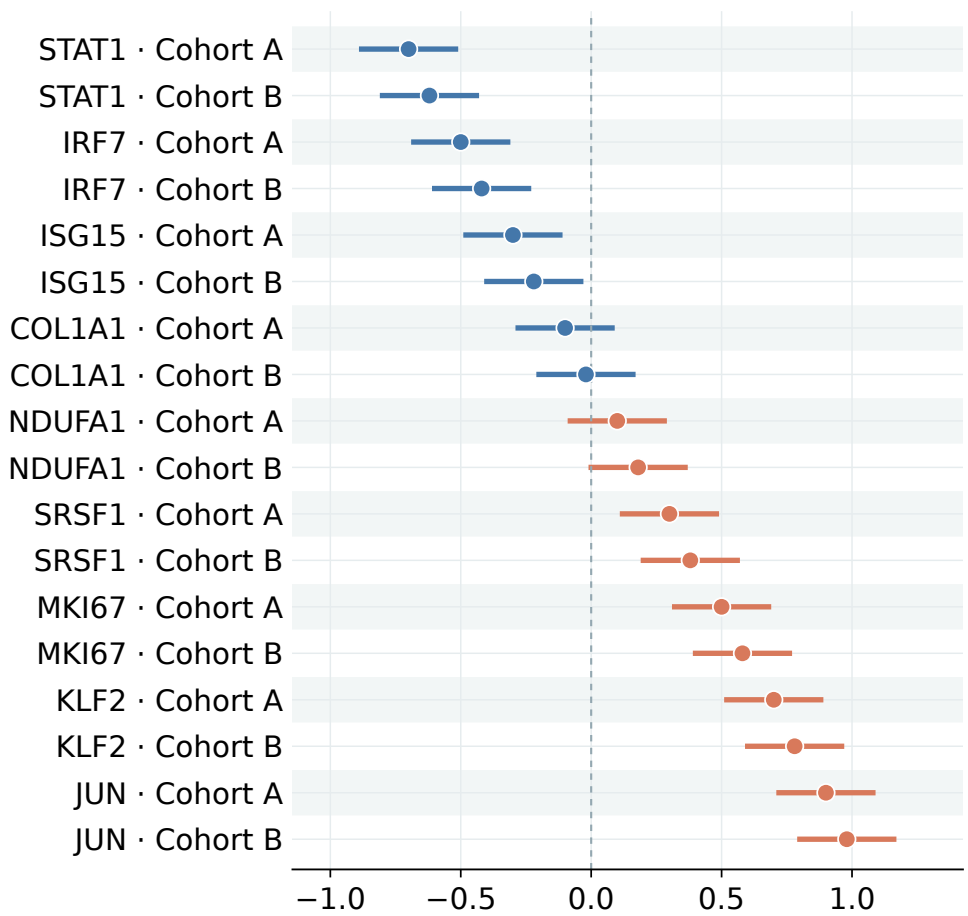


NTC-01	-0.01 [-0.19, 0.17] 0.65
Safe-target-01	0.02 [-0.16, 0.20] 0.65
STAT1-g01	-0.50 [-0.68, -0.32] 0.038
STAT1-g02	-0.45 [-0.63, -0.27] 0.056
IRF7-g01	0.36 [0.18, 0.54] 0.074
IRF7-g02	0.41 [0.23, 0.59] 0.092

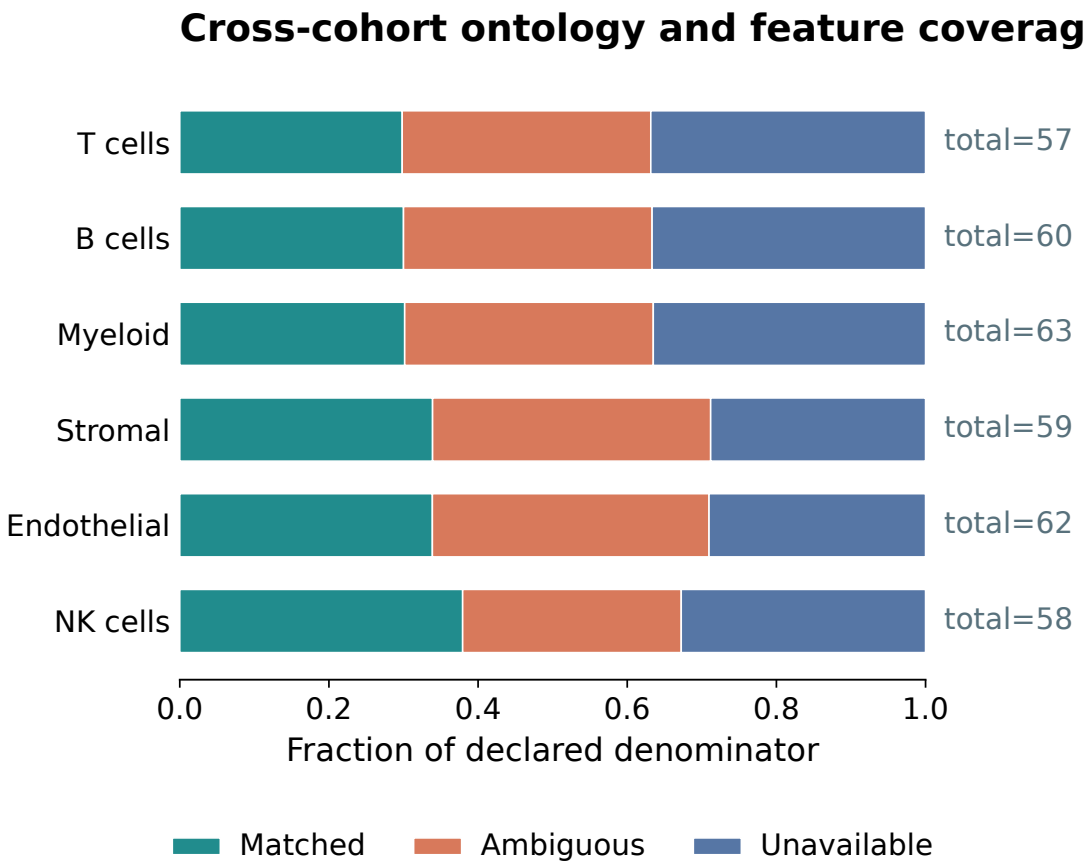
C

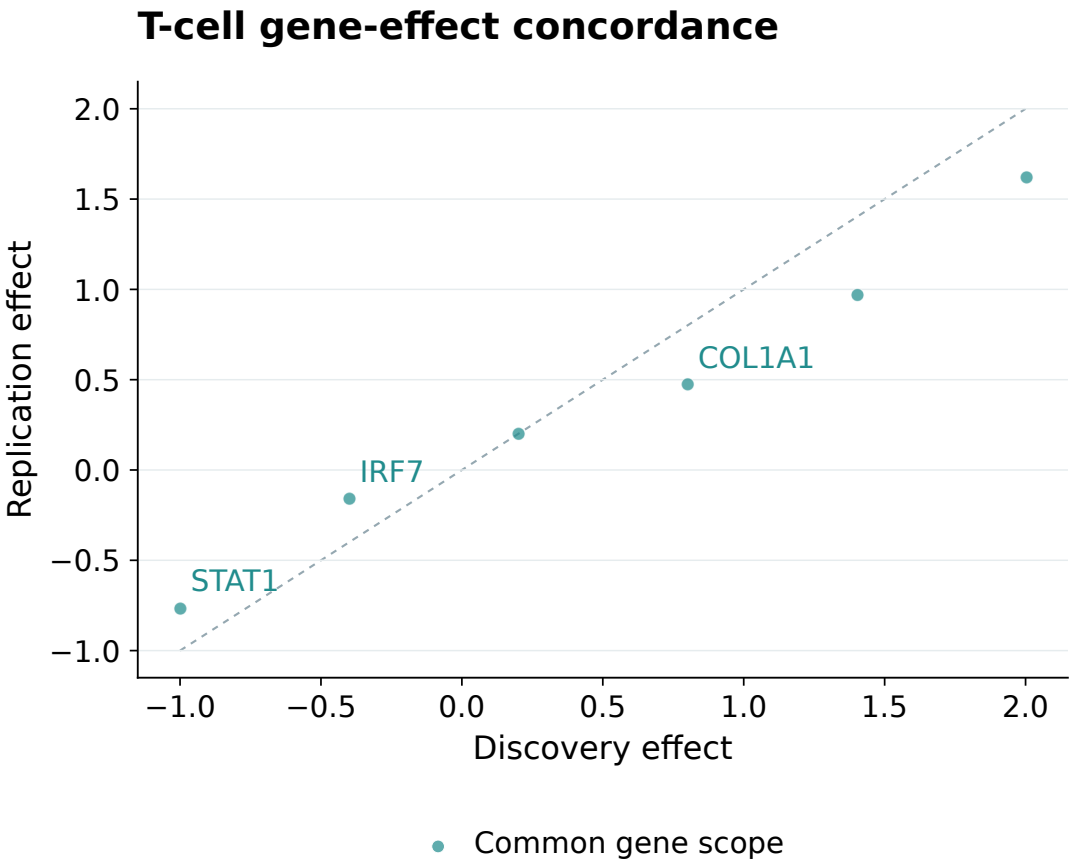
Perturbation response in independent donors

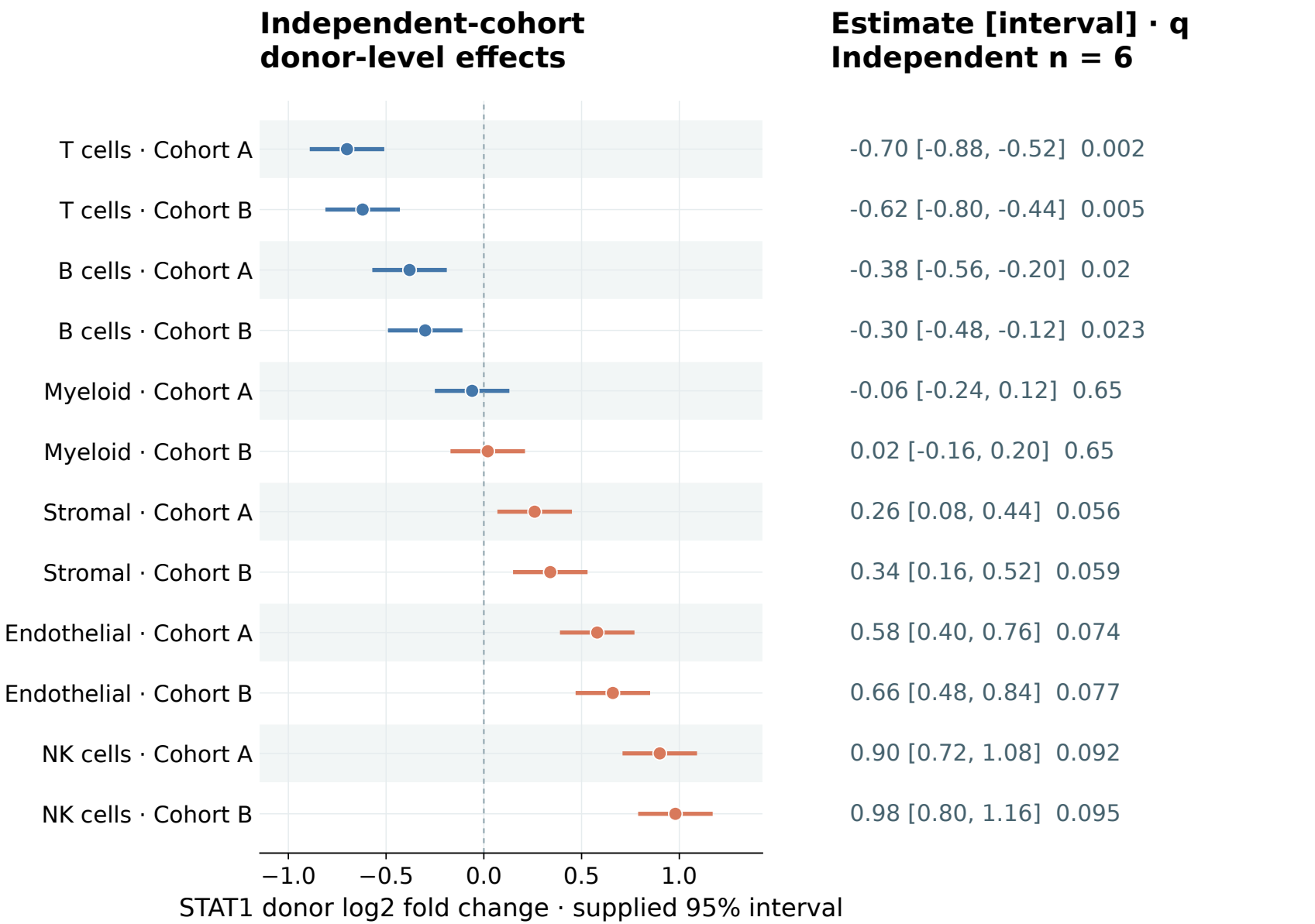
Estimate [interval] · q Independent n = 12

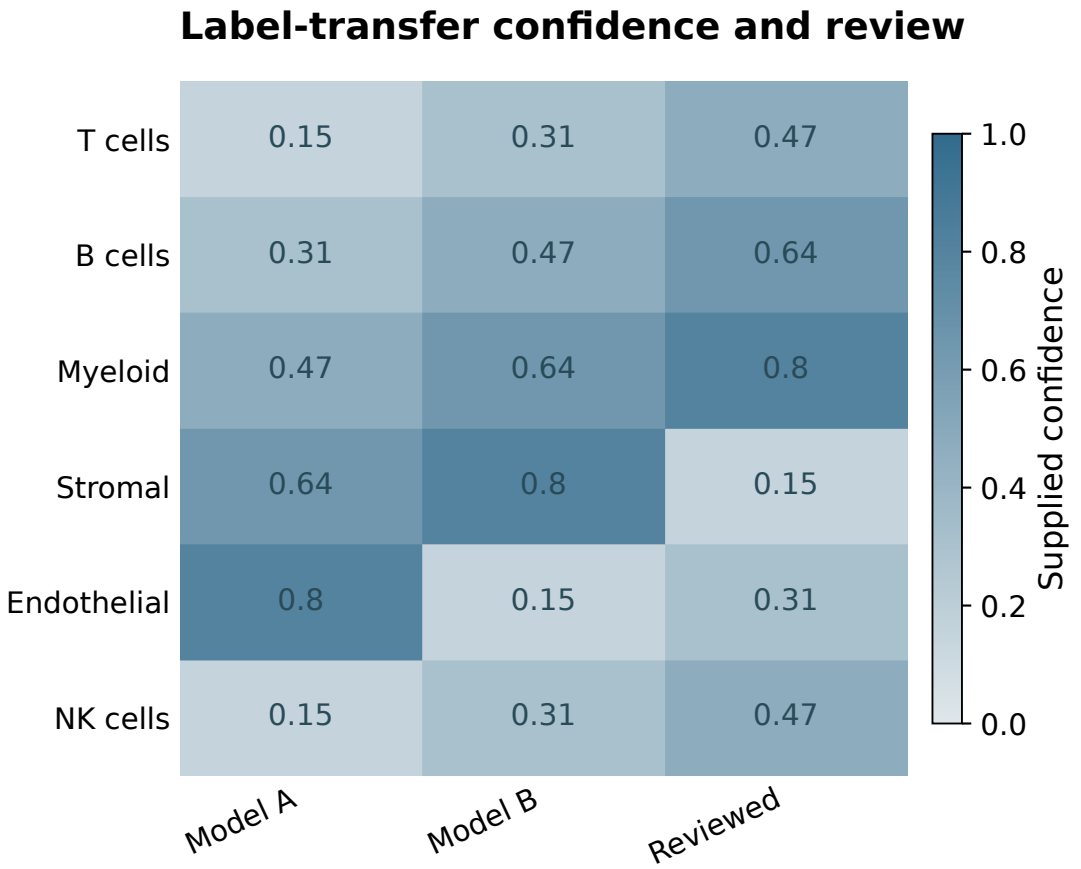


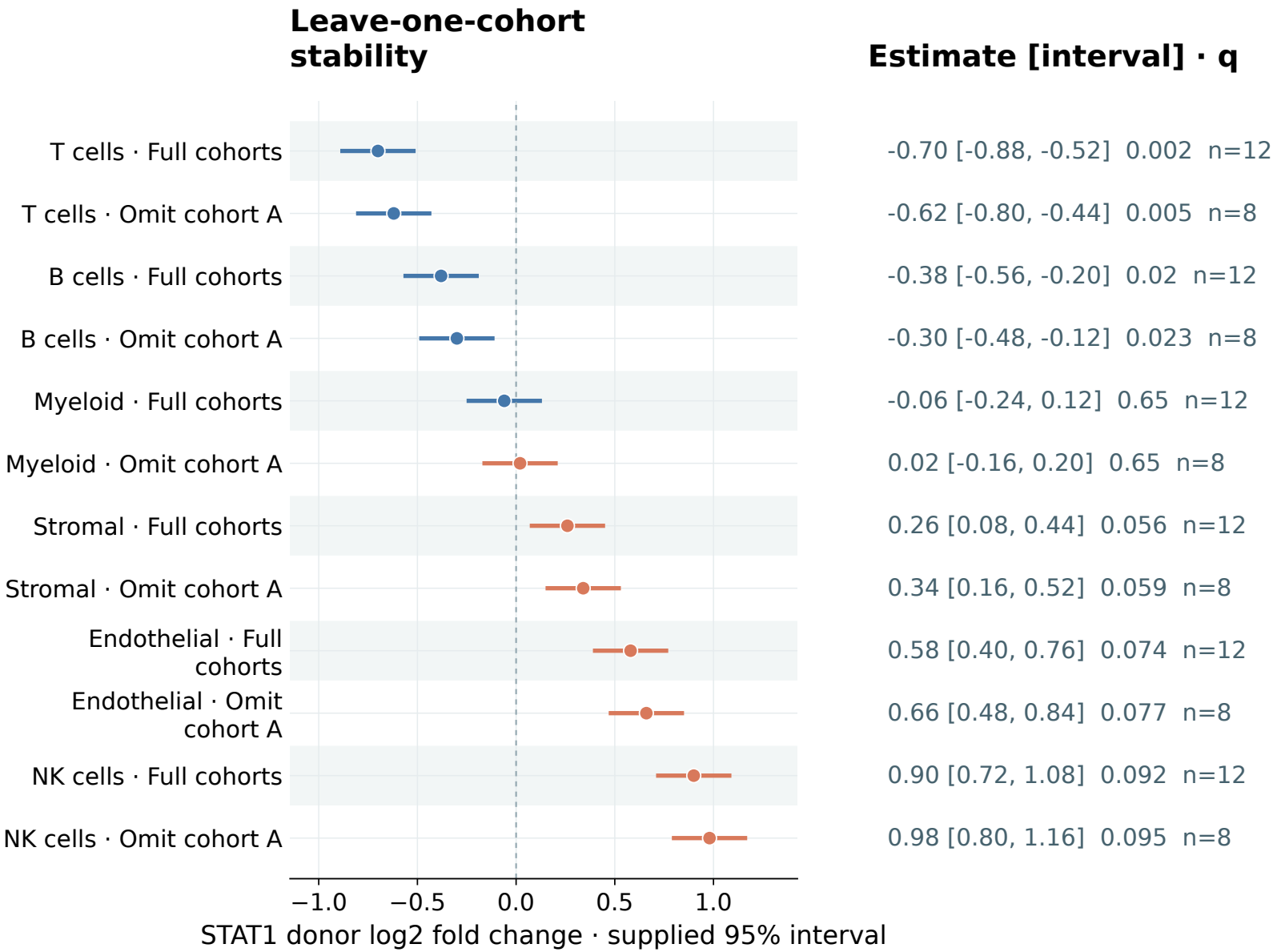
STAT1 · Cohort A	-0.70 [-0.88, -0.52] 0.002
STAT1 · Cohort B	-0.62 [-0.80, -0.44] 0.005
IRF7 · Cohort A	-0.50 [-0.68, -0.32] 0.02
IRF7 · Cohort B	-0.42 [-0.60, -0.24] 0.023
ISG15 · Cohort A	-0.30 [-0.48, -0.12] 0.038
ISG15 · Cohort B	-0.22 [-0.40, -0.04] 0.041
COL1A1 · Cohort A	-0.10 [-0.28, 0.08] 0.65
COL1A1 · Cohort B	-0.02 [-0.20, 0.16] 0.65
NDUFA1 · Cohort A	0.10 [-0.08, 0.28] 0.65
NDUFA1 · Cohort B	0.18 [0.00, 0.36] 0.077
SRSF1 · Cohort A	0.30 [0.12, 0.48] 0.092
SRSF1 · Cohort B	0.38 [0.20, 0.56] 0.095
MKI67 · Cohort A	0.50 [0.32, 0.68] 0.11
MKI67 · Cohort B	0.58 [0.40, 0.76] 0.11
KLF2 · Cohort A	0.70 [0.52, 0.88] 0.13
KLF2 · Cohort B	0.78 [0.60, 0.96] 0.13
JUN · Cohort A	0.90 [0.72, 1.08] 0.15
JUN · Cohort B	0.98 [0.80, 1.16] 0.15


Cross-cohort ontology and feature coverage
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Matched
Ambiguous
Unavailable


T-cell gene-effect concordance
Replication effect
Discovery effect
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
COL1A1
IRF7
STAT1
Common gene scope


Independent-cohort
donor-level effects
Estimate [interval] · q
Independent n = 6
T cells · Cohort A
T cells · Cohort B
B cells · Cohort A
B cells · Cohort B
Myeloid · Cohort A
Myeloid · Cohort B
Stromal · Cohort A
Stromal · Cohort B
Endothelial · Cohort A
Endothelial · Cohort B
NK cells · Cohort A
NK cells · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
STAT1 donor log2 fold change · supplied 95% interval

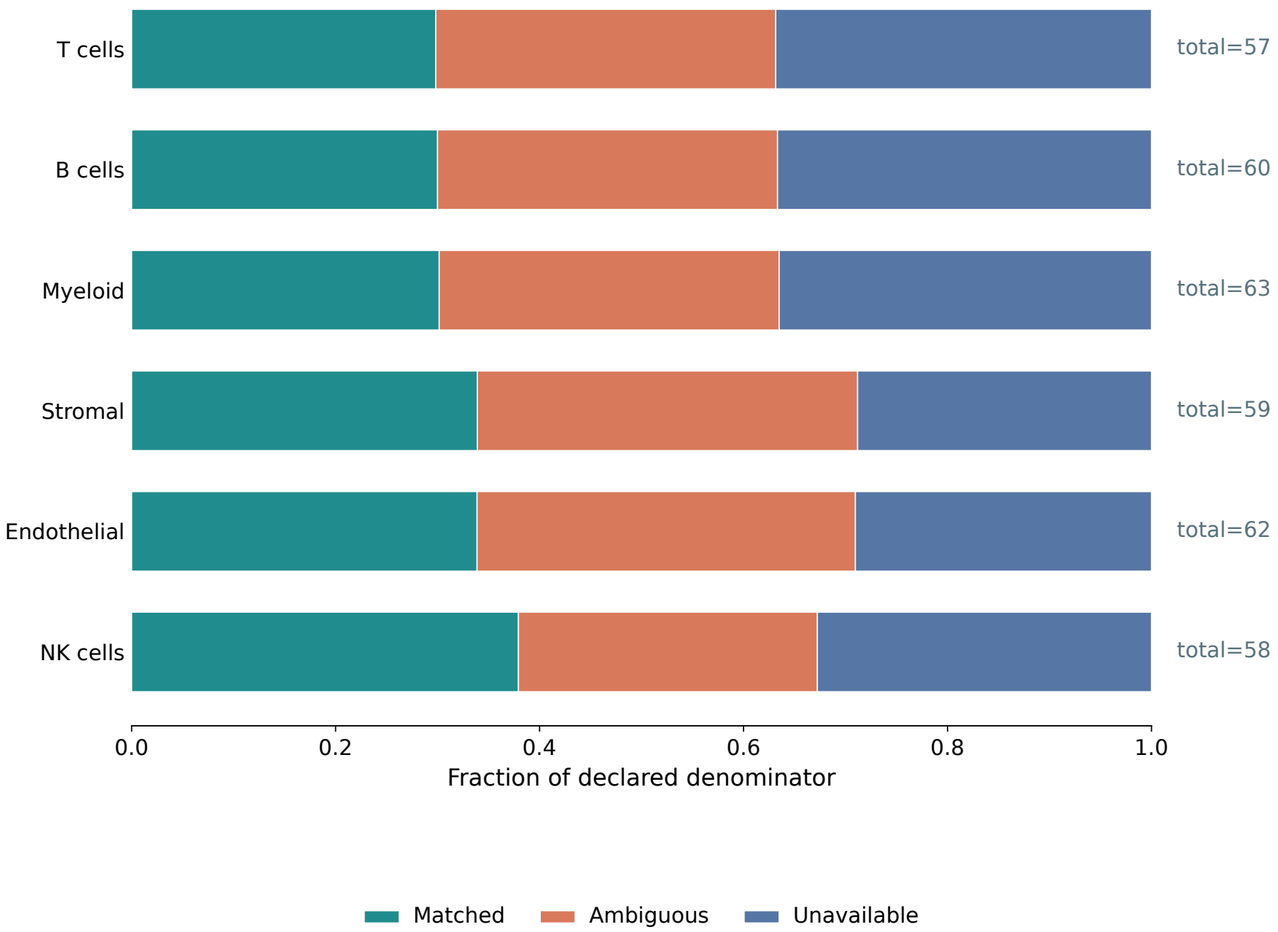

Label-transfer confidence and review
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
0.15
0.31
0.47
0.31
0.47
0.64
0.47
0.64
0.8
0.64
0.8
0.15
0.8
0.15
0.31
0.15
0.31
0.47
Model A
Model B
Reviewed
1.0
0.8
0.6
0.4
0.2
0.0
Supplied confidence


Leave-one-cohort stability
Estimate [interval] · q
T cells · Full cohorts
T cells · Omit cohort A
B cells · Full cohorts
B cells · Omit cohort A
Myeloid · Full cohorts
Myeloid · Omit cohort A
Stromal · Full cohorts
Stromal · Omit cohort A
Endothelial · Full cohorts
Endothelial · Omit cohort A
NK cells · Full cohorts
NK cells · Omit cohort A
-0.70 [-0.88, -0.52] 0.002 n=12
-0.62 [-0.80, -0.44] 0.005 n=8
-0.38 [-0.56, -0.20] 0.02 n=12
-0.30 [-0.48, -0.12] 0.023 n=8
-0.06 [-0.24, 0.12] 0.65 n=12
0.02 [-0.16, 0.20] 0.65 n=8
0.26 [0.08, 0.44] 0.056 n=12
0.34 [0.16, 0.52] 0.059 n=8
0.58 [0.40, 0.76] 0.074 n=12
0.66 [0.48, 0.84] 0.077 n=8
0.90 [0.72, 1.08] 0.092 n=12
0.98 [0.80, 1.16] 0.095 n=8
−1.0
−0.5
0.0
0.5
1.0
STAT1 donor log2 fold change · supplied 95% interval

Single cell · replication — complementary evidence

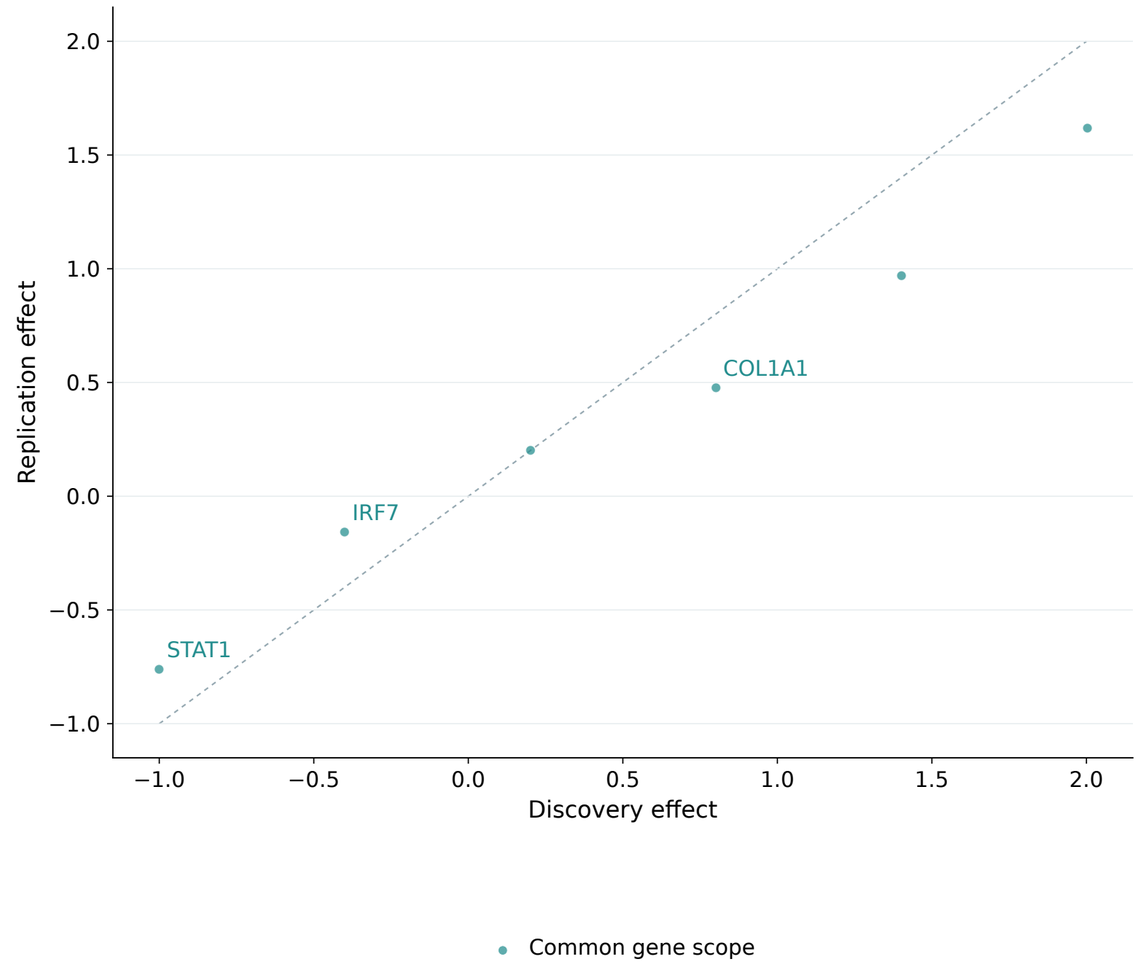
A

Cross-cohort ontology and feature coverage



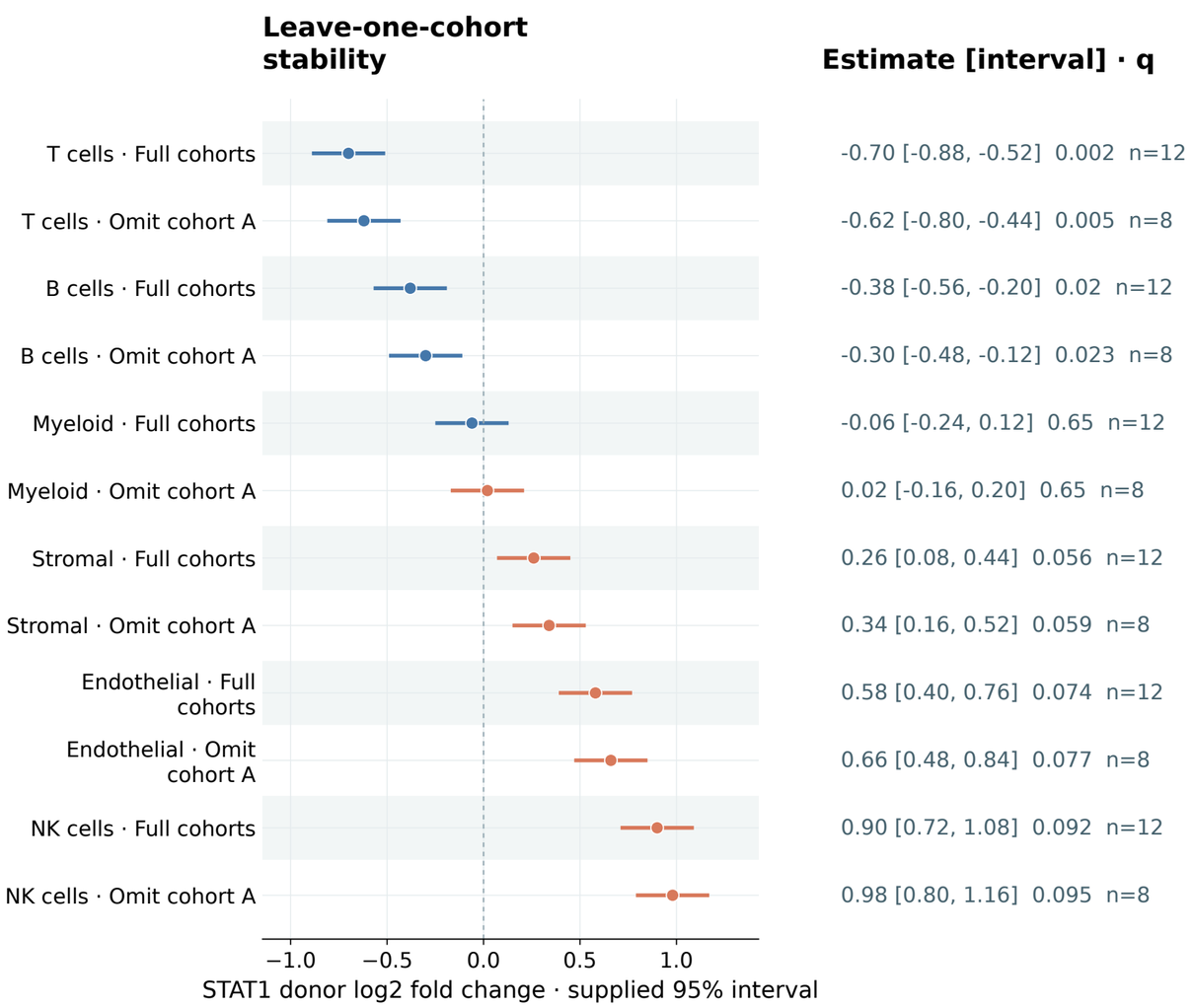
B

T-cell gene-effect concordance



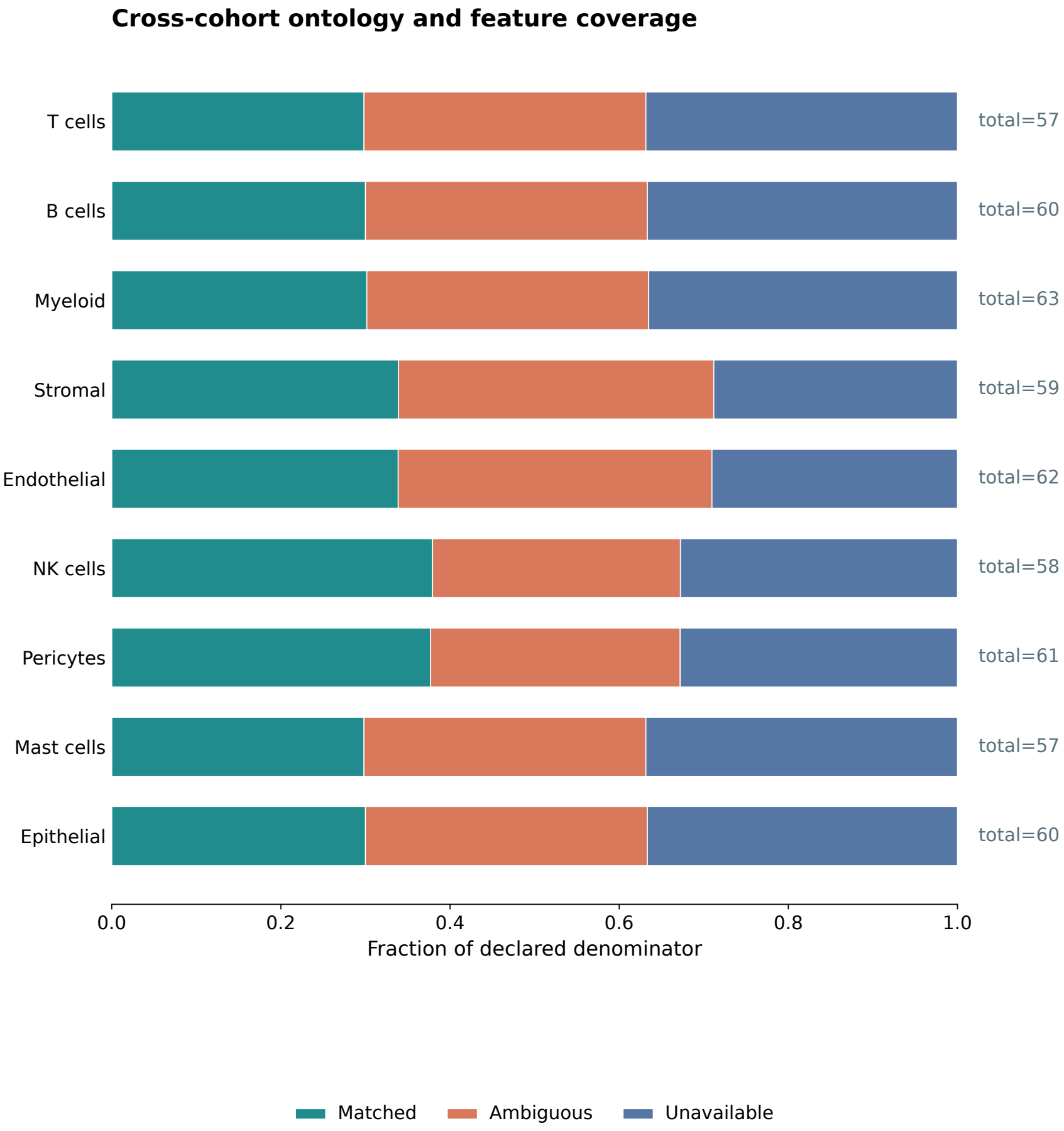
C

Leave-one-cohort stability

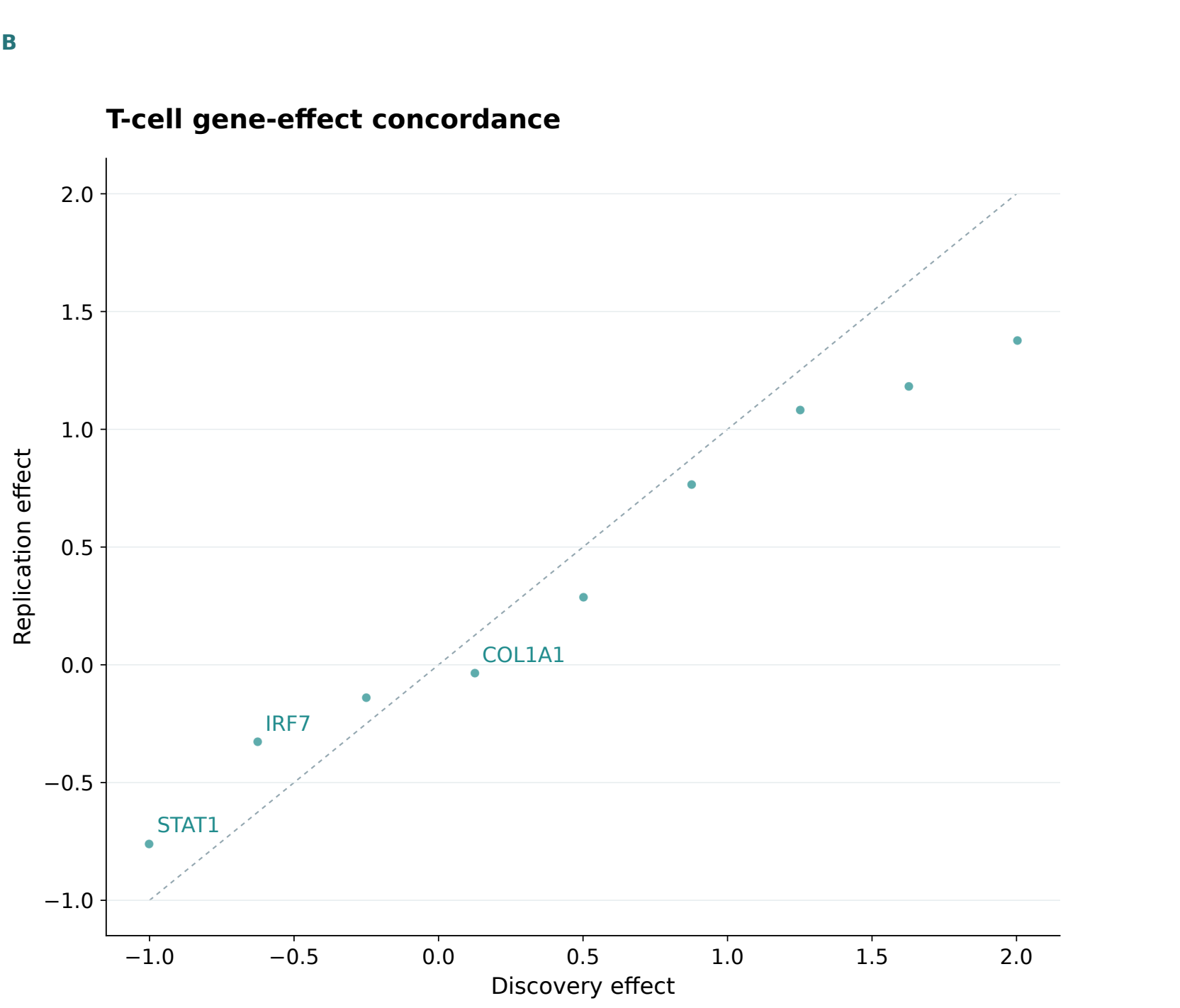


Single cell · replication — complementary evidence

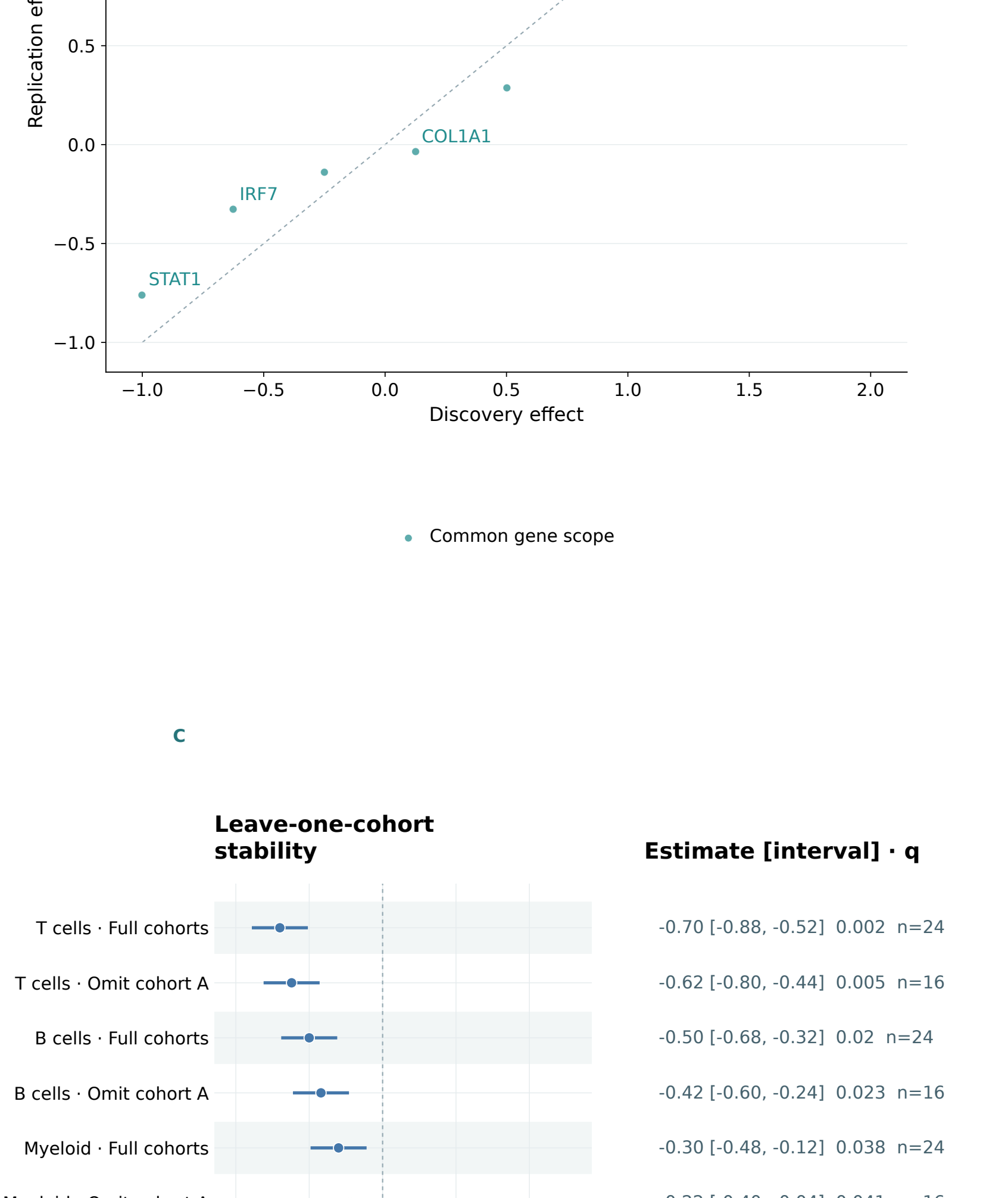
A



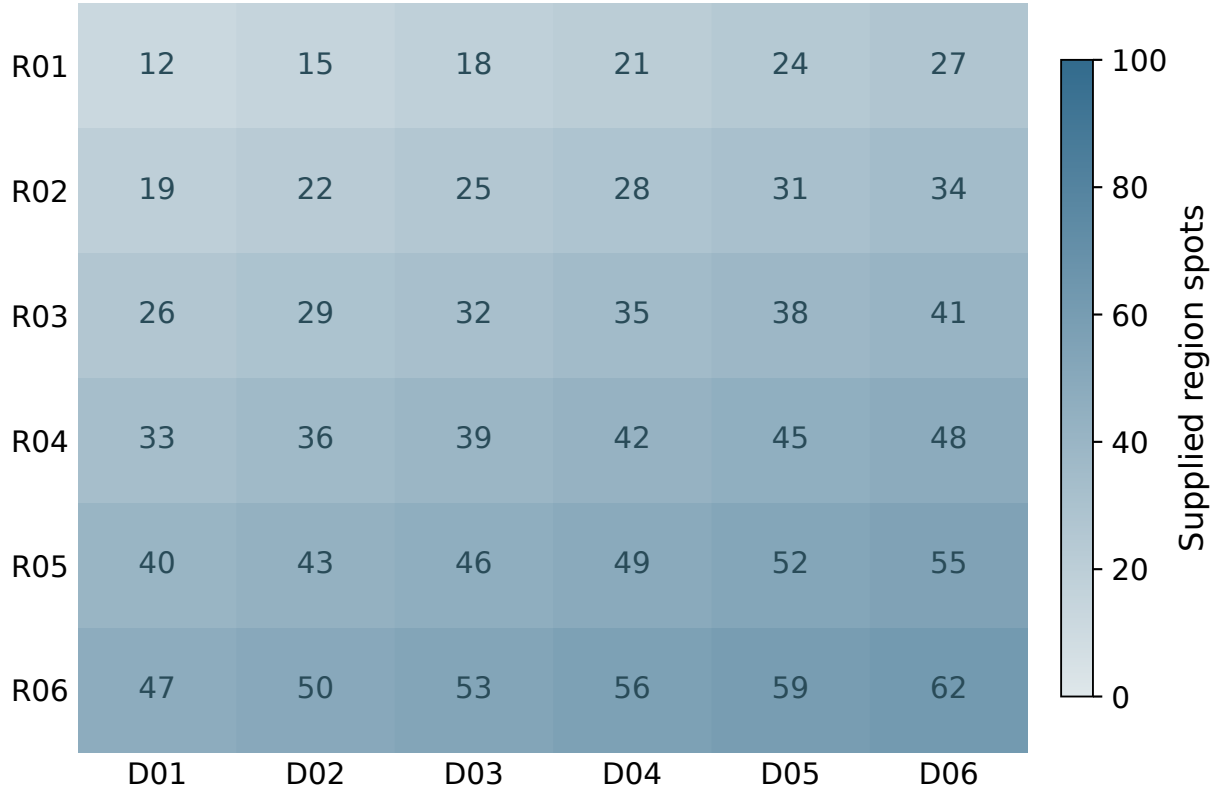
B

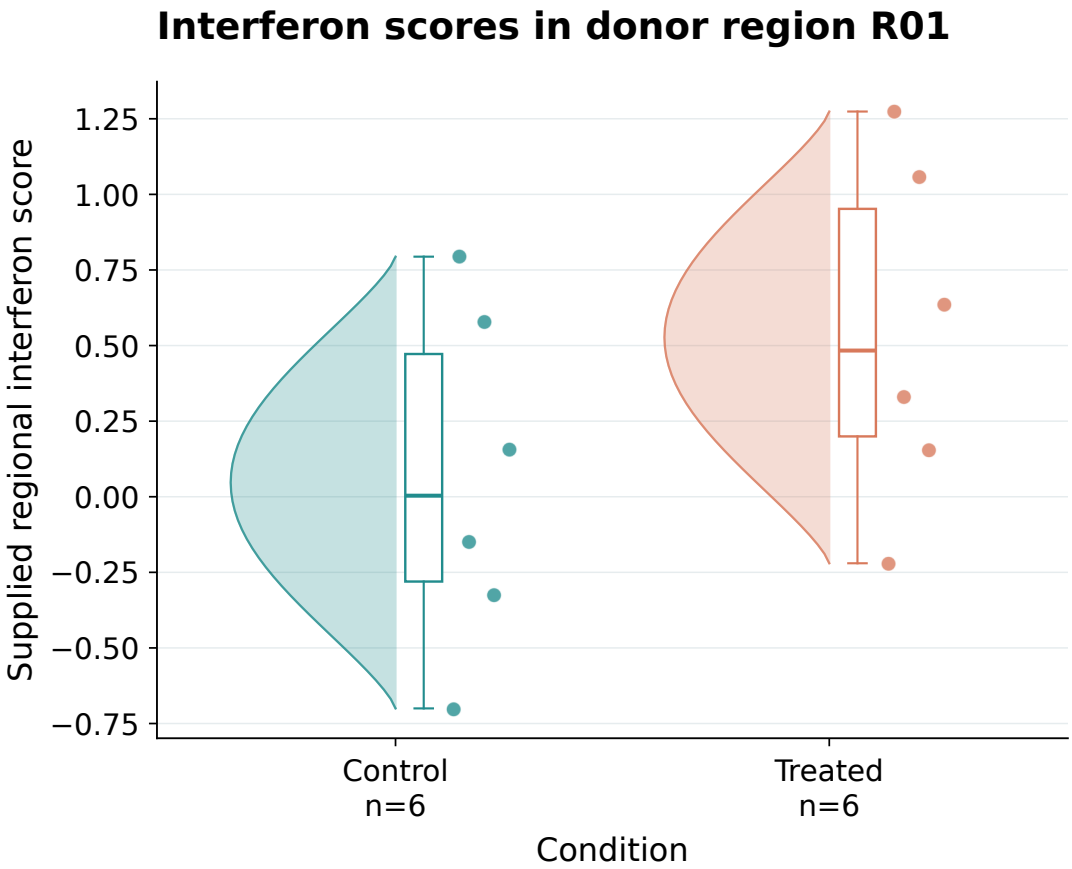


C



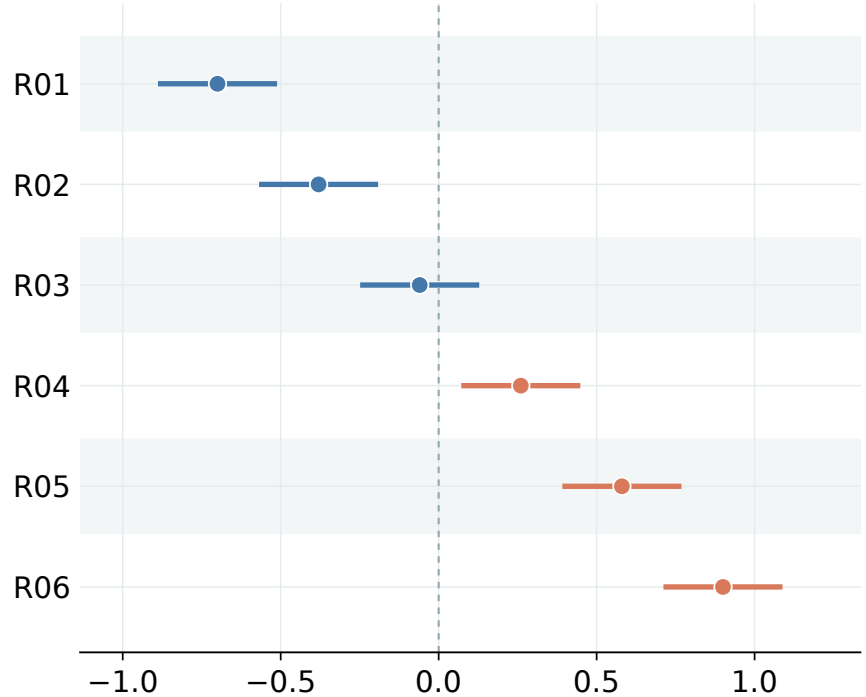
Section and donor support by region





Software test fixture; not biological evidence

Donor-aware region contrasts



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

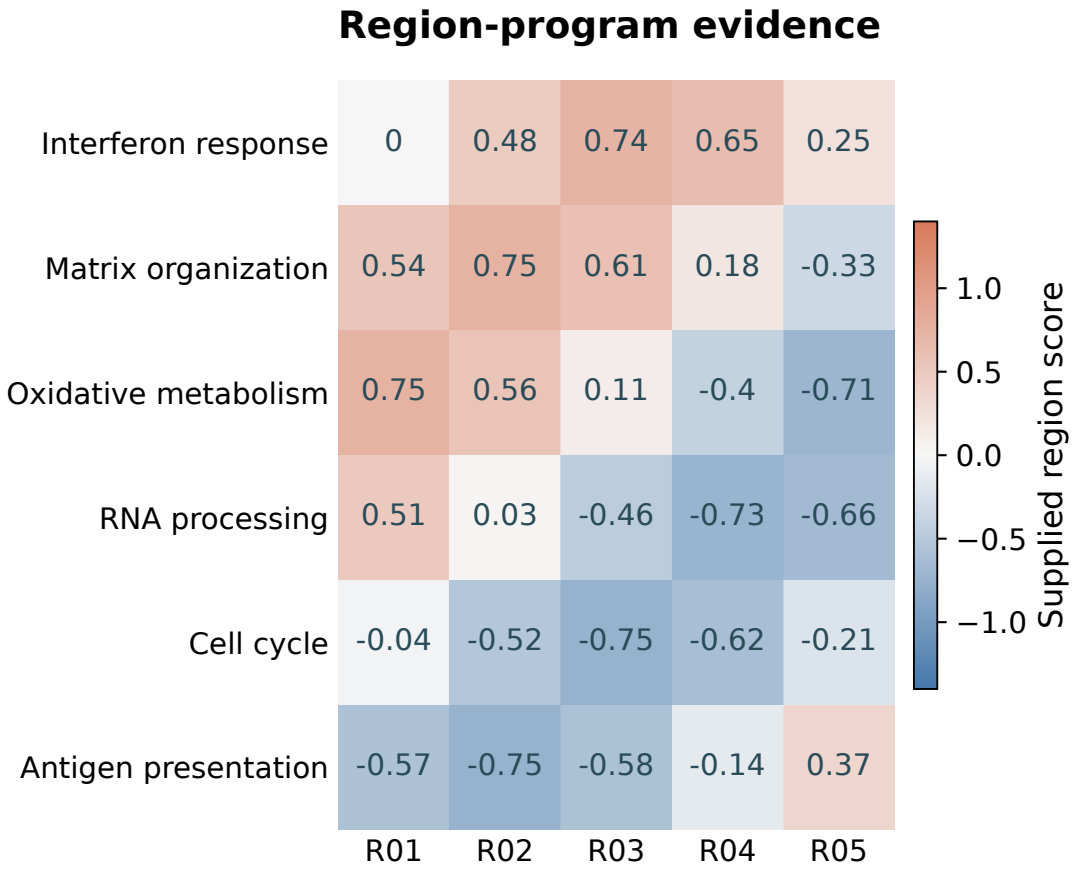
-0.38 [-0.56, -0.20] 0.02

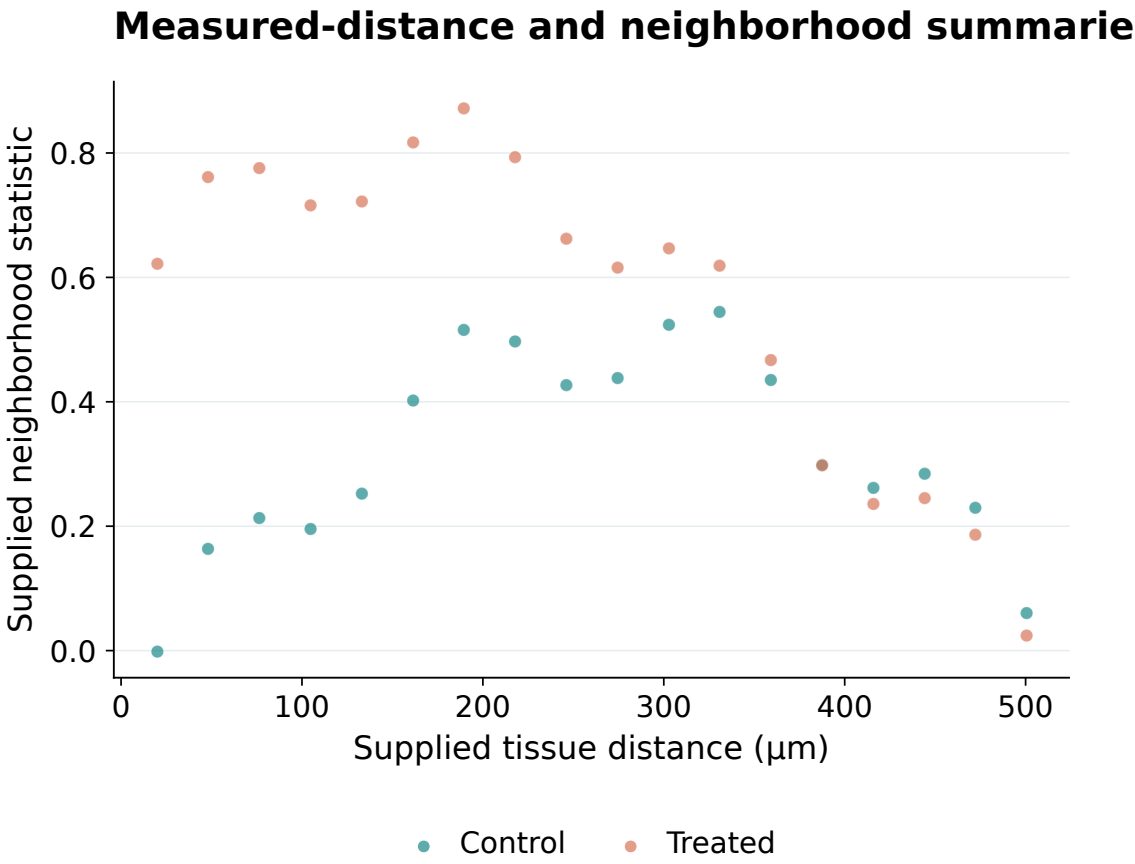
-0.06 [-0.24, 0.12] 0.65

0.26 [0.08, 0.44] 0.056

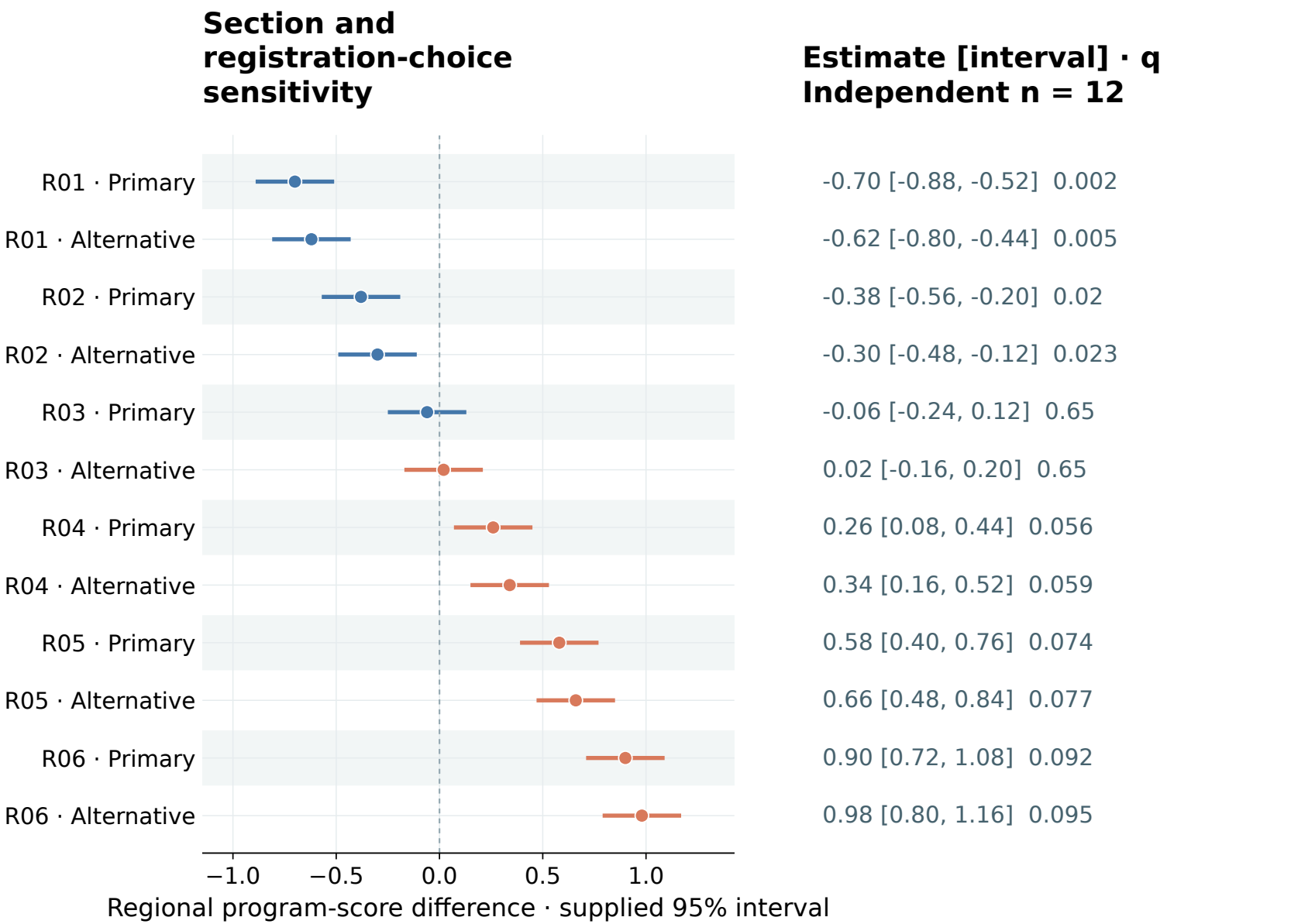
0.58 [0.40, 0.76] 0.074

0.90 [0.72, 1.08] 0.092


Region-program evidence
Interferon response
Matrix organization
Oxidative metabolism
RNA processing
Cell cycle
Antigen presentation
0
0.48
0.74
0.65
0.25
0.54
0.75
0.61
0.18
-0.33
0.75
0.56
0.11
-0.4
-0.71
0.51
0.03
-0.46
-0.73
-0.66
-0.04
-0.52
-0.75
-0.62
-0.21
-0.57
-0.75
-0.58
-0.14
0.37
R01
R02
R03
R04
R05
1.0
0.5
0.0
−0.5
−1.0
Supplied region score



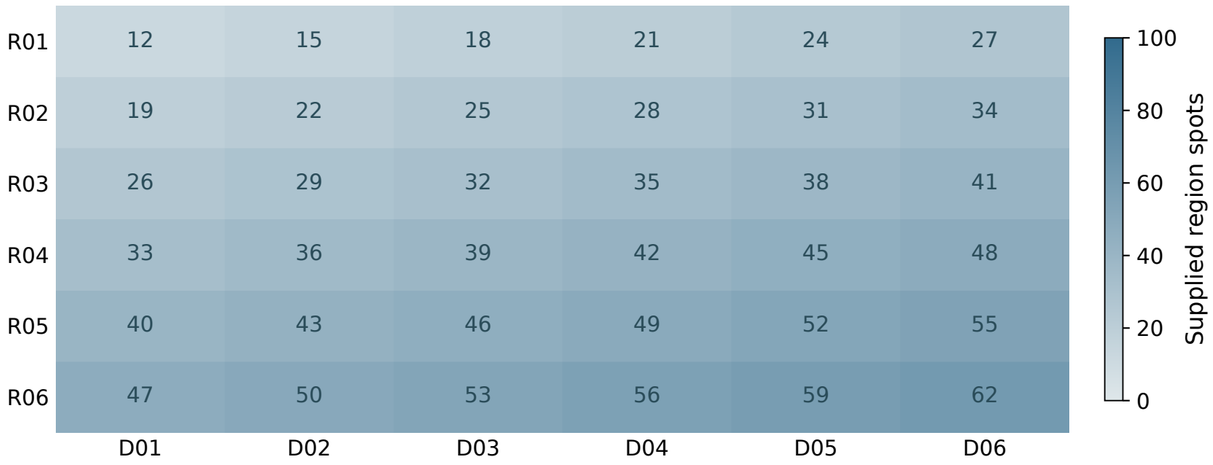
Software test fixture; not biological evidence


Section and registration-choice sensitivity
Estimate [interval] · q
Independent n = 12
R01 · Primary
R01 · Alternative
R02 · Primary
R02 · Alternative
R03 · Primary
R03 · Alternative
R04 · Primary
R04 · Alternative
R05 · Primary
R05 · Alternative
R06 · Primary
R06 · Alternative
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Regional program-score difference · supplied 95% interval

Single cell · spatial — complementary evidence

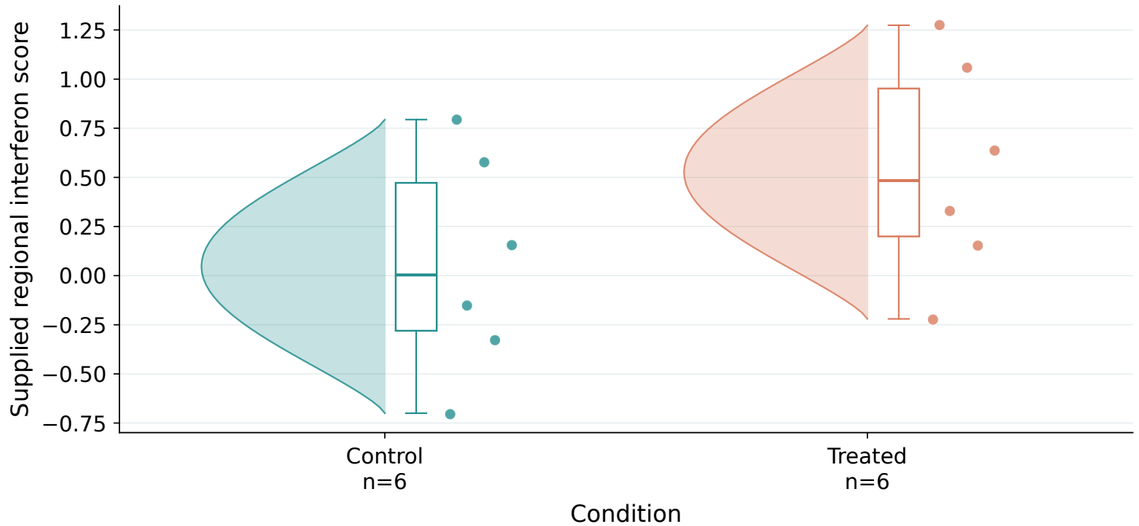
A

Section and donor support by region



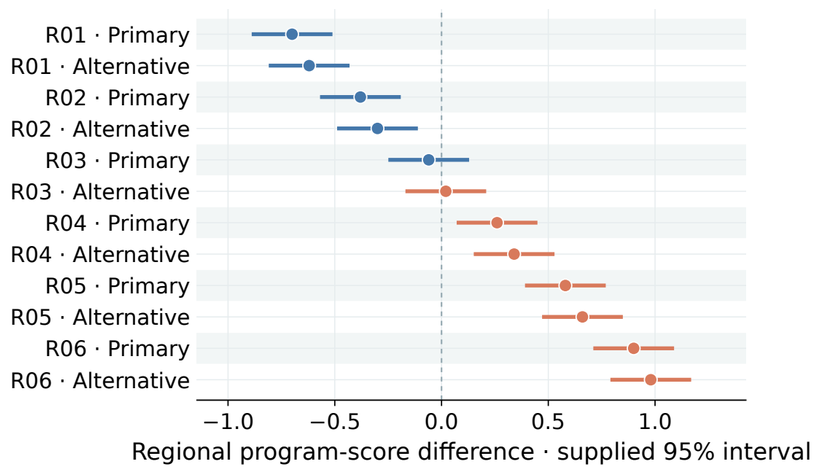
B

Interferon scores in donor region R01



C

Section and registration-choice sensitivity



Estimate [interval] · q
Independent n = 12

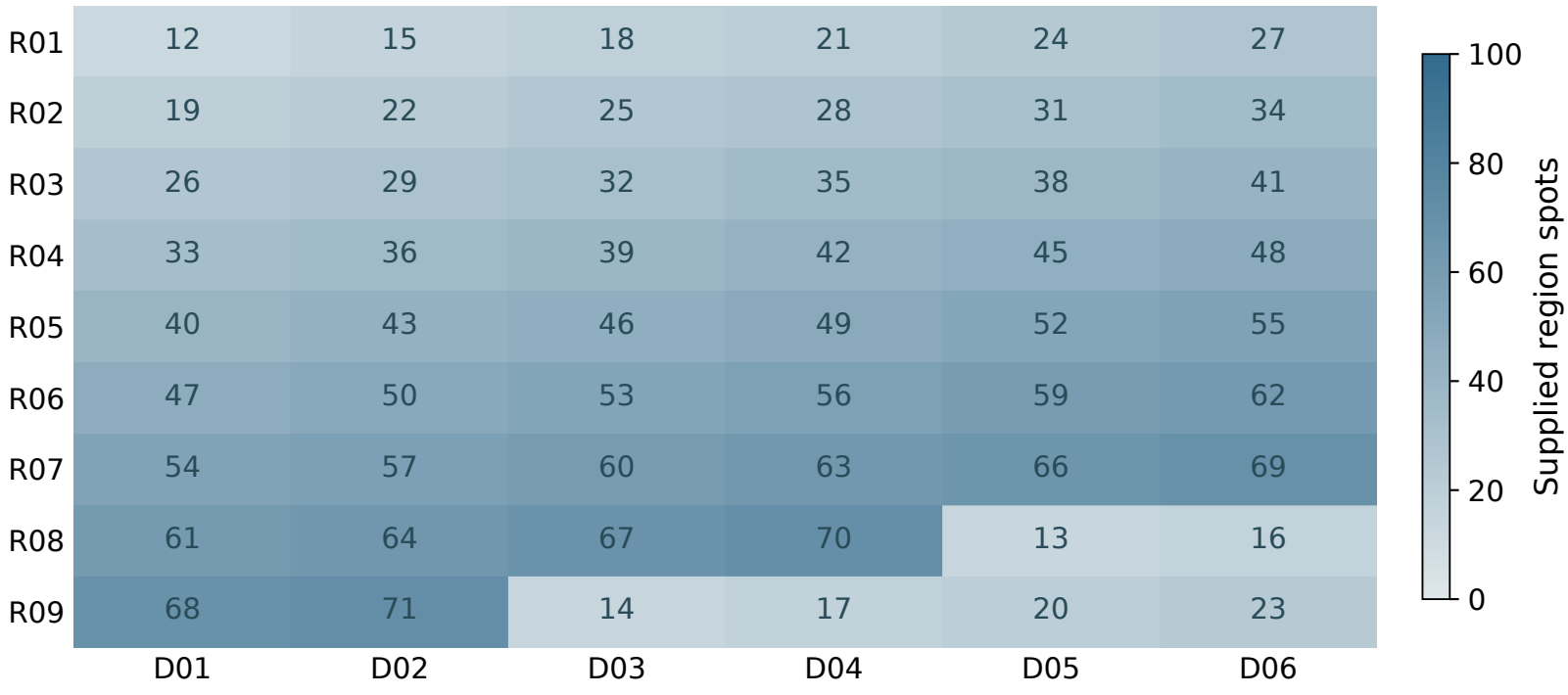
R01 · Primary	-0.70 [-0.88, -0.52] 0.002
R01 · Alternative	-0.62 [-0.80, -0.44] 0.005
R02 · Primary	-0.38 [-0.56, -0.20] 0.02
R02 · Alternative	-0.30 [-0.48, -0.12] 0.023
R03 · Primary	-0.06 [-0.24, 0.12] 0.65
R03 · Alternative	0.02 [-0.16, 0.20] 0.65
R04 · Primary	0.26 [0.08, 0.44] 0.056
R04 · Alternative	0.34 [0.16, 0.52] 0.059
R05 · Primary	0.58 [0.40, 0.76] 0.074
R05 · Alternative	0.66 [0.48, 0.84] 0.077
R06 · Primary	0.90 [0.72, 1.08] 0.092
R06 · Alternative	0.98 [0.80, 1.16] 0.095

Software test fixture; not biological evidence

Single cell · spatial — complementary evidence

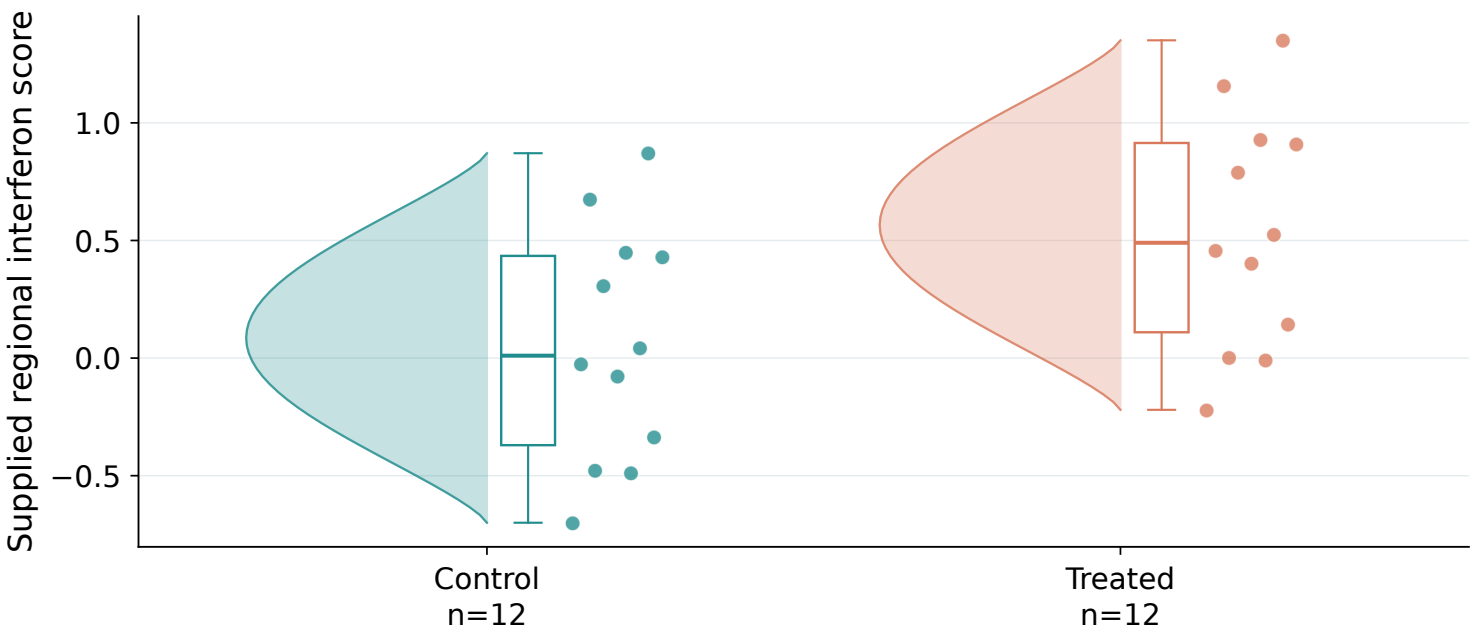
A

Section and donor support by region



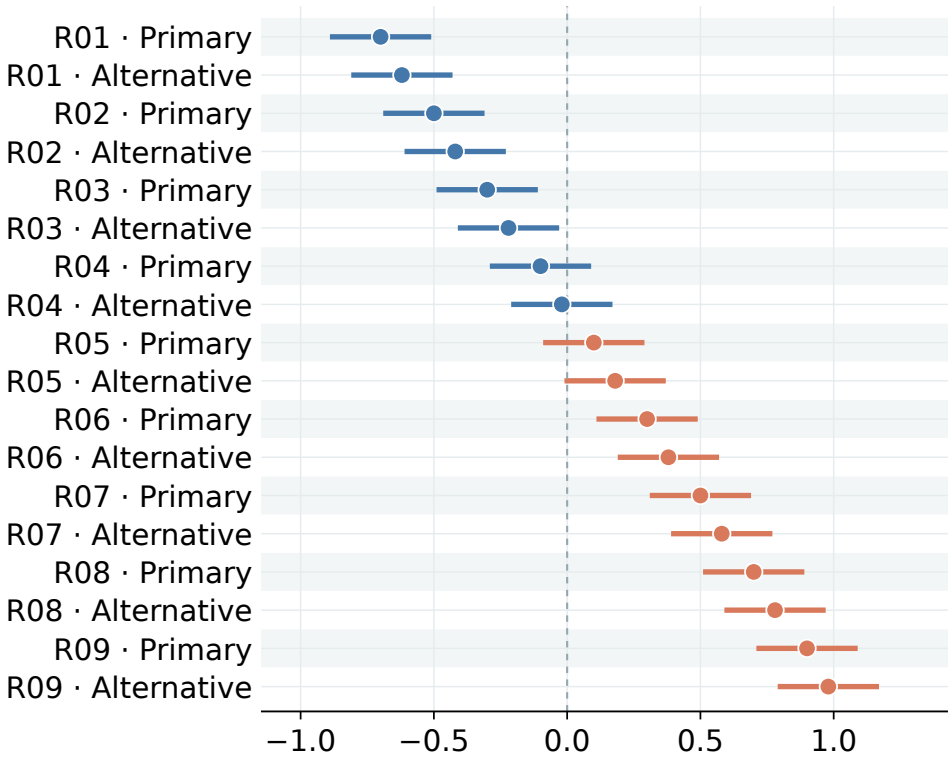
B

Interferon scores in donor region R01



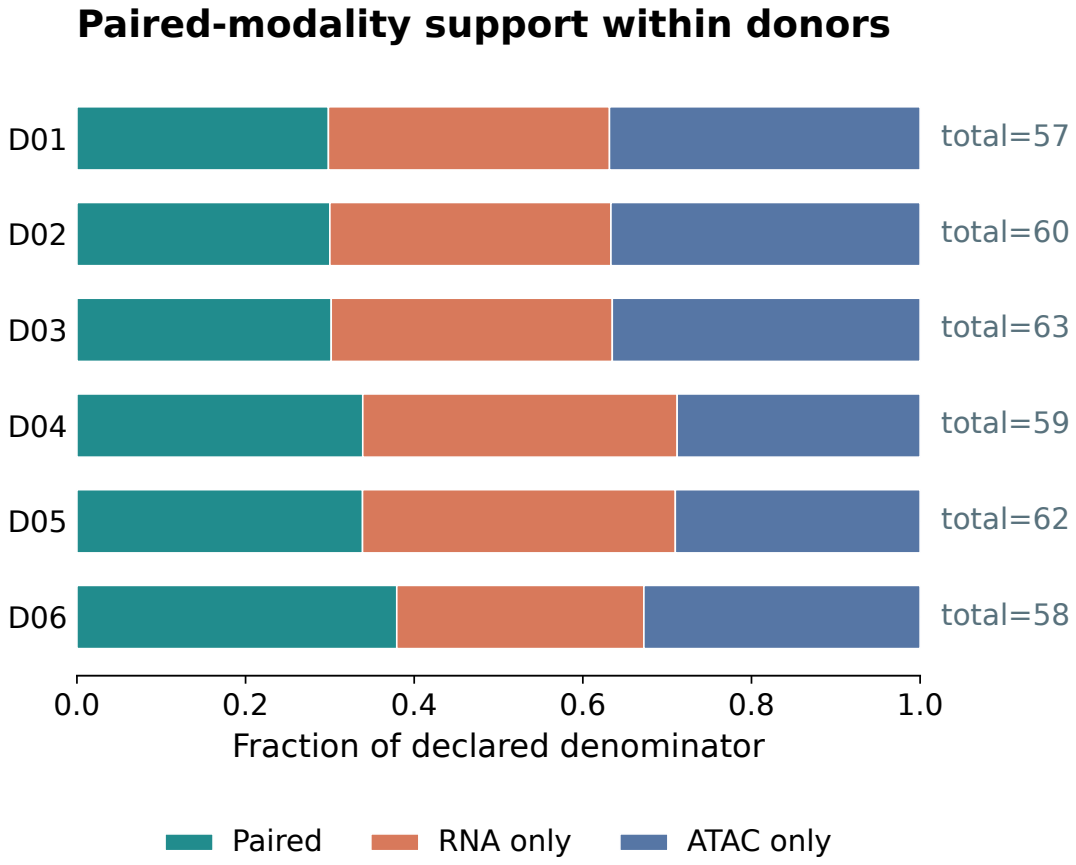
C

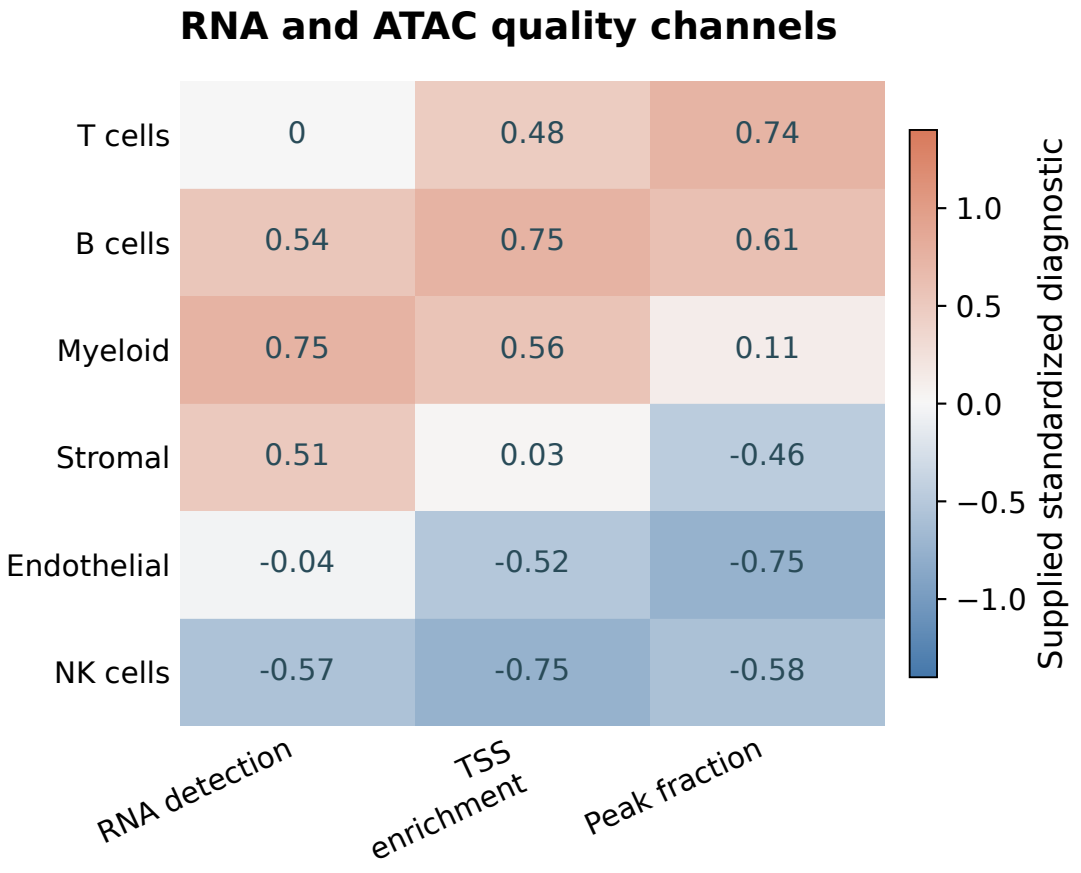
Section and registration-choice sensitivity

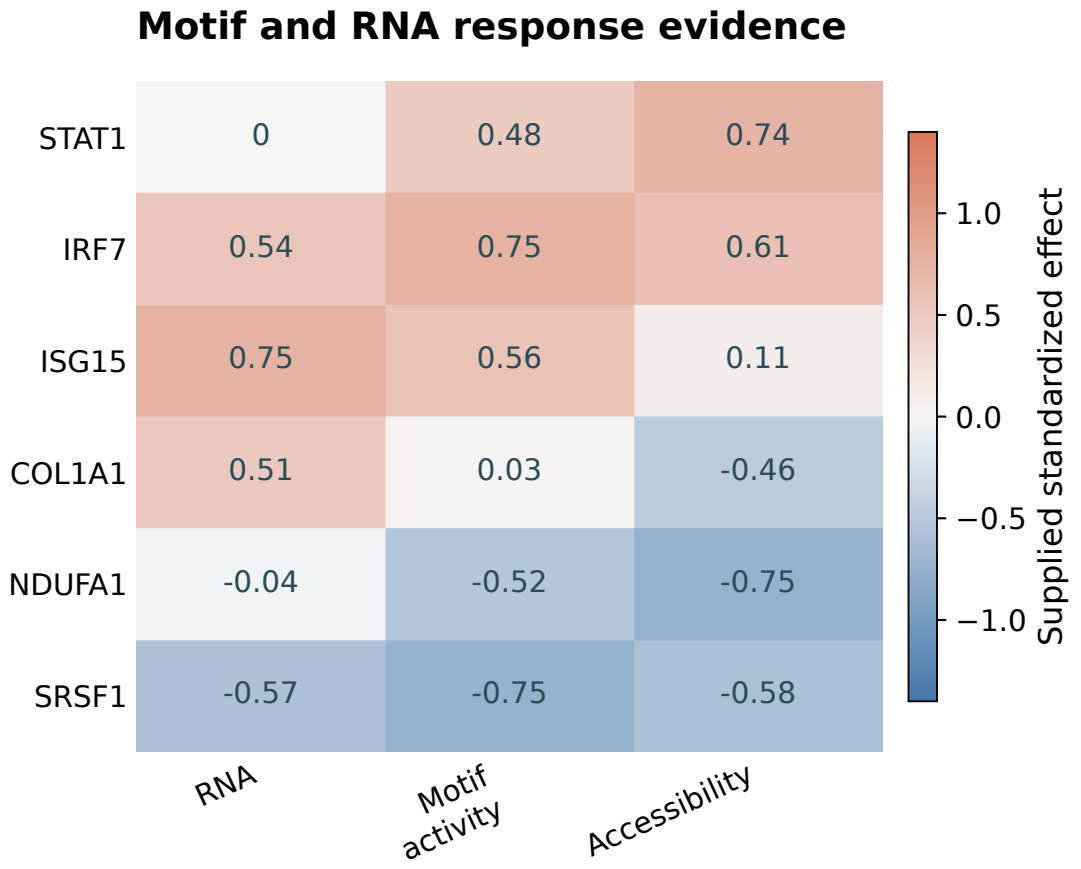


Estimate [interval] · q
Independent n = 24

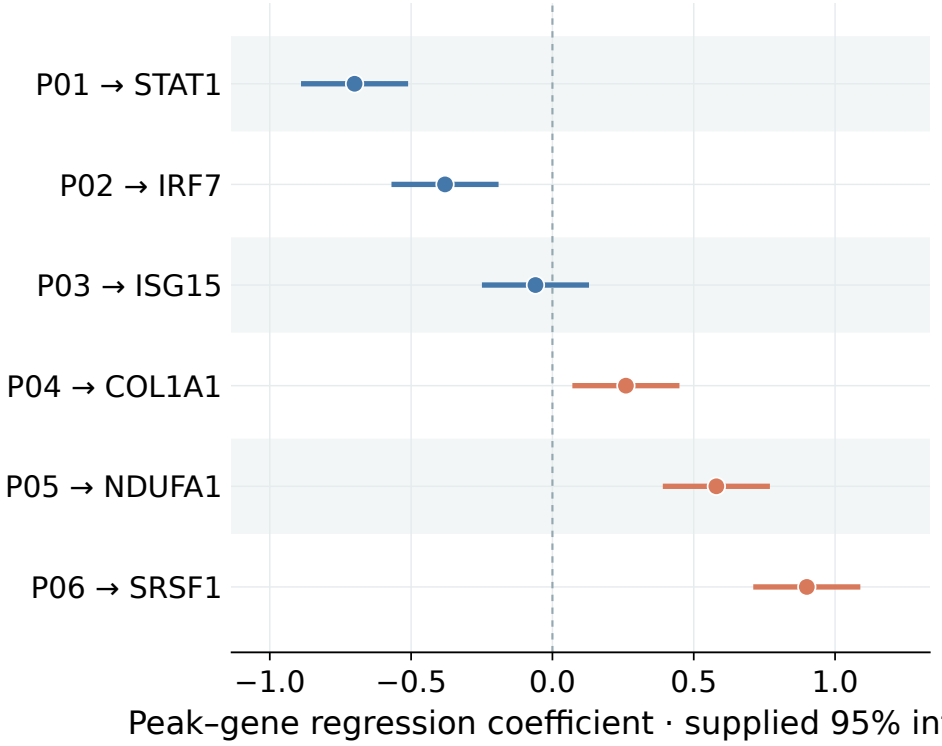
R01 · Primary	-0.70 [-0.88, -0.52] 0.002
R01 · Alternative	-0.62 [-0.80, -0.44] 0.005
R02 · Primary	-0.50 [-0.68, -0.32] 0.02
R02 · Alternative	-0.42 [-0.60, -0.24] 0.023
R03 · Primary	-0.30 [-0.48, -0.12] 0.038
R03 · Alternative	-0.22 [-0.40, -0.04] 0.041
R04 · Primary	-0.10 [-0.28, 0.08] 0.65
R04 · Alternative	-0.02 [-0.20, 0.16] 0.65
R05 · Primary	0.10 [-0.08, 0.28] 0.65
R05 · Alternative	0.18 [0.00, 0.36] 0.077
R06 · Primary	0.30 [0.12, 0.48] 0.092
R06 · Alternative	0.38 [0.20, 0.56] 0.095
R07 · Primary	0.50 [0.32, 0.68] 0.11
R07 · Alternative	0.58 [0.40, 0.76] 0.11
R08 · Primary	0.70 [0.52, 0.88] 0.13
R08 · Alternative	0.78 [0.60, 0.96] 0.13
R09 · Primary	0.90 [0.72, 1.08] 0.15
R09 · Alternative	0.98 [0.80, 1.16] 0.15


Paired-modality support within donors
D01
D02
D03
D04
D05
D06
total=57
total=60
total=63
total=59
total=62
total=58
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of declared denominator
Paired
RNA only
ATAC only


RNA and ATAC quality channels
T cells
B cells
Myeloid
Stromal
Endothelial
NK cells
0
0.48
0.74
0.54
0.75
0.61
0.75
0.56
0.11
0.51
0.03
-0.46
-0.04
-0.52
-0.75
-0.57
-0.75
-0.58
RNA detection
TSS enrichment
Peak fraction
1.0
0.5
0.0
−0.5
−1.0
Supplied standardized diagnostic


Motif and RNA response evidence
STAT1
IRF7
ISG15
COL1A1
NDUFA1
SRSF1
0
0.48
0.74
0.54
0.75
0.61
0.75
0.56
0.11
0.51
0.03
-0.46
-0.04
-0.52
-0.75
-0.57
-0.75
-0.58
RNA
Motif activity
Accessibility
1.0
0.5
0.0
−0.5
−1.0
Supplied standardized effect

Prespecified peak–gene association estimates



Estimate [interval] · q
Independent n = 12

-0.70 [-0.88, -0.52] 0.002

-0.38 [-0.56, -0.20] 0.02

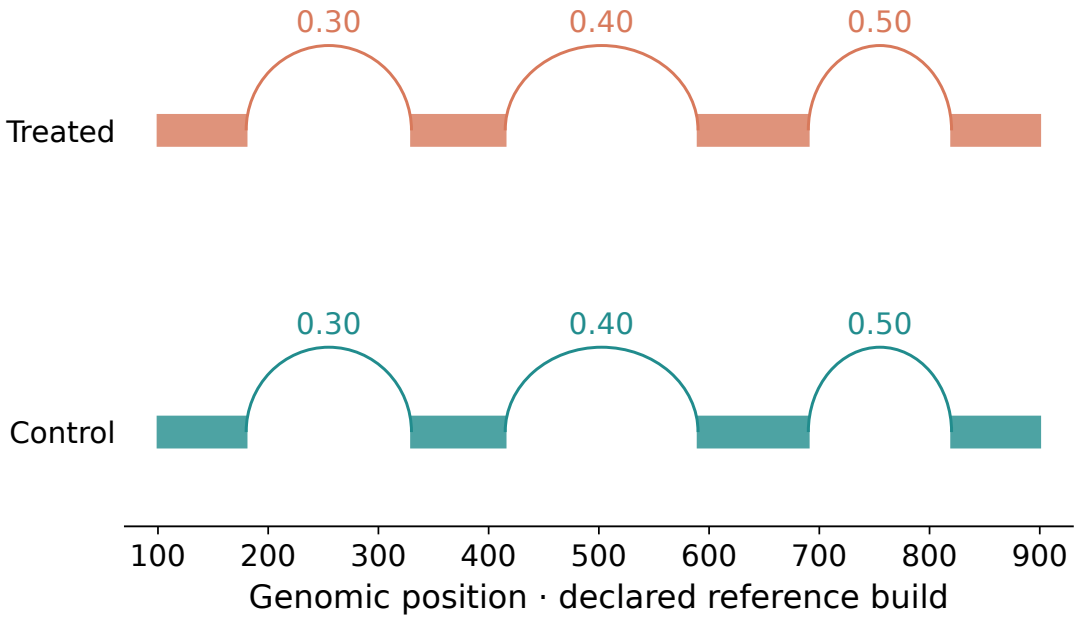
-0.06 [-0.24, 0.12] 0.65

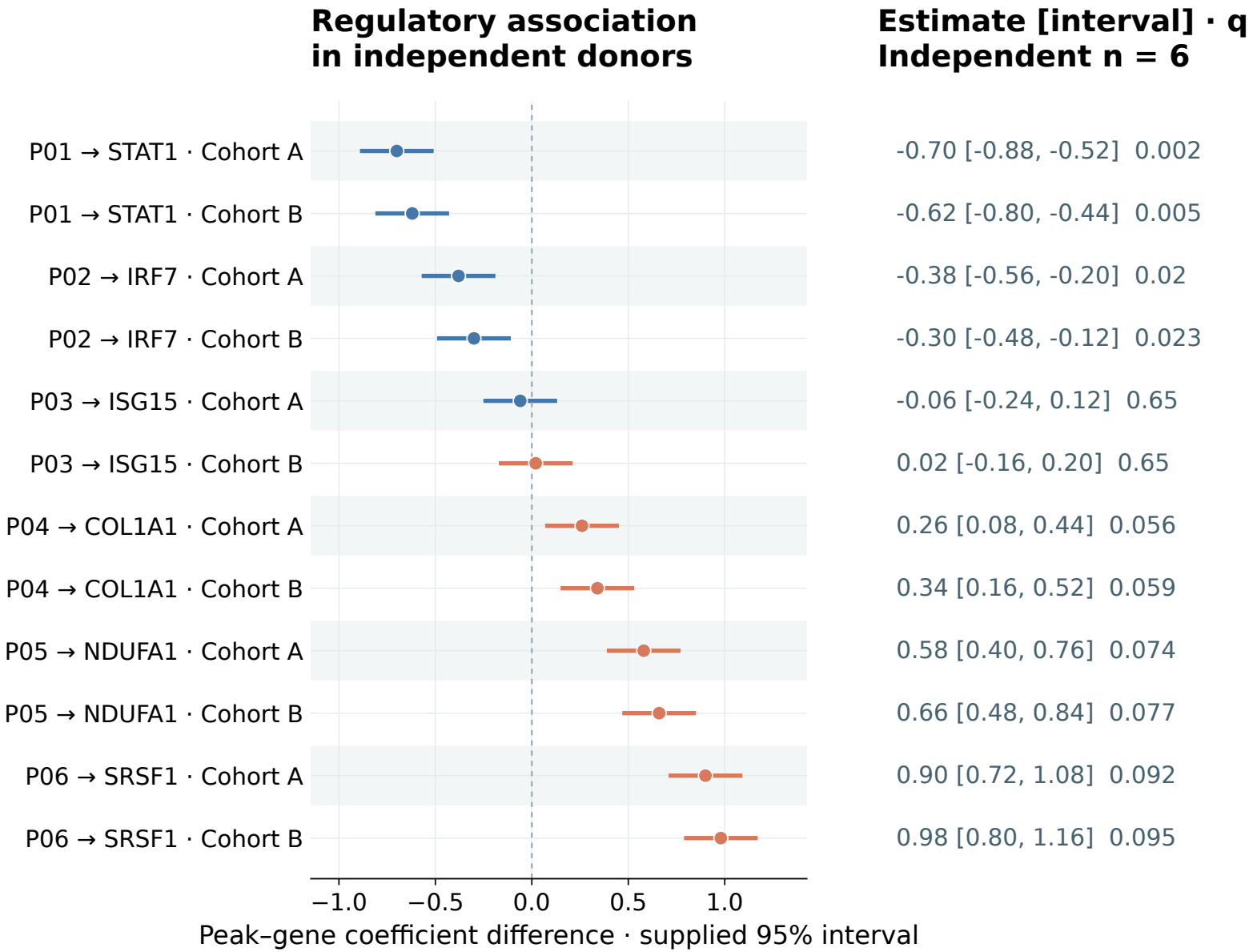
0.26 [0.08, 0.44] 0.056

0.58 [0.40, 0.76] 0.074

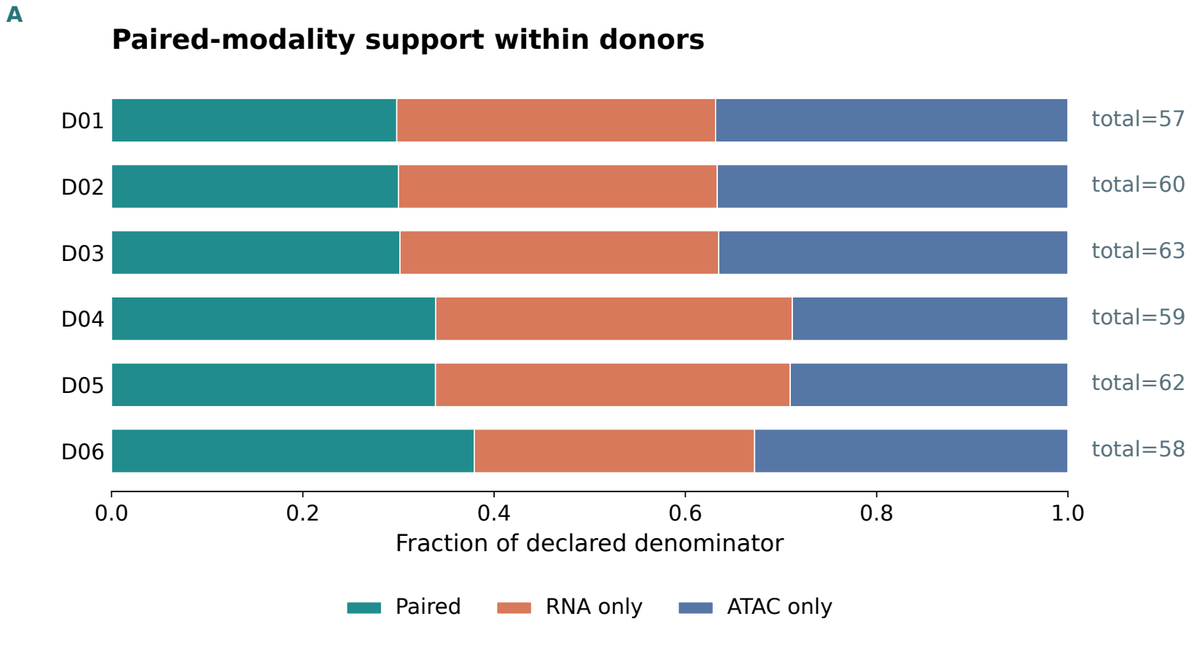
0.90 [0.72, 1.08] 0.092

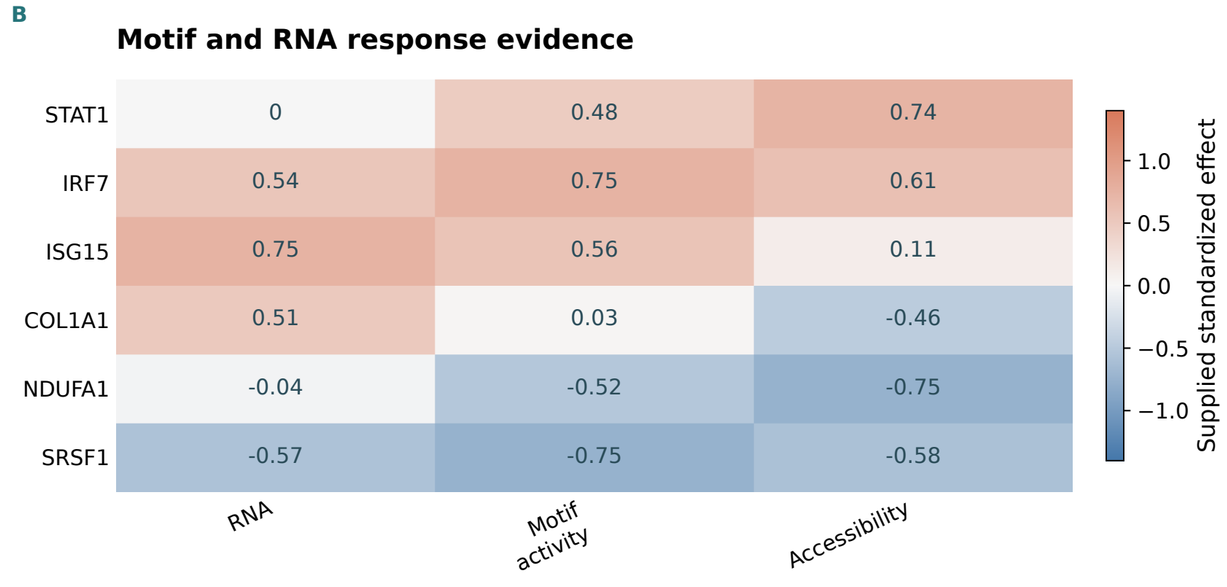
Frozen peak and gene-locus evidence

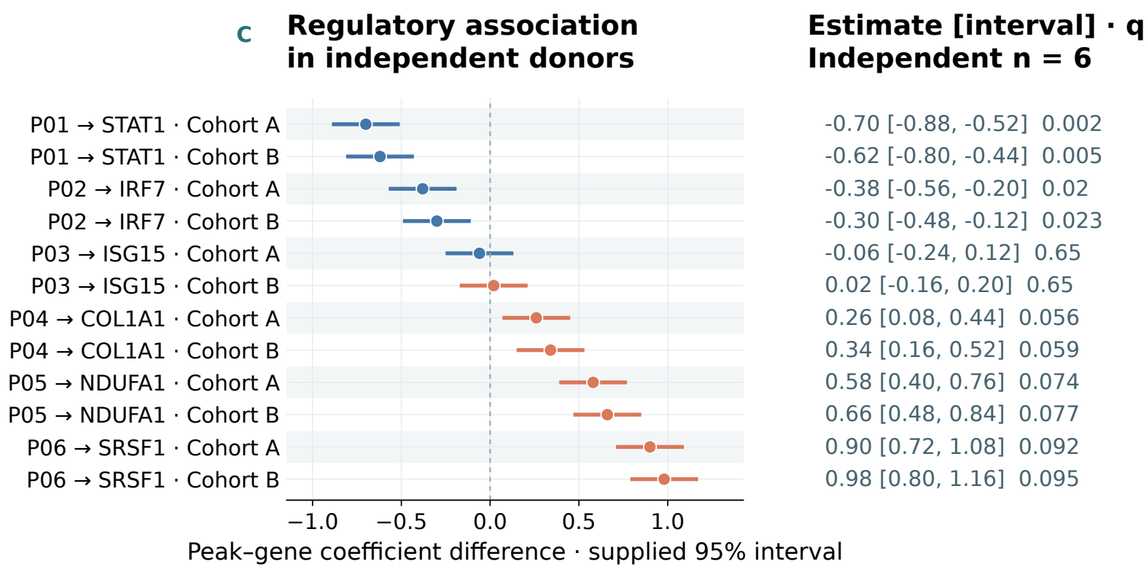



Regulatory association
in independent donors
Estimate [interval] · q
Independent n = 6
P01 → STAT1 · Cohort A
P01 → STAT1 · Cohort B
P02 → IRF7 · Cohort A
P02 → IRF7 · Cohort B
P03 → ISG15 · Cohort A
P03 → ISG15 · Cohort B
P04 → COL1A1 · Cohort A
P04 → COL1A1 · Cohort B
P05 → NDUFA1 · Cohort A
P05 → NDUFA1 · Cohort B
P06 → SRSF1 · Cohort A
P06 → SRSF1 · Cohort B
-0.70 [-0.88, -0.52] 0.002
-0.62 [-0.80, -0.44] 0.005
-0.38 [-0.56, -0.20] 0.02
-0.30 [-0.48, -0.12] 0.023
-0.06 [-0.24, 0.12] 0.65
0.02 [-0.16, 0.20] 0.65
0.26 [0.08, 0.44] 0.056
0.34 [0.16, 0.52] 0.059
0.58 [0.40, 0.76] 0.074
0.66 [0.48, 0.84] 0.077
0.90 [0.72, 1.08] 0.092
0.98 [0.80, 1.16] 0.095
−1.0
−0.5
0.0
0.5
1.0
Peak–gene coefficient difference · supplied 95% interval

Single cell · multiome — complementary evidence



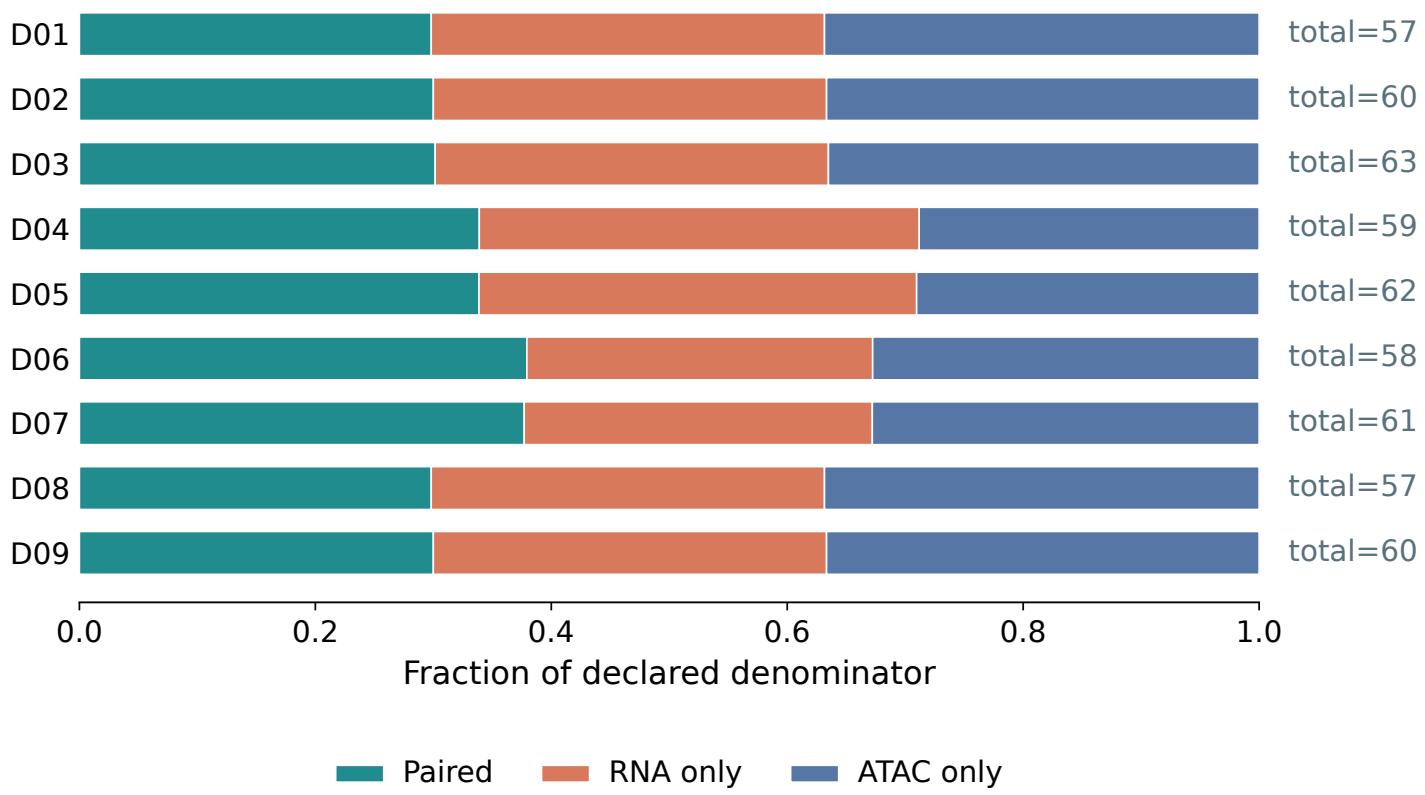




Single cell · multiome — complementary evidence

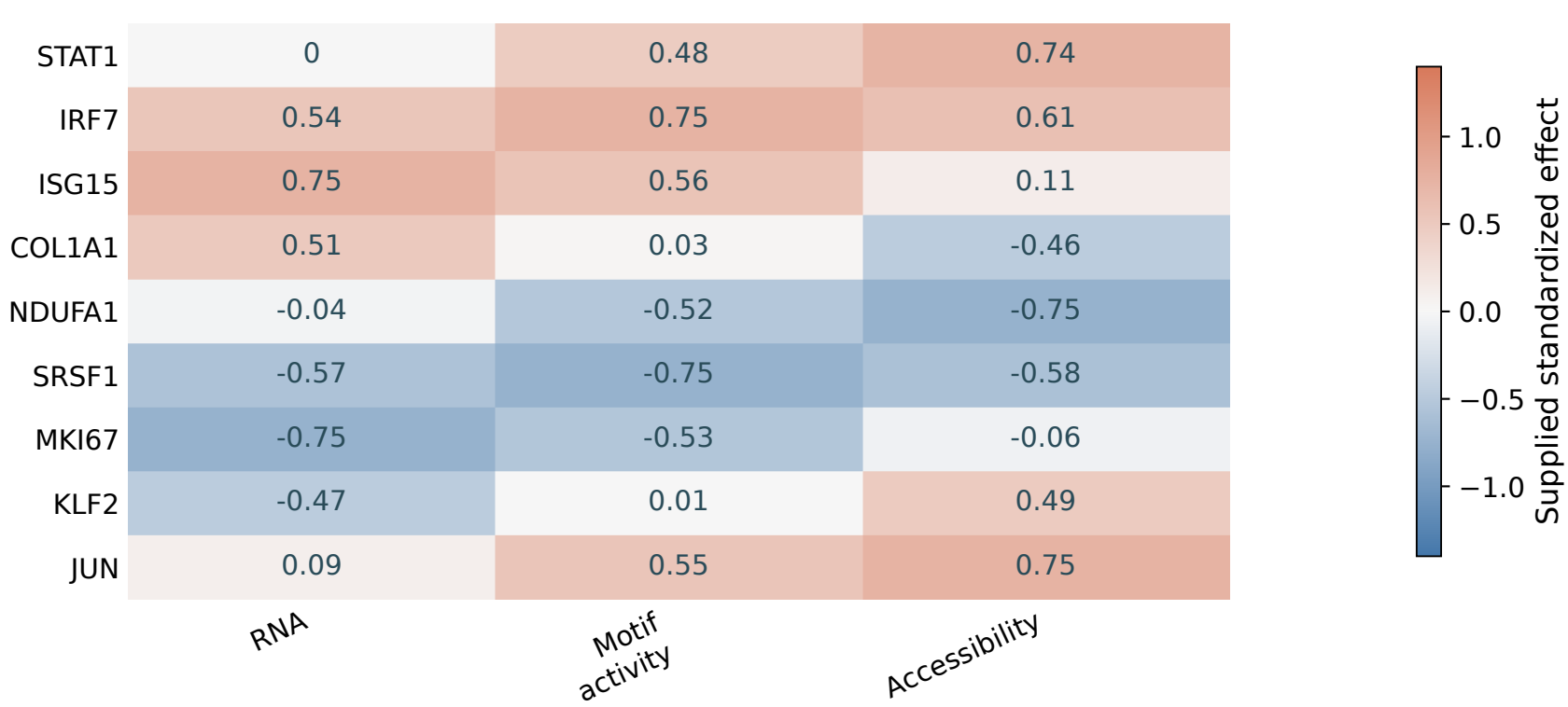
A

Paired-modality support within donors



B

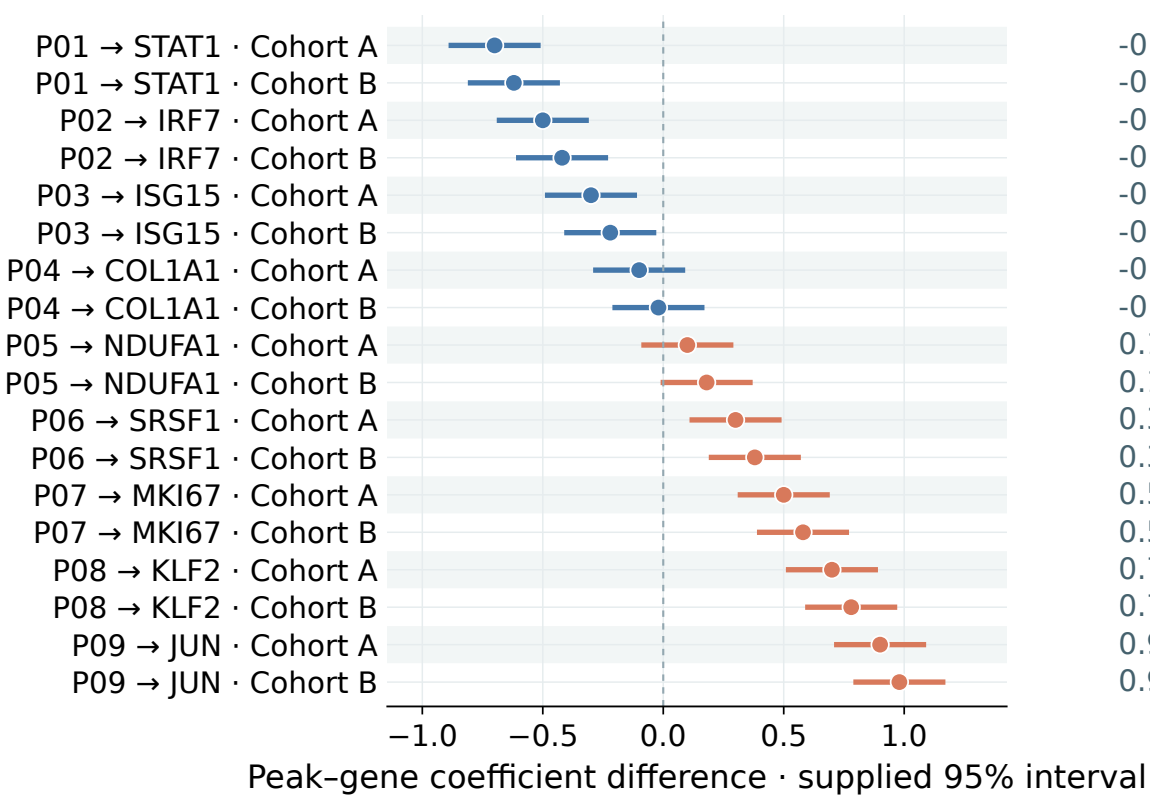
Motif and RNA response evidence



C

Regulatory association in independent donors

Estimate [interval] · q
Independent n = 12

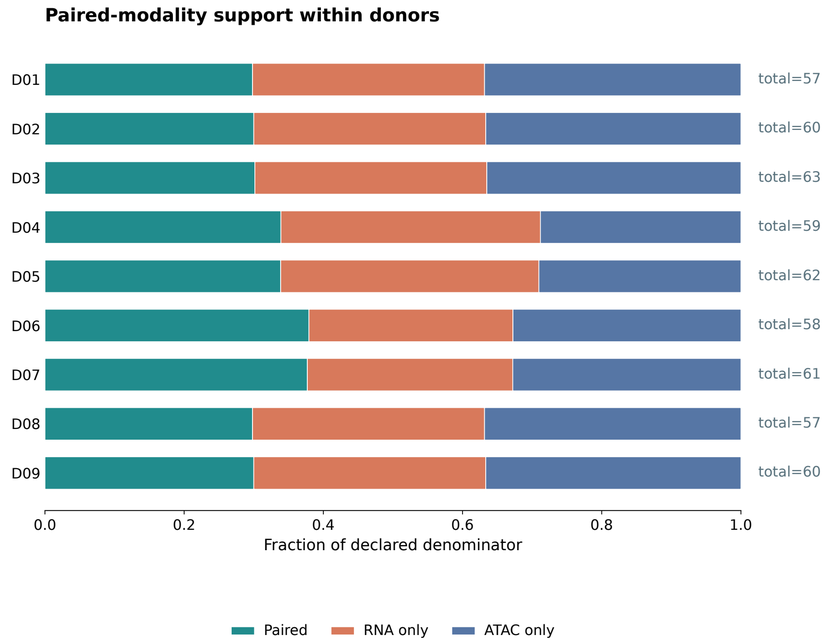


P01 → STAT1 · Cohort A	-0.70 [-0.88, -0.52] 0.002
P01 → STAT1 · Cohort B	-0.62 [-0.80, -0.44] 0.005
P02 → IRF7 · Cohort A	-0.50 [-0.68, -0.32] 0.02
P02 → IRF7 · Cohort B	-0.42 [-0.60, -0.24] 0.023
P03 → ISG15 · Cohort A	-0.30 [-0.48, -0.12] 0.038
P03 → ISG15 · Cohort B	-0.22 [-0.40, -0.04] 0.041
P04 → COL1A1 · Cohort A	-0.10 [-0.28, 0.08] 0.65
P04 → COL1A1 · Cohort B	-0.02 [-0.20, 0.16] 0.65
P05 → NDUFA1 · Cohort A	0.10 [-0.08, 0.28] 0.65
P05 → NDUFA1 · Cohort B	0.18 [0.00, 0.36] 0.077
P06 → SRSF1 · Cohort A	0.30 [0.12, 0.48] 0.092
P06 → SRSF1 · Cohort B	0.38 [0.20, 0.56] 0.095
P07 → MKI67 · Cohort A	0.50 [0.32, 0.68] 0.11
P07 → MKI67 · Cohort B	0.58 [0.40, 0.76] 0.11
P08 → KLF2 · Cohort A	0.70 [0.52, 0.88] 0.13
P08 → KLF2 · Cohort B	0.78 [0.60, 0.96] 0.13
P09 → JUN · Cohort A	0.90 [0.72, 1.08] 0.15
P09 → JUN · Cohort B	0.98 [0.80, 1.16] 0.15

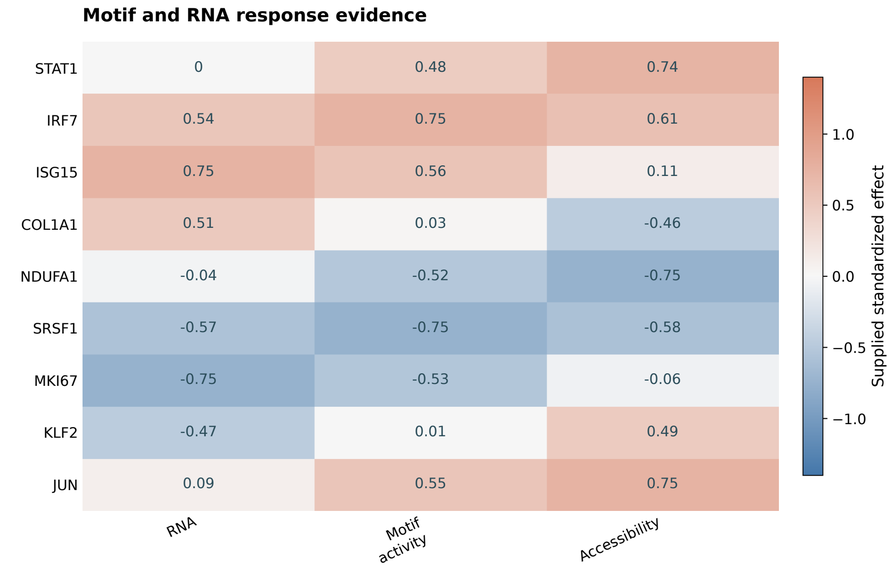
Software test fixture; not biological evidence

Single cell · multiome — complementary evidence

A



B



C

